Local Langlands for $\mathbf{GL}_2(\mathbf{F})$

***EYNTKA* Series**

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0.0.1 Inspiration Very heavily inspired by [2]

Introduction

The Langlands program sets out to understand the absolute Galois group $\mathrm{Gal}(\overline{F}/F)$ of a field. The pursuit has spawned a web of connections linking number theory (Galois representations), harmonic analysis (automorphic forms), and geometry (cohomology of algebraic varieties). At its heart lies a dictionary, the global *Langlands correspondence*, which translates arithmetic data into analytic data and back.

While the global theory is vast and often technically overwhelming, the *Local Langlands Correspondence* provides the essential vocabulary for this dictionary. It asserts that for a local field F (like $\mathbb{Q}_p$), the representation theory of the groups $\mathrm{GL}_n(F)$ for $n \in \mathbb{N}_{>0}$ is mirrored by the representation theory of the absolute Galois group of F.

The Goal The goal of this book is to provide a complete, self-contained, and accessible proof of the Local Langlands Correspondence for the case $n = 2$.

This book follows *The Local Langlands Conjecture for GL(2)* by Colin J. Bushnell and Guy Henniart, and is written in the *EYNTKA* (Everything You Need To Know About) style. The aim is to expose the narrative arc of the subject, to build the background needed for the n-dimensional case, and to prepare the reader for the connection between elliptic curves and modular forms.

The Roadmap

1. *The analytic side* (chapter 1 to chapter 4). The theory of *smooth representations* is established first. Unlike the representation theory of finite groups or of Lie groups, the vector spaces here are infinite dimensional while the groups are totally disconnected. The "easy" representations, the principal series, are classified by parabolic induction. The "hard" ones, the cuspidals, are then constructed explicitly by *compact induction* from open subgroups, and the tame ones are parameterized by *admissible pairs*.

2. *The invariants* (chapter 5). Matching a representation on one side against one on the other

requires quantities that exist in both worlds. These are the local L-function and the local constant, or ε-factor. The chapter closes with the *converse theorem*: if the L- and ε-factors of two representations agree after enough twisting, the representations are isomorphic.

3. *The arithmetic side* (chapter 6). The Galois group is refined to the *Weil group* $\mathcal{W}_F$ and enriched to *Deligne representations*. The enrichment is what accounts for the special (Steinberg) representations on the analytic side.

4. *The correspondence* (chapter 7). The two sides are brought together. The principal series match reducible Weil representations. For the cuspidals, admissible pairs and a rectifying character produce a bijection preserving the arithmetic invariants.

5. *An appendix* (appendix A) proves that irreducible cuspidal representations are projective in the category of smooth representations with a fixed central character, which is the precise form of the "atomicity" the main text appeals to.

Prerequisites

The reader is assumed to be comfortable with:

- the basic theory of local fields (valuations, rings of integers, p-adic numbers), as covered in [9],
- Galois theory, as covered in [4],
- the representation theory of finite groups (characters, induction, Schur's lemma), as covered in [5], and the basics of the representation theory of algebraic groups as covered in [12].
- Topological groups, as covered in [11].
- Basics of Functional Analysis and Fourier Analysis as covered in [15] and [17] respectively.

Familiarity with profinite groups is useful but not assumed. Section 1.1 develops what is needed.

1

Introduction to Smooth Representations

The (classical) Langlands program is, at its core, a bridge between number theory and harmonic analysis. On the number theoretic side, we have Galois groups $\mathrm{Gal}(\overline{F}/F)$, which encode the arithmetic structure of fields. On the analytic side, we have groups of matrices like $\mathrm{GL}_n(F)$. To build this bridge, we must first understand the ground on which the analytic side stands. The goal of this chapter is to establish the foundational language for the "analytic" side of the correspondence: the theory of *smooth representations* of locally profinite groups. Unlike the representation theory of finite groups or Lie groups, the topology of p-adic fields requires a unique approach. We cannot simply look at continuous representations on Hilbert spaces. Instead we must look at representations that respect the totally disconnected topology of $\mathbb{Q}_p$.

We begin in section 1.1 by defining locally profinite groups and exploring the topology of F and $\mathrm{GL}_n(F)$. In section 1.2, we define smooth representations (actions that are locally constant) and show that they satisfy an analogue of Schur's Lemma. In section 1.3, we develop the necessary integration theory (Haar measure) to analyze these groups. Finally, in section 1.4, we introduce the Hecke algebra $\mathcal{H}(G)$, an associative algebra that allows us to translate problems of smooth group representations into the language of modules over a ring, mirroring the relationship between finite groups and the group ring $\mathbb{C}[G]$.

1.1 Locally Profinite Groups

All finite groups will by assumption have the discrete topology. For a deeper review of topological groups, see [11, Topological Group].

The field $\mathbb{Q}_p$ is a *locally profinite groups* with its topology endowed by the metric $|-|_p$. This is a

substantially different topology to $\mathbb{C}$, and hence a moment must be taken to understand better its structure.

Definition 1.1.1: P-adic Topology

Let $\mathbb{Q}_p$ be a p-adic field for a (finite) prime p and let $\mathbb{Z}_p$ be the corresponding ring of integers. Define the neighbourhood basis:

$$\mathbb{Z}_p \supseteq p\mathbb{Z}_p \supseteq p^2\mathbb{Z}_p \supseteq \cdots$$

Giving $\mathbb{Z}_p$ a p-adic topology which can be easily extended to $\mathbb{Q}_p$. As each $p^k\mathbb{Z}_p$ is a closed ball, this matches the metric topology induced by $|-|_p$:

$$B_r(a) = \{x \in \mathbb{Q}_p : |x-a|_p < r\}$$

which extends to $\mathbb{Q}_p$ by defining:

$$\left|\frac{m}{n}\right|_p = \frac{|m|_p}{|n|_p}$$

Lemma 1.1.2: P-adic Field Totally Disconnected

The field $\mathbb{Q}_p$ with the topology induced by the metric $|-|_p$ is totally disconnected. More generally any space X whose topology is induced by an ultra-metric is totally disconnected.

***Proof*:**
Let $a, b \in \mathbb{Q}_p$. As $a \neq b$, there must exist a d such that:

$$0 < |a-b|_p = p^{-k} = d$$

Let:

$$B_d(a) = \{x \in \mathbb{Q}_p : |x-a| < d\}$$

By construction, $a \in B_d(a)$ and $b \notin B_d(a)$.

Let us show that

$$B_d(a)^c = \{x \in \mathbb{Q}_p : |x-a| \geq d\}$$

is also open. Take $x \in B_d(a)^c$ so that $D = |a-x|_p \geq d$ and consider $B_d(x)$ and take $y \in B_d(x)$ so that $\delta = |x-y|_p \leq d$. We want to show that $y \in B_d(a)^c$ so $|y-a|_p \geq d$. Then by the isosceles property of the ultra-metric, for any three points a, x, y two of $d(a,x)$, $d(a,y)$ or $d(x,y)$ must be to the larger one. As $|a-x|_p \geq d$, $|y-a| \geq d$, and so $y \in B_d(a)^c$, showing that $B_d(x) \subseteq B_d(a)^c$ showing that it is *also* open.

Thus, $\mathbb{Q}_p$ is totally disconnected, as we sought to show.

Lemma 1.1.3: P-adic Integers Open Subgroup

The subgroup $\mathbb{Z}_p \subseteq \mathbb{Q}_p$ is an open subgroup with the topology induced by the valuation. Hence, it is also closed

This differs strongly from $\mathbb{C}$: if $H \subseteq \mathbb{C}$ is an open subgroup then as noted earlier it must also be closed, hence connected, and hence $H = \mathbb{C}$. As is $\mathbb{Q}_p$ is totally disconnected, it is possible to have clopen subsets that are not the entire set.

***Proof*:**
First, $\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \leq 1\}$. Take $a \in \mathbb{Z}_p$ (so that $|a| \leq 1$) and consider:

$$B_1(a) = \{x \in \mathbb{Q}_p : |x - a|_p < 1\}$$

We claim $x \in \mathbb{Z}_p$. Indeed:

$$|x|_p = |x + (-a + a)|_p \leq \max(|x - a|_p, |a|_p) = \max(t, 1) = 1$$

where $t < 1$ as $|x - a|_p < 1$. But then $x \in \mathbb{Z}_p$, and hence $B_1(a) \subseteq \mathbb{Z}_p$.
Thus, as $\mathbb{Q}_p$ is a topological group, $\mathbb{Z}_p$ is also closed.

Other groups built from p-adic fields occur throughout, for example finite extension of $\mathbb{Q}_p$, the ring of integers of $L_\mathfrak{p}/\mathbb{Q}_p$, $\overline{\mathbb{Q}}_p$, $\mathbb{C}_p$, $M_n(\mathbb{Q}_p)$, $M_n(\mathcal{O}_{L/\mathbb{Q}_p})$, $M_n(\mathfrak{m}_\mathfrak{p})$, various matrix subgroups of this groups (GL_n, SL_n, PGL_n, Sp_n, O_n, SU_n, etc.) and so forth. All of them carry a topology of the same shape, so the theory is developed once for topological groups in general. Recall the following properties:

- The topology of G is translation invariant: If $U \subseteq G$ is open, then Ux and xU is open for all $x \in G$
- If $H \leqslant G$, G/H has the quotient topology, and if $H \trianglelefteq G$ then G/H is a topological group.
- If K_1, K_2 are compact subset of G, so is K_1K_2
- subgroups in relation to open/closed:
  - if $H \leqslant G$ is open, it must be closed ($\bigcup_{g \in G \setminus H} gH$ is the union of open sets and the complement of H). More strictly, if H is open if and only if it contains an open neighbourhood around the identity (if $e \in U \subseteq H$ with open U, then $hU \subseteq H$ and $H = \bigcup_{h \in H} hU$). If G is connected, it has no non-trivial proper open subgroups. Totally disconnected groups are the ones that matter here.
  - If $H \leqslant G$ is an open subgroup, then the quotient space G/H and $H\backslash G$ each have the discrete topology.
  - A closed subgroup $H \leqslant G$ need not be open (ex. $\mu_n \subseteq \mathbb{C}$). However if $H \leqslant G$ is closed and finite index, it is open (the finite union of closed is closed). If G is first-countable, H is closed if it contains all its limit points (more generally, it it contains all the limits of nets).
  - A subgroup need not be open or closed (ex. $\mathbb{Q} \subseteq \mathbb{R}$).
  - if H is a subgroup of G, so it $\overline{H}$ (the closure under limits/nets).
  - If $G_0 \subseteq G$ is the connected component of the identity, then G_0 is a closed normal subgroup and G/G_0 is the largest totally disconnected quotient. When G is compact, G_0 is the intersection of all *open normal* subgroups of G. The word *open* cannot be dropped: $G = S^1$ is compact with $G_0 = S^1$, while the intersection of all its normal subgroups is trivial.

- symmetric neighbourhood:
  - For every open neighborhood of e, say for example U, there is a symmetric neighborhood (if $A \subseteq G$ with $A = A^{-1}$ is called *symmetric*) V of e such that $V \subseteq U$
  - For every neighborhood U of e, there is a neighborhood V of e with $VV \subseteq U$
- Density: the groups considered here often have dense subsets or subgroups. Sometimes the sequence definition of density will not be equivalent to the definition of density, hence a reminder for those who have internalized the sequence definition to the point that they forgot the general topological definition: A set $S \subseteq X$ is *dense* if it intersects every nonempty open subset of X: $U \cap S \neq \emptyset$ for all open $U \subseteq X$.

Lemma 1.1.4: Characterizing Hausdorffness

Let G be a topological group, $e \in G$ the identity, and $H = \bigcap_{\substack{U \subseteq G \text{ open} \\ e \in U}} U$ Then:

1. H is a subgroup
2. H is the closure of e, i.e. $\overline{e} = H$
3. G/H is Hausdorff
4. G is Hausdorff if and only if $H = \{e\}$

***Proof*:**

1. If $a, b \in H$ and U is an open neighbourhood of e, continuity of $(x, y) \mapsto xy^{-1}$ at (e, e) gives an open neighbourhood V of e with $VV^{-1} \subseteq U$. Since $a, b \in V$ we get $ab^{-1} \in U$. As U was arbitrary, $ab^{-1} \in H$.
2. An element x lies in $\overline{e}$ if and only if every open neighbourhood of x contains e, which by translation invariance holds if and only if every open neighbourhood of e contains x. That is the defining condition for H.
3. By (2), H is closed, so G/H is a topological group in which the class of the identity is closed. A topological group whose identity is closed is Hausdorff: the diagonal is the preimage of the closed set $\{\overline{e}\}$ under the continuous map $(x, y) \mapsto xy^{-1}$.
4. If G is Hausdorff then points are closed, so $\overline{e} = \{e\}$ and $H = \{e\}$ by (2). Conversely if $H = \{e\}$ then $G/H = G$ is Hausdorff by (3), as we sought to show.

Hausdorffness is assumed throughout unless specified otherwise.

Definition 1.1.5: Profinite Group

Take a cofiltered diagram of finite groups, each carrying the discrete topology. A *profinite group* is a topological group isomorphic to the limit of such a system, with the topology induced from the product. Given any topological group G, its *profinite completion* is

$$\widehat{G} = \varprojlim_N G/N$$

where N ranges over the open normal subgroups of finite index, ordered by containment.

Any group G can be given the profinite topology by taking the cosets of finite-index normal subgroups as a basis. Exercise 1.1.1 (1) asks for the check that if G is a topological group and $\widehat{G}$ is its profinite completion, then $\varphi : G \to \widehat{G}$ is a continuous morphism and G is dense in $\widehat{G}$.

Example 1.1.6: Profinite Groups

The following are the "trivial" examples:

1. Let G be finite. Then all normal subgroups have finite index and there are finitely many. It follows that $G \cong \widehat{G}$.
2. Take $G = \mathbb{Q}$. Then $\mathbb{Q}$ has only one finite-index normal subgroup, namely itself. Hence $\widehat{\mathbb{Q}} = \{1\}$.

The following is the "usual" example and is better suited for the intuition of profinite groups. Let $G = \mathbb{Z}$. Then

$$\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$

where $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ is the p-adic completion of $\mathbb{Z}$. The main idea is that for any element $n \in \mathbb{Z}$, in chain of normal subgroups (say $p^k\mathbb{Z} \subseteq p^{k+1}\mathbb{Z} \subseteq \cdots$) the element mod $p^k\mathbb{Z}$ eventually stabilize. For example, take $p = 7$ and $n = 129'654 = 2 \cdot 3^3 \cdot 7^4$. Then the value is zero for the until $\mathbb{Z}/7^5\mathbb{Z}$ where it becomes $12'005$ mod 7^n for $n \in \{5, 6\}$ and then $129'654$ forever onward. $\mathbb{Z}_p$ allows for never-stabilizing chains (mirroring how stable sequences of rational numbers converge to rational numbers).

For Langlands, the important non-abelian example is the absolute Galois group $\mathrm{Gal}(\overline{K}/K) = \varprojlim_L \mathrm{Gal}(L/K)$, the limit taken over the finite Galois extensions L/K. It is profinite outright, not the profinite completion of some larger group. See [4] for the Galois theory and [8] for the profinite refinement.

More generally, profinite groups can be seen as the object that captures all the finite behaviour "up to infinity" of a system. In the $\mathbb{Z}_p$ example, it appeared when taking a solution mod each p^n and for galois it comes from amalgamating all symmetries of roots of some infinite family of (finite) polynomial. Another place this shall show up is in the definition of the étale fundamental group. See *Étale Cohomology* [7].

The above examples show that the map $\varphi : G \to \widehat{G}$ could be injective, surjective, both.

Lemma 1.1.7: Profinite Completion Injective

Let G be a topological group and $\widehat{G}$ its profinite completion. Then $\varphi\colon G \to \widehat{G}$ is injective if and only if the intersection of all open normal subgroups of finite index in G is trivial.

***Proof*:**
The map φ sends g to the compatible family $(gN)_N$, so $g \in \ker\varphi$ precisely when $gN = N$ for every open normal subgroup N of finite index, that is, precisely when g lies in every such N. Therefore $\ker\varphi = \bigcap_N N$, and φ is injective if and only if that intersection is trivial, as we sought to show.

Continuous maps into a profinite group cannot see more of G than its finite quotients do, so they factor uniquely through the completion.

Theorem 1.1.8: Universal Property of Profinite Completion

let $\varphi : G \to H$ be continuous homomorphism with H a profinite group. Then there is a unique continuous homomorphism such that the following diagram commutes:

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & \widehat{G} \\ & \searrow^{\varphi} & \downarrow{\scriptstyle \exists!} \\ & & H \end{array}$$

***Proof*:**
Uniqueness is the density statement: G has dense image in $\widehat{G}$, H is Hausdorff, and two continuous maps into a Hausdorff space agreeing on a dense subset agree everywhere.

For existence, write $H = \varprojlim_i H_i$ with each H_i finite and discrete, and let $p_i\colon H \to H_i$ be the projections. Each composite $p_i \circ \varphi\colon G \to H_i$ is a continuous homomorphism into a finite discrete group, so its kernel is an open normal subgroup N_i of finite index. Hence $p_i \circ \varphi$ factors through G/N_i, and therefore through $\widehat{G} = \varprojlim_N G/N$. These factorizations are compatible with the transition maps of the system defining H, because the corresponding composites with φ are, and G is dense. They therefore assemble into a continuous homomorphism $\widehat{G} \to H$ making the triangle commute, completing the proof.

Being profinite is not an extra structure carried along from the inverse system. It is detected by the topology property:

Theorem 1.1.9: Characterizing Profinite Groups

A topological group is profinite if and only if it is a totally disconnected compact group

***Proof*:**
Let G be profinite. Then it is a closed subgroup of a direct product of finite groups, hence compact and totally disconnected.

Conversely, suppose G is compact and totally disconnected. The key point is that such a group has a basis of open normal subgroups at the identity. In a compact totally disconnected space the connected component of a point equals its quasi-component, so the clopen sets separate points. Hence every neighbourhood of e contains a compact open set C with $e \in C$. Put $U = \{g \in G : gC = C\}$, the stabilizer of C under left translation. It is a subgroup, it is open because C is compact and open, and $U \subseteq C$ since $e \in C$. Intersecting the finitely many conjugates of U, which are finite in number because U is open in a compact group and so of finite index, produces an open *normal* subgroup of finite index inside the original neighbourhood.

Writing $\mathcal{N}$ for the set of these subgroups, the canonical map $G \to \varprojlim_{N \in \mathcal{N}} G/N$ is injective because $\bigcap_{N \in \mathcal{N}} N = \{e\}$ by the previous paragraph, and surjective by compactness: a compatible family determines a family of non-empty closed cosets with the finite intersection property, whose intersection is therefore non-empty. Each G/N is finite and discrete, so G is the limit of a cofiltered system of finite groups, which is what definition 1.1.5 requires, completing the proof.

Corollary 1.1.10: Distinguishing Profinite Completion

Let G be a profinite group. Then G is naturally isomorphic to its profinite completion. More precisely,

$$G \cong \varprojlim_{U} G/U$$

where U ranges over the *open* normal subgroups of G, ordered by containment.
Moreover G is isomorphic to its profinite completion *as an abstract group* (one says G is *strongly complete*) if and only if every subgroup of finite index in G is open.

***Proof*:**
For the first assertion, G is compact by theorem 1.1.9, so every open subgroup has finite index and the two indexing systems in the definition of $\widehat{G}$ and in the display coincide. The canonical map $G \to \varprojlim_{U} G/U$ is injective by lemma 1.1.7, since a compact totally disconnected group has a basis of open normal subgroups at the identity whose intersection is $\{e\}$. It is surjective because G is compact and the maps $G \to G/U$ are surjective, so a compatible family is realized by an element of the intersection of a family of non-empty closed sets with the finite intersection property. A continuous bijection from a compact space to a Hausdorff one is a homeomorphism.

For the second, the profinite completion as an abstract group is the limit over *all* finite-index normal subgroups, whereas G itself is the limit over the *open* ones. The two agree exactly when every finite-index subgroup is open, which is the stated criterion, completing the proof.

Proposition 1.1.11: Dense Subsets of Profinite Group

Let $S \subseteq G$ be a subset of a profinite group. Then S is *dense* if and only if the natural map $S \to G/U$ is surjective for every open normal subgroup $U \subseteq G$.

***Proof*:**
By the proof of theorem 1.1.9, the open normal subgroups form a basis of neighbourhoods of the identity, so the cosets gU with U open normal form a basis for the topology of G.

Suppose S is dense and let U be open normal. Each coset gU is a non-empty open set, so it

meets S. An element of $S \cap gU$ maps to gU, so $S \to G/U$ is surjective.

Conversely, suppose every such map is surjective, and let $V \subseteq G$ be non-empty and open. Pick $g \in V$. Since the cosets of open normal subgroups form a basis, there is an open normal U with $gU \subseteq V$. By hypothesis some $s \in S$ satisfies $sU = gU$, hence $s \in gU \subseteq V$. So S meets every non-empty open set and is dense, as we sought to show.

Definition 1.1.12: Locally Profinite Group

A topological group G is *locally profinite* if it is Hausdorff and every open neighbourhood of the identity contains a compact open subgroup of G.

Notice how there is no mention of an inverse system. The definition implies G is totally disconnected, and hence if it is also compact then a profinite group is recovered. Every *discrete* group is locally profinite, the trivial subgroup being compact and open. Closed subgroups of a locally profinite group are locally profinite, and so are quotients by closed normal subgroups. A locally profinite group is locally compact, and conversely any locally compact totally disconnected group is locally profinite.

The canonical example for these notes is the non-Archimedean local field $(F, +)$, as proven in [14]. The multiplicative group $(F^\times, \times)$ is locally profinite as well, though it is *not* profinite: it is not compact, since the valuation surjects it onto $\mathbb{Z}$. What is profinite is the unit group $U_F = \mathfrak{o}^\times$, and the subgroups

$$U_F^n = 1 + \mathfrak{p}^n$$

form a basis of compact open neighbourhoods of 1 in $F^\times$, which is what local profiniteness asks for.

Being locally profinite is preserved by finite products, so F^n is locally profinite and hence so is $(M_n(F), +) \cong F^{n^2}$. Neither is profinite, for the same reason F is not: they are not compact. The group $\mathrm{GL}_n(F) \subseteq M_n(F)$ is an open topological subgroup under multiplication, and it is locally profinite because $\mathrm{GL}_n(\mathfrak{o})$ is profinite. That subgroup is compact, being homeomorphic to a closed subset of $\mathfrak{o}^{n^2}$, a product of compact groups. It is open because $\mathfrak{o} = \{x \in F : |x|_\mathfrak{p} \leq 1\}$ is open. This was all done in coordinates. More generally the same applies to $\mathrm{End}_F(V)$ and $\mathrm{Aut}_F(V)$ for a finite-dimensional F-vector space V.

Theorem 1.1.13: Continuous Character

let $\varphi : G \to \mathbb{C}^\times$ be a group homomorphism where G is locally profinite. Then the following are equivalent:

1. φ is continuous
2. the kernel of φ is open

If φ satisfies one of these conditions and G is the union of its compact open subgroups (such as $\mathbb{Q}_p$), then $\mathrm{im}\varphi \subseteq S^1 \subseteq \mathbb{C}^\times$.

***Proof*:**
If the kernel is open, then if $U \subseteq \mathbb{C}^\times$ and $K = \ker \varphi$,

$$\varphi^{-1}(U) = \bigcup_{g \in \varphi^{-1}(U)} gK$$

showing φ is continuous. Conversely, Let φ be continuous and U an open neighbourhood of 1. Then $\varphi^{-1}(U)$ is open, and thus by local-profiniteness contains a compact open subgroup $K \subseteq G$. As this applies to any U, we can pick U sufficiently small so that it contains no non-trivial subgroup of $\mathbb{C}^\times$. As the image of group is a group and $K \subseteq \varphi^{-1}(U)$, $K \subseteq \ker\varphi$. As K is open, $\ker\varphi$ contains an open subgroup and hence by topological group theory $\ker\varphi$ is open.

Next, S^1 is the unique maximal compact subgroup of $\mathbb{C}^\times$. If K' is a compact subgroup of G, $\varphi(K') \subseteq S^1$, giving the 2nd result as we sought to show.

This condition has interesting algebraic consequences, for example take the map $\mathbb{Z} \to S^1$ by sending 1 to $e^{2\pi i\theta}$ where θ is irrational. As $\mathbb{Z}$ is dense in $\mathbb{Z}_p$, this map extends to a map $\mathbb{Z}_p \to S^1$. It's kernel is $\{0\}$, i.e. it is not open, and so this map is not continuous with the profinite topology on $\mathbb{Z}_p$.

Corollary 1.1.14: Continuous Maps To General Linear Group

Let $\varphi : G \to \mathrm{GL}_n(\mathbb{C})$ be a group homomorphism where G is locally profinite. Then φ is continuous if and only if the kernel of φ is open.

If moreover $\varphi(G)$ is relatively compact in $\mathrm{GL}_n(\mathbb{C})$, then φ is conjugate to a homomorphism valued in the unitary group $U(n)$.

The second clause cannot be strengthened to "φ takes values in $U(n)$": conjugating a unitary representation by a non-unitary matrix leaves it continuous with the same kernel, so the property is not invariant under change of basis. Nor does the argument of theorem 1.1.13 transfer, since that one turns on S^1 being the *unique* maximal compact subgroup of $\mathbb{C}^\times$, while $U(n)$ for $n \geq 2$ is maximal compact only up to conjugacy and the conjugating element varies with the subgroup.

***Proof*:**
Note that $\mathrm{GL}_n(\mathbb{C})$ is a lie group and that there exists an open neighbourhood of I_n that contains no non-trivial subgroups, the argument follows.

Definition 1.1.15: Continuous Character

A *character* for a locally profinite group G is a continuous homomorphism

$$\varphi : G \to \mathbb{C}^\times$$

Write 1_G, or just 1, for the trivial character (i.e. the constant character).

If $\varphi(G) \subseteq S^1$, φ is called a *unitary character*.

The characters of a local field organize themselves by how deeply ramified they are, and that filtration is what the next section builds on. Write $\widehat{F}$ for the set of all characters of F. As F is a local field, it is the union of its compact open subgroups $\mathfrak{p}^n$ for $n \in \mathbb{Z}$, hence all characters are unitary by theorem 1.1.13. Then if $\varphi \neq 1$ is a character, there exists $d \in \mathbb{Z}$ such that $\mathfrak{p}^d \subseteq \ker(\varphi)$; the least such number is defined as the level of φ .Fixing a d, the set of characters of level $\leq d$ is a subgroup. This defines a duality which extends to the multiplicative set, in other words this is indeed the right notion of character!

Exercise 1.1.1

1. Let G be a topological group and $\widehat{G}$ its profinite completion. Show that the canonical map $\varphi\colon G \to \widehat{G}$ is a continuous homomorphism with dense image, and give an example where φ is not injective.
2. Show that an open subgroup H of a topological group G is closed, and that a closed subgroup of finite index is open. Deduce that a compact open subgroup of a locally profinite group is both.
3. Using the Chinese Remainder Theorem, establish the topological isomorphism $\widehat{\mathbb{Z}} \cong \prod_\ell \mathbb{Z}_\ell$, the product over all primes. Deduce that $\widehat{\mathbb{Z}}$ is uncountable, and exhibit an element of it that is not an integer.
4. Show that $\mathrm{GL}_n(\mathbb{Q}_p)$ is locally profinite, that $\mathrm{GL}_n(\mathbb{Z}_p)$ is a compact open subgroup, and that the subgroups $1 + p^m M_n(\mathbb{Z}_p)$ for $m \geq 1$ form a descending basis of compact open subgroups at the identity.
5. Let G be profinite and let χ be a continuous character of G. Show that χ has finite image. Where does the argument use compactness, and where does it use total disconnectedness?
6. Write $\{x\}_p$ for the p-adic fractional part of $x \in \mathbb{Q}_p$, the unique element of $\mathbb{Z}[1/p]/\mathbb{Z}$ with $x - \{x\}_p \in \mathbb{Z}_p$, and put $\psi_p(x) = e^{2\pi i \{x\}_p}$. Show that ψ_p is a character of $\mathbb{Q}_p$ with kernel $\mathbb{Z}_p$, and that $(y, x) \mapsto \psi_p(yx)$ induces an isomorphism from $\mathbb{Q}_p/\mathbb{Z}_p$ onto the group of continuous characters of $\mathbb{Z}_p$.
7. Show that $\mathbb{Z}_p^\times$ is the unique maximal compact subgroup of $\mathbb{Q}_p^\times$, and that $\mathbb{Q}_p^\times$ is therefore *not* the union of its compact open subgroups. Explain why this is what allows a character of $\mathbb{Q}_p^\times$ to be non-unitary, in contrast with theorem 1.1.13.

1.2 Smooth Representations of Locally Profinite Groups

All representations shall be complex unless specified otherwise. When working with $\mathbb{Q}_p$ representations and more generally representations of locally profinite groups, it is important to take into account the topology, or else we will get representations that give no valuable information:

Example 1.2.1: "Useless" Representation

Take the algebraic closure of $\mathbb{Q}_p$, $\overline{\mathbb{Q}}_p$. By field theory, it embeds into $\mathbb{C}$ (using the axiom of choice). Thus there is an injective ring homomorphism $\mathbb{Q}_p \to \mathbb{C}$ and hence a (valid) representation:

$$\pi : \mathrm{GL}_n(\mathbb{Q}_p) \to \mathrm{GL}_n(\mathbb{C})$$

But this has disregarded all topological (and hence arithmetic) information of $\mathbb{Q}_p$.

The first step to fix this would be to only consider continuous representations as is done in the representation of topological groups. Often when G is not compact, the vector space V is infinite dimensional with which is either a Banach or Hilbert space and $\mathrm{GL}(V)$ is the set of all invertible bounded linear operators. This thread is explored further in [15].

When restricting to locally profinite groups, there is a useful stronger notion called *smooth*. It must be that this notion of smoothness differs from smoothness for manifolds, since the maps in question are $\mathbb{Q}_p \to \mathbb{C}$ which do not share the local topology ($\mathbb{Q}_p$ is also totally disconnected, so there is no way to put a reasonable atlas on it). Instead, the intuition comes from how smoothness (or differentiability) means that zooming in far enough gives the same linear behaviour, or a constant one in suitable coordinates. In terms of locally profinite groups, as they are the union of open compact subgroups, a representation being constant on these groups abstractly shares this same property. This loose analogy turns out to be very practical, hence:

Definition 1.2.2: Smooth Map For Locally Profinite Groups

Let G be a locally profinite group. Then a map $f : G \to \mathbb{C}^n$ is *smooth* if it is locally constant, namely for any $x \in G$ there exists an open neighbourhood U containing x such that $f(x) = f(y)$ for all $x, y \in U$. The collection of all smooth functions $f : G \to \mathbb{C}$ is denoted $C^\infty(G)$.
If the function also has compact support, the space is denoted $C_c^\infty(G)$ and called the *smooth compact support space* or *Schwartz-Bruhat Space*.

1.2.3 Precision in Vocabulary Via Harmonic Analysis There is a more precise comparison when considering the Fourier transform with distributions, so the following assumes [17] for the following.

1. On $\mathbb{R}$: For the Fourier transform, the space of "ideal" functions is the Schwartz space, consisting of functions that are both smooth (C^∞) and rapidly decaying. The Fourier transform is a perfect map on this space: the transform of a smooth, rapidly decaying function is another smooth, rapidly decaying function.

2. On $\mathbb{Q}_p$: There is an analogous theory of Fourier analysis, covered in chapter 5. The "ideal" space of functions (called the Schwartz-Bruhat space) consists of functions that are locally constant and have compact support, namely since such functions under the Fourier transform map back to $C_c^\infty(G)$. Note that in the real case the image of real smooth function with compact support is the Paley-Wiener space which are holomorphic function with a certain decay property, something which does not arise in p-adic analysis. Thus, as Fourier relates smoothness to decay, in p-adic analysis being a compact support function "is" smooth from the Fourier perspective.

Proposition 1.2.4: Equivalence Notion of Smoothness

Let G be a locally profinite group and $f : G \to \mathbb{C}^n$ a map. Then f is smooth if and only if for *any* subset $U \subseteq \mathbb{C}^n$ (which is *not* necessarily open), $f^{-1}(U) \subseteq G$ is open.

Thus, if f is smooth, it is continuous with $\mathbb{C}^n$ having the discrete topology.

***Proof*:**
Suppose f is smooth and let $U \subseteq \mathbb{C}^n$ be any subset. If $x \in f^{-1}(U)$, local constancy gives an open neighbourhood V of x on which f is constant, necessarily equal to $f(x) \in U$. So $V \subseteq f^{-1}(U)$, and $f^{-1}(U)$ is open.

Conversely, suppose every preimage is open and let $x \in G$. Applying the hypothesis to the singleton $U = \{f(x)\}$ shows $f^{-1}(f(x))$ is an open set containing x on which f is constant, so f is locally constant, as we sought to show.

Smoothness would be an empty condition on a connected group, where the only locally constant functions are the constants. On $\mathbb{Q}_p$ there are many:

Example 1.2.5: Smooth Functions On Locally Profinite Groups

1. The p-adic valuation $|-|_p : \mathbb{Q}_p^\times \to \mathbb{R}$ is smooth. To see this, take any $x \in \mathbb{Q}_p$ so that $|x|_p = p^{-n}$ for some $n \in \mathbb{Z}$ giving the range. Then the pre-image of p^{-n} is $p^n\mathbb{Z}_p^\times$ which is open.

   Note that this is not smooth on $\mathbb{Q}_p$ because the pre-image of 0 is 0, i.e. there is no open neighbourhood of 0 on which the absolute value is constant. This can even be stretched to the absolute value on $\mathbb{R}$, when restricting to $\mathbb{R}^\times$ it is smooth, but naturally it is not smooth at 0.

2. The map $\mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{R}$ given by $g \mapsto |\det(g)|_p$ is smooth.

Often, a smooth function should be thought of as being well-defined modulo cosets of an open subgroup.

Identifying $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$ gives the notion of a smooth representation:

Definition 1.2.6: Smooth Representation

Let G be a locally profinite group. Then a *complex representation* (or just *representation*) of G is a pair (π, V) where V is a complex vector space and $\pi : G \to \mathrm{Aut}_{\mathbb{C}}(V)$ a group homomorphism.

A representation (π, V) is called *smooth* if for every $x \in V$, there is a compact open subgroup $K_x \subseteq G$ such that for all $k \in K_x$:

$$\pi(k) \cdot x = x$$

Equivalently, if V^K denotes the space of $\pi(K)$-fixed vector in V,

$$V = \bigcup_K V^K$$

where K ranges over all compact open subgroups of G.

A smooth representation is called *admissible* if V^K is finite dimensional for all compact open $K \subseteq G$.

Proposition 1.2.7: Smooth Representation Alternative

A representation $\pi : G \to \mathrm{Aut}_{\mathbb{C}}(V)$ is smooth if

$$\mathrm{Stab}_G(v) = \{g \in G : \pi(g) \cdot v = v\}$$

is an open subgroup of G for all $v \in V$

***Proof*:**
Suppose $\mathrm{Stab}_G(v)$ is open for every $v \in V$. Fix v. By definition 1.1.12 the open neighbourhood $\mathrm{Stab}_G(v)$ of the identity contains a compact open subgroup K_v of G, and every element of K_v fixes v by construction. That is exactly the condition in definition 1.2.6, so π is smooth.

The converse also holds: if π is smooth then $\mathrm{Stab}_G(v) \supseteq K_v$ for a compact open K_v, and a subgroup containing an open neighbourhood of the identity is open, as we sought to show.

If (π, V) is a smooth representation of G, then any G-stable subspace of V is another smooth representation, and similarly if $U \subseteq V$ is a G-subspace, the induced representation of G on V/U is smooth. A representation (π, V) is *irreducible* if if $V \neq 0$ and V has no nontrivial G-stable subspaces. A G-homomorphism between two representations (π_1, V_1), (π_2, V_2) is a map that commutes with the group action induce by π_i.

Example 1.2.8: Smooth Representations

1. If $(\mathbb{C}, \chi)$ is a one-dimensional representation, by theorem 1.1.13 it is smooth if and only if it is continuous. Hence in one dimensions these two concepts are the same. Indeed, if $\chi(g) \cdot v = v$, as $\chi(g)$ is a complex number and v is nonzero:

   $$\chi(g) = 1$$

   Hence, χ is smooth if and only if the kernel is open (and as G is locally profinite it contains a compact subgroup), which by theorem 1.1.13 is true if and only if χ is continuous.

   By corollary 1.1.14, any finite dimensional continuous representation is smooth.

2. Let $G = \mathbb{Q}_p^\times$ and consider the characters $\psi : \mathbb{Q}_p^\times \to \mathbb{C}^\times$. Note that the unitarity clause of theorem 1.1.13 does *not* apply here: $\mathbb{Q}_p^\times$ is not a union of compact open subgroups, having $\mathbb{Z}_p^\times$ as its unique maximal compact subgroup, and the unramified characters constructed below are not unitary. As $\mathbb{Q}_p^\times \cong \mathbb{Z} \times \mathbb{Z}_p^\times$ (since $x = p^n u$ for $n \in \mathbb{Z}$ and $u \in \mathbb{Z}_p^\times$; namely $n = \nu_p(x)$), and ψ is a group homomorphism, the map is determined by where it sends $(1, 0)$ and $\mathbb{Z}_p^\times$. The two parts separate:

   $$\psi((c, u)) = \psi((c, 1)) \cdot \psi((0, u)) = \psi_1(c) \cdot \psi_2(u)$$

   Thus, for any choice of $s \in \mathbb{C}$, let:

   $$\psi_1(\nu_p(x)) = p^{-s\nu_p(x)}$$

   Next the units. Note that this includes $1 \in \mathbb{Z}_p^\times$, and that the kernel must be open, and as $\varphi(1) = 1$, that means that an open neighbourhood around $1 \in \mathbb{Z}_p^\times$ maps to $1 \in \mathbb{C}^\times$. The open neighborhoods around 1 are:

   $$U_k = 1 + p^k \mathbb{Z}_p = \left\{u \in \mathbb{Z}_p^\times : u \equiv 1 \bmod (p^k)\right\}$$

Thus, for some $k \in \mathbb{N}$ $\psi_2(U_k) = 1$ By the isomorphism theorem the cosets will be:

$$\mathbb{Z}_p^\times / U_k \cong (\mathbb{Z}/p^k\mathbb{Z})^\times$$

But now ψ_2 reduces to $(\mathbb{Z}/p^k\mathbb{Z})^\times \to \mathbb{C}^\times$ which is just the familiar roots of unity maps. Note how this is "ramified". For this reason, ψ_1 is often called the *unramified component* while ψ_2 is called the *ramified component*.

Combining these, with $\psi_1(x) = |x|_p^s$ and $\psi_2(x) = \psi(x)$:

$$\chi(x) = |x|_p^s \psi(x)$$

These classify all continuous/smooth characters of $\mathbb{Q}_p^\times$.

3. In the next dimension up, take $G = \mathrm{GL}_n(\mathbb{Q}_p)$ and $V = \mathbb{C}$, that is smooth characters on $\mathrm{GL}_n(\mathbb{Q}_p)$. Then As $\mathbb{C}^\times$ is commutative, $\chi : \mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{C}^\times$ must be trivial on the commutator. For an *infinite* field F the commutator subgroup of $\mathrm{GL}_n(F)$ is $\mathrm{SL}_n(F)$: the transvections $1+\lambda E_{ij}$ generate $\mathrm{SL}_n(F)$, and each is a commutator, since conjugating $1+\mu E_{ij}$ by $\mathrm{diag}(a_1, \ldots, a_n)$ gives $1 + \mu(a_i/a_j)E_{ij}$, so $[\mathrm{diag}(a), 1 + \mu E_{ij}] = 1 + \mu(a_i/a_j - 1)E_{ij}$ and an infinite F has some $a_i/a_j \neq 1$. Here $F = \mathbb{Q}_p$ is infinite, so $[G, G] = \mathrm{SL}_n(\mathbb{Q}_p)$. As the identity has determinant 1, $I \in \mathrm{SL}_n(\mathbb{Q}_p)$ and as χ is a group homomorphism it must map to 1. Thus $\chi(\mathrm{SL}_n(\mathbb{Q}_p)) = 1$.

   Next, for any $A, B \in \mathrm{GL}_n(\mathbb{Q}_p)$ such that $\det(A) = \det(B)$ so that $AB^{-1} \in \mathrm{SL}_n(\mathbb{Q}_p)$, necessarily $\chi(A) = \chi(B)$, as χ is a homomorphism. But then, χ is only dependent on the determinant of a matrix.

   To construct a character $\chi\colon \mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{C}^\times$, choose a 1-dimensional character $\psi : \mathbb{Q}_p^\times \to \mathbb{C}^\times$ and compose it with the determinant map $A \mapsto \det(A)$ so that:

$$\chi(A) = \psi(\det(A))$$

   Every character of $\mathrm{GL}_2(F)$ is of this form, by [2, §9.2], whose argument is the one just given: the kernel contains $[G, G] = \mathrm{SL}_2(F)$ and det is surjective.

4. Take $G = \mathbb{Q}_p$ and $V = \mathbb{C}$, so the objects are the characters $\chi\colon \mathbb{Q}_p \to \mathbb{C}^\times$. These are all formed by the *standard character*. For any $x \in \mathbb{Q}_p$ the p-adic representation is of the form

$$x = \sum_n^\infty a_i p^i \qquad n \in \mathbb{Z},\ a_i \in \{0, 1, ..., p-1\}$$

   The p-adic fractional part is the sum of the negatively indexed terms. Write $\{x\}_p$ for it. Define:

$$\psi_p : \mathbb{Q}_p \to \mathbb{C}^\times \qquad \psi_p(x) = e^{2\pi i \{x\}_p}$$

   Notice the kernel of this map is $\mathbb{Z}_p$, and hence it is continuous/smooth.

5. Characters of a subgroup arise by restricting characters of $\mathbb{Q}_p$. The dual of the compact subgroup $\mathbb{Z}_p$ is the discrete group $\mathbb{Q}_p/\mathbb{Z}_p$, which is exercise 1.1.1 (6).

6. For a continuous representation that is *not* smooth, let $G = \mathbb{Z}_p$ and let $V = C(\mathbb{Z}_p, \mathbb{C})$ be the space of continuous functions with the uniform-convergence topology, carrying the regular representation

$$(\rho(g) \cdot f)(x) = f(x - g)$$

The function $|-|_p$ lies in V, the norm being continuous. Its stabilizer consists of those $g \in \mathbb{Z}_p$ with $|x-g|_p = |x|_p$ for every $x \in \mathbb{Z}_p$. Taking $x = g$ gives $0 = |g|_p$, so $g = 0$, and $\mathrm{Stab}_{\mathbb{Z}_p}(|-|_p) = \{0\}$ is not open. Hence ρ is continuous but not smooth.

7. (principal series representation) Let:

$$B = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Q}_p) \} \right\}$$

Define a character on B by choosing two characters $\chi_1, \chi_2 : \mathbb{Q}_p^\times \to \mathbb{C}^\times$ given by:

$$\varphi\left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right) = \chi_1(a)\chi_2(c)$$

Now, define the complex vector space V to be smooth complex function $f : \mathrm{GL}_2(\mathbb{Q}_p) \to \mathbb{C}$ such that:

$$f(bg) = \psi(b)f(g) \qquad b \in B,\ g \in \mathrm{GL}_2(\mathbb{Q}_p)$$

Then define the action to be:

$$(\rho(h) \cdot f)(g) = f(gh)$$

Then this is an infinite dimensional smooth representation. For a choice of χ_1, χ_2 denote it $I(\chi_1, \chi_2)$. It is almost always irreducible, but there shall be there are non-reducible representations given the right choices of χ_1, χ_2, see example 1.2.13.

By proposition 1.2.37 below, the abelian groups have no higher dimensional irreducible representations. For higher finite dimensional smooth representations of the non-abelian groups, theorem 2.4.16 shall show that every one of them factors through the determinant, even though these groups are not abelian. This shows that to capture some of their more non-abelian structure of these groups, it will be important to look at (well-behaved) infinite dimensional representations, hence the need to add the smooth condition.

Proposition 1.2.9: Irreducible Smooth Representations of a Compact Group Are Finite Dimensional

Let G be a compact locally profinite group, so that G is profinite, and let (π, V) be an irreducible *smooth* representation of G. Then V is finite dimensional.

Smoothness cannot be dropped. Without it $\mathbb{C}[\mathbb{Z}_p]$ admits simple modules of infinite $\mathbb{C}$-dimension, so the conclusion fails for the compact group $\mathbb{Z}_p$.

***Proof*:**
If $0 \neq v \in V$, then $v \in V^K$ for an open subgroup $K \subseteq G$. As G is profinite, $[G : K]$ is finite and so:

$$\{\pi(g) \cdot v : g \in G/K\}$$

spans V, so $\dim_{\mathbb{C}} V \leq [G : K]$ is already finite. For a cleaner statement of the action, take the smaller $K' = \bigcap_{g \in G/K} gKg^{-1}$, an intersection of finitely many open subgroups and hence an open normal subgroup of G of finite index, on which the action is trivial. Thus V is an irreducible representation of the finite discrete group G/K', completing the proof.

Representations come in many classes distinguished by extra conditions, and the smooth ones need a name of their own:

Definition 1.2.10: Smooth Representation Category

Let G be a locally profinite group. Then $\mathrm{Rep}_{\mathrm{smooth},K}(G)$ is the smooth representation category whose objects are smooth representations (V,π) and morphisms are G-equivariant maps. The representations are always smooth, so the notation is abbreviated to $\mathrm{Rep}_K(G)$ instead of $\mathrm{Rep}_{\mathrm{smooth},K}(G)$. The field K is dropped if it is clear from context.

The field K is $\mathbb{C}$ unless another field is named.

The kernels and cokernels of G-maps between smooth representations are again smooth, so $\mathrm{Rep}(G)$ is an abelian category. For finite groups, representations can be thought of as $\mathbb{C}[G]$-modules. With the smoothness condition imposed, there are more elements in the $\mathbb{C}[G]$-module V than smooth representations. The remedy is to pass to *Hecke algebras*, as discussed in section 1.4.

Semisimplicity has the same three equivalent formulations here as for finite groups, and the proof is the same Zorn's lemma argument, in which neither smoothness nor the topology enters.

Proposition 1.2.11: Semisimple Equivalence

Let G be a locally profinite group, and let (π,V) be a smooth representation of G. The following conditions are equivalent:

1. V is the sum of its irreducible G-subspaces;
2. V is the direct sum of a family of irreducible G-subspaces;
3. any G-subspace of V has a G-complement in V.

***Proof*:**
(1) $\Rightarrow$ (2). Take the set of irreducible G-subspaces U_i of V

$$\{U_i : i \in I\}$$

such that $V = \sum_{i\in I} U_i$. Consider the set $\mathcal{I}$ of subsets J of I such that the sum $\sum_{i\in J} U_i$ is direct. The set $\mathcal{I}$ is non-empty; Indeed, for an induction argument suppose we have a totally ordered set $\{J_a : a \in A\}$ of elements of $\mathcal{I}$. Put $J = \bigcup_{a\in A} J_a$. If the sum $\sum_{j\in J} U_j$ is not direct, there is a finite subset S of J for which $\sum_{j\in S} U_j$ is not direct. Since S must be contained in some J_a, this is impossible. Therefore $J \in \mathcal{I}$. We can now apply Zorn's Lemma to get a maximal element J_0 of $\mathcal{I}$, and it is maximality that delivers (2). Put $V_0 = \bigoplus_{i\in J_0} U_i$. If some U_i were not contained in V_0, then $U_i \cap V_0$ would be a proper G-subspace of the irreducible U_i, hence zero, so the sum $\sum_{j\in J_0\cup\{i\}} U_j$ would be direct and $J_0 \cup \{i\} \in \mathcal{I}$, contradicting maximality. So every U_i lies in V_0, and since the U_i span V by (1),

$$V = \bigoplus_{i\in J_0} U_i,$$

as required for (2).

(2) ⇒ (3). In (3), let W be a G-subspace of V. By (2), we can assume that $V = \bigoplus_{i\in I} U_i$, for a family (U_i) of irreducible G-subspaces of V. We consider the set $\mathcal{J}$ of subsets J of I for which $W \cap \sum_{i\in J} U_i = 0$. Again, the set $\mathcal{J}$ is nonempty and inductively ordered by inclusion. If J is a maximal element of $\mathcal{J}$, the sum $X = W + \bigoplus_{j\in J} U_j$ is direct. If $X \neq V$, there is $i \in I$ with $U_i \not\subset X$, so the sum $X + U_i$ is direct, and $J \cup \{i\} \in \mathcal{J}$, contrary to hypothesis. Thus (2) ⇒ (3).

(3) ⇒ (1) Let V_0 be the sum of all irreducible G-subspaces of V and write $V = V_0 \oplus W$, for some G-subspace W of V. Assume for a contradiction that $W \neq 0$. By its definition, the space W has no irreducible G-subspace. However, there is a non-zero G-subspace W_1 of W which is finitely generated over G. By Zorn's Lemma, W_1 has a maximal G-subspace W_0, and then W_1/W_0 is irreducible. We have $V = V_0 \oplus W_0 \oplus U$, for some G-subspace U of V, and hence a G-projection $V \to U$. The image of W_1 in U is an irreducible G-subspace of U, hence an irreducible subspace of V which is not contained in V_0. This is nonsense, so $V = V_0$ and (3) ⇒ (1).

Definition 1.2.12: Semisimple Representation

A representation (π, V) is *G-semisimple* if it satisfies one of the equivalent conditions of proposition 1.2.11.

Unlike finite groups, locally profinite groups will have (many) non-semisimple complex representations:

Example 1.2.13: Reducible non-semisimple representation

Take $\chi_1 = \chi_2 = 1$, so that $I(1,1)$ is induced from the trivial character of B. The constant functions form a 1-dimensional subrepresentation, on which $\mathrm{GL}_2(\mathbb{Q}_p)$ acts trivially, and it is the only one. Indeed, a 1-dimensional smooth representation of $\mathrm{GL}_2(\mathbb{Q}_p)$ has the form $\theta = \varphi \circ \det$, as example 1.2.8 records, and by theorem 1.2.21 applied to the closed subgroup B it embeds in $I(1,1)$ exactly when $\mathrm{Hom}_B(\theta|_B, 1) \neq 0$. Since θ sends $\left(\begin{smallmatrix} a & b \\ 0 & c \end{smallmatrix}\right)$ to $\varphi(ac)$, that happens only for $\varphi = 1$. The quotient of $I(1,1)$ by the constants is called the *Steinberg representation* and is denoted St, so that

$$0 \to 1_{\mathrm{GL}_2(\mathbb{Q}_p)} \to I(1,1) \to \mathrm{St} \to 0$$

with $1_{\mathrm{GL}_2(\mathbb{Q}_p)}$ the trivial representation. This sequence *does not split*, so $I(1,1)$ is *reducible* but *indecomposable*, hence *not semisimple*. The same sequence reappears as (2.10) once induction from B is treated systematically.

The direction is worth pinning down, since the other natural choice reverses it. For $\chi_1 = |\cdot|_p$ and $\chi_2 = |\cdot|_p^{-1}$ the computation above returns $\mathrm{Hom}_B(\theta|_B, \chi_1 \otimes \chi_2) = 0$ for every φ, because $\varphi(ac) = |a|_p|c|_p^{-1}$ cannot hold identically. So $I(|\cdot|_p, |\cdot|_p^{-1})$ has no 1-dimensional subrepresentation at all: there the trivial character is a quotient and Steinberg is the submodule.

Restricting to a compact open subgroup restores the finite-group picture, because a smooth vector is then fixed by a subgroup of finite index and the usual averaging is available:

Lemma 1.2.14: Compact Open Semistable Representation

Let G be a locally profinite group, and let K be a compact open subgroup of G. Let (π, V) be a smooth representation of G. The space V is the sum of its irreducible K-subspaces:

$$V = \sum_{\substack{W \subseteq V \\ \text{irred. } K\text{-subspace}}} W$$

Such a V is called *K-semisimple.*

***Proof*:**
Let $v \in V$. As in the proof of proposition 1.2.9, v is fixed by an open normal subgroup K' of K, obtained by intersecting the finitely many conjugates of a compact open stabilizer, and it generates a finite-dimensional K-space W on which K' acts trivially. Thus W is effectively a finite-dimensional representation of the finite group K/K' and so is the sum of its irreducible K-subspaces. Since $v \in V$ was chosen at random, the lemma follows.

Definition 1.2.15: Isotypic Component

Let G be a locally profinite group and $K \subseteq G$ a compact open subgroup. Let $\widehat{K}$ denote the set of equivalence classes of irreducible *smooth* representations *of* K. Let $\rho \in \widehat{K}$ and (π, V) be a smooth representation of G. Then:

$$V^\rho = \sum_{\substack{W \subseteq V \\ K\text{-subspace} \\ \text{of class } \rho}} W$$

is called the *ρ-isotypic component* of V.

By definition, V^K is the isotypic subspace for the class of trivial representation 1 of K.

Proposition 1.2.16: K-isotypic Decomposition

Let (π, V) be a smooth representation of G and let K be a compact open subgroup of G.

1. The space V is the direct sum of its K-isotypic components:
$$V = \bigoplus_{\rho \in \widehat{K}} V^\rho$$
2. Let (σ, W) be a smooth representation of G. For any G-homomorphism $f : V \to W$ and $\rho \in \widehat{K}$,
$$f(V^\rho) \subset W^\rho \text{ and } W^\rho \cap f(V) = f(V^\rho)$$

***Proof*:**
By proposition 1.2.11:

$$V = \bigoplus_{i \in I} U_i,$$

for a family of irreducible K-subspaces U_i of V. We let $U(\rho)$ be the sum of those U_i of class ρ. We then have

$$V = \bigoplus_{\rho \in \widehat{K}} U(\rho)$$

If W is an irreducible K-subspace of V of class ρ, then $W \subset U(\rho)$. Otherwise there would be a non-zero K-homomorphism $W \to U_i$, for some U_i of some class $\tau \neq \rho$. We deduce that $V^\rho = U(\rho)$ and (1) follows.

In (2), the image of V^ρ is a sum of irreducible K-subspaces of W, all of class ρ and therefore contained in W^ρ. Moreover, $f(V)$ is the sum of the images $f(V^\tau)$, $\tau \in \widehat{K}$, and $f(V^\tau) \subset W^\tau$. Since the sum of the W^τ is direct, $f(V)$ is the direct sum of the $f(V^\tau)$ and the second assertion follows.

Isotypic decomposition has a consequence that gets used constantly and is easy to overlook: exactness of a sequence of smooth representations can be tested one compact open subgroup at a time. Taking K-fixed vectors is an exact functor, not merely a left exact one, and that is what makes arguments about induced representations reduce to finite-dimensional linear algebra.

Corollary 1.2.17: Exactness reduced to compact subgroup

Let $a : U \to V$, $b : V \to W$ be G-homomorphisms between smooth representations U, V, W of G. The sequence

$$U \xrightarrow{a} V \xrightarrow{b} W$$

is exact if and only if

$$U^K \xrightarrow{a} V^K \xrightarrow{b} W^K$$

is exact, for every compact open subgroup K of G.

***Proof*:**
Both directions rest on proposition 1.2.16: taking K-fixed vectors is the projection onto the isotypic component of the trivial character of K, and by that proposition V is the direct sum of its K-isotypic components, so $V \mapsto V^K$ is an exact functor on smooth representations.

If the original sequence is exact, applying that exact functor gives exactness of the sequence of K-fixed vectors for every compact open K.

Conversely, suppose every K-fixed sequence is exact. Smoothness gives $V = \bigcup_K V^K$, and likewise for U and W. If $v \in \ker b$, choose K fixing v. Then $v \in \ker(b|_{V^K})$, which by hypothesis equals the image of U^K, so $v \in a(U)$. Hence $\ker b \subseteq \mathrm{im} a$, and the reverse inclusion is obtained the same way, as we sought to show.

For a subgroup H of G, define

$$V(H) = \text{the linear span of } \{v - \pi(h)v : v \in V, h \in H\}$$

In particular, $V(H)$ is an H-subspace of V.

Corollary 1.2.18: Splitting Using Compact Subgroups

Let G be a locally profinite group, and let (π, V) be a smooth representation of G. Let K be a compact open subgroup of G. Then

$$V(K) = \bigoplus_{\substack{\rho \in \hat{K} \\ \rho \neq 1}} V^\rho, \quad V = V^K \oplus V(K),$$

and $V(K)$ is the unique K-complement of V^K in V.

***Proof*:**
The sum $W = \bigoplus V^\rho$, with ρ not trivial, is a K-complement of V^K in V. There is therefore a K-surjection $V \to V^K$ with kernel W. Clearly, $V(K)$ is contained in the kernel of any K-homomorphism $V \to V^K$; we conclude that W contains $V(K)$. On the other hand, if U is an irreducible K-space of class $\rho \neq 1$, then $U(K) = U$, so $V^\rho \subset V(K)$.

Recall that in the finite case a subgroup $H \subseteq G$ carrying a representation (W, σ) has a natural induced representation $(\mathbb{C}[G]_{\mathbb{C}[H]}W, \mathrm{Ind}_H^G \sigma)$ where the space $\mathbb{C}[G]_{\mathbb{C}[H]}W$ can be represented as H-equivariant functions and Ind_H^G acts on the set of these functions. The following generalizes this:

Definition 1.2.19: Induced Representation

Let G be a locally profinite group, and let $H \subseteq G$ be a closed subgroup of G so that H is also locally profinite. Let (σ, W) be a smooth representation of H.

Define the space X of functions $f : G \to W$ which satisfy:

1. equivariant: $f(hg) = \sigma(h)f(g)$, $h \in H$, $g \in G$
2. smoothness: there is a compact open subgroup K of G (depending on f) such that $f(gx) = f(g)$, for $g \in G$, $x \in K$ (so f is smooth i.e. continuous with W having the discrete topology)

Then define a homomorphism $\Sigma : G \to \mathrm{Aut}_{\mathbb{C}}(X)$ by

$$\Sigma(g)f : x \longmapsto f(xg), \quad g, x \in G$$

The pair (Σ, X) provides a smooth representation of G and is said to be *smoothly induced by* σ. It is usually denoted

$$(\Sigma, X) = \mathrm{Ind}_H^G \sigma$$

Proposition 1.2.20: Induced Representation is A Functor

The map $\sigma \mapsto \mathrm{Ind}_H^G \sigma$ gives a functor $\mathrm{Rep}(H) \to \mathrm{Rep}(G)$.

***Proof*:**
Let $u\colon (\sigma, W) \to (\sigma', W')$ be an H-homomorphism. Post-composition sends a function $f\colon G \to W$ to $u \circ f\colon G \to W'$. This preserves the two defining conditions of definition 1.2.19: equivariance because $u(\sigma(h)w) = \sigma'(h)u(w)$, and smoothness because $u \circ f$ is fixed by any compact open subgroup fixing f. It also commutes with the action $\Sigma(g)f\colon x \mapsto f(xg)$, so it is a G-homomorphism $\mathrm{Ind}_H^G\sigma \to \mathrm{Ind}_H^G\sigma'$. Functoriality is immediate, since post-composition respects identities and composites, completing the proof.

There is a canonical H-homomorphism

$$\alpha_\sigma : \mathrm{Ind}_H^G\sigma \longrightarrow W, \quad f \longmapsto f(1)$$

The pair $(\mathrm{Ind}_H^G, \alpha)$ has the following fundamental property:

Theorem 1.2.21: Frobenius Reciprocity

Let H be a closed subgroup of a locally profinite group G. For a smooth representation (σ, W) of H and a smooth representation (π, V) of G, the canonical map

$$\begin{aligned} \mathrm{Hom}_G(\pi, \mathrm{Ind}_H^G\sigma) &\longrightarrow \mathrm{Hom}_H(\pi, \sigma) \\ \varphi &\mapsto \alpha_\sigma \circ \varphi, \end{aligned}$$

is an isomorphism that is functorial in both variables π, σ.

***Proof*:**
Write $\Sigma = \mathrm{Ind}_H^G\sigma$. For an H-homomorphism $f\colon V \to W$ define $f_*(v)\colon g \mapsto f(\pi(g)v)$. The proof is that $f \mapsto f_*$ is a two-sided inverse to $\varphi \mapsto \alpha_\sigma \circ \varphi$, which takes four checks.

Step 1: $f_*(v)$ lies in $\mathrm{Ind}_H^G\sigma$. For condition (1) of definition 1.2.19, and $h \in H$,

$$f_*(v)(hg) = f(\pi(h)\pi(g)v) = \sigma(h)\, f(\pi(g)v) = \sigma(h)\, f_*(v)(g)$$

using that f is an H-map. For condition (2), smoothness of π gives a compact open $K \subseteq G$ with $\pi(k)v = v$ for $k \in K$, and then $f_*(v)(gk) = f(\pi(g)\pi(k)v) = f_*(v)(g)$, so $f_*(v)$ is right K-invariant. Smoothness of π is what gives that K. Without it $f_*(v)$ need not lie in $\mathrm{Ind}_H^G\sigma$ at all.

Step 2: f_* is a G-map. For $g, g_0 \in G$,

$$f_*(\pi(g_0)v)(g) = f(\pi(gg_0)v) = f_*(v)(gg_0) = \big(\Sigma(g_0)f_*(v)\big)(g)$$

by the definition of Σ in definition 1.2.19, so $f_* \circ \pi(g_0) = \Sigma(g_0) \circ f_*$.

Step 3: the two maps compose to the identity in one order. Evaluating at 1, $\alpha_\sigma(f_*(v)) = f_*(v)(1) = f(\pi(1)v) = f(v)$, so $\alpha_\sigma \circ f_* = f$.

Step 4: and in the other. Let $\varphi \in \mathrm{Hom}_G(\pi, \Sigma)$ and put $f = \alpha_\sigma \circ \varphi$. For $v \in V$ and $g \in G$,

$$f_*(v)(g) = f(\pi(g)v) = \varphi(\pi(g)v)(1) = \big(\Sigma(g)\varphi(v)\big)(1) = \varphi(v)(g)$$

the third equality because φ is a G-map. Hence $f_* = \varphi$. The two assignments are therefore mutually inverse bijections, and both are visibly $\mathbb{C}$-linear.

Functoriality is immediate from the formulas: a G-map $\pi' \to \pi$ and an H-map $\sigma \to \sigma'$ act on both sides by composition, and α_σ is natural in σ because it is evaluation at 1, completing the proof.

Frobenius reciprocity already makes Ind_H^G a right adjoint, and a right adjoint is automatically left exact. What is less formal is that it is exact on the nose, which is what lets a short exact sequence of H-representations be induced up without losing surjectivity.

Proposition 1.2.22: Induced Representation is Additive and Exact

The functor $\mathrm{Ind}_H^G : \mathrm{Rep}(H) \to \mathrm{Rep}(G)$ is additive and exact.

***Proof*:**
For a smooth representation (σ, W) of H, temporarily let $I(\sigma)$ denote the space of functions $G \to W$ satisfying the first condition $f(hg) = \sigma(h)f(g)$ of the definition above. Thus I is a functor to the category of abstract representations of G; it is clearly additive and exact, while $\mathrm{Ind}_H^G(\sigma) = I(\sigma)^\infty$. Thus Ind_H^G is additive, and left-exactness follows because a left-exact sequence of H-spaces induces one on equivariant function spaces.

To prove it is right-exact, let (σ, W), (τ, U) be smooth representations of H and let $f : W \to U$ be an H-surjection. Take $\varphi \in I(\tau)^\infty$, and choose a compact open subgroup K of G which fixes φ. The support of φ is a union of cosets HgK, and the value $\varphi(g) \in U$ must be fixed by $\tau(H \cap gKg^{-1})$. By corollary 1.2.17 (applied to the group H and its compact open subgroup $H \cap gKg^{-1}$), taking fixed vectors preserves the surjectivity of f, so there exists $w_g \in W$, fixed by $\sigma(H \cap gKg^{-1})$, such that $f(w_g) = \varphi(g)$. We define a function $\Phi : G \to W$ to have the same support as φ and $\Phi(hgk) = \sigma(h)w_g$, for each $g \in H\backslash \mathrm{supp}\varphi / K$. The function Φ is fixed by K, and hence lies in $I(\sigma)^\infty$. Its image in $I(\tau)^\infty$ is φ, as required.

Example 1.2.23: Induced Representation

1. Take $H = G$. Evaluation at the identity, $f \mapsto f(1)$, carries $\mathrm{Ind}_G^G\sigma$ to W. It is injective because $f(g) = \sigma(g)f(1)$ for every g, and surjective because $w \in W$ has open stabilizer, so $g \mapsto \sigma(g)w$ is smooth. Thus $\mathrm{Ind}_G^G\sigma \cong \sigma$, and the map is the α_σ of theorem 1.2.21.

2. Take $H = \{1\}$, which is closed since G is Hausdorff, and σ the trivial character. Condition (1) of definition 1.2.19 is vacuous, so $\mathrm{Ind}_{\{1\}}^G\mathbb{C}$ is the space of locally constant functions $G \to \mathbb{C}$ that are right invariant under *some one* compact open subgroup, with G acting by right translation. This is a proper subspace of the $C^\infty(G)$ of definition 1.2.2, which asks only for local constancy: on $G = \mathbb{Q}_p$ the function $\sum_{n\geq 1} c_n \mathbf{1}_{A_n}$ with $A_n = p^{-n} + p^n\mathbb{Z}_p$ is locally constant, the A_n being disjoint and clopen, yet is invariant under no single $p^N\mathbb{Z}_p$. No support condition appears, and for non-compact G the space is large.

3. Let H be an open subgroup of finite index. A function in $\mathrm{Ind}_H^G\sigma$ is determined by its values on a set of representatives for $H\backslash G$, and those values are unconstrained, so $\dim_{\mathbb{C}} \mathrm{Ind}_H^G\sigma = [G : H] \dim_{\mathbb{C}} W$. In particular the induced representation is finite dimensional exactly when σ is.

4. Let $G = \mathrm{GL}_2(\mathbb{Q}_p)$, let B be its Borel subgroup and let σ be the character $\left(\begin{smallmatrix} a & b \\ 0 & c \end{smallmatrix}\right) \mapsto \chi_1(a)\chi_2(c)$.

Then $\mathrm{Ind}_B^G \sigma$ is the principal series $I(\chi_1, \chi_2)$ already met in example 1.2.8. This is the example the rest of the book is built around.

As the compact open subgroups are the key subgroups to consider for this book, there is the following useful variation of induced representation.

Definition 1.2.24: Compact Induction

Given (σ, W) for a representation of closed $H \subseteq G$ and similarly defined X as in definition 1.2.19, define:

$$X_c = \{f \in X : \text{compactly supported modulo } H\}$$

that is the image of $\mathrm{supp}(f)$ in $H \setminus G$ is compact (equivalently, $\mathrm{supp}(f) \subseteq HC$ for some compact set $C \subseteq G$). The induced functor is denoted $\text{c-Ind}_H^G \sigma$.

Proposition 1.2.25: Compact Induction is Well-defined

The space X_c along with $\text{c-Ind}_H^G \sigma$ is a smooth representation.

***Proof*:**
The space X_c is a subspace of X: the supports of $f_1 + f_2$ and of λf are contained in the union of the supports, and a union of two sets compact modulo H is again compact modulo H.

It is stable under the action. For $g \in G$ the function $\Sigma(g)f : x \mapsto f(xg)$ has support $(\mathrm{supp} f)g^{-1}$, whose image in $H\backslash G$ is the translate by g^{-1} of the image of $\mathrm{supp} f$. Right translation is a homeomorphism of $H\backslash G$, so the translate is again compact.

Smoothness is inherited from X: each $f \in X_c$ is already fixed by a compact open subgroup of G, because it satisfies the second condition of definition 1.2.19. Hence $(\text{c-Ind}_H^G \sigma, X_c)$ is a smooth representation of G, as we sought to show.

Proposition 1.2.26: Compact Induction Is Additive and Exact

The functor $\text{c-Ind}_H^G : \mathrm{Rep}(H) \to \mathrm{Rep}(G)$ is additive and exact.

***Proof*:**
Additivity is clear, since post-composition with a direct sum decomposition splits X_c accordingly.

For exactness, let $0 \to W' \to W \to W'' \to 0$ be exact in $\mathrm{Rep}(H)$. Left exactness is inherited from Ind_H^G (proposition 1.2.22), because $\text{c-Ind}_H^G \sigma$ is a subspace of $\mathrm{Ind}_H^G \sigma$ and the support condition is preserved by an injection.

Right exactness is where the compact support helps rather than hinders. Let $\varphi \in \text{c-Ind}_H^G \sigma''$ and choose a compact open K fixing it. Its support is a finite union of double cosets HgK, since the image of the support in $H\backslash G$ is compact and covered by the open sets HgK. On each such coset pick, using surjectivity of $W \to W''$ and corollary 1.2.17 applied to $H \cap gKg^{-1}$, a vector $w_g \in W$ fixed by $\sigma(H \cap gKg^{-1})$ mapping to $\varphi(g)$. Defining $\Phi(hgk) = \sigma(h)w_g$ on that coset and 0 elsewhere gives a well-defined element of $\text{c-Ind}_H^G \sigma$ with the same, hence compact modulo H,

support, lifting φ. So c-Ind$_H^G$ is exact, completing the proof.

Example 1.2.27: Compact Induction

1. Let K be a compact open subgroup and σ the trivial character of K. A function in c-Ind$_K^G\mathbb{C}$ is constant on each coset Kg and supported on finitely many of them, since a subset of $K\backslash G$ is compact exactly when it is finite, $K\backslash G$ being discrete. So c-Ind$_K^G\mathbb{C}$ is the space of finitely supported functions on $K\backslash G$, with G permuting the cosets. Its $\mathbb{C}$-dimension is the cardinality of $K\backslash G$, countable by box 1.2.31.

2. Take $H = \{1\}$ and σ trivial. Support compact modulo $\{1\}$ is ordinary compact support, so c-Ind$_{\{1\}}^G\mathbb{C} = C_c^\infty(G)$, whereas Ind$_{\{1\}}^G\mathbb{C}$ is the larger space of example 1.2.23. For non-compact G these differ, and they must: $\{1\}\backslash G = G$ is then not compact, which is exactly the condition separating the two functors.

3. When $H\backslash G$ is compact the two functors agree. Openness of H is not needed, which matters here because the case below has H closed and of infinite index. A smooth f is right invariant under some compact open K, so supp$f$ is a union of sets HgK. The image of each in $H\backslash G$ is open, and so is the image of the complement, so the image of supp$f$ is closed in $H\backslash G$ and therefore compact. Every f is thus compactly supported modulo H already. The case that matters here is $G = \mathrm{GL}_2(\mathbb{Q}_p)$ and $H = B$ of example 1.2.8, where $B\backslash G$ is compact.

There is a natural G-embedding:

$$\text{c-Ind}_H^G(\sigma) \to \text{Ind}_H^G(\sigma) \qquad \text{(i.e. a natural transformation c-Ind}_H^G \Rightarrow \text{Ind}_H^G\text{)}$$

The natural transformation c-Ind$_H^G \Rightarrow$ Ind$_H^G$ is an isomorphism of functors if and only if $H\backslash G$ is compact. For a *particular* triple G, H, σ the individual map can be an isomorphism without that, so the quantifier matters. See [2, §11.4].

The map is mostly interesting when $H \subseteq G$ is open in which case there is a natural H-homomorphism:

$$\alpha_\sigma^c : W \to \text{c-Ind}(\sigma) \qquad w \mapsto f_w$$

When H is open, compact induction has an adjunction of its own, and it points the other way from theorem 1.2.21. That theorem makes Ind$_H^G$ a *right* adjoint to restriction; the result below makes c-Ind$_H^G$ a *left* adjoint. The distinction matters in practice, because the maps one wants to count usually go *out* of a compactly induced representation rather than into an induced one.

Theorem 1.2.28: Frobenius Reciprocity for Compact Induction

Let H be an *open* subgroup of a locally profinite group G, let (σ, W) be a smooth representation of H and (π, V) a smooth representation of G. The map

$$\mathrm{Hom}_G\big(\text{c-Ind}_H^G\sigma,\ \pi\big) \longrightarrow \mathrm{Hom}_H\big(\sigma,\ \pi|_H\big), \qquad \Phi \longmapsto \Phi \circ \alpha_\sigma^c$$

is an isomorphism, functorial in both variables.

***Proof*:**
Since H is open, the quotient $H\backslash G$ is discrete, so a subset of G compact modulo H meets only finitely many cosets Hg. That is the only property of openness used.

The inverse map. Let $\varphi \in \mathrm{Hom}_H(\sigma, \pi|_H)$ and $f \in \text{c-Ind}_H^G \sigma$. Define

$$\Psi(\varphi)(f) = \sum_{g \in H\backslash G} \pi(g)^{-1} \varphi\big(f(g)\big)$$

The sum is finite because $\mathrm{supp} f$ is compact modulo H and $H\backslash G$ is discrete. Each term is well defined, that is independent of the representative g: replacing g by hg with $h \in H$ gives

$$\pi(hg)^{-1}\varphi\big(f(hg)\big) = \pi(g)^{-1}\pi(h)^{-1}\varphi\big(\sigma(h)f(g)\big) = \pi(g)^{-1}\varphi\big(f(g)\big)$$

using the equivariance of f and of φ in turn. The map $\Psi(\varphi)$ is G-equivariant: replacing f by $\Sigma(x)f \colon g \mapsto f(gx)$ reindexes the sum by $g \mapsto gx^{-1}$ and produces the factor $\pi(x)$.

The two maps are mutually inverse. For $w \in W$ the function $\alpha_\sigma^c(w) = f_w$ is supported on H with $f_w(h) = \sigma(h)w$, so exactly one coset contributes and $\Psi(\varphi)(f_w) = \varphi(w)$; hence $\Psi(\varphi) \circ \alpha_\sigma^c = \varphi$. Conversely, let $\Phi \in \mathrm{Hom}_G(\text{c-Ind}_H^G \sigma, \pi)$ and put $\varphi = \Phi \circ \alpha_\sigma^c$. Any $f \in \text{c-Ind}_H^G \sigma$ is a finite sum $\sum_g \Sigma(g)^{-1} f_{f(g)}$ over $H\backslash G$, since both sides agree at every point, so applying Φ and using its equivariance gives $\Psi(\varphi)(f) = \Phi(f)$. Functoriality is immediate from the formulas, as we sought to show.

The second tool needed alongside it describes how a compactly induced representation looks when restricted to another subgroup. The answer is indexed by double cosets, and the mechanism is nothing more than sorting functions by their support.

Proposition 1.2.29: Mackey Restriction Formula

Let H be an open subgroup of G, let K be a closed subgroup, and let (σ, W) be a smooth representation of H. For $g \in G$ write σ^g for the representation $x \mapsto \sigma(gxg^{-1})$ of $g^{-1}Hg$. Then restriction to K decomposes as

$$\big(\text{c-Ind}_H^G \sigma\big)\big|_K \;\cong\; \bigoplus_{g \in H\backslash G/K} \text{c-Ind}_{K \cap g^{-1}Hg}^K \big(\sigma^g\big)$$

the sum being over any set of representatives for the double cosets.

***Proof*:**
Each double coset HgK is open, because H is, and the HgK partition G. For $f \in \text{c-Ind}_H^G \sigma$ the support is compact modulo H, so it meets only finitely many of them; hence f is the finite sum of its restrictions $f \cdot \mathbf{1}_{HgK}$, each of which again lies in $\text{c-Ind}_H^G \sigma$ and is K-stable. This gives the direct sum decomposition as K-representations, with the summand indexed by g consisting of the functions supported in HgK.

It remains to identify that summand. Send f supported in HgK to the function $\Phi_g(f) \colon x \mapsto f(gx)$ on K. For $u \in K \cap g^{-1}Hg$ one has $gug^{-1} \in H$, so

$$\Phi_g(f)(ux) = f(gux) = f\big((gug^{-1})gx\big) = \sigma(gug^{-1})f(gx) = \sigma^g(u)\,\Phi_g(f)(x)$$

so $\Phi_g(f)$ is equivariant for σ^g; its support is compact modulo $K \cap g^{-1}Hg$ because $\mathrm{supp} f$ is compact modulo H; and it is smooth because f is. So Φ_g lands in $\text{c-Ind}_{K\cap g^{-1}Hg}^{K}\sigma^g$, and it intertwines the K-actions since both are by right translation. It is bijective, with inverse extending a function on K to HgK by the equivariance rule, completing the proof.

These two results are used together throughout: proposition 1.2.29 turns a restriction into a sum over double cosets, and theorem 1.2.28 evaluates the Hom out of each summand. That combination is what converts an orbit description of a function space into a computable Hom space, and it is used in section 8.1 to prove multiplicity one.

Two induction functors interact with tensor products in one direction only. Induction is not compatible with tensor products in general, but it is compatible with tensoring by a representation of the *large* group, and that one case is used repeatedly: it is what lets a twist be pushed through an induction, and it is the reason $\mathrm{Ind}_{E/F}\Theta \otimes \chi$ can be rewritten as an induction from E in chapter 9 and in chapter 11.

Proposition 1.2.30: Projection Formula for Induction

Let H be a closed subgroup of G, let (θ, W) be a smooth representation of H, and let (σ, U) be a smooth representation of G with U finite dimensional. Then

$$\mathrm{Ind}_H^G\big(\theta \otimes \sigma|_H\big) \cong \big(\mathrm{Ind}_H^G\theta\big) \otimes \sigma$$

naturally in θ and σ, and the same holds with Ind replaced by c-Ind throughout.

***Proof*:**
Since U is finite dimensional, choosing a basis identifies both sides with spaces of smooth functions $G \to W \otimes U$. On the right, $f \otimes u \mapsto \big(g \mapsto f(g) \otimes u\big)$ identifies $(\mathrm{Ind}_H^G\theta) \otimes \sigma$ with

$$\mathcal{B} = \{F\colon G \to W \otimes U \text{ smooth} : F(hg) = (\theta(h) \otimes 1)F(g)\}$$

carrying the action $(x \cdot F)(g) = (1 \otimes \sigma(x))F(gx)$, the two factors of the tensor product acting by right translation and by σ respectively. On the left, $\mathrm{Ind}_H^G(\theta \otimes \sigma|_H)$ is

$$\mathcal{B}' = \{F\colon G \to W \otimes U \text{ smooth} : F(hg) = (\theta(h) \otimes \sigma(h))F(g)\}$$

with $(x \cdot F)(g) = F(gx)$. Define $T\colon \mathcal{B} \to \mathcal{B}'$ by

$$(TF)(g) = \big(1 \otimes \sigma(g)\big)F(g)$$

Step 1: T lands in $\mathcal{B}'$. For $h \in H$,

$$(TF)(hg) = \big(1 \otimes \sigma(hg)\big)F(hg) = \big(1 \otimes \sigma(h)\sigma(g)\big)\big(\theta(h) \otimes 1\big)F(g) = \big(\theta(h) \otimes \sigma(h)\big)\big(1 \otimes \sigma(g)\big)F(g)$$

which is $(\theta(h) \otimes \sigma(h))(TF)(g)$. Smoothness is inherited: F is fixed by some compact open K on the right and σ is smooth, so TF is fixed by a compact open subgroup of K.

Step 2: T is G-equivariant. For $x \in G$,

$$\big(T(x \cdot F)\big)(g) = \big(1 \otimes \sigma(g)\big)\big(1 \otimes \sigma(x)\big)F(gx) = \big(1 \otimes \sigma(gx)\big)F(gx) = (TF)(gx) = \big(x \cdot TF\big)(g)$$

Step 3: T is bijective and preserves supports. The assignment $(T'F)(g) = (1 \otimes \sigma(g)^{-1})F(g)$ is a two-sided inverse by the same computation. Since $\sigma(g)$ is invertible for every g, the functions F and TF vanish at exactly the same points, so $\mathrm{supp}TF = \mathrm{supp}F$. Compact support modulo H is therefore preserved in both directions, which is what makes the statement hold verbatim for c-Ind, completing the proof.

1.2.31 Hypothesis From Now On For any compact open $K \subseteq G$, G/K is countable.

If G/K is countable, so is G/K' for any other compact open. Indeed $K \cap K'$ is compact open and a finite index of K thus

$$\frac{G}{K \cap K'} \to G/K$$

has finite fibers, thus $G/K \cap K'$ and G/K' are both countable. All groups that will be considered in this book have this property. Its use is to bound the size of an irreducible representation: a single vector generates one under the G-action, and if G/K is countable then so is the set of translates, which caps the dimension.

Lemma 1.2.32: Irreducible Smooth Representation At Most Countable

Let (π, V) be a smooth irreducible representation of a locally profinite group G satisfying box 1.2.31. Then $\dim_{\mathbb{C}}(V)$ is countable.

***Proof*:**
Let $0 \neq v \in V$ and choose $K \subseteq G$ such that $v \in V^K$. As V is irreducible,

$$\{\pi(g) \cdot v : g \in G/K\}$$

spans V, and this set is countable completing the proof.

That countability bound is what makes Schur's lemma work in this setting. Over $\mathbb{C}$ the classical argument uses finite dimensionality to produce an eigenvalue. Here the space can be infinite dimensional, and countability of the dimension substitutes for it through a transcendence argument.

Theorem 1.2.33: Schur's Lemma For Smooth Representations

Let (π, V) be a smooth irreducible representation of a locally profinite group G satisfying box 1.2.31. Then

$$\mathrm{End}_G(V) = \mathbb{C}$$

***Proof*:**
Let $0 \neq \varphi \in \mathrm{End}_G(V)$. Its image and kernel are G-subspaces, so irreducibility forces φ to be bijective, and its set-theoretic inverse is again a G-map. Thus $\mathrm{End}_G(V)$ is a division $\mathbb{C}$-algebra.

Next, $\mathrm{End}_G(V)$ has countable dimension over $\mathbb{C}$. Fix $0 \neq v \in V$. The span of $\pi(G)v$ is a non-zero subrepresentation, hence all of V by irreducibility, and a G-map satisfies $\varphi(\pi(g)v) = \pi(g)\varphi(v)$, so φ is determined by $\varphi(v)$ alone. Hence $\varphi \mapsto \varphi(v)$ is an injection of $\mathrm{End}_G(V)$ into V, and $\dim_{\mathbb{C}} V$ is countable by lemma 1.2.32.

Now suppose some $\varphi \in \mathrm{End}_G(V)$ is not a scalar. Then φ is transcendental over $\mathbb{C}$: were it algebraic, its minimal polynomial would split into linear factors over $\mathbb{C}$, and a factor $\varphi - a$ would be a zero divisor in a division algebra, forcing $\varphi = a$. Transcendence makes the elements

$$\{(\varphi - a)^{-1} : a \in \mathbb{C}\}$$

linearly independent over $\mathbb{C}$, since clearing denominators in a linear relation among them would give a polynomial relation for φ. Then $\mathrm{End}_G(V)$ would have uncountable $\mathbb{C}$-dimension, contradicting the previous paragraph. So every element of $\mathrm{End}_G(V)$ is a scalar, as we sought to show.

The converse fails in general: there are reducible smooth π with $\mathrm{End}_G(V) = \mathbb{C}$, and [2, §9.10] exhibits one. It does hold when G is compact, and the reason is complete reducibility. Lemma 1.2.14 with $K = G$ makes every smooth π a direct sum of irreducibles, and a decomposable direct sum has $\dim_{\mathbb{C}} \mathrm{End}_G(V) \geq 2$.

Corollary 1.2.34: Central Character

Let G satisfy box 1.2.31, let (π, V) be an irreducible smooth representation of G, and let $Z = Z(G)$ be the centre of G. Then Z acts on V via the character

$$\omega_\pi : Z \to \mathbb{C}^\times \qquad \text{i.e.} \qquad \pi(z) \cdot v = \omega_\pi(z) \cdot v$$

for all $v \in V$ and $z \in Z$

***Proof*:**
Let $z \in Z$. Since z commutes with every element of G, the operator $\pi(z)$ is a G-endomorphism of V, so $\pi(z) \in \mathrm{End}_G(V) \cong \mathbb{C}$ by theorem 1.2.33. Thus $\pi(z)$ is a scalar multiple of id_V, say $\omega_\pi(z)\mathrm{id}_V$, and $\omega_\pi(zz') = \omega_\pi(z)\omega_\pi(z')$ because π is a homomorphism. To see that ω_π is smooth, take a compact open subgroup K of G with $V^K \neq 0$; then ω_π is trivial on the compact open subgroup $K \cap Z$ of Z. So ω_π is a character of Z, completing the proof.

Definition 1.2.35: Central Character

The function $Z \to \mathbb{C}^\times$ defined in corollary 1.2.34 is called a *central character*.

Definition 1.2.36: Smooth Representation Admitting Central Character

Let (π, V) be a smooth representation of G and χ a central character of $Z(G)$. One says that (π, V) *admits* χ *as a central character* if:

$$\pi(z) \cdot v = \chi(z) \cdot v \qquad \forall v \in V,\ z \in V$$

Proposition 1.2.37: Irreducible Smooth Representation of Abelian Groups

Let G be an abelian locally profinite group. Then an smooth irreducible representation is one-dimensional

***Proof*:**
By corollary 1.2.34, any representation π will act like the character at every element $g \in G$ since $Z(G) = G$. Now, if W is any one-dimensional subspace, as π acts by scaling, W is a G-subspace. Hence, any irreducible subspace must be one-dimensional, completing the proof.

One lemma is needed first. It says that semisimplicity is insensitive to passing to an open subgroup of finite index, which is what lets a statement proved for a small subgroup be promoted to a larger one.

Lemma 1.2.38: Semisimple With Finite-index and Induction

Let G be a locally profinite group, and let H be an open subgroup of G of finite index.

1. If (π, V) is a smooth representation of G, then V is G-semisimple if and only if it is H-semisimple.
2. Let (σ, W) be a semisimple smooth representation of H. The induced representation $\mathrm{Ind}_H^G \sigma$ is G-semisimple.

***Proof*:**
Suppose that V is H-semisimple, and let U be a G-subspace of V. By hypothesis, there is an H-subspace W of V such that $V = U \oplus W$. Let $f : V \to U$ be the H-projection with kernel W. Consider the map

$$f^G : v \mapsto (G : H)^{-1} \sum_{g \in G/H} \pi(g) f(\pi(g)^{-1} v), \quad v \in V$$

The definition is independent of the choice of coset representatives and it follows that f^G is a G-projection $V \to U$. We then have $V = U \oplus \mathrm{Ker}\ f^G$ and $\mathrm{Ker}\ f^G$ is a G-subspace of V. Thus V is G-semisimple by proposition 1.2.11.

Conversely, suppose that V is G-semisimple. Thus V is a direct sum of irreducible G-subspaces by proposition 1.2.11, and it is enough to treat the case where V is irreducible over G. As representation of H, the space V is finitely generated and so admits an irreducible H-quotient U. Suppose for the moment that H is a normal subgroup of G. By Frobenius Reciprocity (theorem 1.2.21), the H-map $V \to U$ gives a non-trivial, hence injective, G-map $V \to \mathrm{Ind}_H^G U$. As representation of H, the induced representation $\mathrm{Ind}_H^G U = \oplus_{g \in G/H} gU$ is a direct sum of G-conjugates of U. These are all irreducible over H, so Ind U is H-semisimple. Proposition 1.2.11 then implies that $V \subset \mathrm{Ind}_H^G U$ is H-semisimple.

In general, we set $H_0 = \bigcap_{g \in G/H} gHg^{-1}$. This is an open normal subgroup of G of finite index. We have just shown that the G-space V is H_0-semisimple; the first part of the proof shows it is H-semisimple.

This completes the proof of (1).

For (2), keep $H_0 = \bigcap_{g \in G/H} gHg^{-1}$, an open normal subgroup of G of finite index contained in H. By (1) it is enough to show that $\mathrm{Ind}_H^G \sigma$ is H_0-semisimple. Since H is open of finite index, a function in $\mathrm{Ind}_H^G \sigma$ is determined by its values on a set of representatives for $H\backslash G$, and the subspace X_g of functions supported on a single coset Hg is stable under H_0: for $h_0 \in H_0$ the function $x \mapsto f(xh_0)$ is supported on Hgh_0^{-1}, which is Hg because $gh_0^{-1}g^{-1} \in H_0 \subseteq H$ by normality. So $\mathrm{Ind}_H^G \sigma = \bigoplus_g X_g$ as H_0-spaces. Evaluation $f \mapsto f(g)$ identifies X_g with W, and

under it h_0 acts by $\sigma(gh_0g^{-1})$, so X_g is the conjugate σ^g of $\sigma|_{H_0}$. Now $\sigma|_{H_0}$ is H_0-semisimple by (1) applied to $H_0 \subseteq H$, conjugating by g is an isomorphism of categories, and a direct sum of semisimple representations is semisimple. Hence $\mathrm{Ind}_H^G\sigma$ is H_0-semisimple, and (1) promotes that to G-semisimplicity.

Proposition 1.2.39: KZ Space

Let (π, V) be a smooth representation of G, admitting χ as a central character. Let K be an open subgroup of G such that KZ/Z is compact. Then:

1. Let $v \in V$. The KZ-space spanned by v is of finite dimension, and is a sum of irreducible KZ-spaces.
2. As representation of KZ, the space V is semisimple.

***Proof*:**

Part (1). Smoothness gives a compact open subgroup $K_0 \subseteq K$ fixing v. Since K_0Z/Z is open in the compact group KZ/Z, the index $[KZ : K_0Z]$ is finite. Now $\pi(kk_0)v = \pi(k)v$ for $k_0 \in K_0$, and Z acts by the scalar χ, so the space W spanned by $\pi(KZ)v$ is spanned by one vector for each class in KZ/K_0Z and has $\dim W \leq [KZ : K_0Z] < \infty$.

That W is a sum of irreducible KZ-spaces takes one more step, and it is where the finite index is used. As W is finite dimensional and π is smooth, each of finitely many basis vectors has open stabilizer, so their intersection contains a compact open $K_1 \subseteq K_0$ acting trivially on all of W. Then K_1Z acts on W through Z alone, hence by scalars, so every subspace of W is a K_1Z-subspace and W is K_1Z-semisimple. The index $[KZ : K_1Z]$ is finite by the argument above, so lemma 1.2.38(1) promotes this to KZ-semisimplicity, which for a finite-dimensional space is the assertion that W is a sum of irreducible KZ-spaces.

Part (2). Every $v \in V$ lies in its own W, so V is the sum of the KZ-subspaces W, each of which is a sum of irreducible KZ-spaces by part (1). Hence V is a sum of irreducible KZ-subspaces, which is condition (1) of proposition 1.2.11 for the group KZ, as we sought to show.

The full linear dual of a smooth representation is almost never smooth: a functional built out of infinitely many coordinates is fixed by no compact open subgroup. The remedy is the same one that produced the definition of a smooth representation in the first place, namely to keep only the vectors that are fixed by something open.

Definition 1.2.40: Smooth Dual (Contragradient)

Let (π, V) be a smooth representation of a locally profinite group G, and let V^* denote the full linear dual of V, carrying the action

$$\langle \pi^*(g)\lambda, v\rangle = \langle \lambda, \pi(g^{-1})v\rangle \qquad g \in G,\ \lambda \in V^*,\ v \in V$$

The *smooth dual* (or *contragredient*) of (π, V) is the subspace of smooth vectors

$$\check{V} = \bigcup_K (V^*)^K$$

where K runs over the compact open subgroups of G, equipped with the restriction $\check{\pi}$ of π^*. It is a smooth representation of G by construction.

The smooth dual is well behaved exactly on the admissible representations, where it becomes an involution. The mechanism is that passing to K-fixed vectors turns it into ordinary linear duality of finite-dimensional spaces, where reflexivity is automatic.

Proposition 1.2.41: Smooth Dual

Let (π, V) be a smooth representation of a locally profinite group G.

1. For every compact open subgroup K of G, restriction of functionals induces an isomorphism $\check{V}^K \cong (V^K)^*$.
2. If π is admissible, so is $\check{\pi}$, and the canonical map $V \to (\check{\check{V}})$ is an isomorphism of representations.
3. If π is admissible, then π is irreducible if and only if $\check{\pi}$ is irreducible.

***Proof*:**

1. By corollary 1.2.18 we have $V = V^K \oplus V(K)$, where $V(K)$ is spanned by the vectors $v - \pi(k)v$. A functional $\lambda \in V^*$ is fixed by K precisely when $\lambda(v - \pi(k)v) = 0$ for all v and all $k \in K$, that is, precisely when λ annihilates $V(K)$. Such a λ is determined by its restriction to V^K, and conversely any functional on V^K extends by zero on $V(K)$ to a K-fixed element of V^*, which lies in $\check{V}$ by definition. The two constructions are mutually inverse.
2. If π is admissible then V^K is finite dimensional, so $\check{V}^K \cong (V^K)^*$ is too by (1), and $\check{\pi}$ is admissible.

   For $v \in V$ let ev_v be evaluation at v. If $v \in V^K$ then ev_v is K-fixed, so the canonical map lands in the smooth dual of $\check{V}$ and is G-equivariant. Taking K-fixed vectors, it becomes the canonical map $V^K \to (\check{V}^K)^* \cong ((V^K)^*)^*$ of a finite-dimensional space into its double dual, which is an isomorphism. Since every vector on either side is fixed by some compact open subgroup, and V^K increases to V as K shrinks, the map is an isomorphism.

3. Suppose π is reducible, say U is a G-subspace with $0 \neq U \neq V$, and set

$$U^\perp = \left\{\lambda \in \check{V} : \lambda|_U = 0\right\}$$

a G-subspace of $\check{V}$. It is non-zero: choose $v \in V \setminus U$ and K fixing v, so that $v \in V^K$ while $v \notin U^K = U \cap V^K$. A functional on V^K vanishing on U^K but not at v then gives, via (1), a non-zero element of $U^\perp$. It is proper: choosing $0 \neq u \in U$ and a compact open K fixing u, a functional on V^K non-zero at u gives an element of $\check{V}$ outside $U^\perp$. Hence $\check{\pi}$ is reducible.

Conversely, if $\check{\pi}$ is reducible then the same argument applied to $\check{\pi}$, which is admissible by (2), shows $(\check{\check{\pi}})$ is reducible. By (2) again that representation is π, completing the proof.

Exercise 1.2.1

1. Let (π, V) be a smooth representation of G and $W \subseteq V$ a G-subspace. Show that W and V/W are smooth. Show also that for an *arbitrary* representation (π, V) the set of vectors with open stabilizer is the largest smooth subrepresentation of V.
2. Let $G = \mathrm{GL}_2(\mathbb{Q}_p)$ and $K = \mathrm{GL}_2(\mathbb{Z}_p)$. Show that $\text{c-Ind}_K^G \mathbb{C}$ is smooth but not admissible, using that the elementary divisors of a matrix over $\mathbb{Z}_p$ are invariants of its K-double coset.
3. Verify that Ind_H^G carries H-maps to G-maps and preserves injections. Show also that the functor I of the proof of proposition 1.2.22, defined by condition (1) of definition 1.2.19 alone, is exact, and locate the one step in that proof where condition (2) has to be restored.
4. Let K be a compact open subgroup of G and σ the trivial character of K. Identify $\text{c-Ind}_K^G \mathbb{C}$ explicitly as a space of functions on $K\backslash G$, describe the G-action, and compute the dimension of its K-fixed subspace.
5. Let H be an open subgroup of G of finite index. Show that the canonical map $\text{c-Ind}_H^G \sigma \to \mathrm{Ind}_H^G \sigma$ is an isomorphism, and identify where the finiteness of the index is used.
6. Let (π, V) be irreducible and smooth and let χ be a smooth character of G. Show that $\chi \otimes \pi$ is irreducible with central character $\chi|_Z \cdot \omega_\pi$, and that $\pi \mapsto \chi \otimes \pi$ is a bijection on isomorphism classes.
7. Proposition 1.2.41(2) makes $V \to (\check{\check{V}})$ an isomorphism for admissible π. Show that the map is injective for every smooth π, and exhibit a smooth π for which it is not surjective. Which hypothesis in the admissible case fails?
8. Let K be compact open and (π, V) smooth. Show that if $V = \bigoplus_{\rho \in \widehat{K}} W_\rho$ with every irreducible K-constituent of W_ρ of class ρ, then $W_\rho = V^\rho$ for every ρ. You will need that a K-subspace of a K-semisimple space is K-semisimple, which follows from condition (3) of proposition 1.2.11.

1.3 Measures and Duality

The space $\mathbb{C}[G]$ is $\mathbb{C}$-algebra isomorphic to $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ where $*$ is convolution. This requires summing over each $g \in G$. In this book, G is a.e. infinite, and so we replace summation with an

appropriate measure on G.

Let G be a locally profinite group, $C_c^\infty(G)$ the space of functions $f : G \to \mathbb{C}$ which are locally constant and of compact support (i.e. smooth and of compact support). Let $f \in C_c^\infty(G)$. Then compact support and smoothness (locally constant), here exists open subgroup $K_1, K_2 \subseteq G$ such that

$$f(k_1 g) = f(g) = f(g k_2) \qquad \forall g \in G,\ k_i \in K_i$$

By letting $K = K_1 \cap K_2$, f must be a finite linear combination of characteristic functions of double cosets KgK. Now, $G \curvearrowright C_c^\infty(G)$ by left translation and right translation:

$$\begin{cases} \lambda : g \mapsto (\lambda_g f : x \mapsto f(g^{-1}x)) \\ \rho : g \mapsto (\rho_g f : x \mapsto f(xg) \end{cases} \qquad x, g \in G,\ f \in C_c^\infty(G)$$

Note that the left action is the one investing since the action is on functions. To see this, apply the operator λ_{g_2} to the function f. This produces a new function:

$$h = \lambda_{g_2} f$$

The definition of this new function h is:

$$h(y) = (\lambda_{g_2} f)(y) = f(g_2 y)$$

Now we apply the operator λ_{g_1} to h; call the new function k:

$$k = \lambda_{g_1} h = \lambda_{g_1}(\lambda_{g_2} f) \qquad \text{so} \qquad k(z) = (\lambda_{g_1} h)(z) = h(g_1 z)$$

therefore:

$$k(z) = f(g_2(g_1 z)) = f(g_2 g_1 z)$$

showing why the left input needs to be the one inverted. Exercise 1.3.1 (1) asks for the check that $(C_c^\infty(G), \lambda)$ and $(C_c^\infty(G), \rho)$ are both smooth G-representation.

Definition 1.3.1: Haar Integral

A *right Haar integral* on G is a non-zero linear functional:

$$I : C_c^\infty(G) \to \mathbb{C}$$

such that

1. $I(\rho_g f) = I(f)$ for all $g \in G,\ f \in C_c^\infty(G)$
2. For all $f \in C_c^\infty(G)$ where $f \geq 0$, $I(f) \geq 0$.

And similarly for the left Haar integral.

Proposition 1.3.2: Existence of Haar Integral

Suppose G has a countable descending basis of compact open subgroups at the identity. Then there exists a right Haar integral $I : C_c^\infty(G) \to \mathbb{C}$, and a linear functional $I' : C_c^\infty(G) \to \mathbb{C}$ is a right Haar integral if and only if $I' = cI$ for some $c > 0$.

The countability assumption is what the construction below uses, and it is not implied by box 1.2.31: the group $\prod_{i\in I}\mathbb{Z}/2$ for uncountable I is locally profinite and has G/K finite for every compact open K, yet no countable family of open subgroups has trivial intersection. Every group this book treats does satisfy it, $\mathrm{GL}_n(F)$ through the chain $1+\mathfrak{p}^n M_n(\mathfrak{o})$. A Haar integral exists on any locally compact group without it, but that is not the route taken here.

***Proof*:**
Let K be a compact open subgroup of G. Denote by ${}^K C_c^\infty(G)$ the space of functions in $C_c^\infty(G)$ fixed by $\lambda(K)$. Then viewing ${}^K C_c^\infty(G)$ as G-space via right translation, it is then identical to the representation of G compactly induced from the trivial representation 1_K of K:

$$ {}^K C_c^\infty(G) = c\text{-}\mathrm{Ind}_K^G 1_K$$

Now, viewing $\mathbb{C}$ as the trivial G-space (and as a consequence K-space), by compact Frobenius reciprocity:

$$\dim_{\mathbb{C}} \mathrm{Hom}_G({}^K C_c^\infty(G), \mathbb{C}) = \dim_{\mathbb{C}} \mathrm{Hom}_K(\mathbb{C}, \mathbb{C}) = 1$$

There exists a non-zero element $I_K \in \mathrm{Hom}_G({}^K C_c^\infty(G), \mathbb{C})$ such that $I_K(f) \geq 0$ whenever $f \geq 0$, and if h_K is the characteristic function of K, then $I_K(h_K) > 0$. Indeed, for $g \in G$, let f_g denote the characteristic function of Kg. The set of functions f_g, $g \in K\backslash G$, then forms a $\mathbb{C}$-basis of the space ${}^K C_c^\infty(G)$. The functional $I_K : f_g \mapsto 1$ has the required properties, noting that $h_K = f_1$.

Now, choose a descending sequence $\{K_n\}_{n\geq 1}$ of compact open subgroups K_n of G such that $\bigcap_n K_n = 1$. Then:

$$C_c^\infty(G) = \bigcup_{n\geq 1} {}^{K_n} C_c^\infty(G)$$

For each $n \geq 1$, there is a unique right G-invariant functional I_n on ${}^{K_n} C_c^\infty(G)$ which maps the characteristic function of K_n to $(K_1 : K_n)^{-1}$. We have $I_{n+1}|_{{}^{K_n} C_c^\infty(G)} = I_n$, and so the family $\{I_n\}$ gives a functional on $C_c^\infty(G)$ of the required kind. The uniqueness statement is immediate, completing the proof.

In the above, think $K = \mathbb{Z}_p$ so that $f_g \to 1$ so $\mu(\mathbb{Z}_p) = 1$. The rest of the compact sets are given by the translation invariance and taking the index $[H : K]$ when $K \subseteq H$. This gives $\mu(a + p^n\mathbb{Z}_p) = \mu(p^n\mathbb{Z}_p) = p^{-n}$. This extends to a left Haar integral on $\mathbb{Q}_p$. For a finite extension $F/\mathbb{Q}_p$, simply replace p with $\mathfrak{p}$ in the above. For $\mathrm{GL}_k(\mathbb{Q}_p)$, recall the chain of compact open subsets:

$$K_m = \{g \in K_0 \big| g \equiv I \pmod{p^m}\} = I + p^m M_k(\mathbb{Z}_p)$$

where $K_0 = \mathrm{GL}_k(\mathbb{Z}_p)$ has measure 1, and the same computation gives $\mu(K_m) = [K_0 : K_m]^{-1}$.

Corollary 1.3.3: Left-Right Haar Integral

For $f \in C_c^\infty(G)$, define $\check{f} \in C_c^\infty(G)$ by $\check{f}(g) = f(g^{-1})$, $g \in G$. The functional

$$I' : C_c^\infty(G) \longrightarrow \mathbb{C},$$

$$I'(f) = I(\check{f}),$$

is a left Haar integral on G. Moreover, any left Haar integral on G is of the form cI', for some $c > 0$.

***Proof*:**
Inversion $g \mapsto g^{-1}$ is a homeomorphism of G carrying compact sets to compact sets and locally constant functions to locally constant functions, so $f \mapsto \check{f}$ is a linear automorphism of $C_c^\infty(G)$, and I' is a non-zero positive linear functional.

For left invariance, let $x \in G$ and write $\lambda_x f \colon g \mapsto f(x^{-1}g)$. Then $(\lambda_x f)^\vee(g) = f(x^{-1}g^{-1}) = \check{f}(gx)$, which is the right translate of $\check{f}$ by x. Since I is a right Haar integral it is invariant under right translation, so $I'(\lambda_x f) = I(\check{f}) = I'(f)$. Thus I' is left invariant.

Uniqueness up to a positive scalar is the uniqueness clause of proposition 1.3.2 applied on the left, as we sought to show.

This defines a measure called the *Haar measure*. Let I be a left Haar integral on G and $\emptyset \neq S \subseteq G$ a compact open subset of a locally profinite group.

$$\mu_G(S) = I(\Gamma_S)$$

where Γ_S is the characteristic function of S. Then $\mu_G(S) > 0$ and μ_G is left-translation invariant:

$$\mu_G(gS) = \mu_G(S) \qquad \forall g \in G$$

That this is indeed a measure is exercise 1.3.1 (5) (see [16] for a review of measure theory)

Definition 1.3.4: Haar Measure

The measure μ_G defined above is called the *Haar measure*.

The function I is often written as:

$$I(f) = \int_G f(g) d\mu_G(g) \qquad f \in C_c^\infty(G)$$

with the usual abbreviation of:

$$\int_S f(x) d\mu_G(x) := \int_G \Gamma_S(x) f(x) d\mu_G(x)$$

Definition 1.3.5: Unimodular Group

Let G be a locally profinite group (more generally a topological group with a Haar measure defined on it). Then G is *unimodular* if any left Haar integral on G is also a right Haar integral.

Most of the groups this book meets are unimodular, and the last item below is the exception that the theory is built to accommodate.

Example 1.3.6: Left and Right Haar Measures

1. An abelian group is unimodular, left and right invariance being the same condition.
2. A compact group G is unimodular. Let I be a left Haar integral and fix $g \in G$. The functional $f \mapsto I(f(-g))$ is again a left Haar integral, so it equals $c(g)I$ for some $c(g) > 0$ by the uniqueness in corollary 1.3.3, which holds for every locally profinite G. Compactness puts the constant function $\mathbf{1}_G$ in $C_c^\infty(G)$, and it is fixed by right translation, so $I(\mathbf{1}_G) = c(g)I(\mathbf{1}_G)$ with $I(\mathbf{1}_G) > 0$. Hence $c(g) = 1$ for every g and I is right invariant. In particular every profinite group is unimodular.
3. $\mathrm{GL}_n(F)$ is unimodular. Let μ be an additive Haar measure on $M_n(F) \cong F^{n^2}$. For $h \in \mathrm{GL}_n(F)$ the map $x \mapsto hx$ is F-linear on $M_n(F)$ with determinant $(\det h)^n$, and likewise for $x \mapsto xh$, so both scale μ by $\|\det h\|^n$. The measure $\|\det x\|^{-n}\, d\mu(x)$ on the open subset $\mathrm{GL}_n(F) \subseteq M_n(F)$ therefore has the two scaling factors cancel on each side, and is invariant under both translations.
4. The Borel subgroup $B \subseteq \mathrm{GL}_2(F)$ of upper triangular matrices is *not* unimodular. Coordinatize $b = \left(\begin{smallmatrix} a & x \\ 0 & d \end{smallmatrix}\right)$ by $a, d \in F^\times$ and $x \in F$, and fix Haar measures $\mu^\times$ on $F^\times$ and μ on F. Multiplying out,
$$\left(\begin{smallmatrix} a_1 & x_1 \\ 0 & d_1 \end{smallmatrix}\right)\left(\begin{smallmatrix} a_2 & x_2 \\ 0 & d_2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} a_1a_2 & a_1x_2+x_1d_2 \\ 0 & d_1d_2 \end{smallmatrix}\right)$$
so left translation moves x by $x \mapsto a_1x + \text{const}$ and right translation by $x \mapsto xd_2 + \text{const}$. Tracking the resulting factor $\|a_1\|$, respectively $\|d_2\|$, against μ shows that
$$d_\ell b = \|a\|^{-1}\, d\mu^\times(a)\, d\mu^\times(d)\, d\mu(x), \qquad d_r b = \|d\|^{-1}\, d\mu^\times(a)\, d\mu^\times(d)\, d\mu(x)$$
are left and right invariant respectively. Substituting $b \mapsto bb_2$ in the first gives $\int f(bb_2)\, d_\ell b = \|a_2/d_2\| \int f(b)\, d_\ell b$, so d_ℓ is not right invariant, and since every left Haar integral is a positive multiple of it, none of them is. The two measures differ by the factor $\|a/d\|$, which is not constant on B.

This is well-defined when working with functions with compact support. However, there are times when one have non-compact induction $\mathrm{Ind}_H^G \sigma$ where $H\backslash G$ is not compact (in this case it is often countable). To that end, extend the domain of integration of the Haar measure. Consider $f : G \to \mathbb{C}$ that is invariant under left translation by a compact subgroup $K \subseteq G$, and let μ_G be a left Haar measure on G. Then if the series:
$$\sum_{g \in K\backslash G} \int_{Kg} |f(x)| d\mu_G(x)$$
converges, so does the version without absolute values, so the following is well defined:
$$\int_G f(x)\mu_G(x) = \sum_{g \in K\backslash G} \int_{Kg} f(x) d\mu_G(x)$$
Notice the cosets are right cosets, because the integral is left-invariant, and hence would have been a sum of constants and hence necessarily diverge. Furthermore, as μ_G is translation invariant this

series is independent of the choice of compact open subgroup, since $K = K_1 \cap K_2$ may be taken, which has finite index with both K_1 and K_2.

Characters are often constructed using tensor products, which requires extending a Haar measure. Let G_1, G_2 be locally profinite groups, and set $G = G_1 \times G_2$. Then G is locally profinite. An element $\sum_{1 \leq i \leq r} f_i^1 \otimes f_i^2$ of the tensor product $C_c^\infty(G_1) \otimes C_c^\infty(G_2)$ gives a function on G by

$$\Phi(g_1, g_2) = \sum_i f_i^1(g_1) f_i^2(g_2)$$

Then $\Phi \in C_c^\infty(G)$. Importantly, this gives an isomorphism

$$C_c^\infty(G_1) \otimes C_c^\infty(G_2) \cong C_c^\infty(G)$$

Let μ_j be a left Haar measure on G_j, $j = 1, 2$. Then there is a unique left Haar measure μ_G on G such that

$$\int_G f_1 \otimes f_2(g)\, d\mu_G(g) = \int_{G_1} f_1(g_1)\, d\mu_1(g_1) \int_{G_2} f_2(g_2)\, d\mu_2(g_2),$$

for $f_i \in C_c^\infty(G_i)$. One writes $\mu_G = \mu_1 \otimes \mu_2$.

For a general $f \in C_c^\infty(G)$, the function

$$f_1(g_1) = \int_{G_2} f(g_1, g_2)\, d\mu_2(g_2)$$

lies in $C_c^\infty(G_1)$, and

$$\int_G f(g)\, d\mu_G(g) = \int_{G_1} f_1(g_1)\, d\mu_1(g_1),$$

and symmetrically: if f is of the form $f_1 \otimes f_2$, this is obvious, and such functions span $C_c^\infty(G)$.

Irreducible representations bring in smooth functions of the form $f \colon G \to V$ that need to be integrated. Let V be a complex vector space, and consider the space $C_c^\infty(G; V)$ of locally constant, compactly supported functions $f : G \to V$. This space is isomorphic to $C_c^\infty(G) \otimes V$ in the obvious way: a tensor $\sum_i f_i \otimes v_i$ gives the function

$$g \mapsto \sum_i f_i(g) v_i$$

If μ_G is a left Haar measure on G, there is a unique linear map $I_V : C_c^\infty(G; V) \to V$ such that

$$I_V(f \otimes v) = \int_G f(g) d\mu_G(g) \cdot v$$

and so define:

$$I_V(\varphi) := \int_G \varphi(g) d\mu_G(g), \quad \varphi \in C_c^\infty(G; V)$$

This has the same invariance properties as the Haar integral on scalar-valued functions.

There are cases where the left and right Haar measures do not agree (recall example 1.3.6). By the uniqueness of the left and right Haar measure up to a constant, one can cleverly equate the left/right Haar integration up to a constant. Let μ_G be a left Haar measure on G. For $g \in G$, consider the functional

$$I_g : C_c^\infty(G) \to \mathbb{C},$$
$$f \mapsto \int_G f(xg)\, d\mu_G(x)$$

Then I_g is a left Haar integral on G, so there is a unique $\delta_G(g) \in \mathbb{R}_+^\times$ such that

$$\delta_G(g) \int_G f(xg)\, d\mu_G(x) = \int_G f(x)\, d\mu_G(x),$$

for all $f \in C_c^\infty(G)$. The function δ_G is a homomorphism $G \to \mathbb{R}_+^\times$. It is trivial if G is abelian. More generally, δ_G is trivial if and only if G is unimodular. If f is the characteristic function of a compact open subgroup K of G, then δ_G is trivial on K and hence a character of G.

Definition 1.3.7: Haar Module of G

The function δ_G as define above is the *module* of G.

Note that if G is compact, then $\delta_G = 1$ and G is unimodular. In the general case, any character $G \to \mathbb{R}_+^*$ is trivial on compact subgroups, since $\mathbb{R}_+^*$ has only the trivial compact subgroup.

Using the module, the functional

$$f \mapsto \int_G \delta_G(x)^{-1} f(x)\, d\mu_G(x), \quad f \in C_c^\infty(G),$$

is a right Haar integral on G. The mnemonic

$$d\mu_G(xg) = \delta_G(g) d\mu_G(x)$$

would help on how the left/right measures "commute".

Next, many important irreducible representations shall be G-subspaces of simpler induced representations, very often by taking a character on a closed subgroup and inducing it. Let H be a closed subgroup of G, with module δ_H. Let $\theta : H \to \mathbb{C}^\times$ be a character of H. Consider the space of functions $f : G \to \mathbb{C}$ which are G-smooth under right translation, compactly supported modulo H, and satisfy the "induction" homogeneity property:

$$f(hg) = \theta(h) f(g), \quad h \in H, g \in G$$

Definition 1.3.8: Induction Space of Smooth Functions

Label this space

$$C_c^\infty(H\backslash G, \theta)$$

and view it as a smooth G-space via right translation ρ.

Note that $C_c^\infty(H\backslash G, \theta) = \text{c-Ind}_H^G \theta$.

Theorem 1.3.9: Compact Induction on Haar Measure

Let $\theta : H \to \mathbb{C}^\times$ be a character of H. The following are equivalent:

1. There exists a non-zero linear functional $I_\theta : C_c^\infty(H\backslash G, \theta) \to \mathbb{C}$ such that $I_\theta(\rho_g f) = I_\theta(f)$, for all $g \in G$.
2. $\theta\delta_H = \delta_G|_H$.

When these conditions hold, the functional I_θ is uniquely determined up to constant factor.

***Proof*:**

(1) $\Rightarrow$ (2) Let μ_G, μ_H be left Haar measures on G, H respectively. Define a G-map

$$C_c^\infty(G) \to C_c^\infty(H\backslash G, \theta) \qquad f \mapsto \tilde{f} \qquad \text{where} \qquad \tilde{f}(g) = \int_H \theta\delta_H(h)^{-1} f(hg)\, d\mu_H(h)$$

This map satisfies

$$\widetilde{\lambda_k f} = \delta_H\theta(k)^{-1}\tilde{f},$$

for $k \in H$ and $f \in C_c^\infty(G)$. If this map is surjective, we shall be able to factor I_ω through this map, allowing to link the two moduli.

To that end, let K be a compact open subgroup of G. The space $C_c^\infty(G)^K$ is spanned by the characteristic functions of the cosets gK, $g \in G/K$: a function in it is right K-invariant, so it is constant on each such coset, and compact support leaves only finitely many cosets on which it is non-zero. On the other hand, each coset HgK supports, at most, a one-dimensional space of functions in $C_c^\infty(H\backslash G, \theta)^K$. These subspaces span $C_c^\infty(H\backslash G, \theta)^K$, so the map $f \mapsto \tilde{f}$ is surjective on K-fixed functions, proving surjectivity.

Suppose that the space $C_c^\infty(H\backslash G, \theta)$ admits a functional I_θ of the required kind. The map $f \mapsto I_\theta(\tilde{f})$ is then a non-trivial G-homomorphism

$$(C_c^\infty(G), \rho) \to \mathbb{C}$$

However, the space $\mathrm{Hom}_G(C_c^\infty(G), \mathbb{C})$ has dimension 1 by Frobenius Reciprocity:

$$\dim_{\mathbb{C}} \mathrm{Hom}_G({}^K C_c^\infty(G), \mathbb{C}) = \dim_{\mathbb{C}} \mathrm{Hom}_K(\mathbb{C}, \mathbb{C}) = 1$$

and is spanned by a right Haar integral. Therefore, the function I_θ exists if and only if the right Haar integral $C_c^\infty(G) \to \mathbb{C}$ factors through the quotient map $C_c^\infty(G) \to C_c^\infty(H\backslash G, \theta)$. When it does hold, I_θ is determined up to a constant factor.

Now, the kernel of the map $f \mapsto \tilde{f}$ contains all functions $\lambda_h f - \delta_H\theta(h)^{-1} f$, for $h \in H$ and $f \in C_c^\infty(G)$. Applying the right Haar integral on G to such functions, we get

$$\int_G (\lambda_h f(g) - \delta_H\theta(h^{-1}) f(g))\delta_G(g)^{-1}\, d\mu_G(g)$$
$$= \left(\delta_G(h^{-1}) - \delta_H\theta(h^{-1})\right) \int_G f(g)\delta_G(g)^{-1}\, d\mu_G(g)$$

This vanishes identically if and only if $\delta_G(h^{-1}) = \delta_H\theta(h^{-1})$, for all $h \in H$. Thus (1) $\Rightarrow$ (2).

(2) $\Rightarrow$ (1) Take a function $f \in C_c^\infty(G)$ such that $\tilde{f} = 0$. The function f is fixed by a compact open subgroup K, and it is enough to treat the case where $\operatorname{supp} f \subset HgK$, for some $g \in G$. Thus f is a finite linear combination of the characteristic function of cosets $h_i gK$, $h_i \in H$. The condition $\tilde{f} = 0$ then amounts to

$$\mu_H(H \cap gKg^{-1}) \sum_i \theta\delta_H(h_i)^{-1} f(h_i g) = 0,$$

since $\theta\delta_H$ is trivial on the compact subgroup $H \cap gKg^{-1}$ of H. On the other hand,

$$\int_G f(x)\delta_G(x)^{-1} d\mu_G(x) = \mu_G(K)\delta_G(g)^{-1} \sum_i \delta_G(h_i)^{-1} f(h_i g)$$

If (2) holds, this will vanish, giving a well-defined non-zero functional I_θ, as we sought to show.

When the conditions of the proposition hold, the character θ takes only positive real values. Let $f \in C_c^\infty(G)$ satisfy $f(g) \geq 0$, for all $g \in G$; then $\tilde{f}(g) \geq 0$ for all g. Consequently:

Corollary 1.3.10: Existence of Compact Induction on Haar Measure

Suppose that the conditions of theorem 1.3.9 hold. There is then a non-zero linear functional I_θ on $C_c^\infty(H\backslash G, \theta)$ such that:

1. $I_\theta(\rho_g f) = I_\theta(f)$, for $f \in C_c^\infty(H\backslash G, \theta), g \in G$;
2. $I_\theta(f) \geq 0$, for $f \in C_c^\infty(H\backslash G, \theta), f \geq 0$.

These conditions determine I_θ uniquely, up to a positive constant factor, and the notation is:

$$I_\theta(f) = \int_{H\backslash G} f(g)\, d\mu_{H\backslash G}(g), \quad f \in C_c^\infty(H\backslash G, \theta)$$

***Proof*:**
Existence of a non-zero G-invariant functional, and its uniqueness up to a scalar, are exactly the content of theorem 1.3.9 under the stated condition $\theta\delta_H = \delta_G|_H$. Only positivity and the sign of the scalar remain.

The averaging map $\pi\colon C_c^\infty(G) \to C_c^\infty(H\backslash G, \theta)$ is the map $F \mapsto \widetilde{F}$ of the proof of theorem 1.3.9,

$$\pi(F)(g) = \int_H (\theta\delta_H)(h)^{-1} F(hg)\, d\mu_H(h)$$

and it is surjective and carries non-negative functions to non-negative functions. The factor δ_H cannot be dropped: with weight $\theta(h)^{-1}$ alone one gets $\pi(F)(h_0 g) = \delta_H(h_0)^{-1}\theta(h_0)\pi(F)(g)$, so $\pi(F)$ would not lie in $C_c^\infty(H\backslash G, \theta)$ unless H were unimodular. Composing a Haar integral on G with a section of π shows that I_θ may be chosen with $I_\theta(\pi(F)) = \int_G F\, d\mu_G$, which is non-negative whenever F is. Since every non-negative element of $C_c^\infty(H\backslash G, \theta)$ is $\pi(F)$ for some non-negative F, this I_θ satisfies (2).

Finally, if I' is another functional satisfying (1) then $I' = cI_\theta$ for some $c \in \mathbb{C}^\times$ by the uniqueness clause, and evaluating both at a non-zero non-negative f shows $c > 0$ precisely when I' also

satisfies (2). So the pair of conditions pins I_θ down up to a positive constant, as we sought to show.

Definition 1.3.11: Semi-invariant Measure

The measure $\mu_{H\backslash G}$ is called a positive *semi-invariant measure* on $H\backslash G$.

Since $\theta = \delta_H^{-1}\delta_G|H$ is uniquely determined, there is no real need to refer to it again, and so it shall be taken as implicit.

The smooth dual of a compact induction is again an induction, with the compactness condition traded for a twist by the modulus:

Theorem 1.3.12: Duality Theorem

Let $\dot{\mu}$ be a positive semi-invariant measure on $H\backslash G$. Let (σ, W) be a smooth representation of H. Then there is a natural isomorphism:

$$(c\text{-Ind}_H^G \sigma)^\vee \cong \mathrm{Ind}_H^G \delta_{H\backslash G} \otimes \check{\sigma},$$

depending only on the choice of $\dot{\mu}$.

***Proof*:**

We view $\delta_{H\backslash G} \otimes \check{\sigma}$ as acting on the same space $\check{W}$ as $\check{\sigma}$. Let $(\check{w}, w) \mapsto \langle \check{w}, w \rangle$ be the evaluation pairing $\check{W} \times W \to \mathbb{C}$. Let $\varphi \in \text{c-Ind}\,\sigma$, $\Phi \in \mathrm{Ind}\,\delta_{H\backslash G} \otimes \check{\sigma}$, and consider the function

$$f : g \mapsto \langle \Phi(g), \varphi(g) \rangle, \quad g \in G$$

This lies in $C_c^\infty(H\backslash G, \delta_{H\backslash G})$, so we have a G-invariant pairing

$$\mathrm{Ind}\,\delta_{H\backslash G} \otimes \check{\sigma} \times \text{c-Ind}\,\sigma \to \mathbb{C},$$
$$(\Phi, \varphi) \mapsto \int_{H\backslash G} \langle \Phi(g), \varphi(g) \rangle \, d\dot{\mu}(g)$$

This induces a G-homomorphism $\mathrm{Ind}\,\delta_{H\backslash G} \otimes \check{\sigma} \to (\text{c-Ind}\,\sigma)^\vee$. It is clearly natural in σ; we show it is an isomorphism.

To do this, we take a compact open subgroup K of G and describe the space $(\text{c-Ind}\,\sigma)^K$ of K-fixed functions in $\text{c-Ind}\,\sigma$. The support of such a function, f say, is a finite union of cosets HgK. The value $f(g) \in W$ is then fixed by the compact open subgroup $H \cap gKg^{-1}$ of H. We choose a set $\mathcal{G}$ of representatives for the coset space $H\backslash G/K$. For each $g \in \mathcal{G}$, we choose a basis $\mathcal{W}_g$ of the space $W^{H \cap gKg^{-1}}$. For each $g \in \mathcal{G}$ and each $w \in \mathcal{W}_g$, there is a unique function $f_{g,w} \in (\text{c-Ind}\,\sigma)^K$, supported on HgK, such that $f_{g,w}(g) = w$. The set

$$\{f_{g,w} : g \in \mathcal{G}, w \in \mathcal{W}_g\}$$

is then a basis of $(\text{c-Ind}\,\sigma)^K$. That is, the space $(\text{c-Ind}\,\sigma)^K$ consists of the finite linear combinations of functions $f_{g,w}$, $g \in \mathcal{G}$, $w \in \mathcal{W}_g$.

Here the argument must not pass through a dual basis. When σ is not admissible the space $W^{H \cap gKg^{-1}}$ is infinite dimensional, and the family dual to a basis of it is linearly independent

without spanning $\mathrm{Hom}_{\mathbb{C}}(W^{H\cap gKg^{-1}},\mathbb{C})$, so a description of $(\mathrm{Ind}\,\delta_{H\backslash G}\otimes\check{\sigma})^K$ by finite combinations of the $f_{g,\check{w}}$ would therefore capture only a proper subspace, and the theorem would be proved for admissible σ alone. Instead evaluate on each double coset separately. Restriction to HgK identifies

$$\left(\mathrm{Ind}\,\delta_{H\backslash G}\otimes\check{\sigma}\right)^K\cong\prod_{g\in\mathcal{G}}\left(W^{H\cap gKg^{-1}}\right)^*$$

with arbitrary, not finitely supported, values in each factor, because a K-fixed function is determined by one value per double coset and those values are unconstrained. That product is exactly the full linear dual of $(\text{c-Ind}_H^G\sigma)^K=\bigoplus_{g\in\mathcal{G}}W^{H\cap gKg^{-1}}$, which is what the pairing must match.

For $g_1,g_2\in\mathcal{G}$, $w\in W^{H\cap g_1Kg_1^{-1}}$ and an arbitrary functional $\check{w}\in\left(W^{H\cap g_2Kg_2^{-1}}\right)^*$, write $f_{g_2,\check{w}}$ for the K-fixed element of $\mathrm{Ind}\,\delta_{H\backslash G}\otimes\check{\sigma}$ supported on Hg_2K with value $\check{w}$ at g_2. Then

$$\langle f_{g_2,\check{w}},f_{g_1,w}\rangle=\begin{cases}\dot{\mu}(Hg_1K)\langle\check{w},w\rangle & \text{if } Hg_1K=Hg_2K,\\ 0 & \text{otherwise,}\end{cases}$$

for some $\dot{\mu}(Hg_1K)>0$. The pairing thus identifies $(\mathrm{Ind}\,\delta_{H\backslash G}\otimes\check{\sigma})^K$ with the linear dual of $(\text{c-Ind}\,\sigma)^K$, and hence $\mathrm{Ind}\,\delta_{H\backslash G}\otimes\check{\sigma}$ with $(\text{c-Ind}\,\sigma)^\vee$, as required.

Exercise 1.3.1

1. Show that $(C_c^\infty(G),\lambda)$ and $(C_c^\infty(G),\rho)$ are both smooth representations of G, where $\lambda_g f(x)=f(g^{-1}x)$ and $\rho_g f(x)=f(xg)$. Verify that λ and ρ commute.
2. Give the argument that a compact locally profinite group is unimodular, and show that a discrete group is unimodular. Where does the compact case use that the constant function lies in $C_c^\infty(G)$?
3. Let μ be an additive Haar measure on $M_n(F)$. Show that $\|\det x\|^{-n}\,d\mu(x)$ is a bi-invariant Haar measure on $\mathrm{GL}_n(F)$, and compute $\mu(\mathrm{GL}_n(\mathfrak{o}))$ relative to $\mu(M_n(\mathfrak{o}))=1$ for $n=2$.
4. For the Borel subgroup $B\subseteq\mathrm{GL}_2(F)$ of upper triangular matrices, verify the two Haar measures computed in example 1.3.6 and show that their ratio at $\left(\begin{smallmatrix}a & x\\ 0 & d\end{smallmatrix}\right)$ is $\|a/d\|$. Deduce that no scalar multiple of a left Haar measure on B is right invariant.
5. Let I be a Haar integral on G and for a compact open $S\subseteq G$ let Γ_S be its characteristic function. Show that $S\mapsto I(\Gamma_S)$ is finitely additive, positive, and invariant under the translation I is invariant under, so that it deserves the name measure. Show also that $\theta=\delta_H^{-1}\delta_G|_H$ takes only positive real values.
6. Apply theorem 1.3.12 with $H=K$ a compact open subgroup and σ trivial to identify $(\text{c-Ind}_K^G\mathbb{C})^\vee$, and check the answer directly against the description of $\text{c-Ind}_K^G\mathbb{C}$ obtained in exercise 1.2.1 (4).

1.4 Hecke Algebra

Linear representations of finite groups are in bijective correspondence to the modules over the group algebra $\mathbb{C}[G]$. A similar conditions holds for smooth representations of locally profinite groups given the suitable generalization of $\mathbb{C}[G]$ to Hecke algebras.

Though not necessary, to avoid technicalities that do not add insight, throughout this entire section the locally profinite group G will be assumed to be unimodular.

Fix a Haar measure μ on G. Then for any $f_1, f_2 \in C_c^\infty(G)$, define the convolution:

$$f_1 * f_2(g) = \int_G f_1(x) f_2(x^{-1}g) d\mu(x)$$

The integrand $(x, g) \mapsto f_1(x)f_2(x^{-1}g)$ lies in $C_c^\infty(G \times G)$, and integrating a locally constant compactly supported function over the compact set $\mathrm{supp} f_1$ leaves a locally constant function whose support lies in $\mathrm{supp} f_1 \cdot \mathrm{supp} f_2$, itself compact; hence $f_1 * f_2 \in C_c^\infty(G)$. This operation is associative: Here's the equation in an align* environment:

$$\begin{aligned} f_1 * (f_2 * f_3)(g) &= \int \int f_1(x) f_2(y) f_3(y^{-1}x^{-1}g) d\mu(y) d\mu(x) \\ &= \int \int f_1(x) f_2(x^{-1}y) f_3(y^{-1}g) d\mu(y) d\mu(x) \\ &= \int \int f_1(x) f_2(x^{-1}y) f_3(y^{-1}g) d\mu(x) d\mu(y) \\ &= (f_1 * f_2) * f_3(g) \end{aligned}$$

Definition 1.4.1: Hecke Algebra

Let G be a unimodular locally profinite group. Then:

$$\mathcal{H}(G) = (C_c^\infty(G), *)$$

is is an associative $\mathbb{C}$-algebra called the *Hecke algebra* of G.

This is an example of a *non-unital* algebra as soon as G is not discrete. If G is commutative, so is $\mathcal{H}(G)$. Note that if μ, ν are two choices of Haar measures giving rise to $\mathcal{H}_\mu(G)$ and $\mathcal{H}_\nu(G)$ on $C_c^\infty(G)$, then there exists a constant $c > 0$ such that $\nu = c\mu$ and a map $f \mapsto c^{-1}f$, and hence

$$\mathcal{H}_\mu(G) \cong \mathcal{H}_\nu(G)$$

Concretely, if $\nu = c\mu$ then $f_1 *_\nu f_2 = c\,(f_1 *_\mu f_2)$, so $f \mapsto cf$ is an isomorphism $\mathcal{H}_\nu(G) \to \mathcal{H}_\mu(G)$. The Hecke algebra is therefore well defined up to canonical isomorphism, and the choice of Haar measure will not be tracked.

$\mathcal{H}(G)$ converts smooth representations into modules, exactly as $\mathbb{C}[G]$ does for a finite group. It has no unit when G is not discrete, and a family of idempotents takes its place.

Definition 1.4.2: Hecke Idempotents

For a compact open subgroup $K \subseteq G$ let $e_K = \mu(K)^{-1}\mathbf{1}_K \in \mathcal{H}(G)$, where $\mathbf{1}_K$ is the characteristic function of K. A module M over $\mathcal{H}(G)$ is called *smooth* if $\mathcal{H}(G) * M = M$.

Each e_K satisfies $e_K * e_K = e_K$, and $e_K * f = f$ precisely when f is invariant under left translation by K; since every $f \in C_c^\infty(G)$ is invariant under some such K, the e_K form an approximate identity.

Theorem 1.4.3: Smooth Representations as Hecke Modules

Let G be a unimodular locally profinite group. The assignments

$$\pi(f)v = \int_G f(g)\,\pi(g)v\,d\mu(g), \qquad \pi(g)(f * m) = (\lambda_g f) * m$$

where $\lambda_g f\colon x \mapsto f(g^{-1}x)$, are mutually inverse and define an equivalence between the category of smooth representations of G and the category of smooth $\mathcal{H}(G)$-modules.

***Proof*:**

From representations to modules. Let (π, V) be smooth and $f \in C_c^\infty(G)$. The integrand $g \mapsto f(g)\pi(g)v$ is locally constant with compact support: f is, and $g \mapsto \pi(g)v$ is locally constant because v has open stabilizer by proposition 1.2.7. So the integral is a finite sum and $\pi(f)v$ is defined. A change of variables gives $\pi(f_1 * f_2) = \pi(f_1)\pi(f_2)$, so V is an $\mathcal{H}(G)$-module. It is smooth: taking K with $v \in V^K$ gives $\pi(e_K)v = v$, so $v \in \mathcal{H}(G) * V$.

From modules to representations. Let M be a smooth $\mathcal{H}(G)$-module. Every $m \in M$ is a finite sum $\sum_i f_i * m_i$, and setting $\pi(g)(f * m) = (\lambda_g f) * m$ extends by linearity. This is well defined: if $\sum_i f_i * m_i = 0$ then $\sum_i (\lambda_g f_i) * m_i = 0$, because λ_g is an automorphism of the left $\mathcal{H}(G)$-module $\mathcal{H}(G)$ and unimodularity makes λ_g compatible with convolution on the right. Since $\lambda_{gh} = \lambda_g \lambda_h$, this is a homomorphism $G \to \mathrm{Aut}_{\mathbb{C}}(M)$.

It is smooth. Given $m = f * m'$, pick a compact open K under which f is left invariant; then $\lambda_k f = f$ for $k \in K$, so $\pi(k)m = m$ and the stabilizer of m is open. Applying proposition 1.2.7 again, (π, M) is a smooth representation.

The two constructions are inverse. Starting from (π, V) and returning, the recovered action sends g to the operator determined by $\pi(\lambda_g f) = \pi(g)\pi(f)$, which is $\pi(g)$ on $\mathcal{H}(G) * V = V$. Starting from M and returning, the recovered module structure is $f * m$ again, since $\int_G f(g)\pi(g)\,d\mu(g)$ applied to $h * m'$ reproduces $(f * h) * m'$ by the associativity computed above. Both assignments carry G-maps to $\mathcal{H}(G)$-maps and conversely, so they give an equivalence of categories, completing the proof.

The classification of smooth representations of G becomes a question about modules over a single associative algebra, and the compact open subgroups enter through the idempotents e_K, whose images $e_K * M = M^K$ are the finite-dimensional pieces an admissible representation is assembled from.

Exercise 1.4.1

1. Show that $\mathcal{H}(G)$ has a unit element if and only if G is discrete, and that for any G the

idempotents e_K form an approximate identity.

2. For a compact open subgroup K and a smooth representation (π, V), show that $\pi(e_K)$ is the projection of V onto V^K with kernel $V(K)$, and identify $e_K * \mathcal{H}(G) * e_K$ as an algebra.

3. Show that $\mathcal{H}(G)$ is commutative when G is abelian. Then, taking $G = \mathrm{GL}_2(\mathbb{Q}_p)$ and $K = \mathrm{GL}_2(\mathbb{Z}_p)$, and granting that every $g \in G$ lies in $K \,\mathrm{diag}(p^a, p^b)\, K$ for some $a \geq b$ (Smith normal form over $\mathbb{Z}_p$), show that transposition fixes every K-double coset and deduce that $e_K * \mathcal{H}(G) * e_K$ is commutative even though $\mathcal{H}(G)$ is not.

2

Smooth Representations of GL2: General Structure and Principal Series

Having established the general framework of smooth representations, we now narrow our focus to the group $G = \mathrm{GL}_2(F)$ for a non-Archimedean field F. This group can be thought of as the group being able to represent the "2D" information of the (local) absolute galois group of F. The goal of this chapter is to classify the "continuous spectrum" of representations of G, known as the *Principal Series*. These are the representations that can be constructed by parabolic induction from characters of the diagonal torus. In doing so, the "discrete spectrum" (the cuspidal representations) is identified by what it is not.

The representation theory of $\mathrm{GL}_2(\mathbb{F}_q)$ comes first, as a finite model for the p-adic case. We then analyze the geometry of $\mathrm{GL}_2(F)$ through its various decompositions (Iwasawa, Cartan, Bruhat). A key tool in our analysis will be the *Mirabolic subgroup* M (section 2.3), whose representation theory is simple enough to serve as a lever for understanding G. Finally, using the Jacquet module (section 2.4), we provide a complete classification of the principal series, determining when they are irreducible and how they decompose when reducible (leading to the Steinberg representation).

2.1 Important Background: Representation of $\mathrm{GL}_2(\mathbb{F}_p)$

When analyzing $\mathrm{GL}_2(\mathbb{Q}_p)$ and related groups, we shall draw inspiration of this special case. We shall first single out the following subgroups:

Definition 2.1.1: Matrix Subgroups

Let F be (an arbitrary) field. Then define the groups:

1. (Borel subgroup) $B = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in \mathrm{GL}_2(F) \right\}$
2. (unipotent subgroup) $N = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2(F) \right\} \cong (F, +)$
3. (torus subgroup) $T = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \in \mathrm{GL}_2(F) \right\}$
4. (center) $Z = \left\{ \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \in \mathrm{GL}_2(F) \right\} \cong F^*$

Note the Borel subgroup is not normal. In fact it is its own normalizer: $N_G(B) = B$.

Lemma 2.1.2: Bruhat Decomposition

Let $G = \mathrm{GL}_2(F)$. Then its *Bruhat decomposition* is:

$$G = B \cup BwB$$

where

$$w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Hence, $\{1, w\}$ is a set of representative of the coset space $B\backslash G/B$.

Furthermore, $B = NT = TN$, hence $B \cong N \rtimes T$

***Proof*:**

Let $M \in GL_2(F)$ so that $\det(M) = ad - bc \neq 0$ if $c = 0$, $ad \neq 0$, so $a, d \neq 0$. Then $M \in B$.

If $c \neq 0$, choose two arbitrary $B_1, B_2 \in B$, calculate $B_1 w B_2$, use the fact that $\det(B_1), \det(B_2) \neq 0$ to get an equation for each entry of M. Note that the bottom-left entry *must* be nonzero and hence $B \cap BwB = \emptyset$, proving the decomposition into disjoint cosets.

For the $B = NT = TN$, notice that:

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \begin{pmatrix} 1 & b/a \\ 0 & 1 \end{pmatrix} \in TN$$

It can also be factored as:

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} 1 & b/d \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \in NT$$

Showing they are a semidirect products completing the proof.

The decomposition is sharper than it looks, because the big cell can be coordinatized. That refinement is used often enough downstream to be worth stating on its own.

Proposition 2.1.3: The Big Cell in Coordinates

Let $w = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ be as in lemma 2.1.2. Then

$$G = B \sqcup BwN$$

the map $B \times N \to BwN$, $(b, n) \mapsto bwn$, is bijective, and BwN is exactly the set of $g \in G$ with $g_{21} \neq 0$.

***Proof*:**
Any representative of BwB has a non-zero entry in the $(2,1)$-position, so BwB is disjoint from B and lemma 2.1.2 gives $G = B \sqcup BwB$. Since $B = TN = NT$ and w normalizes T, one has $BwB = BwTN = BTwN = BwN$.

For the coordinates, take $b = \left(\begin{smallmatrix} \alpha & \beta \\ 0 & \delta \end{smallmatrix}\right) \in B$ and $n = \left(\begin{smallmatrix} 1 & s \\ 0 & 1 \end{smallmatrix}\right) \in N$. Then

$$b\,w\,n = \begin{pmatrix} \beta & \alpha + \beta s \\ \delta & \delta s \end{pmatrix} \tag{2.1}$$

so every element of BwN has $(2,1)$-entry $\delta \neq 0$. Conversely, given $g \in G$ with $g_{21} \neq 0$, the four equations read off from (2.1) force $\beta = g_{11}$, $\delta = g_{21}$, $s = g_{22}/g_{21}$ and, since $\det(bwn) = -\alpha\delta$, also $\alpha = -\det(g)/g_{21}$. Both α and δ are non-zero, so $b \in B$; and the values are determined by g, which is exactly the asserted bijectivity, completing the proof.

Next, For any $M \in \mathrm{GL}_2(F)$, define $\mathrm{ch}_M(t) = \det(tI - M) = t^2 - \mathrm{tr}(M)t + \det(M)$, that is $\mathrm{ch}_M(t)$ is the characteristic polynomial of M, i.e. a monic quadratic F-polynomial and $ch_M(0) = \det(M) \neq 0$.

Proposition 2.1.4: Character Polynomial For $\mathrm{GL}_2(F)$

Let F be an arbitrary field, $M \in \mathrm{GL}_2(F)$, and $f(t) = \mathrm{ch}_M(t) = \det(tI - M)$. Then:

1. If $f(t)$ is irreducible, the F-algebra $F[M] \subseteq M_2(F)$ is a field with $\mathrm{GL}_2(F)$-centralizer $F[M]^*$. Given $f(t) = t^2 + at + b$, then M is $\mathrm{GL}_2(F)$-conjugate to:

$$\begin{pmatrix} 0 & -b \\ 1 & -a \end{pmatrix}$$

   Furthermore, an element $N \in \mathrm{GL}_2(F)$ is GL_2-conjugate to M if and only if $\mathrm{ch}_N(t) = f(t)$.

2. If $f(t)$ has distinct roots $a, b \in F^*$, then M is $\mathrm{GL}_2(F)$-conjugate to:

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

3. If $f(t)$ has a repeated root $a \in F^*$, then M is $\mathrm{GL}_2(F)$-conjugate to exactly one of:

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$$

   In the first case, M lies in the center of $\mathrm{GL}_2(F)$, and in the 2nd the $\mathrm{GL}_2(F)$-centralizer of M is $F[M]^*$.

***Proof*:**

1. As $f(t)$ is irreducible, by Cayley-Hamilton $F[M] \cong F[t]/(f(t))$ where the characteristic polynomial $f(t)$ is also the minimal polynomial. For the conjugacy result, as M is irreducible it has no eigenvalues and so $\{v, Mv\}$ forms a basis for any nonzero $v \in F^2$. Finding the matrix representation of the linear transformation corresponding to M shall give the desired result. For the centralizer, F^2 is a one-dimensional vector space over the field $F[M]$, so $\mathrm{End}_{F[M]}(F^2) = F[M]$, and an element of $\mathrm{GL}_2(F)$ commutes with M exactly when it is an invertible element of $F[M]$.

2. The roots $a \neq b$ both lie in F, so M has an eigenvector v_a for a and an eigenvector v_b for b. Eigenvectors for distinct eigenvalues are linearly independent, so $\{v_a, v_b\}$ is a basis of F^2, and in that basis M is the displayed diagonal matrix.

3. The minimal polynomial of M divides $f(t) = (t-a)^2$. If it is $t - a$ then $M = aI$, which is central. Otherwise it is $(t-a)^2$, so $M - aI \neq 0$ while $(M - aI)^2 = 0$. Choose v with $w := (M - aI)v \neq 0$. Then $(M - aI)w = 0$, and $\{w, v\}$ is a basis: applying $M - aI$ to a relation $\lambda w + \mu v = 0$ gives $\mu w = 0$, hence $\mu = 0$ and then $\lambda = 0$. In that basis $Mw = aw$ and $Mv = w + av$, which is the second displayed matrix. The two cases are exclusive because only the first is central.

   For the centralizer in the second case, N commutes with M exactly when it commutes with $M - aI$, whose matrix in the basis $\{w, v\}$ is $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Writing $N = \begin{pmatrix} x & y \\ z & u \end{pmatrix}$, that

condition reads $z = 0$ and $u = x$, so $N = xI + y(M - aI) \in F[M]$. Since $\det N = x^2$, such an N is invertible exactly when $x \neq 0$. Hence the centralizer is $F[M]^*$.

Now let $F = \mathbb{F}_q$ be a field with $q = p^l$ elements for some $l \in \mathbb{N}$, and let $G = \mathrm{GL}_2(\mathbb{F}_q)$. Let find the irreducible complex representations of G. First off:

Lemma 2.1.5: Order and Irreducible Representations of G

Let $G = \mathrm{GL}_2(\mathbb{F}_q)$. Then:

$$|G| = (q^2 - 1)(q^2 - q)$$

Furthermore, G has $q^2 - 1$ conjugacy classes.

***Proof*:**

Let the first column be any nonzero vector, giving $q^2 - 1$ possibilities. The second column cannot be a scalar multiple of the first. Given a choice v_1 of the first column, the second cannot be kv_1 for all $k \in \mathbb{F}_q$. This gives $q^2 - q$ possibilities. These choices are linearly independent, thus:

$$|G| = (q^2 - 1)(q^2 - q)$$

For the conjugacy classes we repeatedly use proposition 2.1.4. By the first case, as $\mathbb{F}_q$ has exactly $(q^2 - q)/2$ irreducible monic polynomials of degree 2, there are $(q^2 - q)/2$ conjugacy classes. The second case will give $(q - 1)(q - 2)/2$ cases, and the third $2(q - 1)$ cases. Summing these up gives the desired result.

Thus, there are $q^2 - 1$ irreducible representations. As G is finite, character theory applies, so some characters are constructed next. First, note that $N \cong (\mathbb{F}_q, +)$ via $x \mapsto \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$.

Fix a nontrivial character of $(\mathbb{F}_q, +)$ χ (ex. $\chi(x) = e^{2\pi i \mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}(x)/p}$). Then $(a\chi)(x) = \chi(ax)$ ranges over all characters of $(\mathbb{F}_q, +)$. This gives a natural action of conjugation by T (or B) on N. Namely the action to T on N produces an action on the characters:

$$(t\chi)(n) = \chi(t^{-1}nt)$$

when interpreting n as a matrix in N via the isomorphism $N \cong (F_q, +)$ given above. Certainly all non-trivial characters are in the orbit of T, hence the conjugacy classes split into the single trivial character and all other ones.

2.1.6 Non-trivial Characters of N Therefore, all non-trivial characters on N are conjugate.

The trivial one is important in a moment for finding the first irreducible representation.

Let χ_1, χ_2 be characters of $\mathbb{F}_q^*$, and define the character:

$$\chi = \chi_1 \otimes \chi_2 : \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mapsto \chi_1(a)\chi_2(b)$$

This is a character of T, and may be regarded as a character on B trivial on N via the quotient:

$$B \to B/N \cong T$$

This process is given a name:

Definition 2.1.7: Inflation

Let $H \trianglelefteq G$ be a normal subgroup and (π, V) a representation of G/H. Then the *inflation* of π from G/H to G is the representation $\tilde{\pi} : G \to \mathrm{GL}(V)$ such that $\tilde{\pi}(g) = \pi(\overline{g})$, where $\overline{g} = gH$; that is, it is constant on the cosets of H.

After inflating from T to B, form the induced representation of $\mathrm{GL}_2(\mathbb{F}_q) = G$

$$\mathrm{Ind}_B^G \chi$$

and consider its irreducible components:

Lemma 2.1.8: Irreducible Components of G

Let π be an irreducible representation of G. Then the following are equivalent:

($\Rightarrow$) π is equivalent to a G-subspace of $\mathrm{Ind}_B^G \chi$ for a character χ of T seen as a character on B trivial on N.

($\Rightarrow$) π contains the trivial character of N.

***Proof*:**
First, π contains the trivial character of N if and only if it is contains an irreducible representation σ of B containing to the trivial character of N. Indeed, restrict G to B, $\mathrm{Res}_B^G \pi$ and then restrict further to N

$$\mathrm{Res}_N^B(\mathrm{Res}_B^G \pi)$$

By Maschke's theorem breakdown $\mathrm{Res}_B^G \pi$ into irreducibles. Call this irreducible representation of B σ. As Res is additive over direct sums, we see the above if and only if holds. Now suppose σ contains the trivial character of N. The space of N-fixed vectors in σ is then non-zero, and it is B-stable because N is normal in B. By irreducibility it is all of σ, so N acts trivially and σ factors through $B/N \cong T$. As T is abelian and σ is irreducible, σ is one-dimensional, and so σ is the inflation of a character of T. The converse is immediate, as an inflation from T is trivial on N. By Frobenius reciprocity:

$$\mathrm{Hom}_G(\mathrm{Ind}_B^G \chi, \pi) \cong \mathrm{Hom}_B(\chi, \mathrm{Res}_B^G \pi)$$

As π is irreducible we can apply the same ideas as Schur's Lemma to finish the proof.

By the lemma, analyze the induced representations $\mathrm{Ind}_B^G \chi$. To start, define the character of T:

$$\chi^w : \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mapsto \chi_2(a)\chi_1(b)$$

where w is the permutation in G swapping columns. Abusing notation, this is a character of B that is trivial on N.

Proposition 2.1.9: Analyzing Inflation of Torus Character

Let χ, ξ be character of T viewed as characters of B trivial on N. Then:

1. The space $\mathrm{Hom}_G(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\xi))$ is trivial, unless $\chi = \xi$ or $\chi = \xi^w$.
2. The space

$$\mathrm{Hom}_G(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\chi)) \qquad \text{and} \qquad \mathrm{Hom}_G(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\chi^w))$$

have the same dimension: it is 2 if $\chi = \chi^w$ and 1 otherwise.

***Proof*:**
We use the canonical isomorphism of Frobenius Reciprocity

$$\mathrm{Hom}_G(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\xi)) \cong \mathrm{Hom}_B(\mathrm{Res}_B^G(\mathrm{Ind}_B^G(\chi)), \xi) \tag{2.2}$$

The restriction-induction formula of Mackey gives

$$\mathrm{Res}_B^G \mathrm{Ind}_B^G(\chi) \cong \sum_{y \in B\backslash G/B} \mathrm{Ind}_{B\cap B_y}^B \left(\mathrm{Res}^{B_y \cap B}(\chi^y)\right), \tag{2.3}$$

where χ^y denotes the character $b \mapsto \chi(yby^{-1})$ of the group $B^y = y^{-1}By$. By Bruhat decomposition we only have to consider the cases $y = 1$ and $y = w$. The term in (2.3) corresponding to $y = 1$ is just χ, and so contributes a factor $\mathrm{Hom}_T(\chi, \xi)$ to (2.2). The term corresponding to w contributes

$$\mathrm{Hom}_B(\mathrm{Ind}_T^B(\chi^w), \xi) \cong \mathrm{Hom}_T(\chi^w, \xi)$$

Altogether, (2.2) reduces to

$$\mathrm{Hom}_B(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\xi)) \cong \mathrm{Hom}_T(\chi, \xi) \oplus \mathrm{Hom}_T(\chi^w, \xi),$$

and the result follow from Schur's lemma, completing the proof.

Corollary 2.1.10: Irreducible Representations Given Torus Character

Let χ be a character of T viewed as a character of B trivial on N. Then:

1. The representation $\mathrm{Ind}_B^G \chi$ is irreducible if and only if $\chi \neq \chi^w$
2. If $\chi = \chi^w$, then $\mathrm{Ind}_B^G \chi$ has length 2 with distinct component factors

***Proof*:**
The preceding proposition computes $\mathrm{End}_G(\mathrm{Ind}_B^G \chi) \cong \mathrm{Hom}_T(\chi, \chi) \oplus \mathrm{Hom}_T(\chi^w, \chi)$. The first summand is always one-dimensional, and the second is one-dimensional if $\chi = \chi^w$ and zero otherwise, since two characters of T admit a non-zero intertwiner only when they are equal.

If $\chi \neq \chi^w$ the endomorphism algebra is therefore $\mathbb{C}$. By Maschke's theorem $\mathrm{Ind}_B^G \chi$ is a direct sum of irreducibles, and if there were more than one summand, the projection onto one of them

would be an idempotent endomorphism outside $\mathbb{C}$. Hence $\mathrm{Ind}_B^G\chi$ is irreducible, which is (1).

If $\chi = \chi^w$ the endomorphism algebra is two-dimensional, so $\mathrm{Ind}_B^G\chi$ is reducible. By Maschke's theorem it is a direct sum of irreducibles, and the endomorphism algebra of such a sum has dimension $\sum_i m_i^2$, where the m_i are the multiplicities of its distinct irreducible constituents. As the m_i are positive integers, $\sum_i m_i^2 = 2$ forces exactly two constituents with $m_1 = m_2 = 1$, giving (2) and completing the proof.

Example 2.1.11: Induced Character of Trivial Character

Consider the trivial character 1_B of B. The trivial character 1_G occurs in $\mathrm{Ind}_B^G 1_B$, so

$$\mathrm{Ind}_B^G 1_B = 1_G \oplus \mathrm{St}_G,$$

for a unique irreducible representation St_G, called the Steinberg representation. Clearly, $\dim \mathrm{St}_G = q$ since $[G : B] = q + 1$ and $\dim 1_G = 1$.

These irreducible representations are given a name:

Definition 2.1.12: Principal Series Representations

The representation Ind_B^G of a character χ of T is called the *principal series*. The term principal series is often extended to include the irreducible components, namely all irreducible representations of G containing the trivial character of N.

Corollary 2.1.13: Counting Principal Series

Up to isomorphism, the group $\mathrm{GL}_2(F)$ has precisely $\frac{1}{2}(q^2 + q) - 1$ irreducible representations which contain the trivial character of N.

***Proof*:**

This is a counting argument: if $\chi = \chi^w$, there are $2(q-1)$ such characters (2 for the length 2). If they are not equal there are $\frac{1}{2}\cdot[(q-1)^2-(q-1)]$ (where we half since $\mathrm{Ind}_B^G\chi_1\otimes\chi_2) = \mathrm{Ind}_B^G\chi_2\otimes\chi_1)$). Adding them gives the desired result.

Notice this number is less than $q^2 - 1$, meaning not all irreducible representations are of this form. In particular there are $\frac{1}{2}(q^2 - q)$ such irreducible representations. An irreducible representation need not be trivial on N; such representations are given a name:

Definition 2.1.14: Cuspidal Representation

An irreducible representation of $\mathrm{GL}_2(F)$ not containing the trivial character of N is called a *cuspidal representation.*

A cuspidal representation σ must contain some non-trivial character of N. As it is non-trivial on N, any two non-trivial characters of N are B-conjugate (see box 2.1.6), so σ contains all non-trivial characters of N. This adds complexity, in particular the cuspidal representations cannot be constructed directly by induction.

To to that end, let $L/\mathbb{F}_q$ be a quadratic extension so that it has q^2 elements and L^* has q^2-1 elements. Let $F=\mathbb{F}_q$ to simply notation. The characters θ of L^* correspond to the (q^2-1)-th roots of unity, so there are q^2-1 of them. A character is called regular if it is *not* invariant under the Frobenius automorphism, that is $\theta^q \neq \theta$, where $\theta^q(x)=\theta(x^q)$. A character $\theta_j(g^k)=\omega^{jk}$ is *not regular* if $q+1$ divides j, as can be quickly verified. There are $(q^2-1)/(q+1)=q-1$ non-regular characters, and hence $q(q-1)$ regular characters. Choose a basis of L so that $L \cong_F F \oplus F$ and $\mathrm{GL}_2(F)$ is associated with $\mathrm{Aut}_F(L)$. There is a natural action L^* on L, which corresponds to a subgroup of $\mathrm{GL}_2(F) \cong \mathrm{Aut}_F(L)$. Identify:

$$L^* \cong E \subseteq \mathrm{GL}_2(F)$$

Importantly, note that an element of G with irreducible characteristic polynomial is conjugate to an element of E. For let $g \in G$ have ch_g irreducible. Every irreducible monic quadratic over $\mathbb{F}_q$ splits in the quadratic extension L, so ch_g has a root $\lambda \in L^\times = E$, and the other root is its Galois conjugate λ^q. The characteristic polynomial of λ, viewed in $\mathrm{GL}_2(\mathbb{F}_q)$, is therefore ch_g itself, so g and λ are conjugate by proposition 2.1.4(1).

Now, let θ be a regular character of E and ψ a non-trivial character of N. Define θ_ψ of ZN as:

$$\theta_\psi : \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} u \mapsto \theta(a)\psi(u)$$

Note that $\mathrm{Ind}_{ZN}^G \theta_\psi$ is independent of choice of ψ up to equivalence (see box 2.1.6).

Theorem 2.1.15: Cuspidal Representation of Finite Matrix Group

Let θ be a regular character of E and ψ a non-trivial character of N.

1. The virtual representation
$$\pi_\theta = \mathrm{Ind}_{ZN}^G \theta_\psi - \mathrm{Ind}_E^G \theta$$
is an irreducible cuspidal representation of G of dimension $q-1$.
2. Let θ_1, θ_2 be regular characters of E; then $\pi_{\theta_1} \cong \pi_{\theta_2}$ if and only if $\theta_2 = \theta_1$ or $\theta_2 = \theta_1^q$.
3. Every irreducible cuspidal representation of G is of the form π_θ, for some regular character θ of E.

***Proof*:**

1. As we are working with finite groups, representations are characterized by characters. The character of the virtual representation π_θ is given by the difference of the characters of the two induced representations:
$$\mathrm{tr}\pi_\theta(g) = \mathrm{tr}(\mathrm{Ind}_{ZN}^G \theta\psi)(g) - \mathrm{tr}(\mathrm{Ind}_E^G \theta)(g)$$
We use the general formula for the character of an induced representation, namely if (σ, W) is a representation of a subgroup $H \leq G$, then the character of $\Pi = \mathrm{Ind}_H^G \sigma$ is:
$$\mathrm{tr}(\Pi(g)) = \frac{1}{|H|} \sum_{x \in G \text{ s.t. } x^{-1}gx \in H} \sigma(x^{-1}gx)$$

We compute this for each part and for each type of conjugacy class in $G = \mathrm{GL}_2(\mathbb{F}_q)$. First take $\Pi_1 = \mathrm{Ind}_{ZN}^G \theta\psi$ so $H = ZN$ with $|H| = (q-1)q$. The character is $\sigma_1 = \theta\psi$.

- Let $g = z \in Z$. Since z is central, $x^{-1}zx = z$ for all $x \in G$. The element z is in ZN, so the condition is always met.

$$\mathrm{tr}(\Pi_1(z)) = \frac{1}{(q-1)q}\sum_{x\in G}\theta\psi(z) = \frac{|G|}{(q-1)q}\theta(z) = \frac{(q^2-1)(q^2-q)}{(q-1)q}\theta(z) = (q^2-1)\theta(z)$$

- Let $g = zu$ with $z \in Z, u \in N, u \neq 1$. An element with a repeated eigenvalue is only conjugate to other elements with repeated eigenvalues. The conjugates of g that lie in ZN are only those of the form $z'u'$. A more detailed analysis of the character sum shows that the sum over the unipotent parts averages to -1 when paired against a non-trivial character ψ. This yields:

$$\mathrm{tr}(\Pi_1(zu)) = -\theta(z)$$

- Let $g = y \in E \setminus Z$. The characteristic polynomial of y is irreducible over k. Every element of ZN has its eigenvalues in k, so its characteristic polynomial is reducible. Thus, no conjugate of y can be in ZN. The sum in the character formula is empty.

$$\mathrm{tr}(\Pi_1(y)) = 0$$

Next, consider $\Pi_2 = \mathrm{Ind}_E^G \theta$. Here $H = E$ is a subgroup of order q^2-1. The character is $\sigma_2 = \theta$.

- Let $g = z \in Z$. Since $z \in F^\times \subset E$, the condition $x^{-1}zx \in E$ is always met.

$$\mathrm{tr}(\Pi_2(z)) = \frac{1}{q^2-1}\sum_{x\in G}\theta(z) = \frac{|G|}{q^2-1}\theta(z) = (q^2-q)\theta(z)$$

- Let $g = zu$ with $u \neq 1$. The element g has a repeated eigenvalue. No conjugate of g can have an irreducible characteristic polynomial, so no conjugate of g lies in $E \setminus Z$. Conjugation into Z is impossible for $u \neq 1$. The sum is empty.

$$\mathrm{tr}(\Pi_2(zu)) = 0$$

- Let $g = y \in E \setminus Z$. The centralizer of y in G is E. The conjugacy class of y intersects E in exactly two elements: y and its Galois conjugate y^q. So the induced-character sum $\frac{1}{|E|}\sum_{x\,:\,x^{-1}yx\in E}\theta(x^{-1}yx)$ has $|E|$ terms contributing $\theta(y)$ and $|E|$ contributing $\theta(y^q)$, giving:

$$\mathrm{tr}(\Pi_2(y)) = \theta(y) + \theta(y^q) = \theta(y) + \theta^q(y)$$

We now subtract the character values to get:

- For $g = z \in Z$:

$$\mathrm{tr}\pi_\theta(z) = (q^2-1)\theta(z) - (q^2-q)\theta(z) = (q-1)\theta(z)$$

- For $g = zu$, $u \neq 1$:

$$\mathrm{tr}\pi_\theta(zu) = -\theta(z) - 0 = -\theta(z)$$

- For $g = y \in E \setminus Z$:

$$\mathrm{tr}\pi_\theta(y) = 0 - (\theta(y) + \theta^q(y)) = -(\theta(y) + \theta^q(y))$$

This completes the derivation of the character:

Conjugacy Class Representative g	**Character Value** $\mathrm{tr}\pi_\theta(g)$
$z \in Z$	$(q-1)\theta(z)$
zu where $z \in Z, u \in N, u \neq 1$	$-\theta(z)$
$y \in E \setminus Z$	$-(\theta(y) + \theta^q(y))$
g not conjugate to an element of $ZN \cup E$	0

Now, we can take the inner product of the character with itself to get:

$$\frac{1}{|G|}\sum_{g\in G} |\mathrm{tr}\pi_\theta(g)|^2 = 1$$

Now, recall that a representation V is irreducible if given the basis of characters and the inner product with itself, we get 1. In this case, we may have gotten a virtual character, that is the coefficient was negative. This is checked not to be the case by computing that

$$\mathrm{tr}\pi_\theta(e) = q - 1 = \dim(V)$$

If it were virtual, this would be negative, showing it is indeed a character and furthermore π_θ is an irreducible representation. Next, as

$$\sum_{u\in N} \mathrm{tr}\pi_\theta(u) = 0$$

we see that $\pi_\theta|_N$ does not contain the trivial character of N, and hence π_θ is cuspidal.

2. Two representations are isomorphic if they have the same character. It is easy to see from the character table above that $\pi_{\theta_1} \cong \pi_{\theta_2}$ if and only if $\theta_2 = \theta_1$ or $\theta_2 = \theta_1^q$.

3. As L^* has $q^2 - 1$ characters and $q - 1$ of them are non-regular, there are $q^2 - q$ regular characters. By part (2), π_θ and π_{θ^q} are isomorphic, so the regular characters give $(q^2-q)/2$ distinct cuspidal representations. Combining with corollary 2.1.13 gives the total of $\frac{q^2-q}{2} + \frac{q^2+q}{2} - 1 = q^2 - 1$ irreducible representations, and hence we found all irreducible representations by lemma 2.1.5, completing the proof.

The relation between characters and determinants recorded below is presented as an exercise in the source, and it is worth carrying here:

Example 2.1.16: Character and Determinants
Let ψ be a non-trivial character of N, and consider the representation

$$\Pi = \mathrm{Ind}_N^G \psi$$

σ be an irreducible representation of G. Show that:

1. If $\sigma = \varphi \circ \det$, for a character φ of $\mathbb{F}_q^\times$, then σ does not occur in Π.
2. Otherwise, σ occurs in Π with multiplicity one.

2.2 Linear Groups over Local Fields

Given the additional structure, we have some more interesting decompositions. We have B, N, T, and Z as before as these are defined for any field F. Note that $B/N \cong T$ and $B \cong T \ltimes N$ are homeomorphisms.

Proposition 2.2.1: Iwasawa Decomposition

Let B be the Borel subgroup and $K = \mathrm{GL}_2(\mathfrak{o})$. Then:

$$G = BK$$

***Proof*:**
Take $g \in G$; if the $(2,1)$-entry of g is zero, then $g \in B$. Otherwise, post-multiplying by the permutation matrix $w \in K$ if necessary, we can assume $v_F(g_{21}) \geqslant v_F(g_{22})$. We can then post-multiply by a lower triangular matrix in K to achieve $g_{21} = 0$.

Corollary 2.2.2: Compact Quotient

The quotient space

$$B \setminus G$$

is compact

***Proof*:**
$B \setminus G$ is the continuous image of the compact subgroup $K = \mathrm{GL}_2(\mathfrak{o})$.

Proposition 2.2.3: Cartan Decomposition

Let ϖ be a prime element of F. Then the matrices:

$$\begin{pmatrix} \varpi^a & 0 \\ 0 & \varpi^b \end{pmatrix} \qquad a, b \in \mathbb{Z},\ a \leq b$$

form a set of representatives for the coset space $K \backslash G / K$.

***Proof*:**
Permuting rows and columns, using the permutation matrix in K, we can arrange for the largest entry of g (in absolute value) to be in the $(1,1)$-place. Multiplying by elementary matrices from K, we can then arrange for g to be diagonal, and unit factors can be absorbed into K. This gives

$$G = \bigcup_{a \leq b} K \begin{pmatrix} \varpi^a & 0 \\ 0 & \varpi^b \end{pmatrix} K$$

We have to prove this union is disjoint. That is, we have to recover the integers a, b from the coset KgK, where

$$g = \begin{pmatrix} \varpi^a & 0 \\ 0 & \varpi^b \end{pmatrix}$$

First, we have $a + b = v_F(\det h)$, for any $h \in KgK$. Next, the group index $(K : K \cap hKh^{-1})$ depends only on the coset KhK, and

$$(K : K \cap gKg^{-1}) = 1$$

if $b = a$, or

$$(K : K \cap gKg^{-1}) = (q+1)q^{b-a-1}$$

if $b > a$, as we sought to show.

Corollary 2.2.4: Countable Quotient

If K is a compact open subgroup of G, the set G/K is countable.

***Proof*:**
As observed under box 1.2.31, it is enough to show that G/K is countable for one choice of K: we take $K = GL_2(\mathfrak{o})$. The space $K\backslash G/K$ is certainly countable, and each double coset KgK contains only finitely many cosets $g'K$.

The Iwahori decomposition comes next. First, the standard Iwahori subgroup of $\mathrm{GL}_2(F)$ is the compact open subgroup $I \leqslant G$:

$$I = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, d \in U_F,\ b \in \mathfrak{o},\ c \in \mathfrak{p} \right\}$$

Let $N' = N^w$ denote the subgroup of lower triangular unipotent matrices of G. Then:

Proposition 2.2.5: Iwahori Decomposition

If I is the Iwahori subgroup, then:

$$I = (I \cap N')(I \cap T)(I \cap N)$$

More precisely, the product map

$$I \cap N' \times I \cap T \times I \cap N \longrightarrow I$$

is bijective, and a homeomorphism, for any ordering of the factors on the left hand side.

***Proof*:**

Write an element of I as $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a, d \in U_F$, $b \in \mathfrak{o}$ and $c \in \mathfrak{p}$. Since a is a unit, the identity

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ ca^{-1} & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & d - ca^{-1}b \end{pmatrix} \begin{pmatrix} 1 & a^{-1}b \\ 0 & 1 \end{pmatrix}$$

exhibits the factorization. The three factors lie in the asserted groups: $ca^{-1} \in \mathfrak{p}$ so the first is in $I \cap N'$; $a \in U_F$ and $d - ca^{-1}b \in U_F$ because $ca^{-1}b \in \mathfrak{p}$ while $d \in U_F$, so the second is in $I \cap T$; and $a^{-1}b \in \mathfrak{o}$ so the third is in $I \cap N$.

Uniqueness is read off the same formula: the three entries ca^{-1}, $(a, d - ca^{-1}b)$ and $a^{-1}b$ are determined by the original matrix, so the product map is injective, and it is surjective by the computation above. Both the map and its inverse are given by polynomial expressions in the entries and in a^{-1}, hence are continuous, so it is a homeomorphism. Reversing the roles of N and N', which amounts to factoring out d instead of a, gives the statement for any ordering of the factors, completing the proof.

This is useful for the following. Let $K = GL_2(\mathfrak{o})$; under the canonical surjection $K \to GL_2(k)$, the image of I is the standard Borel subgroup of $GL_2(k)$. The Bruhat decomposition for $GL_2(k)$ (lemma 2.1.2) implies

$$K = I \cup IwI$$

Combining this with the Iwasawa decomposition for G gives the more symmetric double-coset decomposition

$$G = BI \cup BwI = B(I \cap N') \cup Bw(I \cap N)$$

The cosets BwI, BI are both open in G.

Furthermore, by letting $L = \mathfrak{o} \oplus \mathfrak{o}$, $L' = \mathfrak{o} \oplus \mathfrak{p}$, the Iwahori subgroup I is then the common G-stabilizer of the two lattices L, L'.

The decompositions above describe the shape of $\mathrm{GL}_2(F)$; what the rest of this section needs is the measure theory that goes with them. Throughout, F remains non-archimedean.

Proposition 2.2.6: Smooth Functions Over F

The vector-space $C_c^\infty(F)$ is spanned by indicator functions (characteristic functions) of sets $a + \mathfrak{p}^n$ for $a \in F$ and $n \in \mathbb{Z}$

***Proof*:**
Certainly the indicator function of $a + \mathfrak{p}^m$ lies in $C_c^\infty(F)$. Conversely, let $\varphi \in C_c^\infty(F)$. Since φ has compact support, there exists $n \in \mathbb{Z}$ such that $\operatorname{supp}\varphi \subset \mathfrak{p}^n$. Also, φ is fixed under translation by a compact open subgroup of F, hence by $\mathfrak{p}^m$, for some $m \in \mathbb{Z}$. Thus φ is a linear combination of characteristic functions of sets $a + \mathfrak{p}^m$, $a \in \mathfrak{p}^n/\mathfrak{p}^m$.

The Haar integral on the relevant groups is defined as follows. Let μ be the Haar measure on F (automatically left and right as F is abelian), and let Φ_0 denote the indicator function (or characteristic function) on $\mathfrak{o}$. Then as $\mathfrak{o}$ is compact:

$$\int_F \Phi_0(x)d\mu(x) = c_0 \qquad c \in \mathbb{R}_{>0}$$

If Φ_b is the indicator function on $a + \mathfrak{p}^b$ for $a \in F$ and $b \in \mathbb{Z}$, by translation invariance:

$$\int_F \Phi_b(x)d\mu(x) = c_0 q^{-b}$$

Since indicator functions span $C_c^\infty(F)$, this integrates any function. This integral has the following "homogeneity" property: if $\Phi \in C_c^\infty(F)$ and $y \in F^*$, then by the above equation it follows that:

$$\int_F \Phi(xy)d\mu(x) = \|y\|^{-1} \int_F \Phi(x)d\mu(x)$$

where $\|y\| = q^{-v_F(y)}$. Thus, define the Haar measure μ^* on F^* given by:

$$\mu^*(x) = \frac{\mu(x)}{\|x\|}$$

Then if $\Phi \in C_c^\infty(F^*)$, the function $x \mapsto \|x\|^{-1}\Phi(x)$ lies in $C_c^\infty(F)$ (and vanishes at 0), so:

$$\int_{F^\times} \Phi(x)d\mu^\times(x) = \int_F \Phi(x)\|x\|^{-1}d\mu(x), \qquad \text{for} \qquad \Phi \in C_c^\infty(F^\times)$$

and hence this certainly defines a Haar integral on F^*.

Viewing $M_2(F)$ under addition, its Haar measure is the product of four copies of F, with the measure given by the tensor product of 4 copies of a Haar measure on F. Let $G = \mathrm{GL}_2(F)$. Then:

Proposition 2.2.7: Haar measure on $\mathrm{GL}_2(F)$

Let μ be a Haar measure on $M_2(F)$. For $\Phi \in C_c^\infty(\mathrm{GL}_2(F))$, the function $x \mapsto \Phi(x)\|\det x\|^{-2}$ (vanishing on $M_2(F) \setminus \mathrm{GL}_2(F)$) lies in $C_c^\infty(M_2(F))$. The functional

$$\Phi \mapsto \int_A \Phi(x)\|\det x\|^{-2}\, d\mu(x), \quad \Phi \in C_c^\infty(G),$$

is a left and right Haar integral on $\mathrm{GL}_2(F)$, i.e. $\mathrm{GL}_2(F)$ is unimodular.

***Proof*:**
Let $A = M_2(F)$ and $G = \mathrm{GL}_2(F)$. Take $g \in G$ and consider the functionals

$$\Phi \mapsto \begin{cases} \int_A \Phi(gx)\, d\mu(x), \\ \int_A \Phi(xg)\, d\mu(x), \end{cases} \qquad \Phi \in C_c^\infty(A)$$

Each is a Haar integral on A and differs from the initial one by a positive constant (depending on g). To evaluate this constant, take Φ to be the indicator function of $\mathfrak{m} = M_2(\mathfrak{o})$. In the first instance, the function $x \mapsto \Phi(gx)$ is the characteristic function of the lattice $\mathfrak{m}' = g^{-1}\mathfrak{m}$. Thus

$$\int_A \Phi(gx)\, d\mu(x) = \mu(\mathfrak{m}') = \mu(\mathfrak{m}) \frac{[\mathfrak{m}' : \mathfrak{m} \cap \mathfrak{m}']}{[\mathfrak{m} : \mathfrak{m} \cap \mathfrak{m}']}$$

This quotient of indices depends only on the double coset KgK, $K = \mathrm{GL}_2(\mathfrak{o})$. Taking g in diagonal form via the Cartan decomposition (proposition 2.2.3), we get

$$\int_A \Phi(gx)\, d\mu(x) = \|\det g\|^{-2} \int_A \Phi(x)\, d\mu(x)$$

The second instance is treated in the same way to get

$$\int_A \Phi(xg)\, d\mu(x) = \|\det g\|^{-2} \int_A \Phi(x)\, d\mu(x)$$

The proposition then follows from simple manipulations. For example, if $\Phi \in C_c^\infty(G)$,

$$\begin{aligned} \int_A \Phi(xg) \|\det x\|^{-2}\, d\mu(x) &= \|\det g\|^{-2} \int_A \Phi(x) \|\det xg^{-1}\|^{-2}\, d\mu(x) \\ &= \int_A \Phi(x) \|\det x\|^{-2}\, d\mu(x), \end{aligned}$$

as we sought to show.

With a Haar measure on $\mathrm{GL}_2(F)$ in hand, turn to the relevant subgroups. Let N, T, B, Z be as in definition 2.1.1. The subgroup N, Z, and T, are taken care of as $N \cong F$, $Z \cong F^*$, and $T \cong F^* \times F^*$, leaving B. As $B \cong T \ltimes N$, define a linear functional on $C_c^\infty(B) := C_c^\infty(T) \otimes C_c^\infty(N)$ by:

$$\Phi \mapsto \int_T \int_N \Phi(tn) d\mu_T(t) d\mu_N(n)$$

where μ_T, μ_N are the Haar measure of T and N respectively induced by the aforementioned isomorphisms. Certainly, this functional is left B-invariant, and hence a left-Haar integral on B. By the above functional, take $\mu_B := \mu_T \otimes \mu_N$. Importantly, this tensor product does not commute, taking into account the semidirect product defining B. In terms of μ_B:

$$\Phi \mapsto \int_B \Phi(b) d\mu_B(b)$$

The two factors do not commute, which reflects the fact that the group B is not unimodular. Its module is:

Proposition 2.2.8: Module of Bruhat Group

The module δ_B of the group B is given by

$$\delta_B : tn \mapsto \|t_2/t_1\|, \quad n \in N, t = \begin{pmatrix} t_1 & 0 \\ 0 & t_2 \end{pmatrix} \in T \tag{2.4}$$

***Proof*:**
Setting

$$c = sm, \quad m \in N, \quad s = \begin{pmatrix} s_1 & 0 \\ 0 & s_2 \end{pmatrix} \in T,$$

we get

$$\int_B \Phi(bc)\, d\mu_B(b) = \int_T \int_N \Phi(ts\, s^{-1} ns\, m)\, d\mu_T(t) d\mu_N(n)$$

We use the obvious isomorphism $N \to F$ to identify μ_N with a certain Haar measure μ_F on F. For $\varphi \in C_c^\infty(N)$, we then have

$$\begin{aligned} \int_N \varphi(s^{-1} ns)\, d\mu_N(n) &= \int_F \varphi\left(\begin{pmatrix} 1 & s_1^{-1} x s_2 \\ 0 & 1 \end{pmatrix}\right) d\mu_F(x) \\ &= \|s_1 s_2^{-1}\| \int_N \varphi(n)\, d\mu_N(n) \end{aligned}$$

By definition,

$$\int_B \Phi(bc)\, d\mu_B(b) = \delta_B(c)^{-1} \int_B \Phi(b)\, d\mu_B(b),$$

completing the proof.

This defines the desired semi-invariant measure:

Corollary 2.2.9: Semi-Invariant Measure on $B\backslash G$

The space $C_c^\infty(B\backslash G, \delta_B^{-1})$ admits a positive semi-invariant measure $\dot{\mu}$, where δ_B is given by proposition 2.2.8. If $K = \mathrm{GL}_2(\mathfrak{o})$, there is a Haar measure μ_K on K such that

$$\int_{B\backslash G} f(g)\, d\dot{\mu}(g) = \int_K f(k)\, d\mu_K(k),$$

for $f \in C_c^\infty(B\backslash G, \delta_B^{-1})$.

***Proof*:**
The character δ_B is trivial on the compact group $K \cap B$. Restriction of functions is an isomorphism $C_c^\infty(B\backslash G, \delta_B^{-1}) \to C_c^\infty(K \cap B\backslash K, 1)$, where 1 denotes the trivial character of $K \cap B$. The semi-invariant measure $\dot{\mu}$ thus restricts to a semi-invariant measure on $C_c^\infty(K \cap B\backslash K, 1)$, but so does any Haar measure on K.

Note that μ_K is effectively the restriction of a Haar measure μ_G on G. It follows that there is a left

Haar measure μ_B on B such that

$$\int_G \varphi(g)\, d\mu_G(g) = \int_K \int_B \varphi(bk)\, d\mu_B(b) d\mu_G(k), \quad \varphi \in C_c^\infty(G) \tag{2.5}$$

showing that the measure and integral "commute" with the decomposition $G = BK$.

2.3 Representation of the Mirabolic Group

A lot of the representation of $\mathrm{GL}_2(F)$ can be simplified by first finding the representation of the following subgroup:

Definition 2.3.1: Mirabolic Subgroup

The group $M \subseteq \mathrm{GL}_2(F) = G$ given by:

$$M = \left\{ \begin{pmatrix} a & x \\ 0 & 1 \end{pmatrix} : a \in F^*,\ x \in F \right\}$$

is called the *mirabolic subgroup* of G. Letting $S = T \cap M \cong F^*$, then $M \cong S \ltimes N$.

My professor once said that mirabolic should be remembered as "miracle-parabolic" group, an apt mnemonic.

Start by analyzing the smooth representations of N, the source of the induction. Let (π, V) be a smooth representation and ϑ a character of N. Let $V(\vartheta)$ be the subspace generated by all $\pi(n)v - \vartheta(n)v$. This is an N subspace, and $V_\vartheta = V/V(\vartheta)$ is well-defined and is, by construction, the maximal N-quotient where the action of N acts via the character ϑ. If ϑ is trivial, then $V(\vartheta) = V(N)$ and so $V_\vartheta = V_N$.

The following lemma generalizes the test in Fourier analysis where $\int f(x) e^{-2\pi i k x}\, dx = 0$:

Lemma 2.3.2: Generalized Harmonic Decomposition

Let μ_N be a Haar measure on N and ϑ a character of N.

1. Let (π, V) be a smooth representation of N and $v \in V$. The vector v lies in $V(\vartheta)$ if and only if there is a compact open subgroup N_0 of N such that

$$\int_{N_0} \vartheta(n)^{-1} \pi(n) v\, d\mu_N(n) = 0 \tag{2.6}$$

2. The process $(\pi, V) \mapsto V_\vartheta$ is an exact functor from $\mathrm{Rep}(N)$ to the category of complex vector spaces.

Think of the integral as the generalized projection operator, $\int_{N_0} \vartheta(n)^{-1} \pi(n) v\, d\mu_N(n) = 0$. Multiplying by $\vartheta(n)^{-1}$ replaces the $e^{-2\pi i k x}$.

***Proof*:**

1. First assume that ϑ is the trivial character. The group $N \cong F$ is the union of an ascending sequence of compact open subgroups. So, if

$$v = \sum_{i=1}^{r} v_i - \pi(n_i)v_i \in V(N),$$

there is a compact open subgroup N_0 of N containing all n_i. Then the relation given by eq. (2.6) holds for this choice of N_0.

Conversely, let $v \in V$ and suppose the integral relation holds. Then there is an open normal subgroup N_1 of N_0 such that $v \in V^{N_1}$. The space V^{N_1} carries a representation of the finite group N_0/N_1. Therefore, in the obvious notation, $V^{N_1} = V^{N_1}(N_0/N_1) \oplus V^{N_0}$ and the map

$$w \mapsto \mu_N(N_0)^{-1} \int_{N_0} \pi(n)w \, d\mu_N(n), \quad w \in V^{N_1},$$

is the N_0-projection $V^{N_1} \to V^{N_0}$. This has kernel $V^{N_1}(N_0/N_1) \subset V(N)$. But then this proves (1) for the trivial character of N.

Now let ϑ be an arbitrary character of N, and consider the representation (π', V') of N, where $V' = V$ and $\pi'(n) = \vartheta(n)^{-1}\pi(n)$. Then $V(\vartheta) = V'(N)$ and so (1) follows in general.

2. right-exactness is clear as we are taking the quotient. For the other way, if $f : U \to V$ is injective we want that the induced $\overline{f}$ is injective. If $\overline{f}(v) = 0$, then for injectivity it must be in $V(\vartheta)$. As it is equal to $\overline{0}$ we get:

$$\int_{N_0} \vartheta(n)^{-1}\pi(n) \cdot (f(v)) \, d\mu_N(n) = 0$$

As f is an N-homomorphism, it commutes with the action given by π

$$\int_{N_0} \vartheta(n)^{-1} f(\pi(n) \cdot v) \, d\mu_N(n) = 0$$

and by linearity f comes out, and by injectivity of f we can drop f, and finally by part 1 we get that $v \in U(\vartheta)$ giving injectivity of $\overline{f}$, completing the proof.

Some important remarks. If (π, V) is a smooth representation of N (or of M), then $V(N)$ is an N- (or M-) subspace of V. The exact sequence

$$0 \to V(N) \to V \to V_N \to 0$$

gives an exact sequence

$$0 \to V(N)_N \to V_N \to V_N \to 0$$

in which the map $V_N \to V_N$ is the identity. Therefore

$$V(N)_N = 0 \quad \text{and} \quad V(N)(N) = V(N) \tag{2.7}$$

If $\vartheta \neq 1$, as N acts trivially on $V/V(N)$, one has $(V/V(N))_\vartheta = 0$ and so the inclusion $V(N) \to V$ induces an isomorphism

$$V(N)_\vartheta \cong V_\vartheta \tag{2.8}$$

Proposition 2.3.3: Character Invariant Subspace Never Whole Space

Let (π, V) be a smooth representation of N and let $0 \neq v \in V$. Then there exists a character ϑ such that $v \notin V(\vartheta)$.

***Proof*:**
We write

$$N_j = \begin{pmatrix} 1 & \mathfrak{p}^j \\ 0 & 1 \end{pmatrix}, \quad j \in \mathbb{Z}$$

Take $v \in V$, $v \neq 0$. We choose $j_0 \in \mathbb{Z}$ such that N_{j_0} fixes v. For $j \leq j_0$, let V_j denote the N_j-space generated by v. This is the direct sum of isotypic components $V_j^{\eta_j}$, as η ranges over the characters of N_j trivial on N_{j_0}. For each $j \leq j_0$, there exists η_j such that $V_j^{\eta_j} \neq 0$; by definition, we have

$$\int_{N_j} \eta_j(n)^{-1} \pi(n) v \, dn \neq 0$$

The N_{j-1}-space generated by $V_j^{\eta_j}$ is contained in V_{j-1}, so we may choose η_{j-1} such that $\eta_{j-1}|_{N_j} = \eta_j$. This satisfies the necessary properties to take the limit of these characters, and as N is a limit of the N_j we get that there exists a character ϑ of N such that, for all $j \leq j_0$, we have

$$\int_{N_j} \vartheta(n)^{-1} \pi(n) v \, dn \neq 0$$

Therefore $v \notin V(\vartheta)$, as required.

Corollary 2.3.4: Trivial If Trivial On Characters

Let (π, V) be a smooth representation of N. If $V_\vartheta = 0$ for all characters ϑ of N, then $V = 0$

***Proof*:**
This is the contrapositive of the preceding proposition. If $V \neq 0$, pick $0 \neq v \in V$; that proposition produces a character ϑ of N with $v \notin V(\vartheta)$, so the quotient $V_\vartheta = V/V(\vartheta)$ is non-zero. Hence if $V_\vartheta = 0$ for every character ϑ then $V = 0$, completing the proof.

Now let (π, V) be a smooth representation of M. Then:

Corollary 2.3.5: Trivial Representation of Mirabolic Group

Let (π, V) be a smooth representation of M. Suppose $V_N = 0$ and for some non-trivial character ϑ of N, $V_\vartheta = 0$. Then $V = 0$.

***Proof*:**
Now let (π, V) be a smooth representation of M. The space $V(N)$ is then an M-subspace of V and V_N carries a natural representation of $M/N = S$: On the other hand, $\pi(S)$ permutes transitively the subspaces $V(\vartheta)$, $\vartheta \neq 1$: for $s \in S$, $\pi(s) \cdot V(\vartheta) = V(\vartheta')$, where $\vartheta'(n) = \vartheta(s^{-1}ns)$. The corollary follows immediately.

Now fix a non-trivial character ϑ of N. Consider first $\mathrm{Ind}_N^M \vartheta$ and $\text{c-Ind}_N^M \vartheta$. As $N \trianglelefteq M$ and all non-trivial characters ϑ, ϑ' of N are conjugate, $\mathrm{Ind}_N^M \vartheta \cong \mathrm{Ind}_N^M \vartheta'$ for any two non-trivial characters ϑ, ϑ' of N, and likewise for the compactly induced representations.

Theorem 2.3.6: c-$\mathrm{Ind}_N^M \vartheta$ is Irreducible

Let ϑ be a non-trivial character of N and set $\mathcal{W} = \mathrm{Ind}_N^M \vartheta$, $\mathcal{W}^c = \text{c-Ind}_N^M \vartheta$. Let $\alpha : \mathcal{W} \to \mathbb{C}$ denote the canonical map $f \mapsto f(1)$.

1. $\mathcal{W}(N) = \mathcal{W}^c(N) = \mathcal{W}^c$ and $(\mathcal{W}/\mathcal{W}^c)(N) = 0$.
2. The map α induces isomorphisms $\mathcal{W}_\vartheta \cong \mathbb{C}$ and $\mathcal{W}_\vartheta^c \cong \mathbb{C}$.

As a consequence, $\text{c-Ind}_N^M \vartheta$ is irreducible over M.

***Proof*:**

1. Let $f \in \mathcal{W}$ and $n \in N$. For $a \in S$, we have $f(an) = \vartheta(ana^{-1})f(a)$. As f is smooth, there is an integer j such that N_j fixes f, and so $f(a) = 0$ if $\|\det a\|$ is sufficiently large. On the other hand, $f(an) = f(a)$ if $\|\det a\|$ is sufficiently small, so $nf - f$ vanishes at a if $\|\det a\|$ is sufficiently small. This implies that $nf - f \in \mathcal{W}^c$. Thus $\mathcal{W}(N) \subset \mathcal{W}^c$ and N acts trivially on $\mathcal{W}/\mathcal{W}^c$.

   Next, to prove that $\mathcal{W}^c(N) = \mathcal{W}^c$, let ψ be the character of F defined by

   $$\psi(x) = \vartheta\left(\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}\right)$$

   Take $a \in F^\times$, $j \in \mathbb{Z}_{\geq 1}$. Let $f_{a,j} \in \mathcal{W}^c$ be the function such that

   $$f_{a,j} : \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}\begin{pmatrix} au & 0 \\ 0 & 1 \end{pmatrix} \mapsto \psi(x), \quad u \in U_F^j,$$

   and which vanishes elsewhere. By similar arguments as done before, the functions $f_{a,j}$ span $\mathcal{W}^c$ over $\mathbb{C}$. We have

   $$\begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix} f_{a,j}\left(\begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix}\right) = \psi(bz) f_{a,j}\left(\begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix}\right)$$

   We deduce that $nf_{a,j}$ has the same support as $f_{a,j}$, $n \in N$. We can certainly find $x \in F$ such that the function $u \mapsto \psi(aux)$, $u \in U_F^j$, is constant, equal to c say, with $c \neq 1$. If $n = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$, then $nf_{a,j} - f_{a,j} = (c-1)f_{a,j}$, whence $f_{a,j} \in \mathcal{W}^c(N)$ and so $\mathcal{W}^c(N) = \mathcal{W}^c$. This implies $\mathcal{W}(N) = \mathcal{W}^c$.

2. The map α induces a surjection $\mathcal{W}_\vartheta \to \mathbb{C}$. On the other hand, since N acts trivially on $\mathcal{W}/\mathcal{W}^c$, the inclusion $\mathcal{W}^c \to \mathcal{W}$ induces an isomorphism $\mathcal{W}_\vartheta^c \cong \mathcal{W}_\vartheta$. To prove (2), therefore, it is enough to show that any $f \in \mathcal{W}^c$ with $f(1) = 0$ belongs to $\mathcal{W}^c(\vartheta)$. A function f, vanishing at 1, is a finite linear combination of functions $f_{a,j}$ with $a \notin U_F^j$, so it is enough to treat such functions. However, as $a \notin U_F^j$, there exists $x \in F$ such that the function

$u \mapsto \psi(aux) - \psi(x)$, $u \in U_F^j$, is a non-zero constant. Taking $n = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$, the same calculation as before shows that $nf_{a,j} - \vartheta(n)f_{a,j}$ is a non-zero constant multiple of $f_{a,j}$, whence $f_{a,j} \in \mathcal{W}^c(\vartheta)$. So $\ker \alpha \subseteq \mathcal{W}^c(\vartheta)$, and since α surjects onto $\mathbb{C}$ this gives $\mathcal{W}^c_\vartheta \cong \mathbb{C}$, which is (2).

Finally, for the irreducibility, let V be an M-subspace of $\mathcal{W}^c$. As $\mathcal{W}^c_N = 0$, the spaces $V_N, (\mathcal{W}^c/V)_N$ are both zero. The sequence

$$0 \to V_\vartheta \to \mathcal{W}^c_\vartheta \to (\mathcal{W}^c/V)_\vartheta \to 0$$

is exact. As $\dim \mathcal{W}^c_\vartheta = 1$, we conclude that $\dim V_\vartheta$ is 0 or 1. In the first case, $V = 0$ by corollary 2.3.5. In the second, $(\mathcal{W}^c/V)_\vartheta = 0$ and so $\mathcal{W}^c = V$, showing it is irreducible and completing the proof.

The following from the textbook [2] summarizes the key points:

1. A function $f \in W$ is determined by its restriction to $S \cong F^\times$. The restriction $f|_{F^\times}$ is a smooth function on $F^\times$.
2. A smooth function φ on $F^\times$ is of the form $\varphi = f|_{F^\times}$, for some $f \in W$, if and only if there exists $c > 0$ such that $\varphi(x) = 0$ for all x satisfying $\|x\| > c$.
3. A function $f \in W$ lies in W^c if and only if $f|_{F^\times} \in C_c^\infty(F^\times)$.

2.3.7 Lack of Irreducibility of Dual The above shows the representation $\mathrm{Ind}_N^M \vartheta$ is never irreducible, for any non-trivial character ϑ of N. The Duality Theorem (theorem 1.3.12) implies

$$(\text{c-Ind}_N^M \vartheta)^\vee \cong \mathrm{Ind}_N^M \check{\vartheta}$$

Thus $\text{c-Ind}_N^M \vartheta$ provides an example of an irreducible smooth representation with reducible contragredient.

The classification of *all* irreducible representations of the mirabolic group M can now be completed. The following theorem is the key technical tool for showing the list is complete. As before let ϑ be a non-trivial character of N. Let (π, V) be a smooth representation of M. Frobenius Reciprocity (theorem 1.2.21) gives a canonical isomorphism

$$\mathrm{Hom}_N(V, V_\vartheta) \cong \mathrm{Hom}_M(V, \mathrm{Ind}_N^M V_\vartheta)$$

Let $q : V \to V_\vartheta$ denote the quotient map and let $q_\star$ be the map $V \to \mathrm{Ind}_N^M V_\vartheta$ corresponding to q under this isomorphism: for $v \in V$,

$$q_\star(v) = (m \mapsto q(\pi(m)v))$$

Theorem 2.3.8: Invariant Space Equivalent to $\text{c-Ind}_N^M V_\vartheta$

Let ϑ be a non-trivial character of N and let (π, V) be a smooth representation of M. The M-homomorphism $q_\star : V \to \mathrm{Ind}_N^M V_\vartheta$ induces an isomorphism $V(N) \cong \text{c-Ind}_N^M V_\vartheta$.

***Proof*:**

$$(\mathrm{Ind}_N^M V_\vartheta)(N) = \text{c-Ind}_N^M V_\vartheta = (\text{c-Ind}_N^M V_\vartheta)(N)$$

Smooth induction does not commute with infinite direct sums, and V_ϑ need not be finite-dimensional, so the identity cannot be assembled from theorem 2.3.6 one summand at a time. The first equality is proved by the two inclusions below; the second then follows, since $(-)(N)$ commutes with $\bigoplus$.

For $\subseteq$, the first paragraph of the proof of theorem 2.3.6(1) applies verbatim to V_ϑ-valued f, since it uses only $f(a)\big(\vartheta(ana^{-1}) - 1\big) = 0$ pointwise. For $\supseteq$, compact induction *does* commute with arbitrary direct sums: a locally constant function whose support is compact mod N takes finitely many values, so it lands in a finite-dimensional subspace of V_ϑ. Writing V_ϑ as a direct sum of copies of ϑ then gives

$$\text{c-Ind}_N^M V_\vartheta = \bigoplus \text{c-Ind}_N^M \vartheta = \bigoplus \big(\text{c-Ind}_N^M \vartheta\big)(N) \subseteq (\mathrm{Ind}_N^M V_\vartheta)(N)$$

the middle equality again by theorem 2.3.6(1). The M-homomorphism $q_\star : V \to \mathrm{Ind}_N^M V_\vartheta$ surely maps $V(N)$ to $(\mathrm{Ind}_N^M V_\vartheta)(N) = \text{c-Ind}_N^M V_\vartheta$.

Let $W = \operatorname{Ker} q_\star \cap V(N)$ and $C = \text{c-Ind}_N^M V_\vartheta / q_\star(V(N))$. By lemma 2.3.2 the natural map $W_N \to V(N)_N$ is injective, so $W_N = 0$. Likewise, $(\text{c-Ind}\, V_\vartheta)_N$ is zero, so $C_N = 0$.

Now, the map $q_\star : V \to \text{c-Ind}\, V_\vartheta$ induces a map

$$q_{\star,\vartheta} : V_\vartheta = V(N)_\vartheta \to (\text{c-Ind}\, V_\vartheta)_\vartheta$$

By theorem 2.3.6(2), the canonical N-map $\mathrm{Ind}\, V_\vartheta \to V_\vartheta$ induces an isomorphism $\alpha_\vartheta : (\text{c-Ind}\, V_\vartheta)_\vartheta \to V_\vartheta$. The composite map $\alpha_\vartheta \circ q_{\star,\vartheta} : V_\vartheta \to V_\vartheta$ is the identity. However, the kernel of this map is W_ϑ and its cokernel is isomorphic to C_ϑ. As the spaces $W_N, W_\vartheta, C_N, C_\vartheta$ are all zero, by corollary 2.3.5 both W and C are trivial and $q_\star : V(N) \to \text{c-Ind}\, V_\vartheta$ is an isomorphism, completing the proof.

Theorem 2.3.9: Classification of Irreducible Representation of Mirabolic Group

Let (π, V) be an irreducible smooth representation of M. Either:

1. $\dim V = 1$ and π is the inflation of a character of $M/N \cong F^\times$
2. $\dim V$ is infinite and $\pi \cong \text{c-Ind}_N^M \vartheta$, for any character $\vartheta \neq 1$ of N.

In case (1), $\dim V_N = 1$ and $V_\vartheta = 0$ for $\vartheta \neq 1$. In case (2), $V_N = 0$ and $\dim V_\vartheta = 1$ for all $\vartheta \neq 1$.

***Proof*:**

If $V(N) = 0$, then N acts trivially on V. The group M/N is abelian, so proposition 1.2.37 implies $\dim V = 1$ and we are in case (1).

If $V(N) \neq 0$, then $V(N) = V$ and $V_N = 0$. Therefore $V_\vartheta \neq 0$ for all characters $\vartheta \neq 1$ of N, and so $\dim V$ is infinite. Theorem 2.3.8 implies that $V = V(N)$ is M-isomorphic to $\text{c-Ind}_N^M V_\vartheta$. The N-space V_ϑ is a direct sum of copies of ϑ, so V is a direct sum of copies of $\text{c-Ind}_N^M \vartheta$ and, since it is irreducible, $V \cong \text{c-Ind}_N^M \vartheta$.

2.4 Jacquet Modules and Induced Representations

We now classify the irreducible smooth representations of $\mathrm{GL}_2(F) = G$. In this section, we deal completely with *principal series representations*, that is irreducible representations where $V_N \neq 0$: This is the equivalent of looking to see if π contains the trivial character of N. The terminology comes from functional analysis when taking the spectrum of a topological group with the continuous spectrum associated to the principal (as in main part) series of characters and discrete spectrum's will be later associated to the cuspidal representations. It shall be the case that much of the important representation theory on the Galois side will be in relation to the cuspidal, but that cannot be properly studied without first grasping the whole picture of the representation of $\mathrm{GL}_2(F)$.

Definition 2.4.1: Principal Series Representation

Let (π, V) be an irreducible smooth representation of $\mathrm{GL}_2(F)$. Then it is called a *principal series* if $V_N \neq 0$.

When $V_N = 0$ these shall be called *cuspidal representations* or *supercuspidal representations*, which could be thought of as the "singular" case (and a singularity in the form of a cusp does appear when looking at the geometric picture of modular forms, for which see *Elliptic Curves* [6]), and are dealt with in chapter 3.

Let (π, V) be a smooth representation of $\mathrm{GL}_2(F)$ and $V(N)$ denote the subspace of V spanned by the vectors $v - \pi(x)v$, for $v \in V$ and $x \in N$. The space $V_N = V/V(N)$ inherits a representation π_N of $B/N = T$, which is smooth. This representation is given a name:

Definition 2.4.2: Jacquet Module

The representation (π_N, V_N) is called the *Jacquet module* of (π, V) at N.
This gives the *Jacquet functor*

$$\begin{aligned} \mathrm{Rep}(G) &\to \mathrm{Rep}(T), \\ (\pi, V) &\mapsto (\pi_N, V_N), \end{aligned}$$

which is exact and additive.

Let (σ, W) be a smooth representation of T. View σ as a smooth representation of B which is trivial on N, and form the smooth induced representation $\mathrm{Ind}_B^G \sigma$. It will often abbreviate $\mathrm{Ind}_B^G \sigma = \mathrm{Ind}\, \sigma$, since B and G are the only groups involved for most of this section. This is often called *parabolic induction*.

If (π, V) is a smooth representation of G, Frobenius Reciprocity (theorem 1.2.21) gives an isomorphism

$$\mathrm{Hom}_G(\pi, \mathrm{Ind}\, \sigma) \cong \mathrm{Hom}_B(\pi, \sigma)$$

However, σ is trivial on N so any B-homomorphism $\pi \to \sigma$ factors through the quotient map $\pi \to \pi_N$. Thus:

$$\mathrm{Hom}_G(\pi, \mathrm{Ind}\, \sigma) \cong \mathrm{Hom}_T(\pi_N, \sigma)$$

Proposition 2.4.3: Jacquet Module and Principal Series Representation

Let (π, V) be an irreducible smooth representation of G. The following are equivalent:

1. The Jacquet module V_N is non-zero.
2. The representation π is isomorphic to a G-subspace of a representation $\mathrm{Ind}_B^G \chi$, for some character χ of T.

***Proof*:**

(1)⇒(2) We have to show that the Jacquet module V_N admits an irreducible T- (or B-) quotient.

Choose $v \in V, v \neq 0$. Since V is irreducible over G, any element of V is a finite linear combination of translates $\pi(g)v$ of v, for various $g \in G$. Write $K = \mathrm{GL}_2(\mathfrak{o})$. The vector v is fixed by a subgroup K' of K of finite index; let $\{v_1, v_2, \ldots, v_r\}$ be the distinct elements of the form $\pi(k)v, k \in K$. In particular, $r \leq (K : K')$. Since $G = BK$, the elements $v_1, \ldots, v_r$ generate V over B, and their images generate V_N over T.

Thus V_N is finitely generated as a representation of T. We choose a minimal generating set $\{u_1, \ldots, u_t\}, t \geq 1$, say. A standard Zorn's Lemma argument shows that V_N has a T-subspace U, containing $u_1, \ldots, u_{t-1}$, and maximal for the property $u_t \notin U$. Then U is a maximal T-subspace of V_N and V_N/U is an irreducible representation of T, hence a character .

(2)⇒(1) As $\mathrm{Hom}_G(\pi, \mathrm{Ind}\, \sigma) \cong \mathrm{Hom}_T(\pi_N, \sigma)$. we get

$$\mathrm{Hom}_T(\pi_N, \chi) \cong \mathrm{Hom}_G(\pi, \mathrm{Ind}\, \chi) \neq 0,$$

so $\pi_N \neq 0$.

Next, let μ_N be a Haar measure on N and let $t \in T$. The measure $S \mapsto \mu_N(t^{-1}St)$ is the Haar measure $\delta_B(t)\mu_N$:

$$\int_N f(txt^{-1})\, d\mu_N(x) = \delta_B(t) \int_N f(x)\, d\mu_N(x), \quad f \in C_c^\infty(N)$$

As before, let w denote the permutation matrix

$$w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

If σ is a smooth representation of T, form the representation $\sigma^w \colon t \mapsto \sigma(wtw^{-1})$, and view it as a representation of B which is trivial on N.

Let α_σ denote the canonical B-map $\mathrm{Ind}\, \sigma \to \sigma$ given by $f \mapsto f(1)$. It induces a canonical T-map $(\mathrm{Ind}\, \sigma)_N \to \sigma$, still denoted α_σ.

Theorem 2.4.4: Restriction-induction Lemma

Let (σ, U) be a smooth representation of T and set $(\Sigma, X) = \mathrm{Ind}_B^G \sigma$. There is an exact sequence of representations of T:

$$0 \to \sigma^w \otimes \delta_B^{-1} \to \Sigma_N \xrightarrow{\alpha_\sigma} \sigma \to 0$$

Moreover the kernel is realized concretely: writing $V = \ker(\alpha_\sigma \colon X \to U)$, one has $V_N \cong \sigma^w \otimes \delta_B^{-1}$.

First, a technical lemma:

Lemma 2.4.5: Technical Lemma

Recall that $G = B \cup BwN$. A function $f \in X$ satisfies $f(1) = 0$ if and only if $\operatorname{supp} f \subset BwN$, that is, if and only if there is a compact open subgroup N_0 of N (depending on f) such that $\operatorname{supp} f \subset BwN_0$.

***Proof*:**
If $\operatorname{supp} f \subset BwN$ then $f(1) = 0$, as $1 \notin BwN$. Conversely, suppose $f(1) = 0$; then f vanishes on all of B, since $f(b) = \sigma(b)f(1)$. Since f is G-smooth, such a function f vanishes on a set BN_0', where N_0' is some compact open subgroup of N'. The identity (for $x \neq 0$)

$$\begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix} = \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -x^{-1} & 0 \\ 0 & x \end{pmatrix} w \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix} \in Bw \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix}$$

implies that $\operatorname{supp} f \subset BwN_0$, for some compact open subgroup N_0 of N, as required.

***Proof*:**
(of theorem 2.4.4)

By definition, X is the space of G-smooth functions $f : G \to U$ such that $f(bg) = \sigma(b)f(g)$, $b \in B, g \in G$. The canonical map $\alpha_\sigma : X \to U$ amounts to restriction of functions to B. Set $V = \operatorname{Ker} \alpha_\sigma$. Thus V provides a smooth representation of B and there is an exact sequence

$$0 \to V_N \to X_N \to U \to 0$$

We have to identify the T-representation V_N.

Let $f \in V$; in view of the preceding lemma, we can define a function $f_N : T \to U$ by

$$f_N(x) = \int_N f(xwn)\, dn = \sigma(x) f_N(1), \quad x \in T$$

By lemma 2.3.2, the kernel of the map $f \mapsto f_N$ is $V(N)$, and so $f \mapsto f_N(1)$ gives a bijective map $V_N \to U$. Taking $t \in T$ and $f \in V$, we have

$$\begin{aligned} (tf)_N(x) &= \int_N f(xwnt)\, dn \\ &= \delta_B(t^{-1}) \int_N f(xt^w wn)\, dn = \delta_B^{-1}(t)(t^w f_N)(x) \end{aligned}$$

Thus $f \mapsto f_N(1)$ is a B-homomorphism $V \to \sigma^w \otimes \delta_B^{-1}$ inducing a T-isomorphism $V_N \cong \sigma^w \otimes \delta_B^{-1}$.

Proposition 2.4.6: Non-cuspidal Irreducible Representation Are Admissible

Let (π, V) be an irreducible smooth representation of $\mathrm{GL}_2(F)$ which is not cuspidal. The representation π is admissible.

***Proof*:**

By definition, $V_N \neq 0$. By proposition 2.4.3, π is equivalent to a sub-representation of $\mathrm{Ind}_B^G \chi$, for some character χ of T. It is enough, therefore, to prove that $\mathrm{Ind}\chi$ is admissible.

We fix a compact open subgroup K of G; shrinking it if necessary, we may assume that $K \subset K_0 = \mathrm{GL}_2(\mathfrak{o})$. The space X^K of K-fixed points in $\mathrm{Ind}\chi$ consists of the functions $f : G \to \mathbb{C}$ satisfying

$$f(bgk) = \chi(b)f(g), \quad b \in B, g \in G, k \in K \tag{2.9}$$

We have $G = BK_0$, so the set $B\backslash G/K$ is finite, and each double coset BgK supports, at most, a one-dimensional space of functions satisfying eq. (2.9). Thus X^K is finite-dimensional, as we sought to show.

Note that irreducible *cuspidal* representations of $\mathrm{GL}_2(F)$ are likewise admissible, and hence all irreducible representations of $\mathrm{GL}_2(F)$ are admissible; the cuspidal case is proposition 3.1.2 with theorem 3.1.3, and the two together give corollary 3.1.5. A more granular classification of the principal series follows. First recall from example 1.2.8 that smooth characters on $\mathrm{GL}_n(F)$ factored via the determinant and a character of F^*. Define the following:

Definition 2.4.7: Twist On Representation

Let (π, V) be a smooth representation of G and φ a character of F^*. A *twist* $\varphi\pi$ is a smooth representation $(\varphi\pi, V)$ of G given by:

$$\varphi\pi(g) = \varphi(\det g)\pi(g), \quad g \in G$$

Similarly for characters of T: if $\chi = \chi_1 \otimes \chi_2$ is a character of T and φ is a character of $F^\times$, put $\varphi \cdot \chi = \varphi\chi_1 \otimes \varphi\chi_2$. Inflating $\varphi \cdot \chi$ to a representation of B trivial on N gives $\varphi \cdot \chi = (\varphi \circ \det|_B) \otimes \chi$. It follows immediately that

$$\mathrm{Ind}_B^G(\varphi \cdot \chi) \cong \varphi \mathrm{Ind}_B^G \chi$$

This allows convenient adjustments to the character χ without changing the essential structure of the induced representation. With it in hand we can give a precise account of the structure of representations of the form $\mathrm{Ind}_B^G \chi$. The main step is:

Theorem 2.4.8: Irreducibility Criterion

Let $\chi = \chi_1 \otimes \chi_2$ be a character of T, and set $(\Sigma, X) = \mathrm{Ind}_B^G \chi$.

1. The representation (Σ, X) is reducible if and only if $\chi_1\chi_2^{-1}$ is either the trivial character or the character $x \mapsto \|x\|^2$ of $F^\times$.
2. Suppose that (Σ, X) is reducible. Then:
   (a) the G-composition length of X is 2;
   (b) one composition factor of X has dimension 1, the other is of infinite dimension;
   (c) X has a 1-dimensional G-subspace if and only if $\chi_1\chi_2^{-1} = 1$;
   (d) X has a 1-dimensional G-quotient if and only if $\chi_1\chi_2^{-1}(x) = \|x\|^2$, $x \in F^\times$.

This refines to a classification of the irreducible principal series representations.

Let:

$$V = \{f \in X : f(1) = 0\}$$

This is a B-subspace of X, and there is an exact sequence

$$0 \to V \to X \to C \to 0,$$

where the one-dimensional space $C = X/V$ carries the character χ of T. By the Restriction-Induction Theorem 2.4.4 gives $V_N \cong \delta_B^{-1}\chi^w$.

Proposition 2.4.9: Irreducible Kernel

Let W be the kernel $V(N)$ of the canonical map $V \to V_N$. The space W is irreducible over B.

First an important technical lemma:

Lemma 2.4.10: Representation of V

For $f \in V$, define a function $f_N \in C_c^\infty(N)$ by $f_N(n) = f(wn)$, $n \in N$. The map

$$\begin{aligned} V &\longrightarrow C_c^\infty(N), \\ f &\longmapsto f_N \end{aligned}$$

is an N-isomorphism.

***Proof*:**
As in lemma 2.4.5, the support of $f \in V$ is contained in $Bw\mathcal{N}$, for some compact open subgroup $\mathcal{N}$ of N. The assertions follow immediately (observing that the notation f_N here is not the same as that in theorem 2.4.4).

***Proof*:**
(of proposition 2.4.9)

We shall prove that W is irreducible as a representation of the mirabolic group M: recall by (2.7) we have $W_N = 0$ and $W = W(N)$. For $\varphi \in C_c^\infty(N)$ and $a \in F^\times$, we define $a\varphi \in C_c^\infty(N)$ by

$$a\varphi \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} = \chi_2(a)\varphi \begin{pmatrix} 1 & a^{-1}x \\ 0 & 1 \end{pmatrix}$$

This gives an action of $F^\times$ on $C_c^\infty(N)$ which we regard as an action of the group S of matrices $\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$. We combine this with the natural action of N to give $C_c^\infty(N)$ the structure of a smooth representation of M. With this structure, the map in lemma 2.4.10 is an M-isomorphism.

Let ϑ be a non-trivial character of N. The map $f \mapsto \vartheta f$ is a linear automorphism of $V = C_c^\infty(N)$ carrying $V(N)$ to $V(\vartheta)$. Since $V/V(N)$ is one dimensional, so is $\dim V_\vartheta = 1$. However, since N acts trivially on V_N, the inclusion $W \to V$ induces an isomorphism $W_\vartheta \cong V_\vartheta$, hence $W_\vartheta \cong \vartheta$. Then apply theorem 2.3.8 to get $W = W(N) \cong \text{c-Ind}_N^M \vartheta$ which, by theorem 2.3.6 is irreducible.

A direct consequence of the proposition:

Corollary 2.4.11: Length of Induced Representation by Character

As a representation of B or of M, $\text{Ind}_B^G \chi$ has composition length 3. Two of the composition factors have dimension one, and the third is of infinite dimension. In particular, the G-composition length of the representation $\text{Ind}_B^G \chi$ is at most 3.

***Proof*:**
Chain the two exact sequences already in play. The first is $0 \to V \to X \to C \to 0$ with C one dimensional, and the second is $0 \to W \to V \to V_N \to 0$ with $W = V(N)$. By theorem 2.4.4, $V_N \cong \delta_B^{-1}\chi^w$ is one dimensional. By proposition 2.4.9, W is irreducible over B; its proof gives more, namely irreducibility over M and the isomorphism $W \cong \text{c-Ind}_N^M \vartheta$, which is infinite dimensional.

Splicing the two sequences exhibits X with the composition series $W \subset V \subset X$, whose factors are W, V_N and C: one of infinite dimension and two of dimension one. So the B-length, and equally the M-length, is exactly 3.

A G-subspace is in particular a B-subspace, so any G-composition series refines to a B-one and the G-length is at most 3, completing the proof.

Proposition 2.4.12: Detecting one-dimensional Subrepresentations

The following are equivalent:

($\Rightarrow$) (1) $\chi_1 = \chi_2$

($\Rightarrow$) (2) X has a one-dimensional N-subspace

When these conditions hold,

1. X has a unique one-dimensional N-subspace X_0;
2. X_0 is a G-subspace of X, and it is not contained in V.

***Proof*:**
If (1) holds, by twisting we may as well take $\chi_1 = \chi_2 = 1$. The (non-zero) constant function then spans a one-dimensional G-subspace of X.

Conversely, let $f \in X$ span an N-stable subspace of dimension 1. Thus N acts on f by right translation as a character. The support of f is left-invariant under B and, since N acts on f by a character, right-invariant under N, so it is a union of B-N double cosets. By proposition 2.1.3 there are only two, B and BwN. It cannot be B itself: f is locally constant while B is the zero set of g_{21} and so has empty interior in G. Hence $\operatorname{supp} f$ is either G or BwN. The second case is impossible since if $f(1) = 0$, by lemma 2.4.5 the support of f is confined to BwN_0, for some compact open subgroup N_0 of N. Thus, $\operatorname{supp} f = G$ and f vanishes nowhere so that $f(1) \neq 0$, and $f \notin V$. Then the canonical N-map $X \to \mathbb{C} = X/V$ identifies the N-space $\mathbb{C}f$ with the trivial N-space $\mathbb{C}$. It follows that N fixes f under right translation. Take $x \in F^\times$, and consider the identity

$$w \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -x^{-1} & 0 \\ 0 & x \end{pmatrix} \begin{pmatrix} 1 & 0 \\ x^{-1} & 1 \end{pmatrix}$$

Taking x so that $\|x\|$ is sufficiently large, then f is fixed under right translation by $\begin{pmatrix} 1 & 0 \\ x^{-1} & 1 \end{pmatrix}$. So, as f is fixed by N, we have

$$f(w) = \chi_1(-1)\chi_1^{-1}(x)\chi_2(x)f(1),$$

for all $x \in F^\times$ of sufficiently large absolute value. Thus $\chi_1 = \chi_2 = \varphi$, say, and $f(g) = \varphi(\det g)f(1)$. We have proved (1) $\Leftrightarrow$ (2), and that the one-dimensional N-subspace is uniquely determined.

The proof of theorem 2.4.8 follows.

***Proof*:**
(of theorem 2.4.8)

Assume X is reducible. By corollary 2.4.11 its G-length is 2 or 3. It follows that X has either a finite-dimensional G-subspace or a finite-dimensional G-quotient: of the three B-composition factors that corollary supplies, only one is infinite-dimensional, so at most one G-composition factor can be infinite-dimensional. The G-length being at least 2, the bottom and top G-

composition factors are distinct, so at least one of them is assembled from the one-dimensional B-factors alone and is therefore finite-dimensional.

Assume the first alternative. Thus X has a one-dimensional N-subspace, and we are in the situation of proposition 2.4.12: X has a one-dimensional G-subspace L, and $\chi_1 = \chi_2 = \varphi$, say. Moreover, G acts on L as the character $\varphi \circ \det$ and $L \cap V = 0$. The quotient $Y = X/L$ is therefore B-isomorphic to V. If X has G-length 3, then Y has G-length 2. However, V has B-length 2 and a unique B-quotient, which is of dimension 1. This gives a G-quotient of Y on which G must act as a character $\varphi' \circ \det$. This would force $\varphi' \otimes \varphi'$ to appear as a factor in the Jacquet module $Y_N \cong \varphi \cdot \delta_B^{-1}$, which it cannot. Thus X has G-length 2 and we are in case (2)(c) of the Irreducibility Criterion (theorem 2.4.8).

In the other alternative, X has a finite-dimensional G-quotient. The representation $\check{X}$ therefore has a finite-dimensional G-subspace, and we are back with the first alternative. By the Duality Theorem (theorem 1.3.12), $\check{X} \cong \mathrm{Ind}_B^G \delta_B^{-1}\check{\chi}$, so we are in the case (2)(d)of the Irreducibility Criterion (theorem 2.4.8).

Thus, in part (1) of the theorem, we have shown that X reducible implies χ has the stated form. The converse is given by proposition 2.4.12 and the dual case. We have also proved statements (a) to (d), completing the proof.

Having fully determined how to get irreducible representations from inductions/inflation of characters, to get a classification of the irreducible non-cuspidal representations of G requires investigating the homomorphisms between induced representations:

Proposition 2.4.13: Homomorphism between induced Representations

Let χ, ξ be characters of T. The space $\mathrm{Hom}_G(\mathrm{Ind}_B^G \chi, \mathrm{Ind}_B^G \xi)$ has dimension 1 if $\xi = \chi$ or $\chi^w \delta_B^{-1}$, zero otherwise.

***Proof*:**
We use Frobenius Reciprocity:

$$\mathrm{Hom}_G(\mathrm{Ind}_B^G \chi, \mathrm{Ind}_B^G \xi) \cong \mathrm{Hom}_T((\mathrm{Ind}\, \chi)_N, \xi)$$

The Jacquet module $(\mathrm{Ind}\, \chi)_N$ fits into the exact sequence

$$0 \to \chi^w \delta_B^{-1} \to (\mathrm{Ind}\, \chi)_N \to \chi \to 0$$

If we assume that $\chi \neq \chi^w \delta_B^{-1}$, then this sequence splits and the result follows immediately. The equation $\chi = \chi^w \delta_B^{-1}$ amounts to $\chi_1(x) = \|x\|\chi_2(x)$, $x \in F^\times$. In this case, $\mathrm{Ind}\, \chi$ is irreducible and the result again follows.

By way of some examples, consider in more detail the case where $\mathrm{Ind}_B^G \chi$ is reducible. Thus there is a character φ of $F^\times$ such that $\chi = \varphi \cdot 1_T$ or $\chi = \varphi \cdot \delta_B^{-1}$. Twisting does not affect the situation materially, so take $\varphi = 1$. Consider first the case $\mathrm{Ind}_B^G 1_T$. The irreducible G-quotient of $\mathrm{Ind}_B^G 1_T$ is called the *Steinberg representation* of G, and is denoted St_G:

$$0 \to 1_G \to \mathrm{Ind}_B^G 1_T \to \mathrm{St}_G \to 0 \tag{2.10}$$

Its dimension is infinite and $(\mathrm{St}_G)_N \cong \delta_B^{-1}$. Likewise, for a character φ of $F^\times$ there is an exact

sequence

$$0 \to \varphi_G \to \operatorname{Ind}_B^G \varphi_T \to \varphi \cdot \operatorname{St}_G \to 0,$$

where $\varphi_G = \varphi \circ \det$ and $\varphi_T = \varphi \otimes \varphi$. (Representations of the form $\varphi \cdot \operatorname{St}_G$ are sometimes called *special*.)

Taking the smooth dual of (2.10) gives an exact sequence

$$0 \to \operatorname{St}_G^\vee \to \operatorname{Ind}_B^G \delta_B^{-1} \to 1_G \to 0 \tag{2.11}$$

The proposition implies

$$\operatorname{St}_G \cong \operatorname{St}_G^\vee \tag{2.12}$$

2.4.14 Remark The proposition also implies that the space $\operatorname{End}_G(\operatorname{Ind} 1_T)$ has dimension 1, while $\operatorname{Ind} 1_T$ is not irreducible. Thus the converse of Schur's Lemma fails in this context, as remarked in theorem 1.2.33.

Observe also the imperfect parallelism between the proposition above and the corresponding result (proposition 2.1.9) for the finite field case.

A new notation is convenient. For a smooth representation σ of T, define

$$\iota_B^G \sigma = \operatorname{Ind}_B^G(\delta_B^{-1/2} \otimes \sigma) \tag{2.13}$$

This provides another exact functor $\operatorname{Rep}(T) \to \operatorname{Rep}(G)$, known as *normalized* or *unitary smooth induction*. It gives rise to more convenient combinatorics, for example:

$$(\iota_B^G \sigma)^\vee \cong \iota_B^G \check{\sigma} \tag{2.14}$$

In this language, the Irreducibility Criterion (theorem 2.4.8) and proposition 2.4.13 say:

Lemma 2.4.15: Normalized Induction Properties

1. Let $\chi = \chi_1 \otimes \chi_2$ be a character of T. The representation $\iota_B^G \chi$ is reducible if and only if $\chi_1 \chi_2^{-1}$ is one of the characters $x \mapsto \|x\|^{\pm 1}$ of $F^\times$ or, equivalently, $\chi = \varphi \cdot \delta_B^{\pm 1/2}$ for some character φ of $F^\times$.
2. Let χ, ξ be characters of T. The space $\operatorname{Hom}_G(\iota_B^G \chi, \iota_B^G \xi)$ is not zero if and only if $\xi = \chi$ or $\xi = \chi^w$.

***Proof*:**

Both parts are the unnormalized statements obtained through the twist $\eta = \delta_B^{-1/2} \chi$ of (2.13), so the whole content is bookkeeping with δ_B. Since $\delta_B(t) = \|t_2/t_1\|$ by proposition 2.2.8, as a character of $T = F^\times \times F^\times$ we have $\delta_B^{-1/2} = \|-\|^{1/2} \otimes \|-\|^{-1/2}$, so writing $\eta = \eta_1 \otimes \eta_2$,

$$\eta_1 = \chi_1 \|-\|^{1/2}, \qquad \eta_2 = \chi_2 \|-\|^{-1/2}, \qquad \text{hence} \qquad \eta_1 \eta_2^{-1} = \chi_1 \chi_2^{-1} \|-\|$$

(1). By theorem 2.4.8(1) the representation $\iota_B^G \chi = \operatorname{Ind}_B^G \eta$ is reducible if and only if $\eta_1 \eta_2^{-1}$ is trivial or equal to $\|-\|^2$. By the display, that happens exactly when $\chi_1 \chi_2^{-1}$ equals $\|-\|^{-1}$ or $\|-\|$ respectively, which is the stated criterion. For the reformulation, $\chi = \varphi \cdot \delta_B^{\pm 1/2}$ says

$\chi_1\chi_2^{-1} = \|-\|^{\mp 1}$, the same condition.

(2). By proposition 2.4.13 the space $\mathrm{Hom}_G(\mathrm{Ind}_B^G\eta, \mathrm{Ind}_B^G\theta)$ is non-zero exactly when $\theta = \eta$ or $\theta = \eta^w\delta_B^{-1}$, where θ is the twist of ξ. The first case gives $\xi = \chi$. For the second, $\eta^w = \eta_2 \otimes \eta_1$ and $\delta_B^{-1} = \|-\| \otimes \|-\|^{-1}$, so

$$\eta^w\delta_B^{-1} = \left(\chi_2\|-\|^{-1/2}\|-\|\right) \otimes \left(\chi_1\|-\|^{1/2}\|-\|^{-1}\right) = \chi_2\|-\|^{1/2} \otimes \chi_1\|-\|^{-1/2}$$

Comparing with $\theta = \xi_1\|-\|^{1/2} \otimes \xi_2\|-\|^{-1/2}$ gives $\xi_1 = \chi_2$ and $\xi_2 = \chi_1$, that is $\xi = \chi^w$. So the normalized statement has the clean form claimed, with the δ_B twist absorbed, completing the proof.

Gathering the earlier arguments and results:

Theorem 2.4.16: Classification Theorem

The following is a complete list of the isomorphism classes of irreducible, non-cuspidal representations of G:

1. the irreducible induced representations $\iota_B^G\chi$, where $\chi \neq \varphi \cdot \delta_B^{\pm 1/2}$ for any character φ of $F^\times$;
2. the one-dimensional representations $\varphi \circ \det$, where φ ranges over the characters of $F^\times$;
3. the special representations $\varphi \cdot \mathrm{St}_G$, where φ ranges over the characters of $F^\times$.

The classes in this list are all distinct except that, in (1), $\iota_B^G\chi \cong \iota_B^G\chi^w$.

***Proof*:**

Proposition 2.4.3 places every irreducible non-cuspidal representation inside some $\mathrm{Ind}_B^G\chi$, and theorem 2.4.8 identifies the two factors when that induced representation is reducible. So every irreducible, non-cuspidal representation of G appears in this list. The relations between the irreducibly induced ones are given by proposition 2.4.13. The same result also implies that no special representation $\varphi \cdot \mathrm{St}_G$ can be equivalent to $\varphi' \cdot \mathrm{St}_G$, $\varphi' \neq \varphi$, or any irreducibly induced representation.

3

Cuspidal Representations

In the previous chapter, we classified the principal series representations. However, this left the *cuspidal* representations. These are the "atomic" particles of the theory; they cannot be constructed by inducing from parabolic subgroups, and they have no analogue in the representation theory of $\mathrm{GL}_n(\mathbb{C})$. They are the representations that carry the "actual" arithmetic information that is used in the Langlands Correspondence. The goal of this chapter is to define, analyze, and provide a general construction for cuspidal representations. Unlike the principal series, which are naturally constructed via the Bruhat decomposition, cuspidal representations are constructed using the compact open subgroups of G.

We start in section 3.1 by giving a more workable definition of cuspidal representations via their matrix coefficients (compact support modulo the center). In section 3.2, we introduce the machinery of *intertwining* and Hecke algebras for compact open subgroups. This leads to the main result of the chapter: a method to construct irreducible cuspidal representations by *compact induction* from representations of open subgroups that contain the center. Section 3.3 then introduces chain orders and *fundamental strata*, the invariant that stratifies these representations by their level of ramification. The last three sections carry that through: section 3.4 classifies the fundamental strata into four types, section 3.5 uses the simple ones to classify the cuspidal representations, and section 3.6 treats the representations with Iwahori-fixed vectors, which are exactly the ones this classification excludes.

3.1 Cuspidal Representations and Coefficients

Our exploration of non-principal series representations interestingly enough starts by revisiting the definition of the cuspidal representation. So far, we have defined it negatively: a smooth representation (π, V) of $\mathrm{GL}_2(F)$ is cuspidal if the Jacquet module (π_N, V_N) is trivial, or equivalently, if it is not isomorphic to a composition factor of an induced representation $\mathrm{Ind}_B^G \chi$. This definition, being the "negative" of non-principal series hides the great importance these representations have in

the Langlands program. Hence, we start by giving a more practical and algebraic definition.

The key insight from this new viewpoint is that irreducible cuspidal representations have an algebraic property: they are *projective objects* in the appropriate subcategory of $\mathrm{Rep}(G)$, as shall be elaborated on in appendix A.

Let (π, V) be a smooth representation of $G = \mathrm{GL}_2(F)$. From vectors $v \in V$ and $\tilde{v} \in \check{V}$, we get a smooth function on G by

$$\gamma_{\tilde{v}\otimes v} : g \mapsto \langle \tilde{v}, \pi(g)v \rangle$$

Let $C(\pi)$ be the vector space spanned by the functions $\gamma_{\tilde{v}\otimes v}$ for $\tilde{v} \in \check{V}, v \in V$. The functions $f \in C(\pi)$ are called the *(matrix) coefficients* of π. The space $\check{V} \otimes V$ carries a smooth representation of the group $G \times G$, while $G \times G$ acts on the function space $C(\pi)$ by translation: the first factor acts by left translation and the second by right translation. The map $\tilde{v} \otimes v \mapsto \gamma_{\tilde{v}\otimes v}$ is then a surjective $G \times G$-homomorphism $\check{V} \otimes V \to C(\pi)$.

The space $C(\pi)$ is primarily, but not exclusively, of interest in the case where π is irreducible. When π is irreducible, the centre Z of G acts on V via the central character ω_π of π and

$$\gamma(zg) = \omega_\pi(z)\gamma(g), \quad z \in Z, g \in G, \gamma \in C(\pi)$$

The support of a coefficient is therefore invariant under translation by Z.

Definition 3.1.1: γ-Cuspidal Representation

Let (π, V) be an irreducible smooth representation of G; one says that π is *γ-cuspidal* if every $\gamma \in C(\pi)$ is compactly supported modulo Z.

The term "γ-cuspidal" is convenient but temporary: it turns out to be equivalent to the standard definition. Some technical control comes first.

Proposition 3.1.2: Admissibility and γ-Cuspidality

1. If (π, V) is an irreducible γ-cuspidal representation of G, then π is admissible.
2. Let (π, V) be an irreducible admissible representation of G, and suppose that some non-zero coefficient of π is compactly supported modulo Z; then π is γ-cuspidal.

***Proof*:**

1. Suppose for a contradiction that π is not admissible. We choose a compact open subgroup K such that V^K has infinite dimension. This dimension is countable (by lemma 1.2.32). The dimension of $\check{V}^K \cong \mathrm{Hom}_{\mathbb{C}}(V^K, \mathbb{C})$ is therefore uncountable.

   We fix a non-zero $v \in V^K$ and consider the map $\Gamma_v : \check{V}^K \to C(\pi)$ given by $\tilde{v} \mapsto \gamma_{\tilde{v}\otimes v}$. Since the translates gv ($g \in G$) span V, the map Γ_v is injective. Its image is a space of functions f on G satisfying

   $$f(zkgk') = \omega_\pi(z)f(g), \quad g \in G, z \in Z, k, k' \in K,$$

   and supported on a finite union of cosets $ZKgK$ (due to the γ-cuspidal property). The dimension of $\Gamma_v(\check{V}^K)$ is therefore at most countable, while Γ_v is injective and $\dim \check{V}^K$ is uncountable. This gives the desired contradiction.

2. The smooth dual $(\check{\pi}, \check{V})$ is irreducible and admissible (proposition 1.2.41). We view the space $\check{V} \otimes V$ as a smooth representation of $G \times G$, and hence as a smooth module over $\mathcal{H}(G \times G) = \mathcal{H}(G) \otimes \mathcal{H}(G)$. If K is a compact open subgroup of G, we have

$$(\check{V} \otimes V)^{K \times K} = (e_K \otimes e_K) * (\check{V} \otimes V) = \check{V}^K \otimes V^K$$

If K is sufficiently small, the spaces $\check{V}^K, V^K$ are finite-dimensional simple modules over $\mathcal{H}(G, K)$, by [2, §4.3], which this book quotes. The Jacobson Density Theorem implies that $\check{V}^K \otimes V^K$ is a simple module over $\mathcal{H}(G, K) \otimes \mathcal{H}(G, K) \cong \mathcal{H}(G \times G, K \times K)$. This holds for all sufficiently small K, so $\check{V} \otimes V$ is an irreducible admissible $G \times G$-space.

The surjective $G \times G$-homomorphism $\gamma : \check{V} \otimes V \to C(\pi)$ is therefore an isomorphism and $C(\pi)$ is irreducible over $G \times G$. If $\gamma \in C(\pi)$ is non-zero, any $\gamma' \in C(\pi)$ is a finite linear combination of functions $(g, h)\gamma$ for $(g, h) \in G \times G$. If γ is compactly supported modulo Z, then so is γ'.

The terminology is justified by the equivalence with the standard definition.

Theorem 3.1.3: Equivalence of Cuspidal Definitions

Let (π, V) be an irreducible smooth representation of G. Then π is cuspidal if and only if it is γ-cuspidal.

***Proof*:**
We first assume that π is cuspidal ($V_N = 0$), and show it is γ-cuspidal. Let ϖ be a prime element of F, and put

$$t = \begin{pmatrix} \varpi & 0 \\ 0 & 1 \end{pmatrix}$$

The set T^+ of powers t^n, $n \geq 0$, provides a family of representatives for $ZK\backslash G/K$, where $K = \mathrm{GL}_2(\mathfrak{o})$ (by Cartan Decomposition, proposition 2.2.3).

Lemma 3.1.4: Vanishing of Coefficients

Let $v \in V, \tilde{v} \in \check{V}$. There exists $m \geq 0$ such that $\gamma_{\tilde{v} \otimes v}(t^n) = 0$ for all $n \geq m$.

***Proof*:**
(of Lemma) We choose a compact open subgroup N_1 of N which fixes $\tilde{v}$. Since $V_N = 0$, we have $v \in V(N)$ and by lemma 2.3.2 there is a compact open subgroup N_2 of N such that $\int_{N_2} \pi(x)v\,dx = 0$. We then have $\int_{N_0} \pi(x)v\,dx = 0$ for any compact open subgroup N_0 containing N_2. However, there exists $m \geq 0$ such that $t^a N_2 t^{-a} \subset N_1$ for all $a \geq m$. For such a we have (for certain positive constants k_1, k_2):

$$\begin{aligned}\langle \tilde{v}, \pi(t^a)v\rangle &= k_1 \int_{N_1} \langle \check{\pi}(x^{-1})\tilde{v}, \pi(t^a)v\rangle\,dx \\ &= k_1 \int_{N_1} \langle \check{\pi}(t^{-a})\tilde{v}, \pi(t^{-a}xt^a)v\rangle\,dx \\ &= k_2 \int_{t^{-a}N_1t^a} \langle \check{\pi}(t^{-a})\tilde{v}, \pi(x)v\rangle\,dx = 0,\end{aligned}$$

since $t^{-a}N_1t^a \supset N_2$.

Returning to the proof of the theorem, fix a non-zero coefficient $f = \gamma_{\tilde{v}\otimes v}$. Write $K = \mathrm{GL}_2(\mathfrak{o})$ and let K' be an open normal subgroup of K fixing both $\tilde{v}$ and v. Let $k_1, \ldots, k_r$ be coset representatives for K/K'. Thus, if $g \in G$, there exists $n \geq 0$ such that $ZKgK = ZKt^nK = \bigcup_{i,j} ZK'k_i^{-1}t^nk_jK'$. It follows that

$$\mathrm{supp} f \subset \bigcup_{1\leq i,j\leq r} ZK'(\mathrm{supp} f_{ij} \cap T^+)K',$$

where f_{ij} denotes the coefficient function $x \mapsto f(k_ixk_j^{-1})$. This set is compact modulo Z by the lemma. Thus π is γ-cuspidal.

Converse: Let π be irreducible and γ-cuspidal. By proposition 3.1.2, π is admissible. By proposition 1.2.41, the dual $(\check{\pi}, \check{V})$ is irreducible and admissible. Let $K_n = 1 + \mathfrak{p}^n M_2(\mathfrak{o})$. Take $v \in V$ and choose $n \geq 1$ so that v is fixed by $\pi(K_n)$. For $\tilde{v} \in \check{V}^{K_n}$, the function $g \mapsto \langle \tilde{v}, \pi(g)v\rangle$ is compactly supported modulo Z; we deduce that $\langle \tilde{v}, \pi(t^a)v\rangle = 0$ for all a sufficiently large. Since $\check{V}^{K_n}$ is finite dimensional, there is a constant c such that this holds for all $\tilde{v} \in \check{V}^{K_n}$ and all $a \geq c$. This implies $\pi(e_{K_n})\pi(t^a)v = 0$ for $a \geq c$. Writing $K_n = N_nT_nN_n'$ (Iwahori decomposition), we set $K_n^{(a)} = t^{-a}K_nt^a$. We have, for $a \geq c$:

$$0 = \pi(e_{K_n})\pi(t^a)v = \pi(t^a)\pi(e_{K_n^{(a)}})v = \pi(t^a) \sum_{x\in N_{n-a}/N_n} \pi(x)\pi(e_{K_n^{(a)}\cap K_n})v$$

However, v is fixed by $K_n^{(a)} \cap K_n$, so this reduces to a sum over N which vanishes. We deduce that $v \in V(N)$. This applies to all $v \in V$, so π is cuspidal.

Combining the previous results gives a structural fact for $\mathrm{GL}_2(F)$.

Corollary 3.1.5: Admissibility of Irreducible Representations

Every irreducible smooth representation of $G = \mathrm{GL}_2(F)$ is admissible.

***Proof*:**
If π is not cuspidal, it is admissible by proposition 2.4.6. If π is cuspidal, it is γ-cuspidal by theorem 3.1.3, and thus admissible by proposition 3.1.2.

3.2 Intertwining and Compact Induction

In this section, we describe a general method for constructing irreducible cuspidal representations of $G = \mathrm{GL}_2(F)$ using compact induction from open subgroups. While the formal framework applies to arbitrary locally profinite groups, its power lies in the specific application to the filtrations of compact subgroups in G.

3.2.1 General Context Throughout this section, G denotes a unimodular locally profinite group satisfying box 1.2.31 and having a countable descending basis of compact open subgroups at the identity, the latter being what proposition 1.3.2 needs to produce the Haar measure used below. Z denotes the centre of G. For a compact open subgroup K, $\widehat{K}$ denotes the set of isomorphism classes of irreducible smooth representations of K.

Definition 3.2.2: Intertwining

For $i = 1, 2$, let K_i be a compact open subgroup of G and let $\rho_i \in \widehat{K}_i$. Let $g \in G$. The element g *intertwines* ρ_1 with ρ_2 if

$$\mathrm{Hom}_{K_1^g \cap K_2}(\rho_1^g, \rho_2) \neq 0,$$

where $K_1^g = g^{-1}K_1 g$ and ρ_1^g denotes the representation $x \mapsto \rho_1(gxg^{-1})$ of K_1^g.

This property depends only on the double coset $K_1 g K_2$. The representations ρ_1 and ρ_2 are said to *intertwine in* G if there exists some $g \in G$ which intertwines them. For a single pair (K, ρ), the element g intertwines ρ if it intertwines ρ with itself.

Definition 3.2.3: Occurrence

Let K be a compact open subgroup of G, and let (π, V) be a smooth representation of G. The representation π is said to *contain* ρ, or ρ *occurs* in π, if

$$\mathrm{Hom}_K(\rho, \pi) \neq 0$$

These definitions extend to the case where K is open and compact modulo Z, provided π admits a central character.

Proposition 3.2.4: Intertwining implies Common Occurrence

For $i = 1, 2$, let K_i be a compact open subgroup of G and let $\rho_i \in \widehat{K}_i$. Let (π, V) be an irreducible smooth representation of G which contains both ρ_1 and ρ_2. Then there exists $g \in G$ which intertwines ρ_1 with ρ_2.

***Proof*:**
For each i, V decomposes into K_i-isotypic components (proposition 1.2.16). The hypothesis is equivalent to $V^{\rho_i} \neq 0$ for $i = 1, 2$. Let e_2 denote the K_2-projection $V \to V^{\rho_2}$. Since π is irreducible and $V^{\rho_1} \neq 0$, the spaces $\pi(g^{-1})V^{\rho_1} = V^{\rho_1^g}$, $g \in G$, span V. Therefore, there exists $g \in G$ such that $e_2 \circ \pi(g^{-1})$ induces a non-zero map $V^{\rho_1} \to V^{\rho_2}$. This is the required element g.

There is a criterion for intertwining involving the Hecke algebra. Recall the idempotent $e_\rho \in \mathcal{H}(G)$ associated to $\rho \in \widehat{K}$ defined by $e_\rho(x) = \frac{\dim \rho}{\mu(K)} \mathrm{tr}\rho(x^{-1})$ for $x \in K$ and 0 otherwise.

Proposition 3.2.5: Hecke Algebra Criterion

Let K be a compact open subgroup of G, let $g \in G$, and $\rho \in \widehat{K}$. The following are equivalent:

1. there exists $f \in e_\rho * \mathcal{H}(G) * e_\rho$ such that $f|_{KgK} \neq 0$;
2. g intertwines ρ.

***Proof*:**
Consider the space $C^\infty(KgK)$ of G-smooth functions on the coset KgK. This carries a smooth representation of $K \times K$ by $(k_1, k_2)f : x \mapsto f(k_1^{-1}xk_2)$. Let H denote the group of pairs $(k, g^{-1}kg) \in K \times K$ for $k \in K \cap gKg^{-1}$. The map $f \mapsto f(g)$ is an H-homomorphism $C^\infty(KgK) \to \mathbb{C}$ (with H acting trivially). The group $K \times K$ acts transitively on the set KgK with the stabilizer of g equal to H, so $KgK \cong H\backslash(K \times K)$ as $K \times K$-sets, giving a $K \times K$-isomorphism

$$C^\infty(KgK) \cong \mathrm{Ind}_H^{K\times K}(1_H)$$

Condition (1) amounts to $e_\rho * C^\infty(KgK) * e_\rho \cong C^\infty(KgK)^{\rho\otimes\check{\rho}} \neq 0$. Equivalently, $\mathrm{Hom}_{K\times K}(\rho \otimes \check{\rho}, C^\infty(KgK)) \cong \mathrm{Hom}_H(\rho \otimes \check{\rho}, 1_H) \neq 0$. This last relation is equivalent to the representation $k \mapsto \rho(k)\otimes\check{\rho}(g^{-1}kg)$ of $K \cap gKg^{-1}$ having a fixed vector, which means $\mathrm{Hom}_{K\cap gKg^{-1}}(\rho, \rho^{g^{-1}}) \neq 0$. This is precisely the condition that g intertwines ρ.

To make this condition operational, one associates a specific algebra of functions to the representation ρ. Let K be an open subgroup of G, containing and compact modulo the centre Z. Let (ρ, W) be an irreducible smooth representation of K. Let $\mathcal{H}(G, \rho)$ be the space of functions $f : G \to \mathrm{End}_{\mathbb{C}}(W)$ which are compactly supported modulo Z and satisfy

$$f(k_1gk_2) = \rho(k_1)f(g)\rho(k_2), \quad k_i \in K, g \in G$$

The support of any $f \in \mathcal{H}(G, \rho)$ is a finite union of double cosets KgK. Under the convolution operation

$$\varphi_1 * \varphi_2(g) = \int_{G/Z} \varphi_1(x)\varphi_2(x^{-1}g)\, d\dot{\mu}(x),$$

$\mathcal{H}(G, \rho)$ becomes an associative $\mathbb{C}$-algebra with identity. It is called the *ρ-spherical Hecke algebra* (or intertwining algebra).

Lemma 3.2.6: Support of Hecke Functions

Let $g \in G$; there exists $\varphi \in \mathcal{H}(G, \rho)$ with support KgK if and only if g intertwines ρ.

***Proof*:**
Let $f \in \mathrm{End}_{\mathbb{C}}(W)$. For a fixed g, the assignment $kgk' \mapsto \rho(k)f\rho(k')$ is well-defined if and only if $f \in \mathrm{Hom}_{K^g \cap K}(\rho, \rho^g)$: writing $k' = g^{-1}k^{-1}g$ and $x = g^{-1}kg$, the condition $\rho(k)f\rho(k') = f$ reads $\rho^g(x) \circ f = f \circ \rho(x)$ for all $x \in K^g \cap K$. Since ρ is semisimple, the spaces $\mathrm{Hom}(\rho^g, \rho)$ and $\mathrm{Hom}(\rho, \rho^g)$ have the same dimension.

This algebra is intimately related to the endomorphisms of the compactly induced representation. Let $(\Sigma, X) = \text{c-Ind}_K^G \rho$. The space X consists of functions $f : G \to W$, compactly supported modulo Z, satisfying $f(kg) = \rho(k)f(g)$. For $\varphi \in \mathcal{H}(G, \rho)$ and $f \in X$, the convolution $\varphi * f$ lies in X. This defines a homomorphism of $\mathbb{C}$-algebras:

$$\mathcal{H}(G, \rho) \longrightarrow \mathrm{End}_G(\text{c-Ind}_K^G \rho)$$

Proposition 3.2.7: Hecke Algebra Isomorphism

The map $\mathcal{H}(G, \rho) \to \mathrm{End}_G(\text{c-Ind}_K^G \rho)$ is an isomorphism of $\mathbb{C}$-algebras.

***Proof*:**
The relation $\mathrm{End}_G(\text{c-Ind}\rho) \cong \mathrm{Hom}_K(\rho, \text{c-Ind}\rho)$ from Frobenius Reciprocity for compact induction (theorem 1.2.28) is used here. Let $\varphi_0 : w \mapsto \varphi_0^w$ be the canonical map $W \to X$ where φ_0^w is supported on K with value $\rho(k)w$. The isomorphism is composition with φ_0. Let $\varphi \in \mathrm{Hom}_K(W, X)$. Define $\Phi(g) \in \mathrm{End}_{\mathbb{C}}(W)$ by $\Phi(g)w = \varphi(w)(g)$. One verifies that $\Phi \in \mathcal{H}(G, \rho)$ and this map provides the inverse.

These algebraic structures provide the necessary foundation for the central result of this section, which establishes a method for constructing irreducible cuspidal representations.

Theorem 3.2.8: Construction of Cuspidal Representations

Let K be an open subgroup of $G = \mathrm{GL}_2(F)$, containing and compact modulo Z. Let (ρ, W) be an irreducible smooth representation of K. Suppose that an element $g \in G$ intertwines ρ if and only if $g \in K$. Then the compactly induced representation $\text{c-Ind}_K^G \rho$ is irreducible and cuspidal.

***Proof*:**
Write $(\Sigma, X) = \text{c-Ind}_K^G \rho$. First, it is shown that Σ has a non-zero coefficient compactly supported modulo Z. Both G and K are unimodular, K being open and compact modulo the centre, so $\delta_{K\backslash G} = 1$ and the Duality Theorem (theorem 1.3.12) gives $\check{X} \cong \mathrm{Ind}_K^G \check{\rho}$. This contains $\text{c-Ind}_K^G \check{\rho}$ as a subspace. Canonical embeddings are used to identify W and $\check{W}$ with functions in X and $\check{X}$ supported on K. For $w \in W, \check{w} \in \check{W}$, the coefficient $\gamma_{\check{w} \otimes w}$ is non-zero and supported in K.

Consequently, it suffices to prove that X is irreducible. If it is, it is admissible (by corollary 3.1.5) and thus γ-cuspidal (by proposition 3.1.2), hence cuspidal by theorem 3.1.3. The centre Z acts on X via the central character ω_ρ. Thus X is the direct sum of its K-isotypic components. Any K-map $W \to X$ has image in X^ρ. Thus:

$$\mathrm{Hom}_K(W, X^\rho) = \mathrm{Hom}_K(W, X) \cong \mathrm{End}_G(X) \cong \mathcal{H}(G, \rho)$$

The condition that g intertwines ρ iff $g \in K$ implies (by lemma 3.2.6) that functions in $\mathcal{H}(G,\rho)$ are supported on K. Thus $\dim \mathcal{H}(G,\rho) = \dim \mathrm{End}_K(\rho) = 1$. Therefore, $\mathrm{End}_G(X) \cong \mathbb{C}$. Since $X^\rho \cong W$ is irreducible as a K-module and generates X as a G-module, X is irreducible. That last step needs an argument, since the converse of Schur's Lemma fails in this setting (theorem 1.2.33 and the remark after it). Let $0 \neq Y \subseteq X$ be a G-subspace. Then

$$0 \neq \mathrm{Hom}_G(Y,X) \subseteq \mathrm{Hom}_G\big(Y, \mathrm{Ind}_K^G \rho\big) \cong \mathrm{Hom}_K(Y,\rho)$$

by Frobenius reciprocity (theorem 1.2.21), and Y is K-semisimple, so $Y^\rho \neq 0$. But $Y^\rho \subseteq X^\rho = W$ with W irreducible, so $Y^\rho = W$, and W generates X. Hence $Y = X$.

This construction principle can be immediately applied to classify the representations of level zero. Let $K = \mathrm{GL}_2(\mathfrak{o})$ and $K_1 = 1 + \mathfrak{p}M_2(\mathfrak{o})$. Let I_1 denote the preimage in K of the group of upper triangular unipotent matrices in $\mathrm{GL}_2(k)$.

Theorem 3.2.9: Classification of Level Zero Representations

Let (π, V) be an irreducible smooth representation of G, and suppose that π contains the trivial character of K_1. Exactly one of the following holds:

1. π contains a representation λ of K, inflated from an irreducible cuspidal representation $\tilde{\lambda}$ of $\mathrm{GL}_2(k)$;
2. π contains the trivial character of I_1.

In the first case, π is cuspidal, and there exists a representation Λ of ZK such that $\Lambda|_K \cong \lambda$ and

$$\pi \cong \text{c-Ind}_{ZK}^G \Lambda$$

***Proof*:**

The group K stabilizes V^{K_1}, so it decomposes into representations of $K/K_1 \cong \mathrm{GL}_2(k)$. Let λ be one such component, inflated from $\tilde{\lambda}$. Either $\tilde{\lambda}$ is cuspidal (in the finite group sense) or it is not. If not, it contains the trivial character of $N(k)$, and hence λ contains the trivial character of I_1.

It must be shown that the cases are mutually exclusive and that the structure in case (1) holds.

Lemma 3.2.10: Intertwining of Level Zero Cuspidals

For $i = 1, 2$, let $\tilde{\rho}_i$ be irreducible representations of $\mathrm{GL}_2(k)$ and ρ_i their inflations to K. Suppose $\tilde{\rho}_1$ is cuspidal.

1. The representations ρ_i intertwine in G if and only if $\tilde{\rho}_1 \cong \tilde{\rho}_2$.
2. An element $g \in G$ intertwines ρ_1 if and only if $g \in ZK$.

***Proof*:**
Let g intertwine ρ_2 with ρ_1. The element g can be taken to be diagonal of the form $\mathrm{diag}(\varpi^a, 1)$ for $a \geq 0$ (Cartan decomposition). If $a \geq 1$, $K_1^g \cap K$ contains the group N_0 of upper triangular unipotent matrices in K. ρ_2^g is trivial on this group. Since $\tilde{\rho}_1$ is cuspidal, it does not contain the trivial character of N_0, so intertwining is impossible for $a \geq 1$. Thus $g \in ZK$.

By the lemma and proposition 3.2.4, the two cases cannot occur together. If $\tilde{\lambda}$ is cuspidal, π contains an extension Λ of λ to ZK. There is a non-trivial map $\text{c-Ind}_{ZK}^{G}\Lambda \to \pi$. By the Lemma part (2) and theorem 3.2.8, c-IndΛ is irreducible. Thus the map is an isomorphism.

3.3 Chain Orders and Fundamental Strata

This section constructs a family of compact open subgroups of $G = \mathrm{GL}_2(F)$, along with specific characters of these subgroups. This structure provides the "genetic code" for admissible representations.

The construction begins with the lattices themselves. Let $V = F \oplus F$ be the standard 2-dimensional vector space over F. $G = \mathrm{Aut}_F(V)$ and $A = \mathrm{End}_F(V) = M_2(F)$.

Definition 3.3.1: Lattice Chain

An $\mathfrak{o}$-lattice chain in V is a non-empty set $\mathcal{L}$ of $\mathfrak{o}$-lattices in V such that:

1. $\mathcal{L}$ is linearly ordered under inclusion (if $L, L' \in \mathcal{L}$, then $L \subseteq L'$ or $L' \subseteq L$).
2. $\mathcal{L}$ is stable under multiplication by $F^\times$ (if $L \in \mathcal{L}$ and $x \in F^\times$, then $xL \in \mathcal{L}$).

Given a lattice chain $\mathcal{L}$, one can index its elements as $\mathcal{L} = \{L_i\}_{i\in\mathbb{Z}}$ such that $L_{i+1} \subsetneq L_i$. Stability under $F^\times$ implies the existence of an integer $e = e(\mathcal{L})$, called the *period* of $\mathcal{L}$, such that $\varpi L_i = L_{i+e}$ for all i.

Proposition 3.3.2: Classification of Lattice Chains

Let $\mathcal{L} = \{L_i\}$ be an $\mathfrak{o}$-lattice chain in V. Then $e(\mathcal{L})$ is either 1 or 2.

1. If $e(\mathcal{L}) = 1$, there exists $g \in G$ such that $gL_i = \mathfrak{p}^i \oplus \mathfrak{p}^i$ for all $i \in \mathbb{Z}$.
2. If $e(\mathcal{L}) = 2$, there exists $g \in G$ such that $gL_{2i} = \mathfrak{p}^i \oplus \mathfrak{p}^i$ and $gL_{2i+1} = \mathfrak{p}^i \oplus \mathfrak{p}^{i+1}$ for all $i \in \mathbb{Z}$.

***Proof*:**
The quotient $L_0/L_e = L_0/\varpi L_0$ is a vector space over the residue field k of dimension 2. The chain $L_0 \supset L_1 \supset \cdots \supset L_e$ defines a flag of subspaces in k^2. The length of a strictly decreasing flag in dimension 2 is at most 2. Thus $e \in \{1, 2\}$. The basis construction follows from the elementary divisor theorem.

Associated to a lattice chain is a ring of endomorphisms.

Definition 3.3.3: Chain Order

Let $\mathcal{L}$ be a lattice chain. The associated *chain order* (or hereditary order) is:

$$\mathfrak{A}_{\mathcal{L}} = \{x \in A : xL \subseteq L \text{ for all } L \in \mathcal{L}\} = \bigcap_{i \in \mathbb{Z}} \mathrm{End}_{\mathfrak{o}}(L_i)$$

The period $e(\mathcal{L})$ is denoted $e_{\mathfrak{A}}$.

By proposition 3.3.2, there are exactly two conjugacy classes of chain orders in $M_2(F)$:

1. $e_{\mathfrak{A}} = 1$: $\mathfrak{A} \cong M_2(\mathfrak{o})$ (The standard maximal order).
2. $e_{\mathfrak{A}} = 2$: $\mathfrak{A} \cong \mathcal{I} = \begin{pmatrix} \mathfrak{o} & \mathfrak{o} \\ \mathfrak{p} & \mathfrak{o} \end{pmatrix}$ (The standard Iwahori order).

Let $\mathfrak{A}$ be a chain order. Its Jacobson radical is denoted by $\mathfrak{P} = \mathrm{rad}(\mathfrak{A})$.

Proposition 3.3.4: Radical of Chain Order

Let $\mathfrak{A}$ be a chain order with period e.

1. $\mathfrak{P} = \{x \in A : xL_i \subseteq L_{i+1} \text{ for all } i\}$.
2. $\mathfrak{P}$ is an invertible two-sided ideal of $\mathfrak{A}$, and $\mathfrak{P}^e = \mathfrak{p}\mathfrak{A}$.
3. There exists $\Pi \in G \cap \mathfrak{A}$ such that $\mathfrak{P} = \Pi\mathfrak{A} = \mathfrak{A}\Pi$. Such an element is called a *prime element* of $\mathfrak{A}$.

***Proof*:**
Write $\mathfrak{P}' = \{x \in A : xL_i \subseteq L_{i+1} \text{ for all } i\}$ for the set described in (1).

(1). $\mathfrak{P}'$ is a two-sided ideal of $\mathfrak{A}$, since composing with an endomorphism preserving the chain keeps the one-step shift. Reduction modulo $\mathfrak{P}'$ identifies $\mathfrak{A}/\mathfrak{P}'$ with $\prod_{i \bmod e} \mathrm{End}_k(L_i/L_{i+1})$, a finite product of matrix algebras over the residue field, which is semisimple; hence $\mathrm{rad}(\mathfrak{A}) \subseteq \mathfrak{P}'$. In the other direction, iterating the defining condition gives $(\mathfrak{P}')^e L_i \subseteq L_{i+e} = \varpi L_i$, so $(\mathfrak{P}')^e \subseteq \mathfrak{p}\mathfrak{A}$ and $\mathfrak{P}'$ is nilpotent modulo the Jacobson radical of $\mathfrak{A}$; a nil ideal lies in the radical, so $\mathfrak{P}' \subseteq \mathrm{rad}(\mathfrak{A})$. Therefore $\mathfrak{P} = \mathrm{rad}(\mathfrak{A}) = \mathfrak{P}'$.

(2). The computation just made gives $\mathfrak{P}^e \subseteq \mathfrak{p}\mathfrak{A}$, and conversely $\mathfrak{p}\mathfrak{A} \subseteq \mathfrak{P}^e$ because $\varpi L_i = L_{i+e}$ realizes multiplication by ϖ as an e-step shift. Hence $\mathfrak{P}^e = \mathfrak{p}\mathfrak{A}$. Invertibility follows from (3): if $\mathfrak{P} = \Pi\mathfrak{A} = \mathfrak{A}\Pi$ with $\Pi \in G$, then $\mathfrak{P}^{-1} = \mathfrak{A}\Pi^{-1}$ is a two-sided $\mathfrak{A}$-module with $\mathfrak{P}\mathfrak{P}^{-1} = \mathfrak{A}$.

(3). By proposition 3.3.2 it is enough to exhibit Π for the two standard orders. If $e_{\mathfrak{A}} = 1$ then $\mathfrak{A} = M_2(\mathfrak{o})$, the chain is $L_i = \mathfrak{p}^i \oplus \mathfrak{p}^i$, and $\Pi = \varpi I$ shifts it by one step. If $e_{\mathfrak{A}} = 2$ then $\mathfrak{A} = \mathcal{I}$ with $L_{2i} = \mathfrak{p}^i \oplus \mathfrak{p}^i$ and $L_{2i+1} = \mathfrak{p}^i \oplus \mathfrak{p}^{i+1}$, and

$$\Pi = \begin{pmatrix} 0 & 1 \\ \varpi & 0 \end{pmatrix}$$

carries L_j onto L_{j+1} for every j, as one checks on the two lattice types. In both cases Π is invertible in A and $\Pi\mathfrak{A} = \mathfrak{A}\Pi$ is exactly the set of elements shifting the chain one step, which is $\mathfrak{P}$ by (1), completing the proof.

For the standard orders:

- If $\mathfrak{A} = M_2(\mathfrak{o})$, $\mathfrak{P} = \varpi M_2(\mathfrak{o})$. A prime element is ϖI.
- If $\mathfrak{A} = \mathcal{I}$, $\mathfrak{P} = \begin{pmatrix} \mathfrak{p} & \mathfrak{o} \\ \mathfrak{p} & \mathfrak{p} \end{pmatrix}$. A prime element is $\Pi = \begin{pmatrix} 0 & 1 \\ \varpi & 0 \end{pmatrix}$.

This structure defines a filtration of compact open subgroups. Let $U_{\mathfrak{A}} = \mathfrak{A}^\times$ be the group of units. For $n \geq 1$, define:

$$U_{\mathfrak{A}}^n = 1 + \mathfrak{P}^n$$

These are compact open normal subgroups of $U_{\mathfrak{A}}$. For integers $n > m \geq 1$ such that $2m \geq n$, the map $x \mapsto 1 + x$ induces an isomorphism $\mathfrak{P}^m/\mathfrak{P}^n \xrightarrow{\sim} U_{\mathfrak{A}}^m/U_{\mathfrak{A}}^n$.

The chain and its order determine each other, and a prime element translates the chain by one step. Both are used whenever an element of G is moved along the chain, so they are recorded before the normalizer is introduced. The treatment follows [2, §12.2].

Proposition 3.3.5: The Lattices Stable Under a Chain Order

Let $\mathfrak{A}$ be a chain order with lattice chain $\mathcal{L} = \{L_i\}_{i \in \mathbb{Z}}$, radical $\mathfrak{P}$, period $e = e_{\mathfrak{A}}$ and prime element Π.

1. $\mathfrak{P}^j = \Pi^j \mathfrak{A} = \mathfrak{A}\Pi^j$ for every $j \in \mathbb{Z}$, where for $j < 0$ the left side is the $|j|$-th power of the inverse of the invertible ideal $\mathfrak{P}$.
2. $\mathfrak{A}L_i = L_i$ and $\mathfrak{P}^j L_i = \Pi^j L_i = L_{i+j}$ for all $i, j \in \mathbb{Z}$.
3. $\dim_k L_i/L_{i+1} = 2/e$ for every i.
4. Every $\mathfrak{o}$-lattice M in V with $\mathfrak{A}M \subseteq M$ equals L_i for exactly one $i \in \mathbb{Z}$.

So $\mathcal{L}$ is precisely the set of $\mathfrak{A}$-stable lattices in V, and $\mathfrak{A}$ determines its chain up to a shift of the index.

***Proof*:**
(1). By proposition 3.3.4(3) there is $\Pi \in G \cap \mathfrak{A}$ with $\mathfrak{P} = \Pi\mathfrak{A} = \mathfrak{A}\Pi$, and induction gives $\mathfrak{P}^j = \Pi^j\mathfrak{A} = \mathfrak{A}\Pi^j$ for $j \geq 0$. Multiplying $\mathfrak{A}\Pi = \Pi\mathfrak{A}$ on both sides by Π^{-1} gives $\Pi^{-1}\mathfrak{A} = \mathfrak{A}\Pi^{-1}$, and $(\Pi^{-1}\mathfrak{A})(\Pi\mathfrak{A}) = \mathfrak{A}$, so $\Pi^{-1}\mathfrak{A}$ is the inverse of $\mathfrak{P}$ given by proposition 3.3.4(2). Taking powers gives the case $j < 0$, and with it $\mathfrak{P}^a\mathfrak{P}^b = \mathfrak{P}^{a+b}$ for all $a, b \in \mathbb{Z}$.

(2). Every $x \in \mathfrak{A}$ has $xL_i \subseteq L_i$ by definition 3.3.3, and $1 \in \mathfrak{A}$, so $\mathfrak{A}L_i = L_i$ and $\mathfrak{P}L_i = \Pi\mathfrak{A}L_i = \Pi L_i$. Also $\Pi \in \mathfrak{P}$, so $\Pi L_i \subseteq L_{i+1}$ by proposition 3.3.4(1).

Fix i and set $S_r = \Pi^{e-r}L_{i+r}$ for $0 \leq r \leq e$. Applying Π^{e-r-1} to $\Pi L_{i+r} \subseteq L_{i+r+1}$ gives $S_r \subseteq S_{r+1}$. At the ends $S_e = L_{i+e}$, while $S_0 = \mathfrak{P}^e L_i = \varpi\mathfrak{A}L_i = \varpi L_i = L_{i+e}$, using $\mathfrak{P}^e = \mathfrak{p}\mathfrak{A}$ from proposition 3.3.4(2) and the period property of definition 3.3.1. A chain of inclusions with equal ends consists of equalities, so $\Pi L_{i+e-1} = L_{i+e}$. As i runs over $\mathbb{Z}$ so does $i + e - 1$, giving $\Pi L_j = L_{j+1}$ for every j. Since Π is invertible in A the identity reads backwards as $\Pi^{-1}L_{j+1} = L_j$, and induction in both directions gives $\Pi^j L_i = L_{i+j}$. Then $\mathfrak{P}^j L_i = \Pi^j\mathfrak{A}L_i = L_{i+j}$.

(3). Each L_j/L_{j+1} is a k-vector space, since $\varpi L_j = L_{j+e} \subseteq L_{j+1}$. The lattice L_i is free of rank two over $\mathfrak{o}$, so $L_i/\varpi L_i = L_i/L_{i+e}$ has k-dimension two, and additivity of length along $L_i \supsetneq \cdots \supsetneq L_{i+e}$ makes the e intermediate dimensions positive integers summing to two. By proposition 3.3.2, e is 1 or 2, so each is $2/e$.

(4). We first claim that

$$\mathfrak{A}m + L_{i+1} = L_i \qquad \text{for } m \in L_i \smallsetminus L_{i+1} \tag{3.1}$$

For $e = 2$ the quotient L_i/L_{i+1} is a line by (3), the image of m is non-zero and so spans it, and $\mathfrak{o} \subseteq \mathfrak{A}$ preserves every lattice. For $e = 1$ we have $L_j = \varpi^{j-i} L_i$, and $\mathrm{End}_{\mathfrak{o}}(\varpi^a L) = \mathrm{End}_{\mathfrak{o}}(L)$, so definition 3.3.3 gives $\mathfrak{A} = \mathrm{End}_{\mathfrak{o}}(L_i)$. An $\mathfrak{o}$-basis of L_i identifies $\mathrm{End}_{\mathfrak{o}}(L_i)$ with $M_2(\mathfrak{o})$ and $\mathrm{End}_k(L_i/L_{i+1})$ with $M_2(k)$ compatibly, and entrywise reduction is surjective, so $\mathfrak{A} \to \mathrm{End}_k(L_i/L_{i+1})$ is onto. Since $\mathrm{End}_k(k^2)v = k^2$ for $v \neq 0$, the claim follows.

Now let M be an $\mathfrak{o}$-lattice with $\mathfrak{A}M \subseteq M$. Any two lattices in V are commensurable, so $L_b \subseteq M \subseteq L_a$ for some $a \leq b$, and the set $\{j : M \subseteq L_j\}$ is non-empty and bounded above by b. Let i be its largest element and pick $m \in M \smallsetminus L_{i+1}$. Then $\mathfrak{A}m \subseteq M$, so (3.1) gives $L_i = M + L_{i+1}$. Inductively $L_i = M + L_{i+r}$ for every $r \geq 0$: given the case r, part (2) and $\mathfrak{P}^r M \subseteq M$ give $L_{i+r} = \mathfrak{P}^r M + \mathfrak{P}^r L_{i+1} \subseteq M + L_{i+r+1}$. Finally $\varpi^N L_i \subseteq M$ for some $N \geq 0$ because both are lattices, and $L_{i+Ne} = \varpi^N L_i$, so $L_i = M + \varpi^N L_i = M$. Uniqueness of i holds because the L_j are pairwise distinct.

The normalizer of the order in G is also needed, together with the valuation that measures how far an element of it moves the chain. The valuation is defined on all of A, because the arguments below apply it to elements that are not invertible.

Definition 3.3.6: The Valuation of a Chain Order

Let $\mathfrak{A}$ be a chain order with radical $\mathfrak{P}$. For $x \in A$ put

$$\nu_{\mathfrak{A}}(x) = \max\left\{j \in \mathbb{Z} : x \in \mathfrak{P}^j\right\}, \qquad \nu_{\mathfrak{A}}(0) = \infty$$

The maximum exists for $x \neq 0$, since the $\mathfrak{P}^j$ decrease, $\bigcap_{n\geq 0} \mathfrak{P}^{ne} = \bigcap_{n\geq 0} \mathfrak{p}^n\mathfrak{A} = 0$ and $\bigcup_{n\geq 0} \mathfrak{P}^{-ne} = \bigcup_{n\geq 0} \varpi^{-n}\mathfrak{A} = A$.

$$\mathcal{K}_{\mathfrak{A}} = \{g \in G : g\mathfrak{A}g^{-1} = \mathfrak{A}\}$$

Proposition 3.3.7: The Valuation on the Normalizer

Let $\mathfrak{A}$ be a chain order with chain $\mathcal{L} = \{L_i\}$, radical $\mathfrak{P}$, period $e = e_{\mathfrak{A}}$ and prime element Π.

1. For each $g \in \mathcal{K}_{\mathfrak{A}}$ there is a unique integer $v_{\mathfrak{A}}(g)$ with $gL_i = L_{i+v_{\mathfrak{A}}(g)}$ for every i, and $\mathcal{K}_{\mathfrak{A}} = \{g \in G : gL \in \mathcal{L} \text{ for all } L \in \mathcal{L}\}$.
2. $v_{\mathfrak{A}} \colon \mathcal{K}_{\mathfrak{A}} \to \mathbb{Z}$ is a surjective homomorphism with kernel $U_{\mathfrak{A}}$, its value $v_{\mathfrak{A}}(g)$ is the unique j with $g\mathfrak{A} = \mathfrak{A}g = \mathfrak{P}^j$, and it is the restriction of $\nu_{\mathfrak{A}}$.
3. $\mathcal{K}_{\mathfrak{A}} = \bigsqcup_{j\in\mathbb{Z}} \Pi^j U_{\mathfrak{A}} = \bigsqcup_{j\in\mathbb{Z}} U_{\mathfrak{A}}\Pi^j$, with $v_{\mathfrak{A}}(\Pi^j u) = j$ for $u \in U_{\mathfrak{A}}$.
4. $v_F(\det g) = \frac{2}{e}\, v_{\mathfrak{A}}(g)$ for $g \in \mathcal{K}_{\mathfrak{A}}$, the factor $2/e$ being an integer by proposition 3.3.2. In particular $v_F(\det u) = 0$ for $u \in U_{\mathfrak{A}}$, and $\nu_{\mathfrak{A}}(x) = e\, v_F(x)$ for $x \in F^\times$.
5. $\nu_{\mathfrak{A}}(gx) = \nu_{\mathfrak{A}}(xg) = v_{\mathfrak{A}}(g) + \nu_{\mathfrak{A}}(x)$ for $g \in \mathcal{K}_{\mathfrak{A}}$, $x \in A$.

***Proof*:**

(1). For $g \in \mathcal{K}_{\mathfrak{A}}$, $\mathfrak{A}(gL_0) = (g\mathfrak{A}g^{-1})gL_0 = g\mathfrak{A}L_0 = gL_0$ by proposition 3.3.5(2), so gL_0 is $\mathfrak{A}$-stable and proposition 3.3.5(4) gives a unique c with $gL_0 = L_c$. Conjugation by g is a ring automorphism carrying $\mathfrak{A}$ onto $\mathfrak{A}$, hence carrying $\mathrm{rad}\mathfrak{A}$ onto itself, so $g\mathfrak{P}^r = \mathfrak{P}^r g$ for $r \geq 0$ and $gL_i = g\mathfrak{P}^i L_0 = \mathfrak{P}^i L_c = L_{c+i}$ for $i \geq 0$. For $i < 0$ pick N with $i + Ne \geq 0$. As ϖ is a scalar it commutes with g, so $gL_i = \varpi^{-N} gL_{i+Ne} = L_{i+c}$. Put $v_{\mathfrak{A}}(g) = c$, unique since the L_j are distinct. This gives $\mathcal{K}_{\mathfrak{A}} \subseteq \mathrm{Aut}_{\mathfrak{o}}(\mathcal{L})$, and conversely if g permutes $\mathcal{L}$ then $g\mathfrak{A}g^{-1} = \bigcap_i \mathrm{End}_{\mathfrak{o}}(gL_i) = \bigcap_j \mathrm{End}_{\mathfrak{o}}(L_j) = \mathfrak{A}$ by definition 3.3.3.

(2) and (3). From $(gh)L_i = L_{i+v_{\mathfrak{A}}(h)+v_{\mathfrak{A}}(g)}$ the map is a homomorphism. If $v_{\mathfrak{A}}(g) = 0$ then g and g^{-1} preserve every L_i, so both lie in $\bigcap_i \mathrm{End}_{\mathfrak{o}}(L_i) = \mathfrak{A}$ and $g \in U_{\mathfrak{A}}$. Conversely a unit of $\mathfrak{A}$ preserves each L_i. Also $\Pi\mathfrak{A}\Pi^{-1} = \mathfrak{A}$, so $\Pi \in \mathcal{K}_{\mathfrak{A}}$ with $v_{\mathfrak{A}}(\Pi) = 1$ by proposition 3.3.5(2), and $v_{\mathfrak{A}}$ is onto with fibres $\Pi^j U_{\mathfrak{A}} = U_{\mathfrak{A}}\Pi^j$. If $g = \Pi^j u$ then $g\mathfrak{A} = \mathfrak{A}g = \mathfrak{P}^j$, and conversely $g\mathfrak{A} = \mathfrak{P}^j$ forces $\Pi^{-j}g \in U_{\mathfrak{A}}$. Finally $g \in \mathfrak{P}^j$ and $g \notin \mathfrak{P}^{j+1}$, since otherwise $\Pi^{-j}g \in \mathfrak{P}$, so $\nu_{\mathfrak{A}}(g) = j$.

(4). Both $g \mapsto v_F(\det g)$ and $v_{\mathfrak{A}}$ are homomorphisms on $\mathcal{K}_{\mathfrak{A}}$, which by (3) is generated by $U_{\mathfrak{A}}$ and Π. For $u \in U_{\mathfrak{A}}$, both u and u^{-1} lie in $\mathfrak{A} \subseteq \mathrm{End}_{\mathfrak{o}}(L_0)$, so their determinants lie in $\mathfrak{o}$ and multiply to 1, giving $\det u \in \mathfrak{o}^\times$. For Π, part (1) of proposition 3.3.5 and proposition 3.3.4(2) give $\Pi^e\mathfrak{A} = \varpi\mathfrak{A}$, so $\Pi^e = \varpi w$ with $w \in U_{\mathfrak{A}}$, and taking determinants gives $e\, v_F(\det \Pi) = 2$. A scalar $x = \varpi^a u$ with $u \in \mathfrak{o}^\times$ has $xL_i = L_{i+ae}$, so $\nu_{\mathfrak{A}}(x) = ae$.

(5). Put $j = v_{\mathfrak{A}}(g)$ and $r = \nu_{\mathfrak{A}}(x)$. By (2), $g\mathfrak{P}^r = g\mathfrak{A}\mathfrak{P}^r = \mathfrak{P}^{j+r}$, so $gx \in \mathfrak{P}^{j+r}$. Membership in $\mathfrak{P}^{j+r+1}$ would give $x \in \mathfrak{P}^{-j}\mathfrak{P}^{j+r+1} = \mathfrak{P}^{r+1}$, which is false. The right-hand version is the same computation with $\mathfrak{P}^r g = \mathfrak{P}^{r+j}$.

If $\mathfrak{A} = M_2(\mathfrak{o})$, $\mathcal{K}_{\mathfrak{A}} = F^\times \mathrm{GL}_2(\mathfrak{o})$. If $\mathfrak{A} = \mathcal{I}$, $\mathcal{K}_{\mathfrak{A}}$ is generated by $F^\times U_{\mathcal{I}}$ and the prime element $\Pi = \begin{pmatrix} 0 & 1 \\ \varpi & 0 \end{pmatrix}$. Note that $\mathcal{K}_{\mathfrak{A}}$ is open, contains Z, and is compact modulo Z.

With the orders and their unit filtrations in place, attention turns to the characters those filtration quotients carry, which is where the arithmetic of a representation is recorded. Fix an additive

character $\psi \in \widehat{F}$ of level one (kernel $\mathfrak{p}$). Define a character on the algebra A by $\psi_A(x) = \psi(\mathrm{tr}(x))$. This induces an isomorphism $A \to \widehat{A}$ via $a \mapsto (x \mapsto \psi_A(ax))$. Under this duality, if $\mathfrak{A}$ is a chain order, the dual lattice of $\mathfrak{P}^n$ is $\mathfrak{P}^{1-n}$.

Definition 3.3.8: Stratum

A *stratum* in A is a triple $(\mathfrak{A}, n, \alpha)$ consisting of:

- A chain order $\mathfrak{A}$;
- An integer n;
- An element $\alpha \in \mathfrak{P}^{-n}$.

Two strata $(\mathfrak{A}, n, \alpha_1)$ and $(\mathfrak{A}, n, \alpha_2)$ are *equivalent* if $\alpha_1 \equiv \alpha_2 \pmod{\mathfrak{P}^{1-n}}$.

If $n \geq 1$, a stratum $(\mathfrak{A}, n, \alpha)$ determines a character ψ_α on the group $U^n_{\mathfrak{A}}$:

$$\psi_\alpha(1+x) = \psi_A(\alpha x), \quad x \in \mathfrak{P}^n$$

Since $\alpha \in \mathfrak{P}^{-n}$ and $x \in \mathfrak{P}^n$, the product is in $\mathfrak{A}$. However, the character is well-defined on the quotient $U^n_{\mathfrak{A}}/U^{n+1}_{\mathfrak{A}}$ because if $y \in \mathfrak{P}^{n+1}$, $\mathrm{tr}(\alpha y) \in \mathrm{tr}(\mathfrak{P}) = \mathfrak{p}$, on which ψ is trivial. The equivalence class of the stratum corresponds bijectively to the character ψ_α.

More is true, and the classification below uses it. Let $m \geq \lfloor n/2 \rfloor$. For $\lfloor n/2 \rfloor \leq m \leq n-1$ the same formula $\psi_\alpha(1+x) = \psi_A(\alpha x)$, now for $x \in \mathfrak{P}^{m+1}$, defines a character of the larger group $U^{m+1}_{\mathfrak{A}}$, trivial on $U^{n+1}_{\mathfrak{A}}$. Multiplicativity is where the bound on m enters: for $x, y \in \mathfrak{P}^{m+1}$ the cross term αxy lies in $\mathfrak{P}^{-n}\mathfrak{P}^{2m+2} \subseteq \mathfrak{P}$, since $2(m+1) \geq n+1$, and ψ_A is trivial there. Two elements give the same character of $U^{m+1}_{\mathfrak{A}}$ exactly when they are congruent modulo $(\mathfrak{P}^{m+1})^* = \mathfrak{P}^{-m}$, so on $U^{m+1}_{\mathfrak{A}}$ the character corresponds to the coset $\alpha + \mathfrak{P}^{-m}$. Taking $m = n-1$ returns the character on $U^n_{\mathfrak{A}}$ and the coset $\alpha + \mathfrak{P}^{1-n}$ just described.

Proposition 3.3.9: Intertwining of Strata

Let $(\mathfrak{A}, n, \alpha)$ and $(\mathfrak{A}, n, \beta)$ be strata with $n \geq 1$, let $m \geq \lfloor n/2 \rfloor$, and let $g \in G$. The following are equivalent:

1. g intertwines the character ψ_α with ψ_β on $U^{m+1}_{\mathfrak{A}}$.
2. The cosets intersect: $g^{-1}(\alpha + \mathfrak{P}^{-m})g \cap (\beta + \mathfrak{P}^{-m}) \neq \emptyset$.

The case $m = n-1$ is the statement on $U^n_{\mathfrak{A}}$ with cosets $\alpha + \mathfrak{P}^{1-n}$.

***Proof*:**

Let $K = U^{m+1}_{\mathfrak{A}}$. The intertwining condition is that $\psi_\alpha(gxg^{-1}) = \psi_\beta(x)$ for $x \in K \cap g^{-1}Kg$. In additive terms, $\psi_A(\alpha g x g^{-1}) = \psi_A(g^{-1}\alpha g x) = \psi_A(\beta x)$ for all $x \in \mathfrak{P}^{m+1} \cap g^{-1}\mathfrak{P}^{m+1}g$. This means $g^{-1}\alpha g - \beta$ lies in the dual of the lattice $\mathfrak{P}^{m+1} \cap g^{-1}\mathfrak{P}^{m+1}g$. By lattice duality properties, $(\mathfrak{P}^{m+1} \cap g^{-1}\mathfrak{P}^{m+1}g)^* = \mathfrak{P}^{-m} + g^{-1}\mathfrak{P}^{-m}g$. Thus $g^{-1}\alpha g - \beta = y_1 + g^{-1}y_2 g$ with $y_i \in \mathfrak{P}^{-m}$, which rearranges to the intersection condition. Nothing in the argument uses $m = n-1$.

Definition 3.3.10: Fundamental Stratum

A stratum $(\mathfrak{A}, n, \alpha)$ is *fundamental* if the coset $\alpha + \mathfrak{P}^{1-n}$ contains no nilpotent element of A.

Proposition 3.3.11: Criterion for Fundamental Strata

A stratum $(\mathfrak{A}, n, \alpha)$ is fundamental if and only if for every $r \geq 1$, $\alpha^r \notin \mathfrak{P}^{1-rn}$.

***Proof*:**

The condition is a statement about the image of α in the graded algebra $\operatorname{gr}\mathfrak{A} = \bigoplus_{j\in\mathbb{Z}} \mathfrak{P}^j/\mathfrak{P}^{j+1}$: the class of α lives in the piece of degree $-n$, its r-th power is the class of α^r in degree $-rn$, and $\alpha^r \in \mathfrak{P}^{1-rn}$ says exactly that this r-th power vanishes. So the stated condition says that the class of α is not nilpotent in $\operatorname{gr}\mathfrak{A}$.

Fundamental implies the condition. Suppose $\alpha^r \in \mathfrak{P}^{1-rn}$ for some $r \geq 1$; we produce a nilpotent element in the coset. Since $A = M_2(F)$, the Cayley-Hamilton theorem lets us work with the characteristic polynomial $T^2 - (\operatorname{tr}\alpha)T + \det\alpha$. First reduce to $r = 2$: the hypothesis says the class $\bar{\alpha}$ of α in $\mathfrak{P}^{-n}/\mathfrak{P}^{1-n}$ satisfies $\bar{\alpha}^{\,r} = 0$ in $\operatorname{gr}\mathfrak{A}$, and $\bar{\alpha}^{\,2} = 0$ follows in each of the two cases of proposition 3.3.2. For $e_{\mathfrak{A}} = 1$ the graded piece is $M_2(k)$, where a nilpotent matrix squares to zero. For $e_{\mathfrak{A}} = 2$ the piece $\mathfrak{A}/\mathfrak{P}$ is $k \times k$ and conjugation by a prime element Π of $\mathfrak{A}$ swaps the two factors, so writing $\bar{\alpha} = (x, y)$ in degree $-n$ gives $\bar{\alpha}^{\,2} = xy$ up to the identification of graded pieces when n is odd, and $\bar{\alpha}^{\,2} = (x^2, y^2)$ when n is even. So $\bar{\alpha}^{\,r} = 0$ forces $xy = 0$ when n is odd and $x = y = 0$ when n is even. Either way $\bar{\alpha}^{\,2} = 0$. With $r = 2$ in hand, the hypothesis forces both coefficients into the ideals $\operatorname{tr}\alpha \in \mathfrak{p}^{\lceil (1-n)/e \rceil}$ and $\det\alpha \in \mathfrak{p}^{\lceil (1-2n)/e \rceil}$. Producing from that an actual nilpotent in $\alpha + \mathfrak{P}^{1-n}$ is the one step taken on citation: see [2, §12.8], which reduces by G-conjugation and by $(\mathfrak{A}, n, a) \rightsquigarrow (\mathfrak{A}, n - e, \varpi a)$ to three cases and exhibits the nilpotent explicitly in each. It is not enough to subtract half the trace: that changes the determinant by $(\operatorname{tr}\alpha)^2/4$, which need not vanish, and in characteristic 2 the element does not exist. Granting the citation, the coset is not free of nilpotents and the stratum is not fundamental.

The condition implies fundamental. Conversely, if some $\alpha_0 \in \alpha + \mathfrak{P}^{1-n}$ is nilpotent then $\alpha_0^2 = 0$ in $M_2(F)$, and since $\alpha \equiv \alpha_0 \pmod{\mathfrak{P}^{1-n}}$ the product expansion gives $\alpha^2 \in \alpha_0^2 + \mathfrak{P}^{1-2n} = \mathfrak{P}^{1-2n}$. Taking $r = 2$ violates the condition, completing the proof.

Note that in rank two only $r \leq 2$ is ever needed, since a nilpotent 2×2 matrix squares to zero; the statement is phrased for all r because that is the form in which it generalizes.

This concept is linked to the normalized level of a representation.

Definition 3.3.12: Normalized Level

Let π be an irreducible smooth representation of G. The *normalized level* is

$$\ell(\pi) = \min\{n/e_{\mathfrak{A}} : (\mathfrak{A}, n) \text{ with } n \geq 0 \text{ such that } \pi \text{ contains the trivial character of } U_{\mathfrak{A}}^{n+1}\}$$

Theorem 3.3.13: Strata and Representation Level

Let π be an irreducible smooth representation. Suppose π contains the character ψ_α associated to a stratum $(\mathfrak{A}, n, \alpha)$ with $n \geq 1$. Then $(\mathfrak{A}, n, \alpha)$ is fundamental if and only if $\ell(\pi) = n/e_{\mathfrak{A}}$.

***Proof Key ideas*:**
If the stratum is not fundamental, one can approximate α by a nilpotent element, or conjugate the order to show that ψ_α is trivial on a subgroup $U^{n'}_{\mathfrak{A}'}$ where $n'/e_{\mathfrak{A}'} < n/e_{\mathfrak{A}}$. If it is fundamental, intertwining arguments show the level cannot be lowered. The full argument is [2, §12.9], whose two supporting lemmas this sketch compresses; this book does not reproduce them.

3.4 Classification of Fundamental Strata

We classify fundamental strata $(\mathfrak{A}, n, \alpha)$ based on the arithmetic of the element α. Let $\nu_{\mathfrak{A}}(\alpha) = -n$.

Definition 3.4.1: Types of Strata

Let $(\mathfrak{A}, n, \alpha)$ be a fundamental stratum.

1. **Essentially Scalar:** $e_{\mathfrak{A}} = 1$ and the image of $\varpi^n\alpha$ in $\mathfrak{A}/\mathfrak{P} \cong M_2(k)$ has a repeated non-zero eigenvalue.
2. **Split:** $e_{\mathfrak{A}} = 1$ and the image of $\varpi^n\alpha$ has distinct eigenvalues in k.
3. **Unramified Simple:** $e_{\mathfrak{A}} = 1$ and the characteristic polynomial of the image of $\varpi^n\alpha$ is irreducible over k.
4. **Ramified Simple:** $e_{\mathfrak{A}} = 2$ and n is odd.

One combination is deliberately absent: $e_{\mathfrak{A}} = 2$ with n even. Such fundamental strata exist, and the classification does not need them, because

$$U^{j+1}_M \subseteq U^{2j+1}_{\mathcal{I}} \subseteq U^{2j}_{\mathcal{I}} \subseteq U^j_M$$

holds for every integer j, which lets one pass to a stratum with $\gcd(n, e_{\mathfrak{A}}) = 1$ ([2, §13.1]). The four cases above therefore exhaust what the classification uses.

The last two cases are collectively called **Simple Strata**. They are characterized by the fact that the algebra $E = F[\alpha]$ is a field, and α is "minimal" in the following sense:

Definition 3.4.2: Minimal Element

An element $\alpha \in G \setminus Z$ is *minimal over* F if $E = F[\alpha]$ is a field, and setting $n = -\nu_E(\alpha)$:

- If E/F is ramified, n is odd.
- If E/F is unramified, the coset $\varpi^n\alpha + \mathfrak{p}_E$ generates the residue extension k_E/k, for a prime element ϖ of F.

Proposition 3.4.3: Properties of Simple Strata

Let $(\mathfrak{A}, n, \alpha)$ be a simple stratum, and write $E = F[\alpha]$. Then:

1. α is minimal over F, and E is a quadratic field extension of F with $v_E(\alpha) = -n$, where v_E is the normalized valuation of E
2. $E^\times \subset \mathcal{K}_\mathfrak{A}$
3. $e(E|F) = e_\mathfrak{A}$

***Proof*:**
By definition 3.4.1 a simple stratum is of one of two shapes, and in dimension two each can be handled explicitly. Throughout, $E = F[\alpha]$.

The unramified case: $e_\mathfrak{A} = 1$ and the characteristic polynomial of the image $\bar{\alpha} \in \mathfrak{A}/\mathfrak{P} \cong M_2(k)$ is irreducible over k. Irreducibility of that polynomial means $k[\bar{\alpha}]$ is the quadratic extension k_2 of k, so $\bar{\alpha}$ generates a field and in particular α satisfies no polynomial over F of degree one modulo $\mathfrak{P}$; hence $E = F[\alpha]$ is a field, quadratic over F. Its residue extension already has degree two, so $f(E|F) = 2$ and therefore $e(E|F) = 1$: the extension is unramified. This gives (3), since $e_\mathfrak{A} = 1$ here. For (1), the unramified clause of definition 3.4.2 asks that the coset $\varpi^n\alpha + \mathfrak{p}_E$ generate k_E/k: the residue of $\varpi^n\alpha$ is the $\bar{\alpha}$ whose characteristic polynomial is irreducible over k, so it generates $k[\bar{\alpha}] = k_E$. As E is a field as just shown, α is minimal over F.

The ramified case: $e_\mathfrak{A} = 2$ and n is odd. Here $\mathfrak{A}$ is conjugate to the Iwahori order and $\mathfrak{P}^2 = \mathfrak{p}\mathfrak{A}$ by proposition 3.3.4(2). A simple stratum is fundamental, and for $e_\mathfrak{A} = 2$ fundamentality pins the determinant. From $\alpha \in \mathfrak{P}^{-n}$ one gets $v_F(\mathrm{tr}\alpha) \geq -(n-1)/2$ and $v_F(\det\alpha) \geq -n$ outright, and failing fundamentality would need both coefficients deeper. That consequence can be seen directly, without pointing into another proof: if $\alpha_0 \in \alpha + \mathfrak{P}^{1-n}$ is nilpotent, write $\alpha = \alpha_0 + z$ with $z \in \mathfrak{P}^{1-n}$, so that $\det\alpha = \det z - \mathrm{tr}(\alpha_0 z)$, and for $e_\mathfrak{A} = 2$ both terms have $v_F \geq 1 - n$. The trace containment is automatic here, so fundamentality forces $v_F(\det\alpha) = -n$ exactly. The characteristic polynomial $T^2 - (\mathrm{tr}\alpha)T + \det\alpha$ then has $v_F(\det\alpha) = -n$ odd and $2\,v_F(\mathrm{tr}\alpha) \geq 1 - n > -n$, so its Newton polygon is a single segment of slope $-n/2 \notin \mathbb{Z}$ and the polynomial is irreducible over F. Hence $E = F[\alpha]$ is a quadratic field extension, and the normalized valuation v_E satisfies $v_E(\alpha) = -n$ with n odd, which is possible only if $e(E|F) = 2$. That is (3), and it also gives (1): definition 3.4.2 in the ramified case asks exactly that n be odd.

Property (2), both cases. Write $e = e_\mathfrak{A}$ and put

$$\alpha_0 = \varpi^n\alpha \quad (e = 1), \qquad \alpha_0 = \varpi^{(n+1)/2}\alpha \quad (e = 2)$$

the exponent in the second case being an integer because n is odd. Each is an $F^\times$-multiple of α, so $E = F[\alpha_0]$ and it is enough to argue with α_0. The chain $\mathcal{L}$ is stable under $F^\times$ by definition 3.3.1, so $F^\times \subseteq \mathcal{K}_\mathfrak{A}$ by proposition 3.3.7(1), and parts (2) and (4) of that result give $v_\mathfrak{A}(\varpi^j) = je$. Part (5) then turns $\nu_\mathfrak{A}(\alpha) = -n$ into

$$\nu_\mathfrak{A}(\alpha_0) = n - n = 0 \quad (e = 1), \qquad \nu_\mathfrak{A}(\alpha_0) = (n+1) - n = 1 \quad (e = 2)$$

In particular $\alpha_0 \in \mathfrak{A}$, so $\mathfrak{o}[\alpha_0] \subseteq \mathfrak{A}$. Since $\mathfrak{A} \subseteq \mathrm{End}_\mathfrak{o}(L_0)$ by definition 3.3.3, the matrix of α_0 in an $\mathfrak{o}$-basis of L_0 has entries in $\mathfrak{o}$, so its characteristic polynomial $g(T) = T^2 - tT + d$ over F has $t = \mathrm{tr}\alpha_0$ and $d = \det\alpha_0$ in $\mathfrak{o}$, and $g(\alpha_0) = 0$ by Cayley-Hamilton.

The determinant of α_0. We claim that

$$v_F(\det \alpha_0) = \tfrac{2}{e}\, \nu_{\mathfrak{A}}(\alpha_0)$$

that is, $v_F(d) = 0$ for $e = 1$ and $v_F(d) = 1$ for $e = 2$.

For $e = 1$ the chain reads $L_j = \varpi^j L_0$, and $\mathrm{End}_{\mathfrak{o}}(\varpi^a L) = \mathrm{End}_{\mathfrak{o}}(L)$, so $\mathfrak{A} = \mathrm{End}_{\mathfrak{o}}(L_0)$ by definition 3.3.3 while $\mathfrak{P} = \mathfrak{p}\mathfrak{A}$ by proposition 3.3.4(2). Reduction modulo $\mathfrak{P}$ is therefore the map $\mathrm{End}_{\mathfrak{o}}(L_0) \to \mathrm{End}_k(L_0/\varpi L_0) \cong M_2(k)$, and $\mathfrak{P}^{-n} = \varpi^{-n}\mathfrak{A}$ makes α_0 the element of $\mathfrak{A}$ whose image under it is the $\bar{\alpha}$ of definition 3.4.1(3). Reducing a matrix reduces its characteristic polynomial, so $\bar{g}$ is the characteristic polynomial of $\bar{\alpha}$, irreducible over k by definition 3.4.1(3). An irreducible quadratic has non-zero constant term, so $\bar{d} \neq 0$ and $v_F(d) = 0$.

For $e = 2$ we use that the stratum is fundamental, as definition 3.4.1 requires. Taking $r = 2$ in proposition 3.3.11 gives $\alpha^2 \notin \mathfrak{P}^{1-2n}$, while $\alpha^2 \in \mathfrak{P}^{-n}\mathfrak{P}^{-n} = \mathfrak{P}^{-2n}$, so $\nu_{\mathfrak{A}}(\alpha^2) = -2n$ and hence $\nu_{\mathfrak{A}}(\alpha_0^2) = \nu_{\mathfrak{A}}(\varpi^{n+1}\alpha^2) = 2(n+1) - 2n = 2$. Next, t lies in $\mathfrak{p}$: an element $x \in \mathfrak{P}$ carries L_0 into L_1 and L_1 into $L_2 = \varpi L_0$ by proposition 3.3.5(2), so the endomorphism it induces on $L_0/\varpi L_0$ has square zero and therefore trace zero in k, and that trace is the reduction of $\mathrm{tr}x$. So $\nu_{\mathfrak{A}}(t\alpha_0) = 2v_F(t) + 1 \geq 3$ by parts (4) and (5) of proposition 3.3.7, the case $t = 0$ being covered by the convention $\nu_{\mathfrak{A}}(0) = \infty$ of definition 3.3.6. Now $g(\alpha_0) = 0$ reads $d = t\alpha_0 - \alpha_0^2$ with $t\alpha_0 \in \mathfrak{P}^3$ and $\alpha_0^2 \in \mathfrak{P}^2 \smallsetminus \mathfrak{P}^3$, so $d \in \mathfrak{P}^2 \smallsetminus \mathfrak{P}^3$ and $\nu_{\mathfrak{A}}(d) = 2$. Since d is a scalar, $\nu_{\mathfrak{A}}(d) = 2v_F(d)$ by proposition 3.3.7(4), and $v_F(d) = 1$.

The valuation of α_0 in E. By (3) the ramification index $e(E|F)$ equals e, so the residue degree $f(E|F)$ is $2/e$ by [9, The Fundamental Identity For Local Fields]. As $[E : F] = 2 = \dim_F V$ and $E \subseteq \mathrm{End}_F(V)$, the space V is one-dimensional over E, and fixing an E-basis identifies V with E so that $x \in E$ acts by multiplication. Then $x \mapsto v_F(\det x)$ is a homomorphism $E^\times \to \mathbb{Z}$. It vanishes on U_E, because $\mathfrak{o}_E$ is an $\mathfrak{o}$-lattice in V under that identification and u, u^{-1} both carry it into itself for $u \in U_E$, forcing $\det u \in \mathfrak{o}^\times$. Writing ϖ_E for a prime element of E, the relation $v_E(\varpi) = e$ gives $\varpi_E^e = \varpi u$ with $u \in U_E$, so $e\, v_F(\det \varpi_E) = v_F(\det \varpi) = v_F(\varpi^2) = 2$. Since $E^\times = \varpi_E^{\mathbb{Z}} U_E$,

$$v_F(\det x) = \tfrac{2}{e}\, v_E(x), \qquad x \in E^\times$$

Comparing with the previous step, $v_E(\alpha_0) = \nu_{\mathfrak{A}}(\alpha_0)$, which is 0 for $e = 1$ and 1 for $e = 2$. In particular $\alpha_0 \in \mathfrak{o}_E$.

The order generated by α_0 is maximal. We show that 1 and α_0 span $\mathfrak{o}_E/\mathfrak{p}\mathfrak{o}_E$ over k. For $e = 1$ the extension $E|F$ is unramified, so $\mathfrak{p}\mathfrak{o}_E = \mathfrak{p}_E$ and $\mathfrak{o}_E/\mathfrak{p}_E$ is the quadratic extension k_2 of k. Reducing $g(\alpha_0) = 0$ shows that the image of α_0 there is a root of $\bar{g}$, which is irreducible over k, so that image lies outside k and $\{1, \alpha_0\}$ is a k-basis of k_2. For $e = 2$ the extension is totally ramified, so $\mathfrak{o}_E/\mathfrak{p}_E = k$ and $\mathfrak{p}\mathfrak{o}_E = \mathfrak{p}_E^2$, while $v_E(\alpha_0) = 1$ makes α_0 a prime element of E. The pieces $\mathfrak{o}_E/\mathfrak{p}_E$ and $\mathfrak{p}_E/\mathfrak{p}_E^2$ are each one-dimensional over k and are spanned by the images of 1 and of α_0, so again $\{1, \alpha_0\}$ spans $\mathfrak{o}_E/\mathfrak{p}\mathfrak{o}_E$. As $\mathfrak{o}_E$ is a finitely generated $\mathfrak{o}$-module, [9, Nakayama's Lemma] gives $\mathfrak{o}_E = \mathfrak{o} + \mathfrak{o}\alpha_0$, and therefore

$$\mathfrak{o}_E = \mathfrak{o}[\alpha_0] \subseteq \mathfrak{A}$$

Neither case assumes that $E|F$ is separable, which the ramified one fails to be when $\mathrm{char}(F) = 2$ and $t = 0$.

Conclusion. Each L_i is an $\mathfrak{o}_E$-module by proposition 3.3.5(2), and it is finitely generated and torsion free over the discrete valuation ring $\mathfrak{o}_E$, hence free of rank $\dim_E V = 1$. Write $L_i = \mathfrak{o}_E w_i$. Then $w_i/w_0 \in E^\times$ and $L_i = \mathfrak{p}_E^{c_i} L_0$ with $c_i = v_E(w_i/w_0)$ and $c_0 = 0$. The inclusions $L_{i+1} \subsetneq L_i$ make (c_i) strictly increasing, and $L_{i+e} = \varpi L_i$ with $v_E(\varpi) = e$ gives $c_{i+e} = c_i + e$. So the e positive integers $c_{i+1} - c_i, \dots, c_{i+e} - c_{i+e-1}$ sum to e and each equals 1, that is $c_i = i$. For $x \in E^\times$ we have $x\mathfrak{o}_E = \mathfrak{p}_E^{v_E(x)}$ and therefore

$$xL_i = \mathfrak{p}_E^{i+v_E(x)} L_0 = L_{i+v_E(x)}$$

So every element of $E^\times$ permutes $\mathcal{L}$, and $E^\times \subseteq \mathcal{K}_\mathfrak{A}$ by proposition 3.3.7(1), as we sought to show.

Corollary 3.4.4: The Normalizer Valuation on $E^\times$

Let $(\mathfrak{A}, n, \alpha)$ be a simple stratum and $E = F[\alpha]$. Then

$$v_\mathfrak{A}(x) = v_E(x), \qquad x \in E^\times$$

where v_E is the normalized valuation of E. Consequently $E^\times \cap U_\mathfrak{A} = U_E$, and a prime element of E is a prime element of $\mathfrak{A}$.

***Proof*:**
$E^\times \subseteq \mathcal{K}_\mathfrak{A}$ is proposition 3.4.3(2) and $e(E|F) = e_\mathfrak{A}$ is (3) of the same result, and we write e for the common value. Both $v_\mathfrak{A}|_{E^\times}$ and v_E are homomorphisms $E^\times \to \mathbb{Z}$, and $E^\times = \varpi_E^{\mathbb{Z}} U_E$ for a prime element ϖ_E of E, so it is enough to compare them on U_E and at ϖ_E.

Since $\dim_F V = 2 = [E : F]$ and $E \subseteq \mathrm{End}_F(V)$, the space V is one-dimensional over E. Fixing an E-basis identifies V with E so that $x \in E$ acts by multiplication. Under that identification $\mathfrak{o}_E$ is an $\mathfrak{o}$-lattice in V, and for $u \in U_E$ both u and u^{-1} carry it into itself, so in an $\mathfrak{o}$-basis of it their matrices have entries in $\mathfrak{o}$ and determinants multiplying to 1. Hence $\det u \in \mathfrak{o}^\times$, and proposition 3.3.7(4) gives $\frac{2}{e} v_\mathfrak{A}(u) = 0$, so $v_\mathfrak{A}(u) = 0 = v_E(u)$.

At ϖ_E, the definition of the ramification index gives $v_E(\varpi) = e(E|F) = e$, so $\varpi_E^e = \varpi u$ with $u \in U_E$. Applying $v_\mathfrak{A}$ and using the last clause of proposition 3.3.7(4) for the scalar ϖ,

$$e\, v_\mathfrak{A}(\varpi_E) = v_\mathfrak{A}(\varpi) + v_\mathfrak{A}(u) = e\, v_F(\varpi) = e$$

so $v_\mathfrak{A}(\varpi_E) = 1 = v_E(\varpi_E)$. The two homomorphisms agree on generators, hence everywhere. Then $E^\times \cap U_\mathfrak{A}$ is the kernel of v_E on $E^\times$, which is U_E. And $v_\mathfrak{A}(\varpi_E) = 1$ gives $\varpi_E \mathfrak{A} = \mathfrak{P}$ by proposition 3.3.7(2), which is the defining property in proposition 3.3.4(3).

3.5 The Classification of Cuspidal Representations

We assume π is an irreducible smooth representation of G.

3.5.1 Exhaustion

Proposition 3.5.1: Principal Series Strata

Let χ be a character of T. Let $\Sigma = \mathrm{Ind}_B^G \chi$.

1. If χ_1 and χ_2 have distinct levels, at least one of them positive, or the same positive level n with $\chi_1\chi_2^{-1}$ non-trivial on U_F^n, then Σ contains a Split fundamental stratum.
2. If χ_1 and χ_2 have the same positive level n and $\chi_1\chi_2^{-1}$ is trivial on U_F^n, Σ contains an Essentially Scalar stratum.
3. If χ_1 and χ_2 both have level zero, Σ contains the trivial character of the Iwahori $U_{\mathcal{I}}^1$

***Proof*:**
The point in every case is that Σ is realized on functions on G determined by their restriction to K (proposition 2.2.1), so the characters of the filtration subgroups occurring in Σ can be read off the torus.

(3). If both χ_i have level zero then χ is trivial on U_T^1. Work with $U_{\mathcal{I}}^1$ throughout: the function supported on $BU_{\mathcal{I}}^1$ and equal to 1 there is well defined and $U_{\mathcal{I}}^1$-fixed, because χ is trivial on

$$B \cap U_{\mathcal{I}}^1 = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : a, d \in U_F^1,\ b \in \mathfrak{o} \right\}$$

which is exactly the level-zero hypothesis. So Σ contains the trivial character of $U_{\mathcal{I}}^1$. Note that fixity under the full Iwahori I would be a stronger statement and is false here: a right-I-invariant function supported on BI, respectively BwI, needs χ trivial on $B \cap I$, respectively $B \cap wIw^{-1}$, and both of those contain $\{\mathrm{diag}(a, d) : a, d \in U_F\}$, so both force χ_1 and χ_2 unramified. Level zero allows them ramified on $\mathfrak{o}^\times$.

(1) and (2). Restrict Σ to K and use $\Sigma|_K \cong \mathrm{Ind}_{B\cap K}^K(\chi|_{B\cap K})$, again by Iwasawa. The characters of $U_{\mathfrak{A}}^n/U_{\mathfrak{A}}^{n+1}$ occurring in this induced representation are, by Frobenius reciprocity (theorem 1.2.21) and Mackey restriction to the subgroup $U_{\mathfrak{A}}^n$, the K-conjugates of the characters obtained by restricting χ along $T \cap U_{\mathfrak{A}}^n \to U_{\mathfrak{A}}^n$. Writing such a character as ψ_α via the duality of definition 3.3.8, the element α can therefore be taken diagonal, $\alpha = \mathrm{diag}(a_1, a_2)$ with a_i determined by the levels of χ_1 and χ_2.

Write n for the larger of the two levels, so $n > 0$ in both cases of (1). If χ_1 and χ_2 have distinct levels, the two diagonal entries have distinct valuations, and the one of smaller level contributes 0 to the reduction; if they share the level n and $\chi_1\chi_2^{-1}$ is non-trivial on U_F^n, then $a_1 \not\equiv a_2 \pmod{\mathfrak{p}^{1-n}}$ by the computation below and the two entries have distinct residues. Either way the image of $\varpi^n\alpha$ in $\mathfrak{A}/\mathfrak{P} \cong M_2(k)$ has distinct eigenvalues in k: that is a Split stratum in the sense of definition 3.4.1(2), and it is fundamental because a diagonal matrix with a non-zero eigenvalue is not nilpotent modulo $\mathfrak{P}$.

If instead χ_1 and χ_2 share a level $n > 0$ and $\chi_1\chi_2^{-1}$ is trivial on U_F^n, then the two entries have the same valuation, and agreement of the χ_i on U_F^n says exactly that $a_1 \equiv a_2 \pmod{\mathfrak{p}^{1-n}}$, so the entries have the same image in k and $\varpi^n\alpha$ reduces to a scalar with a repeated non-zero eigenvalue: that is an Essentially Scalar stratum, definition 3.4.1(1), completing the proof.

This characterizes the cuspidals.

Theorem 3.5.2: Exhaustion Theorem

Let π be an irreducible smooth representation of G. Assume $\ell(\pi) \leq \ell(\chi\pi)$ for all characters χ of $F^\times$ ("minimal level"). The following are equivalent:

1. π is cuspidal.
2. Either:
   (a) $\ell(\pi) = 0$ and π contains a representation of $K = \mathrm{GL}_2(\mathfrak{o})$ inflated from an irreducible *cuspidal* representation of $\mathrm{GL}_2(k)$.
   (b) $\ell(\pi) > 0$ and π contains a *Simple Stratum*.

***Proof Key ideas*:**
Only the direction (2) $\Rightarrow$ (1) is given here, in the contrapositive. The exhaustion direction (1) $\Rightarrow$ (2) (a cuspidal π of minimal level contains a simple stratum or is of level zero) is quoted: it needs that a minimal-level π with $\ell(\pi) > 0$ contains a split, ramified-simple or unramified-simple stratum, and that a representation containing a split fundamental stratum is not cuspidal, neither of which is proved in this book. Both are [2, §13.3, §14.1, §14.5].

If π is not cuspidal, it is a subquotient of some $\Sigma = \mathrm{Ind}_B^G \chi$. Suppose first Σ is irreducible, so $\pi = \Sigma$. Proposition 3.5.1 applies to Σ directly: according to the shape of χ, π contains a split fundamental stratum, an essentially scalar stratum, or the trivial character of $U_{\mathcal{I}}^1$, and in the last case the K-type is a constituent of a principal series of $\mathrm{GL}_2(k)$, not cuspidal. Suppose instead Σ is reducible. Its proper subquotients are not covered by that proposition, and the case is handled directly: π is then $\varphi \circ \det$ or $\varphi\mathrm{St}_G$ for a character φ of $F^\times$, and the minimal-level hypothesis gives $\ell(\pi) = 0$, so π contains the trivial character of $U_{\mathcal{I}}^1$ and its K-type is again not cuspidal. The other two branches are closed on citation, and the minimal-level hypothesis is what closes the first: by [2, §13.3], π contains an essentially scalar stratum exactly when some twist $\chi\pi$ has $\ell(\chi\pi) < \ell(\pi)$, which the hypothesis forbids. And by [2, §13.2], two fundamental strata over M occurring in π must intertwine and so have equal reduced characteristic polynomials, which a ramified simple stratum cannot match. Neither statement is proved in this book.

3.5.2 Cuspidal Types

To generate the representations, we extend the simple stratum character. Let $(\mathfrak{A}, n, \alpha)$ be a simple stratum with $n \geq 1$ and $E = F[\alpha]$. We define a filtration of open subgroups in $\mathcal{K}_{\mathfrak{A}}$:

$$H_\alpha = U_E^1 U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}, \qquad J_\alpha^1 = U_E^1 U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}, \qquad J_\alpha = U_E U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}, \qquad J = E^\times U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}$$

Writing $H_\alpha^1 = H_\alpha$, the pair $J_\alpha^1 \supset H_\alpha^1$ is what chapter 4 uses for the Heisenberg step. The groups J_α^1 and J_α differ by $U_E/U_E^1 \cong k_E^\times$. (the unipotent factors of J_α and J now coincide for every n, so $J = \varpi_E^{\mathbb{Z}} J_\alpha$, the two differing by the prime element, which has $v_{\mathfrak{A}} = 1$ by corollary 3.4.4). The character ψ_α on $U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}$ extends naturally to H_α (using $\psi_A(\alpha\cdot)$).

Theorem 3.5.3: Intertwining of Simple Characters

Let $(\mathfrak{A}, n, \alpha)$ be a simple stratum with $n \geq 1$.

1. An element $g \in G$ intertwines the character ψ_α on $U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}$ if and only if $g \in J$
2. For any irreducible representation Λ of J containing ψ_α, the induced representation $\text{c-Ind}_J^G \Lambda$ is irreducible and cuspidal

***Proof*:**

Write $m = \lfloor n/2 \rfloor$, so the character in question lives on $U_{\mathfrak{A}}^{m+1}$ and $J = E^\times U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}$.

(1), the easy inclusion. If $g \in E^\times$ then g commutes with α, so $\psi_\alpha(gxg^{-1}) = \psi_A(\alpha g x g^{-1}) = \psi_A(g^{-1}\alpha g x) = \psi_A(\alpha x) = \psi_\alpha(x)$, and g intertwines ψ_α with itself. If $g \in U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}$ then $g = 1 + y$ with $y \in \mathfrak{P}^{\lfloor (n+1)/2 \rfloor}$, and for $x \in \mathfrak{P}^{m+1}$ the commutator $\alpha(gxg^{-1} - x)$ lies in $\mathfrak{P}^{-n}\mathfrak{P}^{\lfloor (n+1)/2 \rfloor}\mathfrak{P}^{m+1} \subseteq \mathfrak{P}$, on which ψ_A is trivial. What this shows for each generator is the stronger statement that it *normalizes* ψ_α, and the normalizer is a group, so all of J normalizes and in particular intertwines ψ_α. (The set of elements intertwining a character is not in general a group, so the weaker reading would not close.)

(1), the converse. Suppose g intertwines ψ_α on $U_{\mathfrak{A}}^{m+1}$. By proposition 3.3.9 applied to $(\mathfrak{A}, n, \alpha)$ at this m, which is legitimate since $m = \lfloor n/2 \rfloor$, this says

$$g^{-1}\left(\alpha + \mathfrak{P}^{-m}\right)g \cap \left(\alpha + \mathfrak{P}^{-m}\right) \neq \emptyset$$

so $g^{-1}\alpha g$ and α agree modulo $\mathfrak{P}^{-m}$, that is, to more than half the depth of α. Because α is minimal over F (proposition 3.4.3), the coset $\alpha + \mathfrak{P}^{-m}$ determines $E = F[\alpha]$ and no proper subfield meets it, so this half-depth agreement forces $g^{-1}Eg = E$ and then $g \in E^\times U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor} = J$.

This last forcing step is where the argument leaves what has been developed here: turning agreement to half depth into membership in J is the "intertwining implies conjugacy" mechanism for simple strata, and in the shape needed it rests on the full theory of simple characters rather than on the two-by-two computations above. It is quoted; see [2, §15-§16], where the statement is §15.1 and the proof occupies §16.

(2). Granting (1), the hypothesis of theorem 3.2.8 is met: J is an open subgroup of G containing the centre and compact modulo it (proposition 3.4.3(2) puts $E^\times$ inside $\mathcal{K}_{\mathfrak{A}}$, which is compact modulo Z), Λ is an irreducible smooth representation of J, and an the restriction $\Lambda|_{U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}}$ is a *multiple* of ψ_α: it is semisimple, its constituents are J-conjugate because $U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}$ is normal in J ([5, Clifford's Theorem]), and J normalizes ψ_α by the previous paragraph. Containment alone would not suffice, since then g intertwining Λ would match only *some* constituent of Λ^g with some constituent of Λ. Given the multiplicity, an element of G intertwines Λ only if it intertwines ψ_α, hence only if it lies in J. That theorem then gives that $\text{c-Ind}_J^G \Lambda$ is irreducible and cuspidal, as we sought to show.

Definition 3.5.4: Cuspidal Type

A *cuspidal type* is a triple $(\mathfrak{A}, J, \Lambda)$ where:

1. $\mathfrak{A}$ is a chain order.
2. J is an open subgroup of $\mathcal{K}_{\mathfrak{A}}$ as defined above (associated to a simple stratum or level 0).
3. Λ is an irreducible representation of J constructed as follows:
   - **Level 0:** $J = F^{\times}\mathrm{GL}_2(\mathfrak{o})$, Λ is an extension of a cuspidal rep of $\mathrm{GL}_2(k)$.
   - **Positive Level:** $\Lambda|_{U_{\mathfrak{A}}^{\lfloor n/2 \rfloor+1}}$ is a multiple of ψ_α for a simple stratum $(\mathfrak{A}, n, \alpha)$ with $n \geq 1$.
   - **Twist:** $\Lambda \cong \Lambda_0 \otimes (\chi \circ \det)$ for a triple $(\mathfrak{A}, J, \Lambda_0)$ of one of the two preceding kinds and a character χ of $F^{\times}$.

   The third kind is not decoration. Without it the classification below is false: for π_0 cuspidal of level zero and χ of level $n \geq 1$, the cuspidal $\chi\pi_0$ has positive level but contains an essentially scalar stratum rather than a simple one, so it carries no Λ of the second kind, and $\Lambda_0 \otimes (\chi \circ \det)$ is not trivial on K_1, so none of the first.

Theorem 3.5.5: Classification of Cuspidal Representations

The mapping

$$(\mathfrak{A}, J, \Lambda) \longmapsto \pi_\Lambda = \text{c-Ind}_J^G \Lambda$$

induces a bijection between the set of G-conjugacy classes of cuspidal types and the set of equivalence classes of irreducible cuspidal representations of G.

***Proof*:**

Injectivity: Split by regime. Theorem 3.3.13 fixes the level, so $\ell(\pi_{\Lambda_1}) = \ell(\pi_{\Lambda_2})$, and by the Remark following [2, §15.5] the level together with minimality determines which of the three kinds of definition 3.5.4 a type has: level zero gives the first kind, positive level with $\ell(\pi) \leq \ell(\chi\pi)$ for every χ gives the second, and otherwise the third. The two types therefore have the same kind. For the first kind the statement is theorem 3.2.9. For the third, untwist by definition 3.5.4's third clause and reduce to one of the other two. For the second it is the strong uniqueness property of [2, §15.4], which is stated for simple strata with $n_i \geq 1$ and so applies exactly there: if $\pi_{\Lambda_1} \cong \pi_{\Lambda_2}$ then $n_1 = n_2$ and a single element of G conjugates $(\mathfrak{A}_1, J_1, \Lambda_1)$ to $(\mathfrak{A}_2, J_2, \Lambda_2)$. The level comparison is in-book, $n_i/e_{\mathfrak{A}_i} = \ell(\pi_{\Lambda_i})$ by theorem 3.3.13; the conjugacy argument runs through the ramified/unramified dichotomy of the underlying strata and is not reproduced in this book. **Surjectivity:** First reduce to minimal level. Twisting gives $\text{c-Ind}_J^G(\Lambda_0 \otimes (\chi \circ \det)) \cong \chi \cdot \text{c-Ind}_J^G \Lambda_0$, so by the third kind of definition 3.5.4 it is enough to treat π with $\ell(\pi) \leq \ell(\chi\pi)$ for every character χ of $F^{\times}$, which is exactly the hypothesis theorem 3.5.2 carries. For such a π the Exhaustion Theorem gives a simple stratum (or level 0). If level > 0, the theorem gives ψ_α on $U_{\mathfrak{A}}^n$. Passing to the larger $U_{\mathfrak{A}}^{\lfloor n/2 \rfloor+1}$, what π contains there is a character restricting to ψ_α on $U_{\mathfrak{A}}^n$, hence of the form ψ_β with $\beta \equiv \alpha \pmod{\mathfrak{P}^{1-n}}$. This is a genuine weakening, since $\mathfrak{P}^{1-n} \supsetneq \mathfrak{P}^{-\lfloor n/2 \rfloor}$ once $n \geq 3$, and the stratum $(\mathfrak{A}, n, \beta)$ is

again simple by [2, §13.4, §13.5], so the type produced is built on $F[\beta]^\times U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}$, which is the group J formed from β. One then ascends the hierarchy $H_\alpha \subset J_\alpha \subset J$ using Clifford theory [5, Clifford's Theorem]. For n even a Heisenberg group appears in J_α/H_α. For n odd that quotient is the cyclic $k_E^\times$ and every Λ is a character. This proves π contains some Λ of the required type.

3.6 Representations with Iwahori-Fixed Vectors

We consider representations π such that $V^I \neq 0$, where I is the Iwahori subgroup.

Proposition 3.6.1: Structure of Iwahori-Fixed Representations

Let (π, V) be irreducible. If $V^I \neq 0$, then π is a subquotient of $\mathrm{Ind}_B^G \chi$ where χ is an unramified character of T. Specifically, π is either:

1. An unramified principal series $\mathrm{Ind}_B^G \chi$ (χ regular).
2. An unramified twist of the Steinberg representation $\varphi \otimes \mathrm{St}_G$, for an unramified character φ of $F^\times$.
3. An unramified character $\varphi \circ \det$, for an unramified character φ of $F^\times$.

In particular, such representations are never cuspidal.

***Proof*:**
By Borel's theorem [1], quoted here, the category of smooth representations generated by their I-fixed vectors is equivalent to the category of modules over the Hecke algebra $\mathcal{H}(G, I) = e_I * \mathcal{H}(G) * e_I$. The algebra $\mathcal{H}(G, I)$ is generated by T_w (supported on IwI), T_Π (supported on $I\Pi I$) and T_Π^{-1}, the inverse being needed because the algebra contains the class of ϖ^{-1}. These satisfy the Affine Hecke Algebra relations (e.g., quadratic relation for T_w). The finite dimensional irreducible modules of this algebra are 1 or 2 dimensional. These correspond precisely to the listed representation types via the functor $M \mapsto \mathcal{H}(G) \otimes_{\mathcal{H}(G,I)} M$.

This analysis is crucial for establishing the square-integrability of the Steinberg representation.

Theorem 3.6.2: Square-Integrability of Steinberg

The Steinberg representation St_G is square-integrable modulo the center. That is, for matrix coefficients f,

$$\int_{G/Z} |f(g)|^2 \, d\dot{g} < \infty$$

***Proof*:**
Let $f(g) = \langle \tilde{v}, \pi(g)v \rangle$ where v is I-fixed. Using the Iwahori decomposition and the specific values of the Hecke operators on the unique I-fixed line in St_G, one calculates that $|f(g)| = q^{-\ell(w)}$ where $\ell(w)$ is the length of the element in the affine Weyl group, while $\mu(IgI) = q^{\ell(w)}$. A set of representatives for $ZI\backslash G/I$ is given by the $\Pi^m x$ with $m \in \{0, 1\}$ and x in the affine Weyl

group W_0, which is infinite dihedral with exactly two elements of each length $k \geq 1$. Summing $|f|^2\mu(IgI) = q^{-\ell}$ over those representatives gives

$$\int_{G/Z} |f(g)|^2 \, d\dot{g} = 2 \sum_{x \in W_0} q^{-\ell(x)} = 2 + 4 \sum_{k \geq 1} q^{-k} < \infty$$

4

Quadratic Tame Extensions Parameterizing Tame Cuspidals

We have seen that cuspidal representations are constructed by compact induction, but the construction in the previous chapter was abstract. To establish the Langlands correspondence, we need an explicit parameterization of these representations that can be matched against Galois representations. The goal of this chapter is to provide this explicit parameterization for the "tame" case (when the residual characteristic p is odd). We will show that tame cuspidal representations are in bijection with *Admissible Pairs* $(E/F, \xi)$, consisting of a quadratic extension and a character.

We begin in section 4.1 by defining admissible pairs and analyzing the arithmetic of tame quadratic extensions. In section 4.2, we explicitly construct a cuspidal representation π_ξ for each pair $(E/F, \xi)$ using the machinery of fundamental strata. Finally, in section 4.3, we prove the Parametrization Theorem, showing that this construction yields a bijection between admissible pairs and the set of tame cuspidal representations. This provides the "Group side" of the correspondence that we will later match to the "Galois side."

4.1 Admissible Pairs

We rely on the arithmetic of tame extensions. Throughout, let $\psi \in \widehat{F}$ be a character of level one. As a reminder:

Definition 4.1.1: Tamely Ramified Extension

A finite field extension E/F is *tamely ramified* if $p \nmid e(E|F)$. Equivalently, $\mathrm{tr}_{E/F}(\mathfrak{o}_E) = \mathfrak{o}_F$.

If $p \neq 2$, every quadratic extension is tamely ramified. If $p = 2$, only the unramified quadratic extension is tame.

Lemma 4.1.2: Properties of Tame Extensions

Let E/F be a tamely ramified extension with ramification index e.

1. For $r \in \mathbb{Z}$, $\mathrm{tr}_{E/F}(\mathfrak{p}_E^{1+r}) = \mathfrak{p}_F^{1+\lfloor r/e \rfloor}$.
2. For $m \geq 1$, the norm $N_{E/F}$ induces an isomorphism
$$U_E^{em}/U_E^{em+1} \xrightarrow{\sim} U_F^m/U_F^{m+1}$$
3. The map $N_{E/F} : U_E/U_E^1 \to U_F/U_F^1$ is surjective if E/F is unramified, and has kernel/-cokernel of order $\gcd(e, q-1)$ if totally ramified.

***Proof*:**
Property (1) follows from the fact that the inverse different is $\mathfrak{d}_{E/F}^{-1} = \mathfrak{p}_E^{1-e}$. That is [9, The Different Detects Tameness], stated there for Galois extensions, which covers the quadratic extensions this chapter uses. Property (2) follows from the expansion $N(1+x) = 1 + \mathrm{tr}(x) + \dots$.

Define $\psi_E = \psi \circ \mathrm{tr}_{E/F}$. By the lemma, ψ_E has level one in E.

Proposition 4.1.3: Norm and Character Levels

Let E/F be a tamely ramified extension with ramification index e, let $\psi \in \widehat{F}$ have level one and put $\psi_E = \psi \circ \mathrm{tr}_{E/F}$. Let χ be a character of $F^\times$ of level $m \geq 1$. Let $\chi_E = \chi \circ N_{E/F}$. Then χ_E has level em. If $c \in \mathfrak{p}_F^{-m}$ satisfies $\chi(1+x) = \psi(cx)$ for $x \in \mathfrak{p}_F^{\lfloor m/2 \rfloor + 1}$, then
$$\chi_E(1+y) = \psi_E(cy), \quad y \in \mathfrak{p}_E^{\lfloor em/2 \rfloor + 1}$$

***Proof*:**
For the level, lemma 4.1.2(2) gives an isomorphism $U_E^{em}/U_E^{em+1} \to U_F^m/U_F^{m+1}$ induced by the norm. Since χ has level m it is trivial on U_F^{m+1} and non-trivial on U_F^m, so $\chi_E = \chi \circ N_{E/F}$ is trivial on U_E^{em+1} and non-trivial on U_E^{em}; that is, χ_E has level em.

For the second assertion, expand the norm of a principal unit. Let L/F be a normal closure of E/F and let $\sigma_1, \dots, \sigma_d$ be the distinct F-embeddings $E \to L$, $d = [E:F]$. For $y \in \mathfrak{p}_E$,
$$N_{E/F}(1+y) = \prod_{i=1}^{d} \big(1 + \sigma_i(y)\big) = 1 + \mathrm{tr}_{E/F}(y) + \sum_{k=2}^{d} e_k\big(\sigma_1 y, \dots, \sigma_d y\big) \tag{4.1}$$
where e_k is the k-th elementary symmetric function, so that $e_1 = \mathrm{tr}_{E/F}(y)$ and every e_k lies in F. This is the general expansion; for quadratic E/F it reduces to $(1+y)(1+y^\sigma)$, but E/F here is only assumed tamely ramified.

Take $y \in \mathfrak{p}_E^{\lfloor em/2 \rfloor + 1}$ and put $R = \sum_{k \geq 2} e_k(\sigma_1 y, \dots, \sigma_d y)$. Conjugate elements have equal valuation, so $v_L(\sigma_i y) = v_L(y)$ for every i and each e_k with $k \geq 2$ has $v_L \geq k\, v_L(y) \geq 2 v_L(y)$.

Dividing by $e(L|F)$ to measure in F, and using $v_E(y) \geq \lfloor em/2 \rfloor + 1$,

$$v_F(R) \;\geq\; \frac{2\big(\lfloor em/2 \rfloor + 1\big)}{e} \;>\; m$$

the last inequality because $2(\lfloor em/2 \rfloor + 1) \geq em + 1$. As $v_F(R)$ is an integer this gives $R \in \mathfrak{p}_F^{m+1}$. Writing $1 + \mathrm{tr}_{E/F}(y) + R = (1 + \mathrm{tr}_{E/F}(y))\big(1 + R(1 + \mathrm{tr}_{E/F}(y))^{-1}\big)$, the second factor lies in U_F^{m+1} because $1 + \mathrm{tr}_{E/F}(y)$ is a unit, and χ is trivial there. So the R term may be discarded:

$$\chi_E(1 + y) = \chi\big(1 + \mathrm{tr}_{E/F}(y)\big)$$

By lemma 4.1.2(1), $\mathrm{tr}_{E/F}(y) \in \mathfrak{p}_F^{\lfloor m/2 \rfloor + 1}$, which is exactly the range on which $\chi(1 + x) = \psi(cx)$ holds. Since $c \in F$ the trace is F-linear, so

$$\chi_E(1 + y) = \psi\big(c\,\mathrm{tr}_{E/F}(y)\big) = \psi\big(\mathrm{tr}_{E/F}(cy)\big) = \psi_E(cy)$$

as we sought to show.

Definition 4.1.4: Admissible Pair

A pair $(E/F, \chi)$ is called *admissible* if:

1. E/F is a tamely ramified quadratic extension.
2. χ is a character of $E^\times$.
3. χ does not factor through the norm $N_{E/F}$ (i.e., $\chi \neq \varphi \circ N_{E/F}$ for any $\varphi \in \widehat{F^\times}$).
4. If $\chi|_{U_E^1}$ factors through $N_{E/F}$, then E/F is unramified.

Admissible pairs are isomorphic if there is an F-isomorphism $j : E \to E'$ such that $\chi = \chi' \circ j$. Let $P_2(F)$ denote the set of isomorphism classes.

Definition 4.1.5: Minimality of Pairs

An admissible pair $(E/F, \chi)$ is *minimal* if $\chi|_{U_E^n}$ does not factor through $N_{E/F}$, where n is the level of χ.

Any admissible pair is isomorphic to a twist $(E/F, \chi' \otimes \varphi_E)$ where $(E/F, \chi')$ is minimal.

Proposition 4.1.6: Minimality Criterion

Let $(E/F, \chi)$ be admissible with χ of level $m \geq 1$. Let $\alpha \in \mathfrak{p}_E^{-m}$ satisfy $\chi(1 + x) = \psi_E(\alpha x)$ for $x \in \mathfrak{p}_E^m$. Then $(E/F, \chi)$ is minimal if and only if the element α is *minimal over* F.

***Proof*:**
Both conditions are read off the same congruence, so the argument proves the equivalence in one pass.

The restriction $\chi|_{U_E^m}$ is the character $1+x \mapsto \psi_E(\alpha x)$ on $\mathfrak{p}_E^m$. It factors through the norm exactly when it agrees with $\varphi \circ N_{E/F}$ for some character φ of $F^\times$; by proposition 4.1.3 such a φ, of level m' with $em' = m$, is realized by an element $c \in F$ through $\varphi_E(1+x) = \psi_E(cx)$. So $\chi|_{U_E^m}$ factors through the norm if and only if there is $c \in F$ with

$$\psi_E\big((\alpha - c)x\big) = 1 \qquad \text{for all } x \in \mathfrak{p}_E^m$$

Since ψ_E has level one, this says precisely that $\alpha - c \in \mathfrak{p}_E^{1-m}$, that is, that the coset $\alpha + \mathfrak{p}_E^{1-m}$ meets F.

Now compare with definition 3.4.2. In the ramified case an element of F has even valuation in E, so the coset $\alpha + \mathfrak{p}_E^{1-m}$ meets F exactly when m is even: for m even, writing $\alpha = a + b\varpi_E$ with $a, b \in F$ gives $v_E(a) = -m$ and $\alpha - a = b\varpi_E \in \mathfrak{p}_E^{1-m}$, while for m odd no element of F has valuation $-m$. In the unramified case the coset meets F exactly when the residue of $\varpi^m\alpha$ lies in k, that is, exactly when $\varpi^m\alpha + \mathfrak{p}_E$ fails to generate k_E/k. In both cases meeting F is precisely the failure of definition 3.4.2's condition, and $(E/F, \chi)$ is minimal if and only if α is minimal over F, as we sought to show.

4.2 Construction of Representations

We associate an irreducible cuspidal representation π_χ to each admissible pair $(E/F, \chi)$.

4.2.1 Level Zero Construction

Let $(E/F, \chi)$ be an admissible pair where χ has level zero. By definition, E/F must be unramified.

Lemma 4.2.1: Admissibility at Level Zero

Let E/F be unramified quadratic and χ a character of level zero. Let $\sigma \in \mathrm{Gal}(E/F)$ be the non-trivial element. The following are equivalent:

1. $(E/F, \chi)$ is admissible.
2. $\chi \neq \chi^\sigma$.
3. $\chi|_{U_E} \neq \chi^\sigma|_{U_E}$.

***Proof*:**
Since E/F is unramified, $E^\times = F^\times U_E$. The kernel of the norm consists of x/x^σ. χ factors through the norm iff $\chi(x/x^\sigma) = 1$ iff $\chi = \chi^\sigma$.

Embed E into $A = M_2(F)$ such that $E^\times$ normalizes the maximal order $\mathfrak{A} = M_2(\mathfrak{o})$. This induces an embedding of residue fields $k_E \to M_2(k)$. The character $\chi|_{U_E}$ inflates a character $\tilde{\chi}$ of $k_E^\times$. Since $\tilde{\chi} \neq \tilde{\chi}^q$, this generates a cuspidal representation $\tilde{\pi}$ of $\mathrm{GL}_2(k)$ (see theorem 2.1.15). Let Λ be the extension of the inflation of $\tilde{\pi}$ to $K_{\mathfrak{A}} = F^\times \mathrm{GL}_2(\mathfrak{o})$ such that $\Lambda|_{F^\times}$ is a multiple of $\chi|_{F^\times}$.

Definition 4.2.2: Representation π_χ (Level 0)

For a level zero admissible pair, define $\pi_\chi = \text{c-Ind}_{K_\mathfrak{A}}^G \Lambda$.

Proposition 4.2.3: Bijection at Level Zero

The map $(E/F, \chi) \mapsto \pi_\chi$ induces a bijection between $P_2(F)_0$ and $\mathcal{A}_2^0(F)_0$ (cuspidals of level 0).

***Proof*:**
Both sides are parameterized by the same finite-field data, and the argument is to identify that datum on each side.

Surjectivity. Let π be cuspidal of level zero. By theorem 3.2.9, π contains a representation λ of $K = \mathrm{GL}_2(\mathfrak{o})$ inflated from an irreducible cuspidal representation $\tilde{\lambda}$ of $\mathrm{GL}_2(k)$, and $\pi \cong \text{c-Ind}_{ZK}^G \Lambda$ for an extension Λ of λ to $ZK = K_\mathfrak{A}$. By theorem 2.1.15 the cuspidal representations of $\mathrm{GL}_2(k)$ are parameterized by the characters $\tilde{\chi}$ of $k_E^\times$ with $\tilde{\chi} \neq \tilde{\chi}^q$, two such characters giving the same representation exactly when they are exchanged by $\tilde{\chi} \mapsto \tilde{\chi}^q$. Inflating $\tilde{\chi}$ along $U_E \to k_E^\times$ and extending over $F^\times$ so that the central character matches $\Lambda|_{F^\times}$ produces a character χ of $E^\times = F^\times U_E$ of level zero with $\chi \neq \chi^\sigma$, hence an admissible pair by lemma 4.2.1, and by construction $\pi_\chi \cong \pi$.

Injectivity. If $\pi_{\chi_1} \cong \pi_{\chi_2}$ then the two inducing representations of $K_\mathfrak{A}$ are conjugate, since by theorem 3.2.9 the type is determined by π up to conjugacy; hence $\tilde{\chi}_1 \in \{\tilde{\chi}_2, \tilde{\chi}_2^q\}$. Conjugation by the non-trivial element of $\mathrm{Gal}(E/F)$ realizes $\tilde{\chi} \mapsto \tilde{\chi}^q$ and is an isomorphism of pairs, so $(E/F, \chi_1) \cong (E/F, \chi_2)$ once the central characters are compared: both restrict to ω_π on $F^\times$, which pins the extension from U_E to $E^\times$. Since E/F is forced to be the unramified quadratic extension by lemma 4.2.1, there is no choice of field to make, completing the proof.

4.2.2 Positive Level Construction

Let $(E/F, \chi)$ be a minimal admissible pair with χ of level $n \geq 1$. Choose $\alpha \in \mathfrak{p}_E^{-n}$ realizing χ as a character (via ψ_E). Embed E in A such that $E^\times \subset K_\mathfrak{A}$ for a chain order $\mathfrak{A}$ (where $e_\mathfrak{A} = e(E/F)$). The triple $(\mathfrak{A}, n, \alpha)$ is a simple stratum. Define the groups $J = E^\times U_\mathfrak{A}^{\lfloor (n+1)/2 \rfloor}$ and $J^1 = U_E^1 U_\mathfrak{A}^{\lfloor (n+1)/2 \rfloor}$. We must construct a representation $\Lambda \in C(\psi_\alpha, \mathfrak{A})$ of J.

Case 1: n is odd. In this case, $J = E^\times U_\mathfrak{A}^{(n+1)/2}$. The character ψ_α is defined on $U_\mathfrak{A}^{(n+1)/2}$.

Definition 4.2.4: Representation Λ (Odd)

Define the character Λ of J by:

$$\Lambda|_{U_\mathfrak{A}^{(n+1)/2}} = \psi_\alpha, \qquad \Lambda|_{E^\times} = \chi$$

These are consistent since χ corresponds to α.

Case 2: n is even. Here $n = 2m$. E/F is unramified. $J = E^\times U_{\mathfrak{A}}^m$. Let $H^1 = U_E^1 U_{\mathfrak{A}}^{m+1}$ and $J^1 = U_E^1 U_{\mathfrak{A}}^m$. Define a character θ on H^1 by $\theta(ux) = \chi(u)\psi_\alpha(x)$ for $u \in U_E^1, x \in U_{\mathfrak{A}}^{m+1}$.

Lemma 4.2.5: Heisenberg Extension

There is a unique irreducible representation η of J^1 such that $\eta|_{H^1}$ contains θ.

1. $\dim \eta = q$.
2. $\eta|_{H^1}$ is a multiple of θ.
3. There is a unique extension $\tilde{\eta}$ of η to the group $\mu_E J^1$ (where μ_E are roots of unity) such that $\mathrm{tr}\tilde{\eta}(\zeta) = -\chi(\zeta)$ for $\zeta \in \mu_E \setminus \mu_F$.

***Proof*:**
Everything here is an instance of the group-theoretic analysis carried out in section 4.5, which is stated for an abstract finite p-group with cyclic centre precisely so that it can be applied at this point; the verification that the present groups satisfy its hypotheses is the closing paragraph of that section. Take $G_0 = J^1/\ker\theta$ with centre $Z_0 = H^1/\ker\theta$, and $A = \mu_E/\mu_F$ acting by conjugation.

The quotient $V = G_0/Z_0 \cong J^1/H^1 \cong \mathfrak{p}_F^m/(\mathfrak{p}_E^m + \mathfrak{p}_F^{m+1})$ is elementary abelian of order q^2, and the commutator pairing $(x, y) \mapsto \theta([x, y])$ is a non-degenerate alternating form on it. Existence and uniqueness of η with $\eta|_{Z_0}$ containing θ is then the Stone-von Neumann statement recalled at the head of section 4.5, and $\dim \eta = |V|^{1/2} = q$, which is (1); that $\eta|_{H^1}$ is a multiple of θ is part of the same statement, giving (2).

For (3), A is cyclic of order $q + 1$, has order prime to p, acts trivially on Z_0, and acts on V with no non-trivial fixed point, since k_E is a maximal subfield of $M_2(k)$ and so is its own centralizer. These are exactly hypotheses (a), (b) and (c), so lemma 4.5.2 applies and gives a unique extension $\tilde{\eta}$ of η to $\mu_E J^1$ with $\mathrm{tr}\tilde{\eta}(a) = -1$ for $a \neq 1$ in A. Untwisting by χ, which is what identifies A-characters with characters of μ_E trivial on μ_F, turns this into the normalization $\mathrm{tr}\tilde{\eta}(\zeta) = -\chi(\zeta)$ for $\zeta \in \mu_E \setminus \mu_F$, completing the proof.

Corollary 4.2.6: Representation Λ (Even)

There is a unique irreducible representation Λ of J such that:

1. $\Lambda|_{J^1} \cong \eta$
2. $\Lambda|_{F^\times}$ is a multiple of $\chi|_{F^\times}$
3. For $\zeta \in \mu_E \setminus \mu_F$, $\mathrm{tr}\Lambda(\zeta) = -\chi(\zeta)$

***Proof*:**
Write $J = E^\times U_{\mathfrak{A}}^m$ and $J^1 = U_E^1 U_{\mathfrak{A}}^m$. Since E/F is unramified in the even case, $E^\times = \varpi^{\mathbb{Z}} \mu_E U_E^1$ with ϖ a prime element of F, so

$$J = \varpi^{\mathbb{Z}}\, \mu_E\, J^1$$

and the three groups $\varpi^{\mathbb{Z}}$, μ_E and J^1 generate J with J^1 normal.

Existence. Lemma 4.2.5(3) gives the extension $\tilde{\eta}$ of η to $\mu_E J^1$ normalized by $\mathrm{tr}\tilde{\eta}(\zeta) = -\chi(\zeta)$ on $\mu_E \setminus \mu_F$. The remaining generator ϖ is central in J, so setting $\Lambda(\varpi) = \chi(\varpi) \cdot \mathrm{id}$ and $\Lambda|_{\mu_E J^1} = \tilde{\eta}$ defines a representation of J provided the two prescriptions agree on $\varpi^{\mathbb{Z}} \cap \mu_E J^1$, which they do because $\tilde{\eta}$ restricted to $\mu_F \subset \mu_E$ is a multiple of χ by construction. This Λ satisfies (1), (2) and (3), and it is irreducible because its restriction to J^1 is the irreducible η.

Uniqueness. Suppose Λ' also satisfies the three conditions. By (1) the restrictions to J^1 agree, both being the irreducible η, so $\mathrm{Hom}_{J^1}(\Lambda, \Lambda')$ is one dimensional by Schur's lemma. The group J acts on that line by $(j \cdot \varphi)(v) = \Lambda'(j)\varphi(\Lambda(j)^{-1}v)$, hence through a character ξ of J, and ξ is trivial on J^1 because the elements of the line are J^1-equivariant by definition. So $\Lambda' \cong \xi\Lambda$ with ξ a character of J/J^1: the two differ by a character of that quotient, which is generated by the images of ϖ and of μ_E. Condition (2) forces the two to agree on ϖ, and condition (3) forces the traces to agree on every $\zeta \in \mu_E \setminus \mu_F$; since a character of the cyclic group μ_E/μ_F is determined by its values there, the difference character is trivial and $\Lambda' \cong \Lambda$, completing the proof.

Definition 4.2.7: Representation π_χ (Positive Level)

Let Λ be as defined in Case 1 or Case 2. Define $\pi_\chi = \text{c-Ind}_J^G \Lambda$.

Proposition 4.2.8: Independence of Choices

The equivalence class of π_χ depends only on the isomorphism class of $(E/F, \chi)$.

1. It is independent of the choice of α, ψ, and the embedding $E \to A$.
2. If φ is a character of $F^\times$ such that $(E/F, \chi\varphi_E)$ is minimal, then $\pi_{\chi\varphi_E} = \varphi\pi_\chi$.

***Proof*:**
Any two embeddings are conjugate. Changing α changes ψ_α by a conjugation or a unit, which is absorbed by the uniqueness of the Heisenberg representation.

This defines π_χ for any admissible pair, not necessarily minimal, by twisting: $\pi_\chi = \varphi\pi_{\chi'}$ where $\chi = \chi'\varphi_E$.

4.3 The Parametrization Theorem

We denote $\mathcal{A}_2^0(F)$ the set of irreducible cuspidal representations. $\mathcal{A}_2^{\mathrm{nr}}(F)$ is the subset of unramified representations (twists of level 0 types).

Theorem 4.3.1: Tame Parametrization Theorem

The construction $(E/F, \chi) \mapsto \pi_\chi$ induces a bijection:

$$P_2(F) \xrightarrow{\sim} \begin{cases} \mathcal{A}_2^0(F) & \text{if } p \neq 2 \\ \mathcal{A}_2^{\mathrm{nr}}(F) & \text{if } p = 2 \end{cases}$$

Properties:

1. $\ell(\pi_\chi) = \ell(\chi)/e(E|F)$
2. $\omega_{\pi_\chi} = \chi|_{F^\times}$
3. $\pi_{\check{\chi}} = \check{\pi}_\chi$
4. $\pi_{\chi\varphi_E} = \varphi\pi_\chi$

***Proof*:**

The construction and the two propositions of section 4.4 do the work; what remains is to assemble them and to check that twisting lets the minimal case control the general one.

Step 1: the map is well defined. For a minimal pair this is proposition 4.2.8(1): the class of π_χ does not depend on the choice of α, of ψ, or of the embedding $E \to A$. For a general admissible pair, write $\chi = \chi'\varphi_E$ with $(E/F, \chi')$ minimal, as is always possible, and set $\pi_\chi = \varphi\pi_{\chi'}$. If $\chi = \chi''\varphi'_E$ is a second such factorization then $\chi'(\chi'')^{-1} = (\varphi'\varphi^{-1})_E$ is of the form η_E, so $\pi_{\chi'} = \eta\pi_{\chi''}$ by proposition 4.2.8(2), and the two candidate values agree. Isomorphic pairs give equal representations because an F-isomorphism $j\colon E \to E'$ carries one set of choices to another.

Step 2: injectivity. On minimal pairs this is proposition 4.4.1. In general, suppose $\pi_{\chi_1} \cong \pi_{\chi_2}$ with $\chi_i = \chi_i'(\varphi_i)_{E_i}$ and χ_i' minimal. By definition 3.3.12 a representation and its twists attain a smallest level, and by property (1) below that smallest level is attained exactly at the minimal pairs; twisting both sides by φ_1^{-1} therefore reduces to the case where χ_1 and χ_2 are both minimal, where proposition 4.4.1 applies. Undoing the twist gives $(E_1/F, \chi_1) \cong (E_2/F, \chi_2)$.

Step 3: surjectivity. Let π lie in the target set. Twisting by a character of $F^\times$ we may assume $\ell(\pi) \leq \ell(\varphi\pi)$ for every φ, since the level takes non-negative values in $\frac{1}{2}\mathbb{Z}$ and so attains a minimum on the twist orbit. Proposition 4.4.3 then produces a minimal admissible pair $(E/F, \chi)$ with $\pi \cong \pi_\chi$, and undoing the twist realizes the original π as $\pi_{\chi\varphi_E}$ by property (4).

Step 4: the two cases. By theorem 3.5.5 every irreducible cuspidal representation contains a cuspidal type $(\mathfrak{A}, J, \Lambda)$, and by proposition 3.4.3(3) the chain order satisfies $e_{\mathfrak{A}} = e(E|F)$ for the field $E = F[\alpha]$ of its simple stratum. If $p \neq 2$ both quadratic extensions of F are tamely ramified, so every cuspidal type arises from an admissible pair and the target is all of $\mathcal{A}_2^0(F)$. If $p = 2$ only the unramified quadratic extension is tame, so only chain orders with $e_{\mathfrak{A}} = 1$, that is $\mathfrak{A} \cong M_2(\mathfrak{o})$, are reached; by lemma 4.3.2 these are exactly the representations in $\mathcal{A}_2^{\mathrm{nr}}(F)$.

Step 5: the four properties.

1. The representation π_χ contains the simple stratum $(\mathfrak{A}, n, \alpha)$ with n the level of χ, so $\ell(\pi_\chi) = n/e_{\mathfrak{A}}$ by theorem 3.3.13, simple strata being fundamental (definition 3.3.12 alone gives only the inequality $\leq$), and $e_{\mathfrak{A}} = e(E|F)$ by proposition 3.4.3(3).

2. By construction $\Lambda|_{F^\times}$ is a multiple of $\chi|_{F^\times}$, in both the odd case (definition 4.2.4) and the even case (corollary 4.2.6); the centre of G is $F^\times$ and acts on c-Ind$_J^G\Lambda$ through that character.

3. Both G and J are unimodular, J being open and compact modulo the centre, so the Duality Theorem (theorem 1.3.12) gives $\check{\pi}_\chi \cong \mathrm{Ind}_J^G\check{\Lambda}$. The inducing datum attached to $\check{\chi}$ is $\check{\Lambda}$, so c-Ind$_J^G\check{\Lambda} = \pi_{\check{\chi}}$, and the canonical embedding c-Ind$_J^G\check{\Lambda} \to \mathrm{Ind}_J^G\check{\Lambda}$ is a non-zero map between two irreducible representations: the source is irreducible by theorem 3.2.8 and the target by proposition 1.2.41, since π_χ is irreducible. A non-zero map of irreducibles is an isomorphism.

4. This is proposition 4.2.8(2) for minimal pairs, and the definition of π_χ by twisting in Step 1 for the rest, completing the proof.

Lemma 4.3.2: Unramified Recognition

A cuspidal representation π is unramified (in $\mathcal{A}_2^{\mathrm{nr}}(F)$) if and only if the chain order $\mathfrak{A}$ in its cuspidal type is isomorphic to $M_2(\mathfrak{o})$.

***Proof*:**
By proposition 3.3.2 there are exactly two conjugacy classes of chain order in $M_2(F)$, distinguished by the period: $e_{\mathfrak{A}} = 1$ gives $\mathfrak{A} \cong M_2(\mathfrak{o})$ and $e_{\mathfrak{A}} = 2$ gives the Iwahori order. By proposition 3.4.3(3) the period of the order occurring in a cuspidal type equals the ramification index $e(E|F)$ of the field $E = F[\alpha]$ generated by the stratum. So $\mathfrak{A} \cong M_2(\mathfrak{o})$ if and only if E/F is unramified.

It remains to see that this is what "unramified" means for π. The set $\mathcal{A}_2^{\mathrm{nr}}(F)$ consists of the twists $\varphi\pi_0$ of representations π_0 of level zero. Twisting by a character of $F^\times$ changes neither the chain order nor the field: by proposition 4.2.8(2) it replaces the pair $(E/F, \chi)$ by $(E/F, \chi\varphi_E)$, leaving E untouched. A level-zero representation has E/F unramified by lemma 4.2.1, so every member of $\mathcal{A}_2^{\mathrm{nr}}(F)$ has $\mathfrak{A} \cong M_2(\mathfrak{o})$. Conversely, if $\mathfrak{A} \cong M_2(\mathfrak{o})$ then E/F is unramified, and twisting by a character of $F^\times$ chosen to cancel the restriction of χ to U_E^1 reduces the level of the pair to zero, exhibiting π as a twist of a level-zero representation, as we sought to show.

4.4 Tame Intertwining Properties

We prove the injectivity of the map constructed above.

Proposition 4.4.1: Injectivity of Parametrization

The map $(E/F, \chi) \mapsto \pi_\chi$ is injective on isomorphism classes of minimal pairs.

***Proof*:**
Suppose $\pi_{\chi_1} \cong \pi_{\chi_2}$. Central characters imply $\chi_1|_{F^\times} = \chi_2|_{F^\times}$. The levels must match: write ℓ for the common value. Each representation contains a simple stratum $(\mathfrak{A}_i, n_i, \alpha_i)$, and theorem 3.3.13 gives $n_i/e_{\mathfrak{A}_i} = \ell$ for both. A ramified simple stratum has odd n_i, so the parity of ℓ decides

the order type: half-integral ℓ forces $e_{\mathfrak{A}_1} = e_{\mathfrak{A}_2} = 2$, integral ℓ forces $e_{\mathfrak{A}_1} = e_{\mathfrak{A}_2} = 1$, and in either case $n_1 = n_2 = n$. Chain orders with the same ramification index are conjugate in G by proposition 3.3.2, so after conjugating the first stratum we may take a common order $\mathfrak{A}$. The two strata must then intertwine, meaning the characters ψ_{α_i} are conjugate.

Lemma 4.4.2: Conjugacy of Minimal Elements

Let α_1, α_2 be minimal elements generating extensions E_1, E_2. If their characters intertwine, there exists $u \in U_{\mathfrak{A}}$ such that $uE_1u^{-1} = E_2$ and $u\alpha_1 u^{-1} \equiv \alpha_2 \pmod{U^1_{E_2}}$.

***Proof*:**
(of the Lemma.) By proposition 3.3.9 the hypothesis says there is $g \in G$ with

$$g^{-1}\big(\alpha_1 + \mathfrak{P}^{1-n}\big)g \cap \big(\alpha_2 + \mathfrak{P}^{1-n}\big) \neq \emptyset$$

so after replacing α_1 by $g^{-1}\alpha_1 g$, which changes neither minimality nor the field up to isomorphism, the two cosets $\alpha_1 + \mathfrak{P}^{1-n}$ and $\alpha_2 + \mathfrak{P}^{1-n}$ coincide.

Both strata are simple, so by proposition 3.4.3 each α_i is minimal over F and $E_i^\times \subset \mathcal{K}_{\mathfrak{A}}$. Minimality means the coset $\alpha_i + \mathfrak{P}^{1-n}$ contains no element of a proper subfield, so the coset determines the characteristic polynomial of α_i modulo $\mathfrak{p}$, and hence determines $E_i = F[\alpha_i]$ up to F-isomorphism. Two equal cosets therefore give $E_1 \cong E_2$, and since all embeddings of a quadratic extension into $M_2(F)$ normalizing a fixed chain order are conjugate under $U_{\mathfrak{A}}$ (this is the same statement used in proposition 4.2.8(1)), there is $u \in U_{\mathfrak{A}}$ with $uE_1u^{-1} = E_2$.

Finally, $u\alpha_1 u^{-1}$ and α_2 generate the same field E_2 and lie in the same coset modulo $\mathfrak{P}^{1-n}$, hence agree modulo $\mathfrak{p}_{E_2}^{1-n}$, which for a unit of valuation $-n$ is the statement $u\alpha_1 u^{-1} \equiv \alpha_2 \pmod{U^1_{E_2}}$, as we sought to show.

Using this lemma, we align the fields $E_1 \cong E_2$. The equality of representations implies equality of the inducing types Λ_i. For odd level, $\Lambda = \chi$, so $\chi_1 = \chi_2$. For even level, the trace relation on roots of unity recovers the character on the full group $E^\times$.

Proposition 4.4.3: Minimality Converse

Let $\pi \in \mathcal{A}_2^0(F)$ (or $\mathcal{A}_2^{\mathrm{nr}}(F)$ if $p = 2$). Suppose $\ell(\pi) \leq \ell(\varphi\pi)$ for all φ. Then there exists a minimal pair $(E/F, \chi)$ such that $\pi \cong \pi_\chi$.

***Proof*:**
π contains a cuspidal type $(\mathfrak{A}, J, \Lambda)$. If $\mathfrak{A} = \mathcal{I}$, we are in the ramified case (odd level), and $\chi = \Lambda|_{E^\times}$ works. If $\mathfrak{A} = M_2(\mathfrak{o})$ and level > 0, we are in the even level case. We reconstruct χ from $\Lambda|_{J^1}$ (which gives the restriction to units) and the character formula on roots of unity (which gives the rest).

4.5 A Certain Group Extension

We fix a prime number p. Let G be a finite p-group with cyclic centre Z, such that $V = G/Z$ is an elementary abelian p-group. Let θ be a faithful character of Z. The alternating pairing

$$h : V \times V \longrightarrow \mathbb{C}^\times, \quad (x, y) \longmapsto \theta([x, y]),$$

is non-degenerate. By the Stone-von Neumann theorem, there is a unique irreducible representation η of G such that $\eta|_Z$ contains θ.

We introduce a group of automorphisms. Let A be a finite cyclic group of automorphisms of G, of order relatively prime to p. Assume A acts trivially on Z. Since η is the unique representation with central character θ, and A fixes θ, the representation η is stable under A. Thus, η admits an extension to an irreducible representation of the semi-direct product $AG = A \ltimes G$.

Lemma 4.5.1: Extensions of Heisenberg Representations

Let η_0 be some irreducible representation of AG such that $\eta_0|_G \cong \eta$. Let χ range over the irreducible characters of $AG/G \cong A$. The representations $\chi \otimes \eta_0$ are distinct and

$$\mathrm{Ind}_G^{AG}\eta \cong \bigoplus_{\chi \in \widehat{A}} (\chi \otimes \eta_0)$$

***Proof*:**
The determinant of the representation $\chi \otimes \eta_0$ is $\chi^{\dim \eta} \det \eta_0$. Since $\dim \eta = |V|^{1/2}$ is a power of p and $|A|$ is relatively prime to p, the map $\chi \mapsto \chi^{\dim \eta}$ is a permutation of $\widehat{A}$. Thus the determinants are distinct, implying the representations are distinct. By Frobenius Reciprocity, $\mathrm{Ind}_G^{AG}\eta$ contains each $\chi \otimes \eta_0$. Comparing dimensions ($|A| \dim \eta$ vs $\sum 1 \cdot \dim \eta$), we see they exhaust the induced representation.

Specializing to the case relevant to GL_2 imposes the following hypotheses:

(a) $|V| = q^2$, where q is a power of p.

(b) $|A| = q + 1$.

(c) If $a \in A, a \neq 1$, then a acts on V with only the trivial fixed point (regular action).

Lemma 4.5.2: The Trace -1 Extension

Under hypotheses (a) to (c), there is a unique irreducible representation η_1 of AG such that:

1. $\eta_1|_G \cong \eta$.
2. $\mathrm{tr}\eta_1(a) = -1$ for all $a \in A, a \neq 1$.

***Proof*:**
Let η_i, $1 \leq i \leq q+1$, be the distinct extensions of η to AG. Consider the induced representation

$$\xi = \mathrm{Ind}_{AZ}^{AG}(1_A \otimes \theta),$$

where $1_A \otimes \theta$ is the character $(a, z) \mapsto \theta(z)$ of the abelian group $AZ \cong A \times Z$. We decompose ξ. Since $\xi|_G \cong \mathrm{Ind}_Z^G \theta \cong (\dim \eta)\eta = q\eta$, we must have

$$\xi \cong \bigoplus_{i=1}^{q+1} m_i \eta_i, \quad \text{with } \sum m_i = q$$

We compute the character of ξ restricted to AZ.

$$\xi|_{AZ} \cong \bigoplus_{g \in AZ\backslash AG/AZ} \mathrm{Ind}_{AZ \cap g^{-1}AZg}^{AZ}(1_A \otimes \theta)^g$$

The double coset space $AZ\backslash AG/AZ$ is isomorphic to the orbit space V/A. By hypothesis (c), A acts freely on $V \setminus \{0\}$. Thus we have one trivial orbit (size 1) and $q-1$ regular orbits (size $q+1$). For the trivial orbit ($g = 1$), we get $1_A \otimes \theta$. For non-trivial orbits, the intersection of groups is just Z, so we induce from Z to AZ, obtaining $\mathrm{Reg}_A \otimes \theta$.

$$\xi|_{AZ} \cong (1_A \otimes \theta) \oplus (q-1)(\mathrm{Reg}_A \otimes \theta)$$

Calculating the inner product $\langle \xi, \xi \rangle_{AG} = \langle \xi|_{AZ}, 1_A \otimes \theta \rangle_{AZ} = 1 + (q-1) \cdot 1 = q$. Since $\sum m_i = q$ and $\sum m_i^2 = q$, it must be that one m_i is 0 and the rest are 1. Renumber so $m_1 = 0$. Then $\xi = (\bigoplus \eta_i) - \eta_1$. Since $\bigoplus \eta_i \cong \mathrm{Ind}_G^{AG}\eta$, its trace on any $a \in A \setminus \{1\}$ is 0. Therefore, $\mathrm{tr}\xi(a) = -\mathrm{tr}\eta_1(a)$. From the explicit formula above, $\mathrm{tr}\xi(a) = 1 + (q-1)\mathrm{tr}(\mathrm{Reg}_A)(a) = 1 + 0 = 1$. Thus $\mathrm{tr}\eta_1(a) = -1$.

To confirm this applies in the present context, analyze the conjugacy classes.

Lemma 4.5.3: Conjugacy in AG

Let $a \in A, a \neq 1$.

1. Let $g \in G$; the element ag is G-conjugate to an element of the form az, $z \in Z$.
2. For $z \in Z$, the G-conjugacy class and the AG-conjugacy class of az are the same, and this class has q^2 elements.
3. For $z_1, z_2 \in Z$, the elements az_1, az_2 are AG-conjugate if and only if $z_1 = z_2$.

***Proof*:**
Consider the map $\varphi_a : V \to V$ given by $v \mapsto ava^{-1}v^{-1}$. Since A acts without fixed points on V, this map is injective, hence bijective. This implies every ag can be conjugated by G into aZ, proving (1). Since A is abelian, A centralizes az. Thus the AG-conjugacy class is the same as the G-conjugacy class. The size is $|G|/|C_G(az)|$. The centralizer condition $gag^{-1} = a$ modulo Z is equivalent to g being a fixed point of a in V, which implies $g \in Z$. Thus the centralizer is Z, and the class size is $|G|/|Z| = |V| = q^2$. Part (3) follows because the commutator of AG is

contained in G, so $AG/G \cong A$ is abelian. Conjugacy preserves the projection to A.

The construction in section 4.2 satisfies these hypotheses. Here $n = 2m$.

- $G = J^1_\alpha / \ker \theta$.
- $Z = H^1_\alpha / \ker \theta$.
- θ is the character defined by ψ_α and χ.

Here $V = G/Z \cong J^1_\alpha / H^1_\alpha \cong \mathfrak{p}_F^m / (\mathfrak{p}_E^m + \mathfrak{p}_F^{m+1})$. Identify $A/P \cong M_2(k)$. Since E/F is unramified, the image of $\mathfrak{o}_E$ is the field $k_E \subset M_2(k)$. Then $V \cong M_2(k)/k_E$. This has dimension 2 over k (size q^2). The group $A = \mu_E/\mu_F \cong k_E^\times / k^\times$ has order $(q^2-1)/(q-1) = q+1$. The action is by conjugation. A scalar $x \in k_E^\times$ fixes $y \in M_2(k)$ iff they commute. Since k_E is a maximal subfield, the centralizer is k_E. As x has finite order prime to p, conjugation by x is a semisimple operator on $M_2(k)$, so the fixed space k_E admits an x-invariant complement, on which x has no non-zero fixed vector. Thus the action on the quotient $M_2(k)/k_E$ is free. All hypotheses are satisfied.

5

Characterization by Functional Equations

We have classified representations of G, but to relate them to Galois groups, we need invariant objects that exist on both sides of the fence. In particular, we shall fuse the notion of characters from [5] with the functional theory of [14] to create new invariants. The goal of this chapter is to define these invariants, the local L-function $L(\pi, s)$ and the local constant (or ε-factor) $\varepsilon(\pi, s, \psi)$, and to prove that they completely characterize the representation.

We start in section 5.1 with the thesis of John Tate, establishing the functional equation and ε-factors for $\mathrm{GL}_1(F)$ (characters). We then upgrade this to $\mathrm{GL}_2(F)$ in section 5.2 using the Godement-Jacquet theory of Zeta integrals. We provide explicit formulas for the ε-factors of cuspidal representations using Gauss sums. The key result of this chapter is the *Converse Theorem* (section 5.5), which asserts that if two representations have matching L and ε factors for all twists, they must be isomorphic. This theorem is the engine that makes the Langlands correspondence unique.

5.1 Functional Equation for GL(1)

Recall that if $a \in F$ and $\psi \in \widehat{F}$, then $a\psi$ denotes the character $x \mapsto \psi(ax)$. If $\psi \neq 1$, the map $a \mapsto a\psi$ gives an isomorphism $F \cong \widehat{F}$

Definition 5.1.1: Fourier Transform

Let $\psi \in \widehat{F}$, $\psi \neq 1$ to define an isomorphism $F \cong \hat{F}$, and let μ be a Haar measure on F. For $\Phi \in C_c^\infty(F)$, define the *Fourier transform* $\hat{\Phi}$ of Φ (with respect to to μ and ψ) to be:

$$\hat{\Phi}(x) = \int_F \Phi(y)\psi(xy)\, d\mu(y), \quad x \in F$$

The integrand is locally constant, and hence the integral reduces to a finite sum.

Proposition 5.1.2: Fourier Transform Properties

1. For $\Phi \in C_c^\infty(F)$, the function $\hat{\Phi}$ lies in $C_c^\infty(F)$.
2. There is a positive real number $c = c(\psi, \mu)$ such that
$$\hat{\hat{\Phi}}(x) = c\Phi(-x), \quad \Phi \in C_c^\infty(F),\ x \in F$$
3. For given ψ, there is a unique Haar measure μ_ψ for which $c(\psi, \mu_\psi) = 1$. This measure satisfies $\mu_\psi(\mathfrak{o}) = q^{l/2}$, where l is the level of ψ.
4. For $a \in F^\times$, $\mu_{a\psi} = \|a\|^{\frac{1}{2}}\mu_\psi$.

Notice how (1) is completely false in the real analysis case: By the uncertainly principle it is impossible for f and $\hat{f}$ to be compactly supported, but it *is* the case in p-adic analysis case.

***Proof*:**

1-2. Let l be the level of ψ, so that $\psi(\mathfrak{p}^l) = 1$ and $\psi(\mathfrak{p}^{l-1}) \neq 1$, and let Φ_j be the characteristic function of $\mathfrak{p}^j$. For $a \in F$, the character $a\psi|_{\mathfrak{p}^j}$ is trivial if and only if $a \in \mathfrak{p}^{l-j}$. The support of $\hat{\Phi}_j$ is therefore $\mathfrak{p}^{l-j}$ and

$$\hat{\Phi}_j(x) = \mu(\mathfrak{o})q^{-j}, \quad x \in \mathfrak{p}^{l-j}$$

Therefore, (1) and (2) hold for all functions Φ_j, with $c = \mu(\mathfrak{o})^2 q^{-l}$. Let $\Phi \in C_c^\infty(F)$, $a \in F$, and let Ψ denote the function $x \mapsto \Phi(x - a)$. We then have

$$\hat{\Psi}(x) = \int_F \Phi(y - a)\psi(xy)\, d\mu(y) = a\psi(x)\hat{\Phi}(x)$$

The function $a\psi$ is locally constant, so $\hat{\Psi}$ lies in $C_c^\infty(F)$ provided $\hat{\Phi}$ does. Calculating the Fourier transform again, we get

$$\hat{\hat{\Psi}}(x) = \hat{\hat{\Phi}}(a + x)$$

If parts (1) and (2) of the Lemma hold for Φ, they also therefore hold for Ψ with the same value of c. However, they do hold for the functions Φ_j, and the functions $x \mapsto \Phi_j(x - a)$, $a \in F$, $j \in \mathbb{Z}$, span $C_c^\infty(F)$. The first two parts of the proposition therefore hold for all $\Phi \in C_c^\infty(F)$, with $c = \mu(\mathfrak{o})^2 q^{-l}$, as above.

3. For $b > 0$, we have $c(\psi, b\mu) = b^2 c(\psi, \mu)$; put another way, to achieve $c(\psi, \mu) = 1$, we must have $\mu(\mathfrak{o})^2 q^{-l} = 1$.
4. immediate.

The important concepts from the proof above are worth boxing:

Definition 5.1.3: Self-dual Haar measure

The measure μ_ψ is called *the self-dual Haar measure on F, relative to ψ*. Using μ_ψ to compute the Fourier transform:

$$\hat{\Phi}(x) = \int_F \Phi(y)\psi(xy)\, d\mu_\psi(y),$$

Definition 5.1.4: Fourier Inversion Formula

The 2nd point in proposition 5.1.2 is called the *Fourier inversion formula*

$$\hat{\hat{\Phi}}(x) = \Phi(-x), \quad \Phi \in C_c^\infty(F),\ x \in F \tag{5.1}$$

The multiplicative information is codified by the Mellin transform, i.e. the Gelfand Transform with $(\mathbb{R}_{>0}, \cdot)$ [13].

Definition 5.1.5: Multiplicative Haar Measure

Let μ be a Haar measure on the additive group F. Define a Haar measure μ^* on the multiplicative group $F^\times$ by

$$d\mu^*(x) = \frac{d\mu(x)}{\|x\|}$$

It is easy to show this measure is scaling-invariant. Hence, when integrating over $F^\times$, it is invariant to units. To that end, let χ be a character of $F^\times$, let $\Phi \in C_c^\infty(F)$, and choose a prime element ϖ of F.

Definition 5.1.6: Mellin Transform

Let $\Phi \in C_c^\infty(M)$. Then its *Mellin transform* (with respect to s) is a new function $\mathcal{M}(\Phi, -, s)$ that inputs multiplicative characters and outputs:

$$\mathcal{M}(\Phi, \chi, s) = \int_{F^\times} \Phi(x)\chi(x)\|x\|^s d\mu^*(x)$$

This simplifies considerably. Recall that the multiplicative group is topologically $F^\times \cong \mathbb{Z} \times U_F$, so the field decomposes into "shells" (or annuli) based on valuation. For $m \in \mathbb{Z}$, let e_m denote the characteristic function of the set $\varpi^m U_F = \mathfrak{p}^m \setminus \mathfrak{p}^{m+1}$.

For $\Phi \in C_c^\infty(F)$, the function $\Phi_m = e_m \Phi$ lies in $C_c^\infty(F^\times) \subset C_c^\infty(F)$. It is identically zero for $m \ll 0$, so one may set:

$$z_m = z_m(\Phi, \chi) = \int_{\varpi^m U_F} \Phi(x)\chi(x)\, d\mu^*(x), \quad m \in \mathbb{Z} \tag{5.2}$$

On the shell $\varpi^m U_F$, the norm $\|x\|$ is exactly q^{-m}. Therefore, $\|x\|^s = (q^{-m})^s = (q^{-s})^m$. Thus the value can be pulled out, giving:

$$\mathcal{M}(\Phi, \chi, s) = \sum_{m\in\mathbb{Z}} (q^{-s})^m \underbrace{\left(\int_{\varpi^m U_F} \Phi(x)\chi(x) d\mu^*(x)\right)}_{\text{This is } z_m}$$

The value z_m captures the correlation between the test function Φ and the character χ specifically on the shell where $v_F(x) = m$. These assemble into a generating function, and the q^{-s} vary (as it can depend on the choice of q) by choosing a formal variable X:

Definition 5.1.7: Local Zeta Integral (Formal Series)

Define a formal Laurent series $Z(\Phi, \chi, X) \in \mathbb{C}((X))$ by

$$\mathcal{Z} : C_c^\infty(F) \to \mathbb{C}((X))$$
$$Z(\Phi, \chi, X) = \sum_{m\in\mathbb{Z}} z_m X^m$$

Its image is denoted $\mathcal{Z}(\chi) = \mathcal{Z}(\chi, X) = \{Z(\Phi, \chi, X) : \Phi \in C_c^\infty(F)\}$.

For $a \in F^\times$ and $\Phi \in C_c^\infty(F)$, write $a \cdot \Phi$ for the function $x \mapsto \Phi(a^{-1}x)$. A simple calculation shows:

$$Z(a \cdot \Phi, \chi, X) = \chi(a)X^{v_F(a)}Z(\Phi, \chi, X)$$

Thus $\mathcal{Z}(\chi)$ is a module over the ring $\mathbb{C}[X, X^{-1}]$ of Laurent polynomials.

Recall that on the shell $\varpi^m U_F$, the norm is constant: $\|x\| = q^{-m}$. Formally substituting $X = q^{-s}$ into the series recovers the standard integral notation:

$$Z(\Phi, \chi, q^{-s}) = \sum_{m\in\mathbb{Z}} \int_{\varpi^m U_F} \Phi(x)\chi(x)(q^{-m})^s\, d\mu^*(x) = \int_{F^\times} \Phi(x)\chi(x)\|x\|^s\, d\mu^*(x)$$

The notations $\zeta(\Phi, \chi, s)$ and $Z(\Phi, \chi, q^{-s})$ are used interchangeably. The function z_m acts as the m-th Fourier coefficient of Φ with respect to the valuation dimension of the group $F^\times$.

Proposition 5.1.8: Rationality of Zeta Integrals

Let χ be a character of $F^\times$; then

$$\mathcal{Z}(\chi, X) = P_\chi(X)^{-1}\mathbb{C}[X, X^{-1}],$$

where

$$P_\chi(X) = \begin{cases} 1 - \chi(\varpi)X & \text{if } \chi \text{ is unramified,} \\ 1 & \text{otherwise.} \end{cases}$$

Hence $\mathcal{Z}(\chi)$ is a PID.

***Proof*:**
Suppose first that $\Phi(0) = 0$. Thus $\Phi|_{F^\times}$ lies in $C_c^\infty(F^\times)$ and $Z(\Phi, \chi, X)$ has only finitely many non-zero coefficients, so:

$$Z(\Phi, \chi, X) \in \mathbb{C}[X, X^{-1}]$$

If Φ is the indicator function of a sufficiently small neighbourhood of 1, then $Z(\Phi, \chi, X)$ is a positive constant, and so $1 \in \mathcal{Z}(\chi)$ and

$$\{Z(\Phi, \chi, X) : \Phi \in C_c^\infty(F^\times)\} = \mathbb{C}[X, X^{-1}]$$

Let Φ_0 be the characteristic function of $\mathfrak{o}$. Then:

$$Z(\Phi_0, \chi, X) = \sum_{m \geq 0} \chi(\varpi^m) X^m \int_{U_F} \chi(x)\, d\mu^*(x)$$

The inner integral is $\mu^*(U_F)$ if χ is unramified (trivial on U_F), and is zero otherwise (by orthogonality of characters on the compact group U_F). In other words:

$$\mu^*(U_F)^{-1} Z(\Phi_0, \chi, X) = \begin{cases} (1 - \chi(\varpi)X)^{-1} & \text{if } \chi \text{ is unramified,} \\ 0 & \text{otherwise.} \end{cases}$$

Since $C_c^\infty(F)$ is spanned by Φ_0 and $C_c^\infty(F^\times)$, the result follows.

The polynomial $P_\chi(X)^{-1}$ is given a name:

Definition 5.1.9: Local L-function

The (multiplicative) inverse of the polynomial $P_\chi(X)$ is called the *Local L-function*. It is traditionally is denoted $L(\chi, s)$ when $X = q^{-s}$ so that:

$$L(\chi, s) = \begin{cases} \frac{1}{1 - \chi(\varpi) q^{-s}} & \text{if } \chi \text{ is unramified,} \\ 1 & \text{otherwise.} \end{cases}$$

Corollary 5.1.10: Poles of Unramified L-functions

Let χ_1, χ_2 be unramified characters of $F^\times$. The following are equivalent:

1. The meromorphic functions $L(\chi_1, s)$ and $L(\chi_2, s)$ have a pole in common;
2. The meromorphic functions $L(\chi_1, s)$ and $L(\chi_2, s)$ have the same sets of poles;
3. $\chi_1 = \chi_2$.

***Proof*:**
The implication (3) $\Rightarrow$ (2) is trivial, and (2) $\Rightarrow$ (1) is immediate since the set of poles is non-empty.

We prove (1) $\Rightarrow$ (3). The poles of $L(\chi_i, s)$ are the zeros of the polynomial $1 - \chi_i(\varpi) q^{-s}$. Thus, s is a pole if and only if $q^s = \chi_i(\varpi)$.

Suppose the functions share a pole s_0. Then:

$$\chi_1(\varpi) = q^{s_0} = \chi_2(\varpi)$$

By definition, an unramified character is trivial on the units U_F. Since $F^\times \cong \mathbb{Z} \times U_F$ (generated by ϖ and U_F), an unramified character is completely determined by its value on the prime element ϖ. Therefore $\chi_1 = \chi_2$.

The main result of the local theory is next: the functional equation relating the additive and multiplicative structures. Fix a character $\psi \in \widehat{F}$, $\psi \neq 1$. For $\Phi \in C_c^\infty(F)$, let $\hat{\Phi}$ denote the Fourier transform relative to the self-dual measure μ_ψ.

Theorem 5.1.11: Local Functional Equation (GL(1))

Let χ be a character of $F^\times$. There exists a unique rational function $c(\chi, \psi, X) \in \mathbb{C}(X)$ such that

$$Z(\hat{\Phi}, \check{\chi}, q^{-1}X^{-1}) = c(\chi, \psi, X)Z(\Phi, \chi, X), \tag{5.3}$$

for all $\Phi \in C_c^\infty(F)$. Here $\check{\chi} = \chi^{-1}$.

This equation is often called *Tate's (local) functional equation*

***Proof*:**
Consider the space Λ of linear maps $\lambda : C_c^\infty(F) \to \mathbb{C}(X)$ satisfying the homogeneity condition:

$$\lambda(a \cdot \Phi) = \chi(a)X^{v_F(a)}\lambda(\Phi), \quad \Phi \in C_c^\infty(F), a \in F^\times$$

Then Λ is a vector space over $\mathbb{C}(X)$ and contains the map $\lambda_0 : \Phi \mapsto Z(\Phi, \chi, X)$ which is non-zero by proposition 5.1.8.

For $\Phi \in C_c^\infty(F)$ and $a \in F^\times$, a simple calculation using the properties of the Fourier transform yields:

$$\widehat{a \cdot \Phi}(y) = \|a\|a^{-1} \cdot \hat{\Phi}(y)$$

It follows that the map

$$\lambda_1 : \Phi \mapsto Z(\hat{\Phi}, \check{\chi}, q^{-1}X^{-1})$$

also lies in Λ. If Λ is one-dimensional, we can pick our desired constant.

Lemma 5.1.12: Uniqueness of Functionals

The space Λ has dimension one over $\mathbb{C}(X)$.

***Proof*:**
(of Lemma) Choose $n \geq 0$ such that $U_F^n \subset \operatorname{Ker} \chi$. Let Φ_k denote the characteristic function of U_F^k, $k \geq 1$. We consider the evaluation map $\Lambda \to \mathbb{C}(X)$ given by $\lambda \mapsto \lambda(\Phi_n)$. We show this is injective.

Suppose $\lambda(\Phi_n) = 0$. The defining property of Λ implies $\lambda(a \cdot \Phi_k) = \lambda(\Phi_k)$ for any $a \in U_F^n$ and $k \geq n$. Since the volume of the units scales by indices, we have $\lambda(\Phi_k) = q^{n-k}\lambda(\Phi_n) = 0$ for $k \geq n$.

Next, any $\Phi \in C_c^\infty(F^\times)$ is a finite linear combination of $F^\times$-translates of functions Φ_k for sufficiently large k. Thus, λ vanishes on $C_c^\infty(F^\times)$. For a general $\Phi \in C_c^\infty(F)$, the value $\lambda(\Phi)$ therefore depends only on $\Phi(0)$ (since $\Phi - \Phi(0)\Phi_{\text{small}}$ vanishes at 0). However, $\lambda(a \cdot \Phi) = \chi(a)X^{v(a)}\lambda(\Phi)$ forces $\lambda(\Phi) = 0$ unless λ is identically zero.

Since Λ is one-dimensional and $\lambda_0 \neq 0$, there exists a unique scalar $c(\chi, \psi, X) \in \mathbb{C}(X)$ such that $\lambda_1 = c(\chi, \psi, X)\lambda_0$.

The above is the notation Tate used. The following is the more traditional notation. Let s be a complex variable.

$$\begin{aligned} \zeta(\Phi, \chi, s) &= Z(\Phi, \chi, q^{-s}), \\ L(\chi, s) &= P_\chi(q^{-s})^{-1}, \\ \gamma(\chi, s, \psi) &= c(\chi, \psi, q^{-s}) \end{aligned}$$

The functional equation then reads:

$$\zeta(\hat{\Phi}, \check{\chi}, 1 - s) = \gamma(\chi, s, \psi)\zeta(\Phi, \chi, s)$$

This notation is used from now on.

The structure of $\gamma(\chi, s, \psi)$ deserves a closer look. It is given a name:

Definition 5.1.13: Epsilon Factor

The *epsilon factor* (or local constant) $\varepsilon(\chi, s, \psi)$ defined to be the function $\varepsilon(\chi, s, \psi) \in \mathbb{C}(q^{-s})$ satisfying the relation:

$$\gamma(\chi, s, \psi) = \varepsilon(\chi, s, \psi)\frac{L(\check{\chi}, 1 - s)}{L(\chi, s)}$$

Theorem 5.1.11 also yields a functional equation for the epsilon factor.

Corollary 5.1.14: Functional Equation of the Local Constant

The function $\varepsilon(\chi, s, \psi)$ satisfies the functional equation

$$\varepsilon(\chi, s, \psi)\varepsilon(\check{\chi}, 1 - s, \psi) = \chi(-1)$$

It is of the form

$$\varepsilon(\chi, s, \psi) = q^{(1/2-s)n(\chi,\psi)}\varepsilon(\chi, 1/2, \psi),$$

for some integer $n(\chi, \psi) \in \mathbb{Z}$.

***Proof*:**
The functional equation in theorem 5.1.11 reads

$$\zeta(\hat{\Phi}, \check{\chi}, 1-s) = \gamma(\chi, s, \psi)\zeta(\Phi, \chi, s) \tag{5.4}$$

Applying this twice, we get

$$\zeta(\hat{\hat{\Phi}}, \chi, s) = \gamma(\check{\chi}, 1-s, \psi)\zeta(\hat{\Phi}, \check{\chi}, 1-s) = \gamma(\check{\chi}, 1-s, \psi)\gamma(\chi, s, \psi)\zeta(\Phi, \chi, s)$$

Using the Fourier inversion formula $\hat{\hat{\Phi}}(x) = \Phi(-x)$, we see that

$$\zeta(\hat{\hat{\Phi}}, \chi, s) = \int_{F^\times} \Phi(-x)\chi(x)\|x\|^s \, d\mu^*(x) = \chi(-1)\zeta(\Phi, \chi, s)$$

Comparing the two expressions, we obtain

$$\gamma(\check{\chi}, 1-s, \psi)\gamma(\chi, s, \psi) = \varepsilon(\check{\chi}, 1-s, \psi)\varepsilon(\chi, s, \psi) = \chi(-1),$$

which proves the first relation.

For the second assertion, we can rewrite (5.4) in the form

$$\frac{\zeta(\hat{\Phi}, \check{\chi}, 1-s)}{L(\check{\chi}, 1-s)} = \varepsilon(\chi, s, \psi)\frac{\zeta(\Phi, \chi, s)}{L(\chi, s)}$$

The fraction on either side lies in $\mathbb{C}[q^s, q^{-s}]$ (by the definition of the L-function as the generator of the zeta ideal). As in proposition 5.1.8, we may choose Φ such that $\zeta(\Phi, \chi, s) = L(\chi, s)$. We deduce that $\varepsilon(\chi, s, \psi) \in \mathbb{C}[q^s, q^{-s}]$. Likewise, $\varepsilon(\check{\chi}, 1-s, \psi) \in \mathbb{C}[q^s, q^{-s}]$.

The functional equation implies that $\varepsilon(\chi, s, \psi)$ is an invertible element of the ring $\mathbb{C}[q^s, q^{-s}]$, and hence equal to aq^{ms}, for some $a \in \mathbb{C}^\times, m \in \mathbb{Z}$. This can be written in the required form, completing the proof.

5.1.1 Local Constants

L-functions characterize unramified characters. If the character is ramified, we loose all information:

Lemma 5.1.15: L-function of Ramified Character

If the character χ of $F^\times$ is ramified, then $L(\chi, s) = 1$.

***Proof*:**
By definition, $L(\chi, s)$ is the generator of the ideal $\mathcal{Z}(\chi)$ of all zeta integrals. To prove the generator is 1, it suffices to exhibit a test function $\Phi \in C_c^\infty(F)$ such that the integral $Z(\Phi, \chi, s)$ is a non-zero constant.

Since χ is smooth but ramified, there exists an integer $n \geq 1$ such that χ is non-trivial on U_F but trivial on the subgroup $U_F^n = 1 + \mathfrak{p}^n$. Let Φ be the characteristic function of this set U_F^n. The support of Φ is contained in the units, so $v_F(x) = 0$ (and $\|x\|^s = 1$) for all x in the support.

Computing the integral:

$$Z(\Phi, \chi, s) = \int_{F^\times} \Phi(x)\chi(x)\|x\|^s \, d\mu^*(x) = \int_{U_F^n} 1 \cdot \chi(x) \cdot 1 \, d\mu^*(x)$$

Since χ is trivial on U_F^n, this reduces to:

$$Z(\Phi, \chi, s) = \int_{U_F^n} d\mu^*(x) = \mu^*(U_F^n)$$

This is a non-zero constant (independent of s). The ideal $\mathcal{Z}(\chi)$ therefore contains a unit of the ring $\mathbb{C}[q^s, q^{-s}]$, which implies that the generator $L(\chi, s)$ is 1.

More information is therefore needed to characterize irreducible representations using functionals. Theorem 5.5.2 will show the ε-factors are exactly the required information. It will be essential to have a table of values for the local constant $\varepsilon(\chi, s, \psi)$. Observe first that the dependence on the additive character ψ is transparent.

Lemma 5.1.16: Dependence on Additive Character

Let $a \in F^\times$; then

$$\varepsilon(\chi, s, a\psi) = \chi(a)\|a\|^{s-1/2}\varepsilon(\chi, s, \psi)$$

***Proof*:**
The L-functions $L(\chi, s)$ and $L(\check{\chi}, 1-s)$ depend only on the character χ and not on the additive character ψ. Therefore, the ratio between $\varepsilon(\chi, s, a\psi)$ and $\varepsilon(\chi, s, \psi)$ is the same as the ratio between the factors $\gamma(\chi, s, a\psi)$ and $\gamma(\chi, s, \psi)$. We calculate the latter using the functional equation.

Let $\psi' = a\psi$. From proposition 5.1.2, the self-dual measure scales as $\mu_{\psi'} = \|a\|^{1/2}\mu_\psi$. The Fourier transform relative to ψ' is given by:

$$\begin{aligned}\hat{\Phi}_{\psi'}(x) &= \int_F \Phi(y)\psi(axy) \, d\mu_{\psi'}(y) \\ &= \|a\|^{1/2} \int_F \Phi(y)\psi(axy) \, d\mu_\psi(y) \\ &= \|a\|^{1/2}\hat{\Phi}_\psi(ax)\end{aligned}$$

We substitute this into the zeta integral on the left side of the functional equation:

$$\zeta(\hat{\Phi}_{\psi'}, \check{\chi}, 1-s) = \int_{F^\times} \|a\|^{1/2}\hat{\Phi}_\psi(ax)\check{\chi}(x)\|x\|^{1-s} \, d\mu^*(x)$$

We perform the change of variables $y = ax$. Since μ^* is a Haar measure on $F^\times$, $d\mu^*(x) = d\mu^*(y)$. We observe that $x = a^{-1}y$, so $\check{\chi}(x) = \check{\chi}(a^{-1})\check{\chi}(y) = \chi(a)\check{\chi}(y)$ and $\|x\|^{1-s} = \|a\|^{s-1}\|y\|^{1-s}$. The integral becomes:

$$\begin{aligned}\zeta(\hat{\Phi}_{\psi'}, \check{\chi}, 1-s) &= \|a\|^{1/2}\chi(a)\|a\|^{s-1} \int_{F^\times} \hat{\Phi}_\psi(y)\check{\chi}(y)\|y\|^{1-s} \, d\mu^*(y) \\ &= \chi(a)\|a\|^{s-1/2}\zeta(\hat{\Phi}_\psi, \check{\chi}, 1-s)\end{aligned}$$

Applying the functional equation definition $\zeta(\hat{\Phi}, \check{\chi}, 1-s) = \gamma(\chi, s, \psi)\zeta(\Phi, \chi, s)$ to both sides, we obtain:

$$\gamma(\chi, s, a\psi)\zeta(\Phi, \chi, s) = \chi(a)\|a\|^{s-1/2}\gamma(\chi, s, \psi)\zeta(\Phi, \chi, s)$$

Since this holds for all test functions Φ, the result follows.

The lemma above reduces the computation of $\varepsilon(\chi, s, \psi)$ for characters ψ of level one. The self-dual Haar measure for such a ψ satisfies

$$\int_{\mathfrak{o}} dx = q^{1/2}$$

For the purposes of calculation, it will be convenient to choose the Haar measure d^*x on $F^\times$ such that

$$\int_{U_F} d^*x = 1$$

Proposition 5.1.17: Epsilon Factor for Unramified Characters

Suppose that χ is unramified and that ψ has level one. If ϖ is a prime element of F, then

$$\varepsilon(\chi, s, \psi) = q^{s-1/2}\chi(\varpi)^{-1}$$

***Proof*:**
If Φ denotes the characteristic function of $\mathfrak{o}$, we get $\zeta(\Phi, \chi, s) = L(\chi, s)$, as in the proof of proposition 5.1.8. On the other side, since ψ has level 1, the Fourier transform is supported on $\mathfrak{p}$ and given by $\hat{\Phi}(x) = q^{1/2}\Phi(\varpi^{-1}x)$, so

$$\begin{aligned}
\zeta(\hat{\Phi}, \chi^{-1}, s) &= q^{1/2}\int_{F^\times} \Phi(\varpi^{-1}x)\chi(x)^{-1}\|x\|^s \, d^*x \\
&= q^{1/2-s}\chi(\varpi)^{-1}\int_{F^\times} \Phi(x)\chi(x)^{-1}\|x\|^s \, d^*x \\
&= q^{1/2-s}\chi(\varpi)^{-1}L(\chi^{-1}, s)
\end{aligned}$$

It follows that

$$\varepsilon(\chi, s, \psi) = \frac{\zeta(\hat{\Phi}, \chi^{-1}, 1-s)}{L(\chi^{-1}, 1-s)} = q^{s-1/2}\chi(\varpi)^{-1},$$

as required.

Theorem 5.1.18: Epsilon Factor for Ramified Characters

Suppose that χ has level $n \geq 0$ and is *ramified*. Let $\psi \in \widehat{F}$ have level one. Then

$$\varepsilon(\chi, s, \psi) = q^{n(1/2-s)}\left(\sum_{x \in U_F/U_F^{n+1}} \chi(\alpha x)^{-1}\psi(\alpha x)\right) q^{-(n+1)/2},$$

for any $\alpha \in F^\times$ such that $v_F(\alpha) = -n$.

***Proof*:**
Let Φ be the characteristic function of U_F^{n+1}, so that

$$\zeta(\Phi, \chi, s) = \mu^*(U_F^{n+1})$$

The Fourier transform $\hat{\Phi}$ has support $\mathfrak{p}^{-n}$ and

$$\hat{\Phi}(y) = q^{1/2-n-1}\psi(y), \quad y \in \mathfrak{p}^{-n}$$

Thus

$$\begin{aligned}\zeta(\hat{\Phi}, \chi^{-1}, s) &= q^{1/2-n-1} \int_{\mathfrak{p}^{-n}\setminus\{0\}} \psi(y)\chi(y)^{-1}\|y\|^s \, d^*y \\ &= q^{1/2-n-1} \sum_{m \geq -n} z_m q^{-ms},\end{aligned}$$

where

$$z_m = \int_{\mathfrak{p}^m \setminus \mathfrak{p}^{m+1}} \psi(y)\chi(y)^{-1} \, d^*y$$

We know from corollary 5.1.14 that $\zeta(\hat{\Phi}, \chi^{-1}, s)$ must reduce to a monomial in q^s, so only one of the coefficients z_m is non-zero. The first one is

$$z_{-n} = \int_{U_F} \psi(\alpha u)\chi^{-1}(\alpha u) \, d^*u,$$

for any $\alpha \in F$ with $v_F(\alpha) = -n$. The integrand is constant on cosets of U_F^{n+1} and so the integral reduces to

$$\int_{U_F} \psi(\alpha u)\chi^{-1}(\alpha u) \, d^*u = \mu^*(U_F^{n+1}) \sum_{x \in U_F/U_F^{n+1}} \chi(\alpha x)^{-1}\psi(\alpha x)$$

It remains to show that $z_m = 0$ for every $m \geq 1-n$. To that end, take $\beta \in F$, with $v_F(\beta) = m > -n$, and consider the coefficient

$$z_m = \int_{U_F} \psi(\beta u)\chi(\beta u)^{-1} \, d^*u \tag{5.5}$$

Taking $v \in U_F^n$, we have

$$\int_{U_F} \psi(\beta u)\chi(\beta u)^{-1} \, d^*u = \int_{U_F} \psi(\beta uv)\chi(\beta uv)^{-1} \, d^*u$$

Setting $v = 1 + y$, the first factor reduces to $\psi(\beta u)\psi(\beta uy) = \psi(\beta u)$, since $\beta uy \in \mathfrak{p}^{m+n} \subset \mathfrak{p} \subset \operatorname{Ker}\psi$ (as $m > -n \implies m + n \geq 1$). Thus

$$\int_{U_F} \psi(\beta u)\chi(\beta u)^{-1} \, d^*u = \chi(v)^{-1} \int_{U_F} \psi(\beta u)\chi(\beta u)^{-1} \, d^*u$$

We may certainly choose $v \in U_F^n$ so that $\chi(v) \neq 1$ (by the definition of level), so the integral (5.5) vanishes, completing the proof.

Consider the sum in parentheses, the "trigonometric sum":

$$\sum_{x \in U_F / U_F^{n+1}} \chi(\alpha x)^{-1} \psi(\alpha x)$$

where the name comes from the classical case, in which the image of a character lies in the unit circle. This type of sum, mixing an additive and a multiplicative character, is given a name:

Definition 5.1.19: Gauss Sum

Let χ be a ramified character of $F^\times$ of level $n \geq 0$, and let $\psi \in \widehat{F}$ have level one. Let $c \in F$ satisfy $v_F(c) = -n$. The sum

$$\tau(\chi, \psi) = \sum_{x \in U_F / U_F^{n+1}} \check{\chi}(cx)\psi(cx) \tag{5.6}$$

is called the *Gauss sum* of χ relative to ψ.

Another common convention is to denote the Gauss sum as $\tau(\check{\chi}, \psi)$; the above version is slightly more convenient for the present purposes.

The definition is independent of the choice of coset representative x and element c. Re-writing the equation in theorem 5.1.18 gives:

$$\varepsilon(\chi, s, \psi) = q^{n(1/2-s)} \frac{\tau(\chi, \psi)}{q^{(n+1)/2}} \tag{5.7}$$

The functional equation for the local constant (corollary 5.1.14) gives:

$$\tau(\chi, \psi)\tau(\check{\chi}, \psi) = \chi(-1)q^{n+1} \tag{5.8}$$

The sums of characters can often be simplified (think of the proof of summing the powers of roots of unity). Indeed:

Proposition 5.1.20: Simplification of Gauss Sum

Suppose that χ has level $n \geq 1$. Let $c \in F$ satisfy

$$\chi(1 + x) = \psi(cx), \quad x \in \mathfrak{p}^{[n/2]+1}$$

Then

$$\tau(\chi, \psi) = q^{[(n+1)/2]} \sum_y \check{\chi}(cy)\psi(cy), \tag{5.9}$$

where y ranges over $U_F^{[(n+1)/2]} / U_F^{[n/2]+1}$.

***Proof*:**

In the defining sum of equation (5.6), we write $x = y(1 + z)$, with $y \in U_F / U_F^{[n/2]+1}$ and

$z \in \mathfrak{p}^{[n/2]+1}/\mathfrak{p}^{n+1}$. Then:

$$\check{\chi}(cy(1+z))\psi(cy(1+z)) = \check{\chi}(cy)\psi(cy)\psi(c(y-1)z)$$

Summing:

$$\tau(\chi,\psi) = \sum_y \check{\chi}(cy)\psi(cy)\sum_z \psi(c(y-1)z)$$

Now, the map $z \mapsto \psi(c(y-1)z)$ is a character of $\mathfrak{p}^{[n/2]+1}/\mathfrak{p}^{n+1}$ since $c(y-1) \in \mathfrak{p}^{-n}$. It is the trivial character if and only if $y \equiv 1 \pmod{\mathfrak{p}^{n-[n/2]}}$, i.e. if and only if $y \in U_F^{[(n+1)/2]}$. The inner sum therefore vanishes unless $y \in U_F^{[(n+1)/2]}$ in which case it takes the value $(\mathfrak{p}^{[n/2]+1} : \mathfrak{p}^{n+1}) = q^{[(n+1)/2]}$, as we sought to show.

The formula of proposition 5.1.20 has an important application, which the proof of the Converse Theorem will use:

Theorem 5.1.21: Stability Theorem

Let θ, χ be characters of $F^\times$, of level $l \geq 0, n \geq 1$ respectively. Suppose that $2l < n$. Let $\psi \in \widehat{F}$, $\psi \neq 1$, and let $c \in F$ satisfy $\chi(1+x) = \psi(cx)$, $x \in \mathfrak{p}^{[n/2]+1}$. Then

$$\varepsilon(\theta\chi, s, \psi) = \theta(c)^{-1}\varepsilon(\chi, s, \psi)$$

***Proof*:**
By lemma 5.1.16, it is enough to treat the case where ψ has level one. The character $\theta\chi$ has level n, and agrees with χ on $U_F^{[(n+1)/2]}$. Applying proposition 5.1.20, we get

$$\tau(\theta\chi,\psi) = q^{[(n+1)/2]}\sum_y \check{\theta}\check{\chi}(cy)\psi(cy),$$

where the sum is taken over $y \in U_F^{[(n+1)/2]}/U_F^{[n/2]+1}$. Since θ is trivial on $U_F^{[(n+1)/2]}$, this reduces to $\theta(c)^{-1}\tau(\chi,\psi)$, as required.

5.2 Functional Equation for GL(2)

The transition from the theory for $\mathrm{GL}_1(F)$ (Tate's Thesis) to $\mathrm{GL}_2(F)$ (Godement-Jacquet theory) is not merely a change in dimension, but a structural upgrade of every component of the functional equation. The fundamental philosophy remains identical: an L-function is constructed by integrating a "test function" against a "representation" over the group. However, to accommodate the non-abelian nature of the group, scalars must be replaced by matrices, and characters by matrix coefficients.

The ambient space for the test functions shifts from the field F to the full matrix algebra $A = M_2(F)$. Just as the multiplicative group $F^\times$ is an open dense subset of F, the general linear group $G = \mathrm{GL}_2(F)$ is an open dense subset of the algebra A. Consequently, the test functions Φ are chosen from the space $C_c^\infty(A)$, the space of locally constant, compactly supported functions on the matrix ring. This space is spanned by characteristic functions of lattices, such as $a + \varpi^j M_2(\mathfrak{o})$.

This change in ambient space necessitates a corresponding change in the Fourier transform. Both theories rely on an additive Fourier transform on the underlying linear space. For GL_1, the duality on F was given by the product xy. For the matrix algebra $M_2(F)$, the natural non-degenerate bilinear form is given by the trace of the product. Let $\psi \in \widehat{F}$ be a non-trivial additive character. The character ψ_A of the additive group $A = M_2(F)$ is defined by $\psi_A(x) = \psi(\mathrm{tr}_A(x)) = \psi(x_{11} + x_{22})$. The Haar measure μ on A is normalized to be self-dual relative to ψ_A.

Definition 5.2.1: Matrix Fourier Transform

For a test function $\Phi \in C_c^\infty(A)$, the Fourier transform $\hat{\Phi}$ is defined by

$$\hat{\Phi}(x) = \int_A \Phi(y)\psi_A(xy)\, d\mu(y)$$

Just as in the one-dimensional case, the map $\Phi \mapsto \hat{\Phi}$ is an isomorphism of $C_c^\infty(A)$ onto itself, and the Fourier Inversion Formula holds: $\hat{\hat{\Phi}}(x) = \Phi(-x)$.

The most significant conceptual shift concerns the object being integrated. For GL_1, the representation is a character $\chi : F^\times \to \mathbb{C}^\times$. Since it is one-dimensional, the character itself is the function integrated against Φ. For GL_2, an irreducible representation (π, V) is generally infinite-dimensional. The operator $\pi(g)$ cannot be integrated directly to yield a scalar value. Instead, the theory utilizes *matrix coefficients*.

Definition 5.2.2: Matrix Coefficient

Let (π, V) be a smooth representation of G. For vectors $v \in V$ and $\tilde{v} \in \check{V}$ (the smooth dual), the function

$$f(g) = \gamma_{\tilde{v}\otimes v}(g) = \langle \tilde{v}, \pi(g)v \rangle$$

is called a *matrix coefficient* of π. Let $C(\pi)$ denote the vector space spanned by all such coefficients.

The space $C(\pi)$ carries a smooth representation of $G \times G$ via the action $(g_1, g_2) \cdot f(x) = f(g_1^{-1} x g_2)$. In the context of Zeta integrals, these coefficients replace the character $\chi(x)$. The Zeta integrals are then defined by integrating the test function against the matrix coefficient, weighted by a power of the determinant.

Definition 5.2.3: Zeta Integral for GL_2

Let $\Phi \in C_c^\infty(A)$, $f \in C(\pi)$, and $s \in \mathbb{C}$. The *Zeta integral* is defined as:

$$\zeta(\Phi, f, s) = \int_G \Phi(g) f(g) \|\det g\|^s \, d^\times g,$$

where $d^\times g$ is a Haar measure on G.

Convergence is the first thing to settle, and it is not automatic: G is not compact modulo the centre, so nothing yet prevents the integral from diverging for every s. The situation is better than it looks. Sorting G by the valuation of the determinant turns the zeta integral into a formal Laurent series whose coefficients are integrals over *compact* sets, and whose negative part is finite.

Throughout, write $X = q^{-s}$, so that $\|\det g\|^s = X^{v_F(\det g)}$, and set

$$G_m = \{g \in G : v_F(\det g) = m\} \qquad m \in \mathbb{Z}$$

Lemma 5.2.4: Zeta Integrals as Laurent Series

Let (π, V) be a smooth representation of G, let $\Phi \in C_c^\infty(A)$ and let $f \in C(\pi)$. For $m \in \mathbb{Z}$ put

$$c_m(\Phi, f) = \int_{G_m} \Phi(g) f(g)\, d^\times g$$

1. The set $\mathrm{supp}\Phi \cap G_m$ is a compact subset of G, and $c_m(\Phi, f)$ is an absolutely convergent integral.
2. There is an integer $N \geq 0$, depending only on $\mathrm{supp}\Phi$, such that $c_m(\Phi, f) = 0$ for all $m < -2N$.

Consequently $Z(\Phi, f, X) = \sum_{m\in\mathbb{Z}} c_m(\Phi, f) X^m$ is a well-defined element of $\mathbb{C}((X))$, and $\zeta(\Phi, f, s)$ is its value at $X = q^{-s}$ at any s for which the series converges.

***Proof*:**

1. The set $\varpi^m U_F$ is compact, hence closed, in F, and $\det\colon A \to F$ is a polynomial map, hence continuous. Therefore $G_m = \det^{-1}(\varpi^m U_F)$ is closed in A, and $\mathrm{supp}\Phi \cap G_m$ is a closed subset of the compact set $\mathrm{supp}\Phi$. It is therefore compact, and it lies in G, which carries the topology induced from A since G is open in A.

   Both Φ and f are locally constant, so Φf takes only finitely many values on this compact set. A Haar measure is finite on compact sets, so the integral converges absolutely.

2. The lattices $\varpi^{-N} M_2(\mathfrak{o})$, $N \geq 0$, form an increasing open cover of A, so compactness gives N with $\mathrm{supp}\Phi \subseteq \varpi^{-N} M_2(\mathfrak{o})$. For g in the support we then have $\varpi^N g \in M_2(\mathfrak{o})$, whence

   $$\varpi^{2N} \det g = \det(\varpi^N g) \in \mathfrak{o}$$

   and therefore $v_F(\det g) \geq -2N$, completing the proof.

The classification results already proved reduce the analysis of $Z(\Phi, f, X)$ to two cases, and the two are different in character. A cuspidal representation has coefficients that vanish outside a set compact modulo the centre (theorem 3.1.3), so only the central direction is unbounded, and that direction is governed by Tate's theory applied to the central character. Every other irreducible representation is built from a character of the torus (theorem 2.4.16), and unfolding along the Iwasawa decomposition reduces it to a *pair* of Tate integrals. To pass from the induced representations to their subquotients, the following is needed.

Lemma 5.2.5: Coefficients of a Subquotient

Let (Σ, X) be an admissible smooth representation of G and let π be a subquotient of Σ. Then $C(\pi) \subseteq C(\Sigma)$.

***Proof*:**
It suffices to treat a subrepresentation and a quotient separately, since a subquotient is a quotient of a subrepresentation.

Quotients. Let $\pi = X/Y$ for a G-subspace Y. A smooth functional on X/Y is precisely a smooth functional on X vanishing on Y, so there is a canonical injection $\iota\colon (X/Y)^\vee \to \check{X}$. Given $\tilde{u} \in (X/Y)^\vee$ and $\bar{v} \in X/Y$, lift $\bar{v}$ to $v \in X$; then

$$\langle \tilde{u}, \pi(g)\bar{v}\rangle = \langle \tilde{u}, \overline{\Sigma(g)v}\rangle = \langle \iota(\tilde{u}), \Sigma(g)v\rangle$$

which exhibits the coefficient as one of Σ.

Subrepresentations. Let $\pi = (\Sigma|_Y, Y)$ for a G-subspace $Y \subseteq X$, and let $v \in Y$, $\tilde{v} \in \check{Y}$. Choose a compact open subgroup K fixing both, so $v \in Y^K$ and $\tilde{v} \in \check{Y}^K$. By corollary 1.2.18 applied to Y and to X,

$$Y = Y^K \oplus Y(K), \qquad X = X^K \oplus X(K)$$

The first of these intersections is immediate. For the second, $Y(K) \subseteq Y \cap X(K)$ by definition; conversely, if $y \in Y \cap X(K)$, write $y = y_1 + y_2$ with $y_1 \in Y^K$ and $y_2 \in Y(K)$, so that $y_1 = y - y_2$ lies in $X(K)$ as well as in X^K, forcing $y_1 = 0$ and $y = y_2 \in Y(K)$.

The space $Y(K)$ is spanned by the vectors $y - \pi(k)y$, and $\tilde{v}$ is K-fixed, so $\tilde{v}(y - \pi(k)y) = 0$: thus $\tilde{v}$ annihilates $Y(K)$ and is determined by its restriction to Y^K. Since Σ is admissible, X^K is finite dimensional; extend $\tilde{v}|_{Y^K}$ to a linear functional λ on X^K and let $\tilde{u} \in \check{X}$ be λ composed with the projection $X \to X^K$ along $X(K)$. Then $\tilde{u}$ is K-fixed, hence smooth, and $\tilde{u}|_Y = \tilde{v}$ because the two agree on Y^K and both annihilate $Y(K)$. Therefore

$$\langle \tilde{v}, \pi(g)v\rangle = \langle \tilde{u}, \Sigma(g)v\rangle$$

for every $g \in G$, as we sought to show.

One point of normalization has to be settled before the L-function is defined, because getting it wrong makes the functional equation below fail to be symmetric. For GL_1 the zeta integral is attached to s directly; for GL_n the object with the clean $s \leftrightarrow 1 - s$ symmetry is the integral shifted by $(n-1)/2$, which for $n = 2$ is a shift by $1/2$. The L-function is therefore defined from the shifted family, and this is what makes $\zeta(\Phi, f, s + \frac{1}{2})$, rather than $\zeta(\Phi, f, s)$, the expression appearing throughout the rest of this section.

Proposition 5.2.6: Rationality of Zeta Integrals

Let (π, V) be an irreducible smooth representation of G.

1. The integrals $\zeta(\Phi, f, s)$ represent rational functions of q^{-s}.
2. The $\mathbb{C}[q^s, q^{-s}]$-module generated by the *shifted* integrals $\zeta(\Phi, f, s + \frac{1}{2})$, for $\Phi \in C_c^\infty(A)$ and $f \in C(\pi)$, is a principal fractional ideal generated by a function of the form $P(q^{-s})^{-1}$, where $P(t) \in \mathbb{C}[t]$ and $P(0) = 1$.

This generator is defined to be the *L-function* $L(\pi, s)$.

***Proof*:**
Throughout, π is irreducible, hence admissible by corollary 3.1.5 and equipped with a central character ω_π by corollary 1.2.34.

1. By lemma 5.2.4 the expression $Z(\Phi, f, X)$ is a Laurent series with finitely many negative terms, so the content of the claim is that it is the expansion of a rational function. The argument splits according to whether π is cuspidal, and by theorem 3.1.3 together with theorem 2.4.16 these two cases exhaust the irreducible representations of G. In each case a denominator is produced that depends on π alone and not on Φ or f; this uniformity is what part (2) will use.

   The cuspidal case. By theorem 3.1.3 the representation π is γ-cuspidal, so every $f \in C(\pi)$ is compactly supported modulo Z. Both G (proposition 2.2.7) and $Z \cong F^\times$ are unimodular, so G/Z carries an invariant measure; fix Haar measures with $d^\times g = d^\times a \, d\dot{g}$. Since $\det(ag) = a^2 \det g$ and $f(ag) = \omega_\pi(a) f(g)$ for $a \in F^\times$ identified with $aI \in Z$,

   $$Z(\Phi, f, X) = \int_{G/Z} \left(\int_{F^\times} \Phi(a\dot{g}) \omega_\pi(a) X^{2v_F(a)} \, d^\times a \right) f(\dot{g}) X^{v_F(\det \dot{g})} \, d\dot{g}$$

   Fix $\dot{g} \in G$ and set $\Phi_{\dot{g}}(a) = \Phi(a\dot{g})$ for $a \in F$. The map $a \mapsto a\dot{g}$ is an F-linear injection onto the line $F\dot{g}$, which is closed in A, and it is a homeomorphism onto its image; so $\Phi_{\dot{g}}$ is locally constant with compact support, that is $\Phi_{\dot{g}} \in C_c^\infty(F)$. The inner integral is therefore exactly the Tate zeta integral of definition 5.1.7 attached to $\Phi_{\dot{g}}$, the character ω_π and the variable X^2. By proposition 5.1.8 it converges for $\mathrm{Re}(s)$ large and lies in

   $$P_{\omega_\pi}(X^2)^{-1} \, \mathbb{C}[X^2, X^{-2}]$$

   where $P_{\omega_\pi}(t)^{-1} = L(\omega_\pi, s)$ as in definition 5.1.9.

   For the outer integral, f vanishes off a compact subset of G/Z. There is a compact open subgroup K' of K under which Φ is invariant under right translation: pick N_0 and M with $\mathrm{supp}\Phi \subseteq \varpi^{-N_0} M_2(\mathfrak{o})$ and Φ constant on cosets of $\varpi^M M_2(\mathfrak{o})$, and take $K' = K \cap (1 + \varpi^{M+N_0} M_2(\mathfrak{o}))$. For $y \in \varpi^{-N_0} M_2(\mathfrak{o})$ and $k \in K'$ one has $yk - y = y(k-1) \in \varpi^M M_2(\mathfrak{o})$, so $\Phi(yk) = \Phi(y)$; and for y outside that lattice both values vanish, since k is a unit of $M_2(\mathfrak{o})$ and so preserves it. Shrinking K' further so that it also fixes f, the integrand is constant on the cosets of K' meeting the compact support, of which there are finitely many. The outer integral is thus a finite linear combination, and

   $$Z(\Phi, f, X) \in P_{\omega_\pi}(X^2)^{-1} \, \mathbb{C}[X, X^{-1}]$$

   a rational function of X with denominator depending only on ω_π.

   The non-cuspidal case. By theorem 2.4.16 the representation π is either an irreducible $\iota_B^G \chi$, or $\varphi \circ \det$, or $\varphi \cdot \mathrm{St}_G$; in the latter two cases the exact sequence (2.10) exhibits π as a subquotient of $\mathrm{Ind}_B^G \varphi_T$. In every case π is a subquotient of $\Sigma = \mathrm{Ind}_B^G \chi$ for some character $\chi = \chi_1 \otimes \chi_2$ of T, and Σ is admissible. By lemma 5.2.5 it is enough to prove the claim for $f \in C(\Sigma)$.

   Realize Σ on the space X_χ of smooth $\varphi \colon G \to \mathbb{C}$ with $\varphi(bg) = \chi(b)\varphi(g)$. Since $B\backslash G$ is compact (corollary 2.2.2), the Duality Theorem (theorem 1.3.12) gives $\check{X}_\chi \cong \mathrm{Ind}_B^G \delta_B^{-1} \check{\chi}$ with the pairing $\langle \tilde{\varphi}, \varphi \rangle = \int_K \tilde{\varphi}(k)\varphi(k) \, d\mu_K(k)$, so every $f \in C(\Sigma)$ has the form

   $$f(g) = \int_K \tilde{\varphi}(k) \varphi(kg) \, d\mu_K(k)$$

Substituting this and replacing g by $k^{-1}g$, which leaves $d^\times g$ invariant and does not change $v_F(\det g)$ because $\det k \in U_F$, gives

$$Z(\Phi, f, X) = \int_K \tilde{\varphi}(k) \left(\int_G \Phi^k(g)\varphi(g) X^{v_F(\det g)}\, d^\times g \right) d\mu_K(k), \qquad \Phi^k(g) = \Phi(k^{-1}g)$$

and $\Phi^k \in C_c^\infty(A)$. Apply the Iwasawa decomposition $G = BK$ (proposition 2.2.1) and the associated integration formula (corollary 2.2.9). The resulting integration formula (2.5) reads

$$\int_G F(g)\, d\mu_G(g) = \int_K \int_B F(bk')\, d\mu_B(b)\, d\mu_K(k')$$

for a suitable left Haar measure μ_B on B; the normalizing constant relating μ_G to $d^\times g$ is a positive scalar and affects nothing below, since it does not change the $\mathbb{C}[X, X^{-1}]$-module generated.

Writing $b = tn$ with $t = \operatorname{diag}(t_1, t_2) \in T$ and $n = n(x) \in N$, a left Haar measure on $B = T \ltimes N$ is $d\mu_B(tn) = d^\times t\, d\mu_N(n)$: left translation by $t_0 n_0$ sends (t, n) to $(t_0 t,\ (t^{-1} n_0 t) n)$, which preserves $d^\times t$ in the outer variable and is a translation of N for each fixed t. Since χ is trivial on N and $\det(tn) = t_1 t_2$, the inner integral becomes

$$\int_K \varphi(k') \left(\int_T \int_N \Phi^k(t\, n(x)\, k') \chi_1(t_1)\chi_2(t_2) X^{v_F(t_1)+v_F(t_2)}\, d\mu_N(x)\, d^\times t \right) d\mu_K(k')$$

Fix k, k' and put $\Phi'(y) = \Phi(k^{-1} y k')$, again in $C_c^\infty(A)$. In the inner double integral substitute $y = t_1 x$, so that $d\mu_N(x) = \|t_1\|^{-1} d\mu_N(y)$ and

$$\int_N \Phi' \begin{pmatrix} t_1 & t_1 x \\ 0 & t_2 \end{pmatrix} d\mu_N(x) = \|t_1\|^{-1} \widetilde{\Psi}(t_1, t_2), \qquad \widetilde{\Psi}(t_1, t_2) = \int_F \Phi' \begin{pmatrix} t_1 & y \\ 0 & t_2 \end{pmatrix} d\mu_N(y)$$

The change of variable is what removes the apparent divergence: the function $\widetilde{\Psi}$ lies in $C_c^\infty(F \times F)$. Indeed, choose N_0 with $\operatorname{supp}\Phi' \subseteq \varpi^{-N_0} M_2(\mathfrak{o})$; then the integrand vanishes unless $v_F(t_1), v_F(t_2), v_F(y) \geq -N_0$, so the y-integration runs over the *fixed* compact set $\varpi^{-N_0}\mathfrak{o}$ and $\widetilde{\Psi}$ is supported in $\varpi^{-N_0}\mathfrak{o} \times \varpi^{-N_0}\mathfrak{o}$. Local constancy is inherited from Φ': if Φ' is invariant under translation by $\varpi^M M_2(\mathfrak{o})$ and $t_i \equiv t_i' \bmod \varpi^M \mathfrak{o}$, the two matrices differ by an element of that lattice.

The inner double integral is therefore

$$\int_{F^\times} \int_{F^\times} \widetilde{\Psi}(t_1, t_2)\, \chi_1(t_1) \|t_1\|^{-1} \chi_2(t_2)\, X^{v_F(t_1)+v_F(t_2)}\, d^\times t_1\, d^\times t_2$$

Being locally constant with compact support, $\widetilde{\Psi}$ is constant on the cosets of $\varpi^m\mathfrak{o} \times \varpi^m\mathfrak{o}$ for some m and vanishes outside finitely many of them; the argument of proposition 2.2.6, run on $F \times F$, therefore writes it as a finite sum $\sum_j \widetilde{\Psi}_j^{(1)} \otimes \widetilde{\Psi}_j^{(2)}$ of products of elements of $C_c^\infty(F)$, so the double integral is the finite sum

$$\sum_j Z\big(\widetilde{\Psi}_j^{(1)}, \chi_1 \|-\|^{-1}, X\big)\, Z\big(\widetilde{\Psi}_j^{(2)}, \chi_2, X\big)$$

of products of Tate zeta integrals. By proposition 5.1.8 each summand lies in

$$P_{\chi_1\|-\|^{-1}}(X)^{-1} P_{\chi_2}(X)^{-1}\, \mathbb{C}[X, X^{-1}]$$

All the integrands above are bounded by their absolute values in the same way, and the Tate integrals converge absolutely for $\mathrm{Re}(s)$ large, so Fubini's theorem justifies the interchanges on that half plane; both sides are then rational, so the identity persists. Finally the k and k' integrations run over the compact group K with locally constant integrands, hence contribute finite linear combinations, and $Z(\Phi, f, X)$ lies in the displayed fractional ideal, whose denominator depends only on χ.

2. The shift is harmless and can be transported away at the start. Substituting $X = q^{-s}$, the shifted integral $\zeta(\Phi, f, s + \frac{1}{2})$ is the unshifted series $Z(\Phi, f, -)$ evaluated at $q^{-1/2}X$. The map $X \mapsto q^{-1/2}X$ is a $\mathbb{C}$-algebra automorphism of $\mathbb{C}[X, X^{-1}]$ and of $\mathbb{C}(X)$, so it carries the module generated by the unshifted integrals isomorphically onto the module generated by the shifted ones, carries generators to generators, and fixes constant terms, hence preserves the normalization $P(0) = 1$. It therefore suffices to prove the assertion for the unshifted family, which is what follows.

   Let $\mathcal{Z}(\pi)$ denote the $\mathbb{C}[X, X^{-1}]$-submodule of $\mathbb{C}(X)$ generated by the $Z(\Phi, f, X)$, for $\Phi \in C_c^\infty(A)$ and $f \in C(\pi)$. It is non-zero: choose $f \in C(\pi)$ with $f(1) \neq 0$ and let Φ be the characteristic function of a small enough compact open neighbourhood of 1 in A on which f is constant and which meets only G_0, so that $Z(\Phi, f, X)$ is a non-zero constant.

   By part (1) there is a single polynomial $D \in \mathbb{C}[X]$, depending only on π, with $\mathcal{Z}(\pi) \subseteq D(X)^{-1}\mathbb{C}[X, X^{-1}]$. The ring $\mathbb{C}[X, X^{-1}]$ is a localization of $\mathbb{C}[X]$, hence a principal ideal domain, and $D(X)^{-1}\mathbb{C}[X, X^{-1}]$ is a free module of rank one over it. A submodule of a free rank-one module over a principal ideal domain is again free of rank at most one, so $\mathcal{Z}(\pi)$ is generated by a single non-zero element $F(X) \in \mathbb{C}(X)$.

   It remains to normalize the generator. The units of $\mathbb{C}[X, X^{-1}]$ are exactly the monomials cX^k with $c \in \mathbb{C}^\times$, and multiplying a generator by a unit again gives a generator. Let $c_0 \in \mathbb{C}^\times$ be the constant found above. Since $c_0 \in \mathcal{Z}(\pi) = F\,\mathbb{C}[X, X^{-1}]$, there is $u \in \mathbb{C}[X, X^{-1}]$ with $c_0 = Fu$, so $F = c_0 u^{-1}$. Write $u = X^k \tilde{u}$ with $\tilde{u} \in \mathbb{C}[X]$ and $\tilde{u}(0) \neq 0$; then

$$F = c_0 X^{-k}\, \tilde{u}(X)^{-1}$$

   and since $c_0 X^{-k}$ is a unit, $\tilde{u}(X)^{-1}$ generates $\mathcal{Z}(\pi)$ as well. Put $P = \tilde{u}/\tilde{u}(0)$, which lies in $\mathbb{C}[X]$ and has $P(0) = 1$; it differs from $\tilde{u}$ by the unit $\tilde{u}(0)$, so $P(X)^{-1}$ is again a generator.

   This P is unique. If $P(X)^{-1}$ and $Q(X)^{-1}$ both generate, with $P, Q \in \mathbb{C}[X]$ and $P(0) = Q(0) = 1$, then Q/P is a unit cX^k of $\mathbb{C}[X, X^{-1}]$; both P and Q have non-zero constant term, which forces $k = 0$, and evaluating at $X = 0$ gives $c = 1$. Hence $P = Q$, and $L(\pi, s) = P(q^{-s})^{-1}$ is well defined, completing the proof.

The proof records two facts that the next sections use directly. For π cuspidal the denominator involves only the central character, and for π a subquotient of $\mathrm{Ind}_B^G(\chi_1 \otimes \chi_2)$ it is built from the two GL_1 denominators attached to $\chi_1\|-\|^{-1}$ and χ_2, each taken in the shifted variable. The sharper statements, that $L(\pi, s) = 1$ in the first case and that the L-factor is exactly a product of two GL_1 factors in the second, are proved in section 5.3 and section 5.4 respectively.

The functional equation for GL_2 involves a subtle normalization shift compared to GL_1. In Tate's theory, the equation relates s to $1 - s$. For GL_2, the center of symmetry is shifted. Let $\check{\pi}$ denote the contragredient representation. If $f(g) = \langle \tilde{v}, \pi(g)v \rangle$ is a coefficient of π, then $\check{f}(g) = f(g^{-1}) = \langle v, \check{\pi}(g)\tilde{v} \rangle$ is a coefficient of $\check{\pi}$.

Theorem 5.2.7: Functional Equation (Godement-Jacquet)

Let (π, V) be an irreducible smooth representation. There exists a rational function $\gamma(\pi, s, \psi)$ such that:

$$\zeta\left(\hat{\Phi}, \check{f}, \frac{3}{2} - s\right) = \gamma(\pi, s, \psi)\zeta\left(\Phi, f, s + \frac{1}{2}\right) \tag{5.10}$$

for all $\Phi \in C_c^\infty(A)$ and $f \in C(\pi)$.

The shift of $+1/2$ on the right and the evaluation at $3/2 - s$ on the left (which is $2 - (s + 1/2)$) ensures that the resulting ε-factor satisfies the standard symmetry $s \leftrightarrow 1 - s$.

The proof runs on the same principle as Tate's (theorem 5.1.11): exhibit both sides of the equation as elements of one space of equivariant functionals, and show that space is a line. What plays the role of $F^\times$ acting on F is the group $G \times G$ acting on the matrix algebra A by $(g, h)\colon x \mapsto g^{-1}xh$, and the relevant functionals take values not in $\mathbb{C}$ but in the field $\mathbb{C}(X)$, carrying the action

$$(g, h) \cdot P(X) = X^{v_F(\det(gh^{-1}))}P(X)$$

Write $\mathbb{C}(X)_\Theta$ for $\mathbb{C}(X)$ with this action, and set

$$\mathcal{B}(\pi) = \mathrm{Hom}_{G\times G}\big(C_c^\infty(A) \otimes C(\pi),\ \mathbb{C}(X)_\Theta\big)$$

where $G \times G$ acts on $C_c^\infty(A)$ as above and on $C(\pi)$ by $(g, h)f\colon x \mapsto f(g^{-1}xh)$.

Lemma 5.2.8: Uniqueness of Invariant Bilinear Forms

Let (π, V) be an irreducible smooth representation of G. Then $\dim_{\mathbb{C}(X)} \mathcal{B}(\pi) \leq 1$.

***Proof Key ideas*:**
The argument is the one place in this chapter that reaches beyond the machinery developed here, so the structure is given and the technical input is cited. Stratify A by matrix rank. The rank-two locus is G itself, open in A, and the complement $A \setminus G$ is closed and consists of the rank-one matrices together with 0; extension by zero and restriction give a short exact sequence of smooth $G \times G$-representations

$$0 \to C_c^\infty(G) \to C_c^\infty(A) \to C_c^\infty(A \setminus G) \to 0$$

Applying $\mathrm{Hom}_{G\times G}(- \otimes C(\pi), \mathbb{C}(X)_\Theta)$ turns this into a left exact sequence, so the claim reduces to two statements: that the open stratum contributes at most a line, and that the closed stratum contributes nothing.

For the open stratum, $C_c^\infty(G) \otimes C(\pi)$ has a one-dimensional space of such functionals because $C(\pi) \cong \check{V} \otimes V$ is irreducible over $G \times G$ for irreducible admissible π, exactly as established for cuspidal π in the proof of proposition 3.1.2, and an equivariant functional on $C_c^\infty(G) \otimes C(\pi)$ is determined by integration against a single matrix coefficient.

For the closed stratum, the rank-one matrices form a single $G \times G$-orbit, since over a field every rank-one 2×2 matrix is $g\, e_{11}\, h$ for suitable $g, h \in G$. Hence $C_c^\infty(A \setminus G)$ is built from a representation compactly induced from the stabilizer of e_{11}, and Frobenius reciprocity for compact induction converts the Hom space into one over that stabilizer. The resulting condition forces X to satisfy a non-trivial polynomial equation, which cannot hold identically in $\mathbb{C}(X)$; so

the contribution vanishes.

The step this book does not give is the last one, and it is worth being precise about which step it is. Theorem 1.2.28 gives the adjunction for compact induction from an *open* subgroup, which is not what is needed here: the stabilizer of e_{11} in $G \times G$ is closed but far from open, and the adjunction for a closed subgroup carries a module twist this book does not develop. See [2, §24-§25] for the argument in full.

Granting the lemma, the functional equation is a computation with the two equivariances.

***Proof*:**
(of theorem 5.2.7)

Step 1: the zeta integral is equivariant. Let $\Phi \in C_c^\infty(A)$, $f \in C(\pi)$ and $(g,h) \in G \times G$. Substituting $y = gzh^{-1}$, which preserves $d^\times z$ because G is unimodular (proposition 2.2.7), and using $v_F(\det y) = v_F(\det z) + v_F(\det(gh^{-1}))$,

$$\begin{aligned} Z\big((g,h)\Phi,(g,h)f,X\big) &= \int_G \Phi(g^{-1}yh) f(g^{-1}yh) X^{v_F(\det y)}\, d^\times y \\ &= X^{v_F(\det(gh^{-1}))} \int_G \Phi(z) f(z) X^{v_F(\det z)}\, d^\times z \end{aligned}$$

so $Z(-,-,X)$ lies in $\mathcal{B}(\pi)$. It is non-zero by the argument in part (2) of proposition 5.2.6.

Step 2: the transformed integral has the same equivariance. The map $z \mapsto gzh^{-1}$ is F-linear on the four-dimensional space A with determinant $\det(g)^2 \det(h)^{-2}$, so $d\mu(y) = \|\det(gh^{-1})\|^2 d\mu(z)$. Together with $\operatorname{tr}(x\, gzh^{-1}) = \operatorname{tr}\big((h^{-1}xg)z\big)$ this gives

$$\widehat{(g,h)\Phi} = \|\det(gh^{-1})\|^2\, (h,g)\hat{\Phi}$$

and directly from $\check{f}(x) = f(x^{-1})$,

$$\big((g,h)f\big)^\vee(x) = f\big((h^{-1}xg)^{-1}\big) = \big((h,g)\check{f}\big)(x)$$

Note that both operations *swap* the two factors. Define $W(\Phi,f,X) = Z(\hat{\Phi},\check{f},q^{-2}X^{-1})$ and put $m = v_F(\det(gh^{-1}))$. Applying Step 1 to the pair (h,g), whose determinant valuation is $-m$,

$$W\big((g,h)\Phi,(g,h)f,X\big) = q^{-2m}\left(q^{-2}X^{-1}\right)^{-m} W(\Phi,f,X) = q^{-2m}q^{2m}X^m\, W(\Phi,f,X)$$

The two powers of q cancel exactly, leaving the factor X^m. Thus W lies in $\mathcal{B}(\pi)$ as well: the swap introduced by the Fourier transform is undone by the reflection $X \mapsto q^{-2}X^{-1}$, and the normalization of $\hat{\Phi}$ is what makes the cancellation exact.

Step 3: proportionality. By lemma 5.2.8 the space $\mathcal{B}(\pi)$ has dimension at most one over $\mathbb{C}(X)$, and Step 1 shows it is non-zero. Hence $\mathcal{B}(\pi)$ is a line and W is a $\mathbb{C}(X)$-multiple of Z: there is $\gamma \in \mathbb{C}(X)$ with

$$Z\big(\hat{\Phi},\check{f},q^{-2}X^{-1}\big) = \gamma(X)\, Z(\Phi,f,X)$$

for all Φ and f. Writing $X = q^{-s}$ turns $q^{-2}X^{-1}$ into $q^{-(2-s)}$, so this reads $\zeta(\hat{\Phi},\check{f},2-s) = \gamma\,\zeta(\Phi,f,s)$. Replacing s by $s+1/2$, so that $2-s$ becomes $3/2-s$, gives (5.10) with $\gamma(\pi,s,\psi) = \gamma(q^{-s-1/2})$, completing the proof.

Corollary 5.2.9: Epsilon Factor for GL_2

Define the local constant $\varepsilon(\pi, s, \psi)$ by:

$$\gamma(\pi, s, \psi) = \varepsilon(\pi, s, \psi)\frac{L(\check{\pi}, 1-s)}{L(\pi, s)}$$

Then $\varepsilon(\pi, s, \psi)$ is a monomial of the form cq^{-fs} with $c \in \mathbb{C}^\times$ and $f \in \mathbb{Z}$, and satisfies

$$\varepsilon(\pi, s, \psi)\varepsilon(\check{\pi}, 1-s, \psi) = \omega_\pi(-1)$$

***Proof*:**
The argument is the GL_2 form of corollary 5.1.14. Apply the functional equation twice. Fourier inversion gives $\hat{\hat{\Phi}}(x) = \Phi(-x)$, and $\check{f}^\vee = f$, so

$$\zeta(\hat{\hat{\Phi}}, f, s) = \int_G \Phi(-g)f(g)\|\det g\|^s \, d^\times g = \omega_\pi(-1)\zeta(\Phi, f, s)$$

where the substitution $g \mapsto -g$ leaves $\|\det g\|^s$ unchanged because $\det(-g) = (-1)^2 \det g$ in rank two, and $f(-g) = f((-I)g) = \omega_\pi(-1)f(g)$. Comparing with two applications of (5.10) gives

$$\gamma(\pi, s, \psi)\gamma(\check{\pi}, 1-s, \psi) = \omega_\pi(-1)$$

and substituting the defining relation for ε on each factor, the two L-ratios cancel, leaving the stated identity for ε.

For the shape of ε, rewrite the functional equation as

$$\frac{\zeta(\hat{\Phi}, \check{f}, \frac{3}{2}-s)}{L(\check{\pi}, 1-s)} = \varepsilon(\pi, s, \psi)\frac{\zeta(\Phi, f, s+\frac{1}{2})}{L(\pi, s)}$$

By proposition 5.2.6 the L-function is a generator of the module of zeta integrals, so both fractions lie in $\mathbb{C}[q^s, q^{-s}]$, and there are choices of Φ and f making the right-hand numerator equal to $L(\pi, s)$. Hence $\varepsilon(\pi, s, \psi) \in \mathbb{C}[q^s, q^{-s}]$, and the same argument applied to $\check{\pi}$ puts $\varepsilon(\check{\pi}, 1-s, \psi)$ there too. Their product is the non-zero constant $\omega_\pi(-1)$, so $\varepsilon(\pi, s, \psi)$ is a unit of $\mathbb{C}[q^s, q^{-s}]$. The units of that ring are exactly the monomials, giving $\varepsilon(\pi, s, \psi) = cq^{-fs}$ with $c \in \mathbb{C}^\times$ and $f \in \mathbb{Z}$, as we sought to show.

Summary

Concept	Tate (GL(1))	Godement-Jacquet (GL(2))
Ambient Space	F	$M_2(F)$
Representation	$\chi(x)$	$f(g) = \langle \tilde{v}, \pi(g)v \rangle$
Dual Rep	χ^{-1}	$\check{f}(g) = f(g^{-1})$
Duality Pairing	xy	$\mathrm{tr}(xy)$
Norm Factor	$\|x\|^s$	$\|\det g\|^s$
Symmetry	$s \longleftrightarrow 1-s$	$s+\frac{1}{2} \longleftrightarrow \frac{3}{2}-s$

5.3 Epsilon Factors of Cuspidal Representations

The invariants built in section 5.2 are now computed on the representations constructed in chapter 3 and chapter 4. One fact governs the whole computation: the L-function of a cuspidal representation is trivial. Its zeta integrals have no poles at all, the functional equation loses its denominators, and everything a cuspidal representation contributes to this chapter is concentrated in a single monomial, the ε-factor.

The cuspidal case of the proof of proposition 5.2.6 already isolates the only place a pole can come from. A cuspidal coefficient vanishes off a set compact modulo Z, so the zeta integral becomes an integral over G/Z whose inner factor is a Tate zeta integral for the central character ω_π in the variable X^2. By proposition 5.1.8 such an integral has a pole only when ω_π is unramified, and then only through the value of the test function at 0. What is left is to decide whether that pole occurs. It does not, and the mechanism is orthogonality: the residue is an integral of a matrix coefficient against an unramified character of the determinant, and such an integral is a G-homomorphism from π to a one-dimensional representation.

Proposition 5.3.1: Triviality of the L-function of a Cuspidal Representation

Let (π, V) be an irreducible cuspidal representation of G. Then

$$L(\pi, s) = 1$$

***Proof*:**
Throughout, π is irreducible, hence admissible by corollary 3.1.5 and equipped with a central character ω_π by corollary 1.2.34; being cuspidal it is γ-cuspidal by theorem 3.1.3, so every $f \in C(\pi)$ is compactly supported modulo Z. Fix $\Phi \in C_c^\infty(A)$ and $f \in C(\pi)$, and normalize the measures as in the cuspidal case of the proof of proposition 5.2.6, so that $d^\times g = d^\times a \, d\dot{g}$ with $d\dot{g}$ the invariant measure on G/Z.

Step 1: the zeta integral is a finite sum of Tate integrals. Put

$$H(g) = \left(\int_{F^\times} \Phi(ag)\, \omega_\pi(a)\, X^{2v_F(a)}\, d^\times a \right) f(g)\, X^{v_F(\det g)}$$

Neither factor is invariant under $g \mapsto bg$ for $b \in F^\times$, but their product is. Indeed, substituting $a \mapsto ab^{-1}$ multiplies the bracketed factor by $\omega_\pi(b)^{-1} X^{-2v_F(b)}$, while $f(bg) = \omega_\pi(b) f(g)$ and $\det(bg) = b^2 \det g$ multiply the second factor by $\omega_\pi(b) X^{2v_F(b)}$. Thus H descends to a function on G/Z and

$$Z(\Phi, f, X) = \int_{G/Z} H(\dot{g})\, d\dot{g}$$

Writing $\Phi_{\dot{g}}(a) = \Phi(a\dot{g})$, the proof of proposition 5.2.6 shows $\Phi_{\dot{g}} \in C_c^\infty(F)$, and the bracketed factor is the Tate zeta integral $Z(\Phi_{\dot{g}}, \omega_\pi, X^2)$ of definition 5.1.7.

Choose, as in that proof, a compact open subgroup K' of K under which Φ is invariant on the right and which fixes f. Then H is constant on the cosets $\dot{g}K'$: the function $\Phi_{\dot{g}}$ is unchanged, f is unchanged, and $\det u' \in U_F$ for $u' \in K'$. Since f vanishes off a set compact modulo Z, the image of $\operatorname{supp} f$ in G/Z meets only finitely many such cosets, say those of $\dot{g}_1, \dots, \dot{g}_r$, of measures

$c_1, \dots, c_r > 0$. Hence

$$Z(\Phi, f, X) = \sum_{i=1}^{r} c_i \, Z(\Phi_{\dot g_i}, \omega_\pi, X^2)\, f(\dot g_i)\, X^{v_F(\det \dot g_i)}$$

Step 2: only one denominator is possible. Let $\Phi_{\mathfrak{o}}$ denote the characteristic function of $\mathfrak{o}$. For $\Psi \in C_c^\infty(F)$ the function $\Psi - \Psi(0)\Phi_{\mathfrak{o}}$ vanishes at 0, so by the first paragraph of the proof of proposition 5.1.8 its zeta integral lies in $\mathbb{C}[Y, Y^{-1}]$; and the computation in the same proof gives

$$Z(\Phi_{\mathfrak{o}}, \omega_\pi, Y) = \begin{cases} \kappa\,(1 - \omega_\pi(\varpi)Y)^{-1} & \text{if } \omega_\pi \text{ is unramified,} \\ 0 & \text{otherwise} \end{cases}$$

where $\kappa > 0$ is the volume of U_F in the measure used, a constant independent of Ψ. Consequently $Z(\Psi, \omega_\pi, Y) - \Psi(0) Z(\Phi_{\mathfrak{o}}, \omega_\pi, Y) \in \mathbb{C}[Y, Y^{-1}]$. Apply this with $\Psi = \Phi_{\dot g_i}$ and note that $\Phi_{\dot g_i}(0) = \Phi(0)$ for every i, since $0 \cdot \dot g_i = 0$.

If ω_π is ramified, every summand in Step 1 is a Laurent polynomial in X and Step 4 may be entered at once. Suppose then that ω_π is unramified. Substituting the decomposition into Step 1,

$$Z(\Phi, f, X) = \frac{\kappa\, \Phi(0)}{1 - \omega_\pi(\varpi) X^2}\, D(X) + R(X), \qquad D(X) = \sum_{i=1}^{r} c_i f(\dot g_i) X^{v_F(\det \dot g_i)}$$

with $R \in \mathbb{C}[X, X^{-1}]$.

Step 3: the numerator vanishes at the poles. Let $X_0 \in \mathbb{C}^\times$ satisfy $\omega_\pi(\varpi) X_0^2 = 1$ and let χ be the unramified character of $F^\times$ with $\chi(\varpi) = X_0$; this determines χ, since $F^\times = \varpi^{\mathbb{Z}} \times U_F$ and an unramified character is trivial on U_F. For $b \in F^\times$,

$$f(bg)\,\chi(\det(bg)) = \omega_\pi(b)\chi(b)^2 f(g)\chi(\det g) = \big(\omega_\pi(\varpi) X_0^2\big)^{v_F(b)} f(g)\chi(\det g) = f(g)\chi(\det g)$$

so $g \mapsto f(g)\chi(\det g)$ descends to G/Z. It is locally constant, vanishes off a set compact modulo Z, and is constant on the cosets $\dot g_i K'$ because χ is unramified. Therefore

$$D(X_0) = \int_{G/Z} f(\dot g)\, \chi(\det \dot g)\, d\dot g$$

Fix $\tilde v \in \check V$ and define $\ell \colon V \to \mathbb{C}$ by

$$\ell(v) = \int_{G/Z} \langle \tilde v, \pi(\dot g) v \rangle\, \chi(\det \dot g)\, d\dot g$$

The integrand descends to G/Z by the calculation just made, the integral converges absolutely because π is γ-cuspidal, and ℓ is linear in v. Since $d\dot g$ is invariant under right translation by G, the substitution $\dot g \mapsto \dot g h^{-1}$ gives

$$\ell(\pi(h)v) = \int_{G/Z} \langle \tilde v, \pi(\dot g h) v \rangle \chi(\det \dot g)\, d\dot g = \chi(\det h)^{-1} \ell(v)$$

for every $h \in G$, so ℓ lies in $\mathrm{Hom}_G(\pi, \chi^{-1} \circ \det)$, the target being the one-dimensional smooth representation $g \mapsto \chi(\det g)^{-1}$ of G. Suppose $\ell \neq 0$. Then $\ker \ell$ is a G-subspace of V different

from V, hence zero by irreducibility, and $\pi \cong \chi^{-1} \circ \det$. But N lies in the kernel of det, so N acts trivially on $\chi^{-1} \circ \det$ and the Jacquet module of $\chi^{-1} \circ \det$ is $\chi^{-1} \circ \det$ itself, which is non-zero. (Since χ is unramified one may argue instead from proposition 3.6.1: the representation $\chi^{-1} \circ \det$ is trivial on K, so it has non-zero Iwahori-fixed vectors, and no such representation is cuspidal.) This contradicts the cuspidality of π, so $\ell = 0$.

Every $f \in C(\pi)$ is a finite linear combination of coefficients $\gamma_{\tilde{v} \otimes v}$, and $f \mapsto \int_{G/Z} f(\dot{g}) \chi(\det \dot{g})\, d\dot{g}$ is linear, so $D(X_0) = 0$.

Step 4: conclusion. The polynomial $1 - \omega_\pi(\varpi) X^2$ has two distinct non-zero roots, and Step 3 applies to each. Writing $D(X) = X^{-M} \widetilde{D}(X)$ with $\widetilde{D} \in \mathbb{C}[X]$, the polynomial $\widetilde{D}$ vanishes at both roots and is therefore divisible by the quadratic; hence $D(X) = (1 - \omega_\pi(\varpi) X^2) E(X)$ with $E \in \mathbb{C}[X, X^{-1}]$ and

$$Z(\Phi, f, X) = \kappa\, \Phi(0) E(X) + R(X) \in \mathbb{C}[X, X^{-1}]$$

The same conclusion holds when ω_π is ramified, by Step 2.

Thus $\mathcal{Z}(\pi) \subseteq \mathbb{C}[X, X^{-1}]$. By part (2) of proposition 5.2.6 the module $\mathcal{Z}(\pi)$ is generated by $P(X)^{-1}$ for a unique $P \in \mathbb{C}[X]$ with $P(0) = 1$; in particular $P(X)^{-1}$ lies in $\mathbb{C}[X, X^{-1}]$, so P is a unit of $\mathbb{C}[X, X^{-1}]$ and therefore a monomial cX^n. A polynomial of that shape with $P(0) = 1$ is $P = 1$, so $L(\pi, s) = P(q^{-s})^{-1} = 1$, as we sought to show.

Compactness of the support modulo Z controls nothing along the line $F\dot{g}$: for a of large valuation the matrix $a\dot{g}$ tends to 0 in A, where Φ need not vanish. The tail is a geometric series, not a finite sum, and it contributes exactly the factor $L(\omega_\pi, 2s)$; what kills it is orthogonality against the one-dimensional representations $\chi^{-1} \circ \det$, not any statement about supports.

The L-function is therefore blind to cuspidal representations. It returns the same value, 1, on every one of them, and cannot separate one cuspidal representation from another, nor a cuspidal representation from any of its twists. Whatever arithmetic a cuspidal representation carries has to be read off the ε-factor instead, which is what the rest of this section computes.

With the L-factors gone, the functional equation simplifies. The ratio $L(\check{\pi}, 1 - s)/L(\pi, s)$ that separates γ from ε in corollary 5.2.9 becomes a ratio of two 1's, provided the contragredient of a cuspidal representation is again cuspidal.

Corollary 5.3.2: Gamma and Epsilon Agree for Cuspidal Representations

Let (π, V) be an irreducible cuspidal representation of G. Then $\check{\pi}$ is irreducible and cuspidal, $L(\pi, s) = L(\check{\pi}, s) = 1$,

$$\gamma(\pi, s, \psi) = \varepsilon(\pi, s, \psi)$$

and

$$\varepsilon(\pi, s, \psi)\, \varepsilon(\check{\pi}, 1 - s, \psi) = \omega_\pi(-1)$$

***Proof*:**
The representation π is admissible by corollary 3.1.5, so by proposition 1.2.41 the smooth dual $\check{\pi}$ is admissible, irreducible, and canonically identified with the smooth dual of its own smooth dual. Write e_{K_0} for the projector of V onto V^{K_0} attached to a compact open subgroup K_0, a finite average because π is smooth; it satisfies $\langle \tilde{v}, e_{K_0} v \rangle = \langle e_{K_0} \tilde{v}, v \rangle$, both sides being the average

of $\langle \tilde{v}, \pi(x)v\rangle$ over $x \in K_0$.

A coefficient of $\check{\pi}$ is a finite linear combination of functions $g \mapsto \langle w, \check{\pi}(g)\tilde{v}\rangle$ with $\tilde{v} \in \check{V}$ and w in the smooth dual of $\check{\pi}$. Writing w as the image of $v \in V$ under the isomorphism just established, such a function is $\langle \tilde{v}, \pi(g^{-1})v\rangle = \check{f}(g)$, where $f = \gamma_{\tilde{v}\otimes v}$; this is the identity recorded in section 5.2, taken in the reverse direction. Hence $C(\check{\pi}) = \left\{ \check{f} : f \in C(\pi) \right\}$.

Inversion is a homeomorphism of G carrying Z-stable sets to Z-stable sets, so $\check{f}$ is compactly supported modulo Z whenever f is. By theorem 3.1.3 the representation π is γ-cuspidal, so every coefficient of $\check{\pi}$ is compactly supported modulo Z; thus $\check{\pi}$ is γ-cuspidal, and cuspidal by the same theorem.

Proposition 5.3.1 now applies to π and to $\check{\pi}$, giving $L(\pi, s) = L(\check{\pi}, s) = 1$ and hence $L(\check{\pi}, 1-s) = 1$. The defining relation of corollary 5.2.9 reads

$$\gamma(\pi, s, \psi) = \varepsilon(\pi, s, \psi)\frac{L(\check{\pi}, 1-s)}{L(\pi, s)} = \varepsilon(\pi, s, \psi)$$

and the last identity of the statement is the one already proved in corollary 5.2.9, completing the proof.

The distinction between γ and ε was introduced to strip poles out of the functional equation; for a cuspidal representation there were none to strip. Every statement below can therefore be read with either symbol, and the functional equation of theorem 5.2.7 says, for cuspidal π, that a single non-zero complex monomial converts one Laurent polynomial into another.

That monomial still depends on a choice, namely the additive character ψ used to build the Fourier transform of definition 5.2.1. At GL_1 that dependence was transparent (lemma 5.1.16), and transparency is what allowed the explicit formulas of section 5.1 to be stated for a single normalization of ψ. The same holds here, with the determinant taking over the role that the identity map of F played before.

Lemma 5.3.3: Dependence of the Local Constant on the Additive Character

Let (π, V) be an irreducible smooth representation of G and let $a \in F^\times$. Then

$$\varepsilon(\pi, s, a\psi) = \omega_\pi(a)\,\|a\|^{2(s-1/2)}\,\varepsilon(\pi, s, \psi)$$

***Proof*:**
The L-functions $L(\pi, s)$ and $L(\check{\pi}, 1-s)$ do not involve ψ, so the ratio of the two ε-factors equals the ratio of the two γ-factors, and it suffices to compute the latter. Write $\psi' = a\psi$, so that $\psi'_A(x) = \psi(a\mathrm{tr}x) = \psi_A(ax)$.

Step 1: the self-dual measure on A scales by $\|a\|^2$. For a lattice L in A let $L^* = \{x \in A : \psi_A(xL) = 1\}$ be its dual and Φ_L its characteristic function. Orthogonality on the compact group L gives $\hat{\Phi}_L = \mu(L)\Phi_{L^*}$, and applying the transform again gives $\hat{\hat{\Phi}}_L = \mu(L)\mu(L^*)\Phi_L$, using $L^{**} = L$. The self-dual measure is therefore characterized by $\mu(L)\mu(L^*) = 1$ for every lattice L. Replacing ψ_A by ψ'_A replaces L^* by $a^{-1}L^*$, and A has dimension four over F, so $\mu_{\psi_A}(a^{-1}L^*) = \|a\|^{-4}\mu_{\psi_A}(L^*)$. Writing $\mu_{\psi'_A} = c\,\mu_{\psi_A}$ with $c > 0$ and imposing self-duality yields $c^2\|a\|^{-4} = 1$, that is $c = \|a\|^2$.

Step 2: the Fourier transform. Denoting by $\hat{\Phi}$ the transform relative to ψ and by $\hat{\Phi}^{\psi'}$ the

transform relative to ψ',

$$\hat{\Phi}^{\psi'}(x) = \int_A \Phi(y)\psi'_A(xy)\,d\mu_{\psi'_A}(y) = \|a\|^2 \int_A \Phi(y)\psi_A\big((ax)y\big)\,d\mu_{\psi_A}(y) = \|a\|^2\,\hat{\Phi}(ax)$$

Step 3: change of variable in the zeta integral. Substituting $h = ag$, where a is identified with the scalar matrix $aI \in Z$, one has $\det g = a^{-2}\det h$, so $\|\det g\|^{3/2-s} = \|a\|^{2s-3}\|\det h\|^{3/2-s}$; also $\check{f}(g) = f(ah^{-1}) = \omega_\pi(a)\check{f}(h)$, and $d^\times g = d^\times h$ since $d^\times g$ is a Haar measure. Hence

$$\begin{aligned}\zeta\big(\hat{\Phi}^{\psi'}, \check{f}, \tfrac{3}{2} - s\big) &= \|a\|^2 \int_G \hat{\Phi}(ag)\check{f}(g)\|\det g\|^{3/2-s}\,d^\times g \\ &= \|a\|^2\,\|a\|^{2s-3}\,\omega_\pi(a) \int_G \hat{\Phi}(h)\check{f}(h)\|\det h\|^{3/2-s}\,d^\times h \\ &= \omega_\pi(a)\,\|a\|^{2s-1}\,\zeta\big(\hat{\Phi}, \check{f}, \tfrac{3}{2} - s\big)\end{aligned}$$

The zeta integral $\zeta(\Phi, f, s + \frac{1}{2})$ standing on the right-hand side of (5.10) does not involve ψ, so comparing the functional equation for ψ' with the functional equation for ψ gives

$$\gamma(\pi, s, a\psi)\,\zeta\big(\Phi, f, s + \tfrac{1}{2}\big) = \omega_\pi(a)\|a\|^{2s-1}\gamma(\pi, s, \psi)\,\zeta\big(\Phi, f, s + \tfrac{1}{2}\big)$$

This holds for all Φ and all f, and the zeta integrals are not identically zero, so $\gamma(\pi, s, a\psi) = \omega_\pi(a)\|a\|^{2s-1}\gamma(\pi, s, \psi)$. Since $\|a\|^{2s-1} = \|a\|^{2(s-1/2)}$, this completes the proof.

Two features distinguish the formula from its GL_1 counterpart lemma 5.1.16, and both come from the same source. The scalar matrix aI has determinant a^2, so the change of variable $g \mapsto (aI)g$ moves $\|\det g\|^s$ by $\|a\|^{2s}$ rather than by $\|a\|^s$, which doubles the exponent; and it moves the coefficient f by the central character, which is why ω_π stands where χ stood before.

The formula is consistent with corollary 5.2.9. The contragredient has central character ω_π^{-1}, since $\check{\pi}(zI)\tilde{v} = \tilde{v} \circ \pi(z^{-1}I) = \omega_\pi(z)^{-1}\tilde{v}$, so replacing ψ by $a\psi$ multiplies the left-hand side of the identity $\varepsilon(\pi, s, \psi)\varepsilon(\check{\pi}, 1 - s, \psi) = \omega_\pi(-1)$ by

$$\omega_\pi(a)\|a\|^{2(s-1/2)} \cdot \omega_\pi(a)^{-1}\|a\|^{2((1-s)-1/2)} = \|a\|^{2(s-1/2)+2(1/2-s)} = 1$$

The identity is unchanged by the change of additive character, as it must be, since its right-hand side never mentions ψ. This cancellation forces the exponent to be a multiple of $s - \frac{1}{2}$; that the multiple is 2 rather than 1 is precisely the contribution of the determinant of a rank two matrix.

Corollary 5.2.9 says that $\varepsilon(\pi, s, \psi)$ is a monomial, and its two ingredients play different roles. The complex constant is rigid but hard to compute, while the exponent is a discrete invariant that can be read off the degree of the monomial. Fixing a normalization of ψ turns that exponent into an invariant of π alone.

Definition 5.3.4: Epsilon Exponent

Let (π, V) be an irreducible smooth representation of G and let $\psi \in \widehat{F}$ have level one. By corollary 5.2.9 there are unique $c \in \mathbb{C}^\times$ and $d \in \mathbb{Z}$ with $\varepsilon(\pi, s, \psi) = c\, q^{-ds}$. The integer

$$n(\pi, \psi) = d$$

is called the *epsilon exponent* of π relative to ψ, and equivalently

$$\varepsilon(\pi, s, \psi) = \varepsilon\big(\pi, \tfrac{1}{2}, \psi\big)\, q^{(1/2-s)n(\pi,\psi)}$$

The second display is a rewriting of the first and uses nothing beyond the monomial form: evaluating $c\,q^{-ds}$ at $s = \frac{1}{2}$ gives $c = q^{d/2}\varepsilon(\pi, \frac{1}{2}, \psi)$, and substituting back produces the stated shape. The shape is therefore forced by corollary 5.2.9 rather than assumed, and uniqueness of c and d is immediate, since $cq^{-ds} = c'q^{-d's}$ for all s gives $d = d'$ and then $c = c'$.

Restricting to ψ of level one is what makes $n(\pi, \psi)$ well defined. A character $a\psi$ is trivial on $\mathfrak{p}^m$ exactly when ψ is trivial on $\mathfrak{p}^{m+v_F(a)}$, so the level of $a\psi$ is the level of ψ minus $v_F(a)$. The map $a \mapsto a\psi$ is a bijection of F onto $\widehat{F}$, so two characters of level one differ by an element of U_F, and by lemma 5.3.3 the two ε-factors differ by the constant $\omega_\pi(a)$ alone. The exponent is unchanged, so $n(\pi, \psi)$ does not depend on which character of level one was chosen.

The choice of normalization changes the answer. Against a character ψ' of level l, the same recipe returns the exponent $n(\pi, \psi) + 2(1 - l)$, again by lemma 5.3.3, since $\psi' = a\psi$ forces $v_F(a) = 1 - l$. In particular a character of level zero returns $n(\pi, \psi) + 2$; that is the normalization in which an unramified character of $F^\times$ has ε-factor exactly 1, by proposition 5.1.17 together with lemma 5.1.16. Level one is retained here because every explicit formula of section 5.1 is stated for it.

The exponent is carried with its additive character, and the notation records a real dependence: the integer depends on ψ as well as on π, and lemma 5.3.3 says exactly how. Fixing ψ of level one, it measures how much ramification π carries relative to that choice; it is insensitive to unramified twisting, as the computation below shows, and it vanishes on the cuspidal representations of level zero. The same convention, with the same notation, is used on the Galois side in proposition 6.3.3, which is what allows the two sides to be compared in chapter 7. Note that the classical *conductor* exponent is the value taken against an additive character of level zero, so it differs from $n(\pi, \psi)$ by a shift; nothing below depends on which of the two is used, provided one of them is used consistently. For a cuspidal representation of positive level it grows with the level of the simple stratum contained in π; the closed shape of the answer in terms of the cuspidal type of definition 3.5.4 is theorem 5.3.8 below, which evaluates it as a Gauss sum over a quotient of two unit filtration subgroups of the chain order. That evaluation uses the functional equation and nothing else. A second route runs through the Kirillov model built in chapter 8, where theorem 8.2.4 shows that the local constant of a cuspidal representation is carried by a single operator on it, but the theorem below does not use it.

Two pieces of bookkeeping are used by every computation in this section, and both are recorded before the first of them. One names the matrix coefficient that a compactly induced representation carries on the subgroup it is induced from. The other evaluates the self-dual measure on the powers of the radical of a chain order, for an arbitrary period.

Lemma 5.3.5: Coefficients of a Compactly Induced Representation

Let J be an open subgroup of G containing Z and compact modulo Z, let (Λ, W) be a smooth representation of J on a finite-dimensional space and put $\pi = \text{c-Ind}_J^G \Lambda$. For $w \in W$ and $\tilde{w} \in \check{W}$ the function

$$f_{\tilde{w},w}(g) = \begin{cases} \langle \tilde{w}, \Lambda(g)w \rangle & g \in J \\ 0 & g \notin J \end{cases}$$

is a matrix coefficient of π in the sense of definition 5.2.2, and its contragredient coefficient is

$$\check{f}_{\tilde{w},w}(g) = \begin{cases} \langle \check{\Lambda}(g)\tilde{w}, w \rangle & g \in J \\ 0 & g \notin J \end{cases}$$

where $\check{\Lambda}$ is the contragredient of Λ, so that $\langle \check{\Lambda}(g)\tilde{w}, w \rangle = \langle \tilde{w}, \Lambda(g^{-1})w \rangle$.

***Proof*:**

Realize π on the space X_c of definition 1.2.24: the functions $\varphi \colon G \to W$ with $\varphi(jg) = \Lambda(j)\varphi(g)$ for $j \in J$, right invariant under some compact open subgroup of G and supported on finitely many cosets Jg, with G acting by right translation.

Let φ_w be the function supported on J with $\varphi_w(j) = \Lambda(j)w$ for $j \in J$. It is equivariant by construction, and its support is the single coset J. It is right invariant under the stabilizer of w in J, which is open because Λ is smooth: for $g \in J$ and h in that stabilizer, $\varphi_w(gh) = \Lambda(g)\Lambda(h)w = \varphi_w(g)$, while for $g \notin J$ both sides vanish, since $g \notin J$ and $h \in J$ give $gh \notin J$. So $\varphi_w \in X_c$.

Define $\ell_{\tilde{w}} \colon X_c \to \mathbb{C}$ by $\ell_{\tilde{w}}(\varphi) = \langle \tilde{w}, \varphi(1) \rangle$. It is linear, and for $h \in J$,

$$\big(\check{\pi}(h)\ell_{\tilde{w}}\big)(\varphi) = \ell_{\tilde{w}}\big(\pi(h^{-1})\varphi\big) = \langle \tilde{w}, \varphi(h^{-1}) \rangle = \langle \tilde{w}, \Lambda(h^{-1})\varphi(1) \rangle = \ell_{\check{\Lambda}(h)\tilde{w}}(\varphi)$$

so $\ell_{\tilde{w}}$ is fixed by the stabilizer of $\tilde{w}$ in J, which is again open because $\check{\Lambda}$ is smooth, W being finite dimensional. Hence $\ell_{\tilde{w}}$ lies in the smooth dual of π.

The coefficient attached to this pair is

$$\gamma_{\ell_{\tilde{w}} \otimes \varphi_w}(g) = \big\langle \ell_{\tilde{w}}, \pi(g)\varphi_w \big\rangle = \langle \tilde{w}, \varphi_w(g) \rangle$$

which is $\langle \tilde{w}, \Lambda(g)w \rangle$ for $g \in J$ and 0 otherwise, that is $f_{\tilde{w},w}$. Since J is stable under inversion, $\check{f}_{\tilde{w},w}(g) = f_{\tilde{w},w}(g^{-1})$ vanishes off J and equals $\langle \tilde{w}, \Lambda(g^{-1})w \rangle = \langle \check{\Lambda}(g)\tilde{w}, w \rangle$ on it, as we sought to show.

Lemma 5.3.6: The Self-Dual Measure on a Chain Order

Let $\psi \in \widehat{F}$ have level one, let $\mathfrak{A}$ be a chain order in A with radical $\mathfrak{P}$ and period $e = e_{\mathfrak{A}}$, and let μ be the Haar measure on A that is self-dual for the character $\psi_A(x) = \psi(\mathrm{tr}x)$ of A. Then

$$\big(\mathfrak{P}^j : \mathfrak{P}^{j+1}\big) = q^{4/e} \quad (j \in \mathbb{Z}), \qquad \mu\big(\mathfrak{P}^k\big) = q^{2(1-2k)/e} \quad (k \in \mathbb{Z})$$

In the rest of this section the character ψ_A is written $\psi_{\mathfrak{A}}$.

***Proof*:**
Let Π be a prime element of $\mathfrak{A}$. Multiplication by Π^j is injective on A and carries $\mathfrak{A}$ onto $\mathfrak{P}^j$ and $\mathfrak{P}$ onto $\mathfrak{P}^{j+1}$, by proposition 3.3.5(1), so it induces an isomorphism $\mathfrak{A}/\mathfrak{P} \to \mathfrak{P}^j/\mathfrak{P}^{j+1}$ of $\mathfrak{o}$-modules. The index $(\mathfrak{P}^j : \mathfrak{P}^{j+1})$ is therefore $(\mathfrak{A} : \mathfrak{P})$ for every j.

Only e of the conditions defining $\mathfrak{A}$ in definition 3.3.3 are distinct, since $L_{i+e} = \varpi L_i$ and $\mathrm{End}_{\mathfrak{o}}(\varpi L) = \mathrm{End}_{\mathfrak{o}}(L)$, so $\mathfrak{A} = \bigcap_{0 \le i < e} \mathrm{End}_{\mathfrak{o}}(L_i)$. Any two lattices in V are commensurable, so $\varpi^r \mathrm{End}_{\mathfrak{o}}(L_0) \subseteq \mathrm{End}_{\mathfrak{o}}(L_i)$ for every i once r is large, and $\mathfrak{A}$ is squeezed between $\varpi^r \mathrm{End}_{\mathfrak{o}}(L_0)$ and $\mathrm{End}_{\mathfrak{o}}(L_0)$. Thus $\mathfrak{A}$ is a full $\mathfrak{o}$-lattice in A, free of rank $\dim_F A = 4$ because $\mathfrak{o}$ is a discrete valuation ring, and $(\mathfrak{A} : \varpi\mathfrak{A}) = q^4$. By proposition 3.3.4(2), $\mathfrak{P}^e = \mathfrak{p}\mathfrak{A} = \varpi\mathfrak{A}$, and the chain $\mathfrak{A} \supset \mathfrak{P} \supset \cdots \supset \mathfrak{P}^e$ has e steps of equal index, so $(\mathfrak{A} : \mathfrak{P})^e = q^4$ and $(\mathfrak{A} : \mathfrak{P}) = q^{4/e}$.

Consequently $\mu(\mathfrak{P}^k) = \mu(\mathfrak{A}) q^{-4k/e}$ for every k. For the value of $\mu(\mathfrak{A})$, write $L^* = \{x \in A : \psi_{\mathfrak{A}}(xL) = 1\}$ for the dual of a lattice L and Φ_L for its characteristic function. Orthogonality on the compact group L gives $\hat{\Phi}_L = \mu(L)\Phi_{L^*}$, and a second transform gives $\hat{\hat{\Phi}}_L = \mu(L)\mu(L^*)\Phi_{L^{**}}$, which by Fourier inversion and $L^{**} = L$ forces $\mu(L)\mu(L^*) = 1$. Taking $L = \mathfrak{A}$, whose dual is $\mathfrak{P}$, gives $\mu(\mathfrak{A})^2 q^{-4/e} = 1$, so $\mu(\mathfrak{A}) = q^{2/e}$ and $\mu(\mathfrak{P}^k) = q^{2/e} q^{-4k/e} = q^{2(1-2k)/e}$, as we sought to show.

At $e_{\mathfrak{A}} = 2$, which is the period of the chain order carrying a ramified simple stratum, the formula reads

$$\mu\big(\mathfrak{P}^k\big) = q^{1-2k} \qquad (e_{\mathfrak{A}} = 2) \tag{5.11}$$

and at $e_{\mathfrak{A}} = 1$ with $\mathfrak{A} = M_2(\mathfrak{o})$ and $\mathfrak{P} = M_2(\mathfrak{p})$ it gives $\mu(M_2(\mathfrak{o})) = q^2$ and $\mu(M_2(\mathfrak{p})) = q^{-2}$, the two values the next computation uses.

Example 5.3.7: Epsilon Factor of a Level Zero Cuspidal Representation
Let ψ have level one and let (π, V) be a cuspidal representation of level zero. Level zero is automatically minimal, so theorem 3.5.2 places π in case (2)(a): it contains a representation λ of K inflated from an irreducible cuspidal representation $\tilde{\lambda}$ of $\mathrm{GL}_2(k)$. That is case (1) of theorem 3.2.9, which gives a representation Λ of ZK with $\Lambda|_K \cong \lambda$ and

$$\pi \cong \text{c-Ind}_{ZK}^{G} \Lambda$$

Write $K_1 = 1 + \mathfrak{p}M_2(\mathfrak{o})$, so that λ is trivial on K_1 and $K/K_1 \cong \mathrm{GL}_2(k)$. Let W be the space of λ, and fix $w \in W$ and $\tilde{w} \in \check{W}$ with $\langle \tilde{w}, w \rangle \neq 0$.

1. *A coefficient and a test function.* By lemma 5.3.5 applied with $J = ZK$, the function

$$f(g) = \begin{cases} \langle \tilde{w}, \Lambda(g) w \rangle & g \in ZK \\ 0 & g \notin ZK \end{cases}$$

Since K is open and compact in A, the function Φ defined by $\Phi(u) = \langle \tilde{w}, \lambda(u^{-1}) w \rangle$ for $u \in K$ and $\Phi = 0$ off K lies in $C_c^\infty(A)$, and it is inflated from $\mathrm{GL}_2(k)$ because λ is.

2. *The zeta integral is a non-zero constant.* The supports meet in K, where $\|\det u\| = 1$, so

$$\zeta(\Phi, f, s) = \int_K \langle \tilde{w}, \lambda(u^{-1}) w \rangle \langle \tilde{w}, \lambda(u) w \rangle \, d^\times u = \mathrm{vol}^\times(K_1) \sum_{y \in \mathrm{GL}_2(k)} \langle \tilde{w}, \tilde{\lambda}(y^{-1}) w \rangle \langle \tilde{w}, \tilde{\lambda}(y) w \rangle$$

Schur orthogonality on the finite group $\mathrm{GL}_2(k)$ evaluates the sum as $|\mathrm{GL}_2(k)| \, (\dim \tilde{\lambda})^{-1} \langle \tilde{w}, w \rangle^2$, so $\zeta(\Phi, f, s) = c_0$ is a non-zero constant, independent of s.

3. *The Fourier transform on ZK.* The self-dual measure μ on A relative to ψ_A gives $\mu(M_2(\mathfrak{o})) = q^2$ and $\mu(M_2(\mathfrak{p})) = q^{-2}$, which is lemma 5.3.6 at $\mathfrak{A} = M_2(\mathfrak{o})$, where $e_{\mathfrak{A}} = 1$. The multiplicative cosets of K_1 in K are additive cosets of $M_2(\mathfrak{p})$, because $yK_1 = y + yM_2(\mathfrak{p}) = y + M_2(\mathfrak{p})$ for $y \in K$. Decomposing K accordingly into the sets $y_j + M_2(\mathfrak{p})$, $1 \le j \le |\mathrm{GL}_2(k)|$, on each of which Φ is constant,

$$\hat{\Phi}(x) = \sum_j \Phi(y_j)\,\psi\big(\mathrm{tr}(xy_j)\big) \int_{M_2(\mathfrak{p})} \psi\big(\mathrm{tr}(xm)\big)\,d\mu(m)$$

The character $m \mapsto \psi(\mathrm{tr}(xm))$ of $M_2(\mathfrak{p})$ is trivial precisely when $x \in M_2(\mathfrak{o})$, so the last integral is q^{-2} for such x and 0 otherwise. Now take $x = \varpi^n u$ with $u \in K$ and $n \in \mathbb{Z}$. Since u has an entry in U_F, the matrix $\varpi^n u$ lies in $M_2(\mathfrak{o})$ if and only if $n \ge 0$, and:

- for $n \le -1$, $\hat{\Phi}(\varpi^n u) = 0$;
- for $n \ge 1$, $\mathrm{tr}(\varpi^n u y_j) \in \mathfrak{p} \subseteq \ker\psi$, so $\hat{\Phi}(\varpi^n u) = q^{-2}\sum_j \Phi(y_j) = 0$ by orthogonality of the coefficient of the non-trivial irreducible representation $\tilde{\lambda}$ against the trivial character;
- for $n = 0$, $\hat{\Phi}(u) = q^{-2}\sum_{y\in\mathrm{GL}_2(k)} \langle \tilde{w}, \tilde{\lambda}(y^{-1})w\rangle\,\eta\big(\mathrm{tr}(\bar{u}y)\big)$, where η is the non-trivial character of $k = \mathfrak{o}/\mathfrak{p}$ induced by $\psi|_{\mathfrak{o}}$ and $\bar{u}$ is the image of u in $\mathrm{GL}_2(k)$.

Thus $\hat{\Phi}$ vanishes on $ZK \setminus K$.

4. *The epsilon exponent is zero.* The support of $\check{f}$ is $(ZK)^{-1} = ZK$, so by the previous item

$$\zeta\big(\hat{\Phi}, \check{f}, \tfrac{3}{2} - s\big) = \int_K \hat{\Phi}(u)\check{f}(u)\,d^\times u = c_1$$

is again constant in s. Since $L(\pi, s) = L(\check{\pi}, 1-s) = 1$ by corollary 5.3.2, the functional equation (5.10) reads $c_1 = \varepsilon(\pi, s, \psi)c_0$, so

$$\varepsilon(\pi, s, \psi) = \frac{c_1}{c_0}, \qquad n(\pi, \psi) = 0$$

The ε-factor is constant in s. The computation also forces $c_1 \neq 0$, since $c_0 \neq 0$ and ε is a unit; in particular the finite Fourier transform of a matrix coefficient of a cuspidal representation of $\mathrm{GL}_2(k)$ does not vanish identically on $\mathrm{GL}_2(k)$. Against a character of level zero the exponent is $n(\pi, \psi) + 2 = 2$.

5. *Unramified twisting.* For a character φ of $F^\times$ write $\varphi\pi = (\varphi \circ \det) \otimes \pi$. If φ is unramified then $\varphi = \|-\|^\sigma$ for some $\sigma \in \mathbb{C}$ with $q^{-\sigma} = \varphi(\varpi)$, so $\varphi(\det g)\|\det g\|^s = \|\det g\|^{s+\sigma}$ and

$$\zeta\big(\Phi, (\varphi\circ\det)f, s\big) = \zeta(\Phi, f, s+\sigma)$$

The coefficients of $\varphi\pi$ are exactly the $(\varphi\circ\det)f$ with $f \in C(\pi)$, and $((\varphi\circ\det)f)^\vee = (\varphi^{-1}\circ\det)\check{f}$, a coefficient of $\varphi^{-1}\check{\pi} = (\varphi\pi)^\vee$. Applying (5.10) for π at $s+\sigma$ therefore gives $\gamma(\varphi\pi, s, \psi) = \gamma(\pi, s+\sigma, \psi)$. The representation $\varphi\pi$ is cuspidal, since N lies in the kernel of $\varphi\circ\det$ and the Jacquet module is unchanged, so both γ-factors are ε-factors and

$$\varepsilon(\varphi\pi, s, \psi) = \varepsilon(\pi, s+\sigma, \psi) = \varphi(\varpi)^{n(\pi,\psi)}\varepsilon(\pi, s, \psi)$$

In particular $n(\varphi\pi, \psi) = n(\pi, \psi)$, and for the present π, where $n(\pi, \psi) = 0$, unramified twisting does not change the ε-factor at all.

6. *The constraint from the functional equation.* Since $n(\pi,\psi) = n(\check{\pi},\psi) = 0$, the identity of corollary 5.3.2 reduces to

$$\varepsilon\big(\pi, \tfrac{1}{2}, \psi\big)\,\varepsilon\big(\check{\pi}, \tfrac{1}{2}, \psi\big) = \omega_\pi(-1)$$

The right-hand side is computable inside the finite group: $-1 \in U_F$ and $\omega_\pi|_{U_F}$ is the inflation of the central character of $\tilde{\lambda}$, so $\omega_\pi(-1)$ is the value of that central character at $-1 \in k^\times$.

7. *The remaining constant.* The ratio c_1/c_0 is not evaluated by the argument above, which only shows it is constant in s. The closed form is theorem 5.3.9 below, whose method is the one used here carried out for a stratum of positive level. Let E/F be the unramified quadratic extension and let ξ be the character of $E^\times$ of level zero attached to π by proposition 4.2.3, and put $\psi_E = \psi \circ \mathrm{Tr}_{E/F}$, of level one in E. The answer has the shape

$$\varepsilon(\pi, s, \psi) = \lambda_{E/F}(\psi)\,\frac{\tau(\xi, \psi_E)}{q}$$

where $\tau(\xi,\psi_E)$ is the Gauss sum of definition 5.1.19 formed over E and $\lambda_{E/F}(\psi)$ depends only on E/F and ψ; see [2, §25]. That the right-hand side is constant in s is the consistency check available here: ξ is ramified of level zero and $q_E = q^2$, so theorem 5.1.18 applied over E gives $\varepsilon(\xi, s, \psi_E) = q_E^{-1/2}\tau(\xi,\psi_E)$, with no dependence on s, matching $n(\pi,\psi) = 0$.

Level zero is the case in which the answer is least explicit, because the finite group $\mathrm{GL}_2(k)$ carries the whole computation and its Gauss sum is not evaluated by these methods. At positive level the situation inverts. The local constant becomes a finite sum over a quotient of two unit filtration subgroups of the chain order, and that sum can be written down. It collapses to a single term exactly when the level of the inducing representation is odd, and that is the case chapter 11 compares across a tame extension. The theorem below does the general case, for an arbitrary chain order and an inducing representation of arbitrary dimension, and the one-term case is read off it afterwards.

Theorem 5.3.8: The Local Constant of a Cuspidal Representation as a Gauss Sum

Let $\psi \in \widehat{F}$ have level one, let $\mathfrak{A}$ be a chain order in A with radical $\mathfrak{P}$ and period $e = e_{\mathfrak{A}}$, and let J be an open subgroup of $\mathcal{K}_{\mathfrak{A}}$ containing Z and compact modulo Z. Let (Λ, W) be a smooth representation of J on a finite-dimensional space such that $\pi = \text{c-Ind}_J^G \Lambda$ is irreducible and cuspidal, and write $\check{\Lambda}$ for the contragredient of Λ, so that $\text{tr}\check{\Lambda}(g) = \text{tr}\Lambda(g^{-1})$. Let $N \geq 1$ be an integer and suppose

1. $U_{\mathfrak{A}}^{[(N+1)/2]} \subseteq J$
2. there is $c \in J$ with $c\mathfrak{A} = \mathfrak{P}^{-N}$ and

$$\Lambda(1+x) = \psi_{\mathfrak{A}}(cx) \cdot \text{id}, \qquad x \in \mathfrak{P}^{[N/2]+1}$$

Then N is the least integer for which $U_{\mathfrak{A}}^{N+1} \subseteq \ker \Lambda$, the sum

$$\mathfrak{g}(\Lambda, \psi) = \sum_{x \in U_{\mathfrak{A}}^{[(N+1)/2]} / U_{\mathfrak{A}}^{[N/2]+1}} \text{tr}\check{\Lambda}(cx)\, \psi_{\mathfrak{A}}(cx)$$

is well defined, independent of the choice of c in (2) and non-zero, the epsilon exponent of definition 5.3.4 is $n(\pi, \psi) = 2N/e$, and

$$\varepsilon(\pi, s, \psi) = q^a\, q^{\,n(\pi,\psi)(1/2-s)}\, \mathfrak{g}(\Lambda, \psi) \tag{5.12}$$

where the constant q^a is determined by

$$q^a \dim \Lambda = \begin{cases} 1 & N \text{ odd} \\ (\mathfrak{A} : \mathfrak{P})^{-1/2} = q^{-2/e} & N \text{ even} \end{cases}$$

***Proof*:**

The representation π is irreducible and cuspidal by hypothesis, so corollary 5.3.2 gives $L(\pi, s) = L(\check{\pi}, s) = 1$ and $\gamma(\pi, s, \psi) = \varepsilon(\pi, s, \psi)$. The functional equation (5.10) therefore reads

$$\zeta\big(\hat{\Phi}, \check{f}, \tfrac{3}{2} - s\big) = \varepsilon(\pi, s, \psi)\, \zeta\big(\Phi, f, s + \tfrac{1}{2}\big) \tag{5.13}$$

for every $\Phi \in C_c^\infty(A)$ and every coefficient f of π. The proof evaluates both sides for one test function and for a family of coefficients indexed by a dual basis of W. Throughout, $n = 2N/e$, an integer because $2/e$ is one by proposition 3.3.2. As in the discussion preceding definition 3.3.8, ψ has level one and $\text{tr}_{\mathfrak{A}}(\mathfrak{P}^j) \subseteq \mathfrak{p}$ exactly when $j \geq 1$, so $\psi_{\mathfrak{A}}$ is trivial on $\mathfrak{P}^j$ exactly when $j \geq 1$ and the dual of the lattice $\mathfrak{P}^k$ is $(\mathfrak{P}^k)^* = \mathfrak{P}^{1-k}$.

Step 1: the level of Λ, the coefficient and the test function. Since $c \in J \subseteq \mathcal{K}_{\mathfrak{A}}$ and $c\mathfrak{A} = \mathfrak{P}^{-N}$, proposition 3.3.7(2) gives $v_{\mathfrak{A}}(c) = -N$. The integers satisfy $[(N+1)/2] \leq [N/2] + 1 \leq N$ for $N \geq 1$, so for every $k \geq [N/2] + 1$ the group $U_{\mathfrak{A}}^k$ lies inside $U_{\mathfrak{A}}^{[(N+1)/2]}$ and hence inside J by (1), and hypothesis (2) evaluates Λ on all of these groups.

Take $k = N + 1$. For $x \in \mathfrak{P}^{N+1}$ one has $cx \in (c\mathfrak{A})\mathfrak{P}^{N+1} = \mathfrak{P}^{-N}\mathfrak{P}^{N+1} = \mathfrak{P}$, so $\Lambda(1+x) = \psi_{\mathfrak{A}}(cx) = 1$ and $U_{\mathfrak{A}}^{N+1} \subseteq \ker \Lambda$. Take $k = N$ instead. The same computation gives $c\mathfrak{P}^N = \mathfrak{A}$, on

which $\psi_{\mathfrak{A}}$ is not trivial, so there is $x_0 \in \mathfrak{P}^N$ with

$$\lambda = \psi_{\mathfrak{A}}(cx_0) \neq 1, \qquad u_0 = 1 + x_0 \in U_{\mathfrak{A}}^N \subseteq J \cap U_{\mathfrak{A}}, \qquad \Lambda(u_0) = \lambda \cdot \mathrm{id}$$

For $j < N$ the group $U_{\mathfrak{A}}^{j+1}$ contains u_0, so $U_{\mathfrak{A}}^{j+1} \subseteq \ker \Lambda$ fails and N is the least integer with $U_{\mathfrak{A}}^{N+1} \subseteq \ker \Lambda$.

Fix $w \in W$ and $\tilde{w} \in \check{W}$. By lemma 5.3.5 the function $f = f_{\tilde{w},w}$, supported on J with value $\langle \tilde{w}, \Lambda(g)w \rangle$, is a coefficient of π, and $\check{f}(g) = \langle \check{\Lambda}(g)\tilde{w}, w \rangle$ on J and 0 off J. Take Φ to be the characteristic function of $U_{\mathfrak{A}}^{N+1} = 1 + \mathfrak{P}^{N+1}$, which is a compact open subset of A.

Step 2: the left-hand zeta integral. The support of Φ lies in J, where f is defined by Λ, and Λ is trivial on $U_{\mathfrak{A}}^{N+1}$ by Step 1, so $f = \langle \tilde{w}, w \rangle$ there. Since $v_F(\det g) = 0$ for $g \in U_{\mathfrak{A}}$ by proposition 3.3.7(4),

$$\zeta\big(\Phi, f, s + \tfrac{1}{2}\big) = \int_{U_{\mathfrak{A}}^{N+1}} \langle \tilde{w}, w \rangle \, \| \det g \|^{s+1/2} \, d^\times g = \mathrm{vol}^\times \left(U_{\mathfrak{A}}^{N+1} \right) \langle \tilde{w}, w \rangle$$

independent of s.

Step 3: the Fourier transform of Φ. Splitting $U_{\mathfrak{A}}^{N+1}$ as the additive coset $1 + \mathfrak{P}^{N+1}$,

$$\hat{\Phi}(x) = \psi_{\mathfrak{A}}(x) \int_{\mathfrak{P}^{N+1}} \psi_{\mathfrak{A}}(xz) \, d\mu(z)$$

and the inner integral is $\mu(\mathfrak{P}^{N+1})$ when $x \in (\mathfrak{P}^{N+1})^* = \mathfrak{P}^{-N}$ and 0 otherwise. Thus $\hat{\Phi} = \mu(\mathfrak{P}^{N+1}) \, \psi_{\mathfrak{A}} \cdot \mathbf{1}_{\mathfrak{P}^{-N}}$.

Step 4: the shells of J, and the vanishing of all but one. Sort J by the valuation $v_{\mathfrak{A}}$ of proposition 3.3.7, putting

$$J^0 = J \cap U_{\mathfrak{A}}, \qquad J_j = \{ g \in J : v_{\mathfrak{A}}(g) = j \} \quad (j \in \mathbb{Z})$$

By part (2) of that proposition $v_{\mathfrak{A}}$ restricts to a homomorphism $J \to \mathbb{Z}$ with kernel J^0, so each J_j is either empty or a single coset $g_j J^0$, and in either case it is stable under right translation by J^0. The group J^0 is open in G and closed in the compact group $U_{\mathfrak{A}}$, so $0 < \mathrm{vol}^\times(J^0) < \infty$ and every non-empty shell is compact of that same volume. The shell $J_{-N} = cJ^0$ is not empty. Part (4) of the same proposition gives $v_F(\det g) = (2/e)j$ for $g \in J_j$, so $J_j \subseteq G_{(2/e)j}$ in the notation of lemma 5.2.4 and distinct shells sit in distinct sets G_m.

Since $v_{\mathfrak{A}}$ is the restriction of $\nu_{\mathfrak{A}}$, an element $g \in J$ lies in $\mathfrak{P}^{-N}$ if and only if $v_{\mathfrak{A}}(g) \geq -N$. So $\hat{\Phi}\check{f}$ is supported on $\bigsqcup_{j \geq -N} J_j$, and writing $Z(\hat{\Phi}, \check{f}, Y) = \sum_m b_m Y^m$ for the Laurent series of lemma 5.2.4,

$$b_m = \mu\big(\mathfrak{P}^{N+1}\big) \, S_j \quad \text{for } m = (2/e)j, \ j \geq -N, \qquad S_j = \int_{J_j} \psi_{\mathfrak{A}}(g) \, \big\langle \check{\Lambda}(g)\tilde{w}, w \big\rangle \, d^\times g$$

and $b_m = 0$ for every other m. Each S_j is the integral of a locally constant function over a compact set, hence absolutely convergent.

We claim that $S_j = 0$ for every $j \geq 1 - N$. Right translation by the element $u_0 \in U_{\mathfrak{A}}^N$ of Step 1 carries J_j onto itself, u_0 lying in J^0, and leaves $d^\times g$ unchanged. It multiplies the integrand by the scalar λ^{-1}. Indeed

$$\big\langle \check{\Lambda}(gu_0)\tilde{w}, w \big\rangle = \big\langle \tilde{w}, \Lambda(u_0)^{-1}\Lambda(g^{-1})w \big\rangle = \lambda^{-1} \big\langle \check{\Lambda}(g)\tilde{w}, w \big\rangle$$

because $\Lambda(u_0)$ is the scalar λ, and

$$\psi_{\mathfrak{A}}(gu_0) = \psi_{\mathfrak{A}}(g)\,\psi_{\mathfrak{A}}\big(g(u_0-1)\big) = \psi_{\mathfrak{A}}(g)$$

because $u_0 - 1 = x_0 \in \mathfrak{P}^N$ and proposition 3.3.7(5) gives $\nu_{\mathfrak{A}}(gx_0) = v_{\mathfrak{A}}(g) + \nu_{\mathfrak{A}}(x_0) \geq j + N \geq 1$, so $gx_0 \in \mathfrak{P}$. Substituting $g \mapsto gu_0$ therefore gives $S_j = \lambda^{-1} S_j$, and $\lambda \neq 1$ forces $S_j = 0$. This is the one place where hypothesis (2) is used in its full strength. The operator $\Lambda(u_0)$ has to be a scalar for it to leave the integral, and that is what makes the argument indifferent to $\dim \Lambda$.

Only $m = -n$ is left, so $Z(\hat{\Phi}, \check{f}, Y) = \mu(\mathfrak{P}^{N+1}) S_{-N} Y^{-n}$ in $\mathbb{C}((Y))$. A Laurent series with one term converges at every point, so

$$\zeta\big(\hat{\Phi}, \check{f}, \tfrac{3}{2} - s\big) = \mu\big(\mathfrak{P}^{N+1}\big)\, q^{\,n(3/2-s)}\, S_{-N}, \qquad s \in \mathbb{C}$$

No analytic continuation is needed to reach this. The vanishing was established one Laurent coefficient at a time, and each coefficient is an integral over a compact set.

Step 5: the collapse of the surviving shell. Substituting $g = cu$ with $u \in J^0$ and using invariance of $d^\times g$,

$$S_{-N} = \int_{J^0} \psi_{\mathfrak{A}}(cu)\, \big\langle \check{\Lambda}(cu)\tilde{w}, w \big\rangle\, d^\times u$$

The group $U_{\mathfrak{A}}^{[N/2]+1}$ is open in the compact group J^0, so J^0 is the disjoint union of finitely many cosets $yU_{\mathfrak{A}}^{[N/2]+1}$. On $U_{\mathfrak{A}}^{[N/2]+1}$ the multiplicative and additive measures agree up to a constant κ independent of the level, since $d^\times g = \|\det g\|^{-2} d\mu(g)$ and $\|\det g\| = 1$ there. Write $u = y(1+z)$ with $z \in \mathfrak{P}^{[N/2]+1}$. By hypothesis (2) the operator $\check{\Lambda}(1+z)$ is the scalar $\psi_{\mathfrak{A}}(cz)^{-1}$, and $cu = cy + cyz$, so the integrand is

$$\psi_{\mathfrak{A}}(cu)\, \big\langle \check{\Lambda}(cu)\tilde{w}, w \big\rangle = \psi_{\mathfrak{A}}\big(c(y-1)z\big)\; \psi_{\mathfrak{A}}(cy)\, \big\langle \check{\Lambda}(cy)\tilde{w}, w \big\rangle$$

Integrating over z gives $\kappa\, \mu(\mathfrak{P}^{[N/2]+1})$ when $z \mapsto \psi_{\mathfrak{A}}(c(y-1)z)$ is trivial on $\mathfrak{P}^{[N/2]+1}$, and 0 otherwise. Triviality says $c(y-1) \in (\mathfrak{P}^{[N/2]+1})^* = \mathfrak{P}^{-[N/2]}$, that is $\nu_{\mathfrak{A}}(c(y-1)) \geq -[N/2]$, and proposition 3.3.7(5) rewrites the left side as $v_{\mathfrak{A}}(c) + \nu_{\mathfrak{A}}(y-1) = -N + \nu_{\mathfrak{A}}(y-1)$. Since $[N/2] + [(N+1)/2] = N$, the condition is

$$\nu_{\mathfrak{A}}(y-1) \geq N - [N/2] = \left[\tfrac{N+1}{2}\right], \qquad \text{that is} \qquad y \in U_{\mathfrak{A}}^{[(N+1)/2]}$$

and hypothesis (1) places that group inside J^0. Hence

$$S_{-N} = \kappa\, \mu\big(\mathfrak{P}^{[N/2]+1}\big) \sum_{y} \psi_{\mathfrak{A}}(cy)\, \big\langle \check{\Lambda}(cy)\tilde{w}, w \big\rangle$$

with y running over $U_{\mathfrak{A}}^{[(N+1)/2]} / U_{\mathfrak{A}}^{[N/2]+1}$.

The same computation shows that the summand is well defined on those cosets. Replacing y by $y(1+z)$ multiplies it by $\psi_{\mathfrak{A}}(c(y-1)z)$, which is 1 for exactly the y that are retained. On any larger index set the expression would depend on the representative, which is why the quotient in the statement is the one it is. When N is odd the two indices coincide, the index set is the trivial group, and $y = 1$ may be taken.

Step 6: the constants, and the count over a dual basis. By the comparison of measures used in Step 5, $\mathrm{vol}^\times(U_{\mathfrak{A}}^{N+1}) = \kappa\, \mu(\mathfrak{P}^{N+1})$. Substituting Steps 2, 4 and 5 into (5.13), the factors κ and

$\mu(\mathfrak{P}^{N+1})$ cancel and

$$\varepsilon(\pi,s,\psi)\,\langle\tilde{w},w\rangle = q^{\,n(3/2-s)}\,\mu\big(\mathfrak{P}^{[N/2]+1}\big)\sum_{y}\psi_{\mathfrak{A}}(cy)\,\big\langle\check{\Lambda}(cy)\tilde{w},w\big\rangle$$

for every $w\in W$ and every $\tilde{w}\in\check{W}$. Let $w_1,\dots,w_d$ be a basis of W, where $d=\dim\Lambda$, and let $\tilde{w}_1,\dots,\tilde{w}_d$ be the dual basis of $\check{W}$. Summing the identity over the pairs $(\tilde{w}_i,w_i)$ turns $\langle\tilde{w},w\rangle$ into d, and turns each matrix coefficient into a trace, since $\sum_i\langle T\tilde{w}_i,w_i\rangle=\mathrm{tr}T$ for every $T\in\mathrm{End}_{\mathbb{C}}(\check{W})$. Hence

$$\varepsilon(\pi,s,\psi)\,\dim\Lambda = q^{\,n(3/2-s)}\,\mu\big(\mathfrak{P}^{[N/2]+1}\big)\,\mathfrak{g}(\Lambda,\psi)$$

By lemma 5.3.6, $\mu(\mathfrak{P}^{[N/2]+1})=q^{-2(1+2[N/2])/e}$, and $1+2[N/2]$ is N when N is odd and $N+1$ when N is even. That single arithmetic fact is the whole source of the parity in the statement. For odd N the measure is $q^{-2N/e}=q^{-n}$ and the display reads

$$\varepsilon(\pi,s,\psi)\,\dim\Lambda = q^{\,n(1/2-s)}\,\mathfrak{g}(\Lambda,\psi)$$

For even N the measure is $q^{-2(N+1)/e}=q^{-n}q^{-2/e}$, so the same identity acquires the extra factor $q^{-2/e}$, which is $(\mathfrak{A}:\mathfrak{P})^{-1/2}$ because $(\mathfrak{A}:\mathfrak{P})=q^{4/e}$ by lemma 5.3.6. In both cases this is (5.12) with $q^a\dim\Lambda$ as stated.

Three assertions remain, and all three are read off that identity. By corollary 5.2.9 the local constant is a monomial c_0q^{-fs} with $c_0\in\mathbb{C}^\times$, so it does not vanish, and $\dim\Lambda\geq 1$, whence $\mathfrak{g}(\Lambda,\psi)\neq 0$. Neither side of the identity other than $\mathfrak{g}(\Lambda,\psi)$ mentions c, so the sum takes the same value for every c allowed by (2). And (5.12) exhibits $\varepsilon(\pi,s,\psi)$ as a non-zero constant times q^{-ns}, so definition 5.3.4 gives $n(\pi,\psi)=n=2N/e$, completing the proof.

The constant q^a is fixed by the displayed relation rather than given in closed form, and the relation determines it because $\dim\Lambda$ is a positive integer. The hypothesis $N\geq 1$ keeps the level-zero case outside the theorem, and that is the case in which q^a fails to be an integral power of q, since $\dim\Lambda=q-1$ there.

The statement is [2, 25.5 Corollary] restricted to $N\geq 1$, and $\check{\Lambda}$ is written $\Lambda^\vee$ there. One symbol does not transfer: the n of that corollary is the N here, while $n(\pi,\psi)$ is its $2\ell(\pi)$. The Gauss sum, its index set and the constant $q^a\dim\Lambda$ match as written.

Hypothesis (1) asks for nothing extra when N is odd. The two indices $[(N+1)/2]$ and $[N/2]+1$ are then both equal to $(N+1)/2$, and hypothesis (2) cannot even be stated without $U_{\mathfrak{A}}^{[N/2]+1}\subseteq J$. Only for even N, where the indices differ by one, does (1) ask for a strictly larger group than (2) needs. Theorem 5.3.9 below is the odd case with $\dim\Lambda=1$, so nothing there has to be checked beyond what its own hypotheses give.

The order of the argument matters, and it is not the order the odd one-dimensional case invites. When the index set has a single element the shell can be located the other way round, by comparing the two sides of (5.13) as Laurent series and reading the answer off the one shell whose value is visibly non-zero. That route closes only when the sum has one term. Step 4 identifies the shell first, by a vanishing statement that never evaluates anything, and the non-vanishing of $\mathfrak{g}(\Lambda,\psi)$ then follows from corollary 5.2.9 instead of being an input to the argument. The monomial shape recorded in definition 5.3.4 enters only at the end, to name the exponent.

The exponent identifies $n(\pi,\psi)$ with $2\ell(\pi)$, which is what proposition 10.1.10(1) asserts, and the

identification is a consequence of the theorem rather than a check on it. The triple $(\mathfrak{A}, N, c)$ is a stratum in the sense of definition 3.3.8, and it is fundamental: c normalizes $\mathfrak{A}$, so $\nu_{\mathfrak{A}}(c^r) = -rN$ and $c^r \notin \mathfrak{P}^{1-rN}$ for every $r \geq 1$, which is proposition 3.3.11. Since π contains ψ_c on $U_{\mathfrak{A}}^N$, theorem 3.3.13 gives $\ell(\pi) = N/e$, so $n(\pi, \psi) = 2N/e = 2\ell(\pi)$. That is clause (1) of proposition 10.1.10 for every π of positive level.

For a ramified simple stratum both simplifications are present at once. The period is 2, the level is odd, the inducing representation is a character, and the Gauss sum has a single term whose value is a single character value. The theorem below is that case, and it is the form in which chapter 11 transports the local constant across a tame extension.

Theorem 5.3.9: Local Constant of a Cuspidal Representation at a Ramified Simple Stratum

Let $\psi \in \widehat{F}$ have level one. Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum with $n = 2m+1$ odd, put $E = F[\alpha]$ and $J = E^\times U_{\mathfrak{A}}^{m+1}$, and let Λ be a character of J whose restriction to $U_{\mathfrak{A}}^{m+1}$ is ψ_α. Set $\pi = \text{c-Ind}_J^G \Lambda$. Then

$$\varepsilon(\pi, s, \psi) = q^{n(1/2-s)}\, \Lambda(\alpha)^{-1}\, \psi\big(\mathrm{tr}_{\mathfrak{A}}\alpha\big) \tag{5.14}$$

In particular $n(\pi, \psi) = n$ and $\varepsilon(\pi, \tfrac{1}{2}, \psi) = \Lambda(\alpha)^{-1}\psi(\mathrm{tr}_{\mathfrak{A}}\alpha)$.

***Proof*:**

The stratum is ramified simple, so $e_{\mathfrak{A}} = 2$ and n is odd by definition 3.4.1(4), and $E = F[\alpha]$ is a field with $E^\times \subseteq \mathcal{K}_{\mathfrak{A}}$ and $e(E|F) = e_{\mathfrak{A}}$ by proposition 3.4.3. The proof applies theorem 5.3.8 with $N = n$ and $c = \alpha$.

The subgroup $J = E^\times U_{\mathfrak{A}}^{m+1}$ lies in $\mathcal{K}_{\mathfrak{A}}$, contains $Z = F^\times$ because $F^\times \subseteq E^\times$, is open because $U_{\mathfrak{A}}^{m+1}$ is, and is compact modulo Z because $\mathcal{K}_{\mathfrak{A}}$ is. The representation Λ is a character, so $\dim \Lambda = 1$, and $\pi = \text{c-Ind}_J^G \Lambda$ is irreducible and cuspidal by theorem 3.5.3(2), whose hypothesis holds because Λ restricts to ψ_α on $U_{\mathfrak{A}}^{m+1}$. This is the case of theorem 3.5.5 in which the cuspidal type has positive level.

Since $n = 2m+1$ is odd, $[(n+1)/2] = [n/2] + 1 = m+1$, so hypothesis (1) of theorem 5.3.8 is the containment $U_{\mathfrak{A}}^{m+1} \subseteq J$, which holds by construction. For hypothesis (2), take $c = \alpha$. The element α lies in $E^\times \subseteq J$, and the stratum being fundamental gives $\alpha \notin \mathfrak{P}^{1-n}$ by proposition 3.3.11 at $r = 1$, so $\nu_{\mathfrak{A}}(\alpha) = -n$ and proposition 3.3.7(2) turns this into $\alpha\mathfrak{A} = \mathfrak{P}^{-n}$. The condition on Λ reads $\Lambda(1+x) = \psi_\alpha(1+x) = \psi_{\mathfrak{A}}(\alpha x)$ for $x \in \mathfrak{P}^{m+1}$, and a value of a character is a scalar.

Theorem 5.3.8 therefore applies with $N = n$ and $e = 2$. Its index set $U_{\mathfrak{A}}^{[(n+1)/2]}/U_{\mathfrak{A}}^{[n/2]+1}$ is the trivial group, so the Gauss sum is its single term at $x = 1$,

$$\mathfrak{g}(\Lambda, \psi) = \mathrm{tr}\check{\Lambda}(\alpha)\, \psi_{\mathfrak{A}}(\alpha) = \Lambda(\alpha)^{-1}\, \psi_{\mathfrak{A}}(\alpha)$$

and $q^a = 1$, since n is odd and $\dim \Lambda = 1$. The exponent is $n(\pi, \psi) = 2n/e_{\mathfrak{A}} = n$. Substituting into (5.12),

$$\varepsilon(\pi, s, \psi) = q^{\,n(1/2-s)}\, \Lambda(\alpha)^{-1}\, \psi_{\mathfrak{A}}(\alpha)$$

and $\psi_{\mathfrak{A}}(\alpha) = \psi(\mathrm{tr}_{\mathfrak{A}}\alpha)$ by definition, which is (5.14). Evaluating at $s = \frac{1}{2}$ gives the last assertion, completing the proof.

The exponent agrees with the general check made above. A stratum of level n on a chain order with $e_{\mathfrak{A}} = 2$ has normalized level $\ell(\pi) = n/2$, so proposition 10.1.10(1) gives $n(\pi, \psi) = 2\ell(\pi) = n$.

Two features of (5.14) are worth separating. The dependence on Λ is through the single value $\Lambda(\alpha)$, so two types on the same stratum have local constants differing by a root of unity read off at one point; and the dependence on ψ is through $\psi(\mathrm{tr}_{\mathfrak{A}}\alpha)$ alone. Both are what make the comparison across a tame extension in chapter 11 finite rather than analytic: transporting the formula along K/F substitutes q_K for q, the image of α for α, and ψ_K for ψ.

A statement of a different kind becomes available once twisting is allowed, and it records how little the ε-factor remembers: past a threshold depending on π, the local constant of $\chi\pi$ depends only on χ and on the central character of π.

Theorem 5.3.10: Stability of the Local Constant for GL_2

Let (π, V) be an irreducible smooth representation of G with central character ω_π, and let $\psi \in \widehat{F}$ have level one. There is an integer $m = m(\pi) \geq 1$ with the following property. Let χ be a character of $F^\times$ of level $n \geq m$, and let $c \in F^\times$ satisfy

$$\chi(1 + x) = \psi(cx), \qquad x \in \mathfrak{p}^{[n/2]+1}$$

Then, writing $\chi\pi = (\chi \circ \det) \otimes \pi$,

$$\varepsilon(\chi\pi, s, \psi) = \omega_\pi(c)^{-1}\, \varepsilon(\chi, s, \psi)^2$$

In particular, if (π', V') is a second irreducible smooth representation of G with $\omega_{\pi'} = \omega_\pi$, then $\varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi\pi', s, \psi)$ for every character χ of $F^\times$ of sufficiently large level.

The proof is Deligne's, and it rests on machinery this book does not develop; the theorem is used here as a black box. The GL_1 analogue theorem 5.1.21 *is* proved in full above, and comparing the two shows exactly what is missing.

***Proof Key ideas*:**

At GL_1 the argument is a manipulation of Gauss sums. Theorem 5.1.18 writes $\varepsilon(\chi, s, \psi)$ explicitly in terms of $\tau(\chi, \psi)$, and proposition 5.1.20 collapses the Gauss sum of a character of level n onto a sum over the small group $U_F^{[(n+1)/2]}/U_F^{[n/2]+1}$; and a twisting character θ of level l with $2l < n$ is trivial on that group, so it comes out of the sum as the single scalar $\theta(c)^{-1}$. Every step is available because the GL_1 local constant has a closed formula.

No such formula exists at GL_2. The local constant of a cuspidal representation is not an elementary sum over $F^\times$; the worked example above produced it only as a ratio of two integrals, and even at level zero the constant had to be quoted. The substitute is an analysis of the zeta integral of $\chi\pi$ in the Kirillov model of π, in which the vectors of V are realized as functions on $F^\times$. A twist by a character of very large level makes the integrand oscillate so rapidly that the integral localizes onto a small neighbourhood, and on that neighbourhood the only surviving datum is the action of the centre. The shape of the answer is the one the GL_1 case predicts: a two-dimensional object contributes $\varepsilon(\chi, s, \psi)$ twice, and its determinant, here the central character, contributes the scalar $\omega_\pi(c)^{-1}$.

For the argument in full, see [2, §25].

What the theorem asserts is a limitation. Fix a central character; then all irreducible representations of G with that central character become indistinguishable once they are twisted deeply enough, because their local constants collapse to one expression built out of χ and ω_π. A family of local constants $\{\varepsilon(\chi\pi, s, \psi)\}_\chi$ therefore carries information about π only through the twists of bounded level, together with ω_π. Comparing two cuspidal representations through all their twists looks like an infinite amount of data; past the threshold $m(\pi)$ it is not.

Exercise 5.3.1

1. Let (π, V) be an irreducible smooth representation of G and let χ be a character of $F^\times$. Show that $\mathrm{Hom}_G(\pi, \chi\circ\det) \neq 0$ forces $\pi \cong \chi\circ\det$. Deduce that no irreducible cuspidal representation of G admits a non-zero G-homomorphism onto a one-dimensional representation.

2. Let (π, V) be an irreducible cuspidal representation of G and let χ be any character of $F^\times$. Show that $\chi\pi = (\chi \circ \det) \otimes \pi$ is irreducible and cuspidal, and deduce that $L(\chi\pi, s) = 1$ and $\gamma(\chi\pi, s, \psi) = \varepsilon(\chi\pi, s, \psi)$.

3. Let (π, V) be an irreducible smooth representation of G and let ψ have level one. Show that $n(\pi, \psi) = n(\check\pi, \psi)$ and that $\varepsilon(\pi, \frac{1}{2}, \psi)\varepsilon(\check\pi, \frac{1}{2}, \psi) = \omega_\pi(-1)$. Deduce that if $\pi \cong \check\pi$ and ω_π is trivial, then $\varepsilon(\pi, s, \psi) = \pm q^{(1/2-s)n(\pi,\psi)}$.

4. Let (π, V) be an irreducible smooth representation of G and let ψ have level one.

    1. Show that the level of $a\psi$ is the level of ψ minus $v_F(a)$, for $a \in F^\times$.
    2. Deduce that if ψ' has level l, then $\varepsilon(\pi, s, \psi') = \varepsilon(\pi, \frac{1}{2}, \psi')q^{(1/2-s)(n(\pi,\psi)+2(1-l))}$.
    3. Let π be a cuspidal representation of level zero as in example 5.3.7. Compute $\varepsilon(\pi, s, \varpi\psi)$ in terms of $\varepsilon(\pi, s, \psi)$ and read off the exponent attached to π by an additive character of level zero.

5. Let (π, V) be an irreducible cuspidal representation of G whose central character ω_π is ramified. Give the proof that $L(\pi, s) = 1$ that uses only proposition 5.1.8 together with the expression of $Z(\Phi, f, X)$ as a finite sum of Tate integrals obtained in the proof of proposition 5.3.1, and identify the single point at which the general argument needs ω_π to be unramified.

5.4 Epsilon Factors of Non-Cuspidal Representations

Section 5.3 settled the cuspidal half of the spectrum. What remains is the list of theorem 2.4.16: the irreducible induced representations $\iota_B^G\chi$, the one-dimensional representations $\varphi \circ \det$, and the special representations $\varphi\mathrm{St}_G$. Every coefficient of every one of these is a coefficient of some $\mathrm{Ind}_B^G\chi$, and the GL_2 zeta integral of such a coefficient unfolds into a pair of Tate integrals. The whole of this section is therefore a transcription of the GL_1 theory of section 5.1, with the two normalizations in play kept explicit throughout.

Two normalizations should be kept in view throughout, since almost every error available here is an error in one of them. The first is the shift in the GL_2 functional equation: by proposition 5.2.6 the L-function is the generator of the module of the *shifted* integrals $\zeta(\Phi, f, s + \frac{1}{2})$, which is the normalization that makes (5.10) and corollary 5.2.9 symmetric in $s \leftrightarrow 1 - s$. The generator of the unshifted module is then $L(\pi, s - \frac{1}{2})$, and nothing is lost in passing between the two: $s \mapsto s + \frac{1}{2}$ sends

q^{-s} to $q^{-1/2}q^{-s}$ and so is a ring automorphism of $\mathbb{C}[q^s, q^{-s}]$ carrying one module isomorphically onto the other, generators to generators, and preserving the normalization $P(0) = 1$. The second normalization is the $\delta_B^{-1/2}$ twist of (2.13). Since $\delta_B(t) = \|t_2/t_1\|$ by proposition 2.2.8, as a character of $T = F^\times \times F^\times$ one has $\delta_B^{\pm 1/2} = \|-\|^{\mp 1/2} \otimes \|-\|^{\pm 1/2}$, and therefore

$$\iota_B^G(\chi_1 \otimes \chi_2) = \mathrm{Ind}_B^G\big(\chi_1\|-\|^{1/2} \otimes \chi_2\|-\|^{-1/2}\big), \qquad \mathrm{Ind}_B^G(\chi_1 \otimes \chi_2) = \iota_B^G\big(\chi_1\|-\|^{-1/2} \otimes \chi_2\|-\|^{1/2}\big)$$

The half-integral powers of $\|-\|$ produced by this twist are exactly what the half-integral shift $s+\frac{1}{2} \leftrightarrow \frac{3}{2}-s$ absorbs. Recall for the same purpose that $L(\chi\|-\|^a, s) = L(\chi, s+a)$ and $\zeta(\Phi, \chi\|-\|^a, s) = \zeta(\Phi, \chi, s+a)$, immediate from definition 5.1.6 and definition 5.1.9, and hence $\gamma(\chi\|-\|^a, s, \psi) = \gamma(\chi, s+a, \psi)$ and $\varepsilon(\chi\|-\|^a, s, \psi) = \varepsilon(\chi, s+a, \psi)$ by theorem 5.1.11 and definition 5.1.13, for every $a \in \mathbb{C}$ and every character χ of $F^\times$.

The first family is the irreducible induced representations. Here the answer is the one the notation predicts: the invariants of $\iota_B^G(\chi_1 \otimes \chi_2)$ are the products of the invariants of χ_1 and χ_2, with no shift left over.

Proposition 5.4.1: L-Function and Epsilon Factor of a Principal Series

Let $\chi = \chi_1 \otimes \chi_2$ be a character of T such that $\pi = \iota_B^G\chi$ is irreducible, and let $\psi \in \widehat{F}$ be non-trivial. Then

$$L(\pi, s) = L(\chi_1, s)L(\chi_2, s), \qquad \varepsilon(\pi, s, \psi) = \varepsilon(\chi_1, s, \psi)\varepsilon(\chi_2, s, \psi)$$

and consequently $\gamma(\pi, s, \psi) = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi)$.

***Proof*:**

Step 1: the unfolded zeta integral. Let $\xi = \xi_1 \otimes \xi_2$ be any character of T, inflated to B trivially on N, and write $\Sigma_\xi = \mathrm{Ind}_B^G\xi$, realized on the space of smooth $\varphi\colon G \to \mathbb{C}$ with $\varphi(bg) = \xi(b)\varphi(g)$. The quotient $B\backslash G$ is compact (corollary 2.2.2), so the Duality Theorem (theorem 1.3.12) identifies $\check{\Sigma}_\xi$ with $\mathrm{Ind}_B^G\delta_B^{-1}\check{\xi}$ under the pairing $\langle\tilde{\varphi}, \varphi\rangle = \int_K \tilde{\varphi}(k)\varphi(k)\,d\mu_K(k)$; hence $C(\Sigma_\xi)$ is spanned by the functions

$$f(g) = \int_K \tilde{\varphi}(k)\varphi(kg)\,d\mu_K(k), \qquad \varphi \in \Sigma_\xi,\ \tilde{\varphi} \in \mathrm{Ind}_B^G\delta_B^{-1}\check{\xi}$$

Normalize the Haar measure μ_N on $N \cong F$ entering the decomposition $d\mu_B(tn) = d\mu_T(t)\,d\mu_N(n)$ fixed before proposition 2.2.8 to be the self-dual measure μ_ψ; this changes the left Haar measure μ_B, and hence the constant relating the two sides of (2.5), by a positive scalar and nothing else. For $\Theta \in C_c^\infty(A)$ set

$$\Psi_\Theta(t_1, t_2) = \int_F \Theta\begin{pmatrix} t_1 & y \\ 0 & t_2 \end{pmatrix} d\mu_\psi(y)$$

and for a character ξ of T and $\Psi \in C_c^\infty(F \times F)$ set

$$\Xi_\xi(\Psi, s) = \int_{F^\times}\int_{F^\times} \Psi(t_1, t_2)\,\xi_1(t_1)\|t_1\|^{s-1}\,\xi_2(t_2)\|t_2\|^s\,d^\times t_1\,d^\times t_2$$

The non-cuspidal half of the proof of proposition 5.2.6 is exactly the computation of $\zeta(\Phi, f, s)$ for such an f: substituting the pairing formula for f and replacing g by $k^{-1}g$, applying the Iwasawa decomposition $G = BK$ (proposition 2.2.1) through (2.5), writing $b = tn(x)$ and substituting

$y = t_1 x$ in the N-integral produces

$$\zeta(\Phi, f, s) = c_0 \int_K \int_K \tilde{\varphi}(k)\varphi(k')\, \Xi_\xi\big(\Psi_{\Phi_{k,k'}},\, s\big)\, d\mu_K(k)\, d\mu_K(k'), \qquad \Phi_{k,k'}(y) = \Phi(k^{-1} y k')$$

where $c_0 > 0$ depends only on the chosen Haar measures and not on ξ, Φ or f. That proof also records the two facts making the right-hand side meaningful: $\Psi_\Theta \in C_c^\infty(F \times F)$ for every $\Theta \in C_c^\infty(A)$, and all the interchanges are justified on a half plane $\mathrm{Re}(s) \gg 0$ of absolute convergence, after which both sides are rational in q^{-s} and the identity persists.

Step 2: the partial integral converts the matrix Fourier transform into a two-variable one. For $\Psi \in C_c^\infty(F \times F)$ let $\widehat{\Psi}$ denote the Fourier transform for the character $(y_1, y_2) \mapsto \psi(t_1 y_1 + t_2 y_2)$ and the measure $\mu_\psi \otimes \mu_\psi$, which is self-dual for it. The claim is

$$\Psi_{\hat{\Theta}} = \widehat{\Psi_\Theta}, \qquad \Theta \in C_c^\infty(A)$$

where $\hat{\Theta}$ is the matrix Fourier transform of definition 5.2.1. In coordinates the pairing on A is $\mathrm{tr}(xy) = x_{11}y_{11} + x_{12}y_{21} + x_{21}y_{12} + x_{22}y_{22}$, which is the standard pairing of F^4 with itself after the relabelling that swaps the two off-diagonal coordinates in one of the two arguments; a permutation of coordinates preserves self-duality, so the measure μ on A self-dual for ψ_A is the product $\mu_\psi^{\otimes 4}$ of the four self-dual measures on the entries.

Fix $t_1, t_2 \in F$. Since

$$\mathrm{tr}\left(\begin{pmatrix} t_1 & u \\ 0 & t_2 \end{pmatrix} y\right) = t_1 y_{11} + u y_{21} + t_2 y_{22}$$

the function

$$h(v) = \int_F \int_F \int_F \Theta\begin{pmatrix} y_{11} & y_{12} \\ v & y_{22} \end{pmatrix} \psi(t_1 y_{11} + t_2 y_{22})\, d\mu_\psi(y_{11})\, d\mu_\psi(y_{12})\, d\mu_\psi(y_{22})$$

lies in $C_c^\infty(F)$, and Fubini over the compact support of Θ gives $\hat{\Theta}\left(\begin{smallmatrix} t_1 & u \\ 0 & t_2 \end{smallmatrix}\right) = \hat{h}(u)$. As $\hat{h} \in C_c^\infty(F)$ by proposition 5.1.2, integrating in u against μ_ψ is an absolutely convergent integral, and by the Fourier inversion formula (definition 5.1.4)

$$\Psi_{\hat{\Theta}}(t_1, t_2) = \int_F \hat{h}(u)\, d\mu_\psi(u) = \hat{\hat{h}}(0) = h(0)$$

The value $h(0)$ is by definition $\int_F \int_F \Psi_\Theta(y_{11}, y_{22}) \psi(t_1 y_{11} + t_2 y_{22})\, d\mu_\psi(y_{11}) d\mu_\psi(y_{22}) = \widehat{\Psi_\Theta}(t_1, t_2)$, which is the claim. A scalar change of the measure used in Ψ_Θ multiplies both sides by the same scalar, so the identity is insensitive to the normalization of μ_N fixed in Step 1.

Step 3: unfolding the transformed side. Keep ξ arbitrary and let $f \in C(\Sigma_\xi)$ be as in Step 1. The pairing $\langle -, - \rangle$ is G-invariant, so

$$\check{f}(g) = f(g^{-1}) = \langle \tilde{\varphi}, \Sigma_\xi(g^{-1})\varphi \rangle = \langle \varphi, \check{\Sigma}_\xi(g)\tilde{\varphi} \rangle = \int_K \varphi(k)\tilde{\varphi}(kg)\, d\mu_K(k)$$

Thus $\check{f}$ is a coefficient of Σ_η for $\eta = \delta_B^{-1}\check{\xi}$, built from the pair $(\varphi, \tilde{\varphi})$ with the roles reversed, the outer function being φ; and φ indeed lies in $\mathrm{Ind}_B^G \delta_B^{-1}\check{\eta} = \mathrm{Ind}_B^G \xi$. Applying Step 1 to Σ_η, with test function $\hat{\Phi}$,

$$\zeta(\hat{\Phi}, \check{f}, s') = c_0 \int_K \int_K \varphi(k)\tilde{\varphi}(k')\, \Xi_\eta\big(\Psi_{(\hat{\Phi})_{k,k'}},\, s'\big)\, d\mu_K(k)\, d\mu_K(k')$$

with the *same* constant c_0, since c_0 depends only on the measures. For $k, k' \in K$ the substitution $y \mapsto kyk'^{-1}$ preserves μ (its determinant as an F-linear map on A is $\det(k)^2 \det(k')^{-2}$, of absolute value one) and $\operatorname{tr}(xkyk'^{-1}) = \operatorname{tr}((k'^{-1}xk)y)$, whence $\widehat{\Phi_{k',k}} = (\hat{\Phi})_{k,k'}$. Combining this with Step 2 and relabelling $(k, k') \mapsto (k', k)$,

$$\zeta(\hat{\Phi}, \check{f}, s') = c_0 \int_K \int_K \tilde{\varphi}(k)\varphi(k')\, \Xi_\eta\big(\widehat{\Psi_{\Phi_{k,k'}}},\, s'\big)\, d\mu_K(k)\, d\mu_K(k')$$

so both sides of the functional equation are integrals of the same weight $\tilde{\varphi}(k)\varphi(k')$ against the same family of functions $\Psi_{\Phi_{k,k'}} \in C_c^\infty(F \times F)$.

Step 4: Tate's functional equation in each variable. Since $\delta_B^{-1} = \|-\| \otimes \|-\|^{-1}$, the character $\eta = \delta_B^{-1}\check{\xi}$ has components $\eta_1 = \check{\xi}_1\|-\|$ and $\eta_2 = \check{\xi}_2\|-\|^{-1}$. Put $\sigma = s + \frac{1}{2}$, so that $\frac{3}{2} - s = 2 - \sigma$. Then for $\Psi \in C_c^\infty(F \times F)$,

$$\Xi_\eta(\widehat{\Psi}, 2-\sigma) = \int_{F^\times} \int_{F^\times} \widehat{\Psi}(t_1, t_2)\, \check{\xi}_1(t_1)\|t_1\|^{2-\sigma}\, \check{\xi}_2(t_2)\|t_2\|^{1-\sigma}\, d^\times t_1\, d^\times t_2$$

where the two powers of $\|-\|$ absorbed from η_1 and η_2 have moved the exponents by $+1$ and -1. Comparing with

$$\Xi_\xi(\Psi, \sigma) = \int_{F^\times} \int_{F^\times} \Psi(t_1, t_2)\, \xi_1(t_1)\|t_1\|^{\sigma-1}\, \xi_2(t_2)\|t_2\|^{\sigma}\, d^\times t_1\, d^\times t_2$$

the exponents pair up as $2 - \sigma = 1 - (\sigma - 1)$ in the first variable and $1 - \sigma$ in the second: the transformed integral is Tate's left-hand side for ξ_1 at $u_1 = \sigma - 1$ and for ξ_2 at $u_2 = \sigma$. Both Ψ and $\widehat{\Psi}$ are finite sums of pure tensors, by the argument of proposition 2.2.6 run on $F \times F$, and Fourier transform on $F \times F$ carries $\Psi_1 \otimes \Psi_2$ to $\hat{\Psi}_1 \otimes \hat{\Psi}_2$. On a pure tensor,

$$\Xi_\eta(\widehat{\Psi_1 \otimes \Psi_2}, 2-\sigma) = \zeta(\hat{\Psi}_1, \check{\xi}_1, 1-u_1)\, \zeta(\hat{\Psi}_2, \check{\xi}_2, 1-u_2)$$

so theorem 5.1.11 applied in each variable separately gives

$$\Xi_\eta(\widehat{\Psi}, 2-\sigma) = \gamma(\xi_1, \sigma-1, \psi)\, \gamma(\xi_2, \sigma, \psi)\, \Xi_\xi(\Psi, \sigma)$$

for every $\Psi \in C_c^\infty(F \times F)$, the two γ-factors being independent of Ψ. Feeding this into the two displays of Step 3 and Step 1, the common constant c_0 and the common weight $\tilde{\varphi}(k)\varphi(k')$ cancel and

$$\zeta\Big(\hat{\Phi}, \check{f}, \tfrac{3}{2} - s\Big) = \gamma\big(\xi_1, s - \tfrac{1}{2}, \psi\big)\, \gamma\big(\xi_2, s + \tfrac{1}{2}, \psi\big)\, \zeta\Big(\Phi, f, s + \tfrac{1}{2}\Big) \tag{5.15}$$

for every $\Phi \in C_c^\infty(A)$ and every $f \in C(\operatorname{Ind}_B^G \xi)$. Nothing in the derivation used irreducibility, so (5.15) holds for every character ξ of T.

Step 5: the L-factor. Still with ξ arbitrary, the proof of proposition 5.2.6 bounds the unshifted integrals: for all Φ and all $f \in C(\Sigma_\xi)$,

$$\zeta(\Phi, f, s) \in L(\xi_1, s-1)L(\xi_2, s)\, \mathbb{C}[q^s, q^{-s}]$$

since $L(\xi_1\|-\|^{-1}, s) = L(\xi_1, s-1)$. This bound is attained. Choose compact open subsets S_1, S_2 of F with $\zeta(\mathbf{1}_{S_i}, \xi_i, u) = c_i L(\xi_i, u)$ for constants $c_i > 0$: take $S_i = \mathfrak{o}$ when ξ_i is unramified,

as in the proof of proposition 5.1.8, and $S_i = U_F^{n_i+1}$ with n_i the level of ξ_i when ξ_i is ramified, as in the proof of lemma 5.1.15, where $L(\xi_i, u) = 1$. Let Φ be the characteristic function of

$$\mathcal{U} = \big\{ y \in A : y_{11} \in S_1,\ y_{22} \in S_2,\ y_{12} \in \varpi^{-r}\mathfrak{o},\ y_{21} \in \mathfrak{o} \big\}$$

for any $r \geq 0$; this set is compact and open, so $\Phi \in C_c^\infty(A)$, and since the matrices occurring in Ψ_Φ have vanishing $(2,1)$-entry,

$$\Psi_\Phi = \mu_\psi(\varpi^{-r}\mathfrak{o})\, \mathbf{1}_{S_1} \otimes \mathbf{1}_{S_2}$$

Being locally constant with compact support, Φ is bi-invariant under $K_1 = K \cap (1 + \varpi^M M_2(\mathfrak{o}))$ for M large: if Φ is invariant under translation by $\varpi^{M_0} M_2(\mathfrak{o})$ and supported in $\varpi^{-r_0} M_2(\mathfrak{o})$, then $M = M_0 + r_0$ works, because $k_1 y - y = (k_1 - 1)y$ lies in $\varpi^{M_0} M_2(\mathfrak{o})$ for y in the support, and k_1 preserves the lattice $\varpi^{-r_0} M_2(\mathfrak{o})$, so both values vanish off it. Shrinking K_1 further, assume also $B \cap K_1 \subseteq \ker \xi$, which is possible because ξ is smooth. Now let φ be supported on BK_1 with $\varphi(bk_1) = \xi(b)$, and $\tilde{\varphi}$ supported on BK_1 with $\tilde{\varphi}(bk_1) = \delta_B^{-1}\check{\xi}(b)$. Both are well-defined, since $bk_1 = b'k_1'$ forces $b'^{-1}b \in B \cap K_1 \subseteq \ker \xi$, and both are smooth, being right K_1-invariant with open support.

The weights and the Ξ-values now cancel exactly. Write $b^{-1}\Phi$ and $\Phi b'$ for $y \mapsto \Phi(b^{-1}y)$ and $y \mapsto \Phi(yb')$. For b, b' in B with diagonals (b_1, b_2) and (b_1', b_2'), substituting for the upper-right entry in the defining integral gives

$$\Psi_{b^{-1}\Phi}(t_1, t_2) = \|b_1\|\, \Psi_\Phi(b_1^{-1}t_1, b_2^{-1}t_2), \qquad \Psi_{\Phi b'}(t_1, t_2) = \|b_2'\|^{-1}\, \Psi_\Phi(t_1 b_1', t_2 b_2')$$

Substituting $t_i \mapsto b_i t_i$, respectively $t_i \mapsto t_i b_i'^{-1}$, in Ξ_ξ and using $\|b_1\| = \|b_2\| = \|b_1'\| = \|b_2'\| = 1$ for $b, b' \in B \cap K$ yields

$$\Xi_\xi(\Psi_{b^{-1}\Phi b'}, s) = \xi(b)\, \check{\xi}(b')\, \Xi_\xi(\Psi_\Phi, s), \qquad b, b' \in B \cap K$$

Also $K \cap BK_1 = (B \cap K)K_1$, since $k = bk_1 \in K$ forces $b = kk_1^{-1} \in B \cap K$, and δ_B is trivial on $B \cap K$. At a point $k = bk_1$, $k' = b'k_1'$ of the support the integrand of Step 1 is therefore

$$\check{\xi}(b)\, \xi(b')\, \Xi_\xi(\Psi_{\Phi_{k,k'}}, s) = \check{\xi}(b)\, \xi(b')\, \xi(b)\, \check{\xi}(b')\, \Xi_\xi(\Psi_\Phi, s) = \Xi_\xi(\Psi_\Phi, s)$$

using bi-K_1-invariance of Φ to delete k_1 and k_1'. On the pure tensor Ψ_Φ the double integral Ξ_ξ separates, so

$$\Xi_\xi(\Psi_\Phi, s) = \mu_\psi(\varpi^{-r}\mathfrak{o})\, \zeta(\mathbf{1}_{S_1}, \xi_1, s-1)\, \zeta(\mathbf{1}_{S_2}, \xi_2, s) = c_1 c_2\, \mu_\psi(\varpi^{-r}\mathfrak{o})\, L(\xi_1, s-1) L(\xi_2, s)$$

and therefore

$$\zeta(\Phi, f, s) = c_0\, \mu_K\big((B \cap K)K_1\big)^2\, \Xi_\xi(\Psi_\Phi, s) = c\, L(\xi_1, s-1) L(\xi_2, s)$$

for a constant $c > 0$. The module of unshifted integrals of Σ_ξ is therefore exactly $L(\xi_1, s-1)L(\xi_2, s)\, \mathbb{C}[q^s, q^{-s}]$.

Step 6: specialization to $\pi = \iota_B^G \chi$ and assembly. Take $\xi_1 = \chi_1 \| - \|^{1/2}$ and $\xi_2 = \chi_2 \| - \|^{-1/2}$, so that $\Sigma_\xi = \iota_B^G \chi = \pi$, which by hypothesis is irreducible; then $C(\pi) = C(\Sigma_\xi)$ and Steps 4 and 5 apply verbatim. By Step 5 the generator of the unshifted module is

$$L(\xi_1, s-1) L(\xi_2, s) = L\big(\chi_1, s - \tfrac{1}{2}\big)\, L\big(\chi_2, s - \tfrac{1}{2}\big)$$

so the generator of the shifted module, which is $L(\pi, s)$, is obtained by replacing s by $s + \frac{1}{2}$:

$$L(\pi, s) = L(\chi_1, s)L(\chi_2, s)$$

This is the first place where the two normalizations are seen to match: the $\pm\frac{1}{2}$ carried by the $\delta_B^{-1/2}$ twist and the $+\frac{1}{2}$ carried by the Godement-Jacquet shift cancel, and no half-integral shift survives.

For the ε-factor, $\zeta(\Phi, f, s + \frac{1}{2})$ is not identically zero by part (2) of proposition 5.2.6, so the rational function $\gamma(\pi, s, \psi)$ of theorem 5.2.7 is unique; comparing (5.10) with (5.15),

$$\gamma(\pi, s, \psi) = \gamma\big(\chi_1\|-\|^{1/2}, s - \tfrac{1}{2}, \psi\big)\,\gamma\big(\chi_2\|-\|^{-1/2}, s + \tfrac{1}{2}, \psi\big) = \gamma(\chi_1, s, \psi)\,\gamma(\chi_2, s, \psi)$$

again with the two shifts cancelling. Finally, the contragredient of π is computed from theorem 1.3.12: $\check{\pi} = (\mathrm{Ind}_B^G \delta_B^{-1/2}\chi)^\vee \cong \mathrm{Ind}_B^G \delta_B^{-1}\delta_B^{1/2}\check{\chi} = \iota_B^G \check{\chi}$, so $L(\check{\pi}, 1-s) = L(\check{\chi}_1, 1-s)L(\check{\chi}_2, 1-s)$ by what has just been proved. Substituting the definitions of the two ε factors into

$$\varepsilon(\pi, s, \psi) = \gamma(\pi, s, \psi)\frac{L(\pi, s)}{L(\check{\pi}, 1-s)} = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi)\frac{L(\chi_1, s)L(\chi_2, s)}{L(\check{\chi}_1, 1-s)L(\check{\chi}_2, 1-s)}$$

the four L-factors cancel in pairs against those in $\gamma(\chi_i, s, \psi) = \varepsilon(\chi_i, s, \psi)L(\check{\chi}_i, 1-s)/L(\chi_i, s)$, leaving $\varepsilon(\pi, s, \psi) = \varepsilon(\chi_1, s, \psi)\varepsilon(\chi_2, s, \psi)$, as we sought to show.

The content of the proposition is that a principal series carries no information beyond its inducing characters. Both invariants are computed from χ_1 and χ_2 by the GL_1 recipe, and neither sees anything else about π. On this part of the spectrum a correspondence preserving L and ε is therefore forced before it is constructed: $\iota_B^G(\chi_1 \otimes \chi_2)$ can only be matched with an object whose invariants factor into the GL_1 invariants of χ_1 and χ_2 in the same way, so the matching is determined by the unordered pair $\{\chi_1, \chi_2\}$ and by nothing else. That the unordered pair is the right datum is already true on this side of the fence, since $\iota_B^G\chi \cong \iota_B^G\chi^w$ by theorem 2.4.16. The interesting behaviour is all in the reducible degenerations, which is where the remaining two families live.

Both remaining families sit inside a single reducible induced representation. Fix a character φ of $F^\times$ and recall the non-split exact sequence (2.10), in its twisted form

$$0 \to \varphi_G \to \mathrm{Ind}_B^G \varphi_T \to \varphi\mathrm{St}_G \to 0$$

where $\varphi_G = \varphi \circ \det$ and $\varphi_T = \varphi \otimes \varphi$. Written in the normalized language, $\mathrm{Ind}_B^G\varphi_T = \iota_B^G(\chi_1 \otimes \chi_2)$ with

$$\chi_1 = \varphi\|-\|^{-1/2}, \qquad \chi_2 = \varphi\|-\|^{1/2}$$

so $\chi_1\chi_2^{-1} = \|-\|^{-1}$ and the representation is reducible, exactly as lemma 2.4.15 says. Every coefficient of either constituent is a coefficient of the whole (lemma 5.2.5), so (5.15) applies to both and the γ-factor is the same for both. What is not the same is the L-factor, and the sub-representation is the one that keeps all of it.

Proposition 5.4.2: L-Function and Epsilon Factor of a One-Dimensional Representation

Let φ be a character of $F^\times$ and let $\pi = \varphi_G = \varphi\circ\det$. With $\chi_1 = \varphi\|-\|^{-1/2}$ and $\chi_2 = \varphi\|-\|^{1/2}$ as above,

$$L(\varphi_G, s) = L(\chi_1, s)L(\chi_2, s) = L\big(\varphi, s - \tfrac{1}{2}\big)L\big(\varphi, s + \tfrac{1}{2}\big)$$

and

$$\varepsilon(\varphi_G, s, \psi) = \varepsilon(\chi_1, s, \psi)\,\varepsilon(\chi_2, s, \psi)$$

That is, the one-dimensional constituent has exactly the invariants that the formula of proposition 5.4.1 would assign to the reducible $\iota_B^G(\chi_1 \otimes \chi_2)$ containing it.

***Proof*:**

Step 1: the coefficients and the zeta integral. The space $V = \mathbb{C}$ is one-dimensional and $\check{V} = \mathbb{C}$, so a matrix coefficient of φ_G is $g \mapsto \tilde{v}v\varphi(\det g)$: thus $C(\varphi_G) = \mathbb{C}\varphi_G$, and it is enough to compute $\zeta(\Phi, \varphi_G, s)$ for $\Phi \in C_c^\infty(A)$. By proposition 2.2.7 the Haar integral on G is $\int_G h(g)\,d^\times g = \int_A h(x)\|\det x\|^{-2}\,d\mu(x)$, an identity of Haar integrals which extends to any h for which either side converges absolutely. Hence, on the half plane where the integral converges,

$$\zeta(\Phi, \varphi_G, s) = \int_A \Phi(x)\,\varphi(\det x)\,\|\det x\|^{s-2}\,d\mu(x)$$

Step 2: the upper bound. By (2.10) the representation φ_G is a subrepresentation of $\mathrm{Ind}_B^G\varphi_T$, so $C(\varphi_G) \subseteq C(\mathrm{Ind}_B^G\varphi_T)$ by lemma 5.2.5 and Step 5 of the proof of proposition 5.4.1 bounds the unshifted integrals:

$$\zeta(\Phi, \varphi_G, s) \in L(\varphi, s-1)L(\varphi, s)\,\mathbb{C}[q^s, q^{-s}]$$

Step 3: the ramified case. Suppose φ is ramified of level n. Then $L(\varphi, s-1) = L(\varphi, s) = 1$ by lemma 5.1.15, so Step 2 says the module of unshifted integrals is contained in $\mathbb{C}[q^s, q^{-s}]$. It contains a unit: let Φ be the characteristic function of the compact open set $\big\{g \in K : \det g \in U_F^{n+1}\big\}$, which lies in $C_c^\infty(A)$ because G is open in A. On the support, $\varphi(\det g) = 1$ and $\|\det g\| = 1$, so $\zeta(\Phi, \varphi_G, s)$ is the positive constant $\int_{\mathrm{supp}\Phi} d^\times g$, independent of s. Hence the module is all of $\mathbb{C}[q^s, q^{-s}]$ and $L(\varphi_G, s) = 1 = L(\chi_1, s)L(\chi_2, s)$, the last equality because χ_1 and χ_2 are ramified whenever φ is.

Step 4: the unramified case. Suppose φ is unramified, and let Φ_0 be the characteristic function of $M_2(\mathfrak{o})$. For this computation rescale μ so that $\mu(M_2(\mathfrak{o})) = 1$; a rescaling multiplies every zeta integral by the same positive constant and so does not change the module they generate, which is all that is at stake here. Decompose a matrix $x \in M_2(\mathfrak{o})$ into its rows $r_1, r_2 \in \mathfrak{o}^2$; then μ is the product of the two Haar measures of total mass one on the two row spaces. Let $m = \min\{v_F(x_{11}), v_F(x_{12})\} \geq 0$ be the minimum valuation along the first row. The set of first rows with a given m is $\varpi^m\mathfrak{o}^2 \setminus \varpi^{m+1}\mathfrak{o}^2$, of measure $q^{-2m}(1 - q^{-2})$. For each such r_1 there is $k \in K$ with $r_1 k = (\varpi^m, 0)$; right translation by k preserves μ and multiplies det by the unit $\det k$, and it carries r_2 to a row that is again uniformly distributed over $\mathfrak{o}^2$. Writing $(c, d) = r_2 k$,

the determinant of xk is $\varpi^m d$. Therefore, using that φ is trivial on units,

$$\int_{M_2(\mathfrak{o})} \varphi(\det x)\|\det x\|^z \, d\mu(x) = (1-q^{-2}) \sum_{m\geq 0} \big(\varphi(\varpi) q^{-z-2}\big)^m \int_{\mathfrak{o}} \varphi(d)\|d\|^z \, d\mu_1(d)$$

where μ_1 is the measure of total mass one on $\mathfrak{o}$. Splitting $\mathfrak{o}$ into the shells $\varpi^j U_F$, of measure $q^{-j}(1-q^{-1})$, gives $\int_{\mathfrak{o}} \varphi(d)\|d\|^z d\mu_1(d) = (1-q^{-1}) \sum_{j\geq 0} (\varphi(\varpi) q^{-z-1})^j$. Both geometric series converge for $\mathrm{Re}(z)$ large, and summing them,

$$\int_{M_2(\mathfrak{o})} \varphi(\det x)\|\det x\|^z \, d\mu(x) = (1-q^{-1})(1-q^{-2})\, L(\varphi, z+1)\, L(\varphi, z+2)$$

Setting $z = s-2$ as in Step 1, $\zeta(\Phi_0, \varphi_G, s) = (1-q^{-1})(1-q^{-2}) L(\varphi, s-1) L(\varphi, s)$, a positive constant times the bound of Step 2. The module of unshifted integrals is therefore exactly $L(\varphi, s-1)L(\varphi, s)\mathbb{C}[q^s, q^{-s}]$.

Step 5: assembly. In both cases the generator of the unshifted module is $L(\varphi, s-1)L(\varphi, s)$, so replacing s by $s+\frac{1}{2}$,

$$L(\varphi_G, s) = L\big(\varphi, s-\tfrac{1}{2}\big) L\big(\varphi, s+\tfrac{1}{2}\big) = L(\chi_1, s) L(\chi_2, s)$$

By (5.15) applied to $\xi = \varphi_T$, whose components are both φ, and by uniqueness of γ,

$$\gamma(\varphi_G, s, \psi) = \gamma\big(\varphi, s-\tfrac{1}{2}, \psi\big)\gamma\big(\varphi, s+\tfrac{1}{2}, \psi\big) = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi)$$

Finally $\check{\varphi_G} = \check{\varphi} \circ \det$, so $L(\check{\varphi_G}, 1-s) = L(\check{\chi}_1, 1-s) L(\check{\chi}_2, 1-s)$ by what has just been proved applied to $\check{\varphi}$, and the computation ending the proof of proposition 5.4.1 goes through unchanged, giving $\varepsilon(\varphi_G, s, \psi) = \varepsilon(\chi_1, s, \psi)\varepsilon(\chi_2, s, \psi)$, completing the proof.

The zeta integral of the trivial coefficient is thus a GL_2 analogue of Tate's integral over $\mathfrak{o}$: the lattice $M_2(\mathfrak{o})$ replaces $\mathfrak{o}$, and the two elementary divisors of a matrix contribute one GL_1 factor each. The two factors sit at $s-\frac{1}{2}$ and $s+\frac{1}{2}$, one half-step on either side of the centre of symmetry, which is the pattern the $\delta_B^{\pm 1/2}$ twist was designed to produce. Nothing is lost on restricting from the induced representation to its one-dimensional subspace.

Passing to the quotient is a different matter. The quotient has no constant vectors to integrate, and the coefficient functionals that survive are exactly those annihilating the constants. That single linear condition is what kills one of the two poles.

Proposition 5.4.3: L-Function and Epsilon Factor of a Special Representation

Let φ be a character of $F^\times$, let $\pi = \varphi\mathrm{St}_G$, and keep $\chi_1 = \varphi\|-\|^{-1/2}$, $\chi_2 = \varphi\|-\|^{1/2}$. Then

$$L(\varphi\mathrm{St}_G, s) = L(\chi_2, s) = L\big(\varphi, s+\tfrac{1}{2}\big)$$

and

$$\varepsilon(\varphi\mathrm{St}_G, s, \psi) = \varepsilon(\chi_1, s, \psi)\,\varepsilon(\chi_2, s, \psi)\,\frac{L(\check{\chi}_2, 1-s)}{L(\chi_1, s)}$$

The correcting factor is 1 when φ is ramified, and equals $-\chi_1(\varpi)q^{-s}$ when φ is unramified.

***Proof*:**

Step 1: the γ-factor. By (2.10) the representation $\varphi\mathrm{St}_G$ is a quotient of $\Sigma = \mathrm{Ind}_B^G\varphi_T$, so $C(\varphi\mathrm{St}_G) \subseteq C(\Sigma)$ by lemma 5.2.5, and (5.15) with $\xi = \varphi_T$ together with the uniqueness of γ gives

$$\gamma(\varphi\mathrm{St}_G, s, \psi) = \gamma\big(\varphi, s-\tfrac{1}{2}, \psi\big)\gamma\big(\varphi, s+\tfrac{1}{2}, \psi\big) = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi)$$

the same value as for φ_G.

Step 2: the ramified case. If φ is ramified then so are χ_1 and χ_2, and Step 5 of the proof of proposition 5.4.1 puts the unshifted integrals inside $L(\varphi, s-1)L(\varphi, s)\mathbb{C}[q^s, q^{-s}] = \mathbb{C}[q^s, q^{-s}]$. Writing the generator of the unshifted module as $P(q^{-s})^{-1}$ with $P(0) = 1$ as in proposition 5.2.6, that generator lies in $\mathbb{C}[q^s, q^{-s}]$ and its inverse is a polynomial with constant term 1, which forces $P = 1$ and $L(\varphi\mathrm{St}_G, s) = 1 = L(\chi_2, s)$. The correcting factor is then the ratio of two trivial L-factors, namely 1.

Step 3: the unramified case, and where the pole goes. Let φ be unramified and write $\Sigma = \mathrm{Ind}_B^G\varphi_T$ as in Step 1 of the proof of proposition 5.4.1, with $\xi_1 = \xi_2 = \varphi$. In the variable $X = q^{-s}$ the two Tate denominators are $P_{\varphi\|-\|^{-1}}(X) = 1-\varphi(\varpi)qX$ and $P_\varphi(X) = 1-\varphi(\varpi)X$, with distinct roots

$$X_1 = \big(\varphi(\varpi)q\big)^{-1}, \qquad X_2 = \varphi(\varpi)^{-1}$$

The claim is that no element of the module attached to $\varphi\mathrm{St}_G$ has a pole at X_1.

Let $\Psi \in C_c^\infty(F\times F)$, regard $\Xi_{\varphi_T}(\Psi, s)$ as a rational function of X, and set $\rho(\Psi) = \lim_{X\to X_1}(1-\varphi(\varpi)qX)\Xi_{\varphi_T}(\Psi, s)$. On a pure tensor $\Psi_1\otimes\Psi_2$ the first factor is the Tate series $Z(\Psi_1, \varphi\|-\|^{-1}, X)$; since Ψ_1 is constant near 0, its coefficients satisfy $z_m = \Psi_1(0)\mu^*(U_F)(\varphi(\varpi)q)^m$ for $m \gg 0$, so the series differs from $\Psi_1(0)\mu^*(U_F)(\varphi(\varpi)qX)^{m_0}(1-\varphi(\varpi)qX)^{-1}$ by a Laurent polynomial, and the limit is $\Psi_1(0)\mu^*(U_F)$. The second factor $Z(\Psi_2, \varphi, X)$ has no pole at X_1, because $P_\varphi(X_1) = 1-q^{-1} \neq 0$, and its value there is $\int_{F^\times}\Psi_2(t)\varphi(t)\|t\|^{s_1}d^\times t$ where $q^{-s_1} = X_1$. For unramified φ and t of valuation m one has $\varphi(t)\|t\|^{s_1} = (\varphi(\varpi)X_1)^m = q^{-m} = \|t\|$, so this value is $\int_F \Psi_2(t)\,d\mu(t)$ by definition 5.1.5. Summing over a decomposition of Ψ into pure tensors and using $\sum_j \Psi_1^{(j)}(0)\Psi_2^{(j)} = \Psi(0, -)$,

$$\rho(\Psi) = \mu^*(U_F)\int_F \Psi(0, t)\,d\mu(t)$$

Now specialize to $\Psi = \Psi_{\Phi_{k,k'}}$. The matrices $\left(\begin{smallmatrix} 0 & y \\ 0 & t \end{smallmatrix}\right)$ occurring here are exactly the elements of $\mathfrak{a} = \{x \in A : xe_1 = 0\}$, the left ideal of matrices with vanishing first column, and $dy\,dt$ is a Haar measure on $\mathfrak{a} \cong F^2$. Since $\mathfrak{a}$ is a left ideal, $k^{-1}\mathfrak{a} = \mathfrak{a}$, and the induced map on $\mathfrak{a}$ is $x \mapsto k^{-1}x$, whose Jacobian is $\|\det k\|^{-1} = 1$ for $k \in K$. Hence

$$\rho\big(\Psi_{\Phi_{k,k'}}\big) = \mu^*(U_F)\int_{\mathfrak{a}} \Phi(k^{-1}vk')\,dv = \mu^*(U_F)\int_{\mathfrak{a}} \Phi(vk')\,dv$$

which does not depend on k. The two K-integrations in the display of Step 1 of the proof of proposition 5.4.1 have locally constant integrands on a compact group, hence are finite linear combinations, so ρ passes through them; and since the resulting integrand no longer depends on k, the double integral factors:

$$\lim_{X\to X_1}\big(1-\varphi(\varpi)qX\big)Z(\Phi,f,X) = c_0\mu^*(U_F)\left(\int_K \tilde{\varphi}(k)\,d\mu_K(k)\right)\left(\int_K \varphi(k')\int_{\mathfrak{a}} \Phi(vk')\,dv\,d\mu_K(k')\right)$$

The one-dimensional subspace of Σ is spanned by $\varphi_0(g) = \varphi(\det g)$, and $\varphi(\det k) = 1$ for $k \in K$ because φ is unramified, so $\langle\tilde{\varphi},\varphi_0\rangle = \int_K \tilde{\varphi}(k)\,d\mu_K(k)$. A coefficient of the quotient $\varphi\mathrm{St}_G$ is by definition built from a functional annihilating that subspace, that is from a $\tilde{\varphi}$ with $\int_K \tilde{\varphi} = 0$. Therefore the limit vanishes for every $f \in C(\varphi\mathrm{St}_G)$ and every Φ, and since sums and $\mathbb{C}[X,X^{-1}]$-multiples of functions regular at X_1 are regular at X_1, the generator of the module is regular there as well. Writing $L(\varphi\mathrm{St}_G, s-\frac{1}{2}) = P(X)^{-1}$ with $P(0) = 1$ and P dividing $P_{\varphi\|-\|^{-1}}P_\varphi$, the factor $1-\varphi(\varpi)qX$ cannot divide P, so

$$L(\varphi\mathrm{St}_G, s) = 1 \qquad \text{or} \qquad L(\varphi\mathrm{St}_G, s) = L(\chi_2, s)$$

the second alternative being $P = P_\varphi$ after the shift $s \mapsto s + \frac{1}{2}$.

Step 4: the first alternative is impossible. Two preliminaries about twisting. First, let η be an unramified character of $F^\times$ and let π' be an irreducible representation of G. The coefficients of $\eta\pi'$ are the functions $g \mapsto \eta(\det g)f(g)$ with $f \in C(\pi')$, and $\eta(\det g) = \eta(\varpi)^{v_F(\det g)}$ since η is unramified, so

$$Z(\Phi, \eta f, X) = Z\big(\Phi, f, \eta(\varpi)X\big)$$

The substitution $X \mapsto \eta(\varpi)X$ is a ring automorphism of $\mathbb{C}[X,X^{-1}]$ fixing $X = 0$, so it carries the module attached to π' onto the module attached to $\eta\pi'$, generators to generators, and respects the normalization of proposition 5.2.6: writing the two generators as $P_{\pi'}(X)^{-1}$ and $P_{\eta\pi'}(X)^{-1}$ with $P(0) = 1$, this says $P_{\eta\pi'}(X) = P_{\pi'}(\eta(\varpi)X)$. In particular $L(\eta\pi', s) = 1$ if and only if $L(\pi', s) = 1$. Second, twisting commutes with the contragredient, $(\eta\pi')^\vee = \check{\eta}\,(\pi')^\vee$ directly from the definition of the twist, and $\mathrm{St}_G^\vee \cong \mathrm{St}_G$ by (2.12), so $(\varphi\mathrm{St}_G)^\vee = \check{\varphi}\mathrm{St}_G$.

Now suppose $L(\varphi\mathrm{St}_G, s) = 1$. Since $\check{\varphi}\mathrm{St}_G = \eta\cdot(\varphi\mathrm{St}_G)$ for the unramified character $\eta = \varphi^{-2}$, the first preliminary gives $L(\check{\varphi}\mathrm{St}_G, s) = 1$, and the second turns this into $L((\varphi\mathrm{St}_G)^\vee, 1-s) = 1$. Step 1 then gives

$$\varepsilon(\varphi\mathrm{St}_G, s, \psi) = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi) = \varepsilon(\chi_1, s, \psi)\varepsilon(\chi_2, s, \psi)\,\frac{L(\check{\chi}_1, 1-s)L(\check{\chi}_2, 1-s)}{L(\chi_1, s)L(\chi_2, s)}$$

By corollary 5.2.9 the left-hand side is a monomial in q^{-s}, and the two GL_1 local constants are monomials by corollary 5.1.14; so the remaining quotient must be a monomial. Write $Y = q^{-s}$

and $a_i = \chi_i(\varpi)$, both non-zero. Then $L(\chi_i, s)^{-1} = 1 - a_i Y$ and $L(\check{\chi}_i, 1-s)^{-1} = 1 - a_i^{-1} q^{-1} Y^{-1}$, so the quotient is $N(Y)/D(Y^{-1})$ with

$$N(Y) = (1-a_1Y)(1-a_2Y) = a_1a_2\big(Y-a_1^{-1}\big)\big(Y-a_2^{-1}\big), \qquad D(Y^{-1}) = \big(1-a_1^{-1}q^{-1}Y^{-1}\big)\big(1-a_2^{-1}q^{-1}Y^{-1}\big)$$

The numerator is supported in degrees 0 to 2 in Y and the denominator in degrees -2 to 0, so if $N(Y) = cY^m D(Y^{-1})$ then $m = 2$ and $m - 2 = 0$, and

$$Y^2 D(Y^{-1}) = \big(Y - a_1^{-1}q^{-1}\big)\big(Y - a_2^{-1}q^{-1}\big)$$

Comparing roots forces the multisets $\{a_1^{-1}, a_2^{-1}\}$ and $\{a_1^{-1}q^{-1}, a_2^{-1}q^{-1}\}$ to agree. Matching a_1^{-1} with $a_1^{-1}q^{-1}$ gives $q = 1$, and matching a_1^{-1} with $a_2^{-1}q^{-1}$ and a_2^{-1} with $a_1^{-1}q^{-1}$ gives $a_1 = q^2 a_1$, again $q^2 = 1$. Both are false, so the first alternative cannot hold and $L(\varphi\mathrm{St}_G, s) = L(\chi_2, s)$.

Step 5: the local constant. With $L(\varphi\mathrm{St}_G, s) = L(\chi_2, s)$ the same computation applied to $\check{\varphi}$ gives $L((\varphi\mathrm{St}_G)^\vee, 1-s) = L(\check{\chi}_1, 1-s)$, because the character playing the role of χ_2 for $\check{\varphi}$ is $\check{\varphi}\| - \|^{1/2} = \check{\chi}_1$. Hence

$$\varepsilon(\varphi\mathrm{St}_G, s, \psi) = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi)\frac{L(\chi_2, s)}{L(\check{\chi}_1, 1-s)} = \varepsilon(\chi_1, s, \psi)\varepsilon(\chi_2, s, \psi)\frac{L(\check{\chi}_2, 1-s)}{L(\chi_1, s)}$$

after cancelling $L(\check{\chi}_1, 1-s)$ and $L(\chi_2, s)$. This formula also holds in the ramified case of Step 2, where every L-factor in sight is 1. Finally, when φ is unramified, put $W = \chi_1(\varpi)q^{-s}$; since $\check{\chi}_2(\varpi) = \chi_1(\varpi)^{-1}q$ one has $L(\check{\chi}_2, 1-s)^{-1} = 1 - W^{-1}$ and $L(\chi_1, s)^{-1} = 1 - W$, so the correcting factor is

$$\frac{1-W}{1-W^{-1}} = -W = -\chi_1(\varpi)q^{-s}$$

as we sought to show.

So the special representation keeps the factor $L(\chi_2, s)$ and loses $L(\chi_1, s)$: of the two poles carried by the reducible induced representation, exactly the one belonging to the character $\chi_1 = \varphi\| - \|^{-1/2}$ is annihilated by the condition $\int_K \tilde{\varphi} = 0$. The mechanism behind that condition is the one part of this section that is not bookkeeping: the residue at the lost pole is computed by integrating Φ over the matrices with vanishing first column, and that integral is blind to the left K-variable, so the residue functional factors through the pairing of $\tilde{\varphi}$ against the constant function. Killing the constants kills the residue.

Three formulas are now available for the same short exact sequence, and setting them side by side says something about which of the two invariants carries the content.

5.4.4 Remark: The Invariants Along the Steinberg Sequence Keep $\chi_1 = \varphi\|-\|^{-1/2}$, $\chi_2 = \varphi\|-\|^{1/2}$, so that $\mathrm{Ind}_B^G \varphi_T = \iota_B^G(\chi_1 \otimes \chi_2)$ and (2.10) reads $0 \to \varphi_G \to \iota_B^G(\chi_1 \otimes \chi_2) \to \varphi\mathrm{St}_G \to 0$. Proposition 5.4.2 and proposition 5.4.3 give

$$\gamma(\varphi_G, s, \psi) = \gamma(\varphi\mathrm{St}_G, s, \psi) = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi)$$

but

$$L(\varphi_G, s) = L(\chi_1, s)L(\chi_2, s), \qquad L(\varphi\mathrm{St}_G, s) = L(\chi_2, s)$$

The unnormalized invariant γ is therefore multiplicative along the sequence, in the sense that it is computed from the two GL_1 constituents of the inducing character and is the same for the sub and for the quotient. The L-factor is not: it is not determined by the inducing character alone, and the factor $L(\chi_1, s)$ is present for the subrepresentation and absent for the quotient.

Since $\varepsilon = \gamma \cdot L(\pi, s)/L(\check{\pi}, 1-s)$, the entire difference between the two local constants is the difference between the two L-factors:

$$\frac{\varepsilon(\varphi\mathrm{St}_G, s, \psi)}{\varepsilon(\varphi_G, s, \psi)} = \frac{L(\check{\chi}_2, 1-s)}{L(\chi_1, s)} = \begin{cases} -\chi_1(\varpi)q^{-s} & \text{if } \varphi \text{ is unramified,} \\ 1 & \text{if } \varphi \text{ is ramified} \end{cases}$$

Two consequences, in opposite directions. When φ is unramified and ψ has level one, the explicit values are $\varepsilon(\varphi_G, s, \psi) = \varphi(\varpi)^{-2}q^{2s-1}$ and $\varepsilon(\varphi\mathrm{St}_G, s, \psi) = -\varphi(\varpi)^{-1}q^{s-1/2}$, so the two constituents differ in the exponent of q as well as in the coefficient, and the symmetry of corollary 5.2.9 holds for the second with the two minus signs cancelling: $(-\varphi(\varpi)^{-1}q^{s-1/2})(-\varphi(\varpi)q^{1/2-s}) = 1 = \omega_\pi(-1)$. When φ is ramified, on the other hand, the two constituents have the *same* L-factor, namely 1, and the *same* local constant. A single pair (L, ε) therefore never separates the representations of G on its own.

What separates them is the family of all twists, under which the two invariants behave quite differently. The L-factor does move on the non-cuspidal side: if φ is ramified then $L(\varphi\mathrm{St}_G, s) = 1$, but twisting by $\check{\varphi}$ returns St_G, whose L-factor is $L(\|-\|^{1/2}, s) = (1-q^{-s-1/2})^{-1} \neq 1$. On the cuspidal side it does not move at all: it is 1 for every cuspidal representation by proposition 5.3.1, and a twist of a cuspidal is again cuspidal, since twisting does not change the support of a matrix coefficient (theorem 3.1.3). The family of twisted L-factors therefore detects cuspidality, which is the use section 5.5 makes of it, and says nothing beyond that. The local constant collapses nowhere: it is a monomial cq^{-fs} in which both the coefficient c and the exponent f move as the twist varies, as the formulas above already show on the non-cuspidal side. This is the asymmetry that section 5.5 exploits, and it is why the characterization proved there is by ε with L along for the ride, rather than the other way round.

Everything above becomes explicit when every character in sight is unramified: the poles of the L-function can be located exactly, and the local constant is a single monomial whose coefficient is the central character evaluated at a prime element.

Example 5.4.5: The Unramified Principal Series

Let χ_1, χ_2 be unramified characters of $F^\times$, put $a_i = \chi_i(\varpi) \in \mathbb{C}^\times$, and let $\psi \in \widehat{F}$ have level one. Set $\pi = \iota_B^G(\chi_1 \otimes \chi_2)$.

Irreducibility. By lemma 2.4.15 the representation π is reducible precisely when $\chi_1\chi_2^{-1} = \|-\|^{\pm 1}$. Both sides are unramified, so this is the numerical condition $a_1/a_2 = q^{\mp 1}$, and π is irreducible as soon as $a_1 \neq qa_2$ and $a_2 \neq qa_1$. Note that $a_1 = a_2$ is allowed: the case $\chi_1 = \chi_2 = \varphi$ gives $\chi_1\chi_2^{-1} = 1$, which is not $\|-\|^{\pm 1}$ because $\|\varpi\|^{\pm 1} = q^{\mp 1} \neq 1$.

The L-factor and its poles. By definition 5.1.9 and proposition 5.4.1,

$$L(\pi, s) = \frac{1}{(1 - a_1 q^{-s})(1 - a_2 q^{-s})}$$

whose poles are the s with $q^s = a_1$ or $q^s = a_2$, each family being a single point repeated with period $2\pi i/\log q$. Corollary 5.1.10 says the two families are either disjoint or identical, and identical exactly when $\chi_1 = \chi_2$. So $L(\pi, s)$ has two disjoint families of simple poles when $\chi_1 \neq \chi_2$, and one family of double poles when $\chi_1 = \chi_2$; the latter does occur for an irreducible π, by the previous paragraph.

The local constant. Proposition 5.1.17 gives $\varepsilon(\chi_i, s, \psi) = q^{s-1/2}a_i^{-1}$, so by proposition 5.4.1

$$\varepsilon(\pi, s, \psi) = q^{s-1/2}a_1^{-1} \cdot q^{s-1/2}a_2^{-1} = (a_1a_2)^{-1}q^{2s-1}$$

which is the monomial cq^{-fs} of corollary 5.2.9 with $c = (a_1a_2)^{-1}q^{-1}$ and $f = -2$. The central character of π is $\chi_1\chi_2$: for $z = z_0 I$ in the centre the representation acts by right translation, so $\varphi(gz) = \varphi(zg) = (\delta_B^{-1/2}\chi)(z)\varphi(g)$, and $\delta_B(z) = \|z_0/z_0\| = 1$ leaves $\chi_1(z_0)\chi_2(z_0)$. Hence $\omega_\pi(\varpi) = a_1a_2$ and $\varepsilon(\pi, s, \psi) = \omega_\pi(\varpi)^{-1}q^{2s-1}$.

Two checks. First, the symmetry of corollary 5.2.9. Since $\check{\pi} = \iota_B^G(\check{\chi}_1 \otimes \check{\chi}_2)$, the same formula gives $\varepsilon(\check{\pi}, 1-s, \psi) = a_1a_2\, q^{2(1-s)-1}$, and

$$\varepsilon(\pi, s, \psi)\varepsilon(\check{\pi}, 1-s, \psi) = (a_1a_2)^{-1}q^{2s-1} \cdot a_1a_2q^{1-2s} = 1$$

which is $\omega_\pi(-1)$, because $-1 \in U_F$ and ω_π is unramified. Second, the dependence on ψ. Replacing ψ by $\varpi\psi$, lemma 5.1.16 gives $\varepsilon(\chi_i, s, \varpi\psi) = \chi_i(\varpi)\|\varpi\|^{s-1/2}\varepsilon(\chi_i, s, \psi) = a_i q^{1/2-s} \cdot q^{s-1/2}a_i^{-1} = 1$, so $\varepsilon(\pi, s, \varpi\psi) = 1$. The monomial computed above is therefore entirely an artefact of the choice of additive character, and disappears for a suitable one.

A numerical instance. Take $\chi_1 = 1$ and $\chi_2 = \|-\|^{1/2}$, so $a_1 = 1$ and $a_2 = q^{-1/2}$. Then $a_1/a_2 = q^{1/2}$ is neither q nor q^{-1}, so π is irreducible, and

$$L(\pi, s) = \frac{1}{(1 - q^{-s})(1 - q^{-s-1/2})}, \qquad \varepsilon(\pi, s, \psi) = q^{2s-1/2}$$

with simple poles at $q^s = 1$ and at $q^s = q^{-1/2}$.

Exercise 5.4.1

1. Let $\chi = \chi_1 \otimes \chi_2$ be a character of T with $\pi = \iota_B^G\chi$ irreducible, and let η be a character of $F^\times$. Show that $\eta\pi \cong \iota_B^G(\eta\chi_1 \otimes \eta\chi_2)$ and that this representation is again irreducible, and write

down $L(\eta\pi, s)$ and $\varepsilon(\eta\pi, s, \psi)$. Evaluate both when η, χ_1, χ_2 are unramified and ψ has level one.

2. Let π be any irreducible non-cuspidal representation of G and let $a \in F^\times$. Show that
$$\varepsilon(\pi, s, a\psi) = \omega_\pi(a)\,\|a\|^{2s-1}\,\varepsilon(\pi, s, \psi)$$
Treat the three families of theorem 2.4.16 separately, and identify ω_π in each case.

3. Let $\pi = \iota_B^G(\chi_1 \otimes \chi_2)$ be irreducible with χ_1, χ_2 unramified. Show that $L(\pi, s)$ has a double pole at each of its poles if and only if $\chi_1 = \chi_2$, and that this case is not vacuous. In that case, writing $\chi_1 = \chi_2 = \varphi$, compare the three L-factors $L(\pi, s)$, $L(\varphi_G, s)$ and $L(\varphi\mathrm{St}_G, s)$ and show that no two of them agree.

4. Determine exactly which irreducible non-cuspidal representations π of G satisfy $L(\pi, s) = 1$. Combining the answer with proposition 5.3.1, explain why the L-function alone cannot decide whether an irreducible representation is cuspidal.

5. Verify the identity $\varepsilon(\pi, s, \psi)\varepsilon(\check{\pi}, 1-s, \psi) = \omega_\pi(-1)$ of corollary 5.2.9 directly for $\pi = \varphi\mathrm{St}_G$, for an arbitrary character φ, starting from the formula of proposition 5.4.3 and using only corollary 5.1.14.

5.5 The Converse Theorem for GL_2

Everything the argument needs is now on the table: section 5.3 and section 5.4 compute $L(\pi, s)$ for every irreducible smooth representation of G, and $\varepsilon(\pi, s, \psi)$ as far as elementary means reach. This section shows that those invariants, read across all twists $\chi\pi$ rather than at π alone, determine π up to isomorphism, and records the consequence that at most one map from Galois-side parameters to representations of G can preserve them.

The first thing to ask of the invariants is that they see the division the classification is built on. Nothing in the definition of the zeta integral mentions cuspidality, yet the two families feed into it in completely different ways. The coefficients of a cuspidal representation vanish outside a set compact modulo the centre (theorem 3.1.3), so the only unbounded direction is the central one; a non-cuspidal representation is assembled from characters of the torus (theorem 2.4.16), and its zeta integral inherits their poles. The L-factor records exactly that difference, and it records it uniformly in the twist.

Lemma 5.5.1: L-functions Detect Cuspidality

Let (π, V) be an irreducible smooth representation of G. Then
$$L(\chi\pi, s) = 1 \quad \text{for every character } \chi \text{ of } F^\times$$
if and only if π is cuspidal.

***Proof*:**
Step 1: twisting preserves irreducibility, cuspidality and the three non-cuspidal families. Let χ be a character of $F^\times$ and let $\chi\pi$ denote the twist of π by χ, acting on the same space V by $\chi\pi(g) = \chi(\det g)\pi(g)$. For each $g \in G$ the operators $\chi\pi(g)$ and $\pi(g)$ differ by the non-zero scalar

$\chi(\det g)$, so a subspace of V is stable under π exactly when it is stable under $\chi\pi$, and a vector has open stabilizer for one exactly when it does for the other. The two representations therefore have the same subrepresentations, and $\chi\pi$ is smooth and irreducible.

For $n \in N$ one has $\det n = 1$, so $\chi\pi(n) = \pi(n)$: the restrictions of π and $\chi\pi$ to N are equal as representations of N. The subspace $V(N)$ spanned by the vectors $v - \pi(n)v$, $v \in V$, $n \in N$, therefore does not change, and neither does the Jacquet module $V_N = V/V(N)$. Cuspidality is the vanishing of V_N, so π is cuspidal if and only if $\chi\pi$ is. Twisting can move a cuspidal representation around inside the cuspidal family, but never out of it.

The same computation controls the induced representations. If ξ is a character of T, inflated to B, then the map $\Lambda\colon f \mapsto \big(g \mapsto \chi(\det g) f(g)\big)$ sends $\mathrm{Ind}_B^G \xi$ into $\mathrm{Ind}_B^G(\chi \cdot \xi)$, because $\Lambda(f)(bg) = \chi(\det b)\xi(b)\,\Lambda(f)(g)$. Both target actions are by right translation, so for $h \in G$

$$\Lambda\big(\chi \mathrm{Ind}_B^G \xi(h) f\big)(g) = \chi(\det g)\chi(\det h) f(gh) = \Lambda(f)(gh) = \big(\mathrm{Ind}_B^G(\chi \cdot \xi)(h)\,\Lambda f\big)(g)$$

Its inverse is multiplication by $\chi^{-1} \circ \det$, so $\chi\,\mathrm{Ind}_B^G \xi \cong \mathrm{Ind}_B^G(\chi \cdot \xi)$. Since $\chi \cdot (\delta_B^{-1/2} \otimes \sigma) = \delta_B^{-1/2} \otimes (\chi \cdot \sigma)$, the same holds for normalized induction (2.13):

$$\chi\, \iota_B^G(\mu_1 \otimes \mu_2) \cong \iota_B^G(\chi\mu_1 \otimes \chi\mu_2) \tag{5.16}$$

Directly from the same description one also has $\chi(\varphi \circ \det) = (\chi\varphi) \circ \det$, and twisting the exact sequence (2.10) attached to φ by χ gives $\chi(\varphi \cdot \mathrm{St}_G) = (\chi\varphi) \cdot \mathrm{St}_G$. So each of the three families of theorem 2.4.16 is stable under twisting, with the parameter μ_i or φ multiplied by χ.

Step 2: a cuspidal representation has trivial twisted L-factors. If π is cuspidal then $\chi\pi$ is irreducible and cuspidal for every χ, by Step 1. Applying proposition 5.3.1 to $\chi\pi$ gives $L(\chi\pi, s) = 1$ for every χ.

Step 3: a non-cuspidal representation has a twist with a pole. Suppose π is not cuspidal. By theorem 2.4.16 there are three possibilities, and in each one a single twist suffices.

If $\pi \cong \iota_B^G(\mu_1 \otimes \mu_2)$ is an irreducible normalized induction, take $\chi = \mu_1^{-1}\| - \|^{1/2}$. By (5.16) and proposition 5.4.1,

$$L(\chi\pi, s) = L(1, s)\, L(\mu_1^{-1}\mu_2, s)$$

and the first factor is $(1 - q^{-s})^{-1}$ by definition 5.1.9, so $L(\chi\pi, s) \neq 1$.

If $\pi \cong \varphi \circ \det$, take $\chi = \varphi^{-1}$, so that $\chi\pi$ is the trivial representation 1_G. By proposition 5.4.2,

$$L(1_G, s) = L(1, s)\, L(\| - \|^{-1}, s) = \frac{1}{(1 - q^{-s})(1 - q^{1-s})}$$

which is not 1. If $\pi \cong \varphi \cdot \mathrm{St}_G$, take $\chi = \varphi^{-1}$ again; then $\chi\pi \cong \mathrm{St}_G$ and proposition 5.4.3 gives $L(\mathrm{St}_G, s) = L(1, s) = (1 - q^{-s})^{-1} \neq 1$.

In every case some twist has a non-trivial L-factor. The contrapositive of Step 3 together with Step 2 gives the stated equivalence, as we sought to show.

The lemma is the first statement in which the invariants *characterize* rather than accompany. What it detects is intrinsic: by theorem 3.1.3 cuspidality is a property of the matrix coefficients themselves, and the L-factor is built out of those coefficients. The quantifier over all χ is doing real work, and cannot be dropped. A single L-factor sees only the unramified part of the data: if θ is a character with both θ and θ^2 ramified, then $\iota_B^G(\theta \otimes \check{\theta})$ is irreducible by lemma 2.4.15 and non-cuspidal, yet its L-factor is $L(\theta\|-\|^{-1/2}, s)L(\check{\theta}\|-\|^{-1/2}, s) = 1$ by proposition 5.4.1 and lemma 5.1.15, exactly as for a cuspidal representation. Twisting by $\check{\theta}\|-\|^{1/2}$ is what pushes one of the two characters governing the L-factor to an unramified one, where a pole appears; the computation is carried out in example 5.5.4.

With the two families separated, what remains is to show that inside each family the invariants pin down the representation. On the non-cuspidal side the classification gives the parameters and section 5.4 gives the dictionary between parameters and L-factors, so the problem becomes one of reading a multiset of characters off a family of rational functions. The three formulas of section 5.4 have a common shape: to each irreducible non-cuspidal π there is attached a multiset $\mathcal{S}(\pi)$ of characters of $F^\times$ with

$$L(\pi, s) = \prod_{\mu \in \mathcal{S}(\pi)} L(\mu, s) \tag{5.17}$$

namely

$$\mathcal{S}\big(\iota_B^G(\mu_1 \otimes \mu_2)\big) = \{\mu_1\|-\|^{-1/2},\, \mu_2\|-\|^{-1/2}\}, \qquad \mathcal{S}(\varphi \circ \det) = \{\varphi,\, \varphi\|-\|^{-1}\}, \qquad \mathcal{S}(\varphi \cdot \mathrm{St}_G) = \{\varphi\}$$

by proposition 5.4.1, proposition 5.4.2 and proposition 5.4.3 respectively. The half-integral powers record the normalization: $L(\pi, s)$ is the generator attached in proposition 5.2.6 to the unshifted integral $\zeta(\Phi, f, s)$, and the denominator $P_{\chi_1\|-\|^{-1}}P_{\chi_2}$ produced there for a subquotient of $\mathrm{Ind}_B^G(\chi_1 \otimes \chi_2)$ is exactly what (5.17) sharpens. The first multiset is well defined as an *unordered* pair because $\iota_B^G(\mu_1 \otimes \mu_2) \cong \iota_B^G(\mu_2 \otimes \mu_1)$ and no other coincidences occur (lemma 2.4.15, theorem 2.4.16). By Step 1 of the proof above, $\mathcal{S}(\chi\pi) = \chi\mathcal{S}(\pi) = \{\chi\mu \big| \mu \in \mathcal{S}(\pi)\}$ for every character χ.

Theorem 5.5.2: Converse Theorem for GL_2

Let π and π' be irreducible smooth representations of G with the same central character, $\omega_\pi = \omega_{\pi'}$, and let $\psi \in \widehat{F}$, the group of characters of the additive group of F, have level one. Suppose that

$$L(\chi\pi, s) = L(\chi\pi', s) \qquad \text{and} \qquad \varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi\pi', s, \psi)$$

for every character χ of $F^\times$. Then $\pi \cong \pi'$.

Fixing ψ of level one is harmless. It is the normalization under which the explicit local constants are stated, and the exercises below show that the hypothesis holds for one non-trivial additive character exactly when it holds for every other.

Steps 1 and 2 of the argument below are complete proofs. Step 3 reduces the cuspidal case to a single implication, and that implication is proved in this book, as corollary 8.2.5. It rests on the Kirillov model, which section 8.1 constructs, so the reference points forward. The theorem is stated here because chapter 7 needs it, and its cuspidal half waits on a construction made later. The proof of corollary 8.2.5 does not use the present theorem, so there is no circularity in the arrangement.

***Proof*:**
Step 1: the two representations lie in the same family. The hypothesis includes $L(\chi\pi, s) = L(\chi\pi', s)$ for every χ, so the condition of lemma 5.5.1 holds for π exactly when it holds for π'. Hence π and π' are either both cuspidal or both non-cuspidal.

Step 2: the non-cuspidal case. Assume both are non-cuspidal, and let $\mathcal{U}$ denote the group of unramified characters of $F^\times$. The claim is that the multiset $\mathcal{S}(\pi)$ is determined by the family $\{L(\chi\pi, s)\}_\chi$.

Write $L(\pi, s) = P_\pi(q^{-s})^{-1}$ with $P_\pi \in \mathbb{C}[X]$ and $P_\pi(0) = 1$; this normalization determines P_π uniquely by proposition 5.2.6, so two irreducible representations have equal L-factors precisely when they have equal polynomials P. Combining (5.17) with $\mathcal{S}(\chi\pi) = \chi\mathcal{S}(\pi)$ and with the two-case formula of definition 5.1.9 and lemma 5.1.15,

$$P_{\chi\pi}(X) = \prod_{\substack{\mu\in\mathcal{S}(\pi)\\ \chi\mu \text{ unramified}}} \big(1 - \chi\mu(\varpi)X\big) \tag{5.18}$$

the product being taken with multiplicity. A character μ satisfies "$\chi\mu$ unramified" exactly when χ lies in the coset $\mu^{-1}\mathcal{U}$, so the product in (5.18) runs over the elements of $\mathcal{S}(\pi)$ lying in $\chi^{-1}\mathcal{U}$.

Fix a coset $\mathcal{C}$ of $\mathcal{U}$ in the character group of $F^\times$ and choose χ with $\chi^{-1}\mathcal{U} = \mathcal{C}$. The polynomial $P_{\chi\pi}$ then has degree equal to the number of elements of $\mathcal{S}(\pi)$ lying in $\mathcal{C}$, counted with multiplicity, and its roots are the numbers $\chi\mu(\varpi)^{-1}$ for those μ. An unramified character is trivial on U_F and $F^\times$ is generated by ϖ together with U_F, so such a character is determined by its value at ϖ; equivalently, distinct unramified characters have disjoint pole sets (corollary 5.1.10). The multiset $\{\chi\mu \big| \mu \in \mathcal{S}(\pi)\cap\mathcal{C}\}$ is therefore read off from $P_{\chi\pi}$, and multiplying back by χ^{-1} recovers $\mathcal{S}(\pi)\cap\mathcal{C}$, again with multiplicity. Every element of $\mathcal{S}(\pi)$ lies in exactly one coset, so running the recovery over all cosets produces the whole of $\mathcal{S}(\pi)$. The recovery uses nothing but the family $\{L(\chi\pi, s)\}_\chi$, and it is carried out by the same choices of χ for π and for π'; the hypothesis therefore gives

$$\mathcal{S}(\pi) = \mathcal{S}(\pi')$$

It remains to see that $\mathcal{S}$ separates the three families and determines the representation inside each. If $\mathcal{S}(\pi)$ has one element, then π is neither of the first two types, since those have two; so $\pi \cong \varphi \cdot \mathrm{St}_G$ with $\mathcal{S}(\pi) = \{\varphi\}$, and φ is the unique element of $\mathcal{S}(\pi)$. The same applies to π', and $\pi \cong \varphi \cdot \mathrm{St}_G \cong \pi'$.

If $\mathcal{S}(\pi) = \{\nu_1, \nu_2\}$ has two elements, the ratio $\nu_1\nu_2^{-1}$ decides between the two remaining types. For $\pi \cong \varphi \circ \det$ the ratio is $\|-\|^{\pm 1}$. For $\pi \cong \iota_B^G(\mu_1 \otimes \mu_2)$ irreducible the ratio is $\mu_1\mu_2^{-1}$, and lemma 2.4.15 forbids exactly that value: the normalized induction $\iota_B^G(\mu_1 \otimes \mu_2)$ is reducible if and only if $\mu_1\mu_2^{-1} = \|-\|^{\pm 1}$. So the ratio, which is a function of the multiset alone, tells the two types apart. In the first case, relabelling so that $\nu_1\nu_2^{-1} = \|-\|$ gives $\varphi = \nu_1$ and $\pi \cong \varphi \circ \det \cong \pi'$. In the second, $\mu_i = \nu_i\|-\|^{1/2}$ recovers the inducing characters and $\pi \cong \iota_B^G(\mu_1 \otimes \mu_2) \cong \pi'$, the order of the two being immaterial because $\iota_B^G(\mu_1 \otimes \mu_2) \cong \iota_B^G(\mu_2 \otimes \mu_1)$ (lemma 2.4.15). This settles the non-cuspidal case; note that neither the ε-factors nor the equality of central characters was used.

Step 3: the cuspidal case. Assume both are cuspidal. Then $L(\chi\pi, s) = L(\chi\pi', s) = 1$ for every χ by lemma 5.5.1, so the L-hypothesis is vacuous and all the information sits in the local constants. By corollary 5.3.2 the hypothesis is equivalent to $\gamma(\chi\pi, s, \psi) = \gamma(\chi\pi', s, \psi)$ for every χ.

One consequence is immediate. Each $\varepsilon(\chi\pi, s, \psi)$ is a monomial $c\,q^{-ms}$ with $c \in \mathbb{C}^\times$ and $m \in \mathbb{Z}$ by corollary 5.2.9, and that exponent is what definition 5.3.4 records. The hypothesis therefore forces the epsilon exponents of $\chi\pi$ and $\chi\pi'$ to agree for every χ, together with the leading constants.

What remains is the implication that the twisted local constants determine a cuspidal representation, and that is corollary 8.2.5. Its hypotheses are the ones assumed here: π and π' irreducible cuspidal, $\omega_\pi = \omega_{\pi'}$, the character ψ of level one, and $\varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi\pi', s, \psi)$ for every character χ of $F^\times$. Its proof realizes π on the mirabolic subgroup in its Kirillov model, a space of functions on $F^\times$ built in section 8.1. On that model the actions of Z and of the mirabolic subgroup are already fixed by the central character, so the only datum left is the single operator $\pi_K(w)$, and theorem 8.2.4 evaluates it in terms of the twisted local constants. So $\pi \cong \pi'$ in the cuspidal case as well, completing the proof.

The theorem is an injectivity statement and nothing more. It says that the assignment

$$\pi \longmapsto \Big(\omega_\pi,\ \big\{L(\chi\pi, s)\big\}_\chi,\ \big\{\varepsilon(\chi\pi, s, \psi)\big\}_\chi\Big)$$

is injective on isomorphism classes of irreducible smooth representations of G. It says nothing about which families of rational functions arise: there is no claim here that a prescribed system of L- and ε-factors is realized by some representation. Existence is a separate problem, solved by construction rather than by comparison. The proof also distributes the work very unevenly. In the non-cuspidal case the L-factors alone suffice and the local constants are never consulted; in the cuspidal case the L-factors are all equal to 1 and the local constants carry the entire burden.

The role of the quantifier over χ deserves to be stated plainly. Matching invariants for the trivial twist alone is far too weak. The L-factor of a principal series induced from two ramified characters equals 1, exactly as for a cuspidal representation, so the untwisted L-factor cannot even see the division of lemma 5.5.1; and the untwisted ε-factor is a single monomial, which is very little data with which to distinguish an infinite family. The mechanism by which twisting extracts more is visible in (5.18): the L-factor is blind to ramified characters, and a twist by μ^{-1} moves a ramified μ to the trivial character, where the L-factor acquires a pole. Running over all χ interrogates every character of $F^\times$ in turn. Example 5.5.4 exhibits the failure of the untwisted invariants: it produces two non-isomorphic irreducible representations agreeing in central character, in L-factor and in ε-factor.

There is a bound in the other direction too, and it comes from the stability theorem. By theorem 5.3.10 there is an integer n_π such that for every χ of level at least n_π the constant $\varepsilon(\chi\pi, s, \psi)$ depends only on χ, on ψ and on the central character ω_π, and likewise for π' with some $n_{\pi'}$. Since $\omega_\pi = \omega_{\pi'}$, the two sides of the hypothesis agree automatically for every χ of level at least $\max(n_\pi, n_{\pi'})$. The content of the hypothesis is therefore concentrated in the twists of bounded level, of which there are finitely many up to unramified twist. The proof above does not use this, since corollary 8.2.5 reaches the cuspidal case without a bound on the level, but it says what the theorem is comparing.

The immediate payoff is a uniqueness statement about correspondences, and it is purely formal: once

the theorem is in hand, nothing about the Galois side is needed to conclude that a map preserving these invariants is unique if it exists at all.

Corollary 5.5.3: Uniqueness of an Invariant-Preserving Correspondence

Let $\psi \in \widehat{F}$ have level one. Let $\mathcal{P}$ be a set, and suppose that each $\rho \in \mathcal{P}$ is equipped with a character ω_ρ of $F^\times$ and, for each character χ of $F^\times$, with functions $L(\chi\otimes\rho, s)$ and $\varepsilon(\chi\otimes\rho, s, \psi)$. Let

$$\Pi, \Pi' \colon \mathcal{P} \longrightarrow \{\text{isomorphism classes of irreducible smooth representations of } G\}$$

be two maps such that, for every $\rho \in \mathcal{P}$ and every character χ of $F^\times$,

$$\omega_{\Pi(\rho)} = \omega_\rho, \qquad L(\chi\,\Pi(\rho), s) = L(\chi \otimes \rho, s), \qquad \varepsilon(\chi\,\Pi(\rho), s, \psi) = \varepsilon(\chi \otimes \rho, s, \psi)$$

and such that Π' satisfies the same three conditions. Then $\Pi = \Pi'$.

***Proof*:**
Fix $\rho \in \mathcal{P}$ and put $\pi = \Pi(\rho)$, $\pi' = \Pi'(\rho)$. Both are irreducible smooth representations of G, and the first condition applied to each map gives

$$\omega_\pi = \omega_\rho = \omega_{\pi'}$$

For every character χ of $F^\times$ the second condition gives $L(\chi\pi, s) = L(\chi \otimes \rho, s) = L(\chi\pi', s)$, and the third gives $\varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi \otimes \rho, s, \psi) = \varepsilon(\chi\pi', s, \psi)$. The hypotheses of theorem 5.5.2 are met, so $\pi \cong \pi'$. Since ρ was arbitrary and both maps take values in isomorphism classes, $\Pi = \Pi'$, completing the proof.

The set $\mathcal{P}$ is left abstract on purpose: the argument uses only that its elements come with a character of $F^\times$ and a system of twisted invariants, never what they are. Later, in chapter 6, that data is constructed for two-dimensional Deligne representations of the Weil group, and chapter 7 applies the corollary with $\mathcal{P}$ taken to be that set of parameters, which is where the word "the" in "the local Langlands correspondence" comes from. One point of bookkeeping is needed there. The invariants of this chapter are attached to the unshifted integral $\zeta(\Phi, f, s)$ of proposition 5.2.6, whereas the factors usually matched against Galois-side parameters are the shifted ones $L(\pi, s + \frac{1}{2})$ and $\varepsilon(\pi, s + \frac{1}{2}, \psi)$; the two conventions must not be mixed. Nothing in the corollary depends on which is chosen, since it compares two maps into the same set of representations and a shift applied on both sides at once changes neither its hypotheses nor its conclusion. What the corollary does not do is produce the map. Existence requires building, for each parameter, a representation of G with the prescribed invariants, and that construction proceeds family by family: characters of the torus for the reducible parameters, admissible pairs and the cuspidal types of theorem 3.5.5 for the irreducible ones. The corollary guarantees that there is at most one answer, not that there is one.

The promised example shows the necessity of the quantifier. It exhibits two non-isomorphic irreducible representations whose untwisted invariants agree in every respect, and locates the ramified twist at which they part.

Example 5.5.4: Two Principal Series with Equal Untwisted Invariants

Assume $q > 3$, fix $\psi \in \widehat{F}$ of level one, and let η be the unramified character $\eta(x) = (-1)^{v_F(x)}$ of $F^\times$, so that $\eta^2 = 1$, $\eta \neq 1$ and $\eta(\varpi) = -1$.

Choose a character θ of $F^\times$ of level $n \geq 0$ such that both θ and θ^2 are ramified. Such a θ exists: take θ trivial on U_F^1 with $\theta|_{U_F}$ of order at least 3, which is possible because $U_F/U_F^1 \cong k^\times$ is cyclic of order $q-1 \geq 3$; then $\theta^2|_{U_F} \neq 1$. Put

$$\pi = \iota_B^G(\theta \otimes \check{\theta}), \qquad \pi' = \eta\pi \cong \iota_B^G(\eta\theta \otimes \eta\check{\theta})$$

the second isomorphism being (5.16).

Both are irreducible: the relevant ratio is $\theta\check{\theta}^{-1} = \theta^2$ for π and $(\eta\theta)(\eta\check{\theta})^{-1} = \theta^2$ for π', and θ^2 is ramified, hence not equal to $\|-\|^{\pm 1}$, so lemma 2.4.15 applies. They have the same central character: $\delta_B(\mathrm{diag}(a,a)) = \|a/a\| = 1$ by proposition 2.2.8, so the normalizing factor $\delta_B^{-1/2}$ contributes nothing on Z and the central character is the inducing character of T evaluated at $\mathrm{diag}(a,a)$. This gives $\omega_\pi = \theta\check{\theta} = 1$ and $\omega_{\pi'} = \eta^2\theta\check{\theta} = 1$.

They are not isomorphic. By theorem 2.4.16 an isomorphism would force $\{\eta\theta, \eta\check{\theta}\} = \{\theta, \check{\theta}\}$. If $\eta\theta = \theta$ then $\eta = 1$, which is false; if $\eta\theta = \check{\theta}$ then $\theta^2 = \eta^{-1} = \eta$, which is false because θ^2 is ramified and η is not.

Their L-factors agree. Write $\nu_1 = \theta\|-\|^{-1/2}$ and $\nu_2 = \check{\theta}\|-\|^{-1/2}$, so that $\mathcal{S}(\pi) = \{\nu_1, \nu_2\}$ and $\mathcal{S}(\pi') = \{\eta\nu_1, \eta\nu_2\}$ by proposition 5.4.1. All four of these restrict on U_F to $\theta|_{U_F}^{\pm 1}$, because $\|-\|^{-1/2}$ and η are unramified, hence are ramified of level n; so lemma 5.1.15 and (5.17) give

$$L(\pi, s) = L(\nu_1, s)L(\nu_2, s) = 1 = L(\eta\nu_1, s)L(\eta\nu_2, s) = L(\pi', s)$$

Their ε-factors agree as well. Let μ be any ramified character of level n and let $c \in F^\times$ with $v_F(c) = -n$. Since η is trivial on U_F, the value $\check{\eta}(cx) = \eta(\varpi)^n$ is constant as x runs over U_F/U_F^{n+1}, so the Gauss sum of definition 5.1.19 satisfies

$$\tau(\eta\mu, \psi) = \sum_{x \in U_F/U_F^{n+1}} \check{\eta}(cx)\check{\mu}(cx)\psi(cx) = \eta(\varpi)^n\, \tau(\mu, \psi)$$

The character $\eta\mu$ again has level n, so theorem 5.1.18 turns this into $\varepsilon(\eta\mu, s, \psi) = \eta(\varpi)^n\varepsilon(\mu, s, \psi)$. Applying this to $\mu = \nu_1$ and to $\mu = \nu_2$ and multiplying, proposition 5.4.1 gives

$$\varepsilon(\pi', s, \psi) = \varepsilon(\eta\nu_1, s, \psi)\varepsilon(\eta\nu_2, s, \psi) = \eta(\varpi)^{2n}\varepsilon(\nu_1, s, \psi)\varepsilon(\nu_2, s, \psi) = \varepsilon(\pi, s, \psi)$$

because $\eta(\varpi)^2 = 1$.

So π and π' are non-isomorphic irreducible representations with the same central character, the same L-factor and the same ε-factor. The twist that separates them is $\chi = \nu_1^{-1} = \check{\theta}\|-\|^{1/2}$, which is ramified because $\check{\theta}$ is and $\|-\|^{1/2}$ is not. Indeed (5.16) gives $\chi\pi \cong \iota_B^G(\|-\|^{1/2} \otimes \check{\theta}^2\|-\|^{1/2})$ and $\chi\pi' \cong \iota_B^G(\eta\|-\|^{1/2} \otimes \eta\check{\theta}^2\|-\|^{1/2})$, so $\mathcal{S}(\chi\pi) = \{1, \check{\theta}^2\}$ and $\mathcal{S}(\chi\pi') = \{\eta, \eta\check{\theta}^2\}$; both are

irreducible by the ratio test above, and since θ^2 is ramified while 1 and η are not, (5.17) together with definition 5.1.9 and lemma 5.1.15 gives

$$L(\chi\pi, s) = L(1, s)L(\check{\theta}^2, s) = \frac{1}{1 - q^{-s}}, \qquad L(\chi\pi', s) = L(\eta, s)L(\eta\check{\theta}^2, s) = \frac{1}{1 - \eta(\varpi)q^{-s}} = \frac{1}{1 + q^{-s}}$$

These are different, so the hypothesis of theorem 5.5.2 fails at $\chi = \check{\theta}\|-\|^{1/2}$, exactly as it must.

Exercise 5.5.1

1. Step 1 of the proof of lemma 5.5.1 shows that twisting preserves cuspidality by comparing Jacquet modules. Give the argument instead through matrix coefficients: show that $C(\chi\pi) = \{(\chi \circ \det)f \,\big|\, f \in C(\pi)\}$ for any smooth representation π of G and any character χ of $F^\times$, and deduce from definition 3.1.1 and theorem 3.1.3 that, for irreducible π, the representation π is cuspidal if and only if $\chi\pi$ is.

2. Let φ, φ' be characters of $F^\times$, let $\pi = \varphi \circ \det$ and let $\pi' = \varphi' \cdot \mathrm{St}_G$. Show that $L(\chi\pi, s) = L(\chi\pi', s)$ cannot hold for every character χ of $F^\times$. Show also that if $\varphi = \varphi'$ is ramified then the untwisted L-factors do agree, so that a twist is needed to separate these two families.

3. Let μ_1 be an unramified and μ_2 a ramified character of $F^\times$, and put $\pi = \iota_B^G(\mu_1 \otimes \mu_2)$.

    1. Show that π is irreducible.
    2. Determine the set of characters χ for which $L(\chi\pi, s) \neq 1$, and show it is a disjoint union of two cosets of the group $\mathcal{U}$ of unramified characters.
    3. Recover μ_1 and μ_2 explicitly from the family $\{L(\chi\pi, s)\}_\chi$.

4. In the situation of example 5.5.4, show that $L(\chi\pi, s) = L(\chi\pi', s)$ and $\varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi\pi', s, \psi)$ for *every* unramified character χ of $F^\times$. Conclude that no unramified twist detects the difference between π and π', so that the separating twist found in the example had to be ramified.

5. The statement of theorem 5.5.2 fixes a $\psi \in \widehat{F}$ of level one. Show that the hypothesis does not in fact depend on that choice: if π, π' are irreducible with $\omega_\pi = \omega_{\pi'}$ and $\varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi\pi', s, \psi)$ for every χ, then the same equalities hold with ψ replaced by any non-trivial $\psi' \in \widehat{F}$, and conversely. Deduce that the theorem applies as soon as its hypothesis holds for one non-trivial additive character. Recall from section 5.1 that every such ψ' has the form $a\psi$ for a unique $a \in F^\times$, and use lemma 5.3.3.

6

Weil Groups and Their Representations

We now cross the bridge to the other side of the correspondence: the Galois side. The absolute Galois group $\mathrm{Gal}(\overline{F}/F)$ is the key object of study, but its profinite topology is too coarse to see the complex representations that match the smooth representations of $\mathrm{GL}_2(F)$. The goal of this chapter is to refine the Galois group into the *Weil Group* $\mathcal{W}_F$ and to develop its representation theory to a point where it matches the complexity of the $\mathrm{GL}_2(F)$ side.

We begin by defining the Weil group and its topology. We then review Local Class Field Theory (section 6.2), which essentially constitutes the Langlands correspondence for $n = 1$. We define L-functions and ε-factors for representations of $\mathcal{W}_F$, parallel to those defined for G in the previous chapter. Finally, we introduce *Deligne Representations* (section 6.4), which incorporate a monodromy operator and allows us to generalize from complex representations to choosing any algebraically closed field of characteristic different from the residual characteristic of F. This modification is necessary to account for the difference between the semisimple Galois representations and the non-semisimple (Special) representations of G, and to align the complex theory with ℓ-adic geometry.

6.1 Weil Group and Smooth Representations

We start by recalling some features of the Galois theory of F. Throughout, let p be the characteristic of the residue class field $\mathbf{k} = \mathfrak{o}/\mathfrak{p}$. Choose a separable algebraic closure $\overline{F}$ of F. We set $\Omega_F = \mathrm{Gal}(\overline{F}/F)$, and view it as a profinite group with its natural (Krull) topology. Thus

$$\Omega_F = \varprojlim \mathrm{Gal}(E/F),$$

where E/F ranges over finite Galois extensions with $E \subset \overline{F}$. If K/F is a finite field extension, with $K \subset \overline{F}$, denote by Ω_K the open subgroup $\mathrm{Gal}(\overline{F}/K)$ of Ω_F. If $\widetilde{F}/F$ is some other separable algebraic closure of F, an F-isomorphism $\widetilde{F} \cong \overline{F}$ induces an isomorphism $\mathrm{Gal}(\widetilde{F}/F) \cong \Omega_F$. This does depend on the choice of isomorphism $\widetilde{F} \cong \overline{F}$. Fortunately, it is only up to inner automorphism, so in particular, it induces a *canonical* bijection between the sets of equivalence classes of irreducible smooth representations of the two groups $\mathrm{Gal}(\widetilde{F}/F)$, Ω_F.

Fixing an $\overline{F}/F$, consider extensions E/F where $E \subset \overline{F}$. Then the field F admits a unique unramified extension F_m/F of degree m, for each $m \geqslant 1$. Let F_∞ denote the composite of all these fields. Then F_∞/F is the unique maximal unramified extension of F. The extension F_m/F is Galois and $\mathrm{Gal}(F_m/F)$ is cyclic. An F-automorphism of F_m is determined by its action on the residue field $\mathbf{k}_{F_m} \cong \mathbb{F}_{q^m}$ (see [14, section 1.6]). In particular, there is a unique element φ_m of $\mathrm{Gal}(F_m/F)$ which acts on $\mathbf{k}_{F_m}$ as $x \mapsto x^q$, and it generates $\mathrm{Gal}(F_m/F)$. Set $\Phi_m = \varphi_m^{-1}$, so that Φ_m acts on $\mathbf{k}_{F_m}$ as $x \mapsto x^{q^{-1}}$, the exponent being read modulo $q^m - 1$. The inverse is taken because the two conventions of definition 6.1.1 below differ by exactly that step, and it is the geometric one that this book uses throughout. The map $\Phi_m \mapsto 1$ gives a canonical isomorphism $\mathrm{Gal}(F_m/F) \to \mathbb{Z}/m\mathbb{Z}$. Taking the limit over m, we get a canonical isomorphism of topological groups

$$\mathrm{Gal}(F_\infty/F) \cong \varprojlim \mathbb{Z}/m\mathbb{Z} = \widehat{\mathbb{Z}}$$

and a unique element $\Phi_F \in \mathrm{Gal}(F_\infty/F)$ which acts on F_m as Φ_m, for all m. Note that $\langle \Phi_F \rangle \cong \mathbb{Z}$ and hence cannot generate the entire $\mathrm{Gal}(F_\infty/F)$, however it is *dense* (on every finite group the restriction generates the entire group).

Definition 6.1.1: Geometric Frobenius

The element Φ_F is the *geometric Frobenius substitution* on F_∞. Its inverse $\varphi_F = \Phi_F^{-1}$, which acts on the residue field of every F_m as $x \mapsto x^q$, is the *arithmetic Frobenius substitution.*

An element of Ω_F is called a *geometric Frobenius element* (over F) if its image in $\mathrm{Gal}(F_\infty/F)$ is Φ_F. The Chinese Remainder Theorem gives a canonical isomorphism of topological groups

$$\widehat{\mathbb{Z}} \cong \prod_\ell \mathbb{Z}_\ell,$$

where ℓ ranges over all prime numbers and $\mathbb{Z}_\ell$ is the (additive) group of ℓ-adic integers.

Let $\mathcal{I}_F = \mathrm{Gal}(\overline{F}/F_\infty)$ be the *inertia group* of F. This fits in the exact sequence of topological groups

$$1 \to \mathcal{I}_F \to \Omega_F \to \mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) \cong \widehat{\mathbb{Z}} \to 1$$

Now extend to the maximal tamely ramified extensions. For each integer $n \geqslant 1$, $p \nmid n$, the field F_∞ has a unique extension E_n/F_∞ of degree n. If ϖ is a prime element of F, the extension E_n is generated over F_∞ by an n-th root of ϖ. The extension E_n/F_∞ is cyclic, and the Galois group $\mathrm{Gal}(E_n/F_\infty)$ admits a canonical description using Kummer theory (presented in [4, Kummer Pairing]). Choose $\alpha \in E_n$ such that $\alpha^n = \varpi$. The map

$$\begin{aligned} \mathrm{Gal}(E_n/F_\infty) &\longrightarrow \mu_n, \\ \sigma &\longmapsto \sigma(\alpha)/\alpha, \end{aligned} \tag{6.1}$$

gives a canonical isomorphism of $\mathrm{Gal}(E_n/F_\infty)$ with the group μ_n of n-th roots of unity in F_∞. The composite E_∞ of all extensions E_n/F_∞ is the maximal tamely ramified extension of F. Taking the limit of these isomorphisms gives topological isomorphisms

$$\mathrm{Gal}(E_\infty/F_\infty) \cong \varprojlim \mu_n \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$$

In the limit, n ranges over all positive integers not divisible by p; in the product, ℓ ranges over all prime numbers other than p.

Finally, let $\mathcal{P}_F = \mathrm{Gal}(\overline{F}/E_\infty)$ be the *wild inertia group*, completing the tower

$$\overline{F}/E_\infty/F_\infty/F$$

Any finite extension of E_∞ has degree a power of p, so the profinite group $\mathcal{P}_F$ is a pro p-group. That is enough to pin down the quotient, and the statement is used often enough downstream to be worth recording on its own.

Proposition 6.1.2: Structure of the Tame Quotient

The wild inertia group $\mathcal{P}_F$ is the unique pro-p-Sylow subgroup of $\mathcal{I}_F$; in particular it is normal in $\mathcal{I}_F$, and there is a topological isomorphism

$$\mathcal{I}_F/\mathcal{P}_F \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$$

determined uniquely up to multiplication by an element of $\prod_{\ell \neq p} \mathbb{Z}_\ell^\times$. Every finite quotient of $\mathcal{I}_F/\mathcal{P}_F$ is cyclic of order prime to p.

***Proof*:**

The tower $\overline{F}/E_\infty/F_\infty/F$ above identifies $\mathcal{I}_F/\mathcal{P}_F$ with $\mathrm{Gal}(E_\infty/F_\infty)$, and the construction of E_∞ as the union of the extensions $F_\infty(\varpi^{1/n})$ with $p \nmid n$ presents that group as $\varprojlim_n \mu_n$, the limit being over the positive integers prime to p ordered by divisibility. That limit is $\prod_{\ell \neq p} \mathbb{Z}_\ell$, and the only freedom in the identification is the choice of a compatible system of roots of unity, which is exactly multiplication by a unit.

Since $\mathcal{P}_F$ is a pro-p-group and the quotient has order prime to p in every finite layer, $\mathcal{P}_F$ is a pro-p-Sylow subgroup; any other one is conjugate to it, and $\mathcal{P}_F$ is normal because it is the kernel of $\mathcal{I}_F \to \mathrm{Gal}(E_\infty/F_\infty)$, so it is the unique one. Finally, a finite quotient of $\prod_{\ell \neq p} \mathbb{Z}_\ell$ is a finite quotient of a product of procyclic groups of pairwise coprime orders, hence cyclic of order prime to p, completing the proof.

The group $\mathrm{Gal}(F_\infty/F)$ acts on $\mathrm{Gal}(E_\infty/F_\infty)$ by conjugation. An isomorphism transfers this action to one on the group $\prod_{\ell \neq p} \mathbb{Z}_\ell$. This result is boxed for future reference:

Proposition 6.1.3: Tame Inertia Action

If $t_0 : \mathcal{I}_F/\mathcal{P}_F \to \prod_{\ell \neq p} \mathbb{Z}_\ell$ is a topological isomorphism, then

$$t_0(\Phi_F \sigma \Phi_F^{-1}) = q^{-1} t_0(\sigma), \quad \sigma \in \mathcal{I}_F/\mathcal{P}_F$$

***Proof*:**
The tame quotient is described by roots of unity, and the formula is Frobenius acting on them.

The field E_∞ is obtained from F_∞ by adjoining the m-th roots of a prime element for all m prime to p, so restriction gives a canonical isomorphism

$$\mathcal{I}_F/\mathcal{P}_F \xrightarrow{\sim} \varprojlim_{(m,p)=1} \mu_m(\overline{k})$$

under which $\sigma \in \mathcal{I}_F/\mathcal{P}_F$ corresponds to the compatible system $\left(\sigma(\varpi^{1/m})/\varpi^{1/m}\right)_m$ of roots of unity. Identifying $\varprojlim_m \mu_m(\overline{k}) \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$ is the isomorphism t_0.

Now compute the conjugate. Since Φ_F is a geometric Frobenius it acts on $\overline{k}$, hence on each $\mu_m(\overline{k})$, by $\zeta \mapsto \zeta^{q^{-1}}$, and it fixes ϖ. Therefore

$$\frac{\Phi_F \sigma \Phi_F^{-1}(\varpi^{1/m})}{\varpi^{1/m}} = \Phi_F\left(\frac{\sigma(\varpi^{1/m})}{\varpi^{1/m}}\right) = \left(\frac{\sigma(\varpi^{1/m})}{\varpi^{1/m}}\right)^{q^{-1}}$$

for every m prime to p. Passing to the limit and transporting along t_0, raising to the power q^{-1} becomes multiplication by q^{-1} in $\prod_{\ell \neq p} \mathbb{Z}_\ell$, which is the stated identity, completing the proof.

Let ${}_a\mathcal{W}_F$ be defined by:

$${}_a\mathcal{W}_F = \{\sigma \in \Omega_F : \sigma|_{F_\infty} \in \langle \Phi_F \rangle\} = \mathrm{res}^{-1}(\langle \Phi_F \rangle) \subseteq \Omega_F,$$

where $\mathrm{res} : \Omega_F \to \mathrm{Gal}(F_\infty/F)$ is the restriction map. As any $\varphi \in \mathcal{I}_F$ restricts to the identity, $\mathcal{I}_F \subseteq {}_a\mathcal{W}_F$. If E/F is finite Galois, the group $\langle \Phi_F \rangle$ surjects onto $\mathrm{Gal}(E \cap F_\infty/F)$ and $\mathcal{I}_F$ surjects onto $\mathrm{Gal}(E/E \cap F_\infty)$, so ${}_a\mathcal{W}_F$ surjects onto $\mathrm{Gal}(E/F)$. The subgroups Ω_E for such E form a basis of open subgroups of Ω_F, so ${}_a\mathcal{W}_F$ is *dense* in $\mathrm{Gal}(\overline{F}/F)$.

Next, the map:

$$\mathrm{res} : {}_a\mathcal{W}_F \to \mathrm{Gal}(F_\infty/F) \qquad \sigma \mapsto \sigma|_{F_\infty}$$

is by construction surjective onto $\langle \Phi_F \rangle$ with kernel $\mathcal{I}_F$ forming a short exact sequence:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathcal{I}_F & \longrightarrow & \Omega_F & \longrightarrow & \widehat{\mathbb{Z}} \cong \mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) & \longrightarrow & 1 \\ & & \| & & \uparrow & & \uparrow & & \\ 1 & \longrightarrow & \mathcal{I}_F & \longrightarrow & {}_a\mathcal{W}_F & \longrightarrow & \mathbb{Z} & \longrightarrow & 1 \end{array}$$

Notice the difference between these two quotients:

$${}_a\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z} \quad \text{(Generated by Frobenius)} \qquad \text{and} \qquad \mathrm{Gal}(F_\infty/F) \cong \widehat{\mathbb{Z}} \quad \text{(Profinite completion)}$$

Inertia		Total Group		Unramified
$\mathcal{I}_F$	$\subset$	Ω_F (Galois)	$\twoheadrightarrow$	$\widehat{\mathbb{Z}}$
$\|$		$\cup$		$\cup$
$\mathcal{I}_F$	$\subset$	${}_a\mathcal{W}_F$ (Weil)	$\twoheadrightarrow$	$\mathbb{Z}$

For a concrete element of Ω_F that is not in ${}_a\mathcal{W}_F$, work in $\widehat{\mathbb{Z}} \cong \prod_\ell \mathbb{Z}_\ell$ and let α be the element whose ℓ-component is 1 for $\ell = 2$ and 0 for every other ℓ. Then $\alpha \notin \mathbb{Z}$: an integer n with $n = 0$ in $\mathbb{Z}_\ell$ for

all $\ell \neq 2$ is divisible by every power of every odd prime and hence is 0, whereas $\alpha \neq 0$ because its 2-component is not. Restriction $\Omega_F \to \mathrm{Gal}(F_\infty/F)$ is surjective, so some $\sigma \in \Omega_F$ has $\sigma|_{F_\infty} = \Phi_F^\alpha$, and that σ is not in ${}_a\mathcal{W}_F$. A square root of Φ_F would not have served here, since 2 is not invertible in $\mathbb{Z}_2$ and so $\widehat{\mathbb{Z}}$ contains no element β with $2\beta = 1$.

Now, ${}_a\mathcal{W}_F$ is a profinite group, hence not "enough" representations to correspond to irreducible representations mentioned earlier. A better topology is needed to carry the arithmetic information of ${}_a\mathcal{W}_F$, that is a locally profinite topology:

Definition 6.1.4: Weil Group

The Weil group $\mathcal{W}_F$ of F (relative to $\bar{F}/F$) is the topological group, with underlying abstract group $a\mathcal{W}_F$, so that

1. $\mathcal{I}_F$ is an open subgroup of $\mathcal{W}_F$, and
2. the topology on $\mathcal{I}_F$, as subspace of $\mathcal{W}_F$, coincides with its natural topology as $\mathrm{Gal}(\bar{F}/F_\infty) \subset \Omega_F$.

In particular $\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z}$, generated by the class of a Frobenius element, that being the quotient the underlying group ${}_a\mathcal{W}_F$ carries.

As $\mathcal{I}_F$ is open, that quotient must be discrete, and as it is infinite it must be that $\mathcal{W}_F$ is no longer compact, and hence it is *locally profinite* (recall in a profinite group a set is open if and only if it is finite index). The definition of $\mathcal{W}_F$ does depend on the choice of $\overline{F}/F$ up to inner automorphism of Ω_F.

Lemma 6.1.5: Injection is Continuous

The identity map $\iota_F : \mathcal{W}_F \to {}_a\mathcal{W}_F \subset \Omega_F$ is a continuous injection.

***Proof*:**
Injectivity is immediate, since ι_F is the identity on underlying sets: $\mathcal{W}_F$ and ${}_a\mathcal{W}_F$ have the same elements and differ only in their topology.

For continuity, it is enough to check that the preimage of a basic open set of Ω_F is open in $\mathcal{W}_F$. The topology of Ω_F has a basis of cosets of the subgroups $\mathrm{Gal}(\overline{F}/L)$ for L/F finite, and each such subgroup meets $\mathcal{W}_F$ in a set that is open for the Weil topology: by construction $\mathcal{I}_F$ is open in $\mathcal{W}_F$ and carries its natural profinite topology, so the open subgroups of $\mathcal{I}_F$ remain open in $\mathcal{W}_F$. Hence every Ω_F-open set pulls back to a $\mathcal{W}_F$-open set, so ι_F is continuous.

The map is not a homeomorphism: the Weil topology is strictly finer, since $\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z}$ is discrete for it while the corresponding quotient of Ω_F is $\widehat{\mathbb{Z}}$, which is not, as we sought to show.

Let $\nu_F : \mathcal{W}_F \to \mathbb{Z}$ be the canonical map taking a geometric Frobenius element to 1 and define:

$$\|x\| = q^{-\nu_F(x)}, \quad x \in \mathcal{W}_F$$

Proposition 6.1.6: Weil subgroup and Finite Extensions

1. Let E/F be a finite extension, $E \subset \overline{F}$.
   (a) The group $\mathcal{W}_F$ has a unique subgroup $\mathcal{W}_F^E$ such that
   $$\iota_F(\mathcal{W}_F^E) = {}_a\mathcal{W}_F \cap \Omega_E$$
   (b) The subgroup $\mathcal{W}_F^E$ is open and of finite index in $\mathcal{W}_F$; it is normal in $\mathcal{W}_F$ if and only if E/F is Galois.
   (c) The canonical map $\mathcal{W}_F^E \backslash \mathcal{W}_F \to \Omega_E \backslash \Omega_F$ is a bijection.
   (d) The canonical map $\iota_E : \mathcal{W}_E \to \Omega_E$ induces a topological isomorphism $\mathcal{W}_E \cong \mathcal{W}_F^E$.
2. The map $E/F \mapsto \mathcal{W}_F^E$ is a bijection between the set of finite extensions E of F inside $\overline{F}$ and the set of open subgroups of $\mathcal{W}_F$ of finite index.

***Proof*:**
Throughout, ι_F is the continuous injection of lemma 6.1.5, and we identify $\mathcal{W}_F$ with its image when convenient.

(1)(a). Uniqueness is forced because ι_F is injective, so $\mathcal{W}_F^E$ must be the preimage $\iota_F^{-1}({}_a\mathcal{W}_F \cap \Omega_E)$; that preimage is a subgroup because both factors are.

(1)(b). The subgroup Ω_E is open of finite index $[E : F]$ in Ω_F by infinite Galois theory, so its preimage is open of index at most $[E : F]$ in $\mathcal{W}_F$ by continuity. The index is exactly $[E : F]$ because $\mathcal{W}_F$ surjects onto $\Omega_E \backslash \Omega_F$, which is (c). Normality transfers along the preimage, and $\Omega_E \trianglelefteq \Omega_F$ holds if and only if E/F is Galois.

(1)(c). The map is injective by construction. For surjectivity, note that $\mathcal{W}_F$ is dense in Ω_F, since it contains $\mathcal{I}_F = \mathrm{Gal}(\overline{F}/F_\infty)$ and a Frobenius lift, and Ω_E is open; a dense subgroup meets every coset of an open subgroup, so every class in $\Omega_E \backslash \Omega_F$ has a representative in $\mathcal{W}_F$.

(1)(d). Both $\mathcal{W}_E$ and $\mathcal{W}_F^E$ have the same underlying set, namely the elements of $\mathcal{W}_F$ fixing E, and both carry the topology in which the inertia subgroup $\mathcal{I}_E$ is open with its profinite topology and the quotient by it is discrete. So the identity is a topological isomorphism.

(2). Injectivity is Galois theory: E is recovered from $\mathcal{W}_F^E$ as the fixed field of the closure of its image in Ω_F. For surjectivity, let $H \leq \mathcal{W}_F$ be open of finite index. Its closure $\overline{H}$ in Ω_F is open of finite index, hence equals Ω_E for a unique finite E/F, and then $H = \mathcal{W}_F^E$ because both are the full preimage of Ω_E: an open finite-index subgroup of $\mathcal{W}_F$ contains an open subgroup of $\mathcal{I}_F$ and is therefore determined by its closure, completing the proof.

Henceforth $\mathcal{W}_E$ is identified with the subgroup $\mathcal{W}_F^E$ of $\mathcal{W}_F$.

Attention now turns to the smooth representations of $\mathcal{W}_F$ themselves. Note that as Ω_F is profinite, smooth representations must be semi-simple, while $\mathcal{W}_F$ being locally profinite (as it admits a discrete quotient $\mathbb{Z}$) must admit non-semisimple representations. As will be seen, the irreducible representations of $\mathcal{W}_F$ are closely related to the irreducible representations of Ω_F, but the reducible ones exhibit a greater variety, namely the principal series shall be associated to them, and these are "hidden" from the point of view of Ω_F. Just as in the previous chapters, the focus is on *complex* representations. The ℓ-adic ones are treated in section 6.4.2.

Lemma 6.1.7: Irreducible Representations of Weil Group is Finite

Let (ρ, V) be an irreducible representation of $\mathcal{W}_F$. Then V is finite dimensional.

***Proof*:**
Let $v \in V$. As ρ is smooth it is fixed by a compact open subgroup $K \subseteq \mathcal{W}_F$. Let $\mathcal{J} = K \cap \mathcal{I}_F$ so that v is fixed by an open subgroup of $\mathcal{I}_F$. As $\mathcal{I}_F$ has the same topology as the subspace topology of $\mathcal{I}_F \subseteq \Omega_F$ and $\mathcal{J}$ is open, it corresponds to a finite extension E/F. By taking the Galois closure of E/F, we get a smaller group that is still in the stabilizer of v, hence we may assume E/F is Galois hence $\mathcal{J}$ is normal in $\mathcal{W}_F$. It is an exercise in classical representation theory to show that if ρ is irreducible and $N \trianglelefteq G$ is normal, if N fixes a point v, N acts trivially on all of V. Hence, $\mathcal{J} \subseteq \ker \rho$.

Now, as $\mathcal{J}$ is an open subgroup of $\mathcal{I}_F$, the quotient $\mathcal{I}_F/\mathcal{J}$ is finite. Let $\Phi \in \mathcal{W}_F$ be a Frobenius element, that is, an element with $\nu_F(\Phi) = 1$. Both $\mathcal{I}_F$ and $\mathcal{J}$ are normal in $\mathcal{W}_F$, the second because E/F was arranged to be Galois, so Φ acts by conjugation on $\mathcal{I}_F/\mathcal{J}$, and as this group is finite some Φ^d acts trivially on it. Thus:

$$\Phi^d \cdot x \cdot \Phi^{-d} = xj$$

for appropriate $j \in \mathcal{J}$. Applying ρ:

$$\rho(\Phi^d)\rho(x)\rho(\Phi^{-d}) = \rho(x)\rho(j)$$

As $\mathcal{J} \subseteq \ker \rho$:

$$\rho(\Phi^d)\rho(x)\rho(\Phi^{-d}) = \rho(x) \cdot I = \rho(x)$$

Moving terms around:

$$\rho(\Phi^d)\rho(x) = \rho(x)\rho(\Phi^d)$$

Thus the map $\rho(\Phi^d)$ is in the center of $\rho(\mathcal{W}_F) \subseteq \mathrm{Aut}(V)$, and as ρ is irreducible this means $\rho(\Phi^d)$ acts like scalar multiplication.

Using this, we claim that for the v that was fixed at the very beginning of the proof, the space

$$U = \{\rho(g) \cdot v : g \in \mathcal{W}_F\}$$

is finite dimensional. By definition of $\mathcal{W}_F$, $g = \Phi^k t$ for $t \in \mathcal{I}_F$. By taking the finite $\mathcal{I}_F/\mathcal{J}$ we have coset representatives $r_1, ..., r_n$ so that:

$$\rho(g) \cdot v = \rho(\Phi^k)\rho(r_\alpha)\rho(j) \cdot v$$

where $j \in \mathcal{J}$ is any element. As $\mathcal{J} \subseteq \ker \rho$ we simplify to

$$\rho(g) \cdot v = \rho(\Phi^k)\rho(r_\alpha) \cdot v$$

Next, write $\Phi^k = \Phi^{ad+c}$ where $0 \leq c < d$ so that we can use the fact that $\rho(\Phi^d)$ is multiplication by a scalar λ to get

$$\rho(\Phi^k) = \lambda^a \rho(\Phi^c)$$

But now, we get that there are finitely many possible vectors that are used to span U

$$\rho(\Phi^c)\rho(r_\alpha) \cdot v \qquad 0 \leq c < d, \text{ and } 1 \leq \alpha \leq n$$

showing it is finite. Certainly this space is $\mathcal{W}_F$-stable, and as V is irreducible it must be that $U = V$, as we sought to show.

Hence the representations of $\mathcal{W}_F$ of interest are the finite-dimensional ones. Let ρ be a finite-dimensional smooth representation of $\mathcal{W}_F$. Then the restriction to $\mathcal{I}_F$ is semisimple by lemma 1.2.14 applied to the compact open subgroup $\mathcal{I}_F$ of $\mathcal{W}_F$, and $\rho(\mathcal{I}_F)$ is finite. Therefore, all the complexity comes from the Frobenius elements of $\mathcal{W}_F$.

Proposition 6.1.8: Irreducible in Galois, then in Weil

Let $\mathcal{W}_F$ be the Weil group, $\iota_F : \mathcal{W}_F \to {}_a\mathcal{W}_F$ the identity map (a continuous but not homeomorphic bijection), (ρ, V) a smooth representation of Ω_F, the absolute galois group of F so that $\rho \circ \iota_F$ can be thought of as the restriction to $\mathcal{W}_F$. Then:

1. if ρ is irreducible (on Ω_F), then $\rho \circ \iota_F$ is irreducible (on $\mathcal{W}_F$)
2. If ρ_1, ρ_2 are irreducible on Ω_F, then $\rho_1 \cong \rho_2$ if and only if $\rho_1 \circ \iota_F \cong \rho_2 \circ \iota_F$.

***Proof*:**

1. Take $\ker \rho \subseteq \Omega_F$. This is an open subset. Let $K = \ker \rho$. Then gK is open for all $g \in \Omega_F$. As $\mathcal{W}_F$ is dense, $gK \cap \mathcal{W}_F \neq \emptyset$. Then $gk = w$ for some $k \in K$ and $w \in \mathcal{W}_F$, in particular:

$$g = wk^{-1} \qquad w \in \mathcal{W}_F,\ k^{-1} \in K$$

   As this is true for all $g \in \Omega_F$:

$$\Omega_F = \ker(\rho) \cdot \mathcal{W}_F$$

   Taking ρ of both sides we get:

$$\rho(\Omega_F) = \rho(\mathcal{W}_F)$$

   Thus, if there are no nontrivial $\rho(\Omega_F)$-invariant subspaces, there are no nontrivial $\rho(\mathcal{W}_F)$-invariant subspaces.

2. If $\rho_1 \to \rho_2$ is an Ω_F-isomorphism, restriction certainly makes it a $\mathcal{W}_F$-isomorphism, and the equality $\rho(\Omega_F) = \rho(\mathcal{W}_F)$ proven above shows the other way is true too, completing the proof.

The converse is true *up to tensoring* with an unramified character:

Definition 6.1.9: Unramified Character

Let χ be a (complex) character on Ω_F or $\mathcal{W}_F$. Then χ is *unramified* if it is trivial on $\mathcal{I}_F$ ($\chi|_{\mathcal{I}_F}$ is trivial).

If ρ is a smooth, finite-dimensional representation of Ω_F or $\mathcal{W}_F$, the map $x \mapsto \det(\rho(x))$ is a character, $\det \rho$.

Proposition 6.1.10: Irreducible and Finite in Weil, then in Galois

Let τ be an irreducible smooth representation of $\mathcal{W}_F$. The following are equivalent:

1. the group $\tau(\mathcal{W}_F)$ is finite;
2. $\tau \cong \rho \circ \iota_F$, for some irreducible smooth representation ρ of Ω_F;
3. the character $\det \tau$ has finite order.

For any irreducible smooth representation τ of $\mathcal{W}_F$, there is an unramified character χ of $\mathcal{W}_F$ such that $\chi \otimes \tau$ satisfies (1) to (3).

***Proof*:**

(1) $\Rightarrow$ (2) We need to find the appropriate representation ρ of Ω_F. Let $\Phi \in \mathcal{W}_F$ be a Frobenius element. Then $\tau(\Phi)$ has finite order, say d. Since $\widehat{\mathbb{Z}}/d\widehat{\mathbb{Z}} \cong \mathbb{Z}/d\mathbb{Z}$, every $a \in \widehat{\mathbb{Z}}$ has an integer $\overline{a}$ with $\overline{a} \equiv a \bmod d\widehat{\mathbb{Z}}$, unique modulo $d\mathbb{Z}$, so $\tau(\Phi)^{\overline{a}}$ does not depend on that choice. Now, as shown earlier any $w \in \Omega_F$ can be written as

$$w = \Phi^a \sigma \qquad a \in \hat{\mathbb{Z}},\ \sigma \in \mathcal{I}_F$$

Define ρ of Ω_F as :

$$\rho(w) = \rho(\Phi^a \sigma) = \tau(\Phi)^{\overline{a}} \tau(\sigma)$$

Certainly $\tau = \rho \circ \iota_F$ and ρ is irreducible

(2) $\Rightarrow$ (3) Let τ be an irreducible representation that extends to ρ so that $\det \tau = (\det \rho)|_{\mathcal{W}_F}$. Let $\chi = \det \rho$. Then it is a *character*:

$$\chi : \Omega_F \to \mathbb{C}^\times$$

As χ is smooth, its kernel K is open and as Ω_F is compact Ω_F/K is finite. Thus χ factors through Ω_F/K, which is a finite group, hence $\chi^n = 1$ for an appropriate n, and $\det \tau = \chi|_{\mathcal{W}_F}$ has finite order as well.

(3) $\Rightarrow$ (1) Assume $\det \tau$ has finite order. First, as $\mathcal{I}_F$ is compact, $\tau(\mathcal{I}_F)$ is finite, so conjugation by $\tau(\Phi)$ permutes that finite set and some power $\tau(\Phi)^k$ acts on it trivially. It commutes with $\tau(\Phi)$ as well, hence with all of $\tau(\mathcal{W}_F)$, and so by Schur's lemma $\tau(\Phi)^k = \lambda I$.

Now, letting $n = \dim \tau$ (which exists as τ is irreducible by lemma 6.1.7), $\det(\tau(\Phi)^k) = \det(\lambda I) = \lambda^n$. As $\det \tau$ has finite order (say m):

$$\det(\tau(\Phi)^k)^m = \lambda^{nm} = 1$$

Hence λ is a root of unity, and $\tau(\Phi)^k = \lambda I$ has finite order. As $\mathcal{W}_F = \langle \Phi \rangle \cdot \mathcal{I}_F$ and τ has finite image on each of the two factors on the right hand side, $\tau(\mathcal{W}_F)$ is finite.

For the last part, notice that the argument producing the scalar $\tau(\Phi)^k$ used no hypothesis beyond the compactness of $\mathcal{I}_F$. Say $\tau(\Phi)^k = cI$. Using this c, pick $z_0 \in \mathbb{C}$ with $z_0^k = c$ and define:

$$\chi : \mathcal{W}_F \to \mathbb{C}^\times \qquad \chi(\Phi^m \sigma) = z_0^m, \quad m \in \mathbb{Z},\ \sigma \in \mathcal{I}_F$$

This is certainly a smooth character and unramified by construction. Then letting $\tau' = \chi^{-1} \otimes \tau$ we get:

$$\tau'(\Phi)^k = (\chi^{-1}(\Phi)\tau(\Phi))^k = \chi(\Phi)^{-k}\tau(\Phi)^k = c^{-1} \cdot (c \cdot I) = I$$

Hence $\tau'(\Phi)$ has finite order, and as in the proof of (3) $\Rightarrow$ (1), since $\mathcal{W}_F = \langle \Phi \rangle \cdot \mathcal{I}_F$ and τ' has finite image on both pieces, $\tau'(\mathcal{W}_F)$ is finite, satisfying the first condition and completing the proof.

This result is as good as it can get, that is there exist irreducible representations τ of $\mathcal{W}_F$ whose image $\tau(\mathcal{W}_F)$ is infinite:

Example 6.1.11: Irreducible in Weil But Not Defined in Galois Group

Let $\chi : \mathcal{W}_F \to \mathbb{C}^\times$ be $\chi(w) = \|w\| = q^{-\nu_F(w)}$. This is certainly a smooth one-dimensional irreducible representation, and $\chi(\mathcal{W}_F) = q^{\mathbb{Z}}$ is infinite. This map simply does not exist over Ω_F, whose image under any character must be compact.

Next, relate the representations of $\mathcal{W}_F$ and $\mathcal{W}_E$ where E/F is a finite-separable/galois extension where $E \subseteq \overline{F}$.

Definition 6.1.12: Induction On Weil Representation

Let E/F be a finite separable extension where $E \subseteq \overline{F}$. Then write:

$$\mathrm{Ind}_{E/F} := \mathrm{Ind}_{\mathcal{W}_E}^{\mathcal{W}_F}$$

for the functor of smooth induction from $\mathcal{W}_E$ to $\mathcal{W}_F$, and:

$$\mathrm{Res}_{E/F} := \mathrm{Res}_{\mathcal{W}_E}^{\mathcal{W}_F}$$

From now on, if ρ is a smooth representation of $\mathcal{W}_F$, $\rho|_E = \mathrm{Res}_{E/F}\rho = \rho|_{\mathcal{W}_E}$. Recall lemma 1.2.38, for ease of reference the following translates this lemma with the groups of interest:

Lemma 6.1.13: Weil Representation of Finite Extension

Let E/F be a finite separable field extension, $E \subset \bar{F}$.

1. Let ρ be a smooth representation of $\mathcal{W}_F$. Then ρ is semisimple if and only if $\rho|_E$ is semisimple.
2. Let τ be a smooth representation of $\mathcal{W}_E$. Then τ is semisimple if and only if $\mathrm{Ind}_{E/F}\tau$ is semisimple.

***Proof*:**

Part (1) and the forward implication of (2) are lemma 1.2.38(1) and (2), applied to the open finite-index subgroup $\mathcal{W}_E \subseteq \mathcal{W}_F$ of proposition 6.1.6.

The converse of (2) does not follow from that lemma and takes one more step. Suppose $\mathrm{Ind}_{E/F}\tau$ is semisimple. By (1) it is then semisimple over $\mathcal{W}_E$, and τ embeds in its restriction as the

subspace of functions supported on $\mathcal{W}_E$. A subrepresentation of a semisimple representation is semisimple, by condition (3) of proposition 1.2.11, so τ is semisimple, completing the proof.

Definition 6.1.14: Class of Semisimple Weil Representations

Let F be a local field. For each integer $n \geq 1$, let

$$\mathcal{G}_n^{\mathrm{ss}}(F)$$

denote the set of isomorphism classes of semisimple smooth representations of $\mathcal{W}_F$ of dimension n. Let

$$\mathcal{G}_n^{0}(F)$$

denote the set of isomorphism classes of *irreducible* smooth representations of $\mathcal{W}_F$ of dimension n.

Let E/F be a finite extension with $E \subseteq \overline{F}$. Then lemma 6.1.13 gives the induction and restriction maps

$$\mathrm{Ind}_{E/F} : \mathcal{G}_n^{\mathrm{ss}}(E) \to \mathcal{G}_{nd}^{\mathrm{ss}}(F),$$
$$\mathrm{Res}_{E/F} : \mathcal{G}_n^{\mathrm{ss}}(F) \to \mathcal{G}_n^{\mathrm{ss}}(E),$$

where $d = [E : F]$.

Two choices went into the definition of the Weil group, the separable closure and the Frobenius element, and neither is canonical. What survives is that $\mathcal{W}_F$ is determined up to an inner automorphism of Ω_F: a second pair of choices differs from the first by conjugation by some $\tau \in \Omega_F$, and $\rho \circ \mathrm{Int}(\tau) \cong \rho$ for every $\rho \in \mathcal{G}_n^{\mathrm{ss}}(F)$. The sets $\mathcal{G}_n^{0}(F)$, $\mathcal{G}_n^{\mathrm{ss}}(F)$ are therefore independent of the choices, and so are the two maps above; in particular induction and restriction along a finite separable E/F do not depend on the F-embedding of E in $\bar{F}$. The isomorphism $\rho \circ \mathrm{Int}(\tau) \cong \rho$ is not formal, since the proof uses the density of $\mathcal{W}_F$ in Ω_F and the smoothness of ρ, and we defer it to lemma 11.2.1(4), which establishes it in the generality section 11.2 needs, for an isomorphism between two different local fields. Nothing before that section uses more than the statement.

Proposition 6.1.15: Semisimple Weil Representation

Let (ρ, V) be a smooth representation of $\mathcal{W}_F$ of finite dimension, and let $\Phi \in \mathcal{W}_F$ be a Frobenius element. The following are equivalent:

(1) the representation ρ is semisimple;

(2) the automorphism $\rho(\Phi) \in \mathrm{Aut}_{\mathbb{C}}(V)$ is semisimple;

(3) the automorphism $\rho(\Psi) \in \mathrm{Aut}_{\mathbb{C}}(V)$ is semisimple, for every element Ψ of $\mathcal{W}_F$.

***Proof*:**
(3) $\Rightarrow$ (2) is obvious, so let's prove (2) $\Rightarrow$ (1) and then (1) $\Rightarrow$ (3)

(2) $\Rightarrow$ (1) Recall a semisimple operator means it is diagonalizable. As $\rho(\Phi)$ is semisimple, so is $\rho(\Phi)^d$ for any d. Pick d so that $\rho(\Phi)^d$ commutes with $\rho(\mathcal{I}_F)$ (as it is finite, such a d exists). Then as Φ^d and $\mathcal{I}_F$ commute, $\rho|_H$ is semisimple, where

$H = \langle \Phi^d, \mathcal{I}_F \rangle$ is the set of elements of $\mathcal{W}_F$ whose valuation lies in $d\mathbb{Z}$. That subgroup is $\mathcal{W}_E$ for the unramified E/F of degree d, by proposition 6.1.6, so lemma 6.1.13(1) applies and ρ is semisimple

(1) $\Rightarrow$ (3) Let ρ be semisimple and pick any $\Psi \in \mathcal{W}_F$. If $\Psi \in \mathcal{I}_F$, then $\rho(\Psi)$ is semisimple as $\mathcal{I}_F$ is profinite, so assume $\Psi \notin \mathcal{I}_F$. Now, let $n = \nu_F(\Psi)$ which is nonzero as $\Psi \notin \mathcal{I}_F$. Choose an unramified degree n extension E/F (such an extension exists, see [9, Characterizing Unramified Extension Of Local Fields]) and consider $\mathcal{W}_E$ which by proposition 6.1.6 is an open subgroup of $\mathcal{W}_F$ of finite index and consists of elements whose valuation is divisible by n. Certainly Ψ belongs in this group as:

$$\nu_E(\Psi) = \frac{1}{n}\nu_F(\Psi) = \frac{1}{n} \cdot n = 1$$

and $\mathcal{I}_E = \mathcal{I}_F$, since E/F is unramified. For the generation notice that:

$$1 \to \mathcal{I}_E \to \mathcal{W}_E \xrightarrow{\nu_E} \langle \Psi \rangle \to 1$$

And hence $\mathcal{W}_E = \langle \Psi, \mathcal{I}_E \rangle$. As ρ is assumed to be semisimple and E/F is finite, by lemma 6.1.13(1) $\rho|_E$ is semisimple. As we want to show $\rho(\Psi)$ is semisimple, we may focus on E, or equivalently we may as well assume $E = F$. As we want to prove it is semisimple, we need to prove each block-diagonal is diagonalizable, so we may further reduce to assuming ρ is irreducible. But now, by proposition 6.1.10, any irreducible representation can be twisted by an unramified character χ to get a finite image. By untwisting ($c = \chi(\Psi)^{-1}$), we get:

$$\rho(\Psi) = c\varphi$$

where φ has finite order. But then by regular Maschke's theorem $\rho(\Psi)$ is semisimple, as we sought to show.

A two-dimensional representation of $\mathcal{W}_F$ is pinned down far more tightly by its *projective* image than by its image, because the projective image is a finite subgroup of $\mathrm{PGL}_2(\mathbb{C})$ and those admit a complete list. The list is what separates the representations induced from a quadratic extension from the ones that are not, which is the division the correspondence in chapter 7 and the dyadic analysis of chapter 9 both run on.

Theorem 6.1.16: Finite Subgroups of $\mathrm{PGL}_2(\mathbb{C})$

Let G be a finite subgroup of $\mathrm{PGL}_2(\mathbb{C})$ of order $N > 1$, acting on $\mathbb{P}^1(\mathbb{C})$. Call $x \in \mathbb{P}^1(\mathbb{C})$ a *pole* if its stabilizer G_x is non-trivial. Then every G_x is cyclic, the poles form two or three G-orbits, and writing $n_1 \le \cdots \le n_k$ for the stabilizer orders along those orbits, exactly one of the following holds.

1. $k = 2$, $n_1 = n_2 = N$, and G is cyclic of order N.
2. $k = 3$ and $(n_1, n_2, n_3) = (2, 2, n)$ with $N = 2n$, and G is dihedral of order $2n$.
3. $k = 3$ and (n_1, n_2, n_3) is one of $(2, 3, 3)$, $(2, 3, 4)$, $(2, 3, 5)$, with $N = 12$, 24, 60 respectively.

In case (3) the groups are A_4, S_4 and A_5. That identification is not proved here and is not used below; for it see [2, §42], whose Comment records the classification. Everything used below is the signature.

***Proof*:**

Step 0: fixed points and stabilizers. A non-identity element of $\mathrm{PGL}_2(\mathbb{C})$ is represented by a non-scalar $A \in \mathrm{GL}_2(\mathbb{C})$, and its fixed points on $\mathbb{P}^1(\mathbb{C})$ are the eigenlines of A. An element of G has finite order m, so it satisfies $T^m - 1$, which is separable over $\mathbb{C}$; hence A is diagonalizable with two distinct eigenvalues, and the element has *exactly two* fixed points. (A parabolic element, with a single Jordan block, has only one fixed point; finite order is what excludes it.) Consequently every non-identity element of G lies in G_x for exactly two poles x. Each G_x fixes two points of $\mathbb{P}^1(\mathbb{C})$, so after conjugating it into the diagonal torus it is a finite subgroup of $\mathbb{C}^\times$, hence cyclic.

Step 1: the counting identity. Count pairs (g, x) with $g \neq 1$ and $gx = x$. Summing over g gives $2(N - 1)$ by Step 0. Summing over x gives $\sum_x (|G_x| - 1)$ over the poles. The poles are stable under G and stabilizers along an orbit are conjugate, so with orbits $O_1, \ldots, O_k$ and stabilizer orders n_i one has $|O_i| = N/n_i$ and

$$2 - \frac{2}{N} = \sum_{i=1}^{k} \left(1 - \frac{1}{n_i}\right) \tag{6.2}$$

Step 2: two or three orbits. Each $n_i \ge 2$, so each summand lies in $[\frac{1}{2}, 1)$. The left side is at least 1 because $N \ge 2$, and is less than 2. Hence $k \ge 2$ and $k \le 3$.

Step 3: two orbits give the cyclic case. Here (6.2) reads $1/n_1 + 1/n_2 = 2/N$. Since $n_i | N$ we have $n_i \le N$, so $1/n_1 + 1/n_2 \ge 2/N$ with equality only when $n_1 = n_2 = N$. Both orbits are then single points, both fixed by all of G, so G lies in the pointwise stabilizer of two points, which is conjugate to the diagonal torus; a finite subgroup of it is cyclic.

Step 4: three orbits. Now $1/n_1 + 1/n_2 + 1/n_3 = 1 + 2/N$ with $n_1 \le n_2 \le n_3$. If $n_1 \ge 3$ the left side is at most 1, impossible; so $n_1 = 2$. If then $n_2 \ge 4$ the left side is at most 1, impossible; so $n_2 \in \{2, 3\}$. For $n_2 = 2$ the equation gives $n_3 = N/2$, which is case (2). For $n_2 = 3$ it gives $1/n_3 = 1/6 + 2/N > 1/6$, so $n_3 \in \{3, 4, 5\}$ and correspondingly $2/N = 1/6, 1/12, 1/30$, that is $N = 12, 24, 60$. That is case (3).

Step 5: the case $(2, 2, n)$ is dihedral. The third orbit has $|O_3| = N/n_3 = 2$, say $O_3 = \{x, y\}$, and $C = G_x$ is cyclic of order $n = N/2$. Since $gG_xg^{-1} = G_{gx}$ and G permutes $\{x, y\}$, the subgroup C is normal of index two, and $C = G_y$ as well, so C fixes both x and y. Conjugate so that $x = 0$

and $y = \infty$; then C is the group of $z \mapsto \zeta z$ with ζ an n-th root of unity. An element $g \in G \smallsetminus C$ swaps x and y, so it is $z \mapsto c/z$ for some $c \in \mathbb{C}^\times$, and $g^2 = 1$ in $\mathrm{PGL}_2(\mathbb{C})$. So G is generated by a cyclic group of order n and an involution inverting it, which is the dihedral group of order $2n$. The involution is what distinguishes this from the dicyclic groups, which also have a cyclic subgroup of index two inverted by an outside element, and it is the computation $g^2 = 1$ that gives it, completing the proof.

The three exceptional signatures are not identified above, and the corollary this section needs does not require it: metacyclicity can be read off the signature alone.

Corollary 6.1.17: Metacyclic Subgroups Are Cyclic or Dihedral

A finite subgroup of $\mathrm{PGL}_2(\mathbb{C})$ that is metacyclic, that is, one having a cyclic normal subgroup with cyclic quotient, is cyclic or dihedral.

***Proof*:**
Let G be such a subgroup, $C \trianglelefteq G$ cyclic with G/C cyclic of order m, so $N = |C|m$. By theorem 6.1.16 it suffices to exclude the three signatures of case (3), and for those $\max_i n_i$ is 3, 4 and 5 respectively.

Two bounds. First, C is cyclic, so it has an element of order $|C|$; by Step 0 of theorem 6.1.16 every non-identity element of G lies in some G_x, which is cyclic of order n_i, so every element order divides some n_i and in particular $|C| \leq \max_i n_i$. Second, let $g \in G$ map to a generator of G/C; the order of g is divisible by m, so likewise $m \leq \max_i n_i$.

Hence $N = |C|m \leq (\max_i n_i)^2$. For the three signatures that reads $12 \leq 9$, $24 \leq 16$ and $60 \leq 25$, each false. So a metacyclic G has signature $(2, 2, n)$ or two orbits, and is dihedral or cyclic, completing the proof.

6.2 Weil Group and Local Class Field Theory

Let $\mathcal{W}_F^{\mathrm{der}}$ denote the closure of the commutator subgroup of $\mathcal{W}_F$ and let

$$\mathcal{W}_F^{\mathrm{ab}} = \mathcal{W}_F / \mathcal{W}_F^{\mathrm{der}}$$

The group $\mathcal{W}_F^{\mathrm{ab}}$ is a locally profinite abelian group (defined independently of the choices made in the definition of $\mathcal{W}_F$). The following summarizes the core result of CFT:

Theorem 6.2.1: Local Class Field Theory

There is a canonical continuous group homomorphism

$$\mathbf{a}_F \colon \mathcal{W}_F \longrightarrow F^\times$$

Then:

1. The map $\mathbf{a}_F$ induces a topological isomorphism $\mathcal{W}_F^{\mathrm{ab}} \cong F^\times$.
2. An element $x \in \mathcal{W}_F$ is a geometric Frobenius if and only if $\mathbf{a}_F(x)$ is a prime element of F.
3. $\mathbf{a}_F(\mathcal{I}_F) = U_F$ and $\mathbf{a}_F(\mathcal{P}_F) = U_F^1$.
4. If E/F is a finite separable extension, the diagram

$$\begin{array}{ccc} \mathcal{W}_E & \xrightarrow{\mathbf{a}_E} & E^\times \\ \downarrow & & \downarrow{\scriptstyle N_{E/F}} \\ \mathcal{W}_F & \xrightarrow{\mathbf{a}_F} & F^\times \end{array}$$

   commutes.

5. Let $\alpha \colon F \to F'$ be an isomorphism of fields. The map α induces an isomorphism $\alpha \colon \mathcal{W}_F^{\mathrm{ab}} \to \mathcal{W}_{F'}^{\mathrm{ab}}$, and the diagram

$$\begin{array}{ccc} \mathcal{W}_F^{\mathrm{ab}} & \xrightarrow{\alpha} & \mathcal{W}_{F'}^{\mathrm{ab}} \\ {\scriptstyle \mathbf{a}_F}\downarrow & & \downarrow{\scriptstyle \mathbf{a}_{F'}} \\ F^\times & \xrightarrow{\alpha} & F'^\times \end{array}$$

   commutes.

The map $\mathbf{a}_F$ is the Artin Reciprocity Map.

***Proof*:**
This is the main theorem of local class field theory (see [14, Local Artin Reciprocity])

Theorem 6.2.2: Local Class Theory: Norm

1. The map $E/F \mapsto N_{E/F}(E^\times)$ gives a bijection between the set of finite abelian extensions E/F, $E \subset \bar{F}$, and the set of open subgroups of $F^\times$ of finite index.
2. The map $\mathbf{a}_F$ induces an isomorphism $\mathrm{Gal}(E/F) \cong F^\times / N_{E/F}(E^\times)$, for any finite abelian extension E/F.
3. Let E/F be a finite separable extension. If E_1/F is the maximal abelian sub-extension of E/F, then $N_{E/F}(E^\times) = N_{E_1/F}(E_1^\times)$.

***Proof*:**
This is the main theorem of local class field theory (see [14, Local Artin Reciprocity])

There is also the need for "lifting" a representation given E/F. To that end, let G be some group and H a subgroup of G of finite index. Let $D(G)$ be the commutator subgroup of G (denoted $[G,G]$ in [14] and [8] but abbreviated to $D(G)$ for the derived subgroup). Choose a section $t\colon G/H \to G$ of the quotient map $G \to G/H$. For $x \in G/H$ and $g \in G$, define an element $h_{x,g} \in H$ by $gt(x) = t(gx)h_{x,g}$. Then define the map:

Definition 6.2.3: Verlagerung Map

$$\begin{aligned} \mathrm{ver}_{G/H}\colon G/D(G) &\longrightarrow H/D(H), \\ g\,D(G) &\longmapsto \prod_{x\in G/H} h_{x,g}\,D(H) \end{aligned}$$

The map $\mathrm{ver}_{G/H}$ is called the *transfer*, or the *Verlagerung*, and it is a homomorphism of abelian groups.

If G is locally compact, write G^{der} for the closure of $D(G)$ in G and $G^{\mathrm{ab}} = G/G^{\mathrm{der}}$. If H is a closed subgroup of G of finite index, the transfer then induces a continuous homomorphism $G^{\mathrm{ab}} \to H^{\mathrm{ab}}$, still denoted $\mathrm{ver}_{G/H}$. In particular, if E/F is a finite separable field extension, the transfer gives a map $\mathrm{ver}_{E/F}\colon \mathcal{W}_F^{\mathrm{ab}} \to \mathcal{W}_E^{\mathrm{ab}}$.

Theorem 6.2.4: Transfer Theorem

If E/F is a finite separable field extension, the diagram

$$\begin{array}{ccc} \mathcal{W}_E^{\mathrm{ab}} & \xrightarrow{\mathbf{a}_E} & E^\times \\ {\scriptstyle \mathrm{ver}_{E/F}}\uparrow & & \uparrow \\ \mathcal{W}_F^{\mathrm{ab}} & \xrightarrow[\mathbf{a}_F]{} & F^\times \end{array}$$

commutes, where the map $F^\times \to E^\times$ is inclusion.

The compatibility of the transfer with the reciprocity map is a theorem of local class field theory, and this book's prerequisite *Class Field Theory* [14] proves it in full; by the de-duplication convention of the series the statement is used here and the proof is cited rather than repeated. What matters downstream is the shape of the diagram: it says that restricting a character of $F^\times$ along the inclusion into $E^\times$ corresponds, on the Weil side, to composing with the transfer, and that is exactly the input proposition 6.2.7 needs.

This links to representations as follows. As $\mathcal{W}^{\mathrm{ab}}$ is abelian, all representations are characters. By theorem 6.2.1, there is an isomorphism:

$$\mathrm{Hom}_{\mathbf{Top}}(F^\times, \mathbb{C}^\times) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Top}}(\mathcal{W}_F^{\mathrm{ab}}, \mathbb{C}^\times) \qquad \chi \mapsto \chi \circ \mathbf{a}_F$$

The isomorphism preserves unramified characters, and maps tamely ramified characters of $F^\times$ to characters of $\mathcal{W}_F$ trivial on $\mathcal{P}_F$. Thus one may simply refer to characters of $\mathcal{W}_F$ without composing

with the Artin map.

An important character to consider is $\det\rho$ for $\rho\in\mathcal{G}_d^{ss}(F)$, viewed as a character of $F^\times$. It comes up often, being the natural way to reduce a d-dimensional representation to a character.

Definition 6.2.5: The Character $\varkappa_{K/F}$

Let K/F be a separable extension of finite degree d, and let 1_K denote the trivial character of $\mathcal{W}_K$. Write

$$R_{K/F}=\mathrm{Ind}_{K/F}1_K\in\mathcal{G}_d^{\mathrm{ss}}(F)$$

for the representation of $\mathcal{W}_F$ induced from it, and

$$\varkappa_{K/F}=\det R_{K/F}$$

for its determinant, regarded as a character of $F^\times$ through theorem 6.2.1.

Proposition 6.2.6: The Induced Determinant Is a Sign

Let K/F be finite separable of degree d. The representation $R_{K/F}$ is the permutation representation of $\mathcal{W}_F$ on the coset space $\mathcal{W}_F/\mathcal{W}_K$, and $\varkappa_{K/F}(g)$ is the sign of the permutation that g induces on those d cosets. In particular $\varkappa_{K/F}$ takes values in $\{\pm1\}$ and $\varkappa_{K/F}^2=1$.

***Proof*:**
By definition 6.2.5, $R_{K/F}=\mathrm{Ind}_{K/F}1_K$, and inducing the trivial representation from an open subgroup realizes $\mathcal{W}_F$ on the space of functions on $\mathcal{W}_F/\mathcal{W}_K$, which is the permutation representation on that finite set of d elements. In a basis of characteristic functions of the cosets, $R_{K/F}(g)$ is the permutation matrix of the action of g, whose determinant is the sign of that permutation. A sign is ±1, so $\varkappa_{K/F}^2=1$.

Only when K/F is Galois is $R_{K/F}$ the regular representation of $\mathrm{Gal}(K/F)$ inflated to $\mathcal{W}_F$. For a general separable K/F only the permutation description applies.

A warning on notation, since the two symbols differ by one letter. This book writes $\kappa_{E/F}$ for the quadratic character of $F^\times$ attached to a *quadratic* extension E/F by local class field theory, and $\varkappa_{K/F}$ for the determinant above. The two agree exactly when $[K:F]=2$, where $R_{K/F}=1_F\oplus\kappa_{K/F}$ forces $\varkappa_{K/F}=\kappa_{K/F}$. In higher degree $\kappa_{K/F}$ is not defined at all, while $\varkappa_{K/F}$ still is.

The character $\varkappa_{K/F}$ arises in the following context:

Proposition 6.2.7: Determinant of Inflation

Let K/F be a separable extension of degree d, let $\rho\in\mathcal{G}_m^{\mathrm{ss}}(K)$ and put $\sigma=\mathrm{Ind}_{K/F}\rho\in\mathcal{G}_{md}^{\mathrm{ss}}(F)$. Then

$$\det\sigma=\varkappa_{K/F}^m\otimes\det\rho|_{F^\times}$$

Lemma 6.2.8: Determinant of an Induced Representation

Let G be a group, let H be a subgroup of G of finite index, and ρ a finite dimensional representation of H. Set $\sigma = \mathrm{Ind}_H^G \rho$. Then

$$\det \sigma = \det(R_{G/H})^{\dim \rho} \otimes (\det \rho \circ \mathrm{ver}_{G/H})$$

where $R_{G/H} = \mathrm{Ind}_H^G 1_H$.

***Proof*:**

Fix coset representatives $t(x)$, $x \in G/H$, and realize $\mathrm{Ind}_H^G \rho$ on $\bigoplus_x t(x) \otimes W$ where W is the space of ρ. For $g \in G$ the action permutes the summands according to $x \mapsto gx$ and acts on the summand indexed by x through $\rho(h_{x,g})$, where $h_{x,g} \in H$ is the element defined by $g\,t(x) = t(gx)h_{x,g}$, exactly as in definition 6.2.3.

The matrix of $\sigma(g)$ in this decomposition is therefore a block permutation matrix whose non-zero blocks are the $\rho(h_{x,g})$. Its determinant splits into two contributions. The first is the sign of the permutation of the blocks, each counted $\dim \rho$ times, which is precisely the determinant of the permutation representation $R_{G/H}$ raised to the power $\dim \rho$. The second is the product of the determinants of the blocks,

$$\prod_{x \in G/H} \det \rho(h_{x,g}) = \det \rho\Big(\prod_{x \in G/H} h_{x,g} \Big) = \det \rho\big(\mathrm{ver}_{G/H}(g)\big)$$

where the first equality uses that $\det \rho$ is a homomorphism into an abelian group, so the order of the factors is immaterial, and the second is the definition of the transfer. Multiplying the two contributions gives the stated formula, as we sought to show.

***Proof*:**

(of proposition 6.2.7)

Combining lemma 6.2.8 with the Transfer Theorem (theorem 6.2.4): the lemma gives $\det \sigma = \varkappa_{K/F}^m \otimes (\det \rho \circ \mathrm{ver}_{K/F})$ with $m = \dim \rho$, and the transfer theorem identifies $\det \rho \circ \mathrm{ver}_{K/F}$ with the restriction of $\det \rho$ to $F^\times$ along the inclusion $F^\times \hookrightarrow K^\times$, completing the proof.

6.3 L-functions and Epsilon-Factors

We now turn to the construction of the fundamental invariants, L-functions and ε-factors, attached to finite-dimensional, semisimple, smooth representations of the Weil group $\mathcal{W}_F$. This theory generalizes the Tate theory for characters of $F^\times$ (which corresponds to the 1-dimensional case via Class Field Theory), but requires significantly more machinery to establish existence in higher dimensions.

Take the Artin reciprocity map $\mathbf{a}_F : \mathcal{W}_F \to F^\times$ and identify characters of $F^\times$ with 1-dimensional representations of $\mathcal{W}_F$. For such a character χ, we have already defined the local L-function $L(\chi, s)$

and the local constant $\varepsilon(\chi, s, \psi)$ in section 5.1. We translate these to $\mathcal{W}_F$ by setting:

$$L(\chi \circ \mathbf{a}_F, s) = L(\chi, s), \qquad \varepsilon(\chi \circ \mathbf{a}_F, s, \psi) = \varepsilon(\chi, s, \psi)$$

The objective is to extend these definitions to $\sigma \in \mathcal{G}_n^{\mathrm{ss}}(F)$.

The extension of the L-function is straightforward. For an irreducible representation $\sigma \in \mathcal{G}_n^0(F)$, the definition divides based on dimension. If $n = 1$, it is given by the formula above. If $n \geq 2$:

$$L(\sigma, s) = 1$$

We then extend this to semisimple representations $\sigma = \bigoplus_i \sigma_i$ by additivity (multiplicativity of the function):

$$L(\sigma, s) = \prod_i L(\sigma_i, s)$$

6.3.1 Artin L-functions There is a more uniform description. Let (σ, V) be a representation of $\mathcal{W}_F$. Since $\mathcal{I}_F$ is normal in $\mathcal{W}_F$, the space $V^{\mathcal{I}_F}$ of inertia-fixed vectors is a subrepresentation, and it carries a representation of $\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z}$ generated by a geometric Frobenius Φ. The Artin L-function is defined as:

$$L(\sigma, s) = \det\left(1 - \sigma(\Phi)|_{V^{\mathcal{I}_F}}\, q^{-s}\right)^{-1}$$

the determinant being taken on $V^{\mathcal{I}_F}$, so that it is 1 when that space is zero. This matches the definition given above. If σ is irreducible then $V^{\mathcal{I}_F}$, being a subrepresentation, is 0 or all of V. In the second case σ is trivial on $\mathcal{I}_F$, hence factors through $\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z}$ and has dimension 1. So for irreducible σ of dimension at least 2 the space is zero and the determinant is 1.

The definition of the local constant is far more involved. It is specified axiomatically by its behavior under induction.

Theorem 6.3.2: Existence of the Langlands-Deligne Local Constant

Let $\psi \in \widehat{F}, \psi \neq 1$. Let E/F range over finite extensions inside $\bar{F}$, and let $\psi_E = \psi \circ \mathrm{tr}_{E/F}$. There exists a unique family of functions

$$\mathcal{G}^{\mathrm{ss}}(E) \to \mathbb{C}[q_E^s, q_E^{-s}]^\times, \qquad \rho \mapsto \varepsilon(\rho, s, \psi_E),$$

satisfying the following properties:

1. **Dimension 1:** If χ is a character of $E^\times$, then $\varepsilon(\chi \circ \mathbf{a}_E, s, \psi_E) = \varepsilon(\chi, s, \psi_E)$.
2. **Additivity:** If $\rho_1, \rho_2 \in \mathcal{G}^{\mathrm{ss}}(E)$, then
$$\varepsilon(\rho_1 \oplus \rho_2, s, \psi_E) = \varepsilon(\rho_1, s, \psi_E)\varepsilon(\rho_2, s, \psi_E)$$
3. **Inductivity:** If $\rho \in \mathcal{G}_n^{\mathrm{ss}}(E)$ and $E \supset K \supset F$, then
$$\varepsilon(\mathrm{Ind}_{E/K}\rho, s, \psi_K) = \varepsilon(\rho, s, \psi_E)\left(\frac{\varepsilon(R_{E/K}, s, \psi_K)}{\varepsilon(1_E, s, \psi_E)}\right)^n, \tag{6.3}$$
where $R_{E/K} = \mathrm{Ind}_{E/K} 1_E$ is the induced representation of definition 6.2.5.

The goal for the rest of this section is to establish this theorem. The uniqueness half is local and is proved here in full; the existence half rests on one global input, isolated as theorem 6.3.7 and cited there.

***Proof Key ideas*:**
Uniqueness. Properties (1) to (3) say precisely that $\rho \mapsto \varepsilon(\rho, s, \psi)$ is an induction constant in the sense of definition 6.3.4, whose boundary is the division $(E, \chi) \mapsto \varepsilon(\chi, s, \psi_E)$ fixed by property (1). By lemma 6.3.5 an induction constant is determined by its boundary, so at most one such family exists. This half uses only Brauer induction and is complete.

Existence. By lemma 6.3.6 it suffices to produce, for each finite Galois L/F, an induction constant on $\mathrm{Gal}(L/F)$ with the prescribed boundary, that is, to show the division $D^\psi_{L/F}$ is pre-inductive. That is theorem 6.3.7, and it is the step requiring global methods; lemma 6.3.8 gives the local input it needs. Granting it, the constants glue over the tower of finite Galois extensions and the family exists.

The dependence on ψ is then removed by the argument at the end of this section, which multiplies the division $D^\psi_{L/F}$ by the explicit pre-inductive division $(E, \chi) \mapsto \chi(a)\|a\|_E^{s-1/2}$ attached to $a \in F^\times$, so a family constructed for one $\psi \neq 1$ determines one for every other.

Before turning to the machinery, here are the properties the rest of the book uses. The removal of the restriction on ψ just described is deliberately not one of them. Proposition 6.3.3(2) below records the same dependence, but its own proof is allowed to assume the existence theorem, so it cannot be the step that completes it.

Proposition 6.3.3: Properties of the Local Constant

Let $\psi \in \widehat{F}, \psi \neq 1$, and let $\rho \in \mathcal{G}^{\mathrm{ss}}(F)$.

1. There is an integer $n(\rho, \psi)$ such that
$$\varepsilon(\rho, s, \psi) = q^{n(\rho,\psi)(1/2-s)}\varepsilon(\rho, 1/2, \psi)$$
2. Let $a \in F^\times$. Then:
$$\varepsilon(\rho, s, a\psi) = \det \rho(a)\|a\|^{\dim(\rho)(s-1/2)}\varepsilon(\rho, s, \psi)$$
Consequently, $n(\rho, a\psi) = n(\rho, \psi) + v_F(a) \dim \rho$.
3. The local constants satisfy the functional equation
$$\varepsilon(\rho, s, \psi)\varepsilon(\check{\rho}, 1 - s, \psi) = \det \rho(-1)$$
4. There is an integer n_ρ such that, if χ is a character of $F^\times$ of level $k \geq n_\rho$, then
$$\varepsilon(\chi \otimes \rho, s, \psi) = \det \rho(c(\chi))^{-1}\varepsilon(\chi, s, \psi)^{\dim \rho},$$
for any $c(\chi) \in F^\times$ such that $\chi(1 + x) = \psi(c(\chi)x)$ for $x \in \mathfrak{p}^{[k/2]+1}$.

***Proof*:**
Granting theorem 6.3.2, every assertion reduces to the one-dimensional case already settled in section 5.1. The reduction is the same each time and worth stating once. By the Brauer induction theorem, as used in lemma 6.3.5, the class of ρ minus $\dim \rho$ copies of the trivial representation is a $\mathbb{Z}$-linear combination of terms $\mathrm{Ind}_{H_i}^{H}\chi_i - 1_{H_i}$ with each χ_i a character. Since ε is additive (axiom (2)) and inductive in degree zero (axiom (3)), any identity that holds for characters and is multiplicative in the appropriate sense propagates from those generators to ρ.

1. By theorem 6.3.2 the value $\varepsilon(\rho, s, \psi)$ lies in $\mathbb{C}[q^s, q^{-s}]^\times$. The units of that ring are exactly the monomials cq^{-ms} with $c \in \mathbb{C}^\times$ and $m \in \mathbb{Z}$, so writing the monomial in the form $q^{n(1/2-s)}$ times its value at $s = 1/2$ defines the integer $n(\rho, \psi)$ and gives the stated shape.
2. For a character χ this is lemma 5.1.16, which gives $\varepsilon(\chi, s, a\psi) = \chi(a)\|a\|^{s-1/2}\varepsilon(\chi, s, \psi)$. Both sides of the asserted identity are multiplicative under direct sums, and the correction factor $\det \rho(a)\|a\|^{\dim \rho(s-1/2)}$ is multiplicative in ρ because both det and dim are additive on direct sums; the induction step is compatible because replacing ψ by $a\psi$ replaces ψ_E by $(a\psi)_E = a\psi_E$ for $a \in F^\times$. So the character case propagates, and specializing to $\dim \rho = 1$ recovers the familiar form. The consequence for $n(\rho, a\psi)$ follows by comparing the exponents of q on the two sides, using $\|a\| = q^{-v_F(a)}$.
3. For a character this is corollary 5.1.14, which reads $\varepsilon(\chi, s, \psi)\varepsilon(\check{\chi}, 1 - s, \psi) = \chi(-1)$. Again both sides are multiplicative under direct sums, and $\det \rho(-1)$ is multiplicative in ρ, so the identity propagates from characters to all of $\mathcal{G}^{\mathrm{ss}}(F)$.
4. This is the Galois-side stability statement, and for a character it is theorem 5.1.21: if θ has level l and χ has level $k > 2l$ then $\varepsilon(\theta\chi, s, \psi) = \theta(c(\chi))^{-1}\varepsilon(\chi, s, \psi)$. Take n_ρ larger

than twice the level of every character occurring in a Brauer decomposition of ρ, which is possible because that decomposition is finite; lemma 6.3.8 is what guarantees such a bound exists uniformly over the intermediate fields. For χ of level $k \geq n_\rho$ the character case applies term by term, and multiplying up gives $\varepsilon(\chi \otimes \rho, s, \psi) = \det \rho(c(\chi))^{-1} \varepsilon(\chi, s, \psi)^{\dim \rho}$, completing the proof.

The proof of the existence theorem itself requires more induction machinery, which is developed next in the abstract before being applied to Galois groups.

Let G be a profinite group. Let $K_0 G$ be the Grothendieck group of G, the free abelian group generated by isomorphism classes of irreducible smooth representations. Define $\mathcal{K}_0 G = \bigoplus_H K_0 H$, where H ranges over open subgroups of G. Let $\Gamma(G) = \bigoplus_H \mathrm{Hom}(H, \mathbb{C}^\times)$ be the set of pairs (H, χ) of open subgroups and characters.

Definition 6.3.4: Induction Constant and Division

Let A be an abelian group.

1. An *induction constant* on G is a function $\mathcal{F} : \mathcal{K}_0 G \to A$ such that:
   (a) For each open H, $\mathcal{F}|_{K_0 H}$ is a homomorphism.
   (b) If $H \subset J$ are open and ρ is a representation of H of dimension 0 (virtual), then $\mathcal{F}(J, \mathrm{Ind}_H^J \rho) = \mathcal{F}(H, \rho)$.
2. A *division* on G is a function $D : \Gamma(G) \to A$.

An induction constant defines a division $\partial\mathcal{F}$ by restriction; $\partial\mathcal{F}$ is called the *boundary* of $\mathcal{F}$. Conversely, a division D is *pre-inductive* if $D = \partial\mathcal{F}$ for some induction constant.

Lemma 6.3.5: Uniqueness of Induction Constants

Let $\mathcal{F}$ be an induction constant on G. The division $\partial\mathcal{F}$ determines $\mathcal{F}$ uniquely.

***Proof*:**
It suffices to treat the case where G is finite. Let ρ be an irreducible representation of $H \subset G$ of dimension $m \geq 1$. Since H is then finite, Brauer's induction theorem [18, Brauer's induction theorem] applies: the virtual characters induced from one-dimensional characters of subgroups of H generate the character ring of H over $\mathbb{Z}$. Subtracting m copies of the trivial character from both sides of such an expression for $[\rho]$, and noting that each $\mathrm{Ind}_{H_i}^H[\chi_i]$ contributes its own dimension, gives characters χ_i of subgroups $H_i \subset H$ and integers n_i with

$$[\rho] - m[1_H] = \sum_i n_i (\mathrm{Ind}_{H_i}^H[\chi_i] - [1_{H_i}]) \tag{6.4}$$

Writing A multiplicatively, axiom (b) of the definition implies:

$$\mathcal{F}(H, \rho) = \mathcal{F}(H, 1_H)^m \prod_i \left(\frac{\mathcal{F}(H_i, \chi_i)}{\mathcal{F}(H_i, 1_{H_i})} \right)^{n_i}$$

All terms on the RHS are values of the division $\partial\mathcal{F}$.

Furthermore, one can reduce to finite groups.

Lemma 6.3.6: Reduction to Finite Quotients

If $\mathcal{H}$ is a family of open normal subgroups such that $G \cong \varprojlim G/H$, and a division D restricts to a pre-inductive division on every quotient G/H, then $\overleftarrow{D}$ is pre-inductive on G.

***Proof*:**
For each $H \in \mathcal{H}$ let $\mathcal{F}_H$ be an induction constant on G/H with $\partial\mathcal{F}_H = D|_{G/H}$; by lemma 6.3.5 it is unique. Uniqueness gives compatibility: if $H' \subseteq H$ then inflating along $G/H' \to G/H$ carries $\mathcal{F}_{H'}$ to an induction constant on G/H with the same boundary, hence equal to $\mathcal{F}_H$.

Every smooth representation of G is trivial on some $H \in \mathcal{H}$, because G is the limit of the quotients and the representation is smooth; so $\mathcal{K}_0 G$ is the colimit of the groups $\mathcal{K}_0(G/H)$ along inflation. The compatible family $(\mathcal{F}_H)$ therefore defines a single function $\mathcal{F}$ on $\mathcal{K}_0 G$. Both axioms of definition 6.3.4 are checked one level at a time and hold on each G/H, so they hold for $\mathcal{F}$; and $\partial\mathcal{F} = D$ because that holds at every level, completing the proof.

This applies to the Galois group of a finite Galois extension L/F. Let $G = \mathrm{Gal}(L/F)$. Via Class Field Theory, $\Gamma(G)$ corresponds to pairs (E, χ) where $F \subset E \subset L$ and χ is a character of $E^\times$ trivial on $N_{L/E}(L^\times)$.

Theorem 6.3.7: Existence of Pre-Inductive Division

Let L/F be a finite Galois extension and $G = \mathrm{Gal}(L/F)$. There exists $\psi \in \widehat{F}, \psi \neq 1$, such that the division

$$D^{\psi}_{L/F} : (E, \chi) \mapsto \varepsilon(\chi, s, \psi_E)$$

is pre-inductive on G.

This is the one step of the construction that leaves the local world, and it is the reason theorem 6.3.2 is hard. The local constants of characters are defined by Tate's local functional equation, which knows nothing about how the fields E sit inside one another; producing an induction constant whose boundary is $D^{\psi}_{L/F}$ means finding a coherent system across all of them, and the known route is to realize L/F as the completion of a global extension and to use the functional equation of a global Hecke L-function, whose ε-factor is by construction a product over all places. The global machinery is outside this book's scope; the argument is sketched at the end of this section and the details are cited. See [2, §29-§30].

Before turning to it, a technical local lemma on levels is needed.

Lemma 6.3.8: Level Separation

Let L/F be a finite Galois extension. There exists an integer $n_{L/F} \geq 1$ such that if α is a character of $F^\times$ of level $n_{L/F}$ and $(E, \chi) \in \Gamma(G)$, the level of $\alpha \circ N_{E/F}$ is greater than twice the level of χ.

***Proof*:**

For each intermediate field E, there is a level $\ell(E)$ such that $U_E^{\ell(E)} \subset N_{L/E}(L^\times)$. If $(E,\chi) \in \Gamma(G)$, the level of χ is bounded by $\ell(E)$. Set $m(E) = 2\ell(E)+1$. Choose $n(E)$ such that $U_F^{n(E)} \subset N_{E/F}(U_E^{m(E)})$. The maximum of these $n(E)$ suffices.

Assuming theorem 6.3.7, the main results follow, and the restriction on ψ can be removed at the same time. First, the restriction on ψ is removed. By lemma 6.3.5, the division $\varepsilon(\chi,s,\psi_E)$ defines an induction constant $\varepsilon(\rho,s,\psi_E)$ on Ω_F. For $a \in F^\times$, consider the function

$$(E,\rho) \mapsto \det\rho(a)\|a\|^{(s-1/2)\dim\rho_E}$$

This is clearly an induction constant. Its boundary is the division $(E,\chi) \mapsto \chi(a)\|a\|_E^{s-1/2}$. Recall from lemma 5.1.16 that $\varepsilon(\chi,s,a\psi_E) = \chi(a)\|a\|_E^{s-1/2}\varepsilon(\chi,s,\psi_E)$. Since the product of pre-inductive divisions is pre-inductive, the division $(E,\chi) \mapsto \varepsilon(\chi,s,a\psi_E)$ is pre-inductive. Thus the theorem holds for all ψ.

Properties (1) and (2) of proposition 6.3.3 now hold for Galois representations. The extension to the Weil group uses the unramified twist.

Proposition 6.3.9: Unramified Twist of Epsilon Factor

Let $(\Omega_E,\rho) \in \mathcal{K}_0\Omega_F$, and let φ be an unramified character of $F^\times$ of finite order. Let $\varphi(\varpi) = q^{-s(\varphi)}$. Then $\varepsilon(\varphi_E\otimes\rho,s,\psi_E) = \varepsilon(\rho,s+s(\varphi),\psi_E)$.

***Proof*:**

Both sides define induction constants (as functions of ρ) with the same boundary on 1-dimensional characters. By uniqueness, they are equal.

Next, define the following:

Definition 6.3.10: Langlands Constant

Let E/F be a finite extension, let 1_E be the trivial character of $\mathcal{W}_E$ and set $R_{E/F} = \mathrm{Ind}_{E/F}1_E$. Define the *Langlands Constant*:

$$\lambda_{E/F}(s,\psi) = \frac{\varepsilon(R_{E/F},s,\psi)}{\varepsilon(1_E,s,\psi_E)}$$

Corollary 6.3.11: Langlands Constant Property

The function $\lambda_{E/F}(s,\psi)$ is constant in s.

***Proof*:**

Using the proposition above, $\lambda_{E/F}(s+s(\varphi),\psi) = \lambda_{E/F}(s,\psi)$. Since φ can be any character of finite order, this function is periodic with dense periods, and thus must be constant in s. We write $\lambda_{E/F}(\psi)$.

Three properties of $\lambda_{E/F}(\psi)$ are recorded here, because the comparisons of local constants carried out in chapter 11 rest on all three and none is available from definition 6.3.10 alone. The first evaluates the constant in the one case where it can be computed outright, the second says how it behaves in a tower, and the third bounds its order. In the notation of definition 6.2.5, (6.3) reads

$$\varepsilon(\mathrm{Ind}_{E/K}\rho, s, \psi_K) = \varepsilon(\rho, s, \psi_E)\,\lambda_{E/K}(\psi_K)^n, \qquad \rho \in \mathcal{G}_n^{\mathrm{ss}}(E) \tag{6.5}$$

which is the form used throughout what follows.

Proposition 6.3.12: Langlands Constant of an Unramified Extension

Let K/F be unramified of degree d and let $\psi \in \widehat{F}$ have level one. Then

$$\lambda_{K/F}(\psi) = (-1)^{d-1}$$

In particular $\lambda_{K/F}(\psi) = 1$ when d is odd.

***Proof*:**
An unramified extension of degree d is cyclic, with $\mathrm{Gal}(K/F)$ generated by the Frobenius substitution. Under the isomorphism of theorem 6.2.1, the characters of $\mathcal{W}_F$ trivial on $\mathcal{W}_K$ correspond to the characters of $F^\times$ trivial on $N_{K/F}(K^\times)$, and for K/F unramified these are exactly the unramified characters μ with $\mu^d = 1$. Decomposing the permutation representation of proposition 6.2.6 into characters therefore gives

$$R_{K/F} = \bigoplus_{j=0}^{d-1} \mu^j$$

where μ is unramified with $\mu(\varpi) = \zeta$, a primitive d-th root of unity.

Because K/F is unramified its different is trivial, so $\psi_K = \psi \circ \mathrm{tr}_{K/F}$ again has level one and proposition 5.1.17 applies over both fields. Additivity of the local constant and proposition 5.1.17 give

$$\varepsilon(R_{K/F}, s, \psi) = \prod_{j=0}^{d-1} \varepsilon(\mu^j, s, \psi) = \prod_{j=0}^{d-1} q^{s-1/2}\mu^j(\varpi)^{-1} = q^{d(s-1/2)}\,\zeta^{-d(d-1)/2}$$

while over K, where the residue field has $q_K = q^d$ elements,

$$\varepsilon(1_K, s, \psi_K) = q_K^{s-1/2} = q^{d(s-1/2)}$$

Dividing, $\lambda_{K/F}(\psi) = \zeta^{-d(d-1)/2}$.

It remains to evaluate that root of unity. If d is odd then $(d-1)/2$ is an integer and $d(d-1)/2 = d\cdot\frac{d-1}{2}$ is a multiple of d, so $\zeta^{-d(d-1)/2} = 1$. If d is even then $\zeta^{d/2} = -1$, and $d(d-1)/2 = \frac{d}{2}(d-1)$ gives $\zeta^{-d(d-1)/2} = (\zeta^{d/2})^{-(d-1)} = (-1)^{d-1} = -1$, since $d-1$ is odd. Both cases are summarized by $(-1)^{d-1}$, as we sought to show.

The constant is therefore invisible in odd unramified degree, which is exactly the case chapter 11 needs: the descent there runs along a cubic extension, and a cubic unramified step contributes nothing to the comparison.

Proposition 6.3.13: Tower Formula for the Langlands Constant

Let $F \subset K \subset L$ be finite separable extensions. Then

$$\lambda_{L/F}(\psi) = \lambda_{L/K}(\psi_K)\,\lambda_{K/F}(\psi)^{[L:K]}$$

***Proof*:**
Induction in stages gives $R_{L/F} = \mathrm{Ind}_{L/F}1_L = \mathrm{Ind}_{K/F}\big(\mathrm{Ind}_{L/K}1_L\big) = \mathrm{Ind}_{K/F}R_{L/K}$. Apply (6.5) to the extension K/F and the representation $\rho = R_{L/K} \in \mathcal{G}^{\mathrm{ss}}_{[L:K]}(K)$:

$$\varepsilon(R_{L/F}, s, \psi) = \varepsilon(R_{L/K}, s, \psi_K)\,\lambda_{K/F}(\psi)^{[L:K]}$$

By definition 6.3.10 the two ε-factors appearing here are $\varepsilon(R_{L/F}, s, \psi) = \lambda_{L/F}(\psi)\,\varepsilon(1_L, s, \psi_L)$ and $\varepsilon(R_{L/K}, s, \psi_K) = \lambda_{L/K}(\psi_K)\,\varepsilon(1_L, s, \psi_L)$. Substituting both and cancelling $\varepsilon(1_L, s, \psi_L)$, which is a unit in $\mathbb{C}[q^s, q^{-s}]$ by theorem 6.3.2, completes the proof.

Corollary 6.3.14: The Langlands Constant Is a Fourth Root of Unity

For any finite separable E/F and any $\psi \in \widehat{F}$ with $\psi \neq 1$,

$$\lambda_{E/F}(\psi)^2 = \varkappa_{E/F}(-1) \tag{6.6}$$

In particular $\lambda_{E/F}(\psi)^4 = 1$.

***Proof*:**
The character of $R_{E/F}$ counts fixed points on the coset space $\mathcal{W}_F/\mathcal{W}_E$, and g and g^{-1} fix the same cosets, so that character takes the same value at g and at g^{-1}. The character of $\check{R}_{E/F}$ is $g \mapsto \mathrm{tr}R_{E/F}(g^{-1})$, so the two characters agree. Both representation $R_{E/F}$ is semisimple by lemma 6.1.13(2), hence so is its contragredient, and a semisimple representation is determined by its character, so $\check{R}_{E/F} \cong R_{E/F}$.

Apply proposition 6.3.3(3) twice. Over F, to $\rho = R_{E/F}$, using the self-duality just established and $\det R_{E/F} = \varkappa_{E/F}$:

$$\varepsilon(R_{E/F}, s, \psi)\,\varepsilon(R_{E/F}, 1-s, \psi) = \varkappa_{E/F}(-1)$$

Over E, to $\rho = 1_E$, whose determinant is trivial:

$$\varepsilon(1_E, s, \psi_E)\,\varepsilon(1_E, 1-s, \psi_E) = 1$$

Dividing the first display by the second and using definition 6.3.10 on each factor gives $\lambda_{E/F}(s, \psi)\,\lambda_{E/F}(1-s, \psi) = \varkappa_{E/F}(-1)$. By corollary 6.3.11 both factors equal $\lambda_{E/F}(\psi)$, which is (6.6). Finally $\varkappa^2_{E/F} = 1$ by proposition 6.2.6, so $\varkappa_{E/F}(-1) = \pm 1$ and $\lambda_{E/F}(\psi)^4 = 1$, as we sought to show.

The last statement is what makes the congruences of chapter 11 possible at all: a Langlands constant is a fourth root of unity, so in residual characteristic 2 it lies in the group $\mu_{\mathbb{C}}(2^\infty)$ of roots of unity of

2-power order, and any comparison of local constants taken modulo that group may ignore it entirely.

Now let ρ be an irreducible representation of $\mathcal{W}_E$. There is an unramified character φ for which $\varphi_E \otimes \rho$ factors through a representation ρ_0 of the Galois group Ω_E (see proposition 6.1.10). Define:

$$\varepsilon(\rho, s, \psi_E) = \varepsilon(\rho_0, s - s(\varphi), \psi_E)$$

This definition is well-defined and satisfies the inductive properties.

The remaining properties of proposition 6.3.3 follow.

- **Functional Equation (3):** The maps $\tau \mapsto \varepsilon(\tau, s, \psi)\varepsilon(\check{\tau}, 1-s, \psi)$ and $\tau \mapsto \det\tau(-1)$ are both induction constants. Their boundaries are equal by the functional equation for GL(1) (corollary 5.1.14). Thus the maps are equal.
- **Stability (4):** It suffices to consider ρ irreducible. Use the Brauer relation $[\rho] - m[1_F] = \sum(\mathrm{Ind}\varphi_i - 1)$, which gives

  $$\frac{\varepsilon(\chi \otimes \rho, s, \psi)}{\varepsilon(\chi, s, \psi)^m} = \prod \frac{\varepsilon(\chi_{E_i}\varphi_i, s, \psi_{E_i})}{\varepsilon(\chi_{E_i}, s, \psi_{E_i})}$$

  If χ has sufficiently large level (using lemma 6.3.8), the Stability Theorem for GL(1) (theorem 5.1.21) applies to each term in the product. The RHS becomes $\prod \varphi_i(c(\chi))^{-1}$. Taking the determinant of the Brauer relation yields $\det\rho = \prod \varphi_i \circ \mathrm{ver}$, which implies the result.

What remains is the global input. The following sketches the proof of theorem 6.3.7. This requires global methods (see [14] for the number field case). Let K be a global field and $\mathbb{A}_K$ its adele ring. For a representation ρ of Ω_K, the global Artin L-function is defined as the product of local L-functions $L(\rho, s) = \prod_v L(\rho_v, s)$. This satisfies a global functional equation $L(\rho, s) = \varepsilon(\rho, s)L(\check{\rho}, 1-s)$, where the global epsilon factor is the product of local factors $\varepsilon(\rho, s) = \prod_v \varepsilon(\rho_v, s, \psi_v)$. Critically, $\varepsilon(\mathrm{Ind}_{L/K}\tau, s) = \varepsilon(\tau, s)$.

Lemma 6.3.15: Properties of Global L-functions

1. Let L/K be a finite extension, $L \subset \bar{K}$, and let τ be a smooth representation of $\Omega_L = \mathrm{Gal}(\bar{K}/L)$ of finite dimension. Then

   $$L(\mathrm{Ind}_{L/K}\tau, s) = L(\tau, s)$$

2. Let ρ be a finite-dimensional smooth representation of Ω_K. The product defining the global L-function converges in some half-plane and admits analytic continuation to a meromorphic function on the whole s-plane. It satisfies a functional equation

   $$L(\rho, s) = \varepsilon(\rho, s)L(\check{\rho}, 1-s), \tag{6.7}$$

   for a uniquely determined exponential function $\varepsilon(\rho, s)$ having neither zeros nor poles.

3. In the situation of (1),

   $$\varepsilon(\mathrm{Ind}_{L/K}\tau, s) = \varepsilon(\tau, s) \tag{6.8}$$

***Proof*:**
Part (1) is given by an elementary argument, based on the definitions of the local factors (which align with the definition of Artin L-functions). Part (2) then follows from the Brauer Induction Theorem (as in lemma 6.3.5) and the known case for 1-dimensional characters (Hecke L-functions). Part (3) is an immediate consequence of (1) and the functional equation (2).

Lemma 6.3.16: Global Realization

There exists a finite Galois extension $\mathcal{L}/\mathcal{F}$ of global fields and a place v_0 of $\mathcal{F}$ such that:

1. The completion $\mathcal{F}_{v_0} \cong F$.
2. There is a unique place u_0 of $\mathcal{L}$ above v_0, and $\mathcal{L}_{u_0} \cong L$ (the local field).
3. $\mathcal{F}$ has no real embeddings.

***Proof*:**
This follows from the Weak Approximation Theorem. If $\mathrm{char}F = 0$, we approximate local polynomials to ensure the local behavior at v_0 and full decomposition at real places.

Define local integers n_v for places $v \neq v_0$ based on the ramification of $\mathcal{L}/\mathcal{F}$.

Lemma 6.3.17: Existence of Global Character

By the Grunwald-Wang Theorem, there exists a global character α of the idele class group $C_{\mathcal{F}}$ such that:

1. $\alpha_{v_0} = 1$, and
2. the level l_v of α_v satisfies $l_v \geq n_v$, for every non-Archimedean $v \neq v_0$.

Finally:

***Proof*:**
(of theorem 6.3.7 and theorem 6.3.2)
Let Ψ be a global additive character. Set $\psi = \Psi_{v_0}$. For any (E, χ) in the local Galois group $\mathrm{Gal}(L/F)$, we lift it to the global setting (using lemma 6.3.16). We analyze the product formula for the global ε-factor:

$$\varepsilon(\chi\alpha_E, s) = \prod_w \varepsilon((\chi\alpha_E)_w, s, \Psi_w)$$

We analyze the factors at the different places w:

- At w_0 (the unique place above v_0): We constructed α such that $\alpha_{v_0} = 1$. Thus, this factor is simply $\varepsilon(\chi, s, \psi_E)$, which is the local term we wish to define.
- At $w \neq w_0$: The level of α is sufficiently high (by construction in lemma 6.3.17). We apply the 1-dimensional Stability Theorem (theorem 5.1.21) to substitute these factors with terms depending only on α_E, effectively removing χ.

We now rearrange the global functional equation. By lemma 6.3.15(3), the global epsilon factor is

inductive: $\varepsilon(\mathrm{Ind}(\chi\alpha)_E) = \varepsilon((\chi\alpha)_E)$. We can therefore express the unknown local term $\varepsilon(\chi, s, \psi_E)$ as a quotient of terms that are known to be inductive (global epsilon factors, character values, and local constants of α):

$$\varepsilon(\chi, s, \psi_E) = \frac{\varepsilon(\mathrm{Ind}(\chi\alpha_E))}{\prod_{w \neq w_0} \varepsilon((\chi\alpha_E)_w, s, \Psi_w)}$$

Since the right-hand side is built from inductive terms, the local division $D^{\psi}_{L/F}$ is pre-inductive. By lemma 6.3.5 and lemma 6.3.6, this uniquely defines the induction constant for the local field, proving the existence theorem.

6.4 Deligne Representations

We now encounter an interesting friction in the theory due to our choice to work with *complex* representations. Our goal is to relate representations of $G = \mathrm{GL}_2(F)$ to representations of the Weil group $\mathcal{W}_F$. On the group side, we have encountered representations like the Steinberg representation St_G, which is an irreducible quotient of a reducible principal series. On the Galois side, however, representations of $\mathcal{W}_F$ are often constructed from geometry (for example étale cohomology), naturally appearing as representations on $\overline{\mathbb{Q}}_\ell$-vector spaces rather than $\mathbb{C}$.

These ℓ-adic representations exhibit behaviour, specifically regarding continuity and image size, that does not align with the smooth complex representations we have studied so far. To resolve this, and to ensure our theory is as independent of the coefficient field as possible, we introduce *Deligne Representations*. These objects decouple the "semisimple part" of the action from the "monodromy" (nilpotent) part, allowing us to classify representations over $\mathbb{C}$ that correspond to continuous representations over $\overline{\mathbb{Q}}_\ell$.

6.4.1 Definition and Algebraic Structure

A Deligne representation enriches a smooth representation with a nilpotent operator that tracks how the norm scales under conjugation.

Definition 6.4.1: Deligne Representation

A *Deligne representation* of $\mathcal{W}_F$ is a triple $(\rho, V, \mathfrak{n})$, where:

1. (ρ, V) is a finite-dimensional, smooth representation of $\mathcal{W}_F$ on a complex vector space V.
2. $\mathfrak{n} \in \mathrm{End}_{\mathbb{C}}(V)$ is a nilpotent endomorphism.
3. The operators satisfy the compatibility relation:

$$\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}, \quad \forall w \in \mathcal{W}_F \tag{6.9}$$

A Deligne representation is called *semisimple* if the underlying smooth representation ρ is semisimple. Write $\mathcal{G}_n(F)$ for the set of equivalence classes of n-dimensional, semisimple Deligne representations.

Note that one can identify the "ordinary" smooth representations $\mathcal{G}_n^{\mathrm{ss}}(F)$ with the subset of triples where $\mathfrak{n} = 0$.

Example 6.4.2: The Special Representation $Sp(n)$

Let $V = \mathbb{C}^n$ with basis $v_0, \dots, v_{n-1}$. Let $\mathfrak{n}$ be the standard principal nilpotent element (Jordan block): $\mathfrak{n}v_{n-1} = 0$ and $\mathfrak{n}v_i = v_{i+1}$. Define a smooth representation $\rho_0(w)v_i = \|w\|^i v_i$, and set $\rho(w) = \|w\|^{(1-n)/2}\rho_0(w)$. The triple $(\rho, \mathbb{C}^n, \mathfrak{n})$ is a semisimple Deligne representation denoted $Sp(n)$ (not to be confused with the spin group!)

The standard operations on representations must be adapted to the operator $\mathfrak{n}$. These definitions are chosen to mirror the behaviour of ℓ-adic representations (as will see in section 6.4.2).

Definition 6.4.3: Operations on Deligne Representations

Let $\sigma_1 = (\rho_1, V_1, \mathfrak{n}_1)$ and $\sigma_2 = (\rho_2, V_2, \mathfrak{n}_2)$ be Deligne representations.

1. **Direct Sum:** $\sigma_1 \oplus \sigma_2 = (\rho_1 \oplus \rho_2, V_1 \oplus V_2, \mathfrak{n}_1 \oplus \mathfrak{n}_2)$.
2. **Tensor Product:** $\sigma_1 \otimes \sigma_2 = (\rho_1 \otimes \rho_2, V_1 \otimes V_2, \mathfrak{n}_1 \otimes 1 + 1 \otimes \mathfrak{n}_2)$.
3. **Dual:** $\check{\sigma}_1 = (\check{\rho}_1, \check{V}_1, -\check{\mathfrak{n}}_1)$.

6.4.4 Note: Classification of the Indecomposables The special representations account for all of the indecomposable semisimple Deligne representations: every one is of the form $\rho \otimes Sp(n)$ for some $n \geq 1$ and some $\rho \in \mathcal{G}_m^0(F)$. The mechanism is that the relation $\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}$ makes $\mathfrak{n}$ shift the $\|-\|$-grading by one step, so a semisimple ρ forces V to break into $\mathfrak{n}$-strings, and indecomposability leaves exactly one.

6.4.2 Relation with ℓ-adic Representations

We now motivate the definition above. Let ℓ be a prime with $\ell \neq p$. In nature (arithmetic geometry), representations of $\mathcal{W}_F$ act on vector spaces over $\overline{\mathbb{Q}}_\ell$. These are continuous in the ℓ-adic topology, but almost never smooth (the image of inertia is usually infinite).

Fix a continuous surjection $t_\ell : \mathcal{I}_F \to \mathbb{Z}_\ell$ (this is unique up to a unit in $\mathbb{Z}_\ell^\times$). It satisfies the conjugation relation $t_\ell(gxg^{-1}) = \|g\| t_\ell(x)$.

Theorem 6.4.5: Grothendieck's Monodromy Theorem

Let (σ, V) be a finite-dimensional, continuous representation of $\mathcal{W}_F$ over $\overline{\mathbb{Q}}_\ell$. There exists a unique nilpotent endomorphism $\mathfrak{n}_\sigma \in \mathrm{End}(V)$ such that, for x in some open subgroup of $\mathcal{I}_F$:

$$\sigma(x) = \exp(t_\ell(x)\mathfrak{n}_\sigma)$$

***Proof*:**
Uniqueness is straightforward. For existence, we view σ as a homomorphism into $\mathrm{GL}_d(\overline{\mathbb{Q}}_\ell)$. The kernel of t_ℓ has pro-order prime to ℓ, so its image must be finite (and eventually trivial in the pro-ℓ subgroups of GL_d). Thus σ factors through t_ℓ on an open subgroup. The image of inertia must satisfy $\sigma(\Phi h \Phi^{-1})^q = \sigma(h)$. This forces the eigenvalues of $\sigma(h)$ to be roots of unity that are permuted by the q-power map. Since the image is in a pro-ℓ group, we use the following lemma:

Lemma 6.4.6: Integrality of Roots

If $\alpha \in \overline{\mathbb{Q}}_\ell$ is a root of unity such that $(\alpha - 1)/\ell^2$ is integral, then $\alpha = 1$.

***Proof*:**
If α has order ℓ^r, the valuation of $\alpha - 1$ is $1/(\ell^r - \ell^{r-1})$, which is too small. If the order is prime to ℓ, $\alpha - 1$ is a unit.

Using the matrix logarithm (which converges on the pro-ℓ group), we define $\mathfrak{n} = t_\ell(h_0)^{-1} \log \sigma(h_0)$ for a suitable h_0.

This theorem strips away the non-smooth part of the representation. Define a new representation σ_Φ by:

$$\sigma_\Phi(\Phi^a x) = \sigma(\Phi^a x) \exp(-t_\ell(x)\mathfrak{n}_\sigma), \quad a \in \mathbb{Z}, x \in \mathcal{I}_F$$

This representation σ_Φ is trivial on an open subgroup of inertia, hence *smooth*. The triple $(\sigma_\Phi, V, \mathfrak{n}_\sigma)$ is a Deligne representation over $\overline{\mathbb{Q}}_\ell$.

Theorem 6.4.7: Equivalence of Categories (ℓ-adic to Deligne)

Let $\mathrm{Rep}^f_{\overline{\mathbb{Q}}_\ell}(\mathcal{W}_F)$ be the category of finite-dimensional continuous ℓ-adic representations. The functor

$$(\sigma, V) \longmapsto (\sigma_\Phi, V, \mathfrak{n}_\sigma)$$

is an equivalence of categories $\mathrm{Rep}^f_{\overline{\mathbb{Q}}_\ell}(\mathcal{W}_F) \xrightarrow{\sim} \text{D-Rep}_{\overline{\mathbb{Q}}_\ell}(\mathcal{W}_F)$.

***Proof*:**
We sketched the construction of the object. The inverse functor is given by $\tau(g) = \tau_\Phi(g) \exp(t_\ell(x)\mathfrak{n})$. The check that this is well-defined and functorial relies on the conjugation relation $\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}$, which corresponds exactly to the relation satisfied by the Monodromy operator.

Finally, the notion of semisimplicity needs clarifying.

Proposition 6.4.8: Semisimplicity of ℓ-adic Representations

Let (σ, V) be a continuous ℓ-adic representation. The following are equivalent:

1. The associated Deligne representation is semisimple (i.e., σ_Φ is semisimple).
2. There is a Frobenius element Ψ such that $\sigma(\Psi)$ is semisimple.
3. $\sigma(g)$ is semisimple for every $g \in \mathcal{W}_F \setminus \mathcal{I}_F$.

***Proof*:**
Write $(\rho, V, \mathfrak{n})$ for the Deligne representation attached to σ by theorem 6.4.7, so that $\sigma(g) = \rho(g)\exp(t_\ell(x)\mathfrak{n})$ for $g = \Phi^a x$ in the notation of that theorem, and $\rho = \sigma_\Phi$.

A conjugation. The whole proof rests on one computation: for $g \notin \mathcal{I}_F$ the operator $\sigma(g)$ is conjugate to $\rho(g)$. Write $g = \Phi^a x$, put $N = t_\ell(x)\mathfrak{n}$ and $c = \|g\|$, which is not 1 precisely because g lies outside inertia. The relation (6.9) gives $\rho(g)N\rho(g)^{-1} = cN$, hence $\rho(g)^{-1}N\rho(g) = c^{-1}N$ and therefore $\rho(g)^{-1}\exp(-\lambda N)\rho(g) = \exp(-\lambda c^{-1}N)$ for every scalar λ. Since every exponential appearing here is a polynomial in the single nilpotent N, they all commute with one another. Setting $u = \exp(\lambda N)$,

$$u^{-1}\sigma(g)\,u = \exp(-\lambda N)\rho(g)\exp(N)\exp(\lambda N) = \rho(g)\exp\left((1+\lambda-\lambda c^{-1})N\right)$$

Choosing $\lambda = -c/(c-1)$, which is legitimate as $c \neq 1$, makes the exponent vanish and gives $u^{-1}\sigma(g)u = \rho(g)$. So $\sigma(g)$ is semisimple if and only if $\rho(g)$ is, for every $g \in \mathcal{W}_F \setminus \mathcal{I}_F$.

That reduces all three conditions to statements about ρ, which proposition 6.1.15 already settles. That proposition is stated over $\mathbb{C}$, but its proof uses only that the coefficient field is algebraically closed of characteristic zero, so it applies verbatim over $\overline{\mathbb{Q}}_\ell$.

(1) implies (3). If $\rho = \sigma_\Phi$ is semisimple then $\rho(g)$ is semisimple for every $g \in \mathcal{W}_F$ by proposition 6.1.15(3), so $\sigma(g)$ is semisimple for $g \notin \mathcal{I}_F$.

(3) implies (2). Immediate: a Frobenius element lies in $\mathcal{W}_F \setminus \mathcal{I}_F$.

(2) implies (1). If $\sigma(\Psi)$ is semisimple for a Frobenius element Ψ, the conjugation makes $\rho(\Psi)$ semisimple, and proposition 6.1.15, in the implication (2) $\Rightarrow$ (1), gives that ρ is semisimple, completing the proof.

Such representations are called Φ-semisimple.

Corollary 6.4.9: Bijection of Classes

The isomorphism classes of n-dimensional Φ-semisimple representations over $\overline{\mathbb{Q}}_\ell$ are in canonical bijection with isomorphism classes of n-dimensional semisimple Deligne representations over $\mathbb{C}$ (once an isomorphism $\mathbb{C} \cong \overline{\mathbb{Q}}_\ell$ is chosen).

***Proof*:**
Theorem 6.4.7 already gives an equivalence between continuous ℓ-adic representations of $\mathcal{W}_F$ and Deligne representations over $\overline{\mathbb{Q}}_\ell$, and it preserves dimension since the underlying space is unchanged. By proposition 6.4.8 that equivalence matches the Φ-semisimple objects on one side

with the semisimple Deligne representations on the other, so it restricts to a bijection on those classes.

It remains to transport from $\overline{\mathbb{Q}}_\ell$ to $\mathbb{C}$. A Deligne representation is a triple $(\rho, V, \mathfrak{n})$ with ρ smooth, so ρ has open kernel and its matrix entries generate a finite extension of $\mathbb{Q}$; the datum is therefore defined over a field embeddable in both $\mathbb{C}$ and $\overline{\mathbb{Q}}_\ell$, and a choice of isomorphism $\mathbb{C} \cong \overline{\mathbb{Q}}_\ell$ carries triples to triples, preserving semisimplicity and dimension. Hence the classes correspond, as we sought to show.

The bijection depends on the chosen isomorphism, which is exactly the ambiguity that theorem 7.4.2 removes by twisting.

6.4.3 L-functions and Epsilon-factors

We extend the definitions from section 6.3 to Deligne representations. The presence of the operator $\mathfrak{n}$ modifies the L-function by restricting to the kernel of the monodromy.

Let $\sigma = (\rho, V, \mathfrak{n})$ be a semisimple Deligne representation. Let $V_\mathfrak{n} = \ker(\mathfrak{n})$. This subspace is invariant under ρ (due to the relation (6.9)). We define:

$$L(\sigma, s) = L(\rho|_{V_\mathfrak{n}}, s) \tag{6.10}$$

For the epsilon factor, the two L-factors of σ take the place of those of ρ:

$$\varepsilon(\sigma, s, \psi) = \varepsilon(\rho, s, \psi)\, \frac{L(\check{\rho}, 1-s)}{L(\rho, s)}\, \frac{L(\sigma, s)}{L(\check{\sigma}, 1-s)} \tag{6.11}$$

where $\check{\sigma} = (\check{\rho}, \check{V}, -\check{\mathfrak{n}})$ is the dual of definition 6.4.3, so that $L(\check{\sigma}, 1-s) = L(\check{\rho}|_{\ker \check{\mathfrak{n}}}, 1-s)$. Rearranged as

$$\varepsilon(\sigma, s, \psi)\, \frac{L(\check{\sigma}, 1-s)}{L(\sigma, s)} = \varepsilon(\rho, s, \psi)\, \frac{L(\check{\rho}, 1-s)}{L(\rho, s)}$$

the definition says that the monodromy operator does not change the γ-factor, and moves only the L-factor. That is the Galois-side counterpart of box 5.4.4, where φ_G and $\varphi\mathrm{St}_G$ were found to have the same γ and different L.

Parts (2) and (3) of proposition 6.3.3 are stated for semisimple representations of $\mathcal{W}_F$, and they carry over verbatim to $\mathcal{G}_n(F)$, which is how they are used below. The correction factors in (6.11) involve no ψ, so the dependence of $\varepsilon(\sigma, s, a\psi)$ on a is that of $\varepsilon(\rho, s, a\psi)$, and $\det \sigma = \det \rho$, $\dim \sigma = \dim \rho$. In the product $\varepsilon(\sigma, s, \psi)\varepsilon(\check{\sigma}, 1-s, \psi)$ those same factors appear once each way up and cancel, leaving $\det \rho(-1) = \det \sigma(-1)$.

Example 6.4.10: L-function of $Sp(2)$

Consider $\sigma = Sp(2)$. Here $\rho = \|\cdot\|^{-1/2} \oplus \|\cdot\|^{1/2}$. The kernel of $\mathfrak{n}$ corresponds to the subrepresentation $\|\cdot\|^{1/2}$. Let $\zeta_F(s) = (1-q^{-s})^{-1}$. Then:

$$L(Sp(2), s) = L(\|\cdot\|^{1/2}, s) = \zeta_F(s + 1/2)$$

In contrast, $L(\rho, s) = \zeta_F(s-1/2)\zeta_F(s+1/2)$. The Monodromy operator "kills" the pole at $s = 1/2$. Now the epsilon factor, assuming ψ has level 1. Proposition 5.1.17 gives $\varepsilon(\rho, s, \psi) = q^{s-1/2}q^{-1/2} \cdot q^{s-1/2}q^{1/2} = q^{2s-1}$ and $L(\check{\rho}, 1-s) = \zeta_F(3/2-s)\zeta_F(1/2-s)$. The kernel of $-\check{\mathfrak{n}}$ is the annihilator of $\mathrm{im}\mathfrak{n} = \mathbb{C}v_1$, namely $\mathbb{C}v_0^*$, on which $\check{\rho}$ acts by $\|\cdot\|^{1/2}$, so $L(\check{\sigma}, 1-s) = \zeta_F(3/2-s)$.

Then (6.11) reads

$$\varepsilon(Sp(2), s, \psi) = q^{2s-1} \frac{\zeta_F(3/2-s)\zeta_F(1/2-s)}{\zeta_F(s-1/2)\zeta_F(s+1/2)} \frac{\zeta_F(s+1/2)}{\zeta_F(3/2-s)} = q^{2s-1} \frac{\zeta_F(1/2-s)}{\zeta_F(s-1/2)} = -q^{s-1/2}$$

the last step because $\zeta_F(1/2-s)/\zeta_F(s-1/2) = (1-q^{1/2-s})/(1-q^{s-1/2}) = -q^{1/2-s}$. This is exactly the value $\varepsilon(\mathrm{St}_G, s, \psi) = -q^{s-1/2}$ that proposition 5.4.3 gives for the Steinberg representation of $\mathrm{GL}_2(F)$ at $\varphi = 1$.

7

The Langlands Correspondence

We have now assembled all the necessary components to state and prove the local Langlands correspondence for $\mathrm{GL}_2(F)$. This correspondence establishes a canonical bijection between two-dimensional representations of the Weil group $\mathcal{W}_F$ and irreducible smooth representations of $G = \mathrm{GL}_2(F)$. It is determined by the preservation of arithmetic invariants, specifically, L-functions and ε-factors, which we explored in chapter 5.

In this chapter, we first state the general correspondence. We then handle the "reducible" cases (where the Galois representation is a sum of characters) which correspond to the Principal Series representations. Then, we look at the irreducible cases. For odd residual characteristic ($p \neq 2$), we establish the correspondence explicitly using the method of Admissible Pairs from chapter 4. The dyadic case ($p = 2$) is significantly more involved. The existence theorem stated here covers it, and chapter 8 settles it on the parameters induced from a quadratic extension by a construction that never assumes tameness; what remains open there is the primitive part.

7.1 Statement of the Correspondence

Recall the notations for our categories of representations.

Definition 7.1.1: Representation Sets

Let F be a non-Archimedean local field.

1. Let $\mathcal{G}_2(F)$ denote the set of equivalence classes of 2-dimensional, semisimple, Deligne representations of the Weil group $\mathcal{W}_F$.
2. Let $\mathcal{A}_2(F)$ denote the set of equivalence classes of irreducible smooth representations of $G = \mathrm{GL}_2(F)$.

As usual, a character χ of $F^\times$ is identified via local class field theory with a character of $\mathcal{W}_F$, denoted $\chi \circ \mathbf{a}_F$ or simply χ.

Theorem 7.1.2: The Local Langlands Correspondence

Let $\psi \in \widehat{F}$ be a non-trivial additive character ($\psi \neq 1$). There exists a unique map:

$$\boldsymbol{\pi}_F : \mathcal{G}_2(F) \to \mathcal{A}_2(F)$$

such that for all $\rho \in \mathcal{G}_2(F)$ and all characters χ of $F^\times$:

$$L(\chi \otimes \boldsymbol{\pi}_F(\rho), s) = L(\chi \otimes \rho, s), \tag{7.1}$$
$$\varepsilon(\chi \otimes \boldsymbol{\pi}_F(\rho), s, \psi) = \varepsilon(\chi \otimes \rho, s, \psi) \tag{7.2}$$

Furthermore:

1. The map $\boldsymbol{\pi}_F$ is a bijection.
2. The defining relations hold for *all* $\psi \neq 1$.
3. The central character matches the determinant: $\omega_{\boldsymbol{\pi}_F(\rho)} = \det \rho$.
4. The correspondence commutes with twisting: $\boldsymbol{\pi}_F(\chi \otimes \rho) = \chi \boldsymbol{\pi}_F(\rho)$.
5. The correspondence commutes with contragredients: $\boldsymbol{\pi}_F(\check{\rho}) = \boldsymbol{\pi}_F(\rho)^\vee$.

***Proof*:**

Step 1: uniqueness. Suppose $\boldsymbol{\pi}_F$ and $\boldsymbol{\pi}'_F$ both satisfy the two displayed relations. Fix $\rho \in \mathcal{G}_2(F)$ and put $\pi = \boldsymbol{\pi}_F(\rho)$, $\pi' = \boldsymbol{\pi}'_F(\rho)$. Both are irreducible smooth representations of G, and for every character χ of $F^\times$ the relations give

$$L(\chi\pi, s) = L(\chi \otimes \rho, s) = L(\chi\pi', s), \qquad \varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi \otimes \rho, s, \psi) = \varepsilon(\chi\pi', s, \psi)$$

Write $\psi = a\psi'$ with ψ' of level one and $a \in F^\times$. By lemma 5.3.3 applied to $\chi\pi$ and $\chi\pi'$, whose central characters are $\chi^2\omega_\pi$ and $\chi^2\omega_{\pi'}$, the equality of ε-factors becomes $\omega_\pi(a)\,\varepsilon(\chi\pi, s, \psi') = \omega_{\pi'}(a)\,\varepsilon(\chi\pi', s, \psi')$ after cancelling the common factor $\chi(a)^2\|a\|^{2(s-1/2)}$. Now take χ of level $n \geq \max(m(\pi), m(\pi'))$ with parameter c as in theorem 5.3.10: both ε-factors evaluate to $\varepsilon(\chi, s, \psi')^2$ times $\omega_\pi(c)^{-1}$ and $\omega_{\pi'}(c)^{-1}$ respectively, so the character $\delta = \omega_\pi\omega_{\pi'}^{-1}$ satisfies $\delta(c) = \delta(a)$. Every c of valuation $-n$ arises from some χ of level n: the recipe $\chi(1+x) = \psi'(cx)$ defines a character of $1 + \mathfrak{p}^{[n/2]+1}$, and any extension to $F^\times$ has level n. So δ is constant on all elements of sufficiently negative valuation. Comparing c with uc for $u \in U_F$ and with ϖc

shows δ is trivial on U_F and at ϖ, so $\omega_\pi = \omega_{\pi'}$: the two representations have the same central character. Theorem 5.5.2 then gives $\pi \cong \pi'$. This is the step that was unavailable until the converse theorem was proved, and it is the whole content of uniqueness.

Step 2: the two halves are separately determined. By lemma 5.5.1 a representation $\pi \in \mathcal{A}_2(F)$ is cuspidal if and only if $L(\chi\pi, s) = 1$ for every χ. The Galois side has its own reason: for ρ irreducible of dimension two every twist $\chi \otimes \rho$ is again irreducible of dimension two, so $L(\chi \otimes \rho, s) = 1$ by the definition in section 6.3. For $\rho = (\chi_1 \oplus \chi_2, \mathfrak{n})$ the twist by χ_2^{-1} has $L = \zeta_F(s) \neq 1$. So the same criterion characterizes $\mathcal{G}_2^0(F)$. Since the defining relations equate the two families of L-functions, any map satisfying them restricts to maps $\mathcal{G}_2^1(F) \to \mathcal{A}_2^1(F)$ and $\mathcal{G}_2^0(F) \to \mathcal{A}_2^0(F)$.

Step 3: existence and bijectivity on the reducible half. This is theorem 7.2.1, which constructs a bijection $\mathcal{G}_2^1(F) \to \mathcal{A}_2^1(F)$ preserving both invariants under all twists.

Step 4: existence and bijectivity on the irreducible half. For $p \neq 2$ this is theorem 7.3.2, which uses the parametrization of both sides by admissible pairs together with the rectifier of definition 7.3.1. For $p = 2$ the construction is carried out in chapter 8 on the imprimitive parameters, by theorem 8.5.4, which assumes no tameness. The primitive dyadic parameters are not reached in this book. The four tools of chapter 11 are built for the Existence Theorem that reaches them, and the remaining assembly is carried out in [2, §50-§52]. Chapter 12 records the gap. The statement of the present theorem is made for all residue characteristics, and what its proof takes from the source is three things: the ε-comparison of [2, §34] for the tame half, the presentation of every $\rho \in \mathcal{G}_2^0(F)$ as an induced representation, and the whole primitive dyadic case. Only the last is confined to $p = 2$.

Step 5: the listed properties. Given the map, (1) is Steps 3 and 4 combined with Step 2. For (3), the central character and the determinant are both recovered from the ε-factors of twists by the same recipe, so they agree. Theorem 7.3.2 also checks it directly on the irreducible half. Property (4) holds because twisting acts compatibly on both sides in each of the three reducible families and, in the cuspidal case, because the parametrization by admissible pairs is twist-equivariant. Property (5) follows from (4) and (3) alone, using two rank-two identities. On the Galois side $\check{\rho} \cong \rho \otimes (\det \rho)^{-1}$ for ρ a two-dimensional smooth representation of $\mathcal{W}_F$, which is exercise 8.5.1 (4). For a Deligne representation the dual is $(\check{\rho}, \check{V}, -\check{\mathfrak{n}})$ of definition 6.4.3, and conjugating by a w with $\|w\| \neq 1$ carries $-\check{\mathfrak{n}}$ to $\mathfrak{n}$, so the identity persists with the nilpotent in place. On the group side $\pi^\vee \cong \omega_\pi^{-1}\pi$ for π irreducible admissible of $\mathrm{GL}_2(F)$, which refines $\omega_{\check{\pi}} = \omega_\pi^{-1}$. Granting these,

$$\boldsymbol{\pi}_F(\check{\rho}) = \boldsymbol{\pi}_F\big(\rho \otimes (\det \rho)^{-1}\big) = (\det \rho)^{-1}\boldsymbol{\pi}_F(\rho) = \omega_{\boldsymbol{\pi}_F(\rho)}^{-1}\boldsymbol{\pi}_F(\rho) = \boldsymbol{\pi}_F(\rho)^\vee$$

The functional equations of corollary 5.2.9 and proposition 6.3.3 give the ε-half of (5) on their own, but not the L-half, so the argument is run this way round. For (2), lemma 5.3.3 and proposition 6.3.3 give the dependence of each side on the additive character as $\omega_\pi(a)\|a\|^{2(s-1/2)}$ and $\det \rho(a)\|a\|^{\dim \rho(s-1/2)}$ respectively; since $\dim \rho = 2$ and $\omega_\pi = \det \rho$ by (3), the two agree, so the relations hold for one $\psi \neq 1$ if and only if they hold for all, completing the proof.

The correspondence respects the natural partition of each side into a reducible and an irreducible part.

- *Galois side:* $\mathcal{G}_2(F) = \mathcal{G}_2^1(F) \sqcup \mathcal{G}_2^0(F)$, where $\mathcal{G}_2^1(F)$ consists of the triples $(\rho, V, \mathfrak{n})$ whose underlying smooth representation ρ is *reducible*, hence a sum of two characters, and $\mathcal{G}_2^0(F)$ of

those with ρ irreducible. In the second case $\mathfrak{n} = 0$ automatically: the relation $\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}$ makes $\ker \mathfrak{n}$ a $\mathcal{W}_F$-subspace, so irreducibility forces $\mathfrak{n}$ to be zero or invertible, and a nilpotent endomorphism cannot be invertible.

- GL_2 *side:* $\mathcal{A}_2(F) = \mathcal{A}_2^1(F) \sqcup \mathcal{A}_2^0(F)$, the non-cuspidal and the cuspidal representations. By theorem 2.4.16 the first consists of the irreducible $\iota_B^G\chi$, the one-dimensional $\varphi \circ \det$, and the special representations $\varphi\mathrm{St}_G$.

By lemma 5.5.1, $\pi \in \mathcal{A}_2^0(F)$ if and only if $L(\chi\pi, s) = 1$ for every character χ of $F^\times$. The same criterion characterizes $\mathcal{G}_2^0(F)$, for the reason given in Step 2 of theorem 7.1.2: irreducible two-dimensional twists have $L = 1$ by definition, while a reducible $(\chi_1 \oplus \chi_2, \mathfrak{n})$ is detected by the twist χ_2^{-1}. Any map satisfying the conditions of theorem 7.1.2 therefore carries $\mathcal{G}_2^1(F)$ into $\mathcal{A}_2^1(F)$ and $\mathcal{G}_2^0(F)$ into $\mathcal{A}_2^0(F)$, and the two halves can be treated separately.

Both parts of the reducible side are needed, and the nilpotent is what separates them. Fix a character φ of $F^\times$ and put $\chi_1 = \varphi\|-\|^{-1/2}$, $\chi_2 = \varphi\|-\|^{1/2}$, so that $\chi_1\chi_2^{-1} = \|-\|^{-1}$ and $\rho = \chi_1 \oplus \chi_2$ is reducible with $\iota_B^G(\chi_1 \otimes \chi_2)$ reducible as well. There are exactly two triples in $\mathcal{G}_2(F)$ with this underlying ρ, namely $\mathfrak{n} = 0$ and $\mathfrak{n} \neq 0$, the latter being $\varphi \otimes Sp(2)$; and there are exactly two irreducible representations of G built from that induced representation, namely $\varphi \circ \det$ and $\varphi\mathrm{St}_G$. The invariants computed in section 5.4 match them up: by proposition 5.4.2 the one-dimensional representation carries the *full* product $L(\chi_1, s)L(\chi_2, s)$, which is what ρ with $\mathfrak{n} = 0$ has, while by proposition 5.4.3 the special representation carries only $L(\chi_2, s)$, which is what the nilpotent cuts the ρ-side L-factor down to. So the one-dimensional representations do lie in the image, paired with $\mathfrak{n} = 0$, and it is the special representations that need $\mathfrak{n} \neq 0$.

7.2 The Reducible Correspondence

The reducible half is settled entirely by the computations of section 5.4, because on both sides the objects are built from a pair of characters of $F^\times$ and the invariants are built from the GL_1 invariants of that pair. What has to be got right is the bookkeeping at the reducible locus, where two Galois-side triples and two group-side representations share the same pair of characters.

Theorem 7.2.1: Correspondence for $\mathcal{G}_2^1(F)$

There is a unique bijection $\boldsymbol{\pi}_1 \colon \mathcal{G}_2^1(F) \to \mathcal{A}_2^1(F)$ preserving L-functions and ε-factors under all twists, for every $\psi \in \widehat{F}$ with $\psi \neq 1$. Writing an element of $\mathcal{G}_2^1(F)$ as $(\chi_1 \oplus \chi_2, \mathfrak{n})$, it is given by:

1. if $\chi_1\chi_2^{-1} \neq \|-\|^{\pm 1}$, then necessarily $\mathfrak{n} = 0$ and $\boldsymbol{\pi}_1$ sends it to the irreducible principal series $\iota_B^G(\chi_1 \otimes \chi_2)$;
2. if $\chi_1\chi_2^{-1} = \|-\|^{\pm 1}$, so that $\{\chi_1, \chi_2\} = \{\varphi\|-\|^{-1/2}, \varphi\|-\|^{1/2}\}$ for a character φ of $F^\times$, and $\mathfrak{n} = 0$, then $\boldsymbol{\pi}_1$ sends it to the one-dimensional representation $\varphi \circ \det$;
3. in the same situation with $\mathfrak{n} \neq 0$, that is for $\varphi \otimes Sp(2)$, then $\boldsymbol{\pi}_1$ sends it to the special representation $\varphi\mathrm{St}_G$.

***Proof*:**

Step 1: the three cases are exhaustive and disjoint on both sides. An element of $\mathcal{G}_2^1(F)$ has $\rho = \chi_1 \oplus \chi_2$ with χ_1, χ_2 characters of $\mathcal{W}_F$, identified with characters of $F^\times$ by local class field theory (theorem 6.2.1). If $\chi_1\chi_2^{-1} \neq \|-\|^{\pm 1}$ then $\mathfrak{n} = 0$: the relation $\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}$ says that $\mathfrak{n}$ intertwines ρ with $\rho \otimes \|-\|$, and for a sum of two characters that forces $\{\chi_1, \chi_2\}$ and $\{\chi_1\|-\|, \chi_2\|-\|\}$ to meet, which is exactly the excluded relation. In the remaining case the two values $\mathfrak{n} = 0$ and $\mathfrak{n} \neq 0$ give two distinct classes, and rescaling shows all non-zero $\mathfrak{n}$ give the same class. On the other side, theorem 2.4.16 lists $\mathcal{A}_2^1(F)$ as the irreducible $\iota_B^G\chi$, the one-dimensional $\varphi \circ \det$, and the special $\varphi\mathrm{St}_G$, and by lemma 2.4.15 the first family is irreducible precisely when $\chi_1\chi_2^{-1} \neq \|-\|^{\pm 1}$. The three cases therefore biject with the three families, once each case is shown to preserve the invariants.

Step 2: case (1). By proposition 5.4.1,

$$L\big(\iota_B^G(\chi_1 \otimes \chi_2), s\big) = L(\chi_1, s)L(\chi_2, s), \qquad \varepsilon\big(\iota_B^G(\chi_1 \otimes \chi_2), s, \psi\big) = \varepsilon(\chi_1, s, \psi)\,\varepsilon(\chi_2, s, \psi)$$

On the Galois side $\rho = \chi_1 \oplus \chi_2$ with $\mathfrak{n} = 0$ has, by additivity of the Artin L-function and of the local constant (proposition 6.3.3), exactly the same two products. Twisting by a character χ replaces χ_i by $\chi\chi_i$ on both sides simultaneously, so the agreement persists under all twists.

Step 3: case (2). Take $\chi_1 = \varphi\|-\|^{-1/2}$ and $\chi_2 = \varphi\|-\|^{1/2}$. By proposition 5.4.2 the one-dimensional representation $\varphi \circ \det$ has $L(\varphi \circ \det, s) = L(\chi_1, s)L(\chi_2, s)$ and $\varepsilon(\varphi \circ \det, s, \psi) = \varepsilon(\chi_1, s, \psi)\varepsilon(\chi_2, s, \psi)$, which is again the invariant of $\chi_1 \oplus \chi_2$ with $\mathfrak{n} = 0$. This is the case the naive count overlooks: the reducible locus carries a one-dimensional representation as well as a special one, and it is the one-dimensional representation that pairs with the vanishing nilpotent.

Step 4: case (3). For a Deligne representation the L-factor is computed on the subspace where both inertia and $\mathfrak{n}$ act trivially, so passing from $\mathfrak{n} = 0$ to $\mathfrak{n} \neq 0$ deletes the factor belonging to χ_1 and leaves $L(\chi_2, s)$. By proposition 5.4.3 that is precisely $L(\varphi\mathrm{St}_G, s)$, and the same proposition gives the matching correcting factor in the ε-factor. So $\varphi \otimes Sp(2)$ corresponds to $\varphi\mathrm{St}_G$.

Step 5: uniqueness. Let $\boldsymbol{\pi}_1$ and $\boldsymbol{\pi}_1'$ both satisfy the conditions, fix $\rho \in \mathcal{G}_2^1(F)$, and put $\pi = \boldsymbol{\pi}_1(\rho)$, $\pi' = \boldsymbol{\pi}_1'(\rho)$. Both are irreducible smooth representations of G with the same twisted L- and ε-factors. Their central characters agree, by the argument of Step 1 of theorem 7.1.2: since the conditions hold for every $\psi \neq 1$, lemma 5.3.3 makes the ψ-dependence of $\varepsilon(\chi\pi, s, \psi)$ a function of $\omega_{\chi\pi} = \chi^2\omega_\pi$, and comparing the two sides forces $\omega_\pi = \omega_{\pi'}$. The hypotheses of theorem 5.5.2 are therefore met at a ψ of level one, which gives $\pi \cong \pi'$. Since the three cases exhaust both sides and each is a bijection onto its family, $\boldsymbol{\pi}_1$ is a bijection, as we sought to show.

The asymmetry between $\chi_1\chi_2^{-1} = \|-\|$ and $\chi_1\chi_2^{-1} = \|-\|^{-1}$ is only apparent: the two are exchanged by swapping χ_1 and χ_2, which does not change the isomorphism class of $\chi_1 \oplus \chi_2$, and by lemma 2.4.15 it does not change $\iota_B^G\chi$ either.

7.3 The Tame Correspondence ($p \neq 2$)

We now turn to the heart of the matter: matching irreducible representations of $\mathcal{W}_F$ ($\mathcal{G}_2^0(F)$) with supercuspidal representations of G ($\mathcal{A}_2^0(F)$).

When the residual characteristic p is odd, we can use the classification via *Admissible Pairs* from chapter 4. Recall that $P_2(F)$ is the set of isomorphism classes of admissible pairs $(E/F, \xi)$, where

E/F is a quadratic extension and ξ is a character of $E^\times$ not factoring through the norm.

We have two "naive" maps from $P_2(F)$:

1. *To Galois:* $(E/F, \xi) \mapsto \rho_\xi = \mathrm{Ind}_{E/F}\xi \in \mathcal{G}_2^0(F)$.
2. *To GL(2):* $(E/F, \xi) \mapsto \pi_\xi$, the cuspidal representation constructed via compact induction in chapter 4.

If $p \neq 2$, both maps are bijections. However, the map $\rho_\xi \mapsto \pi_\xi$ is *not* the Langlands correspondence. There is a subtle mismatch in the central characters.

$$\det(\rho_\xi) = \kappa_{E/F} \cdot (\xi|_{F^\times}) \quad \text{but} \quad \omega_{\pi_\xi} = \xi|_{F^\times}$$

the first by proposition 6.2.7. Here $\kappa_{E/F}$ is the quadratic character of $F^\times$ cutting out the norms from E, which exists by theorem 6.2.2(2). To fix this, we must twist by a "rectifier" character.

Definition 7.3.1: Rectifier Character Δ_ξ

For an admissible pair $(E/F, \xi)$, define a character Δ_ξ of $E^\times$ as follows:

1. If E/F is unramified, Δ_ξ is the unique unramified character of $E^\times$ of order 2.
2. If E/F is totally *tamely* ramified and the pair $(E/F, \xi)$ is minimal, fix $\psi \in \widehat{F}$ of level one and let $\lambda_{E/F}(\psi)$ be the Langlands constant defined in definition 6.3.10. Let α be an element relating ξ to the additive character ψ_E, and ϖ_E a prime element of E. Then Δ_ξ is the unique character such that:

$$\Delta_\xi|_{U_E^1} = 1, \quad \Delta_\xi|_{F^\times} = \kappa_{E/F}, \quad \Delta_\xi(\varpi_E) = \kappa_{E/F}(\zeta(\alpha, \varpi_E))\lambda_{E/F}(\psi)^n$$

   where n is the level of ξ (odd, for a minimal totally ramified pair) and $\zeta(\alpha, \varpi_E)$ is a root of unity built from α and ϖ_E.
3. If E/F is totally tamely ramified and $(E/F, \xi)$ is not minimal, write $\xi = \xi'\chi_E$ for a character χ of $F^\times$, where $\chi_E = \chi \circ N_{E/F}$ and $(E/F, \xi')$ is minimal, and set $\Delta_\xi = \Delta_{\xi'}$.

Within level one the recipe in (2) is independent of the choices of ψ and α: replacing ψ by $u\psi$ with $u \in U_F$ changes the right-hand side of the displayed formula by $\kappa_{E/F}(u)^{n-1}$, since α is replaced by $u^{-1}\alpha$ and $\lambda_{E/F}(u\psi) = \kappa_{E/F}(u)\lambda_{E/F}(\psi)$. That factor is trivial because n is odd. So Δ_ξ depends only on the isomorphism class of the pair. The restriction to level one is not cosmetic: at other levels the corrections to α and to $\lambda_{E/F}(\psi)$ no longer cancel.[a]

[a] Case (2) needs E/F tame because $\zeta(\alpha, \varpi_E)$ is extracted from the decomposition $U_E = \mu_F U_E^1$, which fails at $p = 2$. Case (1) carries no such restriction, which is what the dyadic comparison following example 8.5.3 needs: there $E = \mathbb{Q}_2(\zeta)$ is unramified.

Theorem 7.3.2: The Tame Correspondence

Let $p \neq 2$ and let $\psi \in \widehat{F}$ be non-trivial. There is a unique bijection

$$\boldsymbol{\pi}_F^0 \colon \mathcal{G}_2^0(F) \longrightarrow \mathcal{A}_2^0(F)$$

such that $\varepsilon(\chi \otimes \rho, s, \psi) = \varepsilon(\chi\, \boldsymbol{\pi}_F^0(\rho), s, \psi)$ for every $\rho \in \mathcal{G}_2^0(F)$ and every character χ of $F^\times$. On admissible pairs it is given by

$$\boldsymbol{\pi}_F^0(\rho_\xi) = \pi_{\xi'} \qquad \text{where } \xi' = \xi \cdot \Delta_\xi$$

The central characters can be matched outright, and the surjectivity of $P_2(F) \to \mathcal{G}_2^0(F)$ is taken from the source. What is missing beyond that is the ε-factor identity, and it is worth saying exactly which part. A formula for each side exists in this book: the Galois side by inductivity of the local constant, and the group side by theorem 5.3.8 at positive level. What is not proved is that the two agree once ξ is twisted by the rectifier. The proof is therefore given in full except for those two deferrals, both of which are cited.

***Proof Key ideas*:**

Step 0: uniqueness. Two bijections satisfying the displayed identity agree. Their values at a given ρ are cuspidal, so every twisted L-factor on the group side is 1 by proposition 5.3.1, and the argument of Step 1 of theorem 7.1.2 then applies verbatim: from the twisted ε-identity at the one ψ of the statement, together with theorem 5.3.10, it forces the two central characters to agree, after which theorem 5.5.2 at a ψ of level one gives the isomorphism. That argument uses no existence statement, so citing it here is not circular.

Step 1: bijectivity. For $p \neq 2$ the two maps out of $P_2(F)$ are bijections onto $\mathcal{G}_2^0(F)$ and $\mathcal{A}_2^0(F)$ respectively. The second is theorem 4.3.1, proved in this book. The first (that every $\rho \in \mathcal{G}_2^0(F)$ is $\mathrm{Ind}_{E/F}\xi$ for an admissible pair, uniquely up to isomorphism of pairs) is [2, p. 34.1]. Its surjectivity half runs through the decomposition of $\rho|_{\mathcal{P}_F}$ over a tame Galois extension, which this book does not carry out. The rectifier attaches to each admissible pair $(E/F, \xi)$ the pair $(E/F, \xi\Delta_\xi)$, and since Δ_ξ depends only on the isomorphism class of the pair and is invertible, this is a bijection of $P_2(F)$ onto itself. The composite $\rho_\xi \mapsto \pi_{\xi\Delta_\xi}$ is therefore a bijection $\mathcal{G}_2^0(F) \to \mathcal{A}_2^0(F)$.

Step 2: central characters, in full. On the Galois side, the determinant of an induced representation is $\det(\mathrm{Ind}_{E/F}\xi) = \kappa_{E/F} \cdot (\xi|_{F^\times})$ by proposition 6.2.7, where $\kappa_{E/F}$ is the quadratic character of $F^\times$ cutting out the norms from E (theorem 6.2.2(2)). On the group side $\omega_{\pi_{\xi'}} = \xi'|_{F^\times}$, and by construction $\Delta_\xi|_{F^\times} = \kappa_{E/F}$ (definition 7.3.1), so

$$\omega_{\pi_{\xi\Delta_\xi}} = (\xi\Delta_\xi)|_{F^\times} = (\xi|_{F^\times})\,\kappa_{E/F} = \det(\rho_\xi)$$

which is property (3) of theorem 7.1.2. This also shows why the naive map $\rho_\xi \mapsto \pi_\xi$ cannot be the correspondence: it fails this identity by exactly the factor $\kappa_{E/F}$.

Step 3: the ε-factor comparison, cited. Inductivity of the local constant in λ form, (6.5) with $n = 1$, gives for $\rho_\xi = \mathrm{Ind}_{E/F}\xi$,

$$\varepsilon(\rho_\xi, s, \psi) = \lambda_{E/F}(\psi)\,\varepsilon(\xi, s, \psi_E)$$

with $\lambda_{E/F}(\psi)$ the Langlands constant of definition 6.3.10. The corresponding computation on the group side expresses $\varepsilon(\pi_{\xi'}, s, \psi)$ in terms of $\varepsilon(\xi', s, \psi_E)$, and the definition of Δ_ξ is arranged

so that the discrepancy between the two is exactly $\lambda_{E/F}(\psi)$. That discrepancy is what is not proved here. It is not the existence of a group-side formula that is missing: theorem 5.3.8 gives one, as a Gauss sum over a quotient of two unit filtration subgroups of the chain order, for a cuspidal representation of positive level compactly induced from a character in the shape that theorem asks for. See [2, §34] for the comparison. Granting it, the twisted ε-factors agree for all twists, and the L-factors agree trivially because both sides are 1: on the group side $L(\chi\pi_{\xi'}, s) = 1$ by proposition 5.3.1, since $\chi\pi_{\xi'}$ is again cuspidal by Step 1 of lemma 5.5.1. On the Galois side $L(\chi \otimes \rho_\xi, s) = 1$ by the definition in section 6.3, the twist being again irreducible of dimension two.

One later result should not be mistaken for a discharge of that deferral. Corollary 8.5.5 identifies the correspondence just stated with the partial correspondence $\boldsymbol{\pi}^{\mathrm{im}}$ constructed further on, but its proof takes the present theorem as an input, so it inherits the deferral rather than removing it.

7.4 The ℓ-adic Correspondence

So far, we have worked with representations over $\mathbb{C}$. However, in arithmetic geometry (e.g., the cohomology of modular curves), representations naturally arise over $\overline{\mathbb{Q}}_\ell$, where $\ell \neq p$.

An isomorphism $\iota\colon \mathbb{C} \cong \overline{\mathbb{Q}}_\ell$ exists, but using it directly is problematic because the correspondence depends on $\sqrt{q}$.

Both sides carry an action of $\mathrm{Aut}(\mathbb{C})$. For $\sigma \in \mathrm{Aut}(\mathbb{C})$ and a $\mathbb{C}$-vector space V, write $V^\sigma = V \otimes_{\mathbb{C},\sigma} \mathbb{C}$. Given a Deligne representation $(\rho, V, \mathfrak{n}) \in \mathcal{G}_2(F)$, let ρ^σ act on V^σ through $\rho \otimes 1$. The triple $(\rho^\sigma, V^\sigma, \mathfrak{n} \otimes 1)$ is again a Deligne representation: the relation $\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}$ survives because $\|w\| = q^{-v_F(w)}$ is rational and so is fixed by σ, and $\mathfrak{n} \otimes 1$ is still nilpotent. Given $(\pi, W) \in \mathcal{A}_2(F)$, let $\mathrm{GL}_2(F)$ act on W^σ through $\pi \otimes 1$. Smoothness and irreducibility survive because base change along a field automorphism is an equivalence on $\mathbb{C}$-vector spaces, so it preserves stabilizers and the lattice of submodules. Write $\sigma\rho$ and $\sigma\pi$ for the results, and see [2, §32] for both constructions in full.

The correspondence $\boldsymbol{\pi}_F$ is *not* $\mathrm{Aut}(\mathbb{C})$-equivariant: there are σ and ρ for which $\boldsymbol{\pi}_F(\sigma\rho) \ncong \sigma\boldsymbol{\pi}_F(\rho)$.

To fix this, we introduce a normalization.

Definition 7.4.1: Normalized Galois Representation

For $\rho \in \mathcal{G}_2(F)$, define the twist:

$$\tilde{\rho} = \rho \otimes \| - \|^{-1/2}$$

(Note: This is a twist by a character of the Weil group).

Theorem 7.4.2: The ℓ-adic Correspondence

Define a modified correspondence $\Pi_F : \mathcal{G}_2(F) \to \mathcal{A}_2(F)$ by:

$$\Pi_F(\rho) = \boldsymbol{\pi}_F(\tilde{\rho}) = \boldsymbol{\pi}_F(\rho \otimes \| - \|^{-1/2})$$

This map Π_F commutes with the action of $\mathrm{Aut}(\mathbb{C})$. Consequently, for any $\ell \neq p$ and any field isomorphism $\iota\colon \mathbb{C} \to \overline{\mathbb{Q}}_\ell$, transporting Π_F along ι gives a bijection, independent of the choice of ι, between the 2-dimensional semisimple Deligne representations of $\mathcal{W}_F$ over $\overline{\mathbb{Q}}_\ell$ and the irreducible smooth $\overline{\mathbb{Q}}_\ell$-representations of $\mathrm{GL}_2(F)$.

***Proof Key ideas*:**
The $\sqrt{q}$ enters at GL_1, through the self-dual Haar measure on F attached to ψ, which is irrational exactly when q is not a square and ψ has odd level. Writing k for the level of ψ and

$$\alpha(\sigma, \psi) \;=\; \big(\sigma(\sqrt{q}) \,/\, \sqrt{q}\big)^k$$

that measure contributes a factor $\alpha(\sigma,\psi)^{\dim \rho}$ to $\varepsilon(\sigma\rho, s, \sigma \circ \psi)$. At GL_2 nothing of the kind appears: the self-dual measure on the matrix algebra takes rational values on compact open subgroups, for every non-trivial ψ.

In dimension two the GL_1 factor cancels: $\alpha(\sigma,\psi)^2 = (\sigma(q)/q)^k = 1$, because q is a rational integer and every field automorphism fixes it. So each side of the correspondence is separately $\mathrm{Aut}(\mathbb{C})$-equivariant, and the failure of $\boldsymbol{\pi}_F$ is not an ambiguity in $\sqrt{q}$ at all. What remains is that the two sides are equivariant in variables differing by $\frac{1}{2}$: the GL_2 theory is built on the Godement-Jacquet zeta integral $\zeta(\Phi, f, s + \frac{1}{2})$ of (5.10), whose variable is shifted by $\frac{1}{2}$ from the one natural on the Galois side. Twisting by $\| - \|^{-1/2}$ shifts the Galois variable by exactly that $\frac{1}{2}$ and aligns the two, which is what makes Π_F equivariant. Given equivariance, transporting along any ι produces the same bijection, since two choices of ι differ by an element of $\mathrm{Aut}(\mathbb{C})$.

The verification is a computation with the explicit local constants on both sides, and at level zero the group-side constant is quoted in this book and not derived (section 5.3). It is therefore stated here and not carried out. See [2, §35].

7.5 Summary

We have established the Local Langlands Correspondence for $\mathrm{GL}_2(F)$:

7.5.1 The Correspondence Maps

- *Irreducible Principal Series* $\iota_B^G(\chi_1 \otimes \chi_2)$, for $\chi_1\chi_2^{-1} \neq \|-\|^{\pm 1} \longleftrightarrow \chi_1 \oplus \chi_2$ (Reducible, decomposable).
- *One-Dimensional Representations* $\varphi \circ \det \longleftrightarrow \varphi\|-\|^{-1/2} \oplus \varphi\|-\|^{1/2}$ with $\mathfrak{n} = 0$ (Reducible, decomposable, the case the naive count overlooks).
- *Special Representations* $\varphi\mathrm{St}_G \longleftrightarrow \varphi \otimes Sp(2)$ (Reducible, indecomposable Deligne rep).
- *Supercuspidal Representations* $\pi \longleftrightarrow \rho$ (Irreducible).

For the supercuspidal case with $p \neq 2$:

1. Take an admissible pair $(E/F, \xi)$.
2. Construct the Weil representation $\rho = \mathrm{Ind}_{E/F}\xi$.
3. Construct the rectifying character Δ_ξ.
4. The matching GL(2) representation is the compactly induced representation $\pi_{\xi\Delta_\xi}$ constructed in chapter 4.

This satisfies the preservation of L-functions (which are 1 for cuspidals) and ε-factors (which encode the arithmetic information, including Gauss sums).

8

The Weil Representation

A reader heading for Chapter 9 needs only section 8.5 and the groups of section 8.3 from this chapter. The Weil representation itself is built here for its own sake and for the odd-residual-characteristic correspondence, and nothing in Chapter 9 calls on that construction.

The correspondence of chapter 7 is proved by construction wherever the residual characteristic is odd. Theorem 7.3.2 builds a cuspidal representation from an admissible pair and matches it with an induced parameter, and that is enough to settle every cuspidal class when $p \neq 2$. For $p = 2$ the route closes. An admissible pair requires a tamely ramified extension, and the ramified quadratic extensions of a dyadic field are not tame, so the parametrization loses exactly the classes it most needs to reach.

This chapter builds the replacement. The tool is a representation of a metaplectic group attached to a quadratic extension E/F, the *Weil representation*, which manufactures a cuspidal representation of G from a character of $E^\times$ with no tameness hypothesis anywhere. Reaching it takes two preliminary constructions. Section 8.1 realizes an irreducible representation of infinite dimension canonically as a space of functions, and section 8.2 shows that for a cuspidal representation the entire local constant is carried by a single operator on that space. Those two sections are of independent use: between them they prove several statements that earlier chapters could only quote. The chapter ends with theorem 8.5.4, a correspondence valid in every residual characteristic on the parameters induced from a quadratic extension, and with an account of what the dyadic case still leaves unsettled.

8.1 Whittaker and Kirillov Models

The aim of this section is to realize an irreducible smooth representation of infinite dimension concretely, as a space of functions, in a way that is canonical rather than a matter of choice. Fix throughout a non-trivial character ϑ of the group N of upper triangular unipotent matrices. The object that makes everything canonical is the space $V_\vartheta = V/V(\vartheta)$ of ϑ-coinvariants introduced in

section 2.3, where $V(\vartheta)$ is the subspace spanned by the vectors $\pi(n)v - \vartheta(n)v$, and the fact that it is as small as it could be without vanishing.

Theorem 8.1.1: Uniqueness of the Whittaker Functional

Let ϑ be a non-trivial character of N and let (π, V) be an irreducible smooth representation of G of infinite dimension. Then

$$\dim_{\mathbb{C}} V_\vartheta = 1$$

The hypothesis of infinite dimension is not a technicality. If $\dim V = 1$ then $\pi = \varphi \circ \det$ for a character φ of $F^\times$ by theorem 2.4.16, so N acts trivially, and $V_\vartheta = 0$ for every non-trivial ϑ. Everything below starts from a non-zero V_ϑ, so none of it applies to the one-dimensional representations.

***Proof*:**

By theorem 2.4.16 together with proposition 2.4.3, an irreducible smooth representation of G is either cuspidal or a constituent of a representation induced from the Borel. The two cases are genuinely different and are treated separately.

Step 1: the induced case. Let χ be a character of T and set $(\Sigma, X) = \mathrm{Ind}_B^G \chi$. Let $W = X(N)$ be the kernel of the canonical map $X \to X_N$. By theorem 2.4.4 the space X_N has dimension two. This makes W the space $V(N)$ of proposition 2.4.9: writing $X^\circ \subseteq X$ for the kernel of restriction to B, the group N acts trivially on $X/X^\circ \cong \chi$, so $X(N) \subseteq X^\circ$, while $X^\circ(N) \subseteq X(N)$ always. Both have codimension two in X, the first since $\dim X_N = 2$ and the second since $\dim X_N^\circ + \dim(X/X^\circ) = 1 + 1$, so $X(N) = X^\circ(N)$. On X_N the group N acts trivially, so a non-trivial ϑ has no coinvariants there. Hence the inclusion $W \to X$ induces an isomorphism

$$W_\vartheta \xrightarrow{\sim} X_\vartheta$$

By proposition 2.4.9 the space W is irreducible as a representation of M, so theorem 2.3.9 identifies it with $\text{c-Ind}_N^M \vartheta$. For that representation theorem 2.3.6 gives $\dim(\text{c-Ind}_N^M \vartheta)_\vartheta = 1$. Therefore $\dim X_\vartheta = 1$.

Now let (π, V) be a G-composition factor of X of infinite dimension. If Σ is irreducible then $V = X$ and there is nothing more to say. Otherwise theorem 2.4.8 gives Σ composition length two, with one factor of dimension one and the other of infinite dimension. The one-dimensional factor contributes no ϑ-coinvariants, and ϑ-coinvariants are exact on this short exact sequence because taking coinvariants is right exact and the outer term vanishes. Hence $\dim V_\vartheta = \dim X_\vartheta = 1$.

Step 2: the cuspidal case, and what has to be shown. Let π be cuspidal, so $V_N = 0$. The statement is insensitive to twisting by characters of $F^\times$, since twisting changes neither N nor ϑ, so assume $\ell(\pi) \leq \ell(\chi\pi)$ for every character χ of $F^\times$. By theorem 3.5.5 there is a cuspidal type $(\mathfrak{A}, J, \Lambda)$ with $\pi \cong \text{c-Ind}_J^G \Lambda$. Everything now reduces to counting how ϑ meets Λ.

Step 3: the double cosets are carried by the torus. In every cuspidal type there is a quadratic field extension E/F with $E^\times \subseteq J$, by proposition 3.4.3 at positive level and by construction at level zero. The group $E^\times$ acts on F^2, viewed as E itself, transitively on the non-zero vectors. Given $g \in G$, choose $x \in E^\times$ with $xge_1 = e_1$, where e_1 is the first standard basis vector; then xg stabilizes the line Fe_1, so $xg \in B$. Hence

$$G = E^\times B = E^\times TN$$

and since $J \supseteq E^\times$ the natural map

$$(T \cap J)\backslash T \longrightarrow J\backslash G/N, \qquad t \longmapsto JtN$$

is surjective. So the double cosets that will index the coming decomposition may all be taken to have representatives in T.

Step 4: decomposing the restriction to N. The group J is open in G and N is closed, so proposition 1.2.29 sorts the functions in $\text{c-Ind}_J^G \Lambda$ by support and gives

$$\pi|_N \cong \bigoplus_{g \in J\backslash G/N} \text{c-Ind}_{N \cap g^{-1}Jg}^N \Lambda^g$$

and therefore, by theorem 1.2.21,

$$\text{Hom}_G(\pi, \text{Ind}_N^G \vartheta) \cong \text{Hom}_N(\pi, \vartheta) \cong \bigoplus_{g \in J\backslash G/N} \text{Hom}_N\big(\text{c-Ind}_{N \cap g^{-1}Jg}^N \Lambda^g, \vartheta\big)$$

By Step 3 the representatives g may be taken in T. For $t \in T$ the group $N \cap t^{-1}Jt$ is open in N, so theorem 1.2.28 applies and gives

$$\text{Hom}_N\big(\text{c-Ind}_{N \cap t^{-1}Jt}^N \Lambda^t, \vartheta\big) \cong \text{Hom}_{N \cap t^{-1}Jt}\big(\Lambda^t, \vartheta\big) \cong \text{Hom}_{N \cap J}\big(\Lambda, \vartheta^{t^{-1}}\big)$$

the second isomorphism being conjugation by t, which carries $N \cap t^{-1}Jt$ onto $N \cap J$ and Λ^t onto Λ. So the dimension being computed is the total multiplicity, over the T-orbit of ϑ, of the restriction $\vartheta|_{N \cap J}$ inside Λ. Two things must therefore hold: that multiplicity is one, and the relevant characters form a single orbit.

Step 5: locating $N \cap J$ and the characters that occur. Write $\mathfrak{P} = \text{rad}\mathfrak{A}$ and, for $j \in \mathbb{Z}$, let N_j denote the subgroup of N with upper entry in $\mathfrak{p}^j$.

Suppose first $\ell(\pi) = 0$. Then $J = F^\times \text{GL}_2(\mathfrak{o})$ and $N \cap J = N_0$. By theorem 3.2.9 the restriction of Λ to $\text{GL}_2(\mathfrak{o})$ is inflated from an irreducible cuspidal representation of $\text{GL}_2(k)$, and by theorem 2.1.15 the restriction of such a representation to the unipotent subgroup of $\text{GL}_2(k)$ is the sum of *all* its non-trivial characters, each once. Pulling back, $\Lambda|_{N_0}$ is the sum of the non-trivial characters of N_0 trivial on N_1, each with multiplicity one. Any two characters of N that are trivial on N_1 but not on N_0 are conjugate under $T \cap \text{GL}_2(\mathfrak{o})$, because conjugation by $\text{diag}(a, 1)$ scales the upper entry by a and a ranges over all of U_F. Both requirements of Step 4 hold.

Now suppose $\ell(\pi) > 0$ and fix ψ of level one. There is a simple stratum $(\mathfrak{A}, n, \alpha)$ with $n \geq 1$ and Λ containing ψ_α, and $J = E^\times U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}$. Three observations settle the case.

First, $N \cap J = N \cap U_{\mathfrak{A}}^m$ where $m = \lfloor (n+1)/2 \rfloor$. Indeed a unipotent element of $J = E^\times U_{\mathfrak{A}}^m$ lies in $U_E U_{\mathfrak{A}}^m$: writing it as ϵu and taking determinants, $N_{E/F}(\epsilon) \det u = 1$ with $\det u$ a unit, so $v_E(\epsilon) = 0$. Hence such an element is $1 + x$ with $x \in \mathfrak{o}_E + \mathfrak{P}^m$, and we may decompose $x = y + z$ with $y \in \mathfrak{o}_E$, $z \in \mathfrak{P}^m$. Recall $\mathfrak{P}^i \cap E = \mathfrak{p}_E^i$, so the $\mathfrak{P}$-valuation of a non-zero element of $\mathfrak{o}_E$ is its E-valuation. Suppose $y \neq 0$ and set $j = v_E(y)$ and $k = v_{\mathfrak{P}}(z) \geq m$. If $j < k$ then in

$$x^2 = y^2 + yz + zy + z^2$$

the first term has $\mathfrak{P}$-valuation exactly $2j$ while the other three have valuation at least $j + k > 2j$, so $x^2 \in \mathfrak{P}^{2j} \setminus \mathfrak{P}^{2j+1}$ and in particular $x^2 \neq 0$. That contradicts $x^2 = 0$, which holds because x is the

strictly upper triangular part of a unipotent matrix. Hence $j \geq k \geq m$, so $y \in \mathfrak{p}_E^m = \mathfrak{P}^m \cap E \subseteq \mathfrak{P}^m$ and therefore $x = y + z \in \mathfrak{P}^m$.

Second, any character ϑ of N whose restriction to $N \cap J$ occurs in Λ must agree with ψ_α on $N \cap U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}$, because Λ restricted to that group is by construction a multiple of ψ_α. And ψ_α is non-trivial on $U_{\mathfrak{A}}^n \cap N$, since a simple element is not congruent to a scalar.

Third, any two characters of N agreeing with ψ_α on $N \cap U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}$ are conjugate under $T \cap U_{\mathfrak{A}}^m \subseteq T \cap J$. This is a valuation count, and since it comes out tight it is worth doing rather than asserting. Identify N with F by $x \mapsto n(x)$ and write characters of N as $\psi_b \colon x \mapsto \psi(bx)$; as ψ has level one, ψ_b is trivial on N_j exactly when $v_F(b) \geq 1 - j$. Since $\operatorname{tr}\big(\alpha(n(x) - 1)\big) = \alpha_{21} x$, the restriction $\psi_\alpha|_N$ is $\psi_{\alpha_{21}}$, and conjugation by $\operatorname{diag}(a, 1)$ carries ψ_b to ψ_{ab}.

Define c, d and c_0 by

$$N \cap U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1} = N_c, \qquad \big\{ a \in F^\times : \operatorname{diag}(a, 1) \in U_{\mathfrak{A}}^m \big\} = U_F^d, \qquad N \cap U_{\mathfrak{A}}^n = N_{c_0}$$

If ψ_b and $\psi_{b'}$ both agree with $\psi_{\alpha_{21}}$ on N_c then $v_F(b - \alpha_{21})$ and $v_F(b' - \alpha_{21})$ are at least $1 - c$; granted $v_F(\alpha_{21}) < 1 - c$ this forces $v_F(b) = v_F(b') = v_F(\alpha_{21})$, and $\psi_{b'} = \psi_b^{\operatorname{diag}(a,1)}$ for $a = b'/b$, which lies in U_F^d precisely when $v_F(b' - b) \geq d + v_F(\alpha_{21})$. Since $v_F(b' - b) \geq 1 - c$, the orbit claim holds as soon as

$$v_F(\alpha_{21}) \;\leq\; 1 - c - d \tag{8.1}$$

Now ψ_α is non-trivial on N_{c_0}, which says exactly $v_F(\alpha_{21}) \leq -c_0$. So (8.1) follows from the numerical identity $c + d = c_0 + 1$, and that identity is read off from the shape of $\mathfrak{P}$. When $\mathfrak{A}$ is maximal, $N \cap U_{\mathfrak{A}}^j = N_j$ and $\operatorname{diag}(a, 1) \in U_{\mathfrak{A}}^j$ if and only if $a \in U_F^j$; when $\mathfrak{A}$ is Iwahori, $N \cap U_{\mathfrak{A}}^j = N_{\lfloor j/2 \rfloor}$ and $\operatorname{diag}(a, 1) \in U_{\mathfrak{A}}^j$ if and only if $a \in U_F^{\lceil j/2 \rceil}$. Hence

- $\mathfrak{A}$ maximal and $n = 2m$: here $c = m + 1$, $d = m$, $c_0 = 2m$, and $c + d = 2m + 1 = c_0 + 1$
- $\mathfrak{A}$ maximal and $n = 2r + 1$: here $c = r + 1$, $d = r + 1$, $c_0 = 2r + 1$, and $c + d = 2r + 2 = c_0 + 1$
- $\mathfrak{A}$ Iwahori and $n = 2r + 1$: here $c = \lfloor (r + 1)/2 \rfloor$, $d = \lceil (r + 1)/2 \rceil$, $c_0 = r$, and $c + d = r + 1 = c_0 + 1$

In each case the identity holds with no room to spare, which is why the estimate had to be made exactly.

Step 6: multiplicity one. It remains to check that $\vartheta|_{N \cap J}$ occurs in $\Lambda|_{N \cap J}$ with multiplicity at most one. If $\dim \Lambda = 1$ this is immediate. Otherwise $n = 2m$ is even and $\mathfrak{A} \cong M_2(\mathfrak{o})$, so $\dim \Lambda = q$ by lemma 4.2.5, and $J = E^\times U_{\mathfrak{A}}^m$ with $N \cap J = N_m$.

Let θ be a character of N_m occurring in $\Lambda|_{N_m}$, and let $x = \operatorname{diag}(a, 1) \in T \cap U_{\mathfrak{A}}^m$, so $a \in U_F^m$. Then $x \in J$ and x normalizes $N_m = N \cap U_{\mathfrak{A}}^m$, so if w is a vector with $\Lambda(u)w = \theta(u)w$ for all $u \in N_m$, then

$$\Lambda(u)\Lambda(x)w = \Lambda(x)\Lambda(x^{-1}ux)w = \theta\big(x^{-1}ux\big)\Lambda(x)w$$

so $\theta^{x^{-1}}$ occurs in $\Lambda|_{N_m}$ as well. As x runs over $T \cap U_{\mathfrak{A}}^m$ so does x^{-1}, so the entire orbit of θ occurs.

That orbit has exactly q elements. In the notation of Step 5, $\theta = \psi_b|_{N_m}$ with $v_F(b) = v_F(\alpha_{21}) = -n = -2m$, and conjugation by $\operatorname{diag}(a, 1)$ with $a \in U_F^m = 1 + \mathfrak{p}^m$ carries b to ab, so the orbit of the parameter is $b + b\mathfrak{p}^m = b + \mathfrak{p}^{-m}$. Two parameters restrict to the same character of $N_m = \mathfrak{p}^m$

exactly when they differ by an element of $\mathfrak{p}^{1-m}$, so the orbit of θ is in bijection with $\mathfrak{p}^{-m}/\mathfrak{p}^{1-m}$, which has q elements. Since $\dim \Lambda = q$ and all q conjugates occur, each occurs exactly once and $\Lambda|_{N\cap J}$ is multiplicity free.

Combining Steps 4 to 6, the sum in Step 4 has exactly one non-zero summand and that summand has dimension one, completing the proof.

The theorem is the reason everything that follows is canonical rather than chosen. A functional on V transforming under N by ϑ exists, and it is unique up to a scalar; so any construction that depends only on such a functional depends on nothing else. This is the property usually called *multiplicity one* for GL_2, and it is what will let a cuspidal representation be pinned down, in section 8.2, by a single operator.

Corollary 8.1.2: The Whittaker Model

Let (π, V) be an irreducible smooth representation of G of infinite dimension. There is a non-zero G-homomorphism

$$\varphi \colon V \longrightarrow \mathrm{Ind}_N^G \vartheta$$

unique up to a scalar factor, and it is injective. Its image is therefore the unique G-subspace of $\mathrm{Ind}_N^G \vartheta$ isomorphic to π; it is written $W(\pi, \vartheta)$ and called the *Whittaker model* of π.

***Proof*:**
Frobenius reciprocity (theorem 1.2.21), applied first to $N \subseteq M$ and then to $N \subseteq G$, gives canonical isomorphisms

$$\mathrm{Hom}_N(V_\vartheta, \vartheta) \cong \mathrm{Hom}_M(V, \mathrm{Ind}_N^M \vartheta) \cong \mathrm{Hom}_G(V, \mathrm{Ind}_N^G \vartheta)$$

By theorem 8.1.1 the first space has dimension one, so the last does too: a non-zero φ exists and is unique up to a scalar. Its kernel is a proper G-subspace of the irreducible V, hence zero, so φ is injective and $\mathrm{im}\varphi \cong \pi$. Uniqueness of the image follows from uniqueness of φ up to scalar, since scaling does not change the image, as we sought to show.

One fact has to be isolated before going further, because two separate steps below rest on it and it is not a formality: an infinite-dimensional irreducible representation has *no* vectors fixed by N at all.

Lemma 8.1.3: No Vector Is Fixed by N

Let (π, V) be an irreducible smooth representation of G of infinite dimension. Then $V^N = 0$; equivalently, no non-zero subspace of V, G-stable or not, has N acting trivially on it.

***Proof*:**
Suppose $0 \neq v \in V$ satisfies $\pi(n)v = v$ for every $n \in N$.

First, $v \notin V(N)$. For every compact open subgroup N_0 of N one has $\int_{N_0} \pi(n)v \, d\mu_N(n) = \mu_N(N_0)v \neq 0$, whereas by the integral criterion of lemma 2.3.2 a vector of $V(N)$ is annihilated by one such integral.

So the image of v in V_N is non-zero and π is not cuspidal (definition 2.4.1). By theorem 2.4.16

a non-cuspidal irreducible π of infinite dimension is either an irreducible $\iota_B^G\chi = \mathrm{Ind}_B^G\eta$ with $\eta = \delta_B^{-1/2}\chi$, or a special representation $\varphi\mathrm{St}_G$. In the second case (2.12) and (2.11), twisted by $\varphi\circ\det$ (which passes inside the induction as the character φ_T of T), realize $\varphi\mathrm{St}_G$ as a G-subspace of $\mathrm{Ind}_B^G\varphi_T\delta_B^{-1}$. Either way $\mathbb{C}v$ is a one-dimensional N-subspace of some $\mathrm{Ind}_B^G\xi$ with ξ a character of T, so proposition 2.4.12 makes $\mathbb{C}v$ a G-subspace, and it lies inside π. Irreducibility then gives $V = \mathbb{C}v$, against $\dim V = \infty$, completing the proof.

The middle term of that chain is the more useful one, because it lives on the mirabolic group, where section 2.3 already determined everything. Reading it out gives a sharper statement, and one that detects cuspidality.

Corollary 8.1.4: The Image of the Mirabolic Model

Let (π, V) be an irreducible smooth representation of G of infinite dimension.

1. There is a non-zero M-homomorphism $f\colon V \to \mathrm{Ind}_N^M\vartheta$, unique up to a scalar factor
2. The map f is injective
3. The image of f contains $\text{c-Ind}_N^M\vartheta$, and f induces an M-isomorphism

$$V_N \xrightarrow{\sim} \mathrm{im}f/\text{c-Ind}_N^M\vartheta$$

In particular π is cuspidal if and only if $\mathrm{im}f = \text{c-Ind}_N^M\vartheta$.

***Proof*:**

Part (1) is the middle isomorphism of the chain in the proof of corollary 8.1.2.

For (2), write $U = \ker f$. The composite $V \to \mathrm{Ind}_N^M\vartheta \to \vartheta$ induces an isomorphism $V_\vartheta \cong \vartheta$, since by theorem 8.1.1 both sides have dimension one and the map is non-zero; hence $U_\vartheta = 0$. Note that U is only an M-subspace of V, because $\mathrm{Ind}_N^M\vartheta$ carries no action of G and f is equivariant for M alone, so irreducibility of π says nothing about it directly. The argument runs instead through the mirabolic group.

Apply corollary 2.3.5 to $U(N)$, which is an M-subspace of U because U is M-stable. Both hypotheses hold: $U(N)_N = 0$ by (2.7), and $U(N)_\vartheta \cong U_\vartheta = 0$ by (2.8). Hence $U(N) = 0$, which says precisely that N acts trivially on U. By lemma 8.1.3 that forces $U = 0$, so f is injective.

For (3), note first that $V(N) \neq 0$: were it zero, N would act trivially on the whole of V, against lemma 8.1.3. So injectivity gives $f(V(N)) \neq 0$, and $f(V(N)) \subseteq (\mathrm{Ind}_N^M\vartheta)(N) = \text{c-Ind}_N^M\vartheta$. The latter is irreducible over M by theorem 2.3.6, and $f(V(N))$ is a non-zero M-subspace of it, so the two are equal. Hence $\mathrm{im}f \supseteq \text{c-Ind}_N^M\vartheta$, and f induces an isomorphism $V_N = V/V(N) \cong \mathrm{im}f/\text{c-Ind}_N^M\vartheta$.

The final assertion is now immediate: π is cuspidal precisely when $V_N = 0$ (definition 2.4.1), which by (3) happens precisely when $\mathrm{im}f = \text{c-Ind}_N^M\vartheta$, completing the proof.

Definition 8.1.5: Kirillov Model

Let (π, V) be an irreducible smooth representation of G of infinite dimension. Write $K(\pi,\vartheta) = \mathrm{im} f$ for the image of the map f of corollary 8.1.4, and let π_K be the action of G transported from V along the isomorphism $f\colon V \to K(\pi,\vartheta)$, that is $\pi_K(g) = f \circ \pi(g) \circ f^{-1}$. The pair $\big(\pi_K, K(\pi,\vartheta)\big)$ is the *Kirillov model* of π.

Neither half of that is a choice, and it is worth recording precisely why, because what gets cited downstream is the rigidity rather than the construction.

Lemma 8.1.6: The Kirillov Model Is Canonical

With the notation above:

1. $K(\pi,\vartheta)$ is the unique M-subspace of $\mathrm{Ind}_N^M \vartheta$ that is M-isomorphic to V
2. π_K is the unique homomorphism $G \to \mathrm{Aut}_{\mathbb{C}}\big(K(\pi,\vartheta)\big)$ that restricts on M to the natural action of M on $K(\pi,\vartheta)$ and satisfies $\pi_K \cong \pi$

***Proof*:**

For (1), let V' be an M-subspace of $\mathrm{Ind}_N^M \vartheta$ admitting an M-isomorphism $u\colon V \to V'$. Composing with the inclusion makes u a non-zero element of $\mathrm{Hom}_M(V, \mathrm{Ind}_N^M \vartheta)$, so $u = \lambda f$ for some $\lambda \in \mathbb{C}^\times$ by corollary 8.1.4(1). Hence $V' = \mathrm{im} u = \mathrm{im} f$.

For (2), existence is the definition: π_K is a homomorphism because f is bijective onto its image by corollary 8.1.4(2), it satisfies $\pi_K \cong \pi$ by construction, and it restricts on M to the natural action because f is M-equivariant.

Uniqueness rests on $\mathrm{End}_M\big(K(\pi,\vartheta)\big) = \mathbb{C}$, which we check first. Let A be an M-endomorphism of $K = K(\pi,\vartheta)$. Since A commutes with the action of N it preserves $K(N)$, which is $\text{c-Ind}_N^M \vartheta$ by the proof of corollary 8.1.4(3). That space is irreducible over M by theorem 2.3.6, so theorem 1.2.33 makes A act on it by a scalar λ. Then $A - \lambda$ kills $\text{c-Ind}_N^M \vartheta$ and so induces an M-map $K/\text{c-Ind}_N^M \vartheta \to K$. Its source is V_N by corollary 8.1.4(3), on which N acts trivially, so its image is a subspace of K with N acting trivially. As $K \cong V$ as M-representations, lemma 8.1.3 makes that image zero, so $A = \lambda$.

Now let ρ be another such homomorphism. Since $\rho \cong \pi \cong \pi_K$ there is a G-isomorphism $u\colon (\pi_K, K) \to (\rho, K)$. Both actions restrict on M to the same one, so u is an M-endomorphism of K, hence $u = \lambda$ for some $\lambda \in \mathbb{C}^\times$ by the previous paragraph. Equivariance then reads $\lambda \pi_K(g) = \rho(g)\lambda$ for every g, that is $\rho = \pi_K$, as we sought to show.

What has been gained is worth stating precisely, because it is the whole point of the construction. The space $K(\pi,\vartheta)$ is described purely in terms of the mirabolic group: by corollary 8.1.4 it sits between $\text{c-Ind}_N^M \vartheta$ and $\mathrm{Ind}_N^M \vartheta$, and for cuspidal π it *is* $\text{c-Ind}_N^M \vartheta$, a space that section 2.3 understands completely. Yet it carries an action of the whole of G. The extra information beyond the mirabolic action is therefore concentrated in the part of G that M misses, and the Bruhat decomposition says that is a single element. Extracting it is the business of section 8.2.

Corollary 8.1.7: Restriction Identifies the Two Models

Restriction of functions from G to M induces an M-isomorphism

$$W(\pi,\vartheta) \xrightarrow{\sim} K(\pi,\vartheta)$$

and hence a G-isomorphism $\big(\pi, W(\pi,\vartheta)\big) \cong \big(\pi_K, K(\pi,\vartheta)\big)$.

***Proof*:**

The isomorphism $\mathrm{Hom}_M(V, \mathrm{Ind}_N^M \vartheta) \cong \mathrm{Hom}_G(V, \mathrm{Ind}_N^G \vartheta)$ of theorem 1.2.21 is realized concretely by restricting functions from G to M: a G-equivariant map into $\mathrm{Ind}_N^G\vartheta$ composed with restriction is an M-equivariant map into $\mathrm{Ind}_N^M\vartheta$, and the inverse sends an M-map f to the G-map $v \mapsto \big(g \mapsto f(\pi(g)v)(1)\big)$. Applying this to the maps φ and f of corollary 8.1.2 and corollary 8.1.4, which correspond to one another under the chain, identifies their images. The G-structure transports because the correspondence is natural, which is exactly the uniqueness clause in lemma 8.1.6(2), as we sought to show.

Example 8.1.8: The Kirillov Model of a Principal Series

Take $\chi = \chi_1 \otimes \chi_2$ with $\chi_1\chi_2^{-1} \neq \|-\|^{\pm 1}$, so that $\pi = \iota_B^G \chi$ is irreducible by lemma 2.4.15, of infinite dimension. This is the case in which the inclusion of corollary 8.1.4(3) is strict, and the example is worth carrying because it shows what cuspidality gives.

Realize π on $X = \mathrm{Ind}_B^G \eta$ with $\eta = \delta_B^{-1/2}\chi$. By theorem 2.4.4 the Jacquet module X_N sits in an exact sequence with outer terms $\eta^w\delta_B^{-1}$ and η, both one-dimensional characters of T, so $\dim X_N = 2$. By corollary 8.1.4(3),

$$\dim\big(K(\pi,\vartheta)/\text{c-Ind}_N^M\vartheta\big) = \dim X_N = 2$$

So the Kirillov model of an irreducible principal series is strictly larger than $\text{c-Ind}_N^M\vartheta$, by exactly two dimensions.

Concretely, restricting functions to the subgroup S of matrices $\mathrm{diag}(x,1)$ identifies $\text{c-Ind}_N^M\vartheta$ with $C_c^\infty(F^\times)$, as in the discussion following theorem 2.3.6. Under the same identification, $K(\pi,\vartheta)$ becomes the space of functions on $F^\times$ that agree with an element of $C_c^\infty(F^\times)$ away from 0, but which near 0 are allowed two extra degrees of freedom, the two dimensions counted above. The Jacquet module says what those degrees of freedom are, since the isomorphism of corollary 8.1.4(3) is one of M-modules and M acts on X_N through $M \to S$: the germ at 0 of a function in $K(\pi,\vartheta)$ transforms under S by a character occurring in X_N. By the computation in the proof of lemma 2.4.15 the two characters in question are

$$\eta = \chi_1\|-\|^{1/2} \otimes \chi_2\|-\|^{-1/2}, \qquad \eta^w\delta_B^{-1} = \chi_2\|-\|^{1/2} \otimes \chi_1\|-\|^{-1/2}$$

so on S they read $x \mapsto \chi_1(x)\|x\|^{1/2}$ and $x \mapsto \chi_2(x)\|x\|^{1/2}$.

When $\chi_1 \neq \chi_2$ those two characters are distinct, X_N is their direct sum, and a function in $K(\pi,\vartheta)$ behaves near 0 like $c_1\chi_1(x)\|x\|^{1/2} + c_2\chi_2(x)\|x\|^{1/2}$. The hypothesis $\chi_1\chi_2^{-1} \neq \|-\|^{\pm 1}$ also permits $\chi_1 = \chi_2$, and by the display the two characters then coincide, so a single character occurs in X_N and the extension is non-split. Indeed theorem 1.2.21 gives $\mathrm{Hom}_T(X_N,\eta) \cong \mathrm{Hom}_B(X,\eta) \cong \mathrm{Hom}_G(X, \mathrm{Ind}_B^G\eta) = \mathrm{End}_G(X)$, of dimension one by proposition 2.4.13, whereas $X_N \cong \eta \oplus \eta$ would

make it two-dimensional. The second degree of freedom is then a valuation-twisted term rather than a second character, and a function in $K(\pi,\vartheta)$ behaves near 0 like $\big(c_1 + c_2 v_F(x)\big)\chi_1(x)\|x\|^{1/2}$.

For cuspidal π the count is $\dim V_N = 0$, and the model collapses onto $C_c^\infty(F^\times)$ exactly. That is the equality case of corollary 8.1.4, and it is the situation the next section works in.

Exercise 8.1.1

1. Let φ be a character of $F^\times$ and let $\pi = \varphi\circ\det$. Show directly that $V_\vartheta = 0$ for every non-trivial character ϑ of N, and explain where the proof of theorem 8.1.1 uses the hypothesis that $\dim V = \infty$.
2. Let φ be a character of $F^\times$ and consider the exact sequence $0 \to \varphi_G \to \mathrm{Ind}_B^G \varphi_T \to \varphi\mathrm{St}_G \to 0$ of (2.10). Using Step 1 of the proof of theorem 8.1.1, compute $\dim(\varphi\mathrm{St}_G)_\vartheta$ and $\dim(\varphi_G)_\vartheta$, and confirm that the three dimensions are consistent.
3. Show that the group $T \cap \mathrm{GL}_2(\mathfrak{o})$ acts transitively on the set of characters of N that are trivial on N_1 but not on N_0. How many such characters are there?
4. Deduce from corollary 8.1.4 that an irreducible smooth representation of G of infinite dimension is cuspidal if and only if its Kirillov model is $\text{c-Ind}_N^M \vartheta$. Then show that this gives a criterion for cuspidality that makes no reference to Jacquet modules.
5. Let ϑ and ϑ' be two non-trivial characters of N. Show that there is $t \in T$ with $\vartheta' = \vartheta^t$, and deduce that $W(\pi,\vartheta)$ and $W(\pi,\vartheta')$ are carried to one another by the action of t. Conclude that the Whittaker model is independent of the choice of ϑ up to a canonical identification.

8.2 Manifestation of the Local Constant

Section 8.1 put an irreducible cuspidal representation on a space of functions on $F^\times$ and left exactly one thing undetermined there. The mirabolic group acts by formulas that do not mention π, and the Bruhat decomposition reduces every element outside the mirabolic group to the central character together with the single operator $\pi_K(w)$. This section computes that operator, and the local constants of corollary 5.2.9 are what comes out: $\pi_K(w)$ permutes a natural basis of $C_c^\infty(F^\times)$, scaling each basis vector by the local constant of a twist of π.

Throughout this section (π, V) is an irreducible cuspidal representation of G with central character ω_π, and $\psi \in \widehat{F}$ is a fixed additive character of level one, so ψ is trivial on $\mathfrak{p}$ and non-trivial on $\mathfrak{o}$. This is the normalization under which the epsilon exponent $n(\pi,\psi)$ of definition 5.3.4 is defined. The character ϑ of N is the one attached to ψ, namely $\vartheta\left(\begin{smallmatrix}1 & x\\ 0 & 1\end{smallmatrix}\right) = \psi(x)$, and since ϑ is determined by ψ the models of section 8.1 are written $W(\pi,\psi) = W(\pi,\vartheta)$ and $K(\pi,\psi) = K(\pi,\vartheta)$. Three families of matrices occur constantly, and abbreviating them keeps the computations readable:

$$\sigma(x) = \begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix}, \qquad u(b) = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, \qquad \bar{u}(c) = \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}$$

for $x \in F^\times$ and $b, c \in F$, so that $S = \{\sigma(x) : x \in F^\times\}$, $N = \{u(b) : b \in F\}$ and $M = NS$. Finally $w = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, an element of determinant 1 with $w^2 = -I$.

Definition 8.2.1: The Kirillov Model on $C_c^\infty(F^\times)$

Let (π, V) be an irreducible cuspidal representation of G. Restriction of functions to S, followed by the identification $\sigma(x) \mapsto x$ of S with $F^\times$, is a linear isomorphism

$$j\colon K(\pi,\psi) \longrightarrow C_c^\infty(F^\times), \qquad j(W)(x) = W(\sigma(x))$$

Transporting the action π_K of definition 8.1.5 along j gives a smooth representation of G on $C_c^\infty(F^\times)$, isomorphic to π, again written π_K and again called the *Kirillov model* of π. For $\varphi \in C_c^\infty(F^\times)$, $x, y \in F^\times$, $b \in F$ and $z \in F^\times$ it satisfies

$$\big(\pi_K(\sigma(y))\varphi\big)(x) = \varphi(xy) \tag{8.2}$$

$$\big(\pi_K(u(b))\varphi\big)(x) = \psi(bx)\,\varphi(x) \tag{8.3}$$

$$\pi_K(zI)\varphi = \omega_\pi(z)\,\varphi \tag{8.4}$$

Each assertion is a short computation. For j, every element of M factors uniquely as $u(b)\sigma(x)$, since $\left(\begin{smallmatrix} x & b \\ 0 & 1 \end{smallmatrix}\right) = u(b)\sigma(x)$ and the two factors are recovered from the entries. A function W in $\mathrm{Ind}_N^M \vartheta$ therefore satisfies $W(u(b)\sigma(x)) = \psi(b)W(\sigma(x))$ and is determined by its restriction to S. That restriction is locally constant exactly when W is smooth, and has compact support in $F^\times$ exactly when W has compact support modulo N. Both equivalences are the description of $\mathrm{Ind}_N^M \vartheta$ recorded after theorem 2.3.6. Since π is cuspidal, corollary 8.1.4 gives $K(\pi,\psi) = \text{c-Ind}_N^M \vartheta$, so j is a bijection onto $C_c^\infty(F^\times)$. For (8.2), the action of M on $K(\pi,\psi)$ is by right translation, and $\sigma(x)\sigma(y) = \sigma(xy)$, so $j(\pi_K(\sigma(y))W)(x) = W(\sigma(x)\sigma(y)) = j(W)(xy)$. For (8.3), the identity $\sigma(x)u(b) = u(xb)\sigma(x)$ gives $W(\sigma(x)u(b)) = \psi(xb)W(\sigma(x))$. For (8.4), π_K is isomorphic to π by definition 8.1.5, and $\pi(zI)$ is the scalar $\omega_\pi(z)$ by corollary 1.2.34. An isomorphism carries a scalar operator to the same scalar.

The right-hand sides of (8.2) and (8.3) are the same for every cuspidal π, so the restriction of π_K to M carries no information about π, and (8.4) contributes only ω_π. By lemma 2.1.2 one has $G = B \cup BwB$, since the element w used here differs from the representative there by the diagonal matrix $\mathrm{diag}(1,-1) \in T$ and BwB depends only on the coset TwT. Every element of B factors as $zI \cdot m$ with $z \in F^\times$ and $m \in M$, by $\left(\begin{smallmatrix} t_1 & y \\ 0 & t_2 \end{smallmatrix}\right) = t_2 I \cdot u(y/t_2)\sigma(t_1/t_2)$, so $B = ZM$ and

$$G = ZM \cup ZMwM$$

Consequently $\pi_K(g)$ is a product of operators of the three shapes above together with at most one factor $\pi_K(w)$: the representation π_K, and hence π itself, is determined by the pair $\big(\omega_\pi, \pi_K(w)\big)$. Computing $\pi_K(w)$ is therefore the same as computing π.

Before computing it, one compatibility is needed. Twisting π by a character χ of $F^\times$ produces another cuspidal representation, with its own Kirillov model on the same space $C_c^\infty(F^\times)$, and the two models are related in the simplest possible way.

Lemma 8.2.2: Twisting the Kirillov Model

Let (π, V) be an irreducible cuspidal representation of G and let χ be a character of $F^\times$. Then $\chi\pi = (\chi \circ \det) \otimes \pi$ is irreducible and cuspidal, and the map

$$t_\chi \colon C_c^\infty(F^\times) \longrightarrow C_c^\infty(F^\times), \qquad t_\chi(\varphi) = \chi^{-1}\varphi$$

is an isomorphism of representations $(\chi\pi)_K \to \chi\,\pi_K$, that is

$$t_\chi \circ (\chi\pi)_K(g) = \chi(\det g)\,\pi_K(g) \circ t_\chi, \qquad g \in G$$

***Proof*:**
For $n \in N$ one has $\det n = 1$, so $\chi\pi(n) = \pi(n)$ and the subspace $V(N)$ spanned by the vectors $v - \pi(n)v$ is unchanged, so $(\chi\pi)_N = V_N = 0$ and $\chi\pi$ is cuspidal. The operators $\chi\pi(g)$ and $\pi(g)$ differ by a non-zero scalar, so the two representations have the same invariant subspaces and the same open stabilizers, whence $\chi\pi$ is smooth and irreducible. This is the argument already recorded in the proof of lemma 5.5.1.

Work first in the induced space. Define

$$T_\chi \colon \mathrm{Ind}_N^G \vartheta \longrightarrow \mathrm{Ind}_N^G \vartheta, \qquad T_\chi(F)(x) = \chi^{-1}(\det x)\,F(x)$$

This lands in $\mathrm{Ind}_N^G\vartheta$ because $\det(nx) = \det x$ for $n \in N$, so $T_\chi(F)(nx) = \vartheta(n)T_\chi(F)(x)$. It preserves smoothness and is bijective, with inverse $F \mapsto (\chi \circ \det)F$. Writing ρ for the action of G on $\mathrm{Ind}_N^G\vartheta$ by right translation,

$$T_\chi\big(\rho(g)F\big)(x) = \chi^{-1}(\det x)F(xg), \qquad \rho(g)\big(T_\chi F\big)(x) = \chi^{-1}(\det g)\chi^{-1}(\det x)F(xg)$$

so that $T_\chi \circ \rho(g) = \chi(\det g)\,\rho(g) \circ T_\chi$ for every $g \in G$.

Apply this on the subspace $W(\chi\pi, \psi)$, where ρ realizes $\chi\pi$ by corollary 8.1.2. The displayed relation shows that $T_\chi\big(W(\chi\pi,\psi)\big)$ is again ρ-stable, and that $\rho(g)$ acts on it as $\chi^{-1}(\det g)\cdot\chi\pi(g) = \pi(g)$ transported by T_χ. So $T_\chi\big(W(\chi\pi,\psi)\big)$ is a G-subspace of $\mathrm{Ind}_N^G\vartheta$ isomorphic to π, and the uniqueness clause of corollary 8.1.2 forces

$$T_\chi\big(W(\chi\pi,\psi)\big) = W(\pi,\psi)$$

Finally transport to $C_c^\infty(F^\times)$. By corollary 8.1.7 the passage from $W(\pi,\psi)$ to $K(\pi,\psi)$ is restriction of functions, and j of definition 8.2.1 evaluates at $\sigma(x)$, where $\det\sigma(x) = x$. Hence T_χ becomes $\varphi \mapsto \chi^{-1}\varphi$, which is t_χ, and the intertwining relation becomes the one in the statement, as we sought to show.

The functions on which the answer will be expressed are the ones adapted to the decomposition of $F^\times$ into the level sets of the valuation. Each of those sets is a coset of U_F, and a smooth function on a compact abelian group is a combination of characters, so the following functions are the natural coordinates on $C_c^\infty(F^\times)$.

Definition 8.2.3: The Standard Functions $\xi_{\chi,k}$

For a character χ of $F^\times$ and $k \in \mathbb{Z}$, define $\xi_{\chi,k}\colon F^\times \to \mathbb{C}$ by

$$\xi_{\chi,k}(x) = \begin{cases} \chi(x) & v_F(x) = k \\ 0 & \text{otherwise} \end{cases}$$

Then $\xi_{\chi,k} \in C_c^\infty(F^\times)$, and the set of all such functions spans $C_c^\infty(F^\times)$.

That $\xi_{\chi,k}$ lies in $C_c^\infty(F^\times)$ is immediate from the definition: its support is the compact open set $\varpi^k U_F$, and on that set it agrees with the locally constant function χ. For the spanning claim, let $\varphi \in C_c^\infty(F^\times)$. Its support is compact and $F^\times = \bigsqcup_k \varpi^k U_F$ is a partition into open sets, so $\operatorname{supp}\varphi$ meets only finitely many of them, and φ is a finite sum of functions each supported on a single $\varpi^k U_F$. Fix such a summand and call it φ_k. Being smooth, φ_k is invariant under right translation by U_F^m for some $m \geq 1$, so $u \mapsto \varphi_k(\varpi^k u)$ is a function on the finite abelian group U_F/U_F^m. The characters of a finite abelian group form a basis of its space of complex-valued functions: there are $|A|$ of them by [5, Abelian Groups And Degree 1 Decomposition], and they are orthonormal, hence independent, by [5, Row Orthogonality]. So $\varphi_k(\varpi^k u) = \sum_\theta c_\theta\, \theta(u)$ for finitely many characters θ of U_F/U_F^m, inflated to U_F. Each such θ extends to a character of $F^\times$: the decomposition $F^\times = \varpi^{\mathbb{Z}} \times U_F$ lets one set $\chi_\theta(\varpi) = 1$ and $\chi_\theta|_{U_F} = \theta$. Then $\xi_{\chi_\theta,k}(\varpi^k u) = \theta(u)$, so $\varphi_k = \sum_\theta c_\theta\, \xi_{\chi_\theta,k}$.

Two of these functions coincide precisely when the two characters agree on $\varpi^k U_F$, so the indexing is redundant. Choosing one χ for each character of U_F and each k produces an actual basis, since functions with different k have disjoint supports and distinct characters of U_F are linearly independent. Nothing below needs that refinement, only the spanning property.

The action of the mirabolic group on these functions can be written down at once. By (8.2),

$$\big(\pi_K(\sigma(y))\xi_{\chi,k}\big)(x) = \xi_{\chi,k}(xy) = \chi(y)\,\xi_{\chi,k-v_F(y)}(x)$$

so S permutes the lines $\mathbb{C}\xi_{\chi,k}$, shifting k by $-v_F(y)$, and by (8.4) the centre acts on each of them by ω_π. The subgroup N acts by multiplication by the function $x \mapsto \psi(bx)$ of (8.3), and nothing below has to evaluate that action on the $\xi_{\chi,k}$: the theorem evaluates $\pi_K(w)$ on each $\xi_{\chi,k}$ separately, and that, together with the mirabolic formulas, is a complete description of π_K.

Theorem 8.2.4: The Weyl Element Acts by Local Constants

Let (π, V) be an irreducible cuspidal representation of G with central character ω_π, let ψ have level one, and let χ be a character of $F^\times$. Then, for every $k \in \mathbb{Z}$,

$$\pi_K(w)\,\xi_{\chi,k} = \varepsilon\big(\chi^{-1}\pi, \tfrac{1}{2}, \psi\big)\,\xi_{\chi^{-1}\omega_\pi,\, -n(\chi^{-1}\pi,\psi)-k}$$

where ε is the local constant of corollary 5.2.9 and $n(-,\psi)$ the epsilon exponent of definition 5.3.4.

***Proof*:**

Normalize the Haar measure $d^\times x$ on G as in proposition 2.2.7, so that $\int_G h(x)\, d^\times x = \int_A h(x)\|\det x\|^{-2}\, d\mu(x)$ with μ the self-dual measure on A relative to ψ_A used in definition 5.2.1. Nothing depends on this choice: rescaling $d^\times x$ multiplies both sides of the functional equation (5.10) by the same constant, so it leaves γ, and hence ε, unchanged. Write μ_F for the self-dual measure on F relative

to ψ, and $d^\times a = \|a\|^{-1} d\mu_F(a)$ for the Haar measure it determines on $F^\times$, as in the construction of the Haar integral on $F^\times$ preceding proposition 2.2.7. The pairing $\psi_A(xy) = \psi(\mathrm{tr}(xy))$ matches x_{11} with y_{11}, x_{22} with y_{22}, x_{12} with y_{21} and x_{21} with y_{12}, so it is the product of four copies of the pairing on F, composed with a permutation of coordinates. A product of self-dual measures is self-dual for a product pairing, and permuting coordinates does not change a product measure, so μ is the product of four copies of μ_F.

Step 1: zeta integrals attached to a Whittaker function. Let $W \in \mathcal{W}(\pi, \psi)$ and let H be a compact open subgroup of G. The linear form

$$\lambda_H \colon \mathcal{W}(\pi,\psi) \longrightarrow \mathbb{C}, \qquad \lambda_H(W) = \mathrm{vol}(H)^{-1} \int_H W(h)\, d^\times h$$

is well defined, because W is locally constant and H is compact, so the integrand takes finitely many values on a set of finite measure. It is smooth: for $h_0 \in H$ the group G is unimodular (proposition 2.2.7) and $Hh_0 = H$, so $\lambda_H\big(\rho(h_0^{-1})W\big) = \mathrm{vol}(H)^{-1} \int_{Hh_0^{-1}} W(h)\, d^\times h = \lambda_H(W)$, and the stabilizer of λ_H contains the open subgroup H. By corollary 8.1.2 the space $\mathcal{W}(\pi,\psi)$ carries π, so λ_H is an element of the smooth dual, and

$$f_{W,H}(g) = \big\langle \lambda_H, \rho(g)W \big\rangle = \mathrm{vol}(H)^{-1} \int_H W(hg)\, d^\times h$$

is a matrix coefficient of π in the sense of definition 5.2.2.

Now let $\Phi \in C_c^\infty(A)$ and choose H fixing Φ under both left and right translation. Such an H exists: take N_0 and M_0 with $\mathrm{supp}\Phi \subseteq \varpi^{-N_0} M_2(\mathfrak{o})$ and Φ constant on cosets of $\varpi^{M_0} M_2(\mathfrak{o})$. Increasing N_0 enlarges the first lattice and increasing M_0 shrinks the second, so neither condition is lost when the two integers are increased, and one may assume $M_0 + N_0 \geq 1$. The subgroup $H = 1 + \varpi^{M_0+N_0} M_2(\mathfrak{o})$ is then a compact open subgroup of G, and it works, by the computation carried out for right translation in the proof of proposition 5.2.6 and its transpose on the left. Every $h \in H$ has $\det h \in U_F$, because $\det H$ is a compact subgroup of $F^\times$, so left or right multiplication by h preserves each set G_m of lemma 5.2.4. Substituting $y = hx$ and using left invariance,

$$\int_{G_m} \Phi(x) W(hx)\, d^\times x = \int_{G_m} \Phi(h^{-1}y) W(y)\, d^\times y = \int_{G_m} \Phi(y) W(y)\, d^\times y$$

and averaging over $h \in H$ gives, for the coefficients c_m of lemma 5.2.4,

$$c_m(\Phi, f_{W,H}) = \int_{G_m} \Phi(x) W(x)\, d^\times x \tag{8.5}$$

The right-hand side does not mention H. The same computation applies to the contragredient side, once one knows that H fixes $\hat{\Phi}$ as well. Write $\Phi^h(y) = \Phi(yh)$. The substitution $y = zh^{-1}$ in definition 5.2.1 multiplies μ by $\|\det h\|^{-2}$, because $z \mapsto zh^{-1}$ is an F-linear automorphism of the four-dimensional space A of determinant $\det(h)^{-2}$, and $\mathrm{tr}(xzh^{-1}) = \mathrm{tr}\big((h^{-1}x)z\big)$, so

$$\widehat{\Phi^h}(x) = \|\det h\|^{-2}\, \hat{\Phi}(h^{-1}x) = \hat{\Phi}(h^{-1}x)$$

for $h \in H$. Right invariance of Φ under H is therefore left invariance of $\hat{\Phi}$, and the same computation with the two sides exchanged gives the converse, so H fixes $\hat{\Phi}$ under both translations.

Now $\check{f}_{W,H}(x) = \mathrm{vol}(H)^{-1}\int_H W(hx^{-1})\,d^\times h$, and for fixed $h \in H$ the substitution $x = yh$ turns hx^{-1} into y^{-1}, preserves $d^\times x$ by right invariance and preserves G_m, so $\int_{G_m}\hat{\Phi}(x)W(hx^{-1})\,d^\times x = \int_{G_m}\hat{\Phi}(yh)W(y^{-1})\,d^\times y$, which is independent of h. Averaging over H gives, with $W^\vee(g) = W(g^{-1})$,

$$c_m\big(\hat{\Phi}, \check{f}_{W,H}\big) = \int_{G_m} \hat{\Phi}(x)\, W(x^{-1})\, d^\times x \tag{8.6}$$

Write $Z(\Phi, W, X)$ and $Z(\hat{\Phi}, W^\vee, X)$ for the two Laurent series with these coefficients.

Step 2: the functional equation in the variable X. Since π is cuspidal, corollary 5.3.2 gives $L(\pi, s) = L(\check{\pi}, 1-s) = 1$ and $\gamma(\pi, s, \psi) = \varepsilon(\pi, s, \psi)$, so by the proof of corollary 5.2.9 every $Z(\Phi, f, X)$ is a Laurent polynomial. Put $X = q^{-s-1/2}$, so that $\zeta(\Phi, f, s+\frac{1}{2}) = Z(\Phi, f, X)$ and $\zeta(\hat{\Phi}, \check{f}, \frac{3}{2}-s) = Z(\hat{\Phi}, \check{f}, q^{-2}X^{-1})$. With this substitution (5.10) reads

$$Z\big(\hat{\Phi}, W^\vee, q^{-2}X^{-1}\big) = \varepsilon_\pi(X)\, Z(\Phi, W, X) \tag{8.7}$$

where $\varepsilon_\pi(X)$ is $\varepsilon(\pi, s, \psi)$ rewritten in X. By definition 5.3.4, $\varepsilon(\pi, s, \psi) = \varepsilon(\pi, \frac{1}{2}, \psi)q^{(1/2-s)n(\pi,\psi)}$, and $q^{1/2-s} = q^{1/2}\cdot q^{-s} = q^{1/2}\cdot q^{1/2}X = qX$, so

$$\varepsilon_\pi(X) = \varepsilon\big(\pi, \tfrac{1}{2}, \psi\big)\,(qX)^{n(\pi,\psi)} \tag{8.8}$$

a single monomial in X.

Step 3: two Mellin transforms. For $\varphi \in C_c^\infty(F^\times)$ set

$$\Psi(\varphi, X) = \int_{F^\times} \varphi(a)\, X^{v_F(a)}\, d^\times a, \qquad \widetilde{\Psi}(\varphi, X) = \int_{F^\times} \omega_\pi(a)^{-1}\varphi(a)\, X^{v_F(a)}\, d^\times a$$

Both are Laurent polynomials in X: the integrand vanishes outside the compact set $\mathrm{supp}\varphi$, which meets only finitely many sets $\varpi^m U_F$, and on each of those $X^{v_F(a)}$ is the constant X^m. The two transforms evaluate the standard functions of definition 8.2.3 to monomials. Indeed $\omega_\pi^{-1}\xi_{\omega_\pi,k} = \xi_{1,k}$, the characteristic function of $\varpi^k U_F$, so

$$\Psi(\xi_{1,m}, X) = \mathrm{vol}^\times(U_F)\, X^m, \qquad \widetilde{\Psi}(\xi_{\omega_\pi,k}, X) = \mathrm{vol}^\times(U_F)\, X^k \tag{8.9}$$

where $\mathrm{vol}^\times(U_F) = \int_{U_F} d^\times a > 0$.

Step 4: a test function adapted to the big cell. Let $r \geq 1$ be an integer and let $L_r \subseteq A$ be the lattice with entries

$$L_r = \begin{pmatrix} \mathfrak{p}^{1-r} & \mathfrak{p}^{1-r} \\ \mathfrak{p}^r & \mathfrak{p}^r \end{pmatrix}$$

Define $\Phi_r \in C_c^\infty(A)$ by $\Phi_r(y) = \psi(-y_{12})$ for $y \in L_r + e_{22}$ and $\Phi_r = 0$ elsewhere, where e_{ij} are the matrix units. This lies in $C_c^\infty(A)$, because $L_r + e_{22}$ is compact open and ψ is locally constant.

The parameterization

$$(b, z, a, c) \longmapsto x = zI \cdot u(b)\,\sigma(a)\,\bar{u}(c) = z\begin{pmatrix} a+bc & b \\ c & 1 \end{pmatrix}$$

is a bijection from $F \times F^\times \times F^\times \times F$ onto the set of $x \in G$ with $x_{22} \neq 0$: given such an x, the four parameters are recovered as $z = x_{22}$, $b = x_{12}/z$, $c = x_{21}/z$ and $a = \det x / z^2$. Its restriction to $\mathfrak{p}^{1-r} \times U_F^r \times (\mathfrak{p}^{1-r} \setminus \{0\}) \times \mathfrak{p}^r$ is a bijection onto $\operatorname{supp}\Phi_r \cap G$: the four entry conditions read $z \in U_F^r$, $b \in \mathfrak{p}^{1-r}$, $c \in \mathfrak{p}^r$ and $a + bc \in \mathfrak{p}^{1-r}$, and $bc \in \mathfrak{p} \subseteq \mathfrak{p}^{1-r}$, so the last is $a \in \mathfrak{p}^{1-r}$.

Two consequences follow, both using that ψ has level one. First $\Phi_r(x) = \psi(-zb) = \psi(-b)$, because $(z-1)b \in \mathfrak{p}^r\mathfrak{p}^{1-r} = \mathfrak{p} \subseteq \ker\psi$. Second, let $W \in W(\pi, \psi)$ be fixed under right translation by $K_r = 1 + \varpi^r M_2(\mathfrak{o})$, which holds for all large r since W is smooth. Then $\bar{u}(c) \in K_r$ and $zI \in K_r$, so

$$W(x) = W\big(zI\, u(b)\sigma(a)\bar{u}(c)\big) = \psi(b)\, W(\sigma(a)), \qquad \Phi_r(x)W(x) = W(\sigma(a))$$

using $W(u(b)g) = \vartheta(u(b))W(g) = \psi(b)W(g)$.

The measure transports as follows. Integrate over A in the order $x_{22}, x_{21}, x_{12}, x_{11}$ and substitute in that order: $x_{22} = z$, then $x_{21} = zc$, then $x_{12} = zb$, then $x_{11} = z(a + bc)$. Each of the last three substitutions is, with the earlier variables held fixed, a scaling by z followed by a translation, so it multiplies μ_F by $\|z\|$. Hence $d\mu(x) = \|z\|^3\, d\mu_F(z)\, d\mu_F(c)\, d\mu_F(b)\, d\mu_F(a)$. Since $\det x = z^2 a$ one has $\|\det x\|^{-2} = \|z\|^{-4}\|a\|^{-2}$, so

$$d^\times x = \|a\|^{-1}\, d\mu_F(b)\, d^\times z\, d^\times a\, d\mu_F(c) \tag{8.10}$$

Since $v_F(\det x) = v_F(z^2 a) = v_F(a)$, summing (8.5) over m against X^m and collecting the three inert integrations into the positive constant $k_r = \mu_F(\mathfrak{p}^{1-r})\, \mu_F(\mathfrak{p}^r)\, \mathrm{vol}^\times(U_F^r)$ gives

$$Z(\Phi_r, W, X) = k_r \int_{\mathfrak{p}^{1-r}\setminus\{0\}} W(\sigma(a))\, X^{v_F(a)}\, \|a\|^{-1}\, d^\times a$$

Write $\varphi = j(W)$ for the Kirillov function of W, so $\varphi(a) = W(\sigma(a))$ and $\varphi \in C_c^\infty(F^\times)$. On $\varpi^m U_F$ one has $\|a\|^{-1} = q^m$, so the integral is $\Psi(\varphi, qX)$ truncated to $v_F(a) \geq 1 - r$, and since $\operatorname{supp}\varphi$ is compact in $F^\times$ that truncation is vacuous once r is large:

$$Z(\Phi_r, W, X) = k_r\, \Psi(\varphi, qX) \tag{8.11}$$

Step 5: the Fourier transform of that test function. Substituting $y \mapsto y + e_{22}$ in definition 5.2.1 and using $\mathrm{tr}(xe_{22}) = x_{22}$ and $\mathrm{tr}(e_{21}y) = y_{12}$,

$$\hat{\Phi}_r(x) = \int_{L_r} \psi(-y_{12})\, \psi\big(\mathrm{tr}(x(y + e_{22}))\big)\, d\mu(y) = \psi(x_{22}) \int_{L_r} \psi\big(\mathrm{tr}((x - e_{21})y)\big)\, d\mu(y)$$

For $\alpha \in A$ the map $y \mapsto \psi(\mathrm{tr}(\alpha y))$ is a character of the compact group L_r, trivial exactly when α lies in the dual lattice L_r^*, so the remaining integral is $\mu(L_r)$ if $x - e_{21} \in L_r^*$ and 0 otherwise. This is the orthogonality computation carried out in the proof of lemma 5.3.3. The dual lattice is read from the pairing entry by entry, x_{11} against y_{11}, x_{12} against y_{21}, x_{21} against y_{12} and x_{22} against y_{22}, and ψ is trivial exactly on $\mathfrak{p}$:

$$L_r^* = \begin{pmatrix} \mathfrak{p}^r & \mathfrak{p}^{1-r} \\ \mathfrak{p}^r & \mathfrak{p}^{1-r} \end{pmatrix}$$

The lattices L_r and L_r^* are products of the same four subgroups of F in a different order, so $\mu(L_r) = \mu(L_r^*)$, and self-duality gives $\mu(L_r)\mu(L_r^*) = 1$ by the characterization in that same proof. Hence $\mu(L_r) = 1$ and

$$\hat{\Phi}_r(x) = \psi(x_{22}) \quad \text{for } x \in \Sigma_r = L_r^* + e_{21}, \qquad \hat{\Phi}_r = 0 \text{ elsewhere}$$

The set Σ_r has $x_{21} \in 1+\mathfrak{p}^r$, so it meets G inside the cell where $x_{21} \neq 0$, which is parameterized by

$$(b,z,a,c) \longmapsto x = zI \cdot w^{-1}\,\bar{u}(c)\,\mathrm{diag}(1,a)\,u(b) = z\begin{pmatrix} -c & -a-bc \\ 1 & b \end{pmatrix}$$

Exactly as in Step 4 this is a bijection from $F \times F^\times \times F^\times \times F$ onto $\{x \in G : x_{21} \neq 0\}$, with $z = x_{21}$, $c = -x_{11}/z$, $b = x_{22}/z$ and $a = \det x/z^2$, and $\det x = z^2 a$ again. The four entry conditions defining Σ_r read $z \in U_F^r$, $c \in \mathfrak{p}^r$, $b \in \mathfrak{p}^{1-r}$ and $a+bc \in \mathfrak{p}^{1-r}$, the last of which is $a \in \mathfrak{p}^{1-r}$ because $bc \in \mathfrak{p}$, so the domain is again $\mathfrak{p}^{1-r} \times U_F^r \times (\mathfrak{p}^{1-r} \setminus \{0\}) \times \mathfrak{p}^r$. Integrating over A in the order $x_{21}, x_{11}, x_{22}, x_{12}$ and substituting $x_{21} = z$, $x_{11} = -zc$, $x_{22} = zb$, $x_{12} = -z(a+bc)$ repeats the computation of Step 4 line for line, so (8.10) holds in these coordinates too, with the same constant k_r.

On this set $x_{22} = zb$, so $\hat{\Phi}_r(x) = \psi(zb) = \psi(b)$ by the same estimate $(z-1)b \in \mathfrak{p}$ as in Step 4, and

$$x^{-1} = u(-b)\,\mathrm{diag}(1,a^{-1})\,\bar{u}(-c)\,w\,z^{-1}I$$

Three reductions evaluate $W(x^{-1})$. The factor $u(-b)$ stands on the left, so it contributes $\psi(-b)$ by the transformation law of W. Next $\bar{u}(-c)w = w\,u(c)$, a direct multiplication of matrices, and $u(c) \in K_r$ because $c \in \mathfrak{p}^r$, so that factor may be dropped, and likewise $z^{-1}I \in K_r$. Finally $\mathrm{diag}(1,a^{-1}) = (a^{-1}I)\sigma(a)$, which contributes $\omega_\pi(a)^{-1}$. Therefore

$$W(x^{-1}) = \psi(-b)\,\omega_\pi(a)^{-1}\,W(\sigma(a)w), \qquad \hat{\Phi}_r(x)\,W(x^{-1}) = \omega_\pi(a)^{-1}\,W(\sigma(a)w)$$

The function $a \mapsto W(\sigma(a)w)$ is $j(\rho(w)W)$, and j carries ρ to π_K by definition 8.2.1 and corollary 8.1.7, so it is $\pi_K(w)\varphi$. Summing (8.6) over m against X^m and repeating the computation of Step 4 with the factor $\omega_\pi(a)^{-1}$ inserted therefore gives, once r is large enough that $\mathrm{supp}\big(\pi_K(w)\varphi\big) \subseteq \{a : v_F(a) \geq 1-r\}$,

$$Z\big(\hat{\Phi}_r, W^\vee, X\big) = k_r\,\widetilde{\Psi}\big(\pi_K(w)\varphi,\ qX\big) \tag{8.12}$$

Step 6: the case $\chi = 1$. Fix $m \in \mathbb{Z}$ and let $W \in W(\pi,\psi)$ be the Whittaker function with $\varphi = j(W) = \xi_{1,m}$, which exists because j is a bijection. Choose r large enough for Steps 4 and 5 to apply to W and to $\pi_K(w)\varphi$, and substitute (8.11) and (8.12) into (8.7). The constant $k_r > 0$ cancels, leaving

$$\widetilde{\Psi}\big(\pi_K(w)\varphi,\ q^{-1}X^{-1}\big) = \varepsilon_\pi(X)\,\Psi(\varphi, qX) \tag{8.13}$$

Write $\varphi' = \pi_K(w)\xi_{1,m}$. The function φ' transforms under U_F by ω_π. Indeed $\sigma(u)w = w\,(uI)\,\sigma(u^{-1})$ for $u \in U_F$, a direct multiplication of matrices, and $\pi_K(\sigma(u^{-1}))\xi_{1,m} = \xi_{1,m}$ by (8.2), since $\xi_{1,m}$ is the characteristic function of $\varpi^m U_F$, so

$$\varphi'(xu) = \big(\pi_K(\sigma(u))\varphi'\big)(x) = \omega_\pi(u)\,\varphi'(x)$$

Consequently φ' restricted to $\varpi^k U_F$ equals $c_k\,\xi_{\omega_\pi,k}$ with $c_k = \varphi'(\varpi^k)\omega_\pi(\varpi^k)^{-1}$, and $\varphi' = \sum_k c_k \xi_{\omega_\pi,k}$, a finite sum because φ' has compact support. Now evaluate both sides of (8.13) using (8.9) and (8.8):

$$\mathrm{vol}^\times(U_F)\sum_k c_k\,q^{-k}X^{-k} = \varepsilon\big(\pi, \tfrac{1}{2}, \psi\big)(qX)^{n(\pi,\psi)}\,\mathrm{vol}^\times(U_F)\,(qX)^m$$

Cancel $\mathrm{vol}^\times(U_F) \neq 0$ and compare the two Laurent polynomials term by term. The right-hand side is a single monomial of degree $n(\pi,\psi)+m$, so $c_k = 0$ unless $-k = n(\pi,\psi)+m$, and for that one k the right-hand side reads $\varepsilon(\pi,\frac{1}{2},\psi)\,q^{-k}X^{-k}$, giving $c_k = \varepsilon(\pi,\frac{1}{2},\psi)$. Thus

$$\pi_K(w)\,\xi_{1,m} = \varepsilon\big(\pi,\tfrac{1}{2},\psi\big)\,\xi_{\omega_\pi,\,-n(\pi,\psi)-m} \tag{8.14}$$

which is the assertion for $\chi = 1$.

Step 7: the general case. Let χ be a character of $F^\times$. The representation $\chi^{-1}\pi$ is irreducible and cuspidal by lemma 8.2.2, and its central character is $\omega_{\chi^{-1}\pi} = \chi^{-2}\omega_\pi$, since $\chi^{-1}\pi(zI) = \chi^{-1}(z^2)\omega_\pi(z)$. Applying (8.14) to $\chi^{-1}\pi$ gives

$$(\chi^{-1}\pi)_K(w)\,\xi_{1,k} = \varepsilon\big(\chi^{-1}\pi,\tfrac{1}{2},\psi\big)\,\xi_{\chi^{-2}\omega_\pi,\,-n(\chi^{-1}\pi,\psi)-k}$$

Apply lemma 8.2.2 with the character χ and the representation $\chi^{-1}\pi$, whose twist by χ is π itself. Since $\det w = 1$, the intertwining relation at $g = w$ reads $t_\chi \circ \pi_K(w) = (\chi^{-1}\pi)_K(w) \circ t_\chi$. Because $t_\chi \xi_{\chi,k} = \chi^{-1}\xi_{\chi,k} = \xi_{1,k}$,

$$t_\chi\big(\pi_K(w)\xi_{\chi,k}\big) = (\chi^{-1}\pi)_K(w)\,\xi_{1,k} = \varepsilon\big(\chi^{-1}\pi,\tfrac{1}{2},\psi\big)\,\xi_{\chi^{-2}\omega_\pi,\,-n(\chi^{-1}\pi,\psi)-k}$$

Applying the inverse of t_χ, which is multiplication by χ, and using $\chi\,\xi_{\chi^{-2}\omega_\pi,l} = \xi_{\chi^{-1}\omega_\pi,l}$, gives

$$\pi_K(w)\,\xi_{\chi,k} = \varepsilon\big(\chi^{-1}\pi,\tfrac{1}{2},\psi\big)\,\xi_{\chi^{-1}\omega_\pi,\,-n(\chi^{-1}\pi,\psi)-k}$$

completing the proof.

The formula says three separate things. The character index moves by $\chi \mapsto \chi^{-1}\omega_\pi$, an involution of the set of characters, since applying it twice returns χ. It has to be an involution, because $w^2 = -I$ acts by the scalar $\omega_\pi(-1)$ and therefore preserves every line $\mathbb{C}\xi_{\chi,k}$. The valuation index moves by $k \mapsto -k - n(\chi^{-1}\pi,\psi)$, a reflection displaced by the epsilon exponent, so the integer that definition 5.3.4 extracted from the degree of a monomial is here the displacement of a reflection on $\mathbb{Z}$. The scalar is the value at $s = \frac{1}{2}$ of the local constant of the twisted representation, which is the centre of symmetry of the functional equation. Every twist of π contributes, but not to a single line. Neither $\varepsilon(\chi^{-1}\pi,\frac{1}{2},\psi)$ nor $n(\chi^{-1}\pi,\psi)$ depends on k, so the single twist $\chi^{-1}\pi$ governs the whole family $\xi_{\chi,k}$, $k \in \mathbb{Z}$, at once. The constant is attached to the labelled function and not to the line it spans. For unramified ν the function $\xi_{\chi\nu,k} = \nu(\varpi)^k\xi_{\chi,k}$ spans that same line, while the theorem itself forces $\varepsilon\big((\chi\nu)^{-1}\pi,\frac{1}{2},\psi\big) = \nu(\varpi)^{-n(\chi^{-1}\pi,\psi)}\,\varepsilon\big(\chi^{-1}\pi,\frac{1}{2},\psi\big)$, as the exercises below check. The formula stays consistent because the target function is rescaled by the matching factor.

The same formula, read as a statement about the local constants rather than about the operator, says that the family $\varepsilon(\chi\pi,s,\psi)$, taken over all characters χ, records the matrix entries of the one operator that the formulas of definition 8.2.1 leave undetermined. That is enough to recover π.

Corollary 8.2.5: The Cuspidal Converse Theorem from the Kirillov Model

Let π and π' be irreducible cuspidal representations of G with $\omega_\pi = \omega_{\pi'}$, and let ψ have level one. Suppose

$$\varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi\pi', s, \psi)$$

for every character χ of $F^\times$. Then $\pi \cong \pi'$.

***Proof*:**

Fix a character χ of $F^\times$ and apply the hypothesis to χ^{-1}. By corollary 5.2.9 both sides are monomials cq^{-ds}, and such a monomial determines c and d uniquely, as recorded after definition 5.3.4. Hence

$$\varepsilon\big(\chi^{-1}\pi, \tfrac{1}{2}, \psi\big) = \varepsilon\big(\chi^{-1}\pi', \tfrac{1}{2}, \psi\big), \qquad n(\chi^{-1}\pi, \psi) = n(\chi^{-1}\pi', \psi)$$

Since $\omega_\pi = \omega_{\pi'}$, the two right-hand sides in theorem 8.2.4 agree for π and for π', so

$$\pi_K(w)\,\xi_{\chi,k} = \pi'_K(w)\,\xi_{\chi,k}$$

for every χ and every k. These functions span $C_c^\infty(F^\times)$ by definition 8.2.3, and both operators are linear, so $\pi_K(w) = \pi'_K(w)$.

The two representations π_K and π'_K act on the same space $C_c^\infty(F^\times)$. On M they agree, because (8.2) and (8.3) do not involve the representation and $M = NS$; on Z they agree because $\omega_\pi = \omega_{\pi'}$ in (8.4); and at w they agree by the previous paragraph. By $G = ZM \cup ZMwM$, every $g \in G$ is a product of elements of Z and M with at most one factor w, and π_K, π'_K are both group homomorphisms, so $\pi_K(g) = \pi'_K(g)$ for all $g \in G$. Finally $\pi \cong \pi_K = \pi'_K \cong \pi'$ by lemma 8.1.6, as we sought to show.

This discharges the cuspidal case of theorem 5.5.2. That theorem's L-hypothesis is vacuous for cuspidal π, since $L(\chi\pi, s) = 1$ for every twist by lemma 5.5.1, so the two statements have the same content there and Step 3 of its proof invokes the corollary above. The route needs nothing beyond the model built in section 8.1 and the functional equation. In particular the stability theorem theorem 5.3.10 is not needed, so theorem 5.5.2 does not rest on it.

Example 8.2.6: The Weyl Operator for a Level Zero Cuspidal Representation

Let π be cuspidal of level zero and let ψ have level one. Example 5.3.7 computes $n(\pi, \psi) = 0$ and shows that $\varepsilon(\pi, s, \psi)$ is the constant c_1/c_0 formed from two finite sums over $\mathrm{GL}_2(k)$, so $\varepsilon(\pi, \frac{1}{2}, \psi) = c_1/c_0$. With $\chi = 1$, theorem 8.2.4 reads

$$\pi_K(w)\,\xi_{1,k} = \frac{c_1}{c_0}\,\xi_{\omega_\pi,\,-k}$$

so on these functions the operator is a pure reflection $k \mapsto -k$ of the valuation, scaled by one complex number, with no displacement at all. Since $\xi_{1,k}$ is the characteristic function of $\varpi^k U_F$, the functions $\xi_{1,k}$ span the U_F-invariant part of $C_c^\infty(F^\times)$, and $\pi_K(w)$ carries that subspace onto the subspace of functions transforming under U_F by ω_π.

The unramified twists give a check on the formula. Let χ be unramified with $\chi(\varpi) = \lambda$. Then

$\chi(x) = \lambda^{v_F(x)}$, so directly from definition 8.2.3,

$$\xi_{\chi,k} = \lambda^k\, \xi_{1,k}, \qquad \xi_{\chi^{-1}\omega_\pi,\,-k} = \lambda^k\, \xi_{\omega_\pi,\,-k}$$

Item (5) of example 5.3.7 gives $n(\chi^{-1}\pi,\psi) = n(\pi,\psi) = 0$ and $\varepsilon(\chi^{-1}\pi,s,\psi) = \chi^{-1}(\varpi)^{n(\pi,\psi)}\varepsilon(\pi,s,\psi) = \varepsilon(\pi,s,\psi)$. So the theorem applied to χ predicts

$$\pi_K(w)\,\xi_{\chi,k} = \frac{c_1}{c_0}\,\xi_{\chi^{-1}\omega_\pi,\,-k} = \frac{c_1}{c_0}\,\lambda^k\,\xi_{\omega_\pi,\,-k}$$

and applying $\pi_K(w)$ to $\xi_{\chi,k} = \lambda^k\xi_{1,k}$ by linearity gives the same value. The two computations agree, and they agree because the exponent vanishes. For a twist by a ramified χ the displacement $n(\chi^{-1}\pi,\psi)$ need not be zero, and the reflection is then shifted off centre.

Exercise 8.2.1

1. Let π be irreducible cuspidal. Using definition 8.2.1, show that $\pi_K(\sigma(y))\xi_{\chi,k} = \chi(y)\,\xi_{\chi,k-v_F(y)}$ for $y \in F^\times$ and that $\pi_K(zI)\xi_{\chi,k} = \omega_\pi(z)\xi_{\chi,k}$ for $z \in F^\times$. Deduce that the subgroup ZS permutes the lines $\mathbb{C}\xi_{\chi,k}$. Then show that $\pi_K(u(b))\xi_{\chi,k} = \xi_{\chi,k}$ holds if and only if $b \in \mathfrak{p}^{1-k}$.

2. The element w satisfies $w^2 = -I$. Apply theorem 8.2.4 twice to $\xi_{\chi,k}$ and compare with (8.4) to deduce the two identities

$$n\big(\chi\omega_\pi^{-1}\pi,\psi\big) = n\big(\chi^{-1}\pi,\psi\big), \qquad \varepsilon\big(\chi^{-1}\pi,\tfrac12,\psi\big)\,\varepsilon\big(\chi\omega_\pi^{-1}\pi,\tfrac12,\psi\big) = \omega_\pi(-1)$$

   valid for every character χ of $F^\times$. (Two functions $\xi_{\chi,j}$ and $\xi_{\chi,k}$ with $j \neq k$ have disjoint supports.)

3. Let χ be an unramified character of $F^\times$ with $\chi(\varpi) = \lambda$. Show first that $\xi_{\chi,k} = \lambda^k\xi_{1,k}$ and $\xi_{\chi^{-1}\omega_\pi,l} = \lambda^{-l}\xi_{\omega_\pi,l}$. Then evaluate $\pi_K(w)\xi_{\chi,k}$ in two ways, once by linearity from the case $\chi = 1$ of theorem 8.2.4 and once by the general case, and conclude that

$$n(\chi^{-1}\pi,\psi) = n(\pi,\psi), \qquad \varepsilon\big(\chi^{-1}\pi,\tfrac12,\psi\big) = \lambda^{-n(\pi,\psi)}\,\varepsilon\big(\pi,\tfrac12,\psi\big)$$

   Compare with item (5) of example 5.3.7.

4. Let π and π' be irreducible cuspidal representations of G with $\omega_\pi = \omega_{\pi'}$ and with the same operator $\pi_K(w) = \pi'_K(w)$ on $C_c^\infty(F^\times)$. Show that $\pi \cong \pi'$, using only definition 8.2.1 and the decomposition $G = ZM \cup ZMwM$. Where in the argument is cuspidality used?

5. Suppose the operator $\pi_K(w)$ is given as a linear map of $C_c^\infty(F^\times)$, but π is not. Explain how to recover, for every character χ of $F^\times$, the integer $n(\chi^{-1}\pi,\psi)$, the number $\varepsilon(\chi^{-1}\pi,\frac12,\psi)$, and hence the whole function $\varepsilon(\chi^{-1}\pi,s,\psi)$ of s. Which result gives the last step?

6. For $\varphi \in C_c^\infty(F^\times)$ let $\Psi(\varphi,X)$ and $\widetilde{\Psi}(\varphi,X)$ be the two transforms defined in Step 3 of the proof of theorem 8.2.4. Show that for $y \in F^\times$

$$\Psi\big(\pi_K(\sigma(y))\varphi,\ X\big) = X^{-v_F(y)}\,\Psi(\varphi,X), \qquad \widetilde{\Psi}\big(\pi_K(\sigma(y))\varphi,\ X\big) = \omega_\pi(y)\,X^{-v_F(y)}\,\widetilde{\Psi}(\varphi,X)$$

   and explain why these identities make the monomial shape of the answer in theorem 8.2.4 inevitable once the answer is known on a single $\xi_{1,m}$.

7. Fix $k \in \mathbb{Z}$ and $m \geq 1$, and let $E_{k,m}$ be the space of functions in $C_c^\infty(F^\times)$ supported in $\varpi^k U_F$ and invariant under right translation by U_F^m. Show that the functions $\xi_{\chi,k}$ with $\chi|_{U_F^m} = 1$ span $E_{k,m}$, that distinct restrictions $\chi|_{U_F}$ give linearly independent functions, and that $\dim_{\mathbb{C}} E_{k,m} = (q-1)q^{m-1}$.

8.3 A Metaplectic Representation

Fix a separable quadratic extension E/F. This section attaches to E/F a locally profinite group $\mathcal{G}_{E/F}$, assembled from G and $E^\times$ by a determinant condition, and builds on the space $C_c^\infty(E)$ a smooth representation of that group depending on a non-trivial additive character ψ of F. The group is a central extension of an index-two subgroup of G by a compact subgroup of $E^\times$, and cutting the representation down to a single character of that compact subgroup is what converts a character of $E^\times$ into a cuspidal representation of G.

Three symbols shaped like N occur together from here on, and they are kept apart by their decorations. The unadorned N is the group of upper triangular unipotent matrices in G, as throughout the book. The norm and trace of E/F always carry their subscript, $N_{E/F}$ and $\mathrm{tr}_{E/F}$. Script letters $\mathcal{N}$, $\mathcal{A}$, $\mathcal{Z}$, $\mathcal{B}$ denote subgroups of $\mathcal{G}_{E/F}$.

The character of $F^\times$ that governs the whole construction is the one cutting out the norms from E. Its existence is a statement of local class field theory rather than an elementary one, and the book already has it. By theorem 6.2.2(2) the map $\mathbf{a}_F$ induces an isomorphism $\mathrm{Gal}(E/F) \cong F^\times/N_{E/F}(E^\times)$; as E/F is quadratic this reads

$$[F^\times : N_{E/F}(E^\times)] = 2$$

so $N_{E/F}(E^\times)$ is the kernel of a unique character of $F^\times$ of order two. By theorem 6.2.2(1) norm groups of finite abelian extensions are precisely the open subgroups of finite index, so that kernel is open and the character is continuous.

Definition 8.3.1: The Metaplectic Group of a Quadratic Extension

Let E/F be a separable quadratic extension with $\mathrm{Gal}(E/F) = \{1, \sigma\}$, and write $\kappa = \kappa_{E/F}$ for the character of $F^\times$ of order two with kernel $N_{E/F}(E^\times)$. Set

$$G^\kappa = \{g \in G : \kappa(\det g) = 1\}$$

$$\mathcal{G}_{E/F} = \left\{(g,h) : g \in G,\ h \in E^\times,\ \det g = N_{E/F}(h)^{-1}\right\} \subseteq G \times E^\times$$

$$\mathcal{K}_{E/F} = \ker\left(N_{E/F} \colon E^\times \longrightarrow F^\times\right)$$

The group $\mathcal{G}_{E/F}$ is the *metaplectic group* of E/F, and $\mathcal{K}_{E/F}$ is its *norm-one subgroup*.

For $x \in E$ the abbreviation x^σ is the image of x under σ, and $x^{-\sigma}$ means $(x^{-1})^\sigma = (x^\sigma)^{-1}$ for $x \in E^\times$. The normalized absolute value of E is written $\|\cdot\|_E$, so that $\|x\|_E$ is the module of the additive automorphism $y \mapsto xy$ of E.

The word *metaplectic* records a structural fact proved below: the first projection makes $\mathcal{G}_{E/F}$ a central extension of G^κ by the compact group $\mathcal{K}_{E/F}$, so a representation of $\mathcal{G}_{E/F}$ on which $\mathcal{K}_{E/F}$ acts trivially is the same thing as a representation of G^κ. Since $\|N_{E/F}(x)\|$ is the module of multiplication

by x on the two-dimensional F-vector space E, and the determinant of that multiplication over F is $N_{E/F}(x)$, one has

$$\|N_{E/F}(x)\| = \|x\|_E, \qquad x \in E^\times \tag{8.15}$$

which is the compatibility used repeatedly below.

The unramified extension is the case where every ingredient can be named explicitly, and it will serve as the running illustration for the whole section.

Example 8.3.2: The Unramified Quadratic Extension

Let E/F be the unramified quadratic extension, so $e(E|F) = 1$, $f(E|F) = 2$, the prime ϖ of F is also a prime of E, the residue field of E is k_E with q^2 elements, and σ induces the q-power map on k_E.

1. *The character κ is unramified.* The inertia subgroup $\mathcal{I}_F \subseteq \mathcal{W}_F$ acts trivially on every unramified extension of F, so $\mathcal{I}_F \subseteq \mathcal{W}_E$. Combining the identification $N_{E/F}(E^\times) = \mathbf{a}_F(\mathcal{W}_E)$, which follows from theorem 6.2.1(1) applied to E together with the compatibility $\mathbf{a}_F|_{\mathcal{W}_E} = N_{E/F} \circ \mathbf{a}_E$ of part (4), with theorem 6.2.1(3), which gives $\mathbf{a}_F(\mathcal{I}_F) = U_F$, yields $U_F \subseteq N_{E/F}(E^\times)$. Thus κ is trivial on U_F. As $\kappa \neq 1$ it must satisfy $\kappa(\varpi) = -1$, so

$$\kappa(a) = (-1)^{v_F(a)}, \qquad N_{E/F}(E^\times) = \big\{a \in F^\times : v_F(a) \in 2\mathbb{Z}\big\}$$

   and $G^\kappa = \{g \in G : v_F(\det g) \in 2\mathbb{Z}\}$.

2. *The norm is surjective on units.* If $u \in U_F$ then $u \in N_{E/F}(E^\times)$ by (1), say $u = N_{E/F}(x)$. Then $\|x\|_E = \|u\| = 1$ by (8.15), so $x \in U_E$. Hence $N_{E/F}(U_E) = U_F$.

3. *Roots of unity inside $\mathcal{K}_{E/F}$.* Let $\mu \subseteq U_E$ be the group of roots of unity of order dividing $q^2 - 1$, cyclic of that order, mapping isomorphically onto $k_E^\times$ under reduction. Reduction commutes with σ and is injective on μ, so $\omega^\sigma = \omega^q$ for $\omega \in \mu$, giving $N_{E/F}(\omega) = \omega^{1+q}$. Therefore

$$\mu \cap \mathcal{K}_{E/F} = \big\{\omega \in \mu : \omega^{q+1} = 1\big\}$$

   is cyclic of order $q + 1$. In particular $\mathcal{K}_{E/F} \neq \{1\}$.

4. *Elements of $\mathcal{G}_{E/F}$.* Taking $x = \varpi$, then $x = \omega \in \mu \cap \mathcal{K}_{E/F}$, then $\zeta = \varpi$ in the notation of proposition 8.3.3 below produces

$$\left(\begin{pmatrix} \varpi^2 & 0 \\ 0 & 1 \end{pmatrix}, \varpi^{-1}\right), \qquad \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \omega\right), \qquad \left(\begin{pmatrix} \varpi & 0 \\ 0 & \varpi \end{pmatrix}, \varpi^{-1}\right)$$

   In each case the determinant of the matrix is the inverse norm of the second entry, since $N_{E/F}(\varpi^{-1})^{-1} = \varpi^2$ for the first and third, and $N_{E/F}(\omega)^{-1} = 1$ for the second.

The subgroups of $\mathcal{G}_{E/F}$ that matter are the ones lying over the subgroups T, N and Z of G, and the statement that organizes them is a Bruhat decomposition matching the one for G itself. Everything later in the section is a computation inside the decomposition below.

Proposition 8.3.3: Structure of $\mathcal{G}_{E/F}$

Let E/F be a separable quadratic extension.

1. $\mathcal{G}_{E/F}$ is a closed subgroup of $G \times E^\times$, hence locally profinite. The two projections give exact sequences

$$1 \longrightarrow \mathcal{K}_{E/F} \longrightarrow \mathcal{G}_{E/F} \longrightarrow G^\kappa \longrightarrow 1, \qquad 1 \longrightarrow \mathrm{SL}_2(F) \longrightarrow \mathcal{G}_{E/F} \longrightarrow E^\times \longrightarrow 1$$

where the first map of the first sequence is $u \mapsto (1, u)$. The group $\mathcal{K}_{E/F}$ is contained in U_E and is compact.

2. The elements

$$n(t) = \left(\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, 1\right), \quad a(x) = \left(\begin{pmatrix} N_{E/F}(x) & 0 \\ 0 & 1 \end{pmatrix}, x^{-\sigma}\right), \quad z(\zeta) = \left(\begin{pmatrix} \zeta & 0 \\ 0 & \zeta \end{pmatrix}, \zeta^{-1}\right)$$

lie in $\mathcal{G}_{E/F}$ for $t \in F$, $x \in E^\times$, $\zeta \in F^\times$, and the three maps $t \mapsto n(t)$, $x \mapsto a(x)$, $\zeta \mapsto z(\zeta)$ are injective homomorphisms from $(F, +)$, $E^\times$ and $F^\times$ respectively. Write $\mathcal{N}$, $\mathcal{A}$, $\mathcal{Z}$ for their images and $\mathcal{A}_1 = \{a(u) : u \in \mathcal{K}_{E/F}\}$. Then $a(u) = (1, u)$ for $u \in \mathcal{K}_{E/F}$, the subgroup $\mathcal{Z}$ is central in $\mathcal{G}_{E/F}$, and

$$a(x)\, n(t)\, a(x)^{-1} = n\big(t\, N_{E/F}(x)\big), \qquad t \in F,\ x \in E^\times \tag{8.16}$$

The centre of $\mathcal{G}_{E/F}$ is $\mathcal{Z}\mathcal{A}_1$.

3. The product map $\mathcal{Z} \times \mathcal{A} \times \mathcal{N} \to \mathcal{G}_{E/F}$ is a homeomorphism onto the closed subgroup $\mathcal{B} = \mathcal{G}_{E/F} \cap (B \times E^\times)$, and

$$z(\zeta)a(x)n(t) = \left(\begin{pmatrix} \zeta N_{E/F}(x) & \zeta N_{E/F}(x)\, t \\ 0 & \zeta \end{pmatrix}, \zeta^{-1} x^{-\sigma}\right) \tag{8.17}$$

The group $\mathcal{N}$ is normal in $\mathcal{B}$, and the multiplication rule in these coordinates is

$$z(\zeta)a(x)n(t) \cdot z(\zeta')a(x')n(t') = z(\zeta\zeta')\, a(xx')\, n\big(t\, N_{E/F}(x')^{-1} + t'\big) \tag{8.18}$$

4. Put $w = \left(\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, 1\right) \in \mathcal{G}_{E/F}$. Then

$$\mathcal{G}_{E/F} = \mathcal{B} \sqcup \mathcal{B}w\mathcal{N}$$

and the map $\mathcal{B} \times \mathcal{N} \to \mathcal{B}w\mathcal{N}$, $(\mathfrak{b}, \mathfrak{n}) \mapsto \mathfrak{b}w\mathfrak{n}$, is a bijection.

***Proof*:**

1. The map $\Delta\colon G \times E^\times \to F^\times$, $\Delta(g, h) = \det(g)\, N_{E/F}(h)$, is a continuous homomorphism, being the product of the two continuous homomorphisms det and $N_{E/F}$ into the abelian group $F^\times$. Its kernel is $\mathcal{G}_{E/F}$, which is therefore a closed subgroup. Both G and $E^\times$ are locally profinite, so $G \times E^\times$ has a basis of neighbourhoods of the identity consisting of

compact open subgroups. Intersecting each such subgroup with the closed subgroup $\mathcal{G}_{E/F}$ produces a compact open subgroup of $\mathcal{G}_{E/F}$, so $\mathcal{G}_{E/F}$ is locally profinite.

Consider the second projection $p_2(g,h) = h$. Given $h \in E^\times$, the element $\operatorname{diag}(N_{E/F}(h)^{-1}, 1)$ of G has determinant $N_{E/F}(h)^{-1}$, so p_2 is surjective, and its kernel is $\{(g,1) : \det g = 1\} = \mathrm{SL}_2(F) \times \{1\}$. For the first projection $p_1(g,h) = g$, the pair (g,h) lies in $\mathcal{G}_{E/F}$ exactly when $\det g = N_{E/F}(h^{-1})$, so the image of p_1 is the set of g with $\det g \in N_{E/F}(E^\times) = \ker \kappa$, which is G^κ. The kernel of p_1 is $\{(1,h) : N_{E/F}(h) = 1\} = \{1\} \times \mathcal{K}_{E/F}$.

For the last assertion, σ fixes F, hence preserves the integral closure $\mathfrak{o}_E$ of $\mathfrak{o}$ in E and its unique maximal ideal, so $v_E \circ \sigma = v_E$. Consequently $v_E(N_{E/F}(h)) = v_E(h) + v_E(h^\sigma) = 2v_E(h)$. If $N_{E/F}(h) = 1$ then $v_E(h) = 0$, that is $h \in U_E$. So $\mathcal{K}_{E/F} \subseteq U_E$, and $\mathcal{K}_{E/F}$ is closed in $E^\times$ as the kernel of the continuous map $N_{E/F}$, hence closed in the compact group U_E and therefore compact.

2. Each of the three elements satisfies the defining condition. For $n(t)$ this reads $\det \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right) = 1 = N_{E/F}(1)^{-1}$. For $a(x)$, the norm is σ-invariant, so $N_{E/F}(x^{-\sigma}) = N_{E/F}(x)^{-1}$ and therefore $N_{E/F}(x^{-\sigma})^{-1} = N_{E/F}(x) = \det \operatorname{diag}(N_{E/F}(x), 1)$. For $z(\zeta)$, one has $N_{E/F}(\zeta^{-1}) = \zeta^{-2}$ for $\zeta \in F^\times$, whose inverse is $\zeta^2 = \det(\zeta \cdot 1)$.

Multiplying matrices gives $n(t)n(t') = n(t+t')$ and $z(\zeta)z(\zeta') = z(\zeta\zeta')$, while

$$\operatorname{diag}(N_{E/F}(x),1)\operatorname{diag}(N_{E/F}(x'),1) = \operatorname{diag}(N_{E/F}(xx'),1), \qquad x^{-\sigma}(x')^{-\sigma} = (xx')^{-\sigma}$$

so $a(x)a(x') = a(xx')$. Each map is injective: $n(t)$ determines t from the upper right entry, $z(\zeta)$ determines ζ from the diagonal, and $a(x)$ determines $x^{-\sigma}$, hence x.

If $u \in \mathcal{K}_{E/F}$ then $uu^\sigma = 1$, so $u^\sigma = u^{-1}$ and $u^{-\sigma} = u$. Also $N_{E/F}(u) = 1$, so $a(u) = (1,u)$ as claimed. Every $z(\zeta)$ is central in $\mathcal{G}_{E/F}$ because its first coordinate is a scalar matrix, which is central in G, and its second coordinate lies in the abelian group $E^\times$. For (8.16), the first coordinate of $a(x)n(t)a(x)^{-1}$ is

$$\operatorname{diag}(N_{E/F}(x),1)\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}\operatorname{diag}(N_{E/F}(x),1)^{-1} = \begin{pmatrix} 1 & t\,N_{E/F}(x) \\ 0 & 1 \end{pmatrix}$$

and the second is $x^{-\sigma} \cdot 1 \cdot (x^{-\sigma})^{-1} = 1$.

Now let (g,h) be central in $\mathcal{G}_{E/F}$. For every $s \in \mathrm{SL}_2(F)$ the pair $(s,1)$ lies in $\mathcal{G}_{E/F}$, and $(g,h)(s,1) = (gs,h)$ while $(s,1)(g,h) = (sg,h)$, so g commutes with every element of $\mathrm{SL}_2(F)$. Writing $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ and taking $s = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ gives

$$gs = \begin{pmatrix} a & a+b \\ c & c+d \end{pmatrix}, \qquad sg = \begin{pmatrix} a+c & b+d \\ c & d \end{pmatrix}$$

so $c = 0$ and $a = d$. Taking $s = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ with $g = \left(\begin{smallmatrix} a & b \\ 0 & a \end{smallmatrix}\right)$ gives

$$gs = \begin{pmatrix} a+b & b \\ a & a \end{pmatrix}, \qquad sg = \begin{pmatrix} a & b \\ a & a+b \end{pmatrix}$$

so $b = 0$ and $g = a \cdot 1$ is scalar, say $g = \zeta \cdot 1$ with $\zeta \in F^\times$. Conversely any $(\zeta \cdot 1, h) \in \mathcal{G}_{E/F}$ is central, since scalar matrices are central in G and $E^\times$ is abelian. Such a pair satisfies $\zeta^2 N_{E/F}(h) = 1$, and

$$(\zeta \cdot 1, h) = z(\zeta)\,(1, \zeta h), \qquad N_{E/F}(\zeta h) = \zeta^2 N_{E/F}(h) = 1$$

so $(1, \zeta h) = a(\zeta h) \in \mathcal{A}_1$. Hence the centre is contained in $\mathcal{Z}\mathcal{A}_1$. The reverse inclusion holds because $\mathcal{Z}$ is central and $\mathcal{A}_1 = \{1\} \times \mathcal{K}_{E/F}$ is the kernel of p_1, whose elements have trivial first coordinate and therefore commute with everything.

3. Multiplying out, $z(\zeta)a(x) = \big(\mathrm{diag}(\zeta N_{E/F}(x), \zeta),\ \zeta^{-1}x^{-\sigma}\big)$, and multiplying on the right by $n(t)$ gives (8.17). The right side lies in $B \times E^\times$, so the product map lands in $\mathcal{B}$. It is injective: from (8.17) the entry in position $(2,2)$ is ζ, the entry in position $(1,1)$ then determines $N_{E/F}(x)$, the entry in position $(1,2)$ determines t, and the second coordinate determines $x^{-\sigma}$ and hence x.

   It is surjective onto $\mathcal{B}$. Let $\left(\left(\begin{smallmatrix}\alpha & \beta \\ 0 & \delta\end{smallmatrix}\right), h\right) \in \mathcal{B}$, so $\alpha\delta = N_{E/F}(h)^{-1}$. Put $\zeta = \delta$, $x = (\delta h)^{-\sigma}$ and $t = \beta/\alpha$. Then

$$N_{E/F}(x) = N_{E/F}(\delta h)^{-1} = \big(\delta^2 N_{E/F}(h)\big)^{-1} = \delta^{-2}\alpha\delta = \alpha\delta^{-1}$$

   so $\zeta N_{E/F}(x) = \alpha$ and $\zeta N_{E/F}(x)t = \alpha \cdot \beta/\alpha = \beta$, while $\zeta^{-1}x^{-\sigma} = \delta^{-1}(\delta h) = h$ because $u \mapsto u^{-\sigma}$ is an involution of $E^\times$. Comparison with (8.17) gives the required factorization. The formulas just written for ζ, x and t in terms of α, β, δ, h are continuous, so the inverse of the product map is continuous and the map is a homeomorphism. The subgroup $\mathcal{B}$ is closed because B is closed in G.

   The multiplication rule (8.18) follows by multiplying two matrices of the shape appearing in (8.17):

$$\begin{pmatrix} \zeta N_{E/F}(x) & \zeta N_{E/F}(x)t \\ 0 & \zeta \end{pmatrix} \begin{pmatrix} \zeta' N_{E/F}(x') & \zeta' N_{E/F}(x')t' \\ 0 & \zeta' \end{pmatrix} = \begin{pmatrix} \zeta\zeta' N_{E/F}(xx') & \zeta\zeta' N_{E/F}(x)\,\big(N_{E/F}(x')t' + t\big) \\ 0 & \zeta\zeta' \end{pmatrix}$$

   Matching against (8.17) with parameters $\zeta'' = \zeta\zeta'$ and $N_{E/F}(x'') = N_{E/F}(xx')$ forces $\zeta\zeta' N_{E/F}(xx')t'' = \zeta\zeta' N_{E/F}(x)\,\big(N_{E/F}(x')t' + t\big)$, that is $t'' = t' + t\,N_{E/F}(x')^{-1}$, while the second coordinates multiply to $(\zeta\zeta')^{-1}(xx')^{-\sigma}$, giving $x'' = xx'$. Finally $\mathcal{N}$ is normalized by $\mathcal{A}$ by (8.16), by $\mathcal{Z}$ because $\mathcal{Z}$ is central, and by itself because it is a group, so $\mathcal{N} \trianglelefteq \mathcal{B}$.

4. First, $\det\left(\begin{smallmatrix}0 & 1 \\ -1 & 0\end{smallmatrix}\right) = 1 = N_{E/F}(1)^{-1}$, so $w \in \mathcal{G}_{E/F}$. Write $w_0 = \left(\begin{smallmatrix}0 & 1 \\ -1 & 0\end{smallmatrix}\right)$ for its first coordinate. The G-level statement is proposition 2.1.3, stated there for the Weyl element $w' = \left(\begin{smallmatrix}0 & 1 \\ 1 & 0\end{smallmatrix}\right)$ of lemma 2.1.2, and transporting it to w_0 costs one line: $w_0 = \mathrm{diag}(1,-1)\,w'$ with $\mathrm{diag}(1,-1) \in T \subseteq B$, so $Bw_0N = Bw'N$, and $(b,n) \mapsto bw_0n$ is the composite of the bijection $(b,n) \mapsto \big(b\mathrm{diag}(1,-1), n\big)$ of $B \times N$ with the bijection of proposition 2.1.3. Hence $G = B \sqcup Bw_0N$, the factorization $g = bw_0n$ is unique, and Bw_0N is exactly the set of g with $g_{21} \neq 0$.

   Lift this to $\mathcal{G}_{E/F}$. Let $(g,h) \in \mathcal{G}_{E/F}$. If $g \in B$ then $(g,h) \in \mathcal{B}$. Otherwise $g = bw_0n$ uniquely with $b \in B$, $n = \left(\begin{smallmatrix}1 & s \\ 0 & 1\end{smallmatrix}\right) \in N$, and then

$$(g,h) = (b,h) \cdot w \cdot n(s)$$

because w and $n(s)$ have second coordinate 1. Moreover $\det b = \det g$, since $\det w_0 = \det n = 1$, so $(b,h) \in \mathcal{G}_{E/F} \cap (B \times E^\times) = \mathcal{B}$. Uniqueness of (b,h) and s follows from uniqueness of (b,n) in G, the second coordinate h being forced. Finally $\mathcal{B}$ and $\mathcal{B}w\mathcal{N}$ are disjoint because the first coordinates of their elements lie in the disjoint sets B and Bw_0N, completing the proof.

Part (1) is the statement the rest of the section runs on. On one side $\mathcal{G}_{E/F}$ surjects onto G^κ with kernel $\mathcal{A}_1 = \{1\} \times \mathcal{K}_{E/F}$, which part (2) shows is central, so $\mathcal{G}_{E/F}/\mathcal{A}_1 \cong G^\kappa$ and a representation of $\mathcal{G}_{E/F}$ on which $\mathcal{A}_1$ acts trivially is a representation of G^κ. On the other side $\mathcal{G}_{E/F}$ surjects onto $E^\times$ with kernel $\mathrm{SL}_2(F)$, so a character of $E^\times$ pulls back to a character of $\mathcal{G}_{E/F}$. Twisting a representation of $\mathcal{G}_{E/F}$ by the inverse of such a character adjusts how $\mathcal{A}_1$ acts without disturbing the restriction to $\mathrm{SL}_2(F)$, and that is the mechanism by which a character of $E^\times$ will select a representation of G^κ.

Part (4) is what makes a representation of $\mathcal{G}_{E/F}$ specifiable by finitely much data. A representation is determined by its restriction to $\mathcal{B}$, which by part (3) means three commuting families of formulas, together with the single operator attached to w. The construction below writes down exactly that.

The space carrying the representation is $C_c^\infty(E)$, the locally constant compactly supported functions on the additive group of E. Three of the four required formulas are elementary: $\mathcal{N}$ acts by multiplication by an additive character built from the norm form, $\mathcal{A}$ acts by dilation, and $\mathcal{Z}$ acts by the scalar κ. The fourth, at w, is a Fourier transform, and the constant standing in front of it is the Langlands constant of definition 6.3.10.

Since E is itself a non-archimedean local field and $\psi_E = \psi \circ \mathrm{tr}_{E/F}$ is a non-trivial character of E (the trace is surjective because E/F is separable, and $\psi \neq 1$), the whole of definition 5.1.1 and proposition 5.1.2 applies verbatim over E. In particular there is a unique Haar measure on E that is self-dual for ψ_E, and the Fourier transform taken with respect to it satisfies $\hat{\hat{f}}(x) = f(-x)$ and carries $C_c^\infty(E)$ into itself.

Definition 8.3.4: The Standard Representation $\eta_\psi^{\mathcal{G}}$

Let $\psi \in \widehat{F}$, $\psi \neq 1$, put $\psi_E = \psi \circ \mathrm{tr}_{E/F}$, and let $f \mapsto \hat{f}$ be the Fourier transform on $C_c^\infty(E)$ taken with respect to ψ_E and the Haar measure on E self-dual for ψ_E. Let $\lambda_{E/F}(\psi)$ be the Langlands constant of definition 6.3.10, which is independent of s by corollary 6.3.11. The *standard representation* $\eta = \eta_\psi^{\mathcal{G}}$ of $\mathcal{G}_{E/F}$ on $C_c^\infty(E)$ is the representation determined by the relations

$$\begin{aligned} \eta\big(n(t)\big)f &: y \longmapsto \psi\big(t\,N_{E/F}(y)\big)\,f(y), && t \in F \\ \eta\big(a(x)\big)f &: y \longmapsto \|x\|_E^{1/2}\,f(xy), && x \in E^\times \\ \eta\big(z(\zeta)\big)f &: y \longmapsto \kappa(\zeta)\,f(y), && \zeta \in F^\times \\ \eta(w)f &: y \longmapsto \lambda_{E/F}(\psi)\,\hat{f}(y^\sigma) \end{aligned} \tag{8.19}$$

for $f \in C_c^\infty(E)$ and $y \in E$. That these relations determine one and only one representation, and that it is smooth, is proposition 8.3.5.

Each of the four operators in (8.19) is written down explicitly, so all four can be evaluated before any existence question is settled. Take E/F unramified as in example 8.3.2, let ψ have level zero, and let $f = \mathbf{1}_{\mathfrak{o}_E}$ be the characteristic function of $\mathfrak{o}_E$. Because E/F is unramified it is tamely ramified, so $\mathrm{tr}_{E/F}(\mathfrak{o}_E) = \mathfrak{o}$ by lemma 4.1.2. Hence ψ_E is trivial on $\mathfrak{o}_E$, and since $\mathfrak{p}_E^{-1} = \varpi^{-1}\mathfrak{o}_E$ gives $\mathrm{tr}_{E/F}(\mathfrak{p}_E^{-1}) = \mathfrak{p}^{-1}$, on which ψ is non-trivial, the character ψ_E has level zero as well. By

proposition 5.1.2(3) over E the self-dual measure then satisfies $\mu_{\psi_E}(\mathfrak{o}_E) = 1$. Now

$$\hat{f}(x) = \int_{\mathfrak{o}_E} \psi_E(xv)\, dv$$

and the character $v \mapsto \psi_E(xv)$ of the group $\mathfrak{o}_E$ is trivial exactly when $x\mathfrak{o}_E \subseteq \mathfrak{o}_E$, that is when $x \in \mathfrak{o}_E$. For $x \notin \mathfrak{o}_E$ the integral of a non-trivial character over the compact group $\mathfrak{o}_E$ vanishes. Hence $\hat{f} = \mathbf{1}_{\mathfrak{o}_E} = f$. Since $\mathfrak{o}_E^\sigma = \mathfrak{o}_E$, the four operators give

$$\eta\big(n(t)\big)f = f \ \ (t \in \mathfrak{o}), \qquad \eta\big(a(x)\big)f = f \ \ (x \in U_E), \qquad \eta\big(z(\zeta)\big)f = (-1)^{v_F(\zeta)}f, \qquad \eta(w)f = \lambda_{E/F}(\psi)\, f$$

The first holds because $N_{E/F}(\mathfrak{o}_E) \subseteq \mathfrak{o}$ and ψ is trivial on $\mathfrak{o}$, the second because $\|x\|_E = 1$ and $x\mathfrak{o}_E = \mathfrak{o}_E$ for $x \in U_E$, the third by example 8.3.2(1). Taking $x = \varpi$ instead of a unit gives a different function,

$$\eta\big(a(\varpi)\big)f \colon y \longmapsto q^{-1}\mathbf{1}_{\mathfrak{o}_E}(\varpi y) = q^{-1}\mathbf{1}_{\mathfrak{p}_E^{-1}}(y)$$

using $\|\varpi\|_E = q^{-2}$. So $\mathbf{1}_{\mathfrak{o}_E}$ is an eigenvector for w and for the maximal compact subgroup of $\mathcal{A}$, and dilation by ϖ carries it to a multiple of $\mathbf{1}_{\mathfrak{p}_E^{-1}}$.

The four formulas of (8.19) prescribe an operator for each element of a generating set of $\mathcal{G}_{E/F}$. Whether those operators are compatible with the relations among the generators is a separate question, and it is the substance of the following proposition.

Proposition 8.3.5: Existence and Properties of $\eta_\psi^{\mathcal{G}}$

Let E/F be a separable quadratic extension and let $\psi \in \widehat{F}$, $\psi \neq 1$.

1. There is exactly one representation $\eta = \eta_\psi^{\mathcal{G}}$ of $\mathcal{G}_{E/F}$ on $C_c^\infty(E)$ satisfying (8.19), and it is smooth. On $\mathcal{B}$ it is given by

   $$\eta\big(z(\zeta)a(x)n(t)\big)f \colon y \longmapsto \kappa(\zeta)\,\|x\|_E^{1/2}\,\psi\big(t\,N_{E/F}(xy)\big)\,f(xy)$$

   for $\zeta \in F^\times$, $x \in E^\times$, $t \in F$ and $f \in C_c^\infty(E)$.

2. $\eta(w)^2 f \colon y \mapsto \kappa(-1)f(-y)$. In particular $\eta(w)^4$ is the identity, so $\eta(w)$ is invertible.

3. For $u \in \mathcal{K}_{E/F}$ one has $\eta\big(a(u)\big)f \colon y \mapsto f(uy)$, and $\mathcal{A}_1$ is central in $\mathcal{G}_{E/F}$. For a smooth character θ of $\mathcal{K}_{E/F}$ set

   $$C_c^\infty(E,\theta) = \big\{f \in C_c^\infty(E) : f(uy) = \theta(u)f(y) \text{ for all } u \in \mathcal{K}_{E/F},\ y \in E\big\}$$

   Then each $C_c^\infty(E,\theta)$ is a $\mathcal{G}_{E/F}$-subspace of $C_c^\infty(E)$.

4. The sum $\bigoplus_\theta C_c^\infty(E,\theta)$ over all smooth characters θ of $\mathcal{K}_{E/F}$ is direct and equals $C_c^\infty(E)$, and $C_c^\infty(E,\theta) \subseteq C_c^\infty(E^\times)$ whenever $\theta \neq 1$.

One step of the proof below is taken on citation rather than carried out. The relations listed in Step 3 are verified in full, and so is the action of the Fourier operator on the first three of them. What is deferred is the fourth relation, whose verification rests on a twisted Fourier inversion formula over E whose constant is computed by a Gauss sum. The structure of that computation is given in Step 5, and the details are in [2, §38]. The presentation statement quoted in Step 3 is deferred on the same terms.

***Proof*:**
Step 1: uniqueness. By proposition 8.3.3(3) and (4) every element of $\mathcal{G}_{E/F}$ is either $z(\zeta)a(x)n(t)$ or $z(\zeta)a(x)n(t)\,w\,n(s)$. Both expressions are products of elements drawn from $\mathcal{Z}\cup\mathcal{A}\cup\mathcal{N}\cup\{w\}$, so two representations of $\mathcal{G}_{E/F}$ agreeing on those four families are equal. Since (8.19) prescribes the operator attached to each element of those four families, at most one representation satisfies it.

Step 2: the first three formulas define a smooth representation of $\mathcal{B}$. Each of the three prescribed operators carries $C_c^\infty(E)$ into itself. For $\eta(a(x))$ the function $y\mapsto f(xy)$ is locally constant because f is and $y\mapsto xy$ is continuous, and its support is $x^{-1}\mathrm{Supp}f$, which is compact. For $\eta(z(\zeta))$ the operator is multiplication by a scalar. For $\eta(n(t))$ we must see that $y\mapsto\psi(tN_{E/F}(y))$ is locally constant. From $N_{E/F}(y+z)=N_{E/F}(y)+\mathrm{tr}_{E/F}(yz^\sigma)+N_{E/F}(z)$ and the continuity of $\mathrm{tr}_{E/F}$ and $N_{E/F}$, for y in a fixed compact set and z in a small enough $\mathfrak{p}_E^n$ the difference $N_{E/F}(y+z)-N_{E/F}(y)$ lies in any prescribed neighbourhood of 0 in F, in particular in the open subgroup on which $t\psi$ is trivial. So $\psi(tN_{E/F}(\cdot))$ is locally constant, and the product of a locally constant function with an element of $C_c^\infty(E)$ lies in $C_c^\infty(E)$.

By proposition 8.3.3(3) every element of $\mathcal{B}$ is uniquely $z(\zeta)a(x)n(t)$, so

$$R(\zeta,x,t)f\colon y\longmapsto\kappa(\zeta)\|x\|_E^{1/2}\psi\big(t\,N_{E/F}(x)N_{E/F}(y)\big)\,f(xy)$$

is a well-defined map from $\mathcal{B}$ to the invertible linear operators on $C_c^\infty(E)$, and it is the only candidate compatible with the first three lines of (8.19), since composing those three operators in the order $n(t)$, then $a(x)$, then $z(\zeta)$ produces exactly $R(\zeta,x,t)$. We check that R is a homomorphism. On one hand,

$$\begin{aligned}R(\zeta,x,t)R(\zeta',x',t')f(y)&=\kappa(\zeta)\|x\|_E^{1/2}\psi\big(tN_{E/F}(x)N_{E/F}(y)\big)\,\big(R(\zeta',x',t')f\big)\,(xy)\\&=\kappa(\zeta\zeta')\|xx'\|_E^{1/2}\psi\big(\big(tN_{E/F}(x)+t'N_{E/F}(x')N_{E/F}(x)\big)N_{E/F}(y)\big)\,f(x'xy)\end{aligned}$$

using $\kappa(\zeta)\kappa(\zeta')=\kappa(\zeta\zeta')$, $\|x\|_E^{1/2}\|x'\|_E^{1/2}=\|xx'\|_E^{1/2}$ and $N_{E/F}(xy)=N_{E/F}(x)N_{E/F}(y)$. On the other hand (8.18) gives

$$R\big(\zeta\zeta',\,xx',\,tN_{E/F}(x')^{-1}+t'\big)\,f(y)=\kappa(\zeta\zeta')\|xx'\|_E^{1/2}\psi\big(\big(tN_{E/F}(x)+t'N_{E/F}(x)N_{E/F}(x')\big)N_{E/F}(y)\big)\,f(xx'y)$$

since $\big(tN_{E/F}(x')^{-1}+t'\big)\,N_{E/F}(xx')=tN_{E/F}(x)+t'N_{E/F}(x)N_{E/F}(x')$. The two agree because $E^\times$ is abelian, so R is a homomorphism on $\mathcal{B}$.

Smoothness. Fix $f\in C_c^\infty(E)$. Being locally constant with compact support, f is fixed by translation by some $\mathfrak{p}_E^n$ and supported in some $\mathfrak{p}_E^{-m}$. For $x\in U_E^{n+m}$ and $y\in\mathfrak{p}_E^{-m}$ one has $xy-y=(x-1)y\in\mathfrak{p}_E^n$, so $f(xy)=f(y)$. For $y\notin\mathfrak{p}_E^{-m}$ both $f(y)$ and $f(xy)$ vanish, because $\|xy\|_E=\|y\|_E$. As $\|x\|_E=1$ on U_E, the subgroup $\mathcal{A}_0=\big\{a(x):x\in U_E^{n+m}\big\}$ fixes f. Next, $N_{E/F}(\mathfrak{p}_E^{-m})$ is compact, hence contained in some $\mathfrak{p}^{-M}$. If l is the level of ψ, then for $t\in\mathfrak{p}^{M+l}$ and $y\in\mathfrak{p}_E^{-m}$ we get $tN_{E/F}(y)\in\mathfrak{p}^l$, where ψ is trivial, so $\mathcal{N}_0=\big\{n(t):t\in\mathfrak{p}^{M+l}\big\}$ fixes f. Finally $\mathcal{Z}_0=\{z(\zeta):\zeta\in\ker\kappa\}$ fixes f, and $\ker\kappa$ is open. The set $\mathcal{Z}_0\mathcal{A}_0\mathcal{N}_0$ is a subgroup, because $\mathcal{Z}_0$ is central and $a(x)n(t)a(x)^{-1}=n(tN_{E/F}(x))$ with $N_{E/F}(x)\in U_F$ for $x\in U_E$. It is open, because the product map of proposition 8.3.3(3) is a homeomorphism. So f is fixed by an open subgroup of $\mathcal{B}$.

Step 3: the relations at w. The following identities hold in $\mathcal{G}_{E/F}$:

$$\begin{aligned} w^2 &= z(-1)a(-1) \\ w\,z(\zeta) &= z(\zeta)\,w, && \zeta \in F^\times \\ w\,a(x) &= z\big(N_{E/F}(x)\big)\,a\big(x^{-\sigma}\big)\,w, && x \in E^\times \\ w\,n(t)\,w &= z(-t)\,a(-t^{-1})\,n(-t)\,w\,n(-t^{-1}), && t \in F^\times \end{aligned} \tag{8.20}$$

For the first, $w_0^2 = -1$ and the second coordinate of w^2 is 1, while $z(-1) = (-1,-1)$ and $a(-1) = (1,-1)$ because $N_{E/F}(-1) = 1$ and $(-1)^{-\sigma} = -1$, and their product is $(-1,1)$. The second relation holds because $\mathcal{Z}$ is central by proposition 8.3.3(2). For the third, the left side has first coordinate $w_0\mathrm{diag}(N_{E/F}(x),1) = \left(\begin{smallmatrix} 0 & 1 \\ -N_{E/F}(x) & 0 \end{smallmatrix}\right)$ and second coordinate $x^{-\sigma}$, whereas

$$z\big(N_{E/F}(x)\big)\,a\big(x^{-\sigma}\big) = \big(\mathrm{diag}\big(N_{E/F}(x),N_{E/F}(x)\big)\,\mathrm{diag}\big(N_{E/F}(x)^{-1},1\big)\,,\ N_{E/F}(x)^{-1}x\big) = \big(\mathrm{diag}\big(1,N_{E/F}(x)\big)\,,\ x^{-\sigma}$$

using $\big(x^{-\sigma}\big)^{-\sigma} = x$ and $N_{E/F}(x)^{-1}x = (xx^\sigma)^{-1}x = x^{-\sigma}$. Multiplying on the right by w reproduces the left side. For the fourth, the first coordinate of the left side is

$$w_0\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}w_0 = \begin{pmatrix} 0 & 1 \\ -1 & -t \end{pmatrix}w_0 = \begin{pmatrix} -1 & 0 \\ t & -1 \end{pmatrix}$$

while on the right $z(-t)a(-t^{-1}) = \big(\mathrm{diag}(-t,-t)\mathrm{diag}(t^{-2},1),\,(-t)^{-1}(-t)\big) = \big(\mathrm{diag}(-t^{-1},-t),\,1\big)$, so that

$$\begin{pmatrix} -t^{-1} & 0 \\ 0 & -t \end{pmatrix}\begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix}w_0\begin{pmatrix} 1 & -t^{-1} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & -t^{-1} \\ t & 0 \end{pmatrix}\begin{pmatrix} 1 & -t^{-1} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ t & -1 \end{pmatrix}$$

and every second coordinate on the right is 1.

The Bruhat decomposition of proposition 8.3.3(4) is what makes (8.20), taken together with the relations holding inside $\mathcal{B}$, a presentation of $\mathcal{G}_{E/F}$: an element is either $\mathfrak{b}$ or $\mathfrak{b}w\mathfrak{n}$, and (8.20) is exactly what is needed to rewrite any product of such expressions in one of those two forms. Consequently, if ρ is a representation of $\mathcal{B}$ and A is an invertible operator on the same space satisfying

$$\begin{aligned} A^2 &= \rho\big(z(-1)a(-1)\big) \\ A\,\rho\big(z(\zeta)\big) &= \rho\big(z(\zeta)\big)A, && \zeta \in F^\times \\ A\,\rho\big(a(x)\big) &= \rho\big(z\big(N_{E/F}(x)\big)\,a\big(x^{-\sigma}\big)\big)\,A, && x \in E^\times \\ A\,\rho\big(n(t)\big)A &= \rho\big(z(-t)a(-t^{-1})n(-t)\big)\,A\,\rho\big(n(-t^{-1})\big)\,, && t \in F^\times \end{aligned} \tag{8.21}$$

then there is a representation of $\mathcal{G}_{E/F}$ restricting to ρ on $\mathcal{B}$ and sending w to A. The rewriting argument is long and mechanical, and it is the one part of this proof carried by citation rather than by computation: see [2, §38].

Step 4: the operator A satisfies the first three conditions. Let $Af\colon y \mapsto \lambda\hat{f}(y^\sigma)$ with $\lambda = \lambda_{E/F}(\psi)$, and let $\rho = R$ be the representation of $\mathcal{B}$ from Step 2. Write $[f](y) = \hat{f}(y^\sigma)$, so $A = \lambda[\,\cdot\,]$. The operator A carries $C_c^\infty(E)$ into itself, by proposition 5.1.2(1) over E together with the fact that σ is a homeomorphism of E.

The map σ preserves the self-dual measure: it is an F-linear automorphism of E with $\sigma^2 = 1$, so $\det_F(\sigma)^2 = 1$ and its module $\|\det_F(\sigma)\|$ equals 1. Hence, for $u \in E$,

$$\widehat{[f]}(u) = \int_E \hat{f}(v^\sigma)\psi_E(uv)\,dv = \int_E \hat{f}(v)\psi_E(uv^\sigma)\,dv = \int_E \hat{f}(v)\psi_E(u^\sigma v)\,dv = \hat{\hat{f}}(u^\sigma) = f(-u^\sigma)$$

where the second equality is the substitution $v \mapsto v^\sigma$, the third uses $\mathrm{tr}_{E/F}(uv^\sigma) = \mathrm{tr}_{E/F}((uv^\sigma)^\sigma) = \mathrm{tr}_{E/F}(u^\sigma v)$, and the last is Fourier inversion for the self-dual measure. Therefore $[[f]](y) = \widehat{[f]}(y^\sigma) = f(-y)$, and

$$A^2 f\colon y \longmapsto \lambda^2 f(-y)$$

Meanwhile $R(-1,-1,0)f(y) = \kappa(-1)\|-1\|_E^{1/2} f(-y) = \kappa(-1)f(-y)$. So the first condition in (8.21) amounts to

$$\lambda_{E/F}(\psi)^2 = \kappa_{E/F}(-1) \tag{8.22}$$

which we now derive from the properties of local constants. Put $R_{E/F} = \mathrm{Ind}_{E/F}1_E$ as in definition 6.3.10. Since $\mathcal{W}_E$ has index two in $\mathcal{W}_F$, the representation $R_{E/F}$ is the permutation representation of $\mathcal{W}_F$ on the two-element set $\mathcal{W}_F/\mathcal{W}_E$, which is $1_F \oplus \kappa$ with κ the character of order two attached to E/F. Hence $\check{R}_{E/F} = 1_F \oplus \kappa^{-1} = R_{E/F}$ and $\det R_{E/F} = \kappa$. Applying proposition 6.3.3(3) to $R_{E/F}$ over F and to 1_E over E gives

$$\varepsilon(R_{E/F}, s, \psi)\,\varepsilon(R_{E/F}, 1-s, \psi) = \kappa(-1), \qquad \varepsilon(1_E, s, \psi_E)\,\varepsilon(1_E, 1-s, \psi_E) = 1$$

the second because 1_E is self-dual with trivial determinant. By corollary 6.3.11 the quotient $\lambda_{E/F}(s,\psi)$ is independent of s, so

$$\lambda_{E/F}(\psi)^2 = \frac{\varepsilon(R_{E/F}, s, \psi)}{\varepsilon(1_E, s, \psi_E)} \cdot \frac{\varepsilon(R_{E/F}, 1-s, \psi)}{\varepsilon(1_E, 1-s, \psi_E)} = \frac{\kappa(-1)}{1}$$

which is (8.22). The second condition in (8.21) holds because $R(\zeta, 1, 0)$ is multiplication by the scalar $\kappa(\zeta)$, which commutes with every linear operator.

For the third condition, substituting $v \mapsto x^{-1}v$ in the defining integral gives $\widehat{f(x\,\cdot)}(u) = \|x\|_E^{-1}\hat{f}(x^{-1}u)$, so

$$A\,R(1,x,0)f\colon y \longmapsto \lambda\|x\|_E^{1/2}\|x\|_E^{-1}\hat{f}(x^{-1}y^\sigma) = \lambda\|x\|_E^{-1/2}\hat{f}(x^{-1}y^\sigma)$$

On the other side, $Af\colon y \mapsto \lambda\hat{f}(y^\sigma)$ and hence

$$R\big(N_{E/F}(x), x^{-\sigma}, 0\big)\,Af\colon y \longmapsto \kappa\big(N_{E/F}(x)\big)\,\big\|x^{-\sigma}\big\|_E^{1/2}\,\lambda\,\hat{f}\big(\big(x^{-\sigma}y\big)^\sigma\big) = \lambda\|x\|_E^{-1/2}\hat{f}\big(x^{-1}y^\sigma\big)$$

using $\kappa(N_{E/F}(x)) = 1$, then $\|x^{-\sigma}\|_E = \|x\|_E^{-1}$ because σ preserves v_E, and finally $(x^{-\sigma}y)^\sigma = x^{-1}y^\sigma$. The two agree.

Step 5: the fourth condition, and where it is deferred. Set

$$\mathbf{q}(x) = \psi\big(N_{E/F}(x)\big), \qquad \langle x, y\rangle = \psi_E(xy^\sigma), \qquad x, y \in E$$

Expanding $N_{E/F}(x+y) = N_{E/F}(x) + \mathrm{tr}_{E/F}(xy^\sigma) + N_{E/F}(y)$ gives $\mathbf{q}(x+y) = \mathbf{q}(x)\mathbf{q}(y)\langle x, y\rangle$, so $\mathbf{q}$ fails to be a character of E by exactly the pairing $\langle\cdot,\cdot\rangle$. The fourth condition in (8.21) at $t = 1$ unwinds, using the first three conditions, to the identity

$$\int_E \mathbf{q}(x)[f](x)\,dx = \lambda_{E/F}(\psi)\int_E \mathbf{q}(x)^{-1}f(x)\,dx, \qquad f \in C_c^\infty(E)$$

a twisted form of Fourier inversion. Its proof has two halves. First, the linear functional $f \mapsto \int_E \mathbf{q}\,[\mathbf{q}f]$ is invariant under translation $f \mapsto f(\cdot + y)$, by the displayed relation between $\mathbf{q}$ and the pairing, hence is a scalar multiple $\gamma_{E/F}(\psi)$ of the Haar integral. The scalar is non-zero because $f \mapsto \mathbf{q}[\mathbf{q}f]$ is bijective on $C_c^\infty(E)$. Second, evaluating that functional on the characteristic function of $\mathfrak{p}_E^d$ for large d reduces the scalar to an integral $\int_{\mathfrak{p}^k} \kappa(y)\psi(y)\,dy$, which is the Gauss sum computing $\varepsilon(\kappa, 1, \psi)$ up to a positive real factor, and the functional equation for GL_1 identifies that with $\lambda_{E/F}(\psi)$ up to a positive real factor. Since both $\gamma_{E/F}(\psi)$ and $\lambda_{E/F}(\psi)$ have absolute value 1, they are equal.

The general case $t \neq 1$ follows by running the same argument with ψ replaced by the character $t\psi\colon x \mapsto \psi(tx)$, for which $(t\psi)_E = t\psi_E$ and

$$\lambda_{E/F}(t\psi) = \frac{\varepsilon(R_{E/F}, s, t\psi)}{\varepsilon(1_E, s, t\psi_E)} = \frac{\kappa(t)\|t\|^{2(s-1/2)}\varepsilon(R_{E/F}, s, \psi)}{\|t\|_E^{s-1/2}\varepsilon(1_E, s, \psi_E)} = \kappa(t)\lambda_{E/F}(\psi)$$

by proposition 6.3.3(2) applied over F to $R_{E/F}$, whose determinant is κ and whose dimension is two, and over E to 1_E, together with $\|t\|_E = \|N_{E/F}(t)\| = \|t\|^2$ for $t \in F^\times$. These are the explicit cocycle computations, and [2, §38] carries them out in full.

Step 6: existence and smoothness. By Steps 4 and 5 the operator A satisfies (8.21) for the representation R of $\mathcal{B}$, and A is invertible because A^2 is $\kappa(-1)$ times the involution $f \mapsto f(-\cdot)$, so $A^4 = 1$. Step 3 therefore produces a representation η of $\mathcal{G}_{E/F}$ with $\eta|_{\mathcal{B}} = R$ and $\eta(w) = A$, and this is precisely (8.19). For smoothness, let $f \in C_c^\infty(E)$. By Step 2 there is an open subgroup $H_1 \subseteq \mathcal{B}$ fixing f and an open subgroup $H_2 \subseteq \mathcal{B}$ fixing Af. Put $H = H_1 \cap H_2$, an open subgroup of $\mathcal{B}$. Then H fixes f, and so does $w^{-1}Hw$, since for $h \in H$ we get $\eta(w)^{-1}\eta(h)\eta(w)f = \eta(w)^{-1}\eta(h)Af = \eta(w)^{-1}Af = f$. The subgroup of $\mathcal{G}_{E/F}$ generated by H and $w^{-1}Hw$ is open and fixes f, so f is a smooth vector. With Step 1 this proves (1).

Step 7: the remaining assertions. The formula on $\mathcal{B}$ in part (1) is the formula for $R(\zeta, x, t)$ from Step 2, rewritten using $N_{E/F}(x)N_{E/F}(y) = N_{E/F}(xy)$. Part (2) is the computation of A^2 in Step 4 combined with (8.22), and $\eta(w)^4 = 1$ follows because $f \mapsto f(-\cdot)$ is an involution and $\kappa(-1)^2 = 1$. For part (3), if $u \in \mathcal{K}_{E/F}$ then $\|u\|_E = 1$ by (8.15), so $\eta(a(u))f(y) = f(uy)$. The group $\mathcal{A}_1$ is central by proposition 8.3.3(2).

Fix a smooth character θ of $\mathcal{K}_{E/F}$. The condition $f(uy) = \theta(u)f(y)$ says exactly that $\eta(a(u))f = \theta(u)f$ for all $u \in \mathcal{K}_{E/F}$, so $C_c^\infty(E, \theta)$ is the θ-eigenspace of the central subgroup $\mathcal{A}_1$, and is therefore stable under every $\eta(\mathfrak{g})$: for $\mathfrak{g} \in \mathcal{G}_{E/F}$ and $f \in C_c^\infty(E, \theta)$,

$$\eta\big(a(u)\big)\eta(\mathfrak{g})f = \eta(\mathfrak{g})\eta\big(a(u)\big)f = \theta(u)\,\eta(\mathfrak{g})f$$

That proves part (3). For part (4), distinct characters give independent eigenspaces, since eigenvectors of a commuting family of operators for distinct systems of eigenvalues are linearly

independent, so the sum of the $C_c^\infty(E,\theta)$ is direct. It exhausts $C_c^\infty(E)$: given f, smoothness gives an open subgroup $H \subseteq \mathcal{K}_{E/F}$ fixing f, and H is normal because $\mathcal{K}_{E/F}$ is abelian, so the span W of the finite set $\{\eta(a(u))f : u \in \mathcal{K}_{E/F}/H\}$ is a finite-dimensional space on which $\mathcal{K}_{E/F}$ acts through the finite abelian quotient $\mathcal{K}_{E/F}/H$. Commuting operators of finite order on a finite-dimensional complex vector space are simultaneously diagonalizable, so W, and with it f, decomposes into eigenvectors for the characters of $\mathcal{K}_{E/F}/H$.

Finally suppose $\theta \neq 1$ and $f \in C_c^\infty(E,\theta)$. Taking $y = 0$ in the defining condition gives $f(0) = \theta(u)f(0)$ for every $u \in \mathcal{K}_{E/F}$, and $\theta \neq 1$ forces $f(0) = 0$. As f is locally constant it vanishes on a neighbourhood of 0, so its support is a compact subset of E disjoint from 0, hence a compact subset of $E^\times$. Therefore $f \in C_c^\infty(E^\times)$, completing the proof.

The decomposition in part (4) is what the construction was built for. The group $\mathcal{K}_{E/F}$ is compact and central in $\mathcal{G}_{E/F}$, so it cannot act irreducibly on anything larger than a line, and $C_c^\infty(E)$ is far from irreducible: example 8.3.2(3) exhibits $q+1$ distinct characters of $\mathcal{K}_{E/F}$ already among the roots of unity, in the unramified case. Selecting one character θ cuts out a $\mathcal{G}_{E/F}$-stable subspace on which $a(u)$ acts by the scalar $\theta(u)$, and by the remark following proposition 8.3.3 such a subspace, after twisting by the inverse of a character Θ of $E^\times$ with $\Theta|_{\mathcal{K}_{E/F}} = \theta$, which makes $\mathcal{A}_1$ act trivially, descends to a representation of $\mathcal{G}_{E/F}/\mathcal{A}_1 \cong G^\kappa$. No extension theorem is needed for that: the construction of section 8.4 starts from a character Θ of $E^\times$ and takes θ to be its restriction. The final clause of part (4) says that this subspace consists of functions supported away from the origin, which is what allows the resulting representation of G^κ to be transported to a space of functions on $\ker\kappa \subseteq F^\times$.

Part (2) is a smaller point with a concrete use. It says $\eta(w)$ has order dividing four, and that its square is the scalar $\kappa(-1)$ times the reflection $f \mapsto f(-\cdot)$. Combined with proposition 8.3.3(4), which makes η determined by $\eta|_{\mathcal{B}}$ and $\eta(w)$, this says that once the three formulas on $\mathcal{B}$ are fixed, the operator at w must square to a prescribed involution, and (8.22) is the statement that the Langlands constant meets that constraint.

Exercise 8.3.1

1. Let $(g,h) \in \mathcal{G}_{E/F}$ with $g = \left(\begin{smallmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{smallmatrix}\right)$ and $g_{21} \neq 0$. Using proposition 8.3.3(3) and (4), write $(g,h) = z(\zeta)a(x)n(t)\,w\,n(s)$ and express ζ, x, t, s explicitly in terms of the entries of g and of h. Then verify that the value of $N_{E/F}(x)$ obtained from your formula for x agrees with the one forced by the identity $\det g = N_{E/F}(h)^{-1}$.

2. Keep the unramified setting of example 8.3.2 and let $\mu \subseteq U_E$ be the group of roots of unity of order dividing q^2-1, so that $\omega^\sigma = \omega^q$ for $\omega \in \mu$.

    1. Show that $\{\omega^{1-q} : \omega \in \mu\} = \mu \cap \mathcal{K}_{E/F}$.
    2. Show that $\mu \cap F^\times$ is the group of roots of unity of order dividing $q-1$, and deduce that $\mu \cap \mathcal{K}_{E/F} \cap F^\times = \{\pm 1\}$ when q is odd and is trivial when q is even.

3. Keep the unramified setting of example 8.3.2, take ψ of level zero, fix $m \in \mathbb{Z}$, and let $f = \mathbf{1}_{\mathfrak{p}_E^m}$.

    1. Show that $\hat{f} = q^{-2m}\mathbf{1}_{\mathfrak{p}_E^{-m}}$.
    2. Deduce that $\eta_\psi^{\mathcal{G}}(w)f = \lambda_{E/F}(\psi)\,q^{-2m}\mathbf{1}_{\mathfrak{p}_E^{-m}}$.

3. Compute $\eta_\psi^{\mathcal{G}}(w)^2 f$ directly from (2) and check that the answer agrees with proposition 8.3.5(2).

4. Show that $\{\mathfrak{g} \in \mathcal{G}_{E/F} : \mathfrak{g}\mathcal{N}\mathfrak{g}^{-1} = \mathcal{N}\} = \mathcal{B}$. (For the inclusion that is not immediate from proposition 8.3.3(3), identify the subspace of F^2 fixed by every element of N and show that a matrix normalizing N must preserve it.)

5. With $\mathbf{q}(x) = \psi(N_{E/F}(x))$ and $\langle x, y\rangle = \psi_E(xy^\sigma)$ for $x, y \in E$, establish the following.

   1. $\mathbf{q}(x+y) = \mathbf{q}(x)\mathbf{q}(y)\langle x, y\rangle$, and $\langle \cdot, \cdot \rangle$ is symmetric and bi-additive in the multiplicative sense that $\langle x + x', y\rangle = \langle x, y\rangle\langle x', y\rangle$. It is the composite of ψ with the symmetric F-bilinear form $(x, y) \mapsto \mathrm{tr}_{E/F}(xy^\sigma)$ on E, and it is non-degenerate: if $\langle x, y\rangle = 1$ for all $y \in E$ then $x = 0$.
   2. $\mathbf{q}(-x) = \mathbf{q}(x)$ and $\overline{\mathbf{q}(x)} = \mathbf{q}(x)^{-1}$.
   3. $\mathbf{q}$ is locally constant, so that $\mathbf{q}f \in C_c^\infty(E)$ for every $f \in C_c^\infty(E)$.

8.4 The Weil Representation

The representation $\eta_\psi^{\mathcal{G}}$ of definition 8.3.4 lives on $\mathcal{G}_{E/F}$, and G enters only as a quotient of a subgroup of it. This section converts it into a representation of G. A character Θ of $E^\times$ selects a subspace of $C_c^\infty(E)$ on which the twisted action of $\mathcal{K}_{E/F}$ is trivial, so that what acts there is G^κ, and induction to G produces an irreducible cuspidal representation attached to the pair $(E/F, \Theta)$. Its ε-factors, twisted by every character of $F^\times$, are those of the Weil group representation $\mathrm{Ind}_{E/F}\Theta$.

Which characters Θ are usable is settled first. Proposition 8.3.5(3) attaches to every character θ of $\mathcal{K}_{E/F}$ the subspace

$$C_c^\infty(E, \theta) = \{f \in C_c^\infty(E) : f(zy) = \theta(z)f(y) \text{ for all } z \in \mathcal{K}_{E/F},\ y \in E\}$$

stable under $\eta_\psi^{\mathcal{G}}$. Only the case $\theta \neq 1$ is of use here. If $\theta = 1$ the subspace contains every $\mathcal{K}_{E/F}$-invariant function, including functions that are non-zero at 0, whereas the construction below has to land inside $C_c^\infty(E^\times)$: that is where the mirabolic formulas of section 8.1 live, and the construction works by matching them. The hypothesis is therefore that Θ be non-trivial on $\mathcal{K}_{E/F}$, and that condition has a Galois-theoretic name.

Definition 8.4.1: Regular Character

Let E/F be a separable quadratic extension with $\mathrm{Gal}(E/F) = \{1, \sigma\}$, and let Θ be a character of $E^\times$. Write Θ^σ for the character $y \mapsto \Theta(y^\sigma)$. The character Θ is *E/F-regular* if

$$\Theta^\sigma \neq \Theta$$

Regularity is the same condition as non-triviality on $\mathcal{K}_{E/F}$. It is worth recording as a lemma, because both this section and section 8.5 turn on it.

Lemma 8.4.2: Regularity Is Non-Triviality on $\mathcal{K}_{E/F}$

Let Θ be a smooth character of $E^\times$. Then $\Theta^\sigma = \Theta$ if and only if Θ is trivial on $\mathcal{K}_{E/F}$. Equivalently, Θ is E/F-regular in the sense of definition 8.4.1 if and only if $\Theta|_{\mathcal{K}_{E/F}} \neq 1$.

***Proof*:**
In one direction, $N_{E/F}(y/y^\sigma) = N_{E/F}(y)/N_{E/F}(y^\sigma) = 1$, so $y/y^\sigma \in \mathcal{K}_{E/F}$ for every $y \in E^\times$. If Θ is trivial on $\mathcal{K}_{E/F}$ then $\Theta(y)\Theta^\sigma(y)^{-1} = \Theta(y/y^\sigma) = 1$ for all y, that is $\Theta = \Theta^\sigma$. In the other direction, every $z \in \mathcal{K}_{E/F}$ has the form y/y^σ. Indeed, for $c \in E$ put $y = c + zc^\sigma$. Then

$$z\,y^\sigma = zc^\sigma + zz^\sigma c = y$$

because $zz^\sigma = N_{E/F}(z) = 1$, so $z = y/y^\sigma$ provided $y \neq 0$. Were $c + zc^\sigma = 0$ for every $c \in E$, the choice $c = 1$ would give $z = -1$ and the equation would then read $c = c^\sigma$ for all $c \in E$, which is false since $E \neq F$. So a σ-invariant Θ is trivial on $\mathcal{K}_{E/F}$, and the two conditions agree, completing the proof.

One consequence is recorded now because every ε-factor computation below uses it. A regular Θ is ramified: each $z \in \mathcal{K}_{E/F}$ has $\|z\|_E = \|N_{E/F}(z)\|_F = 1$ by the norm-module identity of section 8.3, so $\mathcal{K}_{E/F} \subseteq U_E$, and a character trivial on U_E is trivial on $\mathcal{K}_{E/F}$. Hence $L(\Theta, s) = 1 = L(\Theta^{-1}, 1-s)$ by lemma 5.1.15, and $\gamma(\Theta, s, \psi_E) = \varepsilon(\Theta, s, \psi_E)$ by definition 5.1.13.

Regular characters are plentiful, and in the unramified case they can be counted. Let E/F be unramified and let Θ have level zero, meaning that Θ is trivial on U_E^1 and not on U_E, so that $\Theta|_{U_E}$ is inflated from a character $\overline{\Theta}$ of $k_E^\times = U_E/U_E^1$. The image of $\mathcal{K}_{E/F}$ in $k_E^\times$ is the kernel of the norm $N_{k_E/k}$. It is contained in that kernel because reduction carries $N_{E/F}$ to $N_{k_E/k}$, the automorphism σ inducing $\alpha \mapsto \alpha^q$ on k_E by example 8.3.2. It contains that kernel because a class killed by $N_{k_E/k}$ lifts to an element ζ of the group μ_{q^2-1} of roots of unity of order prime to p, for which $N_{E/F}(\zeta) = \zeta^{1+q}$ lies in μ_{q-1} and reduces to 1, hence is 1. So Θ is regular exactly when $\overline{\Theta}$ is non-trivial on $\ker N_{k_E/k}$, that is, exactly when $\overline{\Theta}$ does not factor through $N_{k_E/k}$. That kernel has order $(q^2-1)/(q-1) = q+1$, since the norm of finite fields is surjective, and the characters of the cyclic group $k_E^\times$ trivial on it number $(q^2-1)/(q+1) = q-1$. So exactly $q^2 - q$ of the $q^2 - 1$ characters of $k_E^\times$ are regular.

Fix now a regular Θ and set $\theta = \Theta|_{\mathcal{K}_{E/F}}$, a non-trivial character of $\mathcal{K}_{E/F}$. The first gain from $\theta \neq 1$ is the containment promised above. If $f \in C_c^\infty(E, \theta)$ then $f(0) = f(z \cdot 0) = \theta(z)f(0)$ for every $z \in \mathcal{K}_{E/F}$, so $f(0) = 0$. Being locally constant, f then vanishes on a neighbourhood of 0, so its support is a compact subset of $E^\times$, giving

$$C_c^\infty(E, \theta) \subseteq C_c^\infty(E^\times)$$

The space is not zero. Write $\Theta^{-\sigma}$ for the character $y \mapsto \Theta(y^\sigma)^{-1}$. The function f equal to $\Theta^{-\sigma}$ on U_E and to 0 off U_E is smooth and compactly supported, and it satisfies $f(zy) = \Theta(z^\sigma)^{-1}f(y) = \theta(z)f(y)$ for $z \in \mathcal{K}_{E/F}$, since $z^\sigma = z^{-1}$ there.

The character Θ enters a second time, as a twist. For $\mathbf{g} = (g, h) \in \mathcal{G}_{E/F}$ set

$$\xi(\Theta, \psi)(\mathbf{g}) = \Theta(h)^{-1}\,\eta_\psi^{\mathcal{G}}(\mathbf{g})$$

This is again a representation of $\mathcal{G}_{E/F}$, because $(g, h) \mapsto \Theta(h)^{-1}$ is a character of $\mathcal{G}_{E/F}$, and it is smooth for the same reason $\eta_\psi^{\mathcal{G}}$ is. It preserves $C_c^\infty(E, \theta)$, on which it does something the untwisted

representation does not: it kills $\mathcal{K}_{E/F}$. For $z \in \mathcal{K}_{E/F}$ the element $(1,z)$ of $\mathcal{G}_{E/F}$ is $a(z)$, and $\eta_\psi^{\mathcal{G}}(a(z))f\colon y \mapsto f(zy) = \theta(z)f(y)$ by proposition 8.3.5(3), so

$$\xi(\Theta,\psi)\big((1,z)\big)f = \Theta(z)^{-1}\theta(z)f = f, \qquad z \in \mathcal{K}_{E/F},\ f \in C_c^\infty(E,\theta)$$

A representation trivial on $\mathcal{K}_{E/F}$ is a representation of the quotient, which proposition 8.3.3(1) identifies with G^κ through the first projection pr_1. Smoothness passes to the quotient provided pr_1 is open, and it is. Writing $f_{E/F}$ for the residue degree of E/F, a norm $N_{E/F}(x)$ lying in U_F has $v_F(N_{E/F}(x)) = f_{E/F}\, v_E(x) = 0$ and hence $x \in U_E$; conversely $\|N_{E/F}(u)\|_F = \|u\|_E = 1$ for $u \in U_E$. Since $N_{E/F}(E^\times) = \ker\kappa$,

$$N_{E/F}(U_E) = U_F \cap \ker\kappa \tag{8.23}$$

The right-hand side is the intersection of two open subgroups of $F^\times$, hence open. The subgroup $N_{E/F}(U_E^n)$ has finite index in it, because U_E^n has finite index in U_E, and is closed, because U_E^n is compact, so it too is open. Now let $(g,h) \in \mathcal{G}_{E/F}$ and let $\mathcal{G}_{E/F} \cap (V \times hU_E^n)$ be a basic neighbourhood of it. If $g' \in V$ satisfies $\det g' \in \det g \cdot N_{E/F}(U_E^n)$, write $\det g' = \det g\, N_{E/F}(u)$ with $u \in U_E^n$. Then $N_{E/F}(hu^{-1})^{-1} = N_{E/F}(h)^{-1}N_{E/F}(u) = \det g'$, so (g', hu^{-1}) lies in the neighbourhood and maps to g'. The image of the neighbourhood therefore contains the open set $V \cap \det{}^{-1}\left(\det g \cdot N_{E/F}(U_E^n)\right)$, which proves openness. A vector fixed by an open subgroup $\mathcal{U}$ of $\mathcal{G}_{E/F}$ is consequently fixed by the open subgroup $\mathrm{pr}_1(\mathcal{U})$ of G^κ, so the induced representation of G^κ on $C_c^\infty(E,\theta)$ is smooth.

What remains is to see that representation in coordinates, and this is where the norm map turns $C_c^\infty(E,\theta)$ into a space of functions on $F^\times$. The map $N_{E/F}$ carries $E^\times$ onto $\ker\kappa$ with kernel $\mathcal{K}_{E/F}$, so it induces a continuous bijective homomorphism $E^\times/\mathcal{K}_{E/F} \to \ker\kappa$. That map is a homeomorphism. Since $\mathcal{K}_{E/F} \subseteq U_E$, the image of U_E in the quotient is a compact open subgroup, and the induced map carries it bijectively and continuously onto the open subgroup $N_{E/F}(U_E)$ of (8.23); a continuous bijection from a compact space to a Hausdorff space is a homeomorphism, and both groups are the union of the translates of those two subgroups. Write

$$F_\kappa^\times = \ker\kappa = N_{E/F}(E^\times)$$

for the target, an open subgroup of index two in $F^\times$.

The transport of functions is by the prescription

$$\Lambda f\big(N_{E/F}(y)\big) = \|y\|_E^{1/2}\,\Theta^\sigma(y)\,f(y), \qquad y \in E^\times \tag{8.24}$$

which is consistent: replacing y by zy with $z \in \mathcal{K}_{E/F}$ leaves $\|y\|_E$ unchanged and multiplies the remaining two factors by $\Theta(z^\sigma) = \Theta(z)^{-1}$ and by $\theta(z) = \Theta(z)$ respectively. Because $E^\times/\mathcal{K}_{E/F} \cong F_\kappa^\times$ as topological groups and $\mathcal{K}_{E/F}$ is compact, Λ is a linear isomorphism

$$\Lambda\colon C_c^\infty(E,\theta) \xrightarrow{\sim} C_c^\infty(F_\kappa^\times)$$

its inverse sending φ to $y \mapsto \|y\|_E^{-1/2}\Theta^\sigma(y)^{-1}\varphi(N_{E/F}(y))$: local constancy passes both ways because $N_{E/F}$ is continuous and open onto $F_\kappa^\times$, and compactness of supports passes both ways because the preimage of a compact subset of $E^\times/\mathcal{K}_{E/F}$ under the quotient map is compact.

Transporting $\xi(\Theta,\psi)$ along Λ gives a smooth representation $\pi_\kappa = \pi_\kappa(\Theta,\psi)$ of G^κ on $C_c^\infty(F_\kappa^\times)$, and the three generating families of proposition 8.3.3(2) act by explicit operators. The images under pr_1 of $n(t)$, $z(\zeta)$ and $a(x)$ are $\left(\begin{smallmatrix}1 & t\\ 0 & 1\end{smallmatrix}\right)$, ζI and $\mathrm{diag}(N_{E/F}(x),1)$, and (8.19) gives, for $f \in C_c^\infty(E,\theta)$ and

$y \in E^\times$,

$$\begin{aligned}
\xi(\Theta,\psi)\big(n(t)\big)f\colon y &\longmapsto \psi\big(t\,N_{E/F}(y)\big)f(y)\\
\xi(\Theta,\psi)\big(z(\zeta)\big)f &= \kappa(\zeta)\,\Theta(\zeta)\,f\\
\xi(\Theta,\psi)\big(a(x)\big)f\colon y &\longmapsto \Theta^\sigma(x)\,\|x\|_E^{1/2}\,f(xy)
\end{aligned}$$

the second because $z(\zeta) = (\zeta I, \zeta^{-1})$ contributes the scalar $\Theta(\zeta^{-1})^{-1}$, and the third because $a(x) = (\mathrm{diag}(N_{E/F}(x),1), x^{-\sigma})$ contributes $\Theta(x^{-\sigma})^{-1} = \Theta^\sigma(x)$. Applying Λ to each line turns them into operators on $C_c^\infty(F_\kappa^\times)$. The first is unchanged, since $\psi(tN_{E/F}(y))$ depends only on $N_{E/F}(y)$; the second is a scalar; and for the third,

$$\Lambda\big(\xi(\Theta,\psi)(a(x))f\big)\big(N_{E/F}(y)\big) = \|y\|_E^{1/2}\Theta^\sigma(y)\,\Theta^\sigma(x)\|x\|_E^{1/2}f(xy) = \|xy\|_E^{1/2}\Theta^\sigma(xy)f(xy) = \Lambda f\big(N_{E/F}(xy)\big)$$

so the operator is translation by $N_{E/F}(x)$. Collecting the three, and writing u for the variable in $F_\kappa^\times$,

$$\begin{aligned}
\pi_\kappa\left(\begin{pmatrix}1 & t\\ 0 & 1\end{pmatrix}\right)\varphi\colon u &\longmapsto \psi(tu)\varphi(u), && t \in F\\
\pi_\kappa\big(\zeta I\big)\varphi &= \kappa(\zeta)\Theta(\zeta)\varphi, && \zeta \in F^\times\\
\pi_\kappa\left(\begin{pmatrix}a & 0\\ 0 & 1\end{pmatrix}\right)\varphi\colon u &\longmapsto \varphi(au), && a \in F_\kappa^\times
\end{aligned} \tag{8.25}$$

Every Θ has vanished from the first and third lines. The half-power of $\|y\|_E$ and the factor Θ^σ built into Λ are what remove it, and what is left is the pair of mirabolic formulas of section 2.3 with $F^\times$ replaced by its open subgroup $F_\kappa^\times$. The character Θ survives only in the central line.

Definition 8.4.3: The Weil Representation

Let E/F be a separable quadratic extension, let Θ be an E/F-regular character of $E^\times$ and let $\psi \in \widehat{F}$, $\psi \neq 1$. Write $\pi_\kappa(\Theta,\psi)$ for the smooth representation of G^κ on $C_c^\infty(F_\kappa^\times)$ obtained from $\xi(\Theta,\psi)$ by transport along (8.24). The *Weil representation* attached to $(E/F,\Theta)$ and ψ is

$$\pi(\Theta,\psi) = \mathrm{Ind}_{G^\kappa}^{G}\pi_\kappa(\Theta,\psi)$$

Since G^κ has index two in G, this is also the compact induction of definition 1.2.24. Its isomorphism class does not depend on ψ, by theorem 8.4.5(3), and is written $\pi_{E/F}(\Theta)$.

The unramified case shows the shape of the definition. There $\kappa(u) = (-1)^{v_F(u)}$ by example 8.3.2, so $F_\kappa^\times$ is the set of elements of even valuation and G^κ the set of g with $v_F(\det g)$ even. The space $C_c^\infty(F_\kappa^\times)$ is then the space of functions on $F^\times$ supported on the even-valuation part, the Borel operators of (8.25) move that support only by even amounts, and the two-dimensional induction to G restores the odd valuations. In the ramified case κ is ramified, $F_\kappa^\times$ contains $\varpi^{\mathbb{Z}}$ up to a choice of prime element, and it is U_F that gets halved.

Reproducing the mirabolic formulas is what forces the induced representation of G to be cuspidal. Nothing about Θ beyond regularity enters the argument.

Proposition 8.4.4: The Weil Representation is Cuspidal

Let Θ be an E/F-regular character of $E^\times$ and $\psi \in \widehat{F}$, $\psi \neq 1$. Write $\pi = \pi(\Theta, \psi)$.

1. π is an irreducible cuspidal representation of G, of infinite dimension, and $\pi|_M \cong \text{c-Ind}_N^M \vartheta$.
2. For every character χ of $F^\times$, writing $\chi_E = \chi \circ N_{E/F}$,
$$\pi(\chi_E \Theta, \psi) \cong \chi \pi(\Theta, \psi)$$
In particular $\kappa\, \pi(\Theta, \psi) \cong \pi(\Theta, \psi)$.
3. The central character of π is $\omega_\pi = \kappa \cdot \Theta|_{F^\times}$.

***Proof*:**
Step 1: the restriction to the mirabolic group. Write $M^\kappa = M \cap G^\kappa$, the group of matrices $\left(\begin{smallmatrix} a & b \\ 0 & 1 \end{smallmatrix}\right)$ with $a \in F_\kappa^\times$, and $s(a) = \mathrm{diag}(a, 1)$. Every element of M^κ factors uniquely as $n\, s(a)$ with $n \in N$, so restriction of functions to $\{s(a) : a \in F_\kappa^\times\}$ identifies $\text{c-Ind}_N^{M^\kappa} \vartheta$ with $C_c^\infty(F_\kappa^\times)$, by the computation made for M itself in section 2.3 with $F^\times$ replaced by its open subgroup $F_\kappa^\times$. Write $\varphi(u) = \Psi(s(u))$ for the function attached to $\Psi \in \text{c-Ind}_N^{M^\kappa} \vartheta$. Under that identification the two generating families act by right translation as

$$\big(\Psi \cdot s(a)\big)\big(s(u)\big) = \Psi\big(s(ua)\big) = \varphi(ua), \qquad \big(\Psi \cdot n(t)\big)\big(s(u)\big) = \Psi\big(n(ut)s(u)\big) = \psi(ut)\varphi(u)$$

the second using $s(u)n(t) = n(ut)s(u)$ together with the N-equivariance of Ψ. These are the first and third lines of (8.25). Hence

$$\pi_\kappa|_{M^\kappa} \cong \text{c-Ind}_N^{M^\kappa} \vartheta$$

Now $G = G^\kappa M$: given $g \in G$, choose $m \in M$ with $\kappa(\det m) = \kappa(\det g)$, which is possible because det maps M onto $F^\times$, and then $gm^{-1} \in G^\kappa$. Also $G^\kappa \cap M = M^\kappa$. Restriction of functions from G to M therefore carries $\mathrm{Ind}_{G^\kappa}^G \pi_\kappa$ into $\mathrm{Ind}_{M^\kappa}^M(\pi_\kappa|_{M^\kappa})$ injectively, since $G = G^\kappa M$ and a function in the induced space is determined by its values on M. It is surjective, since a function Ψ on M with the M^κ-equivariance extends by $\Phi(hm) = \pi_\kappa(h)\Psi(m)$, this being well defined because $hm = h'm'$ forces $h'^{-1}h = m'm^{-1} \in G^\kappa \cap M = M^\kappa$ and Ψ is M^κ-equivariant. Smoothness is automatic on both sides: G^κ and M^κ have finite index, so a function satisfying the equivariance is smooth as soon as its values $\Phi(g_i)$ at coset representatives are smooth vectors, because $\bigcap_i g_i^{-1}\mathrm{Stab}(\Phi(g_i))g_i$ is then open. So

$$\pi|_M \cong \mathrm{Ind}_{M^\kappa}^M \text{c-Ind}_N^{M^\kappa} \vartheta$$

The right-hand side is $\text{c-Ind}_N^M \vartheta$. The map sending Φ in it to the function $m \mapsto \Phi(m)(1)$ on M is the isomorphism: it lands in $\mathrm{Ind}_N^M \vartheta$ because for $n \in N \subseteq M^\kappa$ one has $\Phi(nm) = \pi_\kappa(n)\Phi(m)$, whose value at 1 is $\Phi(m)(n) = \vartheta(n)\Phi(m)(1)$; it commutes with right translation, the image of $\Phi \cdot m_0$ being the function $m \mapsto \Phi(mm_0)(1)$; it is injective because $\Phi(m)(m') = \Phi(m'm)(1)$

recovers Φ; and the support conditions match, both sides consisting of the functions supported on finitely many cosets Nm. Hence $\pi|_M \cong \text{c-Ind}_N^M \vartheta$.

Step 2: irreducibility, cuspidality and part (1). The representation $\text{c-Ind}_N^M \vartheta$ is irreducible as a representation of M by theorem 2.3.6, and it is of infinite dimension, being isomorphic to $C_c^\infty(F^\times)$. A non-zero G-subspace of π is in particular an M-subspace of $\pi|_M$, hence everything, so π is irreducible and of infinite dimension. Corollary 8.1.4 applies to it and produces an injective M-map $f\colon \pi \to \text{Ind}_N^M \vartheta$ whose image contains $\text{c-Ind}_N^M \vartheta$. That image is M-isomorphic to $\pi|_M$, which is irreducible, and $\text{c-Ind}_N^M \vartheta$ is a non-zero subspace of it, so $\text{im} f = \text{c-Ind}_N^M \vartheta$. By the last assertion of corollary 8.1.4 this says exactly that π is cuspidal.

Step 3: twisting, part (2). The character $\chi_E = \chi \circ N_{E/F}$ is trivial on $\mathcal{K}_{E/F}$, so $\chi_E\Theta$ has the same restriction θ to $\mathcal{K}_{E/F}$ as Θ does, and it is regular because $(\chi_E\Theta)^\sigma(\chi_E\Theta)^{-1} = \Theta^\sigma\Theta^{-1} \neq 1$, using $N_{E/F}(y^\sigma) = N_{E/F}(y)$. Both constructions therefore start from the same space $C_c^\infty(E,\theta)$. On it, for $\mathbf{g} = (g,h)$ with $\det g = N_{E/F}(h)^{-1}$,

$$\xi(\chi_E\Theta,\psi)(\mathbf{g}) = \chi\big(N_{E/F}(h)\big)^{-1}\Theta(h)^{-1}\eta_\psi^{\mathcal{G}}(\mathbf{g}) = \chi(\det g)\,\xi(\Theta,\psi)(\mathbf{g})$$

The two transport maps differ by a multiplication operator. Writing Λ' for the map of (8.24) formed with $\chi_E\Theta$, and using $(\chi_E\Theta)^\sigma(y) = \chi(N_{E/F}(y))\Theta^\sigma(y)$,

$$\Lambda' f\big(N_{E/F}(y)\big) = \chi\big(N_{E/F}(y)\big)\Lambda f\big(N_{E/F}(y)\big)$$

that is $\Lambda' = m_\chi \circ \Lambda$ with $m_\chi\varphi = \chi\varphi$. Conjugating,

$$\pi_\kappa(\chi_E\Theta,\psi)(\mathbf{g}) = \Lambda'\,\xi(\chi_E\Theta,\psi)(\mathbf{g})\,\Lambda'^{-1} = \chi(\det g)\,m_\chi\,\pi_\kappa(\Theta,\psi)(\mathbf{g})\,m_\chi^{-1}$$

so m_χ is an isomorphism $\pi_\kappa(\chi_E\Theta,\psi) \cong (\chi\circ\det)\otimes\pi_\kappa(\Theta,\psi)$ of representations of G^κ. It remains to move the twist through the induction, and that is a general fact. Let $\mathcal{H}$ be a locally profinite group, H an open subgroup of finite index, μ a character of $\mathcal{H}$ and (ϱ,V) a smooth representation of H. The map $T_\mu\colon \Phi \mapsto \big(x \mapsto \mu(x)^{-1}\Phi(x)\big)$ carries $\text{Ind}_H^{\mathcal{H}}(\mu|_H \otimes \varrho)$ bijectively onto $\text{Ind}_H^{\mathcal{H}}\varrho$, since the two equivariance conditions differ by exactly the factor $\mu(h)$, and

$$T_\mu\big(\varrho^{\mathcal{H}}(x_0)\Phi\big)(x) = \mu(x)^{-1}\Phi(xx_0) = \mu(x_0)\,\big(\varrho^{\mathcal{H}}(x_0)T_\mu\Phi\big)(x)$$

where $\varrho^{\mathcal{H}}$ denotes right translation. So T_μ is an isomorphism $\text{Ind}_H^{\mathcal{H}}(\mu|_H\otimes\varrho) \cong \mu\otimes\text{Ind}_H^{\mathcal{H}}\varrho$. Applying it with $\mathcal{H} = G$, $H = G^\kappa$ and $\mu = \chi\circ\det$ gives $\pi(\chi_E\Theta,\psi) \cong \chi\pi(\Theta,\psi)$. For the last assertion, $\kappa_E = \kappa\circ N_{E/F} = 1$ because $\ker\kappa = N_{E/F}(E^\times)$.

Step 4: the central character, part (3). The centre Z of G lies in G^κ, since $\kappa(\det\zeta I) = \kappa(\zeta)^2 = 1$, and the second line of (8.25) says that $\pi_\kappa(\zeta I)$ is the scalar $\kappa(\zeta)\Theta(\zeta)$. For Φ in the induced space,

$$\big(\pi(\zeta I)\Phi\big)(g) = \Phi(g\,\zeta I) = \Phi(\zeta I\,g) = \pi_\kappa(\zeta I)\Phi(g) = \kappa(\zeta)\Theta(\zeta)\Phi(g)$$

so $\omega_\pi = \kappa\cdot\Theta|_{F^\times}$, completing the proof.

Part (1) already places $\pi_{E/F}(\Theta)$ inside the classification of theorem 3.5.5: it is a cuspidal representation, so it is compactly induced from a cuspidal type, and it has a Kirillov model in the sense

of definition 8.1.5. Nothing so far, however, connects it to Θ quantitatively. Part (3) is the first such connection, and it is exactly the identity the Galois side predicts, since $\det \mathrm{Ind}_{E/F}\Theta = \kappa \cdot \Theta|_{F^\times}$ by proposition 6.2.7. The theorem below strengthens that agreement of central characters to an agreement of all twisted local constants.

The mechanism is the one section 8.2 isolated. A cuspidal representation is determined by its central character together with the single operator $\pi_K(w)$ on its Kirillov model, and theorem 8.2.4 evaluates that operator in terms of ε-factors. On the other side, $\pi_\kappa(w)$ is computable directly from definition 8.3.4, because w acts there through a Fourier transform on E and a Fourier transform against a multiplicative character is a Gauss sum. Comparing the two evaluations of the same operator is the proof.

Theorem 8.4.5: Local Constants of the Weil Representation

Let Θ be an E/F-regular character of $E^\times$, let $\psi \in \widehat{F}$, $\psi \neq 1$, and put

$$\pi = \pi(\Theta, \psi), \qquad \rho = \mathrm{Ind}_{E/F}\Theta, \qquad \psi_E = \psi \circ \mathrm{Tr}_{E/F}$$

1. $n(\pi, \psi) = f_{E/F}\, n(\Theta, \psi_E)$ and $\varepsilon\big(\pi, \tfrac{1}{2}, \psi\big) = \lambda_{E/F}(\psi)\, \varepsilon\big(\Theta, \tfrac{1}{2}, \psi_E\big)$.
2. For every character χ of $F^\times$ and every $\psi' \in \widehat{F}$, $\psi' \neq 1$,
$$\varepsilon(\chi\pi, s, \psi') = \varepsilon(\chi \otimes \rho, s, \psi') \tag{8.26}$$
3. $\pi(\Theta, \psi) \cong \pi(\Theta, \psi')$ for all $\psi, \psi' \in \widehat{F}$ different from 1.

***Proof*:**

Throughout, $n(\pi, \psi)$ denotes the exponent in the monomial form $\varepsilon(\pi, s, \psi) = \varepsilon(\pi, \tfrac{1}{2}, \psi) q^{(1/2-s)n(\pi,\psi)}$ given by corollary 5.2.9, which agrees with definition 5.3.4 when ψ has level one. The corresponding exponent for a character of $E^\times$ against ψ_E comes from corollary 5.1.14 applied over E, with q replaced by $q_E = q^{f_{E/F}}$.

Step 1: a Gauss sum on E. Let g be the function equal to Θ^{-1} on U_E and to 0 off U_E, and let $\hat{g}$ be its Fourier transform with respect to ψ_E and the self-dual measure, as in definition 5.1.1. Set $n_\Theta = n(\Theta, \psi_E)$. The claim is

$$\hat{g}(x) = \begin{cases} \varepsilon(\Theta, 1, \psi_E)\, \Theta(x) & v_E(x) = -n_\Theta \\ 0 & \text{otherwise} \end{cases} \tag{8.27}$$

For $u \in U_E$, substituting $y \mapsto u^{-1}y$, which preserves U_E and the measure,

$$\hat{g}(ux) = \int_{U_E} \Theta^{-1}(y)\psi_E(uxy)\, dy = \int_{U_E} \Theta^{-1}(u^{-1}y)\psi_E(xy)\, dy = \Theta(u)\hat{g}(x)$$

so $\Theta^{-1}\hat{g}$ is constant on each set $S_m = \{x \in E^\times : v_E(x) = m\}$, say with value c_m there. Since Θ is ramified, $\hat{g}(0) = \int_{U_E} \Theta^{-1}(y)\, dy = 0$, and $\hat{g}$ lies in $C_c^\infty(E)$ by proposition 5.1.2(1), so it vanishes near 0 and off a compact set: only finitely many c_m are non-zero. Fix a Haar measure $\mu^\times$ on $E^\times$, write $d^\times x$ for integration against it, and put $\alpha = \mu^\times(U_E)$, so that $\mu^\times(S_m) = \alpha$ for

every m. Writing $\zeta(\,\cdot\,,\,\cdot\,,s)$ for the GL_1 zeta integral of section 5.1, formed over E,

$$\zeta(g,\Theta,s)=\int_{E^\times} g(x)\Theta(x)\|x\|_E^s\,d^\times x=\alpha,\qquad \zeta(\hat g,\Theta^{-1},1-s)=\alpha\sum_m c_m q_E^{-m}q_E^{ms}$$

The functional equation of theorem 5.1.11 over E, together with $\gamma(\Theta,s,\psi_E)=\varepsilon(\Theta,s,\psi_E)$ for ramified Θ, gives

$$\alpha\sum_m c_m q_E^{-m}q_E^{ms}=\varepsilon(\Theta,s,\psi_E)\,\alpha=\alpha\,q_E^{n_\Theta/2}\varepsilon\big(\Theta,\tfrac12,\psi_E\big)q_E^{-sn_\Theta}$$

Comparing coefficients of the powers q_E^{ms}, which are linearly independent functions of s, leaves $c_m=0$ for $m\neq -n_\Theta$ and $c_{-n_\Theta}=q_E^{-n_\Theta/2}\varepsilon(\Theta,\frac12,\psi_E)=\varepsilon(\Theta,1,\psi_E)$, the last equality by corollary 5.1.14 over E. This is (8.27).

Step 2: the operator $\pi_\kappa(w)$. Let φ be the characteristic function of $U_F\cap F_\kappa^\times$. By (8.23) and (8.24), $\varphi=\Lambda f$ where f is the function equal to $\Theta^{-\sigma}$ on U_E and 0 elsewhere: for $y\in U_E$ one has $\|y\|_E^{1/2}\Theta^\sigma(y)\Theta^{-\sigma}(y)=1$. The element $w=(w,1)$ of $\mathcal{G}_{E/F}$ lies over $w\in G^\kappa$ and contributes no twist, so $\xi(\Theta,\psi)(w)=\eta_\psi^{\mathcal{G}}(w)$, and by definition 8.3.4,

$$\eta_\psi^{\mathcal{G}}(w)f(x)=\lambda_{E/F}(\psi)\,\hat f(x^\sigma)$$

Now $\hat f(x^\sigma)=\hat g(x)$: substituting $y\mapsto y^\sigma$, which preserves the self-dual measure because σ is an isometry of E fixing ψ_E,

$$\hat f(x^\sigma)=\int_E f(y)\psi_E(x^\sigma y)\,dy=\int_E f(y^\sigma)\psi_E\big((xy)^\sigma\big)\,dy=\int_E g(y)\psi_E(xy)\,dy$$

using $f(y^\sigma)=\Theta(y)^{-1}\mathbf{1}_{U_E}(y)=g(y)$ and $\psi_E\circ\sigma=\psi_E$. So by (8.27), $\eta_\psi^{\mathcal{G}}(w)f$ is supported on $\{x:v_E(x)=-n_\Theta\}$, where it takes the value $\lambda_{E/F}(\psi)\varepsilon(\Theta,1,\psi_E)\Theta(x)$. Transporting by Λ and using $v_F(N_{E/F}(x))=f_{E/F}\,v_E(x)$, so that $\|N_{E/F}(x)\|_F^{1/2}=q^{f_{E/F}n_\Theta/2}=q_E^{n_\Theta/2}$ on that support,

$$\Lambda\big(\eta_\psi^{\mathcal{G}}(w)f\big)\big(N_{E/F}(x)\big)=\|x\|_E^{1/2}\Theta^\sigma(x)\lambda_{E/F}(\psi)\varepsilon(\Theta,1,\psi_E)\Theta(x)=\lambda_{E/F}(\psi)\,\varepsilon\big(\Theta,\tfrac12,\psi_E\big)\,\Theta\big(N_{E/F}(x)\big)$$

the last equality because $\Theta^\sigma(x)\Theta(x)=\Theta(N_{E/F}(x))$ and $\|x\|_E^{1/2}\varepsilon(\Theta,1,\psi_E)=q_E^{n_\Theta/2}q_E^{-n_\Theta/2}\varepsilon(\Theta,\frac12,\psi_E)$. Writing $n_\pi=f_{E/F}n_\Theta$ and recalling that $N_{E/F}$ carries $\{x:v_E(x)=-n_\Theta\}$ onto $\{u\in F_\kappa^\times:v_F(u)=-n_\pi\}$,

$$\pi_\kappa(w)\varphi(u)=\begin{cases}\lambda_{E/F}(\psi)\,\varepsilon\big(\Theta,\frac12,\psi_E\big)\,\Theta(u) & u\in F_\kappa^\times,\ v_F(u)=-n_\pi\\ 0 & \text{otherwise}\end{cases}\tag{8.28}$$

Step 3: the same operator inside the Kirillov model. By proposition 8.4.4(1) the representation π is irreducible, cuspidal and of infinite dimension, so it has a Kirillov model (definition 8.1.5), realized on $C_c^\infty(F^\times)$ by definition 8.2.1. The isomorphism $\pi|_M\cong\text{c-Ind}_N^M\vartheta$ built in Step 1 of the proof of proposition 8.4.4 is an M-isomorphism onto $K(\pi,\vartheta)$, and transporting π through it gives a G-action on $K(\pi,\vartheta)$ that extends the natural M-action and is isomorphic to π. By the uniqueness clause of lemma 8.1.6(2) that action is π_K. Composing that isomorphism with

restriction of functions to $S = \{s(u) : u \in F^\times\}$ sends Φ in the induced space to the function $u \mapsto \Phi(s(u))(1)$ on $F^\times$. If $\operatorname{supp}\Phi \subseteq G^\kappa$ this vanishes for $u \notin F^\times_\kappa$, while for $u \in F^\times_\kappa$,

$$\Phi\big(s(u)\big)(1) = \big(\pi_\kappa(s(u))\Phi(1)\big)(1) = \Phi(1)(u)$$

by the third line of (8.25). So the subspace of Φ supported in G^κ corresponds to $C_c^\infty(F^\times_\kappa) \subseteq C_c^\infty(F^\times)$, the correspondence being $\Phi \mapsto \Phi(1)$, and that subspace is stable under G^κ because right translation by G^κ preserves the support condition. Since $(\pi(h)\Phi)(1) = \Phi(h) = \pi_\kappa(h)\Phi(1)$ for $h \in G^\kappa$, the action of G^κ there is π_κ. As $\det w = 1$ puts w in G^κ,

$$\pi_K(w)\big|_{C_c^\infty(F^\times_\kappa)} = \pi_\kappa(w)$$

Step 4: comparison, and part (1). Recall from definition 8.2.3 that $\xi_{\chi,k}$ is the function equal to χ on $\{x \in F^\times : v_F(x) = k\}$ and to 0 elsewhere. The function φ of Step 2 is

$$\varphi = \tfrac{1}{2}\big(\xi_{1,0} + \xi_{\kappa,0}\big)$$

since $\frac{1}{2}(1+\kappa(u))$ is 1 for $u \in U_F \cap F^\times_\kappa$ and 0 for the remaining units, an identity that also covers the case $\kappa|_{U_F} = 1$, where the two basis functions coincide and $U_F \subseteq F^\times_\kappa$.

Theorem 8.2.4 is stated for ψ of level one, and that hypothesis can be dropped once $n(\,\cdot\,,\psi)$ is read as the exponent of corollary 5.2.9, which is the reading in use here. Write $\psi = a\psi_0$ with $\psi_0 \in \widehat{F}$ of level one and $a \in F^\times$, possible by definition 5.1.1, and put $\vartheta_0 = \psi_0|_N$. The map $T\varphi\colon u \mapsto \varphi(au)$ turns the ϑ_0-equivariance into ϑ-equivariance and commutes with the operators $\pi_K(s(b))$, so it is an M-isomorphism of $K(\pi,\vartheta_0)$ onto $K(\pi,\vartheta)$, and by the uniqueness clause of lemma 8.1.6(2) it carries the operator $\pi_K(w)$ formed over ψ_0 to the one formed over ψ. Since $T\xi_{\chi,k} = \chi(a)\xi_{\chi,k-v_F(a)}$, applying T to the formula of theorem 8.2.4 over ψ_0 at the index $k + v_F(a)$ gives

$$\pi_K(w)\xi_{\chi,k} = \varepsilon\big(\chi^{-1}\pi, \tfrac{1}{2}, \psi\big)\, \xi_{\chi^{-1}\omega_\pi,\, -n(\chi^{-1}\pi,\psi)-k}$$

over ψ itself: the scalar because $\chi(a)^{-1}(\chi^{-1}\omega_\pi)(a) = \omega_{\chi^{-1}\pi}(a)$ and $\varepsilon(\chi^{-1}\pi, \frac{1}{2}, a\psi_0) = \omega_{\chi^{-1}\pi}(a)\varepsilon(\chi^{-1}\pi, \frac{1}{2}, \psi_0)$, the index because $n(\chi^{-1}\pi, a\psi_0) = n(\chi^{-1}\pi,\psi_0) + 2v_F(a)$, both by lemma 5.3.3.

Apply that formula to each summand of φ, at $k = 0$. With $\chi = 1$ it gives $\pi_K(w)\xi_{1,0} = \varepsilon(\pi, \frac{1}{2}, \psi)\xi_{\omega_\pi, -n(\pi,\psi)}$. With $\chi = \kappa$, and using $\kappa^{-1} = \kappa$ together with $\kappa\pi \cong \pi$ from proposition 8.4.4(2), which forces $\varepsilon(\kappa\pi, s, \psi) = \varepsilon(\pi, s, \psi)$ and hence $n(\kappa\pi,\psi) = n(\pi,\psi)$, it gives $\pi_K(w)\xi_{\kappa,0} = \varepsilon(\pi, \frac{1}{2}, \psi)\xi_{\kappa\omega_\pi, -n(\pi,\psi)}$. Adding,

$$\pi_K(w)\varphi(u) = \varepsilon\big(\pi, \tfrac{1}{2}, \psi\big)\, \omega_\pi(u)\, \frac{1+\kappa(u)}{2} \qquad \text{for } v_F(u) = -n(\pi,\psi)$$

and 0 for all other u. This function is supported on $\{u \in F^\times_\kappa : v_F(u) = -n(\pi,\psi)\}$ and non-zero there, since $\varepsilon(\pi, \frac{1}{2}, \psi) \in \mathbb{C}^\times$ by corollary 5.2.9. By Step 3 it equals (8.28), which is non-zero exactly on $\{u \in F^\times_\kappa : v_F(u) = -n_\pi\}$. Two non-empty sets of that shape agree only if the valuations agree, so

$$n(\pi,\psi) = n_\pi = f_{E/F}\, n(\Theta, \psi_E)$$

Evaluating both expressions at a single u in the common support, where $\omega_\pi(u) = \kappa(u)\Theta(u) = \Theta(u) \neq 0$ by proposition 8.4.4(3), gives $\varepsilon(\pi, \frac{1}{2}, \psi) = \lambda_{E/F}(\psi)\varepsilon(\Theta, \frac{1}{2}, \psi_E)$. This is part (1).

Step 5: the untwisted identity. Inductivity in degree zero, axiom (3) of theorem 6.3.2, applied to the virtual representation $[\Theta] - [1_E]$ of dimension zero, reads

$$\frac{\varepsilon(\mathrm{Ind}_{E/F}\Theta, s, \psi)}{\varepsilon(R_{E/F}, s, \psi)} = \frac{\varepsilon(\Theta, s, \psi_E)}{\varepsilon(1_E, s, \psi_E)}$$

so $\varepsilon(\rho, s, \psi) = \lambda_{E/F}(\psi)\varepsilon(\Theta, s, \psi_E)$ by definition 6.3.10 and corollary 6.3.11. Now use part (1) twice. Since $q_E^{n_\Theta} = q^{f_{E/F}n_\Theta} = q^{n(\pi,\psi)}$,

$$\varepsilon(\pi, s, \psi) = q^{(1/2-s)n(\pi,\psi)}\varepsilon\big(\pi, \tfrac{1}{2}, \psi\big) = \lambda_{E/F}(\psi)\, q_E^{(1/2-s)n_\Theta}\varepsilon\big(\Theta, \tfrac{1}{2}, \psi_E\big) = \varepsilon(\rho, s, \psi)$$

Step 6: general χ. Both sides of (8.26) are computed by applying Step 5 to a different regular character. On the automorphic side $\chi\pi(\Theta, \psi) \cong \pi(\chi_E\Theta, \psi)$ by proposition 8.4.4(2). On the Galois side, the computation in Step 3 of the proof of proposition 8.4.4 was made for an arbitrary character of the ambient group and an arbitrary open subgroup of finite index. Carried out for $\mathcal{W}_E \subseteq \mathcal{W}_F$ and the character $\chi \circ \mathbf{a}_F$, it gives

$$\mathrm{Ind}_{\mathcal{W}_E}^{\mathcal{W}_F}\big((\chi \circ \mathbf{a}_F)|_{\mathcal{W}_E} \otimes \Theta\big) \cong \chi \otimes \mathrm{Ind}_{E/F}\Theta$$

and $(\chi \circ \mathbf{a}_F)|_{\mathcal{W}_E} = \chi \circ N_{E/F} \circ \mathbf{a}_E = \chi_E \circ \mathbf{a}_E$ by theorem 6.2.1(4). Hence $\chi \otimes \rho \cong \mathrm{Ind}_{E/F}(\chi_E\Theta)$. Since $\chi_E\Theta$ is again regular (Step 3 of the proof of proposition 8.4.4), Step 5 applied to it gives $\varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi \otimes \rho, s, \psi)$.

Step 7: general ψ', and part (2). By definition 5.1.1 the map $a \mapsto a\psi$ is a bijection of F onto $\widehat{F}$, so every $\psi' \neq 1$ is $a\psi$ for a unique $a \in F^\times$. Lemma 5.3.3 and proposition 6.3.3(2), the latter with $\dim \rho = 2$, give

$$\varepsilon(\chi\pi, s, a\psi) = \omega_{\chi\pi}(a)\|a\|^{2(s-1/2)}\varepsilon(\chi\pi, s, \psi), \qquad \varepsilon(\chi\otimes\rho, s, a\psi) = \det(\chi\otimes\rho)(a)\|a\|^{2(s-1/2)}\varepsilon(\chi\otimes\rho, s, \psi)$$

The two correction factors coincide. Indeed $\omega_{\chi\pi} = \chi^2\omega_\pi$ and $\det(\chi \otimes \rho) = \chi^2 \det\rho$, while $\det\rho = \varkappa_{E/F} \cdot \Theta|_{F^\times}$ by proposition 6.2.7 and $\varkappa_{E/F} = \kappa$: the representation $R_{E/F}$ is the permutation representation of $\mathcal{W}_F$ on the two-element set $\mathcal{W}_F/\mathcal{W}_E$, so it is the sum of the constants and of the line on which $\mathcal{W}_F$ acts through the non-trivial character of $\mathcal{W}_F/\mathcal{W}_E$, which is $\kappa \circ \mathbf{a}_F$ by the identification of $\mathcal{W}_F/\mathcal{W}_E$ with $F^\times/N_{E/F}(E^\times)$ made in section 8.3, so $\det R_{E/F} = \kappa$. Hence $\det\rho = \kappa \cdot \Theta|_{F^\times} = \omega_\pi$ by proposition 8.4.4(3), and (8.26) at ψ propagates to every ψ'.

Step 8: part (3). Let $\psi' \neq 1$ and put $\pi' = \pi(\Theta, \psi')$. Both π and π' are cuspidal with the same central character $\kappa \cdot \Theta|_{F^\times}$, by proposition 8.4.4. For any character χ, the representations $\chi\pi$ and $\chi\pi'$ are cuspidal as well, since $\chi \circ \det$ is trivial on N and the Jacquet module is therefore unchanged, so $L(\chi\pi, s) = 1 = L(\chi\pi', s)$ by proposition 5.3.1. Fix $\psi_0 \in \widehat{F}$ of level one. Part (2), applied to π and then to π', gives

$$\varepsilon(\chi\pi, s, \psi_0) = \varepsilon(\chi \otimes \rho, s, \psi_0) = \varepsilon(\chi\pi', s, \psi_0)$$

for every χ. Theorem 5.5.2 now gives $\pi \cong \pi'$, completing the proof.

Part (3) is what justifies the notation $\pi_{E/F}(\Theta)$ of definition 8.4.3: the isomorphism class depends only on the pair $(E/F, \Theta)$, the additive character entering the construction but not its outcome. What the pair produces is a cuspidal representation of G whose family of twisted local constants is that of $\mathrm{Ind}_{E/F}\Theta$, for every additive character at once.

The Galois-side object is moreover irreducible, so it lies in the set $\mathcal{G}_2^0(F)$ that theorem 7.1.2 matches with the cuspidal representations. Suppose $\rho = \mathrm{Ind}_{E/F}\Theta$ were reducible. Being two-dimensional, it would then contain a one-dimensional subrepresentation μ, and $\mathrm{Hom}_{\mathcal{W}_F}(\mu, \mathrm{Ind}_{\mathcal{W}_E}^{\mathcal{W}_F}\Theta) \cong \mathrm{Hom}_{\mathcal{W}_E}(\mu|_{\mathcal{W}_E}, \Theta)$ by theorem 1.2.21, so $\Theta \circ \mathbf{a}_E = \mu|_{\mathcal{W}_E}$. The restriction of a character of $\mathcal{W}_F$ to the normal subgroup $\mathcal{W}_E$ is invariant under conjugation by any element of $\mathcal{W}_F$, and under $\mathbf{a}_E$ that conjugation action is $\Theta \mapsto \Theta^\sigma$. Two things go into that last identification: the functoriality of theorem 6.2.1(5), applied to the F-automorphism σ of E, and the compatibility asserting that the automorphism of $\mathcal{W}_E^{\mathrm{ab}}$ induced by σ is conjugation by any lift $\tau \in \mathcal{W}_F \setminus \mathcal{W}_E$, for which see [14]. So $\Theta^\sigma = \Theta$, contradicting regularity. Hence $\rho \in \mathcal{G}_2^0(F)$ whenever Θ is regular.

Example 8.4.6: The Weil Representation of a Level Zero Character

Let E/F be the unramified quadratic extension, let ψ have level one, and let Θ be a regular character of $E^\times$ of level zero, so that Θ is trivial on U_E^1 and its restriction to U_E is inflated from a character $\overline{\Theta}$ of $k_E^\times$ with $\overline{\Theta}^q \neq \overline{\Theta}$. Write $\pi = \pi_{E/F}(\Theta)$.

The additive character on E. Since E/F is unramified it is tamely ramified, so $\mathrm{Tr}_{E/F}(\mathfrak{o}_E) = \mathfrak{o}$ by the computation recorded after definition 8.3.4. Hence ψ_E is trivial on $\mathfrak{p}_E = \mathfrak{p}\mathfrak{o}_E$, because $\mathrm{Tr}_{E/F}(\mathfrak{p}\mathfrak{o}_E) = \mathfrak{p}$, and non-trivial on $\mathfrak{o}_E$, that is, ψ_E has level one over E.

The exponent. Theorem 5.1.18, applied over E to the ramified character Θ of level 0 against ψ_E of level one, gives

$$\varepsilon(\Theta, s, \psi_E) = q_E^{-1/2}\,\tau(\Theta, \psi_E) = q^{-1}\tau(\Theta, \psi_E)$$

with τ the Gauss sum of definition 5.1.19 formed over E, since $q_E = q^2$. The right-hand side does not involve s, so $n(\Theta, \psi_E) = 0$, and theorem 8.4.5(1) gives

$$n(\pi, \psi) = f_{E/F} \cdot 0 = 0$$

The Weil representation of a level zero character is therefore unramified in the sense measured by the epsilon exponent, matching the vanishing recorded after definition 5.3.4 for cuspidal representations of level zero.

The Langlands constant. The decomposition $R_{E/F} = 1_F \oplus \kappa$ established in Step 7 of the proof of theorem 8.4.5, together with additivity, axiom (2) of theorem 6.3.2, gives $\varepsilon(R_{E/F}, s, \psi) = \varepsilon(1_F, s, \psi)\varepsilon(\kappa, s, \psi)$. Here κ is unramified with $\kappa(\varpi) = -1$ (example 8.3.2), and 1_E is unramified over E with $\varpi_E = \varpi$, so proposition 5.1.17 evaluates all three factors:

$$\varepsilon(1_F, s, \psi) = q^{s-1/2}, \qquad \varepsilon(\kappa, s, \psi) = -q^{s-1/2}, \qquad \varepsilon(1_E, s, \psi_E) = q_E^{s-1/2} = q^{2s-1}$$

Therefore

$$\lambda_{E/F}(\psi) = \frac{\varepsilon(R_{E/F}, s, \psi)}{\varepsilon(1_E, s, \psi_E)} = \frac{-q^{2s-1}}{q^{2s-1}} = -1$$

consistent with $\lambda_{E/F}(\psi)^2 = \kappa(-1) = 1$, which is (8.22) from the proof of proposition 8.3.5.

The local constant. Combining the last two computations with theorem 8.4.5(1),

$$\varepsilon(\pi, s, \psi) = \varepsilon\big(\pi, \tfrac{1}{2}, \psi\big) = \lambda_{E/F}(\psi)\,\varepsilon\big(\Theta, \tfrac{1}{2}, \psi_E\big) = -\frac{\tau(\Theta, \psi_E)}{q}$$

Example 5.3.7 asserts the same shape $\lambda_{E/F}(\psi)\tau(\xi, \psi_E)/q$ for an arbitrary cuspidal representation of level zero, with ξ the character of $E^\times$ attached to it by proposition 4.2.3, and cites it to an external source for want of a Kirillov model. The computation above is an independent derivation of that shape, but it is not yet a proof of the statement there. Matching the two would need $\pi_{E/F}(\Theta)$ to be of level zero in the sense of definition 3.3.12, whereas what has been shown is $n(\pi, \psi) = 0$, which is unramifiedness as measured by the epsilon exponent; and it would need the character attached to $\pi_{E/F}(\Theta)$ by proposition 4.2.3 to be Θ itself. That second point is a matching statement of the same kind as theorem 7.3.2, and nothing so far establishes it.

The central character. By proposition 8.4.4(3), $\omega_\pi = \kappa \cdot \Theta|_{F^\times}$. On U_F this is $\Theta|_{U_F}$, the inflation of $\overline{\Theta}|_{k^\times}$, since κ is trivial on U_F, and $\omega_\pi(\varpi) = -\Theta(\varpi)$, the sign being the value of κ at a prime element.

Exercise 8.4.1

1. Let E/F be the unramified quadratic extension and let Θ range over the characters of $E^\times$ that are trivial on U_E^1 and on $\varpi^{\mathbb{Z}}$. Show that Θ is E/F-regular if and only if the induced character $\overline{\Theta}$ of $k_E^\times$ does not factor through the norm $N_{k_E/k}$, and determine how many of the $q^2 - 1$ characters of $k_E^\times$ are regular.

2. Let Θ and Θ' be characters of $E^\times$. Show that $\Theta'|_{\mathcal{K}_{E/F}} = \Theta|_{\mathcal{K}_{E/F}}$ if and only if $\Theta' = \chi_E\Theta$ for some character χ of $F^\times$, where $\chi_E = \chi \circ N_{E/F}$. (Use the topological isomorphism $E^\times/\mathcal{K}_{E/F} \cong F_\kappa^\times$, and the fact that a character of a subgroup of finite index of an abelian group extends to the whole group.) Combining with proposition 8.4.4(2), deduce that the isomorphism class of $\pi(\Theta, \psi)$ up to twisting by characters of $F^\times$ depends only on the restriction $\Theta|_{\mathcal{K}_{E/F}}$.

3. Let Θ be any character of $E^\times$ and put $\rho = \mathrm{Ind}_{E/F}\Theta$. Using theorem 1.2.28 and the description of $\rho|_{\mathcal{W}_E}$ as $\Theta \oplus \Theta^\sigma$, compute $\dim_{\mathbb{C}} \mathrm{End}_{\mathcal{W}_F}(\rho)$ in the two cases $\Theta^\sigma \neq \Theta$ and $\Theta^\sigma = \Theta$.

4. Let Θ be E/F-regular. Show that the contragredient of $\pi_{E/F}(\Theta)$ is $\pi_{E/F}(\Theta^{-1})$. (Check first that Θ^{-1} is regular and that, writing $\rho = \mathrm{Ind}_{E/F}\Theta$, the contragredient satisfies $\check{\rho} \cong \mathrm{Ind}_{E/F}\Theta^{-1}$, and then compare the two functional equations corollary 5.2.9 and proposition 6.3.3(3) using theorem 8.4.5, and finish with theorem 5.5.2.)

5. In the situation of example 8.4.6, show that $|\varepsilon(\pi, \frac{1}{2}, \psi)| = 1$. (The values $\Theta(u)$ for $u \in U_E$ are roots of unity, so complex conjugation of the Gauss sum can be computed by the substitution $x \mapsto -x$. Then use the identity relating $\tau(\Theta, \psi_E)$ and $\tau(\Theta^{-1}, \psi_E)$ that follows from corollary 5.1.14 over E.)

6. Let $u \in N_{E/F}(E^\times)$ and choose $x \in E^\times$ with $N_{E/F}(x) = u$. Prove directly, without using theorem 8.4.5, that $\pi(\Theta, u\psi) \cong \pi(\Theta, \psi)$. (The content is the identity

$$\eta_{u\psi}^{\mathcal{G}}(\mathfrak{g}) = \eta_\psi^{\mathcal{G}}\big(a(x)\big)\,\eta_\psi^{\mathcal{G}}(\mathfrak{g})\,\eta_\psi^{\mathcal{G}}\big(a(x)\big)^{-1}, \qquad \mathfrak{g} \in \mathcal{G}_{E/F}$$

Check it on the four generators of (8.19). Three are immediate; at w use $a(x)\,w\,a(x)^{-1} = z(u^{-1})a(u)w$ together with the rescaling of the self-dual measure for $(u\psi)_E$ by $\|u\|_E^{1/2}$, and note that the identity closes only because $\lambda_{E/F}(u\psi) = \kappa(u)\lambda_{E/F}(\psi) = \lambda_{E/F}(\psi)$, u being a norm, established in Step 5 of the proof of proposition 8.3.5.)

8.5 A Partial Correspondence

The construction of section 8.4 starts from a pair $(E/F, \Theta)$ and returns a cuspidal representation of G. We now read it as a map defined on Galois parameters, and determine which parameters lie in its domain.

Three pieces of notation are used throughout. $\mathcal{G}_2^0(F)$ is the set of equivalence classes of irreducible two-dimensional smooth representations of $\mathcal{W}_F$, sitting inside $\mathcal{G}_2(F)$ (definition 7.1.1) as the Deligne representations with irreducible underlying representation and vanishing nilpotent. $\mathcal{A}_2^0(F)$ is the set of classes of irreducible cuspidal representations of G. And $\mathrm{Ind}_{E/F}\Theta$ is the smooth representation of $\mathcal{W}_F$ induced from the character $\Theta \circ \mathbf{a}_E$ of $\mathcal{W}_E$, the notation of section 7.3.

A parameter $\rho \in \mathcal{G}_2^0(F)$ is available to definition 8.4.3 only when it is presented as a representation induced from a quadratic extension. Whether such a presentation exists is a property of ρ alone, and that property defines the domain of the map built below.

Definition 8.5.1: Imprimitive and Primitive Representations

Let $\rho \in \mathcal{G}_2^0(F)$. Then ρ is *imprimitive* if there are a separable quadratic extension E/F and a smooth character Θ of $E^\times$ with

$$\rho \cong \mathrm{Ind}_{E/F}\Theta$$

and *primitive* otherwise. Write $\mathcal{G}_2^{\mathrm{im}}(F)$ for the set of imprimitive classes in $\mathcal{G}_2^0(F)$, and $\mathcal{G}_2^{\mathrm{nr}}(F)$ for the set of those classes admitting such a presentation with E/F unramified.

The unramified quadratic extension of F is unique, so the extension E occurring in such a presentation is determined; the character is determined only up to its $\mathrm{Gal}(E/F)$-orbit.

Which characters can occur in definition 8.5.1 is settled by a computation inside the induced model, and the answer is the class of E/F-regular characters of definition 8.4.1. Fix a separable quadratic extension E/F. Being separable of degree two it is Galois, so $\mathrm{Gal}(E/F) = \{1, \sigma\}$ and $\mathcal{W}_E$ is a normal subgroup of index two in $\mathcal{W}_F$. Choose $\tau \in \mathcal{W}_F \setminus \mathcal{W}_E$, write Θ also for the character $\Theta \circ \mathbf{a}_E$ of $\mathcal{W}_E$, and realize $\rho = \mathrm{Ind}_{E/F}\Theta$ on

$$V = \{f\colon \mathcal{W}_F \to \mathbb{C} : f(hg) = \Theta(h)f(g) \text{ for all } h \in \mathcal{W}_E,\ g \in \mathcal{W}_F\}$$

with $\mathcal{W}_F$ acting by right translation, $(\rho(x)f)(g) = f(gx)$. As $\mathcal{W}_F = \mathcal{W}_E \sqcup \mathcal{W}_E\tau$, an $f \in V$ is determined by the two values $f(1)$ and $f(\tau)$, so $\dim_{\mathbb{C}} V = 2$. Let f_1, f_2 be the elements of V with $f_1(1) = 1$, $f_1(\tau) = 0$ and $f_2(1) = 0$, $f_2(\tau) = 1$. Write Θ^σ for the character $h \mapsto \Theta(\tau h \tau^{-1})$ of $\mathcal{W}_E$. Transported through $\mathbf{a}_E$ this is the character $\Theta \circ \sigma$ of $E^\times$, by the functoriality of theorem 6.2.1(5) applied to the F-automorphism σ of E, together with the compatibility identifying the automorphism of $\mathcal{W}_E^{\mathrm{ab}}$ that σ induces with conjugation by a lift $\tau \in \mathcal{W}_F \setminus \mathcal{W}_E$, for which see [14]. For $h \in \mathcal{W}_E$ the two computations

$$(\rho(h)f_1)(1) = f_1(h) = \Theta(h), \qquad (\rho(h)f_2)(\tau) = f_2(\tau h) = f_2\big((\tau h \tau^{-1})\tau\big) = \Theta^\sigma(h)$$

together with the normality of $\mathcal{W}_E$, which keeps $\rho(h)f_i$ supported on the coset carrying f_i, give

$$\rho(h)f_1 = \Theta(h)f_1, \qquad \rho(h)f_2 = \Theta^\sigma(h)f_2 \qquad (h \in \mathcal{W}_E) \tag{8.29}$$

The element τ exchanges the two lines: $(\rho(\tau)f_1)(g) = f_1(g\tau)$ vanishes unless $g \in \mathcal{W}_E\tau^{-1} = \mathcal{W}_E\tau$, the two cosets being equal because $\tau^2 \in \mathcal{W}_E$, and $(\rho(\tau)f_2)(g) = f_2(g\tau)$ vanishes unless $g \in \mathcal{W}_E$. So $\rho(\tau)f_1 \in \mathbb{C}f_2$ and $\rho(\tau)f_2 \in \mathbb{C}f_1$.

Two consequences follow, and they are the criterion the rest of this section uses. Suppose first that Θ is trivial on $\mathcal{K}_{E/F} = \ker N_{E/F}$ (definition 8.3.1), so that $\Theta^\sigma = \Theta$ by lemma 8.4.2. Then (8.29) says that $\mathcal{W}_E$ acts on V through the scalar character Θ, so every line in V is $\mathcal{W}_E$-stable. The linear map $\rho(\tau)$ of the two-dimensional complex space V has an eigenvector $v \neq 0$, and $\mathbb{C}v$ is then stable under $\mathcal{W}_E$ and under τ, hence under the group these generate, which is $\mathcal{W}_F$. So ρ is reducible. Suppose instead that $\Theta^\sigma \neq \Theta$. The two characters occurring in (8.29) are then distinct, so $\mathbb{C}f_1$ and $\mathbb{C}f_2$ are the only $\mathcal{W}_E$-stable lines in V, and neither is stable under $\rho(\tau)$ by the exchange just computed. A non-zero $\mathcal{W}_F$-stable subspace of V is $\mathcal{W}_E$-stable, hence is one of $\mathbb{C}f_1$, $\mathbb{C}f_2$, V, and the first two are excluded: ρ is irreducible.

An imprimitive class is therefore always induced from a character that is E/F-regular in the sense of definition 8.4.1.

Lemma 8.5.2: Irreducibility of an Induced Character

Let E/F be a separable quadratic extension and Θ a character of $E^\times$. Then $\mathrm{Ind}_{E/F}\Theta$ is irreducible if and only if Θ is E/F-regular, that is if and only if $\Theta^\sigma \neq \Theta$, equivalently $\Theta|_{\mathcal{K}_{E/F}} \neq 1$.

***Proof*:**
Irreducibility of ρ forces $\Theta^\sigma \neq \Theta$ by the first consequence computed above, and a regular Θ has $\Theta^\sigma \neq \Theta$, so the second consequence applies and gives the converse. That $\Theta^\sigma \neq \Theta$ is regularity is definition 8.4.1, and lemma 8.4.2 converts it into $\Theta|_{\mathcal{K}_{E/F}} \neq 1$.

Example 8.5.3: Imprimitive Classes over $\mathbb{Q}_2$

Let $F = \mathbb{Q}_2$, so $q = 2$ and F is dyadic. Let $E = \mathbb{Q}_2(\zeta)$ where ζ is a primitive cube root of unity, the unramified quadratic extension of $\mathbb{Q}_2$: the polynomial $X^2 + X + 1$ is irreducible modulo 2, so E/F is unramified with residue field k_E of four elements, and 2 is a prime element of E. Hensel's lemma lifts $k_E^\times$, cyclic of order three, to the group $\mu_3 = \{1, \zeta, \zeta^2\} \subseteq U_E$, and

$$E^\times = 2^{\mathbb{Z}} \times \mu_3 \times U_E^1$$

The non-trivial element σ of $\mathrm{Gal}(E/F)$ carries ζ to a primitive cube root of unity other than ζ, since $\zeta \notin F$, so $\sigma(\zeta) = \zeta^2$.

Let $\theta\colon \mu_3 \to \mathbb{C}^\times$ be an injective character and let Θ be the character of $E^\times$ with

$$\Theta(2) = 1, \qquad \Theta|_{\mu_3} = \theta, \qquad \Theta|_{U_E^1} = 1$$

It is smooth, being trivial on the open subgroup U_E^1. Its restriction to μ_3 detects regularity at once: for $\zeta' \in \mu_3$,

$$N_{E/F}(\zeta') = \zeta'\sigma(\zeta') = \zeta'(\zeta')^2 = 1$$

so $\mu_3 \subseteq \mathcal{K}_{E/F}$, while $\Theta|_{\mu_3} = \theta \neq 1$. Thus $\Theta|_{\mathcal{K}_{E/F}} \neq 1$ and Θ is E/F-regular. Directly, $\Theta^\sigma(\zeta) = \Theta(\zeta^2) = \theta(\zeta)^2 \neq \theta(\zeta)$, because $\theta(\zeta)$ is a primitive cube root of unity, so $\Theta^\sigma \neq \Theta$ and the criterion above makes $\rho = \mathrm{Ind}_{E/F}\Theta$ irreducible. Hence $\rho \in \mathcal{G}_2^{\mathrm{nr}}(\mathbb{Q}_2) \subseteq \mathcal{G}_2^{\mathrm{im}}(\mathbb{Q}_2)$.

The pair $(E/F, \Theta)$ is also admissible in the sense of definition 4.1.4: the extension is unramified, hence tamely ramified, which is condition (1). Condition (2) holds by hypothesis, the pair consisting of a quadratic extension and a smooth character of $E^\times$. Next, Θ does not factor through the norm, since $\Theta = \varphi \circ N_{E/F}$ would give $\Theta|_{\mu_3} = \varphi(1) = 1$, which is condition (3). And condition (4) is met because E/F is unramified. So this single pair satisfies the hypotheses of both constructions, the tame parametrization of theorem 4.3.1 and the Weil representation of definition 8.4.3, and the two cuspidal representations they produce have different central characters, as the computation after theorem 8.5.4 shows.

For $p = 2$ the example lies in the part of the imprimitive locus already handled by chapter 4. The rest of that locus is not handled there: a quadratic extension of a dyadic field is tamely ramified only when it is unramified, so a ramified quadratic E/F produces imprimitive classes for which theorem 4.3.1 gives nothing. The theorem below treats all of them uniformly, and does so for every residual characteristic, because the Weil representation was built without a tameness hypothesis.

The two ingredients are already in place. Theorem 8.4.5 says that $\pi_{E/F}(\Theta)$ has the twisted local constants of $\mathrm{Ind}_{E/F}\Theta$, which is existence. Theorem 5.5.2 says that twisted local constants determine an irreducible smooth representation, which is uniqueness. What the theorem adds is the check that the two combine into a well-defined map: a class ρ admits many presentations $\mathrm{Ind}_{E/F}\Theta$, and the value must not depend on which one is chosen.

Theorem 8.5.4: The Imprimitive Correspondence

Let $\psi \in \widehat{F}$ be non-trivial. There is exactly one map

$$\boldsymbol{\pi}^{\mathrm{im}} \colon \mathcal{G}_2^{\mathrm{im}}(F) \longrightarrow \mathcal{A}_2^0(F)$$

such that, for every $\rho \in \mathcal{G}_2^{\mathrm{im}}(F)$, the central character of $\boldsymbol{\pi}^{\mathrm{im}}(\rho)$ is $\det \rho$ and

$$\varepsilon(\chi \otimes \rho, s, \psi) = \varepsilon\big(\chi\, \boldsymbol{\pi}^{\mathrm{im}}(\rho), s, \psi\big) \tag{8.30}$$

for every character χ of $F^\times$. It is given by

$$\boldsymbol{\pi}^{\mathrm{im}}\big(\mathrm{Ind}_{E/F}\Theta\big) = \pi_{E/F}(\Theta)$$

and it has the following properties, for every $\rho \in \mathcal{G}_2^{\mathrm{im}}(F)$ and every character χ of $F^\times$.

1. Identity (8.30) holds for every non-trivial $\psi \in \widehat{F}$, so $\boldsymbol{\pi}^{\mathrm{im}}$ does not depend on the choice of ψ.
2. $L\big(\chi\, \boldsymbol{\pi}^{\mathrm{im}}(\rho), s\big) = 1$.
3. $\chi \otimes \rho$ lies in $\mathcal{G}_2^{\mathrm{im}}(F)$ and $\boldsymbol{\pi}^{\mathrm{im}}(\chi \otimes \rho) = \chi\, \boldsymbol{\pi}^{\mathrm{im}}(\rho)$.

***Proof*:**
Step 1: each presentation of ρ gives a candidate. Let $\rho \in \mathcal{G}_2^{\mathrm{im}}(F)$ and fix a separable quadratic E/F and a character Θ of $E^\times$ with $\rho \cong \mathrm{Ind}_{E/F}\Theta$. Since ρ is irreducible, the criterion established after definition 8.5.1 gives $\Theta^\sigma \neq \Theta$ and hence $\Theta|_{\mathcal{K}_{E/F}} \neq 1$, so Θ is E/F-regular (definition 8.4.1) and definition 8.4.3 attaches to $(E/F, \Theta)$ the Weil representation $\pi_{E/F}(\Theta)$. By proposition 8.4.4 it is irreducible and cuspidal, so it lies in $\mathcal{A}_2^0(F)$, and by theorem 8.4.5

$$\varepsilon(\chi \otimes \rho, s, \psi') = \varepsilon\big(\chi\, \pi_{E/F}(\Theta), s, \psi'\big) \tag{8.31}$$

for every character χ of $F^\times$ and every non-trivial $\psi' \in \widehat{F}$.

Step 2: matching local constants at two additive characters forces the central character. Let $\pi \in \mathcal{A}_2^0(F)$ satisfy $\varepsilon(\rho, s, \psi') = \varepsilon(\pi, s, \psi')$ for the two characters $\psi' = \psi$ and $\psi' = a\psi$, where $a \in F^\times$. On the group side lemma 5.3.3 gives

$$\varepsilon(\pi, s, a\psi) = \omega_\pi(a)\, \|a\|^{2(s-1/2)}\, \varepsilon(\pi, s, \psi)$$

and on the Galois side proposition 6.3.3(2), with $\dim \rho = 2$, gives

$$\varepsilon(\rho, s, a\psi) = \det \rho(a)\, \|a\|^{2(s-1/2)}\, \varepsilon(\rho, s, \psi)$$

The left-hand sides agree by hypothesis, as do the factors $\varepsilon(\pi, s, \psi)$ and $\varepsilon(\rho, s, \psi)$. That common factor is non-zero: by proposition 6.3.3(3), $\varepsilon(\rho, s, \psi)\varepsilon(\check{\rho}, 1-s, \psi) = \det \rho(-1)$, and the right-hand side is a root of unity. Dividing the two displays by it gives $\omega_\pi(a) = \det \rho(a)$, and $a \in F^\times$ was arbitrary, so $\omega_\pi = \det \rho$.

Step 3: the assignment is independent of the presentation. Suppose $\rho \cong \mathrm{Ind}_{E/F}\Theta$ and $\rho \cong \mathrm{Ind}_{E'/F}\Theta'$, and put $\pi = \pi_{E/F}(\Theta)$ and $\pi' = \pi_{E'/F}(\Theta')$. Both lie in $\mathcal{A}_2^0(F)$ by Step 1, and (8.31) applied to each presentation gives

$$\varepsilon(\chi\pi, s, \psi') = \varepsilon(\chi \otimes \rho, s, \psi') = \varepsilon(\chi\pi', s, \psi')$$

for all χ and all non-trivial ψ'. Taking χ trivial and applying Step 2 to each of π and π' gives $\omega_\pi = \det \rho = \omega_{\pi'}$. Both representations are cuspidal, so lemma 5.5.1 gives $L(\chi\pi, s) = 1 = L(\chi\pi', s)$ for every χ. It remains to place the additive character at level one. For $b \in F^\times$ and $m \in \mathbb{Z}$, the character $b\psi$ is trivial on $\mathfrak{p}^m$ exactly when ψ is trivial on $b\mathfrak{p}^m = \mathfrak{p}^{m+v_F(b)}$, so the level of $b\psi$ is the level of ψ minus $v_F(b)$. Choosing b with $v_F(b)$ equal to the level of ψ minus one produces a character $\psi_1 = b\psi$ of level one, and the displayed identities hold at ψ_1. The hypotheses of theorem 5.5.2 are met by π and π' relative to ψ_1, so $\pi \cong \pi'$.

Setting $\boldsymbol{\pi}^{\mathrm{im}}(\rho) = \pi_{E/F}(\Theta)$ for any one presentation is therefore unambiguous. Step 1 gives (8.30) for the fixed ψ and property (1) for every other non-trivial additive character, Step 2 gives the central character, and lemma 5.5.1 applied to the cuspidal representation $\boldsymbol{\pi}^{\mathrm{im}}(\rho)$ gives property (2).

Step 4: uniqueness. Let $\boldsymbol{\pi}^\dagger$ be a second map $\mathcal{G}_2^{\mathrm{im}}(F) \to \mathcal{A}_2^0(F)$ whose values have central character $\det \rho$ and satisfy (8.30) for the given ψ. Fix ρ and put $\pi = \boldsymbol{\pi}^{\mathrm{im}}(\rho)$, $\pi^\dagger = \boldsymbol{\pi}^\dagger(\rho)$. Both are cuspidal with central character $\det \rho$, so lemma 5.5.1 makes all their twisted L-functions equal

to 1, and the two instances of (8.30) give $\varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi\pi^\dagger, s, \psi)$ for every χ. The scalar matrix zI acts on the twist $\chi\pi$ by $\chi(\det(zI))\omega_\pi(z) = \chi(z)^2\omega_\pi(z)$, so $\omega_{\chi\pi} = \chi^2\omega_\pi = \chi^2 \det\rho = \omega_{\chi\pi^\dagger}$. Applying lemma 5.3.3 to $\chi\pi$ and to $\chi\pi^\dagger$ therefore multiplies the two sides of the last identity by the same factor $\omega_{\chi\pi}(b)\|b\|^{2(s-1/2)}$, so the identity persists at $b\psi$ for every $b \in F^\times$, and in particular at a character of level one, built as in Step 3. Now theorem 5.5.2 gives $\pi \cong \pi^\dagger$, so $\boldsymbol{\pi}^\dagger = \boldsymbol{\pi}^{\mathrm{im}}$.

Step 5: compatibility with twisting. Let χ be a character of $F^\times$ and write $\tilde{\chi} = \chi \circ \mathbf{a}_F$ for the corresponding character of $\mathcal{W}_F$. With V as in the model preceding example 8.5.3, the map $(Tf)(g) = \tilde{\chi}(g)f(g)$ satisfies

$$(Tf)(hg) = \tilde{\chi}(h)\tilde{\chi}(g)\Theta(h)f(g) = \big(\Theta \cdot \tilde{\chi}|_{\mathcal{W}_E}\big)(h)\,(Tf)(g) \qquad (h \in \mathcal{W}_E)$$

and

$$\big(T((\chi \otimes \rho)(x)f)\big)(g) = \tilde{\chi}(x)\tilde{\chi}(g)f(gx) = \tilde{\chi}(gx)f(gx) = (Tf)(gx)$$

so T is an isomorphism from $\chi \otimes \rho$ onto the representation induced from $\Theta \cdot \tilde{\chi}|_{\mathcal{W}_E}$. By theorem 6.2.1(4) the restriction $\tilde{\chi}|_{\mathcal{W}_E}$ is $\chi \circ N_{E/F} \circ \mathbf{a}_E$, so

$$\chi \otimes \mathrm{Ind}_{E/F}\Theta \cong \mathrm{Ind}_{E/F}\big(\Theta \cdot (\chi \circ N_{E/F})\big)$$

Twisting by a character does not change which subspaces are stable, so $\chi \otimes \rho$ is irreducible, and the display puts it in $\mathcal{G}_2^{\mathrm{im}}(F)$. Now compare $\chi\,\boldsymbol{\pi}^{\mathrm{im}}(\rho)$ with $\boldsymbol{\pi}^{\mathrm{im}}(\chi \otimes \rho)$. Writing $\pi = \boldsymbol{\pi}^{\mathrm{im}}(\rho)$, the twist $\chi\pi$ is cuspidal, since $L(\mu(\chi\pi), s) = L((\mu\chi)\pi, s) = 1$ for every character μ of $F^\times$ by property (2), and lemma 5.5.1 converts that into cuspidality. Its central character is $\chi^2 \det\rho$, computed above, which equals $\det(\chi \otimes \rho)$ because $\chi \otimes \rho$ has matrices $\tilde{\chi}(g)\rho(g)$ of size two. Finally, for every character μ of $F^\times$ and every non-trivial ψ',

$$\varepsilon\big(\mu(\chi\pi), s, \psi'\big) = \varepsilon\big((\mu\chi)\pi, s, \psi'\big) = \varepsilon\big((\mu\chi) \otimes \rho, s, \psi'\big) = \varepsilon\big(\mu \otimes (\chi \otimes \rho), s, \psi'\big)$$

the middle equality by property (1) for ρ. So $\chi\pi$ and $\boldsymbol{\pi}^{\mathrm{im}}(\chi \otimes \rho)$ are cuspidal representations with the same central character, with all their twisted L-functions equal to 1 by lemma 5.5.1, and with the same twisted local constants. Applying theorem 5.5.2 at a character of level one, built as in Step 3, gives $\chi\pi \cong \boldsymbol{\pi}^{\mathrm{im}}(\chi \otimes \rho)$, completing the proof.

The theorem is a correspondence in the sense theorem 7.1.2 asks for, restricted to the imprimitive parameters and valid for every residual characteristic. Its two characterizing conditions are not independent of each other: imposing (8.30) for every non-trivial ψ already forces the central character, by Step 2 of the proof, so the map is equally well described as the unique one preserving twisted local constants for all additive characters at once. Fixing a single ψ and adding the central-character condition is the sharper statement, and it is the one theorem 5.5.2 applies to without further work.

What the theorem does not assert is that $\boldsymbol{\pi}^{\mathrm{im}}$ is injective or that its image is all of $\mathcal{A}_2^0(F)$. Proving injectivity from the ingredients assembled here would need a converse theorem on the Galois side, matching representations of $\mathcal{W}_F$ by their twisted local constants, and surjectivity would need every cuspidal representation to arise from a pair $(E/F, \Theta)$. Neither ingredient is available here, and the two assertions are not on the same footing: surjectivity fails for $p = 2$, since by corollary 8.5.5(2) the primitive classes lie outside the domain and the cuspidal representations matching them lie outside the image, whereas injectivity holds for every residual characteristic once the full correspondence is granted. Theorem 7.1.2 asserts that $\boldsymbol{\pi}_F$ is a bijection, and the comparison run in the proof of

corollary 8.5.5(1) below identifies $\boldsymbol{\pi}^{\mathrm{im}}$ with the restriction of $\boldsymbol{\pi}_F$ to $\mathcal{G}_2^{\mathrm{im}}(F)$, by an ε-comparison using no parity hypothesis once ρ is imprimitive and $\boldsymbol{\pi}_F(\rho)$ is known to be cuspidal. For $p = 2$ that appeals to a correspondence this book does not construct, so injectivity there rests on [2, §50-§52] along with the correspondence itself.

The two constructions can now be compared on the dyadic example. Keep the notation of example 8.5.3, so $F = \mathbb{Q}_2$, $E = \mathbb{Q}_2(\zeta)$ is unramified and Θ has $\Theta(2) = 1$, $\Theta|_{\mu_3} = \theta$ and $\Theta|_{U_E^1} = 1$. Recall from section 7.3 the determinant of an induced representation,

$$\det\left(\mathrm{Ind}_{E/F}\Theta\right) = \kappa_{E/F} \cdot \left(\Theta|_{F^\times}\right)$$

with $\kappa_{E/F}$ the quadratic character of $F^\times$ cutting out the norms from E (theorem 6.2.2(2)), the determinant itself being proposition 6.2.7. Every unit of $\mathbb{Z}_2$ has trivial image in $k_E^\times$, since that image lies in the residue field of F, which has two elements, so $\mathbb{Z}_2^\times \subseteq U_E^1$ and $\Theta|_{\mathbb{Z}_2^\times} = 1$. Together with $\Theta(2) = 1$ this gives $\Theta|_{F^\times} = 1$, hence

$$\omega_{\pi_{E/F}(\Theta)} = \det\rho = \kappa_{E/F}$$

by theorem 8.5.4. The pair $(E/F, \Theta)$ is admissible, so theorem 4.3.1 attaches to it a second cuspidal representation π_Θ, of level zero, with central character $\omega_{\pi_\Theta} = \Theta|_{F^\times} = 1$ by property (2) of that theorem. The two central characters differ by $\kappa_{E/F}$, which has order two, so

$$\pi_{E/F}(\Theta) \ncong \pi_\Theta$$

The discrepancy is exactly the one definition 7.3.1 corrects, and it appears here for $p = 2$, where theorem 7.3.2 does not apply.

That observation generalizes, and it is what identifies the new map with the old one when p is odd.

Corollary 8.5.5: Comparison with the Tame Correspondence

1. Suppose $p \neq 2$. Then $\mathcal{G}_2^{\mathrm{im}}(F) = \mathcal{G}_2^0(F)$, and $\boldsymbol{\pi}^{\mathrm{im}}$ is the bijection $\mathcal{G}_2^0(F) \to \mathcal{A}_2^0(F)$ of theorem 7.3.2. In particular, for every admissible pair $(E/F, \xi)$,

   $$\pi_{E/F}(\xi) \cong \pi_{\xi\Delta_\xi}$$

   with Δ_ξ the rectifier of definition 7.3.1.

2. Suppose $p = 2$. Then $\mathcal{G}_2^{\mathrm{nr}}(F) \subseteq \mathcal{G}_2^{\mathrm{im}}(F) \subseteq \mathcal{G}_2^0(F)$, and both inclusions are strict. Every class in $\mathcal{G}_2^{\mathrm{nr}}(F)$ is $\mathrm{Ind}_{E/F}\xi$ for an admissible pair $(E/F, \xi)$ (definition 4.1.4), and every admissible pair has E/F unramified. The classes in $\mathcal{G}_2^{\mathrm{im}}(F) \setminus \mathcal{G}_2^{\mathrm{nr}}(F)$ are induced from ramified quadratic extensions.

***Proof*:**

1. The inclusion $\mathcal{G}_2^{\mathrm{im}}(F) \subseteq \mathcal{G}_2^0(F)$ holds by definition 8.5.1. For the reverse inclusion, theorem 7.3.2 presents every element of $\mathcal{G}_2^0(F)$ as $\rho_\xi = \mathrm{Ind}_{E/F}\xi$ for an admissible pair, which is exactly the assertion that every irreducible two-dimensional smooth representation of $\mathcal{W}_F$ is induced from a character of a quadratic extension when p is odd. That assertion is an input here, on the same terms as in section 7.3. Its proof decomposes $\rho|_{\mathcal{P}_F}$, the restriction to the wild inertia subgroup, into characters and constructs the quadratic extension from

their stabilizers, and it is taken on faith from [2, §34]. Granting it, $\mathcal{G}_2^0(F) = \mathcal{G}_2^{\mathrm{im}}(F)$.

Now fix $\rho \in \mathcal{G}_2^0(F)$ and compare $\pi_1 = \boldsymbol{\pi}_F(\rho)$ with $\pi_2 = \boldsymbol{\pi}^{\mathrm{im}}(\rho)$. By Step 2 of theorem 7.1.2 the correspondence $\boldsymbol{\pi}_F$ carries $\mathcal{G}_2^0(F)$ into $\mathcal{A}_2^0(F)$, so both representations are cuspidal and lemma 5.5.1 makes all their twisted L-functions equal to 1. Their central characters agree, both being $\det\rho$: for π_1 this is property (3) of theorem 7.1.2 and for π_2 it is theorem 8.5.4. Their twisted local constants agree, since property (2) of theorem 7.1.2 gives $\varepsilon(\chi\pi_1, s, \psi) = \varepsilon(\chi \otimes \rho, s, \psi)$ for every character χ and every non-trivial ψ, which is (8.30) for π_2. Applying theorem 5.5.2 at a ψ of level one gives $\pi_1 \cong \pi_2$. The two maps therefore coincide, and theorem 7.3.2 both records that this map is a bijection onto $\mathcal{A}_2^0(F)$ and evaluates it on ρ_ξ as $\pi_{\xi\Delta_\xi}$, which is the displayed isomorphism.

2. The two inclusions hold by definition 8.5.1. That both are strict is an input here, on the same terms as the classification quoted in (1): for $p = 2$ there exist primitive classes, so $\mathcal{G}_2^{\mathrm{im}}(F) \neq \mathcal{G}_2^0(F)$, and there exist imprimitive classes admitting no presentation with E/F unramified, so $\mathcal{G}_2^{\mathrm{nr}}(F) \neq \mathcal{G}_2^{\mathrm{im}}(F)$. Both non-emptiness statements come from sources this book does not develop, and they are not the same source. That $\mathcal{G}_2^{\mathrm{nr}}(F) \neq \mathcal{G}_2^{\mathrm{im}}(F)$ is dyadic arithmetic, [2, §41]. That $\mathcal{G}_2^{\mathrm{im}}(F) \neq \mathcal{G}_2^0(F)$, the existence of primitive classes, is not proved there at all: [2, Ch. 10, introduction] declines it explicitly and recovers it only once the correspondence itself is available, in [2, §§46-52]. Let $\rho = \mathrm{Ind}_{E/F}\xi \in \mathcal{G}_2^{\mathrm{nr}}(F)$ with E/F unramified. Then $e(E|F) = 1$ is prime to p, so E/F is tamely ramified, giving condition (1) of definition 4.1.4. Condition (2) holds by hypothesis, the pair consisting of a quadratic extension and a smooth character of $E^\times$. If $\xi = \varphi \circ N_{E/F}$ for a character φ of $F^\times$ then ξ is trivial on $\mathcal{K}_{E/F}$, so $\xi^\sigma = \xi$ and ρ is reducible by the criterion after definition 8.5.1, contradicting $\rho \in \mathcal{G}_2^0(F)$, which gives condition (3). And condition (4) is satisfied because E/F is unramified. So $(E/F, \xi)$ is admissible. Conversely an admissible pair has E/F tamely ramified, and a quadratic extension with $e(E|F) = 2 = p$ is not tamely ramified, so $e(E|F) = 1$ and E/F is unramified. The last assertion is definition 8.5.1: a class outside $\mathcal{G}_2^{\mathrm{nr}}(F)$ admits no presentation with E/F unramified, while being imprimitive it admits one with E/F quadratic, necessarily ramified, as we sought to show.

For p odd the corollary is a second construction of the tame correspondence, running through the Weil representation instead of through admissible pairs and compact induction, and the two agree because the converse theorem forces them to agree. The identification $\pi_{E/F}(\xi) \cong \pi_{\xi\Delta_\xi}$ is the concrete content: the rectifier, which section 7.3 introduced to repair the central character of the naive matching, is exactly the difference between compact induction from the pair $(E/F, \xi)$ and the Weil representation attached to it.

For $p = 2$ the corollary measures how far the correspondence has now been carried. Everything induced from the unramified quadratic extension was already within reach of chapter 4, up to the rectification computed above; everything induced from a ramified quadratic extension is new.

8.5.6 Note: What Remains in the Dyadic Case Let $p = 2$. Theorem 8.5.4 determines a cuspidal representation $\boldsymbol{\pi}^{\mathrm{im}}(\rho)$ for every imprimitive parameter ρ, characterized by its central character and its twisted local constants. Three things are not settled by it.

1. *The domain is not described.* Corollary 8.5.5(2) locates $\mathcal{G}_2^{\mathrm{im}}(F)$ strictly between $\mathcal{G}_2^{\mathrm{nr}}(F)$ and $\mathcal{G}_2^{0}(F)$. Making the middle term explicit means classifying the ramified quadratic extensions of a dyadic field, deciding which characters of $E^\times$ are E/F-regular, and deciding when two pairs induce isomorphic representations. That is the arithmetic carried out in [2, §41], with the resulting characterization of the image in [2, §§44-45].
2. *The primitive classes are untouched.* A class in $\mathcal{G}_2^{0}(F) \setminus \mathcal{G}_2^{\mathrm{im}}(F)$ is induced from no character of any quadratic extension, so no construction beginning with a pair $(E/F, \Theta)$ reaches it, and definition 8.4.3 has nothing to say about it. Matching these requires a construction of a different shape, see [2, §§46-52].
3. *The group side is not identified.* Theorem 3.5.5 builds every cuspidal representation of G from a cuspidal type, and theorem 8.5.4 produces $\boldsymbol{\pi}^{\mathrm{im}}(\rho)$ without saying which cuspidal type it contains. For $p \neq 2$ that identification is theorem 7.3.2. For $p = 2$ it is available only on the cuspidal representations attached to pairs with E/F unramified, through theorem 4.3.1.

Exercise 8.5.1

1. Let E/F be a separable quadratic extension with $\mathrm{Gal}(E/F) = \{1, \sigma\}$, and let Θ be an unramified character of $E^\times$, that is, one trivial on U_E. Show that $v_E(\sigma(x)) = v_E(x)$ for every $x \in E^\times$, deduce that $\mathcal{K}_{E/F} \subseteq U_E$, and conclude that $\mathrm{Ind}_{E/F}\Theta$ is reducible. Hence no class in $\mathcal{G}_2^{\mathrm{im}}(F)$ is induced from an unramified character.
2. Let $\rho \in \mathcal{G}_2^{\mathrm{im}}(F)$, let $\pi = \boldsymbol{\pi}^{\mathrm{im}}(\rho)$ and let χ be a character of $F^\times$. Show that $\det(\chi \otimes \rho) = \chi^2 \det \rho$ and that $\omega_{\chi\pi} = \chi^2 \omega_\pi$, and check that the two computations make properties (3) of theorem 8.5.4 and the central-character condition consistent with one another.
3. Let $\rho \in \mathcal{G}_2^{\mathrm{im}}(F)$ and $\pi = \boldsymbol{\pi}^{\mathrm{im}}(\rho)$. By corollary 5.2.9 one may write $\varepsilon(\pi, s, \psi') = c\, q^{-ds}$ with $c \in \mathbb{C}^\times$ and $d \in \mathbb{Z}$, for each non-trivial $\psi' \in \widehat{F}$. Write $d = n(\pi, \psi')$, which for ψ' of level one is the epsilon exponent of definition 5.3.4. Show that $n(\pi, \psi') = n(\rho, \psi')$ for every non-trivial ψ', where $n(\rho, \psi')$ is the integer of proposition 6.3.3(1), and that $n(\pi, a\psi') = n(\pi, \psi') + 2v_F(a)$ for $a \in F^\times$. Conclude that the parity of $n(\pi, \psi')$ is the same for every non-trivial choice of ψ', so that it is an invariant of ρ.
4. Let ρ be a two-dimensional smooth representation of $\mathcal{W}_F$, with contragredient $\check{\rho}(g) = \rho(g^{-1})^{\mathsf{T}}$. Using $w = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, verify the matrix identity $A^{-\mathsf{T}} = (\det A)^{-1}\, wAw^{-1}$ for $A \in \mathrm{GL}_2(\mathbb{C})$ and deduce that $\check{\rho} \cong (\det \rho)^{-1} \otimes \rho$. Now let $\rho \in \mathcal{G}_2^{\mathrm{im}}(F)$ and show that $\check{\rho} \in \mathcal{G}_2^{\mathrm{im}}(F)$ with $\boldsymbol{\pi}^{\mathrm{im}}(\check{\rho}) \cong (\det \rho)^{-1} \boldsymbol{\pi}^{\mathrm{im}}(\rho)$.
5. Keep the model of $\mathrm{Ind}_{E/F}\Theta$ used after definition 8.5.1, with $\tau \in \mathcal{W}_F \setminus \mathcal{W}_E$ and $\Theta^\sigma(h) = \Theta(\tau h \tau^{-1})$. Show that $(Sf)(g) = f(\tau g)$ defines an isomorphism $\mathrm{Ind}_{E/F}\Theta \to \mathrm{Ind}_{E/F}\Theta^\sigma$ of representations of $\mathcal{W}_F$. Deduce that if Θ is E/F-regular then $\pi_{E/F}(\Theta) \cong \pi_{E/F}(\Theta^\sigma)$, so the

Weil representation depends only on the $\mathrm{Gal}(E/F)$-orbit of Θ.

9

Arithmetic of Dyadic Fields

The residual characteristic of F is 2 throughout this chapter. That hypothesis is what makes the chapter necessary and what makes it delicate: every quadratic extension E/F other than the unramified one is now wildly ramified, so the tame arithmetic of chapter 4 says nothing about it, and the invariants have to be recomputed from the ground up.

The target is a converse theorem on the Galois side. Theorem 8.5.4 produced a map $\boldsymbol{\pi}^{\mathrm{im}}$ from imprimitive parameters to cuspidal representations, in every residual characteristic, but section 8.5 had to leave it possibly non-injective: separating two parameters by their twisted local constants is a statement about $\mathcal{W}_F$, and nothing available at that point could prove it. Theorem 9.3.10 is that statement, and corollary 9.3.11 is the consequence. Reaching it takes one section of arithmetic first: section 9.1 computes the discriminant, trace and norm invariants of a wildly ramified quadratic extension and reads off the group $T(\rho)$ of characters fixing a parameter under twisting, and the converse theorem draws on that section alone. Section 9.2, between them, is the structure theory of the parameters that are not induced at all. Later chapters need it, and the converse theorem does not.

One thing is deliberately not proved. That primitive parameters *exist* in residual characteristic 2 is not established here, so section 9.2 reads as conditional structure throughout. The reason is not that it is hard but that it is out of order: the cleanest argument runs through the correspondence itself, which is the business of a later chapter. Box 9.2.3 says exactly what is owed.

9.1 Imprimitive Representations

Throughout this chapter the residual characteristic of F is 2, so q is a power of 2 and a quadratic extension of F is either unramified or wildly ramified. The tame parametrization of chapter 4 reaches only the unramified ones, and the invariant that replaces tameness for the rest is the discriminant exponent $d(E|F)$. This section computes it from a minimal polynomial, works out the trace and

norm filtrations it governs, and applies the results to the parameters in $\mathcal{G}_2^0(F)$.

The invariant that organizes the parameters is the group $T(\rho)$ of characters χ of $F^\times$ with $\chi \otimes \rho \cong \rho$. Its order is 1, 2 or 4. It is trivial exactly when ρ is primitive in the sense of definition 8.5.1. When it is not trivial its non-trivial elements are the characters $\kappa_{E/F}$ attached to the quadratic extensions from which ρ is induced. The main result bounds the levels of those characters against the level of ρ by a factor of three. The configurations it forces are what section 9.2 uses to pin down the cubic extension attached to a primitive parameter.

For a character χ of $F^\times$ the *level* $\ell(\chi)$ is the least integer $n \geq 0$ with $\chi(U_F^{n+1}) = 1$, and χ is *unramified* when $\chi(U_F) = 1$. Here $U_F^0 = U_F$ and $U_F^m = 1 + \mathfrak{p}^m$ for $m \geq 1$. The same convention is used over E. For $\psi \in \widehat{F}$, $\psi \neq 1$, the level of ψ is the least integer m with $\psi(\mathfrak{p}^m) = 1$, so a character of level one has kernel exactly $\mathfrak{p}$. Finally χ_E denotes $\chi \circ N_{E/F}$, a character of $E^\times$. By theorem 6.2.1(4) the character of $\mathcal{W}_E$ attached to χ_E is the restriction to $\mathcal{W}_E$ of the character $\chi \circ \mathbf{a}_F$ of $\mathcal{W}_F$.

A separable quadratic extension E/F carries a relative discriminant $\mathfrak{d}_{E/F}$, an ideal of $\mathfrak{o}$. Its existence, and the fact that it is generated by the discriminant of any $\mathfrak{o}$-basis of $\mathfrak{o}_E$, are local field theory, developed in *Class Field Theory* [14] and used here without proof; the determinant formula for the discriminant of a tuple is [3, Discriminant Via Trace-Pairing]. This chapter uses only the exponent of $\mathfrak{p}$ in that ideal.

Definition 9.1.1: Discriminant Exponent of a Quadratic Extension

Let E/F be a separable quadratic extension. The *discriminant exponent* $d(E|F)$ is the integer determined by

$$\mathfrak{d}_{E/F} = \mathfrak{p}^{\,d(E|F)+1}$$

where $\mathfrak{d}_{E/F}$ is the relative discriminant of E/F. For a finite separable extension M/L of any degree we write $d(M|L)$ for the integer determined the same way by the *different*:

$$\mathfrak{D}_{M/L} = \mathfrak{p}_M^{\,d(M|L)+1}$$

The two agree when $[M : L] = 2$, because $\mathfrak{d}_{M/L} = \mathrm{Nm}_{M/L}\big(\mathfrak{D}_{M/L}\big)$ gives $v_L(\mathfrak{d}_{M/L}) = f(M|L)\, v_M(\mathfrak{D}_{M/L})$, and a ramified quadratic extension has $f = 1$ while an unramified one has both exponents zero. The shift by one is what makes $d(E|F)$ equal the level of $\kappa_{E/F}$, and what puts lemma 9.1.19 in the form $\ell(\rho) = \ell(\xi) + d$. Read against the unshifted convention, in which the different of $\mathbb{Q}_2(\sqrt{2})/\mathbb{Q}_2$ has exponent 3 rather than 2, every formula below is off by one.

For E/F unramified $\mathfrak{d}_{E/F} = \mathfrak{o}$ and $d(E|F) = -1$. For $F = \mathbb{Q}_2$ and $E = \mathbb{Q}_2(\sqrt{2})$ one has $\mathfrak{d}_{E/F} = 8\mathbb{Z}_2 = \mathfrak{p}^3$, so $d(E|F) = 2$, both computations being carried out in example 9.1.3. The content of the definition is in the ramified case, and there the exponent is computed by the minimal polynomial of a prime element. Since $[E : F] = 2$, a ramified E/F is totally ramified, and it is wildly so because $p = 2$.

Proposition 9.1.2: Discriminant of a Totally Ramified Quadratic Extension

Let E/F be quadratic, separable and totally ramified, let ϖ_E be a prime element of E, and let $f(X) = X^2 + aX + b$ be the minimal polynomial of ϖ_E over F. Then

1. $v_F(b) = 1$, $v_F(a) \geq 1$ and $\mathfrak{o}_E = \mathfrak{o}[\varpi_E] = \mathfrak{o} \oplus \mathfrak{o}\varpi_E$;
2. $d(E|F) = \min\{2v_F(a) - 1,\ v_F(4)\}$, and $d(E|F) \leq v_F(4)$, with $d(E|F) \geq 1$ when the residual characteristic is 2;
3. $d(E|F)$ is odd unless $d(E|F) = v_F(4)$.

***Proof*:**
Write $\varpi = \varpi_E$, σ for the non-trivial element of $\mathrm{Gal}(E/F)$ and $d = d(E|F)$. Since $v_E(\varpi) = 1$ is odd and $v_E(F^\times) = 2\mathbb{Z}$, the element ϖ does not lie in F, so $E = F(\varpi)$ and f has degree 2 with ϖ^σ as its second root. Thus

$$\varpi + \varpi^\sigma = -a, \qquad \varpi\varpi^\sigma = b$$

Step 1: the two valuations and the ring of integers. The automorphism σ preserves v_E, so $v_E(\varpi^\sigma) = 1$ and $v_E(b) = 2$, that is $v_F(b) = 1$. Also $v_E(a) = v_E(\varpi + \varpi^\sigma) \geq 1$, and $v_E(a) = 2v_F(a)$ because $a \in F$. Hence $v_F(a) \geq 1/2$ and therefore $v_F(a) \geq 1$, a being an element of F with integral valuation. (If $a = 0$ read $v_F(a) = \infty$. This happens only when $\mathrm{char}(F) \neq 2$, since $f'(X) = 2X + a$ vanishes identically when $\mathrm{char}(F) = 2$ and $a = 0$, contradicting separability.)

Every $z \in E$ is $z = c_0 + c_1\varpi$ with $c_0, c_1 \in F$, and $v_E(c_0) = 2v_F(c_0)$ is even while $v_E(c_1\varpi) = 2v_F(c_1) + 1$ is odd. Two elements of distinct valuation have sum of the smaller valuation, so

$$v_E(z) = \min\{2v_F(c_0),\ 2v_F(c_1) + 1\}$$

Hence $v_E(z) \geq 0$ if and only if $v_F(c_0) \geq 0$ and $2v_F(c_1) \geq -1$, the latter being $v_F(c_1) \geq 0$. So $\mathfrak{o}_E = \mathfrak{o} \oplus \mathfrak{o}\varpi$, and this equals $\mathfrak{o}[\varpi]$ because $\varpi^2 = -a\varpi - b$ keeps all higher powers inside $\mathfrak{o} \oplus \mathfrak{o}\varpi$. This proves (1).

Step 2: the discriminant. By the convention fixed above, together with [3, Discriminant Via Trace-Pairing], the discriminant $\mathfrak{d}_{E/F}$ is the ideal generated by $\det\left(\mathrm{tr}_{E/F}(e_ie_j)\right)$ for any $\mathfrak{o}$-basis (e_1, e_2) of $\mathfrak{o}_E$. Take $(e_1, e_2) = (1, \varpi)$, legitimate by (1). From $\mathrm{tr}_{E/F}(1) = 2$, $\mathrm{tr}_{E/F}(\varpi) = -a$ and $\varpi^2 = -a\varpi - b$, which gives $\mathrm{tr}_{E/F}(\varpi^2) = -a\mathrm{tr}_{E/F}(\varpi) - 2b = a^2 - 2b$, the matrix is

$$\begin{pmatrix} 2 & -a \\ -a & a^2 - 2b \end{pmatrix}$$

with determinant $2(a^2 - 2b) - a^2 = a^2 - 4b$. So $\mathfrak{d}_{E/F} = (a^2 - 4b)\mathfrak{o}$.

Step 3: the valuation of $a^2 - 4b$. Suppose first $\mathrm{char}(F) \neq 2$, so that $v_F(4) = 2v_F(2)$ is finite. Then $v_F(a^2) = 2v_F(a)$ is even while $v_F(4b) = v_F(4) + 1$ is odd, by $v_F(b) = 1$. The two are therefore never equal, and

$$v_F(a^2 - 4b) = \min\{2v_F(a),\ v_F(4) + 1\}$$

If instead $\mathrm{char}(F) = 2$, which the standing hypothesis on the residual characteristic does not exclude, then $4 = 0$, so $v_F(4) = \infty$ and $a^2 - 4b = a^2$. The displayed formula holds again, both sides being $2v_F(a)$, which is finite by Step 1. Subtracting 1 gives $d = \min\{2v_F(a) - 1,\ v_F(4)\}$, which is (2). The bounds follow: $2v_F(a) - 1 \geq 1$ by Step 1, and the minimum is at most its second entry, so $d \leq v_F(4)$. For the lower bound, $p = 2$ gives $v_F(4) \geq 1$, and both entries of the minimum are then at least 1, so $d \geq 1$. For $p \neq 2$ the bound fails, and indeed $v_F(4) = 0$ forces $d = 0$, as the remark below records. For (3), the first entry $2v_F(a) - 1$ is odd and the second is $v_F(4)$, so d fails to be odd only when the minimum is attained at $v_F(4)$. In characteristic 2 that exception is empty, $v_F(4) = \infty$ forcing $d = 2v_F(a) - 1$, which is odd. This completes the proof.

The exponent $d(E|F)$ measures the wild ramification of E/F: it equals -1 for the unramified extension, and $\mathfrak{d}_{E/F} = \mathfrak{p}^{d+1}$ for a ramified one. Nothing in the argument used $p = 2$ beyond the fact that 4 is not a unit, and when $p \neq 2$ the same computation returns $v_F(4) = 0$ and hence $d(E|F) = 0$ for every ramified quadratic extension. When $p = 2$ the value ranges over $1 \leq d \leq v_F(4)$, and every construction below depends on which value occurs.

Example 9.1.3: Quadratic Extensions of $\mathbb{Q}_2$

Let $F = \mathbb{Q}_2$, so $q = 2$, $v_F(2) = 1$ and $v_F(4) = 2$. By proposition 9.1.2 a ramified quadratic extension has $d(E|F) \in \{1, 2\}$.

1. $E = \mathbb{Q}_2(\sqrt{2})$. The element $\varpi_E = \sqrt{2}$ has $N_{E/F}(\varpi_E) = -2$ of valuation 1, so it is a prime element, and its minimal polynomial is $X^2 - 2$: here $a = 0$ and $b = -2$. Then $v_F(b) = 1$ and $d(E|F) = \min\{\infty, 2\} = 2$. Directly, $a^2 - 4b = 8$ and $\mathfrak{d}_{E/F} = \mathfrak{p}^3 = \mathfrak{p}^{d(E|F)+1}$, which returns $d(E|F) = 2$ through definition 9.1.1.

2. $E = \mathbb{Q}_2(i)$. Now i is a unit, but $\varpi_E = 1 + i$ has $N_{E/F}(1 + i) = 2$, so it is a prime element. From $(\varpi_E - 1)^2 = -1$ the minimal polynomial is $X^2 - 2X + 2$, so $a = -2$, $b = 2$, $v_F(a) = 1$, and $d(E|F) = \min\{2 \cdot 1 - 1,\ 2\} = 1$. Directly $a^2 - 4b = -4$, of valuation 2.

3. $E = \mathbb{Q}_2(\sqrt{5})$. Since $X^2 - X - 1$ is irreducible modulo 2 and has a root $(1 + \sqrt{5})/2$ in E, this extension is unramified, so $\mathfrak{d}_{E/F} = \mathfrak{o}$ and $d(E|F) = -1$.

Both values allowed by the bound $1 \leq d \leq v_F(4) = 2$ therefore occur over $\mathbb{Q}_2$, and the odd value $d = 1$ occurs with $2v_F(a) - 1 = 1$ while the even value $d = 2 = v_F(4)$ occurs with the minimum attained at $v_F(4)$, matching proposition 9.1.2(3).

The exponent also controls how the trace map moves the filtration of E down to the filtration of F. Every level count in this chapter passes through that computation, in the sharp form below rather than in the crude bound $\mathrm{tr}_{E/F}(\mathfrak{p}_E^j) \subseteq \mathfrak{p}_E^j \cap F$.

Proposition 9.1.4: The Trace on the Filtration

Let E/F be quadratic and totally ramified, with $d = d(E|F)$ and σ the non-trivial element of $\mathrm{Gal}(E/F)$. Then

1. $\mathrm{tr}_{E/F}\left(\mathfrak{p}_E^j\right) = \mathfrak{p}^{\lfloor (j+d+1)/2 \rfloor}$ for every $j \in \mathbb{Z}$;
2. for every $m \in \mathbb{Z}$ the trace induces an isomorphism of k-vector spaces
$$\mathfrak{p}_E^{2m-d}/\mathfrak{p}_E^{2m-d+1} \xrightarrow{\sim} \mathfrak{p}^m/\mathfrak{p}^{m+1}$$
3. $v_E(\varpi - \varpi^\sigma) = d+1$ for every prime element ϖ of E.

***Proof*:**

Fix a prime element ϖ of E with minimal polynomial $X^2 + aX + b$ over F, and keep the notation of proposition 9.1.2.

Step 1: $\mathrm{tr}_{E/F}(\mathfrak{p}_E^j)$ is a fractional ideal, and a recursion. By proposition 9.1.2(1), $\mathfrak{p}_E^j = \varpi^j \mathfrak{o}_E = \mathfrak{o}\varpi^j + \mathfrak{o}\varpi^{j+1}$, so
$$\mathrm{tr}_{E/F}\left(\mathfrak{p}_E^j\right) = \mathfrak{o}\,\mathrm{tr}_{E/F}(\varpi^j) + \mathfrak{o}\,\mathrm{tr}_{E/F}(\varpi^{j+1})$$
using $\mathfrak{o}$-linearity of the trace. This is a finitely generated $\mathfrak{o}$-submodule of F, and it is non-zero: since $\mathfrak{o}_E = \mathfrak{o} \oplus \mathfrak{o}\varpi$ one has $E = F \oplus F\varpi$, and multiplication by ϖ^j is an F-automorphism of E, so the two generators ϖ^j and ϖ^{j+1} form an F-basis of E. Separability of E/F makes $\mathrm{tr}_{E/F}\colon E \to F$ not identically zero, so it cannot vanish at both of them. Hence $\mathrm{tr}_{E/F}(\mathfrak{p}_E^j) = \mathfrak{p}^{t_j}$ for an integer t_j. Since $\varpi_F \mathfrak{o}_E = \mathfrak{p}_E^2$ for a prime element ϖ_F of F, and $\mathrm{tr}_{E/F}(\varpi_F z) = \varpi_F \mathrm{tr}_{E/F}(z)$,
$$t_{j+2} = t_j + 1 \qquad (j \in \mathbb{Z})$$
so it suffices to compute t_0 and t_1.

Step 2: t_0 and t_1. With $j = 0$ the displayed formula gives $\mathrm{tr}_{E/F}(\mathfrak{o}_E) = 2\mathfrak{o} + a\mathfrak{o}$, whence $t_0 = \min\{v_F(2), v_F(a)\}$. With $j = 1$ it gives $\mathrm{tr}_{E/F}(\mathfrak{p}_E) = a\mathfrak{o} + (a^2 - 2b)\mathfrak{o}$. Since $v_F(a^2) = 2v_F(a) \geq v_F(a)$, the ideal $a\mathfrak{o} + (a^2 - 2b)\mathfrak{o}$ equals $a\mathfrak{o} + 2b\mathfrak{o}$, so $t_1 = \min\{v_F(a),\ v_F(2) + 1\}$, using $v_F(b) = 1$.

Compare with the asserted values. Proposition 9.1.2(2) splits into two cases. If $d = 2v_F(a) - 1$, which by the shape of the minimum means $v_F(a) \leq v_F(2)$, then $\lfloor (d+1)/2 \rfloor = v_F(a) = t_0$ and $\lfloor (d+2)/2 \rfloor = v_F(a) = t_1$, the second because $v_F(a) \leq v_F(2) \leq v_F(2) + 1$. If instead $d = v_F(4) = 2v_F(2)$, which forces $2v_F(a) - 1 > v_F(4)$ and hence $v_F(a) \geq v_F(2) + 1$, then $\lfloor (d+1)/2 \rfloor = \lfloor (2v_F(2) + 1)/2 \rfloor = v_F(2) = t_0$ and $\lfloor (d+2)/2 \rfloor = v_F(2) + 1 = t_1$. In both cases $t_0 = \lfloor (0 + d + 1)/2 \rfloor$ and $t_1 = \lfloor (1 + d + 1)/2 \rfloor$. Since $\lfloor (j + d + 1)/2 \rfloor$ also satisfies the recursion of Step 1, (1) holds for all $j \in \mathbb{Z}$.

Step 3: the graded map. Put $j = 2m - d$, so that by (1) $t_j = \lfloor (2m+1)/2 \rfloor = m$ and $t_{j+1} = \lfloor (2m+2)/2 \rfloor = m + 1$. Thus $\mathrm{tr}_{E/F}(\mathfrak{p}_E^j) = \mathfrak{p}^m$ and $\mathrm{tr}_{E/F}(\mathfrak{p}_E^{j+1}) = \mathfrak{p}^{m+1}$, so the trace carries $\mathfrak{p}_E^{j+1}$ into $\mathfrak{p}^{m+1}$ and induces a map $\mathfrak{p}_E^j/\mathfrak{p}_E^{j+1} \to \mathfrak{p}^m/\mathfrak{p}^{m+1}$. It is surjective because the

image of $\mathfrak{p}_E^j$ is all of $\mathfrak{p}^m$. Both sides are one-dimensional over $k = \mathfrak{o}/\mathfrak{p}$: the target because $\mathfrak{p}^m/\mathfrak{p}^{m+1} \cong \mathfrak{o}/\mathfrak{p}$, the source because $\mathfrak{p}_E^j/\mathfrak{p}_E^{j+1} \cong \mathfrak{o}_E/\mathfrak{p}_E = k$, the residue field of E being k by total ramification. The map is k-linear since $\mathrm{tr}_{E/F}$ is $\mathfrak{o}$-linear. A surjection between one-dimensional k-spaces is an isomorphism, which is (2).

Step 4: the difference of conjugates. From $\varpi + \varpi^\sigma = -a$,

$$\varpi - \varpi^\sigma = 2\varpi + a$$

If $\mathrm{char}(F) = 0$ then $v_E(2\varpi) = 2v_F(2) + 1 = v_F(4) + 1$ is odd while $v_E(a) = 2v_F(a)$ is even, so the two valuations differ. If $\mathrm{char}(F) = 2$ then $2\varpi = 0$, so $v_E(2\varpi) = \infty = v_F(4) + 1$ while $v_E(a) = 2v_F(a)$ is finite, and again they differ. Either way

$$v_E(\varpi - \varpi^\sigma) = \min\{v_F(4) + 1,\ 2v_F(a)\} = \min\{v_F(4),\ 2v_F(a) - 1\} + 1 = d + 1$$

by proposition 9.1.2(2). A second prime element of E is $u\varpi$ with $u \in U_E$, and the same computation applies to its own minimal polynomial, so the value is independent of the choice. This gives (3), as we sought to show.

Example 9.1.5: The Trace Filtration for $\mathbb{Q}_2(\sqrt{2})$

Take $F = \mathbb{Q}_2$ and $E = \mathbb{Q}_2(\sqrt{2})$, so $d = 2$ by example 9.1.3. Proposition 9.1.4(1) reads $\mathrm{tr}_{E/F}(\mathfrak{p}_E^j) = \mathfrak{p}^{\lfloor (j+3)/2 \rfloor}$. Three values, checked by hand against $\mathfrak{o}_E = \mathbb{Z}_2 \oplus \mathbb{Z}_2\sqrt{2}$, $\mathrm{tr}_{E/F}(1) = 2$ and $\mathrm{tr}_{E/F}(\sqrt{2}) = 0$:

$$\mathrm{tr}_{E/F}(\mathfrak{o}_E) = 2\mathbb{Z}_2 = \mathfrak{p}, \qquad \mathrm{tr}_{E/F}(\mathfrak{p}_E) = \mathbb{Z}_2\mathrm{tr}_{E/F}(\sqrt{2}) + \mathbb{Z}_2\mathrm{tr}_{E/F}(2) = 4\mathbb{Z}_2 = \mathfrak{p}^2, \qquad \mathrm{tr}_{E/F}(\mathfrak{p}_E^2) = 2\mathfrak{p} = \mathfrak{p}^2$$

against the predicted exponents $\lfloor 3/2 \rfloor = 1$, $\lfloor 4/2 \rfloor = 2$ and $\lfloor 5/2 \rfloor = 2$. The graded isomorphism of proposition 9.1.4(2) occurs at $j = 2m - 2$, that is at even j, and indeed the two exponents coincide at $j = 1, 2$, so no well-defined map on graded pieces exists at the odd index $j = 1$.

The norm behaves less uniformly than the trace: it is an isomorphism on graded pieces away from one index, and at that index it has kernel and cokernel of order 2. That order-2 discrepancy is what produces the quadratic character $\kappa_{E/F}$ and fixes its level. Locating the kernel of the norm inside the unit filtration comes first.

Lemma 9.1.6: Norm-One Elements and the Filtration

Let E/F be quadratic and totally ramified with $d = d(E|F)$, let σ generate $\mathrm{Gal}(E/F)$, and write $\widetilde{N}(y) = y/y^{\sigma}$ for $y \in E^{\times}$. Then:

1. $\widetilde{N}(U_E^j) \subseteq U_E^{j+d}$ for $j \geq 1$, and $\widetilde{N}(U_E) \subseteq U_E^{d+1}$;
2. for every integer $m \geq 0$ the map $\widetilde{N}$ induces an isomorphism
$$U_E^{2m+1}/U_E^{2m+2} \xrightarrow{\sim} U_E^{2m+d+1}/U_E^{2m+d+2}$$
3. $\mathcal{K}_{E/F} \subseteq U_E^d$, and the image of $\mathcal{K}_{E/F}$ in U_E^d/U_E^{d+1} has order 2, generated by the class of ϖ/ϖ^{σ} for any prime element ϖ of E.

***Proof*:**
Fix a prime element ϖ of E, so that $\mathfrak{o}_E = \mathfrak{o} \oplus \mathfrak{o}\varpi$ by proposition 9.1.2(1) and $v_E(\varpi - \varpi^{\sigma}) = d+1$ by proposition 9.1.4(3).

Step 1: the estimate (1). Let $z \in \mathfrak{p}_E^j$ with $j \geq 1$ and write $z = c + c'\varpi$ with $c, c' \in F$. Then $z - z^{\sigma} = c'(\varpi - \varpi^{\sigma})$, so $v_E(z - z^{\sigma}) = 2v_F(c') + d + 1$. From the valuation formula in the proof of proposition 9.1.2 one has $v_E(z) \leq 2v_F(c') + 1$, hence $2v_F(c') \geq v_E(z) - 1 \geq j - 1$ and $v_E(z - z^{\sigma}) \geq j + d$. For $y = 1 + z \in U_E^j$,
$$\frac{y}{y^{\sigma}} = 1 + \frac{z - z^{\sigma}}{y^{\sigma}}$$
and $y^{\sigma} \in U_E$, so $\widetilde{N}(y) \in U_E^{j+d}$. For the second assertion, the residue field of E is k, so the reduction $U_E \to k^{\times}$ is surjective already on U_F, giving $U_E = U_F U_E^1$. Now $\widetilde{N}(U_F) = 1$ and $\widetilde{N}(U_E^1) \subseteq U_E^{1+d}$ by the case $j = 1$ just proved, and $\widetilde{N}$ is a homomorphism because $E^{\times}$ is abelian. Hence $\widetilde{N}(U_E) \subseteq U_E^{d+1}$.

Step 2: the isomorphism (2). Fix $m \geq 0$. Since $\mathfrak{p}_E^{2m+1} = \varpi_F^m \varpi \mathfrak{o}_E = \mathfrak{p}^m\varpi + \mathfrak{p}^m\varpi^2$ and $v_E(\mathfrak{p}^m\varpi^2) \geq 2m + 2$, the elements $1 + y\varpi$ with y running over a set of representatives for $\mathfrak{p}^m/\mathfrak{p}^{m+1}$ represent the cosets of U_E^{2m+1}/U_E^{2m+2}. For such an element,
$$\frac{1 + y\varpi}{1 + y\varpi^{\sigma}} = 1 + \frac{y(\varpi - \varpi^{\sigma})}{1 + y\varpi^{\sigma}}$$
and $(1 + y\varpi^{\sigma})^{-1} \in 1 + \mathfrak{p}_E^{2m+1}$, so the quotient lies in $1 + y(\varpi - \varpi^{\sigma}) + \mathfrak{p}_E^{2m+d+2}$ because $v_E\big(y(\varpi - \varpi^{\sigma})\big) = 2m+d+1$ and the correction term has valuation at least $(2m+d+1)+(2m+1) \geq 2m + d + 2$. The same argument that identified representatives above, applied to the element $\varpi - \varpi^{\sigma}$ of valuation $d+1$ in place of ϖ, shows that the elements $1 + y(\varpi - \varpi^{\sigma})$ with y running over representatives of $\mathfrak{p}^m/\mathfrak{p}^{m+1}$ represent the cosets of $U_E^{2m+d+1}/U_E^{2m+d+2}$. So $\widetilde{N}$ meets every coset exactly once, and it remains only to see the induced map is well defined: $\mathfrak{p}_E^{2m+2} = \mathfrak{p}^{m+1}\mathfrak{o}_E$ gives $U_E^{2m+2} = U_F^{m+1}U_E^{2m+3}$ by the same representative computation, and $\widetilde{N}(U_F^{m+1}) = 1$ while $\widetilde{N}(U_E^{2m+3}) \subseteq U_E^{2m+d+3}$ by (1). Hence $\widetilde{N}(U_E^{2m+2}) \subseteq U_E^{2m+d+2}$, and the induced map is a bijective homomorphism.

Step 3: the kernel of the norm (3). Hilbert's Theorem 90 for the cyclic extension E/F [14, The Cyclic Norm Form] says $\mathcal{K}_{E/F} = \widetilde{N}(E^\times)$. Now $E^\times = \varpi^{\mathbb{Z}} U_F U_E^1$, and $\widetilde{N}$ kills U_F, sends U_E^1 into U_E^{d+1} by (1), and sends ϖ to an element of valuation-difference d: from $\varpi/\varpi^\sigma = 1 + (\varpi - \varpi^\sigma)/\varpi^\sigma$ and $v_E\big((\varpi - \varpi^\sigma)/\varpi^\sigma\big) = d$, the element ϖ/ϖ^σ lies in $U_E^d \setminus U_E^{d+1}$. Therefore $\mathcal{K}_{E/F} = \langle \varpi/\varpi^\sigma \rangle \, \widetilde{N}(U_E^1) \subseteq U_E^d$, and its image in U_E^d/U_E^{d+1} is generated by the non-trivial class of ϖ/ϖ^σ. That class has order 2 because $U_E^d/U_E^{d+1} \cong \mathfrak{p}_E^d/\mathfrak{p}_E^{d+1} \cong k$ is an elementary abelian 2-group, k having characteristic 2. A second prime element of E is $u\varpi$ with $u \in U_E$, and $u\varpi/(u\varpi)^\sigma = \widetilde{N}(u) \cdot (\varpi/\varpi^\sigma)$ with $\widetilde{N}(u) \in U_E^{d+1}$ by (1), so the class is the same for every choice of ϖ. This completes the proof.

With the kernel of the norm located, the norm itself can be compared filtration step by filtration step: expanding $N_{E/F}(1+x)$ turns the trace computation above into a statement about graded pieces, bijective at every index but one.

Lemma 9.1.7: The Norm on the Unit Filtrations

Let E/F be quadratic, separable and totally ramified with $d = d(E|F)$, and define

$$\varphi(j) = \begin{cases} j & 0 \le j \le d \\ 2j - d & j \ge d \end{cases}$$

Then, for every integer $j \ge 0$,

1. $N_{E/F}\big(U_E^{\varphi(j)}\big) \subseteq U_F^j$ and $N_{E/F}\big(U_E^{\varphi(j)+1}\big) \subseteq U_F^{j+1}$, so that $N_{E/F}$ induces a homomorphism
$$\nu_j \colon U_E^{\varphi(j)}/U_E^{\varphi(j)+1} \longrightarrow U_F^j/U_F^{j+1}$$
2. ν_j is an isomorphism for every $j \neq d$;
3. $N_{E/F}\big(U_E^{d+1}\big) = U_F^{d+1}$;
4. ν_d has kernel and cokernel of order 2, and its kernel is the image of $\mathcal{K}_{E/F}$.

***Proof*:**
For $x \in \mathfrak{p}_E^m$ with $m \ge 1$,

$$N_{E/F}(1+x) = (1+x)(1+x^\sigma) = 1 + \mathrm{tr}_{E/F}(x) + N_{E/F}(x) \tag{9.1}$$

with $v_F\big(N_{E/F}(x)\big) = v_E(x) \ge m$ (total ramification makes $v_F \circ N_{E/F} = v_E$) and $v_F\big(\mathrm{tr}_{E/F}(x)\big) \ge \lfloor (m+d+1)/2 \rfloor$ by proposition 9.1.4(1).

Step 1: the containments (1). Let $m = \varphi(j)$. If $j \le d$ then $m = j$, and $\lfloor (j+d+1)/2 \rfloor \ge j$ because $j + d + 1 \ge 2j$ for $j \le d+1$. If $j \ge d$ then $m = 2j - d \ge j$ and $\lfloor (2j+1)/2 \rfloor = j$. Either way both terms on the right of (9.1) lie in $\mathfrak{p}^j$, giving the first containment. For the second take $m = \varphi(j) + 1$. If $j \le d-1$ then $m = j+1$ and $\lfloor (j+d+2)/2 \rfloor \ge j+1$ because $j + d + 2 \ge 2j + 2$. If $j \ge d$ then $m = 2j - d + 1 \ge j + 1$ and $\lfloor (2j+2)/2 \rfloor = j+1$. Both terms lie in $\mathfrak{p}^{j+1}$.

Step 2: ν_j for $j = 0$ and for $1 \le j \le d-1$. For $j = 0$ the reduction $U_E/U_E^1 \cong k^\times \cong U_F/U_F^1$

identifies ν_0 with $u \mapsto uu^\sigma$ on $k^\times$. Since σ fixes $\mathfrak{o}$ pointwise and $k_E = k$ is generated by the image of $\mathfrak{o}$, the automorphism induced by σ on k is the identity, so ν_0 is $u \mapsto u^2$. The group $k^\times$ has odd order $q-1$, so squaring is bijective on it.

Let $1 \le j \le d-1$ and $x \in \mathfrak{p}_E^j$. Then $\lfloor (j+d+1)/2 \rfloor \ge j+1$, since $j+d+1 \ge 2j+2$ is exactly $j \le d-1$. So (9.1) gives $N_{E/F}(1+x) \equiv 1 + N_{E/F}(x) \bmod \mathfrak{p}^{j+1}$. The bijections $U_E^j/U_E^{j+1} \cong \mathfrak{p}_E^j/\mathfrak{p}_E^{j+1}$ and $U_F^j/U_F^{j+1} \cong \mathfrak{p}^j/\mathfrak{p}^{j+1}$ sending $1+x$ to x turn ν_j into $x \mapsto N_{E/F}(x)$. Representatives of $\mathfrak{p}_E^j/\mathfrak{p}_E^{j+1}$ are $u\varpi^j$ with u running over representatives of k, and $N_{E/F}(u\varpi^j) = u^2 N_{E/F}(\varpi)^j$ with $v_F\big(N_{E/F}(\varpi)^j\big) = j$. So under the identifications with k, the map ν_j is $\bar{u} \mapsto \bar{u}^2$ times a fixed non-zero scalar. Squaring is the Frobenius of the finite field k of characteristic 2, hence bijective, so ν_j is an isomorphism.

Step 3: ν_j for $j \ge d+1$. Now $\varphi(j) = 2j-d$ and, for $x \in \mathfrak{p}_E^{2j-d}$, $v_F\big(N_{E/F}(x)\big) = 2j-d \ge j+1$, so (9.1) gives $N_{E/F}(1+x) \equiv 1 + \mathrm{tr}_{E/F}(x) \bmod \mathfrak{p}^{j+1}$. Under the same identifications $U_E^{2j-d}/U_E^{2j-d+1} \cong \mathfrak{p}_E^{2j-d}/\mathfrak{p}_E^{2j-d+1}$ and $U_F^j/U_F^{j+1} \cong \mathfrak{p}^j/\mathfrak{p}^{j+1}$, the map ν_j becomes the graded trace map, which is an isomorphism by proposition 9.1.4(2) with $m = j$. This proves (2).

Step 4: surjectivity of the norm on U_E^{d+1}. The containment $N_{E/F}(U_E^{d+1}) \subseteq U_F^{d+1}$ is Step 1 with $j = d$, since $\varphi(d)+1 = d+1$. For the reverse, let $u \in U_F^{d+1}$. Using surjectivity of ν_j for $j \ge d+1$, construct inductively elements $w_j \in U_E^{2j-d}$, $j \ge d+1$, with

$$N_{E/F}(w_{d+1} w_{d+2} \cdots w_j) \equiv u \bmod U_F^{j+1}$$

at each stage: given the congruence at $j-1$, the element $u\, N_{E/F}(w_{d+1}\cdots w_{j-1})^{-1}$ lies in U_F^j and ν_j is onto, so a suitable $w_j \in U_E^{2j-d}$ exists. As $j \to \infty$ the exponents $2j-d$ tend to infinity, so the partial products form a Cauchy sequence in the compact group U_E^{d+2} and converge to some $w \in U_E^{d+2}$. The norm is continuous, so $N_{E/F}(w) = u$. Hence $U_F^{d+1} \subseteq N_{E/F}(U_E^{d+2}) \subseteq N_{E/F}(U_E^{d+1})$, which is (3).

Step 5: the exceptional index (4). By lemma 9.1.6(3) the group $\mathcal{K}_{E/F}$ lies in U_E^d and has image of order 2 in U_E^d/U_E^{d+1}. That image lies in $\ker \nu_d$ because $N_{E/F}$ is trivial on $\mathcal{K}_{E/F}$. Conversely, let $u \in U_E^d$ with $N_{E/F}(u) \in U_F^{d+1}$. By (3) there is $z \in U_E^{d+1}$ with $N_{E/F}(z) = N_{E/F}(u)$, and then $y = uz^{-1} \in U_E^d$ satisfies $N_{E/F}(y) = 1$, so $y \in \mathcal{K}_{E/F}$ and $u \in \mathcal{K}_{E/F} U_E^{d+1}$. Hence $\ker \nu_d$ is exactly the image of $\mathcal{K}_{E/F}$, of order 2. Finally U_E^d/U_E^{d+1} and U_F^d/U_F^{d+1} both have order q, so the image of ν_d has order $q/2$ and its cokernel has order 2, completing the proof.

The character $\kappa_{E/F}$ of definition 8.3.1 was defined by its kernel $N_{E/F}(E^\times)$, with no indication of how deep into the filtration of $F^\times$ that kernel reaches. Lemma 9.1.7 answers this, and the answer is $d(E|F)$: the discriminant exponent is at the same time the ramification datum of the extension and the level of $\kappa_{E/F}$.

Proposition 9.1.8: The Level of $\kappa_{E/F}$

Let E/F be a separable quadratic extension. Then $\kappa_{E/F}$ is unramified if and only if E/F is unramified. If E/F is ramified, then $\kappa_{E/F}$ has level $d(E|F) \geq 1$.

***Proof*:**
Write $\kappa = \kappa_{E/F}$, so $\ker \kappa = N_{E/F}(E^\times)$ by definition 8.3.1.

Step 1: the unramified case. Suppose E/F is unramified. Then E lies in the maximal unramified extension of F, so the inertia group $\mathcal{I}_F$ acts trivially on E and $\mathcal{I}_F \subseteq \mathcal{W}_E$. By theorem 6.2.1(4) the restriction of $\mathbf{a}_F$ to $\mathcal{W}_E$ is $N_{E/F} \circ \mathbf{a}_E$, and $\mathbf{a}_E$ is surjective onto $E^\times$ by part (1) of the same theorem. Hence $\mathbf{a}_F(\mathcal{W}_E) = N_{E/F}(E^\times) = \ker \kappa$. Applying κ to theorem 6.2.1(3), which gives $\mathbf{a}_F(\mathcal{I}_F) = U_F$, we get $\kappa(U_F) = \kappa\big(\mathbf{a}_F(\mathcal{I}_F)\big) \subseteq \kappa\big(\mathbf{a}_F(\mathcal{W}_E)\big) = 1$. So κ is unramified.

Step 2: the ramified case, upper bound. Suppose E/F is ramified, hence totally ramified, and put $d = d(E|F) \geq 1$ by proposition 9.1.2(2). By lemma 9.1.7(3), $U_F^{d+1} = N_{E/F}(U_E^{d+1}) \subseteq \ker \kappa$, so κ has level at most d.

Step 3: the ramified case, lower bound. It suffices to produce an element of U_F^d outside $\ker \kappa$. First, if $u \in U_E$ satisfies $N_{E/F}(u) \in U_F^d$ then $u \in U_E^d$: indeed, if $u \in U_E^j$ with $0 \leq j \leq d-1$ then $N_{E/F}(u)$ lies in $U_F^d \subseteq U_F^{j+1}$, so $\nu_j(\bar{u}) = 1$, and ν_j is injective by lemma 9.1.7(2), whence $u \in U_E^{j+1}$. Starting from $j = 0$ and iterating d times gives the claim. Next, an element of $N_{E/F}(E^\times)$ lying in U_F has the form $N_{E/F}(z)$ with $v_E(z) = v_F(N_{E/F}(z)) = 0$, that is with $z \in U_E$. Combining the two,

$$\ker \kappa \cap U_F^d = N_{E/F}(E^\times) \cap U_F^d = N_{E/F}\big(U_E^d\big)$$

By lemma 9.1.7(3) and (4), $N_{E/F}(U_E^d)$ contains U_F^{d+1} and its image in U_F^d/U_F^{d+1} has index 2, so it is a proper subgroup of U_F^d. Any $u \in U_F^d$ outside it has $\kappa(u) \neq 1$, so κ has level exactly d, completing the proof.

Two corollaries of proposition 9.1.8 are used constantly below. The first is that over a dyadic field no quadratic character is tamely ramified: a ramified $\kappa_{E/F}$ has level $d \geq 1$, never level 0. The second is the following bookkeeping, which converts the level of a character of $F^\times$ into the level of its pullback to $E^\times$.

Lemma 9.1.9: Levels Under Pullback by the Norm

Let E/F be quadratic and totally ramified with $d = d(E|F)$, let χ be a character of $F^\times$, and put $\chi_E = \chi \circ N_{E/F}$. Let φ be as in lemma 9.1.7.

1. If χ is unramified, so is χ_E.
2. If χ is ramified of level j with $j \neq d$, then χ_E is ramified of level $\varphi(j)$.
3. If χ has level d, then χ_E has level d when $\chi|_{U_F^d} \neq \kappa_{E/F}|_{U_F^d}$, and $\chi_E = (\kappa_{E/F}\chi)_E$ with $\kappa_{E/F}\chi$ of level at most $d-1$ otherwise.

***Proof*:**
Part (1) holds because $N_{E/F}(U_E) \subseteq U_F$: if χ is trivial on U_F then χ_E is trivial on U_E.

For (2), let $j = \ell(\chi) \neq d$. By lemma 9.1.7(1) the group $N_{E/F}(U_E^{\varphi(j)+1})$ lies in $U_F^{j+1} \subseteq \ker\chi$, so χ_E is trivial on $U_E^{\varphi(j)+1}$. By lemma 9.1.7(2) the map ν_j is surjective, so $N_{E/F}(U_E^{\varphi(j)})U_F^{j+1} = U_F^j$. As χ is non-trivial on U_F^j and trivial on U_F^{j+1}, it is non-trivial on $N_{E/F}(U_E^{\varphi(j)})$. Hence $\ell(\chi_E) = \varphi(j)$, and χ_E is ramified because $\varphi(j) \geq 0$ with χ_E non-trivial on $U_E^{\varphi(j)}$.

For (3), let $\ell(\chi) = d$. Then χ kills $U_F^{d+1} = N_{E/F}(U_E^{d+1})$ by lemma 9.1.7(3), so χ_E is trivial on U_E^{d+1} and its restriction to U_E^d is the composite of ν_d with the character $\bar{\chi}$ that χ induces on U_F^d/U_F^{d+1}. So $\chi_E|_{U_E^d} = 1$ if and only if $\bar{\chi}$ is trivial on the image of ν_d, a subgroup of index 2 by lemma 9.1.7(4). The characters of U_F^d/U_F^{d+1} trivial on a subgroup of index 2 form a group of order 2. One of its two elements is trivial and the other is the class of $\kappa_{E/F}$, since $(\kappa_{E/F})_E = 1$ makes $\kappa_{E/F}$ trivial on the image of ν_d while $\ell(\kappa_{E/F}) = d$ by proposition 9.1.8 makes it non-trivial on U_F^d. As $\bar{\chi} \neq 1$, it follows that $\chi_E|_{U_E^d} = 1$ exactly when $\chi|_{U_F^d} = \kappa_{E/F}|_{U_F^d}$, and in that case $\kappa_{E/F}\chi$ is trivial on U_F^d, hence of level at most $d-1$. In the remaining case χ_E is non-trivial on U_E^d and trivial on U_E^{d+1}, so it has level d, completing the proof.

Recall from definition 8.5.1 that $\rho \in \mathcal{G}_2^0(F)$ is imprimitive when it is induced from a character of a separable quadratic extension, and that $\mathcal{G}_2^{\mathrm{nr}}(F)$ collects the classes admitting such a presentation with E/F unramified. Induced parameters interact with twisting in a way that exhibits both the presentations and their multiplicity at once.

Lemma 9.1.10: Twisting an Induced Parameter

Let E/F be a separable quadratic extension with $\mathrm{Gal}(E/F) = \{1, \sigma\}$, let ξ, ξ' be characters of $E^\times$, and let χ be a character of $F^\times$. Then

1. $\chi \otimes \mathrm{Ind}_{E/F}\xi \cong \mathrm{Ind}_{E/F}(\chi_E\xi)$;
2. $\mathrm{Ind}_{E/F}\xi^\sigma \cong \mathrm{Ind}_{E/F}\xi$;
3. if $\mathrm{Ind}_{E/F}\xi \cong \mathrm{Ind}_{E/F}\xi'$, then $\xi' \in \{\xi, \xi^\sigma\}$.

***Proof*:**
Realise $\mathrm{Ind}_{E/F}\xi$ on the space $V_\xi = \{f\colon \mathcal{W}_F \to \mathbb{C} : f(hg) = \xi(h)f(g) \text{ for } h \in \mathcal{W}_E,\ g \in \mathcal{W}_F\}$ with $\mathcal{W}_F$ acting by right translation, as in section 8.5, and fix $\tau \in \mathcal{W}_F \setminus \mathcal{W}_E$.

1. Write $\widetilde{\chi} = \chi \circ \mathbf{a}_F$, a character of $\mathcal{W}_F$ whose restriction to $\mathcal{W}_E$ is the character attached to χ_E, by theorem 6.2.1(4). Define $\Phi\colon V_\xi \to V_{\chi_E\xi}$ by $(\Phi f)(g) = \widetilde{\chi}(g)f(g)$. It is well defined because $(\Phi f)(hg) = \widetilde{\chi}(h)\widetilde{\chi}(g)\xi(h)f(g) = (\chi_E\xi)(h)(\Phi f)(g)$ for $h \in \mathcal{W}_E$, and it is a linear bijection with inverse $f \mapsto \widetilde{\chi}^{-1}f$. For $x \in \mathcal{W}_F$,
$$\big(\Phi(\rho_\xi(x)f)\big)(g) = \widetilde{\chi}(g)f(gx), \qquad \big(\rho_{\chi_E\xi}(x)\Phi f\big)(g) = \widetilde{\chi}(gx)f(gx)$$
so $\rho_{\chi_E\xi}(x) \circ \Phi = \widetilde{\chi}(x)\,\Phi \circ \rho_\xi(x)$, that is Φ intertwines $\widetilde{\chi} \otimes \rho_\xi$ with $\rho_{\chi_E\xi}$.
2. Define $S\colon V_\xi \to V_{\xi^\sigma}$ by $(Sf)(g) = f(\tau g)$, where $\xi^\sigma(h) = \xi(\tau h\tau^{-1})$. Then $(Sf)(hg) = f\big((\tau h\tau^{-1})\tau g\big) = \xi^\sigma(h)(Sf)(g)$, so S is well defined, and it is bijective with inverse $f \mapsto \big(g \mapsto f(\tau^{-1}g)\big)$. It intertwines the two actions because $\big(S(\rho_\xi(x)f)\big)(g) = f(\tau gx) = (Sf)(gx)$.
3. Restrict to $\mathcal{W}_E$. By (8.29) the restriction of $\mathrm{Ind}_{E/F}\xi$ to $\mathcal{W}_E$ is the direct sum of the characters ξ and ξ^σ, and likewise for ξ'. Isomorphic representations have isomorphic restrictions, and a direct sum of two characters determines the pair of characters occurring in it up to order, so $\{\xi, \xi^\sigma\} = \{\xi', \xi'^\sigma\}$ and in particular $\xi' \in \{\xi, \xi^\sigma\}$.

This settles all three parts, completing the proof.

Part (1) of lemma 9.1.10 shows that twisting a parameter induced from E produces another parameter induced from E. The characters that return ρ itself are therefore the ones to name.

Definition 9.1.11: The Twisting Group $T(\rho)$

Let $\rho \in \mathcal{G}_2^0(F)$. The *twisting group* of ρ is
$$T(\rho) = \{\chi : \chi \otimes \rho \cong \rho\}$$
where χ runs over the characters of $F^\times$, each acting on ρ through $\chi \circ \mathbf{a}_F$. The parameter ρ is *simply imprimitive* when $|T(\rho)| = 2$ and *triply imprimitive* when $|T(\rho)| = 4$.

The set $T(\rho)$ is a group: if χ, χ' lie in it then $\chi\chi' \otimes \rho \cong \chi \otimes \rho \cong \rho$, and $\chi^{-1} \otimes \rho \cong \rho$ follows by twisting the defining isomorphism by χ^{-1}. Proposition 9.1.15 below shows the only possible orders are 1, 2 and 4, so the two names just introduced exhaust the imprimitive case. Lemma 9.1.10 converts membership in $T(\rho)$ into a condition on characters of $E^\times$, which is how every computation of $T(\rho)$ below proceeds.

Lemma 9.1.12: Criterion for Membership in $T(\rho)$

Let E/F be a separable quadratic extension, let ξ be a character of $E^\times$ with $\rho = \mathrm{Ind}_{E/F}\xi \in \mathcal{G}_2^0(F)$, and let η be a character of $F^\times$. Then $\eta \in T(\rho)$ if and only if $\eta_E = 1$ or $\eta_E = \xi^\sigma/\xi$.

***Proof*:**
By lemma 9.1.10(1), $\eta \otimes \rho \cong \mathrm{Ind}_{E/F}(\eta_E \xi)$, so $\eta \in T(\rho)$ says $\mathrm{Ind}_{E/F}(\eta_E\xi) \cong \mathrm{Ind}_{E/F}\xi$. By lemma 9.1.10(3) this holds if and only if $\eta_E\xi \in \{\xi, \xi^\sigma\}$, that is $\eta_E = 1$ or $\eta_E = \xi^\sigma/\xi$. Conversely each of those two conditions returns $\mathrm{Ind}_{E/F}(\eta_E\xi) \cong \mathrm{Ind}_{E/F}\xi$, the second by lemma 9.1.10(2), as we sought to show.

Example 9.1.13: A Simply Imprimitive Parameter over $\mathbb{Q}_2$
Return to the parameter of example 8.5.3: $F = \mathbb{Q}_2$, $E = \mathbb{Q}_2(\zeta)$ the unramified quadratic extension with ζ a primitive cube root of unity, $E^\times = 2^{\mathbb{Z}} \times \mu_3 \times U_E^1$, and Θ the character with $\Theta(2) = 1$, $\Theta|_{\mu_3} = \theta$ injective and $\Theta|_{U_E^1} = 1$. There $\rho = \mathrm{Ind}_{E/F}\Theta$ was shown to lie in $\mathcal{G}_2^{\mathrm{nr}}(\mathbb{Q}_2)$.

Compute $T(\rho)$ from lemma 9.1.12. First, $\Theta^\sigma(\zeta) = \Theta(\zeta^2) = \theta(\zeta)^2$, so $(\Theta^\sigma/\Theta)(\zeta) = \theta(\zeta)$, while Θ^σ/Θ is trivial on 2 and on U_E^1 because Θ is and σ fixes 2 and preserves U_E^1. Thus $\Theta^\sigma/\Theta = \Theta$, which is non-trivial on μ_3. On the other hand $N_{E/F}(\zeta') = 1$ for every $\zeta' \in \mu_3$, so $\eta_E|_{\mu_3} = 1$ for every character η of $F^\times$. No η can satisfy $\eta_E = \Theta^\sigma/\Theta$.

The remaining possibility is $\eta_E = 1$, that is η trivial on $N_{E/F}(E^\times)$, a subgroup of index 2 by theorem 6.2.2(2). Such η are 1 and $\kappa_{E/F}$. Hence $T(\rho) = \{1, \kappa_{E/F}\}$ and ρ is simply imprimitive.

The twisting group in that example consists of 1 and the character $\kappa_{E/F}$ of the extension from which ρ was induced. That is the general situation, and it holds in both directions.

Lemma 9.1.14: Induction from E Detected by $\kappa_{E/F}$

Let $\rho \in \mathcal{G}_2^0(F)$ and let E/F be a separable quadratic extension. The following are equivalent:

1. $\kappa_{E/F} \in T(\rho)$;
2. $\rho \cong \mathrm{Ind}_{E/F}\xi$ for some character ξ of $E^\times$.

When they hold, ξ is E/F-regular and is determined by ρ up to replacing it by ξ^σ.

***Proof*:**
Write $\kappa = \kappa_{E/F}$ and $\widetilde{\kappa} = \kappa \circ \mathbf{a}_F$. As in Step 1 of the proof of proposition 9.1.8, $\mathbf{a}_F(\mathcal{W}_E) = N_{E/F}(E^\times) = \ker\kappa$, so $\mathcal{W}_E \subseteq \ker\widetilde{\kappa}$. Both subgroups have index 2 in $\mathcal{W}_F$, the first because $[E:F] = 2$ and the second because $\widetilde{\kappa}$ is a non-trivial character of order 2. Hence

$$\ker\widetilde{\kappa} = \mathcal{W}_E \tag{9.2}$$

(2) implies (1). If $\rho \cong \mathrm{Ind}_{E/F}\xi$ then $\kappa_E = \kappa \circ N_{E/F} = 1$, because $\ker\kappa = N_{E/F}(E^\times)$, so lemma 9.1.12 puts κ in $T(\rho)$. Regularity of ξ and its uniqueness up to σ are the content of lemma 8.5.2 and lemma 9.1.10(3) respectively: irreducibility of ρ forces $\xi^\sigma \neq \xi$, and any second presentation uses ξ or ξ^σ.

(1) implies (2). Let (ρ, V) with $\dim_{\mathbb{C}} V = 2$ and suppose $\kappa \otimes \rho \cong \rho$, so there is $A \in \mathrm{GL}(V)$ with

$$A\rho(x)A^{-1} = \widetilde{\kappa}(x)\rho(x) \qquad (x \in \mathcal{W}_F) \tag{9.3}$$

Step 1: the restriction to $\mathcal{W}_E$ is reducible. Suppose it were irreducible. By (9.2), $\widetilde{\kappa}(x) = 1$ for $x \in \mathcal{W}_E$, so (9.3) makes A commute with $\rho(x)$ for every $x \in \mathcal{W}_E$. Schur's lemma applied to the irreducible restriction forces A to be a scalar, and then (9.3) reads $\rho(x) = \widetilde{\kappa}(x)\rho(x)$ for all $x \in \mathcal{W}_F$, so $\widetilde{\kappa} = 1$. This contradicts $\kappa \neq 1$.

Step 2: the restriction is a sum of two distinct characters. Let $L \subseteq V$ be a $\mathcal{W}_E$-stable line, which Step 1 provides. Since $\mathcal{W}_E$ is normal in $\mathcal{W}_F$, the line $\rho(\tau)L$ is again $\mathcal{W}_E$-stable for $\tau \in \mathcal{W}_F \setminus \mathcal{W}_E$. If $\rho(\tau)L = L$ then L is stable under $\mathcal{W}_E$ and under τ, hence under the group they generate, which is $\mathcal{W}_F = \mathcal{W}_E \sqcup \mathcal{W}_E\tau$. This contradicts irreducibility of ρ. Therefore $V = L \oplus \rho(\tau)L$. Let ξ be the character of $\mathcal{W}_E$ on L. For $v \in L$ and $h \in \mathcal{W}_E$,

$$\rho(h)\rho(\tau)v = \rho(\tau)\rho(\tau^{-1}h\tau)v = \xi(\tau^{-1}h\tau)\rho(\tau)v$$

and $\xi(\tau^{-1}h\tau) = \xi(\tau h\tau^{-1}) = \xi^{\sigma}(h)$, because $\tau^2 \in \mathcal{W}_E$ acts trivially by conjugation on $\mathcal{W}_E^{\mathrm{ab}}$, through which the character ξ factors. So $\mathcal{W}_E$ acts on V by $\xi \oplus \xi^{\sigma}$. If $\xi = \xi^{\sigma}$ then $\mathcal{W}_E$ acts by the scalar character ξ. By lemma 8.4.2 such a ξ is trivial on $\mathcal{K}_{E/F}$, hence factors as $\xi = \eta_E$ for a character η of $F^{\times}$ (the induced character of $E^{\times}/\mathcal{K}_{E/F} \cong N_{E/F}(E^{\times})$ extends to $F^{\times}$ because $\mathbb{C}^{\times}$ is divisible, and the extension is smooth because $N_{E/F}(E^{\times})$ is open by theorem 6.2.2(1)). Then $\eta^{-1} \otimes \rho$ is trivial on $\mathcal{W}_E$, so it factors through the group $\mathcal{W}_F/\mathcal{W}_E$ of order 2 and is a sum of two characters, contradicting irreducibility of ρ. Hence $\xi \neq \xi^{\sigma}$.

Step 3: identifying ρ. Let $\lambda \in V^*$ be the linear form that is the composite of the projection $V \to L$ along $\rho(\tau)L$ with an isomorphism $L \cong \mathbb{C}$. Since $\rho(h)$ preserves both summands and acts on L by $\xi(h)$, one has $\lambda(\rho(h)w) = \xi(h)\lambda(w)$ for $h \in \mathcal{W}_E$ and $w \in V$. Define $\Phi\colon V \to V_{\xi}$ by $(\Phi w)(g) = \lambda(\rho(g)w)$. Then $(\Phi w)(hg) = \xi(h)(\Phi w)(g)$, so Φ lands in V_{ξ}, and $(\Phi(\rho(x)w))(g) = \lambda(\rho(gx)w) = (\Phi w)(gx)$, so Φ is a $\mathcal{W}_F$-map. It is non-zero because $\lambda \neq 0$. As ρ is irreducible, Φ is injective, and both spaces have dimension 2, so Φ is an isomorphism $\rho \cong \mathrm{Ind}_{E/F}\xi$. This completes the proof.

Each non-trivial element of $T(\rho)$ is thus attached to a quadratic extension from which ρ is induced, and counting the possibilities bounds the order of the group.

Proposition 9.1.15: The Order of the Twisting Group

Let $\rho \in \mathcal{G}_2^0(F)$. Then $T(\rho)$ is an elementary abelian 2-group of order 1, 2 or 4.

***Proof*:**
Step 1: exponent 2. Let $\chi \in T(\rho)$. Taking determinants in $\chi \otimes \rho \cong \rho$ and using $\dim \rho = 2$ gives $\chi^2 \det \rho = \det \rho$, so $\chi^2 = 1$. Thus every element of the group $T(\rho)$ has order dividing 2, and $T(\rho)$ is elementary abelian.

Step 2: the order. If $T(\rho)$ is trivial there is nothing to prove, so let $\kappa \in T(\rho)$ with $\kappa \neq 1$.

Being smooth, κ is trivial on some U_F^n, so $\ker\kappa$ is open, and it has index 2 by Step 1. By theorem 6.2.2(1) it is the norm group $N_{E/F}(E^\times)$ of a finite abelian extension E/F, and by theorem 6.2.2(2) that extension has $[E:F] = |\mathrm{Gal}(E/F)| = 2$. Hence $\kappa = \kappa_{E/F}$ in the notation of definition 8.3.1, and lemma 9.1.14 presents $\rho \cong \mathrm{Ind}_{E/F}\xi$ for a character ξ of $E^\times$.

By lemma 9.1.12 an element $\eta \in T(\rho)$ satisfies $\eta_E = 1$ or $\eta_E = \xi^\sigma/\xi$. The characters η of $F^\times$ with $\eta_E = 1$ are those trivial on $N_{E/F}(E^\times)$, a subgroup of index 2, so there are exactly two of them, namely 1 and κ. If η and η' both satisfy $\eta_E = \eta'_E = \xi^\sigma/\xi$, then $(\eta/\eta')_E = 1$, so $\eta/\eta' \in \{1, \kappa\}$ and there are at most two such η. Hence $|T(\rho)| \le 4$, and being an elementary abelian 2-group containing $\{1, \kappa\}$ its order is 2 or 4, completing the proof.

The group $T(\rho)$ therefore records exactly the presentations of ρ as an induced representation: its non-trivial elements are the characters $\kappa_{E/F}$ for the quadratic extensions E from which ρ can be induced, and there are none, one or three of them. The next proposition states that correspondence, and with it the criterion for imprimitivity in terms of $T(\rho)$.

Proposition 9.1.16: Imprimitivity Read from the Twisting Group

Let $\rho \in \mathcal{G}_2^0(F)$. Then

1. ρ is imprimitive if and only if $T(\rho) \neq \{1\}$, and primitive if and only if $T(\rho) = \{1\}$;
2. $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$ if and only if $T(\rho)$ contains a non-trivial unramified character.

***Proof*:**
In (1), if ρ is imprimitive then $\rho \cong \mathrm{Ind}_{E/F}\Theta$ for some separable quadratic E/F, and $\kappa_{E/F}$ is a non-trivial element of $T(\rho)$ by lemma 9.1.14. Conversely, if $T(\rho)$ contains $\kappa \neq 1$ then Step 2 of the proof of proposition 9.1.15 produces a separable quadratic E/F with $\kappa = \kappa_{E/F}$ and a presentation $\rho \cong \mathrm{Ind}_{E/F}\xi$, so ρ is imprimitive. The statement about primitivity is the negation, definition 8.5.1 defining primitive as not imprimitive.

In (2), suppose $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$, so $\rho \cong \mathrm{Ind}_{E/F}\Theta$ with E/F unramified. Then $\kappa_{E/F} \in T(\rho)$ by lemma 9.1.14, it is non-trivial, and it is unramified by proposition 9.1.8. Conversely, let $\chi \in T(\rho)$ be non-trivial and unramified. As in Step 2 of the proof of proposition 9.1.15, $\chi = \kappa_{E/F}$ for a separable quadratic E/F, and E/F is unramified by proposition 9.1.8, since a ramified E/F would give $\kappa_{E/F}$ level $d(E|F) \ge 1$. By lemma 9.1.14, $\rho \cong \mathrm{Ind}_{E/F}\xi$ with E/F unramified, that is $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$, completing the proof.

The parameters in $\mathcal{G}_2^{\mathrm{nr}}(F)$ are the ones already reached by the tame parametrization of chapter 4. Those outside it occupy the rest of the chapter, so they get a name.

Definition 9.1.17: Totally Ramified Parameter

A parameter $\rho \in \mathcal{G}_2^0(F)$ is *totally ramified* if $\rho \notin \mathcal{G}_2^{\mathrm{nr}}(F)$, that is, by proposition 9.1.16(2), if $T(\rho)$ contains no non-trivial unramified character.

By proposition 9.1.16(1) a primitive parameter has trivial twisting group, so it is totally ramified. The interesting content of definition 9.1.17 concerns the imprimitive classes, where it says the inducing

extension is forced to be ramified. The parameter of example 9.1.13 is not totally ramified, its twisting group containing the unramified character $\kappa_{E/F}$ for the unramified E.

The level of a character measures how deep in the filtration its non-triviality begins. The corresponding invariant for a two-dimensional parameter is the exponent of q in its local constant, normalized by fixing the additive character.

Definition 9.1.18: Level and Minimality of a Parameter

Let $\rho \in \mathcal{G}_2^0(F)$ and let $\psi \in \widehat{F}$ have level one. The *level* of ρ is

$$\ell(\rho) = n(\rho, \psi)$$

with $n(\rho,\psi)$ the exponent of proposition 6.3.3(1). The parameter ρ is *minimal* if $\ell(\rho) \le \ell(\chi\otimes\rho)$ for every character χ of $F^\times$. For a separable quadratic extension E/F, a character ξ of $E^\times$ is *minimal over* F if $\ell(\xi) \le \ell(\chi_E\xi)$ for every character χ of $F^\times$.

The definition does not depend on which ψ of level one is used. Any two non-trivial additive characters of F differ by multiplication by an element of $F^\times$, by the self-duality of F established in section 5.1, and $a\psi$ has level $1 - v_F(a)$. So two characters of level one differ by an $a \in U_F$, and $n(\rho, a\psi) = n(\rho,\psi) + v_F(a)\dim\rho = n(\rho,\psi)$ by proposition 6.3.3(2).

The word *level* is already in use for characters of $F^\times$ and for the strata of definition 3.3.12; definition 9.1.18 extends the first of these, since for a ramified character χ of $F^\times$ regarded as an element of $\mathcal{G}_1(F)$ one has $n(\chi,\psi) = \ell(\chi)$ by theorem 5.1.18. So a χ trivial on U_F^3 but not on U_F^2 has $n(\chi,\psi) = 2$, its level in either reading. The stratum-theoretic level, by contrast, is an invariant of representations of $\mathrm{GL}_2(F)$ and is unrelated to the present one except through the correspondence.

Lemma 9.1.19: The Level of an Induced Parameter

Let E/F be quadratic and totally ramified with $d = d(E|F)$, let ξ be an E/F-regular character of $E^\times$, and put $\rho = \mathrm{Ind}_{E/F}\xi$. Let $\psi \in \widehat{F}$ have level one. Then $\psi_E = \psi\circ\mathrm{tr}_{E/F}$ has level $1-d$, and

$$\ell(\rho) = \ell(\xi) + d$$

***Proof*:**

Step 1: the level of ψ_E. By proposition 9.1.4(1), $\psi_E(\mathfrak{p}_E^j) = \psi\big(\mathfrak{p}^{\lfloor (j+d+1)/2\rfloor}\big)$, which is trivial exactly when $\lfloor (j+d+1)/2\rfloor \ge 1$, that is when $j+d+1 \ge 2$, that is when $j \ge 1-d$. The least such j is $1-d$.

Step 2: $n(\rho,\psi) = n(\xi,\psi_E)$. Apply (6.3) with $n=1$ and $K=F$:

$$\varepsilon\big(\mathrm{Ind}_{E/F}\xi, s, \psi\big) = \varepsilon(\xi, s, \psi_E)\,\lambda_{E/F}(s,\psi)$$

with $\lambda_{E/F}$ the Langlands constant of definition 6.3.10. By corollary 6.3.11 the factor $\lambda_{E/F}(s,\psi)$ does not depend on s. The residue field of E is k, so $q_E = q$, and both ε-factors are monomials in q^{-s} with exponents recorded by proposition 6.3.3(1). Comparing the exponents of q^{-s} on the two sides gives $n(\rho,\psi) = n(\xi,\psi_E)$.

Step 3: $n(\xi,\psi_E)=\ell(\xi)+d$. First, ξ is ramified: an unramified ξ is trivial on U_E, hence on $\mathcal{K}_{E/F}\subseteq U_E$, hence satisfies $\xi^\sigma=\xi$ by lemma 8.4.2, contradicting regularity. Choose $\psi'_E\in\widehat{E}$ of level one. By the self-duality of E there is $a\in E^\times$ with $\psi_E=a\psi'_E$, and comparing levels gives $1-v_E(a)=1-d$, so $v_E(a)=d$. Now proposition 6.3.3(2), applied over E to the one-dimensional ξ, gives $n(\xi,\psi_E)=n(\xi,\psi'_E)+v_E(a)$, and $n(\xi,\psi'_E)=\ell(\xi)$ by theorem 5.1.18 applied over E. Hence $n(\xi,\psi_E)=\ell(\xi)+d$, which with Step 2 completes the proof.

The formula $\ell(\rho)=\ell(\xi)+d$ turns every question about the level of ρ into a question about the level of ξ. Which levels a regular ξ can have, and which of those are minimal, is settled by the filtration results of the first half of this section.

Lemma 9.1.20: Levels of Regular and Minimal Characters

Let E/F be quadratic and totally ramified with $d=d(E|F)\geq 1$, and let ξ be a character of $E^\times$.

1. If ξ is E/F-regular, then $\ell(\xi)\geq d$.
2. If ξ is E/F-regular with $\ell(\xi)=d$, then ξ^σ/ξ is unramified and non-trivial, and $\mathrm{Ind}_{E/F}\xi\in\mathcal{G}_2^{\mathrm{nr}}(F)$.
3. If $\ell(\xi)\geq d+1$, then ξ is minimal over F if and only if $\ell(\xi)\not\equiv d \bmod 2$.

***Proof*:**

(1). By lemma 8.4.2 regularity means $\xi|_{\mathcal{K}_{E/F}}\neq 1$, and $\mathcal{K}_{E/F}\subseteq U_E^d$ by lemma 9.1.6(3). So ξ is non-trivial on U_E^d. A character of level $n\leq d-1$ is trivial on $U_E^{n+1}\supseteq U_E^d$, so $\ell(\xi)\geq d$.

(2). Let $\ell(\xi)=d$, so ξ is trivial on U_E^{d+1}. For $y\in U_E^1$ the element y/y^σ lies in U_E^{1+d} by lemma 9.1.6(1), so $(\xi^\sigma/\xi)(y)=\xi(y^\sigma/y)=1$. And $y/y^\sigma=1$ for $y\in U_F$. Since $U_E=U_FU_E^1$, as noted in the proof of lemma 9.1.6, the character ξ^σ/ξ is trivial on U_E, that is unramified. It is non-trivial because ξ is regular. An unramified character η of $E^\times$ is χ_E for an unramified character χ of $F^\times$: choose χ unramified with $\chi(\varpi_F)=\eta(\varpi_E)$, where ϖ_E is a prime element of E. Then $\chi_E(\varpi_E)=\chi(N_{E/F}\varpi_E)=\chi(\varpi_F)$ because $v_F(N_{E/F}\varpi_E)=1$ and χ kills U_F, and both characters are unramified, hence determined by their value at ϖ_E. The representation $\mathrm{Ind}_{E/F}\xi$ is irreducible because ξ is regular, by lemma 8.5.2, so lemma 9.1.12 applies to it. Taking $\eta=\xi^\sigma/\xi$ it puts the non-trivial unramified χ in $T(\mathrm{Ind}_{E/F}\xi)$, so $\mathrm{Ind}_{E/F}\xi\in\mathcal{G}_2^{\mathrm{nr}}(F)$ by proposition 9.1.16(2).

(3). Write $l=\ell(\xi)\geq d+1$ and let φ be as in lemma 9.1.7. The character ξ fails to be minimal over F exactly when some χ has $\ell(\chi_E\xi)<l$, which makes $\chi_E\xi$ trivial on U_E^l and forces χ_E to agree with ξ^{-1} on U_E^l and to be trivial on U_E^{l+1}. Since $\xi|_{U_E^l}\neq 1$, in particular $\ell(\chi_E)=l$.

Suppose first $l\not\equiv d \bmod 2$. By lemma 9.1.9 the possible values of $\ell(\chi_E)$ are: the value 0 for unramified χ, since then χ_E is unramified. The value $\varphi(j)=j\leq d-1$ when $\ell(\chi)=j\leq d-1$. The value $\varphi(j)=2j-d$ when $\ell(\chi)=j\geq d+1$. And, when $\ell(\chi)=d$, either d or a value at most $d-1$. All values that exceed d are of the form $2j-d\equiv d \bmod 2$, so none of them equals l. No value at most d equals $l\geq d+1$. Hence no χ lowers the level and ξ is minimal.

Suppose instead $l \equiv d \bmod 2$, so $l \geq d+2$ and $j = (l+d)/2$ is an integer with $j \geq d+1$ and $\varphi(j) = l$. By lemma 9.1.7(2) the map $\nu_j \colon U_E^l/U_E^{l+1} \to U_F^j/U_F^{j+1}$ is an isomorphism. The restriction $\xi^{-1}|_{U_E^l}$ is trivial on U_E^{l+1}, so it induces a character $\bar{\xi}^{-1}$ of U_E^l/U_E^{l+1}; let $\bar{\eta} = \bar{\xi}^{-1} \circ \nu_j^{-1}$, a character of U_F^j/U_F^{j+1}, inflate it to U_F^j and extend it to a character χ of $F^\times$ trivial on U_F^{j+1}, which is possible because $\mathbb{C}^\times$ is divisible and U_F^{j+1} is open. Then χ_E and ξ^{-1} agree on U_E^l by construction, so $\chi_E\xi$ is trivial on U_E^l and $\ell(\chi_E\xi) \leq l-1 < l$. Hence ξ is not minimal over F, and the two implications together give (3), completing the proof.

Transported to the induced parameter, these constraints decide membership in $\mathcal{G}_2^{\mathrm{nr}}(F)$, and over a range of levels they force the twisting group to be as large as possible.

Proposition 9.1.21: Levels of a Minimal Induced Parameter

Let E/F be quadratic and totally ramified with $d = d(E|F)$, let ξ be an E/F-regular character of $E^\times$ that is minimal over F, of level l, and put $\rho = \mathrm{Ind}_{E/F}\xi \in \mathcal{G}_2^0(F)$.

1. $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$ if and only if $l = d$.
2. If $d < l < 2d$, then ρ is triply imprimitive and $T(\rho)$ contains a character of level $l-d$.

***Proof*:**
By lemma 9.1.20(1), regularity gives $l \geq d$.

Step 1: the level of ξ^σ/ξ when $l > d$. Minimality and lemma 9.1.20(3) give $l \not\equiv d \bmod 2$, so $l-d$ is an odd positive integer, say $l-d = 2m+1$. By lemma 9.1.6(2) the map $y \mapsto y/y^\sigma$ induces an isomorphism $U_E^{l-d}/U_E^{l-d+1} \to U_E^l/U_E^{l+1}$. Consequently, for $y \in U_E^{l-d+1}$ one has $y/y^\sigma \in U_E^{l+1}$ and hence $(\xi^\sigma/\xi)(y) = \xi(y^\sigma/y) = 1$, while surjectivity of the induced map and $\xi|_{U_E^l} \neq 1$ produce $y \in U_E^{l-d}$ with $(\xi^\sigma/\xi)(y) \neq 1$. Therefore

$$\ell\left(\xi^\sigma/\xi\right) = l - d \geq 1$$

so ξ^σ/ξ is ramified.

Step 2: part (1). If $l = d$ then $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$ by lemma 9.1.20(2). Suppose conversely $l \neq d$, so $l > d$ and Step 1 applies. If ρ were in $\mathcal{G}_2^{\mathrm{nr}}(F)$, then by proposition 9.1.16(2) there would be a non-trivial unramified $\chi \in T(\rho)$, and lemma 9.1.12 would give $\chi_E = 1$ or $\chi_E = \xi^\sigma/\xi$. The first forces $\chi \in \{1, \kappa_{E/F}\}$ as in the proof of proposition 9.1.15, and $\kappa_{E/F}$ has level $d \geq 1$ by proposition 9.1.8, so is ramified: both options contradict χ being unramified and non-trivial. The second is impossible because χ_E is unramified by lemma 9.1.9(1) while ξ^σ/ξ is ramified by Step 1. Hence $\rho \notin \mathcal{G}_2^{\mathrm{nr}}(F)$.

Step 3: part (2). Assume $d < l < 2d$ and put $d' = l - d$, so $1 \leq d' \leq d-1$. By Step 1, ξ^σ/ξ has level $d' < d$, hence is trivial on $U_E^d \supseteq \mathcal{K}_{E/F}$ using lemma 9.1.6(3). Being trivial on $\mathcal{K}_{E/F} = \ker N_{E/F}$, it factors through $E^\times/\mathcal{K}_{E/F} \cong N_{E/F}(E^\times)$; extending along the open subgroup $N_{E/F}(E^\times) \subseteq F^\times$, which is possible because $\mathbb{C}^\times$ is divisible, gives a character η of $F^\times$ with $\eta_E = \xi^\sigma/\xi$.

The level of η is d', after replacing η by $\kappa_{E/F}\eta$ if necessary. Indeed η is ramified, since an unramified η would make η_E unramified by lemma 9.1.9(1) while $\ell(\eta_E) = d' \geq 1$. Write $j = \ell(\eta)$. If $j \leq d-1$ then $\ell(\eta_E) = j$ by lemma 9.1.9(2), so $j = d'$. If $j \geq d+1$ then $\ell(\eta_E) = 2j - d \geq d+2$, contradicting $\ell(\eta_E) = d' \leq d-1$. If $j = d$, then $\ell(\eta_E) = d' < d$ forces, by lemma 9.1.9(3), that $\kappa_{E/F}\eta$ has level at most $d-1$. Since $(\kappa_{E/F}\eta)_E = \eta_E$ still has level d', the case $j \leq d-1$ applied to $\kappa_{E/F}\eta$ gives $\ell(\kappa_{E/F}\eta) = d'$, and $\kappa_{E/F}\eta$ replaces η.

By lemma 9.1.12, $\eta \in T(\rho)$. It is not 1, having level $d' \geq 1$, and it is not $\kappa_{E/F}$, which has level $d > d'$ by proposition 9.1.8. So $T(\rho)$ contains the four distinct elements 1, $\kappa_{E/F}$, η, $\kappa_{E/F}\eta$, and proposition 9.1.15 caps its order at 4. Hence $|T(\rho)| = 4$, so ρ is triply imprimitive and $T(\rho)$ contains the character η of level $l - d$, as we sought to show.

Example 9.1.22: A Minimal Totally Ramified Parameter over $\mathbb{Q}_2$

Let $F = \mathbb{Q}_2$ and $E = \mathbb{Q}_2(\sqrt{2})$, so $d = d(E|F) = 2$ by example 9.1.3. Every character ξ of $E^\times$ of level 3 is E/F-regular. Indeed, by lemma 9.1.6(2) with $m = 0$ the map $y \mapsto y/y^\sigma$ induces an isomorphism $U_E^1/U_E^2 \to U_E^3/U_E^4$, so choosing $y \in U_E^1 \setminus U_E^2$ produces an element y/y^σ of $\mathcal{K}_{E/F}$ lying in $U_E^3 \setminus U_E^4$. And $U_E^3/U_E^4 \cong k$ has order 2, so a character of level 3, being non-trivial on U_E^3 and trivial on U_E^4, is non-trivial on that element. Lemma 8.4.2 then gives regularity.

Fix such a ξ and put $\rho = \mathrm{Ind}_{E/F}\xi$. Since $\ell(\xi) = 3 \geq d+1 = 3$ and $3 \not\equiv 2 \bmod 2$, the character ξ is minimal over F by lemma 9.1.20(3). As $\ell(\xi) = 3 \neq 2 = d$, proposition 9.1.21(1) puts ρ outside $\mathcal{G}_2^{\mathrm{nr}}(F)$, so ρ is totally ramified. By lemma 9.1.19,

$$\ell(\rho) = \ell(\xi) + d = 3 + 2 = 5$$

The inequalities $d < \ell(\xi) < 2d$ read $2 < 3 < 4$ and hold, so proposition 9.1.21(2) applies: ρ is triply imprimitive and $T(\rho)$ contains a character of level $\ell(\xi) - d = 1$ alongside $\kappa_{E/F}$, which has level 2.

Minimality of a totally ramified imprimitive parameter can now be decided by a parity condition on $\ell(\rho)$ alone, with no reference to the inducing extension or character.

Lemma 9.1.23: Minimality Is Oddness of the Level

Let $\rho \in \mathcal{G}_2^0(F)$ be totally ramified and imprimitive. Then ρ is minimal if and only if $\ell(\rho)$ is odd.

***Proof*:**

By proposition 9.1.16(1) and lemma 9.1.14 there are a separable quadratic E/F and a character ξ of $E^\times$, necessarily E/F-regular, with $\rho \cong \mathrm{Ind}_{E/F}\xi$. The extension E/F is not unramified, since that would place ρ in $\mathcal{G}_2^{\mathrm{nr}}(F)$ against definition 9.1.17. So E/F is totally ramified and $d = d(E|F) \geq 1$ by proposition 9.1.2.

Step 1: minimality of ρ is minimality of ξ. For a character χ of $F^\times$, lemma 9.1.10(1) gives $\chi \otimes \rho \cong \mathrm{Ind}_{E/F}(\chi_E\xi)$, which is irreducible because ρ is, so $\chi_E\xi$ is E/F-regular by lemma 8.5.2.

Lemma 9.1.19 therefore applies to both ξ and $\chi_E\xi$ and gives

$$\ell(\rho) = \ell(\xi) + d, \qquad \ell(\chi \otimes \rho) = \ell(\chi_E\xi) + d$$

So $\ell(\rho) \leq \ell(\chi \otimes \rho)$ for all χ if and only if $\ell(\xi) \leq \ell(\chi_E\xi)$ for all χ.

Step 2: the parity. By lemma 9.1.20(1), $\ell(\xi) \geq d$. And $\ell(\xi) = d$ is impossible, since lemma 9.1.20(2) would then place ρ in $\mathcal{G}_2^{\mathrm{nr}}(F)$. Hence $\ell(\xi) \geq d+1$ and lemma 9.1.20(3) applies: ξ is minimal over F if and only if $\ell(\xi) \not\equiv d \bmod 2$, that is if and only if $\ell(\xi) + d = \ell(\rho)$ is odd. With Step 1 this completes the proof.

The results above bound the levels available to the twisting group. A minimal totally ramified imprimitive parameter always admits a twist by a character of level at most $\ell(\rho)/3$. What uses that bound downstream is corollary 9.1.25, whose two level configurations section 9.2 uses when it restricts a primitive parameter to a cubic extension.

Theorem 9.1.24: Level Bound for the Twisting Group

Let $\rho \in \mathcal{G}_2^0(F)$ be totally ramified, minimal and imprimitive. Then $T(\rho)$ contains a character χ of level $m \geq 1$ with

$$3m \leq \ell(\rho)$$

***Proof*:**
Write $n = \ell(\rho)$. By proposition 9.1.16(1) the group $T(\rho)$ contains some $\chi \neq 1$, and by Step 2 of the proof of proposition 9.1.15 that χ is $\kappa_{E/F}$ for a separable quadratic E/F, with $\rho \cong \mathrm{Ind}_{E/F}\xi$ for a character ξ of $E^\times$ furnished by lemma 9.1.14. The extension E/F is ramified, since an unramified one would place ρ in $\mathcal{G}_2^{\mathrm{nr}}(F)$ against definition 9.1.17. So $d = d(E|F) \geq 1$ by proposition 9.1.2(2) and $\ell(\chi) = d$ by proposition 9.1.8.

By lemma 9.1.19, $\ell(\xi) = n - d$, and by Step 1 of the proof of lemma 9.1.23 the minimality of ρ is the minimality of ξ over F. Step 2 of that proof gives $\ell(\xi) \geq d+1$, so $\ell(\xi) > d$.

If $\ell(\xi) \geq 2d$, then $n = \ell(\xi) + d \geq 3d$ and the character χ itself, of level $m = d \geq 1$, satisfies the bound. Otherwise $d < \ell(\xi) < 2d$, and proposition 9.1.21(2) provides $\eta \in T(\rho)$ of level $m = \ell(\xi) - d$, with $1 \leq m \leq d-1$. Then

$$n = \ell(\xi) + d = m + 2d > m + 2m = 3m$$

using $d > m$. So η satisfies the bound, completing the proof.

In the triply imprimitive case the argument says more: it pins the level of ρ to the levels of the three non-trivial twists, and leaves exactly two configurations.

Corollary 9.1.25: Level Configurations in the Triply Imprimitive Case

Let $\rho \in \mathcal{G}_2^0(F)$ be totally ramified, triply imprimitive and minimal, of level n, and let χ_1, χ_2, χ_3 be the non-trivial elements of $T(\rho)$, of levels $m_i = \ell(\chi_i)$. After renumbering, exactly one of the following holds.

1. $m_1 < m_2 = m_3$ and $n = 2m_2 + m_1$. In particular $3m_2 > n > 3m_1$.
2. $m_1 = m_2 = m_3 = m$ and $n = 3m$, and the restrictions $\chi_i|_{U_F^m}$ are pairwise distinct.

***Proof*:**
By proposition 9.1.15 the group $T(\rho)$ is elementary abelian of order 4, so $\chi_3 = \chi_1\chi_2$ and likewise for any renumbering. Each χ_i equals $\kappa_{E_i/F}$ for a separable quadratic E_i/F, by Step 2 of the proof of proposition 9.1.15, and E_i/F is ramified because ρ is totally ramified. So $m_i = d(E_i|F) \geq 1$ by proposition 9.1.8.

Step 1: only two patterns of levels. Order the indices so that $m_1 \leq m_2 \leq m_3$. If $m_1 < m_2$, then $\chi_1\chi_2$ is non-trivial on $U_F^{m_2}$ exactly when χ_2 is, because χ_1 is trivial there. Hence $\ell(\chi_1\chi_2) = m_2$ and, $\chi_1\chi_2$ being χ_3, the pattern is $m_1 < m_2 = m_3$. Otherwise $m_1 = m_2 \leq m_3$, and here $m_3 > m_2$ is impossible: the same argument applied to the pair χ_2, χ_3 would give $\ell(\chi_2\chi_3) = m_3$, while $\chi_2\chi_3 = \chi_1$ has level $m_1 = m_2 < m_3$. So $m_3 = m_1$ and all three levels are equal.

Step 2: the key level computation. Fix i and write $E = E_i$, $d = m_i$. By lemma 9.1.14 there is an E/F-regular ξ with $\rho \cong \mathrm{Ind}_{E/F}\xi$, and by Steps 1 and 2 of the proof of lemma 9.1.23 the character ξ is minimal over F with $\ell(\xi) = n - d \geq d + 1$. Step 1 of the proof of proposition 9.1.21 then gives

$$\ell\big(\xi^\sigma/\xi\big) = \ell(\xi) - d = n - 2d$$

For any $j \neq i$ the character χ_j lies in $T(\rho)$ and is neither 1 nor $\kappa_{E/F} = \chi_i$, so $(\chi_j)_E \neq 1$ by the computation of $\ker(\eta \mapsto \eta_E)$ in the proof of proposition 9.1.15, and lemma 9.1.12 forces $(\chi_j)_E = \xi^\sigma/\xi$. Hence $\ell\big((\chi_j)_E\big) = n - 2m_i$ for every $j \neq i$.

Step 3: the case $m_1 < m_2 = m_3$. Take $i = 1$ and $j = 2$, so $d = m_1$ and $\ell(\chi_2) = m_2 > m_1 = d$. By lemma 9.1.9(2), $\ell\big((\chi_2)_E\big) = 2m_2 - m_1$. Comparing with Step 2 gives $n - 2m_1 = 2m_2 - m_1$, that is $n = 2m_2 + m_1$. Then $3m_2 - n = m_2 - m_1 > 0$ and $n - 3m_1 = 2(m_2 - m_1) > 0$, which is the stated pair of inequalities.

Step 4: the case $m_1 = m_2 = m_3 = m$. The restrictions $\chi_i|_{U_F^m}$ are pairwise distinct: if χ_1 and χ_2 agreed on U_F^m, then $\chi_3 = \chi_1\chi_2$ would be trivial on U_F^m and so have level at most $m - 1$, against $m_3 = m$. Now take $i = 1$ and $j = 2$, so $E = E_1$ has $d(E|F) = m = \ell(\chi_2)$. Since $\chi_2|_{U_F^m} \neq \chi_1|_{U_F^m} = \kappa_{E/F}|_{U_F^m}$, lemma 9.1.9(3) gives $\ell\big((\chi_2)_E\big) = m$. Comparing with Step 2 gives $n - 2m = m$, that is $n = 3m$. The two cases are mutually exclusive by Step 1, completing the proof.

The corollary identifies the extremal case of theorem 9.1.24 among triply imprimitive parameters. In configuration (1) the smallest twist level satisfies $3m_1 < n$ strictly, while in configuration (2) all three twist levels equal $n/3$. So a triply imprimitive ρ attains the equality $3m = \ell(\rho)$ exactly when

its three non-trivial twists have a common level.

Equality is not confined to that case, and the corollary does not characterize it. Whenever ρ is minimal and totally ramified with $\rho \cong \mathrm{Ind}_{E/F}\xi$ and $\ell(\xi) = 2d$, $d = d(E|F)$, the first case in the proof of theorem 9.1.24 returns $\ell(\rho) = 3d$ together with the twist $\kappa_{E/F}$ of level d, and such a ρ may be only simply imprimitive. Take $F = \mathbb{Q}_2$, $E = \mathbb{Q}_2(i)$ with $d = 1$ by example 9.1.3, and ξ an E/F-regular character of level 2: it is minimal over F by lemma 9.1.20(3), and ρ is totally ramified of level 3 by proposition 9.1.21(1) and lemma 9.1.19, hence minimal by lemma 9.1.23. Step 1 of the proof of proposition 9.1.21 gives $\ell(\xi^\sigma/\xi) = \ell(\xi) - d = 1$. Since $q = 2$ the group U_E^1/U_E^2 has order 2 and is therefore the whole image of $\mathcal{K}_{E/F}$ given by lemma 9.1.6(3), so ξ^σ/ξ, being non-trivial on U_E^1 and trivial on U_E^2, is non-trivial on $\mathcal{K}_{E/F} = \ker N_{E/F}$ and is not η_E for any character η of $F^\times$. By lemma 9.1.12, $T(\rho) = \{1, \kappa_{E/F}\}$, and the bound $3m \leq \ell(\rho)$ is attained with a single non-trivial twist.

Exercise 9.1.1

1. Let $F = \mathbb{Q}_2$. Show that $\sqrt{3} - 1$ is a prime element of $E = \mathbb{Q}_2(\sqrt{3})$ and that $\sqrt{6}$ is a prime element of $E' = \mathbb{Q}_2(\sqrt{6})$. Compute the minimal polynomials of these two elements, and deduce $d(E|F)$, $d(E'|F)$, and the levels of $\kappa_{E/F}$ and $\kappa_{E'/F}$.

2. Let $F = \mathbb{Q}_2$ and $E = \mathbb{Q}_2(i)$, so that $d(E|F) = 1$ by example 9.1.3. Compute $\mathrm{tr}_{E/F}(\mathfrak{p}_E^j)$ for $j = -1, 0, 1, 2$, both from proposition 9.1.4(1) and directly from the $\mathbb{Z}_2$-basis $(1, i)$ of $\mathfrak{o}_E$, and say for which j the trace induces an isomorphism $\mathfrak{p}_E^j/\mathfrak{p}_E^{j+1} \to \mathfrak{p}^m/\mathfrak{p}^{m+1}$.

3. Let χ be a character of $F^\times$ with $\chi^2 = 1$. Show that χ is either unramified or of level at least 1, so that no quadratic character of a dyadic field is ramified of level 0. Deduce that in proposition 9.1.8 the value $d(E|F) \geq 1$ could not have been 0.

4. Let $\rho \in \mathcal{G}_2^0(F)$ and let χ be a character of $F^\times$. Show that $T(\chi \otimes \rho) = T(\rho)$. Deduce that whether ρ is primitive, simply imprimitive or triply imprimitive, and whether ρ is totally ramified, depend only on the family $\{\chi \otimes \rho : \chi \text{ a character of } F^\times\}$.

5. Let $F = \mathbb{Q}_2$, $E = \mathbb{Q}_2(\sqrt{2})$, and let ξ be an E/F-regular character of $E^\times$ of level 5. Show that ξ is minimal over F, that $\rho = \mathrm{Ind}_{E/F}\xi$ is totally ramified of level 7, and that ρ is minimal. Show that proposition 9.1.21(2) does not apply here, and exhibit an element of $T(\rho)$ realizing the bound of theorem 9.1.24.

6. Let E/F be a separable quadratic extension. Show that $\ker\left(\kappa_{E/F} \circ \mathbf{a}_F\right) = \mathcal{W}_E$. Deduce that if E and E' are separable quadratic extensions of F inside the fixed separable closure with $\kappa_{E/F} = \kappa_{E'/F}$, then $\mathcal{W}_E = \mathcal{W}_{E'}$.

7. In the situation of corollary 9.1.25, order the levels so that $m_1 \leq m_2 \leq m_3$. Show that $n = 2m_3 + m_1$ in both cases, and that m_1 is odd. Show further that in case (2) all three levels are odd.

8. Let $\rho \in \mathcal{G}_2^0(F)$ be totally ramified and imprimitive, and let E/F be a quadratic extension with $\rho \cong \mathrm{Ind}_{E/F}\xi$. Show that $\ell(\rho) \geq 2d(E|F) + 1$, with equality if and only if $\ell(\xi) = d(E|F) + 1$.

9.2 Primitive Representations

Section 9.1 read the imprimitive classes through the group $T(\rho)$, and everything there began from a quadratic extension E/F and a character of $\mathcal{W}_E$. The complementary class is the one admitting no such presentation, so that $T(\rho)$ is trivial and carries no information at all. Recall from section 9.1 that $\rho \in \mathcal{G}_2^0(F)$ is *totally ramified* when $\rho \notin \mathcal{G}_2^{\mathrm{nr}}(F)$, and that $T(\rho)$ is regarded as a group of characters of $F^\times$ through the reciprocity map $\mathbf{a}_F$ of theorem 6.2.1.

The substitute for a quadratic presentation is a cubic one, taken in the opposite direction. For a finite extension K/F write $\rho_K = \rho|_{\mathcal{W}_K}$. This section produces, for each primitive ρ, a cubic extension K/F for which ρ_K *is* imprimitive, shows that K is determined by ρ up to F-isomorphism, and identifies the finite group through which the associated projective representation of $\mathcal{W}_F$ factors: it is A_4 or S_4, according to whether K/F is Galois. Throughout, S_n denotes the symmetric group on n letters and A_n the alternating subgroup.

Definition 9.2.1: Primitive Parameter

Let $\rho \in \mathcal{G}_2^0(F)$. Following definition 8.5.1, ρ is *primitive* when it is not imprimitive: there is no separable quadratic extension E/F and character Θ of $E^\times$ with $\rho \cong \mathrm{Ind}_{E/F}\Theta$. Equivalently, ρ lies in $\mathcal{G}_2^0(F) \setminus \mathcal{G}_2^{\mathrm{im}}(F)$.

The definition is a negation. What makes it usable is the following translation into a statement about $T(\rho)$, the invariant this chapter computes with.

Proposition 9.2.2: Primitivity and the Twisting Group

Let $\rho \in \mathcal{G}_2^0(F)$.

1. ρ is primitive if and only if $T(\rho) = \{1\}$.
2. A primitive ρ is totally ramified.
3. If ρ is totally ramified, then every non-trivial element of $T(\rho)$ is wildly ramified, that is, non-trivial on U_F^1.

***Proof*:**

1. Primitivity is the negation of imprimitivity, so this is proposition 9.1.16(1): the class ρ is imprimitive if and only if $T(\rho) \neq \{1\}$. Recall from the proof there that a non-trivial $\chi \in T(\rho)$ is $\kappa_{E/F}$ for the unique separable quadratic E/F with $\ker\chi = N_{E/F}(E^\times)$.
2. A class in $\mathcal{G}_2^{\mathrm{nr}}(F)$ is by definition 8.5.1 of the form $\mathrm{Ind}_{E/F}\Theta$ with E/F unramified quadratic, hence imprimitive. So a primitive ρ lies outside $\mathcal{G}_2^{\mathrm{nr}}(F)$, which is the assertion.
3. Let $\chi \in T(\rho)$ with $\chi \neq 1$, so $\chi^2 = 1$ by proposition 9.1.15. If χ were unramified, then $\chi = \kappa_{E/F}$ with E/F unramified quadratic, by the identification in (1) together with proposition 9.1.8, and lemma 9.1.14 would present ρ as $\mathrm{Ind}_{E/F}\xi$ with E/F unramified, placing ρ in $\mathcal{G}_2^{\mathrm{nr}}(F)$ and contradicting total ramification. So $\chi|_{U_F} \neq 1$. Suppose χ were trivial on U_F^1. Then $\chi|_{U_F}$ factors through $U_F/U_F^1 \cong k^\times$, a cyclic group of order $q-1$.

Since $p = 2$, the integer q is a power of two and $q - 1$ is odd, so a character of $k^\times$ satisfying $\chi^2 = 1$ is trivial. That contradicts $\chi|_{U_F} \neq 1$, so $\chi|_{U_F^1} \neq 1$, completing the proof.

The trichotomy of section 9.1 is therefore complete: $|T(\rho)|$ is 1, 2 or 4, and the first case is exactly the primitive one. Everything the previous section proves about levels and twisting is silent here, since it starts from a non-trivial twisting character. A different mechanism is needed, and it is developed under a hypothesis that this chapter does not discharge.

9.2.3 Note: Primitive Classes Are Not Shown To Exist Nothing in this chapter proves that $\mathcal{G}_2^0(F) \setminus \mathcal{G}_2^{\mathrm{im}}(F)$ is non-empty. Every result of this section is conditional in form: *if* ρ is primitive, *then* its restrictions behave as described. By proposition 9.2.17 a primitive ρ must have projective image A_4 or S_4, so producing one requires at least a two-dimensional representation of $\mathcal{W}_F$ with that projective image; whether every such representation is primitive is a converse this section does not prove, so exhibiting one is necessary rather than known to be sufficient. Either way it requires knowing which such extensions of a dyadic field occur. The Langlands correspondence settles the question by transporting it to the cuspidal side, where the classification of cuspidal representations establishes the count; see [2, §§46-52]. The structure theory below is what that later argument rests on, so it is developed first.

Two of the three main arguments below produce an induced presentation of a two-dimensional representation out of a normal subgroup acting by two distinct characters. In that situation the quadratic extension and the character to induce from are always available.

Lemma 9.2.4: Induction from a Normal Subgroup

Let K/F be a finite separable extension, let π be an irreducible smooth representation of $\mathcal{W}_K$ on a two-dimensional complex space V, and let N be a closed normal subgroup of $\mathcal{W}_K$ such that $\pi|_N$ is the direct sum of two *distinct* characters of N. Then there are a separable quadratic extension K_1/K with $N \subseteq \mathcal{W}_{K_1}$ and a character Θ of $K_1^\times$ with

$$\pi \cong \mathrm{Ind}_{K_1/K}\Theta$$

***Proof*:**
Write $\pi|_N = \alpha_1 \oplus \alpha_2$ with $\alpha_1 \neq \alpha_2$, and let $\mathbb{C}v_1, \mathbb{C}v_2$ be the corresponding eigenlines, so $V = \mathbb{C}v_1 \oplus \mathbb{C}v_2$ and these are the only N-stable lines in V.

For $g \in \mathcal{W}_K$ and $h \in N$ one has $\pi(h)\pi(g)v_i = \pi(g)\pi(g^{-1}hg)v_i = \alpha_i(g^{-1}hg)\,\pi(g)v_i$, using $g^{-1}Ng = N$. So $\pi(g)\mathbb{C}v_i$ is again an N-stable line, and $\mathcal{W}_K$ permutes the two-element set $\{\mathbb{C}v_1, \mathbb{C}v_2\}$. If both lines were fixed by every g, then $\mathbb{C}v_1$ would be a proper non-zero $\mathcal{W}_K$-subspace, contradicting irreducibility. Let $S \subseteq \mathcal{W}_K$ be the stabiliser of $\mathbb{C}v_1$. It is then a subgroup of index two. It is open: π is smooth, so $\ker\pi$ is open, and the permutation of $\{\mathbb{C}v_1, \mathbb{C}v_2\}$ induced by g depends only on the coset $g\ker\pi$, so S is a union of cosets of $\ker\pi$. An open subgroup of index two in $\mathcal{W}_K$ is $\mathcal{W}_{K_1}$ for a separable quadratic extension K_1/K, and $N \subseteq S = \mathcal{W}_{K_1}$ because N fixes each eigenline.

The group $\mathcal{W}_{K_1}$ fixes $\mathbb{C}v_1$, hence also $\mathbb{C}v_2$, so it acts on $\mathbb{C}v_1$ by a smooth character θ, and the

projection $V \to \mathbb{C}v_1$ with kernel $\mathbb{C}v_2$ is a non-zero $\mathcal{W}_{K_1}$-map $\pi|_{\mathcal{W}_{K_1}} \to \theta$. Frobenius reciprocity (theorem 1.2.21) converts it into a non-zero $\mathcal{W}_K$-map $\pi \to \mathrm{Ind}_{K_1/K}\theta$, which is injective because π is irreducible. Both spaces have dimension two, so the map is an isomorphism. Finally θ is a smooth character of $\mathcal{W}_{K_1}$, hence of the form $\Theta \circ \mathbf{a}_{K_1}$ for a character Θ of $K_1^\times$ by theorem 6.2.1(1), as we sought to show.

The next lemma is what makes tameness the right hypothesis everywhere below. It says that total ramification of ρ is visible already on the wild inertia subgroup $\mathcal{P}_F$, and $\mathcal{P}_F$ is precisely the part of $\mathcal{W}_F$ that a tamely ramified extension leaves untouched.

Lemma 9.2.5: Total Ramification Detected on Wild Inertia

Let $\rho \in \mathcal{G}_2^0(F)$. Then ρ is totally ramified if and only if $\rho|_{\mathcal{P}_F}$ is irreducible.

***Proof*:**
Suppose first that $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$, so $\rho \cong \mathrm{Ind}_{E/F}\Theta$ with E/F unramified quadratic by definition 8.5.1. By (8.29) the restriction $\rho|_{\mathcal{W}_E}$ is the sum of the two characters Θ and Θ^σ of $\mathcal{W}_E$. The field E is unramified over F, hence contained in the maximal tamely ramified extension E_∞ of F, so $\mathcal{P}_F = \mathrm{Gal}(\overline{F}/E_\infty) \subseteq \mathcal{W}_E$. Restricting further, $\rho|_{\mathcal{P}_F}$ is a sum of two characters and is reducible.

Conversely, suppose $\rho|_{\mathcal{P}_F}$ is reducible. The claim is that $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$. The group $\mathcal{P}_F$ is compact and ρ is smooth, so $\ker\rho$ is open and $\rho(\mathcal{P}_F)$ is finite. A complex representation of a finite group is semisimple, so $\rho|_{\mathcal{P}_F} = \beta_1 \oplus \beta_2$ for characters β_i of $\mathcal{P}_F$. The subgroup $\mathcal{P}_F = \mathrm{Gal}(\overline{F}/E_\infty)$ is normal in $\mathcal{W}_F$, since E_∞/F is Galois.

Case 1: $\beta_1 \neq \beta_2$. Lemma 9.2.4, applied with $K = F$ and $N = \mathcal{P}_F$, gives a separable quadratic E/F with $\mathcal{P}_F \subseteq \mathcal{W}_E$ and a character Θ of $E^\times$ with $\rho \cong \mathrm{Ind}_{E/F}\Theta$. From $\mathcal{P}_F = \mathrm{Gal}(\overline{F}/E_\infty) \subseteq \mathcal{W}_E$ one gets $E \subseteq E_\infty$, so E/F is tamely ramified and $e(E|F)$ divides 2 while being prime to $p = 2$. Hence $e(E|F) = 1$, the extension E/F is unramified, and $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$.

Case 2: $\beta_1 = \beta_2$. Then $\rho(\mathcal{P}_F)$ consists of scalars. Write $\overline{\rho}$ for the composite of ρ with the quotient map $\mathrm{GL}_2(\mathbb{C}) \to \mathrm{PGL}_2(\mathbb{C})$; thus $\overline{\rho}(\mathcal{P}_F) = 1$, and $\overline{\rho}(\mathcal{I}_F)$ is a quotient of $\mathcal{I}_F/\mathcal{P}_F$. That group is isomorphic to $\prod_{\ell\neq p}\mathbb{Z}_\ell$ by proposition 6.1.2, and $\rho(\mathcal{I}_F)$ is finite by the compactness argument above, so $\overline{\rho}(\mathcal{I}_F)$ is a finite quotient of $\prod_{\ell\neq p}\mathbb{Z}_\ell$ by an open subgroup. Any such quotient is cyclic: an open subgroup contains all but finitely many of the factors $\mathbb{Z}_\ell$, the image of each remaining factor is a quotient of $\mathbb{Z}_\ell$ and so is cyclic of ℓ-power order, and these images commute and generate. A product of finitely many cyclic groups of pairwise coprime orders is cyclic, by the Chinese remainder theorem. Choose $a \in \rho(\mathcal{I}_F)$ whose image generates $\overline{\rho}(\mathcal{I}_F)$. Then $\rho(\mathcal{I}_F)$ is generated by a together with the scalars it contains, so $\rho(\mathcal{I}_F)$ is abelian. Being finite abelian, it acts diagonalisably, and $\rho|_{\mathcal{I}_F} = \alpha_1 \oplus \alpha_2$ for characters α_i of $\mathcal{I}_F$, which is normal in $\mathcal{W}_F$.

If $\alpha_1 \neq \alpha_2$, then lemma 9.2.4 with $N = \mathcal{I}_F$ produces a separable quadratic E/F with $\mathcal{I}_F = \mathrm{Gal}(\overline{F}/F_\infty) \subseteq \mathcal{W}_E$ and $\rho \cong \mathrm{Ind}_{E/F}\Theta$. Now $E \subseteq F_\infty$, so E/F is unramified and $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$.

If $\alpha_1 = \alpha_2$, then $\rho(\mathcal{I}_F)$ consists of scalars. The quotient $\mathcal{W}_F/\mathcal{I}_F$ is infinite cyclic, generated by the class of a geometric Frobenius Φ, so $\rho(\mathcal{W}_F)$ is generated by $\rho(\Phi)$ together with a group of scalars and is therefore abelian. Two commuting operators on $\mathbb{C}^2$ have a common eigenvector, so ρ has a $\mathcal{W}_F$-stable line, contradicting $\rho \in \mathcal{G}_2^0(F)$. This case does not occur, and the proof is

complete.

With this in hand the behaviour of ρ under restriction is settled. Part (1) below holds for every finite extension and is recorded because the identification $\chi|_{\mathcal{W}_K} = \chi \circ N_{K/F}$ is used constantly. The remaining parts are where tameness enters.

Proposition 9.2.6: Restriction to a Tamely Ramified Extension

Let $\rho \in \mathcal{G}_2^0(F)$ and let K/F be a finite separable extension.

1. The representation ρ_K is semisimple, and for a character χ of $F^\times$ the restriction of χ to $\mathcal{W}_K$ is the character $\chi_K := \chi \circ N_{K/F}$ of $K^\times$. In particular $\chi_K = 1$ if and only if $\chi|_{\mathcal{W}_K} = 1$.

Suppose from now on that ρ is totally ramified and that K/F is tamely ramified. Then:

2. ρ_K lies in $\mathcal{G}_2^0(K)$ and is totally ramified.
3. The map $\chi \mapsto \chi_K$ is an injective group homomorphism $T(\rho) \to T(\rho_K)$.
4. If $\rho' \in \mathcal{G}_2^0(F)$ satisfies $\rho'_K \cong \rho_K$, then $\rho' \cong \chi \otimes \rho$ for a unique character χ of $F^\times$ trivial on $N_{K/F}(K^\times)$.

***Proof*:**

1. The subgroup $\mathcal{W}_K$ is open of finite index in $\mathcal{W}_F$, so lemma 1.2.38(1) says that a smooth representation of $\mathcal{W}_F$ is $\mathcal{W}_F$-semisimple exactly when it is $\mathcal{W}_K$-semisimple. The irreducible ρ is the former, hence ρ_K is semisimple. For the second assertion, apply χ to the commuting square of theorem 6.2.1(4): for $x \in \mathcal{W}_K$ one has $\mathbf{a}_F(x) = N_{K/F}(\mathbf{a}_K(x))$, so $\chi(\mathbf{a}_F(x)) = (\chi \circ N_{K/F})(\mathbf{a}_K(x))$. By theorem 6.2.1(1) the map $\mathbf{a}_K$ has image all of $K^\times$, so the character $\chi|_{\mathcal{W}_K}$ of $\mathcal{W}_K$ is $\chi \circ N_{K/F}$ transported through $\mathbf{a}_K$, and one is trivial exactly when the other is.

2. Since K/F is tamely ramified, $K \subseteq E_\infty$, and the maximal tamely ramified extension of K is again E_∞: it contains E_∞ because E_∞/K is tamely ramified, and it is contained in E_∞ because ramification indices multiply in towers, so an extension of K tamely ramified over K is tamely ramified over F. Hence $\mathcal{P}_K = \mathrm{Gal}(\overline{F}/E_\infty) = \mathcal{P}_F$, and in particular $\mathcal{P}_F \subseteq \mathcal{W}_K$. By lemma 9.2.5 the restriction $\rho|_{\mathcal{P}_F}$ is irreducible, so ρ_K is irreducible, and $\rho_K|_{\mathcal{P}_K} = \rho|_{\mathcal{P}_F}$ is irreducible, so lemma 9.2.5 applied over K makes ρ_K totally ramified.

3. If $\chi \otimes \rho \cong \rho$, then restriction to $\mathcal{W}_K$ gives $\chi_K \otimes \rho_K \cong \rho_K$ by (1), so the map is well defined, and it is a homomorphism because restriction of characters is. Let $\chi \in T(\rho)$ with $\chi \neq 1$ and $\chi_K = 1$. By (1) this means $\chi|_{\mathcal{W}_K} = 1$, and proposition 9.1.15 gives $\chi^2 = 1$, so $\ker\chi$ is an open subgroup of $\mathcal{W}_F$ of index two containing $\mathcal{W}_K$. Thus $\ker\chi = \mathcal{W}_M$ for a separable quadratic extension M/F with $M \subseteq K$. As a subextension of the tamely ramified K/F, the extension M/F is tamely ramified, so $e(M|F)$ divides two and is prime to $p = 2$, forcing $e(M|F) = 1$: the extension M/F is unramified. By theorem 6.2.2(2) the quotient $F^\times/N_{M/F}(M^\times)$ has order two, so there are exactly two characters of $F^\times$ trivial on $N_{M/F}(M^\times)$. By (1) these are exactly the two characters of $\mathcal{W}_F$ trivial on $\mathcal{W}_M$, and the non-trivial one is $\kappa_{M/F}$. Hence $\chi = \kappa_{M/F}$, which is unramified by proposition 9.1.8,

contradicting proposition 9.2.2(3). So $\chi = 1$ and the map is injective.

4. Assume first that K/F is Galois. Since $\rho'_K \cong \rho_K$, one may replace ρ' by an isomorphic representation and assume that ρ and ρ' act on the same space V with $\rho(h) = \rho'(h)$ for all $h \in \mathcal{W}_K$. For $g \in \mathcal{W}_F$ and $h \in \mathcal{W}_K$ the element ghg^{-1} lies in $\mathcal{W}_K$ by normality, so

$$\rho(g)\rho(h)\rho(g)^{-1} = \rho(ghg^{-1}) = \rho'(ghg^{-1}) = \rho'(g)\rho(h)\rho'(g)^{-1}$$

and therefore $\rho(g)^{-1}\rho'(g)$ commutes with $\rho(h)$ for every $h \in \mathcal{W}_K$. The representation ρ_K is irreducible by (2), so theorem 1.2.33 makes $\rho(g)^{-1}\rho'(g)$ a scalar $\chi(g)$. Comparing $\rho'(g_1g_2) = \chi(g_1g_2)\rho(g_1g_2)$ with $\rho'(g_1)\rho'(g_2) = \chi(g_1)\chi(g_2)\rho(g_1g_2)$ shows that χ is a homomorphism $\mathcal{W}_F \to \mathbb{C}^\times$. It is trivial on the open subgroup $\mathcal{W}_K$, hence smooth, and $\rho' = \chi \otimes \rho$. By (1) the corresponding character of $F^\times$ is trivial on $N_{K/F}(K^\times)$.

For general K/F let L/F be the normal closure of K/F inside $\overline{F}$. It is the compositum of the F-conjugates of K, each tamely ramified over F, and a compositum of tamely ramified extensions is tamely ramified because each conjugate lies in E_∞. So L/F is a tamely ramified Galois extension. Restricting the isomorphism $\rho'_K \cong \rho_K$ to $\mathcal{W}_L \subseteq \mathcal{W}_K$ gives $\rho'_L \cong \rho_L$, so the Galois case produces a character χ of $F^\times$, trivial on $N_{L/F}(L^\times)$, with $\rho' \cong \chi \otimes \rho$. Restricting to $\mathcal{W}_K$ gives $\rho_K \cong \rho'_K \cong \chi_K \otimes \rho_K$, so $\chi_K \in T(\rho_K)$. Now χ is trivial on $\mathcal{W}_L$, hence on $\mathcal{P}_F = \mathcal{P}_L \subseteq \mathcal{W}_L$, so by theorem 6.2.1(3) it is trivial on $\mathbf{a}_F(\mathcal{P}_F) = U_F^1$. The same computation over K, using $\mathcal{P}_K = \mathcal{P}_F \subseteq \mathcal{W}_L$ and theorem 6.2.1(3) for K, shows χ_K is trivial on U_K^1. By (2) the representation ρ_K is totally ramified, so proposition 9.2.2(3) applied over K forces $\chi_K = 1$, which by (1) says χ is trivial on $N_{K/F}(K^\times)$.

Uniqueness holds in both cases. If χ and χ' are two such characters, then $\chi\chi'^{-1} \in T(\rho)$, and both are trivial on $N_{K/F}(K^\times)$, hence on $\mathcal{W}_K \supseteq \mathcal{P}_F$ by (1), hence on U_F^1 by theorem 6.2.1(3). Proposition 9.2.2(3) then leaves $\chi\chi'^{-1} = 1$, completing the proof.

The tameness hypothesis in proposition 9.2.6(2) cannot be dropped. Over a dyadic field a ramified quadratic extension is wildly ramified, and restricting to it destroys irreducibility in the most direct way available.

Example 9.2.7: Restriction to a Ramified Quadratic Extension

Let E/F be a ramified separable quadratic extension, so $e(E|F) = 2$ and E/F is wildly ramified because $p = 2$. Let Θ be a character of $E^\times$ with $\Theta^\sigma \neq \Theta$, so that $\rho = \mathrm{Ind}_{E/F}\Theta$ lies in $\mathcal{G}_2^0(F)$ by the irreducibility criterion lemma 8.5.2. Then $\rho_E = \Theta \oplus \Theta^\sigma$ by (8.29): the restriction is a sum of two characters, hence reducible, and $\rho_E \notin \mathcal{G}_2^0(E)$.

Concretely, take $F = \mathbb{Q}_2$ and $E = \mathbb{Q}_2(\sqrt{2})$ and let ξ be a level-3 character of $E^\times$. Example 9.1.22 proves every such ξ is E/F-regular, so $\rho = \mathrm{Ind}_{E/F}\xi$ is irreducible, and proves ρ totally ramified of level 5. Both hypotheses of proposition 9.2.6(2) are therefore in force except tameness, which is what makes this a test of that hypothesis: $\rho_E = \xi \oplus \xi^\sigma$ is reducible all the same. Nothing in proposition 9.2.6 is contradicted: E/F is not tamely ramified, so $\mathcal{P}_E$ is a proper subgroup of $\mathcal{P}_F$ and the wild inertia of F is not available inside $\mathcal{W}_E$ to keep ρ irreducible.

The route to a cubic extension runs through the following question. Given a Galois extension and a twisting character upstairs that the Galois action fixes, does the character come from downstairs? For cyclic groups the answer is yes, and the reason is that an invariant irreducible representation of a normal subgroup with cyclic quotient always extends.

Lemma 9.2.8: Extension along a Cyclic Quotient

Let K/F be a finite cyclic extension of degree n inside $\overline{F}$ and let τ be an irreducible smooth representation of $\mathcal{W}_K$ with $\tau^\gamma \cong \tau$ for every $\gamma \in \mathrm{Gal}(K/F)$. Then there is an irreducible smooth representation ρ of $\mathcal{W}_F$ with $\rho|_{\mathcal{W}_K} \cong \tau$. If τ is a character, so is ρ.

***Proof*:**
Write τ on the space V. Choose $g_0 \in \mathcal{W}_F$ whose image generates $\mathrm{Gal}(K/F) = \mathcal{W}_F/\mathcal{W}_K$, and call that image γ_0. The hypothesis $\tau^{\gamma_0} \cong \tau$ gives $B \in \mathrm{GL}(V)$ with

$$B\,\tau(h)\,B^{-1} = \tau(g_0 h g_0^{-1}) \qquad (h \in \mathcal{W}_K) \tag{9.4}$$

Put $z = g_0^n \in \mathcal{W}_K$. Iterating (9.4) n times gives $B^n\tau(h)B^{-n} = \tau(g_0^n h g_0^{-n}) = \tau(z)\tau(h)\tau(z)^{-1}$, so $\tau(z)^{-1}B^n$ commutes with $\tau(\mathcal{W}_K)$ and is a scalar c by theorem 1.2.33. That is, $B^n = c\,\tau(z)$. Choose $\lambda \in \mathbb{C}^\times$ with $\lambda^n = c^{-1}$ and replace B by λB. This changes neither (9.4) nor the scalar-commuting relations, and now $B^n = \tau(z)$.

Every element of $\mathcal{W}_F$ is $g_0^i h$ with $i \in \mathbb{Z}$ and $h \in \mathcal{W}_K$. Set $\rho(g_0^i h) = B^i\tau(h)$. This is well defined: if $g_0^i h = g_0^{i'} h'$ then $g_0^{i-i'} \in \mathcal{W}_K$, so $i - i' = nm$ for some $m \in \mathbb{Z}$ and $h' = z^m h$, whence $B^{i'}\tau(h') = B^{i-nm}\tau(z)^m\tau(h) = B^i\tau(h)$. It is a homomorphism: rewriting (9.4) as $\tau(g_0^{-j} h g_0^j) = B^{-j}\tau(h)B^j$,

$$\rho\big((g_0^i h)(g_0^j h')\big) = \rho\big(g_0^{i+j}(g_0^{-j} h g_0^j)h'\big) = B^{i+j}B^{-j}\tau(h)B^j\tau(h') = \big(B^i\tau(h)\big)\big(B^j\tau(h')\big)$$

Its restriction to the open subgroup $\mathcal{W}_K$ is τ, which is smooth, so ρ is smooth. And a $\mathcal{W}_F$-stable subspace is $\mathcal{W}_K$-stable, so ρ is irreducible because τ is. If $\dim V = 1$ then $\dim \rho = 1$, completing the proof.

Applied to twisting characters, this yields the descent statement the rest of the section runs on. For finite extensions $F \subseteq K_1 \subseteq K_2$ with K_2/K_1 Galois, the group $\mathrm{Gal}(K_2/K_1) = \mathcal{W}_{K_1}/\mathcal{W}_{K_2}$ acts on the characters of $\mathcal{W}_{K_2}$ by $\chi^g(h) = \chi(ghg^{-1})$, matching the orientation fixed for quadratic extensions in section 9.1, and this depends only on the class of g modulo $\mathcal{W}_{K_2}$ because a character of $\mathcal{W}_{K_2}$ kills commutators. The action preserves $T(\rho_{K_2})$ for any $\rho \in \mathcal{G}_2^0(F)$, since $\chi^g \otimes \rho_{K_2} \cong (\chi \otimes \rho_{K_2})^g \cong \rho_{K_2}^g \cong \rho_{K_2}$, the last isomorphism being implemented by $\rho(g)$.

Lemma 9.2.9: Descent of Twisting Characters

Let $\rho \in \mathcal{G}_2^0(F)$ be totally ramified and let $F \subseteq K_1 \subseteq K_2$ be finite tamely ramified extensions of F.

1. Suppose K_2/K_1 is Galois. If $\chi = \psi_{K_2}$ for some character ψ of $\mathcal{W}_{K_1}$, then χ is fixed by $\mathrm{Gal}(K_2/K_1)$.
2. Suppose K_2/K_1 is cyclic. Then a character $\chi \in T(\rho_{K_2})$ is fixed by $\mathrm{Gal}(K_2/K_1)$ if and only if $\chi = \psi_{K_2}$ for some $\psi \in T(\rho_{K_1})$, and such a ψ is then unique.

Only (2) needs cyclicity; (1) uses no more than the normality of $\mathcal{W}_{K_2}$ in $\mathcal{W}_{K_1}$, which is why it is stated separately, since the extensions it is applied to below are Galois but need not be cyclic.

***Proof*:**
For (1), let $\chi = \psi_{K_2}$ with ψ a character of $\mathcal{W}_{K_1}$. For $g \in \mathcal{W}_{K_1}$ and $h \in \mathcal{W}_{K_2}$ one has $\chi^g(h) = \psi(ghg^{-1}) = \psi(h) = \chi(h)$, the middle equality because ψ is a character and so kills commutators. Hence χ is fixed. Nothing here uses $\mathrm{Gal}(K_2/K_1)$ cyclic; normality of $\mathcal{W}_{K_2}$ in $\mathcal{W}_{K_1}$ is what makes χ^g a character of $\mathcal{W}_{K_2}$ at all.

For (2), one direction is (1). Conversely let $\chi \in T(\rho_{K_2})$ be $\mathrm{Gal}(K_2/K_1)$-fixed. A character is an irreducible smooth representation, and being fixed is exactly the hypothesis of lemma 9.2.8 for the cyclic extension K_2/K_1, so there is a character $\widetilde{\chi}$ of $\mathcal{W}_{K_1}$ with $\widetilde{\chi}_{K_2} = \chi$. Then $(\widetilde{\chi} \otimes \rho_{K_1})|_{\mathcal{W}_{K_2}} = \chi \otimes \rho_{K_2} \cong \rho_{K_2}$. The class ρ_{K_1} lies in $\mathcal{G}_2^0(K_1)$ and is totally ramified by proposition 9.2.6(2), and K_2/K_1 is tamely ramified because K_2/F is, so proposition 9.2.6(4) applied over the base K_1 gives a character θ of $K_1^\times$, trivial on $N_{K_2/K_1}(K_2^\times)$, with $\widetilde{\chi} \otimes \rho_{K_1} \cong \theta \otimes \rho_{K_1}$. Set $\psi = \theta^{-1}\widetilde{\chi}$. Then $\psi \in T(\rho_{K_1})$, and $\psi_{K_2} = \theta_{K_2}^{-1}\chi = \chi$ because $\theta_{K_2} = 1$ by proposition 9.2.6(1). Uniqueness is the injectivity of $T(\rho_{K_1}) \to T(\rho_{K_2})$, which is proposition 9.2.6(3), completing the proof.

The Galois groups that arise below are all of one shape: a cyclic inertia subgroup with cyclic quotient, a shape inherited by every subgroup.

Lemma 9.2.10: Galois Groups of Tamely Ramified Extensions

Let L/F be a finite tamely ramified Galois extension and put $\Gamma = \mathrm{Gal}(L/F)$. Let $I \trianglelefteq \Gamma$ be its inertia subgroup. Then I is cyclic, Γ/I is cyclic, L/L^I is cyclic and totally ramified, and L^I/F is the maximal unramified subextension of L/F. Every subgroup $J \leq \Gamma$ contains a cyclic normal subgroup with cyclic quotient, namely $J \cap I$.

***Proof*:**
Since L/F is tamely ramified, $L \subseteq E_\infty$ and $\mathcal{P}_F = \mathrm{Gal}(\overline{F}/E_\infty)$ acts trivially on L. The subgroup I is the image of $\mathcal{I}_F$ in Γ, so it is a quotient of $\mathcal{I}_F/\mathcal{P}_F \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$ by proposition 6.1.2. A finite quotient of that group is cyclic, by the argument in Case 2 of the proof of lemma 9.2.5. By the definition of the inertia subgroup, L^I/F is the maximal unramified subextension, $\Gamma/I \cong \mathrm{Gal}(L^I/F)$ is the Galois group of an unramified extension of local fields and is cyclic,

and $\mathrm{Gal}(L/L^I) = I$ makes L/L^I cyclic and totally ramified. For the last assertion, $J \cap I$ is a subgroup of the cyclic group I, hence cyclic, and it is normal in J because I is normal in Γ. And $J/(J \cap I) \cong JI/I$ embeds in the cyclic group Γ/I, hence is cyclic, completing the proof.

Two lemmas now carry the main theorem. The first says that primitivity of ρ forces a rigid statement about the Galois action on the twisting group of a restriction. The second says that some tamely ramified restriction is imprimitive at all.

Lemma 9.2.11: No Fixed Twisting Character

Let $\rho \in \mathcal{G}_2^0(F)$ be primitive and let L/F be a finite tamely ramified Galois extension with ρ_L imprimitive. Then:

1. the only element of $T(\rho_L)$ fixed by $\mathrm{Gal}(L/F)$ is the trivial one;
2. ρ_L is triply imprimitive, so $T(\rho_L)$ has order four;
3. if $F \subseteq E \subseteq L$ and $\mathrm{Gal}(L/E)$ acts trivially on $T(\rho_L)$, then $T(\rho_E) \to T(\rho_L)$ is an isomorphism.

***Proof*:**
Throughout, ρ is totally ramified by proposition 9.2.2(2), so proposition 9.2.6 applies to every tamely ramified extension of F, and every intermediate field of L/F is tamely ramified over F.

1. Let $E_0 = L^I$, so that E_0/F is the maximal unramified subextension of L/F and L/E_0 is cyclic and E_0/F is unramified, hence cyclic, by lemma 9.2.10. Let $\xi \in T(\rho_L)$ be fixed by $\mathrm{Gal}(L/F)$. It is in particular fixed by $\mathrm{Gal}(L/E_0)$, so lemma 9.2.9(2), applied with $K_1 = E_0$ and $K_2 = L$, produces a unique $\chi \in T(\rho_{E_0})$ with $\chi_L = \xi$.

   Let $\sigma \in \mathrm{Gal}(E_0/F)$ and lift it to $\widetilde{\sigma} \in \mathrm{Gal}(L/F)$, with $g \in \mathcal{W}_F$ a lift of $\widetilde{\sigma}$. Then $\chi^\sigma \in T(\rho_{E_0})$, and for $h \in \mathcal{W}_L$ one computes $(\chi^\sigma)_L(h) = \chi(ghg^{-1}) = \widetilde{\xi^\sigma}(h) = \xi(h)$, the last equality because ξ is fixed by $\mathrm{Gal}(L/F)$. So $(\chi^\sigma)_L = \chi_L$, and the injectivity in proposition 9.2.6(3) gives $\chi^\sigma = \chi$. Thus χ is fixed by $\mathrm{Gal}(E_0/F)$, and lemma 9.2.9(2) applied with $K_1 = F$ and $K_2 = E_0$ gives $\psi \in T(\rho)$ with $\psi_{E_0} = \chi$. Since ρ is primitive, $T(\rho) = \{1\}$ by proposition 9.2.2(1), so $\psi = 1$, hence $\chi = 1$ and $\xi = \chi_L = 1$. When $E_0 = F$ the middle paragraph is vacuous and the argument reads $\xi = \chi_L$ with $\chi \in T(\rho) = \{1\}$.

2. By proposition 9.1.15 the order of $T(\rho_L)$ divides four, and it is not one, because ρ_L is imprimitive and proposition 9.2.2(1) applies over L. If the order were two, then $T(\rho_L) = \{1, \kappa\}$, and a group action fixing 1 must fix the remaining element κ, contradicting (1). So the order is four.

3. Injectivity is proposition 9.2.6(3). For surjectivity, put $J = \mathrm{Gal}(L/E)$ and let $E' = L^{J \cap I}$, where I is the inertia subgroup of $\mathrm{Gal}(L/F)$. By lemma 9.2.10 the group $J \cap I = \mathrm{Gal}(L/E')$ is cyclic and normal in J, and $\mathrm{Gal}(E'/E) \cong J/(J \cap I)$ is cyclic. Both L/E' and E'/E are Galois. Let $\chi \in T(\rho_L)$. It is fixed by $\mathrm{Gal}(L/E') \leq J$, so lemma 9.2.9(2) with $K_1 = E'$, $K_2 = L$ gives $\xi' \in T(\rho_{E'})$ with $\xi'_L = \chi$. For $\gamma \in \mathrm{Gal}(E'/E)$ lift to $\widetilde{\gamma} \in J$. Then $(\xi'^\gamma)_L = (\xi'_L)^{\widetilde{\gamma}} = \chi^{\widetilde{\gamma}} = \chi = \xi'_L$, because J acts trivially on $T(\rho_L)$, and injectivity gives $\xi'^\gamma = \xi'$. So ξ' is fixed by $\mathrm{Gal}(E'/E)$, and lemma 9.2.9(2) with $K_1 = E$, $K_2 = E'$ gives $\xi \in T(\rho_E)$ with $\xi_{E'} = \xi'$, whence $\xi_L = \xi'_L = \chi$, completing the proof.

The extension in the second lemma is cut out by the projective image of ρ, which is finite. A Sylow 2-subgroup of that image is what makes the degree odd.

Lemma 9.2.12: A Tamely Ramified Extension Making ρ Imprimitive

Let $\rho \in \mathcal{G}_2^0(F)$ be primitive. There is a finite tamely ramified extension K_0/F of odd degree such that ρ_{K_0} is imprimitive.

***Proof*:**
Write $\overline{\rho}$ for the composite of ρ with $\mathrm{GL}_2(\mathbb{C}) \to \mathrm{PGL}_2(\mathbb{C})$.

Step 1: the projective image is finite. The group $\mathcal{I}_F$ is compact and $\ker\rho$ is open, so $N = \rho(\mathcal{I}_F)$ is finite, and $\mathrm{Aut}(N)$ is finite. Fix a geometric Frobenius $\Phi \in \mathcal{W}_F$. Conjugation by $\rho(\Phi)$ is an automorphism of N, because $\mathcal{I}_F$ is normal in $\mathcal{W}_F$, so $\rho(\Phi)^r$ centralises N for some $r \geq 1$. It also commutes with $\rho(\Phi)$, and $\mathcal{W}_F$ is generated by Φ together with $\mathcal{I}_F$, so $\rho(\Phi)^r$ commutes with all of $\rho(\mathcal{W}_F)$ and is a scalar by theorem 1.2.33. Hence $\overline{\rho}(\Phi)^r = 1$ and $\overline{\rho}(\mathcal{W}_F) = \bigcup_{i=0}^{r-1} \overline{\rho}(\Phi)^i\,\overline{\rho}(\mathcal{I}_F)$ is finite.

Step 2: a field cut out by the projective image. The kernel of $\overline{\rho}$ contains $\ker\rho$, hence is open, and it is normal of finite index by Step 1. An open normal subgroup of finite index in $\mathcal{W}_F$ is $\mathcal{W}_{F'}$ for a finite Galois extension F'/F, and $\overline{G} := \overline{\rho}(\mathcal{W}_F) \cong \mathrm{Gal}(F'/F)$. Let $\overline{H}$ be a Sylow 2-subgroup of $\overline{G}$ and let $K_0 \subseteq F'$ be the fixed field of the corresponding subgroup of $\mathrm{Gal}(F'/F)$, so that $\mathcal{W}_{K_0} = \overline{\rho}^{-1}(\overline{H})$ and $[K_0 : F] = [\overline{G} : \overline{H}]$ is odd. The ramification index $e(K_0|F)$ divides $[K_0 : F]$ and is therefore odd, so K_0/F is tamely ramified.

Step 3: the restriction is imprimitive. By proposition 9.2.2(2) the class ρ is totally ramified, so $\rho_{K_0} \in \mathcal{G}_2^0(K_0)$ by proposition 9.2.6(2). Put $H = \rho(\mathcal{W}_{K_0})$. Every element of the centre $Z(H)$ commutes with the irreducible ρ_{K_0}, hence is a scalar by theorem 1.2.33; conversely the scalars in H are central. So $Z(H) = H \cap \mathbb{C}^\times$ and $H/Z(H) = \overline{\rho}(\mathcal{W}_{K_0}) = \overline{H}$. If $\overline{H}$ were trivial, then H would be abelian and ρ_{K_0} reducible. So $\overline{H}$ is a non-trivial finite 2-group and its centre is non-trivial. Pick an element of order two in $Z(\overline{H})$, let $\overline{A}$ be the subgroup it generates and let $A \subseteq H$ be its preimage. Then A is normal in H, because $\overline{A}$ is central in $\overline{H}$. It is abelian, being generated by the central $Z(H)$ together with one further element. And $A \neq Z(H)$, so A contains a non-scalar element.

The group A acts diagonalisably on $\mathbb{C}^2$: it is $Z(H)\langle a\rangle$ with a non-scalar and $a^2 \in Z(H) \subseteq \mathbb{C}^\times$, so a has minimal polynomial $X^2 - c$ with $c \neq 0$, which is separable. (Finiteness of A is not available here: $\rho(\mathcal{W}_F)$ is finite only when $\det\rho$ has finite order, by proposition 6.1.10.) Its two eigencharacters are distinct, since equal ones would make A act by scalars. Let $N_0 \subseteq \mathcal{W}_{K_0}$ be the preimage of A under ρ. It is a closed normal subgroup of $\mathcal{W}_{K_0}$ on which ρ acts by two distinct characters, so lemma 9.2.4 produces a separable quadratic extension K_1/K_0 and a character Θ of $K_1^\times$ with $\rho_{K_0} \cong \mathrm{Ind}_{K_1/K_0}\Theta$. By definition 8.5.1 the class ρ_{K_0} is imprimitive, completing the proof.

The two lemmas combine into the structure theorem. The odd-degree extension of lemma 9.2.12 is far from unique and can be large. The content of the theorem is that it may always be replaced by a cubic one, and that the two possible shapes of that cubic extension are governed by whether it is

Galois.

Theorem 9.2.13: Cubic Resolution of a Primitive Parameter

Let $\rho \in \mathcal{G}_2^0(F)$ be primitive. There is a cubic extension K/F such that ρ_K is imprimitive. Moreover, for any such K:

1. if K/F is Galois, hence cyclic, then ρ_K is triply imprimitive;
2. if K/F is not cyclic, let L/F be its normal closure and let E/F be the maximal unramified subextension of L/F. Then $\mathrm{Gal}(L/F) \cong S_3$, the extension E/F is unramified quadratic, K/F is totally ramified, L/K is unramified quadratic, and

$$\rho_K \text{ is simply imprimitive,} \qquad \rho_L \text{ is triply imprimitive,} \qquad \rho_E \text{ is primitive}$$

***Proof*:**

Step 1: a tamely ramified Galois extension on which ρ is triply imprimitive. Let K_0/F be as in lemma 9.2.12 and let L_0/F be the normal closure of K_0/F. As a compositum of F-conjugates of K_0, each contained in E_∞, the extension L_0/F is tamely ramified and Galois. The class ρ_{K_0} is imprimitive and totally ramified (proposition 9.2.6(2)), and L_0/K_0 is tamely ramified, so proposition 9.2.6(3) applied over K_0 embeds the non-trivial group $T(\rho_{K_0})$ into $T(\rho_{L_0})$. Hence $T(\rho_{L_0}) \neq \{1\}$ and ρ_{L_0} is imprimitive by proposition 9.2.2(1). By lemma 9.2.11(2) it is triply imprimitive.

Step 2: shrinking to a Galois group of order three or six. The group $T(\rho_{L_0})$ is elementary abelian of order four by Step 1, so an automorphism of it permutes its three non-trivial elements, and conversely every permutation of those three elements is additive, because the sum of two distinct non-trivial elements is the third. Thus $\mathrm{Aut}(T(\rho_{L_0}))$ is the group of permutations of a three-element set, isomorphic to S_3. Let $\Gamma \subseteq S_3$ be the image of $\mathrm{Gal}(L_0/F)$ under the resulting action and let J be its kernel. By lemma 9.2.11(1) no non-trivial element of $T(\rho_{L_0})$ is fixed by all of $\mathrm{Gal}(L_0/F)$, so Γ fixes none of the three points. A trivial subgroup fixes all three, and a subgroup of order two is generated by a transposition, which fixes one. So Γ has order three or six, and in both cases Γ acts transitively on the three points.

Let $L = L_0^J$. It is Galois over F because J is normal in $\mathrm{Gal}(L_0/F)$, and it is tamely ramified over F. The group $\mathrm{Gal}(L_0/L) = J$ acts trivially on $T(\rho_{L_0})$, so lemma 9.2.11(3) makes $T(\rho_L) \to T(\rho_{L_0})$ an isomorphism. In particular ρ_L is triply imprimitive. That isomorphism is equivariant for $\mathrm{Gal}(L_0/F)$, which acts on $T(\rho_L)$ through $\mathrm{Gal}(L/F)$, so $\mathrm{Gal}(L/F) \cong \Gamma$ acts faithfully on $T(\rho_L)$ and $[L : F]$ is three or six.

Step 3: producing the cubic extension. Suppose first $[L : F] = 3$. Then $K := L$ is a cubic extension and $\rho_K = \rho_L$ is imprimitive.

Suppose next $[L : F] = 6$, so $\mathrm{Gal}(L/F) \cong S_3$ acting faithfully on $T(\rho_L)$. Let $\varsigma \in \mathrm{Gal}(L/F)$ correspond to a transposition. It fixes exactly one non-trivial element χ of $T(\rho_L)$. Put $K = L^{\langle\varsigma\rangle}$, a cubic extension of F. The extension L/K is quadratic, hence cyclic, and χ is fixed by $\mathrm{Gal}(L/K)$, so lemma 9.2.9(2) produces a non-trivial element of $T(\rho_K)$ and ρ_K is imprimitive by proposition 9.2.2(1). This proves the existence assertion in both cases.

Step 4: the cyclic case. Let now K/F be any cubic extension with ρ_K imprimitive. Since $[K:F]=3$ is odd, $e(K|F)$ is odd and K/F is tamely ramified. If K/F is Galois, then $\mathrm{Gal}(K/F)$ has order three and is cyclic, and lemma 9.2.11(2), applied with the tamely ramified Galois extension K/F in place of L/F, shows that ρ_K is triply imprimitive. This is (1).

Step 5: the non-cyclic case. Suppose K/F is not cyclic and let L/F be its normal closure, so $\mathrm{Gal}(L/F)\cong S_3$ and $[L:F]=6$. As a compositum of conjugates of the tamely ramified K/F, the extension L/F is tamely ramified, and ρ_L is imprimitive by the argument of Step 1 with K in place of K_0; lemma 9.2.11(2) makes ρ_L triply imprimitive. As in Step 2, the image of $\mathrm{Gal}(L/F)$ in $\mathrm{Aut}(T(\rho_L))\cong S_3$ has order three or six. That image is a quotient of S_3, and the quotients of S_3 have orders 1, 2 and 6, so the image has order six and $\mathrm{Gal}(L/F)$ acts faithfully on $T(\rho_L)$.

By lemma 9.2.10 the inertia subgroup $I\trianglelefteq\mathrm{Gal}(L/F)\cong S_3$ is cyclic. The normal cyclic subgroups of S_3 are the trivial one and A_3. If I were trivial, then L/F would be unramified and $\mathrm{Gal}(L/F)$ cyclic, which it is not. So $I=A_3$, the field $E=L^{A_3}$ is the maximal unramified subextension of L/F and is quadratic over F, and $e(L|F)=|I|=3$. Writing $K=L^{\langle\varsigma\rangle}$ for an involution ς, the subgroups $\langle\varsigma\rangle$ and A_3 generate S_3, so $K\cap E=F$. The maximal unramified subextension of K/F is contained in $L\cap F_\infty=E$, hence in $K\cap E=F$, so K/F is totally ramified. Finally KE lies in L and strictly contains K, so $KE=L$, and L/K is unramified quadratic because E/F is unramified.

It remains to identify $T(\rho_K)$ and $T(\rho_E)$. The map $T(\rho_K)\to T(\rho_L)$ is injective by proposition 9.2.6(3). If it were surjective, every element of $T(\rho_L)$ would be ψ_L with $\psi\in T(\rho_K)$, hence fixed by $\mathrm{Gal}(L/K)=\langle\varsigma\rangle$ by lemma 9.2.9(1). That would make ς act trivially on $T(\rho_L)$, contradicting faithfulness. So $|T(\rho_K)|<|T(\rho_L)|=4$, while $T(\rho_K)\neq\{1\}$ by hypothesis and proposition 9.1.15 makes $|T(\rho_K)|$ a power of two. Hence $|T(\rho_K)|=2$ and ρ_K is simply imprimitive. For E, suppose $T(\rho_E)\neq\{1\}$ and take $\psi\in T(\rho_E)$ non-trivial. Then $\psi_L\neq 1$ by proposition 9.2.6(3), and ψ_L is fixed by $\mathrm{Gal}(L/E)=A_3$ by lemma 9.2.9(1). But A_3 acts on the three non-trivial elements of $T(\rho_L)$ as a three-cycle, by faithfulness, and fixes none of them. Hence $T(\rho_E)=\{1\}$ and ρ_E is primitive by proposition 9.2.2(1), completing the proof.

The theorem converts a question about ρ into a question about the cubic extensions of F, of which a local field has very few. The following instance is small enough to list completely, and it exhibits both cases of the theorem as the only shapes available.

Example 9.2.14: Cubic Extensions of $\mathbb{Q}_2$

Let $F=\mathbb{Q}_2$, so $q=2$. There is exactly one unramified cubic extension of F, namely $\mathbb{Q}_2(\zeta_7)$, since 2 has order three modulo 7. It is cyclic over F.

A ramified cubic extension K/F has $e(K|F)=3$, so it is totally and tamely ramified, and is therefore $K=F(\varpi^{1/3})$ for a prime element ϖ of F ([9, Characterizing Totally Tamely Ramified Extensions]). Every prime element of $\mathbb{Q}_2$ is $2u$ with $u\in\mathbb{Z}_2^\times$, and cubing is a bijection of $\mathbb{Z}_2^\times$: that group is the product of $\{\pm1\}$ with the pro-2 group $1+4\mathbb{Z}_2$, and raising to the third power is bijective on a group of order two and on a pro-2 group, since 3 is odd. Writing $u=w^3$ gives $(2u)^{1/3}=2^{1/3}w$ for a suitable choice of cube roots, so $F(\varpi^{1/3})=F(2^{1/3})$. Thus $\mathbb{Q}_2$ has exactly one ramified cubic extension up to $\mathbb{Q}_2$-isomorphism.

The extension $\mathbb{Q}_2(2^{1/3})/\mathbb{Q}_2$ is not Galois: its conjugates are $\zeta_3^i\, 2^{1/3}$, and $\zeta_3 \notin \mathbb{Q}_2$ because μ_3 would inject into $k^\times$, a group of order $q-1=1$. Its normal closure is $L=\mathbb{Q}_2(2^{1/3},\zeta_3)$, of degree six with $\mathrm{Gal}(L/\mathbb{Q}_2)\cong S_3$, and the maximal unramified subextension of $L/\mathbb{Q}_2$ is $\mathbb{Q}_2(\zeta_3)$, the unramified quadratic extension, with $e(L|\mathbb{Q}_2)=3$. This is exactly the configuration of theorem 9.2.13(2).

So a primitive $\rho\in\mathcal{G}_2^0(\mathbb{Q}_2)$, should one exist, has its cubic resolution equal to $\mathbb{Q}_2(\zeta_7)$ or to $\mathbb{Q}_2(2^{1/3})$, and no other possibility arises.

A first consequence records what the transitivity of the Galois action costs the levels of the twisting characters. It uses the two level configurations of corollary 9.1.25 and selects the second of them.

Corollary 9.2.15: Levels in a Triply Imprimitive Restriction

Let $\rho\in\mathcal{G}_2^0(F)$ be primitive and let K/F be a finite tamely ramified Galois extension such that ρ_K is imprimitive and minimal. Then the three non-trivial elements of $T(\rho_K)$ all have the same level $d\geq 1$, and $\ell(\rho_K)=3d$.

***Proof*:**
By lemma 9.2.11(2) the class ρ_K is triply imprimitive, so $T(\rho_K)$ has three non-trivial elements, and by proposition 9.2.6(2) it is totally ramified. By lemma 9.2.11(1) the image of $\mathrm{Gal}(K/F)$ in the group of permutations of those three elements fixes none of them, so as in Step 2 of the proof of theorem 9.2.13 it has order three or six and is transitive. An element $\gamma\in\mathrm{Gal}(K/F)$ carries U_K^m onto U_K^m for every m, so χ^γ and χ have the same level; transitivity therefore makes the three levels equal, say to d, and $d\geq 1$ by proposition 9.2.2(3) applied over K. Now ρ_K is totally ramified, triply imprimitive and minimal, so corollary 9.1.25 applies: either exactly one non-trivial element of $T(\rho_K)$ has strictly least level, which the equality of the three levels excludes, or all three levels are equal and $\ell(\rho_K)=3d$. The second alternative holds, as we sought to show.

The cubic extension produced in theorem 9.2.13 came out of a construction with several choices in it, so it is worth checking that the result does not depend on them. It does not, and the same argument identifies every tamely ramified extension over which ρ becomes imprimitive.

Proposition 9.2.16: Uniqueness of the Cubic Resolution

Let $\rho\in\mathcal{G}_2^0(F)$ be primitive.

1. The cubic extension K/F with ρ_K imprimitive is unique up to F-isomorphism.
2. Let E/F be a finite tamely ramified extension. Then ρ_E is imprimitive if and only if E contains an F-conjugate of K.

***Proof*:**
Fix K/F as in theorem 9.2.13, let L/F be its normal closure and put $\Gamma=\mathrm{Gal}(L/F)$. By theorem 9.2.13 and Steps 4 and 5 of its proof, Γ is cyclic of order three or isomorphic to S_3, it acts faithfully on $T(\rho_L)$, and ρ_L is triply imprimitive. In the cyclic case the faithfulness is automatic: the image of Γ in $\mathrm{Aut}(T(\rho_L))\cong S_3$ fixes none of the three non-trivial elements

by lemma 9.2.11(1), so it has order three or six by the argument of Step 2 of the proof of theorem 9.2.13, and a group of order three has only the trivial proper quotient of that size.

Step 1: a tamely ramified Galois extension with imprimitive restriction contains L. Let E/F be a finite tamely ramified Galois extension with ρ_E imprimitive, and suppose $L \not\subseteq E$. The field $E \cap L$ is Galois over F and corresponds to a normal subgroup of Γ. If Γ is cyclic of order three, the only proper subfield of L containing F is F itself, so $E \cap L = F$. If $\Gamma \cong S_3$, the proper subfields of L that are Galois over F are F and $L_0 := L^{A_3}$, the unramified quadratic subextension identified in theorem 9.2.13(2), so $E \cap L \in \{F, L_0\}$.

Both ρ_E and ρ_L are triply imprimitive by lemma 9.2.11(2), so $T(\rho_E)$ and $T(\rho_L)$ have order four. The compositum EL is tamely ramified and Galois over F, and the maps $T(\rho_L) \to T(\rho_{EL})$ and $T(\rho_E) \to T(\rho_{EL})$ are injective by proposition 9.2.6(3). Since $|T(\rho_{EL})|$ divides four (proposition 9.1.15), both are isomorphisms. By lemma 9.2.9(1), every element of $T(\rho_{EL})$ coming from $T(\rho_E)$ is fixed by $\mathrm{Gal}(EL/E)$, and every element coming from $T(\rho_L)$ is fixed by $\mathrm{Gal}(EL/L)$. By surjectivity, both groups act trivially on $T(\rho_{EL})$. Since E and L are Galois over $E \cap L$, restriction gives an isomorphism $\mathrm{Gal}(EL/E \cap L) \to \mathrm{Gal}(L/E \cap L) \times \mathrm{Gal}(E/E \cap L)$ whose two factors are the images of $\mathrm{Gal}(EL/E)$ and $\mathrm{Gal}(EL/L)$, so $\mathrm{Gal}(EL/E \cap L)$ acts trivially on $T(\rho_{EL})$. Lemma 9.2.11(3) now gives $T(\rho_{E\cap L}) \cong T(\rho_{EL})$, of order four. If $E \cap L = F$, this says $|T(\rho)| = 4$, contradicting primitivity (proposition 9.2.2(1)). If $E \cap L = L_0$, it says ρ_{L_0} is imprimitive, contradicting theorem 9.2.13(2). So $L \subseteq E$.

Step 2: uniqueness. Let K'/F be a cubic extension with $\rho_{K'}$ imprimitive and let L'/F be its normal closure, so $[L' : F]$ is three or six. As in Step 1 of the proof of theorem 9.2.13, the extension L'/F is tamely ramified and Galois and $\rho_{L'}$ is imprimitive, so Step 1 gives $L \subseteq L'$. Suppose $[L : F] = 6$. Then $[L' : F] = 6$ as well and $L' = L$, so K' is a cubic subfield of L. The cubic subfields of L correspond to the subgroups of order two of $\mathrm{Gal}(L/F) \cong S_3$, and those three subgroups are conjugate, so K' is an F-conjugate of K. Suppose instead $[L : F] = 3$. If $[L' : F] = 3$ then $L' = L$ and $K' = L' = L = K$. If $[L' : F] = 6$ then $\mathrm{Gal}(L'/F) \cong S_3$ contains $\mathrm{Gal}(L'/L)$ as a normal subgroup of order two, and S_3 has no such subgroup, so this case does not occur.

Step 3: which tamely ramified extensions see the imprimitivity. Suppose $E \supseteq K^\gamma$ for an F-embedding γ of K into $\overline{F}$, and extend γ to an automorphism of $\overline{F}$ over F. Transport of structure along γ carries ρ_K to ρ_{K^γ}, because ρ is a representation of $\mathcal{W}_F$ and γ fixes F, so ρ_{K^γ} is imprimitive. It is totally ramified by proposition 9.2.6(2). As E/K^γ is tamely ramified, proposition 9.2.6(3) embeds the non-trivial group $T(\rho_{K^\gamma})$ into $T(\rho_E)$, so ρ_E is imprimitive by proposition 9.2.2(1).

Conversely let E/F be finite tamely ramified with ρ_E imprimitive, and let E'/F be the normal closure of E/F, again tamely ramified and Galois. As in Step 1 of the proof of theorem 9.2.13, $T(\rho_E)$ embeds in $T(\rho_{E'})$, so $\rho_{E'}$ is imprimitive and Step 1 above gives $L \subseteq E'$. Both $T(\rho_L)$ and $T(\rho_{E'})$ have order four and the map between them is injective, hence an isomorphism, and it is equivariant for $\mathrm{Gal}(E'/F)$, which acts on $T(\rho_L)$ through $\mathrm{Gal}(L/F) = \Gamma$. Take a non-trivial $\psi \in T(\rho_E)$. Its image in $T(\rho_{E'})$ is non-trivial and is fixed by $H := \mathrm{Gal}(E'/E)$, by lemma 9.2.9(1). So the image $\overline{H}$ of H in Γ fixes a non-trivial element of $T(\rho_L)$.

If Γ is cyclic of order three, it acts on the three non-trivial elements of $T(\rho_L)$ as a three-cycle, so the only subgroup fixing one of them is trivial: $\overline{H} = 1$, that is $H \subseteq \mathrm{Gal}(E'/L)$, and taking

fixed fields gives $L \subseteq E$. Here $K = L$, so E contains K itself. If $\Gamma \cong S_3$, the subgroups of Γ fixing a non-trivial element of $T(\rho_L)$ are the trivial group and the three subgroups generated by an involution, by faithfulness. So $\overline{H} \subseteq \langle \varsigma \rangle$ for some involution ς, hence H is contained in the preimage of $\langle \varsigma \rangle$ in $\mathrm{Gal}(E'/F)$, whose fixed field is $L^{\langle \varsigma \rangle}$. Taking fixed fields reverses inclusions, so $E = (E')^H \supseteq L^{\langle \varsigma \rangle}$, and $L^{\langle \varsigma \rangle}$ is one of the three F-conjugate cubic subfields of L, hence an F-conjugate of K. This completes the proof.

The last structural question is which finite group a primitive ρ comes from. Twisting ρ by a character changes neither $T(\rho)$ nor any restriction ρ_K up to a twist, so the invariant attached to ρ itself is its projective image.

Proposition 9.2.17: The Projective Image of a Primitive Parameter

Let $\rho \in \mathcal{G}_2^0(F)$ be primitive, let K/F be its cubic resolution and let L/F be the normal closure of K/F, so $L = K$ when K/F is cyclic. Write $\overline{\rho}$ for the composite of ρ with $\mathrm{GL}_2(\mathbb{C}) \to \mathrm{PGL}_2(\mathbb{C})$. Then $\ker \overline{\rho} = \mathcal{W}_M$ for a finite Galois extension M/F containing L, with $\mathrm{Gal}(M/L)$ elementary abelian of order four, and

$$\overline{\rho}(\mathcal{W}_F) \cong \mathrm{Gal}(M/F) \cong \begin{cases} A_4 & \text{if } K/F \text{ is cyclic,} \\ S_4 & \text{if } K/F \text{ is not cyclic} \end{cases}$$

***Proof*:**
By theorem 9.2.13 and the first paragraph of the proof of proposition 9.2.16, the class ρ_L is triply imprimitive, L/F is a tamely ramified Galois extension, and $\Gamma := \mathrm{Gal}(L/F)$ acts faithfully on $T(\rho_L)$. It is cyclic of order three in the first case and isomorphic to S_3 in the second.

Step 1: the field M. Let χ_1, χ_2, χ_3 be the three non-trivial elements of $T(\rho_L)$. Each has order two by proposition 9.1.15, so as in the proof of proposition 9.1.16(1) each is $\kappa_{M_i/L}$ for a unique quadratic extension M_i/L, with $\ker \chi_i = N_{M_i/L}(M_i^\times)$. Put $M = M_1 M_2$. The fields M_1 and M_2 are distinct because $\chi_1 \neq \chi_2$, so $[M : L] = 4$. By theorem 6.2.2(1) and (2) the group $N_{M/L}(M^\times)$ has index four in $L^\times$ and is contained in $\ker \chi_1 \cap \ker \chi_2$, which also has index four, so the two coincide. Hence $\mathrm{Gal}(M/L) \cong L^\times / N_{M/L}(M^\times)$ is the character-dual of the elementary abelian group $T(\rho_L)$ and is itself elementary abelian of order four, and $M \supseteq M_3$ as well, since $\chi_3 = \chi_1\chi_2$ is trivial on $N_{M/L}(M^\times)$.

Step 2: ρ is scalar on $\mathcal{W}_M$. By lemma 9.1.14 there is a character ξ of $\mathcal{W}_{M_1}$ with $\rho_L \cong \mathrm{Ind}_{M_1/L}\xi$. Let σ generate $\mathrm{Gal}(M_1/L)$ and apply lemma 9.1.12 over the base L to $\chi_2 \in T(\rho_L)$. If $(\chi_2)_{M_1}$ were trivial, then by proposition 9.2.6(1) the character χ_2 would be trivial on $N_{M_1/L}(M_1^\times) = \ker \chi_1$, a subgroup of index two, and the only such characters are 1 and χ_1. Both are excluded. Hence $(\chi_2)_{M_1} = \xi^\sigma/\xi$. Restrict to $\mathcal{W}_M$: since $M \supseteq M_2$ and $\chi_2 = \kappa_{M_2/L}$ is trivial on $N_{M_2/L}(M_2^\times)$, proposition 9.2.6(1) makes χ_2 trivial on $\mathcal{W}_{M_2} \supseteq \mathcal{W}_M$, so $\xi^\sigma|_{\mathcal{W}_M} = \xi|_{\mathcal{W}_M}$. By (8.29) the restriction $\rho|_{\mathcal{W}_{M_1}}$ is $\xi \oplus \xi^\sigma$, so $\rho|_{\mathcal{W}_M}$ is the character $\xi|_{\mathcal{W}_M}$ taken twice, that is, scalar. Hence $\mathcal{W}_M \subseteq \ker \overline{\rho}$.

Step 3: M/F is Galois, and the extension of Galois groups. The group $\mathrm{Gal}(\overline{F}/F)$ fixes L setwise and permutes $T(\rho_L)$, hence permutes $\{M_1, M_2, M_3\}$, by the uniqueness in Step 1 and the functoriality of theorem 6.2.1(5). So $M = M_1M_2M_3$ is stable and M/F is Galois. Conjugation

makes $\mathrm{Gal}(M/F)$ act on the abelian normal subgroup $V := \mathrm{Gal}(M/L)$, trivially on V itself, so the action factors through Γ. Under the duality of Step 1 that action is dual to the action of Γ on $T(\rho_L)$, which is faithful. An automorphism of a finite abelian group is trivial exactly when its dual is, so Γ acts faithfully on V. Thus

$$1 \longrightarrow V \longrightarrow \mathrm{Gal}(M/F) \longrightarrow \Gamma \longrightarrow 1$$

is an extension of Γ by an elementary abelian group of order four on which Γ acts faithfully. In particular Γ embeds in $\mathrm{Aut}(V)$, the group of permutations of the three non-trivial elements of V.

Step 4: the extension splits. Extensions of Γ by V with the given action are classified by $H^2(\Gamma, V)$, the split one corresponding to the zero class ([8, Classification of Extensions by H^2]). Since V has exponent two, multiplication by 2 annihilates the cochain complex and hence $H^2(\Gamma, V)$. If Γ is cyclic of order three, then $H^2(\Gamma, V)$ is also annihilated by $|\Gamma| = 3$ ([8, Consequences of $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$]), and a group annihilated by two coprime integers is trivial. If $\Gamma \cong S_3$, then $H^2(\Gamma, V)$ is a 2-group, so restriction to a Sylow 2-subgroup $\langle \varsigma \rangle$ is injective on it ([8, Consequences of $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$]). Here ς acts on V by interchanging two non-trivial elements and fixing the third, say e, and for the cyclic group $\langle \varsigma \rangle$ one has $H^2(\langle \varsigma \rangle, V) \cong V^{\varsigma}/(1+\varsigma)V$ ([8, Cohomology of Cyclic Groups]). Both groups are $\{0, e\}$: the fixed points are 0 and e, while $(1+\varsigma)v$ is 0 for $v \in \{0, e\}$ and is e for each of the two interchanged elements. So $H^2(\langle \varsigma \rangle, V) = 0$, hence $H^2(\Gamma, V) = 0$. In both cases $\mathrm{Gal}(M/F) \cong V \rtimes \Gamma$.

Step 5: identifying the group. Let $V \rtimes \Gamma$ act on the four-element set V by $(t, \gamma) \cdot v = \gamma(v) + t$. This is an action, and it is faithful: if (t, γ) acts trivially, then evaluating at $v = 0$ gives $t = 0$, and then γ fixes every element of V, so $\gamma = \mathrm{id}$ by the faithfulness of Step 3. So $\mathrm{Gal}(M/F)$ embeds into the group of permutations of a four-element set, which is S_4. Its order is $4|\Gamma|$, that is 12 when Γ is cyclic of order three and 24 when $\Gamma \cong S_3$. In the second case the embedding is onto S_4. In the first case the image is a subgroup of S_4 of index two, hence normal with quotient of order two, so it contains every square in S_4 and in particular every 3-cycle. The 3-cycles generate A_4, which has order 12, so the image is A_4.

Step 6: the kernel is exactly $\mathcal{W}_M$. By Step 2 the kernel contains $\mathcal{W}_M$, so it equals $\mathcal{W}_{F'}$ for a field F' with $F \subseteq F' \subseteq M$, Galois over F because $\ker \overline{\rho}$ is normal. The subgroup $\overline{\rho}(\mathcal{W}_L)$ is a quotient of $\mathrm{Gal}(M/L) = V$. It is not cyclic, since a cyclic $\overline{\rho}(\mathcal{W}_L)$ would make $\rho(\mathcal{W}_L)$ generated by one element together with scalars, hence abelian, hence ρ_L reducible. As V has order four, the quotient must be V itself, so $\ker \overline{\rho} \cap \mathcal{W}_L = \mathcal{W}_M$ and $\mathrm{Gal}(M/F') \cap V = 1$ inside $\mathrm{Gal}(M/F)$. Now $\mathrm{Gal}(M/F')$ is normal in $\mathrm{Gal}(M/F)$, which is A_4 or S_4. The conjugacy class sizes are $1, 3, 4, 4$ in A_4 and $1, 6, 3, 8, 6$ in S_4, and a normal subgroup is a union of classes containing 1 whose size divides the group order; running through the possibilities leaves 1, V and A_4 inside A_4, and 1, V, A_4 and S_4 inside S_4. In every case a non-trivial normal subgroup contains V. Since $\mathrm{Gal}(M/F')$ meets V trivially, it is trivial, so $F' = M$ and $\overline{\rho}(\mathcal{W}_F) \cong \mathrm{Gal}(M/F)$, completing the proof.

Which of the two groups occurs is the traditional name of a primitive ρ.

Definition 9.2.18: Tetrahedral and Octahedral Parameters

Let $\rho \in \mathcal{G}_2^0(F)$ be primitive and let M/F be the extension of proposition 9.2.17. Then ρ is *tetrahedral* when $\mathrm{Gal}(M/F) \cong A_4$, and *octahedral* when $\mathrm{Gal}(M/F) \cong S_4$; equivalently, by proposition 9.2.17, according as the cubic resolution of ρ is cyclic or not. The names are those of the rotation groups of the corresponding solids.

Proposition 9.2.17 also says what would have to be produced to settle the existence question of box 9.2.3: a Galois extension M/F of a dyadic field with group A_4 or S_4, carrying a two-dimensional projective representation that lifts to a representation of $\mathcal{W}_F$.

The final result of the section runs the construction backwards. Given a cubic K/F and a class over K, it says exactly when that class is a restriction, which is the form in which the analysis is used when a correspondence over K is transported back to F.

Proposition 9.2.19: Descent along a Cubic Extension

Let K/F be a cubic extension and let $\tau \in \mathcal{G}_2^0(K)$ be totally ramified.

1. Suppose K/F is cyclic, with $\mathrm{Gal}(K/F) = \langle\sigma\rangle$. Then $\tau \cong \rho_K$ for some $\rho \in \mathcal{G}_2^0(F)$ if and only if $\tau^\sigma \cong \tau$.
2. Suppose K/F is not cyclic and let L/F be its normal closure. Then $\tau \cong \rho_K$ for some $\rho \in \mathcal{G}_2^0(F)$ if and only if $\tau_L \cong \rho'_L$ for some $\rho' \in \mathcal{G}_2^0(F)$.
3. In the situation of (2), let $\vartheta \in \mathcal{G}_2^0(L)$ be totally ramified. Then $\vartheta \cong \rho_L$ for some $\rho \in \mathcal{G}_2^0(F)$ if and only if $\vartheta^\gamma \cong \vartheta$ for every $\gamma \in \mathrm{Gal}(L/F)$.

***Proof*:**

First record the shape of the extensions involved. As $[K : F] = 3$ is odd, $e(K|F)$ is odd and K/F is tamely ramified. If K/F is not cyclic then $\mathrm{Gal}(L/F) \cong S_3$ and L/F is tamely ramified, being a compositum of conjugates of K. By lemma 9.2.10 the inertia subgroup I of $\mathrm{Gal}(L/F)$ is cyclic and normal, so I is trivial or A_3. It is not trivial, since L/F unramified would make $\mathrm{Gal}(L/F)$ cyclic. So $I = A_3$, the field $E := L^{A_3}$ is unramified quadratic over F, the extension L/E is cyclic and totally ramified of degree three, and $K \cap E = F$ because $\langle\varsigma\rangle$ and A_3 generate S_3 for the involution ς with $K = L^{\langle\varsigma\rangle}$. As in Step 5 of the proof of theorem 9.2.13, K/F is then totally ramified and $L = KE$ is unramified quadratic over K.

1. If $\tau \cong \rho_K$ then, for a lift $g \in \mathcal{W}_F$ of σ, conjugation by $\rho(g)$ gives $\rho_K^\sigma \cong \rho_K$, so $\tau^\sigma \cong \tau$. Conversely, $\mathrm{Gal}(K/F)$ is cyclic of order three generated by σ, and $\tau^{\sigma^i} \cong \tau$ for every i follows from $\tau^\sigma \cong \tau$, so lemma 9.2.8 produces an irreducible smooth representation ρ of $\mathcal{W}_F$ with $\rho|_{\mathcal{W}_K} \cong \tau$. It is two-dimensional, hence lies in $\mathcal{G}_2^0(F)$.
2. Necessity is immediate with $\rho' = \rho$. For sufficiency, suppose $\tau_L \cong \rho'_L$ with $\rho' \in \mathcal{G}_2^0(F)$. Since L/K is unramified, hence tamely ramified, and τ is totally ramified, proposition 9.2.6(2) makes τ_L irreducible and totally ramified. Therefore ρ'_L is irreducible, so ρ'_K is irreducible as well. It is also totally ramified: otherwise $\rho'_K \cong \mathrm{Ind}_{L'/K}\Theta$ with L'/K unramified quadratic by definition 8.5.1, and $L' = L$ is the unique unramified quadratic extension of K, so (8.29) would make ρ'_L reducible. Now proposition 9.2.6(4), applied over the base K to the tamely

ramified extension L/K and to the isomorphism $\tau_L \cong (\rho'_K)_L$, gives a character φ of $K^\times$ trivial on $N_{L/K}(L^\times)$ with $\tau \cong \varphi \otimes \rho'_K$. By theorem 6.2.2(2) there are exactly two such characters, namely 1 and $\kappa_{L/K}$, and $\kappa_{L/K}$ is unramified by proposition 9.1.8. If $\varphi = 1$, take $\rho = \rho'$. Otherwise let χ be the unramified character of $F^\times$ with $\chi(\varpi) = -1$ for a prime element ϖ of F. Since K/F is totally ramified, $f(K|F) = 1$ and $v_F(N_{K/F}(x)) = v_K(x)$ for $x \in K^\times$, so $N_{K/F}$ carries a prime element of K to a prime element of F. Hence χ_K is unramified with $\chi_K(\varpi_K) = -1$, so $\chi_K = \kappa_{L/K} = \varphi$. Then $\tau \cong \chi_K \otimes \rho'_K = (\chi \otimes \rho')_K$ and $\rho = \chi \otimes \rho'$ serves.

3. Necessity is immediate, since $\rho_L^\gamma \cong \rho_L$ for every $\gamma \in \mathrm{Gal}(L/F)$ by conjugating with the value of ρ at a lift of γ. For sufficiency, assume $\vartheta^\gamma \cong \vartheta$ for all γ. The extension L/E is cyclic, so lemma 9.2.8 gives an irreducible smooth representation μ of $\mathcal{W}_E$, necessarily two-dimensional, with $\mu_L \cong \vartheta$.

   The class μ is totally ramified. Otherwise $\mu \cong \mathrm{Ind}_{E'/E}\Theta$ with E'/E unramified quadratic, so $\kappa_{E'/E} \in T(\mu)$ by lemma 9.1.14 and $\kappa_{E'/E}$ is unramified by proposition 9.1.8. Restricting, $(\kappa_{E'/E})_L \in T(\mu_L)$, and since L/E is totally ramified one has $f(L|E) = 1$, so $N_{L/E}$ carries a prime element of L to a prime element of E and $(\kappa_{E'/E})_L$ is unramified with value -1 there, hence non-trivial. That contradicts proposition 9.2.2(3) applied to the totally ramified $\vartheta = \mu_L$.

   Let X be the set of classes $\mu' \in \mathcal{G}_2^0(E)$ with $\mu'_L \cong \vartheta$. By proposition 9.2.6(4), applied over E to the tamely ramified extension L/E, the set X consists exactly of the twists $\chi \otimes \mu$ by the characters χ of $E^\times$ trivial on $N_{L/E}(L^\times)$. Each such twist does lie in X, since $\chi_L = 1$ by proposition 9.2.6(1). There are three such characters, because $E^\times/N_{L/E}(L^\times) \cong \mathrm{Gal}(L/E)$ has order three by theorem 6.2.2(2). The three twists are pairwise non-isomorphic: $\det(\chi \otimes \mu) = \chi^2 \det \mu$, and squaring is a bijection of a group of order three, so the three determinants are distinct. Thus X has exactly three elements.

   The group $\mathrm{Gal}(E/F)$ acts on X: for $\sigma \in \mathrm{Gal}(E/F)$ with lift $g \in \mathcal{W}_F$ and $\mu' \in X$, the class $(\mu')^g$ satisfies $((\mu')^g)_L \cong (\mu'_L)^\gamma \cong \vartheta^\gamma \cong \vartheta$, where $\gamma \in \mathrm{Gal}(L/F)$ is the image of g. Since E/F is unramified quadratic, $\mathrm{Gal}(E/F)$ has order two, and its orbits on X have size one or two, so the odd cardinality of X forces a fixed point. Choose a fixed $\mu_0 \in X$, so $\mu_0^\sigma \cong \mu_0$ for the generator σ of the cyclic group $\mathrm{Gal}(E/F)$. By lemma 9.2.8 there is $\rho \in \mathcal{G}_2^0(F)$ with $\rho_E \cong \mu_0$, and then $\rho_L \cong (\mu_0)_L \cong \vartheta$, the last step because $\mu_0 \in X$, completing the proof.

Taken together, theorem 9.2.13, proposition 9.2.16 and proposition 9.2.19 describe the primitive classes entirely in terms of imprimitive ones over cubic extensions: each primitive ρ determines a cubic K/F, the restriction ρ_K is a class already covered by section 9.1, and proposition 9.2.19 says which classes over K arise in this way. What no argument here produces is a single primitive ρ on which to run this, which is the deferral of box 9.2.3.

Exercise 9.2.1

1. Let F be a dyadic field and let χ be a character of $F^\times$ with $\chi^2 = 1$ and $\chi \neq 1$. Show that either χ is unramified, in which case $\chi = \kappa_{E/F}$ for the unramified quadratic extension E/F and there is exactly one such χ, or else χ is non-trivial on U_F^1.

2. Let $\rho \in \mathcal{G}_2^0(F)$ be totally ramified and let K/F be a finite extension of odd degree. Show that K/F is tamely ramified, and that if $\rho' \in \mathcal{G}_2^0(F)$ satisfies $\rho'_K \cong \rho_K$ and $\det \rho' = \det \rho$, then $\rho' \cong \rho$.

3. Let $\rho \in \mathcal{G}_2^0(F)$. Show that $\rho|_{\mathcal{I}_F}$ is irreducible if and only if $\rho|_{\mathcal{P}_F}$ is irreducible, and that both are equivalent to ρ being totally ramified.

4. Let F be a dyadic field. Show that every cubic extension K/F is tamely ramified, and that K/F is Galois if and only if either K/F is unramified, or $K = F(\varpi^{1/3})$ for a prime element ϖ of F with $\mu_3 \subseteq F$, which happens exactly when 3 divides $q-1$. You may use that a totally tamely ramified extension of degree e is generated by an e-th root of a prime element.

5. Let F be a dyadic field with $3|q-1$, let ϖ be a prime element of F and put $K = F(\varpi^{1/3})$. Show that K/F is cyclic, cubic and totally ramified. Show that for every totally ramified $\rho \in \mathcal{G}_2^0(F)$ the map $T(\rho) \to T(\rho_K)$ is injective, and that if in addition ρ is primitive and ρ_K is imprimitive, then ρ_K is triply imprimitive.

6. Let $\rho \in \mathcal{G}_2^0(F)$ be primitive and let E/F be any separable quadratic extension. Show that ρ_E is irreducible. Note that E/F may be wildly ramified, so proposition 9.2.6(2) does not apply; argue instead from the two ways in which ρ_E could fail to be irreducible.

7. Let $\rho \in \mathcal{G}_2^0(F)$ be primitive and write $\overline{\rho}$ for the composite of ρ with $\mathrm{GL}_2(\mathbb{C}) \to \mathrm{PGL}_2(\mathbb{C})$. Show that $\overline{\rho}(\mathcal{W}_F)$ has trivial centre, and deduce that the centre of $\rho(\mathcal{W}_F)$ is exactly the group of scalar matrices it contains.

8. Let $\rho \in \mathcal{G}_2^0(F)$ be primitive with non-cyclic cubic resolution K/F, let L/F be the normal closure of K/F and let E/F be the unramified quadratic extension. Show that E is the maximal unramified subextension of L/F, that $\rho_E \in \mathcal{G}_2^0(E)$ is primitive, and that the cubic resolution of ρ_E over E is $L = KE$. Show finally that L/E is cyclic, so that ρ_E falls into case (1) of theorem 9.2.13.

9.3 A Converse Theorem

The family of local constants $\varepsilon(\chi \otimes \rho, s, \psi)$, with χ running over the characters of $F^\times$, recovers an imprimitive parameter ρ that is totally ramified. The mechanism is already at work for a single character. If θ is a ramified character of $F^\times$ of level l and ψ has level one, theorem 5.1.18 writes $\varepsilon(\theta, \frac{1}{2}, \psi)$ as a sum of the values of $\check{\theta}$ against ψ over the elements of valuation $-l$. Twisting θ by all characters trivial on U_F^{l+1}, and letting ψ vary over its unit multiples, produces the whole Fourier transform of $\check{\theta}$ on a finite quotient of $F^\times$, and Fourier inversion returns $\check{\theta}$. For $\rho = \mathrm{Ind}_{E/F}\xi$ the same sums appear, by inductivity of the local constant, but with two changes. The sums now run over $E^\times$, and the twists available are only the $\chi_E = \chi \circ N_{E/F}$, which depend on the inducing character ξ only through its values on norms. The additive character makes up the difference: varying ψ by units of F pairs ξ against the *trace* of its argument, and the pair $(N_{E/F}(y), \mathrm{tr}_{E/F}(y))$ is the characteristic polynomial of y over F, which determines y up to $\mathrm{Gal}(E/F)$. Lemma 9.3.8 shows that this remains true at the precision the local constants reach, and proposition 9.3.9 turns it into an inversion formula. The arithmetic of section 9.1 is what yields that precision.

Throughout, E/F is a separable quadratic extension with $\mathrm{Gal}(E/F) = \{1, \sigma\}$, and $\psi_E = \psi \circ \mathrm{tr}_{E/F}$. For a ramified character θ of $E^\times$, the *level* $\ell(\theta)$ is the least integer $l \geq 0$ with θ trivial on U_E^{l+1}, as in theorem 5.1.18. The level of an additive character is the least l with ψ trivial on $\mathfrak{p}^l$, so that a character of level one has kernel $\mathfrak{p}$. For $j \in \mathbb{Z}$ write

$$E\langle j \rangle = \left\{ y \in E^\times : v_E(y) = -j \right\}$$

and likewise $F\langle j\rangle$, and let Γ_j be the group of characters of $F^\times$ that are trivial on U_F^j, so that Γ_0 is the group of unramified characters. Finally $\check{\theta} = \theta^{-1}$.

Every computation below rests on one formula, which is theorem 5.1.18 with the restriction to additive characters of level one removed. It is recorded once, over a general local field, because it is applied over E as often as over F.

Lemma 9.3.1: Gauss Sum at an Arbitrary Additive Character

Let K be a non-archimedean local field with residue field of q_K elements, let θ be a ramified character of $K^\times$ of level $l \geq 0$, and let $\mu \in \widehat{K}$ be non-trivial of level λ. Put $N_0 = l + 1 - \lambda$. Then $n(\theta, \mu) = N_0$ and

$$\varepsilon\big(\theta, \tfrac{1}{2}, \mu\big) = q_K^{-(l+1)/2} \sum_{y \in K\langle N_0\rangle / U_K^{l+1}} \check{\theta}(y)\, \mu(y) \tag{9.5}$$

the sum being over any set of representatives for the cosets of U_K^{l+1} in $K\langle N_0\rangle$.

***Proof*:**
Theorem 5.1.18 and lemma 5.1.16 are proved for F, but their statements and proofs use nothing about F beyond its being a non-archimedean local field with a finite residue field, so both hold verbatim over K.

Fix $\mu_1 \in \widehat{K}$ of level one. Since μ has level λ and the level of $a\mu_1$ is $1 - v_K(a)$, the element $a \in K^\times$ with $\mu = a\mu_1$ has $v_K(a) = 1 - \lambda$. Take $\alpha \in K^\times$ with $v_K(\alpha) = -l$. Theorem 5.1.18 at $s = \frac{1}{2}$ gives

$$\varepsilon\big(\theta, \tfrac{1}{2}, \mu_1\big) = q_K^{-(l+1)/2} \sum_{x \in U_K / U_K^{l+1}} \check{\theta}(\alpha x)\, \mu_1(\alpha x)$$

and the elements αx run through a set of representatives for $K\langle l\rangle / U_K^{l+1}$. Lemma 5.1.16 at $s = \frac{1}{2}$ reads $\varepsilon(\theta, \frac{1}{2}, a\mu_1) = \theta(a)\varepsilon(\theta, \frac{1}{2}, \mu_1)$, so

$$\varepsilon\big(\theta, \tfrac{1}{2}, \mu\big) = q_K^{-(l+1)/2} \sum_{z \in K\langle l\rangle / U_K^{l+1}} \theta(a)\check{\theta}(z)\, \mu_1(z)$$

Substituting $z = ay$ turns $\theta(a)\check{\theta}(z)$ into $\check{\theta}(y)$ and $\mu_1(z)$ into $\mu_1(ay) = \mu(y)$, while $v_K(y) = -l - v_K(a) = -N_0$. And $z \mapsto a^{-1}z$ is a bijection of $K\langle l\rangle / U_K^{l+1}$ onto $K\langle N_0\rangle / U_K^{l+1}$. This is (9.5). The summand is well defined on cosets of U_K^{l+1}: $\check{\theta}$ is trivial there by the definition of the level, and for $y \in K\langle N_0\rangle$ and $t \in \mathfrak{p}_K^{l+1}$ one has $yt \in \mathfrak{p}_K^{l+1-N_0} = \mathfrak{p}_K^{\lambda}$, on which μ is trivial.

For the exponent, theorem 5.1.18 shows $n(\theta, \mu_1) = l$, and proposition 6.3.3(2) gives $n(\theta, a\mu_1) = n(\theta, \mu_1) + v_K(a) = l + 1 - \lambda$, completing the proof.

One feature of (9.5) is used constantly: the value is non-zero, since proposition 6.3.3(3) makes $\varepsilon(\theta, s, \mu)$ invertible.

Local constants convert the level of a character of $E^\times$ into the level of the induced parameter, and

the conversion factor is the discriminant exponent: for totally ramified E/F, lemma 9.1.19 reads $\ell(\mathrm{Ind}_{E/F}\xi) = \ell(\xi) + d(E|F)$. The unramified case is its complement, and everything numerical in this section comes from the two together.

Lemma 9.3.2: Level of a Parameter Induced from an Unramified Extension

Let E/F be the unramified quadratic extension, let ξ be a character of $E^\times$ with $\xi^\sigma \neq \xi$, and put $\rho = \mathrm{Ind}_{E/F}\xi$. Then ξ is ramified, ψ_E has level one for every $\psi \in \widehat{F}$ of level one, and $\ell(\rho) = 2\ell(\xi)$. In particular $\ell(\rho)$ is a non-negative even integer whenever $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$.

***Proof*:**
Suppose ξ were unramified. Any conjugate ϖ_E^σ is again a prime element of E, so $\varpi_E^\sigma = \varpi_E u$ with $u \in U_E$, whence $\xi^\sigma(\varpi_E) = \xi(\varpi_E u) = \xi(\varpi_E)$; and $\xi^\sigma = \xi$ on U_E because both are trivial there. As $E^\times$ is generated by ϖ_E and U_E, this gives $\xi^\sigma = \xi$, against the hypothesis. So ξ is ramified and $\ell(\xi) \geq 0$.

Inductivity, (6.3) applied with $n = 1$ to the tower $E \supset F \supset F$, reads

$$\varepsilon\big(\mathrm{Ind}_{E/F}\xi, s, \psi\big) = \varepsilon(\xi, s, \psi_E)\frac{\varepsilon(R_{E/F}, s, \psi)}{\varepsilon(1_E, s, \psi_E)} = \lambda_{E/F}(\psi)\,\varepsilon(\xi, s, \psi_E)$$

by definition 6.3.10, the constant being independent of s by corollary 6.3.11. Here $\varepsilon(\xi \circ \mathbf{a}_E, s, \psi_E) = \varepsilon(\xi, s, \psi_E)$ by theorem 6.3.2(1). Both sides are monomials in $q^{\pm s}$ by proposition 6.3.3(1), the left carrying $q^{\ell(\rho)(1/2-s)}$ and the right $q_E^{n(\xi,\psi_E)(1/2-s)}$, and $q_E = q^{f_{E/F}}$. Comparing the two shapes gives

$$\ell(\rho) = f_{E/F} \cdot n(\xi, \psi_E) \tag{9.6}$$

It remains to compute the level of ψ_E and to apply lemma 9.3.1. The character ψ_E is trivial on $\mathfrak{p}_E^j$ precisely when $\mathrm{tr}_{E/F}(\mathfrak{p}_E^j) \subseteq \ker\psi = \mathfrak{p}$. The different of an unramified extension is the unit ideal by [9, The Different Detects Tameness], and the discussion following [9, Different and Codifferent] turns that into $\mathrm{tr}_{E/F}(\mathfrak{o}_E) = \mathfrak{o}$; as E/F is unramified, $\mathfrak{p}_E^j = \mathfrak{p}^j\mathfrak{o}_E$ and $\mathfrak{o}$-linearity of the trace gives $\mathrm{tr}_{E/F}(\mathfrak{p}_E^j) = \mathfrak{p}^j$. Then $\mathfrak{p}^j \subseteq \mathfrak{p}$ exactly when $j \geq 1$. So ψ_E has level one, lemma 9.3.1 gives $n(\xi, \psi_E) = \ell(\xi)$, and $f_{E/F} = 2$ turns (9.6) into $\ell(\rho) = 2\ell(\xi)$.

This value is non-negative because $\ell(\xi) \geq 0$. Finally, a class $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$ is by definition 8.5.1 of the form $\mathrm{Ind}_{E/F}\xi$ with E/F unramified, and its irreducibility forces $\xi^\sigma \neq \xi$ by lemma 8.5.2, so the computation above applies to it and $\ell(\rho) = 2\ell(\xi)$ is even, as we sought to show.

The last sentence gives a parity test for total ramification: an imprimitive class of odd level cannot lie in $\mathcal{G}_2^{\mathrm{nr}}(F)$. That test is how the hypothesis of theorem 9.3.10 passes from ρ_1 to ρ_2.

The twisting and conjugation formulas $\chi \otimes \mathrm{Ind}_{E/F}\xi \cong \mathrm{Ind}_{E/F}(\chi_E\xi)$ and $\mathrm{Ind}_{E/F}\xi^\sigma \cong \mathrm{Ind}_{E/F}\xi$ are lemma 9.1.10(1) and (2). Two complements to them are used repeatedly below. The second turns an unramified σ-quotient into membership of $\mathcal{G}_2^{\mathrm{nr}}(F)$, and it is the only way total ramification is ever contradicted in this section.

Lemma 9.3.3: Twists and Conjugate Quotients of an Induced Parameter

Let E/F be a separable quadratic extension and let ξ be a character of $E^\times$ with $\xi^\sigma \neq \xi$. Write $\rho = \mathrm{Ind}_{E/F}\xi$.

1. $(\chi_E\xi)^\sigma \neq \chi_E\xi$ for every character χ of $F^\times$.
2. Suppose E/F is totally ramified and ξ^σ/ξ is unramified. Then $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$.

***Proof*:**

1. The character $\chi_E = \chi \circ N_{E/F}$ satisfies $\chi_E^\sigma = \chi_E$ because $N_{E/F} \circ \sigma = N_{E/F}$, so $(\chi_E\xi)^\sigma/(\chi_E\xi) = \xi^\sigma/\xi \neq 1$.
2. Since E/F is totally ramified, $v_F(N_{E/F}(\varpi_E)) = 1$, so $N_{E/F}(\varpi_E)$ is a prime element of F. Let η be the unramified character of $F^\times$ with $\eta(N_{E/F}(\varpi_E)) = (\xi^\sigma/\xi)(\varpi_E)$. Then η_E is unramified, because $N_{E/F}(U_E) \subseteq U_F$, and it agrees with the unramified character ξ^σ/ξ at ϖ_E, so $\eta_E = \xi^\sigma/\xi$. It is non-trivial, since $\xi^\sigma \neq \xi$, hence $\eta \neq 1$.

   Now $\eta_E\xi = \xi^\sigma$, so lemma 9.1.10(1) and (2) give $\eta \otimes \rho \cong \mathrm{Ind}_{E/F}\xi^\sigma \cong \rho$. Comparing determinants in $\eta \otimes \rho \cong \rho$ gives $\eta^2 \det\rho = \det\rho$, so η has order two. By theorem 6.2.2 the kernel of η, an open subgroup of index two, is the norm group of a unique quadratic extension E_0/F, so $\eta = \kappa_{E_0/F}$, and E_0/F is unramified because η is, by proposition 9.1.8. Lemma 9.1.14 then presents ρ as induced from E_0, which by definition 8.5.1 places ρ in $\mathcal{G}_2^{\mathrm{nr}}(F)$, completing the proof.

The next lemma records how far an element of $E^\times$ moves under σ, in a totally ramified quadratic extension. It has two uses: it makes the fibres of the map Ψ_E below consist of exactly two elements rather than one, and it identifies the non-trivial class in the kernel of the norm on the one graded piece where that kernel is not trivial.

Lemma 9.3.4: Conjugate Quotients in a Totally Ramified Quadratic Extension

Let E/F be totally ramified quadratic with $d = d(E|F)$, let ϖ_E be a prime element of E and put $w = \varpi_E/\varpi_E^\sigma$. Then:

1. $u^\sigma/u \in U_E^{d+1}$ for every $u \in U_E$;
2. $w \in U_E^d \setminus U_E^{d+1}$ and $w^2 \in U_E^{d+1}$;
3. if $y \in E^\times$ has $v_E(y)$ odd, then $y/y^\sigma \in wU_E^{d+1}$. In particular $y/y^\sigma \in U_E^d \setminus U_E^{d+1}$.

***Proof*:**

1. This is lemma 9.1.6(1), which gives $u/u^\sigma \in U_E^{d+1}$ for $u \in U_E$. The subgroup U_E^{d+1} is closed under inversion, so u^σ/u lies in it too.
2. That $w \in U_E^d \setminus U_E^{d+1}$ is the last clause of lemma 9.1.6(3): the class of ϖ_E/ϖ_E^σ generates

the order-two image of $\mathcal{K}_{E/F}$ in U_E^d/U_E^{d+1}, so it lies in U_E^d and is non-trivial modulo U_E^{d+1}. For the square, applying σ gives $w^\sigma = \varpi_E^\sigma/\varpi_E = w^{-1}$, so $w^\sigma/w = w^{-2}$, and part (1) applied to the unit w gives $w^{-2} \in U_E^{d+1}$, hence $w^2 \in U_E^{d+1}$.

3. Write $y = \varpi_E^v u$ with $v = v_E(y)$ odd and $u \in U_E$. Then $y/y^\sigma = w^v\,(u/u^\sigma)$. Part (1) puts u/u^σ in U_E^{d+1}, and $w^v = w\,(w^2)^{(v-1)/2}$ lies in wU_E^{d+1} by part (2). So $y/y^\sigma \in wU_E^{d+1}$, which is disjoint from U_E^{d+1} because $w \notin U_E^{d+1}$, and is contained in U_E^d because $w \in U_E^d$, completing the proof.

The converse theorem is proved by comparing two presentations $\rho_i \cong \mathrm{Ind}_{E/F}\xi_i$ over a common E. The next proposition records the numerical constraints those presentations satisfy, and they are exactly the hypotheses under which Ψ_E is injective.

Proposition 9.3.5: Inducing Data of a Minimal Parameter

Let $\rho \in \mathcal{G}_2^{\mathrm{im}}(F)$ be totally ramified and minimal, put $n = \ell(\rho)$, and let $\kappa_{E/F}$ be a non-trivial element of $T(\rho)$. Then:

1. E/F is totally ramified and $d := d(E|F) \geq 1$;
2. $\rho \cong \mathrm{Ind}_{E/F}\xi$ for a character ξ of $E^\times$ with $\xi^\sigma \neq \xi$, ramified of level $n-d$, and ξ is minimal over F, that is $\ell(\xi) \leq \ell(\chi_E\xi)$ for every character χ of $F^\times$;
3. n is odd and $n > 2d$.

***Proof*:**

1. By lemma 9.1.14 there is a character ξ of $E^\times$ with $\rho \cong \mathrm{Ind}_{E/F}\xi$. If E/F were unramified, then ρ would lie in $\mathcal{G}_2^{\mathrm{nr}}(F)$ by definition 8.5.1, against the hypothesis that ρ is totally ramified. A quadratic extension that is not unramified has $e(E|F) = 2$, hence is totally ramified, and $d \geq 1$ by proposition 9.1.2.

2. Keep the ξ produced in (1). Since ρ is irreducible, $\xi^\sigma \neq \xi$ by lemma 8.5.2, so lemma 9.1.19 applies and gives $\ell(\xi) = n - d$. And ξ is ramified: it is E/F-regular, so lemma 8.4.2 makes ξ itself non-trivial on $\mathcal{K}_{E/F}$, and $\mathcal{K}_{E/F} \subseteq U_E$ because $zz^\sigma = 1$ forces $v_E(z) = 0$. So ξ cannot be trivial on U_E.

   Let χ be a character of $F^\times$. By lemma 9.1.10(1), $\chi\otimes\rho \cong \mathrm{Ind}_{E/F}(\chi_E\xi)$, with $(\chi_E\xi)^\sigma \neq \chi_E\xi$ by lemma 9.3.3(1), so lemma 9.1.19 applies to $\chi_E\xi$ as well and gives $\ell(\chi\otimes\rho) = \ell(\chi_E\xi) + d$. Minimality of ρ says $\ell(\rho) \leq \ell(\chi\otimes\rho)$, that is $\ell(\xi) + d \leq \ell(\chi_E\xi) + d$, so $\ell(\xi) \leq \ell(\chi_E\xi)$.

3. That n is odd is lemma 9.1.23, applicable because ρ is totally ramified, imprimitive and minimal. For $n > 2d$ it suffices, by (2), to show $l := \ell(\xi)$ satisfies $l > d$, and this is done by excluding $l < d$ and $l = d$ in turn.

   The case $l < d$ does not arise. By (2) the character ξ satisfies $\xi^\sigma \neq \xi$, so it is E/F-regular (definition 8.4.1), and by (1) the extension E/F is totally ramified with $d \geq 1$. Lemma 9.1.20(1) then gives $l = \ell(\xi) \geq d$ outright.

The case $l = d$. For $u \in U_E$ one has $(\xi^\sigma/\xi)(u) = \xi(u^\sigma/u)$, and $u^\sigma/u \in U_E^{d+1}$ by lemma 9.3.4(1), a group on which ξ is trivial because $\ell(\xi) = d$. So ξ^σ/ξ is trivial on U_E, that is unramified, and lemma 9.3.3(2) then places ρ in $\mathcal{G}_2^{\mathrm{nr}}(F)$, contradicting total ramification.

Hence $l > d$, and $l = n - d$ gives $n > 2d$, completing the proof.

Fix now, for the rest of the section, a totally ramified separable quadratic extension E/F with $d = d(E|F) \geq 1$, an odd integer $n = 2m+1$ with $n > 2d$, and a character ξ of $E^\times$, minimal over F, of level $n - d$. By proposition 9.3.5 this is exactly the data attached to a totally ramified minimal imprimitive parameter of level n together with a non-trivial element of its twisting group. Two inequalities are used constantly: $m \geq d$, which follows from $n \geq 2d+1$, and $n - d + 1 \geq d + 2$, which is the same inequality rearranged.

The elements of $E\langle n\rangle$ are the arguments at which (9.5) evaluates $\check{\xi}$, and U_E^{n-d+1} is the precision at which $\check{\xi}$ is constant. The next definition records what the local constants can detect of such an element.

Definition 9.3.6: The Norm-Trace Map Ψ_E

With E/F, d and $n = 2m+1$ as above, the *norm-trace map* of E/F at level n is

$$\Psi_E \colon \mathrm{Gal}(E/F)\backslash E\langle n\rangle/U_E^{n-d+1} \longrightarrow F\langle n\rangle/U_F^{m+1} \times \mathfrak{p}^{-m}/\mathfrak{p}, \qquad y \longmapsto \big(N_{E/F}(y),\ \mathrm{tr}_{E/F}(y)\big)$$

That Ψ_E is defined at all is four computations. Let $y \in E\langle n\rangle$ and $u \in U_E^{n-d+1}$.

First, $v_F(N_{E/F}(y)) = f_{E/F}\, v_E(y) = -n$, so $N_{E/F}(y) \in F\langle n\rangle$. Second, $\mathrm{tr}_{E/F}(y) \in \mathfrak{p}^{\lfloor(-n+d+1)/2\rfloor} = \mathfrak{p}^{\lfloor d/2\rfloor - m}$ by proposition 9.1.4, and $\lfloor d/2\rfloor \geq 0$, so $\mathrm{tr}_{E/F}(y) \in \mathfrak{p}^{-m}$. Third, $N_{E/F}(yu) = N_{E/F}(y)N_{E/F}(u)$, and $N_{E/F}(u) \in U_F^{m+1}$: for $x \in \mathfrak{p}_E^i$ with $i \geq d$ one has $v_F(N_{E/F}(x)) = i$ and $v_F(\mathrm{tr}_{E/F}(x)) \geq \lfloor(i+d+1)/2\rfloor$, the latter being at most i because $i \geq d$, so the identity $N_{E/F}(1+x) = 1 + \mathrm{tr}_{E/F}(x) + N_{E/F}(x)$ gives

$$N_{E/F}\big(U_E^i\big) \subseteq U_F^{\lfloor(i+d+1)/2\rfloor} \qquad (i \geq d) \tag{9.7}$$

and at $i = n-d+1 = 2m-d+2$, which is at least d because $m \geq d$, the exponent is $\lfloor(2m+3)/2\rfloor = m+1$, exactly what is wanted. Fourth, $\mathrm{tr}_{E/F}(yu) - \mathrm{tr}_{E/F}(y) = \mathrm{tr}_{E/F}(y(u-1))$ with $y(u-1) \in \mathfrak{p}_E^{1-d}$, whose trace lies in $\mathfrak{p}^{\lfloor(1-d+d+1)/2\rfloor} = \mathfrak{p}$. Finally both coordinates are constant on $\mathrm{Gal}(E/F)$-orbits, the norm and the trace being σ-invariant by definition.

Example 9.3.7: The Norm-Trace Map for $\mathbb{Q}_2(\sqrt{2})$

Let $F = \mathbb{Q}_2$ and $E = \mathbb{Q}_2(\sqrt{2})$, so $q = 2$ and $\varpi_E = \sqrt{2}$. Here $d(E|F) = 2$ by example 9.1.3, and $\varpi_E - \varpi_E^\sigma = 2\sqrt{2}$ has $v_E = 3 = d+1$ by proposition 9.1.4(3), so $w = \varpi_E/\varpi_E^\sigma = -1$ lies in $U_E^2 \setminus U_E^3$ because $-1 = 1 - 2$ and $v_E(2) = 2$.

Take $n = 5$, the smallest odd integer exceeding $2d = 4$. Then $m = 2$ and $n - d + 1 = 4$. Put $\gamma = \varpi_E^{-5} = \sqrt{2}/8$. Then

$$N_{E/F}(\gamma) = \frac{\sqrt{2}}{8} \cdot \frac{-\sqrt{2}}{8} = -\frac{1}{32}, \qquad \mathrm{tr}_{E/F}(\gamma) = \frac{\sqrt{2}}{8} - \frac{\sqrt{2}}{8} = 0$$

so $v_F(N_{E/F}(\gamma)) = -5$ as predicted, and $\mathrm{tr}_{E/F}(\gamma) = 0$ lies in $\mathfrak{p}^{-2}$.

Compare γ with $\gamma' = \gamma(1+2\sqrt{2})$. Here $v_E(2\sqrt{2}) = 3$, so $1+2\sqrt{2}$ lies in U_E^3 but not in U_E^4, and the two elements are distinct in the source of Ψ_E. The norm coordinate does not detect this: $N_{E/F}(1+2\sqrt{2}) = 1-8 = -7$, which lies in $U_F^3 = 1+8\mathbb{Z}_2$, so $N_{E/F}(\gamma')$ and $N_{E/F}(\gamma)$ agree modulo $U_F^{m+1} = U_F^3$. The trace coordinate does: $\gamma\sqrt{2} = 1/4$, so $\mathrm{tr}_{E/F}(\gamma\sqrt{2}) = 1/2$ and

$$\mathrm{tr}_{E/F}(\gamma') = \mathrm{tr}_{E/F}(\gamma) + 2\mathrm{tr}_{E/F}(\gamma\sqrt{2}) = 0 + 1$$

which is not in $\mathfrak{p} = 2\mathbb{Z}_2$. So $\Psi_E(\gamma') \neq \Psi_E(\gamma)$, and it is the trace that separates them. Multiplying instead by $1+4\sqrt{2} \in U_E^4$ changes the norm by $1-32 \in U_F^3$ and the trace by $4 \cdot \frac{1}{2} = 2 \in \mathfrak{p}$, so the class is unchanged, as it must be.

In the example each coordinate settles a case the other misses. The next lemma shows that together they settle every case.

Lemma 9.3.8: Injectivity of the Norm-Trace Map

The map Ψ_E of definition 9.3.6 is injective. Moreover the action of $\mathrm{Gal}(E/F)$ on $E\langle n\rangle/U_E^{n-d+1}$ is free, so each non-empty fibre of Ψ_E, read in $E\langle n\rangle/U_E^{n-d+1}$, consists of exactly two classes $\{y, y^\sigma\}$.

***Proof*:**

Freeness. Let $y \in E\langle n\rangle$, so $v_E(y) = -n$ is odd. By lemma 9.3.4(3), $y/y^\sigma \notin U_E^{d+1}$, and $n-d+1 \geq d+2 > d+1$ gives $U_E^{n-d+1} \subseteq U_E^{d+1}$. So y and y^σ are distinct modulo U_E^{n-d+1}.

Injectivity. Let $\gamma_1, \gamma_2 \in E\langle n\rangle$ with $\Psi_E(\gamma_1) = \Psi_E(\gamma_2)$, that is

$$N_{E/F}(\gamma_1)N_{E/F}(\gamma_2)^{-1} \in U_F^{m+1}, \qquad \mathrm{tr}_{E/F}(\gamma_1) - \mathrm{tr}_{E/F}(\gamma_2) \in \mathfrak{p}$$

The goal is $\gamma_1 \in \{\gamma_2, \gamma_2^\sigma\}\, U_E^{n-d+1}$. Both conditions are unchanged if γ_2 is replaced by γ_2^σ, so that replacement is available at any point.

Step 1: the quotient is a principal unit. Put $u = \gamma_1\gamma_2^{-1} \in U_E$. Since E/F is totally ramified, $k_E = k$ and σ acts trivially on the residue field, so the residue of $N_{E/F}(u) = uu^\sigma$ is $\bar{u}^2$. The residual characteristic is 2, so squaring is injective on the finite field k. From $N_{E/F}(u) \in U_F^{m+1} \subseteq U_F^1$ we get $\bar{u}^2 = 1$, hence $\bar{u} = 1$ and $u \in U_E^1$. Write $\gamma_1 = \gamma_2(1+x)$ with $x \in \mathfrak{p}_E$. If $x = 0$ there is nothing left to prove, so assume $x \neq 0$ and set $i = v_E(x) \geq 1$.

Step 2: the case $i \not\equiv d \pmod 2$. Put $z = \gamma_2 x$, so that $\mathrm{tr}_{E/F}(\gamma_1) - \mathrm{tr}_{E/F}(\gamma_2) = \mathrm{tr}_{E/F}(z)$, and $i_0 := v_E(z) = i-n$. Since n is odd, $i_0 \equiv i+1 \equiv d \pmod 2$. Writing $i_0 = 2j-d$, proposition 9.1.4 provides an isomorphism $\mathfrak{p}_E^{2j-d}/\mathfrak{p}_E^{2j-d+1} \to \mathfrak{p}^j/\mathfrak{p}^{j+1}$ induced by the trace, and z is non-zero in the source, so

$$v_F\big(\mathrm{tr}_{E/F}(z)\big) = j = \frac{i-n+d}{2}$$

exactly. The hypothesis $\mathrm{tr}_{E/F}(z) \in \mathfrak{p}$ forces $(i-n+d)/2 \geq 1$, that is $i \geq n-d+2$. In particular

$1 + x \in U_E^{n-d+1}$ and $\gamma_1 \in \gamma_2 U_E^{n-d+1}$.

Step 3: the case $i \equiv d \pmod 2$ and $i \neq d$. Here $v_F(N_{E/F}(x)) = i$, and proposition 9.1.4 applied as in Step 2, now to x itself, gives $v_F(\mathrm{tr}_{E/F}(x)) = (i+d)/2$ exactly. If $i < d$ then $(i+d)/2 > i$, so the identity

$$N_{E/F}(1+x) - 1 = \mathrm{tr}_{E/F}(x) + N_{E/F}(x) \tag{9.8}$$

gives $v_F(N_{E/F}(1+x) - 1) = i$; but $N_{E/F}(1+x) \in U_F^{m+1}$ requires $i \geq m+1$, while $i < d \leq m$. So $i > d$, and then $(i+d)/2 < i$, so (9.8) gives $v_F(N_{E/F}(1+x) - 1) = (i+d)/2$. The hypothesis forces $(i+d)/2 \geq m+1$, that is $i \geq 2m + 2 - d = n - d + 1$, which is again the conclusion.

Step 4: the case $i = d$. By lemma 9.1.7 the map $U_E^d/U_E^{d+1} \to U_F^d/U_F^{d+1}$ induced by the norm has kernel of order two. That kernel contains the class of $w = \varpi_E/\varpi_E^\sigma$, since $N_{E/F}(w) = ww^\sigma = 1$, and that class is non-trivial by lemma 9.3.4(2). So the kernel is exactly $\{1, w\}U_E^{d+1}/U_E^{d+1}$. Now $1 + x \in U_E^d$ and $N_{E/F}(1+x) \in U_F^{m+1} \subseteq U_F^{d+1}$, using $m \geq d$, so $1 + x$ lies in that kernel. And $1 + x \notin U_E^{d+1}$ because $v_E(x) = d$. Hence $1 + x \in wU_E^{d+1}$.

Replace γ_2 by γ_2^σ. By lemma 9.3.4(3), $\gamma_2/\gamma_2^\sigma \in wU_E^{d+1}$, so

$$\frac{\gamma_1}{\gamma_2^\sigma} = \frac{\gamma_1}{\gamma_2} \cdot \frac{\gamma_2}{\gamma_2^\sigma} \in w^2 U_E^{d+1} = U_E^{d+1}$$

by lemma 9.3.4(2). So after the replacement the integer i is at least $d+1$, and Steps 2 and 3 apply to the new pair and give $\gamma_1 \in \gamma_2^\sigma U_E^{n-d+1}$.

In every case γ_1 and γ_2 agree modulo U_E^{n-d+1} up to conjugation, which is what injectivity of Ψ_E asserts, as we sought to show.

The three cases divide as follows. In Step 2 the conclusion comes from the trace condition, in Step 3 from the norm condition, and Step 4 is the single layer U_E^d at which the norm has a kernel and neither condition separates γ_2 from γ_2^σ. That layer is why the map is injective on $\mathrm{Gal}(E/F)$-orbits and not on classes.

The identity below is what the converse theorem uses. It expresses the two conjugate values of $\check{\xi}$ at a point of $E\langle n\rangle$ as a finite sum of local constants, so that the family of local constants determines the pair $\{\xi, \xi^\sigma\}$ on $E\langle n\rangle$.

Proposition 9.3.9: Local Constants Recover the Inducing Character

Keep the standing data E/F, d, $n = 2m+1 > 2d$ and ξ minimal over F of level $n-d$, and fix $\psi \in \widehat{F}$ of level one. Put

$$C_{n,d} = \frac{q^{(n-d+1)/2}}{(q-1)\,q^{n}}$$

Then for every $x \in E\langle n\rangle$,

$$\check{\xi}(x) + \check{\xi}^{\sigma}(x) = C_{n,d} \sum_{\substack{\chi \in \Gamma_{m+1}/\Gamma_0 \\ c \in U_F/U_F^{m+1}}} \varepsilon\big(\chi_E\xi, \tfrac{1}{2}, c\psi_E\big)\, \chi\big(N_{E/F}(x)\big)\, \psi\big(-c\mathrm{tr}_{E/F}(x)\big) \tag{9.9}$$

and every term of the sum is well defined.

***Proof*:**

Step 1: the level of the twisted character. Let $\chi \in \Gamma_{m+1}$, so χ is trivial on U_F^{m+1}. By (9.7) with $i = 2m-d+1$, which satisfies $i \geq d$ because $n = 2m+1 > 2d$, one has $N_{E/F}(U_E^{2m-d+1}) \subseteq U_F^{\lfloor (2m+2)/2\rfloor} = U_F^{m+1}$. So χ_E is trivial on U_E^{2m-d+1} and has level at most $2m-d = n-1-d$. Since ξ has level exactly $n-d$, the product $\chi_E\xi$ is a ramified character of $E^\times$ of level exactly $n-d$.

Step 2: the local constant as a sum over $E\langle n\rangle$. For $c \in U_F$ the additive character $c\psi_E$ has the same level as ψ_E, namely $1-d$: by proposition 9.1.4, $\mathrm{tr}_{E/F}(\mathfrak{p}_E^{r}) = \mathfrak{p}^{\lfloor (r+d+1)/2\rfloor}$, which lies in $\ker\psi = \mathfrak{p}$ exactly when $r \geq 1-d$. Applying lemma 9.3.1 over $K = E$, with $l = n-d$, $\lambda = 1-d$ and hence $N_0 = (n-d+1)-(1-d) = n$, and using $q_E = q$ because E/F is totally ramified,

$$\varepsilon\big(\chi_E\xi, \tfrac{1}{2}, c\psi_E\big) = q^{-(n-d+1)/2}\, G(\chi, c), \qquad G(\chi,c) := \sum_{y \in E\langle n\rangle / U_E^{n-d+1}} \check{\xi}(y)\, \check{\chi}\big(N_{E/F}(y)\big)\, \psi\big(c\mathrm{tr}_{E/F}(y)\big) \tag{9.10}$$

using $(\check{\chi_E\xi})(y) = \check{\xi}(y)\check{\chi}(N_{E/F}(y))$ and $(c\psi_E)(y) = \psi(c\mathrm{tr}_{E/F}(y))$. The right-hand side of (9.10) makes sense for every $c \in \mathfrak{o}$, and depends only on c modulo $\mathfrak{p}^{m+1}$: indeed $\mathrm{tr}_{E/F}(y) \in \mathfrak{p}^{-m}$ for $y \in E\langle n\rangle$, so $c \mapsto \psi(c\mathrm{tr}_{E/F}(y))$ is trivial on $\mathfrak{p}^{m+1}$.

Step 3: $G(\chi, c) = 0$ for $c \in \mathfrak{p}$. Write $\theta = \chi_E\xi$, of level $l = n-d$ by Step 1, and set

$$j = \begin{cases} l & \text{if } c = 0 \\ \max\{1,\ n+1-d-v_F(c)\} & \text{if } c \neq 0 \end{cases}$$

Then $1 \leq j \leq l$: for $c \neq 0$ the bound $j \leq l$ holds because $v_F(c) \geq 1$, and $j \geq 1$ by construction. The function $y \mapsto \psi(c\mathrm{tr}_{E/F}(y))$ is constant on the cosets yU_E^{j} inside $E\langle n\rangle$. For $c = 0$ this is trivial. For $c \neq 0$ and $v \in U_E^{j}$, the difference $y(v-1)$ lies in $\mathfrak{p}_E^{j-n}$, so $\mathrm{tr}_{E/F}(y(v-1)) \in \mathfrak{p}^{\lfloor (j-n+d+1)/2\rfloor}$ by proposition 9.1.4; when $j = n+1-d-v_F(c)$ this exponent is $1-\lceil v_F(c)/2\rceil$, so $c\mathrm{tr}_{E/F}(y(v-1)) \in \mathfrak{p}^{\,1+\lfloor v_F(c)/2\rfloor} \subseteq \mathfrak{p}$, and when $j = 1 > n+1-d-v_F(c)$ the same estimate at $j=1$ gives the exponent $v_F(c) + \lfloor (d+2-n)/2\rfloor$, which is at least 1: writing $t = n-d \geq 1$ and using $v_F(c) \geq t+1$, it is at least $t+2-\lceil t/2\rceil \geq (t+3)/2 \geq 2$. Either way $\psi(c\mathrm{tr}_{E/F}(y(v-1))) = 1$.

Since θ has level $l \geq j$, it is non-trivial on U_E^j. Choose $v \in U_E^j$ with $\check{\theta}(v) \neq 1$. Multiplication by v permutes $E\langle n\rangle/U_E^{l+1}$, so replacing y by yv in (9.10) leaves $G(\chi, c)$ unchanged while multiplying every summand by $\check{\theta}(v)$. Hence $\check{\theta}(v)G(\chi, c) = G(\chi, c)$ and $G(\chi, c) = 0$.

Step 4: two orthogonality relations. Restriction to U_F identifies Γ_{m+1}/Γ_0 with the full character group of U_F/U_F^{m+1}: the kernel of restriction is Γ_0 by definition, and every character of the finite abelian group U_F/U_F^{m+1} extends to $F^\times$ by prescribing a value at a prime element. So $|\Gamma_{m+1}/\Gamma_0| = (q-1)q^m$ and, for $u \in U_F$,

$$\sum_{\chi \in \Gamma_{m+1}/\Gamma_0} \chi(u) = \begin{cases} (q-1)q^m & u \in U_F^{m+1} \\ 0 & \text{otherwise} \end{cases} \tag{9.11}$$

Next, the pairing $(c, t) \mapsto \psi(ct)$ on $\mathfrak{o} \times \mathfrak{p}^{-m}$ is trivial on $\mathfrak{p}^{m+1} \times \mathfrak{p}^{-m}$ and on $\mathfrak{o} \times \mathfrak{p}$, since ψ has kernel $\mathfrak{p}$; and its two kernels are no larger, because $c\mathfrak{p}^{-m} \subseteq \mathfrak{p}$ forces $v_F(c) \geq m+1$ and $t\mathfrak{o} \subseteq \mathfrak{p}$ forces $t \in \mathfrak{p}$. Both quotients $\mathfrak{o}/\mathfrak{p}^{m+1}$ and $\mathfrak{p}^{-m}/\mathfrak{p}$ have q^{m+1} elements, so the pairing is perfect and

$$\sum_{c \in \mathfrak{o}/\mathfrak{p}^{m+1}} \psi(ct) = \begin{cases} q^{m+1} & t \in \mathfrak{p} \\ 0 & \text{otherwise} \end{cases} \qquad (t \in \mathfrak{p}^{-m}) \tag{9.12}$$

Step 5: the computation. Denote by $S(x)$ the sum on the right of (9.9) without the factor $C_{n,d}$. By Step 3 the terms with $c \in \mathfrak{p}/\mathfrak{p}^{m+1}$ contribute nothing to the corresponding sum over $\mathfrak{o}/\mathfrak{p}^{m+1}$, so (9.10) gives

$$S(x) = q^{-(n-d+1)/2} \sum_{\substack{\chi \in \Gamma_{m+1}/\Gamma_0 \\ c \in \mathfrak{o}/\mathfrak{p}^{m+1}}} G(\chi, c)\, \chi\big(N_{E/F}(x)\big)\, \psi\big(-c\mathrm{tr}_{E/F}(x)\big)$$

Substituting the definition of G and exchanging the order of summation,

$$S(x) = q^{-(n-d+1)/2} \sum_{y \in E\langle n\rangle/U_E^{n-d+1}} \check{\xi}(y) \left(\sum_{\chi} \chi\Big(\frac{N_{E/F}(x)}{N_{E/F}(y)}\Big) \right) \left(\sum_{c} \psi\big(c(\mathrm{tr}_{E/F}(y) - \mathrm{tr}_{E/F}(x))\big) \right)$$

The argument of χ lies in U_F, both norms having valuation $-n$, and the argument of ψ lies in $\mathfrak{p}^{-m}$, both traces lying there. So (9.11) and (9.12) apply and leave only those y with $\Psi_E(y) = \Psi_E(x)$, each contributing $(q-1)q^m \cdot q^{m+1} = (q-1)q^n$.

By lemma 9.3.8 those y are exactly the two classes x and x^σ, and $\check{\xi}(x^\sigma) = \check{\xi}^\sigma(x)$. Hence

$$S(x) = q^{-(n-d+1)/2}(q-1)q^n\big(\check{\xi}(x) + \check{\xi}^\sigma(x)\big)$$

Multiplying by $C_{n,d} = q^{(n-d+1)/2}/((q-1)q^n)$ gives (9.9).

Step 6: the terms are well defined. In the variable c this is Step 2: if $c' = c(1+e)$ with $e \in \mathfrak{p}^{m+1}$ then $c' - c \in \mathfrak{p}^{m+1}$, so $G(\chi, c') = G(\chi, c)$ and, since $\mathrm{tr}_{E/F}(x) \in \mathfrak{p}^{-m}$, also $\psi(-c'\mathrm{tr}_{E/F}(x)) = \psi(-c\mathrm{tr}_{E/F}(x))$.

In the variable χ, let $\eta \in \Gamma_0$ be unramified and compare the terms at χ and at $\eta\chi$. Every y occurring in $G(\eta\chi, c)$ has $v_E(y) = -n$, so $\check{\eta}_E(y) = \eta_E(\varpi_E)^n$ is one and the same constant, and (9.10) gives

$$\varepsilon\big((\eta\chi)_E\xi, \tfrac{1}{2}, c\psi_E\big) = \eta_E(\varpi_E)^n\,\varepsilon\big(\chi_E\xi, \tfrac{1}{2}, c\psi_E\big)$$

Take the prime element $\varpi = N_{E/F}(\varpi_E)$ of F, legitimate because $v_F(N_{E/F}(\varpi_E)) = 1$; then $\eta_E(\varpi_E) = \eta(\varpi)$, while $v_F(N_{E/F}(x)) = -n$ and the triviality of η on U_F give $\eta(N_{E/F}(x)) = \eta(\varpi)^{-n}$. The two factors $\eta_E(\varpi_E)^n$ and $\eta(N_{E/F}(x))$ therefore cancel, so each term of (9.9) depends only on the coset $\chi\Gamma_0$, completing the proof.

The left side of (9.9) is a value of the inducing character and the right side is built from local constants of ρ alone. Two parameters induced from the same E with the same twisted local constants therefore have inducing characters whose $\mathrm{Gal}(E/F)$-symmetrizations agree on $E\langle n\rangle$. Turning that into an isomorphism is the last step, and it uses the determinant to spread the agreement from $E\langle n\rangle$ to all of $E^\times$.

Theorem 9.3.10: Galois Converse Theorem

Let $\rho_1, \rho_2 \in \mathcal{G}_2^{\mathrm{im}}(F)$ and suppose ρ_1 is totally ramified. If there is a non-trivial $\psi \in \widehat{F}$ with

$$\varepsilon(\chi \otimes \rho_1, s, \psi) = \varepsilon(\chi \otimes \rho_2, s, \psi)$$

for every character χ of $F^\times$, then $\rho_1 \cong \rho_2$.

***Proof*:**

Step 1: the determinants agree. Let $n_0 = \max\{n_{\rho_1}, n_{\rho_2}, 2\}$ with n_{ρ_i} the integers of proposition 6.3.3(4), and let $t \geq n_0$. For a character χ of $F^\times$ of level t, that part gives

$$\varepsilon(\chi \otimes \rho_i, s, \psi) = \det \rho_i\big(c(\chi)\big)^{-1}\varepsilon(\chi, s, \psi)^2, \qquad i = 1, 2$$

for any $c(\chi)$ with $\chi(1+y) = \psi(c(\chi)y)$ on $\mathfrak{p}^{[t/2]+1}$. Its hypotheses hold because $t \geq n_{\rho_i}$ and $\dim \rho_i = 2$. Since $\varepsilon(\chi, s, \psi)$ is invertible by proposition 6.3.3(3), the hypothesis of the theorem gives $\det \rho_1(c(\chi)) = \det \rho_2(c(\chi))$.

Write λ for the level of ψ, so that ψ is trivial on $\mathfrak{p}^{\lambda}$ and not on $\mathfrak{p}^{\lambda-1}$. The normalization of ψ carried out in Step 2 depends on the conclusion of the present step, so it is not available here. Write also $j = [t/2]+1$, so $2j \geq t+1$. The map $1+y \mapsto y$ is an isomorphism $U_F^j/U_F^{t+1} \to \mathfrak{p}^j/\mathfrak{p}^{t+1}$, because $(1+y)(1+y') = 1+y+y'+yy'$ with $yy' \in \mathfrak{p}^{2j} \subseteq \mathfrak{p}^{t+1}$. Given any $c \in F^\times$ with $v_F(c) = \lambda - 1 - t$, the assignment $1+y \mapsto \psi(cy)$ is therefore a character of U_F^j/U_F^{t+1}: it is well defined because $c\mathfrak{p}^{t+1} = \mathfrak{p}^{\lambda} \subseteq \ker\psi$. Extend it to U_F/U_F^{t+1} and then to $F^\times$ by a value at a prime element. The resulting χ is trivial on U_F^{t+1} and non-trivial on U_F^t, since $c\mathfrak{p}^t = \mathfrak{p}^{\lambda-1}$ and ψ is non-trivial there, so it has level t and $c(\chi) = c$ is a legitimate choice. Hence $\theta := \det\rho_1(\det\rho_2)^{-1}$ is trivial on $F\langle t-\lambda+1\rangle$ for every $t \geq n_0$, that is on $F\langle r\rangle$ for every $r \geq n_0 - \lambda + 1$.

Fix such an r. Every $u \in U_F$ is a quotient of two elements of $F\langle r\rangle$, so $\theta|_{U_F} = 1$. A quotient of an element of $F\langle r\rangle$ by one of $F\langle r+1\rangle$ has valuation $-r + (r+1) = 1$, so θ is trivial at a prime element too. As $F^\times$ is generated by U_F and a prime element, $\theta = 1$, that is $\det\rho_1 = \det\rho_2$.

Step 2: the hypothesis is independent of ψ, and both parameters may be taken minimal. For any χ, $\det(\chi \otimes \rho_i) = \chi^2 \det \rho_i$, and Step 1 makes these equal for $i = 1, 2$. Proposition 6.3.3(2) then gives, for every $a \in F^\times$,

$$\varepsilon(\chi \otimes \rho_i, s, a\psi) = \chi^2 \det \rho_1(a)\, \|a\|^{2(s-1/2)}\, \varepsilon(\chi \otimes \rho_i, s, \psi)$$

with the same prefactor for both i. So the hypothesis holds for every non-trivial additive character once it holds for one, and ψ may be taken of level one from now on. In particular the exponents agree:

$$\ell(\chi \otimes \rho_1) = \ell(\chi \otimes \rho_2) \qquad \text{for every character } \chi \tag{9.13}$$

Twisting does not change the twisting group, since $\eta \otimes (\chi \otimes \rho) \cong \chi \otimes \rho$ if and only if $\eta \otimes \rho \cong \rho$; so $T(\chi \otimes \rho) = T(\rho)$. By lemma 9.1.14 and theorem 6.2.2, a class in $\mathcal{G}_2^0(F)$ is imprimitive exactly when T is non-trivial, and lies in $\mathcal{G}_2^{\mathrm{nr}}(F)$ exactly when T contains the unramified character of order two, that character being $\kappa_{E_0/F}$ for the unramified quadratic E_0/F by proposition 9.1.8. Both properties are therefore preserved by twisting.

By lemmas 9.1.19 and 9.3.2 the level takes non-negative integer values on $\mathcal{G}_2^{\mathrm{im}}(F)$, so $\{\ell(\chi \otimes \rho_1)\}$ has a least element. Choose χ_0 attaining it and replace ρ_i by $\chi_0 \otimes \rho_i$. The hypothesis survives, since $\chi \otimes (\chi_0 \otimes \rho_i) = (\chi\chi_0) \otimes \rho_i$, and by (9.13) both new parameters are minimal. They are still imprimitive, and ρ_1 is still totally ramified. It suffices to prove $\chi_0 \otimes \rho_1 \cong \chi_0 \otimes \rho_2$, so assume from now on that ρ_1 and ρ_2 are minimal.

Step 3: the case of a common twisting character. Suppose $T(\rho_1) \cap T(\rho_2)$ contains a non-trivial element. By theorem 6.2.2 and lemma 9.1.14 that element is $\kappa_{E/F}$ for a quadratic extension E/F. Set $n = \ell(\rho_1)$, which equals $\ell(\rho_2)$ by (9.13) at $\chi = 1$. Proposition 9.3.5, applied to ρ_1, shows that E/F is totally ramified with $d = d(E|F) \geq 1$, that $n = 2m + 1$ is odd with $n > 2d$, and that $\rho_1 \cong \mathrm{Ind}_{E/F}\xi_1$ with ξ_1 minimal over F of level $n - d$.

The same holds for ρ_2. First ρ_2 is totally ramified: were it in $\mathcal{G}_2^{\mathrm{nr}}(F)$, lemma 9.3.2 would make $\ell(\rho_2) = n$ even. Since $\kappa_{E/F} \in T(\rho_2)$ and ρ_2 is minimal and imprimitive, proposition 9.3.5 applies to it with the same E and gives $\rho_2 \cong \mathrm{Ind}_{E/F}\xi_2$ with ξ_2 minimal over F of level $\ell(\rho_2) - d = n - d$. So ξ_1 and ξ_2 are two characters to which proposition 9.3.9 applies, with the same E, d, n and ψ.

Step 4: the twisted local constants of the ξ_i agree. Let χ be a character of $F^\times$ and $c \in U_F$. By lemma 9.1.10(1), $\chi \otimes \rho_i \cong \mathrm{Ind}_{E/F}(\chi_E \xi_i)$, and inductivity as in the proof of lemma 9.3.2 gives

$$\varepsilon(\chi \otimes \rho_i, s, c\psi) = \lambda_{E/F}(c\psi)\, \varepsilon\big(\chi_E \xi_i, s, c\psi_E\big)$$

using $(c\psi)_E = c\psi_E$, which holds because $\mathrm{tr}_{E/F}(cy) = c\,\mathrm{tr}_{E/F}(y)$ for $c \in F$. By Step 2 the hypothesis holds at the additive character $c\psi$, and $\lambda_{E/F}(c\psi)$ is a non-zero constant depending on neither i nor s. Cancelling it and evaluating at $s = \frac{1}{2}$,

$$\varepsilon\big(\chi_E \xi_1, \tfrac{1}{2}, c\psi_E\big) = \varepsilon\big(\chi_E \xi_2, \tfrac{1}{2}, c\psi_E\big)$$

for every character χ of $F^\times$ and every $c \in U_F$. The right-hand sides of (9.9) formed with ξ_1 and with ξ_2 are therefore equal, and so

$$\check{\xi}_1(x) + \check{\xi}_1^{\sigma}(x) = \check{\xi}_2(x) + \check{\xi}_2^{\sigma}(x) \qquad (x \in E\langle n \rangle) \tag{9.14}$$

Step 5: from $E\langle n\rangle$ to $E^\times$. By proposition 6.2.7 with $m=1$, $\det\rho_i=\kappa_{E/F}\cdot(\xi_i|_{F^\times})$, so Step 1 gives $\xi_1|_{F^\times}=\xi_2|_{F^\times}$. Let $a\in F^\times$ and $x\in E\langle n\rangle$. Since $a^\sigma=a$,

$$\check\xi_i(ax)+\check\xi_i^\sigma(ax)=\check\xi_i(a)\big(\check\xi_i(x)+\check\xi_i^\sigma(x)\big)$$

and $\check\xi_1(a)=\check\xi_2(a)$, so (9.14) extends from $E\langle n\rangle$ to $F^\times E\langle n\rangle$. That set is $\{y\in E^\times : v_E(y)\text{ odd}\}$: valuations $v_E(ax)=2v_F(a)-n$ are odd, and conversely y with $v_E(y)=2j-n$ is $\varpi^j(\varpi^{-j}y)$ with ϖ a prime element of F, which has $v_E(\varpi)=2$.

Let φ be the unramified character of $E^\times$ of order two, $\varphi(y)=(-1)^{v_E(y)}$. The function $1-\varphi$ vanishes at even valuations and equals 2 at odd ones, so multiplying (9.14) by $\frac12(1-\varphi(y))$ produces an identity valid at every $y\in E^\times$:

$$\big(\check\xi_1-\varphi\check\xi_1\big)+\big(\check\xi_1^\sigma-\varphi\check\xi_1^\sigma\big)=\big(\check\xi_2-\varphi\check\xi_2\big)+\big(\check\xi_2^\sigma-\varphi\check\xi_2^\sigma\big)\tag{9.15}$$

both sides being zero at even valuations and twice the two sides of (9.14) at odd ones.

Step 6: linear independence. Rewriting (9.15) with all signs positive,

$$\check\xi_1+\check\xi_1^\sigma+\varphi\check\xi_2+\varphi\check\xi_2^\sigma=\check\xi_2+\check\xi_2^\sigma+\varphi\check\xi_1+\varphi\check\xi_1^\sigma$$

an identity between two sums of four characters of the group $E^\times$. Distinct characters of a group are linearly independent over $\mathbb{C}$, so the two multisets of four characters coincide. In particular $\check\xi_1$ belongs to $\{\check\xi_2,\check\xi_2^\sigma,\varphi\check\xi_1,\varphi\check\xi_1^\sigma\}$. It is not $\varphi\check\xi_1$, since $\varphi\neq1$. It is not $\varphi\check\xi_1^\sigma$ either: that would read $\xi_1^\sigma/\xi_1=\varphi$, an unramified σ-quotient, and lemma 9.3.3(2) would then place ρ_1 in $\mathcal{G}_2^{\mathrm{nr}}(F)$, against the hypothesis.

So $\xi_1=\xi_2$ or $\xi_1=\xi_2^\sigma$. In the first case $\rho_1\cong\rho_2$ directly, and in the second $\rho_1\cong\mathrm{Ind}_{E/F}\xi_2^\sigma\cong\mathrm{Ind}_{E/F}\xi_2\cong\rho_2$ by lemma 9.1.10(2).

Step 7: the twisting groups always meet. It remains to show that $T(\rho_1)\cap T(\rho_2)$ is non-trivial. Since ρ_2 is imprimitive, $T(\rho_2)$ contains a non-trivial κ. For every character χ of $F^\times$,

$$\varepsilon(\chi\otimes\rho_1,s,\psi)=\varepsilon(\chi\otimes\rho_2,s,\psi)=\varepsilon\big(\chi\kappa\otimes\rho_2,s,\psi\big)=\varepsilon\big(\chi\kappa\otimes\rho_1,s,\psi\big)$$

the first and third equalities by hypothesis, applied at χ and at $\chi\kappa$, and the middle one because $\kappa\otimes\rho_2\cong\rho_2$. So ρ_1 and $\kappa\otimes\rho_1$ satisfy the hypothesis of the theorem. Both are imprimitive and both are totally ramified, by the invariance of T and of membership in $\mathcal{G}_2^{\mathrm{nr}}(F)$ under twisting recorded in Step 2. Both are minimal as well, though not because the level is twist-invariant, which it is not: comparing the exponents in the display above, at the level-one ψ fixed in Step 2, gives $\ell(\chi\otimes\rho_1)=\ell(\chi\kappa\otimes\rho_1)$ for every χ, which at $\chi=1$ reads $\ell(\kappa\otimes\rho_1)=\ell(\rho_1)$; so $\ell\big(\chi\otimes(\kappa\otimes\rho_1)\big)=\ell(\chi\otimes\rho_1)\geq\ell(\rho_1)=\ell(\kappa\otimes\rho_1)$ for every χ, the inequality being the minimality of ρ_1. Finally $T(\kappa\otimes\rho_1)=T(\rho_1)$ is non-trivial, so Steps 3 to 6 apply to the pair $(\rho_1,\kappa\otimes\rho_1)$ and give $\kappa\otimes\rho_1\cong\rho_1$. Hence $\kappa\in T(\rho_1)\cap T(\rho_2)$, and Steps 3 to 6 applied to (ρ_1,ρ_2) give $\rho_1\cong\rho_2$, completing the proof.

Step 7 is not circular: it applies the special case of Steps 3 to 6 to the pair $(\rho_1,\kappa\otimes\rho_1)$, whose twisting

groups are equal and hence certainly meet, and only then to (ρ_1, ρ_2). Read forwards, the theorem says that on the totally ramified part of $\mathcal{G}_2^{\mathrm{im}}(F)$ the family of twisted local constants is a complete invariant, which is the Galois-side counterpart of what theorem 5.5.2 asserts on the group side.

Section 8.5 constructed the map $\boldsymbol{\pi}^{\mathrm{im}}$ and stated plainly what was missing: injectivity would need a converse theorem on the Galois side, matching representations of $\mathcal{W}_F$ by their twisted local constants. Theorem 9.3.10 is that matching, for pairs of which one member is totally ramified.

Corollary 9.3.11: Injectivity of the Imprimitive Correspondence

Let $\rho_1, \rho_2 \in \mathcal{G}_2^{\mathrm{im}}(F)$, at least one of them totally ramified, and suppose $\boldsymbol{\pi}^{\mathrm{im}}(\rho_1) \cong \boldsymbol{\pi}^{\mathrm{im}}(\rho_2)$. Then $\rho_1 \cong \rho_2$. In particular the map $\boldsymbol{\pi}^{\mathrm{im}} \colon \mathcal{G}_2^{\mathrm{im}}(F) \to \mathcal{A}_2^0(F)$ of theorem 8.5.4 is injective on the totally ramified classes.

***Proof*:**

Fix a non-trivial $\psi \in \widehat{F}$. Identity (8.30) gives, for every character χ of $F^\times$,

$$\varepsilon(\chi \otimes \rho_i, s, \psi) = \varepsilon\big(\chi\, \boldsymbol{\pi}^{\mathrm{im}}(\rho_i), s, \psi\big), \qquad i = 1, 2$$

and the two right-hand sides are equal because the two representations are isomorphic. So ρ_1 and ρ_2 satisfy the hypothesis of theorem 9.3.10 at ψ.

If ρ_1 is totally ramified, theorem 9.3.10 gives $\rho_1 \cong \rho_2$ at once. If ρ_2 is totally ramified, the hypothesis is symmetric in the two parameters, so the theorem applies to the pair (ρ_2, ρ_1) and gives the same conclusion.

One of ρ_1, ρ_2 is totally ramified by hypothesis, so one of the two cases occurs, as we sought to show.

For two classes both in $\mathcal{G}_2^{\mathrm{nr}}(F)$, theorem 9.3.10 says nothing: its proof runs through lemma 9.3.8, every step of which uses $d(E|F) \geq 1$, while the unramified quadratic extension has $d(E|F) = -1$. What would settle that case is the injectivity of $\mathrm{Ind}_{E_0/F}\xi \mapsto \pi_{E_0/F}(\xi)$ for the unramified quadratic E_0/F, which theorem 7.3.2 gives when $p \neq 2$; for $p = 2$ it is treated in [2, §34], and nothing in this section rests on it.

Two consequences follow. First, $\boldsymbol{\pi}^{\mathrm{im}}$ is a bijection of the totally ramified classes in $\mathcal{G}_2^{\mathrm{im}}(F)$ onto their image, so the cuspidal representations they produce are pairwise non-isomorphic and are counted by those parameters. Second, the gap named in the closing discussion of section 8.5 is closed for those classes: injectivity was what the arithmetic of this chapter could reach. It is worth being exact about what this does *not* settle. The three items of box 8.5.6, namely describing the domain, reaching the primitive classes, and identifying the group side, are all still open, and none of them is injectivity. The first two need section 9.2 together with constructions that follow it, and the third needs the classification theory.

Exercise 9.3.1

1. Let E/F be a separable quadratic extension and let ξ_1, ξ_2 be characters of $E^\times$ with $\xi_i^\sigma \neq \xi_i$. Show that $\varepsilon(\mathrm{Ind}_{E/F}\xi_1, s, \psi) = \varepsilon(\mathrm{Ind}_{E/F}\xi_2, s, \psi)$ if and only if $\varepsilon(\xi_1, s, \psi_E) = \varepsilon(\xi_2, s, \psi_E)$.

Deduce that, for two parameters induced from the same E, the hypothesis of theorem 9.3.10 is equivalent to $\varepsilon(\chi_E\xi_1, s, \psi_E) = \varepsilon(\chi_E\xi_2, s, \psi_E)$ for every character χ of $F^\times$.

2. Keep the notation of example 9.3.7, so $F = \mathbb{Q}_2$, $E = \mathbb{Q}_2(\sqrt{2})$, $d = 2$, $n = 5$ and $\gamma = \sqrt{2}/8$. Compute $N_{E/F}(\gamma(1+\sqrt{2}))$ and $\mathrm{tr}_{E/F}(\gamma(1+\sqrt{2}))$, and decide which of the two coordinates of Ψ_E distinguishes $\gamma(1+\sqrt{2})$ from γ. Then verify directly that $\Psi_E(\gamma^\sigma) = \Psi_E(\gamma)$ while $\gamma^\sigma \neq \gamma$ modulo U_E^4, as lemma 9.3.8 predicts.

3. Let E/F be the *unramified* quadratic extension. Show that $v_E(y^\sigma) = v_E(y)$ for every $y \in E^\times$, hence that $y/y^\sigma \in U_E$ always. Then show that for every $j \geq 1$ there is $y \in E^\times$ with $y/y^\sigma \in U_E^j \setminus U_E^{j+1}$. Contrast this with lemma 9.3.4(3), and identify which step of the proof of lemma 9.3.8 fails in the unramified case.

4. Let ρ_1, ρ_2 satisfy the hypotheses of theorem 9.3.10. Using proposition 6.3.3(3) and Step 1 of that theorem's proof, show that $\check{\rho}_1$ and $\check{\rho}_2$ satisfy the same hypotheses at the same ψ. Show also that $T(\check{\rho}) = T(\rho)$ for every $\rho \in \mathcal{G}_2^0(F)$, and deduce that $\check{\rho}_1$ is again imprimitive and totally ramified.

5. With the standing data of definition 9.3.6, show that $E\langle n\rangle/U_E^{n-d+1}$ has $(q-1)q^{n-d}$ elements and that the target of Ψ_E has $(q-1)q^n$ elements. Using the freeness in lemma 9.3.8, compute the proportion of the target that lies in the image of Ψ_E, and check that it is independent of n.

6. Let ρ be totally ramified, minimal and imprimitive of level n, and suppose $T(\rho)$ has order two, say $T(\rho) = \{1, \kappa_{E/F}\}$. Combine theorem 9.1.24 with proposition 9.1.8 to show $3d(E|F) \leq n$, and deduce that the character ξ of proposition 9.3.5 has level at least $2d(E|F)$.

7. Keep the standing data of proposition 9.3.9. Show that $\sum_x \check{\xi}(x) = 0$, the sum running over a set of representatives for $E\langle n\rangle/U_E^{n-d+1}$, and deduce that $\sum_x \big(\check{\xi}(x) + \check{\xi}^\sigma(x)\big) = 0$ as well.

8. Let $F = \mathbb{Q}_2$ and $E = \mathbb{Q}_2(\sqrt{2})$, so $d(E|F) = 2$ by example 9.1.3. Determine which integers n can be the level of a totally ramified minimal imprimitive ρ with $\kappa_{E/F} \in T(\rho)$, and check that the smallest is $n = 5$. Then compute the orders of U_E/U_E^4 and U_E/U_E^3 and count the characters of U_E of level exactly 3 that are trivial on U_E^4.

9. Let $\rho \in \mathcal{G}_2^{\mathrm{im}}(F)$ be totally ramified and let χ be a character of $F^\times$. Using property (3) of theorem 8.5.4 together with corollary 9.3.11, show that $\chi\boldsymbol{\pi}^{\mathrm{im}}(\rho) \cong \boldsymbol{\pi}^{\mathrm{im}}(\rho)$ if and only if $\chi \in T(\rho)$. Conclude that the number of characters χ fixing $\boldsymbol{\pi}^{\mathrm{im}}(\rho)$ is 2 or 4.

10

Ordinary Representations

Two chapters have now been spent on the map $\boldsymbol{\pi}^{\mathrm{im}}$. Chapter 8 built it, and chapter 9 showed it separates the totally ramified parameters it is defined on. What neither settled is the reverse question: given a cuspidal representation of G, is it in the image at all?

The criterion refers only to the stratum a cuspidal representation contains, in the sense of chapter 3, and not to the Weil representation, to metaplectic groups, or to any part of the construction, so a reader who skipped chapter 8 entirely could still apply it. Section 10.1 sets up the correspondence between strata and the arithmetic data that produce them, and proves that being in the image is a property of the stratum rather than of the individual representation: either every representation containing a given ramified simple stratum lies in the image, or none does. Section 10.2 then makes the criterion explicit, and it turns out to be a question about a cubic polynomial having a root in F.

Throughout the chapter the residual characteristic of F is 2, and every representation considered is cuspidal, minimal and totally ramified. Those hypotheses hold from here on and are not repeated in each statement.

10.1 Ordinary Representations and Strata

Two standing hypotheses run through this chapter and are not repeated after this paragraph. The residual characteristic of F is 2, so that every ramified quadratic extension of F is wildly ramified. And every representation of $\mathrm{GL}_2(F)$ that appears is irreducible and cuspidal, while every Weil parameter that appears is totally ramified and minimal. We also fix once and for all an additive character $\psi \in \widehat{F}$ of level one, write $A = M_2(F)$ and $G = \mathrm{GL}_2(F)$, and use the character $\psi_A(x) = \psi(\mathrm{tr}x)$ of A from definition 3.3.8.

The cuspidal representations that lie in the image of $\boldsymbol{\pi}^{\mathrm{im}}$ can be named in the language of strata. The answer is a pair of congruences between the trace and determinant of a stratum element and the

trace and norm of an element of a quadratic extension, and the proof is a count: the strata attached to the possible quadratic data are produced first, and then the two sides are shown to have the same number of orbits under twisting by tame characters.

Definition 10.1.1: Ordinary and Exceptional Representations

Let $\pi \in \mathcal{A}_2^0(F)$. Then π is *ordinary* if $\pi \cong \boldsymbol{\pi}^{\mathrm{im}}(\rho)$ for some $\rho \in \mathcal{G}_2^{\mathrm{im}}(F)$, and *exceptional* otherwise.

Nothing in definition 10.1.1 is checkable as it stands: it asks whether π lies in the image of a map built in section 8.5 out of metaplectic data, and that construction plays no further part here. What is used of $\boldsymbol{\pi}^{\mathrm{im}}$ is only what theorem 8.5.4 records about it, namely the identity (8.30) for every character χ of $F^\times$ and every non-trivial additive character, the central-character identity $\omega_{\boldsymbol{\pi}^{\mathrm{im}}(\rho)} = \det \rho$, the twisting rule $\boldsymbol{\pi}^{\mathrm{im}}(\chi \otimes \rho) = \chi\, \boldsymbol{\pi}^{\mathrm{im}}(\rho)$, and the injectivity on totally ramified classes proved in corollary 9.3.11. Everything below is deduced from those four facts and from the classification of cuspidal representations in chapter 3.

The classification enters through the set of cuspidal representations attached to a fixed simple stratum. Theorem 3.5.2 attaches at least one simple stratum to every cuspidal representation of positive level that is minimal among its twists, and the next definition turns that relation around and asks which representations a given stratum collects.

Definition 10.1.2: The Cuspidals Attached to a Stratum

Let L/F be a finite field extension and $n \in \mathbb{Z}$. The *level-n shell* of L is

$$L\langle n \rangle = \{x \in L : v_L(x) = -n\}$$

Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum in A, in the sense of definition 3.4.1(4), so that $e_{\mathfrak{A}} = 2$ and n is odd. Write $n = 2m+1$. Then $\mathcal{A}_2^0(F;\alpha)$ denotes the set of $\pi \in \mathcal{A}_2^0(F)$ whose restriction to $U_{\mathfrak{A}}^{m+1}$ contains the character $\psi_\alpha \colon 1 + x \mapsto \psi_A(\alpha x)$.

The convention $n = 2m+1$ applies for the rest of the section: whenever a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ appears, m denotes $(n-1)/2$. Note that ψ_α is a character of $U_{\mathfrak{A}}^{m+1}$ and not only of $U_{\mathfrak{A}}^{n}$: for $x \in \mathfrak{P}^{m+1}$ one has $\alpha x \in \mathfrak{P}^{-n+m+1} = \mathfrak{P}^{-m}$, and for $x, y \in \mathfrak{P}^{m+1}$ the product xy lies in $\mathfrak{P}^{2m+2} = \mathfrak{P}^{n+1}$, so $\alpha xy \in \mathfrak{P}$ and ψ_A kills it. Thus $1 + x \mapsto \psi_A(\alpha x)$ is multiplicative on $U_{\mathfrak{A}}^{m+1}$. Two elements α, α' of $\mathfrak{P}^{-n}$ define the same character of $U_{\mathfrak{A}}^{m+1}$ exactly when $\alpha - \alpha' \in \mathfrak{P}^{-m}$, since $\mathfrak{P}^{-m}$ is the dual lattice of $\mathfrak{P}^{m+1}$ for ψ_A.

Proposition 10.1.3: Properties of the Cuspidals Attached to a Stratum

Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum, $n = 2m+1 \geq 1$, and put $E = F[\alpha]$ and $J = E^\times U_{\mathfrak{A}}^{m+1}$. Let $C(\psi_\alpha)$ be the set of characters Λ of J with $\Lambda|_{U_{\mathfrak{A}}^{m+1}} = \psi_\alpha$.

1. $C(\psi_\alpha)$ is non-empty, and $\Lambda \mapsto \text{c-Ind}_J^G \Lambda$ is a bijection of $C(\psi_\alpha)$ onto $\mathcal{A}_2^0(F;\alpha)$.
2. Every $\pi \in \mathcal{A}_2^0(F;\alpha)$ has normalized level $\ell(\pi) = n/2$.
3. If $(\mathfrak{A}', n', \alpha')$ is a second ramified simple stratum with $n' \geq 1$, then $\mathcal{A}_2^0(F;\alpha')$ and $\mathcal{A}_2^0(F;\alpha)$ are either equal or disjoint.

***Proof*:**
By proposition 3.4.3 the algebra $E = F[\alpha]$ is a field with $e(E|F) = e_{\mathfrak{A}} = 2$, so E/F is totally ramified quadratic, and $E^\times \subseteq \mathcal{K}_{\mathfrak{A}}$. Since $\mathfrak{P}^j \cap E = \mathfrak{p}_E^j$ for every j, one has $E^\times \cap U_{\mathfrak{A}}^j = U_E^j$, and $J/U_{\mathfrak{A}}^{m+1} \cong E^\times/U_E^{m+1}$ is abelian.

Step 1: $C(\psi_\alpha)$ is non-empty. The map $1 + x \mapsto \psi_E(\alpha x)$, with $\psi_E = \psi \circ \text{tr}_{E/F}$, is a character of U_E^{m+1}, and it agrees there with ψ_α because $\text{tr}(\alpha x) = \text{tr}_{E/F}(\alpha x)$ for $x \in E$, the characteristic polynomial of an element of the quadratic field E being its minimal polynomial. The group $\mathbb{C}^\times$ is divisible and U_E^{m+1} is open in $E^\times$, so this character extends to a character Λ_0 of $E^\times$. Setting Λ_0 on $E^\times$ and ψ_α on $U_{\mathfrak{A}}^{m+1}$ defines a character of J, the two prescriptions agreeing on the intersection U_E^{m+1}.

Step 2: the map lands in $\mathcal{A}_2^0(F;\alpha)$ and is injective. For $\Lambda \in C(\psi_\alpha)$ the group J is exactly $E^\times U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}$, since n is odd, so theorem 3.5.3(2) applies and $\pi_\Lambda = \text{c-Ind}_J^G \Lambda$ is irreducible and cuspidal. It contains Λ, hence ψ_α, so $\pi_\Lambda \in \mathcal{A}_2^0(F;\alpha)$. Suppose $\pi_\Lambda \cong \pi_{\Lambda'}$ with $\Lambda, \Lambda' \in C(\psi_\alpha)$. Both $(\mathfrak{A}, J, \Lambda)$ and $(\mathfrak{A}, J, \Lambda')$ are cuspidal types in the sense of definition 3.5.4, the positive-level case, so theorem 3.5.5 makes them G-conjugate: there is $g \in G$ with $g\mathfrak{A}g^{-1} = \mathfrak{A}$, $gJg^{-1} = J$ and $\Lambda^g \cong \Lambda'$. From $g\mathfrak{A}g^{-1} = \mathfrak{A}$ we get $gU_{\mathfrak{A}}^{m+1}g^{-1} = U_{\mathfrak{A}}^{m+1}$ and, restricting the last isomorphism, $\psi_\alpha^g = \psi_\alpha$ on $U_{\mathfrak{A}}^{m+1}$. Thus g intertwines ψ_α with itself, so $g \in J$ by theorem 3.5.3(1), and $\Lambda' = \Lambda^g = \Lambda$ because Λ is a character of J.

Step 3: the map is onto $\mathcal{A}_2^0(F;\alpha)$. Let $\pi \in \mathcal{A}_2^0(F;\alpha)$ and let $V^{\psi_\alpha} \neq 0$ be the ψ_α-eigenspace of $U_{\mathfrak{A}}^{m+1}$ in the space of π. The group J normalizes $U_{\mathfrak{A}}^{m+1}$ and fixes ψ_α, by theorem 3.5.3(1) again, so J acts on V^{ψ_α}. That action is smooth, and J contains $U_{\mathfrak{A}}^{m+1}F^\times$ with finite quotient $E^\times/F^\times U_E^{m+1}$, so V^{ψ_α} contains an irreducible J-subrepresentation Λ. Its restriction to $U_{\mathfrak{A}}^{m+1}$ is a multiple of ψ_α, so $\Lambda \otimes \Lambda_0^{-1}$, with Λ_0 from Step 1, is an irreducible representation of the abelian group $J/U_{\mathfrak{A}}^{m+1}$ and therefore has dimension one. Hence $\Lambda \in C(\psi_\alpha)$. Since J is open in G, compact induction is left adjoint to restriction, so the non-zero J-map $\Lambda \to \pi|_J$ produces a non-zero G-map $\text{c-Ind}_J^G \Lambda \to \pi$. Both sides are irreducible, so it is an isomorphism.

Step 4: the level, part (2). A ramified simple stratum is fundamental, being one of the four types listed in definition 3.4.1 for fundamental strata. A representation containing ψ_α on $U_{\mathfrak{A}}^{m+1}$ contains its restriction to the subgroup $U_{\mathfrak{A}}^n$, which is the character attached to the stratum $(\mathfrak{A}, n, \alpha)$ by definition 3.3.8. So theorem 3.3.13 applies and gives $\ell(\pi) = n/e_{\mathfrak{A}} = n/2$.

Step 5: part (3). Suppose $\pi \in \mathcal{A}_2^0(F;\alpha) \cap \mathcal{A}_2^0(F;\alpha')$. By part (2), $n/2 = \ell(\pi) = n'/2$, so $n' = n$ and $m' = m$. By Step 3 there are cuspidal types $(\mathfrak{A}, J, \Lambda)$ and $(\mathfrak{A}', J', \Lambda')$, attached to α and to α' respectively, with $\pi \cong \text{c-Ind}_J^G \Lambda \cong \text{c-Ind}_{J'}^G \Lambda'$. Theorem 3.5.5 makes the two types G-conjugate, so there is $g \in G$ with $g\mathfrak{A}g^{-1} = \mathfrak{A}'$ and $\Lambda^g \cong \Lambda'$. Restricting to $gU_{\mathfrak{A}}^{m+1}g^{-1} = U_{\mathfrak{A}'}^{m+1}$ gives $\psi_{\alpha'} = \psi_\alpha^g = \psi_{g\alpha g^{-1}}$ as characters of $U_{\mathfrak{A}'}^{m+1}$, the last equality because $\psi_\alpha(g^{-1}(1+y)g) = \psi_A(\alpha g^{-1}yg) = \psi_A(g\alpha g^{-1}y)$. Finally, for any $\sigma \in \mathcal{A}_2^0(F)$ the operator $\sigma(g)$ carries the ψ_α-eigenspace of $U_{\mathfrak{A}}^{m+1}$ onto the $\psi_{g\alpha g^{-1}}$-eigenspace of $U_{\mathfrak{A}'}^{m+1}$, so $\sigma \in \mathcal{A}_2^0(F;\alpha)$ if and only if $\sigma \in \mathcal{A}_2^0(F; g\alpha g^{-1}) = \mathcal{A}_2^0(F;\alpha')$. The two sets are therefore equal, completing the proof.

Part (1) says that $\mathcal{A}_2^0(F;\alpha)$ is as large and as explicit as the character group of $E^\times/U_E^{m+1}$ allows, and part (3) says that the sets $\mathcal{A}_2^0(F;\alpha)$ partition the cuspidal representations of half-integral positive level. So a criterion for membership in the image of $\boldsymbol{\pi}^{\text{im}}$ that is stated in terms of α alone will automatically be a statement about a whole block of the partition, and that is the shape theorem 10.1.17 takes.

The character ψ_α depends only on the coset $\alpha + \mathfrak{P}^{-m}$, and that coset is pinned down by two congruences.

Lemma 10.1.4: Trace and Determinant Determine a Ramified Simple Stratum

Let $(\mathfrak{A}, n, \alpha)$ and $(\mathfrak{A}, n, \alpha')$ be ramified simple strata in A with $n = 2m+1 \geq 1$. Then

$$\text{tr}\big(\alpha + \mathfrak{P}^{-m}\big) \subseteq \text{tr}\alpha + \mathfrak{p}^{-\lfloor m/2 \rfloor}, \qquad \det\big(\alpha + \mathfrak{P}^{-m}\big) \subseteq \det\alpha \cdot U_F^{\lfloor m/2 \rfloor + 1}$$

and the following are equivalent.

1. $\det\alpha' \equiv \det\alpha \bmod U_F^{\lfloor m/2 \rfloor+1}$ and $\text{tr}\alpha' \equiv \text{tr}\alpha \bmod \mathfrak{p}^{-\lfloor m/2 \rfloor}$.
2. The cosets $\alpha + \mathfrak{P}^{-m}$ and $\alpha' + \mathfrak{P}^{-m}$ are $\mathcal{K}_{\mathfrak{A}}$-conjugate.

When they hold, $\mathcal{A}_2^0(F;\alpha) = \mathcal{A}_2^0(F;\alpha')$.

***Proof*:**
By proposition 3.3.2 there are exactly two conjugacy classes of chain order in A, so all ramified ones are G-conjugate and there is no loss in taking $\mathfrak{A}$ to be the standard Iwahori order $\mathcal{I}$, with $\mathfrak{A} = \begin{pmatrix} \mathfrak{o} & \mathfrak{o} \\ \mathfrak{p} & \mathfrak{o} \end{pmatrix}$ and $\mathfrak{P} = \begin{pmatrix} \mathfrak{p} & \mathfrak{o} \\ \mathfrak{p} & \mathfrak{p} \end{pmatrix}$. For an even exponent, $\mathfrak{P}^{2j} = \varpi^j\mathfrak{A}$. For an odd one, $\mathfrak{P}^{2j+1} = \varpi^j\mathfrak{P}$. Reading off the diagonal entries in both cases gives

$$\text{tr}\big(\mathfrak{P}^j\big) = \mathfrak{p}^{\lceil j/2 \rceil} \tag{10.1}$$

for every $j \in \mathbb{Z}$, and in particular $\text{tr}(\mathfrak{P}^{-m}) = \mathfrak{p}^{-\lfloor m/2 \rfloor}$ and $\text{tr}(\mathfrak{P}^{m+1}) = \mathfrak{p}^{\lfloor m/2 \rfloor+1}$, since $\lceil -m/2 \rceil = -\lfloor m/2 \rfloor$ and $\lceil (m+1)/2 \rceil = \lfloor m/2 \rfloor + 1$.

Step 1: the two inclusions. Let $x \in \mathfrak{P}^{-m}$. The first inclusion is $\text{tr}(\alpha + x) = \text{tr}\alpha + \text{tr}x$ together with $\text{tr}x \in \mathfrak{p}^{-\lfloor m/2 \rfloor}$. For the second, α is invertible with $\alpha^{-1} \in \mathfrak{P}^n$, because $\alpha \in \mathfrak{P}^{-n} \setminus \mathfrak{P}^{1-n}$ generates the field $E = F[\alpha]$ and $v_E(\alpha) = -n$. Hence $y := \alpha^{-1}x \in \mathfrak{P}^{n-m} = \mathfrak{P}^{m+1}$ and $\det(\alpha + x) = \det\alpha \cdot \det(1+y)$ with $\det(1+y) = 1 + \text{tr}y + \det y$. Here $\text{tr}y \in \mathfrak{p}^{\lfloor m/2 \rfloor+1}$ by (10.1), and $\det y \in \mathfrak{p}^{m+1} \subseteq \mathfrak{p}^{\lfloor m/2 \rfloor+1}$, because $y \cdot \text{adj}(y) = \det(y) \cdot 1$ with $\text{adj}(y) = (\text{tr}y)1 - y \in \mathfrak{P}^{m+1}$, so $\det(y) \cdot 1 \in \mathfrak{P}^{2m+2} = \varpi^{m+1}\mathfrak{A}$. Both inclusions follow, and both quantities are unchanged by

conjugation, so (2) implies (1).

Step 2: (1) implies that some $\alpha_2 \in \alpha + \mathfrak{P}^{-m}$ *has* $\mathrm{tr}\alpha_2 = \mathrm{tr}\alpha'$ *and* $\det\alpha_2 = \det\alpha'$. Diagonal matrices in $\mathfrak{P}^{m+1}$ are exactly the $\mathrm{diag}(u_1, u_2)$ with $u_i \in \mathfrak{p}^{\lfloor m/2\rfloor+1}$, and matrices $\begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix}$ in $\mathfrak{P}^{m+1}$ are exactly those with $c \in \mathfrak{p}^{\lfloor (m+1)/2\rfloor+1}$. Both follow from $\mathfrak{P}^{2j} = \varpi^j\mathfrak{A}$ and $\mathfrak{P}^{2j+1} = \varpi^j\mathfrak{P}$. Put $u = \det\alpha'/\det\alpha - 1$, which lies in $\mathfrak{p}^{\lfloor m/2\rfloor+1}$ by hypothesis, and set $\alpha_1 = \alpha(1 + \mathrm{diag}(u, 0))$. Then $\alpha_1 \in \alpha\, U_{\mathfrak{A}}^{m+1} = \alpha + \mathfrak{P}^{-m}$ and $\det\alpha_1 = \det\alpha\,(1+u) = \det\alpha'$, while $\mathrm{tr}\alpha_1 \in \mathrm{tr}\alpha + \mathfrak{p}^{-\lfloor m/2\rfloor}$ by Step 1.

Next, the $(1,2)$-entry of α has valuation exactly $-m-1$: the entries of $\mathfrak{P}^{-n} = \varpi^{-m-1}\mathfrak{P}$ lie in $\mathfrak{p}^{-m}$ except the $(1,2)$-entry, which lies in $\mathfrak{p}^{-m-1}$, whereas every entry of $\mathfrak{P}^{1-n} = \varpi^{-m}\mathfrak{A}$ lies in $\mathfrak{p}^{-m}$; since $\alpha \notin \mathfrak{P}^{1-n}$, only the $(1,2)$-entry can be responsible. The same holds for α_1, which differs from α by an element of $\mathfrak{P}^{-m} \subseteq \mathfrak{P}^{1-n}$. For $c \in \mathfrak{p}^{\lfloor (m+1)/2\rfloor+1}$ set $w_c = \begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix} \in \mathfrak{P}^{m+1}$. Then $\det(1+w_c) = 1$ and $\mathrm{tr}(\alpha_1 w_c) = c\,(\alpha_1)_{12}$, which runs over $\mathfrak{p}^{\lfloor (m+1)/2\rfloor+1-m-1} = \mathfrak{p}^{-\lfloor m/2\rfloor}$ as c runs over $\mathfrak{p}^{\lfloor (m+1)/2\rfloor+1}$, because $\lfloor (m+1)/2\rfloor - m = -\lfloor m/2\rfloor$. Choose c so that $\mathrm{tr}\alpha_1 + \mathrm{tr}(\alpha_1 w_c) = \mathrm{tr}\alpha'$, which is possible since $\mathrm{tr}\alpha' - \mathrm{tr}\alpha_1 \in \mathfrak{p}^{-\lfloor m/2\rfloor}$, and put $\alpha_2 = \alpha_1(1+w_c)$. Then $\alpha_2 \in \alpha\, U_{\mathfrak{A}}^{m+1}$, $\det\alpha_2 = \det\alpha_1 = \det\alpha'$ and $\mathrm{tr}\alpha_2 = \mathrm{tr}\alpha'$.

Step 3: α_2 *and* α' *are* $\mathcal{K}_{\mathfrak{A}}$*-conjugate.* Write $t = \mathrm{tr}\alpha'$ and $D = \det\alpha'$. Since $E' := F[\alpha']$ is a totally ramified quadratic extension with $v_{E'}(\alpha') = -n$, we have $v_F(D) = v_F(N_{E'/F}\alpha') = -n$, and $t \in \mathfrak{p}^{-m}$ by (10.1). The polynomial $X^2 - tX + D$ is irreducible over F: two roots in F would have valuations summing to the odd number $-n$, hence distinct valuations, so the valuation of their sum t would be the smaller of the two, at most $-m-1$, against $t \in \mathfrak{p}^{-m}$. Both α_2 and α' have this irreducible characteristic polynomial, so neither is scalar and each is conjugate in G to its companion matrix. Hence $\alpha' = g\alpha_2 g^{-1}$ for some $g \in G$.

It remains to see $g \in \mathcal{K}_{\mathfrak{A}}$. The element α_2 lies in $\mathfrak{P}^{-n} \setminus \mathfrak{P}^{1-n}$ and $F[\alpha_2]$ is the field $F[X]/(X^2 - tX + D)$, totally ramified quadratic since $v_F(D) = -n$ is odd. So $(\mathfrak{A}, n, \alpha_2)$ is a stratum. It is fundamental because $\alpha_2^2 = t\alpha_2 - D$ has $\mathfrak{P}$-valuation exactly $-2n$ (the term $t\alpha_2$ lying in $\mathfrak{P}^{1-2n}$ while $D \cdot 1$ does not) and $\alpha_2 \notin \mathfrak{P}^{1-n}$, so proposition 3.3.11 applies, and it is ramified simple by definition 3.4.1(4). Now $\mathfrak{A}$ is the ring of endomorphisms of a chain of $\mathfrak{o}$-lattices in F^2, and by proposition 3.4.3(2) both $F[\alpha_2]^\times$ and $F[\alpha']^\times$ normalize $\mathfrak{A}$. A lattice chain normalized by $F[\alpha_2]^\times$ is a chain of $\mathfrak{o}_{F[\alpha_2]}$-lattices in the one-dimensional $F[\alpha_2]$-space F^2, and those are exactly the fractional ideals of $\mathfrak{o}_{F[\alpha_2]}$, a single chain. So $\mathfrak{A}$ is the unique chain order normalized by $F[\alpha_2]^\times$, and likewise for $F[\alpha']^\times$. Conjugating by g carries $F[\alpha_2]$ to $F[\alpha']$, hence carries $\mathfrak{A}$ to the unique chain order normalized by $F[\alpha']^\times$, which is $\mathfrak{A}$ again. Thus $g \in \mathcal{K}_{\mathfrak{A}}$ and $g(\alpha_2 + \mathfrak{P}^{-m})g^{-1} = \alpha' + \mathfrak{P}^{-m}$, while $\alpha_2 + \mathfrak{P}^{-m} = \alpha + \mathfrak{P}^{-m}$ by Step 2. This is (2).

Step 4: the last assertion. If $g \in \mathcal{K}_{\mathfrak{A}}$ carries $\alpha + \mathfrak{P}^{-m}$ to $\alpha' + \mathfrak{P}^{-m}$, then $\psi_{\alpha'} = \psi_{g\alpha g^{-1}}$ on $U_{\mathfrak{A}}^{m+1}$, and the eigenspace computation at the end of the proof of proposition 10.1.3 gives $\mathcal{A}_2^0(F;\alpha') = \mathcal{A}_2^0(F;\alpha)$, completing the proof.

So a ramified simple stratum contributes exactly two numbers to the classification, and they are the coefficients of a quadratic polynomial.

Example 10.1.5: A Ramified Simple Stratum of Level Three over $\mathbb{Q}_2$

Let $F = \mathbb{Q}_2$, so $q = 2$, $\varpi = 2$ and $U_F = U_F^1 = 1 + 2\mathbb{Z}_2$. Take $n = 3$, so $m = 1$ and $\lfloor m/2 \rfloor = 0$, and let

$$\alpha = \begin{pmatrix} 0 & \frac{1}{4} \\ \frac{1}{2} & 0 \end{pmatrix}$$

inside $\mathfrak{A} = \mathcal{I}$: the entries place α in $\mathfrak{P}^{-3} = \varpi^{-2}\mathfrak{P}$ and not in $\mathfrak{P}^{-2} = \varpi^{-1}\mathcal{I}$, the $(1,2)$-entry having valuation -2. Then $\mathrm{tr}\alpha = 0$ and $\det\alpha = -\frac{1}{8}$, of valuation $-3 = -n$. The characteristic polynomial $X^2 - \frac{1}{8}$ is irreducible over $\mathbb{Q}_2$ because $\frac{1}{8}$ has odd valuation, so $E = F[\alpha]$ is ramified quadratic and $v_E(\alpha) = -3$. The stratum $(\mathfrak{A}, 3, \alpha)$ is therefore ramified simple. By lemma 10.1.4 the character ψ_α of $U^2_{\mathfrak{A}}$ is determined, up to $\mathcal{K}_{\mathfrak{A}}$-conjugacy, by

$$\mathrm{tr}\alpha \bmod \mathfrak{o}, \qquad \det\alpha \bmod U_F$$

Over $\mathbb{Q}_2$ the second condition is empty beyond $v_F(\det\alpha) = -3$, since $U_F^{\lfloor m/2 \rfloor + 1} = U_F^1 = U_F$. And the first says $\mathrm{tr}\alpha \in \mathfrak{o}$, which excludes exactly the strata of trace-valuation -1. So at level three over $\mathbb{Q}_2$ there are precisely two $\mathcal{K}_{\mathfrak{A}}$-conjugacy classes of ramified simple strata for each value of $\det\alpha$ modulo U_F: those with $\mathrm{tr}\alpha \in \mathfrak{o}$, of which the present α is one, and those with $v_F(\mathrm{tr}\alpha) = -1$.

The parameters to be matched against such strata are carried by three pieces of data: an extension, a level and an element of the shell at that level, constrained by the factor-of-three bound that theorem 9.1.24 produced for the twisting group.

Definition 10.1.6: Admissible Wild Triple

An *admissible wild triple* $(E/F, n, \beta)$ consists of a totally ramified separable quadratic extension E/F inside the separable closure used to define $\mathcal{W}_F$, an odd integer $n = 2m + 1 \geq 1$, and an element $\beta \in E\langle n \rangle$, subject to

$$n \geq 3\, d(E|F)$$

Two pieces of notation attached to E/F are used constantly below. The first is the character $\kappa_{E/F}$ of $F^\times$ with kernel $N_{E/F}(E^\times)$, which by proposition 9.1.8 has level $d = d(E|F) \geq 1$. Because $2 + 2\lfloor d/2 \rfloor \geq d + 1$, the map $x \mapsto \kappa_{E/F}(1 + x)$ is a character of the additive group $\mathfrak{p}^{1+\lfloor d/2 \rfloor}$, trivial on $\mathfrak{p}^{d+1}$, and since ψ has level one the dual of $\mathfrak{p}^{1+\lfloor d/2 \rfloor}$ is $\mathfrak{p}^{-\lfloor d/2 \rfloor}$ while the annihilator of $\mathfrak{p}^{d+1}$ is $\mathfrak{p}^{-d}$. That duality is what makes the next definition legitimate.

Definition 10.1.7: Shift Element of a Ramified Quadratic Extension

Let E/F be totally ramified quadratic and $d = d(E|F)$. A *shift element* for E/F is an element $\delta_{E/F} \in F$ with

$$\kappa_{E/F}(1 + x) = \psi(x\, \delta_{E/F}), \qquad x \in \mathfrak{p}^{1+\lfloor d/2 \rfloor} \tag{10.2}$$

One exists, any two differ by an element of $\mathfrak{p}^{-\lfloor d/2 \rfloor}$, and every one of them has $v_F(\delta_{E/F}) = -d$.

Existence and uniqueness modulo $\mathfrak{p}^{-\lfloor d/2 \rfloor}$ are the duality just recorded, which also places $\delta_{E/F}$ in $\mathfrak{p}^{-d}$. The valuation is exactly $-d$: if $v_F(\delta_{E/F}) \geq 1 - d$ then $x\delta_{E/F} \in \mathfrak{p}$ for every $x \in \mathfrak{p}^d$, and $d \geq 1 + \lfloor d/2 \rfloor$, so $\kappa_{E/F}$ would be trivial on U_F^d, against proposition 9.1.8. The second piece of notation is $\psi_E = \psi \circ \mathrm{tr}_{E/F}$, of level $1 - d$ by lemma 9.1.19.

Definition 10.1.8: Parameters Attached to a Wild Triple

Let $(E/F, n, \beta)$ be an admissible wild triple and $d = d(E|F)$. Then $\mathcal{G}_2^0(F;\beta)$ denotes the set of classes $\rho = \mathrm{Ind}_{E/F}\xi \in \mathcal{G}_2^0(F)$ for which the character ξ of $E^\times$ satisfies

$$\xi(1+x) = \psi_E(\beta x), \qquad x \in \mathfrak{p}_E^{1+\lfloor (n-d)/2 \rfloor} \tag{10.3}$$

Before any counting can start, the set a triple collects has to be known to be non-empty and to consist of parameters minimal of level n: that is the level at which they will be matched against the cuspidals of a ramified simple stratum, and a set that was empty or carried the wrong level would make the two counts incomparable.

Proposition 10.1.9: Parameters of a Wild Triple Are Minimal of Level n

Let $(E/F, n, \beta)$ be an admissible wild triple, $d = d(E|F)$, and let ξ be a character of $E^\times$ satisfying (10.3). Then ξ is ramified of level $n-d$, it is E/F-regular and minimal over F, and $\rho = \mathrm{Ind}_{E/F}\xi$ is irreducible, totally ramified and minimal of level $\ell(\rho) = n$. In particular $\mathcal{G}_2^0(F;\beta)$ is non-empty.

***Proof*:**
Write $l = n - d$. Admissibility gives $n \geq 3d$ and $d \geq 1$, hence $l \geq 2d \geq d+1 \geq 2$ and $m \geq d$, the latter because $2m+1 \geq 3d$ forces $m \geq (3d-1)/2 \geq d$.

Step 1: existence and level of ξ. Since $l \geq 2$ we have $l \geq 1 + \lfloor l/2 \rfloor$, so U_E^l and U_E^{l+1} are both contained in the group on which (10.3) prescribes ξ. And that prescription is a character of $U_E^{1+\lfloor l/2 \rfloor}$ because $2(1+\lfloor l/2 \rfloor) \geq l+1$ makes the cross term $\psi_E(\beta xy)$ trivial. It extends to $E^\times$ by divisibility of $\mathbb{C}^\times$. For $v_E(x) \geq l+1$ we get $v_E(\beta x) \geq 1-d$, so ξ is trivial on U_E^{l+1}. For $v_E(x) = l$ we get $v_E(\beta x) = -d$, and ψ_E has level $1-d$, so ξ is non-trivial on U_E^l. Hence $\ell(\xi) = l$.

Step 2: ξ is regular. Suppose $\xi^\sigma = \xi$, σ generating $\mathrm{Gal}(E/F)$. Then ξ is trivial on $\{y^\sigma/y : y \in E^\times\} = \mathcal{K}_{E/F}$, so it factors through $E^\times/\mathcal{K}_{E/F} \cong N_{E/F}(E^\times)$ and, that group being open in $F^\times$ and $\mathbb{C}^\times$ divisible, $\xi = \chi_E$ for a character χ of $F^\times$. By lemma 9.1.9 the level of a non-trivial χ_E is either at most d or of the form $2j-d$ with $j \geq d+1$, hence congruent to d modulo 2. But $l = n-d > d$ and $l - d = n - 2d$ is odd, n being odd and $2d$ even. So no such χ exists and $\xi^\sigma \neq \xi$, which is regularity by lemma 8.4.2, and lemma 8.5.2 makes $\rho = \mathrm{Ind}_{E/F}\xi$ irreducible.

Step 3: minimality and level. Lemma 9.1.20(3) applies because $\ell(\xi) = l \geq d+1$, and $l \not\equiv d \bmod 2$ as just noted, so ξ is minimal over F. Proposition 9.1.21(1) then gives $\rho \notin \mathcal{G}_2^{\mathrm{nr}}(F)$, since $l \neq d$. That is, ρ is totally ramified in the sense of definition 9.1.17. Lemma 9.1.19 gives $\ell(\rho) = \ell(\xi) + d = n$, which is odd, so ρ is minimal by lemma 9.1.23, as we sought to show.

Part of the proposition below is proved in this book and part is quoted, and the line between them is worth drawing precisely. Clause (1) at positive level is theorem 5.3.8 together with theorem 3.3.13: the first gives $n(\pi,\psi) = 2N/e_{\mathfrak{A}}$ for a representation induced from a character of the shape considered there, and the second identifies $N/e_{\mathfrak{A}}$ with $\ell(\pi)$. What remains quoted is clause (2), and clause (1) for π of level zero, which theorem 5.3.8 does not reach because its hypotheses ask for $N \geq 1$.

Those are quoted from [2, §25]: clause (1) at level zero from 25.2 and 25.4, clause (2) from the 25.5 Corollary the proof below names. Everything else in this section is proved in full.

Proposition 10.1.10: Local Constants of a Cuspidal Representation Against Its Stratum

Let $\psi \in \widehat{F}$ have level one.

1. For every $\pi \in \mathcal{A}_2^0(F)$ one has $n(\pi, \psi) = 2\,\ell(\pi)$.
2. Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum with $n = 2m+1 \geq 1$, let $\pi \in \mathcal{A}_2^0(F;\alpha)$ and let χ be a character of $F^\times$ of level at most $\lfloor m/2 \rfloor$. Then

$$\varepsilon\big(\chi\pi, \tfrac{1}{2}, \psi\big) = \chi(\det \alpha)^{-1}\, \varepsilon\big(\pi, \tfrac{1}{2}, \psi\big)$$

***Proof Key ideas*:**
Both parts are read off one computation: the value of the operator $\pi_K(w)$ of theorem 8.2.4 on the standard functions $\xi_{\chi,k}$ of definition 8.2.3 when π is compactly induced from a cuspidal type $(\mathfrak{A}, J, \Lambda)$ as in definition 3.5.4. That value is a Gauss sum formed from Λ over J, and it is a monomial in q^{-s}. Part (1) is the degree of that monomial, which definition 5.3.4 names $n(\pi, \psi)$, and the degree comes out as twice the normalized level of π. Part (2) is the effect on the same sum of replacing Λ by $\Lambda \cdot (\chi \circ \det)|_J$, which is the inducing datum of $\chi\pi$: a character χ of level at most $\lfloor m/2 \rfloor$ is constant on the cosets the sum runs over, so the sum is unchanged apart from the single scalar $\chi(\det \alpha)^{-1}$. The computation itself is long and elementary, and it is the part this book quotes: it is [2, 25.5 Corollary], in the chapter on cuspidal local constants, for the level-zero case of (1) and for (2). Clause (1) at positive level is not quoted; it is proved above, by theorem 5.3.8 with theorem 3.3.13.

It is worth saying exactly what is borrowed, because it is not a new debt. This is the deferral section 5.3 already declares: the Kirillov model the evaluation runs through is built in chapter 8, and theorem 8.2.4 shows that the local constant of a cuspidal representation is carried by the single operator $\pi_K(w)$, but the closed-form evaluation of that operator against the cuspidal type is performed nowhere in this book. This chapter consumes that evaluation and adds nothing to what is owed. Everything else in section 10.1 is proved in full.

A totally ramified minimal parameter ρ can now be compared with a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ for which $\boldsymbol{\pi}^{\mathrm{im}}(\rho)$ lies in $\mathcal{A}_2^0(F;\alpha)$, and the comparison is made twice: once through the central character, which sees the trace, and once through the local constants of twists, which see the determinant.

Lemma 10.1.11: The Congruences Attached to a Wild Triple

Let $\rho \in \mathcal{G}_2^{\mathrm{im}}(F)$ be totally ramified and minimal, write $n = \ell(\rho)$ and let $\kappa_{E/F}$ be a non-trivial element of $T(\rho)$, so that $\rho \cong \mathrm{Ind}_{E/F}\xi$ as in proposition 9.3.5. Put $d = d(E|F)$ and assume $n \geq 3d$, and let $\beta \in E\langle n\rangle$ satisfy (10.3). Then $n = 2m+1$ is odd, there is a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ with $\boldsymbol{\pi}^{\mathrm{im}}(\rho) \in \mathcal{A}_2^0(F;\alpha)$, and every such α satisfies

$$\det\alpha \equiv N_{E/F}(\beta) \bmod U_F^{\lfloor m/2\rfloor+1}, \qquad \mathrm{tr}\alpha \equiv \delta_{E/F} + \mathrm{tr}_{E/F}(\beta) \bmod \mathfrak{p}^{-\lfloor m/2\rfloor} \tag{10.4}$$

***Proof*:**
By proposition 9.3.5(3) the integer n is odd, so $n = 2m+1$, and $\ell(\xi) = n - d$ by part (2) of the same proposition. The hypothesis $n \geq 3d$ gives $m \geq d$ as in the proof of proposition 10.1.9, hence $\lfloor m/2\rfloor \geq \lfloor d/2\rfloor$. Write $\pi = \boldsymbol{\pi}^{\mathrm{im}}(\rho)$.

Step 1: π lies in $\mathcal{A}_2^0(F;\alpha)$ for a ramified simple stratum of level n. Taking χ trivial in (8.30) gives $\varepsilon(\rho, s, \psi) = \varepsilon(\pi, s, \psi)$, and comparing the exponents of q^{-s} in the two monomials, which is what proposition 6.3.3(1) and definition 5.3.4 record, gives $n(\pi,\psi) = n(\rho,\psi) = \ell(\rho) = n$. For an arbitrary character χ of $F^\times$, (8.30) applied to χ gives in the same way $n(\chi\pi,\psi) = n(\chi\otimes\rho,\psi) = \ell(\chi\otimes\rho) \geq \ell(\rho) = n$, the inequality being minimality of ρ. By proposition 10.1.10(1) this reads $2\ell(\chi\pi) \geq 2\ell(\pi)$, so π has minimal level among its twists and $\ell(\pi) = n/2 > 0$. By theorem 3.5.5 the representation π is $\mathrm{c\text{-}Ind}_J^G\Lambda$ for a cuspidal type $(\mathfrak{A}, J, \Lambda)$, necessarily of positive level, so definition 3.5.4 attaches to it a simple stratum $(\mathfrak{A}, n_0, \alpha)$ with $\Lambda|_{U_{\mathfrak{A}}^{\lfloor n_0/2\rfloor+1}}$ a multiple of ψ_α. Then π contains ψ_α on $U_{\mathfrak{A}}^{n_0}$, and theorem 3.3.13 gives $n_0/e_{\mathfrak{A}} = \ell(\pi) = n/2$. As n is odd this forces $e_{\mathfrak{A}} = 2$ and $n_0 = n$, so the stratum is ramified simple by definition 3.4.1(4) and $\pi \in \mathcal{A}_2^0(F;\alpha)$.

Step 2: the trace congruence. Let $(\mathfrak{A}, n, \alpha)$ be any ramified simple stratum with $\pi \in \mathcal{A}_2^0(F;\alpha)$. The scalar matrices in $U_{\mathfrak{A}}^{m+1}$ are the $1+x$ with $x \in \mathfrak{p}^{\lceil (m+1)/2\rceil} = \mathfrak{p}^{\lfloor m/2\rfloor+1}$, because $c\cdot 1 \in \mathfrak{P}^{m+1}$ if and only if $c \in \mathfrak{p}^{\lceil (m+1)/2\rceil}$. On such an element π acts by $\omega_\pi(1+x)$, while a vector in the ψ_α-eigenspace is multiplied by $\psi_\alpha(1+x) = \psi(\mathrm{tr}(\alpha x)) = \psi(x\mathrm{tr}\alpha)$. Hence

$$\omega_\pi(1+x) = \psi(x\mathrm{tr}\alpha), \qquad x \in \mathfrak{p}^{\lfloor m/2\rfloor+1}$$

On the other side $\omega_\pi = \det\rho$ by theorem 8.5.4, and $\det\rho = \varkappa_{E/F}\cdot(\xi|_{F^\times})$ by proposition 6.2.7. Here $\varkappa_{E/F} = \det\mathrm{Ind}_{E/F}1$ is the non-trivial character of $\mathrm{Gal}(E/F)$ transported to $F^\times$, so its kernel is $N_{E/F}(E^\times)$ by theorem 6.2.2 and $\varkappa_{E/F} = \kappa_{E/F}$. For $x \in \mathfrak{p}^{\lfloor m/2\rfloor+1}$ we have $x \in \mathfrak{p}^{1+\lfloor d/2\rfloor}$, since $\lfloor m/2\rfloor \geq \lfloor d/2\rfloor$, so $\kappa_{E/F}(1+x) = \psi(x\delta_{E/F})$ by (10.2). And $v_E(x) = 2v_F(x) \geq 2\lfloor m/2\rfloor + 2 \geq m+1 \geq 1 + \lfloor (n-d)/2\rfloor$, the last inequality because $\lfloor (n-d)/2\rfloor \leq (n-1)/2 = m$ as $d \geq 1$. So (10.3) applies and $\xi(1+x) = \psi_E(\beta x) = \psi(\mathrm{tr}_{E/F}(\beta x)) = \psi(x\mathrm{tr}_{E/F}\beta)$, using that x lies in F. Combining,

$$\psi\big(x(\mathrm{tr}\alpha - \delta_{E/F} - \mathrm{tr}_{E/F}\beta)\big) = 1, \qquad x \in \mathfrak{p}^{\lfloor m/2\rfloor+1}$$

and the annihilator of $\mathfrak{p}^{\lfloor m/2\rfloor+1}$ for ψ of level one is $\mathfrak{p}^{-\lfloor m/2\rfloor}$. That is the second congruence in (10.4).

Step 3: the determinant congruence. Let χ be a character of $F^\times$ of level $j \leq \lfloor m/2\rfloor$. The pullback χ_E is unramified when χ is, and otherwise has level at most $\varphi(j)$ in the notation of lemma 9.1.7,

by lemma 9.1.9. In either case $\ell(\chi_E) \le \max\{j, 2j-d\}$. Both bounds satisfy $2\ell(\chi_E) < n-d$: if $j \le d$ then $2j \le 2\lfloor m/2 \rfloor \le m$ and $m < n-d$ because $n-d-m = m+1-d > 0$. If $j > d$ then $2(2j-d) \le 4\lfloor m/2 \rfloor - 2d \le 2m-2d < 2m+1-d$. So theorem 5.1.21, the GL_1 stability theorem underlying proposition 6.3.3(4) and the form of it that carries an explicit level bound, applied over the local field E with $\theta = \chi_E$ of level $\ell(\chi_E)$, with the more ramified character ξ of level $n-d$, and with the additive character ψ_E, gives

$$\varepsilon(\chi_E\xi, s, \psi_E) = \chi_E(\beta)^{-1}\,\varepsilon(\xi, s, \psi_E)$$

the element β playing the role of the element c of that theorem, since (10.3) prescribes $\xi(1+x) = \psi_E(\beta x)$ on $\mathfrak{p}_E^{1+\lfloor (n-d)/2 \rfloor}$, which is the exponent it requires. By lemma 9.1.10(1) we have $\chi \otimes \rho \cong \mathrm{Ind}_{E/F}(\chi_E\xi)$, so dividing the two instances of the inductivity relation (6.3) used in the proof of lemma 9.1.19, one for $\chi_E\xi$ and one for ξ, the Langlands constant cancels and

$$\varepsilon(\chi \otimes \rho, s, \psi) = \chi\big(N_{E/F}\beta\big)^{-1}\,\varepsilon(\rho, s, \psi)$$

using $\chi_E(\beta) = \chi(N_{E/F}\beta)$. Now (8.30) turns both sides into local constants of π and its twist, and proposition 10.1.10(2) evaluates the left-hand side at $s = \frac{1}{2}$ as $\chi(\det\alpha)^{-1}\varepsilon(\pi, \frac{1}{2}, \psi)$. The factor $\varepsilon(\pi, \frac{1}{2}, \psi)$ is non-zero, being a unit in $\mathbb{C}[q^s, q^{-s}]$ evaluated at $s = \frac{1}{2}$, so

$$\chi(\det\alpha) = \chi\big(N_{E/F}\beta\big)$$

for every character χ of $F^\times$ of level at most $\lfloor m/2 \rfloor$. Those characters are exactly the characters of $F^\times$ trivial on $U_F^{\lfloor m/2 \rfloor + 1}$, and they separate the points of $F^\times / U_F^{\lfloor m/2 \rfloor + 1}$, so $\det\alpha \equiv N_{E/F}(\beta) \bmod U_F^{\lfloor m/2 \rfloor + 1}$, completing the proof.

The lemma is the whole of one direction of theorem 10.1.17, and it also gives the other direction its starting point, because it can be read as producing a stratum from a triple rather than congruences from a stratum.

Proposition 10.1.12: Every Admissible Wild Triple Produces a Stratum

Let $(E/F, n, \beta)$ be an admissible wild triple, $n = 2m+1$. There is a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ satisfying (10.4), its coset $\alpha + \mathfrak{P}^{-m}$ is determined up to $\mathcal{K}_{\mathfrak{A}}$-conjugacy by those congruences, and

$$\boldsymbol{\pi}^{\mathrm{im}}\big(\mathcal{G}_2^0(F;\beta)\big) \subseteq \mathcal{A}_2^0(F;\alpha)$$

***Proof*:**

By proposition 10.1.9 there is a character ξ of $E^\times$ satisfying (10.3), and $\rho = \mathrm{Ind}_{E/F}\xi$ is irreducible, totally ramified and minimal of level n. It is imprimitive, being induced, so lemma 9.1.14 places $\kappa_{E/F}$ in $T(\rho)$, and $\kappa_{E/F} \ne 1$ because E/F is ramified. The hypotheses of lemma 10.1.11 are therefore met, and it produces a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ with $\boldsymbol{\pi}^{\mathrm{im}}(\rho) \in \mathcal{A}_2^0(F;\alpha)$ and with (10.4) satisfied. Uniqueness of the coset up to $\mathcal{K}_{\mathfrak{A}}$-conjugacy is lemma 10.1.4, since (10.4) prescribes $\det\alpha$ modulo $U_F^{\lfloor m/2 \rfloor + 1}$ and $\mathrm{tr}\alpha$ modulo $\mathfrak{p}^{-\lfloor m/2 \rfloor}$.

Now let $\rho' \in \mathcal{G}_2^0(F;\beta)$ be arbitrary, say $\rho' = \mathrm{Ind}_{E/F}\xi'$ with ξ' satisfying (10.3). The same argument applies to ρ' and yields a ramified simple stratum $(\mathfrak{A}', n, \alpha')$ with $\boldsymbol{\pi}^{\mathrm{im}}(\rho') \in \mathcal{A}_2^0(F;\alpha')$

and with α' satisfying (10.4) for the same E and the same β.

The two strata sit on chain orders that need not be equal, and lemma 10.1.4 compares only strata on a common one, so the orders must be matched first. By proposition 3.3.2 the ramified chain orders form a single G-conjugacy class, so $\mathfrak{A}' = h^{-1}\mathfrak{A}h$ for some $h \in G$. Then $(\mathfrak{A}, n, h\alpha' h^{-1})$ is again a ramified simple stratum, and conjugation does not move the two invariants in play: $\mathrm{tr}(h\alpha' h^{-1}) = \mathrm{tr}\alpha'$ and $\det(h\alpha' h^{-1}) = \det\alpha'$. Moreover $\psi_{h\alpha' h^{-1}}$ is $\psi_{\alpha'}$ transported along h, so conjugation carries the $\psi_{\alpha'}$-eigenspace of $U^{m+1}_{\mathfrak{A}'}$ onto the $\psi_{h\alpha' h^{-1}}$-eigenspace of $U^{m+1}_{\mathfrak{A}}$, whence $\mathcal{A}^0_2(F;\alpha') = \mathcal{A}^0_2(F;h\alpha' h^{-1})$ by the transport recorded in Step 5 of the proof of proposition 10.1.3.

Now α and $h\alpha' h^{-1}$ are ramified simple strata on the same $\mathfrak{A}$ satisfying the same congruences, so they satisfy hypothesis (1) of lemma 10.1.4, and that lemma gives $\mathcal{A}^0_2(F;h\alpha' h^{-1}) = \mathcal{A}^0_2(F;\alpha)$. Combining the two displays, $\mathcal{A}^0_2(F;\alpha') = \mathcal{A}^0_2(F;\alpha)$, as we sought to show.

A stratum satisfying the congruences therefore collects at least one ordinary representation. That it then collects nothing else is a matter of counting, and the count is run after dividing out the twists that move no stratum. Write Γ for the group of characters of $F^\times$ trivial on U^1_F, so that Γ consists of the unramified characters together with those of level zero.

Lemma 10.1.13: Two Counting Estimates

1. Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum, $n = 2m+1 \geq 1$. Then $\mathcal{A}^0_2(F;\alpha)$ is stable under $\pi \mapsto \chi\pi$ for $\chi \in \Gamma$, and the orbit set $\Gamma\backslash\mathcal{A}^0_2(F;\alpha)$ has q^m elements.
2. Let $(E/F, n, \beta)$ be an admissible wild triple, $d = d(E|F)$. Then $\mathcal{G}^0_2(F;\beta)$ is stable under $\rho \mapsto \chi \otimes \rho$ for $\chi \in \Gamma$, and

$$\left|\Gamma\backslash\mathcal{G}^0_2(F;\beta)\right| = \begin{cases} q^{\lfloor (n-d)/2 \rfloor} & n > 3d \\ \frac{1}{2} q^{\lfloor (n-d)/2 \rfloor} & n = 3d \end{cases}$$

***Proof*:**

Step 1: stability in (1). For $1 + x \in U^{m+1}_{\mathfrak{A}}$ one has $\det(1+x) = 1 + \mathrm{tr}x + \det x$, and $\mathrm{tr}x \in \mathfrak{p}^{\lfloor m/2 \rfloor + 1} \subseteq \mathfrak{p}$ while $\det x \in \mathfrak{p}^{m+1} \subseteq \mathfrak{p}$, both by the computation in Step 1 of the proof of lemma 10.1.4. So $\det(U^{m+1}_{\mathfrak{A}}) \subseteq U^1_F$ and $\chi \circ \det$ is trivial on $U^{m+1}_{\mathfrak{A}}$ for $\chi \in \Gamma$. Hence $\chi\pi$ contains ψ_α whenever π does, and $\chi\pi$ is again irreducible cuspidal.

Step 2: the count in (1). Keep $E = F[\alpha]$, $J = E^\times U^{m+1}_{\mathfrak{A}}$ and $C(\psi_\alpha)$ as in proposition 10.1.3. Since $\text{c-Ind}^G_J(\Lambda \cdot (\chi\circ\det)|_J) \cong \chi\,\text{c-Ind}^G_J\Lambda$, the bijection of that proposition turns the Γ-action into translation of $C(\psi_\alpha)$ by the characters $(\chi\circ\det)|_J$. Now $C(\psi_\alpha)$ is a torsor under the character group of $J/U^{m+1}_{\mathfrak{A}} \cong E^\times/U^{m+1}_E$, so

$$\left|\Gamma\backslash\mathcal{A}^0_2(F;\alpha)\right| = \left|\,\widehat{E^\times/U^{m+1}_E}/H\,\right|, \qquad H = \{(\chi\circ\det)|_J : \chi \in \Gamma\}$$

On $E^\times$ the determinant is $N_{E/F}$, so $(\chi\circ\det)|_{E^\times} = \chi_E$. If $\chi \in \Gamma$ then χ_E is trivial on U^1_E, because $N_{E/F}(U^1_E) \subseteq U^1_F$ by lemma 9.1.7(1) with $j = 1 \leq d$. Conversely let η be a character of $E^\times$ trivial on U^1_E. The induced map $\nu_0 \colon U_E/U^1_E \to U_F/U^1_F$ is $\bar{u} \mapsto \bar{u}^2$ on $k^\times$, by Step 2 of the

proof of lemma 9.1.7, and squaring is bijective on the group $k^\times$ of odd order $q-1$. So there is a character χ_0 of U_F, trivial on U_F^1, with $\chi_0(N_{E/F}u) = \eta(u)$ for $u \in U_E$. Extend χ_0 to $F^\times$ by choosing its value at ϖ so that $\chi_0(N_{E/F}\varpi_E) = \eta(\varpi_E)$ for a prime element ϖ_E of E. The resulting χ lies in Γ and $\chi_E = \eta$, both characters being determined by their values on U_E and at ϖ_E. So H is exactly the group of characters of $E^\times/U_E^{m+1}$ trivial on U_E^1/U_E^{m+1}, which is the kernel of the restriction map to $\widehat{U_E^1/U_E^{m+1}}$. That map is surjective because $\mathbb{C}^\times$ is divisible. Hence the quotient has order $|U_E^1/U_E^{m+1}| = q^m$, each graded piece U_E^j/U_E^{j+1} for $1 \le j \le m$ being isomorphic to k.

Step 3: stability in (2). Put $l = n-d$ and $e = \lfloor l/2 \rfloor$. For $\chi \in \Gamma$ the character χ_E is trivial on U_E^1, hence on U_E^{1+e}, so $\chi_E\xi$ satisfies (10.3) whenever ξ does. And $\chi \otimes \mathrm{Ind}_{E/F}\xi \cong \mathrm{Ind}_{E/F}(\chi_E\xi)$ by lemma 9.1.10(1).

Step 4: the count in (2). By lemma 9.1.10(2) and (3), two characters ξ_1, ξ_2 of $E^\times$ satisfying (10.3) give the same class in $\mathcal{G}_2^0(F;\beta)$ exactly when $\xi_2 \in \{\xi_1, \xi_1^\sigma\}$. Two such characters differ by an element of Γ exactly when their restrictions to U_E^1 agree: if $\xi_2 = \chi_E\xi_1$ with $\chi \in \Gamma$ the restrictions agree by Step 3, and conversely ξ_2/ξ_1 is then a character of $E^\times$ trivial on U_E^1, hence of the form χ_E with $\chi \in \Gamma$ by Step 2. Therefore $\Gamma\backslash\mathcal{G}_2^0(F;\beta)$ is the set of characters η of U_E^1 with $\eta|_{U_E^{1+e}} \colon 1+x \mapsto \psi_E(\beta x)$, modulo the action $\eta \mapsto \eta^\sigma$ where it is defined. There are $[U_E^1 : U_E^{1+e}] = q^e$ such η.

The action is defined exactly when ξ^σ again satisfies (10.3), that is when $\psi_E(\beta^\sigma x) = \psi_E(\beta x)$ for $v_E(x) \ge 1+e$. The annihilator of $\mathfrak{p}_E^{1+e}$ for ψ_E, of level $1-d$, is $\mathfrak{p}_E^{-d-e}$, so the condition reads $\beta - \beta^\sigma \in \mathfrak{p}_E^{-d-e}$, that is $v_E(\beta/\beta^\sigma - 1) \ge n-d-e = \lceil l/2 \rceil$. By lemma 9.3.4(3), applied to β of odd valuation $-n$, the quotient β/β^σ lies in $U_E^d \setminus U_E^{d+1}$, so $v_E(\beta/\beta^\sigma - 1) = d$ and the condition is $\lceil l/2 \rceil \le d$, that is $n-d \le 2d$, that is $n \le 3d$. Admissibility gives $n \ge 3d$, so the action is defined precisely when $n = 3d$.

If $n > 3d$ the q^e characters η are pairwise inequivalent and the count is q^e. If $n = 3d$ then σ acts on the set of q^e characters, and it acts without fixed points: a fixed η would make ξ^σ/ξ trivial on U_E^1, hence unramified, whereas Step 1 of the proof of proposition 9.1.21, applicable because ξ is regular and minimal over F of level $l \ge d+1$ by proposition 10.1.9, gives $\ell(\xi^\sigma/\xi) = l-d = n-2d = d \ge 1$. So the orbits have two elements each and the count is $q^e/2$, completing the proof.

Comparing the two estimates needs to know how many admissible wild triples share a stratum. The valuation of the trace of an element of the shell is what controls that, and it determines the discriminant exponent outright.

Lemma 10.1.14: The Trace of a Shell Element Determines the Discriminant Exponent

Let E/F be a *separable* totally ramified quadratic extension with $d = d(E|F)$, let $n = 2m+1 \geq 1$ be odd and let $\beta \in E\langle n \rangle$. Separability is not automatic here: in characteristic 2 an element of trace zero generates an inseparable quadratic extension, for which $\mathrm{tr}_{E/F}$ vanishes identically and definition 9.1.1 assigns no exponent at all. Then

$$v_F\big(\mathrm{tr}_{E/F}\beta\big) \geq \lfloor d/2 \rfloor - m$$

with equality when d is odd, and

$$d = \min\left\{\, 2\, v_F(\mathrm{tr}_{E/F}\beta) + n,\ v_F(4) \,\right\}$$

***Proof*:**

By proposition 9.1.4(1), $\mathrm{tr}_{E/F}(\mathfrak{p}_E^{-n}) = \mathfrak{p}^{\lfloor (d+1-n)/2 \rfloor}$, and $\lfloor (d+1-n)/2 \rfloor = \lfloor (d-2m)/2 \rfloor = \lfloor d/2 \rfloor - m$. Since $\beta \in \mathfrak{p}_E^{-n}$, the inequality follows.

Equality for odd d. If d is odd then $-n \equiv -d \bmod 2$, so $-n = 2r - d$ with $r = (d-n)/2 = \lfloor d/2 \rfloor - m$, and proposition 9.1.4(2) makes $\mathrm{tr}_{E/F}$ induce an isomorphism $\mathfrak{p}_E^{-n}/\mathfrak{p}_E^{-n+1} \to \mathfrak{p}^{r}/\mathfrak{p}^{r+1}$. The class of β in the source is non-zero because $v_E(\beta) = -n$ exactly, so $\mathrm{tr}_{E/F}\beta \notin \mathfrak{p}^{r+1}$ and $v_F(\mathrm{tr}_{E/F}\beta) = r$.

The formula. Suppose first d is odd. Then $2v_F(\mathrm{tr}_{E/F}\beta) + n = 2\big((d-1)/2 - m\big) + 2m + 1 = d$, and $d < v_F(4)$: by proposition 9.1.2(2) one has $d \leq v_F(4)$, and $v_F(4) = 2v_F(2)$ is even when $\mathrm{char}(F) = 0$ and infinite when $\mathrm{char}(F) = 2$, so it is never equal to the odd number d. The minimum is therefore d.

Suppose instead d is even. By proposition 9.1.2(3) this forces $d = v_F(4)$, and the inequality already proved gives $2v_F(\mathrm{tr}_{E/F}\beta) + n \geq 2(d/2 - m) + 2m + 1 = d + 1 > v_F(4)$. The minimum is again $v_F(4) = d$, completing the proof.

Two admissible wild triples can perfectly well produce the same stratum, and the next proposition says how far apart they can then be. The first two parts say that the coarse congruences already pin the extension down to its discriminant exponent and its shift element. The third says that the parameters two triples produce are either identical or have nothing in common, and identifies the finer congruences that the first case forces.

Proposition 10.1.15: Wild Triples Meeting in One Stratum

Let $(E/F, n, \beta)$ be an admissible wild triple, $n = 2m+1$, $d = d(E|F)$. Let E'/F be a totally ramified separable quadratic extension and $\beta' \in E'\langle n\rangle$ with

$$N_{E'/F}(\beta') \equiv N_{E/F}(\beta) \bmod U_F^{\lfloor m/2\rfloor+1}, \qquad \mathrm{tr}_{E'/F}(\beta') \equiv \mathrm{tr}_{E/F}(\beta) \bmod \mathfrak{p}^{-\lfloor m/2\rfloor} \tag{10.5}$$

Put $s = \lceil (n-d)/2\rceil$. Then

1. $d(E'|F) = d$, and $(E'/F, n, \beta')$ is again an admissible wild triple;
2. $\delta_{E'/F} \equiv \delta_{E/F} \bmod \mathfrak{p}^{-\lfloor m/2\rfloor}$;
3. if $\mathcal{G}_2^0(F;\beta) \cap \mathcal{G}_2^0(F;\beta') \neq \emptyset$, then $E' = E$ and $\beta' \in \beta^\tau U_E^s$ for some $\tau \in \mathrm{Gal}(E/F)$. In that case $\mathcal{G}_2^0(F;\beta') = \mathcal{G}_2^0(F;\beta)$ and

$$N_{E/F}(\beta') \in N_{E/F}(\beta)\, N_{E/F}\big(U_E^s\big), \qquad \mathrm{tr}_{E/F}(\beta') \in \mathrm{tr}_{E/F}(\beta) + \mathrm{tr}_{E/F}\big(\mathfrak{p}_E^{-n+s}\big)$$

***Proof*:**

Throughout, $a = \lfloor m/2\rfloor + 1$, $T = \mathrm{tr}_{E/F}\beta$, $T' = \mathrm{tr}_{E'/F}\beta'$ and $d' = d(E'|F)$. Admissibility gives $m \geq d$, as in the proof of proposition 10.1.9, and hence $n - 2\lfloor m/2\rfloor \geq n - m = m+1 > d$.

Step 1: part (1). Lemma 10.1.14 gives $d = \min\{2v_F(T)+n,\ v_F(4)\}$ and $d' = \min\{2v_F(T')+n,\ v_F(4)\}$. If $d < v_F(4)$, the first identity forces $2v_F(T)+n = d$, and $d < n - 2\lfloor m/2\rfloor$ gives $v_F(T) < -\lfloor m/2\rfloor$; the trace congruence in (10.5) then gives $v_F(T') = v_F(T)$ and hence $d' = d$. If instead $d = v_F(4)$, then $2v_F(T)+n \geq d$, and the trace congruence gives $v_F(T') \geq \min\{v_F(T),\ -\lfloor m/2\rfloor\}$, whence $2v_F(T')+n \geq \min\{2v_F(T)+n,\ n-2\lfloor m/2\rfloor\} \geq d$ and again $d' = d$. Since n is odd, $\beta' \in E'\langle n\rangle$ and $n \geq 3d = 3d'$, the triple $(E'/F, n, \beta')$ is admissible.

Step 2: part (2) reduces to a statement about $\kappa_{E/F}$ and $\kappa_{E'/F}$. Both characters have level d by proposition 9.1.8 and part (1). Since $m \geq d$ we have $a \geq \lfloor d/2\rfloor + 1$, so (10.2) applies on $\mathfrak{p}^a$ to both. If $\kappa_{E/F}$ and $\kappa_{E'/F}$ agree on U_F^a then $\psi(x(\delta_{E/F} - \delta_{E'/F})) = 1$ for all $x \in \mathfrak{p}^a$, so $\delta_{E/F} - \delta_{E'/F} \in \mathfrak{p}^{1-a} = \mathfrak{p}^{-\lfloor m/2\rfloor}$, which is (2). If $a \geq d+1$ the two characters agree on U_F^a trivially, both being trivial there. So assume $a \leq d$ for the rest of the argument.

Step 3: the two characters agree on U_F^a. Write $\kappa = \kappa_{E/F}$ and $\kappa' = \kappa_{E'/F}$. Both are non-trivial on U_F^a, having level $d \geq a$, and both are quadratic, so each has kernel of index two in U_F^a. A character of order at most two is determined by its kernel, so it suffices to prove $\ker\kappa \cap U_F^a \subseteq \ker\kappa'$.

Let $z \in E^\times$ with $N_{E/F}(z) \in U_F^a$. Then $v_E(z) = v_F(N_{E/F}z) = 0$, so $z \in U_E$. And for $0 \leq j \leq a-1 < d$ the map ν_j of lemma 9.1.7 is injective with $\varphi(j) = j$, so $N_{E/F}(z) \in U_F^{j+1}$ and $z \in U_E^j$ force $z \in U_E^{j+1}$. Iterating from $j = 0$ gives $z \in U_E^a$. Every element of $E = F[\beta]$ is $x + y\beta$ with $x, y \in F$, since $v_E(\beta) = -n$ is odd and so $\beta \notin F$; writing $z = x + y\beta$, the two summands of $z - 1 = (x-1) + y\beta$ have valuations $2v_F(x-1)$ and $2v_F(y) - n$, of opposite parity, so both are at least a. Hence $x \in U_F$ and, with $w = y/x$,

$$z = x(1 + w\beta), \qquad 2v_F(w) \geq n + a$$

Since κ' is quadratic, $\kappa'(N_{E/F}z) = \kappa'(x^2)\,\kappa'(N_{E/F}(1+w\beta)) = \kappa'(N_{E/F}(1+w\beta))$.

Now compare $N_{E/F}(1+w\beta) = 1+wT+w^2N_{E/F}\beta$ with $N_{E'/F}(1+w\beta') = 1+wT'+w^2N_{E'/F}\beta'$. For the trace terms,

$$v_F\big(w(T-T')\big) \geq \tfrac{n+a}{2} - \lfloor m/2 \rfloor = \tfrac{n+a}{2} - a + 1 = \tfrac{n-a}{2} + 1 \geq d+1$$

the last step because $n \geq 3d$ and $a \leq d$ give $n-a \geq 2d$. For the norm terms, $N_{E'/F}\beta' - N_{E/F}\beta$ lies in $N_{E/F}(\beta)\mathfrak{p}^a$, of valuation at least $-n+a$, so $v_F(w^2(N_{E/F}\beta - N_{E'/F}\beta')) \geq (n+a)-n+a = 2a \geq d+1$, using $2a = 2\lfloor m/2 \rfloor + 2 \geq m+1 \geq d+1$. So the two norms differ by an element of $\mathfrak{p}^{d+1}$ and, both being units, their ratio lies in $U_F^{d+1} \subseteq \ker\kappa'$. Since $N_{E'/F}(1+w\beta') \in \ker\kappa'$, we get $\kappa'(N_{E/F}(1+w\beta)) = 1$ and therefore $\kappa'(N_{E/F}z) = 1$. As $\ker\kappa \cap U_F^a = N_{E/F}(E^\times) \cap U_F^a$, this is the required inclusion and finishes (2).

Step 4: part (3), the extensions coincide. Let $\rho \in \mathcal{G}_2^0(F;\beta) \cap \mathcal{G}_2^0(F;\beta')$. By proposition 10.1.9, applied to each of the two triples, ρ is totally ramified, imprimitive and minimal with $\ell(\rho) = n$. Lemma 9.1.14 places $\kappa_{E/F}$ and $\kappa_{E'/F}$ in $T(\rho)$, and they are distinct if $E \neq E'$, distinct quadratic extensions having distinct norm groups by theorem 6.2.2. Suppose $E \neq E'$. Then $|T(\rho)| \geq 3$, so $|T(\rho)| = 4$ by proposition 9.1.15 and ρ is triply imprimitive, and corollary 9.1.25 applies with the three levels equal to the three discriminant exponents, by proposition 9.1.8. Two of those exponents equal d, by part (1). In configuration (1) of that corollary the smallest level occurs once and the other occurs twice, so the repeated level is d and $n = 2d + m_1 < 3d$, against admissibility. So configuration (2) holds: all three levels equal d and $n = 3d$, and the restrictions of the three characters to U_F^d are pairwise distinct. But $m = (3d-1)/2$ gives $\lfloor m/2 \rfloor = \lfloor (3d-1)/4 \rfloor \leq d-1$, so part (2) gives $\delta_{E'/F} \equiv \delta_{E/F} \bmod \mathfrak{p}^{1-d}$, and then for $x \in \mathfrak{p}^d \subseteq \mathfrak{p}^{1+\lfloor d/2 \rfloor}$ the product $x(\delta_{E/F} - \delta_{E'/F})$ lies in $\mathfrak{p}$ and (10.2) gives $\kappa_{E/F}(1+x) = \kappa_{E'/F}(1+x)$. The two restrictions to U_F^d therefore agree, a contradiction. Hence $E' = E$.

Step 5: part (3), the finer congruences. Write $\rho \cong \mathrm{Ind}_{E/F}\xi \cong \mathrm{Ind}_{E/F}\xi'$ with ξ and ξ' satisfying (10.3) for β and for β'. By lemma 9.1.10(3) there is $\tau \in \mathrm{Gal}(E/F)$ with $\xi' = \xi^\tau$. Put $e = \lfloor (n-d)/2 \rfloor$, so that $s = n-d-e$. For $v_E(x) \geq 1+e$,

$$\psi_E(\beta' x) = \xi'(1+x) = \xi(1+x^\tau) = \psi_E(\beta x^\tau) = \psi_E(\beta^\tau x)$$

using $\psi_E \circ \tau = \psi_E$, which holds because $\mathrm{tr}_{E/F}$ is $\mathrm{Gal}(E/F)$-invariant. The annihilator of $\mathfrak{p}_E^{1+e}$ for ψ_E, of level $1-d$, is $\mathfrak{p}_E^{-d-e} = \mathfrak{p}_E^{-n+s}$, so $\beta' - \beta^\tau \in \mathfrak{p}_E^{-n+s}$, that is $\beta'/\beta^\tau \in U_E^s$. Applying $N_{E/F}$ and $\mathrm{tr}_{E/F}$ to $\beta' = \beta^\tau(\beta'/\beta^\tau)$ and to $\beta' = \beta^\tau + (\beta' - \beta^\tau)$ gives the two displayed relations, since $N_{E/F}$ and $\mathrm{tr}_{E/F}$ are unchanged by τ. Finally β' and β^τ define the same character of U_E^{1+e}, so $\mathcal{G}_2^0(F;\beta') = \mathcal{G}_2^0(F;\beta^\tau)$, and $\xi \mapsto \xi^\tau$ is a bijection between the characters satisfying (10.3) for β and those satisfying it for β^τ, under which $\mathrm{Ind}_{E/F}\xi$ is unchanged by lemma 9.1.10(2); so $\mathcal{G}_2^0(F;\beta^\tau) = \mathcal{G}_2^0(F;\beta)$, completing the proof.

The stratum carries its own quadratic extension $F[\alpha]$, and the congruences force its discriminant exponent into a narrow relation with that of E.

Proposition 10.1.16: Triple and Stratum Compared

Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum with $n = 2m+1 \geq 1$, and let $(E/F, n, \beta)$ be an admissible wild triple satisfying (10.4). Put $d = d(E|F)$ and $d_\alpha = d(F[\alpha]|F)$. Then

$$n > 3d \iff n > 3d_\alpha, \qquad n = 3d \iff n \leq 3d_\alpha$$

and in the first case $d_\alpha = d$. Consequently, if $n \leq 3d_\alpha$ then $n = 3d$, so $n \equiv 0 \bmod 3$.

***Proof*:**

By proposition 3.4.3 the field $F[\alpha]$ is totally ramified quadratic over F with $v_{F[\alpha]}(\alpha) = -n$, and $\mathrm{tr}\alpha = \mathrm{tr}_{F[\alpha]/F}(\alpha)$. The extension is separable: its minimal polynomial is $X^2 - \mathrm{tr}(\alpha)X + \det(\alpha)$, of derivative $2X - \mathrm{tr}\alpha$, so inseparability would force $\mathrm{char}(F) = 2$ and $\mathrm{tr}\alpha = 0$, and then d_α is not defined, which is why that case is excluded by hypothesis. So lemma 10.1.14 applies to α inside $F[\alpha]$ and gives $d_\alpha = \min\{2v_F(\mathrm{tr}\alpha) + n,\ v_F(4)\}$. Admissibility gives $m \geq d$ and $n - 2\lfloor m/2 \rfloor \geq m + 1 > d$, as before. Write $T = \mathrm{tr}_{E/F}\beta$ and $\delta = \delta_{E/F}$, so that $v_F(\delta) = -d$ and, by (10.4), $\mathrm{tr}\alpha \equiv \delta + T \bmod \mathfrak{p}^{-\lfloor m/2 \rfloor}$.

Step 1: $n > 3d$ with d odd. Lemma 10.1.14 gives $v_F(T) = (d-1)/2 - m$. From $3d < n = 2m+1$ we get $(3d-1)/2 < m$, that is $(d-1)/2 - m < -d = v_F(\delta)$, so $v_F(\delta + T) = (d-1)/2 - m$. Moreover $(d-1)/2 \leq (m-1)/2 < \lceil m/2 \rceil$, using $d \leq m$, so $(d-1)/2 - m < -\lfloor m/2 \rfloor$ and the congruence transfers the valuation: $v_F(\mathrm{tr}\alpha) = (d-1)/2 - m$. Hence $2v_F(\mathrm{tr}\alpha) + n = d$, and $d < v_F(4)$ as in the proof of lemma 10.1.14, so $d_\alpha = d$ and $n > 3d_\alpha$.

Step 2: $n > 3d$ with d even. Then $d = v_F(4)$ by proposition 9.1.2(3), and $v_F(T) \geq d/2 - m = (d-n+1)/2 > (d-n)/2$, while $v_F(\delta) = -d \geq (d-n)/2$ because $n \geq 3d$, and $-\lfloor m/2 \rfloor \geq (d-n)/2$ because $n - 2\lfloor m/2 \rfloor > d$. So $v_F(\mathrm{tr}\alpha) \geq (d-n)/2$, giving $2v_F(\mathrm{tr}\alpha) + n \geq d = v_F(4)$ and therefore $d_\alpha = v_F(4) = d$ and $n > 3d_\alpha$.

Step 3: $n = 3d$. Here n odd forces d odd, hence $d < v_F(4)$, and lemma 10.1.14 gives $v_F(T) = (d-n)/2 = -d = v_F(\delta)$. Also $\lfloor m/2 \rfloor \leq d$, since $m = (3d-1)/2$. So every term in the congruence has valuation at least $-d$ and $v_F(\mathrm{tr}\alpha) \geq -d$, whence $2v_F(\mathrm{tr}\alpha) + n \geq -2d + 3d = d$ and $d_\alpha = \min\{2v_F(\mathrm{tr}\alpha) + n,\ v_F(4)\} \geq \min\{d, v_F(4)\} = d$. Thus $3d_\alpha \geq 3d = n$.

Step 4: the equivalences. Admissibility says $n \geq 3d$, so exactly one of $n > 3d$ and $n = 3d$ holds. Steps 1 and 2 show that the first implies $n > 3d_\alpha$, and Step 3 shows that the second implies $n \leq 3d_\alpha$. The two conclusions exclude each other, so each implication reverses. The final assertion is the contrapositive of the first equivalence together with $3 \mid 3d$, completing the proof.

The congruences (10.4) are a condition on the stratum alone, so if they detect one ordinary representation in $\mathcal{A}_2^0(F;\alpha)$ they cannot distinguish between the members of that set, and the theorem says that the expected conclusion holds: the property of being ordinary is constant on each block of the partition of proposition 10.1.3(3).

Theorem 10.1.17: The Congruence Criterion for Ordinary Representations

Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum in A with $n = 2m+1 \geq 1$. The following are equivalent.

1. The set $\mathcal{A}_2^0(F;\alpha)$ contains an ordinary representation.
2. Every $\pi \in \mathcal{A}_2^0(F;\alpha)$ is ordinary.
3. There are a totally ramified separable quadratic extension E/F and an element $\beta \in E\langle n\rangle$ with $n \geq 3d(E|F)$ and

$$\det\alpha \equiv N_{E/F}(\beta) \bmod U_F^{\lfloor m/2\rfloor+1}, \qquad \mathrm{tr}\alpha \equiv \delta_{E/F} + \mathrm{tr}_{E/F}(\beta) \bmod \mathfrak{p}^{-\lfloor m/2\rfloor}$$

Replacing ψ by $u\psi$ with $u \in U_F$, which is the general change of additive character of level one, replaces α by $u^{-1}\alpha$, $\delta_{E/F}$ by $u^{-1}\delta_{E/F}$ and β by $u^{-1}\beta$. The direction matters and is worth checking: $(u\psi)\big(\mathrm{tr}(u^{-1}\alpha x)\big) = \psi\big(\mathrm{tr}(\alpha x)\big)$, so it is $u^{-1}\alpha$, not $u\alpha$, that carries the same character of $U_{\mathfrak{A}}^{m+1}$ over to the new additive character. Under that rule both sides of the first congruence scale by u^{-2} and both sides of the second by u^{-1}, so each condition is carried to the corresponding condition for the transported data, and the criterion does not depend on which character of level one was fixed.

***Proof*:**
Write d, $e = \lfloor (n-d)/2\rfloor$, $s = \lceil (n-d)/2\rceil$ and $a = \lfloor m/2\rfloor + 1$ as the proof proceeds. Implication (2) implies (1) is immediate, since $\mathcal{A}_2^0(F;\alpha)$ is non-empty by proposition 10.1.3(1).

Step 1: (3) implies (1). Let $(E/F, n, \beta)$ be as in (3), an admissible wild triple by definition 10.1.6. Proposition 10.1.12 produces a ramified simple stratum $(\mathfrak{A}_0, n, \alpha_0)$ satisfying (10.4) for that triple, with $\boldsymbol{\pi}^{\mathrm{im}}(\mathcal{G}_2^0(F;\beta)) \subseteq \mathcal{A}_2^0(F;\alpha_0)$, and $\mathcal{G}_2^0(F;\beta)$ is non-empty by proposition 10.1.9. Since α and α_0 satisfy the same two congruences, lemma 10.1.4 gives $\mathcal{A}_2^0(F;\alpha) = \mathcal{A}_2^0(F;\alpha_0)$, which therefore contains an ordinary representation.

Step 2: (1) implies (3), and the parameter is totally ramified. Let $\pi \in \mathcal{A}_2^0(F;\alpha)$ be ordinary, say $\pi \cong \boldsymbol{\pi}^{\mathrm{im}}(\rho)$ with $\rho \in \mathcal{G}_2^{\mathrm{im}}(F)$. Suppose ρ were not totally ramified. Then definition 9.1.17 and proposition 9.1.16(2) give a non-trivial unramified $\chi_0 \in T(\rho)$, so $\chi_0 \otimes \rho \cong \rho$ and theorem 8.5.4(3) gives $\chi_0\pi \cong \pi$. Writing $\pi = \text{c-Ind}_J^G\Lambda$ with $\Lambda \in C(\psi_\alpha)$ as in proposition 10.1.3(1), and noting $\chi_0 \in \Gamma$, we get $\chi_0\pi = \text{c-Ind}_J^G(\Lambda \cdot (\chi_0 \circ \det)|_J)$ in $\mathcal{A}_2^0(F;\alpha)$ by lemma 10.1.13(1); injectivity of the parametrization forces $(\chi_0 \circ \det)|_J = 1$. Taking a prime element ϖ_E of $E = F[\alpha]$, which lies in J, gives $\chi_0(N_{E/F}\varpi_E) = 1$, and $v_F(N_{E/F}\varpi_E) = 1$ with χ_0 unramified gives $\chi_0(\varpi) = 1$. So $\chi_0 = 1$, a contradiction. Hence ρ is totally ramified.

Step 3: (1) implies (3), the triple. By proposition 10.1.3(2) and proposition 10.1.10(1) we have $n(\pi,\psi) = 2\ell(\pi) = n$, and taking χ trivial in (8.30) gives $\ell(\rho) = n(\rho,\psi) = n(\pi,\psi) = n$, which is odd. So ρ is minimal by lemma 9.1.23. Theorem 9.1.24 gives a character in $T(\rho)$ of level $j \geq 1$ with $3j \leq n$, and by Step 2 of the proof of proposition 9.1.15 that character is $\kappa_{E/F}$ for a separable quadratic E/F, with $j = d(E|F) =: d$ by proposition 9.1.8. So $n \geq 3d$, and

proposition 9.3.5 gives that E/F is totally ramified and $\rho \cong \mathrm{Ind}_{E/F}\xi$ with $\ell(\xi) = n - d$.

Since $n - d \geq 2d \geq 2$, the exponent $1 + \lfloor (n-d)/2 \rfloor$ is at most $n - d$, so ξ is prescribed on U_E^{1+e} and, ψ_E having level $1 - d$, there is $\beta \in E$ with $\xi(1+x) = \psi_E(\beta x)$ for $v_E(x) \geq 1 + e$, determined modulo $\mathfrak{p}_E^{-d-e}$. Its valuation is $-n$: triviality of ξ on U_E^{n-d+1} forces $v_E(\beta) + (n - d + 1) \geq 1 - d$, and non-triviality on U_E^{n-d} forces $v_E(\beta) + (n - d) \leq -d$. So $(E/F, n, \beta)$ is an admissible wild triple, and lemma 10.1.11 applies to ρ and gives (10.4) for every ramified simple stratum whose set of cuspidals contains π, in particular for α. This is (3).

Step 4: the family of triples attached to α. Assume (1), and keep $(E/F, n, \beta)$ from Step 3. Let $\mathcal{S}$ be the set of pairs $(E'/F, \beta')$ with E'/F totally ramified separable quadratic and $\beta' \in E'\langle n \rangle$ satisfying (10.5). By proposition 10.1.15(1) each such pair is an admissible wild triple with $d(E'|F) = d$, and by proposition 10.1.15(2) together with (10.5) each satisfies (10.4) relative to α. Proposition 10.1.12 and lemma 10.1.4 then give $\boldsymbol{\pi}^{\mathrm{im}}(\mathcal{G}_2^0(F; \beta')) \subseteq \mathcal{A}_2^0(F; \alpha)$ for every $(E'/F, \beta') \in \mathcal{S}$.

Define Φ on $\mathcal{S}$ by

$$\Phi(E'/F, \beta') = \Big(N_{E'/F}(\beta') \bmod N_{E/F}\big(U_E^s\big),\ \mathrm{tr}_{E'/F}(\beta') \bmod \mathrm{tr}_{E/F}\big(\mathfrak{p}_E^{-n+s}\big) \Big)$$

Both subgroups depend only on d, so they are the same for every member of $\mathcal{S}$, but the two halves are not equally immediate and the second needs the boundary case handled separately. For the trace, $d(E'|F) = d$ and proposition 9.1.4(1) gives $\mathrm{tr}_{E'/F}(\mathfrak{p}_{E'}^{-n+s}) = \mathfrak{p}^{\lfloor (s+d+1-n)/2 \rfloor}$, an ideal named by d, n and s alone. For the norm, when $s > d$ the map ν_s of lemma 9.1.7 is surjective and $N_{E'/F}(U_{E'}^s) = U_F^{\lceil (s+d)/2 \rceil}$, again independent of E'. When $n = 3d$, however, $s = d$, and there lemma 9.1.7(4) says only that $N_{E'/F}(U_{E'}^d)$ has index two in U_F^d, which subgroup of index two is not settled by d on its own. Two facts close that gap. First, Step 3 of the proof of proposition 9.1.8 identifies the subgroup: $N_{E'/F}(U_{E'}^d) = \ker \kappa_{E'/F} \cap U_F^d$. Second, $n = 3d$ forces $a = \lfloor m/2 \rfloor + 1 \leq d$, so Step 3 of the proof of proposition 10.1.15 applies and gives $\kappa_{E'/F} = \kappa_{E/F}$ on $U_F^a \supseteq U_F^d$. The two kernels therefore meet U_F^d in the same subgroup, and $N_{E'/F}(U_{E'}^d) = N_{E/F}(U_E^d)$ for every member of $\mathcal{S}$. With both subgroups pinned down, Φ takes values in $\big(N_{E/F}(\beta)U_F^a / N_{E/F}(U_E^s)\big) \times \big((\mathrm{tr}_{E/F}\beta + \mathfrak{p}^{1-a})/\mathrm{tr}_{E/F}(\mathfrak{p}_E^{-n+s})\big)$, a set with

$$r = \big[U_F^a : N_{E/F}(U_E^s)\big] \cdot \big[\mathfrak{p}^{1-a} : \mathrm{tr}_{E/F}(\mathfrak{p}_E^{-n+s})\big]$$

elements. By proposition 10.1.15(3), two members of $\mathcal{S}$ with distinct Φ-values have disjoint sets of parameters.

Step 5: Φ is surjective. Let $D \in N_{E/F}(\beta)U_F^a$ and $T \in \mathrm{tr}_{E/F}(\beta) + \mathfrak{p}^{1-a}$. Then $v_F(D) = v_F(N_{E/F}\beta) = -n$, and $v_F(T) \geq -m$: indeed $v_F(\mathrm{tr}_{E/F}\beta) \geq \lfloor d/2 \rfloor - m \geq -m$ by lemma 10.1.14 and $1 - a \geq -m$. The polynomial $f(X) = X^2 - TX + D$ is irreducible over F, by the valuation argument in Step 3 of the proof of lemma 10.1.4. It is also separable: if $\mathrm{char}(F) \neq 2$ this is automatic, and if $\mathrm{char}(F) = 2$ then $v_F(4) = \infty$, so lemma 10.1.14 forces $2v_F(\mathrm{tr}_{E/F}\beta) + n = d$, that is $v_F(\mathrm{tr}_{E/F}\beta) = (d - n)/2 < -\lfloor m/2 \rfloor = 1 - a$ because $n - 2\lfloor m/2 \rfloor \geq m + 1 > d$. Hence $v_F(T) = (d - n)/2$ is finite, $T \neq 0$ and $f' = -T \neq 0$. Put $E' = F[X]/(f)$ and let β' be the class of X, so $N_{E'/F}(\beta') = D$ and $\mathrm{tr}_{E'/F}(\beta') = T$. From $2v_{E'}(\beta') = v_{E'}(D) = -n\, e(E'|F)$ and n odd

we get $e(E'|F) = 2$ and $v_{E'}(\beta') = -n$, so E'/F is totally ramified and $(E'/F, \beta') \in \mathcal{S}$ with the prescribed image. So Φ takes exactly r values on $\mathcal{S}$.

Step 6: the index r. By lemma 9.1.7(3), $N_{E/F}(U_E^{d+1}) = U_F^{d+1}$. For $b > d$ and $x \in \mathfrak{p}_E^b$ the expansion $N_{E/F}(1+x) = 1 + \mathrm{tr}_{E/F}(x) + N_{E/F}(x)$ has $v_F(N_{E/F}x) = b \geq \lfloor (b+d+1)/2 \rfloor$ and $v_F(\mathrm{tr}_{E/F}x) \geq \lfloor (b+d+1)/2 \rfloor = \lceil (b+d)/2 \rceil$ by proposition 9.1.4(1), so $N_{E/F}(U_E^b) \subseteq U_F^{\lceil (b+d)/2 \rceil}$. For the reverse inclusion, Step 4 of the proof of lemma 9.1.7 shows that $N_{E/F}(U_E^{2j-d}) = U_F^j$ for every $j \geq d+1$, its successive-approximation argument applying verbatim once ν_j is surjective; taking $j = \lceil (b+d)/2 \rceil$, so that $2j - d$ is b or $b+1$, gives $U_F^{\lceil (b+d)/2 \rceil} \subseteq N_{E/F}(U_E^b)$. Hence $N_{E/F}(U_E^b) = U_F^{\lceil (b+d)/2 \rceil}$ for $b > d$. For $b = d$, lemma 9.1.7(4) makes $N_{E/F}(U_E^d)$ a subgroup of U_F^d of index two containing U_F^{d+1}.

If $n > 3d$ then $n - d > 2d$ and $s > d$, so $[U_F^a : N_{E/F}(U_E^s)] = q^{\lceil (s+d)/2 \rceil - a}$. If $n = 3d$ then $s = d$ and the same index is $2q^{\lceil (s+d)/2 \rceil - a}$. By proposition 9.1.4(1), $\mathrm{tr}_{E/F}(\mathfrak{p}_E^{-n+s}) = \mathfrak{p}^{\lfloor (s+d+1-n)/2 \rfloor}$, so the second index is $q^{\lfloor (s+d+1-n)/2 \rfloor + a - 1}$. Writing $c = n - d$, so that $s = \lceil c/2 \rceil$, $e = \lfloor c/2 \rfloor$, $s + e = c$ and $d = 2m + 1 - c$, we get

$$\left\lceil \tfrac{s+d}{2} \right\rceil = \left\lceil \tfrac{2m+1-e}{2} \right\rceil = m + \left\lceil \tfrac{1-e}{2} \right\rceil, \qquad \left\lfloor \tfrac{s+d+1-n}{2} \right\rfloor = \left\lfloor \tfrac{1-e}{2} \right\rfloor$$

and therefore, using $\lceil k/2 \rceil + \lfloor k/2 \rfloor = k$ for $k \in \mathbb{Z}$,

$$r = q^{\,m+(1-e)-1} \cdot \begin{cases} 1 & n > 3d \\ 2 & n = 3d \end{cases} = q^{\,m-e} \cdot \begin{cases} 1 & n > 3d \\ 2 & n = 3d \end{cases}$$

Step 7: (1) implies (2). Choose $(E_i/F, \beta_i) \in \mathcal{S}$, $1 \leq i \leq r$, one for each value of Φ. The sets $\mathcal{G}_2^0(F; \beta_i)$ are pairwise disjoint by Step 4, and every parameter in them is totally ramified by proposition 10.1.9, so corollary 9.3.11 applies to every pair drawn from their union, its total-ramification hypothesis being discharged, and $\boldsymbol{\pi}^{\mathrm{im}}$ is injective there. Hence the sets $\boldsymbol{\pi}^{\mathrm{im}}(\mathcal{G}_2^0(F; \beta_i))$ are pairwise disjoint subsets of $\mathcal{A}_2^0(F; \alpha)$, and each is stable under Γ, by theorem 8.5.4(3) and lemma 10.1.13. Since $\boldsymbol{\pi}^{\mathrm{im}}$ commutes with twisting and is injective here, it carries Γ-orbits to Γ-orbits injectively, so lemma 10.1.13(2) gives

$$\left| \Gamma \backslash \bigsqcup_{i=1}^{r} \boldsymbol{\pi}^{\mathrm{im}}\big(\mathcal{G}_2^0(F; \beta_i)\big) \right| = \begin{cases} r\, q^e = q^{m-e} q^e = q^m & n > 3d \\ r\, q^e / 2 = 2q^{m-e} q^e / 2 = q^m & n = 3d \end{cases}$$

By lemma 10.1.13(1) the ambient set $\mathcal{A}_2^0(F; \alpha)$ has exactly q^m orbits under Γ, so the disjoint union above meets every orbit and hence equals $\mathcal{A}_2^0(F; \alpha)$. Every $\pi \in \mathcal{A}_2^0(F; \alpha)$ is therefore ordinary, which is (2).

Step 8: independence of ψ. Let $u \in U_F$ and $\psi' = u\psi$, which is the general character of level one, two such differing by a unit. Then $\psi'_A(\alpha x) = \psi_A(u\alpha x)$, so the set attached to α and ψ' by definition 10.1.2 is the set attached to $u\alpha$ and ψ. And $\psi'_E(y) = \psi_E(uy)$, so the element attached to ξ and ψ' by (10.3) is $u^{-1}\beta$, while (10.2) gives $u^{-1}\delta_{E/F}$ for the shift element. Multiplying the first congruence of (3) by u^2 and the second by u turns the conditions for (α, ψ') into the conditions for $(u\alpha, \psi)$, since $\det(u\alpha) = u^2 \det \alpha$, $\mathrm{tr}(u\alpha) = u\mathrm{tr}\alpha$, $N_{E/F}(u\beta') = u^2 N_{E/F}(\beta')$ and $u\mathfrak{p}^{-\lfloor m/2 \rfloor} = \mathfrak{p}^{-\lfloor m/2 \rfloor}$. So all three conditions transport together, completing the proof.

The criterion is arithmetic rather than representation-theoretic: it asks whether the pair $(\mathrm{tr}\alpha, \det\alpha)$, read to the precision $\mathfrak{p}^{-\lfloor m/2\rfloor}$ and $U_F^{\lfloor m/2\rfloor+1}$, can be matched by the trace and norm of a single element of a totally ramified quadratic extension whose discriminant exponent is small enough. What it does not yet do is answer that question, since it quantifies over all such extensions and all elements of the relevant shell.

Exercise 10.1.1

1. Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum with $n = 2m+1 \geq 1$. Show that $v_F(\det\alpha) = -n$ and $v_F(\mathrm{tr}\alpha) \geq -m$. Using only these two facts, show that $X^2 - (\mathrm{tr}\alpha)X + \det\alpha$ is irreducible over F.

2. Let $F = \mathbb{Q}_2$ and $E = \mathbb{Q}_2(i)$, and put $\beta = -(1+i)/4$. Compute $v_E(\beta)$, $N_{E/F}(\beta)$ and $\mathrm{tr}_{E/F}(\beta)$, and use lemma 10.1.14 with $n = 3$ to recover $d(E|F)$. Decide whether $(E/F, 3, \beta)$ is an admissible wild triple, and whether it falls in the case $n > 3d$ or the case $n = 3d$.

3. Let $F = \mathbb{Q}_2$ and $E = \mathbb{Q}_2(\sqrt{2})$, so that $d(E|F) = 2$ by example 9.1.3. Show that no admissible wild triple $(E/F, n, \beta)$ has $n < 7$, and exhibit one with $n = 7$.

4. Let $(\mathfrak{A}, 1, \alpha)$ be a ramified simple stratum, so $n = 1$ and $m = 0$. Show that $\mathcal{A}_2^0(F;\alpha)$ is a single orbit under twisting by the characters of $F^\times$ trivial on U_F^1, and that every member of it is exceptional.

5. Let E/F be totally ramified quadratic with $d = d(E|F)$. Show that (10.2) determines $\delta_{E/F}$ modulo $\mathfrak{p}^{-\lfloor d/2\rfloor}$, and deduce that for an admissible wild triple $(E/F, n, \beta)$ the second congruence in (10.4) does not depend on which shift element was chosen.

6. Keep $F = \mathbb{Q}_2$ and $E = \mathbb{Q}_2(\sqrt{2})$, $d = 2$, and take $n = 7$. Compute $\mathrm{tr}_{E/F}(\beta)$ for $\beta = \sqrt{2}/16$ and for $\beta' = \sqrt{2}/16 + 1/8$, and check both against lemma 10.1.14: the inequality is strict for one of them and an equality for the other, while the formula for d returns 2 in both cases.

7. Let $F = \mathbb{Q}_2$, let $(\mathfrak{A}, 3, \alpha)$ be a ramified simple stratum whose set $\mathcal{A}_2^0(F;\alpha)$ contains an ordinary representation, and let $(E/F, 3, \beta)$ be the triple of exercise 10.1.1 (2), assumed to satisfy (10.4) for α. Compute the two orbit counts of lemma 10.1.13 and the index r of Step 6 of the proof of theorem 10.1.17 directly from lemma 9.1.7 and proposition 9.1.4, and check that the three numbers fit together as that step requires.

8. Let $(\mathfrak{A}, n, \alpha)$ and $(\mathfrak{A}, n, \alpha')$ be ramified simple strata on the same chain order, with $n = 2m+1$, and suppose $\mathcal{A}_2^0(F;\alpha) \cap \mathcal{A}_2^0(F;\alpha')$ is non-empty. Show that $\det\alpha' \equiv \det\alpha \bmod U_F^{\lfloor m/2\rfloor+1}$ and $\mathrm{tr}\alpha' \equiv \mathrm{tr}\alpha \bmod \mathfrak{p}^{-\lfloor m/2\rfloor}$.

9. Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum with $n \leq 3d(F[\alpha]|F)$ and $n \not\equiv 0 \bmod 3$. Show that every $\pi \in \mathcal{A}_2^0(F;\alpha)$ is exceptional.

10.2 Exceptional Representations and Strata

Section 10.1 characterized the ordinary ramified simple strata $(\mathfrak{A}, n, \alpha)$ by asking for an admissible wild triple $(E/F, n, \beta)$ whose invariants match those of α to the precision recorded in (10.4). That criterion quantifies over all totally ramified separable quadratic extensions of F and all elements of

$E\langle n\rangle$, so a stratum presented in matrix form does not visibly satisfy or fail it. This section removes the quantifier. The two invariants of α occurring in (10.4) are $\operatorname{tr}\alpha$ and $\det\alpha$, and they combine into a single cubic polynomial over F whose reducibility settles the question.

The cubic depends on which additive character ψ was used to attach α to a representation, and only one normalization of ψ makes the criterion correct.

The residue field k has characteristic 2, so the map $y \mapsto y + y^2$ is additive on k. This map governs the norm computation of lemma 10.2.9 below, and the characters of F that are trivial on its image are the ones to work with.

Lemma 10.2.1: A Character Killing $y + y^2$

Write $\wp(y) = y + y^2$. There exists a character $\psi \in \widehat{F}$ such that

1. ψ has level one, and
2. $\psi\big(\wp(y)\big) = 1$ for every $y \in \mathfrak{o}$.

A character $\psi' \in \widehat{F}$ has properties (1) and (2) if and only if $\psi' = u\psi$ for some $u \in U_F^1$, where $u\psi$ denotes the character $x \mapsto \psi(ux)$.

***Proof*:**
Throughout, "ψ has level one" means that ψ is trivial on $\mathfrak{p}$ and non-trivial on $\mathfrak{o}$.

Step 1: the residue field. The map $\wp\colon k \to k$ is additive, since $(y+z)+(y+z)^2 = y+z+y^2+z^2+2yz$ and $2 = 0$ in k. Its kernel is $\{y \in k : y^2 = y\} = \mathbb{F}_2$, so $\wp(k)$ is an $\mathbb{F}_2$-subspace of index 2 in k. A character of the additive group k is trivial on $\wp(k)$ exactly when it lies in the annihilator of $\wp(k)$, a group of order $[k : \wp(k)] = 2$. Hence there is exactly one non-trivial character ψ_0 of k with $\psi_0(\wp(y)) = 1$ for all $y \in k$.

Step 2: existence. Pick any $\psi_1 \in \widehat{F}$ of level one. Its restriction to $\mathfrak{o}$ is trivial on $\mathfrak{p}$ and so is inflated from a non-trivial character χ_1 of k. Every non-trivial character of k has the form $y \mapsto \chi_1(ay)$ for a unique $a \in k^\times$. Choose a with $\psi_0 = \chi_1(a\,\cdot)$ and lift a to $u_1 \in U_F$. Then $\psi = u_1\psi_1$ has level one, because u_1 is a unit, and $\psi|_{\mathfrak{o}}$ is the inflation of ψ_0. For $y \in \mathfrak{o}$ the element $\wp(y)$ lies in $\mathfrak{o}$ and reduces to $\wp(\bar{y})$, so $\psi(\wp(y)) = \psi_0(\wp(\bar{y})) = 1$. This gives (1) and (2).

Step 3: uniqueness. Let ψ' satisfy (1) and (2). Then $\psi'|_{\mathfrak{o}}$ is inflated from a non-trivial character of k killing $\wp(k)$, which is ψ_0 by Step 1, so $\psi'|_{\mathfrak{o}} = \psi|_{\mathfrak{o}}$. Since $\widehat{F}$ is free of rank one over F under $(x\psi)(y) = \psi(xy)$, there is a unique $u \in F^\times$ with $\psi' = u\psi$, and $u \in U_F$ because ψ and ψ' have the same level. Now $\psi'|_{\mathfrak{o}} = \psi|_{\mathfrak{o}}$ says $\psi\big((u-1)x\big) = 1$ for all $x \in \mathfrak{o}$. As ψ is trivial on $\mathfrak{p}$ and non-trivial on $\mathfrak{o}$, the largest fractional ideal on which ψ vanishes is $\mathfrak{p}$, so $(u-1)\mathfrak{o} \subseteq \mathfrak{p}$ and $u \in U_F^1$. Conversely, if $u \in U_F^1$ then $(u-1)x \in \mathfrak{p}$ for $x \in \mathfrak{o}$, so $u\psi$ agrees with ψ on $\mathfrak{o}$ and therefore inherits (1) and (2), completing the proof.

The argument in Step 1 used only that $\wp$ is additive with a kernel of order 2, and additivity persists on a bounded neighbourhood of $\mathfrak{o}$ inside F: for $y, z \in \mathfrak{p}^{-j}$ the error term $2yz$ lies in $\mathfrak{p}^{v_F(2)-2j}$, which sits inside $\mathfrak{p}$ as soon as $2j < v_F(2)$. Lemma 10.2.1 therefore has a graded refinement. It is recorded

because it measures how much room the normalization of ψ still leaves, and because Step 3 of its proof is reused in the answer to exercise 10.2.1 (1). The main theorem of the section needs only the case $j = 0$.

Definition 10.2.2: Special Character

Let $j \geq 0$ be an integer. A character $\psi \in \widehat{F}$ is *special of depth* j if ψ has level one and $\psi(y + y^2) = 1$ for every $y \in \mathfrak{p}^{-j}$. A character special of depth 0 is called simply *special*.

Such characters exist for the reason they do at depth 0: the image $\wp(\mathfrak{p}^{-j})$, larger though it is than $\wp(k)$, still fails to contain $\mathfrak{o}/\mathfrak{p}$, which leaves room for a character trivial on the image and non-trivial on $\mathfrak{o}$.

Lemma 10.2.3: Special Characters of Positive Depth

Let j be an integer with $0 \leq j$ and $2j \leq v_F(2) - 1$. Then a character of F special of depth j exists, and it is special of depth j' for every $0 \leq j' \leq j$.

***Proof*:**

Step 1: $\wp$ is additive on $\mathfrak{p}^{-j}$ modulo $\mathfrak{p}$. For $y, z \in \mathfrak{p}^{-j}$,

$$\wp(y + z) - \wp(y) - \wp(z) = 2yz$$

and $v_F(2yz) \geq v_F(2) - 2j \geq 1$ by hypothesis. So $y \mapsto \wp(y) + \mathfrak{p}$ is a homomorphism from $\mathfrak{p}^{-j}$ to $F/\mathfrak{p}$, and its image W is a subgroup.

Step 2: W does not contain $\mathfrak{o}/\mathfrak{p}$. If $y \in \mathfrak{p}^{-j}$ has $v_F(y) < 0$ then $v_F(y^2) = 2v_F(y) < v_F(y)$, so $v_F(\wp(y)) = 2v_F(y) < 0$ and $\wp(y) \notin \mathfrak{o}$. Hence an element of W meeting $\mathfrak{o}/\mathfrak{p}$ comes from some $y \in \mathfrak{o}$, and the elements of W lying in $\mathfrak{o}/\mathfrak{p}$ are exactly the classes $\wp(y) + \mathfrak{p}$ with $y \in \mathfrak{o}$, that is, the image $\wp(k) \subseteq k$. By Step 1 of the proof of lemma 10.2.1, $\wp(k)$ has index 2 in k, so $W \cap (\mathfrak{o}/\mathfrak{p}) \neq \mathfrak{o}/\mathfrak{p}$.

Step 3: existence. The group $\Gamma = \mathfrak{p}^{-2j}/\mathfrak{p}$ is finite and contains both W and $\mathfrak{o}/\mathfrak{p}$. By Step 2 the image of $\mathfrak{o}/\mathfrak{p}$ in Γ/W is non-trivial, so some character of Γ/W is non-trivial on it; pulling back gives a character χ of Γ with $\chi|_W = 1$ and $\chi|_{\mathfrak{o}/\mathfrak{p}} \neq 1$. The quotient $F/\mathfrak{p}$ is discrete, being a quotient by an open subgroup, so χ extends to a character of $F/\mathfrak{p}$; inflating to F produces $\psi \in \widehat{F}$ trivial on $\mathfrak{p}$, non-trivial on $\mathfrak{o}$, and trivial on $\wp(\mathfrak{p}^{-j})$. That is a character special of depth j.

Step 4: smaller depths. If $0 \leq j' \leq j$ then $\mathfrak{p}^{-j'} \subseteq \mathfrak{p}^{-j}$, so a character trivial on $\wp(\mathfrak{p}^{-j})$ is trivial on $\wp(\mathfrak{p}^{-j'})$, and the depth-j condition implies the depth-j' one, completing the proof.

No uniqueness statement accompanies lemma 10.2.3, and the argument above supplies none. The uniqueness clause of lemma 10.2.1 came from computing the annihilator of $\wp(k)$ inside $\mathfrak{o}/\mathfrak{p}$, that is, in the residue field. For $j \geq 1$ the set $\wp(\mathfrak{p}^{-j})$ is a group only modulo $\mathfrak{p}$ and is not an $\mathfrak{o}$-module, so that computation is unavailable, and the condition on u is left as membership in the annihilator of that group. Whether the characters special of depth j form a single coset of a unit group is not

settled here, and nothing below needs it.

Example 10.2.4: Special Characters over $\mathbb{Q}_2$ and over an Unramified Quadratic Extension

1. Let $F = \mathbb{Q}_2$, so $q = 2$ and $v_F(2) = 1$. Here $\wp(y) = y(1+y)$ is a product of two consecutive 2-adic integers for $y \in \mathfrak{o}$, hence lies in $\mathfrak{p}$. Every character of level one is therefore trivial on $\wp(\mathfrak{o})$, and *every* level one character of $\mathbb{Q}_2$ is special. This matches the uniqueness clause of lemma 10.2.1: $U_F = U_F^1$ when $q = 2$. Concretely, $\psi(x) = e^{2\pi i\{x/2\}}$, where $\{\cdot\}$ denotes the 2-adic fractional part, is trivial on $2\mathbb{Z}_2 = \mathfrak{p}$ and non-trivial on $\mathbb{Z}_2$, so it is special. The bound $2j \le v_F(2) - 1 = 0$ permits only $j = 0$, so lemma 10.2.3 adds nothing here.
2. Let $F = \mathbb{Q}_2(\omega)$ with $\omega^2 + \omega + 1 = 0$, the unramified quadratic extension of $\mathbb{Q}_2$: now $q = 4$ and $k = \mathbb{F}_4$. Here $\wp(\mathbb{F}_4) = \{0, 1\} = \mathbb{F}_2$, since $\wp(0) = \wp(1) = 0$ and $\wp(\omega) = \omega + \omega^2 = 1$. Of the four characters of $\mathbb{F}_4$, exactly two are trivial on $\mathbb{F}_2$, namely the trivial one and $\psi_0 = \eta \circ \mathrm{tr}_{\mathbb{F}_4/\mathbb{F}_2}$ with η the non-trivial character of $\mathbb{F}_2$. So of the three non-trivial characters of k, only one inflates to a special character of F: a level one character of $\mathbb{Q}_2(\omega)$ is special for exactly one of the three cosets of U_F^1 in U_F.

Definition 10.1.1 calls a representation $\pi \in \mathcal{A}_2^0(F)$ ordinary when it lies in the image of $\boldsymbol{\pi}^{\mathrm{im}}$. Theorem 10.1.17 shows that the property is constant on each set $\mathcal{A}_2^0(F;\alpha)$, so it descends to the stratum.

Definition 10.2.5: Ordinary and Exceptional Strata

A ramified simple stratum $(\mathfrak{A}, n, \alpha)$ with $n \ge 1$ is *ordinary* if $\mathcal{A}_2^0(F;\alpha)$ contains an ordinary representation, and *exceptional* otherwise. By theorem 10.1.17, $(\mathfrak{A}, n, \alpha)$ is ordinary if and only if $\mathcal{A}_2^0(F;\alpha) \subseteq \boldsymbol{\pi}^{\mathrm{im}}\big(\mathcal{G}_2^{\mathrm{im}}(F)\big)$, and it is exceptional if and only if every $\pi \in \mathcal{A}_2^0(F;\alpha)$ is exceptional.

Definition 10.2.5 sorts the ramified simple strata by whether they contribute to the image of $\boldsymbol{\pi}^{\mathrm{im}}$, which theorem 8.5.4 does not determine: the exceptional strata are exactly those whose representations lie outside $\boldsymbol{\pi}^{\mathrm{im}}(\mathcal{G}_2^{\mathrm{im}}(F))$.

The invariant that settles the matter is the following cubic, whose coefficients are the two invariants of α occurring in (10.4). Its constant term is $\det\alpha$ rather than $-\det\alpha$ because the congruences pair $\det\alpha$ with $N_{E/F}(\beta)$ and $\mathrm{tr}\alpha$ with $\delta_{E/F} + \mathrm{tr}_{E/F}(\beta)$, and lemma 10.2.9 converts the second pairing into the identity $(\mathrm{tr}\alpha - \gamma)\gamma^2 = \det\alpha$ satisfied by a root γ.

Theorem 10.2.6: Ordinary Strata and the Cubic

Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum with $n = 2m+1 \geq 1$, and assume $\mathrm{tr}\alpha \neq 0$ in case $\mathrm{char}(F) = 2$. Put $d_\alpha = d(F[\alpha]|F)$, and assume the character ψ used to form ψ_α, $\mathcal{A}_2^0(F;\alpha)$ and $\delta_{E/F}$ is special. Set

$$C_\alpha(X) = X^3 - \mathrm{tr}(\alpha)X^2 + \det(\alpha)$$

The following are equivalent.

1. The stratum $(\mathfrak{A}, n, \alpha)$ is ordinary.
2. Either $n > 3d_\alpha$, or $C_\alpha(X)$ has a root in F.

Both hypotheses are mild, and both are needed. A special character exists by lemma 10.2.1, and no depth beyond 0 is asked for. The proviso on $\mathrm{tr}\alpha$ is what makes d_α exist at all: the minimal polynomial of α over F is $X^2 - \mathrm{tr}(\alpha)X + \det(\alpha)$, whose derivative is $2X - \mathrm{tr}\alpha$, so $F[\alpha]/F$ is inseparable exactly when $\mathrm{char}(F) = 2$ and $\mathrm{tr}\alpha = 0$. In that case $\mathrm{tr}_{F[\alpha]/F}$ vanishes identically, the discriminant $\mathrm{tr}(\alpha)^2 - 4\det(\alpha)$ is 0, and definition 9.1.1, which is stated for separable quadratic extensions, and reads $d(E|F)$ off the exponent of a non-zero ideal, assigns no value to d_α. Both proposition 9.1.4 and definition 10.1.6, which the proof feeds $F[\alpha]$ to, fail for the same reason, so the proviso cannot be dropped by rewriting the argument. The same proviso appears in lemma 10.2.8(2). When $\mathrm{char}(F) = 0$ the hypothesis is vacuous, so nothing is lost over $\mathbb{Q}_2$ and its finite extensions.

10.2.7 Warning: The Normalization of ψ Is Not Removable The conditions of theorem 10.1.17 do not depend on which character of level one is used, but the cubic criterion does. Replacing ψ by $u^{-1}\psi$ with $u \in U_F$ replaces the character ψ_α of $U_{\mathfrak{A}}^{m+1}$ by the character attached to $u\alpha$, since $(u^{-1}\psi)\big(\mathrm{tr}(u\alpha x)\big) = \psi\big(\mathrm{tr}(\alpha x)\big)$. The set of representations is unchanged, in the new normalization it is $\mathcal{A}_2^0(F;u\alpha)$, as in Step 8 of the proof of theorem 10.1.17, but the cubic attached to it is $C_{u\alpha}$, not C_α. Taken in the other direction, the same identity says that passing from ψ to $u\psi$ replaces the element attached to a fixed set of representations by $u^{-1}\alpha$, not by $u\alpha$: $(u\psi)\big(\mathrm{tr}(u^{-1}\alpha x)\big) = \psi\big(\mathrm{tr}(\alpha x)\big)$. That is the transport rule matching $\delta_{E/F} \mapsto u^{-1}\delta_{E/F}$ and $\beta \mapsto u^{-1}\beta$, under which both congruences of (10.4) scale by u^{-2} and u^{-1} respectively and so transport intact. Indeed $\mathrm{tr}(u\alpha) = u\mathrm{tr}\alpha$ and $\det(u\alpha) = u^2\det\alpha$, so

$$C_{u\alpha}(uX) = u^3\Big(X^3 - \mathrm{tr}(\alpha)X^2 + u^{-1}\det(\alpha)\Big)$$

and $C_{u\alpha}$ has a root in F if and only if $X^3 - \mathrm{tr}(\alpha)X^2 + u^{-1}\det(\alpha)$ does. The constant term has moved. Exercise 10.2.1 (6) exhibits a residue field over which the root behaviour changes under such a scaling, so the hypothesis on ψ in theorem 10.2.6 cannot be dropped.

The first lemma pins the valuation of any root of C_α, and in doing so shows that $3 \big| n$ is forced before a root can exist. The second turns a prescribed trace and determinant into an actual quadratic extension, which is how a root of the cubic is converted into an admissible wild triple.

Lemma 10.2.8: Roots of the Cubic, and Quadratic Extensions with Prescribed Invariants

Let $n \geq 1$ be an integer.

1. Let $a, b \in F$ with $v_F(b) = -n$ and $3v_F(a) \geq -n$. If $X^3 + aX^2 + b$ has a root $\gamma \in F$, then $3 \big| n$ and $v_F(\gamma) = -n/3$.

2. Suppose n is odd, $n = 2m + 1$. Let $D \in F^\times$ with $v_F(D) = -n$ and let $\epsilon \in F$ with $v_F(\epsilon) \geq -m$, and assume $\epsilon \neq 0$ if $\mathrm{char}(F) = 2$. Then $X^2 - \epsilon X + D$ is irreducible and separable over F; writing E for the field it generates and $\beta \in E$ for a root,

$$\mathrm{tr}_{E/F}(\beta) = \epsilon, \qquad N_{E/F}(\beta) = D, \qquad v_E(\beta) = -n$$

the extension E/F is totally ramified quadratic, and

$$d(E|F) = \min\{\, 2v_F(\epsilon) + n,\ v_F(4) \,\}$$

***Proof*:**

1. Write $g = v_F(\gamma)$ and use $\gamma^3 + a\gamma^2 = -b$, so that $v_F(\gamma^3 + a\gamma^2) = -n$. Three cases exhaust the possibilities.

   If $g < v_F(a)$ then $v_F(\gamma^3) = 3g$ is strictly smaller than $v_F(a\gamma^2) \geq v_F(a) + 2g$, so $v_F(\gamma^3 + a\gamma^2) = 3g$ and $3g = -n$. Thus $3 \big| n$ and $g = -n/3$.

   If $g > v_F(a)$ then $v_F(a\gamma^2) = v_F(a) + 2g < 3g = v_F(\gamma^3)$, so $v_F(a) + 2g = -n$, that is $v_F(a) = -n - 2g$. Combining with $g > v_F(a)$ gives $3g > -n$, whence $3v_F(a) = -3n - 6g < -3n + 2n = -n$, contradicting $3v_F(a) \geq -n$. So this case is empty.

   If $g = v_F(a)$ then both terms have valuation $3g$, so $-n = v_F(\gamma^3 + a\gamma^2) \geq 3g$, giving $3g \leq -n$. And $3v_F(a) = 3g \geq -n$ gives $3g \geq -n$. Hence $3g = -n$, so $3 \big| n$ and $g = -n/3$. In the two surviving cases the conclusion is the stated one.

2. *Irreducibility and separability.* Suppose $X^2 - \epsilon X + D$ had roots $\lambda_1, \lambda_2 \in F$. Then $v_F(\lambda_1) + v_F(\lambda_2) = v_F(D) = -n$ is odd, so the two valuations differ and $v_F(\epsilon) = v_F(\lambda_1 + \lambda_2) = \min\{v_F(\lambda_i)\}$. That minimum is at most $-n/2$, hence at most $-m - 1$ since it is an integer and $n = 2m + 1$. This contradicts $v_F(\epsilon) \geq -m$. So the polynomial is irreducible. Its derivative is $2X - \epsilon$, which is the zero polynomial only if $\mathrm{char}(F) = 2$ and $\epsilon = 0$, excluded by hypothesis. So it is separable.

   The valuations. Let $E = F[\beta]$ with β a root. Then $\mathrm{tr}_{E/F}(\beta) = \epsilon$ and $N_{E/F}(\beta) = D$ are the coefficients of the minimal polynomial. Since $N_{E/F}(\beta) = \beta\beta^\sigma$ with σ the non-trivial F-automorphism, and σ preserves v_E, we get $2v_E(\beta) = v_E(D) = e(E|F)\, v_F(D) = -e(E|F)\, n$. If E/F were unramified this would read $2v_E(\beta) = -n$ with n odd, which is impossible. So $e(E|F) = 2$, the extension is totally ramified, and $v_E(\beta) = -n$.

   The discriminant exponent. The element $\varpi_E = \varpi^{m+1}\beta$ has $v_E(\varpi_E) = -n + 2(m + 1) = 1$, so it is a prime element of E. Its minimal polynomial is $X^2 + a'X + b'$ with

$$a' = -\varpi^{m+1}\epsilon, \qquad b' = \varpi^{n+1}D$$

and $v_F(b') = n+1-n = 1$, as proposition 9.1.2(1) requires. By proposition 9.1.2(2),

$$d(E|F) = \min\{2v_F(a') - 1,\ v_F(4)\} = \min\{2(m+1) + 2v_F(\epsilon) - 1,\ v_F(4)\}$$

and $2(m+1) - 1 = 2m + 1 = n$, which is the asserted formula.

Both parts are established, completing the proof.

Part (2) is the mechanism by which a root of the cubic becomes a wild triple. Its last formula says that, for an element with prescribed determinant of valuation $-n$, the wild ramification of the field it generates is a function of the valuation of its trace alone, saturating at $v_F(4)$. Over $F = \mathbb{Q}_2$, where $v_F(4) = 2$, an element α with $v_F(\det\alpha) = -5$ generates an extension with $d = 1$ when $v_F(\mathrm{tr}\alpha) = -2$ and with $d = 2$ when $v_F(\mathrm{tr}\alpha) \geq -1$, and those are the only options because $\mathrm{tr}\alpha \in \mathfrak{p}^{-2}$.

The congruences (10.4) compare $\mathrm{tr}\alpha$ with $\delta_{E/F} + \mathrm{tr}_{E/F}(\beta)$, so applying theorem 10.1.17 in either direction requires knowing $\delta_{E/F}$ to precision $\mathfrak{p}^{-\lfloor m/2 \rfloor}$. The next lemma gives it in closed form from β itself. This is where the normalization of ψ enters, and it is the only place it does.

Lemma 10.2.9: A Closed Form for $\delta_{E/F}$

Let E/F be a totally ramified separable quadratic extension with $d = d(E|F)$ odd, let $\beta \in E$ satisfy $v_E(\beta) = -3d$, and let ψ be special. Then $v_F(\mathrm{tr}_{E/F}(\beta)) = -d$, and

$$\delta_{E/F}\,\frac{\mathrm{tr}_{E/F}(\beta)^2}{N_{E/F}(\beta)} \in U_F^1$$

Equivalently, $\delta_{E/F} \equiv N_{E/F}(\beta)\,\mathrm{tr}_{E/F}(\beta)^{-2}$ modulo $\mathfrak{p}^{1-d}$.

***Proof*:**
Write $t = \mathrm{tr}_{E/F}(\beta)$, $N = N_{E/F}(\beta)$ and $\delta = \delta_{E/F}$.

Step 1: the valuations. Since E/F is totally ramified, $v_F \circ N_{E/F} = v_E$, so $v_F(N) = v_E(\beta) = -3d$. Because d is odd, the integer $\mu = (d-3d)/2 = -d$ satisfies $2\mu - d = -3d$, so proposition 9.1.4(2) gives an isomorphism $\mathfrak{p}_E^{-3d}/\mathfrak{p}_E^{-3d+1} \to \mathfrak{p}^{-d}/\mathfrak{p}^{-d+1}$ induced by $\mathrm{tr}_{E/F}$. The class of β in the source is non-zero, hence so is the class of t in the target, and $v_F(t) = -d$ exactly. In particular $t \neq 0$. Consequently

$$c := \delta\, t^2 N^{-1}, \qquad v_F(c) = -d + 2(-d) - (-3d) = 0$$

so $c \in U_F$, the valuation $v_F(\delta) = -d$ being part of definition 10.1.7.

Step 2: a family of norms. Fix $y \in \mathfrak{o}$ and set $x = y\,t/N \in F$, so that

$$v_F(x) = v_F(y) - d + 3d = v_F(y) + 2d \geq 2d$$

Then $v_E(x\beta) = 2v_F(x) - 3d \geq 4d - 3d = d \geq 1$. Hence $1 + x\beta \in U_E$ and

$$N_{E/F}(1 + x\beta) = (1 + x\beta)(1 + x\beta^\sigma) = 1 + xt + x^2 N$$

Substituting $x = yt/N$ gives $xt = y\,t^2 N^{-1}$ and $x^2 N = y^2\,t^2 N^{-1}$, so

$$N_{E/F}(1 + x\beta) = 1 + \wp(y)\,t^2 N^{-1}, \qquad \wp(y) = y + y^2$$

Step 3: applying the defining formula for δ. We have $\wp(y) \in \mathfrak{o}$, so

$$v_F\big(\wp(y)\, t^2 N^{-1}\big) \geq 0 - 2d + 3d = d$$

and $d \geq 1 + \lfloor d/2 \rfloor$, since d is odd gives $d - \lfloor d/2 \rfloor = \lfloor d/2 \rfloor + 1$. Thus $N_{E/F}(1 + x\beta)$ lies in $U_F^{1+\lfloor d/2 \rfloor}$ and

$$\kappa_{E/F}\big(N_{E/F}(1 + x\beta)\big) = \psi\big(\delta\, \wp(y)\, t^2 N^{-1}\big) = \psi\big(c\, \wp(y)\big)$$

The left side equals 1, because $\kappa_{E/F}$ is trivial on $N_{E/F}(E^\times)$ by definition 8.3.1 and $1 + x\beta \in E^\times$. Hence $\psi(c\,\wp(y)) = 1$ for every $y \in \mathfrak{o}$.

Step 4: reading off c. Since ψ is special we also have $\psi(\wp(y)) = 1$ for $y \in \mathfrak{o}$, so $\psi\big((c-1)\wp(y)\big) = 1$ for all $y \in \mathfrak{o}$. Both $c - 1$ and $\wp(y)$ lie in $\mathfrak{o}$, and ψ is trivial on $\mathfrak{p}$, so $x \mapsto \psi\big((c-1)x\big)$ is inflated from the character $y \mapsto \psi_0\big(\overline{c-1}\, y\big)$ of k, where ψ_0 is the character of k inflating to $\psi|_{\mathfrak{o}}$ and $\overline{c-1}$ is the residue class of $c - 1$. What the previous paragraph shows is that this character of k kills $\wp(k)$. By Step 1 of the proof of lemma 10.2.1 the only characters of k trivial on $\wp(k)$ are the trivial one and ψ_0 itself, and $y \mapsto \psi_0(ay)$ determines $a \in k$ uniquely. Hence $\overline{c-1} \in \{0, 1\}$. The value $\overline{c-1} = 1$ would give $\overline{c} = 1 + 1 = 0$ in k, contradicting $c \in U_F$. So $\overline{c-1} = 0$, that is $c \in U_F^1$. Multiplying that relation by Nt^{-2}, whose valuation is $-3d + 2d = -d$, turns it into $\delta - Nt^{-2} \in \mathfrak{p}^{-d}\mathfrak{p} = \mathfrak{p}^{1-d}$, completing the proof.

The precision $\mathfrak{p}^{1-d}$ is the exact strength of the argument, and taking ψ special of a higher depth does not improve it by this route. Running Steps 2 and 3 with $y \in \mathfrak{p}^{-j}$ instead of $y \in \mathfrak{o}$ does produce the relation $\psi\big((c-1)\wp(y)\big) = 1$ on that larger set, but the resulting condition on $c - 1$ is membership in the annihilator of $\wp(\mathfrak{p}^{-j})$, which is not an $\mathfrak{o}$-module, so Step 4 has no residue-field computation to read a sharper bound off. The extra precision needed in theorem 10.2.6 is obtained instead by lemma 10.2.10 below.

On both routes from condition (2) of theorem 10.2.6 back to condition (1) the admissible wild triple has to be built by successive approximation: a first approximation to $\delta_{E/F}$ is available, from lemma 10.2.9 when a root of the cubic is at hand, and from α itself when $n > 3d_\alpha$, and the precision demanded by (10.4) is reached by improving it. Each step perturbs the trace of the generator, and the effect on $\delta_{E/F}$ has to be controlled. The following lemma is the contraction estimate, and it uses nothing beyond the norm computation of lemma 9.1.7.

Lemma 10.2.10: Perturbing the Trace Moves $\delta_{E/F}$ Slightly

Let $n = 2m+1 \geq 1$, let $D \in F^\times$ with $v_F(D) = -n$, and let $\epsilon, \epsilon' \in F$ with $v_F(\epsilon), v_F(\epsilon') \geq -m$, both non-zero if $\mathrm{char}(F) = 2$. Write $E = F[\beta]$, $E' = F[\beta']$ for the extensions of lemma 10.2.8(2) attached to (ϵ, D) and (ϵ', D), and $d = d(E|F)$. Assume

$$n \geq 2d + v + 1, \qquad \epsilon' \equiv \epsilon \pmod{\mathfrak{p}^{-v}}, \qquad \lfloor d/2 \rfloor < v \leq d$$

Then $d(E'|F) = d$ and $\delta_{E'/F} \equiv \delta_{E/F} \pmod{\mathfrak{p}^{1-v}}$.

***Proof*:**
Step 1: $d(E'|F) = d$. By lemma 10.2.8(2), $d = \min\{2v_F(\epsilon) + n, v_F(4)\}$ and $d(E'|F) = \min\{2v_F(\epsilon') + n, v_F(4)\}$. Note first that $d + 2v < n$: the hypothesis gives $n \geq 2d + v + 1$, and $v \leq d$ turns this into $n \geq d + 2v + 1$.

Suppose d is odd. Then $d \neq v_F(4)$, because $v_F(4) = 2v_F(2)$ is even or infinite while $d \leq v_F(4)$ by proposition 9.1.2(2). Hence $2v_F(\epsilon) + n = d$ and $v_F(\epsilon) = (d - n)/2$. From $d + 2v < n$ we get $(d - n)/2 < -v$, so $v_F(\epsilon) < -v$ and therefore $v_F(\epsilon') = v_F(\epsilon)$, whence $d(E'|F) = d$.

Suppose instead d is even. Then $d = v_F(4)$ by proposition 9.1.2(3), and $2v_F(\epsilon) + n \geq d$. Also $2(-v) + n > d$, again from $d + 2v < n$. Since $v_F(\epsilon') \geq \min\{v_F(\epsilon), -v\}$, we get $2v_F(\epsilon') + n \geq d$ and hence $d(E'|F) = \min\{2v_F(\epsilon') + n, v_F(4)\} = v_F(4) = d$.

Step 2: the kernel of $\kappa_{E/F}$ *inside* U_F^v. Write $\kappa = \kappa_{E/F}$ and $\kappa' = \kappa_{E'/F}$. Both have level d by proposition 9.1.8. We claim

$$\ker \kappa \cap U_F^v = N_{E/F}\big(U_E^v\big)$$

An element of $N_{E/F}(E^\times)$ lying in U_F is $N_{E/F}(z)$ with $v_E(z) = v_F(N_{E/F}(z)) = 0$, that is $z \in U_E$. So $\ker \kappa \cap U_F^v$ consists of the norms $N_{E/F}(z)$, $z \in U_E$, that lie in U_F^v. If $z \in U_E^i$ with $0 \leq i \leq v - 1$ and $N_{E/F}(z) \in U_F^v \subseteq U_F^{i+1}$, then the map ν_i of lemma 9.1.7 kills the class of z. As $i < v \leq d$, ν_i is injective by lemma 9.1.7(2), so $z \in U_E^{i+1}$. Iterating from $i = 0$ gives $z \in U_E^v$. The reverse containment is lemma 9.1.7(1) with $j = v \leq d$, where $\varphi(v) = v$. This proves the claim, and since κ is non-trivial on U_F^v (its level is $d \geq v$) the subgroup $N_{E/F}(U_E^v)$ has index 2 in U_F^v. The same argument applies to E', using $d(E'|F) = d$.

Step 3: κ' *kills* $N_{E/F}(U_E^v)$. Every element of E is uniquely $x + y\beta$ with $x, y \in F$, and $v_E(x) = 2v_F(x)$ is even while $v_E(y\beta) = 2v_F(y) - n$ is odd. Hence $x + y\beta \in U_E^v$ if and only if $v_F(x - 1) \geq \lceil v/2 \rceil$ and $v_F(y) \geq \lceil (v + n)/2 \rceil$, that is

$$U_E^v = U_F^{\lceil v/2 \rceil}\big(1 + \mathfrak{p}^W \beta\big), \qquad W = \lceil (v + n)/2 \rceil$$

Applying $N_{E/F}$, the group $N_{E/F}(U_E^v)$ is generated by the squares u^2 with $u \in U_F^{\lceil v/2 \rceil}$ and by the elements $N_{E/F}(1 + w\beta) = 1 + w\epsilon + w^2 D$ with $w \in \mathfrak{p}^W$. The character κ' is quadratic, so it kills every square. For the second family, put

$$P = 1 + w\epsilon + w^2 D, \qquad Q = 1 + w\epsilon' + w^2 D = N_{E'/F}(1 + w\beta')$$

Both are units, $\kappa'(Q) = 1$ because Q is a norm from E', and $P - Q = w(\epsilon - \epsilon')$ has $v_F(P - Q) \geq W - v$. Now

$$W - v \geq \tfrac{v+n}{2} - v = \tfrac{n-v}{2}$$

and we check $W - v \geq d + 1$ in both parities. If $v + n$ is odd then $W - v = (n - v + 1)/2 \geq d + 1$ is exactly $n \geq v + 2d + 1$, which is the hypothesis. If $v + n$ is even then $W - v = (n - v)/2 \geq d + 1$ is exactly $n \geq v + 2d + 2$. Here $n \geq v + 2d + 1$ as before, and equality $n = v + 2d + 1$ would make $v + n = 2v + 2d + 1$ odd, so in fact $n \geq v + 2d + 2$. Either way $P/Q \in U_F^{d+1} \subseteq \ker \kappa'$, so $\kappa'(P) = \kappa'(Q) = 1$. Hence $N_{E/F}(U_E^v) \subseteq \ker \kappa'$.

Step 4: conclusion. By Steps 2 and 3, $N_{E/F}(U_E^v)$ and $\ker \kappa' \cap U_F^v$ are subgroups of U_F^v of index 2, one contained in the other, so they coincide with $\ker \kappa \cap U_F^v$. Two non-trivial characters of

U_F^v with the same kernel of index 2 are equal, so κ and κ' agree on U_F^v. Finally $v \geq \lfloor d/2 \rfloor + 1$, so for $x \in \mathfrak{p}^v$ both defining formulas apply and

$$\psi\big(x\delta_{E/F}\big) = \kappa(1+x) = \kappa'(1+x) = \psi\big(x\delta_{E'/F}\big)$$

Thus $\psi\big(x(\delta_{E/F} - \delta_{E'/F})\big) = 1$ for all $x \in \mathfrak{p}^v$, and since ψ is trivial exactly on $\mathfrak{p}$ this gives $\delta_{E/F} - \delta_{E'/F} \in \mathfrak{p}^{1-v}$, as we sought to show.

The hypothesis $n \geq 2d + v + 1$ is implied by $n > 3d$ whenever $v \leq d$, which is how the lemma is applied when $n > 3d$. At $n = 3d$ it still allows every $v \leq d - 1$, and that is what the boundary case of theorem 10.2.6 needs.

The proof of theorem 10.2.6 can now be given. It has two implications to establish, and the second splits according to which of the two alternatives in condition (2) is available.

***Proof*:**
Write $d_\alpha = d(F[\alpha]|F)$, which is defined because $F[\alpha]/F$ is separable: either $\mathrm{char}(F) \neq 2$, or $\mathrm{tr}\alpha \neq 0$ by hypothesis and the minimal polynomial $X^2 - \mathrm{tr}(\alpha)X + \det(\alpha)$ of α has non-zero derivative. Since $(\mathfrak{A}, n, \alpha)$ is a ramified simple stratum, $F[\alpha]$ is a totally ramified quadratic extension of F with $v_{F[\alpha]}(\alpha) = -n$ by proposition 3.4.3, and the matrix trace and determinant of α are its field trace and norm: $\mathrm{tr}\alpha = \mathrm{tr}_{F[\alpha]/F}(\alpha)$, $\det\alpha = N_{F[\alpha]/F}(\alpha)$.

Step 1: a bound on $v_F(\mathrm{tr}\alpha)$ when $n \leq 3d_\alpha$. Assume $n \leq 3d_\alpha$ and put $s = \lfloor n/3 \rfloor$, so $d_\alpha \geq \lceil n/3 \rceil$. By proposition 9.1.4(1),

$$v_F(\mathrm{tr}\alpha) \geq \left\lfloor \tfrac{-n+d_\alpha+1}{2} \right\rfloor \geq \left\lfloor \tfrac{-n+\lceil n/3 \rceil+1}{2} \right\rfloor$$

Writing $n = 3s + r$ with $r \in \{0, 1, 2\}$ and evaluating the three cases gives the value $-s$ each time: for $r = 0$ the bracket is $\lfloor (1-2s)/2 \rfloor = -s$, for $r = 1$ it is $\lfloor (1-2s)/2 \rfloor = -s$, and for $r = 2$ it is $\lfloor -2s/2 \rfloor = -s$. Hence $v_F(\mathrm{tr}\alpha) \geq -s$ and therefore $3v_F(\mathrm{tr}\alpha) \geq -3s \geq -n$.

Step 2: (1) implies (2). Suppose $(\mathfrak{A}, n, \alpha)$ is ordinary and $n \leq 3d_\alpha$. We produce a root of C_α in F. By theorem 10.1.17, (1) implies (3), so there are a totally ramified separable quadratic E/F and $\beta \in E\langle n \rangle$ with $n \geq 3d$, $d = d(E|F)$, satisfying (10.4). By proposition 10.1.16, the alternative $n \leq 3d_\alpha$ forces $n = 3d$. Since n is odd, $d = n/3$ is odd, and $v_E(\beta) = -n = -3d$.

Set $t = \mathrm{tr}_{E/F}(\beta)$, $N = N_{E/F}(\beta)$ and $\delta = \delta_{E/F}$. Lemma 10.2.9 applies, d being odd, $v_E(\beta) = -3d$ and ψ special: it gives $v_F(t) = -d$ and $c := \delta t^2 N^{-1} \in U_F^1$.

Now evaluate C_α at t. The congruences (10.4) say $\det\alpha = N(1 + \eta_2)$ with $v_F(\eta_2) \geq \lfloor m/2 \rfloor + 1$, and $\mathrm{tr}\alpha = \delta + t + \eta_1$ with $v_F(\eta_1) \geq -\lfloor m/2 \rfloor$. Hence

$$C_\alpha(t) = t^3 - (\delta + t + \eta_1)t^2 + N + N\eta_2 = N(1-c) - \eta_1 t^2 + N\eta_2$$

using $\delta t^2 = cN$. The three terms are estimated separately. First $v_F(N(1-c)) \geq -n + 1$ because $c \in U_F^1$. Second, $\lfloor m/2 \rfloor \leq d - 1$: indeed $m = (3d-1)/2$, so $\lfloor m/2 \rfloor = \lfloor (3d-1)/4 \rfloor$, and $\lfloor (3d-1)/4 \rfloor \leq d-1$ holds because $(3d-1)/4 < d$ for $d \geq 1$. Therefore $v_F(\eta_1 t^2) \geq -\lfloor m/2 \rfloor - 2d \geq 1 - d - 2d = 1 - n$. Third $v_F(N\eta_2) \geq -n + \lfloor m/2 \rfloor + 1 \geq 1 - n$. So $v_F(C_\alpha(t)) \geq 1 - n$.

Rescale. Write $\det\alpha = \zeta\varpi^{-3d}$ with $\zeta \in U_F$, set $s_0 = \varpi^d t$, which lies in U_F because $v_F(t) = -d$, and put

$$c_\alpha(X) = X^3 - \varpi^d \mathrm{tr}(\alpha)X^2 + \zeta$$

Its middle coefficient is integral: $v_F(\mathrm{tr}\alpha) \geq -s = -d$ by Step 1, since $n = 3d$ gives $s = d$. A direct computation gives $c_\alpha(Y) = \varpi^{3d} C_\alpha(\varpi^{-d}Y)$, so $c_\alpha(s_0) = \varpi^n C_\alpha(t)$ has $v_F \geq n+(1-n) = 1$, that is $c_\alpha(s_0) \in \mathfrak{p}$. Its derivative at s_0 is $c'_\alpha(s_0) = 3s_0^2 - 2\varpi^d \mathrm{tr}(\alpha)s_0$, where $v_F(3s_0^2) = 0$ because 3 is a unit in the dyadic field F and $s_0 \in U_F$, while $v_F(2\varpi^d \mathrm{tr}(\alpha)s_0) \geq v_F(2) \geq 1$; so $c'_\alpha(s_0) \in U_F$. Since $v_F(c_\alpha(s_0)) \geq 1 > 0 = 2v_F(c'_\alpha(s_0))$, Hensel's lemma ([9, Hensel's Lemma – Lifting Roots]) produces a root $s_1 \in \mathfrak{o}$ of c_α, and $\gamma = \varpi^{-d}s_1$ is a root of C_α in F. So (2) holds.

Step 3: (2) with a root implies (1). Suppose $n \leq 3d_\alpha$ and C_α has a root $\gamma \in F$. By Step 1, $3v_F(\mathrm{tr}\alpha) \geq -n$, so lemma 10.2.8(1) applies with $a = -\mathrm{tr}\alpha$ and $b = \det\alpha$: it gives $3 \mid n$ and $v_F(\gamma) = -n/3$. Put $d = n/3$, an odd integer since n is odd, and note $d \leq d_\alpha \leq v_F(4)$ by $n \leq 3d_\alpha$ and proposition 9.1.2(2). In particular $\gamma \neq 0$.

From $C_\alpha(\gamma) = 0$ we get $(\mathrm{tr}\alpha - \gamma)\gamma^2 = \det\alpha$. Set $\delta = \mathrm{tr}\alpha - \gamma$. Then $v_F(\delta) = v_F(\det\alpha) - 2v_F(\gamma) = -3d + 2d = -d$, and $\delta \neq 0$. Apply lemma 10.2.8(2) with $\epsilon = \gamma$ and $D = \det\alpha$. The hypothesis $v_F(\gamma) = -d \geq -m$ holds because $2d \leq 3d - 1 = n - 1 = 2m$, and $\gamma \neq 0$ covers the characteristic 2 case. This yields a totally ramified separable quadratic E/F and $\beta \in E$ with

$$\mathrm{tr}_{E/F}(\beta) = \gamma, \qquad N_{E/F}(\beta) = \det\alpha, \qquad v_E(\beta) = -n$$

and $d(E|F) = \min\{2(-d) + 3d,\ v_F(4)\} = \min\{d, v_F(4)\} = d$. So $(E/F, n, \beta)$ is an admissible wild triple in the sense of definition 10.1.6: $n = 3d = 3d(E|F)$.

It remains to verify (10.4), and the triple just built is in general not the one that does so. The norm congruence is an equality, $N_{E/F}(\beta) = \det\alpha$, and for the trace congruence one would want $\delta \equiv \delta_{E/F}$ modulo $\mathfrak{p}^{-\lfloor m/2 \rfloor}$; but lemma 10.2.9 controls $\delta_{E/F}$ only modulo $\mathfrak{p}^{1-d}$, and $1 - d \geq -\lfloor m/2 \rfloor$ fails as soon as $d \geq 5$, since $\lfloor m/2 \rfloor = \lfloor (3d-1)/4 \rfloor$ is then smaller than $d - 1$. The trace is therefore corrected repeatedly, the extension changing at each correction.

Put $\delta_{-1} = \delta = \mathrm{tr}\alpha - \gamma$ and define, for $0 \leq i \leq \lfloor d/2 \rfloor$,

$$\epsilon_i = \mathrm{tr}\alpha - \delta_{i-1}$$

together with the extension $E_i = F[\beta_i]$ that lemma 10.2.8(2) attaches to the pair $(\epsilon_i, \det\alpha)$, so that $\mathrm{tr}_{E_i/F}(\beta_i) = \epsilon_i$, $N_{E_i/F}(\beta_i) = \det\alpha$ and $v_{E_i}(\beta_i) = -n$. Write $\delta_i = \delta_{E_i/F}$. The first of these is the triple already constructed: $\epsilon_0 = \gamma$, $E_0 = E$, $\beta_0 = \beta$.

We claim that $\delta_i \equiv \delta_{i-1}$ modulo $\mathfrak{p}^{i+1-d}$ for $0 \leq i \leq \lfloor d/2 \rfloor$. The construction is legitimate as far as the claim has been established: granting it for all indices $< i$, the successive differences $\epsilon_j - \epsilon_{j-1} = \delta_{j-2} - \delta_{j-1}$ for $1 \leq j \leq i$ lie in $\mathfrak{p}^{j-d} \subseteq \mathfrak{p}^{1-d}$, so $\epsilon_i \equiv \epsilon_0 = \gamma$ modulo $\mathfrak{p}^{1-d}$ and hence $v_F(\epsilon_i) = -d$, the correction having valuation $\geq 1 - d > -d$. In particular $\epsilon_i \neq 0$, which covers the characteristic 2 proviso, and $v_F(\epsilon_i) = -d \geq -m$. So ϵ_i is a legitimate input for lemma 10.2.8(2), which also gives $d(E_i|F) = \min\{2(-d) + n,\ v_F(4)\} = \min\{d, v_F(4)\} = d$.

The case $i = 0$ of the claim is lemma 10.2.9 applied to E_0 and β_0:

$$\delta_0 \equiv N_{E/F}(\beta)\mathrm{tr}_{E/F}(\beta)^{-2} = \det\alpha \cdot \gamma^{-2} = \delta = \delta_{-1} \pmod{\mathfrak{p}^{1-d}}$$

the middle equality being $(\mathrm{tr}\alpha - \gamma)\gamma^2 = \det\alpha$ again. For the inductive step, let $0 \le i \le \lfloor d/2 \rfloor - 1$ and suppose $\delta_i - \delta_{i-1} \in \mathfrak{p}^{i+1-d}$. Then $\epsilon_{i+1} - \epsilon_i = \delta_{i-1} - \delta_i \in \mathfrak{p}^{i+1-d}$, and lemma 10.2.10 applies to $(\epsilon_i, \epsilon_{i+1}, \det\alpha)$ with $v = d - i - 1$. Its two constraints hold: $i \le \lfloor d/2 \rfloor - 1 = (d-3)/2$ gives $d - i - 1 \ge (d+1)/2 > \lfloor d/2 \rfloor$, and $n \ge 2d + v + 1 = 3d - i$ holds because $n = 3d$ and $i \ge 0$. The conclusion is $\delta_{i+1} \equiv \delta_i$ modulo $\mathfrak{p}^{1-(d-i-1)} = \mathfrak{p}^{i+2-d}$, which is the claim for $i+1$.

Now take $i = \lfloor d/2 \rfloor$. Since d is odd, $i + 1 - d = -\lfloor d/2 \rfloor$, so

$$\delta_{E_i/F} + \mathrm{tr}_{E_i/F}(\beta_i) = \delta_i + \epsilon_i = \mathrm{tr}\alpha + (\delta_i - \delta_{i-1}) \equiv \mathrm{tr}\alpha \pmod{\mathfrak{p}^{-\lfloor d/2 \rfloor}}$$

while $N_{E_i/F}(\beta_i) = \det\alpha$ exactly. As $d \le m$, we have $\lfloor d/2 \rfloor \le \lfloor m/2 \rfloor$ and the trace congruence holds a fortiori modulo $\mathfrak{p}^{-\lfloor m/2 \rfloor}$. Since $d(E_i|F) = d$ and $n = 3d$, the triple $(E_i/F, n, \beta_i)$ is admissible in the sense of definition 10.1.6 and satisfies (10.4). By theorem 10.1.17, (3) implies (1), so $(\mathfrak{A}, n, \alpha)$ is ordinary.

Step 4: $n > 3d_\alpha$ implies (1). Write $d = d_\alpha$ and $D = \det\alpha$, so $v_F(D) = -n$. Note $d < n/3 \le m$, since $n = 2m + 1$ and $m \ge 1$ (the inequality $n > 3d \ge 3$ rules out $m = 0$). Every totally ramified quadratic E'/F occurring below has $v_F(\delta_{E'/F}) = -d(E'|F)$ by definition 10.1.7. Two cases.

Suppose first $\lfloor m/2 \rfloor \ge d$. Take $E = F[\alpha]$, which is separable over F by the opening paragraph, and $\beta = \alpha$. Then $N_{E/F}(\alpha) = \det\alpha$ and $\mathrm{tr}_{E/F}(\alpha) = \mathrm{tr}\alpha$, and $v_F(\delta_{E/F}) = -d \ge -\lfloor m/2 \rfloor$, so $\mathrm{tr}\alpha \equiv \delta_{E/F} + \mathrm{tr}_{E/F}(\alpha)$ modulo $\mathfrak{p}^{-\lfloor m/2 \rfloor}$. The triple $(E/F, n, \alpha)$ is admissible because $n > 3d$, and (10.4) holds. By theorem 10.1.17, the stratum is ordinary.

Suppose instead $\lfloor m/2 \rfloor < d$, and set $h = d - \lfloor m/2 \rfloor \ge 1$. We build elements $\epsilon_0, \epsilon_1, \ldots, \epsilon_h$ of F and, through lemma 10.2.8(2) applied to (ϵ_i, D), extensions $E_i = F[\beta_i]$ with $\mathrm{tr}_{E_i/F}(\beta_i) = \epsilon_i$, $N_{E_i/F}(\beta_i) = D$. Write $\delta_i = \delta_{E_i/F}$. Start with $\epsilon_0 = \mathrm{tr}\alpha$, so that $E_0 = F[\alpha]$ and $\beta_0 = \alpha$, and set

$$\epsilon_{i+1} = \mathrm{tr}\alpha - \delta_i, \qquad 0 \le i \le h - 1$$

Each ϵ_i is admissible input for lemma 10.2.8(2): by induction $d(E_i|F) = d$, so $v_F(\delta_i) = -d \ge -m$, while $v_F(\mathrm{tr}\alpha) \ge -m$ by proposition 9.1.4(1). Hence $v_F(\epsilon_{i+1}) \ge -m$. If $\mathrm{char}(F) = 2$ then d is odd by proposition 9.1.2(3), so $v_F(\mathrm{tr}\alpha) = -m + (d-1)/2 < -d$ by proposition 9.1.4(2) and the inequality $2m > 3d - 1$. As $v_F(\delta_i) = -d$, this forces $v_F(\epsilon_{i+1}) = v_F(\mathrm{tr}\alpha)$ and in particular $\epsilon_{i+1} \neq 0$.

We claim $\delta_{i+1} \equiv \delta_i$ modulo $\mathfrak{p}^{i+1-d}$ for $0 \le i \le h - 1$, and $d(E_i|F) = d$ throughout. For $i = 0$ we have $\epsilon_1 - \epsilon_0 = -\delta_0 \in \mathfrak{p}^{-d}$, so lemma 10.2.10 applies with $v = d$ (legitimate: $\lfloor d/2 \rfloor < d$, and $n > 3d$ gives $n \ge 2d + v + 1$), giving $d(E_1|F) = d$ and $\delta_1 \equiv \delta_0$ modulo $\mathfrak{p}^{1-d}$. Inductively, if $\delta_i - \delta_{i-1} \in \mathfrak{p}^{i-d}$ then $\epsilon_{i+1} - \epsilon_i = \delta_{i-1} - \delta_i \in \mathfrak{p}^{i-d}$, so lemma 10.2.10 applies with $v = d - i$, again with $n > 3d \ge 2d + v$. The constraint $\lfloor d/2 \rfloor < d - i$ holds because $i \le h - 1$ gives $d - i \ge \lfloor m/2 \rfloor + 1 > \lfloor d/2 \rfloor$, using $\lfloor m/2 \rfloor \ge \lfloor d/2 \rfloor$ from $m > d$. The conclusion is $d(E_{i+1}|F) = d$ and $\delta_{i+1} \equiv \delta_i$ modulo $\mathfrak{p}^{1-(d-i)} = \mathfrak{p}^{i+1-d}$, which is the claim.

Take $i = h-1$: then $\delta_h \equiv \delta_{h-1}$ modulo $\mathfrak{p}^{h-d} = \mathfrak{p}^{-\lfloor m/2 \rfloor}$. Since $\epsilon_h = \mathrm{tr}\alpha - \delta_{h-1}$,

$$\delta_{E_h/F} + \mathrm{tr}_{E_h/F}(\beta_h) = \delta_h + \epsilon_h = \mathrm{tr}\alpha + (\delta_h - \delta_{h-1}) \equiv \mathrm{tr}\alpha \pmod{\mathfrak{p}^{-\lfloor m/2 \rfloor}}$$

while $N_{E_h/F}(\beta_h) = D = \det \alpha$ exactly. As $d(E_h|F) = d$ and $n > 3d$, the triple $(E_h/F, n, \beta_h)$ is admissible and satisfies (10.4). By theorem 10.1.17, $(\mathfrak{A}, n, \alpha)$ is ordinary.

Step 5: assembling. Step 2 shows that (1) implies (2), since if $n > 3d_\alpha$ then (2) holds outright and otherwise a root is produced. Steps 3 and 4 show that each of the two alternatives in (2) implies (1), completing the proof.

The criterion is now checkable. Given α in matrix form, compute $\mathrm{tr}\alpha$ and $\det \alpha$, read d_α off lemma 10.2.8(2), and either observe $n > 3d_\alpha$ or test the cubic for a root. The second test is itself finite: by lemma 10.2.8(1) a root must have valuation $-n/3$, so it can only exist when $3 \mid n$, and then Hensel's lemma reduces the search to the residue field.

Corollary 10.2.11: Exceptional Strata

Let $(\mathfrak{A}, n, \alpha)$, $n = 2m+1 \geq 1$, be a ramified simple stratum satisfying the hypotheses of theorem 10.2.6: $\mathrm{tr}\alpha \neq 0$ in case $\mathrm{char}(F) = 2$, and ψ special. The following are equivalent.

1. The stratum $(\mathfrak{A}, n, \alpha)$ is exceptional.
2. $n \leq 3d_\alpha$ and $C_\alpha(X)$ is irreducible over F.

In particular, if $n \leq 3d_\alpha$ and $n \not\equiv 0 \bmod 3$, then $(\mathfrak{A}, n, \alpha)$ is exceptional, and every $\pi \in \mathcal{A}_2^0(F; \alpha)$ is an exceptional representation.

***Proof*:**
A cubic over a field is reducible if and only if it has a root in that field, since a factorization into two non-constant factors must have a linear factor. So (2) is the negation of condition (2) of theorem 10.2.6, and the equivalence is that theorem together with definition 10.2.5. For the last assertion, suppose $n \leq 3d_\alpha$ and $n \not\equiv 0 \bmod 3$. Step 1 of the proof of theorem 10.2.6 gives $3v_F(\mathrm{tr}\alpha) \geq -n$, so lemma 10.2.8(1) shows that a root of C_α in F would force $3 \mid n$. There is therefore no root, C_α is irreducible, and the stratum is exceptional. The statement about representations is definition 10.2.5, completing the proof.

Example 10.2.12: An Exceptional Stratum over $\mathbb{Q}_2$
Let $F = \mathbb{Q}_2$, so $q = 2$, $v_F(2) = 1$ and $v_F(4) = 2$. By example 10.2.4(1) every character of level one is special, so the normalization in theorem 10.2.6 is automatic here, and $\mathrm{char}(F) = 0$ makes its proviso on $\mathrm{tr}\alpha$ vacuous.

Take $n = 5$, so $m = 2$, and let $\alpha \in M_2(\mathbb{Q}_2)$ have

$$\mathrm{tr}\alpha = 0, \qquad \det \alpha = 2^{-5}$$

Then $F[\alpha] = \mathbb{Q}_2\big(\sqrt{-2^{-5}}\big) = \mathbb{Q}_2(\sqrt{-2})$, since $-2^{-5} = -2 \cdot 2^{-6}$ and 2^{-6} is a square. By lemma 10.2.8(2), $d_\alpha = \min\{2v_F(0) + 5,\ 2\} = 2$. This agrees with the direct computation:

the prime element $\sqrt{-2}$ has minimal polynomial X^2+2, of discriminant $0^2-4\cdot 2=-8$, so $\mathfrak{d}_{F[\alpha]/F}=8\mathbb{Z}_2=\mathfrak{p}^3$, and $\mathfrak{p}^3=\mathfrak{p}^{d_\alpha+1}$ is what definition 9.1.1 requires of $d_\alpha=2$. The extension is totally ramified, so $(\mathfrak{A},5,\alpha)$ is a ramified simple stratum with $v_{F[\alpha]}(\alpha)=-5$.

Now $n=5\le 6=3d_\alpha$ and $5\not\equiv 0 \bmod 3$, so corollary 10.2.11 declares the stratum exceptional. The cubic confirms it directly: $C_\alpha(X)=X^3+2^{-5}$, and a root would satisfy $3v_F(X)=-5$, impossible for $v_F(X)\in\mathbb{Z}$. Every $\pi\in\mathcal{A}_2^0(\mathbb{Q}_2;\alpha)$ therefore lies outside the image of $\boldsymbol{\pi}^{\mathrm{im}}$.

Changing n to 7 with the same recipe, $\det\alpha=2^{-7}$ and $\mathrm{tr}\alpha=0$, gives $F[\alpha]=\mathbb{Q}_2(\sqrt{-2})$ again and $d_\alpha=2$, but now $n=7>6=3d_\alpha$ and the stratum is ordinary. The two strata differ only in the depth n at which they sit.

Exercise 10.2.1

1. Let $\wp(y)=y+y^2$ on the residue field k of F. Show that $\wp$ is $\mathbb{F}_2$-linear with kernel $\mathbb{F}_2$, and deduce that when $q=2$ every character of F of level one is special. Show also that when $q>2$ there exists a character of F of level one that is not special.

2. Let $(\mathfrak{A},n,\alpha)$ be a ramified simple stratum with $\mathrm{tr}\alpha\neq 0$ in case $\mathrm{char}(F)=2$, so that d_α is defined, and suppose $n\le 3d_\alpha$. Show that $n\le 3v_F(4)$. Deduce that if $v_F(2)=1$, as for $F=\mathbb{Q}_2$, then every ramified simple stratum with $n\ge 7$ is ordinary.

3. Let $F=\mathbb{Q}_2$ and let $(\mathfrak{A},n,\alpha)$ be a ramified simple stratum. Using lemma 10.2.8(2) and theorem 10.2.6, decide the ordinary or exceptional status of $(\mathfrak{A},n,\alpha)$ for every odd n and every value of d_α. The pairs with $n=3$ are the only ones the first alternative of theorem 10.2.6(2) and corollary 10.2.11 leave open; settle them by testing C_α for a root directly, and conclude that over $\mathbb{Q}_2$ the status is a function of the pair (n,d_α) alone.

4. Let $F=\mathbb{Q}_2$ and let $(\mathfrak{A},5,\alpha)$ be a ramified simple stratum. Determine d_α in terms of $v_F(\mathrm{tr}\alpha)$, and conclude that $(\mathfrak{A},5,\alpha)$ is ordinary when $v_F(\mathrm{tr}\alpha)=-2$ and exceptional when $v_F(\mathrm{tr}\alpha)\ge -1$.

5. Let $F=\mathbb{Q}_2$ and let α have $\mathrm{tr}\alpha=0$ and $\det\alpha=2^{-1}$, so $n=1$ and $m=0$. Show that $(\mathfrak{A},1,\alpha)$ is a ramified simple stratum and that it is exceptional, in two ways: by corollary 10.2.11, and by exhibiting directly that C_α has no root in $\mathbb{Q}_2$.

6. Let F be a non-archimedean local field whose residue field has $q=4$ elements. Show that there are $b\in F^\times$ and $u\in U_F$ such that X^3+b has a root in F while X^3+ub does not. Explain why this makes the hypothesis on ψ in theorem 10.2.6 indispensable, by reference to box 10.2.7.

11

Tame Lifting

Chapter 10 decided which cuspidal representations lie in the image of $\boldsymbol{\pi}^{\mathrm{im}}$, and the answer was a condition on a stratum. What that leaves is the other half of the dyadic problem. Theorem 7.1.2 states the correspondence for every residual characteristic, but Step 4 of its proof still defers the construction on the primitive parameters when $p = 2$, and nothing so far reaches them: a primitive parameter is induced from no quadratic extension, so every construction that begins with a pair $(E/F, \Theta)$ misses it by definition.

The way in is to change the field. A primitive parameter over F becomes imprimitive over a suitable cubic extension K, by theorem 9.2.13, and over K the machinery of the preceding chapters applies. What is then needed is a way back down. This chapter builds it, in four pieces. Section 11.1 constructs *tame lifting*, which carries a cuspidal representation of $\mathrm{GL}_2(F)$ to one over K and is compatible with the classification by types, and identifies its image as a set of fixed points. Section 11.2 shows the correspondence commutes with isomorphisms of the base field, which is what makes those fixed points recognizable on both sides at once. Section 11.3 and section 11.4 give the numerical control: a local constant is too rigid to survive lifting intact, but its class modulo the roots of unity of 2-power order does survive, and that is exactly the precision the descent argument needs.

None of this proves the correspondence. The four tools are assembled here; the Existence Theorem that combines them and reaches the primitive parameters is [2, §50-§52], and chapter 12 records that this book stops short of it. Throughout, the residual characteristic of F is 2 unless a statement says otherwise.

11.1 Lifting of Strata and Types

Throughout this chapter the residual characteristic of F is 2, unless a statement says otherwise.

The task of this section is a single construction. Given a finite tamely ramified extension K/F we

build a map

$$\mathrm{Lft}_{K/F}\colon \mathcal{A}_2^{\mathrm{wr}}(F) \longrightarrow \mathcal{A}_2^{\mathrm{wr}}(K)$$

on totally ramified cuspidal representations, and we establish enough of its formal properties (functoriality in K, its effect on central characters and on twists, and the exact size of its fibres) to make it usable. The construction runs through the classification by cuspidal types of theorem 3.5.5: a totally ramified π is, after a twist, the representation attached to a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ and a character Λ of the associated group J, and the lift is obtained by lifting the stratum and then the character. Neither step is forced on us by a universal property, so most of the work is showing that the answer does not depend on the choices made along the way.

The section closes with the descent statement the construction was built for: for cyclic tame K/F the image of $\mathrm{Lft}_{K/F}$ is exactly the set of $\mathrm{Gal}(K/F)$-fixed points of $\mathcal{A}_2^{\mathrm{wr}}(K)$, together with its two consequences: the cubic case and the case of the normal closure of a non-cyclic cubic.

Three conventions are fixed once, here.

11.1.1 Three Conventions, Fixed Here *The sign.* Definition 6.2.5 attaches to a finite separable K/F the character $\varkappa_{K/F} = \det R_{K/F}$ of $F^\times$, where $R_{K/F} = \mathrm{Ind}_{K/F} 1_K$, and proposition 6.2.6 identifies its values as signs of permutations, so $\varkappa_{K/F}^2 = 1$. This book *also* writes $\kappa_{E/F}$ for the quadratic class-field character of a quadratic extension E/F. The two agree only when $[K : F] = 2$. Below, $\varkappa$ always means $\det R_{K/F}$ and κ always means the quadratic character. No proof here passes between them.
The direction of transport. The lifted character Λ_K of theorem 11.1.10 is defined by pulling Λ back along the norm $\mathrm{Nm}_{EK/E}$, from the larger field to the smaller, and *not* by pushing forward. Every formula below is written in that direction, and the twisting rule $\chi_K = \chi \circ \mathrm{Nm}_{K/F}$ of proposition 11.1.12(2) is the corresponding statement for characters of $F^\times$. The direction is restated at each point of use.
The Galois superscript. Let γ be a field automorphism of a finite extension L of F; it acts on $M_2(L)$ entry by entry, hence on $\mathrm{GL}_2(L)$. Transport along γ is always *push-forward* (the word said here of the automorphism γ, not of the norm of the previous paragraph) and is written with a superscript: $x^\gamma = \gamma(x)$ for an element, $S^\gamma = \gamma(S)$ for a subset such as a chain order $\mathfrak{A}$ or a group $U_{\mathfrak{A}}^j$, and

$$\varphi^\gamma = \varphi \circ \gamma^{-1}$$

for a function φ on a GL_2-group or on $M_2(L)$: a character, a representation, or the class of one. So Θ^σ means $\Theta \circ \sigma^{-1}$, never $\Theta \circ \sigma$. The two differ, and this is the choice for which the superscript is a left action, since $(\varphi^\sigma)^\gamma = \varphi \circ \sigma^{-1} \circ \gamma^{-1} = \varphi^{\gamma\sigma}$, and the one for which $\varphi^\gamma(x^\gamma) = \varphi(x)$. For a group Σ of such automorphisms and a set X carrying the resulting action we write $X^\Sigma = \{x \in X : x^\sigma = x \text{ for all } \sigma \in \Sigma\}$ for the fixed set; $\mathcal{A}_2^{\mathrm{wr}}(L)^\Sigma$ is formed this way, from the action on isomorphism classes.

Fix once and for all a character $\psi \in \widehat{F}$ of level one. All strata and all sets $\mathcal{C}(\psi_\alpha, \mathfrak{A})$, $\mathcal{A}_2^0(F; \alpha)$ are formed with respect to it. The independence of the construction from this choice is proposition 11.1.15(5).

A ramified simple stratum $(\mathfrak{A}, n, \alpha)$ in $M_2(F)$ has $e_{\mathfrak{A}} = 2$ and $n = 2m + 1 \geq 1$ odd, which is what definition 3.4.1(4) asks of it, while $E = F[\alpha]$ is a field with $E^\times \subseteq \mathcal{K}_{\mathfrak{A}}$ and $e(E|F) = e_{\mathfrak{A}}$ by proposition 3.4.3, hence a totally ramified quadratic extension of F. The element α has $v_E(\alpha) = -n$, by proposition 3.3.11 at $r = 1$ and corollary 3.4.4. Since $p = 2$, such an E/F is totally *wildly* ramified. Lemma 11.1.2 discharges, once, the hypothesis under which the results that follow are posed: that

EK is a field and EK/K is totally ramified quadratic.

Lemma 11.1.2: The Compositum of a Wild Quadratic and a Tame Extension

Let E/F be a totally ramified quadratic extension and let K/F be a finite tamely ramified extension. Write $e_K = e(K|F)$ and $f_K = f(K|F)$. Then e_K is odd, the K-algebra $EK = E \otimes_F K$ is a field, and EK/K is totally ramified quadratic. The valuation of EK satisfies

$$v_{EK}|_E = e_K\, v_E, \qquad v_{EK}|_K = 2\, v_K$$

In particular, if $v_E(\alpha) = -n$ with n odd then $v_{EK}(\alpha) = -e_K n$ is odd, so α is minimal over K.

***Proof*:**
Tameness of K/F means $p \nmid e_K$, and $p = 2$, so e_K is odd.

EK is a field. If E/F is separable then $[E : F] = 2$ makes E/F Galois, and $E \otimes_F K$ is a field if and only if $E \cap K = F$. If E/F is inseparable (possible only when $\mathrm{char}(F) = 2$) then $E = F(\lambda)$ with $\lambda^2 \in F$, and $E \otimes_F K$ is a field if and only if λ^2 is not a square in K, that is, if and only if $E \not\subseteq K$. In both cases it suffices to rule out $E \subseteq K$. But $E \subseteq K$ would give $2 = e(E|F) \big| e_K$, and e_K is odd. So EK is a field, and $[EK : K] = 2$.

EK/K is totally ramified. Multiplicativity of e in the towers $F \subseteq K \subseteq EK$ and $F \subseteq E \subseteq EK$ gives $e(EK|K)\, e_K = e(EK|F) = e(EK|E) \cdot 2$, so $2 \big| e(EK|K) e_K$. As e_K is odd, $2 \big| e(EK|K)$. Since $[EK : K] = 2$ this forces $e(EK|K) = 2$, so EK/K is totally ramified, and then $f(EK|E)\, f(E|F) = f(EK|K) f_K$ reads $f(EK|E) = f_K$ while $e(EK|E) = e(EK|F)/2 = e_K$.

The valuations. $v_{EK}|_E = e(EK|E) v_E = e_K v_E$ and $v_{EK}|_K = e(EK|K) v_K = 2 v_K$. If $v_E(\alpha) = -n$ then $v_{EK}(\alpha) = -e_K n$, which is odd because both e_K and n are. An element of a ramified quadratic extension of K whose valuation is prime to $e(EK|K) = 2$ is minimal over K, completing the proof.

The next lemma is the uniqueness statement that makes "the" chain order attached to a minimal element well posed. Proposition 3.4.3 asserts existence only, and the statement is needed over K as much as over F, so it is proved here for a general base.

Lemma 11.1.3: A Minimal Element Determines Its Chain Order

Let L be a non-archimedean local field and let $\alpha \in M_2(L)$ be such that $E = L[\alpha]$ is a totally ramified quadratic extension of L with $v_E(\alpha) = -n$ odd. Then there is exactly one $\mathfrak{o}_L$-chain order $\mathfrak{A}$ in $M_2(L)$ with $\alpha \in \mathcal{K}_{\mathfrak{A}}$. Its lattice chain is $(\mathfrak{p}_E^i)_{i \in \mathbb{Z}}$ under the identification of L^2 with E, it has $e_{\mathfrak{A}} = 2$, and $v_{\mathfrak{A}}(\alpha) = -n$. Moreover $\mathcal{K}_{\mathfrak{A}} = \varpi_E^{\mathbb{Z}} U_{\mathfrak{A}}$ for any prime element ϖ_E of E.

***Proof*:**
The space $V = L^2$ is a module over the field $E = L[\alpha] \subseteq M_2(L)$, and $\dim_L V = 2 = [E : L]$, so V is a one-dimensional E-vector space. Fix an E-basis and identify V with E.

Existence. The family $(\mathfrak{p}_E^i)_{i\in\mathbb{Z}}$ is an $\mathfrak{o}_L$-lattice chain with $\varpi_L\,\mathfrak{p}_E^i = \mathfrak{p}_E^{i+2}$, so its chain order $\mathfrak{A}$ has $e_{\mathfrak{A}} = 2$; and $\alpha\,\mathfrak{p}_E^i = \mathfrak{p}_E^{i-n}$, so α normalizes the chain and lies in $\mathcal{K}_{\mathfrak{A}}$ with $v_{\mathfrak{A}}(\alpha) = -n$.

Uniqueness. Let $\mathfrak{B}$ be an $\mathfrak{o}_L$-chain order with $\alpha \in \mathcal{K}_{\mathfrak{B}}$ and let (M_i) be its lattice chain. By proposition 3.3.7(1) an element of $\mathcal{K}_{\mathfrak{B}}$ translates the chain by its $\mathfrak{B}$-valuation, so $\alpha M_i = M_{i+v_{\mathfrak{B}}(\alpha)}$, and likewise $\varpi_L M_i = M_{i+e_{\mathfrak{B}}}$, the latter by the last clause of proposition 3.3.7(4). Set

$$\varpi_E = \alpha\,\varpi_L^{(n+1)/2} \in E^\times$$

Then $v_E(\varpi_E) = -n + 2\cdot\frac{n+1}{2} = 1$, so ϖ_E is a prime element of E, and $\varpi_E \in \mathcal{K}_{\mathfrak{B}}$ because α and ϖ_L both are.

Write $b = v_{\mathfrak{B}}(\varpi_E)$, so $\varpi_E M_0 = M_b$. Comparing indices pins b. Each quotient M_i/M_{i+1} is a non-zero k_L-vector space, because $\varpi_L M_i = M_{i+e_{\mathfrak{B}}} \subseteq M_{i+1}$, and the $e_{\mathfrak{B}}$ of them lying between M_0 and $M_{e_{\mathfrak{B}}} = \varpi_L M_0$ have dimensions summing to $\dim_{k_L} M_0/\varpi_L M_0 = 2$. So for $e_{\mathfrak{B}} = 2$ each of the two steps has index q_L, while for $e_{\mathfrak{B}} = 1$ the single step has index q_L^2. In both cases $[M_0 : M_b] = q_L^{2b/e_{\mathfrak{B}}}$, while $[M_0 : \varpi_E M_0] = q_L^{v_L(\det \varpi_E)} = q_L$ because $\det \varpi_E = \mathrm{Nm}_{E/L}(\varpi_E)$ has valuation 1. The index formula used there is the elementary divisor theorem over the discrete valuation ring $\mathfrak{o}_L$: for $g \in \mathrm{GL}_2(L)$ with $gM_0 \subseteq M_0$ there is a basis of M_0 in which gM_0 is $\mathfrak{p}_L^{a_1} \oplus \mathfrak{p}_L^{a_2}$, and $a_1 + a_2 = v_L(\det g)$, so the index is $q_L^{v_L(\det g)}$. Hence $2b = e_{\mathfrak{B}}$. As $e_{\mathfrak{B}} \le 2$ and b is a positive integer, $b = 1$ and $e_{\mathfrak{B}} = 2$.

Since E/L is quadratic and totally ramified, $\mathfrak{o}_E = \mathfrak{o}_L[\varpi_E]$ by proposition 9.1.2(1), whose statement names only the minimal polynomial of a prime element and so applies over L as it stands. The lattice M_0 is an $\mathfrak{o}_L$-module with $\varpi_E M_0 = M_1 \subseteq M_0$, hence is stable under $\mathfrak{o}_L[\varpi_E] = \mathfrak{o}_E$. That is, M_0 is an $\mathfrak{o}_E$-lattice in the one-dimensional E-space V, and therefore $M_0 = \mathfrak{p}_E^k$ for some $k \in \mathbb{Z}$. Then $M_i = \varpi_E^i M_0 = \mathfrak{p}_E^{k+i}$ for every i, so the chain of $\mathfrak{B}$ is the chain of $\mathfrak{A}$ up to indexing and $\mathfrak{B} = \mathfrak{A}$.

The normalizer. By corollary 3.4.4 the element ϖ_E has $v_{\mathfrak{A}}(\varpi_E) = v_E(\varpi_E) = 1$, so it is a prime element of $\mathfrak{A}$, and proposition 3.3.7(3) gives $\mathcal{K}_{\mathfrak{A}} = \varpi_E^{\mathbb{Z}} U_{\mathfrak{A}}$, completing the proof.

We record a numerical identity used three times below and once more in lemma 11.1.6. For integers x and positive integers a, b,

$$\left\lceil \tfrac{1}{b}\left\lceil \tfrac{x}{a}\right\rceil\right\rceil = \left\lceil \tfrac{x}{ab}\right\rceil, \qquad \left\lfloor \tfrac{1}{b}\left\lfloor \tfrac{x}{a}\right\rfloor\right\rfloor = \left\lfloor \tfrac{x}{ab}\right\rfloor \tag{11.1}$$

For the first: an integer c satisfies $\lceil x/a\rceil \le bc$ if and only if $x/a \le bc$, because bc is an integer, if and only if $x \le abc$. So both sides are the least integer c with $x \le abc$. The second follows by applying the first to $-x$.

Lifting a representation begins with lifting the data that carries it, and the first half of that data is the stratum. What has to be produced over K is a chain order and a level, and what has to be

proved is that the passage is canonical in both directions: the element α alone pins the chain order over K, and the stratum over F is recovered from its lift by intersecting with $M_2(F)$. That recovery is what later lets a character of the lifted group be compared with the character it came from.

Proposition 11.1.4: Lifting a Ramified Simple Stratum

Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum in $M_2(F)$ with $n = 2m+1 \geq 1$, and let K/F be a finite tamely ramified extension with $e_K = e(K|F)$. Regard $M_2(F)$ as embedded in $M_2(K)$ in the obvious way, and write $\mathfrak{P} = \mathrm{rad}\mathfrak{A}$.

1. There is exactly one $\mathfrak{o}_K$-chain order $\mathfrak{A}_K$ in $M_2(K)$ with $\alpha \in \mathcal{K}_{\mathfrak{A}_K}$, and $(\mathfrak{A}_K, n_K, \alpha)$ is a ramified simple stratum in $M_2(K)$, where $n_K = e_K n = 2m_K + 1$.
2. Write $\mathfrak{P}_K = \mathrm{rad}\mathfrak{A}_K$ and let $T\colon M_2(K) \to M_2(F)$ be the map applying $\mathrm{tr}_{K/F}$ to each entry. Then, for every $j \in \mathbb{Z}$,
$$\mathfrak{P}_K^j \cap M_2(F) \;=\; T\big(\mathfrak{P}_K^j\big) \;=\; \mathfrak{P}^{\lceil j/e_K \rceil}$$
In particular $\mathfrak{A} = \mathfrak{A}_K \cap M_2(F)$ and $\mathfrak{P} = \mathfrak{P}_K \cap M_2(F)$.
3. $\mathcal{K}_{\mathfrak{A}} \subseteq \mathcal{K}_{\mathfrak{A}_K}$, and $\mathfrak{A}_K$ is the unique $\mathfrak{o}_K$-chain order in $M_2(K)$ with that property.

***Proof*:**

Part (1). By lemma 11.1.2, $EK = K[\alpha]$ is a totally ramified quadratic extension of K and $v_{EK}(\alpha) = -e_K n = -n_K$ is odd. Lemma 11.1.3, applied over K, produces a unique $\mathfrak{o}_K$-chain order $\mathfrak{A}_K$ with $\alpha \in \mathcal{K}_{\mathfrak{A}_K}$, with $e_{\mathfrak{A}_K} = 2$ and $v_{\mathfrak{A}_K}(\alpha) = -n_K$. That is precisely the assertion that $(\mathfrak{A}_K, n_K, \alpha)$ is a ramified simple stratum; and n_K is odd, so $m_K = (n_K - 1)/2$ is an integer, explicitly $m_K = e_K m + (e_K - 1)/2$.

Coordinates. Put $\varpi_E = \alpha\varpi^{(n+1)/2}$, a prime element of E as in lemma 11.1.3, with minimal polynomial $X^2 - tX + N$ over F. Then $v_F(N) = v_E(\varpi_E) = 1$, and $v_F(t) \geq 1$: otherwise the Newton polygon of $X^2 - tX + N$ would have two distinct slopes and the polynomial would split over F, contradicting $[E : F] = 2$. Identify $V = F^2$ with E and use $\{1, \varpi_E\}$ as F-basis. Then $\mathfrak{o}_E = \mathfrak{o} \oplus \mathfrak{o}\varpi_E$ and, since $\varpi_E^2 = t\varpi_E - N$ and $N\mathfrak{o} = \mathfrak{p}$,
$$\mathfrak{p}_E = \mathfrak{o}\varpi_E + \mathfrak{o}\varpi_E^2 = \mathfrak{p} \oplus \mathfrak{o}\varpi_E$$
so in these coordinates the lattices $L_i = \mathfrak{p}_E^i$ of lemma 11.1.3 read $L_0 = \mathfrak{o} \oplus \mathfrak{o}$ and $L_1 = \mathfrak{p} \oplus \mathfrak{o}$, whence, using $L_{i+2} = \varpi L_i$,
$$L_i = \mathfrak{p}^{\lceil i/2 \rceil} \oplus \mathfrak{p}^{\lceil (i-1)/2 \rceil}, \qquad \mathfrak{P}^j = \begin{pmatrix} \mathfrak{p}^{\lceil j/2 \rceil} & \mathfrak{p}^{\lceil (j+1)/2 \rceil} \\ \mathfrak{p}^{\lceil (j-1)/2 \rceil} & \mathfrak{p}^{\lceil j/2 \rceil} \end{pmatrix} \tag{11.2}$$
the second display being the set of y with $yL_i \subseteq L_{i+j}$ for all i, computed entry by entry from the first.

The chain over K. Set $s = (e_K - 1)/2$, an integer, and define $\mathfrak{o}_K$-lattices in $V \otimes_F K$ by
$$L_j^K = \mathfrak{p}_K^{\lceil j/2 \rceil} \oplus \mathfrak{p}_K^{\lceil (j-e_K)/2 \rceil}, \qquad j \in \mathbb{Z}$$

so that $L^K_{j+2} = \varpi_K L^K_j$ and $L^K_0 = \mathfrak{o}_K \oplus \mathfrak{p}_K^{-s}$. We claim $\varpi_E L^K_j = L^K_{j+e_K}$. Writing $A = \lceil j/2 \rceil$ and $B = \lceil (j-e_K)/2 \rceil$, one has $A - B \in \{s, s+1\}$ according to the parity of j, and $\varpi_E(x,y) = (-Ny,\ x+ty)$. Now $v_K(N) = e_K$, so $-Ny \in \mathfrak{p}_K^{B+e_K} = \mathfrak{p}_K^{\lceil (j+e_K)/2 \rceil}$; and $v_K(t) = e_K v_F(t) \geq e_K \geq s+1 \geq A - B$, so $x + ty \in \mathfrak{p}_K^A$. Thus $\varpi_E L^K_j \subseteq L^K_{j+e_K}$. Both are $\mathfrak{o}_K$-lattices, and $[L^K_j : \varpi_E L^K_j] = q_K^{v_K(\det \varpi_E)} = q_K^{e_K}$ because $\det \varpi_E = N$, while $[L^K_j : L^K_{j+e_K}] = q_K^{e_K}$ because each step of a chain of period 2 in a two-dimensional space has index q_K. Hence $\varpi_E L^K_j = L^K_{j+e_K}$, so ϖ_E, and with it $\alpha = \varpi_E \varpi^{-(n+1)/2}$, normalizes the chain (L^K_j). By the uniqueness in part (1), $\mathfrak{A}_K$ is the chain order of (L^K_j), and the same entrywise computation as over F gives

$$\mathfrak{P}^j_K = \begin{pmatrix} \mathfrak{p}_K^{\lceil j/2 \rceil} & \mathfrak{p}_K^{\lceil (j+e_K)/2 \rceil} \\ \mathfrak{p}_K^{\lceil (j-e_K)/2 \rceil} & \mathfrak{p}_K^{\lceil j/2 \rceil} \end{pmatrix} \tag{11.3}$$

Part (2). For $z \in F$ one has $v_K(z) = e_K v_F(z)$, so $\mathfrak{p}^a_K \cap F = \mathfrak{p}^{\lceil a/e_K \rceil}$. Since K/F is tamely ramified its different is $\mathfrak{p}_K^{e_K - 1}$, and lemma 4.1.2(1) gives $\mathrm{tr}_{K/F}(\mathfrak{p}^a_K) = \mathfrak{p}^{\lceil a/e_K \rceil}$ as well. Both operations therefore act on the displayed matrix of $\mathfrak{P}^j_K$ by replacing each exponent a by $\lceil a/e_K \rceil$, and both act entry by entry, T because it is applied entrywise and the module is a product of its entries. Writing $J = \lceil j/e_K \rceil$, the three exponents to be matched are

$$\left\lceil \tfrac{1}{e_K} \left\lceil \tfrac{j}{2} \right\rceil \right\rceil = \left\lceil \tfrac{j}{2e_K} \right\rceil = \left\lceil \tfrac{J}{2} \right\rceil, \qquad \left\lceil \tfrac{1}{e_K} \left\lceil \tfrac{j \pm e_K}{2} \right\rceil \right\rceil = \left\lceil \tfrac{j \pm e_K}{2e_K} \right\rceil = \left\lceil \tfrac{J \pm 1}{2} \right\rceil$$

each equality being an instance of (11.1), the last one after rewriting $\lceil (J \pm 1)/2 \rceil = \lceil \lceil (j \pm e_K)/e_K \rceil /2 \rceil$. Comparing with the matrix of $\mathfrak{P}^J$ gives the assertion. Taking $j = 0$ and $j = 1$ yields $\mathfrak{A} = \mathfrak{A}_K \cap M_2(F)$ and $\mathfrak{P} = \mathfrak{P}_K \cap M_2(F)$.

Part (3). By lemma 11.1.3, $\mathcal{K}_{\mathfrak{A}} = \varpi_E^{\mathbb{Z}} U_{\mathfrak{A}}$. Now $\varpi_E \in \mathcal{K}_{\mathfrak{A}_K}$ was shown above, and $U_{\mathfrak{A}} \subseteq U_{\mathfrak{A}_K}$ because $\mathfrak{A} = \mathfrak{A}_K \cap M_2(F) \subseteq \mathfrak{A}_K$ and the inverse of a unit of $\mathfrak{A}$ lies in $\mathfrak{A}$. Hence $\mathcal{K}_{\mathfrak{A}} \subseteq \mathcal{K}_{\mathfrak{A}_K}$. If $\mathfrak{B}$ is any $\mathfrak{o}_K$-chain order with $\mathcal{K}_{\mathfrak{A}} \subseteq \mathcal{K}_{\mathfrak{B}}$ then $\alpha \in \mathcal{K}_{\mathfrak{B}}$, so $\mathfrak{B} = \mathfrak{A}_K$ by part (1), completing the proof.

Keep $(\mathfrak{A}, n, \alpha)$ and K/F as above, together with T, $\mathfrak{A}_K$ and $n_K = 2m_K + 1$. By proposition 11.1.4(2),

$$T\big(\mathfrak{P}_K^{n_K+1}\big) = \mathfrak{P}^{n+1}, \qquad T\big(\mathfrak{P}_K^{m_K+1}\big) = \mathfrak{P}^{m+1}$$

the second because $\lceil (m_K+1)/e_K \rceil = m+1$: indeed $2m_K+1 = n_K = e_K(2m+1)$ gives $m_K = e_K m + s_0$ with $s_0 = (e_K - 1)/2$, so that $0 \leq s_0 < e_K$. Consequently $1 + x \mapsto 1 + T(x)$ is a well-defined surjective homomorphism

$$T^* \colon U^{m_K+1}_{\mathfrak{A}_K} / U^{n_K+1}_{\mathfrak{A}_K} \longrightarrow U^{m+1}_{\mathfrak{A}} / U^{n+1}_{\mathfrak{A}} \tag{11.4}$$

This map is a homomorphism because $2(m_K + 1) = n_K + 1$ makes the source abelian and kills the cross terms, and surjective by the displayed equality.

The character $\psi_K = \psi \circ \mathrm{tr}_{K/F}$ of K has level one, by lemma 4.1.2(1). Write $\psi_\alpha(1+x) = \psi(\mathrm{tr}(\alpha x))$ on $U^{m+1}_{\mathfrak{A}}$ and $\psi^K_\alpha(1+y) = \psi_K(\mathrm{tr}(\alpha y))$ on $U^{m_K+1}_{\mathfrak{A}_K}$. Then

$$\psi^K_\alpha = \psi_\alpha \circ T^* \tag{11.5}$$

since tr is applied entrywise and $\psi_K = \psi \circ \mathrm{tr}_{K/F}$.

The next lemma is the bridge that chapter 10 did not write down. Lemma 10.1.4 computes the $\mathcal{K}_{\mathfrak{A}}$-conjugacy class of ψ_α for a *fixed* chain order. What is needed here, and again in theorem 11.1.19, is a comparison of two strata carried by two *different* chain orders, under conjugacy by the whole of GL_2. The step across is proposition 3.3.2.

Lemma 11.1.5: Conjugacy of Stratum Characters Across Chain Orders

Let L be a non-archimedean local field, let $\eta \in \widehat{L}$ have level one, and let $(\mathfrak{A}, n, \alpha)$, $(\mathfrak{B}, n, \beta)$ be ramified simple strata in $M_2(L)$ with $n = 2m+1 \geq 1$. The following are equivalent.

1. There is $g \in \mathrm{GL}_2(L)$ with $\mathfrak{B} = g^{-1}\mathfrak{A}g$ and $\eta_\beta(u) = \eta_\alpha(gug^{-1})$ for all $u \in U_{\mathfrak{B}}^{m+1}$.
2. $\det\beta \in \det\alpha\, U_L^{\lfloor m/2\rfloor+1}$ and $\mathrm{tr}\beta \in \mathrm{tr}\alpha + \mathfrak{p}_L^{-\lfloor m/2\rfloor}$.

***Proof*:**
Both chain orders have $e = 2$, so proposition 3.3.2 provides $h \in \mathrm{GL}_2(L)$ with $\mathfrak{B} = h^{-1}\mathfrak{A}h$. Put $\beta' = h\beta h^{-1}$. Then $(\mathfrak{A}, n, \beta')$ is a ramified simple stratum, $\mathrm{tr}\beta' = \mathrm{tr}\beta$ and $\det\beta' = \det\beta$, so condition (2) is unchanged on replacing β by β'; and $\eta_{\beta'}(1+hxh^{-1}) = \eta(\mathrm{tr}(\beta' hxh^{-1})) = \eta(\mathrm{tr}(\beta x)) = \eta_\beta(1+x)$ with $hU_{\mathfrak{B}}^{m+1}h^{-1} = U_{\mathfrak{A}}^{m+1}$, so condition (1) is unchanged as well. We may therefore assume $\mathfrak{B} = \mathfrak{A}$, and then any g as in (1) satisfies $g\mathfrak{A}g^{-1} = \mathfrak{A}$, that is $g \in \mathcal{K}_{\mathfrak{A}}$.

Two steps remain, and it is the first that carries the argument from characters to cosets. For α', α'' with $(\mathfrak{A}, n, \alpha')$, $(\mathfrak{A}, n, \alpha'')$ ramified simple,

$$\eta_{\alpha'} = \eta_{\alpha''} \text{ on } U_{\mathfrak{A}}^{m+1} \iff \alpha' - \alpha'' \in \mathfrak{P}^{-m} \tag{11.6}$$

Indeed $\eta_{\alpha'}(1+x) = \eta(\mathrm{tr}(\alpha' x))$ for $x \in \mathfrak{P}^{m+1}$, so the two agree exactly when η kills $\mathrm{tr}\big((\alpha' - \alpha'')\mathfrak{P}^{m+1}\big)$. As η has level one its kernel is $\mathfrak{p}_L$, so this says that $\alpha' - \alpha''$ lies in the dual of $\mathfrak{P}^{m+1}$ under the pairing $(a, x) \mapsto \eta(\mathrm{tr}(ax))$, and that dual is $\mathfrak{P}^{-m}$.

Now $\eta_\alpha(gug^{-1}) = \eta(\mathrm{tr}(\alpha gug^{-1})) = \eta_{g^{-1}\alpha g}(u)$, so (1) says $\eta_\beta = \eta_{g^{-1}\alpha g}$ on $U_{\mathfrak{A}}^{m+1}$, which by (11.6) is $\beta + \mathfrak{P}^{-m} = g^{-1}(\alpha + \mathfrak{P}^{-m})g$. That is exactly $\mathcal{K}_{\mathfrak{A}}$-conjugacy of the two cosets, and lemma 10.1.4 says that this holds if and only if (2) does. That result is stated over the book's base field, but its statement names only tr, det, $\mathfrak{P}$ and the unit filtration and no additive character at all, so it applies verbatim over L. This completes the proof.

The bridge decides conjugacy by two congruences, on the determinant and on the trace, and those congruences are conditions on elements of the base field. Lifting changes the field in which they are read but not the elements, and the next lemma is the resulting comparison: over F and over K the two conditions say the same thing.

Lemma 11.1.6: Lifting Preserves the Conjugacy of Stratum Characters

Let $(\mathfrak{A}, n, \alpha)$ and $(\mathfrak{B}, n, \beta)$ be ramified simple strata in $M_2(F)$, $n = 2m+1 \geq 1$, and let K/F be finite tamely ramified. Write $(\mathfrak{A}_K, n_K, \alpha)$ and $(\mathfrak{B}_K, n_K, \beta)$ for their lifts. Then $\psi_\alpha\big|U_{\mathfrak{A}}^{m+1}$ and $\psi_\beta\big|U_{\mathfrak{B}}^{m+1}$ are conjugate in $G = \mathrm{GL}_2(F)$, in the sense of lemma 11.1.5(1), if and only if $\psi_\alpha^K\big|U_{\mathfrak{A}_K}^{m_K+1}$ and $\psi_\beta^K\big|U_{\mathfrak{B}_K}^{m_K+1}$ are conjugate in $G_K = \mathrm{GL}_2(K)$.

***Proof*:**
By lemma 11.1.5, applied over F with $\eta = \psi$ and over K with $\eta = \psi_K$, each side is equivalent to a pair of congruences, and the two pairs read

$$\det\beta \in \det\alpha\, U_F^{\lfloor m/2\rfloor+1}, \quad \mathrm{tr}\beta \in \mathrm{tr}\alpha + \mathfrak{p}^{-\lfloor m/2\rfloor} \qquad \text{and} \qquad \det\beta \in \det\alpha\, U_K^{\lfloor m_K/2\rfloor+1}, \quad \mathrm{tr}\beta \in \mathrm{tr}\alpha + \mathfrak{p}_K^{-\lfloor m_K/2\rfloor}$$

All four elements $\det\alpha, \det\beta, \mathrm{tr}\alpha, \mathrm{tr}\beta$ lie in F, so it suffices to prove

$$U_K^{\lfloor m_K/2\rfloor+1} \cap F = U_F^{\lfloor m/2\rfloor+1}, \qquad \mathfrak{p}_K^{-\lfloor m_K/2\rfloor} \cap F = \mathfrak{p}^{-\lfloor m/2\rfloor}$$

Write $e = e_K$, so $m_K = (en-1)/2$ and $m = (n-1)/2$, and put $A = \lfloor m_K/2\rfloor = \lfloor (en-1)/4\rfloor$, $B = \lfloor m/2\rfloor = \lfloor (n-1)/4\rfloor$.

$\lfloor A/e\rfloor = B$. By (11.1), $\lfloor A/e\rfloor = \lfloor (en-1)/(4e)\rfloor$. Write $n = 4c+r$ with $r \in \{1,3\}$, both possible values being odd. If $r = 1$ then $(en-1)/(4e) = c + \frac{1}{4} - \frac{1}{4e}$ and $0 \le \frac{1}{4} - \frac{1}{4e} < 1$, so the floor is $c = B$. If $r = 3$ then $(en-1)/(4e) = c + \frac{3}{4} - \frac{1}{4e}$ and $0 < \frac{3}{4} - \frac{1}{4e} < 1$, so the floor is again c, while $B = \lfloor (4c+2)/4\rfloor = c$.

The two intersections. For $z \in F$, $z \in \mathfrak{p}_K^a$ if and only if $v_F(z) \ge \lceil a/e\rceil$, so $\mathfrak{p}_K^{-A} \cap F = \mathfrak{p}^{\lceil -A/e\rceil} = \mathfrak{p}^{-\lfloor A/e\rfloor} = \mathfrak{p}^{-B}$, which is the second identity. For the first, $u \in F$ lies in U_K^{A+1} if and only if $u - 1 \in \mathfrak{p}_K^{A+1} \cap F = \mathfrak{p}^{\lceil (A+1)/e\rceil}$. Write $A = Be + r$ with $0 \le r \le e-1$, possible by the previous paragraph. Then $\lceil (A+1)/e\rceil = B + \lceil (r+1)/e\rceil = B+1$ because $1 \le r+1 \le e$. Hence $U_K^{A+1} \cap F = U_F^{B+1}$, completing the proof.

The lifted type carries a sign, and the sign is attached to the extension rather than to the stratum. It is the value at a prime element of the determinant character of definition 6.2.5, and the point of the next lemma is that for tame K/F this value is well defined at all.

Lemma 11.1.7: The Determinant Character of a Tame Extension Is Unramified

Let K/F be a finite tamely ramified extension and let $\varkappa_{K/F} = \det R_{K/F}$ be the character of $F^\times$ of definition 6.2.5.

1. $\varkappa_{K/F}$ is unramified, and $\varepsilon_{K/F} := \varkappa_{K/F}(\varpi) \in \{\pm 1\}$ does not depend on the choice of prime element ϖ of F.
2. If L/K is a further finite tamely ramified extension, then $\varepsilon_{L/F} = \varepsilon_{K/F}^{[L:K]}\, \varepsilon_{L/K}$.

***Proof*:**
Part (1). Regard $\varkappa_{K/F}$ as a character of $\mathcal{W}_F$ through theorem 6.2.1. The representation $R_{K/F} = \mathrm{Ind}_{K/F} 1_K$ is the permutation representation of $\mathcal{W}_F$ on $\mathcal{W}_F/\mathcal{W}_K$, so $\varkappa_{K/F}(g)$ is the sign of the permutation induced by g, and $\varkappa_{K/F}^2 = 1$. Both are proposition 6.2.6. Since K/F is tamely ramified it lies in the maximal tamely ramified extension E_∞ of F, so $\mathcal{P}_F = \mathrm{Gal}(\overline{F}/E_\infty) \subseteq \mathrm{Gal}(\overline{F}/K) = \Omega_K$ straight from the definition of the wild inertia group; and $\mathcal{P}_F \subseteq \mathcal{I}_F \subseteq \mathcal{W}_F$, so $\mathcal{P}_F \subseteq \mathcal{W}_F \cap \Omega_K = \mathcal{W}_K$. That group is also normal in $\mathcal{W}_F$, because E_∞/F is Galois. For $x\mathcal{W}_K \in \mathcal{W}_F/\mathcal{W}_K$ and $w \in \mathcal{P}_F$ we then have $w\,x\mathcal{W}_K = x(x^{-1}wx)\mathcal{W}_K = x\mathcal{W}_K$, so $\mathcal{P}_F$ acts trivially and $\varkappa_{K/F}$ is trivial on $\mathcal{P}_F$. Through class field theory this says that $\varkappa_{K/F}$ is trivial

on U_F^1. Its restriction to U_F therefore factors through $U_F/U_F^1 \cong k^\times$, a group of odd order $q-1 = 2^f - 1$. A character of order dividing 2 on a group of odd order is trivial, so $\varkappa_{K/F}$ is trivial on U_F, that is, unramified. Any two prime elements differ by a unit, so $\varepsilon_{K/F} = \varkappa_{K/F}(\varpi)$ is well defined, and it is a square root of 1.

Part (2). Write $W = \mathcal{W}_F$, $M = \mathcal{W}_K$, $H = \mathcal{W}_L$, $X = W/H$, $Y = W/M$, and let $p\colon X \to Y$ be the projection, whose fibres have $c = [L:K]$ elements. Fix $g \in W$ and decompose Y into $\langle g\rangle$-orbits. Let O be such an orbit, of length ℓ, with base point $y_0 = w_0 M$. Then g^ℓ fixes y_0, so $h_O := w_0^{-1} g^\ell w_0 \in M$, and g permutes the ℓc points of $p^{-1}(O)$ in the following way: each cycle of length k of h_O acting on $M/H \cong p^{-1}(y_0)$, the identification being $mH \mapsto w_0 mH$, produces one cycle of length $k\ell$. Indeed g carries $p^{-1}(g^i y_0)$ bijectively onto $p^{-1}(g^{i+1}y_0)$, so a g-orbit inside $p^{-1}(O)$ meets $p^{-1}(y_0)$ in a single orbit of g^ℓ and is ℓ times as long as that orbit; and on $p^{-1}(y_0)$ the map g^ℓ reads $g^\ell w_0 mH = w_0 h_O mH$, which is left translation by h_O on M/H. If h_O has $c(h_O)$ cycles on the c points, the sign of g on $p^{-1}(O)$ is therefore $(-1)^{c\ell - c(h_O)} = \big((-1)^{\ell-1}\big)^c (-1)^{c - c(h_O)}$, that is, the sign of g on O raised to the power c, times the sign of h_O on M/H. Multiplying over the orbits,

$$\varkappa_{L/F}(g) \;=\; \varkappa_{K/F}(g)^{\,c} \prod_{O} \varkappa_{L/K}(h_O)$$

Now take g with $d_F(g) = 1$, where $d_F\colon \mathcal{W}_F \to \mathbb{Z}$ is the map recording the power of Frobenius. Under class field theory g corresponds to a prime element of F, so $\varkappa_{L/F}(g) = \varepsilon_{L/F}$ and $\varkappa_{K/F}(g) = \varepsilon_{K/F}$. On $\mathcal{W}_K$ one has $d_F = f_K\, d_K$ with $f_K = f(K|F)$, so $d_K(h_O) = \ell/f_K$. In particular $f_K \big| \ell$, since $h_O \in M$. Because L/K is tamely ramified, part (1) applies to it and $\varkappa_{L/K}$ is unramified, hence $\varkappa_{L/K}(h_O) = \varepsilon_{L/K}^{\ell/f_K}$. Summing the orbit lengths,

$$\prod_{O} \varkappa_{L/K}(h_O) = \varepsilon_{L/K}^{|Y|/f_K} = \varepsilon_{L/K}^{[K:F]/f_K} = \varepsilon_{L/K}^{e_K} = \varepsilon_{L/K}$$

the last step because e_K is odd and $\varepsilon_{L/K} = \pm 1$. This gives $\varepsilon_{L/F} = \varepsilon_{K/F}^{c}\varepsilon_{L/K}$, completing the proof.

The other input to the construction is a surjectivity statement for norms. Lemma 4.1.2(2) records the graded form of it. What follows is the passage from the graded form to the assertion about the groups themselves. Clause (2) is used at full strength: theorem 11.1.19 needs the norm of a unit group high in the filtration of EK, which the case $j = 1$ does not give.

Lemma 11.1.8: Norms Are Surjective on the Unit Filtration in a Tame Extension

Let L/M be a finite tamely ramified extension of non-archimedean local fields, and let $e = e(L|M)$.

1. $\mathrm{Nm}_{L/M}\colon U_L^1 \to U_M^1$ is surjective.
2. More precisely $\mathrm{Nm}_{L/M}\big(U_L^j\big) = U_M^{\lceil j/e \rceil}$ for every $j \geq 1$.

***Proof*:**

The inclusion $\subseteq$. Let $u = 1 + z$ with $z \in \mathfrak{p}_L^j$, $j \geq 1$, and let w be the unique extension of v_M to an algebraic closure $\overline{M}$, so that $w|_L = v_L/e$ and $w(z) \geq j/e$. Writing m_z for multiplication by z on L regarded as an M-vector space and $d = [L : M]$,

$$\mathrm{Nm}_{L/M}(u) = \det(1 + m_z) = 1 + \sum_{i=1}^{d} c_i$$

where $c_i \in M$ is the i-th elementary symmetric function of the roots of the characteristic polynomial of m_z. That polynomial is a power of the minimal polynomial of z over M, whose roots form a single orbit under $\mathrm{Aut}(\overline{M}/M)$; and w, being the unique extension of v_M, is $\mathrm{Aut}(\overline{M}/M)$-invariant. So every root λ has $w(\lambda) = w(z) \geq j/e$, whence $w(c_i) \geq ij/e \geq j/e$. As $c_i \in M$ the value $v_M(c_i) = w(c_i)$ is an integer, so $v_M(c_i) \geq \lceil j/e \rceil$ and $\mathrm{Nm}_{L/M}(U_L^j) \subseteq U_M^{\lceil j/e \rceil}$.

Reduction for the reverse inclusion. Let M_0 be the maximal unramified sub-extension of L/M, so that L/M_0 is totally tamely ramified of degree e, the extension M_0/M is unramified, and $\mathrm{Nm}_{L/M} = \mathrm{Nm}_{M_0/M} \circ \mathrm{Nm}_{L/M_0}$. It suffices to prove $\mathrm{Nm}_{L/M_0}(U_L^j) \supseteq U_{M_0}^{\lceil j/e \rceil}$ and $\mathrm{Nm}_{M_0/M}(U_{M_0}^i) \supseteq U_M^i$ for all $j, i \geq 1$.

L/M_0 totally tamely ramified of degree e. Then $\gcd(e, p) = 1$. Put $i = \lceil j/e \rceil$, so $ei \geq j$. Since $v_L(x - 1) = e\, v_{M_0}(x - 1)$ for $x \in M_0$, we have $U_{M_0}^i \subseteq U_L^{ei} \subseteq U_L^j$, and for such x, $\mathrm{Nm}_{L/M_0}(x) = x^e$. The group $U_{M_0}^i$ is pro-p, so raising to the e-th power is a bijection of it. Hence $\mathrm{Nm}_{L/M_0}(U_L^j) \supseteq (U_{M_0}^i)^e = U_{M_0}^i$.

M_0/M unramified of degree f. Let $i \geq 1$ and $u \in U_M^i$. We construct $x_r \in U_{M_0}^i$ with $\mathrm{Nm}_{M_0/M}(x_r) \equiv u$ modulo U_M^r for $r \geq i$, starting from $x_i = 1$. Given x_r, write $u\,\mathrm{Nm}_{M_0/M}(x_r)^{-1} = 1 + c$ with $c \in \mathfrak{p}_M^r$. The map induced by $\mathrm{tr}_{M_0/M}$ on $\mathfrak{p}_{M_0}^r/\mathfrak{p}_{M_0}^{r+1} \to \mathfrak{p}_M^r/\mathfrak{p}_M^{r+1}$ is, after dividing by ϖ_M^r, the trace $\mathrm{tr}_{k_{M_0}/k_M}$, which is surjective because k_{M_0}/k_M is separable. Choose $b \in \mathfrak{p}_{M_0}^r$ with $\mathrm{tr}_{M_0/M}(b) \equiv c$ modulo $\mathfrak{p}_M^{r+1}$, and set $x_{r+1} = x_r(1 + b)$. Since $\mathrm{Nm}_{M_0/M}(1 + b) = \prod_\sigma (1 + \sigma b) \equiv 1 + \mathrm{tr}_{M_0/M}(b)$ modulo $\mathfrak{p}_M^{2r} \subseteq \mathfrak{p}_M^{r+1}$, we get $\mathrm{Nm}_{M_0/M}(x_{r+1}) \equiv u$ modulo U_M^{r+1}. The sequence (x_r) is Cauchy, because $x_{r+1}x_r^{-1} \in U_{M_0}^r$, so it converges to some $x \in U_{M_0}^i$. The norm being continuous, $\mathrm{Nm}_{M_0/M}(x) = u$.

Combining the four paragraphs (the last three give $\mathrm{Nm}_{L/M}(U_L^j) = \mathrm{Nm}_{M_0/M}\big(\mathrm{Nm}_{L/M_0}(U_L^j)\big) \supseteq U_M^{\lceil j/e \rceil}$, and the first the opposite inclusion) yields $\mathrm{Nm}_{L/M}(U_L^j) = U_M^{\lceil j/e \rceil}$ for every $j \geq 1$, which is (2); and (1) is the case $j = 1$, since $\lceil 1/e \rceil = 1$, completing the proof.

We come to the central construction. Recall that the group $J = E^\times U_{\mathfrak{A}}^{m+1}$ attached to a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ is the set of elements of G intertwining $\psi_\alpha \big| U_{\mathfrak{A}}^{m+1}$, by theorem 3.5.3(1), and that $\mathcal{C}(\psi_\alpha, \mathfrak{A})$ is the set of characters of J restricting to ψ_α on $U_{\mathfrak{A}}^{m+1}$, the set $C(\psi_\alpha)$ of proposition 10.1.3. In particular J and $\mathcal{C}(\psi_\alpha, \mathfrak{A})$ depend on α only through the character ψ_α of $U_{\mathfrak{A}}^{m+1}$. Over K we write $J_K = (EK)^\times U_{\mathfrak{A}_K}^{m_K+1}$ and $\mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$ for the corresponding objects.

The local constant enters at one point only, and it is convenient to isolate that point before the theorem it serves.

Lemma 11.1.9: The Local Constant of a Lifted Type

Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum in $M_2(F)$, $n = 2m+1 \geq 1$, let K/F be finite tamely ramified, and let $\Lambda \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$. Suppose $\Theta \in \mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$ satisfies

$$\Theta(y) = \varepsilon_{K/F}^{v_{EK}(y)}\, \Lambda\big(\mathrm{Nm}_{EK/E}(y)\big), \qquad y \in (EK)^\times$$

Put $\pi = \text{c-Ind}_J^G \Lambda$ and $\Pi = \text{c-Ind}_{J_K}^{G_K} \Theta$. Then

$$\varepsilon(\Pi, s, \psi_K) = \varepsilon_{K/F}\, \varepsilon(\pi, s, \psi)^{[K:F]} \tag{11.7}$$

***Proof*:**
Both π and Π are cuspidal, attached to the ramified simple strata $(\mathfrak{A}, n, \alpha)$ and $(\mathfrak{A}_K, n_K, \alpha)$ of odd levels n and n_K, with one-dimensional inducing characters, and ψ, ψ_K have level one. Theorem 5.3.9 therefore applies to each and gives

$$\varepsilon(\pi, s, \psi) = q^{\,n(1/2-s)}\, \Lambda(\alpha)^{-1} \psi(\mathrm{tr}\alpha), \qquad \varepsilon(\Pi, s, \psi_K) = q_K^{\,n_K(1/2-s)}\, \Theta(\alpha)^{-1} \psi_K(\mathrm{tr}\alpha)$$

Now $q_K = q^{f_K}$ and $n_K = e_K n$, so $q_K^{\,n_K(1/2-s)} = \big(q^{\,n(1/2-s)}\big)^{[K:F]}$. Next $\mathrm{tr}\alpha \in F$, so $\mathrm{tr}_{K/F}(\mathrm{tr}\alpha) = [K:F]\,\mathrm{tr}\alpha$ and $\psi_K(\mathrm{tr}\alpha) = \psi(\mathrm{tr}\alpha)^{[K:F]}$. Finally $\alpha \in E^\times$, so $\mathrm{Nm}_{EK/E}(\alpha) = \alpha^{[EK:E]} = \alpha^{[K:F]}$, while $v_{EK}(\alpha) = -e_K n$ is odd, so $\varepsilon_{K/F}^{\,v_{EK}(\alpha)} = \varepsilon_{K/F}$. Hence $\Theta(\alpha)^{-1} = \varepsilon_{K/F}\Lambda(\alpha)^{-[K:F]}$, using $\varepsilon_{K/F} = \pm 1$. Multiplying the three factors gives (11.7), completing the proof.

Theorem 11.1.10: The Lifted Type

Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum in $M_2(F)$ with $n = 2m+1 \geq 1$, let $E = F[\alpha]$ and $J = E^\times U_{\mathfrak{A}}^{m+1}$, and let K/F be a finite tamely ramified extension, with $\mathfrak{A}_K$, $n_K = 2m_K + 1$, $EK = K[\alpha]$ and $J_K = (EK)^\times U_{\mathfrak{A}_K}^{m_K+1}$ as above. For $\Lambda \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$ there is exactly one character

$$\Lambda_K = \mathrm{Lft}_{K/F}^{\alpha}(\Lambda) \in \mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$$

such that

$$\Lambda_K(y) = \varepsilon_{K/F}^{\,v_{EK}(y)}\, \Lambda\big(\mathrm{Nm}_{EK/E}(y)\big), \qquad y \in (EK)^\times \tag{11.8}$$

Moreover:

1. For $\Lambda, \Lambda' \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$ one has $\Lambda'_K = \Lambda_K$ if and only if $\Lambda' = (\chi \circ \det)\Lambda$ for a character χ of $F^\times$ trivial on $\mathrm{Nm}_{K/F}(K^\times)$.
2. If L/K is a further finite tamely ramified extension, then $\mathrm{Lft}_{L/F}^{\alpha} = \mathrm{Lft}_{L/K}^{\alpha} \circ \mathrm{Lft}_{K/F}^{\alpha}$.
3. If $(\mathfrak{A}, n, \beta)$ is a ramified simple stratum with $\psi_\beta = \psi_\alpha$ on $U_{\mathfrak{A}}^{m+1}$, then $\mathcal{C}(\psi_\beta, \mathfrak{A}) = \mathcal{C}(\psi_\alpha, \mathfrak{A})$, the lift of $(\mathfrak{A}, n, \beta)$ is carried by the same chain order $\mathfrak{A}_K$, $\mathcal{C}(\psi_\beta^K, \mathfrak{A}_K) = \mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$ and $\mathrm{Lft}_{K/F}^{\beta} = \mathrm{Lft}_{K/F}^{\alpha}$.
4. On the centre, $\Lambda_K \big| K^\times = (\Lambda \big| F^\times) \circ \mathrm{Nm}_{K/F}$.

The map $\mathrm{Lft}^{\alpha}_{K/F}$ therefore depends only on the character ψ_α of $U^{m+1}_{\mathfrak{A}}$, and clause (3) is what justifies the notation $\mathrm{Lft}_{K/F}$ without a superscript from here on. Clause (3) is the only one whose proof passes outside the theory of types, through the local constant of lemma 11.1.9, and it is proved last.

***Proof*:**

Existence and uniqueness. The chain order $\mathfrak{A}_K$ is the one attached to the lattice chain $(\mathfrak{p}^i_{EK})$ by proposition 11.1.4(1), so $\mathfrak{P}^j_K \cap EK$ is the set of $z \in EK$ with $v_{EK}(z) \geq j$, and therefore $(EK)^\times \cap U^j_{\mathfrak{A}_K} = U^j_{EK}$ for every $j \geq 1$, and likewise over F. Total ramification of EK/K is what makes that chain the right one, but it is the description of $\mathfrak{P}_K$ that gives the intersection. So $(EK)^\times \cap U^{m_K+1}_{\mathfrak{A}_K} = U^{m_K+1}_{EK}$. As J_K is the product of $(EK)^\times$ and $U^{m_K+1}_{\mathfrak{A}_K}$, a character of J_K is determined by its restrictions to the two factors. It can also be assembled from them whenever they agree on the intersection. For $(EK)^\times \subseteq \mathcal{K}_{\mathfrak{A}_K}$ by proposition 3.4.3(2) applied to the ramified simple stratum $(\mathfrak{A}_K, n_K, \alpha)$ over K, so conjugation by $w \in (EK)^\times$ preserves $U^{m_K+1}_{\mathfrak{A}_K}$. It also fixes ψ^K_α, since w commutes with α and so $\mathrm{tr}(\alpha\, w^{-1}yw) = \mathrm{tr}(w\alpha w^{-1}y) = \mathrm{tr}(\alpha y)$ for $1 + y \in U^{m_K+1}_{\mathfrak{A}_K}$; given a character Ξ of $(EK)^\times$ agreeing with ψ^K_α on the intersection, the function $wu \mapsto \Xi(w)\,\psi^K_\alpha(u)$ on J_K is therefore well defined, two decompositions of the same element differing by a member of the intersection, and it is multiplicative because $(w_1u_1)(w_2u_2) = w_1w_2\,(w_2^{-1}u_1w_2)\,u_2$. The right-hand side of (11.8) is a character of $(EK)^\times$, being a product of a homomorphism to $\{\pm1\}$ and a composite of homomorphisms. It remains to check that on $U^{m_K+1}_{EK}$ it agrees with ψ^K_α.

Identify $M_2(K) = M_2(F) \otimes_F K$. Then $T = \mathrm{id} \otimes \mathrm{tr}_{K/F}$, so on $EK = E \otimes_F K$ the map T restricts to $\mathrm{tr}_{EK/E}$. Let $u = 1 + z$ with $z \in \mathfrak{p}^{m_K+1}_{EK}$. Then

$$\mathrm{Nm}_{EK/E}(u) \in \big(1 + \mathrm{tr}_{EK/E}(z)\big)\, U^{n+1}_E \tag{11.9}$$

because the remaining terms of the product $\prod(1+\sigma z)$ lie in $\mathfrak{p}^{2m_K+2}_{EK} \cap E = \mathfrak{p}^{n_K+1}_{EK} \cap E = \mathfrak{p}^{n+1}_E$, the last equality by $e(EK|E) = e_K$ and $\lceil (e_K n + 1)/e_K \rceil = n + 1$. Since Λ restricts to ψ_α on $U^{m+1}_{\mathfrak{A}}$, and ψ_α is trivial on $U^{n+1}_{\mathfrak{A}} \supseteq U^{n+1}_E$, we get $\Lambda(\mathrm{Nm}_{EK/E}(u)) = \psi_\alpha(1 + T(z)) = \psi^K_\alpha(u)$ by (11.5); and $v_{EK}(u) = 0$, so the sign contributes nothing. Thus Λ_K exists, and it is unique because (11.8) pins it on $(EK)^\times$ while membership of $\mathcal{C}(\psi^K_\alpha, \mathfrak{A}_K)$ pins it on $U^{m_K+1}_{\mathfrak{A}_K}$.

Part (1). Two elements of $\mathcal{C}(\psi_\alpha, \mathfrak{A})$ agree on $U^{m+1}_{\mathfrak{A}}$, so $\theta = \Lambda'/\Lambda$ is a character of $J/U^{m+1}_{\mathfrak{A}}$, determined by its restriction to $E^\times$. Suppose $\Lambda'_K = \Lambda_K$. By (11.8), θ is trivial on $\mathrm{Nm}_{EK/E}\big((EK)^\times\big)$. The extension EK/E is tamely ramified, of degree $[K:F]$ and ramification index e_K prime to p, so lemma 11.1.8(1) gives $\mathrm{Nm}_{EK/E}(U^1_{EK}) = U^1_E$ and θ is trivial on U^1_E. Since E/F is totally ramified, $E^\times/U^1_E \cong \varpi_E^{\mathbb{Z}} \times \mu_{q-1}$ and $\mathrm{Nm}_{E/F}$ induces on this quotient the map sending ϖ_E to an element of valuation 1 and acting on μ_{q-1} by $u \mapsto u^2$, a bijection because $q-1$ is odd. Hence $\mathrm{Nm}_{E/F}$ induces an isomorphism $E^\times/U^1_E \to F^\times/U^1_F$, and $\theta = \chi \circ \mathrm{Nm}_{E/F}$ for a unique character χ of $F^\times$ trivial on U^1_F. As $\det U^{m+1}_{\mathfrak{A}} \subseteq U^1_F$, the character $\chi \circ \det$ is trivial on $U^{m+1}_{\mathfrak{A}}$ and restricts to $\chi \circ \mathrm{Nm}_{E/F}$ on $E^\times$, so $\Lambda' = (\chi \circ \det)\Lambda$.

It remains to identify the condition on χ. Write $\chi_K = \chi \circ \mathrm{Nm}_{K/F}$. Triviality of θ on $\mathrm{Nm}_{EK/E}((EK)^\times)$ says that $\chi \circ \mathrm{Nm}_{EK/F} = \chi_K \circ \mathrm{Nm}_{EK/K}$ is the trivial character of $(EK)^\times$. Since $[EK:K] = 2$, the norm $\mathrm{Nm}_{EK/K}$ restricts on $K^\times$ to $z \mapsto z^2$, so $\chi_K^2 = 1$. Also χ_K is trivial on U^1_K, because $\mathrm{Nm}_{K/F}(U^1_K) = U^1_F$ by lemma 11.1.8(1) and χ is trivial on U^1_F. Its restriction

to $U_K/U_K^1 \cong k_K^\times$ is then a character of order dividing 2 of a group of odd order, hence trivial. So χ_K is unramified. Let ϖ_{EK} be a prime element of EK. Since EK/K is totally ramified, $f(EK|K) = 1$ and $v_K\big(\mathrm{Nm}_{EK/K}(\varpi_{EK})\big) = f(EK|K)\, v_{EK}(\varpi_{EK}) = 1$, so $\mathrm{Nm}_{EK/K}(\varpi_{EK})$ is a prime element of K, and χ_K takes the value 1 on it. An unramified character trivial at one prime element is trivial on $K^\times = \varpi_K^{\mathbb{Z}} U_K$. Therefore $\chi_K = 1$, that is, χ is trivial on $\mathrm{Nm}_{K/F}(K^\times)$. The argument stays on the K-side deliberately: when E/F is purely inseparable (possible in characteristic 2, and allowed for in the proof of lemma 11.1.2) the extension EK/K is purely inseparable as well, so $\mathcal{W}_{EK}$, $\mathrm{Gal}(EK/K)$ and the quadratic class-field character of EK/K do not exist and class field theory over EK is unavailable. Conversely, if χ is trivial on $\mathrm{Nm}_{K/F}(K^\times)$ then $\chi \circ \mathrm{Nm}_{EK/F} = \chi_K \circ \mathrm{Nm}_{EK/K}$ is trivial, so Λ and $(\chi \circ \det)\Lambda$ have the same lift.

Part (2). For $y \in (EL)^\times$, using (11.8) twice,

$$\mathrm{Lft}^\alpha_{L/K}\big(\Lambda_K\big)(y) = \varepsilon_{L/K}^{\,v_{EL}(y)}\, \varepsilon_{K/F}^{\,v_{EK}(\mathrm{Nm}_{EL/EK}(y))}\, \Lambda\big(\mathrm{Nm}_{EL/E}(y)\big)$$

and $v_{EK}(\mathrm{Nm}_{EL/EK}(y)) = f(EL|EK)\, v_{EL}(y)$. Since E/F is totally ramified the residue fields of EK and EL are those of K and L, so $f(EL|EK) = f(L|K)$, and $[L:K] = e(L|K)f(L|K)$ with $e(L|K)$ odd. Hence $f(EL|EK) \equiv [L:K]$ modulo 2 and, the signs being square roots of 1, $\varepsilon_{K/F}^{f(EL|EK)} = \varepsilon_{K/F}^{[L:K]}$. By lemma 11.1.7(2) the total sign is $\varepsilon_{L/K}\varepsilon_{K/F}^{[L:K]} = \varepsilon_{L/F}$, so $\mathrm{Lft}^\alpha_{L/K}(\Lambda_K)$ satisfies the defining formula (11.8) for L/F. It also lies in $\mathcal{C}(\psi_\alpha^L, \mathfrak{A}_L)$, so by uniqueness it equals $\mathrm{Lft}^\alpha_{L/F}(\Lambda)$.

Part (4). For $y \in K^\times$ the valuation $v_{EK}(y) = 2v_K(y)$ is even, so the sign in (11.8) contributes nothing; and $\mathrm{Nm}_{EK/E}(y) = \mathrm{Nm}_{K/F}(y)$ by base change of the multiplication matrix. So (11.8) reads $\Lambda_K(y) = \Lambda\big(\mathrm{Nm}_{K/F}(y)\big)$ on $K^\times$.

Part (3), first assertions. If $\psi_\beta = \psi_\alpha$ on $U_{\mathfrak{A}}^{m+1}$ then the groups J and the sets $\mathcal{C}$ attached to α and β coincide, both depending on α only through ψ_α. Also $\beta \in \mathcal{K}_{\mathfrak{A}} \subseteq \mathcal{K}_{\mathfrak{A}_K}$ by proposition 11.1.4(3), so the chain order attached to β over K is again $\mathfrak{A}_K$, by the uniqueness in proposition 11.1.4(1); and $\psi_\beta^K = \psi_\beta \circ T^* = \psi_\alpha \circ T^* = \psi_\alpha^K$ by (11.5), so J_K and $\mathcal{C}(\psi_\beta^K, \mathfrak{A}_K) = \mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$ also coincide.

Part (3) for cyclic K/F. Let $\Gamma = \mathrm{Gal}(K/F)$ be cyclic of order d with generator γ, acting on $M_2(K)$ coefficient by coefficient. Then $M_2(F)$ is the fixed set, $\mathfrak{A}_K$ is Γ-stable by the uniqueness in proposition 11.1.4(1), and J_K is Γ-stable. Form the semi-direct product $\Gamma \ltimes J_K$ and put $N_\gamma(x) = (\gamma x)^d$ for $x \in J_K$. This lies in J_K, and explicitly $N_\gamma(x) = \gamma^{d-1}(x) \cdots \gamma(x)\, x$.

For $x = 1 + a$ with $a \in \mathfrak{P}_K^{m_K+1}$ we get, modulo $\mathfrak{P}_K^{2m_K+2} = \mathfrak{P}_K^{n_K+1}$,

$$N_\gamma(x) \equiv 1 + \sum_{j=0}^{d-1} \gamma^j(a) = 1 + T(a)$$

so N_γ carries $U_{\mathfrak{A}_K}^{m_K+1}$ into $U_{\mathfrak{A}}^{m+1} U_{\mathfrak{A}_K}^{n_K+1}$ and induces, by (11.4), a surjective homomorphism $U_{\mathfrak{A}_K}^{m_K+1}/U_{\mathfrak{A}_K}^{n_K+1} \to U_{\mathfrak{A}}^{m+1}/U_{\mathfrak{A}}^{n+1}$ which agrees with T^*. By (11.5) it satisfies $\psi_\alpha^K(x) = \psi_\alpha(N_\gamma(x))$. Writing $X = \ker \psi_\alpha$ and $X_K = \ker \psi_\alpha^K$, the map N_γ therefore induces an isomorphism $U_{\mathfrak{A}_K}^{m_K+1}/X_K \to U_{\mathfrak{A}}^{m+1}/X$.

Let now β be as in (3). Since $J_K = K[\beta]^\times U_{\mathfrak{A}_K}^{m_K+1}$, write $x = zu$ with $z \in K[\beta]^\times$ and $u \in U_{\mathfrak{A}_K}^{m_K+1}$.

Both γ and conjugation by elements of $K[\beta]^\times$ fix ψ_α^K, hence fix X_K, so

$$N_\gamma(x) \equiv \mathrm{Nm}_{K[\beta]/F[\beta]}(z)\, N_\gamma(u) \pmod{X_K}$$

and N_γ induces a homomorphism $J_K/X_K \to J/X$ which does not refer to β at all. Its definition $N_\gamma(x) = (\gamma x)^d$, taken in $\Gamma \ltimes J_K$, names only γ and d; and its source and target were pinned without β as well, since J_K, J, $X_K = \ker \psi_\alpha^K$ and $X = \ker \psi_\alpha$ are the same for β as for α by the first assertions of (3).

Write $\overline{N}_\gamma \colon J_K/X_K \to J/X$ for that induced homomorphism. Three things make it well posed, and they are worth separating because the composite below is otherwise ill-typed: $N_\gamma(y)$ lies in J_K, not in J, so Λ cannot be applied to it directly. First, the recipe does not depend on the decomposition $x = zu$: if $zu = z'u'$ then $w = z^{-1}z' = u(u')^{-1}$ lies in $K[\beta]^\times \cap U_{\mathfrak{A}_K}^{m_K+1} = U_{K[\beta]}^{m_K+1}$, and there the two clauses agree, since γ restricted to $K[\beta]$ generates $\mathrm{Gal}(K[\beta]|F[\beta])$ and so $N_\gamma(w) = \prod_j \gamma^j(w) = \mathrm{Nm}_{K[\beta]/F[\beta]}(w)$. Second, it kills X_K, by the isomorphism $U_{\mathfrak{A}_K}^{m_K+1}/X_K \to U_{\mathfrak{A}}^{m+1}/X$ just established. Third, it is multiplicative, because conjugation by an element of $K[\beta]^\times$ fixes ψ_α^K and hence preserves X_K, so the two factors of J_K may be commuted past one another modulo X_K.

One caution, since the notation invites the opposite reading: $\overline{N}_\gamma$ does *not* restrict to the identity on J. An element y of J is fixed by γ, so $N_\gamma(y) = y^d$, and the composite below reads $\Lambda(y)^d$ there rather than $\Lambda(y)$. Since Λ is trivial on X it descends to J/X, and the composite

$$y \longmapsto \varepsilon_{K/F}^{v_K(\det y)}\, \Lambda\big(\overline{N}_\gamma(y)\big), \qquad y \in J_K$$

is therefore a character of J_K, and it is independent of β. For $y \in (EK)^\times$ one has $N_\gamma(y) = \mathrm{Nm}_{EK/E}(y)$ and $v_K(\det y) = v_K(\mathrm{Nm}_{EK/K}(y)) = v_{EK}(y)$, so this character satisfies (11.8); and for $y \in U_{\mathfrak{A}_K}^{m_K+1}$ it equals $\psi_\alpha^K(y)$, so it lies in $\mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$. By uniqueness it is $\mathrm{Lft}_{K/F}^\alpha(\Lambda)$, and equally $\mathrm{Lft}_{K/F}^\beta(\Lambda)$. This proves (3) for cyclic K/F.

Part (3) for Galois K/F. A finite tamely ramified Galois extension has metacyclic Galois group: by lemma 9.2.10 the inertia subgroup is cyclic and the quotient is cyclic. So $F \subseteq K_0 \subseteq K$ with K_0/F and K/K_0 cyclic and tamely ramified, and (3) follows from the cyclic case together with the transitivity (2).

Part (3), the general case. Let K/F be tamely ramified, not necessarily Galois. Writing $F \subseteq K_0 \subseteq K$ with K_0/F unramified, hence cyclic, and K/K_0 totally tamely ramified, the transitivity (2) and the case already proved reduce us to the situation where K/F is totally tamely ramified, hence of odd degree.

Set $J^1 = J \cap U_{\mathfrak{A}}^1 = U_E^1 U_{\mathfrak{A}}^{m+1}$, the second equality because $E^\times \cap U_{\mathfrak{A}}^1 = U_E^1$. The group $J \cap U_{\mathfrak{A}} = \mu_{q-1} U_E^1 U_{\mathfrak{A}}^{m+1}$ is strictly larger as soon as $q > 2$, and it is the group cut out by $U_{\mathfrak{A}}^1$ that the argument below needs. Let $\mathcal{C}_1(\psi_\alpha, \mathfrak{A})$ be the set of characters of J^1 restricting to ψ_α on $U_{\mathfrak{A}}^{m+1}$, with the analogous notation over other base fields. Restriction of (11.8) to J_K^1 gives a map $\mathrm{Lft}_{K'/F}^{1,\alpha} \colon \mathcal{C}_1(\psi_\alpha, \mathfrak{A}) \to \mathcal{C}_1(\psi_\alpha^{K'}, \mathfrak{A}_{K'})$ for any tame K'/F, and it is injective: if two elements of $\mathcal{C}_1(\psi_\alpha, \mathfrak{A})$ have the same image, their ratio is trivial on $\mathrm{Nm}_{EK'/E}(U_{EK'}^1) = U_E^1$ by lemma 11.1.8(1), hence trivial on J^1.

Let L/F be the normal closure of K/F. It is tamely ramified. Then $\Lambda_K \big| J_K^1$ is the unique element of $\mathcal{C}_1(\psi_\alpha^K, \mathfrak{A}_K)$ whose image under $\mathrm{Lft}_{L/K}^{1,\alpha}$ is $\mathrm{Lft}_{L/F}^{1,\alpha}(\Lambda \big| J^1)$, by (2) and the injectivity

just proved. For β the same characterization holds with $\mathrm{Lft}^{1,\beta}_{L/K}$ and $\mathrm{Lft}^{1,\beta}_{L/F}$ in place of these, so two things have to be matched, not one: the targets and the maps.

The targets agree because L/F is Galois, where $\mathrm{Lft}^{\alpha}_{L/F}$ is already known to be independent of α. The maps agree for the same reason applied one field up. The extension L/K is Galois, since L/F is. The assertions of (3) already proved make $(\mathfrak{A}_K, n_K, \beta)$ a ramified simple stratum carried by the same chain order $\mathfrak{A}_K$ as $(\mathfrak{A}_K, n_K, \alpha)$, with $\mathcal{C}(\psi^K_\beta, \mathfrak{A}_K) = \mathcal{C}(\psi^K_\alpha, \mathfrak{A}_K)$. That equality of sets says exactly that $\psi^K_\beta = \psi^K_\alpha$ on $U^{m_K+1}_{\mathfrak{A}_K}$, which is the hypothesis of (3) over the base K. The Galois case of (3), applied over K to the extension L/K, therefore gives $\mathrm{Lft}^{\alpha}_{L/K} = \mathrm{Lft}^{\beta}_{L/K}$, hence $\mathrm{Lft}^{1,\alpha}_{L/K} = \mathrm{Lft}^{1,\beta}_{L/K}$ on restriction to J^1_K. This step cannot be skipped: injectivity of one map says nothing about preimages under another, and the two recipes are built over K from the different subfields $K[\alpha]$ and $K[\beta]$ of $M_2(K)$. With one map and one target, its injectivity gives $\Lambda_K\big|J^1_K = \Lambda'_K\big|J^1_K$, so $\Lambda_K\big|J^1_K$ does not depend on α. Neither does $\Lambda_K\big|K^\times$, which equals $(\Lambda\big|F^\times) \circ \mathrm{Nm}_{K/F}$ by part (4).

Write $\Lambda'_K = \mathrm{Lft}^{\beta}_{K/F}(\Lambda)$. The two characters Λ_K and Λ'_K of J_K agree on $J^1_K K^\times$, so their ratio is a character of $J_K/J^1_K K^\times$. Since EK/K is totally ramified, $(EK)^\times$ is generated by a prime element ϖ_{EK}, the roots of unity of order prime to p (which lie in $K^\times$) and $U^1_{EK} \subseteq J^1_K$; and $\varpi^2_{EK} \in K^\times U^1_{EK}$. So $J_K/J^1_K K^\times$ is generated by the class of ϖ_{EK} and has order dividing 2, and the ratio is either trivial or the character $\chi_0 \circ \det$, where χ_0 is the unramified quadratic character of $K^\times$: for $\chi_0 \circ \det$ is trivial on J^1_K, trivial on $K^\times$ because $\det y = y^2$ there, and equal to -1 at ϖ_{EK} because $v_K(\det \varpi_{EK}) = 1$.

Suppose the ratio were $\chi_0 \circ \det$. Put $\pi = \text{c-Ind}^G_J \Lambda$, $\Pi = \text{c-Ind}^{G_K}_{J_K} \Lambda_K$ and $\Pi' = \text{c-Ind}^{G_K}_{J_K} \Lambda'_K = \chi_0 \Pi$, the last equality by proposition 1.2.30. The hypotheses of lemma 11.1.9 hold for α and for β alike: by the first assertions of (3) the two strata carry the same J and the same set $\mathcal{C}(\psi_\alpha, \mathfrak{A}) \ni \Lambda$, hence the same $\pi = \text{c-Ind}^G_J \Lambda$, and also the same $\mathfrak{A}_K$ and $\mathcal{C}(\psi^K_\alpha, \mathfrak{A}_K)$, while Λ_K and Λ'_K satisfy (11.8) for α and for β respectively by construction. Applying the lemma twice gives $\varepsilon(\Pi, s, \psi_K) = \varepsilon(\Pi', s, \psi_K)$. On the other hand $\Pi' = \chi_0 \Pi$ is attached to the same stratum $(\mathfrak{A}_K, n_K, \alpha)$, χ_0 being unramified, so theorem 5.3.9 gives

$$\varepsilon(\Pi', s, \psi_K) = \chi_0(\det \alpha)^{-1} \varepsilon(\Pi, s, \psi_K) = -\varepsilon(\Pi, s, \psi_K)$$

because $v_K(\det \alpha) = v_{EK}(\alpha) = -n_K$ is odd. This is a contradiction, so the ratio is trivial and $\mathrm{Lft}^{\beta}_{K/F} = \mathrm{Lft}^{\alpha}_{K/F}$, completing the proof.

Example 11.1.11: A Level One Stratum and Its Cubic Lift over $\mathbb{Q}_2$

Let $F = \mathbb{Q}_2$ and $E = F(\sqrt{2})$, with $\varpi_E = \sqrt{2}$, whose minimal polynomial is $X^2 - 2$, so that $t = 0$ and $N = -2$ in the coordinates of the proof of proposition 11.1.4. Put $\alpha = \varpi_E/2$, so that $v_E(\alpha) = -1$ and $n = 1$, $m = 0$. In the basis $\{1, \varpi_E\}$,

$$\alpha = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & 0 \end{pmatrix}, \qquad \mathfrak{A} = \begin{pmatrix} \mathfrak{o} & \mathfrak{p} \\ \mathfrak{o} & \mathfrak{o} \end{pmatrix}$$

Let $K = F(2^{1/3})$, totally ramified of degree 3 over F and tamely ramified because $p = 2$. Then $e_K = 3$, $n_K = 3$ and $m_K = 1$, and (11.3) gives

$$\mathfrak{A}_K = \begin{pmatrix} \mathfrak{o}_K & \mathfrak{p}^2_K \\ \mathfrak{p}^{-1}_K & \mathfrak{o}_K \end{pmatrix}, \qquad \mathfrak{P}_K = \begin{pmatrix} \mathfrak{p}_K & \mathfrak{p}^2_K \\ \mathfrak{p}^{-1}_K & \mathfrak{p}_K \end{pmatrix}$$

Intersecting entry by entry with $M_2(F)$ returns $\mathfrak{A}$ and $\mathfrak{P}$, as proposition 11.1.4(2) asserts: $\mathfrak{p}_K^2 \cap F = \mathfrak{p}$ and $\mathfrak{p}_K^{-1} \cap F = \mathfrak{o}$. The type moves from $J = E^\times U_{\mathfrak{A}}^1$ to $J_K = (EK)^\times U_{\mathfrak{A}_K}^2$, with $EK = K(\sqrt{2})$, and on $(EK)^\times$ the lifted character is $\Lambda_K(y) = \varepsilon_{K/F}^{v_{EK}(y)}\, \Lambda(\mathrm{Nm}_{EK/E}(y))$.

The lift was built from a stratum and a character, and the two operations one performs on such data are conjugation in G and twisting by a character of $F^\times$. Both pass through it, the second at the cost of moving α and replacing χ by $\chi \circ \mathrm{Nm}_{K/F}$.

Proposition 11.1.12: Lifting Respects the Relations Between Strata

Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum in $M_2(F)$, $n = 2m+1 \geq 1$, and let K/F be finite tamely ramified.

1. For $g \in G$ and $\Lambda \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$, the stratum $(g^{-1}\mathfrak{A}g, n, g^{-1}\alpha g)$ is ramified simple, its lift is $g^{-1}\mathfrak{A}_K g$, and
$$\mathrm{Lft}_{K/F}^{g^{-1}\alpha g}\big(\Lambda^g\big) = \big(\mathrm{Lft}_{K/F}^{\alpha}(\Lambda)\big)^g$$
where $\Lambda^g(x) = \Lambda(gxg^{-1})$.

2. Let χ be a character of $F^\times$ of level l with $2l < n$, and choose $c \in F$ with $\chi(1+x) = \psi(cx)$ for $x \in \mathfrak{p}^{\lfloor l/2 \rfloor + 1}$, taking $c = 0$ if $l = 0$. Then $(\mathfrak{A}, n, \alpha + c)$ is a ramified simple stratum, $(\chi \circ \det)\Lambda \in \mathcal{C}(\psi_{\alpha+c}, \mathfrak{A})$, and
$$\mathrm{Lft}_{K/F}^{\alpha+c}\big((\chi \circ \det)\Lambda\big) = (\chi_K \circ \det)\, \mathrm{Lft}_{K/F}^{\alpha}(\Lambda), \qquad \chi_K = \chi \circ \mathrm{Nm}_{K/F}$$

***Proof*:**

Part (1). Conjugation by g is an automorphism of $M_2(F)$ fixing F pointwise and extending to $M_2(K)$. It carries chain orders to chain orders, preserves valuations, traces and determinants, and carries $\mathcal{K}_{\mathfrak{A}}$ to $\mathcal{K}_{g^{-1}\mathfrak{A}g}$. So $g^{-1}\mathfrak{A}_K g$ is an $\mathfrak{o}_K$-chain order containing $g^{-1}\alpha g$ in its normalizer, hence is the lift of $g^{-1}\mathfrak{A}g$ by the uniqueness in proposition 11.1.4(1). Since $g^{-1}Eg = F[g^{-1}\alpha g]$ and $g^{-1}(EK)g = K[g^{-1}\alpha g]$, and N and v transport, both sides of the asserted identity are elements of $\mathcal{C}(\psi^K_{g^{-1}\alpha g}, g^{-1}\mathfrak{A}_K g)$ satisfying (11.8), so they agree by uniqueness.

Part (2). If $l = 0$ then $c = 0$ and the assertions are trivial, so assume $l \geq 1$. Then $v_F(c) = -l$: the defining relation $\chi(1+x) = \psi(cx)$ on $\mathfrak{p}^{\lfloor l/2 \rfloor + 1}$ makes $x \mapsto \psi(cx)$ non-trivial on $\mathfrak{p}^l$ and trivial on $\mathfrak{p}^{l+1}$, because χ has level exactly l, and ψ has level one. Hence $v_{\mathfrak{A}}(c) = -2l > -n$, so $v_{\mathfrak{A}}(\alpha + c) = -n$; and $F[\alpha + c] = E$ with $v_E(\alpha + c) = -n$ odd, so $\alpha + c$ is minimal and $(\mathfrak{A}, n, \alpha + c)$ is a ramified simple stratum. For $1 + x \in U_{\mathfrak{A}}^{m+1}$ we have $\det(1+x) = 1 + \mathrm{tr}x + \det x$. By (11.2), $\mathrm{tr}\mathfrak{P}^{m+1} \subseteq \mathfrak{p}^{\lceil (m+1)/2 \rceil}$, while the two diagonal entries of $x \in \mathfrak{P}^{m+1}$ lie in $\mathfrak{p}^{\lceil (m+1)/2 \rceil}$ and the two off-diagonal ones in $\mathfrak{p}^{\lceil (m+2)/2 \rceil}$ and $\mathfrak{p}^{\lceil m/2 \rceil}$. So the product of the diagonal entries has valuation at least $2\lceil (m+1)/2 \rceil \geq m+1$, that of the off-diagonal ones at least $\lceil (m+2)/2 \rceil + \lceil m/2 \rceil \geq m+1$, and $v_F(\det x) \geq m+1 > l$. The shorter route through $\mathfrak{P}^{m+1} \subseteq \varpi^{\lceil (m+1)/2 \rceil}\mathfrak{A}$ is not available: that inclusion runs the wrong way when $m+1$ is odd, since at $m+1 = 3$ one has $\varpi^2\mathfrak{A} = \mathfrak{P}^4 \subsetneq \mathfrak{P}^3$. As $2l < n = 2m+1$ gives $l \leq m$, we have $\lfloor l/2 \rfloor + 1 \leq \lceil (m+1)/2 \rceil$, so $\mathrm{tr}x + \det x$ lies in $\mathfrak{p}^{\lfloor l/2 \rfloor + 1}$ and
$$\chi(\det(1+x)) = \psi\big(c(\mathrm{tr}x + \det x)\big) = \psi(c\,\mathrm{tr}x) = \psi(\mathrm{tr}(cx))$$
the middle equality because $v_F(c \det x) \geq 1$. Hence $(\chi \circ \det)\Lambda$ restricts to $\psi_{\alpha+c}$ on $U_{\mathfrak{A}}^{m+1}$ and

lies in $\mathcal{C}(\psi_{\alpha+c}, \mathfrak{A})$.

For $y \in (EK)^\times$, using (11.8) for $\alpha + c$ (note that $F[\alpha + c] = E$, so the same norm map occurs) and $\det \mathrm{Nm}_{EK/E}(y) = \mathrm{Nm}_{E/F}(\mathrm{Nm}_{EK/E}(y)) = \mathrm{Nm}_{EK/F}(y) = \mathrm{Nm}_{K/F}(\mathrm{Nm}_{EK/K}(y)) = \mathrm{Nm}_{K/F}(\det y)$,

$$\mathrm{Lft}^{\alpha+c}_{K/F}\big((\chi \circ \det)\Lambda\big)(y) = \varepsilon_{K/F}^{v_{EK}(y)}\chi\big(\mathrm{Nm}_{K/F}(\det y)\big)\Lambda\big(\mathrm{Nm}_{EK/E}(y)\big) = \chi_K(\det y)\,\Lambda_K(y)$$

Both sides of the asserted identity therefore agree on $(EK)^\times$. Both lie in $\mathcal{C}(\psi^K_{\alpha+c}, \mathfrak{A}_K)$, the right-hand one by the computation of the previous paragraph carried out over K with χ_K in place of χ; the element attached to χ_K there is again c, by proposition 4.1.3 applied to the tame extension K/F. Uniqueness in theorem 11.1.10 finishes the proof.

Two more ingredients are needed before the definition: a name for the sets involved, and the decomposition of an arbitrary totally ramified representation into a twist of a minimal one.

Recall from lemma 4.3.2 that a cuspidal $\pi \in \mathcal{A}_2^0(F)$ contains a cuspidal type carried by the chain order $M_2(\mathfrak{o})$, that is by a chain order with $e_{\mathfrak{A}} = 1$, exactly when π is unramified. Otherwise it contains one carried by a chain order with $e_{\mathfrak{A}} = 2$, since proposition 3.3.2 allows only $e_{\mathfrak{A}} \in \{1, 2\}$. Recall also that $\ell(\pi)$ denotes the level of π, and that $n(\pi, \psi) = 2\ell(\pi)$ by proposition 10.1.10(1), so that $2\ell(\pi)$ is a non-negative integer. Call $\pi \in \mathcal{A}_2^0(F)$ *minimal* if $\ell(\pi) \leq \ell(\chi\pi)$ for every character χ of $F^\times$. That is, if no twist of π has smaller level.

Lemma 11.1.13: The Minimal-Twist Decomposition

Let $\pi \in \mathcal{A}_2^0(F)$ be cuspidal and not unramified.

1. There is a character χ of $F^\times$ such that $\pi = \chi\pi'$ with π' minimal and not unramified.
2. If π is minimal and not unramified, it contains a character $\Lambda \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$ for a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ with $n = 2m + 1 \geq 1$, and then $\ell(\pi) = n/2$.
3. Let π' be minimal and not unramified, attached as in (2) to a stratum of level n, and let χ have level l. Then
$$\chi\pi' \text{ is minimal} \iff 2l < n$$
and in that case proposition 11.1.12(2) applies to the pair (χ, π').
4. π is minimal if and only if $\ell(\pi) \notin \mathbb{Z}$. A minimal π has $\ell(\pi) = n/2$ with n odd, and a non-minimal one has $\ell(\pi) = l$, the level of the character χ of (1).

***Proof*:**

Part (1). The set $\{2\ell(\chi\pi) : \chi\}$ consists of non-negative integers, so it has a least element. Choose χ_0 attaining it and put $\pi' = \chi_0\pi$. For every χ, $\ell(\chi\pi') = \ell(\chi\chi_0\pi) \geq \ell(\chi_0\pi) = \ell(\pi')$, so π' is minimal, and $\pi = \chi_0^{-1}\pi'$. The set $\mathcal{A}_2^{\mathrm{nr}}(F)$ is by definition the set of twists $\varphi\pi_0$ of the representations π_0 of level zero, so it is stable under twisting by a character of $F^\times$. Were π' unramified, $\pi = \chi_0^{-1}\pi'$ would be unramified as well. So π' is not unramified either.

Part (2). This is the content of the classification by cuspidal types: by theorem 3.5.5 and

definition 3.5.4, a minimal cuspidal representation contains a type attached to a simple stratum, and by lemma 4.3.2 with proposition 3.3.2 the chain order has $e_{\mathfrak{A}} = 2$, so the stratum is ramified simple and its level n is odd. That $\pi \in \mathcal{A}_2^0(F;\alpha)$ then has $\ell(\pi) = n/2$ is proposition 10.1.3(2).

Part (3). Let $\pi' = \text{c-Ind}_J^G \Lambda$ with $\Lambda \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$ as in (2), so $\chi\pi' = \text{c-Ind}_J^G \Lambda_2$ with $\Lambda_2 = (\chi \circ \det)\Lambda$, by proposition 1.2.30. Since $e_{\mathfrak{A}} = 2$ we have $\mathfrak{P}^{2t} = \varpi^t \mathfrak{A}$ and $\det U_{\mathfrak{A}}^{2t} = U_F^t$, while $\mathrm{tr}\mathfrak{P}^{2t+1} = \mathrm{tr}(\varpi^t \mathfrak{P}) \subseteq \mathfrak{p}^{t+1}$ gives $\det U_{\mathfrak{A}}^{2t+1} \subseteq U_F^{t+1}$: for $x \in \mathfrak{P}^{2t+1}$ one has $\det(1+x) = 1 + \mathrm{tr}x + \det x$, the middle term lies in $\mathfrak{p}^{t+1}$ by the display just made, and $v_F(\det x) \geq 2t+1 \geq t+1$ because $x\,\mathrm{adj}(x) = \det(x)\,1$ with both x and $\mathrm{adj}(x)$ in $\mathfrak{P}^{2t+1}$. So $\chi \circ \det$ is non-trivial on $U_{\mathfrak{A}}^{2l}$ and trivial on $U_{\mathfrak{A}}^{2l+1}$.

Suppose $2l > n$, that is $2l \geq n+1$. Then $l \geq 1$. The character Λ is trivial on $U_{\mathfrak{A}}^{n+1} \supseteq U_{\mathfrak{A}}^{2l}$, so Λ_2 agrees with $\chi \circ \det$ on $U_{\mathfrak{A}}^{2l}$, where it is non-trivial, and is trivial on $U_{\mathfrak{A}}^{2l+1}$. Identify that character: let $c \in F$ have $v_F(c) = -l$ and $\chi(1+z) = \psi(cz)$ for $z \in \mathfrak{p}^{\lfloor l/2 \rfloor + 1}$. For $1 + x \in U_{\mathfrak{A}}^{2l} = 1 + \varpi^l \mathfrak{A}$ one has $\det(1+x) = 1 + \mathrm{tr}x + \det x$ with $\mathrm{tr}x \in \varpi^l \mathrm{tr}\mathfrak{A} = \mathfrak{p}^l$ and $v_F(\det x) \geq 2l$, both exponents at least $\lfloor l/2 \rfloor + 1$, while $v_F(c \det x) \geq l \geq 1$. So $\chi(\det(1+x)) = \psi(c\,\mathrm{tr}x) = \psi_c(1+x)$. Thus $\chi\pi'$ contains the character ψ_c of the stratum $(\mathfrak{A}, 2l, c)$, and that stratum is fundamental by proposition 3.3.11, since $v_{\mathfrak{A}}(c^r) = -2lr$ for every $r \geq 1$. So theorem 3.3.13 gives $\ell(\chi\pi') = 2l/e_{\mathfrak{A}} = l$. Now π' is a twist of $\chi\pi'$, namely $\pi' = \chi^{-1}(\chi\pi')$, so if $\chi\pi'$ were minimal its level would be at most $\ell(\pi') = n/2$, giving $2l \leq n$ against $2l \geq n+1$. So $\chi\pi'$ is not minimal. This route avoids asking what chain order carries a type for the non-minimal $\chi\pi'$, which is a question theorem 3.5.5 does not answer.

Suppose instead $2l < n$. Then proposition 11.1.12(2) applies and attaches $\chi\pi'$ to the ramified simple stratum $(\mathfrak{A}, n, \alpha + c)$, of the same level n on the same chain order, so $\ell(\chi\pi') = n/2 = \ell(\pi')$ by (2). Minimality of π' says $\ell(\pi')$ is least among the levels of the twists of π', and $\chi\pi'$ has the same twists as π' and attains that least value. So $\chi\pi'$ is minimal too. The remaining case $2l = n$ does not occur, n being odd, so the two implications are exhaustive and give the stated equivalence.

Consequently, if both are minimal then $2l \leq n$; and n is odd, so $2l < n$, which is the hypothesis of proposition 11.1.12(2).

Part (4). A minimal π has $\ell(\pi) = n/2$ by (2), with n odd, so $\ell(\pi)$ is not an integer. A non-minimal π is $\chi\pi'$ with π' minimal by (1), and (3) gives $2l > n$ for the level l of χ, and the case $2l > n$ of part (3) computes $\ell(\chi\pi') = 2l/e_{\mathfrak{A}} = l$. The two cases are exhaustive, and the first value is a half-integer while the second is an integer, which is the stated equivalence and completes the proof.

Definition 11.1.14: Totally Ramified Classes, and the Tame Lifting Map

Write $\mathcal{A}_2^{\mathrm{nr}}(F)$ for the set of classes in $\mathcal{A}_2^0(F)$ of unramified cuspidal representations, in the sense of lemma 4.3.2, and set

$$\mathcal{A}_2^{\mathrm{wr}}(F) = \mathcal{A}_2^0(F) \smallsetminus \mathcal{A}_2^{\mathrm{nr}}(F), \qquad \mathcal{G}_2^{\mathrm{wr}}(F) = \mathcal{G}_2^0(F) \smallsetminus \mathcal{G}_2^{\mathrm{nr}}(F)$$

Elements of $\mathcal{A}_2^{\mathrm{wr}}(F)$ are called *totally ramified*. By definition 9.1.17, $\mathcal{G}_2^{\mathrm{wr}}(F)$ is exactly the set of totally ramified parameters. No minimality condition is imposed on either set.
Let K/F be a finite tamely ramified extension. The lifting map $\mathrm{Lft}_{K/F}\colon \mathcal{A}_2^{\mathrm{wr}}(F) \to \mathcal{A}_2^{\mathrm{wr}}(K)$ is defined in two clauses.

1. *The minimal case.* Let $\pi' \in \mathcal{A}_2^{\mathrm{wr}}(F)$ be minimal and choose, by lemma 11.1.13(2), a type $\Lambda \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$ contained in π', with $(\mathfrak{A}, n, \alpha)$ a ramified simple stratum. Set
$$\mathrm{Lft}_{K/F}(\pi') = \pi'_K = \text{c-Ind}_{J_K}^{G_K} \mathrm{Lft}_{K/F}(\Lambda) \in \mathcal{A}_2^{\mathrm{wr}}(K)$$
2. *The twist clause.* For general $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F)$, choose by lemma 11.1.13(1) a decomposition $\pi = \chi\pi'$ with π' minimal, and set
$$\mathrm{Lft}_{K/F}(\pi) = \pi_K = \chi_K\, \pi'_K, \qquad \chi_K = \chi \circ \mathrm{Nm}_{K/F}$$
with π'_K given by (1).

The representation π_K is called the K/F-lift of π. That (2) does not depend on the decomposition chosen in it is proposition 11.1.15.

Three choices went into that definition: the minimal twist, the type inside it, and the character ψ. None of them leaves a trace on π_K, and once that is known the formal properties of the map follow from the corresponding properties of the lift of types. In the situation of example 11.1.11 the definition returns, for a minimal π' containing $\Lambda \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$, the representation $\pi'_K = \text{c-Ind}_{J_K}^{G_K} \Lambda_K$ induced from $J_K = (EK)^\times U^2_{\mathfrak{A}_K}$ over $K = \mathbb{Q}_2(2^{1/3})$, with Λ_K the character displayed there.

Proposition 11.1.15: The Lift Is Well Defined, and Its Basic Properties

Definition 11.1.14 does not depend on any of the choices made in it, and the resulting map $\mathrm{Lft}_{K/F}\colon \mathcal{A}_2^{\mathrm{wr}}(F) \to \mathcal{A}_2^{\mathrm{wr}}(K)$ satisfies, for $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F)$ and K/F finite tamely ramified:

1. $\omega_{\pi_K} = \omega_\pi \circ \mathrm{Nm}_{K/F}$ for the central characters;
2. $(\chi\pi)_K = \chi_K\, \pi_K$ for every character χ of $F^\times$;
3. $\mathrm{Lft}_{L/F} = \mathrm{Lft}_{L/K} \circ \mathrm{Lft}_{K/F}$ for a further finite tamely ramified L/K;
4. for $\pi, \pi' \in \mathcal{A}_2^{\mathrm{wr}}(F)$ one has $\pi'_K \cong \pi_K$ if and only if $\pi' \cong \chi\pi$ for a character χ of $F^\times$ trivial on $\mathrm{Nm}_{K/F}(K^\times)$;
5. $\mathrm{Lft}_{K/F}$ does not depend on the choice of the level one character ψ.

***Proof*:**
Well-definedness for minimal π'. Suppose π' contains $\Lambda \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$ and $\Lambda' \in \mathcal{C}(\psi_\beta, \mathfrak{B})$. By theorem 3.5.5 the two cuspidal types are G-conjugate: there is $g \in G$ with $\mathfrak{B} = g^{-1}\mathfrak{A}g$ and $\Lambda' = \Lambda^g$, and then ψ_β agrees on $U_{\mathfrak{B}}^{m+1}$ with the character attached to $g^{-1}\alpha g$. By theorem 11.1.10(3) the maps $\mathrm{Lft}_{K/F}$ attached to $g^{-1}\alpha g$ and to β coincide, and by proposition 11.1.12(1) $\mathrm{Lft}_{K/F}(\Lambda') = \mathrm{Lft}_{K/F}(\Lambda)^g$. Compactly inducing a character and its g-conjugate from conjugate groups gives isomorphic representations, so π'_K does not depend on the type chosen. It lies in $\mathcal{A}_2^{\mathrm{wr}}(K)$ because $e_{\mathfrak{A}_K} = 2$.

Well-definedness in general. Let $\chi_1\pi_1 = \chi_2\pi_2$ with π_1, π_2 minimal. Then $\pi_2 = \chi\pi_1$ with $\chi = \chi_1\chi_2^{-1}$, and by lemma 11.1.13(3) the level of χ satisfies $2l < n$, so proposition 11.1.12(2) gives $(\pi_2)_K = \chi_K(\pi_1)_K$. Hence $\chi_{2K}(\pi_2)_K = \chi_{2K}\chi_K(\pi_1)_K = \chi_{1K}(\pi_1)_K$, as required.

Part (2). If $\pi = \chi'\pi'$ with π' minimal, then $\chi\pi = (\chi\chi')\pi'$ is a legitimate decomposition of $\chi\pi$, and $(\chi\chi')_K = \chi_K\chi'_K$, so $(\chi\pi)_K = \chi_K\chi'_K\pi'_K = \chi_K\pi_K$.

Part (1). For minimal π', the central character of $\text{c-Ind}_{J_K}^{G_K}\Lambda_K$ is $\Lambda_K\big|K^\times$, which equals $(\Lambda\big|F^\times)\circ \mathrm{Nm}_{K/F} = \omega_{\pi'} \circ \mathrm{Nm}_{K/F}$ by theorem 11.1.10(4). In general $\pi = \chi\pi'$ gives $\omega_\pi = \chi^2\omega_{\pi'}$ and $\omega_{\pi_K} = \chi_K^2\omega_{\pi'_K} = (\chi^2\omega_{\pi'}) \circ \mathrm{Nm}_{K/F}$.

Part (3). For minimal π' this is theorem 11.1.10(2), since $(\mathfrak{A}_K)_L = \mathfrak{A}_L$ by the uniqueness in proposition 11.1.4(1). In general write $\pi = \chi\pi'$ and use part (2) together with $(\chi \circ \mathrm{Nm}_{K/F}) \circ \mathrm{Nm}_{L/K} = \chi \circ \mathrm{Nm}_{L/F}$.

Part (4). Suppose first π, π' minimal, attached to $(\mathfrak{A}, n, \alpha)$ and $(\mathfrak{B}, n', \beta)$, and $\pi_K \cong \pi'_K$. Then π_K contains types in $\mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$ and in $\mathcal{C}(\psi_\beta^K, \mathfrak{B}_K)$, which are G_K-conjugate by theorem 3.5.5. In particular $n_K = n'_K$, so $n = n'$, and ψ_α^K, ψ_β^K are G_K-conjugate. By lemma 11.1.6 ψ_α and ψ_β are G-conjugate, so after conjugating π' (which changes nothing) we may assume $\mathfrak{B} = \mathfrak{A}$ and $\psi_\beta = \psi_\alpha$. Then theorem 11.1.10(3) makes the two lifting maps equal, and $\Lambda_K = \Lambda'_K$, so theorem 11.1.10(1) gives $\Lambda' = (\chi \circ \det)\Lambda$ with χ trivial on $\mathrm{Nm}_{K/F}(K^\times)$, whence $\pi' = \chi\pi$ by proposition 1.2.30. Conversely, let χ be trivial on $\mathrm{Nm}_{K/F}(K^\times)$. Then χ is trivial on $\mathrm{Nm}_{K/F}(U_K^1) = U_F^1$ by lemma 11.1.8(1), so χ has level 0 and $\chi_K = \chi \circ \mathrm{Nm}_{K/F} = 1$. Part (2) gives $(\chi\pi)_K = \chi_K\pi_K = \pi_K$, and this direction used nothing about minimality.

For the forward direction at general π, π', write $\pi = \chi_1\pi_1$ and $\pi' = \chi_2\pi_2$ with π_1, π_2 minimal, by lemma 11.1.13(1). Part (2) turns $\pi'_K \cong \pi_K$ into $(\pi_1)_K \cong (\eta\pi_2)_K$ with $\eta = \chi_2\chi_1^{-1}$. The minimal case does not apply yet, because $\eta\pi_2$ need not be minimal, and that has to be earned rather than assumed. Let n_2 be the level of the stratum attached to π_2 and $n_{2K} = e_K n_2$ that of its lift, by proposition 11.1.4(1). Now $(\pi_1)_K$ is attached to a ramified simple stratum of odd level, so $\ell\big((\pi_1)_K\big) \notin \mathbb{Z}$ by proposition 10.1.3(2), and lemma 11.1.13(4) over K turns that into minimality. So $(\pi_1)_K$, and with it $(\eta\pi_2)_K = \eta_K(\pi_2)_K$, is minimal. Applying lemma 11.1.13(3) over K to the minimal $(\pi_2)_K$ and the character η_K gives $2\,\ell(\eta_K) < n_{2K}$, and $\ell(\eta_K) = e_K\,\ell(\eta)$ by proposition 4.1.3 when $\ell(\eta) \geq 1$, the case $\ell(\eta) = 0$ being immediate. Dividing by e_K gives $2\,\ell(\eta) < n_2$, so lemma 11.1.13(3) over F makes $\eta\pi_2$ minimal. The minimal case now applies to the pair π_1, $\eta\pi_2$ and returns $\eta\pi_2 \cong \chi_0\pi_1$ with χ_0 trivial on $\mathrm{Nm}_{K/F}(K^\times)$. Since $\chi_2 = \chi_1\eta$, this gives $\pi' = \chi_2\pi_2 = \chi_1(\eta\pi_2) \cong \chi_1\chi_0\pi_1 = \chi_0(\chi_1\pi_1) = \chi_0\pi$, which is the assertion.

Part (5). Any other level one character is $u\psi$ with $u \in U_F$. The identity $(u\psi)(\mathrm{tr}(u^{-1}\alpha x)) = \psi(\mathrm{tr}(\alpha x))$ shows that $\mathcal{C}(\psi_\alpha, \mathfrak{A}) = \mathcal{C}((u\psi)_{u^{-1}\alpha}, \mathfrak{A})$ as sets of characters of the same group J (note $F[u^{-1}\alpha] = E$ and $\mathfrak{A}$ is unchanged) and likewise over K, since $(u\psi)_K = u\psi_K$. The recipe (11.8) refers only to $\varepsilon_{K/F}$, to v_{EK} and to $\mathrm{Nm}_{EK/E}$, none of which involves ψ. So the same character Λ receives the same lift under either normalization, completing the proof.

The construction is natural for isomorphisms of the base field. Section 11.2 treats that naturality in general. The special case needed here is the following.

Lemma 11.1.16: Lifting Commutes with Automorphisms of the Ambient Field

Let $F \subseteq E \subseteq L$ be finite extensions with L/E tamely ramified, and let γ be an automorphism of L over F with $\gamma(E) = E$. Then for every $\zeta \in \mathcal{A}_2^{\mathrm{wr}}(E)$,

$$\big(\zeta^{\gamma}\big)_L = \big(\zeta_L\big)^{\gamma}$$

where the superscripts are the push-forward of box 11.1.1: on the left $\zeta^\gamma = \zeta \circ (\gamma|E)^{-1}$, transported along the automorphism of $\mathrm{GL}_2(E)$ induced by $\gamma|E$, and on the right $(\zeta_L)^\gamma = \zeta_L \circ \gamma^{-1}$ on $\mathrm{GL}_2(L)$. In particular, if L/E is Galois with group Σ then $\mathrm{Lft}_{L/E}\big(\mathcal{A}_2^{\mathrm{wr}}(E)\big) \subseteq \mathcal{A}_2^{\mathrm{wr}}(L)^{\Sigma}$.

***Proof*:**
The automorphism γ carries a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ in $M_2(E)$ to the ramified simple stratum $(\mathfrak{A}^\gamma, n, \alpha^\gamma)$, carries $\mathfrak{A}_L$ to $(\mathfrak{A}^\gamma)_L$ by the uniqueness in proposition 11.1.4(1), and carries the field $E[\alpha]L$ generated by the stratum to $\gamma(E[\alpha])L = E[\alpha^\gamma]L$, compatibly with $\mathrm{Nm}_{E[\alpha]L/E[\alpha]}$ and $v_{E[\alpha]L}$. These are the fields the recipe uses; the compositum EL is L, since $E \subseteq L$, and plays no part. It fixes $\varepsilon_{L/E}$, because γ maps L to L and E to E and $\varkappa_{L/E}$ is defined from the pair. Hence γ carries the recipe (11.8) for $(\mathfrak{A}, n, \alpha)$ to the recipe for $(\mathfrak{A}^\gamma, n, \alpha^\gamma)$, and by proposition 11.1.15(5) the intervening change of ψ to $\psi \circ \gamma^{-1}$ is immaterial. Compact induction commutes with transport, and definition 11.1.14(2) transports because $(\chi \circ \gamma^{-1}) \circ \mathrm{Nm}_{L/E} = (\chi \circ \mathrm{Nm}_{L/E}) \circ \gamma^{-1}$. The last assertion is the case $E = F$, where $\zeta^\gamma = \zeta$ for $\gamma \in \Sigma$, completing the proof.

The descent theorem needs two further inputs. The first is a vanishing statement for the cohomology of the unit and ideal filtrations of a tame extension. The case $j = 1$ of the multiplicative filtration is [8, Cohomology of Tamely Ramified Extensions], and what follows is the rest.

Lemma 11.1.17: Vanishing for the Filtrations of a Tame Galois Extension

Let K/F be a finite tamely ramified Galois extension with group Σ. Then

$$H^1\big(\Sigma,\, U_K^j\big) = 0 \quad (j \geq 1), \qquad H^1\big(\Sigma,\, \mathfrak{p}_K^i\big) = 0 \quad (i \in \mathbb{Z})$$

***Proof*:**
Step 1: the graded pieces are semilinear lines. Let $\Sigma_0 \subseteq \Sigma$ be the inertia subgroup, of order e_K. It acts trivially on k_K and $\Sigma/\Sigma_0 \cong \mathrm{Gal}(k_K/k)$. For $i \in \mathbb{Z}$ put $M = \mathfrak{p}_K^i/\mathfrak{p}_K^{i+1}$, and for $j \geq 1$ put $M = U_K^j/U_K^{j+1}$, the latter written additively through $1 + x \mapsto x$. In both cases M is a one-dimensional k_K-vector space, the module structure coming from multiplication by $\mathfrak{o}_K$, and

Σ acts on M semilinearly over its action on k_K.

Step 2: $H^1(\Sigma, M) = 0$. Since Σ_0 acts trivially on k_K, it acts k_K-linearly on the line M, hence through a character $\Sigma_0 \to k_K^\times$; and M is p-torsion while $|\Sigma_0| = e_K$ is prime to p, so $H^1(\Sigma_0, M) = 0$ by the restriction-corestriction argument ([8, Consequences of Cor $\circ$Res $= [G : H]$]). The inflation-restriction sequence therefore gives $H^1(\Sigma, M) \cong H^1\big(\mathrm{Gal}(k_K/k),\, M^{\Sigma_0}\big)$. If the character of Σ_0 is non-trivial then $M^{\Sigma_0} = 0$ and there is nothing to prove. Otherwise $M^{\Sigma_0} = M$ is a one-dimensional k_K-vector space with a semilinear action of $\mathrm{Gal}(k_K/k)$. Since $H^1(\mathrm{Gal}(k_K/k), k_K^\times) = 1$, such a line has a basis of invariants and $M \cong k_K$ as a $\mathrm{Gal}(k_K/k)$-module. Finally $H^1(\mathrm{Gal}(k_K/k), k_K) = 0$ by the normal basis theorem [4, Normal Basis Theorem], which makes k_K a free $k[\mathrm{Gal}(k_K/k)]$-module and hence cohomologically trivial.

Step 3: dévissage and passage to the limit. Fix $j \geq 1$ and $N \geq 1$. From the exact sequences $1 \to U_K^{j+1}/U_K^{j+N} \to U_K^j/U_K^{j+N} \to U_K^j/U_K^{j+1} \to 1$ of finite Σ-modules and induction on N, using Step 2 at each stage, we get $H^1(\Sigma, U_K^j/U_K^{j+N}) = 0$ for all N. The groups U_K^j/U_K^{j+N} are finite, and the transition maps of the inverse system are surjective, so the system satisfies the Mittag-Leffler condition and $H^1(\Sigma, U_K^j) = H^1(\Sigma, \varprojlim_N U_K^j/U_K^{j+N}) = \varprojlim_N H^1(\Sigma, U_K^j/U_K^{j+N}) = 0$, the first equality being the completeness of K.

The middle equality is worth doing rather than quoting, because the statement wanted here is not the one this series stocks: [8, Cohomology as a Colimit] is a colimit over the open subgroups of a profinite group at fixed coefficients, whereas what is needed is a limit in the *coefficients* of a *finite* group. For finite Σ the cochain groups $C^n(\Sigma, M) = \mathrm{Map}(\Sigma^n, M)$ are finite products of copies of M indexed by the finite set Σ^n, so $C^n(\Sigma, \varprojlim_N M_N) = \varprojlim_N C^n(\Sigma, M_N)$, and the differentials match. A limit of finite groups along surjective transition maps is exact, so taking $\varprojlim_N$ commutes with passing to kernels and images of the differentials, which gives $H^1(\Sigma, \varprojlim_N M_N) = \varprojlim_N H^1(\Sigma, M_N)$ for the system at hand. The same argument, with $\mathfrak{p}_K^i/\mathfrak{p}_K^{i+N}$ in place of U_K^j/U_K^{j+N}, gives the additive statement for every $i \in \mathbb{Z}$, completing the proof.

The second input converts a prescribed trace and determinant into a stratum carrying them.

Lemma 11.1.18: A Ramified Simple Stratum with Prescribed Invariants

Let $n = 2m + 1 \geq 1$, let $x \in F^\times$ with $v_F(x) = -n$, and let $y \in \mathfrak{p}^{-m}$. Then there is a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ in $M_2(F)$ with $\det\alpha = x$ and $\mathrm{tr}\alpha = y$, and $\mathfrak{A}$ is uniquely determined by α.

***Proof*:**
Consider $P(X) = X^2 - yX + x$. If P had a root in F, its two roots λ_1, λ_2 would satisfy $v_F(\lambda_1) + v_F(\lambda_2) = v_F(x) = -n$, which is odd, so the two valuations differ, and then $v_F(y) = v_F(\lambda_1 + \lambda_2) = \min_i v_F(\lambda_i) \leq \lfloor -n/2 \rfloor = -m - 1$, contradicting $y \in \mathfrak{p}^{-m}$. So P is irreducible and $E = F[X]/(P)$ is a quadratic field extension of F. No separability is needed for this.

Let $\alpha \in E$ be a root. Then $\mathrm{Nm}_{E/F}(\alpha) = x$ and $\mathrm{tr}_{E/F}(\alpha) = y$, and $2v_E(\alpha) = v_E(x) = e(E|F)v_F(x) = -e(E|F)n$. If E/F were unramified this would read $2v_E(\alpha) = -n$ with n odd, which is impossible. So E/F is totally ramified and $v_E(\alpha) = -n$, an odd integer prime to

$e(E|F) = 2$, so α is minimal over F. Regard E as a two-dimensional F-vector space and α as the F-linear map given by multiplication. Choosing a basis realizes α inside $M_2(F)$ with matrix trace y and matrix determinant x. Lemma 11.1.3 produces a unique $\mathfrak{o}$-chain order $\mathfrak{A}$ with $\alpha \in \mathcal{K}_{\mathfrak{A}}$, and $v_{\mathfrak{A}}(\alpha) = -n$, so $(\mathfrak{A}, n, \alpha)$ is a ramified simple stratum, completing the proof.

One inclusion of the descent theorem is lemma 11.1.16: a lift from F is fixed by $\Sigma = \mathrm{Gal}(K/F)$. The content is the reverse inclusion, and the proof descends a Σ-fixed representation in stages: first its trace and determinant, which are cocycles killed by lemma 11.1.17, then the stratum they determine, and finally the type.

Theorem 11.1.19: Cyclic Descent

Let K/F be a finite, cyclic, tamely ramified extension and put $\Sigma = \mathrm{Gal}(K/F)$. Then

$$\mathrm{Lft}_{K/F}\big(\mathcal{A}_2^{\mathrm{wr}}(F)\big) = \mathcal{A}_2^{\mathrm{wr}}(K)^{\Sigma}$$

More precisely, for every $\tau \in \mathcal{A}_2^{\mathrm{wr}}(K)^{\Sigma}$ there is $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F)$ with $\tau = \pi_K$, and π is determined up to twisting by a character of $F^\times$ trivial on $\mathrm{Nm}_{K/F}(K^\times)$.

***Proof*:**
The inclusion $\subseteq$ is lemma 11.1.16, and the final clause is proposition 11.1.15(4). It remains to produce π.

Step 1: τ minimal. Choose, by lemma 11.1.13(2), a ramified simple stratum $(\mathfrak{B}, s, \beta)$ in $M_2(K)$ with $s = 2r + 1$ and a type $\Theta \in \mathcal{C}(\psi_\beta^K, \mathfrak{B})$ contained in τ, where ψ_β^K is formed with ψ_K. Every superscript below is the push-forward of box 11.1.1: $\beta^\sigma = \sigma(\beta)$ on elements, $\Theta^\sigma = \Theta \circ \sigma^{-1}$ on characters. For $\sigma \in \Sigma$ the character Θ^σ is also a type contained in $\tau^\sigma = \tau$, so Θ and Θ^σ are G_K-conjugate by theorem 3.5.5. Since ψ_K is Σ-invariant (because $\mathrm{tr}_{K/F} \circ \sigma = \mathrm{tr}_{K/F}$) we have $(\psi_\beta^K)^\sigma = \psi_{\beta^\sigma}^K$, so ψ_β^K and $\psi_{\beta^\sigma}^K$ are G_K-conjugate and lemma 11.1.5 gives

$$\det \beta^\sigma \in \det \beta \, U_K^{\lfloor r/2 \rfloor + 1}, \qquad \mathrm{tr}\beta^\sigma \in \mathrm{tr}\beta + \mathfrak{p}_K^{-\lfloor r/2 \rfloor}$$

Step 2: descending the invariants. The first congruence says that $\sigma \mapsto \det \beta^\sigma / \det \beta$ is a 1-cocycle of Σ with values in $U_K^{\lfloor r/2 \rfloor + 1}$, and the second that $\sigma \mapsto \mathrm{tr}\beta^\sigma - \mathrm{tr}\beta$ is one with values in $\mathfrak{p}_K^{-\lfloor r/2 \rfloor}$. By lemma 11.1.17 both are coboundaries, so there are $u \in U_K^{\lfloor r/2 \rfloor + 1}$ and $w \in \mathfrak{p}_K^{-\lfloor r/2 \rfloor}$ with $x := u \det \beta$ and $y := w + \mathrm{tr}\beta$ fixed by Σ, that is, lying in F. Then $v_K(x) = v_K(\det \beta) = -s$, so $e_K | s$. Put $n = s/e_K$, which is odd because s is, and write $n = 2m + 1$. Then $v_F(x) = -n$. Also $y \in F \cap \mathfrak{p}_K^{-r}$: indeed $\beta \in \mathfrak{P}_{\mathfrak{B}}^{-s}$, and (11.2), read over K, gives $\mathrm{tr}\mathfrak{P}_{\mathfrak{B}}^{-s} \subseteq \mathfrak{p}_K^{\lceil -s/2 \rceil} = \mathfrak{p}_K^{-r}$, while $v_K(w) \geq -\lfloor r/2 \rfloor \geq -r$; and $F \cap \mathfrak{p}_K^{-r} = \mathfrak{p}^{\lceil -r/e_K \rceil} = \mathfrak{p}^{-m}$, because $-r/e_K = -\frac{n}{2} + \frac{1}{2e_K}$ lies in $(-m - \frac{1}{2}, -m]$. So $y \in \mathfrak{p}^{-m}$.

Step 3: replacing the stratum. By lemma 11.1.18 there is a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ in $M_2(F)$ with $\det \alpha = x$ and $\mathrm{tr}\alpha = y$. Its lift $(\mathfrak{A}_K, n_K, \alpha)$ has $n_K = e_K n = s$ and $m_K = r$. By the choice of x and y,

$$\det \alpha \in \det \beta \, U_K^{\lfloor r/2 \rfloor + 1}, \qquad \mathrm{tr}\alpha \in \mathrm{tr}\beta + \mathfrak{p}_K^{-\lfloor r/2 \rfloor}$$

so $\psi_\alpha^K|U_{\mathfrak{A}_K}^{r+1}$ and $\psi_\beta^K|U_{\mathfrak{B}}^{r+1}$ are G_K-conjugate, by lemma 11.1.5 over K. Conjugating the type inside its class we may therefore assume $(\mathfrak{B}, s, \beta) = (\mathfrak{A}_K, n_K, \alpha)$ and $\Theta \in \mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$.

Step 4: descending the type. The chain order $\mathfrak{A}_K$ and the character ψ_α^K are Σ-invariant, the first by the uniqueness in proposition 11.1.4(1) and the second because $\alpha \in M_2(F)$ and ψ_K is Σ-invariant. So Θ^σ again lies in $\mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$ and is contained in τ. Only one element of that set is, since proposition 10.1.3(1) applied over K makes $\Theta \mapsto \text{c-Ind}_{J_K}^{G_K}\Theta$ a bijection of $\mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$ onto $\mathcal{A}_2^0(K;\alpha)$. Concretely, if $\Theta'' \in \mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$ occurs in $\tau = \text{c-Ind}_{J_K}^{G_K}\Theta$, then $\Theta'' = \Theta^g$ for some $g \in G_K$ intertwining the two, and such a g intertwines ψ_α^K with itself, hence lies in J_K by theorem 3.5.3(1). Conjugation by J_K fixes characters of J_K, so $\Theta'' = \Theta$. Therefore $\Theta^\sigma = \Theta$ for all σ. Put $\Theta'(z) = \varepsilon_{K/F}^{-v_{EK}(z)}\Theta(z)$ for $z \in (EK)^\times$. As v_{EK} is Σ-invariant, Θ' is a Σ-invariant character of $(EK)^\times$. The extension EK/E is cyclic, with group identified with Σ by restriction. Hilbert's Theorem 90 ([14, Hilbert's Theorem 90]) gives $H^1\big(\Sigma, (EK)^\times\big) = 1$, and for a cyclic Σ the Tate groups are periodic of period two, so $\hat{H}^{-1}\big(\Sigma, (EK)^\times\big) = H^1\big(\Sigma, (EK)^\times\big) = 1$. That is the identification wanted, since $\hat{H}^{-1}$ of a cyclic group is by definition the kernel of the norm modulo the augmentation image: the kernel of $\text{Nm}_{EK/E}$ is exactly $\big\{z^{\sigma-1} : z \in (EK)^\times\big\}$, on which Θ' is trivial. Step 5 below runs the same periodicity argument for U_K^t and states it there. It is the same one step. Hence $\Theta' = \Lambda \circ \text{Nm}_{EK/E}$ for a character Λ of the subgroup $\text{Nm}_{EK/E}((EK)^\times)$ of $E^\times$, which extends to a character Λ of $E^\times$ because $\mathbb{C}^\times$ is divisible. That is, Θ satisfies (11.8) for this Λ.

It remains to see that Λ lies in $\mathcal{C}(\psi_\alpha, \mathfrak{A})$. On $U_{EK}^{m_K+1}$ we have $\Theta = \psi_\alpha^K$ and, by (11.9) together with (11.5), $\psi_\alpha^K(u) = \psi_\alpha(\text{Nm}_{EK/E}(u))$ modulo the ambiguity U_E^{n+1}, on which ψ_α is trivial. So Λ agrees with ψ_α on $\text{Nm}_{EK/E}(U_{EK}^{m_K+1})$, and that subgroup is the whole of U_E^{m+1}: the extension EK/E is tamely ramified with $e(EK|E) = e_K$, so lemma 11.1.8(2) gives

$$\text{Nm}_{EK/E}\big(U_{EK}^{m_K+1}\big) = U_E^{\lceil (m_K+1)/e_K \rceil} = U_E^{m+1}$$

the exponent because $2m_K + 1 = n_K = e_K(2m+1)$ gives $m_K = e_K m + s_0$ with $s_0 = (e_K - 1)/2$, so that $0 \le s_0 < e_K$. Moreover $U_E^{m+1} \subseteq U_E^1 = \text{Nm}_{EK/E}(U_{EK}^1)$ already lies in the subgroup on which Λ was pinned, so every extension of Λ to $E^\times$ has the same restriction to U_E^{m+1}. Since $E^\times \cap U_{\mathfrak{A}}^{m+1} = U_E^{m+1}$, the character Λ of $E^\times$ and the character ψ_α of $U_{\mathfrak{A}}^{m+1}$ agree on the intersection of their domains, and so assemble to an element of $\mathcal{C}(\psi_\alpha, \mathfrak{A})$. The assembly is multiplicative for the reason given in the existence step of theorem 11.1.10, that conjugation by an element of $E^\times$ preserves $U_{\mathfrak{A}}^{m+1}$ and fixes ψ_α. Set $\pi = \text{c-Ind}_J^G\Lambda$, so that $\pi \in \mathcal{A}_2^{\text{wr}}(F)$.

Before reading off $\pi_K = \tau$ from definition 11.1.14(1), note that its hypothesis holds: π is minimal. Indeed π is attached to the ramified simple stratum $(\mathfrak{A}, n, \alpha)$ of odd level n, so $\ell(\pi) = n/2 \notin \mathbb{Z}$ by proposition 10.1.3(2), and lemma 11.1.13(4) turns that into minimality. So definition 11.1.14(1) applies and gives $\pi_K = \text{c-Ind}_{J_K}^{G_K}\Lambda_K = \tau$.

Step 5: τ arbitrary. Write $\tau = \chi\tau'$ with τ' minimal, by lemma 11.1.13(1). Let $(\mathfrak{B}, s, \beta)$ be a ramified simple stratum with $\tau' \in \mathcal{A}_2^0(K;\beta)$. Only the G_K-conjugacy class of $\mathfrak{B}$ is determined, and by proposition 3.3.2 every $\mathfrak{o}_K$-chain order with $e = 2$ is G_K-conjugate to $\mathfrak{A}_K$ for an $\mathfrak{o}$-chain order $\mathfrak{A}$ in $M_2(F)$. So we may take $\mathfrak{B}$ to be Σ-stable. Let t be the level of χ. If τ is minimal we are in Step 1, so assume not, and then $2t > s$ by lemma 11.1.13(3). Since $2t \ge s+1$ and τ' contains the trivial character of $U_{\mathfrak{B}}^{s+1}$ (it contains a type in $\mathcal{C}(\psi_\beta^K, \mathfrak{B})$, and ψ_β^K is trivial on

$U_{\mathfrak{B}}^{s+1}$) the representation $\tau = \chi\tau'$ contains the character $\chi \circ \det$ of the Σ-stable group $U_{\mathfrak{B}}^{2t}$, and, $\sigma \in \Sigma$ fixing $\mathfrak{B}$ and τ, it contains $\chi^\sigma \circ \det$ there as well. Neither restriction is isotypic, and it is scalars rather than triviality that has to be excluded: were τ to act by scalars on an open subgroup U, it would act by scalars on every G_K-conjugate of U, and those generate $\mathrm{SL}_2(K)$, so $\tau(\mathrm{SL}_2(K))$ would consist of scalars and hence be abelian. But $\mathrm{SL}_2(K)$ is its own commutator subgroup, K being infinite, so τ would be trivial on $\mathrm{SL}_2(K)$, would factor through det and would be one-dimensional. The comparison of χ with χ^σ is therefore made through levels. Transport along σ permutes the $\mathfrak{o}_K$-chain orders of $M_2(K)$, preserving $e_{\mathfrak{A}}$ and carrying $U_{\mathfrak{A}}^j$ to $U_{\mathfrak{A}^\sigma}^j$, so $\ell(\varrho^\sigma) = \ell(\varrho)$ for every irreducible smooth representation ϱ of G_K, by definition 3.3.12, while $(\xi\varrho)^\sigma = \xi^\sigma\varrho^\sigma$ for every character ξ of $K^\times$. Hence $(\tau')^\sigma$ is again minimal, and comparing $\tau = \tau^\sigma = \chi^\sigma(\tau')^\sigma$ with $\tau = \chi\tau'$ gives $(\tau')^\sigma = \eta\tau'$, where $\eta = \chi(\chi^\sigma)^{-1}$. So τ' and $\eta\tau'$ are both minimal, and lemma 11.1.13(3) forces the level l of η to satisfy $2l < s < 2t$. Thus $l < t$, the character η is trivial on U_K^t, and $\chi = \chi^\sigma$ on U_K^t.

Thus $\chi\big|U_K^t$ is Σ-invariant, hence trivial on $\left\{u^{\sigma-1} : u \in U_K^t\right\}$. For the cyclic group Σ, periodicity of Tate cohomology and lemma 11.1.17 give $\hat{H}^{-1}(\Sigma, U_K^t) \cong \hat{H}^1(\Sigma, U_K^t) = 0$, that is, the kernel of $\mathrm{Nm}_{K/F}$ on U_K^t is exactly $\left\{u^{\sigma-1} : u \in U_K^t\right\}$. So $\chi\big|U_K^t$ factors through $\mathrm{Nm}_{K/F}$, and its image is $\mathrm{Nm}_{K/F}(U_K^t) = U_F^{\lceil t/e_K \rceil}$ by lemma 11.1.8(2). The resulting character of $U_F^{\lceil t/e_K \rceil}$ extends to a character χ_0 of $F^\times$, since $\mathbb{C}^\times$ is divisible and the extension is taken along the chain of subgroups between $U_F^{\lceil t/e_K \rceil}$ and $F^\times$. Then $\chi_{0,K} = \chi_0 \circ \mathrm{Nm}_{K/F}$ agrees with χ on U_K^t. Replacing τ by $\chi_{0,K}^{-1}\tau$ (which is again Σ-fixed, and which by proposition 11.1.15(2) has τ in the image of $\mathrm{Lft}_{K/F}$ if and only if it does) strictly lowers the level of the twisting character. The iteration terminates: the level drops by at least one at each pass, and once it reaches 0 the inequality $2t < s$ holds, so τ is minimal by lemma 11.1.13(3) and Step 1 applies. That completes the proof.

Read backwards, theorem 11.1.19 says that a totally ramified representation of $\mathrm{GL}_2(K)$ is defined over F exactly when it is Σ-fixed, and that when it is, the representations of $\mathrm{GL}_2(F)$ producing it form a single twist orbit of at most $[K : F]$ elements. An invariance hypothesis, which can be checked, has become an existence statement, which is what the correspondence will need.

The two consequences the next chapter uses are now short. The first needs the criterion of theorem 10.2.6, whose hypotheses have to be discharged over K rather than over F. The transport of the normalization of ψ is the content of the next lemma. Box 10.2.7 shows that renormalizing by a unit instead is not an option: the constant term of the cubic moves and the conclusion can flip.

Lemma 11.1.20: A Special Character Stays Special Under a Tame Extension

Let $\psi \in \widehat{F}$ be special in the sense of definition 10.2.2 and let K/F be finite tamely ramified. Then $\psi_K = \psi \circ \mathrm{tr}_{K/F}$ is special.

***Proof*:**
That ψ_K has level one is lemma 4.1.2(1). Let $y \in \mathfrak{o}_K$ and write $\wp(y) = y + y^2$. Since K/F is tamely ramified, $\mathrm{tr}_{K/F}(\mathfrak{o}_K) = \mathfrak{o}$ and the induced map on residue fields is $\bar{z} \mapsto \overline{e_K}\,\mathrm{tr}_{k_K/k}(\bar{z})$. Here $\overline{e_K} = 1$ in k because e_K is odd and k has characteristic 2. So reduction intertwines $\mathrm{tr}_{K/F}$ with $\mathrm{tr}_{k_K/k}$. In characteristic 2 squaring is a field endomorphism commuting with the Galois

trace, so $\mathrm{tr}_{k_K/k}(\bar{y}^2) = \big(\mathrm{tr}_{k_K/k}(\bar{y})\big)^2$, and therefore

$$\mathrm{tr}_{K/F}\big(\wp(y)\big) \equiv \wp\big(\mathrm{tr}_{K/F}(y)\big) \pmod{\mathfrak{p}}$$

As ψ is trivial on $\mathfrak{p}$ and special, and $\mathrm{tr}_{K/F}(y) \in \mathfrak{o}$, we get $\psi_K(\wp(y)) = \psi(\wp(\mathrm{tr}_{K/F}(y))) = 1$, completing the proof.

Corollary 11.1.21: Cubic Descent for Representations

Let ψ be special, let $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F)$, and assume, when $\mathrm{char}(F) = 2$, that some ramified simple stratum $(\mathfrak{A}, n, \alpha)$ attached to a minimal twist of π has $\mathrm{tr}\alpha \neq 0$. Then there is a tamely ramified extension K/F with $[K : F] \in \{1, 3\}$ such that π_K is ordinary.

***Proof*:**
By lemma 11.1.13(1) write $\pi = \chi\pi'$ with π' minimal, attached to a ramified simple stratum $(\mathfrak{A}, n, \alpha)$, $n = 2m + 1$. When $\mathrm{char}(F) = 2$, take for π' and $(\mathfrak{A}, n, \alpha)$ the minimal twist and the stratum named in the hypothesis, so that $\mathrm{tr}\alpha \neq 0$. The choice has to be made here, because the trace is not an invariant of the stratum attached to π': that stratum is pinned only up to G-conjugacy and the coset $\alpha + \mathfrak{P}^{-m}$, and $\mathrm{tr}(\alpha + \mathfrak{P}^{-m}) = \mathrm{tr}\alpha + \mathrm{tr}\mathfrak{P}^{-m} = \mathrm{tr}\alpha + \mathfrak{p}^{-\lfloor m/2 \rfloor}$, the inclusion $\subseteq$ being lemma 10.1.4 and the reverse holding because (11.2) puts $\mathrm{diag}(z, 0)$ in $\mathfrak{P}^{-m}$ for every $z \in \mathfrak{p}^{-\lfloor m/2 \rfloor}$, a computation tr carries across the G-conjugacy of proposition 3.3.2. So when $\mathrm{tr}\alpha \in \mathfrak{p}^{-\lfloor m/2 \rfloor}$ the same coset also contains ramified simple representatives of trace 0. Being ordinary means lying in $\boldsymbol{\pi}^{\mathrm{im}}(\mathcal{G}_2^{\mathrm{im}}(F))$, and that set is stable under twisting by theorem 8.5.4. By proposition 11.1.15(2) it therefore suffices to treat π'.

If $(\mathfrak{A}, n, \alpha)$ is ordinary, take $K = F$. Otherwise theorem 10.2.6 (whose two hypotheses hold, ψ being special and $\mathrm{tr}\alpha \neq 0$ by the choice just made when $\mathrm{char}(F) = 2$) gives $n \leq 3d_\alpha$ and shows the cubic $C_\alpha(X) = X^3 - \mathrm{tr}(\alpha)X^2 + \det(\alpha)$ has no root in F. That cubic is separable, which in residual characteristic 2 is worth a line: a repeated root is a common root of C_α and $C'_\alpha = X\big(3X - 2\mathrm{tr}(\alpha)\big)$, and $X = 0$ is excluded because $C_\alpha(0) = \det\alpha \neq 0$, so a repeated root would be $2\mathrm{tr}(\alpha)/3 \in F$, against the hypothesis that C_α has no root in F. Let γ be a root of C_α in a separable closure and put $K = F[\gamma]$, so $[K : F] = 3$. Since $p = 2$ the extension K/F is tamely ramified.

By definition 11.1.14(1), $\pi'_K \in \mathcal{A}_2^0(K; \alpha)$, attached to the lifted stratum $(\mathfrak{A}_K, n_K, \alpha)$. Its invariants over K are the same elements $\mathrm{tr}\alpha$ and $\det\alpha$, so the cubic attached to $(\mathfrak{A}_K, n_K, \alpha)$ over K is again C_α, and it has the root $\gamma \in K$. The character ψ_K is special by lemma 11.1.20, and $\mathrm{tr}\alpha \neq 0$ persists, so theorem 10.2.6 applies over K and makes $(\mathfrak{A}_K, n_K, \alpha)$ ordinary. Hence π'_K, and with it π_K, is ordinary, completing the proof.

The last statement of the section is the one the Existence Theorem of the next chapter runs on: descent along the normal closure of a cubic extension that is not cyclic. It is a corollary of theorem 11.1.19, obtained by descending twice, and the arithmetic input is the identification of the norm group of L/F with that of its maximal abelian sub-extension.

Corollary 11.1.22: Descent Along the Normal Closure of a Cubic Extension

Let K/F be a cubic extension, let L/F be its normal closure and write $\Sigma = \mathrm{Gal}(L/F)$. Let $\tau \in \mathcal{A}_2^{\mathrm{wr}}(L)^{\Sigma}$. Then there is $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F)$ with $\pi_L = \tau$, and π is determined up to twisting by a character of $F^\times$ trivial on $\mathrm{Nm}_{L/F}(L^\times)$.

***Proof*:**
Since $p = 2$ and $[K : F] = 3$, the extension K/F is tamely ramified, and so is L/F. If K/F is cyclic then $L = K$ and the assertion is theorem 11.1.19. Assume therefore $\Sigma \cong S_3$.

The tower. The inertia subgroup of Σ is cyclic, L/F being tame (lemma 9.2.10), and normal. The cyclic normal subgroups of S_3 are the trivial one and A_3. It is not trivial, since L/F unramified would make Σ cyclic. So the inertia subgroup is A_3 and $E := L^{A_3}$ is the maximal unramified sub-extension, with $[E : F] = 2$ and L/E totally ramified cyclic of degree 3. Both L/E and E/F are tamely ramified.

First descent. As τ is $\mathrm{Gal}(L/E)$-fixed, theorem 11.1.19 produces $\zeta \in \mathcal{A}_2^{\mathrm{wr}}(E)$ with $\zeta_L = \tau$, and the set S of all such ζ is the orbit of one of them under twisting by characters of $E^\times$ trivial on $\mathrm{Nm}_{L/E}(L^\times)$. By theorem 6.2.2(2) that group of characters is dual to $E^\times/\mathrm{Nm}_{L/E}(L^\times) \cong \mathrm{Gal}(L/E)$ and has order 3; and distinct twists give distinct representations, because $\chi\zeta \cong \zeta$ forces $\chi^2\omega_\zeta = \omega_\zeta$, hence $\chi^2 = 1$, hence $\chi = 1$ for χ of order dividing 3. So $|S| = 3$. By lemma 11.1.16, applied to $F \subseteq E \subseteq L$ and to a lift to Σ of a generator γ of $\mathrm{Gal}(E/F)$, the set S is $\mathrm{Gal}(E/F)$-stable. A permutation of order dividing 2 of a set of three elements has a fixed point, so there is $\zeta^0 \in S$ with $(\zeta^0)^\gamma = \zeta^0$.

Second descent. The extension E/F is unramified quadratic, hence cyclic and tame, and ζ^0 is $\mathrm{Gal}(E/F)$-fixed; theorem 11.1.19 produces $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F)$ with $\pi_E = \zeta^0$. By proposition 11.1.15(3), $\pi_L = (\pi_E)_L = \zeta^0_L = \tau$.

Uniqueness. By proposition 11.1.15(4) applied to L/F, the fibre of $\mathrm{Lft}_{L/F}$ through π is the set of twists $\chi\pi$ with χ trivial on $\mathrm{Nm}_{L/F}(L^\times)$. Since $\Sigma \cong S_3$ has abelianization of order 2 and $E = L^{[\Sigma,\Sigma]}$ is the maximal abelian sub-extension of L/F, theorem 6.2.2(3) gives $\mathrm{Nm}_{L/F}(L^\times) = \mathrm{Nm}_{E/F}(E^\times)$, so the ambiguity is the group of order two generated by the character of $F^\times/\mathrm{Nm}_{E/F}(E^\times)$. This is the asserted uniqueness, completing the proof.

The section's contract is discharged by the two descent statements together. The next chapter meets a totally ramified representation over an extension of F that is fixed by the Galois group and has to produce one over F lifting to it, with the ambiguity known exactly; the extensions it meets are the cyclic ones and the normal closure of a cubic, and those are theorem 11.1.19 and corollary 11.1.22.

Exercise 11.1.1

1. Let E/F be a totally ramified quadratic extension. Show that if K/F is a finite extension with $2 \big| e(K|F)$ then $E \otimes_F K$ need not be a field, by exhibiting an example, and explain which step of lemma 11.1.2 fails. Show also that $E \otimes_F K$ is a field whenever K/F is unramified,

without assuming tameness.

2. Let K/F be unramified, so $e_K = 1$. Show that in the coordinates of the proof of proposition 11.1.4 one has $\mathfrak{A}_K = \mathfrak{o}_K\mathfrak{A}$ and $\mathfrak{P}_K^j = \mathfrak{o}_K\mathfrak{P}^j$ for all j, and deduce proposition 11.1.4(2) in this case directly, without (11.1). What does lemma 11.1.6 assert here?

3. Let K/F be a tamely ramified cubic extension. Show that $\varepsilon_{K/F} = 1$ if K/F is cyclic and $\varepsilon_{K/F} = -1$ if it is not. Deduce that the sign in (11.8) cannot be dropped from the definition of the lifted type.

4. Give the argument of proposition 11.1.15(5) in detail: show that replacing ψ by $u\psi$, $u \in U_F$, replaces the element attached to a fixed character of $U_{\mathfrak{A}}^{m+1}$ by $u^{-1}\alpha$ and leaves $\mathfrak{A}$, $\mathfrak{A}_K$ and Λ unchanged, and hence leaves Λ_K unchanged. Where in the argument is it used that u is a unit?

5. Let K/F be a finite cyclic tamely ramified extension. Show that every non-empty fibre of $\mathrm{Lft}_{K/F}$ has exactly $[K:F]$ elements when $[K:F]$ is odd, and exactly 2 elements when K/F is the unramified quadratic extension. (For the second case, show first that the two elements have the *same* central character, so that central characters cannot separate them, and separate them by their local constants instead.)

6. Prove the two identities used in lemma 11.1.6, namely $U_K^{\lfloor m_K/2\rfloor+1} \cap F = U_F^{\lfloor m/2\rfloor+1}$ and $\mathfrak{p}_K^{-\lfloor m_K/2\rfloor} \cap F = \mathfrak{p}^{-\lfloor m/2\rfloor}$. Show next that the last step of that proof, $U_K^{A+1} \cap F = U_F^{\lfloor A/e_K\rfloor+1}$, uses no parity at all: it holds for every $e_K \geq 1$ and every $A \geq 0$. Where, then, is the oddness of e_K used, and what becomes of the two displayed identities when e_K is even?

7. Let π' be minimal and totally ramified, attached to a ramified simple stratum $(\mathfrak{A}, n, \alpha)$, and let χ have level l with $2l > n$. Show that $\chi\pi'$ contains a character of $U_{\mathfrak{A}}^{2l}$ that is non-trivial and factors through det, that $\ell(\chi\pi') = l$, and that $\chi\pi'$ is therefore not minimal. Deduce that the level of a non-minimal totally ramified representation is an integer.

11.2 Interior Actions

Section 11.1 showed that the image of $\mathrm{Lft}_{K/F}$ consists of $\mathrm{Gal}(K/F)$-fixed points whenever K/F is Galois, and identified it with the full fixed set in two cases: K/F cyclic (theorem 11.1.19) and the normal closure of a non-cyclic cubic (corollary 11.1.22). Any such description presupposes an action constructed on both sides of the correspondence and shown to commute with it. The action is transport along a field isomorphism: an automorphism γ of K moves a representation of $\mathrm{GL}_2(K)$ to another one, and moves a representation of $\mathcal{W}_K$ to another one, and the question is whether the correspondence between the two sides is preserved by the move. This section builds the action on both sides and proves that it does.

Almost nothing below is dyadic, and it is worth saying exactly how much. Everything up to and including proposition 11.2.9, together with clause (1) of corollary 11.2.10, holds for an arbitrary residual characteristic: the inputs are the characterization of the local constant by its three axioms, theorem 6.3.2, the Converse Theorem, theorem 5.5.2, and the imprimitive correspondence with its ε-identity, theorem 8.5.4 and (8.30), none of which restricts the residual characteristic. Two statements are exceptions, and for them the chapter's standing hypothesis $p = 2$ applies. Clause (2) of corollary 11.2.10 is proved by invoking corollary 9.3.11, which chapter 9 proves for $p = 2$ only;

and proposition 11.2.11 is a statement about $\mathrm{Lft}_{K/F}$, which exists in this book only as the map constructed in section 11.1 under that same hypothesis. Both are flagged again where they occur.

Throughout, an *isomorphism of local fields* $\varphi\colon F \to F'$ means an isomorphism of fields with $\varphi(\mathfrak{o}) = \mathfrak{o}'$, where $\mathfrak{o}$ and $\mathfrak{o}'$ are the discrete valuation rings of F and F'. The condition is automatic in the only case this chapter applies the theory to, namely $\varphi \in \mathrm{Gal}(K/F)$ acting on a finite Galois extension K of F: there $\mathfrak{o}_K$ is the integral closure of $\mathfrak{o}_F$ in K, and φ fixes F pointwise and so preserves that integral closure.

Lemma 11.2.1: Transport of Galois and Weil Groups

Let $\varphi\colon F \to F'$ be an isomorphism of local fields.

1. $\varphi(\mathfrak{p}) = \mathfrak{p}'$ and φ induces an isomorphism $k \to k'$ of residue fields. In particular $q = q'$, and $v_{F'}(\varphi x) = v_F(x)$, $\|\varphi x\|_{F'} = \|x\|_F$ for $x \in F^\times$. A character $\eta \in \widehat{F'}$ and the character $\eta \circ \varphi \in \widehat{F}$ have the same level.

2. Fix separable closures $\overline{F}/F$ and $\overline{F'}/F'$. Then φ extends to an isomorphism $\tilde{\varphi}\colon \overline{F} \to \overline{F'}$, and any two such extensions differ by $\tilde{\varphi} \mapsto \tau\tilde{\varphi}$ with $\tau \in \Omega_{F'}$.

3. The map
$$\varphi_*\colon \Omega_F \longrightarrow \Omega_{F'}, \qquad \sigma \longmapsto \tilde{\varphi} \circ \sigma \circ \tilde{\varphi}^{-1}$$
is an isomorphism of topological groups carrying $\mathcal{I}_F$ onto $\mathcal{I}_{F'}$, $\mathcal{P}_F$ onto $\mathcal{P}_{F'}$ and $\mathcal{W}_F$ onto $\mathcal{W}_{F'}$, and carrying geometric Frobenius elements to geometric Frobenius elements. It restricts to a homeomorphic isomorphism $\mathcal{W}_F \to \mathcal{W}_{F'}$ for the Weil topologies. For a finite separable E/F inside $\overline{F}$, writing $E' = \tilde{\varphi}(E)$, it carries Ω_E onto $\Omega_{E'}$ and $\mathcal{W}_E$ onto $\mathcal{W}_{E'}$.

4. Let ρ be a smooth finite-dimensional representation of $\mathcal{W}_{F'}$ and let $\tau \in \Omega_{F'}$. Then $\rho \circ \mathrm{Int}(\tau) \cong \rho$ as representations of $\mathcal{W}_{F'}$, where $\mathrm{Int}(\tau)(x) = \tau x \tau^{-1}$. Consequently the isomorphism class of $\rho \circ \varphi_*$ does not depend on the choice of $\tilde{\varphi}$ made in (2).

***Proof*:**
(1). The maximal ideal is the unique maximal ideal of the valuation ring, so $\varphi(\mathfrak{p}) = \mathfrak{p}'$ and $\varphi(\mathfrak{p}^j) = \mathfrak{p}'^j$ for all $j \in \mathbb{Z}$. Passing to quotients gives $k \cong k'$, whence $q = q'$; and $v_{F'}(\varphi x) = v_F(x)$ because φ matches the filtrations by powers of the maximal ideals. The level of a character is measured by the largest power of the maximal ideal on which it is trivial, and $\eta \circ \varphi$ is trivial on $\mathfrak{p}^j$ exactly when η is trivial on $\mathfrak{p}'^j$.

(2). Regard $\overline{F'}$ as an extension of F through φ, so that both closures sit over the same base field and the identity on F is available as the map to extend. Fix algebraic closures containing $\overline{F}$ and $\overline{F'}$ respectively. By [4, Uniqueness Of Algebraic Closure] the two algebraic closures admit an isomorphism extending the identity on F. Such an isomorphism carries an element separable over F to an element separable over F, since it fixes F and therefore matches minimal polynomials, and separability is a property of the minimal polynomial. It consequently restricts to an isomorphism of the subfields of separable elements, which are $\overline{F}$ and $\overline{F'}$. The restriction to separable elements is needed because F^{sep} is a proper subfield of the algebraic closure whenever F is imperfect, so uniqueness of the algebraic closure does not by itself deliver uniqueness of the

separable one. The result is an isomorphism $\tilde{\varphi}\colon \overline{F} \to \overline{F'}$ restricting to φ on F. If $\tilde{\varphi}_1$ is another, then $\tau = \tilde{\varphi}_1\tilde{\varphi}^{-1}$ is an automorphism of $\overline{F'}$ fixing F' pointwise, that is, an element of $\Omega_{F'}$.

(3). Conjugation by a field isomorphism is a group isomorphism $\mathrm{Aut}(\overline{F}/F) \to \mathrm{Aut}(\overline{F'}/F')$. It is a homeomorphism because it carries the basic open subgroup Ω_E onto $\Omega_{E'}$ for every finite E/F, and these form a neighbourhood basis of the identity on each side. That observation is also the last clause of (3).

Since $\tilde{\varphi}(\mathfrak{o}_F) = \mathfrak{o}_{F'}$ and valuation rings in algebraic extensions are integral closures, $\tilde{\varphi}$ carries $\mathfrak{o}_E$ onto $\mathfrak{o}_{E'}$ for every finite E/F, so it preserves ramification indices and residue degrees and carries the maximal unramified extension F_∞ of F onto F'_∞. Hence $\varphi_*(\mathcal{I}_F) = \varphi_*\mathrm{Gal}(\overline{F}/F_\infty) = \mathrm{Gal}(\overline{F'}/F'_\infty) = \mathcal{I}_{F'}$. By proposition 6.1.2 the group $\mathcal{P}_F$ is the unique pro-p-Sylow subgroup of $\mathcal{I}_F$, and $p = p'$ by (1), so $\varphi_*(\mathcal{P}_F) = \mathcal{P}_{F'}$.

For the Frobenius statement, let $\overline{k}$ and $\overline{k'}$ be the residue fields of $\overline{F}$ and $\overline{F'}$. The map $\tilde{\varphi}$ induces an isomorphism $\overline{k} \to \overline{k'}$ intertwining the residual action of σ with that of $\varphi_*\sigma$. As $q = q'$, the element σ acts on $\overline{k}$ by $x \mapsto x^{q^{-1}}$ if and only if $\varphi_*\sigma$ acts on $\overline{k'}$ in the same way. So φ_* induces the identity on $\Omega_F/\mathcal{I}_F \cong \widehat{\mathbb{Z}} \cong \Omega_{F'}/\mathcal{I}_{F'}$, and $\mathcal{W}_F$, the preimage of $\mathbb{Z}$, goes onto $\mathcal{W}_{F'}$. The Weil topology is determined by declaring the inertia subgroup open with its profinite topology (definition 6.1.4), and φ_* is a homeomorphism $\mathcal{I}_F \to \mathcal{I}_{F'}$, so it is one for the Weil topologies.

(4). Write $\Omega = \Omega_{F'}$, $\mathcal{W} = \mathcal{W}_{F'}$, $\mathcal{I} = \mathcal{I}_{F'}$.

First, $\mathcal{W}$ is dense in Ω: its closure is a closed subgroup containing $\mathcal{I}$ whose image in $\Omega/\mathcal{I} \cong \widehat{\mathbb{Z}}$ is closed and contains $\mathbb{Z}$, hence is all of $\widehat{\mathbb{Z}}$.

Put $H = \ker\rho \cap \mathcal{I}$. Smoothness and finite-dimensionality make $\ker\rho$ open in $\mathcal{W}$, so H is an open, hence closed, subgroup of the compact group $\mathcal{I}$, and it is normalized by $\mathcal{W}$. Its normalizer in Ω is $\bigcap_{h\in H}\left\{g \in \Omega : ghg^{-1} \in H\right\}$, an intersection of closed sets and therefore closed. It contains the dense subgroup $\mathcal{W}$, so it is all of Ω and H is normal in Ω.

Fix a geometric Frobenius $\Phi \in \mathcal{W}$ and let D be the closure of $\langle\Phi\rangle$ in Ω. The image of D in $\Omega/\mathcal{I} \cong \widehat{\mathbb{Z}}$ is closed and contains $\mathbb{Z}$, so it is everything, and $\Omega = \mathcal{I}D$. Write $\tau = i\theta$ with $i \in \mathcal{I} \subseteq \mathcal{W}$ and $\theta \in D$. Then $\rho \circ \mathrm{Int}(i) \cong \rho$ by way of the intertwiner $\rho(i)$, so it suffices to treat $\tau = \theta$.

The conjugation action of D on the finite group $\mathcal{I}/H$ is a continuous homomorphism into a finite discrete group, so its kernel U is open in D. Since $\langle\Phi\rangle$ is dense in D, the open set θU contains some Φ^b with $b \in \mathbb{Z}$, and then $\mathrm{Int}(\theta)$ and $\mathrm{Int}(\Phi^b)$ agree on $\mathcal{I}/H$. They also agree on Φ, since D is abelian and so $\theta\Phi\theta^{-1} = \Phi = \Phi^b\Phi\Phi^{-b}$. Every element of $\mathcal{W}$ has the form $x\Phi^k$ with $x \in \mathcal{I}$ and $k \in \mathbb{Z}$, and

$$\mathrm{Int}(\theta)(x\Phi^k) = (\theta x\theta^{-1})\Phi^k, \qquad \mathrm{Int}(\Phi^b)(x\Phi^k) = (\Phi^b x\Phi^{-b})\Phi^k$$

with $\theta x\theta^{-1}$ and $\Phi^b x\Phi^{-b}$ in the same coset of H. As ρ is trivial on H, the two composites $\rho \circ \mathrm{Int}(\theta)$ and $\rho \circ \mathrm{Int}(\Phi^b)$ are equal, and the second is isomorphic to ρ through $\rho(\Phi^b)$.

Finally, replacing $\tilde{\varphi}$ by $\tau\tilde{\varphi}$ replaces φ_* by $\mathrm{Int}(\tau) \circ \varphi_*$, so it replaces $\rho \circ \varphi_*$ by $(\rho \circ \mathrm{Int}(\tau)) \circ \varphi_*$, which is isomorphic to $\rho \circ \varphi_*$ by the case already treated, completing the proof.

Clause (4) read with $F' = F$ and φ the identity is the statement deferred in chapter 6: conjugation by an element of Ω_F does not move the isomorphism class of a smooth semisimple representation of $\mathcal{W}_F$, which is what makes $\mathcal{G}_n^{\mathrm{ss}}(F)$ independent of the separable closure and the Frobenius element

chosen to define $\mathcal{W}_F$. The transport of that independence to a second field, and its compatibility with induction, is what the rest of this section is for.

The lemma moves the groups. The objects attached to them move by composition with the maps it produces. Additive characters, characters of the multiplicative group, Galois parameters and cuspidal representations are all functions on something that φ or φ_* acts on, so all four transport by the same device, and the four rules are worth recording together under one name.

Definition 11.2.2: Interior Action

Let $\varphi\colon F \to F'$ be an isomorphism of local fields and fix an extension $\tilde{\varphi}$ and the induced φ_* of lemma 11.2.1. The *interior action of* φ is the following collection of maps, all written $(\,\cdot\,)^{\varphi}$.

1. For $\eta \in \widehat{F'}$, put $\eta^{\varphi} = \eta \circ \varphi \in \widehat{F}$.
2. For a smooth character χ of $F'^{\times}$, put $\chi^{\varphi} = \chi \circ \varphi$, a smooth character of $F^{\times}$.
3. For $\rho \in \mathcal{G}^{\mathrm{ss}}(F')$, put $\rho^{\varphi} = \rho \circ \varphi_*$, a representation of $\mathcal{W}_F$ of the same dimension. This gives bijections $\mathcal{G}_n^{\mathrm{ss}}(F') \to \mathcal{G}_n^{\mathrm{ss}}(F)$ and $\mathcal{G}_n^{0}(F') \to \mathcal{G}_n^{0}(F)$.
4. Apply φ to matrix entries to get an isomorphism of topological groups $\mathrm{GL}_2(F) \to \mathrm{GL}_2(F')$, again written φ. For $\pi \in \mathcal{A}_2^0(F')$, put $\pi^{\varphi} = \pi \circ \varphi$. This gives a bijection $\mathcal{A}_2^0(F') \to \mathcal{A}_2^0(F)$.

Clause (3) is well posed on isomorphism classes by lemma 11.2.1(4), and it does send semisimple to semisimple and irreducible to irreducible, because φ_* is an isomorphism of topological groups and so matches subrepresentations on the two sides. The same remark applies to (4).

The direction of transport is fixed once here and used without further comment.

11.2.3 Convention: the direction of transport The interior action is *contravariant*: for $\varphi\colon F \to F'$, the operation $(\,\cdot\,)^{\varphi}$ carries objects over F' to objects over F. If $\varphi\colon F \to F'$ and $\theta\colon F' \to F''$ are isomorphisms of local fields, then

$$(\,\cdot\,)^{\theta\circ\varphi} = \big((\,\cdot\,)^{\theta}\big)^{\varphi}$$

on each of the four kinds of object, as one sees at once from $\chi \circ (\theta\varphi) = (\chi \circ \theta) \circ \varphi$ and from $(\theta\varphi)_* = \theta_* \circ \varphi_*$ for compatible choices of extensions.
In particular, for a finite Galois extension K/F with $\Sigma = \mathrm{Gal}(K/F)$, the assignment $\gamma \mapsto (\,\cdot\,)^{\gamma}$ is a *right* action of Σ on each of $\widehat{K}$, the characters of $K^{\times}$, $\mathcal{G}^{\mathrm{ss}}(K)$ and $\mathcal{A}_2^0(K)$: $(\rho^{\gamma})^{\gamma'} = \rho^{\gamma\gamma'}$. An object is Σ-*fixed* if $\rho^{\gamma} \cong \rho$ for every $\gamma \in \Sigma$, and $\mathcal{G}^{\mathrm{ss}}(K)^{\Sigma}$, $\mathcal{A}_2^0(K)^{\Sigma}$ denote the sets of Σ-fixed classes.
Section 11.1 writes the same superscript for the opposite motion, and the two must not be run together. There $(\,\cdot\,)^{\gamma}$ is the *push-forward* along γ, on objects over a single field and its extensions rather than between two fields: a character goes to $\chi\circ\gamma^{-1}$, a stratum $(\mathfrak{A}, n, \alpha)$ goes to $(\gamma\mathfrak{A}, n, \gamma\alpha)$, and ψ goes to $\psi\circ\gamma^{-1}$, so that $\gamma \mapsto (\,\cdot\,)^{\gamma}$ is there a *left* action. The dictionary is one line and nothing more: that section's $(\,\cdot\,)^{\gamma}$ is this section's $(\,\cdot\,)^{\gamma^{-1}}$, the left action $\gamma\cdot(\,\cdot\,) = (\,\cdot\,)^{\gamma^{-1}}$ associated with the right action above. In particular the two conventions have the same fixed sets, since $\gamma \mapsto \gamma^{-1}$ is a bijection of Σ, so $\mathcal{A}_2^{\mathrm{wr}}(K)^{\Sigma}$ in theorem 11.1.19 is the set of Σ-fixed classes in the sense used here. Which of the two motions is called $(\,\cdot\,)^{\gamma}$ is a matter of convention. From here to the end of this section, the contravariant rule of the first paragraph is the one in use.

In the Galois case the ambiguity in $\tilde{\gamma}$ can be cut down to something visibly harmless. Let K/F be finite Galois and $\gamma \in \Sigma = \mathrm{Gal}(K/F)$. The restriction map $\mathcal{W}_F \to \mathrm{Gal}(K/F)$ is surjective, because $\mathcal{W}_F$ is dense in Ω_F and $\mathrm{Gal}(K/F)$ is a discrete quotient of it, so γ has an extension $\tilde{\gamma} \in \mathcal{W}_F$; and $\mathcal{W}_K = \mathcal{W}_F \cap \Omega_K$ is normal in $\mathcal{W}_F$, so $\gamma_* = \mathrm{Int}(\tilde{\gamma})|_{\mathcal{W}_K}$. Two such choices $\tilde{\gamma}_1$ and $\tilde{\gamma}_2$ differ by the element $\tau = \tilde{\gamma}_1\tilde{\gamma}_2^{-1}$, which lies in $\mathcal{W}_K$, and conjugation by an element of $\mathcal{W}_K$ changes a representation ρ of $\mathcal{W}_K$ only by the inner intertwiner $\rho(\tau)$. So in the Galois case one never has to appeal to lemma 11.2.1(4).

Example 11.2.4: The interior action through class field theory
Let $\varphi\colon F \to F'$ be an isomorphism of local fields and let χ be a character of $F'^{\times}$, so that $\chi \circ \mathbf{a}_{F'}$ is a character of $\mathcal{W}_{F'}$. Theorem 6.2.1(5) says $\varphi \circ \mathbf{a}_F = \mathbf{a}_{F'} \circ \varphi_*$ on $\mathcal{W}_F^{\mathrm{ab}}$. Hence

$$(\chi \circ \mathbf{a}_{F'})^{\varphi} = \chi \circ \mathbf{a}_{F'} \circ \varphi_* = \chi \circ \varphi \circ \mathbf{a}_F = \chi^{\varphi} \circ \mathbf{a}_F \tag{11.10}$$

so the interior action on one-dimensional Galois parameters is the interior action on characters of $F'^{\times}$, transported by the Artin map. Specializing to $F = F' = K$ and $\varphi = \gamma \in \mathrm{Gal}(K/F)$ gives $(\chi \circ \mathbf{a}_K)^{\gamma} = \chi^{\gamma} \circ \mathbf{a}_K$. This is the mechanism behind every computation below: the two sides of the correspondence are moved by the same field isomorphism, and class field theory is what makes the two motions agree in dimension one.

The local constants are built from Fourier analysis on F, so the first thing to transport is the measure.

Lemma 11.2.5: Transport of the Self-Dual Measure

Let $\varphi\colon F \to F'$ be an isomorphism of local fields and let $\eta \in \widehat{F'}$ be non-trivial. Write $\varphi_*\mu$ for the push-forward $(\varphi_*\mu)(S) = \mu(\varphi^{-1}S)$ of a measure μ on F. Then

$$\varphi_*\big(\mu_{\eta^\varphi}\big) = \mu_\eta$$

where μ_η is the self-dual Haar measure of definition 5.1.3. Moreover, for $\Phi' \in C_c^\infty(F')$ and $\Phi = \Phi' \circ \varphi$,

$$\widehat{\Phi'} \circ \varphi = \widehat{\Phi} \tag{11.11}$$

the transforms being taken relative to (η, μ_η) on the left and $(\eta^\varphi, \mu_{\eta^\varphi})$ on the right, and $\varphi_*(\mu^*) = (\varphi_*\mu)^*$ for the multiplicative measures of definition 5.1.5.

***Proof*:**
The map φ is an isomorphism of topological groups $(F,+) \to (F',+)$, so $\varphi_*\mu$ is a Haar measure on F' whenever μ is one on F, and $\Phi' \mapsto \Phi' \circ \varphi$ is a bijection $C_c^\infty(F') \to C_c^\infty(F)$.

Let μ be any Haar measure on F and compute the Fourier transform of Φ' relative to η and $\varphi_*\mu$. For $x \in F$,

$$\widehat{\Phi'}(\varphi x) = \int_{F'} \Phi'(y')\,\eta\big((\varphi x)y'\big)\,d(\varphi_*\mu)(y') = \int_F \Phi'(\varphi y)\,\eta\big(\varphi(xy)\big)\,d\mu(y) = \int_F \Phi(y)\,\eta^\varphi(xy)\,d\mu(y)$$

which is the Fourier transform of Φ relative to η^φ and μ, evaluated at x. This is (11.11) once the measures are the self-dual ones, and it holds for any μ.

Take $\mu = \mu_{\eta^\varphi}$ and iterate. The display gives $\widehat{\widehat{\Phi'}}(\varphi x) = \widehat{\widehat{\Phi}}(x)$, and Fourier inversion over F (definition 5.1.4) gives $\widehat{\widehat{\Phi}}(x) = \Phi(-x) = \Phi'(-\varphi x)$. So $\widehat{\widehat{\Phi'}}(y') = \Phi'(-y')$ for every $y' \in F'$, that is, the constant $c(\eta, \varphi_*\mu_{\eta^\varphi})$ of proposition 5.1.2(2) equals 1. By proposition 5.1.2(3) there is exactly one Haar measure on F' with that property, namely μ_η, so $\varphi_*\mu_{\eta^\varphi} = \mu_\eta$.

For the last clause, $d\mu^*(x) = d\mu(x)/\|x\|_F$ and $\|\varphi x\|_{F'} = \|x\|_F$ by lemma 11.2.1(1), so pushing forward the additive measure and dividing by the norm can be done in either order, completing the proof.

Lemma 11.2.6: Interior Action and the Local Constant of a Character

Let $\varphi\colon F \to F'$ be an isomorphism of local fields, let χ be a character of $F'^\times$ and let $\eta \in \widehat{F'}$ be non-trivial. Then $\check{\chi}^\varphi = (\chi^\varphi)^\vee$ and

$$L(\chi^\varphi, s) = L(\chi, s), \qquad \varepsilon(\chi^\varphi, s, \eta^\varphi) = \varepsilon(\chi, s, \eta)$$

***Proof*:**
The contragredient of a character is its inverse, and $(\chi^{-1})\circ\varphi = (\chi\circ\varphi)^{-1}$, which is the first assertion.

Step 1: the zeta integrals agree. Fix a prime element ϖ of F. Then $\varphi(\varpi)$ is a prime element of F' and $\varphi\big(\varpi^m U_F\big) = \varphi(\varpi)^m U_{F'}$ by lemma 11.2.1(1). For $\Phi' \in C_c^\infty(F')$ and $\Phi = \Phi'\circ\varphi$, the coefficients (5.2) computed with the measures of lemma 11.2.5 satisfy

$$z_m(\Phi,\chi^\varphi) = \int_{\varpi^m U_F} \Phi'(\varphi x)\,\chi(\varphi x)\,d\mu^*_{\eta^\varphi}(x) = \int_{\varphi(\varpi)^m U_{F'}} \Phi'(y)\,\chi(y)\,d\mu^*_{\eta}(y) = z_m(\Phi',\chi)$$

Hence $Z(\Phi,\chi^\varphi,X) = Z(\Phi',\chi,X)$ for every Φ', and since $\Phi'\mapsto\Phi'\circ\varphi$ is a bijection of $C_c^\infty(F')$ onto $C_c^\infty(F)$, the two $\mathbb{C}[X,X^{-1}]$-modules $\mathcal{Z}(\chi^\varphi)$ and $\mathcal{Z}(\chi)$ of definition 5.1.7 coincide. Their normalized generator is the same polynomial, so $L(\chi^\varphi,s) = L(\chi,s)$. The same applies to $\check{\chi}$.

Step 2: the functional equations correspond. By (11.11), the pair $(\widehat{\Phi'},\widehat{\Phi})$ is again of the form $(\Psi',\Psi'\circ\varphi)$. So Step 1 applies to it as well, and the functional equation of theorem 5.1.11 over F',

$$\zeta(\widehat{\Phi'},\check{\chi},1-s) = \gamma(\chi,s,\eta)\,\zeta(\Phi',\chi,s)$$

becomes, term by term, $\zeta(\widehat{\Phi},(\check{\chi^\varphi}),1-s) = \gamma(\chi,s,\eta)\,\zeta(\Phi,\chi^\varphi,s)$. Choosing Φ' with $\zeta(\Phi',\chi,s)\neq 0$ and comparing with the functional equation over F gives $\gamma(\chi^\varphi,s,\eta^\varphi) = \gamma(\chi,s,\eta)$. Both L-factors in the defining relation of definition 5.1.13 were matched in Step 1, so the ε-factors agree, completing the proof.

The Galois-side statement now follows from the axiomatic characterization of ε rather than from any computation. That is the point of theorem 6.3.2: a family of functions satisfying its three properties is unique, so any competitor built by transport must be the family itself.

Proposition 11.2.7: Interior Action and the Langlands-Deligne Constant

Let $\varphi\colon F\to F'$ be an isomorphism of local fields and let $\eta\in\widehat{F'}$ be non-trivial. Then

$$\varepsilon(\rho^\varphi,s,\eta^\varphi) = \varepsilon(\rho,s,\eta)$$

for every $\rho\in\mathcal{G}^{\mathrm{ss}}(F')$. More generally, for every finite separable E/F inside $\overline{F}$ and every $\sigma\in\mathcal{G}^{\mathrm{ss}}(E')$, where $E' = \tilde{\varphi}(E)$,

$$\varepsilon\big(\sigma^{\varphi_E},s,(\eta_{E'})^{\varphi_E}\big) = \varepsilon(\sigma,s,\eta_{E'}),\qquad \varphi_E = \tilde{\varphi}|_E$$

***Proof*:**
Fix $\tilde{\varphi}$ once and for all and write $\varphi_E = \tilde{\varphi}|_E\colon E\to E'$ for each finite E/F inside $\overline{F}$. This is an isomorphism of local fields extending φ, and $(\varphi_E)_* = \varphi_*|_{\Omega_E}$ by lemma 11.2.1(3). Set $\psi = \eta^\varphi$.

Step 1: the additive characters match up the tower. For $x\in E$,

$$(\eta_{E'})^{\varphi_E}(x) = \eta\big(\mathrm{tr}_{E'/F'}(\varphi_E x)\big) = \eta\big(\varphi(\mathrm{tr}_{E/F}x)\big) = \psi_E(x)$$

the middle equality because φ_E is an isomorphism of the extension E/F onto E'/F' and therefore matches the two trace forms. Also $q_E = q_{E'}$ by lemma 11.2.1(1), so the target ring $\mathbb{C}[q_E^s, q_E^{-s}]^\times$ of theorem 6.3.2 is the same over E as over E'.

Step 2: transport commutes with induction. Let $F \subseteq K \subseteq E$ be finite and let σ be a smooth representation of $\mathcal{W}_{E'}$ on V. Realize $\mathrm{Ind}_{E'/K'}\sigma$ on the space of functions $f\colon \mathcal{W}_{K'} \to V$ with $f(hx) = \sigma(h)f(x)$ for $h \in \mathcal{W}_{E'}$, with $\mathcal{W}_{K'}$ acting by right translation. By lemma 11.2.1(3) the map φ_* carries the pair $(\mathcal{W}_K, \mathcal{W}_E)$ isomorphically onto $(\mathcal{W}_{K'}, \mathcal{W}_{E'})$, so $f \mapsto f \circ \varphi_*$ is a linear isomorphism onto the corresponding space for $\mathrm{Ind}_{E/K}(\sigma^{\varphi_E})$, and it intertwines the two right-translation actions. Hence

$$\left(\mathrm{Ind}_{E'/K'}\sigma\right)^{\varphi_K} = \mathrm{Ind}_{E/K}\left(\sigma^{\varphi_E}\right) \tag{11.12}$$

Taking $\sigma = 1_{E'}$ gives $1_{E'}^{\varphi_E} = 1_E$ and $R_{E'/K'}^{\varphi_K} = R_{E/K}$.

Step 3: the transported family. For each finite E/F inside $\overline{F}$ and each $\tau \in \mathcal{G}^{\mathrm{ss}}(E)$ there is a unique $\sigma \in \mathcal{G}^{\mathrm{ss}}(E')$ with $\tau = \sigma^{\varphi_E}$, by definition 11.2.2(3). Define

$$\varepsilon^\sharp(\tau, s, \psi_E) = \varepsilon(\sigma, s, \eta_{E'})$$

a family of functions $\mathcal{G}^{\mathrm{ss}}(E) \to \mathbb{C}[q_E^s, q_E^{-s}]^\times$ indexed by the finite extensions E/F inside $\overline{F}$, which by Step 1 is a family of the shape considered in theorem 6.3.2 for the character ψ of F.

Dimension 1. Let χ be a character of $E^\times$ and let ξ be the character of $E'^\times$ with $\chi = \xi^{\varphi_E}$. Theorem 6.2.1(5) gives $\varphi_E \circ \mathbf{a}_E = \mathbf{a}_{E'} \circ (\varphi_E)_*$, whence $(\xi \circ \mathbf{a}_{E'})^{\varphi_E} = \chi \circ \mathbf{a}_E$. So

$$\varepsilon^\sharp(\chi \circ \mathbf{a}_E, s, \psi_E) = \varepsilon(\xi \circ \mathbf{a}_{E'}, s, \eta_{E'}) = \varepsilon(\xi, s, \eta_{E'}) = \varepsilon(\chi, s, \psi_E)$$

the second equality by property (1) of theorem 6.3.2 over F' and the third by lemma 11.2.6 together with Step 1.

Additivity. $(\sigma_1 \oplus \sigma_2)^{\varphi_E} = \sigma_1^{\varphi_E} \oplus \sigma_2^{\varphi_E}$, so additivity of $\varepsilon^\sharp$ is additivity of ε over F'.

Inductivity. For $E \supseteq K \supseteq F$ and $\sigma \in \mathcal{G}_n^{\mathrm{ss}}(E')$, Step 2 turns (6.3) over F' into

$$\varepsilon^\sharp\left(\mathrm{Ind}_{E/K}\sigma^{\varphi_E}, s, \psi_K\right) = \varepsilon^\sharp\left(\sigma^{\varphi_E}, s, \psi_E\right)\left(\frac{\varepsilon^\sharp(R_{E/K}, s, \psi_K)}{\varepsilon^\sharp(1_E, s, \psi_E)}\right)^n$$

which is (6.3) for $\varepsilon^\sharp$.

Step 4: uniqueness. The family $\varepsilon^\sharp$ satisfies the three properties that theorem 6.3.2 shows determine a unique family for the character ψ of F. Hence $\varepsilon^\sharp(\tau, s, \psi_E) = \varepsilon(\tau, s, \psi_E)$ for every E and every $\tau \in \mathcal{G}^{\mathrm{ss}}(E)$, which is the second display of the proposition. Taking $E = F$ gives the first, completing the proof.

The group side has no such uniqueness theorem behind it, but it needs none: the Godement-Jacquet construction is built out of the same three ingredients that lemma 11.2.5 already transports, namely the additive character, the self-dual measure and the modulus.

Proposition 11.2.8: Interior Action and the Godement-Jacquet Constant

Let $\varphi\colon F \to F'$ be an isomorphism of local fields, let $\eta \in \widehat{F'}$ be non-trivial, let $\pi \in \mathcal{A}_2^0(F')$ and let χ be a character of $F'^{\times}$. Then

1. $\pi^{\varphi} \in \mathcal{A}_2^0(F)$, $\check{\pi}^{\varphi} = (\check{\pi}^{\varphi})$, and the central characters satisfy $\omega_{\pi^{\varphi}} = (\omega_{\pi})^{\varphi}$;
2. $(\chi\pi)^{\varphi} = \chi^{\varphi}\,\pi^{\varphi}$;
3. $L(\chi^{\varphi}\pi^{\varphi}, s) = L(\chi\pi, s) = 1$ and

$$\varepsilon(\pi^{\varphi}, s, \eta^{\varphi}) = \varepsilon(\pi, s, \eta)$$

***Proof*:**

(1) and (2). Applying φ to matrix entries is an isomorphism of topological groups $G = \mathrm{GL}_2(F) \to G' = \mathrm{GL}_2(F')$, so $\pi^{\varphi} = \pi \circ \varphi$ is smooth and irreducible, and it is cuspidal because φ carries the unipotent radical of a Borel subgroup of G onto one of G' and therefore matches the Jacquet modules. On the centre, $\varphi(zI) = \varphi(z)I$, so $\omega_{\pi^{\varphi}}(z) = \omega_{\pi}(\varphi z)$. Composition with φ commutes with taking contragredients, since $\check{\pi}^{\varphi}$ and $(\check{\pi}^{\varphi})$ act on the same smooth dual by the same formula, and with twisting: $(\chi\pi)^{\varphi}(g) = \chi(\det \varphi g)\pi(\varphi g) = \chi^{\varphi}(\det g)\,\pi^{\varphi}(g)$, using $\det(\varphi g) = \varphi(\det g)$.

(3), the L-factors. Both $\chi\pi$ and $\chi^{\varphi}\pi^{\varphi}$ are irreducible cuspidal, so both L-factors are 1 by proposition 5.3.1.

(3), the ε-factors, Step 1: the matrix algebra transports. Write $A = M_2(F)$ and $A' = M_2(F')$, and let $\varphi\colon A \to A'$ again denote the entrywise map. Then $\mathrm{tr}_{A'}(\varphi x) = \varphi(\mathrm{tr}_A x)$ and $\det(\varphi x) = \varphi(\det x)$, so

$$\eta_{A'} \circ \varphi = (\eta^{\varphi})_A, \qquad \|\det \varphi g\|_{F'} = \|\det g\|_F$$

the second by lemma 11.2.1(1). As topological groups, A and A' are products of four copies of F and of F'. The character $\eta_{A'}$ is *not* the product of four copies of η (it is $\eta(x_{11} + x_{22})$, and is trivial on the off-diagonal coordinates) but the duality pairing it defines is: $\eta_{A'}(xy) = \eta(\mathrm{tr}(xy))$ matches x_{11} with y_{11}, x_{22} with y_{22}, x_{12} with y_{21} and x_{21} with y_{12}, so it is the product of four copies of the pairing $(a, b) \mapsto \eta(ab)$ on F', composed with a permutation of the coordinates; and likewise for $(\eta^{\varphi})_A$ over F. Permuting coordinates changes no product measure and commutes with the entrywise map φ, so the argument of lemma 11.2.5, applied in each coordinate, shows that φ carries the measure on A self-dual for $(\eta^{\varphi})_A$ to the measure on A' self-dual for $\eta_{A'}$, and that $\widehat{\Phi'} \circ \varphi = \widehat{\Phi' \circ \varphi}$ for $\Phi' \in C_c^{\infty}(A')$.

Step 2: the zeta integrals transport. Fix a Haar measure $d^{\times}g$ on G and give G' its push-forward. If $f \in C(\pi)$ is the matrix coefficient $g' \mapsto \langle \tilde{v}, \pi(g')v\rangle$, then $f \circ \varphi$ is the matrix coefficient of π^{φ} attached to the same pair of vectors, and $f \mapsto f \circ \varphi$ is a bijection $C(\pi) \to C(\pi^{\varphi})$ carrying $\check{f}$ to $(f \mathbin{\check{\circ}} \varphi)$, because $\varphi(g^{-1}) = \varphi(g)^{-1}$. So for $\Phi' \in C_c^{\infty}(A')$ and $\Phi = \Phi' \circ \varphi$,

$$\zeta(\Phi, f \circ \varphi, s) = \int_G \Phi'(\varphi g) f(\varphi g) \|\det \varphi g\|^s \, d^{\times}g = \zeta(\Phi', f, s)$$

by Step 1 and the choice of measure on G'.

Step 3: comparison. By Steps 1 and 2 the two sides of the Godement-Jacquet functional equation (5.10) for (π, η) over F' are literally the two sides of the functional equation for $(\pi^{\varphi}, \eta^{\varphi})$ over F, evaluated on the corresponding data. Take $f \in C(\pi)$ with $f(1) \neq 0$ and let Φ' be the characteristic function of a compact open neighbourhood of 1 in A' small enough that f is constant on it and that it meets no G'_m with $m \neq 0$. By lemma 5.2.4 the associated series is then a non-zero constant, so $\zeta(\Phi', f, s + \frac{1}{2}) \neq 0$. Comparing the two functional equations gives $\gamma(\pi^{\varphi}, s, \eta^{\varphi}) = \gamma(\pi, s, \eta)$. All four L-factors entering the defining relation of corollary 5.2.9 equal 1, so the same identity holds for ε. Finally γ, and hence ε, does not depend on the choice of $d^{\times}g$, since rescaling $d^{\times}g$ multiplies both sides of (5.10) by the same positive constant, completing the proof.

The two transport statements are now assembled. Each side of the correspondence has been moved by the same field isomorphism, and the Converse Theorem is what forces the two motions to land on the same representation.

Proposition 11.2.9: The Imprimitive Correspondence is Natural in the Base Field

Let $\varphi\colon F \to F'$ be an isomorphism of local fields and let $\rho \in \mathcal{G}_2^{\mathrm{im}}(F')$. Then $\rho^{\varphi} \in \mathcal{G}_2^{\mathrm{im}}(F)$, and ρ^{φ} is totally ramified if and only if ρ is. Moreover

$$\boldsymbol{\pi}^{\mathrm{im}}\big(\rho^{\varphi}\big) = \boldsymbol{\pi}^{\mathrm{im}}(\rho)^{\varphi}$$

where the left-hand $\boldsymbol{\pi}^{\mathrm{im}}$ is the map of theorem 8.5.4 over F and the right-hand one is the map over F'.

***Proof*:**
Step 1: the two classes are preserved. If $\rho \cong \mathrm{Ind}_{E'/F'}\Theta$ with E'/F' separable quadratic, put $E = \tilde{\varphi}^{-1}(E')$. Then E/F is separable quadratic and, by Step 2 of the proof of proposition 11.2.7, $\rho^{\varphi} \cong \mathrm{Ind}_{E/F}\big(\Theta^{\varphi_E}\big)$, so $\rho^{\varphi} \in \mathcal{G}_2^{\mathrm{im}}(F)$. Since $\tilde{\varphi}$ preserves ramification indices (lemma 11.2.1(3)), E/F is unramified exactly when E'/F' is. As $\Theta \mapsto \Theta^{\varphi_E}$ is a bijection on characters, every presentation of ρ corresponds to a presentation of ρ^{φ} with the same ramification. So $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F')$ if and only if $\rho^{\varphi} \in \mathcal{G}_2^{\mathrm{nr}}(F)$, which by definition 9.1.17 is the assertion about total ramification.

Step 2: the two candidates have the same twisted local constants. Choose $\eta \in \widehat{F'}$ of level one and put $\psi = \eta^{\varphi}$, which has level one by lemma 11.2.1(1). Write $\pi_1 = \boldsymbol{\pi}^{\mathrm{im}}(\rho)^{\varphi}$ and $\pi_2 = \boldsymbol{\pi}^{\mathrm{im}}(\rho^{\varphi})$, both in $\mathcal{A}_2^0(F)$. Let λ be a character of $F^{\times}$ and let χ be the character of $F'^{\times}$ with $\lambda = \chi^{\varphi}$. Every λ arises this way exactly once.

On the one hand, (8.30) over F' and proposition 11.2.8(2), (3) give

$$\varepsilon\big(\lambda\pi_1, s, \psi\big) = \varepsilon\big(\chi^{\varphi}\,\boldsymbol{\pi}^{\mathrm{im}}(\rho)^{\varphi}, s, \eta^{\varphi}\big) = \varepsilon\big(\chi\,\boldsymbol{\pi}^{\mathrm{im}}(\rho), s, \eta\big) = \varepsilon(\chi \otimes \rho, s, \eta)$$

On the other hand, proposition 11.2.7 applied to $\chi \otimes \rho$, together with $(\chi \otimes \rho)^{\varphi} = \chi^{\varphi} \otimes \rho^{\varphi}$ (immediate from definition 11.2.2(3) and (11.10), which is what identifies the transport of the twisting character with χ^{φ}) gives

$$\varepsilon(\chi \otimes \rho, s, \eta) = \varepsilon\big(\lambda \otimes \rho^{\varphi}, s, \psi\big) = \varepsilon\big(\lambda\pi_2, s, \psi\big)$$

the last step by (8.30) over F. So $\varepsilon(\lambda\pi_1, s, \psi) = \varepsilon(\lambda\pi_2, s, \psi)$ for every character λ of $F^{\times}$.

Step 3: the Converse Theorem. Both π_1 and π_2 are irreducible cuspidal, so $L(\lambda\pi_1, s) = L(\lambda\pi_2, s) = 1$ for every λ by proposition 5.3.1. Their central characters agree: by theorem 8.5.4 the central character of $\boldsymbol{\pi}^{\mathrm{im}}(\rho)$ is $\det\rho$, so $\omega_{\pi_1} = (\det\rho)^{\varphi}$ by proposition 11.2.8(1), while $\omega_{\pi_2} = \det(\rho^{\varphi})$; and these two agree because $\det(\rho\circ\varphi_*) = (\det\rho)\circ\varphi_*$, which is $(\det\rho)^{\varphi}$ read as a character of $F^\times$ by (11.10). With ψ of level one, the hypotheses of theorem 5.5.2 are met and $\pi_1 \cong \pi_2$.

The choice of η played no role in the conclusion, as it could not: by theorem 8.5.4(1) the map $\boldsymbol{\pi}^{\mathrm{im}}$ does not depend on the additive character used to define it, completing the proof.

The case that section 11.1 calls for is the one where $F' = F = K$ and φ runs through a Galois group. One hypothesis has to be discharged: corollary 9.3.11 needs a totally ramified parameter, and Step 1 of proposition 11.2.9 is what makes that hypothesis stable under the action.

Corollary 11.2.10: Galois Fixed Points

Let K/F be a finite Galois extension and $\Sigma = \mathrm{Gal}(K/F)$.

1. For every $\rho \in \mathcal{G}_2^{\mathrm{im}}(K)$ and every $\gamma \in \Sigma$,
$$\boldsymbol{\pi}^{\mathrm{im}}\big(\rho^{\gamma}\big) = \boldsymbol{\pi}^{\mathrm{im}}(\rho)^{\gamma}$$
so $\boldsymbol{\pi}^{\mathrm{im}}\colon \mathcal{G}_2^{\mathrm{im}}(K) \to \mathcal{A}_2^0(K)$ is Σ-equivariant for the actions of box 11.2.3.
2. Let $\rho \in \mathcal{G}_2^{\mathrm{im}}(K)$ be totally ramified. Then ρ is Σ-fixed if and only if $\boldsymbol{\pi}^{\mathrm{im}}(\rho)$ is.

***Proof*:**
(1) is proposition 11.2.9 applied to the isomorphism of local fields $\gamma\colon K \to K$.

For (2), suppose first that $\rho^{\gamma} \cong \rho$ for all $\gamma \in \Sigma$. Applying $\boldsymbol{\pi}^{\mathrm{im}}$ and using (1) gives $\boldsymbol{\pi}^{\mathrm{im}}(\rho)^{\gamma} \cong \boldsymbol{\pi}^{\mathrm{im}}(\rho)$ for all γ.

Conversely, suppose $\boldsymbol{\pi}^{\mathrm{im}}(\rho)^{\gamma} \cong \boldsymbol{\pi}^{\mathrm{im}}(\rho)$ for all $\gamma \in \Sigma$. By (1) this reads $\boldsymbol{\pi}^{\mathrm{im}}(\rho^{\gamma}) \cong \boldsymbol{\pi}^{\mathrm{im}}(\rho)$. Both ρ^{γ} and ρ lie in $\mathcal{G}_2^{\mathrm{im}}(K)$ and both are totally ramified, the first by Step 1 of proposition 11.2.9, so the hypothesis of corollary 9.3.11 is met and $\rho^{\gamma} \cong \rho$, completing the proof.

The hypothesis in (2) is needed. Corollary 9.3.11 says nothing about two classes both lying in $\mathcal{G}_2^{\mathrm{nr}}(K)$, so without it the implication from the group side back to the Galois side has no proof here. The warrant is clause (2) of this corollary itself, which carries the total-ramification hypothesis in its statement, together with Step 1 of proposition 11.2.9, which makes that hypothesis stable under the action. Clause (2) is also the point at which the chapter's standing hypothesis enters this section: corollary 9.3.11 rests on the Galois-side converse theorem of chapter 9, which is proved for $p = 2$. Clause (1), and everything before it, is free of that hypothesis.

The last statement of the section says that the construction of section 11.1 is transported by an isomorphism as well. It needs a φ defined on the upper field, since the lift lives there. It is the section's second dyadic statement, for the plain reason that $\mathrm{Lft}_{K/F}$ is built only under the chapter's standing hypothesis $p = 2$. The transport argument itself uses nothing about the residual characteristic beyond what that construction already assumed.

Proposition 11.2.11: Tame Lifting Commutes with the Interior Action

Let K/F and K'/F' be finite tamely ramified extensions and let $\varphi\colon K \to K'$ be an isomorphism of local fields with $\varphi(F) = F'$. Write $\varphi_0 = \varphi|_F\colon F \to F'$. Then, for every $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F')$, one has $\pi^{\varphi_0} \in \mathcal{A}_2^{\mathrm{wr}}(F)$ and

$$\mathrm{Lft}_{K/F}\big(\pi^{\varphi_0}\big) = \big(\mathrm{Lft}_{K'/F'}\,\pi\big)^{\varphi}$$

***Proof*:**

Step 1: φ_0 transports types. The entrywise map $\varphi_0\colon M_2(F) \to M_2(F')$ carries $\mathfrak{o}$-lattice chains in F^2 to $\mathfrak{o}'$-lattice chains in F'^2, hence chain orders to chain orders with the same period, and it carries $\mathcal{K}_{\mathfrak{A}}$ onto $\mathcal{K}_{\varphi_0\mathfrak{A}}$ and $U^j_{\mathfrak{A}}$ onto $U^j_{\varphi_0\mathfrak{A}}$. It carries a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ of definition 3.3.8 to the ramified simple stratum $(\varphi_0\mathfrak{A}, n, \varphi_0\alpha)$, because φ_0 restricts to an isomorphism $F[\alpha] \to F'[\varphi_0\alpha]$ over φ_0 and so preserves the valuation of α and the ramification of the field it generates. The characters ψ_α live on the unit groups, not on the orders, so for $\eta \in \widehat{F'}$ and $x \in \mathfrak{P}^{m+1}$,

$$\eta_{\varphi_0\alpha}(1+\varphi_0 x) = \eta\big(\mathrm{tr}(\varphi_0\alpha\cdot\varphi_0 x)\big) = \eta\big(\varphi_0(\mathrm{tr}(\alpha x))\big) = \big(\eta^{\varphi_0}\big)_\alpha(1+x)$$

so the characters ψ_α of definition 3.5.4 correspond as well. Consequently $(\mathfrak{A}, J, \Lambda) \mapsto (\varphi_0\mathfrak{A}, \varphi_0 J, \Lambda\circ\varphi_0^{-1})$ is a bijection between cuspidal types over F and over F', compatible with G-conjugacy, and compact induction commutes with it:

$$\big(\text{c-Ind}_{J'}^{\mathrm{GL}_2(F')}\Lambda'\big)^{\varphi_0} = \text{c-Ind}_{\varphi_0^{-1}J'}^{\mathrm{GL}_2(F)}\big(\Lambda'\circ\varphi_0\big)$$

since $f \mapsto f\circ\varphi_0$ matches the two spaces of compactly supported equivariant functions.

Step 2: $\mathcal{A}_2^{\mathrm{wr}}$ is preserved. By lemma 4.3.2, a cuspidal representation lies in $\mathcal{A}_2^{\mathrm{nr}}$ exactly when the chain order in its cuspidal type is isomorphic to a matrix algebra over the ring of integers. Step 1 matches the chain orders and their periods, so $\pi \in \mathcal{A}_2^{\mathrm{nr}}(F')$ if and only if $\pi^{\varphi_0} \in \mathcal{A}_2^{\mathrm{nr}}(F)$, and therefore φ_0 carries $\mathcal{A}_2^{\mathrm{wr}}(F')$ bijectively onto $\mathcal{A}_2^{\mathrm{wr}}(F)$ in the notation of definition 11.1.14.

Step 3: the minimal case. Let $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F')$ be minimal, so that $\mathrm{Lft}_{K'/F'}\pi$ is given by definition 11.1.14(1): write $\pi = \text{c-Ind}_{J'}^{\mathrm{GL}_2(F')}\Lambda'$ for a cuspidal type attached to a ramified simple stratum $(\mathfrak{A}', n, \alpha')$, and $\mathrm{Lft}_{K'/F'}\pi = \text{c-Ind}_{J'_{K'}}^{\mathrm{GL}_2(K')}\Lambda'_{K'}$ with $\Lambda'_{K'}$ the unique character of theorem 11.1.10. Minimality is the condition $\ell(\pi) \le \ell(\chi\pi)$ for every character χ of $F'^\times$; Step 1 matches cuspidal types over F with cuspidal types over F' and hence levels with levels, and $\chi \mapsto \chi^{\varphi_0}$ is a bijection on characters, so π^{φ_0} is minimal too.

Now run theorem 11.1.10 over F and over F' simultaneously. The full-scale isomorphism $\varphi\colon K \to K'$ restricts to φ_0 on F, so by Step 1 applied over K it carries the stratum $(\mathfrak{A}_K, e_K n, \alpha)$ of proposition 11.1.4 to $(\mathfrak{A}'_{K'}, e_{K'} n, \alpha')$ (the two ramification indices agree because φ matches the extensions K/F and K'/F') and it carries the norm map $\mathrm{Nm}_{K/F}$ to $\mathrm{Nm}_{K'/F'}$: $\mathrm{Nm}_{K/F}(y) = \det(m_y)$ for the F-linear multiplication operator m_y on K, and φ conjugates m_y into $m_{\varphi y}$, so

$$\varphi_0\big(\mathrm{Nm}_{K/F}(y)\big) = \mathrm{Nm}_{K'/F'}(\varphi y), \qquad y \in K^\times \tag{11.13}$$

The recipe (11.8) defining Λ_K in theorem 11.1.10 reads $\Lambda_K(y) = \varepsilon_{K/F}^{\,v_{EK}(y)}\Lambda\big(\mathrm{Nm}_{EK/E}(y)\big)$, and refers to exactly three things: the sign $\varepsilon_{K/F}$, the valuation v_{EK} and the norm $\mathrm{Nm}_{EK/E}$, which

is what proposition 11.1.15(5) rests on. The character ψ_α is not among them. It enters only through membership in $\mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$, which Step 1 transports. Each of the three does transport. The norm is (11.13) applied over EK, and the valuation is lemma 11.2.1(1). For the sign, take $\sigma = 1_{K'}$ in (11.12) over the pair (F, K) to get $R_{K'/F'}^{\varphi_0} = R_{K/F}$. Taking determinants gives $\varkappa_{K/F} = \varkappa_{K'/F'} \circ \varphi_0$, so

$$\varepsilon_{K/F} = \varkappa_{K/F}(\varpi) = \varkappa_{K'/F'}(\varphi_0 \varpi) = \varepsilon_{K'/F'}$$

because $\varphi_0 \varpi$ is a prime element of F' by lemma 11.2.1(1), and by lemma 11.1.7(1) the value of $\varkappa_{K'/F'}$ at a prime does not depend on which prime is taken. By the uniqueness clause of theorem 11.1.10, $\big(\Lambda'_{K'}\big) \circ \varphi = \mathrm{Lft}_{K/F}\big(\Lambda' \circ \varphi_0\big)$, and applying Step 1 over K to compact induction gives $\mathrm{Lft}_{K/F}(\pi^{\varphi_0}) = (\mathrm{Lft}_{K'/F'}\pi)^{\varphi}$.

Step 4: the general case. Let $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F')$ be arbitrary and write $\pi = \chi\pi_0$ with π_0 minimal, as definition 11.1.14(2) requires. Then $\pi^{\varphi_0} = \chi^{\varphi_0}\pi_0^{\varphi_0}$ by proposition 11.2.8(2), and $\pi_0^{\varphi_0}$ is minimal by Step 3. By definition 11.1.14(2) and proposition 11.1.15,

$$\mathrm{Lft}_{K/F}\big(\chi^{\varphi_0}\pi_0^{\varphi_0}\big) = \big(\chi^{\varphi_0} \circ \mathrm{Nm}_{K/F}\big)\,\mathrm{Lft}_{K/F}\big(\pi_0^{\varphi_0}\big)$$

By (11.13), one has $\chi^{\varphi_0} \circ \mathrm{Nm}_{K/F} = \big(\chi \circ \mathrm{Nm}_{K'/F'}\big)^{\varphi}$, and Step 3 handles the remaining factor. Combining the two gives

$$\mathrm{Lft}_{K/F}\big(\pi^{\varphi_0}\big) = \big(\chi \circ \mathrm{Nm}_{K'/F'}\big)^{\varphi}\big(\mathrm{Lft}_{K'/F'}\pi_0\big)^{\varphi} = \Big(\big(\chi \circ \mathrm{Nm}_{K'/F'}\big)\mathrm{Lft}_{K'/F'}\pi_0\Big)^{\varphi} = \big(\mathrm{Lft}_{K'/F'}\pi\big)^{\varphi}$$

the middle equality by proposition 11.2.8(2) over K and the last by the twisting clause of definition 11.1.14(2) again. Since $\mathrm{Lft}_{K/F}$ is well defined, the answer does not depend on the minimal twist chosen, completing the proof.

Together, corollary 11.2.10 and proposition 11.2.11 are what make a descent argument possible. The first says that a totally ramified $\rho \in \mathcal{G}_2^{\mathrm{im}}(K)$ is Σ-fixed exactly when $\boldsymbol{\pi}^{\mathrm{im}}(\rho)$ is, so that for a parameter of that kind, and by the paragraph following the corollary for no other, recognizing a fixed point may be done on whichever side is more convenient. The second says that the lift itself commutes with the action, so that the image of $\mathrm{Lft}_{K/F}$ consists of fixed points to begin with.

Exercise 11.2.1

1. Let $\varphi\colon F \to F'$ and $\theta\colon F' \to F''$ be isomorphisms of local fields. Show that $\rho^{\theta\circ\varphi} = (\rho^{\theta})^{\varphi}$ for $\rho \in \mathcal{G}^{\mathrm{ss}}(F'')$ and that $\pi^{\theta\circ\varphi} = (\pi^{\theta})^{\varphi}$ for $\pi \in \mathcal{A}_2^0(F'')$. Deduce that, for K/F finite Galois with $\Sigma = \mathrm{Gal}(K/F)$, the recipe $\gamma \mapsto (\,\cdot\,)^{\gamma}$ is a right action of Σ on $\mathcal{A}_2^0(K)$, and write down the associated left action.

2. Show that an isomorphism of local fields $\varphi\colon F \to F'$ is a homeomorphism, and deduce that the only automorphism of $\mathbb{Q}_p$ as a local field is the identity. Conclude that the interior action of the automorphism group of $\mathbb{Q}_p$ on $\mathcal{A}_2^0(\mathbb{Q}_p)$ is trivial, and explain why this does not make corollary 11.2.10 vacuous for $F = \mathbb{Q}_p$.

3. Let K/F be a finite Galois extension with $\Sigma = \mathrm{Gal}(K/F)$.

    1. Show that $\big(\lambda \circ \mathrm{Nm}_{K/F}\big)^{\gamma} = \lambda \circ \mathrm{Nm}_{K/F}$ for every character λ of $F^\times$ and every $\gamma \in \Sigma$.

2. Show that $\left(\mathrm{Res}_{K/F}\,\sigma\right)^{\gamma} \cong \mathrm{Res}_{K/F}\,\sigma$ for every $\sigma \in \mathcal{G}^{\mathrm{ss}}(F)$ and every $\gamma \in \Sigma$, where $\mathrm{Res}_{K/F}\sigma = \sigma|_{\mathcal{W}_K}$.

So restriction from F lands in the Σ-fixed classes over K. The content of a descent theorem is the converse.

4. Let $\varphi\colon F \to F'$ be an isomorphism of local fields, let $\eta \in \widehat{F'}$ be non-trivial, let $a \in F'^{\times}$ and let $\rho \in \mathcal{G}^{\mathrm{ss}}(F')$ have dimension d. Show that $(a\eta)^{\varphi} = \varphi^{-1}(a)\,\eta^{\varphi}$ and that $\det(\rho^{\varphi}) = (\det\rho)^{\varphi}$. Verify that proposition 11.2.7 is consistent with the ψ-dependence recorded in proposition 6.3.3(2), and deduce $n(\rho^{\varphi},\eta^{\varphi}) = n(\rho,\eta)$.

11.3 The Local Constant Modulo Roots of Unity

Section 11.1 moves a cuspidal representation to a tame extension K, and section 11.2 makes that move compatible with isomorphisms of the base field. To compare the two sides of the correspondence along such a move we need an invariant of a Weil group representation coarse enough to survive restriction to $\mathcal{W}_K$ and still sharp enough to separate representations. This section produces such an invariant, the tame period $c(\rho,\psi)$.

The invariant has two layers. The first is an element $c(\rho,\psi) \in F^{\times}$, canonical modulo U_F^1, which records the effect on $\varepsilon(\rho,s,\psi)$ of twisting ρ by a tamely ramified character. It is a stability statement, but in the direction opposite to proposition 6.3.3(4), where it was the *twisting* character that was allowed to be deep. The second is the observation that, once $p = 2$, this single element already determines $\varepsilon(\rho,\frac{1}{2},\psi)$ modulo roots of unity of 2-power order. Section 11.4 proves the matching statement for the Godement-Jacquet constant.

The first two results below hold in every residual characteristic and are stated that way. From proposition 11.3.8 onwards the chapter's blanket hypothesis $p = 2$ is essential, and we say so at the point of use.

Throughout, ψ is a non-trivial character of F, of any level. This relaxes the standing convention of section 11.1, which fixes ψ of level one because the type apparatus there is built against one fixed ψ. No result below needs that restriction, and two of them are stated at a generality the restriction would hide: proposition 11.3.8(2) compares $\varepsilon(\rho_K,s,\psi_K)$ with $\varepsilon(\rho,s,\psi)$ for tame K/F, and ψ_K has level $e\,n(\psi) - d$ rather than one; and exercise 11.3.1 (1) separates two readings of theorem 11.3.12 by an example that needs $n(\psi) \neq l+1$. Where a result of section 11.1 is quoted below, its level-one hypothesis is quoted with it. We write $n(\psi)$ for the least integer n with $\psi|_{\mathfrak{p}^n} = 1$, so that "$\psi$ has level one" means $n(\psi) = 1$. For a finite extension L/F we write $\psi_L = \psi\circ\mathrm{tr}_{L/F}$ and $\mathrm{Res}_{L/F}\rho = \rho_L = \rho|_{\mathcal{W}_L}$, the restriction of parameters. The subscript therefore means two different things on the two sides of the correspondence, and section 11.4 compares them: ρ_K is a restriction, while $\pi_K = \mathrm{Lft}_{K/F}\pi$ of definition 11.1.14 is a lift. The set $\mathcal{G}_2^{\mathrm{wr}}(F)$ of totally ramified parameters is the one named in definition 11.1.14 and characterized in definition 9.1.17. Recall $\mathrm{tr}_{L/F}(\mathfrak{p}_L^c) = \mathfrak{p}^{\lfloor (c+d+1)/e\rfloor}$ with $e = e(L|F)$ and $d = d(L|F)$ the exponent of definition 9.1.1, so that $n(\psi_L) = e\,n(\psi) - d - 1$. That definition is shifted by one against the exponent of the different, and every displayed formula of this section is written in the shifted convention, as chapter 9 is. We write $\mathcal{G}^{\mathrm{ss}}(F)$ for the equivalence classes of finite-dimensional semisimple smooth representations of $\mathcal{W}_F$ and $\mathcal{G}_n^0(F)$ for the irreducible ones of dimension n, and $\mathcal{P}_F$ for the wild inertia subgroup of $\mathcal{W}_F$.

Definition 11.3.1: The constant modulo roots of unity

Let $\mu_{\mathbb{C}}(2^\infty)$ denote the group of roots of unity in $\mathbb{C}^\times$ of 2-power order.
Let $\rho \in \mathcal{G}^{\mathrm{ss}}(F)$ and let ψ be non-trivial. An element $c \in F^\times$ is a *tame period* for (ρ, ψ) if

$$\varepsilon(\chi \otimes \rho, s, \psi) = \chi(c)^{-1}\, \varepsilon(\rho, s, \psi) \tag{11.14}$$

for every tamely ramified character χ of $F^\times$. A tame period is unique modulo U_F^1 when it exists, since the tamely ramified characters of $F^\times$ are exactly those trivial on U_F^1 and so separate the cosets of U_F^1. We then write $c(\rho, \psi)$, or c_ρ when ψ is fixed, for any representative, and read every identity involving it modulo U_F^1.

Two conventions have to be pinned before anything is proved, because this book already carries a second normalisation of the same element.

11.3.2 Two normalisations of one element Theorem 5.1.21 and proposition 6.3.3(4) attach to a character χ of $F^\times$ of level $l \geq 1$ the element c determined by $\chi(1+x) = \psi(cx)$ for $x \in \mathfrak{p}^{[l/2]+1}$. In this section, and in the Gauss-sum computations below, we attach to χ the element δ determined by the same formula on the smaller ideal $\mathfrak{p}^l \subseteq \mathfrak{p}^{[l/2]+1}$, and call δ the *element attached to* χ relative to ψ. Within this section that is the sense the phrase carries; α keeps throughout the chapter the meaning section 11.1 gives it, the element of a simple stratum $(\mathfrak{A}, n, \alpha)$, and never denotes an element attached to a character.
The second condition is the weaker one, so any c meeting the first meets the second. Both elements have valuation $n(\psi) - l - 1$, and an element of that valuation satisfying the second condition is unique modulo U_F^1: if $\psi(\delta x) = \psi(\delta' x)$ for all $x \in \mathfrak{p}^l$ then $v_F(\delta - \delta') \geq n(\psi) - l > v_F(\delta)$. Hence $c \equiv \delta \pmod{U_F^1}$. A tamely ramified character cannot tell them apart, and that is the accuracy at which every statement of this section is made. Neither element is ever used more finely. The same remark applies verbatim over any finite extension L/F with ψ_L in place of ψ.
For Gauss sums we use definition 5.1.19: for χ of level $l \geq 1$ and $c \in F^\times$ with $c\mathfrak{o} = \mathfrak{p}^{-l}$,

$$\tau(\chi, \psi) = \sum_{x \in U_F / U_F^{l+1}} \chi(cx)^{-1} \psi(cx)$$

which is independent of the choice of c, since replacing c by cu permutes the summation. When ψ has level one, lemma 9.3.1 gives $\varepsilon(\chi, \frac{1}{2}, \psi) = q^{-(l+1)/2} \tau(\chi, \psi)$.

The next lemma is the one elementary compatibility we use repeatedly: pulling a character back by the norm does not move the element attached to it, provided the level of the character is large enough against the different. Along a tame extension the proviso is automatic, and that is the case we use most; theorem 11.3.12 will meet an extension where it is not automatic, and where the proof has to carry it as a hypothesis it does not establish.

For tame M/L this is proposition 4.1.3, which gives the same conclusion at the $\mathfrak{p}^{\lfloor m/2 \rfloor + 1}$ pinning rather than at the $\mathfrak{p}^m$ one that box 11.3.2 adopts. The two normalisations agree modulo U^1, so nothing below distinguishes them, and where only the tame case is used one may read that result instead. The lemma is stated here because the primitive case of theorem 11.3.12 needs it for a wildly ramified M/L, which proposition 4.1.3 does not cover, and needs the hypothesis in the explicit form (11.15) in order to see what fails there.

Lemma 11.3.3: Norms and the attached element

Let M/L be a finite separable extension of local fields, $e = e(M|L)$ and $d = d(M|L)$. Let ξ be a character of $L^\times$ of level $l_\xi \geq 1$, with $\xi(1+y) = \psi_L(\delta_\xi y)$ for $y \in \mathfrak{p}_L^{l_\xi}$, and put $\chi = \xi \circ \mathrm{Nm}_{M/L}$ and $l = e\,l_\xi + e - d - 2$. Assume

$$2\,l > e\,l_\xi \tag{11.15}$$

Then χ has level l and

$$\chi(1+x) = \psi_M(\delta_\xi x), \qquad x \in \mathfrak{p}_M^{l};$$

that is, δ_ξ is an admissible choice of the element attached to χ, and every other choice lies in $\delta_\xi U_M^1$. Hypothesis (11.15) holds whenever M/L is tamely ramified, and then $d = e - 2$ and $l = e\,l_\xi$. It also holds whenever $e\,l_\xi \geq 2d + 2$.

***Proof*:**

From $n(\psi_M) = e\,n(\psi_L) - d - 1$ and $v_L(\delta_\xi) = n(\psi_L) - l_\xi - 1$ we get $v_M(\delta_\xi) = e\,n(\psi_L) - e\,l_\xi - e = n(\psi_M) - l - 1$, which is the valuation required of the element attached to a character of level l. Note that (11.15) forces $l \geq 1$.

Let $x \in \mathfrak{p}_M^{j}$ with $j \geq l$ and expand $\mathrm{Nm}_{M/L}(1+x) = 1 + \mathrm{tr}_{M/L}(x) + \eta$, where η is the sum of the elementary symmetric functions of the conjugates of x of degree at least 2, so that $v_M(\eta) \geq 2j \geq 2l$ and, η lying in L, $v_L(\eta) \geq \lceil 2l/e \rceil \geq l_\xi + 1$, the last step by (11.15). Also $v_L(\mathrm{tr}_{M/L}x) \geq \lfloor (j+d+1)/e \rfloor \geq \lfloor (l+d+1)/e \rfloor = l_\xi$, because $l + d + 1 = e\,l_\xi + e - 1$. Both $\mathrm{tr}_{M/L}(x)$ and η therefore lie in $\mathfrak{p}_L^{l_\xi}$, and $(1 + \mathrm{tr}_{M/L}(x))(1+\eta)$ differs from $1 + \mathrm{tr}_{M/L}(x) + \eta$ by a factor in $U_L^{2l_\xi} \subseteq U_L^{l_\xi+1}$, on which ξ is trivial. Hence

$$\chi(1+x) = \psi_L\big(\delta_\xi \mathrm{tr}_{M/L}(x)\big)\,\psi_L(\delta_\xi\eta) = \psi_M(\delta_\xi x)\,\psi_L(\delta_\xi\eta)$$

and the correction is trivial: $v_L(\delta_\xi\eta) \geq (n(\psi_L) - l_\xi - 1) + (l_\xi + 1) = n(\psi_L)$. Finally χ has level exactly l, since δ_ξ has the valuation computed above, so that $\psi_M(\delta_\xi\,\cdot\,)$ is non-trivial on $\mathfrak{p}_M^{l}$ and trivial on $\mathfrak{p}_M^{l+1}$. Uniqueness modulo U_M^1 is the argument of box 11.3.2.

For the last assertion: a tamely ramified M/L has different $\mathfrak{p}_M^{e-1}$, so $d = e - 2$ and $l = e\,l_\xi$, giving $2l - e\,l_\xi = e\,l_\xi > 0$. If instead $e\,l_\xi \geq 2d + 2$ then $2l - e\,l_\xi = e\,l_\xi + 2e - 2d - 4 \geq 2e - 2$, which is positive for $e \geq 2$, while for $e = 1$ one has $d = -1$ and $2l - e\,l_\xi = l_\xi > 0$. So (11.15) holds in both of the cases named.

The first of those two cases is the one used throughout theorem 11.3.5, which shows that a tame period exists as soon as $\rho|_{\mathcal{P}_F}$ omits the trivial character. The proof runs through a variant of Brauer induction adapted to a prescribed restriction, which we isolate first.

Lemma 11.3.4: Induction with a prescribed restriction

Let Γ be a finite group, Z a normal subgroup of Γ, and ρ an irreducible representation of Γ whose restriction to Z is a sum of Γ-conjugates of a character φ of Z. Then, in the Grothendieck group of virtual representations of Γ,

$$\rho = \sum_{i=1}^{r} n_i \operatorname{Ind}_{\Delta_i}^{\Gamma} \xi_i, \qquad n_i \in \mathbb{Z} \tag{11.16}$$

where each Δ_i is a subgroup of Γ containing Z and each ξ_i is a character of Δ_i with $\xi_i|Z = \varphi$. In particular $\sum_i n_i[\Gamma : \Delta_i] = \dim \rho$.

***Proof*:**
Let Γ_0 be the stabiliser of φ in Γ and ρ_0 the φ-isotypic subspace of $\rho|Z$, a representation of Γ_0. By [5, Clifford Induction From the Stabilizer], ρ_0 is irreducible and $\rho = \operatorname{Ind}_{\Gamma_0}^{\Gamma} \rho_0$. Induction is transitive, so it suffices to treat Γ_0. Assume therefore that φ is Γ-stable, so that $\rho|Z = \varphi \otimes \mathrm{id}$.

Passing to $\Gamma/\ker\rho$ changes nothing: $\rho(z) = \varphi(z)\mathrm{id}$ for $z \in Z$, so φ is trivial on $Z \cap \ker\rho$ and descends. We may therefore take ρ faithful, and then $\rho(Z)$ consists of scalars, so Z is central in Γ.

Write $\rho = \sum_j m_j \operatorname{Ind}_{H_j}^{\Gamma} \zeta_j$ with ζ_j a character of a subgroup H_j, by Brauer's induction theorem [18, Brauer's induction theorem]. Taking φ-isotypic components for the central subgroup Z is additive, and ρ is its own φ-isotypic component. Because Z is central, the Mackey restriction formula of proposition 1.2.29 gives

$$\left(\operatorname{Ind}_{H}^{\Gamma} \zeta\right)\big|_{Z} = [\Gamma : HZ] \cdot \operatorname{Ind}_{H\cap Z}^{Z} \left(\zeta|_{H\cap Z}\right)$$

so the φ-isotypic component of $\operatorname{Ind}_{H}^{\Gamma}\zeta$ vanishes unless ζ and φ agree on $H \cap Z$. When they do, $\tilde{\zeta}(hz) = \zeta(h)\varphi(z)$ is a well-defined character of HZ restricting to φ on Z, and the φ-isotypic component of $\operatorname{Ind}_{H}^{\Gamma}\zeta$ is $\operatorname{Ind}_{HZ}^{\Gamma}\tilde{\zeta}$, of the same dimension $[\Gamma : HZ]$. Discarding the terms that contribute nothing and setting $\Delta_i = H_j Z$, $\xi_i = \tilde{\zeta}_j$ gives (11.16). Dimension is additive on the Grothendieck group and $\dim \operatorname{Ind}_{\Delta_i}^{\Gamma} \xi_i = [\Gamma : \Delta_i]$, so $\sum_i n_i[\Gamma : \Delta_i] = \dim\rho$. The prescribed restriction $\xi_i|Z = \varphi$ is what Brauer's theorem alone does not give, and it is what both later appeals to the lemma use.

Theorem 11.3.5: Existence of the tame period

Let $\rho \in \mathcal{G}^{\mathrm{ss}}(F)$ and suppose $\rho|_{\mathcal{P}_F}$ does not contain the trivial character. Let ψ be non-trivial.

1. A tame period $c(\rho,\psi)$ exists. Consequently $n(\chi\otimes\rho,\psi) = n(\rho,\psi)$ for every tamely ramified χ, and $v_F\big(c(\rho,\psi)\big) = -n(\rho,\psi)$.
2. If φ is a tamely ramified character of $\mathcal{W}_F$, then $c(\varphi\otimes\rho,\psi) \equiv c(\rho,\psi) \pmod{U_F^1}$.
3. If K/F is a finite tamely ramified extension, then $c(\rho_K,\psi_K) \equiv c(\rho,\psi) \pmod{U_K^1}$.
4. If $\rho = \sum_i n_i \mathrm{Ind}_{L_i/F}\xi_i$ in the Grothendieck group, with each L_i/F finite, each ξ_i a character of $L_i^\times$ of level $l_i \geq 1$, and $\delta_i \in L_i^\times$ chosen with $\xi_i(1+x) = \psi_{L_i}(\delta_i x)$ for $x \in \mathfrak{p}_{L_i}^{l_i}$, then
$$c(\rho,\psi) \equiv \prod_i \mathrm{Nm}_{L_i/F}(\delta_i)^{n_i} \pmod{U_F^1}$$

***Proof*:**

Both sides of (11.14) are multiplicative in ρ, so the class of representations admitting a tame period is closed under direct sums, with $c(\rho_1\oplus\rho_2,\psi) = c(\rho_1,\psi)c(\rho_2,\psi)$, and a summand of ρ again omits the trivial character on $\mathcal{P}_F$. We may therefore assume ρ irreducible. Twisting ρ by an unramified character of $\mathcal{W}_F$ translates the variable s and multiplies both sides of (11.14) by the same factor. Such a twist exists: by proposition 6.1.10 every irreducible smooth representation of $\mathcal{W}_F$ admits an unramified twist with finite image. So we may further assume ρ has finite image. That is, ρ is an irreducible representation of $\Gamma = \mathrm{Gal}(E/F)$ for a finite Galois E/F, and we may take E to be the fixed field of $\ker\rho$.

Let F_1/F be the maximal tamely ramified sub-extension of E/F, so that $\Gamma_1 = \mathrm{Gal}(E/F_1)$ is a finite p-group, normal in Γ, and equal to the image of $\mathcal{P}_F$. By hypothesis $\rho|\Gamma_1$ does not contain the trivial character. In particular $\Gamma_1 \neq 1$, so its centre Z is non-trivial [4, p-Groups Have Nontrivial Center], and Z, being characteristic in Γ_1, is normal in Γ. By [5, Clifford's Theorem], $\rho|Z$ is a sum of Γ-conjugates of one character φ, and $\varphi \neq 1$ because ρ is faithful.

Lemma 11.3.4 gives $\rho = \sum_i n_i \mathrm{Ind}_{\Delta_i}^{\Gamma}\xi_i$ with $Z \subseteq \Delta_i$ and $\xi_i|Z = \varphi$. Put $L_i = E^{\Delta_i}$ and read ξ_i as a character of $L_i^\times$ through theorem 6.2.1. Each ξ_i is wildly ramified, since it restricts to the non-trivial φ on $Z \subseteq \Gamma_1$ and Γ_1 is the image of $\mathcal{P}_F$. Write $l_i \geq 1$ for its level and let δ_i be the attached element, of valuation $n(\psi_{L_i}) - l_i - 1$ and well defined modulo $U_{L_i}^1$.

Consider one term $\theta = \mathrm{Ind}_{L/F}\xi$ with ξ of level $l \geq 1$ and attached element δ. For a tamely ramified character χ of $F^\times$, the projection formula proposition 1.2.30 gives $\chi\otimes\mathrm{Ind}_{L/F}\xi = \mathrm{Ind}_{L/F}(\chi_L\xi)$ with $\chi_L = \chi\circ\mathrm{Nm}_{L/F}$ tamely ramified, and (6.5) gives $\varepsilon(\chi\otimes\theta,s,\psi) = \lambda_{L/F}(\psi)\varepsilon(\chi_L\xi,s,\psi_L)$. By theorem 5.1.21, $\varepsilon(\chi_L\xi,s,\psi_L) = \chi_L(\delta)^{-1}\varepsilon(\xi,s,\psi_L)$. The theorem pins its element on $\mathfrak{p}_L^{[l/2]+1}$ rather than on $\mathfrak{p}_L^l$, and box 11.3.2 records that the two agree modulo U_L^1, which is all a tamely ramified character sees. Dividing by the same computation with $\chi = 1$,

$$\varepsilon(\chi\otimes\theta,s,\psi) = \chi\big(\mathrm{Nm}_{L/F}(\delta)\big)^{-1}\varepsilon(\theta,s,\psi)$$

so θ has tame period $\mathrm{Nm}_{L/F}(\delta)$. Multiplicativity over the expression for ρ gives (1) and (4) at once.

For the supplements in (1): (11.14) says $\varepsilon(\chi\otimes\rho,s,\psi)$ and $\varepsilon(\rho,s,\psi)$ differ by a factor independent

of s, so the exponents in proposition 6.3.3(1) agree and $n(\chi\otimes\rho,\psi)=n(\rho,\psi)$. Taking χ unramified in (11.14) and comparing with $\varepsilon(\chi\otimes\rho,s,\psi)=\chi(\varpi)^{n(\rho,\psi)}\varepsilon(\rho,s,\psi)$, which follows from proposition 6.3.3(1) once an unramified $\chi=\|\cdot\|^z$ is read as the translation $s\mapsto s+z$, gives $\chi(\varpi)^{n(\rho,\psi)+v_F(c_\rho)}=1$ for every unramified χ, whence $v_F(c_\rho)=-n(\rho,\psi)$.

For (2), let φ be tamely ramified. Then $\varepsilon(\chi\varphi\otimes\rho,s,\psi)=\chi(c_\rho)^{-1}\varphi(c_\rho)^{-1}\varepsilon(\rho,s,\psi)$, while $\varepsilon(\chi\otimes(\varphi\otimes\rho),s,\psi)=\chi(c_{\varphi\otimes\rho})^{-1}\varepsilon(\varphi\otimes\rho,s,\psi)$ and $\varepsilon(\varphi\otimes\rho,s,\psi)=\varphi(c_\rho)^{-1}\varepsilon(\rho,s,\psi)$. Comparing, $\chi(c_{\varphi\otimes\rho})=\chi(c_\rho)$ for every tamely ramified χ.

For (3), let K/F be finite and tamely ramified. Enlarging E we may assume $K\subseteq E$, and set $\Sigma=\mathrm{Gal}(E/K)$. Restrict (11.16) to Σ and take one term $\theta=\mathrm{Ind}_\Delta^\Gamma\xi$, with $L=E^\Delta$ and attached element δ. The Mackey restriction formula of proposition 1.2.29 reads, in terms of fields, the double coset of γ contributing the fixed field of $\Delta^\gamma\cap\Sigma$, which is the compositum $L^\gamma K$:

$$\theta|\Sigma=\bigoplus_{\gamma\in\Delta\backslash\Gamma/\Sigma}\mathrm{Ind}_{L^\gamma K/K}\big(\xi^\gamma\circ\mathrm{Nm}_{L^\gamma K/L^\gamma}\big)$$

Each $L^\gamma K/L^\gamma$ is tamely ramified, so by lemma 11.3.3 the element attached to $\xi^\gamma\circ\mathrm{Nm}_{L^\gamma K/L^\gamma}$ may be taken to be $\gamma(\delta)$. Applying (4) over K, and noting that $\mathrm{Nm}_{L^\gamma K/K}(\gamma(\delta))$ is the product of the images of $\gamma(\delta)$ under the K-embeddings of $L^\gamma K$ into E, while those embeddings, collected over the double cosets, restrict to the F-embeddings of L into E exactly once each,

$$c(\theta|\Sigma,\psi_K)\equiv\prod_{\gamma\in\Delta\backslash\Gamma/\Sigma}\mathrm{Nm}_{L^\gamma K/K}\big(\gamma(\delta)\big)=\mathrm{Nm}_{L/F}(\delta)\pmod{U_K^1}$$

So (3) holds for θ, and by multiplicativity for ρ. With (1), (2) and (4) already in hand, this completes the proof.

Part (3) is the one the chapter uses: the tame period is unchanged by the tame base change of section 11.1, so it is an invariant of ρ that is preserved by the move to K, and (4) computes it from any induced expression for ρ. The defining property (11.14) concerns twists by characters. For proposition 11.3.8 we must twist by a representation of higher dimension, provided it is trivial on $\mathcal{P}_F$. The passage rests on the fact that such a representation is induced from a character, which the structure of the tame quotient makes elementary.

Lemma 11.3.6: Representations trivial on wild inertia are monomial

Let θ be an irreducible smooth representation of $\mathcal{W}_F$ trivial on $\mathcal{P}_F$. Then $\theta=\mathrm{Ind}_{L/F}\chi$ for a finite tamely ramified extension L/F and a tamely ramified character χ of $L^\times$.

***Proof*:**
The kernel of θ is open, so $\theta(\mathcal{I}_F)$ is finite and $\theta|\mathcal{I}_F$ is semisimple. By proposition 6.1.2 the quotient $\mathcal{I}_F/\mathcal{P}_F$ is procyclic, hence abelian, so $\theta|\mathcal{I}_F$ is a sum of characters. As $\mathcal{I}_F$ is normal in $\mathcal{W}_F$, [5, Clifford's Theorem] presents this sum as the $\mathcal{W}_F$-orbit of one character φ, with finitely many members because θ is finite dimensional. Let H be the stabiliser of φ, a subgroup containing $\mathcal{I}_F$ whose index is the size of that orbit and so is finite, and let θ_0 be the φ-isotypic subspace. By [5, Clifford Induction From the Stabilizer], whose hypotheses ask only that θ be irreducible on a finite-dimensional space and not that $\mathcal{W}_F$ be finite, $\theta=\mathrm{Ind}_H^{\mathcal{W}_F}\theta_0$ with θ_0 irreducible.

On H the subgroup $\mathcal{I}_F$ acts by the scalars φ, and $H/\mathcal{I}_F$ is a subgroup of $\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z}$, hence cyclic. So $\theta_0(H)$ is generated by scalars together with the image of a single element and is abelian. An irreducible representation with abelian image is one-dimensional, so $\theta_0 = \chi$ is a character. Finally $H = \mathcal{W}_L$ for a finite extension L/F, and $\mathcal{P}_F \subseteq \mathcal{I}_F \subseteq H$ places L inside the maximal tamely ramified extension of F. Thus L/F is tame and χ, trivial on $\mathcal{P}_F = \mathcal{P}_L$, is tamely ramified. So $\theta = \mathrm{Ind}_{L/F}\chi$ is a presentation of the asserted kind.

That reduction is all the twist needs. What corollary 11.3.7 requires of it is the tameness of L/F, since that is what keeps c_ρ available over L. Once θ is induced from a tamely ramified character of a tame extension, inductivity moves the twist to a field where theorem 11.3.5 already applies, and the outcome is (11.17): the twist by θ contributes the single factor $\det\theta(c_\rho)^{-1}$, exactly as the character $\det\theta$ does.

Corollary 11.3.7: Twisting by a representation trivial on wild inertia

Let $\rho \in \mathcal{G}_n^0(F)$ with $\rho|_{\mathcal{P}_F}$ irreducible and non-trivial, and write $c_\rho = c(\rho,\psi)$. Let $\theta \in \mathcal{G}^{\mathrm{ss}}(F)$ be trivial on $\mathcal{P}_F$. Then

$$\varepsilon(\theta\otimes\rho,s,\psi) = \det\theta(c_\rho)^{-1}\,\varepsilon(\rho,s,\psi)^{\dim\theta} \tag{11.17}$$

***Proof*:**

An irreducible non-trivial $\rho|_{\mathcal{P}_F}$ omits the trivial character, so c_ρ exists. The same holds for ρ_L for every tamely ramified L/F, since $\mathcal{P}_L = \mathcal{P}_F$.

For a virtual representation V of $\mathcal{W}_F$ set $\mathcal{F}(V) = \varepsilon(V\otimes\rho,s,\psi)\,\varepsilon(\rho,s,\psi)^{-\dim V}$. Both $V \mapsto V\otimes\rho$ and $V \mapsto \dim V$ are additive and ε is multiplicative, so $\mathcal{F}$ is a homomorphism from the Grothendieck group to $\mathbb{C}^\times$. The assertion is $\mathcal{F}(\theta) = \det\theta(c_\rho)^{-1}$. Both sides are multiplicative, so we may take θ irreducible, hence $\theta = \mathrm{Ind}_{L/F}\chi$ with L/F tame and χ tamely ramified, by lemma 11.3.6.

Step 1: virtual degree zero. Let V be a virtual representation of $\mathcal{W}_L$ of degree 0. The projection formula proposition 1.2.30, extended to the Grothendieck group by additivity of both of its sides, gives $\mathrm{Ind}_{L/F}(V)\otimes\rho = \mathrm{Ind}_{L/F}(V\otimes\rho_L)$, and $V\otimes\rho_L$ again has degree 0, so inductivity in degree zero (6.3) gives $\varepsilon(\mathrm{Ind}_{L/F}(V)\otimes\rho,s,\psi) = \varepsilon(V\otimes\rho_L,s,\psi_L)$. As $\dim\mathrm{Ind}_{L/F}(V) = 0$,

$$\mathcal{F}\big(\mathrm{Ind}_{L/F}V\big) = \varepsilon(V\otimes\rho_L,s,\psi_L)\,\varepsilon(\rho_L,s,\psi_L)^{-\dim V}$$

Taking $V = \chi - 1_L$ and applying theorem 11.3.5(1) over L, then (3) of the same theorem and the tameness of χ,

$$\mathcal{F}\big(\mathrm{Ind}_{L/F}(\chi-1_L)\big) = \frac{\varepsilon(\chi\otimes\rho_L,s,\psi_L)}{\varepsilon(\rho_L,s,\psi_L)} = \chi\big(c(\rho_L,\psi_L)\big)^{-1} = \chi(c_\rho)^{-1} \tag{11.18}$$

Step 2: the permutation representation. We claim $\mathcal{F}(R_{L/F}) = \varkappa_{L/F}(c_\rho)^{-1}$ for every finite tamely ramified L/F, where $R_{L/F} = \mathrm{Ind}_{L/F}1_L$ and $\varkappa_{L/F} = \det R_{L/F}$ (definition 6.2.5). We induct on $[L:F]$, the case $L = F$ being trivial. Note $R_{L/F}$ is trivial on $\mathcal{P}_F$, since $\mathcal{P}_F$ is normal in $\mathcal{W}_F$ and contained in $\mathcal{W}_L$. Decompose $R_{L/F} = \bigoplus_i \theta_i$ into irreducibles and write $\theta_i = \mathrm{Ind}_{M_i/F}\eta_i$ by lemma 11.3.6, with M_i/F tame. The trivial representation occurs in $R_{L/F}$, contributing a term with $M_i = F$. Since $\sum_i [M_i:F] = [L:F]$, every M_i has $[M_i:F] < [L:F]$ once $L \neq F$.

Combining (11.18) with the induction hypothesis,

$$\mathcal{F}(\theta_i) = \mathcal{F}\big(\mathrm{Ind}_{M_i/F}(\eta_i - 1)\big)\,\mathcal{F}\big(R_{M_i/F}\big) = \eta_i(c_\rho)^{-1}\varkappa_{M_i/F}(c_\rho)^{-1} = \det\theta_i(c_\rho)^{-1}$$

the last step because $\det \mathrm{Ind}_{M/F}\eta = \varkappa_{M/F}\cdot(\eta|_{F^\times})$ by proposition 6.2.7 with $m=1$. Multiplying over i gives $\mathcal{F}(R_{L/F}) = \det R_{L/F}(c_\rho)^{-1} = \varkappa_{L/F}(c_\rho)^{-1}$.

Step 3. For $\theta = \mathrm{Ind}_{L/F}\chi$ we have $\theta = \mathrm{Ind}_{L/F}(\chi - 1_L) + R_{L/F}$ in the Grothendieck group, so the two steps give $\mathcal{F}(\theta) = \chi(c_\rho)^{-1}\varkappa_{L/F}(c_\rho)^{-1} = \det\theta(c_\rho)^{-1}$. Unwinding the definition of $\mathcal{F}$, this says $\varepsilon(\theta\otimes\rho,s,\psi) = \det\theta(c_\rho)^{-1}\varepsilon(\rho,s,\psi)^{\dim\theta}$, which is (11.17).

Those three steps are the induction-constant formalism of definition 6.3.4 carried out by hand, and it is worth saying why they are not replaced by an appeal to it. Axiom (a) for $\mathcal{F}$ is the multiplicativity noted at the outset and axiom (b) is Step 1; the boundary $\partial\mathcal{F}\colon (L,\chi)\mapsto\chi(c_\rho)^{-1}$ is also the boundary of $(L,\theta)\mapsto\det\theta(c_\rho)^{-1}$, by proposition 6.2.7, so lemma 6.3.5 would identify the two and give (11.17) at once. But that lemma is proved by Brauer induction on a finite group, and extended by lemma 6.3.6 to profinite ones, while θ here is only asked to be trivial on $\mathcal{P}_F$: an unramified character of infinite order qualifies and factors through no finite quotient of $\mathcal{W}_F$. Steps 2 and 3 establish exactly that generality, replacing Brauer induction by an induction on $[L:F]$ in which the only finiteness used is that of $R_{L/F}$.

From here to the end of the section the standing hypothesis $p=2$ applies and is used. Its first consequence is that the hypothesis of corollary 11.3.7 is automatic for the representations this chapter cares about.

Proposition 11.3.8: What $p=2$ gives

Assume $p=2$ and let $\rho\in\mathcal{G}_2^0(F)$.

1. $\rho\in\mathcal{G}_2^{\mathrm{wr}}(F)$ if and only if $\rho|_{\mathcal{P}_F}$ is irreducible. In particular theorem 11.3.5 and corollary 11.3.7 apply to every $\rho\in\mathcal{G}_2^{\mathrm{wr}}(F)$, and $\rho_K\in\mathcal{G}_2^{\mathrm{wr}}(K)$ for every finite tamely ramified K/F.
2. Let $\rho\in\mathcal{G}_2^{\mathrm{wr}}(F)$, write $c_\rho = c(\rho,\psi)$, and let K/F be finite and tamely ramified. Then

$$\varepsilon(\rho_K,s,\psi_K) = \lambda_{K/F}(\psi)^{-2}\,\varkappa_{K/F}(c_\rho)^{-1}\,\varepsilon(\rho,s,\psi)^{[K:F]} \tag{11.19}$$

***Proof*:**

(1) Suppose $\rho|_{\mathcal{P}_F}$ is reducible. Being the restriction of an irreducible representation to a normal subgroup it is semisimple, so it is a sum $\varphi_1\oplus\varphi_2$ of $\mathcal{W}_F$-conjugate characters of $\mathcal{P}_F$.

If $\varphi_1\neq\varphi_2$, let H be the stabiliser of φ_1, of index 2, so that $\rho=\mathrm{Ind}_{L/F}\xi$ with $[L:F]=2$. Every character of $\mathcal{P}_F$ is trivial on $[\mathcal{P}_F,\mathcal{P}_F]$, so $\mathcal{P}_F$ stabilises φ_1 and $\mathcal{P}_F\subseteq\mathcal{W}_L$. Hence L lies in the maximal tamely ramified extension of F, so L/F is tame. As $p=2$ a ramified quadratic extension is wildly ramified, so L/F is unramified and $\rho\in\mathcal{G}_2^{\mathrm{nr}}(F)$.

If $\varphi_1=\varphi_2$, then $\rho(\mathcal{P}_F)$ consists of scalars, so the projective representation $\mathcal{W}_F\to\mathrm{PGL}_2(\mathbb{C})$ attached to ρ is trivial on $\mathcal{P}_F$ and factors through $\mathcal{W}_F/\mathcal{P}_F$. By proposition 6.1.2 the group $\mathcal{I}_F/\mathcal{P}_F$ is procyclic, and $\mathcal{W}_F/\mathcal{I}_F\cong\mathbb{Z}$ by definition 6.1.4, so $\mathcal{W}_F/\mathcal{P}_F$ is an extension of a procyclic group by a procyclic group and its finite quotients are metacyclic, and the metacyclic finite subgroups of $\mathrm{PGL}_2(\mathbb{C})$ are the cyclic and the dihedral ones (corollary 6.1.17). A cyclic projective

image forces $\rho(\mathcal{W}_F)$ abelian and ρ reducible, which is excluded. So the image is dihedral. Let C be its cyclic subgroup of index two, L/F the quadratic extension with $\mathcal{W}_L$ the preimage of C in $\mathcal{W}_F$, and ξ a character of $\mathcal{W}_L$ with $\rho = \mathrm{Ind}_{L/F}\xi$. The projective image of $\mathcal{P}_F$ is trivial, since $\rho(\mathcal{P}_F)$ consists of scalars, so it lies in C. Again $\mathcal{P}_F \subseteq \mathcal{W}_L$, L/F is unramified and $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$.

Conversely, if $\rho \in \mathcal{G}_2^{\mathrm{nr}}(F)$ then $\rho = \mathrm{Ind}_{L/F}\xi$ with L/F unramified quadratic, so $\mathcal{P}_F = \mathcal{P}_L \subseteq \mathcal{W}_L$ and $\rho|_{\mathcal{P}_F} = \xi|_{\mathcal{P}_F} \oplus \xi^\sigma|_{\mathcal{P}_F}$ is reducible. Since $\rho \in \mathcal{G}_2^{\mathrm{wr}}(F)$ means exactly $\rho \notin \mathcal{G}_2^{\mathrm{nr}}(F)$ (definition 9.1.17), this is the stated equivalence. An irreducible $\rho|_{\mathcal{P}_F}$ is non-trivial, so corollary 11.3.7 applies; and for tame K/F one has $\mathcal{P}_K = \mathcal{P}_F$, so $\rho_K|_{\mathcal{P}_K}$ is irreducible, ρ_K is irreducible, and $\rho_K \in \mathcal{G}_2^{\mathrm{wr}}(K)$.

(2) By proposition 1.2.30, $\mathrm{Ind}_{K/F}(\rho_K) = \rho \otimes R_{K/F}$, and $R_{K/F}$ is trivial on $\mathcal{P}_F$ because $\mathcal{P}_F$ is normal in $\mathcal{W}_F$ and contained in $\mathcal{W}_K$. Inductivity in λ form (6.5) and corollary 11.3.7 with $\theta = R_{K/F}$ give

$$\lambda_{K/F}(\psi)^2\,\varepsilon(\rho_K, s, \psi_K) = \varepsilon(\rho \otimes R_{K/F}, s, \psi) = \varkappa_{K/F}(c_\rho)^{-1}\varepsilon(\rho, s, \psi)^{[K:F]}$$

the exponent 2 on λ being $\dim\rho$. Dividing by $\lambda_{K/F}(\psi)^2$, which is non-zero, gives

$$\varepsilon(\rho_K, s, \psi_K) = \lambda_{K/F}(\psi)^{-2}\,\varkappa_{K/F}(c_\rho)^{-1}\,\varepsilon(\rho, s, \psi)^{[K:F]}$$

which is (11.19).

Part (1) discharges the hypothesis of corollary 11.3.7 for every $\rho \in \mathcal{G}_2^{\mathrm{wr}}(F)$, so that corollary is available unconditionally from here on, and (11.19) is the rule by which the local constant itself, not only the tame period, transports along the base change of section 11.1.

The congruence itself rests on an arithmetic fact about number fields: an element whose absolute value is 1 at every place is a unit lying in the kernel of the logarithmic embedding, and that embedding has discrete image with kernel exactly the roots of unity. It is what turns the Gauss-sum estimates below into a statement about roots of unity of 2-power order.

Lemma 11.3.9: Units of absolute value one

Let $\mathbb{K}$ be a number field and $u \in \mathbb{K}^\times$ with $|u|_v = 1$ for every place v of $\mathbb{K}$, finite or archimedean. Then u is a root of unity.

***Proof*:**

An element with $|u|_v = 1$ at every finite place lies in $\mathfrak{o}_{\mathbb{K}}^\times$. By Dirichlet's unit theorem [3, Dirichlet Unit Theorem] the logarithmic embedding $\mathfrak{o}_{\mathbb{K}}^\times \to \mathbb{R}^{r_1+r_2}$, $x \mapsto (\log|x|_v)_{v\mid\infty}$, has discrete image and kernel exactly the group of roots of unity of $\mathbb{K}$. The hypothesis places u in that kernel, so u is a root of unity. Applied below with $\mathbb{K} = \mathbb{Q}(\zeta_{2^j})$, this returns a root of unity of 2-power order.

The other ingredient is a pair of identities for the Gauss sum of definition 5.1.19: one for the product of the sums attached to χ and to χ^{-1}, one for complex conjugation. Between them they fix the absolute value of the Gauss sum, which is the quantity lemma 11.3.11 divides out.

Lemma 11.3.10: Two Gauss-sum identities

Let ψ have level one, let χ be a character of $F^\times$ of level $l \geq 1$, and let $\delta \in F^\times$ satisfy $\chi(1+x) = \psi(\delta x)$ for $x \in \mathfrak{p}^l$. Write $\check{\chi} = \chi^{-1}$. Then

$$\tau(\chi,\psi)\,\tau(\check{\chi},\psi) = \chi(-1)\,q^{l+1}, \tag{11.20}$$

$$\overline{\tau(\chi,\psi)} = \chi(-1)\,|\chi(\delta)|^{-2}\,\tau(\check{\chi},\psi). \tag{11.21}$$

Consequently $\big|\chi(\delta)\,\tau(\chi,\psi)\big| = q^{(l+1)/2}$.

***Proof*:**
The character $\check{\chi}$ also has level l, and τ does not depend on the choice of the element of valuation $-l$ used to form it (box 11.3.2), so both Gauss sums may be computed with δ. Identity (11.20) is (5.8), established in section 5.1 from the functional equation of the local constant; only (11.21) is new here.

For (11.21), note that $\chi(U_F)$ is a finite subgroup of $\mathbb{C}^\times$, so $|\chi(x)| = 1$ for $x \in U_F$ and $|\chi(\delta x)|$ is the constant $|\chi(\delta)|$ throughout the sum. Using $\bar{z} = |z|^2 z^{-1}$ and $\overline{\psi(y)} = \psi(-y)$,

$$\overline{\tau(\chi,\psi)} = \sum_{x \in U_F/U_F^{l+1}} |\chi(\delta)|^{-2}\chi(\delta x)\,\psi(-\delta x) = |\chi(\delta)|^{-2}\sum_{x}\chi(-\delta x)\,\psi(\delta x)$$

the second equality by $x \mapsto -x$, which permutes U_F/U_F^{l+1}. Pulling out $\chi(-1)$ and recognising $\sum_x \chi(\delta x)\psi(\delta x) = \tau(\check{\chi},\psi)$ gives (11.21). Multiplying it by $\tau(\chi,\psi)$ and inserting (11.20),

$$|\tau(\chi,\psi)|^2 = \chi(-1)|\chi(\delta)|^{-2}\chi(-1)q^{l+1} = |\chi(\delta)|^{-2}q^{l+1}$$

since $\chi(-1)^2 = \chi(1) = 1$. Multiplying by $|\chi(\delta)|^2$ gives $\big|\chi(\delta)\,\tau(\chi,\psi)\big|^2 = q^{l+1}$, and taking the positive square root gives $\big|\chi(\delta)\,\tau(\chi,\psi)\big| = q^{(l+1)/2}$.

That absolute value is the quantity lemma 11.3.11 divides out. When $p = 2$ every value in sight is a root of unity of 2-power order: the values of χ on the pro-2 group U_F^1, and the values of ψ on the 2-groups $\mathfrak{p}^{-m}/\mathfrak{p}$. The content of the next lemma is that, after the Gauss sum is divided by its absolute value, nothing else is left. The argument is a place-by-place one, run in a cyclotomic field where the Galois action can be moved onto the pair (χ,ψ).

Lemma 11.3.11: The abelian constant modulo 2-power roots of unity

Assume $p = 2$. Let ψ be a non-trivial character of F, let χ be a character of $F^\times$ of level $l \geq 1$, and let $\delta \in F^\times$ satisfy $\chi(1+x) = \psi(\delta x)$ for $x \in \mathfrak{p}^l$. Then

$$\chi(\delta)\,\varepsilon(\chi,\tfrac{1}{2},\psi) \in \mu_{\mathbb{C}}(2^\infty)$$

***Proof*:**
Two reductions. Replacing ψ by $a\psi\colon x \mapsto \psi(ax)$ replaces δ by δa^{-1}, while $\varepsilon(\chi,\frac{1}{2},a\psi) = \chi(a)\varepsilon(\chi,\frac{1}{2},\psi)$ by proposition 6.3.3(2). The product is unchanged. Every non-trivial character of F is some $a\psi$, so we may assume $n(\psi) = 1$. Next let η be a tamely ramified character of $F^\times$:

then $\eta\chi$ has level l with the same δ, and δ is a tame period for (χ,ψ) by theorem 5.1.21 and box 11.3.2, so $\varepsilon(\eta\chi,\frac{1}{2},\psi)=\eta(\delta)^{-1}\varepsilon(\chi,\frac{1}{2},\psi)$ and $(\eta\chi)(\delta)\varepsilon(\eta\chi,\frac{1}{2},\psi)=\chi(\delta)\varepsilon(\chi,\frac{1}{2},\psi)$. Writing $F^\times=\varpi^{\mathbb{Z}}\times\mu_{q-1}\times U_F^1$ and taking η trivial on ϖ and on U_F^1 and inverse to χ on μ_{q-1}, we may assume χ trivial on μ_{q-1}.

An algebraic integer. Set

$$\Xi=\chi(\delta)\,\tau(\chi,\psi)=\sum_{x\in U_F/U_F^{l+1}}\chi(x)^{-1}\psi(\delta x)$$

For $x\in U_F$ we now have $\chi(x)\in\chi(U_F^1)$, and U_F^1 is a pro-2 group on which χ is smooth, so $\chi(x)\in\mu_{\mathbb{C}}(2^\infty)$. Likewise $\mathfrak{p}^{-l}/\mathfrak{p}$ is a finite abelian 2-group, so $\psi(\delta x)\in\mu_{\mathbb{C}}(2^\infty)$. Hence Ξ is a sum of roots of unity of 2-power order, so an algebraic integer of $\mathbb{K}=\mathbb{Q}(\zeta)$ for a primitive 2^j-th root of unity ζ, with j large enough to accommodate all the values above. We take $j\geq 3$, so that $\sqrt{2}\in\mathbb{K}$ and hence $q^{(l+1)/2}\in\mathbb{K}$ for $q=2^f$. The same applies to $\check{\Xi}=\check{\chi}(\delta)\tau(\check{\chi},\psi)$, and (11.20) gives

$$\Xi\cdot\check{\Xi}=\tau(\chi,\psi)\tau(\check{\chi},\psi)=\chi(-1)\,q^{\,l+1}\tag{11.22}$$

Places. Put $u=\Xi/q^{(l+1)/2}\in\mathbb{K}^\times$. Let v be a finite place of $\mathbb{K}$ not above 2. Both Ξ and $\check{\Xi}$ are integral at v, and their product has $|\cdot|_v=1$ by (11.22) because q is a power of 2. Hence $|\Xi|_v=1$ and $|u|_v=1$.

Let $\sigma\in\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$. Define χ^σ by $\chi^\sigma(x)=\sigma(\chi(x))$ for $x\in U_F$ and $\chi^\sigma(\varpi)=\chi(\varpi)$. The values of ψ do not all lie in $\mathbb{K}$, so $\sigma\circ\psi$ is not defined; but each $\mathfrak{p}^{-m}/\mathfrak{p}$ is a finite abelian 2-group, so ψ takes its values in $\mu_{\mathbb{C}}(2^\infty)$, and restriction $\mathrm{Gal}(\mathbb{Q}(\mu_{2^\infty})/\mathbb{Q})\to\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is surjective. Fix a lift $\widetilde{\sigma}$ of σ and put $\psi^\sigma=\widetilde{\sigma}\circ\psi$. Since $\widetilde{\sigma}$ restricts to σ on $\mathbb{K}$, it may be read as σ on every value of ψ occurring below, all of which lie in $\mathbb{K}$. As $\widetilde{\sigma}$ is injective on $\mu_{\mathbb{C}}(2^\infty)$, ψ^σ is again a character of F of level one; χ^σ is a character of $F^\times$ of level l, and $\chi^\sigma(1+x)=\sigma(\psi(\delta x))=\psi^\sigma(\delta x)$ for $x\in\mathfrak{p}^l$, so δ serves for the pair $(\chi^\sigma,\psi^\sigma)$. Applying σ term by term,

$$\sigma(\Xi)=\sum_x\chi^\sigma(x)^{-1}\psi^\sigma(\delta x)=\chi^\sigma(\delta)\,\tau(\chi^\sigma,\psi^\sigma)$$

of absolute value $q^{(l+1)/2}$ by lemma 11.3.10. So $|u|_v=1$ at every archimedean place v.

The field $\mathbb{K}=\mathbb{Q}(\zeta_{2^j})$ is totally ramified over $\mathbb{Q}$ at 2, so it carries a single place above 2, and every other factor in the product formula [8, Product Formula and Discreteness] has just been shown to be 1. So the one remaining factor, at that place, is also 1. So u satisfies the hypothesis of lemma 11.3.9 and is a root of unity of $\mathbb{K}$. The roots of unity of $\mathbb{Q}(\zeta_{2^j})$ are the 2^j-th roots of unity, so $u\in\mu_{\mathbb{C}}(2^\infty)$. Finally $u=\chi(\delta)q^{-(l+1)/2}\tau(\chi,\psi)=\chi(\delta)\varepsilon(\chi,\frac{1}{2},\psi)$ by box 11.3.2, so $\chi(\delta)\,\varepsilon(\chi,\frac{1}{2},\psi)\in\mu_{\mathbb{C}}(2^\infty)$. The hypothesis $p=2$ entered twice: once in the values of χ and ψ, once in the choice of $\mathbb{K}$.

Lemma 11.3.11 is the one-dimensional case of the section's main theorem, and the two-dimensional case is where the work is: an imprimitive ρ is induced from a character of a quadratic extension and the lemma applies there, while a primitive ρ has to be broken up by lemma 11.3.4 and reassembled. We write $z\sim w$ for $zw^{-1}\in\mu_{\mathbb{C}}(2^\infty)$.

Theorem 11.3.12: The congruence for $c(\rho,\psi)$

Assume $p=2$. Let $\rho\in\mathcal{G}_2^{\mathrm{wr}}(F)$, let ψ be non-trivial, and write $c_\rho=c(\rho,\psi)$. Then

$$\varepsilon(\rho,\tfrac12,\psi)^2\equiv\det\rho(c_\rho)^{-1}\pmod{\mu_{\mathbb{C}}(2^\infty)}\tag{11.23}$$

which we also write $\varepsilon(\rho,\frac12,\psi)\equiv\det\rho(c_\rho)^{-1/2}$: the square root is ambiguous only up to a sign, and $-1\in\mu_{\mathbb{C}}(2^\infty)$. The exponent 2 on the left of (11.23) is $\dim\rho$, and it must be read as such: for a one-dimensional $\rho=\chi$ the statement of the same shape is $\varepsilon(\chi,\frac12,\psi)\equiv\chi\big(c(\chi,\psi)\big)^{-1}$, which is lemma 11.3.11, whereas (11.23) transcribed to $\dim\rho=1$ with the exponent 2 left in place would be false.

***Proof*:**

Both sides are multiplied by $\nu(c_\rho)^{-2}$ when ρ is replaced by $\nu\otimes\rho$ for an unramified character ν of $\mathcal{W}_F$: indeed $c(\nu\otimes\rho,\psi)\equiv c_\rho$ by theorem 11.3.5(2), $\varepsilon(\nu\otimes\rho,\frac12,\psi)=\nu(c_\rho)^{-1}\varepsilon(\rho,\frac12,\psi)$ by (1) of the same theorem, and $\det(\nu\otimes\rho)=\nu^2\det\rho$. We may therefore twist freely by unramified characters.

Case 1: ρ imprimitive. Write $\rho=\mathrm{Ind}_{E/F}\xi$ with $[E:F]=2$. If E/F were unramified then ρ would lie in $\mathcal{G}_2^{\mathrm{nr}}(F)$, so E/F is ramified, hence totally wildly ramified because $p=2$. In particular $\mathcal{P}_F$ is not contained in $\mathcal{W}_E$, so $\mathcal{P}_E=\mathcal{P}_F\cap\mathcal{W}_E$ has index two in $\mathcal{P}_F$ and $\mathcal{P}_F\mathcal{W}_E=\mathcal{W}_F$. The character ξ has level $k\geq1$: were it tamely ramified, that is trivial on $\mathcal{P}_E$, Mackey's formula would give $\rho|_{\mathcal{P}_F}=\mathrm{Ind}_{\mathcal{P}_E}^{\mathcal{P}_F}\big(\xi|_{\mathcal{P}_E}\big)=1\oplus\nu$, with ν the non-trivial character of $\mathcal{P}_F/\mathcal{P}_E$. So $\rho|_{\mathcal{P}_F}$ would be reducible, contradicting proposition 11.3.8(1). Choose $\delta\in E^\times$ with $\xi(1+x)=\psi_E(\delta x)$ for $x\in\mathfrak{p}_E^k$. Then $c_\rho\equiv\mathrm{Nm}_{E/F}(\delta)$ by theorem 11.3.5(4).

Let σ generate $\mathrm{Gal}(E/F)$, so $\sigma(\delta)-\delta=\mathrm{tr}_{E/F}(\delta)-2\delta$. Here $d=d(E|F)\geq1$ by proposition 9.1.2(2), and $\mathrm{tr}_{E/F}(\mathfrak{p}_E^c)=\mathfrak{p}^{\lfloor(c+d+1)/2\rfloor}$, so $v_E(\mathrm{tr}_{E/F}\delta)\geq2\lfloor(v_E(\delta)+d+1)/2\rfloor\geq v_E(\delta)+d\geq v_E(\delta)+1>v_E(\delta)$. Also $v_E(2\delta)>v_E(\delta)$, the term being zero if $\mathrm{char}\,F=2$. Hence $\sigma(\delta)/\delta\in U_E^1$, and since U_E^1 is pro-2 and ξ is smooth, $\xi(\sigma(\delta)/\delta)\in\mu_{\mathbb{C}}(2^\infty)$. Therefore $\xi(\mathrm{Nm}_{E/F}\delta)=\xi(\delta)\xi(\sigma(\delta))\sim\xi(\delta)^2$.

Now $\det\rho=\varkappa_{E/F}\cdot(\xi|_{F^\times})$ by proposition 6.2.7, with $\varkappa_{E/F}^2=1$ (proposition 6.2.6), so $\varkappa_{E/F}$ takes the values ±1 and $\det\rho(c_\rho)\sim\xi(\mathrm{Nm}_{E/F}\delta)\sim\xi(\delta)^2$. On the other side (6.5) gives $\varepsilon(\rho,\frac12,\psi)=\lambda_{E/F}(\psi)\,\varepsilon(\xi,\frac12,\psi_E)$, and $\lambda_{E/F}(\psi)$ is a fourth root of unity by corollary 6.3.14, hence lies in $\mu_{\mathbb{C}}(2^\infty)$. So lemma 11.3.11 over E, with the character ψ_E, gives $\varepsilon(\rho,\frac12,\psi)\sim\varepsilon(\xi,\frac12,\psi_E)\sim\xi(\delta)^{-1}$. Squaring and comparing yields (11.23).

Case 2: ρ primitive. After an unramified twist, which exists by proposition 6.1.10, we may assume ρ has finite image, so that ρ is a faithful irreducible representation of $\Gamma=\mathrm{Gal}(E/F)$ for a finite Galois E/F. Let F_1/F be the maximal tamely ramified sub-extension of E/F and $\Gamma_1=\mathrm{Gal}(E/F_1)$, a finite 2-group, normal in Γ and equal to the image of $\mathcal{P}_F$. It is non-trivial because $\rho|_{\mathcal{P}_F}$ is irreducible (proposition 11.3.8(1)). Let Z be the centre of Γ_1, non-trivial by [4, p-Groups Have Nontrivial Center] and normal in Γ, being characteristic in Γ_1.

By [5, Clifford's Theorem], $\rho|Z$ is a sum of Γ-conjugates of a character φ, which is faithful on Z because ρ is faithful. If the two constituents were distinct, ρ would be induced from the index-two stabiliser of φ, hence imprimitive. So φ is Γ-stable and $\rho|Z=\varphi\oplus\varphi$.

Set $\widetilde{E}=E^Z$, Galois over F since $Z\trianglelefteq\Gamma$, and regard φ as a character χ of $\widetilde{E}^\times$ through

theorem 6.2.1. The extension $E/\widetilde{E}$ is a non-trivial sub-extension of the totally wildly ramified E/F_1, so χ is wildly ramified. Let $l \geq 1$ be its level and choose $\delta \in \widetilde{E}^\times$ with $\chi(1+x) = \psi_{\widetilde{E}}(\delta x)$ for $x \in \mathfrak{p}_{\widetilde{E}}^l$.

Since φ is Γ-stable we have $\chi^\gamma = \chi$ for every $\gamma \in \mathrm{Gal}(\widetilde{E}/F)$, and $\psi_{\widetilde{E}}$ is γ-invariant. Comparing $\chi(1+\gamma^{-1}x) = \psi_{\widetilde{E}}(\gamma(\delta)x)$ with the definition of δ gives

$$\gamma(\delta)/\delta \in U^1_{\widetilde{E}}, \qquad \gamma \in \Gamma \tag{11.24}$$

Apply lemma 11.3.4 with this Z and φ: in the Grothendieck group $\rho = \sum_i n_i \mathrm{Ind}_{\Delta_i}^\Gamma \xi_i$ with $Z \subseteq \Delta_i$, $\xi_i|Z = \varphi$ and $\sum_i n_i[\Gamma : \Delta_i] = 2$. Put $L_i = E^{\Delta_i} \subseteq \widetilde{E}$ and read ξ_i as a character of $L_i^\times$. Then $\xi_i \circ \mathrm{Nm}_{\widetilde{E}/L_i} = \chi$. Each ξ_i has level $l_i \geq 1$, since $\mathrm{Nm}_{\widetilde{E}/L_i}(U^1_{\widetilde{E}}) \subseteq U^1_{L_i}$ while χ is non-trivial on $U^1_{\widetilde{E}}$. Let δ_i be its attached element. Because φ is faithful on Z we have $\ker \xi_i \cap Z = 1$, and this forces $\Delta_i^u = 1$ for every $u > l_i$. Indeed such a Δ_i^u lies in $\ker \xi_i$, while lower numbering is compatible with subgroups, so $\Delta_i^u = \Delta_i \cap \Gamma_v$, where Γ_v is the v-th ramification group of Γ in the lower numbering and $v > 0$ is the index corresponding to u under the Herbrand correspondence for E/L_i [9, The Herbrand Function]. Now Γ_v is normal in Γ and, v being positive, contained in Γ_1. Were it non-trivial it would meet Z non-trivially: conjugation by Γ_1 permutes Γ_v in orbits of 2-power length whose fixed points are exactly $\Gamma_v \cap Z$, so $|\Gamma_v \cap Z| \equiv |\Gamma_v| \equiv 0 \pmod 2$, and $\Gamma_v \cap Z$ contains 1. Since $Z \subseteq \Delta_i$ that would give $1 \neq \Gamma_v \cap Z \subseteq \Delta_i \cap \Gamma_v = \Delta_i^u \subseteq \ker \xi_i$, against $\ker \xi_i \cap Z = 1$. Hence $\Gamma_v = 1$ and $\Delta_i^u = 1$: every ramification break of E/L_i, and with it every ramification break of $\widetilde{E}/L_i$, is at most l_i.

The extension $\widetilde{E}/L_i$ is in general wildly ramified: its Galois group is Δ_i/Z, whose wild part $(\Delta_i \cap \Gamma_1)/Z$ need not be trivial. So lemma 11.3.3 is used here in its general form. Writing $e_i = e(\widetilde{E}|L_i)$ and $d_i = d(\widetilde{E}|L_i)$, hypothesis (11.15) for $\widetilde{E}/L_i$ reads

$$e_i\, l_i + 2e_i > 2d_i + 4 \tag{11.25}$$

a level-against-different bound. It is what the rest of the primitive case rests on, and we prove it.

Put $H = \Gamma/Z$, $W = \Gamma_1/Z$ and $G = \Delta_i/Z = \mathrm{Gal}(\widetilde{E}/L_i)$. Lower-numbering ramification groups pass to subgroups, $G_w = G \cap H_w$ by [9, Subgroups and the Lower Numbering], and $H_1 = W$, since Γ_0/Γ_1 has odd order and W is therefore the 2-Sylow subgroup of the inertia group of H. Write $V = G_1 = G \cap W$ for the wild inertia subgroup of G, so that $G_w \subseteq V$ for every $w \geq 1$.

Step A: $\Gamma_1/Z \cong V_4$. The projective representation attached to ρ has kernel $Z(\Gamma)$, because ρ is faithful and its centre consists of scalars, so proposition 9.2.17 gives $\Gamma/Z(\Gamma) \cong A_4$ or S_4 with the image of Γ_1 an elementary abelian group of order four. That image is $\Gamma_1 Z(\Gamma)/Z(\Gamma) \cong \Gamma_1/(Z(\Gamma) \cap \Gamma_1)$, and $Z(\Gamma) \cap \Gamma_1 = Z$: one inclusion because $\rho|Z$ is scalar and ρ is faithful, the other because $Z(\Gamma) \cap \Gamma_1$ is central in Γ_1. Hence $\Gamma_1/Z \cong V_4$, and in particular Γ_1/Z is abelian, so $[\Gamma_1, \Gamma_1] \subseteq Z$.

Step B: $\Gamma_1 \not\subseteq \Delta_i$, *so* $|V| \leq 2$. The restriction $\rho|\Gamma_1$ is irreducible of dimension two by proposition 11.3.8(1), so Γ_1 is not abelian and $[\Gamma_1, \Gamma_1] \neq 1$. Were Γ_1 contained in Δ_i, the character ξ_i would be trivial on $[\Delta_i, \Delta_i]$, hence on $[\Gamma_1, \Gamma_1]$, which Step A places inside Z. That would make $\ker \xi_i \cap Z$ non-trivial, against the faithfulness of $\varphi = \xi_i|Z$. So V is a proper subgroup of $\Gamma_1/Z \cong V_4$ and $|V| \in \{1, 2\}$. Equivalently, no L_i produced by lemma 11.3.4 is tamely ramified over F.

Step C: what (11.25) *reduces to.* Since $G_w \subseteq V$ for $w \geq 1$ and $|V| \leq 2$, the filtration of G has

the shape G_0 of order e_i, then $G_1 = \cdots = G_\beta = V$ and $G_{\beta+1} = 1$, where β is the last lower break. By [9, The Different and the Ramification Filtration] and the shifted convention of definition 9.1.1,

$$d_i + 1 = \sum_{w \geq 0} \big(|G_w| - 1\big) = (e_i - 1) + \beta\big(|V| - 1\big)$$

so $2d_i + 4 = 2e_i + 2\beta(|V| - 1)$ and (11.25) is *equivalent* to

$$e_i\, l_i > 2\beta\big(|V| - 1\big)$$

If $|V| = 1$ the right-hand side is zero and $l_i \geq 1$ settles it. If $|V| = 2$ the inequality reads $e_i l_i > 2\beta$, and by [9, Properties of the Herbrand Function] the last upper break of $\widetilde{E}/L_i$ is $\varphi_{\widetilde{E}/L_i}(\beta) = 2\beta/e_i$. So everything reduces to the statement that l_i strictly exceeds the last break of $\widetilde{E}/L_i$.

The bound proved above is not by itself enough, and the reason locates exactly what Step B contributes. That bound gives every break at most l_i, which is the inequality only once $|V| \leq 2$: at $|V| = 4$ the reduced form reads $e_i l_i > 6\beta$ while the break bound yields only $e_i l_i > 2\beta$, and no relation between the two numberings closes the difference. So the exclusion in Step B is what makes the reduction usable.

Step D: the strict inequality. Let t be the last lower break of Γ, so $\Gamma_t \neq 1$ and $\Gamma_{t+1} = 1$, and $t \geq 1$ because $\Gamma_1 \neq 1$. Then Γ_t is normal in Γ by [9, The Ramification Filtration] and abelian, since $\Gamma_t = \Gamma_t/\Gamma_{t+1}$ embeds in a quotient of the unit filtration by [9, Ramification Quotients]. By [5, Clifford's Theorem] the restriction $\rho|\Gamma_t$ is a Γ-orbit of two characters. Were they distinct, ρ would be induced from the index-two stabiliser of one of them and so imprimitive. They therefore coincide, $\rho(\Gamma_t)$ is scalar, and faithfulness places $\Gamma_t \subseteq Z(\Gamma) \cap \Gamma_1 = Z$.

Now $\Gamma_t \subseteq Z \subseteq \Delta_i$, so $(\Delta_i)_t = \Delta_i \cap \Gamma_t = \Gamma_t$ and $(\Delta_i)_w = 1$ for $w > t$, whence $\Delta_i^{\varphi_{E/L_i}(t)} = \Gamma_t$ by [9, Ramification Groups in the Upper Numbering]. Two consequences. First, $\varphi_{E/L_i}(t) \leq l_i$, since $\Delta_i^u = 1$ for every $u > l_i$ by the paragraph above while $\Delta_i^{\varphi_{E/L_i}(t)} = \Gamma_t \neq 1$. Second, $G^u = \Delta_i^u Z/Z$ by [9, Herbrand's Theorem in the Upper Numbering], so the last break of $\widetilde{E}/L_i$ is the largest u with $\Delta_i^u \not\subseteq Z$, and $\Delta_i^u \subseteq Z$ already at $u = \varphi_{E/L_i}(t)$. So that break is strictly smaller than $\varphi_{E/L_i}(t) \leq l_i$, which is what Step C asked for. This proves (11.25).

Two remarks on what the argument uses. It needs no ramification input beyond the inclusion $\Delta_i^u \subseteq \ker \xi_i$ for $u > l_i$ that the paragraph above already invoked, and in particular not the reverse inclusion. That is why the inequality $\varphi_{E/L_i}(t) \leq l_i$ is proved rather than the equality, which would need it. The source leaves this step unproved: B-H assert the conclusion (11.26) with no norm-compatibility hypothesis at all [2, §49].

With (11.25) in hand, lemma 11.3.3 applied to ξ_i along $\widetilde{E}/L_i$ shows that δ_i is an admissible choice of the element attached to $\chi = \xi_i \circ \mathrm{Nm}_{\widetilde{E}/L_i}$ over $\widetilde{E}$:

$$\delta_i/\delta \in U^1_{\widetilde{E}} \tag{11.26}$$

The local constant. By multiplicativity, (6.5), corollary 6.3.14 and lemma 11.3.11,

$$\varepsilon(\rho, \tfrac{1}{2}, \psi) = \prod_i \Big(\lambda_{L_i/F}(\psi)\, \varepsilon(\xi_i, \tfrac{1}{2}, \psi_{L_i})\Big)^{n_i} \sim \prod_i \xi_i(\delta_i)^{-n_i} \tag{11.27}$$

The determinant. By theorem 11.3.5(4), $c_\rho \equiv \prod_i \mathrm{Nm}_{L_i/F}(\delta_i)^{n_i}$. The F-embeddings of L_i into E are the restrictions of coset representatives γ of Γ/Δ_i, all of which preserve $\widetilde{E}$, so (11.26) and (11.24) give

$$\mathrm{Nm}_{L_i/F}(\delta_i) = \prod_{\gamma\in\Gamma/\Delta_i} \gamma(\delta_i) \equiv \delta^{[\Gamma:\Delta_i]} \pmod{U^1_{\widetilde{E}}}$$

whence $c_\rho \equiv \delta^{\sum_i n_i[\Gamma:\Delta_i]} = \delta^2 \pmod{U^1_{\widetilde{E}}}$.

Extend each ξ_i to a smooth character ζ_i of the subgroup $L_i^\times\langle\delta\rangle U^1_{\widetilde{E}}$ of $\widetilde{E}^\times$. The extension has to be smooth, not merely a homomorphism, because the next display evaluates ζ_i on elements congruent to one another only modulo $U^1_{\widetilde{E}}$, so it is built in three steps.

First extend $\xi_i|_{U^1_{L_i}}$ to $U^1_{\widetilde{E}}$. Being smooth, ξ_i is trivial on $U^m_{L_i}$ for some $m \geq 1$, and since the valuation of $\widetilde{E}$ restricts to a multiple of that of L_i there is an n with $U^n_{\widetilde{E}} \cap L_i^\times \subseteq U^m_{L_i}$. So $\xi_i|_{U^1_{L_i}}$ factors through the finite abelian group $U^1_{L_i}/(U^n_{\widetilde{E}} \cap U^1_{L_i})$, which the inclusion embeds into the finite abelian group $U^1_{\widetilde{E}}/U^n_{\widetilde{E}}$. A character of a subgroup of a *finite* abelian group extends to the whole group, since $\mathbb{C}^\times$ is divisible and the induction is over a finite chain. Pulling back gives a character of $U^1_{\widetilde{E}}$ trivial on $U^n_{\widetilde{E}}$, hence smooth.

Next glue. Since $v_{\widetilde{E}}$ restricted to $L_i^\times$ is a positive multiple of v_{L_i}, one has $L_i^\times \cap U^1_{\widetilde{E}} = U^1_{L_i}$, so ξ_i and the character just built agree on the intersection and together define a smooth character of $L_i^\times U^1_{\widetilde{E}}$. Finally extend across $\langle\delta\rangle$, which is cyclic; here divisibility of $\mathbb{C}^\times$ is what is needed and smoothness is preserved because the subgroup already contains the open subgroup $U^1_{\widetilde{E}}$. A smooth character of the pro-2 group $U^1_{\widetilde{E}}$ has open kernel of 2-power index, so it takes values in $\mu_{\mathbb{C}}(2^\infty)$. Then $\zeta_i(c_\rho) \sim \zeta_i(\delta)^2 \sim \zeta_i(\delta_i)^2 = \xi_i(\delta_i)^2$ by (11.26). Since $\det \mathrm{Ind}_{L_i/F}\xi_i = \varkappa_{L_i/F} \cdot (\xi_i|_{F^\times})$ by proposition 6.2.7 and $\varkappa_{L_i/F}$ takes the values ± 1 (proposition 6.2.6),

$$\det \rho(c_\rho) = \prod_i \big(\varkappa_{L_i/F}(c_\rho)\,\xi_i(c_\rho)\big)^{n_i} \sim \prod_i \xi_i(\delta_i)^{2n_i} \tag{11.28}$$

Comparing (11.27) with (11.28) gives $\det\rho(c_\rho) \sim \varepsilon(\rho, \frac{1}{2}, \psi)^{-2}$, which is (11.23).

Theorem 11.3.12 is what this section was after: an invariant of ρ visible through a single element of $F^\times$ modulo U^1_F, which (11.19) transports along a tame extension in a controlled way.

Exercise 11.3.1

1. Let χ be a character of $F^\times$ of level $l \geq 1$ and let δ be as in lemma 11.3.11. Show that δ is a tame period for (χ, ψ), and deduce that lemma 11.3.11 is the congruence of theorem 11.3.12 written for a one-dimensional ρ, that is, with the exponent $\dim\rho$ on the left of (11.23) read as 1. Show also that transcribing (11.23) to $\dim\rho = 1$ with its exponent 2 left in place gives a false statement.

2. Let $p = 2$, let $\rho \in \mathcal{G}_2^{\mathrm{wr}}(F)$ and let χ be a tamely ramified character of $F^\times$. Verify directly that (11.23) for ρ is equivalent to (11.23) for $\chi \otimes \rho$.

3. Let $\rho \in \mathcal{G}^{\mathrm{ss}}(F)$ with $\rho|_{\mathcal{P}_F}$ omitting the trivial character, let $a \in F^\times$ and let $a\psi\colon x \mapsto \psi(ax)$. Show that $c(\rho, a\psi) \equiv a^{-\dim\rho} c(\rho, \psi) \pmod{U^1_F}$.

11.4 The Godement-Jacquet Constant Under Lifting

Section 11.3 attached to a totally ramified parameter ρ an element $c(\rho,\psi)$ of $F^\times$, canonical modulo U_F^1, and showed that it controls both the behaviour of $\varepsilon(\rho,s,\psi)$ under restriction to a tame extension and the class of $\varepsilon(\rho,\frac{1}{2},\psi)$ modulo roots of unity of 2-power order. This section proves the matching statements on the group side. It attaches to a totally ramified cuspidal representation π of G an element $c(\pi,\psi)$ with the same defining property, computes it in the two cases that between them exhaust $\mathcal{A}_2^{\mathrm{wr}}(F)$, and determines $\varepsilon(\pi_K,s,\psi_K)$ in terms of $\varepsilon(\pi,s,\psi)$ for a cubic extension K/F. The last of these, proposition 11.4.11, has the same shape as the Galois-side rule (11.19), and setting the two against each other is corollary 11.4.12, the statement this chapter exists to produce.

The comparison is possible at all because the subscript K means opposite things on the two sides. On the Galois side $\rho_K = \mathrm{Res}_{K/F}\rho$ is a restriction, and on the group side $\pi_K = \mathrm{Lft}_{K/F}\pi$ is the lift constructed in section 11.1. That the two operations obey one rule is what lets a representation recognized over K be pulled back to F. The pulling back itself is carried out in [2, §50], not here.

Four conventions govern what follows, and they are collected here rather than reconstructed piecemeal at their points of use.

11.4.1 Conventions *The additive character.* Section 11.1 fixes once and for all a character $\psi \in \widehat{F}$ of level one, and forms all strata and all sets $\mathcal{C}(\psi_\alpha, \mathfrak{A})$ and $\mathcal{A}_2^0(F;\alpha)$ with respect to it. That convention applies here without exception. For a finite tamely ramified K/F the character $\psi_K = \psi \circ \mathrm{tr}_{K/F}$ again has level one, by lemma 4.1.2(1), so the same apparatus is available over K with no adjustment. Section 11.3 relaxes the restriction and works at an arbitrary non-trivial ψ. Every result of that section quoted below is used only in its level-one specialisation, and this is said again at each point of use. We write $\psi_{\mathfrak{A}}(x) = \psi(\mathrm{tr}x)$ for $x \in A = M_2(F)$, as theorem 5.3.8 does, and $\psi_\alpha(1+x) = \psi_{\mathfrak{A}}(\alpha x)$ on $U_{\mathfrak{A}}^{m+1}$.
The two elements. The symbol α is the element of a simple stratum $(\mathfrak{A}, n, \alpha)$, as section 11.1 fixes it, and δ is the element attached to a character χ of $F^\times$ relative to ψ. The two meet in a single expression in this section, the element $\alpha + \delta$ of lemma 11.4.4, and that expression is the reason the two symbols are kept apart.
The pinning of δ, which is finer here than in section 11.3. For a character χ of $F^\times$ of level $l \geq 1$ we always take $\delta \in F^\times$ with

$$\chi(1+y) = \psi(\delta y), \qquad y \in \mathfrak{p}^{\lfloor l/2 \rfloor + 1} \tag{11.29}$$

which is the normalization of theorem 5.1.21 and proposition 6.3.3(4). Box 11.3.2 pins its δ on the smaller ideal $\mathfrak{p}^l$ instead, which is the weaker of the two conditions because the pinning is asked on fewer elements, notes that, and observes that the two choices agree modulo U_F^1. That is enough for section 11.3, where only tamely ramified characters ever evaluate δ. It is *not* enough here. Lemma 11.4.3 needs $\chi(\det(1+x)) = \psi_{\mathfrak{A}}(\delta x)$ for $x \in \mathfrak{P}^{l+1}$, and $\mathrm{tr}(\mathfrak{P}^{l+1}) = \mathfrak{p}^{\lfloor l/2 \rfloor + 1}$ with equality, so replacing δ by δu with $v_F(u-1) = 1$ leaves (11.29) on $\mathfrak{p}^l$ intact and breaks the determinant identity as soon as $l \geq 3$. The hypothesis $v_F(u-1) = 1$ is needed and not a convenience: the obstruction lies in $\mathfrak{p}^{\lfloor l/2 \rfloor + 1 - l + v_F(u-1)}$, so a u with $v_F(u-1) \geq \lceil l/2 \rceil$ leaves clause (3) standing. Exercise 11.4.1 (3) works out the threshold. Every statement below whose value is read modulo U_F^1 is of course insensitive to the difference, and that includes the tame period itself.
The sign, and the two kappas. Box 11.1.1 fixes $\varkappa_{K/F} = \det R_{K/F}$ and reserves $\kappa_{E/F}$ for the quadratic class-field character of a quadratic E/F. Below, $\varkappa$ always means $\det R_{K/F}$. By lemma 11.1.7(1) the character $\varkappa_{K/F}$ is unramified for tame K/F, and $\varepsilon_{K/F} = \varkappa_{K/F}(\varpi) \in \{\pm 1\}$ is its value at any prime element.

Throughout, K/F is a finite tamely ramified extension, $e_K = e(K|F)$, $f_K = f(K|F)$ and $q_K = q^{f_K}$. For a character χ of $F^\times$ we write $\chi_K = \chi \circ \mathrm{Nm}_{K/F}$, as proposition 11.1.12(2) does. Every result of the earlier chapters quoted below is stated over the base field of the book, but each names only a non-archimedean local field, a character of it of level one, and the objects built from those, so each applies verbatim over K. That is the reading section 11.1 already uses, for instance when it applies theorem 3.5.5 and lemma 11.1.13 over K.

The first step is a count of levels. It is what makes every identity below an identity of complex numbers rather than of functions of s: once the two sides are monomials in q^{-s} of the same degree, checking them at $s = \frac{1}{2}$ settles them everywhere.

Lemma 11.4.2: Levels Under Lifting

Let $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F)$ and let K/F be a finite tamely ramified extension. Then

$$\ell(\pi_K) = e_K\,\ell(\pi), \qquad n(\pi_K,\psi_K) = e_K\,n(\pi,\psi)$$

and consequently

$$q_K^{\,n(\pi_K,\psi_K)} = q^{\,[K:F]\,n(\pi,\psi)} \tag{11.30}$$

***Proof*:**
By proposition 10.1.10(1) one has $n(\varrho,\eta) = 2\ell(\varrho)$ for every cuspidal ϱ and every level one η, over F and over K alike, so the second assertion follows from the first. Given the first, $q_K = q^{f_K}$ and $e_K f_K = [K:F]$ give

$$q_K^{\,n(\pi_K,\psi_K)} = q^{\,f_K\cdot 2e_K\ell(\pi)} = q^{\,[K:F]\cdot 2\ell(\pi)}$$

which is (11.30). So only $\ell(\pi_K) = e_K\ell(\pi)$ is at issue, and by lemma 11.1.13(1) there are two cases.

Step 1: π minimal. By lemma 11.1.13(2) there is a ramified simple stratum $(\mathfrak{A},n,\alpha)$ with $\pi \in \mathcal{A}_2^0(F;\alpha)$, and $\ell(\pi) = n/2$ by proposition 10.1.3(2). Definition 11.1.14(1) makes π_K the representation induced from the lifted type, so $\pi_K \in \mathcal{A}_2^0(K;\alpha)$ for the lifted stratum $(\mathfrak{A}_K,n_K,\alpha)$, and $n_K = e_K n$ by proposition 11.1.4(1). Applying proposition 10.1.3(2) over K gives $\ell(\pi_K) = n_K/2 = e_K n/2 = e_K\ell(\pi)$.

Step 2: the level of a non-minimal representation. Let $(\mathfrak{B},r,\beta)$ be a ramified simple stratum over a local field L with a level one character η, let $\Theta \in \mathcal{C}(\eta_\beta,\mathfrak{B})$, let $\varrho = \text{c-Ind}\Theta$, and let ξ be a character of $L^\times$ of level t with $2t > r$ and attached element δ_ξ pinned by (11.29) over L. Write $\eta_{\mathfrak{B}}(y) = \eta(\mathrm{tr}y)$ and $\eta_\beta(1+x) = \eta_{\mathfrak{B}}(\beta x)$, as box 11.4.1 does over F. We claim $\ell(\xi\varrho) = t$.

Indeed $\xi\varrho = \text{c-Ind}\big((\xi\circ\det)\Theta\big)$ by proposition 1.2.30. Since $e_{\mathfrak{B}} = 2$ one has $\mathfrak{P}_{\mathfrak{B}}^{2t} = \varpi_L^t\mathfrak{B}$, so every $x \in \mathfrak{P}_{\mathfrak{B}}^{2t}$ is $\varpi_L^t x'$ with $x' \in \mathfrak{B}$. The trace and the determinant of an element of an $\mathfrak{o}_L$-order in $M_2(L)$ are integral, being coefficients of its characteristic polynomial, so $\mathrm{tr}x = \varpi_L^t\mathrm{tr}x' \in \mathfrak{p}_L^t$ and $\det x = \varpi_L^{2t}\det x' \in \mathfrak{p}_L^{2t}$. As $t \geq \lfloor t/2\rfloor + 1$ for $t \geq 1$, both $\mathrm{tr}x$ and $\det x$ lie in the range of (11.29), and $v_L(\delta_\xi\det x) \geq -t + 2t \geq 1$, so

$$\xi\big(\det(1+x)\big) = \xi\big(1+\mathrm{tr}x+\det x\big) = \eta\big(\delta_\xi\mathrm{tr}x\big) = \eta_{\mathfrak{B}}\big(\delta_\xi x\big), \qquad x \in \mathfrak{P}_{\mathfrak{B}}^{2t}$$

Now Θ is trivial on $U_{\mathfrak{B}}^{r+1} \supseteq U_{\mathfrak{B}}^{2t}$, so $\xi\varrho$ contains the character η_{δ_ξ} of the stratum $(\mathfrak{B},2t,\delta_\xi)$ on $U_{\mathfrak{B}}^{2t}$. That stratum is fundamental by proposition 3.3.11, since δ_ξ is a scalar and proposition 3.3.7(4) gives $\nu_{\mathfrak{B}}(\delta_\xi^j) = 2v_L(\delta_\xi^j) = -2tj$ for every $j \geq 1$. Hence theorem 3.3.13 gives $\ell(\xi\varrho) = 2t/e_{\mathfrak{B}} = t$, which is the claim.

Step 3: π not minimal. Write $\pi = \chi\pi'$ with π' minimal, by lemma 11.1.13(1), let $(\mathfrak{A},n,\alpha)$ be a ramified simple stratum with $\pi' \in \mathcal{A}_2^0(F;\alpha)$ and let l be the level of χ. Since π is not minimal, lemma 11.1.13(3) gives $2l > n$, and in particular $l \geq 1$. Step 2 over F gives $\ell(\pi) = l$.

Over K we have $\pi_K = \chi_K\pi'_K$ by definition 11.1.14(2), with π'_K attached to $(\mathfrak{A}_K,n_K,\alpha)$ as in Step 1 and χ_K of level $e_K l$ by proposition 4.1.3, which applies because K/F is tamely ramified

and $l \geq 1$. That proposition also names δ itself as an admissible attached element for χ_K relative to ψ_K, pinned on $\mathfrak{p}_K^{\lfloor e_K l/2 \rfloor + 1}$, which is (11.29) over K. Finally $2e_K l > e_K n = n_K$, so Step 2 applies over K to the pair (χ_K, π'_K) and gives $\ell(\pi_K) = e_K l = e_K \ell(\pi)$, completing the proof.

The identity (11.30) is the whole of the reduction to $s = \frac{1}{2}$. Both $\varepsilon(\pi_K, s, \psi_K)$ and $\varepsilon(\pi, s, \psi)^{[K:F]}$ are non-zero constants times $q_K^{-n(\pi_K, \psi_K)s}$ and $q^{-[K:F]n(\pi,\psi)s}$ respectively, by definition 5.3.4, and (11.30) says the two exponentials coincide.

The rest of the section rests on one computation of a local constant, and that computation is the place where the pinning (11.29) is used. We isolate it.

Lemma 11.4.3: The Determinant of a Deep Unit Against a Character

Let $\mathfrak{A}$ be a chain order in A with $e_{\mathfrak{A}} = 2$ and radical $\mathfrak{P}$, let χ be a character of $F^\times$ of level $l \geq 1$, and let $\delta \in F^\times$ satisfy (11.29). Then:

1. $v_F(\delta) = -l$ and $\nu_{\mathfrak{A}}(\delta) = -2l$
2. $\operatorname{tr}(\mathfrak{P}^j) = \mathfrak{p}^{\lceil j/2 \rceil}$ for every $j \in \mathbb{Z}$, and $v_F(\det x) \geq j$ for every $x \in \mathfrak{P}^j$
3. $\chi\big(\det(1+x)\big) = \psi_{\mathfrak{A}}(\delta x)$ for every $x \in \mathfrak{P}^{l+1}$

Clause (2) names only $\mathfrak{A}$, and is quoted below at chain orders carrying no character at all. Moreover, if $u \in U_F^1$ has $v_F(u-1) = 1$ then δu satisfies (11.29) on the smaller ideal $\mathfrak{p}^l$, while clause (3) fails for δu at some $x \in \mathfrak{P}^{l+1}$ as soon as $l \geq 3$.

***Proof*:**

Part (1). The character χ has level exactly l, so $y \mapsto \psi(\delta y)$ is non-trivial on $\mathfrak{p}^l$ and trivial on $\mathfrak{p}^{l+1}$. As ψ has level one this says that $\delta\mathfrak{p}^l$ is not contained in $\mathfrak{p}$ while $\delta\mathfrak{p}^{l+1}$ is, that is $v_F(\delta) = -l$. Then proposition 3.3.7(4) gives $\nu_{\mathfrak{A}}(\delta) = e_{\mathfrak{A}} v_F(\delta) = -2l$.

Part (2), the trace. Since ψ has level one, an $\mathfrak{o}$-submodule M of F satisfies $\psi(M) = 1$ if and only if $M \subseteq \mathfrak{p}$, and $\psi_{\mathfrak{A}}$ is trivial on $\mathfrak{P}^j$ exactly when $j \geq 1$. This is the normalization fixed alongside definition 3.3.8 and used again in theorem 5.3.8. Now $\varpi \in \mathcal{K}_{\mathfrak{A}}$ with $v_{\mathfrak{A}}(\varpi) = 2$ by proposition 3.3.7(4), so $\varpi\mathfrak{A} = \mathfrak{P}^2$ by part (2) of the same proposition, and therefore $\varpi^{1-r}\mathfrak{P}^j = \mathfrak{P}^{j+2-2r}$ for every r. Since tr is $\mathfrak{o}$-linear,

$$\operatorname{tr}(\mathfrak{P}^j) \subseteq \mathfrak{p}^r \iff \operatorname{tr}(\varpi^{1-r}\mathfrak{P}^j) \subseteq \mathfrak{p} \iff \operatorname{tr}(\mathfrak{P}^{j+2-2r}) \subseteq \mathfrak{p} \iff j + 2 - 2r \geq 1$$

the first equivalence because $\varpi^{1-r}M \subseteq \mathfrak{p}$ says exactly $M \subseteq \mathfrak{p}^r$, and the last because $\psi_{\mathfrak{A}}$ is trivial on $\mathfrak{P}^k$ exactly when $k \geq 1$, which is the statement just quoted. The condition $j + 2 - 2r \geq 1$ reads $r \leq (j+1)/2$, and the greatest such integer is $\lfloor (j+1)/2 \rfloor = \lceil j/2 \rceil$. As $\operatorname{tr}(\mathfrak{P}^j)$ is an $\mathfrak{o}$-submodule of F, and the proper non-zero ones are exactly the $\mathfrak{p}^r$, it equals $\mathfrak{p}^{\lceil j/2 \rceil}$.

Part (2), the determinant. For $x \in A$ the Cayley-Hamilton identity in rank two reads $\operatorname{adj}(x) = \operatorname{tr}(x) \cdot 1 - x$. If $x \in \mathfrak{P}^j$ then $\operatorname{tr} x \in \mathfrak{p}^{\lceil j/2 \rceil}$ by the previous paragraph, so $\operatorname{tr}(x) \cdot 1 \in \mathfrak{p}^{\lceil j/2 \rceil}\mathfrak{A} = \mathfrak{P}^{2\lceil j/2 \rceil} \subseteq \mathfrak{P}^j$, and hence $\operatorname{adj}(x) \in \mathfrak{P}^j$. Then $\det(x) \cdot 1 = x \operatorname{adj}(x) \in \mathfrak{P}^{2j}$, so $2v_F(\det x) = \nu_{\mathfrak{A}}(\det(x) \cdot 1) \geq 2j$ and $v_F(\det x) \geq j$.

Part (3). Let $x \in \mathfrak{P}^{l+1}$. By part (2), $\mathrm{tr}x \in \mathfrak{p}^{\lceil (l+1)/2 \rceil} = \mathfrak{p}^{\lfloor l/2 \rfloor + 1}$ and $v_F(\det x) \geq l+1 \geq \lfloor l/2 \rfloor + 1$, so $\mathrm{tr}x + \det x$ lies in the ideal on which (11.29) evaluates χ. Hence

$$\chi\big(\det(1+x)\big) = \chi\big(1 + \mathrm{tr}x + \det x\big) = \psi\big(\delta\,\mathrm{tr}x\big)\,\psi\big(\delta \det x\big) = \psi\big(\mathrm{tr}(\delta x)\big) = \psi_{\mathfrak{A}}(\delta x)$$

the third equality because $v_F(\delta \det x) \geq -l + l + 1 = 1$ and ψ is trivial on $\mathfrak{p}$.

The last assertion. Let $u \in U_F^1$ with $v_F(u-1) = 1$. For $y \in \mathfrak{p}^l$ one has $v_F(\delta(u-1)y) \geq -l + 1 + l = 1$, so $\psi(\delta u y) = \psi(\delta y)$, and $\mathfrak{p}^l \subseteq \mathfrak{p}^{\lfloor l/2 \rfloor + 1}$ because $l \geq \lfloor l/2 \rfloor + 1$ for $l \geq 1$. So δu satisfies (11.29) on $\mathfrak{p}^l$. On the other hand, for $x \in \mathfrak{P}^{l+1}$ the two sides of (3) formed with δu differ by the factor $\psi(\delta(u-1)\mathrm{tr}x)$, and as x runs over $\mathfrak{P}^{l+1}$ the element $\mathrm{tr}x$ runs over all of $\mathfrak{p}^{\lfloor l/2 \rfloor + 1}$, by the equality in part (2). The resulting ideal $\delta(u-1)\mathfrak{p}^{\lfloor l/2 \rfloor + 1} = \mathfrak{p}^{\lfloor l/2 \rfloor - l + 2}$ is contained in $\mathfrak{p}$ if and only if $\lfloor l/2 \rfloor \geq l - 1$, which fails exactly when $l \geq 3$. For such l the character ψ is non-trivial on that ideal, so some x makes the factor differ from 1, completing the proof.

Part (3) is the identity a level-one character sees, and it is worth naming what would go wrong without it. The two normalizations of box 11.3.2 differ by a unit of U_F^1, and U_F^1 is invisible to every tamely ramified character, which is why section 11.3 never has to choose. Here the character being evaluated is χ itself, of level l, and it sees U_F^1 perfectly well.

We can now compute the local constant of a non-minimal totally ramified representation. The datum that carries the computation is not a simple stratum, and it is worth saying so before the statement rather than after: the element $\alpha + \delta$ has even valuation in E, so it is not minimal, and the triple $(\mathfrak{A}, 2l, \alpha + \delta)$ is not a stratum of any of the four types definition 3.4.1 names. Theorem 5.3.8 is stated without any hypothesis of that kind. It asks for an element c of the inducing group with $c\mathfrak{A} = \mathfrak{P}^{-N}$, for the inducing representation to be a scalar multiple of $\psi_{\mathfrak{A}}(c\,\cdot)$ deep enough in the filtration, which is exactly what lemma 11.4.3(3) establishes, and for the containment $U_{\mathfrak{A}}^{[(N+1)/2]} \subseteq J$, checked in the proof below.

Lemma 11.4.4: The Local Constant at an Even Level

Let $(\mathfrak{A}, n, \alpha)$ be a ramified simple stratum in A with $n = 2m+1 \geq 1$, put $E = F[\alpha]$ and $J = E^\times U_{\mathfrak{A}}^{m+1}$, let $\Lambda' \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$ and $\pi' = \text{c-Ind}_J^G \Lambda'$. Let χ be a character of $F^\times$ of level l with $2l > n$, let δ satisfy (11.29), and put $\pi = \chi\pi'$.

1. $\alpha + \delta \in E^\times$, with $v_E(\alpha+\delta) = v_{\mathfrak{A}}(\alpha+\delta) = -2l$ and $(\alpha+\delta)\mathfrak{A} = \mathfrak{P}^{-2l}$. The element $\alpha + \delta$ is not minimal over F, and $(\mathfrak{A}, 2l, \alpha+\delta)$ is a fundamental stratum belonging to none of the four types of definition 3.4.1.

2. $n(\pi, \psi) = 2l$ and

$$\varepsilon\big(\pi, \tfrac{1}{2}, \psi\big) = q^{-1}\,\psi(\mathrm{tr}\alpha)\,\Lambda'(\alpha+\delta)^{-1} \sum_x \chi\big(\det((\alpha+\delta)x)\big)^{-1}\,\psi_{\mathfrak{A}}(\delta x) \tag{11.31}$$

the sum being over $x \in U_{\mathfrak{A}}^l / U_{\mathfrak{A}}^{l+1}$.

***Proof*:**

Part (1). The field $E = F[\alpha]$ contains the scalar δ, so $\alpha + \delta \in E$, and $E^\times \subseteq \mathcal{K}_\mathfrak{A}$ by proposition 3.4.3(2). By corollary 3.4.4 the two valuations agree on $E^\times$, and $v_E(\alpha) = -n$ while $v_E(\delta) = e(E|F)v_F(\delta) = -2l$ by lemma 11.4.3(1). Since $2l > n$ these are distinct, so $v_E(\alpha+\delta) = -2l$, and in particular $\alpha + \delta \neq 0$. Then proposition 3.3.7(2) turns $v_\mathfrak{A}(\alpha+\delta) = -2l$ into $(\alpha+\delta)\mathfrak{A} = \mathfrak{P}^{-2l}$.

Since δ is a scalar, $F[\alpha+\delta] = E$ is a ramified quadratic extension, and definition 3.4.2 calls $\alpha+\delta$ minimal over F exactly when $-v_E(\alpha+\delta)$ is odd. It is $2l$, so $\alpha+\delta$ is not minimal over F. The stratum $(\mathfrak{A}, 2l, \alpha+\delta)$ is fundamental, because $\alpha+\delta$ normalizes $\mathfrak{A}$ and hence $\nu_\mathfrak{A}\big((\alpha+\delta)^r\big) = -2lr$ for every $r \geq 1$, so that $(\alpha+\delta)^r \notin \mathfrak{P}^{1-2lr}$, which is the criterion of proposition 3.3.11. It is of none of the four types of definition 3.4.1: the first three require $e_\mathfrak{A} = 1$, and the fourth requires the level to be odd, while $e_\mathfrak{A} = 2$ and $2l$ is even.

Part (2), the hypotheses of theorem 5.3.8. Apply that theorem with the chain order $\mathfrak{A}$, the group J, the character $\Lambda = (\chi \circ \det)\Lambda'$, the integer $N = 2l$ and the element $c = \alpha + \delta$. The group $J = E^\times U_\mathfrak{A}^{m+1}$ lies in $\mathcal{K}_\mathfrak{A}$ by proposition 3.4.3(2), contains $Z = F^\times$ because $F^\times \subseteq E^\times$, is open because $U_\mathfrak{A}^{m+1}$ is, and is compact modulo Z because $\mathcal{K}_\mathfrak{A}$ is. By proposition 1.2.30, $\text{c-Ind}_J^G \Lambda = \chi\pi' = \pi$, and π is irreducible and cuspidal because twisting by $\chi \circ \det$ changes neither the lattice of G-stable subspaces, the twisting factor being scalar, nor the Jacquet module, since det is trivial on N. The dimension of Λ is one.

Hypothesis (1) of theorem 5.3.8 asks for $U_\mathfrak{A}^{[(N+1)/2]} = U_\mathfrak{A}^l \subseteq J$. From $2l > n = 2m+1$ we get $l \geq m+1$, so $U_\mathfrak{A}^l \subseteq U_\mathfrak{A}^{m+1} \subseteq J$. Hypothesis (2) asks for an element $c \in J$ with $c\mathfrak{A} = \mathfrak{P}^{-N}$ and $\Lambda(1+x) = \psi_\mathfrak{A}(cx)$ for $x \in \mathfrak{P}^{[N/2]+1} = \mathfrak{P}^{l+1}$. Part (1) gives $c = \alpha + \delta \in E^\times \subseteq J$ with $(\alpha+\delta)\mathfrak{A} = \mathfrak{P}^{-2l}$. For the character condition, let $x \in \mathfrak{P}^{l+1}$. Since $l+1 > m+1$ we have $1 + x \in U_\mathfrak{A}^{m+1}$, so $\Lambda'(1+x) = \psi_\alpha(1+x) = \psi_\mathfrak{A}(\alpha x)$, while lemma 11.4.3(3) gives $\chi(\det(1+x)) = \psi_\mathfrak{A}(\delta x)$. Multiplying,

$$\Lambda(1+x) = \psi_\mathfrak{A}(\alpha x)\,\psi_\mathfrak{A}(\delta x) = \psi_\mathfrak{A}\big((\alpha+\delta)x\big)$$

which is the condition.

Part (2), the conclusion. Theorem 5.3.8 now gives $n(\pi,\psi) = 2N/e_\mathfrak{A} = 2l$, and since $N = 2l$ is even and $e_\mathfrak{A} = 2$ the constant there satisfies $q^a \dim \Lambda = (\mathfrak{A} : \mathfrak{P})^{-1/2} = q^{-2/e_\mathfrak{A}} = q^{-1}$, so $q^a = q^{-1}$. The index set $U_\mathfrak{A}^{[(N+1)/2]}/U_\mathfrak{A}^{[N/2]+1}$ is $U_\mathfrak{A}^l/U_\mathfrak{A}^{l+1}$, and $\text{tr}\check{\Lambda}(g) = \Lambda(g)^{-1}$ because Λ is a character. Evaluating (5.12) at $s = \frac{1}{2}$,

$$\varepsilon\big(\pi, \tfrac{1}{2}, \psi\big) = q^{-1} \sum_x \Lambda\big((\alpha+\delta)x\big)^{-1}\,\psi_\mathfrak{A}\big((\alpha+\delta)x\big)$$

with x over $U_\mathfrak{A}^l/U_\mathfrak{A}^{l+1}$.

Part (2), the summand. Fix such an x. Both $\alpha+\delta$ and x lie in J, and Λ is a character of J, so

$$\Lambda\big((\alpha+\delta)x\big) = \chi\big(\det((\alpha+\delta)x)\big)\,\Lambda'(\alpha+\delta)\,\Lambda'(x)$$

Now $U_\mathfrak{A}^l \subseteq U_\mathfrak{A}^{m+1}$, so $\Lambda'(x) = \psi_\alpha(x) = \psi_\mathfrak{A}(\alpha(x-1))$. On the other side $\psi_\mathfrak{A}$ is a character of the additive group A, so

$$\psi_\mathfrak{A}\big((\alpha+\delta)x\big) = \psi_\mathfrak{A}(\alpha x)\,\psi_\mathfrak{A}(\delta x) = \psi_\mathfrak{A}(\alpha)\,\psi_\mathfrak{A}\big(\alpha(x-1)\big)\,\psi_\mathfrak{A}(\delta x)$$

The two factors $\psi_{\mathfrak{A}}(\alpha(x-1))$ cancel, and $\psi_{\mathfrak{A}}(\alpha)=\psi(\mathrm{tr}\alpha)$. What remains of the summand is $\chi(\det((\alpha+\delta)x))^{-1}\Lambda'(\alpha+\delta)^{-1}\psi(\mathrm{tr}\alpha)\psi_{\mathfrak{A}}(\delta x)$, and pulling the three factors that do not depend on x out of the sum gives (11.31), completing the proof.

The sum in (11.31) is evaluated below, in corollary 11.4.8. For the tame period it is not needed: only the way the sum responds to a tamely ramified twist matters, and that is visible without evaluating anything.

Definition 11.4.5: The Tame Period of a Cuspidal Representation

Let $\pi\in\mathcal{A}_2^0(F)$. An element $c\in F^\times$ is a *tame period* for (π,ψ) if

$$\varepsilon(\chi\pi,s,\psi)=\chi(c)^{-1}\,\varepsilon(\pi,s,\psi) \tag{11.32}$$

for every tamely ramified character χ of $F^\times$, where $\chi\pi=(\chi\circ\det)\pi$. A tame period is unique modulo U_F^1 when it exists, since the tamely ramified characters of $F^\times$ are exactly those trivial on U_F^1 and so separate the cosets of U_F^1. We write $c(\pi,\psi)$, or c_π when ψ is fixed, for any representative, and read every identity involving it modulo U_F^1.

This is definition 11.3.1 with a cuspidal representation of G in place of a representation of $\mathcal{W}_F$ and $\chi\pi$ in place of $\chi\otimes\rho$. The parallel is exact and it is the reason both objects are called tame periods, but nothing below transports between them formally. Corollary 11.4.12 compares the two only through the identities each satisfies.

Proposition 11.4.6: Existence and the Two Values of the Tame Period

Let $\pi\in\mathcal{A}_2^{\mathrm{wr}}(F)$.

1. If $\pi\in\mathcal{A}_2^0(F;\alpha)$ for a ramified simple stratum $(\mathfrak{A},n,\alpha)$ with $n\geq 1$, then $\det\alpha$ is a tame period for (π,ψ).
2. If π is not minimal, write $\pi=\chi\pi'$ with π' minimal, as in lemma 11.1.13(1), and let $l\geq 1$ be the level of χ and δ its attached element as in (11.29). Then δ^2 is a tame period for (π,ψ).
3. Consequently $c(\pi,\psi)$ exists for every $\pi\in\mathcal{A}_2^{\mathrm{wr}}(F)$, and $v_F(c(\pi,\psi))=-n(\pi,\psi)$.
4. For every finite tamely ramified K/F,

$$c(\pi_K,\psi_K)\equiv c(\pi,\psi)\pmod{U_K^1} \tag{11.33}$$

***Proof*:**

Part (1). We record first a consequence of lemma 11.4.3(2) used three times below: for every chain order $\mathfrak{A}$ with $e_{\mathfrak{A}}=2$ and every $j\geq 1$,

$$\det U_{\mathfrak{A}}^j\subseteq U_F^1 \tag{11.34}$$

Indeed $\det(1+x)=1+\mathrm{tr}x+\det x$ for $x\in\mathfrak{P}^j$, and clause (2) puts $\mathrm{tr}x$ in $\mathfrak{p}^{\lceil j/2\rceil}\subseteq\mathfrak{p}$ and $\det x$ in $\mathfrak{p}^j\subseteq\mathfrak{p}$.

Now choose $\Lambda\in\mathcal{C}(\psi_\alpha,\mathfrak{A})$ with $\pi=\text{c-Ind}_J^G\Lambda$, which is possible by proposition 10.1.3(1). Let χ

be tamely ramified, so of level zero and trivial on U_F^1. By (11.34) with $j = m+1$ the character $\chi \circ \det$ is trivial on $U_{\mathfrak{A}}^{m+1}$, and $(\chi \circ \det)\Lambda$ again restricts to ψ_α there, that is $(\chi \circ \det)\Lambda \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$. By proposition 1.2.30, $\chi\pi = \text{c-Ind}_J^G\big((\chi \circ \det)\Lambda\big)$, so theorem 5.3.9 applies to π and to $\chi\pi$ against the same stratum and gives

$$\varepsilon(\chi\pi, s, \psi) = q^{\,n(1/2-s)}\big[(\chi \circ \det)\Lambda\big](\alpha)^{-1}\psi(\mathrm{tr}\alpha) = \chi(\det\alpha)^{-1}\,\varepsilon(\pi, s, \psi)$$

which is (11.32) with $c = \det\alpha$.

Part (2). By lemma 11.1.13(2) there is a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ and $\Lambda' \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$ with $\pi' = \text{c-Ind}_J^G\Lambda'$, and by lemma 11.1.13(3) the level l of χ satisfies $2l > n$, so $l \geq 1$. Let η be tamely ramified. Then $\eta\pi = (\eta\chi)\pi'$, and $\eta\chi$ again has level l with the same attached element δ. Indeed η is trivial on U_F^1, which contains $U_F^{\lfloor l/2\rfloor+1}$, so $(\eta\chi)(1+y) = \psi(\delta y)$ for $y \in \mathfrak{p}^{\lfloor l/2\rfloor+1}$, which is (11.29) for $\eta\chi$. That identity holds in particular on $\mathfrak{p}^{\,l}$, where $y \mapsto \psi(\delta y)$ is non-trivial because $v_F(\delta) = -l$, so $\eta\chi$ is non-trivial on U_F^l, while it is trivial on U_F^{l+1} because both factors are. So its level is exactly l.

Apply lemma 11.4.4 to the pair (χ, π') and to the pair $(\eta\chi, \pi')$. Both produce the same element $\alpha + \delta$ and the same index set, and the two right-hand sides of (11.31) differ only in the character applied to $\det((\alpha+\delta)x)$, the extra factor being $\eta(\det((\alpha+\delta)x))^{-1}$. That factor is the same for every x in the index set. For $\det((\alpha+\delta)x) = \det(\alpha+\delta)\det x$, and both factors are pinned modulo U_F^1, on which η is trivial. Indeed $\det x \in \det U_{\mathfrak{A}}^l \subseteq U_F^1$ by (11.34) with $j = l \geq 1$, while $\det(\alpha+\delta) = \delta^2\det(1+\alpha\delta^{-1})$ and $\nu_{\mathfrak{A}}(\alpha\delta^{-1}) = -n + 2l \geq 1$, so $1+\alpha\delta^{-1} \in U_{\mathfrak{A}}^1$ and $\det(1+\alpha\delta^{-1}) \in U_F^1$ by (11.34) with $j = 1$. So the constant $\eta(\delta^2)^{-1}$ comes out of the sum, and

$$\varepsilon\big(\eta\pi, \tfrac{1}{2}, \psi\big) = \eta(\delta^2)^{-1}\,\varepsilon\big(\pi, \tfrac{1}{2}, \psi\big)$$

Both π and $\eta\pi$ are non-minimal twists of π' by characters of level l, so Step 2 of the proof of lemma 11.4.2 gives them the same level l, hence the same epsilon exponent $2l$ by proposition 10.1.10(1). By definition 5.3.4 the identity at $s = \frac{1}{2}$ therefore holds at every s, which is (11.32) with $c = \delta^2$.

Part (3). By lemma 11.1.13(1) every $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F)$ is either minimal, in which case lemma 11.1.13(2) puts it in $\mathcal{A}_2^0(F;\alpha)$ for a ramified simple stratum and part (1) applies, or not minimal, in which case part (2) applies. In the first case $v_F(\det\alpha) = v_{\mathfrak{A}}(\alpha) = -n$ by proposition 3.3.7(4) and corollary 3.4.4, and $n(\pi,\psi) = 2\ell(\pi) = n$ by proposition 10.1.10(1) and proposition 10.1.3(2). In the second $v_F(\delta^2) = -2l$ and $n(\pi,\psi) = 2l$ by lemma 11.4.4(2). Both give $v_F(c_\pi) = -n(\pi,\psi)$.

Part (4). Suppose first π minimal, so $\pi \in \mathcal{A}_2^0(F;\alpha)$ as above. Then $\pi_K \in \mathcal{A}_2^0(K;\alpha)$ for the lifted stratum $(\mathfrak{A}_K, n_K, \alpha)$, by definition 11.1.14(1) and proposition 11.1.4(1). Part (1) over K therefore gives $c(\pi_K, \psi_K) = \det\alpha = c(\pi,\psi)$, the same element of $F^\times$, so (11.33) holds on the nose.

Suppose next π not minimal, and write $\pi = \chi\pi'$ as in part (2). Then $\pi_K = \chi_K\pi'_K$ by definition 11.1.14(2), with $\pi'_K \in \mathcal{A}_2^0(K;\alpha)$ attached to $(\mathfrak{A}_K, n_K, \alpha)$, and χ_K of level $e_K l$ with attached element δ, by proposition 4.1.3. The representation π'_K is minimal, since proposition 10.1.3(2) over K gives it the level $n_K/2 \notin \mathbb{Z}$ and lemma 11.1.13(4) over K turns that into minimality. Since $2e_Kl > e_Kn = n_K$, the representation π_K is not minimal by lemma 11.1.13(3) over K, and part (2) over K gives $c(\pi_K,\psi_K) = \delta^2 = c(\pi,\psi)$, again the same element. In both cases the two

periods are represented by one element of $F^\times$, so they are congruent modulo U_K^1, completing the proof.

Part (4) is the statement that makes the tame period usable across a change of base field, and it is stronger than the congruence recorded: the two periods are literally the same element, and only the group modulo which it is read has changed. The Galois-side twin is theorem 11.3.5(3), where the corresponding statement is a genuine congruence and takes a Mackey computation to prove.

We turn to the sum in (11.31). Evaluating it is a matrix analogue of the computation that produces the abelian Gauss sum of definition 5.1.19, and the answer is that the matrix sum is the square of the abelian one, up to an explicit power of q. The source states the identity and calls it a direct calculation, giving no proof ([2, §25]), so it is proved here.

Lemma 11.4.7: The Matrix Gauss Sum

Let $\mathfrak{A}$ be a chain order in A with $e_{\mathfrak{A}} = 2$ and radical $\mathfrak{P}$, let χ be a character of $F^\times$ of level $l \geq 1$, and let δ satisfy (11.29). Put

$$\tau_{\mathfrak{A}}(\chi, \psi) = \sum_{x \in U_{\mathfrak{A}} / U_{\mathfrak{A}}^{2l+1}} \chi\big(\det \delta x\big)^{-1} \psi_{\mathfrak{A}}(\delta x)$$

1. The summand depends only on the coset of x, and $\tau_{\mathfrak{A}}(\chi, \psi)$ is unchanged if δ is replaced by any $c \in F^\times$ with $v_F(c) = -l$.
2. $\displaystyle\sum_{x \in U_{\mathfrak{A}}^l / U_{\mathfrak{A}}^{l+1}} \chi\big(\det \delta x\big)^{-1} \psi_{\mathfrak{A}}(\delta x) = q^{-l}\, \tau(\chi, \psi)^2$, where $\tau(\chi, \psi)$ is the Gauss sum of definition 5.1.19.
3. $\tau_{\mathfrak{A}}(\chi, \psi)\, \big(\mathfrak{A} : \mathfrak{P}^{2l+1}\big)^{-1/2} = \tau(\chi, \psi)^2\, q^{-(l+1)}$.

***Proof*:**

Step 1: the summand is well defined, and δ may be replaced. Let $x \in U_{\mathfrak{A}}$ and $w \in \mathfrak{P}^{2l+1}$. By lemma 11.4.3(2), $\mathrm{tr} w \in \mathfrak{p}^{\lceil (2l+1)/2 \rceil} = \mathfrak{p}^{l+1}$ and $v_F(\det w) \geq 2l+1 \geq l+1$, so $\det(1+w) = 1 + \mathrm{tr} w + \det w$ lies in U_F^{l+1}, on which χ is trivial. Also δx lies in $\mathcal{K}_{\mathfrak{A}}$ with $v_{\mathfrak{A}}(\delta x) = v_{\mathfrak{A}}(\delta) + v_{\mathfrak{A}}(x) = -2l$, so proposition 3.3.7(5) gives $\nu_{\mathfrak{A}}(\delta x w) = -2l + \nu_{\mathfrak{A}}(w) \geq -2l + (2l+1) = 1$, and therefore $\psi_{\mathfrak{A}}(\delta x w) = 1$ and $\psi_{\mathfrak{A}}(\delta x(1+w)) = \psi_{\mathfrak{A}}(\delta x)$. The summand is therefore constant on cosets of $U_{\mathfrak{A}}^{2l+1}$. If $c = \delta u$ with $u \in U_F$, then $u \in U_{\mathfrak{A}}$ and the substitution $x \mapsto u^{-1}x$ permutes $U_{\mathfrak{A}}/U_{\mathfrak{A}}^{2l+1}$ and carries the summand formed with c to the summand formed with δ. Since every c with $v_F(c) = -l$ is of that form, part (1) holds.

Step 2: the sum collapses onto $U_{\mathfrak{A}}^l / U_{\mathfrak{A}}^{l+1}$. Write each $x \in U_{\mathfrak{A}}$ as $y(1+z)$ with y running over $U_{\mathfrak{A}}/U_{\mathfrak{A}}^{l+1}$ and z over $\mathfrak{P}^{l+1}/\mathfrak{P}^{2l+1}$. The map $1+z \mapsto z$ is an isomorphism $U_{\mathfrak{A}}^{l+1}/U_{\mathfrak{A}}^{2l+1} \to \mathfrak{P}^{l+1}/\mathfrak{P}^{2l+1}$, because $2(l+1) \geq 2l+1$ kills the cross terms. By lemma 11.4.3(3), $\chi(\det(1+z))^{-1} = \psi_{\mathfrak{A}}(\delta z)^{-1}$. Expanding $\det(\delta y(1+z)) = \det(\delta y)\det(1+z)$ and $\psi_{\mathfrak{A}}(\delta y(1+z)) = \psi_{\mathfrak{A}}(\delta y)\psi_{\mathfrak{A}}(\delta y z)$, the summand factors as

$$\chi\big(\det \delta y\big)^{-1} \psi_{\mathfrak{A}}(\delta y) \cdot \psi_{\mathfrak{A}}(\delta z)^{-1} \psi_{\mathfrak{A}}(\delta y z) = \chi\big(\det \delta y\big)^{-1} \psi_{\mathfrak{A}}(\delta y) \cdot \psi_{\mathfrak{A}}\big(\delta(y-1)z\big)$$

For fixed y the map $z \mapsto \psi_{\mathfrak{A}}(\delta(y-1)z)$ is a character of $\mathfrak{P}^{l+1}/\mathfrak{P}^{2l+1}$, well defined because

$\nu_{\mathfrak{A}}(\delta(y-1)z) \geq -2l+0+2l+1 = 1$ for $z \in \mathfrak{P}^{2l+1}$. It is trivial exactly when $\delta(y-1)$ lies in the dual lattice $(\mathfrak{P}^{l+1})^* = \mathfrak{P}^{-l}$, that is when $\nu_{\mathfrak{A}}(\delta(y-1)) = -2l + \nu_{\mathfrak{A}}(y-1) \geq -l$, that is when $y \in U^l_{\mathfrak{A}}$. By orthogonality on the finite group $\mathfrak{P}^{l+1}/\mathfrak{P}^{2l+1}$ the inner sum is $(\mathfrak{P}^{l+1} : \mathfrak{P}^{2l+1})$ for such y and 0 otherwise. Hence

$$\tau_{\mathfrak{A}}(\chi,\psi) = \big(\mathfrak{P}^{l+1} : \mathfrak{P}^{2l+1}\big)\, G_{\mathfrak{A}}, \qquad G_{\mathfrak{A}} = \sum_{y \in U^l_{\mathfrak{A}}/U^{l+1}_{\mathfrak{A}}} \chi\big(\det \delta y\big)^{-1} \psi_{\mathfrak{A}}(\delta y)$$

By lemma 5.3.6 each step of the filtration has index $q^{4/e_{\mathfrak{A}}} = q^2$, so $(\mathfrak{P}^{l+1} : \mathfrak{P}^{2l+1}) = q^{2l}$.

The summand is constant on the cosets the sum is now indexed by, and not only on the finer ones Step 2 produced. Replacing x by xw with $w \in U^{l+1}_{\mathfrak{A}}$ multiplies it by $\psi_{\mathfrak{A}}(\delta(x-1)w)$, and $\nu_{\mathfrak{A}}(\delta(x-1)w) \geq -2l + l + (l+1) = 1$, so that factor is trivial.

Step 3: coordinates. Both $G_{\mathfrak{A}}$ and $\tau_{\mathfrak{A}}$ are unchanged when $\mathfrak{A}$ is replaced by $g\mathfrak{A}g^{-1}$, since tr, det and the filtration all transport and δ is a scalar. By proposition 3.3.2(2) we may therefore take the lattice chain to be $L_{2i} = \mathfrak{p}^i \oplus \mathfrak{p}^i$ and $L_{2i+1} = \mathfrak{p}^i \oplus \mathfrak{p}^{i+1}$, for which

$$\mathfrak{A} = \begin{pmatrix} \mathfrak{o} & \mathfrak{o} \\ \mathfrak{p} & \mathfrak{o} \end{pmatrix}, \qquad \mathfrak{P} = \begin{pmatrix} \mathfrak{p} & \mathfrak{o} \\ \mathfrak{p} & \mathfrak{p} \end{pmatrix}, \qquad \mathfrak{P}^2 = \varpi\mathfrak{A}$$

the first two computed from definition 3.3.3 and proposition 3.3.4(1), that is from $\mathfrak{A} = \{y : yL_i \subseteq L_i \text{ for all } i\}$ and $\mathfrak{P} = \{y : yL_i \subseteq L_{i+1} \text{ for all } i\}$. Let $R \subseteq \mathfrak{o}$ be a set of representatives for $\mathfrak{o}/\mathfrak{p}$. Reading the displayed matrices entry by entry, a complete set of representatives for $\mathfrak{A}/\mathfrak{P}$ is given by the diagonal matrices $\mathrm{diag}(a_1, a_2)$ with $a_1, a_2 \in R$, and one for $\mathfrak{P}/\mathfrak{P}^2$ by the matrices

$$w(b,c) = \begin{pmatrix} 0 & b \\ \varpi c & 0 \end{pmatrix}, \qquad b, c \in R$$

Since $\mathfrak{P}^{2t} = \varpi^t\mathfrak{A}$ and $\mathfrak{P}^{2t+1} = \varpi^t\mathfrak{P}$, and since $1+z \mapsto z$ identifies $U^l_{\mathfrak{A}}/U^{l+1}_{\mathfrak{A}}$ with $\mathfrak{P}^l/\mathfrak{P}^{l+1}$, this describes the index set of $G_{\mathfrak{A}}$ in both parities.

Step 4: $l = 2t+1$ odd. Here $z = \varpi^t w(b,c)$, so $\mathrm{tr} z = 0$ and $\det z = -\varpi^l bc$. Hence $\det(1+z) = 1 - \varpi^l bc$ and $\mathrm{tr}(1+z) = 2$. Now $l \geq \lfloor l/2 \rfloor + 1$, so $\varpi^l bc$ lies in the range of (11.29) and $\chi(\det(1+z))^{-1} = \psi(\gamma bc)$, where $\gamma = \delta\varpi^l \in U_F$. Also $\psi_{\mathfrak{A}}(\delta(1+z)) = \psi(2\delta) = \psi(\delta)^2$. Therefore

$$G_{\mathfrak{A}} = \chi(\delta)^{-2}\,\psi(\delta)^2 \sum_{b,c \in R} \psi(\gamma bc)$$

The values γbc lie in $\mathfrak{o}$, and ψ restricted to $\mathfrak{o}$ factors through $\mathfrak{o}/\mathfrak{p} = k$ as a non-trivial character $\overline{\psi}$, because ψ has level one. For fixed b the map $c \mapsto \overline{\psi}(\overline{\gamma}\,\overline{b}\,\overline{c})$ is a character of k, trivial exactly when $\overline{b} = 0$, so the sum over c is q for the single $b \in R$ with $\overline{b} = 0$ and 0 for the rest. The double sum is therefore q and $G_{\mathfrak{A}} = q\,\chi(\delta)^{-2}\psi(\delta)^2$. On the other side, proposition 5.1.20 with $n = l$ has index set $U_F^{[(l+1)/2]}/U_F^{[l/2]+1}$, which is trivial for odd l, so $\tau(\chi,\psi) = q^{(l+1)/2}\chi(\delta)^{-1}\psi(\delta)$ and $q^{-l}\tau(\chi,\psi)^2 = q\,\chi(\delta)^{-2}\psi(\delta)^2 = G_{\mathfrak{A}}$.

Step 5: $l = 2t$ even. Here $z = \varpi^t\mathrm{diag}(a_1, a_2)$, so $1+z = \mathrm{diag}(1+\varpi^t a_1,\, 1+\varpi^t a_2)$ and both

det and tr split:

$$\det\big(\delta(1+z)\big) = \delta^2 \prod_{i=1,2} \big(1+\varpi^t a_i\big), \qquad \psi_{\mathfrak{A}}\big(\delta(1+z)\big) = \prod_{i=1,2} \psi\big(\delta(1+\varpi^t a_i)\big)$$

the second because $\mathrm{tr}(\delta(1+z)) = \delta(1+\varpi^t a_1)+\delta(1+\varpi^t a_2)$. Writing $y = 1+\varpi^t a$, the elements y obtained as a runs over R form a complete set of representatives for U_F^t/U_F^{t+1}, and the sum factors:

$$G_{\mathfrak{A}} = \left(\sum_{y \in U_F^t/U_F^{t+1}} \chi(\delta y)^{-1} \psi(\delta y) \right)^2$$

The inner expression is exactly the collapsed sum of proposition 5.1.20 with $n = l$, whose index set is $U_F^{[(l+1)/2]}/U_F^{[l/2]+1} = U_F^t/U_F^{t+1}$, and that proposition gives $\tau(\chi,\psi) = q^t$ times it. Hence $G_{\mathfrak{A}} = q^{-2t}\tau(\chi,\psi)^2 = q^{-l}\tau(\chi,\psi)^2$. Steps 4 and 5 together are part (2).

Step 6: part (3). By lemma 5.3.6 again, $(\mathfrak{A} : \mathfrak{P}^{2l+1}) = (q^2)^{2l+1} = q^{4l+2}$, so $(\mathfrak{A} : \mathfrak{P}^{2l+1})^{-1/2} = q^{-(2l+1)}$. Combining with Steps 2 and 5,

$$\tau_{\mathfrak{A}}(\chi,\psi)\,\big(\mathfrak{A} : \mathfrak{P}^{2l+1}\big)^{-1/2} = q^{2l}\, G_{\mathfrak{A}}\, q^{-(2l+1)} = q^{-1}\, q^{-l}\, \tau(\chi,\psi)^2 = \tau(\chi,\psi)^2\, q^{-(l+1)}$$

which is part (3), completing the proof.

Two remarks on the scope of that lemma. It is stated for $e_{\mathfrak{A}} = 2$, which is the only case a case arising in this book, and the only case in which $\mathfrak{A}$ carries a ramified simple stratum. Steps 1 and 2 would go through with $2l$ replaced by $e_{\mathfrak{A}} l$ throughout, given the general-period form $\mathrm{tr}(\mathfrak{P}^j) = \mathfrak{p}^{\lceil j/e_{\mathfrak{A}} \rceil}$ of lemma 11.4.3(2), which is not stated here. The evaluation in Steps 4 and 5 would not. For $e_{\mathfrak{A}} = 1$ and even l the index set is the whole of $M_2(k)$, on which the determinant is a quadratic form and not a product of two independent linear ones, and the sum does not factor.

The second remark concerns the two roles δ plays. Part (1) says $\tau_{\mathfrak{A}}(\chi,\psi)$ does not depend on which element of valuation $-l$ is used to form it, so it is an invariant of the pair (χ,ψ) alone. The evaluation in part (2) nevertheless runs through a δ that satisfies (11.29), because Step 2 needs lemma 11.4.3(3). There is no conflict, and the point is worth separating because the same δ appears in the closed form below as a value of χ, where the pinning cannot be relaxed.

Corollary 11.4.8: The Local Constant at an Even Level, in Closed Form

In the situation of lemma 11.4.4,

$$\varepsilon\big(\pi, \tfrac{1}{2}, \psi\big) = \psi(\mathrm{tr}\alpha)\, \Lambda'(\alpha+\delta)^{-1}\, \chi\big(\det(1+\alpha\delta^{-1})\big)^{-1}\, \tau(\chi,\psi)^2\, q^{-(l+1)} \tag{11.35}$$

***Proof*:**
In (11.31) split the character value as $\chi(\det((\alpha+\delta)x))^{-1} = \chi(\det(\alpha+\delta))^{-1}\chi(\det x)^{-1}$ and write $\det(\alpha+\delta) = \delta^2 \det(1+\alpha\delta^{-1})$, which is legitimate because $\delta \in F^\times$ is a scalar. Then

$$\chi\big(\det((\alpha+\delta)x)\big)^{-1} = \chi\big(\det(1+\alpha\delta^{-1})\big)^{-1}\, \chi\big(\det \delta x\big)^{-1}$$

and the sum in (11.31) becomes $\chi(\det(1+\alpha\delta^{-1}))^{-1}$ times the sum evaluated in lemma 11.4.7(2), which is $q^{-l}\tau(\chi,\psi)^2$. Multiplying by the prefactor q^{-1} gives (11.35), as we sought to show.

The comparison of (11.35) over F with its counterpart over K reduces to a statement about the abelian Gauss sums alone, and that statement in turn needs a fact about the Langlands constant of a cubic extension. That fact is one of the two places where the standing hypothesis $p = 2$ enters this section directly, the other being the observation that a cubic extension of F is tamely ramified.

Lemma 11.4.9: The Langlands Constant of a Tame Extension Is a Square Root of Unity

Let K/F be a finite tamely ramified extension, the residual characteristic of F being 2. Then

$$\lambda_{K/F}(\psi)^2 = 1$$

***Proof*:**
By corollary 6.3.14, $\lambda_{K/F}(\psi)^2 = \varkappa_{K/F}(-1)$, an identity valid for every non-trivial ψ and in particular for the level one character fixed here. By lemma 11.1.7(1) the character $\varkappa_{K/F}$ is unramified, that is trivial on U_F, and $-1 \in U_F$ because $v_F(-1) = 0$. So $\varkappa_{K/F}(-1) = 1$, completing the proof.

The source proves the cubic case of this in three parts, treating unramified, cyclic totally ramified and non-cyclic K/F separately [2, §49]. The case analysis is unnecessary once $\varkappa_{K/F}$ is known to be unramified, which lemma 11.1.7 establishes for every tame K/F and which is where residual characteristic 2 is used: the argument there needs $q - 1$ odd. Exercise 11.4.1 (5) shows that the conclusion genuinely fails at odd residual characteristic.

Lemma 11.4.10: The Abelian Constant Under a Cubic Extension

Let K/F be a cubic extension and let χ be a character of $F^\times$ of level $l \geq 1$ with attached element δ as in (11.29). Then χ_K has level $e_K l$, the same element δ satisfies (11.29) for the pair (χ_K, ψ_K) over K, and

$$\varepsilon\big(\chi_K, \tfrac{1}{2}, \psi_K\big) = \lambda_{K/F}(\psi)^{-1}\, \varkappa_{K/F}(\delta)^{-1}\, \varepsilon\big(\chi, \tfrac{1}{2}, \psi\big)^3 \tag{11.36}$$

Consequently

$$\tau(\chi_K, \psi_K)^2\, q_K^{-(e_K l+1)} = \tau(\chi, \psi)^6\, q^{-3(l+1)} \tag{11.37}$$

***Proof*:**
Step 1: the level and the attached element. Since $[K : F] = 3$ and the residual characteristic is 2, the extension K/F is tamely ramified, and $e_K f_K = 3$. The assertions about the level of χ_K and about δ are proposition 4.1.3, applied to the tame extension K/F with the pinning (11.29).

Step 2: δ is a tame period for (χ, ψ). Let θ be a tamely ramified character of $F^\times$, of level 0. Since $l \geq 1$ we have $2 \cdot 0 < l$, so theorem 5.1.21 applies with the pair (θ, χ) and its element c, which is pinned by exactly the condition (11.29). It gives $\varepsilon(\theta\chi, s, \psi) = \theta(\delta)^{-1}\varepsilon(\chi, s, \psi)$, which is (11.14) for the one-dimensional $\rho = \chi$ with $c = \delta$. So $c(\chi, \psi) \equiv \delta$ modulo U_F^1.

Step 3: corollary 11.3.7 applies with $\rho = \chi$. That corollary asks for $\rho \in \mathcal{G}_n^0(F)$ with $\rho|_{\mathcal{P}_F}$

irreducible and non-trivial. A character is irreducible of dimension one, so $\chi \in \mathcal{G}_1^0(F)$ and $\chi|_{\mathcal{P}_F}$ is a character, hence irreducible. It is non-trivial, and here the level enters: reading χ as a character of $\mathcal{W}_F$ through $\mathbf{a}_F$, theorem 6.2.1(3) gives $\mathbf{a}_F(\mathcal{P}_F) = U_F^1$, so χ is trivial on $\mathcal{P}_F$ if and only if it is trivial on U_F^1. It is not, because $l \geq 1$ makes χ non-trivial on $U_F^l \subseteq U_F^1$.

The twisting representation is $\theta = R_{K/F}$, which is trivial on $\mathcal{P}_F$: the group $\mathcal{P}_F$ is normal in $\mathcal{W}_F$ and, K/F being tamely ramified, contained in $\mathcal{W}_K$, so it acts trivially on the coset space $\mathcal{W}_F/\mathcal{W}_K$. This is the argument recorded in the proof of lemma 11.1.7(1). Since $\det R_{K/F} = \varkappa_{K/F}$ by definition 6.2.5 and $\dim R_{K/F} = 3$, corollary 11.3.7 with Step 2 gives that identity with $\varkappa_{K/F}$ evaluated at the tame period $c(\chi, \psi)$. That period is δ modulo U_F^1 by Step 2, and $\varkappa_{K/F}$ is unramified by lemma 11.1.7(1), hence trivial on U_F^1, so the two evaluations agree and

$$\varepsilon\big(\chi \otimes R_{K/F}, s, \psi\big) = \varkappa_{K/F}(\delta)^{-1}\, \varepsilon(\chi, s, \psi)^3 \tag{11.38}$$

Step 4: inductivity. The commuting square of theorem 6.2.1(4) says $\mathbf{a}_F$ restricted to $\mathcal{W}_K$ is $\mathrm{Nm}_{K/F} \circ \mathbf{a}_K$, so the character $\chi \circ \mathbf{a}_F$ of $\mathcal{W}_F$ restricts on $\mathcal{W}_K$ to $\chi \circ \mathrm{Nm}_{K/F} \circ \mathbf{a}_K$, that is to χ_K. Hence proposition 1.2.30 gives $\mathrm{Ind}_{K/F}\chi_K = \chi \otimes \mathrm{Ind}_{K/F} 1_K = \chi \otimes R_{K/F}$. This is the same step theorem 11.3.5 takes when it moves a tame twist inside an induction. Inductivity in λ form, (6.5) with $n = 1$, then reads

$$\varepsilon\big(\chi \otimes R_{K/F}, s, \psi\big) = \varepsilon(\chi_K, s, \psi_K)\, \lambda_{K/F}(\psi)$$

Comparing this with (11.38), dividing by the non-zero $\lambda_{K/F}(\psi)$ and evaluating at $s = \frac{1}{2}$ gives (11.36).

Step 5: the Gauss-sum form. Square (11.36). By lemma 11.4.9 the factor $\lambda_{K/F}(\psi)^{-2}$ is 1, and by proposition 6.2.6 so is $\varkappa_{K/F}(\delta)^{-2}$. Hence

$$\varepsilon\big(\chi_K, \tfrac{1}{2}, \psi_K\big)^2 = \varepsilon\big(\chi, \tfrac{1}{2}, \psi\big)^6 \tag{11.39}$$

Both χ and χ_K are ramified, of levels l and $e_K l$, and both ψ and ψ_K have level one, so theorem 5.1.18 over F and over K gives $\varepsilon(\chi, \frac{1}{2}, \psi) = q^{-(l+1)/2}\tau(\chi, \psi)$ and $\varepsilon(\chi_K, \frac{1}{2}, \psi_K) = q_K^{-(e_K l+1)/2}\tau(\chi_K, \psi_K)$. Substituting both into (11.39) turns it into (11.37), completing the proof.

Only (11.37) is used below, and it is weaker than (11.36) by exactly the two square roots of unity that lemma 11.4.9 and proposition 6.2.6 remove. That is the reason proposition 11.4.11 can be proved without ever determining $\lambda_{K/F}(\psi)$ itself, which the book does not compute for a ramified cubic extension.

Everything is now in place for the main computation. Following the source, we prove it only for cubic K/F, which is the case the Existence Theorem uses: the descent of section 11.1 runs along a cubic extension and, in the non-cyclic case, along the normal closure of one.

Proposition 11.4.11: The Local Constant Under a Cubic Lift

Let $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F)$, let K/F be a cubic extension and write $c_\pi = c(\pi, \psi)$. Then

$$\varepsilon(\pi_K, s, \psi_K) = \varkappa_{K/F}(c_\pi)^{-1}\, \varepsilon(\pi, s, \psi)^3 \tag{11.40}$$

***Proof*:**

Step 1: the statement is well posed, and reduces to $s = \frac{1}{2}$. Since $[K : F] = 3$ is odd and the residual characteristic is 2, the extension K/F is tamely ramified, $e_K f_K = 3$ and ψ_K has level one by lemma 4.1.2(1). The element c_π is pinned only modulo U_F^1 by definition 11.4.5, and $\varkappa_{K/F}$ is unramified by lemma 11.1.7(1), hence trivial on $U_F \supseteq U_F^1$, so the right-hand side of (11.40) does not depend on the representative chosen.

By definition 5.3.4 both sides of (11.40) are non-zero constants times monomials in q^{-s}, of degrees $n(\pi_K, \psi_K)$ measured in q_K and $3\,n(\pi, \psi)$ measured in q. Lemma 11.4.2, in the form (11.30) with $[K : F] = 3$, says these two monomials are equal. So it suffices to prove (11.40) at $s = \frac{1}{2}$, and by lemma 11.1.13(1) there are two cases.

Step 2: π minimal. By lemma 11.1.13(2) there is a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ with $n = 2m+1$ and $\pi \in \mathcal{A}_2^0(F; \alpha)$, and by proposition 10.1.3(1) a type $\Lambda \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$ with $\pi = \text{c-Ind}_J^G \Lambda$. Theorem 11.1.10 produces $\Lambda_K \in \mathcal{C}(\psi_\alpha^K, \mathfrak{A}_K)$ satisfying (11.8), and definition 11.1.14(1) makes $\pi_K = \text{c-Ind}_{J_K}^{G_K} \Lambda_K$. That is exactly the hypothesis of lemma 11.1.9, whose conclusion (11.7) is

$$\varepsilon(\pi_K, s, \psi_K) = \varepsilon_{K/F}\, \varepsilon(\pi, s, \psi)^3$$

for every s, the exponent being $[K : F] = 3$. It remains to identify the sign. By proposition 11.4.6(1) we have $c_\pi = \det \alpha$, and $v_F(\det \alpha) = v_{\mathfrak{A}}(\alpha) = v_E(\alpha) = -n$ by proposition 3.3.7(4) and corollary 3.4.4. Writing $\det \alpha = \varpi^{-n} u$ with $u \in U_F$ and using that $\varkappa_{K/F}$ is unramified,

$$\varkappa_{K/F}(c_\pi)^{-1} = \varkappa_{K/F}(\varpi)^{\,n} = \varepsilon_{K/F}^{\,n} = \varepsilon_{K/F}$$

because n is odd and $\varepsilon_{K/F} = \pm 1$. So (11.40) holds in this case, at every s and not only at $s = \frac{1}{2}$.

Step 3: π not minimal, the data over F and over K. Write $\pi = \chi\pi'$ with π' minimal, by lemma 11.1.13(1), take a ramified simple stratum $(\mathfrak{A}, n, \alpha)$ with $\pi' \in \mathcal{A}_2^0(F; \alpha)$ by lemma 11.1.13(2) and a type $\Lambda' \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$ with $\pi' = \text{c-Ind}_J^G \Lambda'$ by proposition 10.1.3(1), and let l be the level of χ and δ its attached element as in (11.29). By lemma 11.1.13(3), $2l > n$, so $l \geq 1$ and lemma 11.4.4 applies. Corollary 11.4.8 gives

$$\varepsilon\big(\pi, \tfrac{1}{2}, \psi\big) = \psi(\mathrm{tr}\alpha)\, \Lambda'(\alpha + \delta)^{-1}\, \chi\big(\det(1 + \alpha\delta^{-1})\big)^{-1}\, \tau(\chi, \psi)^2\, q^{-(l+1)}$$

Over K we have $\pi_K = \chi_K \pi'_K$ by definition 11.1.14(2), with π'_K attached to the lifted stratum $(\mathfrak{A}_K, n_K, \alpha)$ and the lifted type Λ'_K, and χ_K of level $l_K = e_K l$ with the same attached element δ, by proposition 4.1.3. The hypotheses of lemma 11.4.4 hold over K: the stratum $(\mathfrak{A}_K, n_K, \alpha)$ is ramified simple by proposition 11.1.4(1), the character ψ_K has level one, and $2l_K = 2e_K l > e_K n = n_K$. Corollary 11.4.8 over K therefore gives

$$\varepsilon\big(\pi_K, \tfrac{1}{2}, \psi_K\big) = \psi_K(\mathrm{tr}\alpha)\, \Lambda'_K(\alpha + \delta)^{-1}\, \chi_K\big(\det(1 + \alpha\delta^{-1})\big)^{-1}\, \tau(\chi_K, \psi_K)^2\, q_K^{-(l_K+1)}$$

Step 4: the four factors. We compare the two displays factor by factor. Since $\mathrm{tr}\alpha \in F$ and $\mathrm{tr}_{K/F}$ restricts on F to multiplication by $[K:F]=3$,

$$\psi_K(\mathrm{tr}\alpha) = \psi\big(\mathrm{tr}_{K/F}(\mathrm{tr}\alpha)\big) = \psi(\mathrm{tr}\alpha)^3$$

Next, $\alpha+\delta$ lies in $E^\times$ by lemma 11.4.4(1), so $\mathrm{Nm}_{EK/E}(\alpha+\delta) = (\alpha+\delta)^{[EK:E]} = (\alpha+\delta)^3$, while $v_{EK}(\alpha+\delta) = e_K\, v_E(\alpha+\delta) = -2e_K l$ by lemma 11.1.2, an even integer. The recipe (11.8) for the lifted type therefore contributes no sign and

$$\Lambda'_K(\alpha+\delta) = \varepsilon_{K/F}^{-2e_K l}\,\Lambda'\big((\alpha+\delta)^3\big) = \Lambda'(\alpha+\delta)^3$$

Third, the matrix $1+\alpha\delta^{-1}$ has entries in F, so its determinant is the same element of $F^\times$ whether computed in $M_2(F)$ or in $M_2(K)$, and $\mathrm{Nm}_{K/F}$ restricted to $F^\times$ is the cube. Hence

$$\chi_K\big(\det(1+\alpha\delta^{-1})\big) = \chi\big(\mathrm{Nm}_{K/F}\det(1+\alpha\delta^{-1})\big) = \chi\big(\det(1+\alpha\delta^{-1})\big)^3$$

Fourth, (11.37) of lemma 11.4.10 says precisely $\tau(\chi_K,\psi_K)^2 q_K^{-(l_K+1)} = \tau(\chi,\psi)^6 q^{-3(l+1)}$. Multiplying the four comparisons,

$$\varepsilon\big(\pi_K,\tfrac12,\psi_K\big) = \varepsilon\big(\pi,\tfrac12,\psi\big)^3$$

Step 5: the sign is trivial in this case. By proposition 11.4.6(2) we have $c_\pi = \delta^2$, so $v_F(c_\pi) = -2l$ is even, and $\varkappa_{K/F}$ is unramified, so $\varkappa_{K/F}(c_\pi) = \varepsilon_{K/F}^{-2l} = 1$. The identity of Step 4 is therefore (11.40) at $s=\frac12$, and Step 1 extends it to every s, completing the proof.

The two branches produce the sign in opposite ways, and the shape of (11.40) is what reconciles them. In the minimal case the sign $\varepsilon_{K/F}$ comes out of the lifted type through lemma 11.1.9, and $\varkappa_{K/F}(\det\alpha)$ reproduces it because $v_F(\det\alpha)$ is odd. In the non-minimal case the lifted type contributes no sign at all, because the valuation of $\alpha+\delta$ is even, and $\varkappa_{K/F}(\delta^2)$ is correspondingly trivial. A formula written with $\varepsilon_{K/F}$ in place of $\varkappa_{K/F}(c_\pi)^{-1}$ would be correct in the first case and false in the second.

We can now set the two sides of the correspondence against each other. Recall that the subscript K means a restriction on the Galois side and a lift on the group side.

Corollary 11.4.12: The Comparison Under a Cubic Extension

Let K/F be a cubic extension, let $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F)$ and let $\rho \in \mathcal{G}_2^{\mathrm{wr}}(F)$. Then

$$\varepsilon(\pi_K, s, \psi_K) = \varkappa_{K/F}(c_\pi)^{-1}\varepsilon(\pi, s, \psi)^3, \qquad \varepsilon(\rho_K, s, \psi_K) = \varkappa_{K/F}(c_\rho)^{-1}\varepsilon(\rho, s, \psi)^3 \tag{11.41}$$

where $c_\pi = c(\pi, \psi)$ and $c_\rho = c(\rho, \psi)$. Suppose in addition that $c_\pi \equiv c_\rho \pmod{U_F^1}$. Then

$$\frac{\varepsilon(\pi_K, s, \psi_K)}{\varepsilon(\rho_K, s, \psi_K)} = \left(\frac{\varepsilon(\pi, s, \psi)}{\varepsilon(\rho, s, \psi)}\right)^3 \tag{11.42}$$

and consequently:

1. if $\varepsilon(\pi, s, \psi) = \varepsilon(\rho, s, \psi)$ for all s, then $\varepsilon(\pi_K, s, \psi_K) = \varepsilon(\rho_K, s, \psi_K)$ for all s
2. if $\varepsilon(\pi_K, s, \psi_K) = \varepsilon(\rho_K, s, \psi_K)$ for all s, then $n(\pi, \psi) = n(\rho, \psi)$ and $\varepsilon(\pi, s, \psi) = \omega\,\varepsilon(\rho, s, \psi)$ for a cube root of unity ω independent of s.

***Proof*:**
The first identity of (11.41) is proposition 11.4.11. The second is proposition 11.3.8(2), which reads $\varepsilon(\rho_K, s, \psi_K) = \lambda_{K/F}(\psi)^{-2}\varkappa_{K/F}(c_\rho)^{-1}\varepsilon(\rho, s, \psi)^{[K:F]}$ for any non-trivial ψ, specialised to the level one character fixed in box 11.4.1, together with $[K : F] = 3$ and $\lambda_{K/F}(\psi)^2 = 1$, which is lemma 11.4.9 applied to the tame extension K/F. Neither ε-factor vanishes, the Galois-side one because theorem 6.3.2 takes values in $\mathbb{C}[q^s, q^{-s}]^\times$ and the group-side one because corollary 5.2.9 makes it a monomial cq^{-fs} with $c \in \mathbb{C}^\times$. So the quotient in (11.42) is defined, and the hypothesis $c_\pi \equiv c_\rho$ makes the two factors $\varkappa_{K/F}(c_\bullet)^{-1}$ equal, since $\varkappa_{K/F}$ is trivial on U_F^1 by lemma 11.1.7(1). Dividing the two identities of (11.41) gives (11.42), and (1) is immediate from it.

For (2), write $\varepsilon(\pi, s, \psi) = A\,q^{n_\pi(1/2-s)}$ and $\varepsilon(\rho, s, \psi) = B\,q^{n_\rho(1/2-s)}$ with $A, B \in \mathbb{C}^\times$, using definition 5.3.4 and proposition 6.3.3(1) and writing $n_\pi = n(\pi, \psi)$, $n_\rho = n(\rho, \psi)$. By hypothesis the left side of (11.42) is 1, so

$$(A/B)^3\,q^{3(n_\pi - n_\rho)(1/2-s)} = 1 \qquad \text{for all } s$$

A non-constant monomial in q^{-s} cannot be identically 1, so $n_\pi = n_\rho$, and then $(A/B)^3 = 1$. Setting $\omega = A/B$ gives the assertion, completing the proof.

Clause (2) is the form the Existence Theorem of [2, §50] uses. That theorem matches a cuspidal representation with a parameter over a cubic extension K, where both are imprimitive and the correspondence is already available, and then has to bring the matching down to F. Clause (2) brings it down up to a cube root of unity, and that ambiguity is not removed here. Theorem 11.3.12 pins the Galois-side constant $\varepsilon(\rho, \frac{1}{2}, \psi)$ modulo $\mu_{\mathbb{C}}(2^\infty)$, but ω is a ratio of a group-side constant to a Galois-side one, and pinning the denominator says nothing about it. What closes the gap is the group-side counterpart, the congruence $\varepsilon(\pi, \frac{1}{2}, \psi)^2 \equiv \omega_\pi(c_\pi)^{-1}$ modulo $\mu_{\mathbb{C}}(2^\infty)$, together with the matching of ω_π with $\det\rho$. Neither is proved in this chapter, and the Existence Theorem of [2, §50] needs both. Exercise 11.4.1 (8) works through what they then give: $\omega^2 \in \mu_{\mathbb{C}}(2^\infty)$ and $\omega^3 = 1$ force $\omega = 1$, the orders 3 and 2^j being coprime.

That is the last of the four tools this chapter was written to build. The construction they are built for

is not carried out here. Section 11.1 constructs the lifting map and the descent statements that invert it, section 11.2 makes the descent recognizable on both sides at once, section 11.3 and this section prove the numerical identities that are preserved by the passage to K, and [2, §50-§52] assembles them into the correspondence on the primitive parameters.

Exercise 11.4.1

1. Let $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F)$ and let K/F be a finite tamely ramified extension. Deduce (11.30) from lemma 11.4.2, and explain why it is exactly the statement that both sides of (11.40) are monomials in q^{-s} of the same degree. Show next that the identity $\ell(\pi_K) = [K:F]\,\ell(\pi)$, which (11.30) might appear to suggest, is false for every unramified K/F of degree greater than one. Identify which of e_K and f_K accounts for each of the two factors of $[K:F]$ in (11.30).

2. In the situation of lemma 11.4.4, show directly that $(\mathfrak{A}, 2l, \alpha+\delta)$ is a fundamental stratum and that $\alpha+\delta$ is not minimal over F, and check against definition 3.4.1 that the triple has no type at all. Then state which hypotheses of theorem 5.3.8 stand in place of simplicity, and locate the step of that theorem's proof at which hypothesis (2) is used at full strength.

3. Let $\mathfrak{A}$ be a chain order with $e_{\mathfrak{A}} = 2$, let χ have level $l \geq 1$ with attached element δ pinned by (11.29), and let $u \in U_F^1$ with $v_F(u-1) = 1$. Show that δu satisfies the weaker condition $\chi(1+y) = \psi(\delta u y)$ for $y \in \mathfrak{p}^l$. Show next that $\chi(\det(1+x)) = \psi_{\mathfrak{A}}(\delta u x)$ holds for every $x \in \mathfrak{P}^{l+1}$ when $l \leq 2$, and fails for some x when $l \geq 3$. Where in the computation is the equality $\mathrm{tr}(\mathfrak{P}^{l+1}) = \mathfrak{p}^{\lfloor l/2 \rfloor + 1}$ used, as opposed to the inclusion?

4. Let $\mathfrak{A}$ have $e_{\mathfrak{A}} = 2$ and let χ have odd level $l = 2t+1 \geq 1$. Using the coordinates of Step 3 of the proof of lemma 11.4.7, show that every coset of $\mathfrak{P}^{l+1}$ in $\mathfrak{P}^l$ has a representative of trace 0 and determinant in $\mathfrak{p}^l$, and evaluate the sum of lemma 11.4.7(2) as $q\,\chi(\delta)^{-2}\psi(\delta)^2$. Check this against $q^{-l}\tau(\chi,\psi)^2$ using proposition 5.1.20, and say why the collapsed index set of that proposition is trivial for odd l but not for even l.

5. Prove lemma 11.4.9 again, tracing at which point the hypothesis that the residual characteristic is 2 is used. Then let F have odd residual characteristic q and let K/F be ramified quadratic. Show that $\lambda_{K/F}(\psi)^2 = \kappa_{K/F}(-1)$, where $\kappa_{K/F}$ is the quadratic class-field character of box 11.1.1, and that $\kappa_{K/F}$ restricted to U_F is the Legendre symbol of the residue field. For the second point, write $K = F(\sqrt{\varpi_0})$ for a prime element ϖ_0 and expand the norm of $a + b\sqrt{\varpi_0}$ modulo $\mathfrak{p}$, using for the converse that U_F^1 is pro-p with p odd, so squaring is bijective on it. Conclude that $\lambda_{K/F}(\psi)^2 = -1$ whenever $q \equiv 3 \pmod 4$, so that lemma 11.4.9 is not a formal consequence of corollary 6.3.14.

6. Take $F = \mathbb{Q}_2$, $E = F(\sqrt{2})$, $\alpha = \sqrt{2}/2$ and $K = F(2^{1/3})$, as in example 11.1.11, and let $\pi \in \mathcal{A}_2^0(F;\alpha)$ contain $\Lambda \in \mathcal{C}(\psi_\alpha, \mathfrak{A})$.

   1. Compute $c(\pi,\psi)$ and check that its valuation is $-n(\pi,\psi)$.
   2. Show that K/F is not Galois, and deduce $\varepsilon_{K/F} = -1$ from exercise 11.1.1 (3).
   3. Evaluate $\varkappa_{K/F}(c(\pi,\psi))$ and write out (11.40) explicitly for this π, using theorem 5.3.9 to compute $\varepsilon(\pi,s,\psi)$.
   4. Verify the answer against lemma 11.1.9.

7. Let $\pi \in \mathcal{A}_2^{\mathrm{wr}}(F)$.

1. Show that $c(\eta\pi,\psi)\equiv c(\pi,\psi)$ modulo U_F^1 for every tamely ramified character η of $F^\times$, directly from definition 11.4.5 and without using proposition 11.4.6.
2. Let $a\in U_F$, so that $a\psi$ again has level one. Show that $c(\pi,a\psi)\equiv a^{-2}c(\pi,\psi)$ modulo U_F^1, and compare the exponent with the one in the Galois-side statement of exercise 11.3.1 (3).
3. Show that $\varkappa_{K/F}$ is trivial on U_F^1 for every finite tamely ramified K/F, and explain why this is exactly what makes the right-hand side of (11.40) well defined, given that c_π is pinned only modulo U_F^1.

8. Let K/F be a cubic extension, let $\pi\in\mathcal{A}_2^{\mathrm{wr}}(F)$ and $\rho\in\mathcal{G}_2^{\mathrm{wr}}(F)$ with $c(\pi,\psi)\equiv c(\rho,\psi)$ modulo U_F^1, and suppose $\varepsilon(\pi_K,s,\psi_K)=\varepsilon(\rho_K,s,\psi_K)$ for every s. Show that $n(\pi,\psi)=n(\rho,\psi)$ and that $\varepsilon(\pi,\frac{1}{2},\psi)/\varepsilon(\rho,\frac{1}{2},\psi)$ is a cube root of unity. Show next that a cube root of unity lying in $\mu_{\mathbb{C}}(2^\infty)$ is 1, and say what statement about $\varepsilon(\pi,\frac{1}{2},\psi)$ would have to be proved, alongside theorem 11.3.12 for ρ, to place that ratio in $\mu_{\mathbb{C}}(2^\infty)$ and force $\varepsilon(\pi,s,\psi)=\varepsilon(\rho,s,\psi)$.

12

What Next?

The correspondence proved here is one case of a programme, and it is a small one: a single group, GL_2, over a single kind of field. What follows sketches the directions that open from it, and where in the series each is taken up.

Larger groups

The obvious next move is to replace GL_2 by GL_n, and then by a general reductive group. Almost every mechanism used here survives, but each becomes harder in a specific way: parabolic induction runs over more parabolic subgroups than the single Borel, so the classification acquires the Bernstein-Zelevinsky combinatorics. Compact induction still produces supercuspidals, but the chain orders of section 3.3 become the much richer theory of Bruhat-Tits buildings, and admissible pairs are replaced by the types of Bushnell-Kutzko. The series takes this up in *Smooth Representations of p-adic Groups*, which generalizes the worked case of this book. For the archimedean analogue, where $F = \mathbb{R}$ or $\mathbb{C}$ and smooth representations give way to $(\mathfrak{g}, K)$-modules, see *Real Reductive Groups*.

The global correspondence

The local statement exists to be assembled. Given a global field K, the local correspondences at all places v glue into a statement about automorphic representations of $\mathrm{GL}_2(\mathbb{A}_K)$, and the invariants of chapter 5 become the local factors of a global L-function. The mechanism that makes the gluing meaningful is the trace formula. The series treats this in *Automorphic Forms and Representations*, with the trace formula itself in a volume of its own.

Worth noticing on the way out: the global argument already appeared here, once. The proof of theorem 6.3.7, the one step of chapter 6 that this book quotes rather than proves, runs through the

functional equation of a global Hecke L-function. The local theory could not close that gap by itself, which is a fair preview of how the two levels depend on each other.

Modularity

The case that motivated much of the subject is the correspondence between elliptic curves over $\mathbb{Q}$ and modular forms. An elliptic curve produces a compatible system of ℓ-adic Galois representations, and section 6.4.2 is where those enter the picture here: Grothendieck's monodromy theorem converts them into the Deligne representations of chapter 6, and the local correspondence then attaches to each place a representation of $\mathrm{GL}_2(\mathbb{Q}_p)$. Modularity is the statement that the resulting object is automorphic. The series develops the curve side in *Elliptic Curves* [6] and the form side in *Modular Forms* [10].

The dyadic case, and p-adic coefficients

Two gaps left open in this book are themselves research-shaped. The correspondence for $p = 2$ is settled here only in part. The tame parametrization of chapter 4 needs quadratic extensions to be tamely ramified, which fails at $p = 2$; chapter 8 works around that with the Weil representation and reaches every parameter induced from a quadratic extension, but a dyadic field also carries *primitive* two-dimensional parameters. Their structure is worked out in [2, §42], but their *existence* is established only once the correspondence is in hand, in [2, §§46-52]. Describing the imprimitive domain explicitly needs the dyadic arithmetic of [2, §41]. Separately, this book works with complex coefficients throughout. Replacing $\mathbb{C}$ by $\overline{\mathbb{Q}}_p$ produces the p-adic local Langlands correspondence, a different subject with a different shape, taken up in *p-adic Local Langlands*.

External reading

Bushnell and Henniart, *The Local Langlands Conjecture for GL(2)* [2], is the source this book follows and the place to go for everything deferred here, in particular the dyadic case and the explicit local constants. Bump, *Automorphic Forms and Representations*, is the standard bridge from the local theory to the global one. For the Galois side in its own right, Tate's *Number Theoretic Background* in the Corvallis proceedings remains the compact reference for Weil-Deligne representations and local constants.

A

The Projectivity Theorem

Throughout the text, we have focused on the construction and classification of representations. In this appendix, we establish a categorical property of the cuspidal representations. The goal of this appendix is to prove that irreducible cuspidal representations are *projective objects* in the category of smooth representations with a fixed central character. This result highlights the "isolated" nature of cuspidal representations: they cannot be deformed or connected to other representations. It also provides a rigorous underpinning for some of the homological algebra used implicitly in the main text. We prove this using the Schur Orthogonality Relations for square-integrable representations.

The main result is:

Theorem A.0.1: Projectivity Theorem

Let G be a unimodular locally profinite group satisfying box 1.2.31 and having a countable descending basis of compact open subgroups at the identity, with centre Z. The second condition is what gives the Haar measures below, and it passes to G/Z through the images of the basis. Assume that any character $\chi : Z \to \mathbb{R}_+^\times$ extends to a character $G \to \mathbb{R}_+^\times$.

Let (π, V) be an irreducible γ-cuspidal representation of G. Let (τ, U) be a smooth representation of G admitting ω_π as central character. Let $f : U \to V$ be a surjective G-homomorphism. Then there exists a G-homomorphism $\varphi : V \to U$ such that $f \circ \varphi = 1_V$ (i.e., π is projective in this category).

The proof requires the orthogonality relations for square-integrable representations. The group G/Z is locally profinite and unimodular. Let $d\dot{g}$ be a Haar measure on G/Z.

Theorem A.0.2: Schur's Orthogonality Relation

Let (π, V) be an irreducible γ-cuspidal representation. For $v_1, v_2 \in V$ and $\tilde{v}_1, \tilde{v}_2 \in \check{V}$,

$$\int_{G/Z} \langle \check{\pi}(g)\tilde{v}_1, v_1\rangle\langle \tilde{v}_2, \pi(g)v_2\rangle \, d\dot{g} = d(\pi)^{-1}\langle \tilde{v}_1, v_2\rangle\langle \tilde{v}_2, v_1\rangle$$

for a constant $d(\pi) > 0$ depending only on π and the measure $d\dot{g}$.

***Proof*:**
Since π is γ-cuspidal, the integrand has compact support in G/Z, so the integral converges. Fixing $\tilde{v}_1$ and v_2, the integral defines a G-invariant pairing $\check{V} \times V \to \mathbb{C}$, given by a homomorphism $\Theta : \check{V} \to \check{V}$. Since π is admissible and irreducible, Schur's Lemma implies Θ is a scalar. Thus the integral is equal to $c(\tilde{v}_1, v_2)\langle \tilde{v}_2, v_1\rangle$. The bilinear form $c(-,-)$ is again G-invariant, hence proportional to the evaluation pairing. This gives the formula. To show $d(\pi) > 0$, one twists to make ω_π unitary, then defines a G-invariant Hermitian form using the integral. Schur's Lemma implies this form is unique up to a scalar, which must be positive definite.

The constant $d(\pi)$ is given a name as it generalizes the notion of dimension:

Definition A.0.3: Formal Degree

The constant $d(\pi)$ is called the *formal degree* of π.

***Proof*:**
of theorem A.0.1

Let χ be a character of $F^\times$. Let $\mathcal{H}_\chi(G)$ be the space of locally constant functions $f : G \to \mathbb{C}$, compactly supported modulo Z, such that $f(zg) = \chi(z)^{-1}f(g)$. This is an algebra under convolution. Let $\omega = \omega_\pi$. Since π is γ-cuspidal, matrix coefficients of π lie in $\mathcal{H}_\omega(G)$. Let $u \in U$ such that $f(u) \neq 0$. The map $H_\omega(G) \to V$ given by $\varphi \mapsto f(\tau(\varphi)u)$ is surjective (using the irreducibility of π). It suffices to show this map splits. We choose $\tilde{v}_0 \in \check{V}$ such that $d(\pi)^{-1}\langle \tilde{v}_0, f(u)\rangle = 1$. The function $\varphi_0(g) = \langle \check{\pi}(g)\tilde{v}_0, f(u)\rangle$ lies in $\mathcal{H}_\omega(G)$. Define $\Phi : V \to H_\omega(G)$ by $v \mapsto \varphi_v$, where $\varphi_v(g) = \langle \check{\pi}(g)\tilde{v}_0, v\rangle$. The composite map $V \to H_\omega(G) \to V$ sends $w \in V$ to

$$\pi(\varphi_w)f(u) = \int_{G/Z} \langle \check{\pi}(g)\tilde{v}_0, w\rangle \pi(g)f(u) \, d\dot{g}$$

By Schur's orthogonality relations, this equals $\langle \tilde{v}_0, f(u)\rangle d(\pi)^{-1}w = w$. Thus the map splits.

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